# NUMERICAL METHODS WITH APPLICATIONS

## Front Cover

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# Dedication

To Sherrie, Candace, Angelie, and Bucky J Barks

## Principal Author

Autar Kaw is a professor of mechanical engineering at the University of South Florida. He is a recipient of the 2012 U.S. Professor of the Year Award from the Council for Advancement and Support of Education and Carnegie Foundation for Advancement of Teaching.

Professor Kaw's primary scholarly interests are in education research methods, open courseware development, flipped and adaptive learning, bascule bridge design, fracture mechanics, composite materials, and the state and future of higher education.

Funded by National Science Foundation (2002-23), under Professor Kaw's leadership, he and his colleagues from around the nation have developed, implemented, refined, and assessed online resources for open courseware in Numerical Methods (http://nm.MathForCollege.com). This courseware annually receives 1,000,000+page views (http://mathforcollege.com), 2,000,000+ views of the YouTube lectures (http://youtube.com/numericalmethodsguy), and 90,000+ visitors to the "numerical methods guy" blog (http://AutarKaw.org). This body of work is also used in the understanding of the impact of the flipped, blended and adaptive settings on cognitive and affective learning gains of engineering students.

Professor Kaw has written more than 100 refereed technical papers and his opinion editorials have appeared in the Tampa Bay Times, Tampa Tribune and Chronicle Vitae. His work has been covered/cited/quoted in many media outlets, including the Chronicle of Higher Education, Inside Higher Education, U.S. Congressional Record, Florida Senate Resolution, ASEE Prism, Times of India, NSF Discovery, and Voice of America.

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# Preface

This textbook is written for a course in numerical methods for engineering undergraduates.

In 2002, National Science Foundation funded a prototype proposal on Holistic Numerical Methods to develop various resources for typical numerical methods topics of interpolation and nonlinear equations. With the success of this proposal, NSF continued to fund the proposal for other topics of numerical methods via three more multi-university grants in 2004-07, 2008-12, 2009-11. This funding has so far resulted in complete resources for a comprehensive course in Numerical Methods. Revision of the resources continued with funding from more NSF grants in 2013-16, 2016-20, 2020-23. These resources include textbook chapters, PowerPoint presentations, worksheets in MATLAB, MATHEMATICA, Maple and MathCAD, multiple-choice tests, experiments, digital audiovisual lectures, and a blog.

Go to the main website http://nm.mathforcollege.com and click on the numerical method of your choice. You will have access to PowerPoint presentations, worksheets, additional examples, and multiple-choice tests. In addition, we have uploaded broadcast quality (https://nm.mathforcollege.com/audiovisual-digital-lectures/) instructional audiovisual content for most of the course content. A blog on numerical methods and MATLAB is also available at http://blog.autarkaw.com.

The textbook is consists of eleven topics:

1. Introduction to Scientific Computing
2. Differentiation
3. Nonlinear Equations
4. Simultaneous Linear Equations
5. Interpolation
6. Regression
7. Integration
8. Ordinary Differential Equations
9. Optimization
10. Partial Differential Equations
11. Fast Fourier Transforms

Each subtopic in these topics is covered in several separate chapters because we decided to keep the chapters concise and independent. This modular nature of chapters allows you to customize the book based on your needs. Supplemental material is always available from the website.

The chapters in the book are numbered as Chapter XX.YY. The XX stands for the main topic, while YY is the chapter number within that topic.

Chapter 01.YY introduces scientific computing by taking a real-life example to show that solving an engineering problem requires one to develop a mathematical model, solve the model, and then implement the corresponding solution. This content is followed by a discussion of sources of numerical error and their measurement, binary, and floating-point representation of numbers, propagation of errors, and Taylor series.

Chapters 02.YY starts with physical applications of numerical differentiation followed by a just-in-time primer on differential calculus, numerical differentiation of continuous functions and functions that are only given at discrete points.

Chapter 03.YY starts with physical applications of solution of nonlinear equations, a primer on quadratic and cubic equations, and numerical methods of solving nonlinear equations, including bisection, Newton Raphson, secant methods. Simultaneous nonlinear equations are covered as well.

Chapter 04.YY starts with physical applications of solution of simultaneous linear equations, a just-in-time primer for matrix algebra, and numerical methods such as Gaussian elimination, LU decomposition and Gauss-Seidel methods.

Chapter 05.YY starts with physical applications of interpolation. It starts with the background of interpolation, followed by numerical methods of the direct method, Newton divided difference interpolation, Lagrange interpolation, and spline interpolation. Also included are chapters on extrapolation being a bad idea and how to numerically calculate the length of a curve.

Chapter 06.YY starts with physical application of regression. Background content for regression, include simple statistics, minimum of functions, and partial derivatives. The content includes linear regression, nonlinear regression and adequacy of linear regression models.

Chapter 07.YY starts with physical applications of numerical integration. It starts with a primer of integral calculus followed by numerical methods of trapezoidal rule, Simpson's 1/3rd rule, Simpson's 3/8 rule, Gauss quadrature rule, and integrating functions that are given at discrete points.

Chapter 08.YY begins with physical applications of numerical solution of ordinary differential equation. It begins with a primer for ordinary differential equations followed by numerical methods of Euler's, Runge-Kutta 2nd order, Runge-Kutta 4th order, shooting method, and finite difference methods.

Chapter 9 on optimization will be added soon.

Chapter 10 on the numerical solutions of partial differential equations will be added soon.

Chapter 11 on the fast Fourier transforms will be added soon.

Numerical methods used to solve the mathematical procedure are shown with complete treatment and examples. Most chapters are followed by a multiple-choice test and a problem set. Comprehensive solutions to individual multiple-choice tests are available via a link at the end of each multiple-choice question set. The final answers to the problem set are given at the end of each problem.

We would like to thank - Sri Harsha Garapati, Luke Snyder, Eric Marvella, Sue Britten, and Matthew Emmons for reformatting and typing the textbook. Sean Rodby's meticulous proofreading has been critical in maintaining the accuracy of the contents of the book. We would like to thank Cuong Nguyen, Praveen Chalasani, Michael Keteltas, and Luke Snyder for their contributions to the textbook. The conversion of the original files from Word to markdown was a considerable task, and Kaw would like to thank Jonas Fernandes, Bharath Pulaparthi, Jayendra Patel, and Gregory Sims for the detailed work involved. Kaw would like to thank his spouse, Sherrie and children Candace and Angelie for their encouragement in writing this textbook.

We would like to thank Professors Melvin Corley of Louisiana Technical University, Tianxia Zhao of Indiana University-Purdue University, Fort Wayne and Xudong Jia of California State Polytechnic University for reviewing the contents of the textbook content of the first edition of the textbook.

We would appreciate feedback, questions, or comments that you may have on the book or the numerical methods project. You can contact the first author, Autar Kaw, via

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# Chapter 01.01: Introduction to Numerical Methods

Lesson: Why Numerical Methods?

## Learning Objectives

*After successful completion of this lesson, you should be able to:*
1) *Enumerate the need for numerical methods.*

## Introduction

Numerical methods are techniques to approximate mathematical processes (examples of mathematical processes are integrals, differential equations, nonlinear equations).

Approximations are needed because

1) we cannot solve the procedure analytically, such as the standard normal cumulative distribution function

$$\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} e^{-t^2/2} dt \qquad (1)$$

or

2) the analytical method is intractable, such as solving a set of a thousand simultaneous linear equations for a thousand unknowns for finding forces in a truss (Figure 1).

**Figure 1.** A truss holding a roof.

In the case of Equation (1), an exact solution is not available for $\Phi(x)$ other than for $x = 0$ and $x \to \infty$. For other values of $x$ where an exact solution is not available, one may solve the problem by using approximate techniques such as the left-hand Reimann sum you were introduced to in the Integral Calculus course.

In the truss problem, one can solve 1000 simultaneous linear equations for 1000 unknowns without using a calculator. One can use fractions, long divisions, and long multiplications to get the exact answer. But just the thought of such a task is laborious. The task may seem less laborious if we are allowed to use a calculator, but it would still fall under the category of an intractable, if not an impossible, problem. So, we need to find a numerical technique and convert it into a computer program that solves a set of $n$ equations and $n$ unknowns.

Again, what are numerical methods? They are techniques to solve a mathematical problem approximately. As we go through the course, you will see that numerical methods let us find solutions close to the exact one, and we can quantify the approximate error associated with the answer. After all, what good is an approximation without quantifying how good the approximation is?

# Chapter 01.01: Introduction to Numerical Methods

Lesson: Steps of Solving an Engineering Problem

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *go through the stages (problem description, mathematical modeling, solving and implementation) of solving a particular physical problem.*

## Introduction

Numerical methods are used by engineers and scientists to solve problems. However, numerical methods are just one step in solving an engineering problem. There are four steps for solving an engineering problem, as shown in Figure 1.

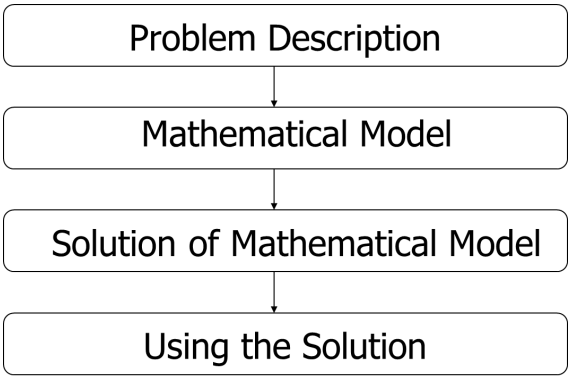



**Figure 1** Steps of solving a problem

The first step is to describe the problem. The description would involve writing the background of the problem and the need for its solution. The second step is developing a mathematical model for the problem, and this could include the use of experiments or/and theory. The third step involves solving the mathematical model. The solution may consist of analytical or/and numerical means. The fourth step is implementing the solution to see if the problem is solved.

Let us see through an example of these four steps of solving an engineering problem.

## Problem Description

To make the fulcrum (Figure 2) of a bascule bridge, a long hollow steel shaft called the trunnion is shrink fit into a steel hub. The resulting steel trunnion-hub assembly is then shrunk-fit into the girder of the bridge.

**Figure 2** Trunnion-Hub-Girder (THG) assembly.

The shrink-fitting is done by first immersing the trunnion in a cold medium such as a dry-ice/alcohol mixture. After the trunnion reaches the steady-state temperature, that is, the temperature of the cold medium, the outer diameter of the trunnion contracts. The trunnion is taken out of the medium and slid through the hole of the hub (Figure 3).

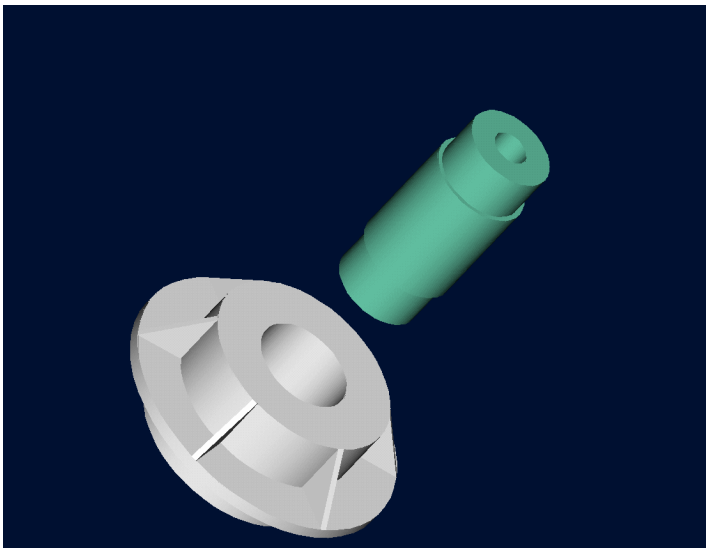


**Figure 3** Trunnion slided through the hub after contracting

When the trunnion heats up, it expands and creates an interference fit with the hub. In 1995, on one of the bridges in Florida, this assembly procedure did not work as designed. Before the trunnion could be inserted fully into the hub, the trunnion got stuck. Luckily, the trunnion was taken out before it got stuck permanently. Otherwise, a new trunnion and hub would need to be ordered at the cost of $\$50,000$. Coupled with construction delays, the total loss could have been more than a hundred thousand dollars.

Why did the trunnion get stuck - because the trunnion had not contracted enough to slide through the hole. Can you find out why?

## Simple Mathematical Model

A hollow trunnion of an outside diameter $12.363''$ is to be fitted in a hub of inner diameter $12.358''$. The trunnion was put in a dry ice/alcohol mixture (temperature of the fluid - dry-ice/alcohol mixture is $-108°$F) to contract the trunnion so that it can be slid through the hole of the hub. To slide the trunnion without sticking, a diametrical clearance of at least $0.01''$ is required between the trunnion and the hub. Assuming the room temperature is $80°$F, is immersing the trunnion in a dry-ice/alcohol mixture a correct decision?

To calculate the contraction in the diameter of the trunnion, the thermal expansion coefficient at room temperature is used. In that case, the reduction $\Delta D$ in the outer diameter of the trunnion is

$$\Delta D = D\alpha\Delta T \qquad (1)$$

where

$$D = \text{ outer diameter of the trunnion,}$$

$$\alpha = \text{ coefficient of thermal expansion coefficient at room temperature, and}$$

$$\Delta T = \text{change in temperature.}$$

## Solution to Simple Mathematical Model

Given

$$D = 12.363''$$

$$\alpha = 6.47 \times 10^{-6} \text{ in/in/}^{\circ}\text{F at } 80^{\circ}\text{F}$$

$$\begin{aligned} \Delta T &= T_{fluid} - T_{room} \\ &= -108 - 80 \\ &= -188^{\circ} \text{ F} \end{aligned}$$

where

$$T_{fluid} = \text{ temperature of dry-ice/alcohol mixture}$$

$$T_{room} = \text{ room temperature}$$

the reduction in the outer diameter of the trunnion from Equation (1) hence is given by

$$\begin{aligned}\Delta D &= (12.363)\left(6.47 \times 10^{-6}\right)(-188) \\ &= -0.01504''\end{aligned}$$

So the trunnion is predicted to reduce in diameter by $0.01504''$. But is this enough reduction in diameter? As per specifications, the trunnion diameter needs to change by

$$\begin{aligned}&= \text{trunnion outside diameter} - \text{hub inner diameter} + \text{diametric clearance} \\ &= 12.363 - 12.358 + 0.01 \\ &= -0.015''\end{aligned}$$

So, according to his calculations, immersing the steel trunnion in dry-ice/alcohol mixture gives the desired contraction of greater than $0.015''$ as the predicted contraction is $0.01504''$. But, when the steel trunnion was put in the hub, it got stuck. Why did this happen? Was our mathematical model adequate for this problem, or did we create a mathematical error?

## Accurate Mathematical Model

As shown in Figure 4 and Table 1, the thermal expansion coefficient of steel decreases with temperature and is not constant over the range of temperature the trunnion goes through. Hence, Equation (1) would overestimate the thermal contraction.

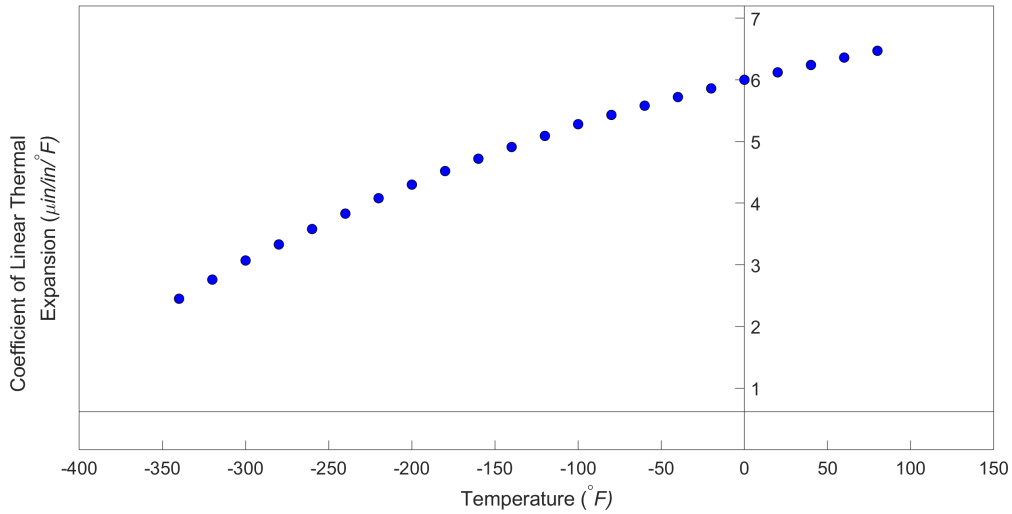



**Figure 4** Varying thermal expansion coefficient as a function of temperature for cast steel.

The contraction in the diameter of the trunnion for which the thermal expansion coefficient varies as a function of temperature is given by

$$\Delta D = D \int_{T_{room}}^{T_{fluid}} \alpha dT \qquad (2)$$

## Solution to More Accurate Mathematical Model

So, one needs to curve fit the data to find the coefficient of thermal expansion as a function of temperature. This curve is found by regression where we best fit a function to the data given in Table 1. In this case, we may fit a second-order polynomial

$$\alpha = a_0 + a_1 T + a_2 T^2 \qquad (3)$$

**Table 1** Instantaneous thermal expansion coefficient as a function of temperature.

| Temperature | Instantaneous Thermal Expansion |
|---|---|
| °F | $\mu$in/in/°F |
| 80 | 6.47 |
| 60 | 6.36 |
| 40 | 6.24 |
| 20 | 6.12 |
| 0 | 6.00 |
| −20 | 5.86 |
| −40 | 5.72 |
| −60 | 5.58 |
| −80 | 5.43 |
| −100 | 5.28 |
| −120 | 5.09 |
| −140 | 4.91 |
| −160 | 4.72 |
| −180 | 4.52 |
| −200 | 4.30 |
| −220 | 4.08 |
| −240 | 3.83 |
| −260 | 3.58 |
| −280 | 3.33 |
| −300 | 3.07 |
| −320 | 2.76 |
| −340 | 2.45 |

The values of the coefficients in the above Equation (3) will be found by polynomial regression (we will learn how to do this later in the chapter on Nonlinear Regression). At this point, we are just going to give you these values, and they are

$$\begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 6.0150 \times 10^{-6} \\ 6.1946 \times 10^{-9} \\ -1.2278 \times 10^{-11} \end{bmatrix}$$

to give the polynomial regression model (Figure 5) as

$$\begin{aligned} \alpha &= a_0 + a_1 T + a_2 T^2 \\ &= 6.0150 \times 10^{-6} + 6.1946 \times 10^{-9} T - 1.2278 \times 10^{-11} T^2 \end{aligned} \qquad (4)$$

Knowing the values of $a_0$, $a_1$, and $a_2$, we can then find the contraction in the trunnion diameter from Equations (2) and (3) as

$$\begin{aligned}\Delta D &= D\int_{T_{room}}^{T_{fluid}} (a_0 + a_1 T + a_2 T^2)dT \\ &= D\left[a_0 T + a_1\frac{T^2}{2} + a_2\frac{T^3}{3}\right]_{T_{room}}^{T_{fluid}} \\ &= D[a_0(T_{fluid} - T_{room}) + a_1\frac{({T_{fluid}}^2 - {T_{room}}^2)}{2} \\ &\quad + a_2\frac{({T_{fluid}}^3 - {T_{room}}^3)}{3}] \qquad (5)\end{aligned}$$

Substituting the values of the variables gives

$$\begin{aligned}\Delta D &= 12.363\begin{bmatrix} 6.0150\times10^{-6}\times(-108-80) \\ +6.1946\times10^{-9}\dfrac{((-108)^2-(80)^2)}{2} \\ -1.2278\times10^{-11}\dfrac{((-108)^3-(80)^3)}{3}\end{bmatrix} \\ &= -0.013689''\end{aligned}$$

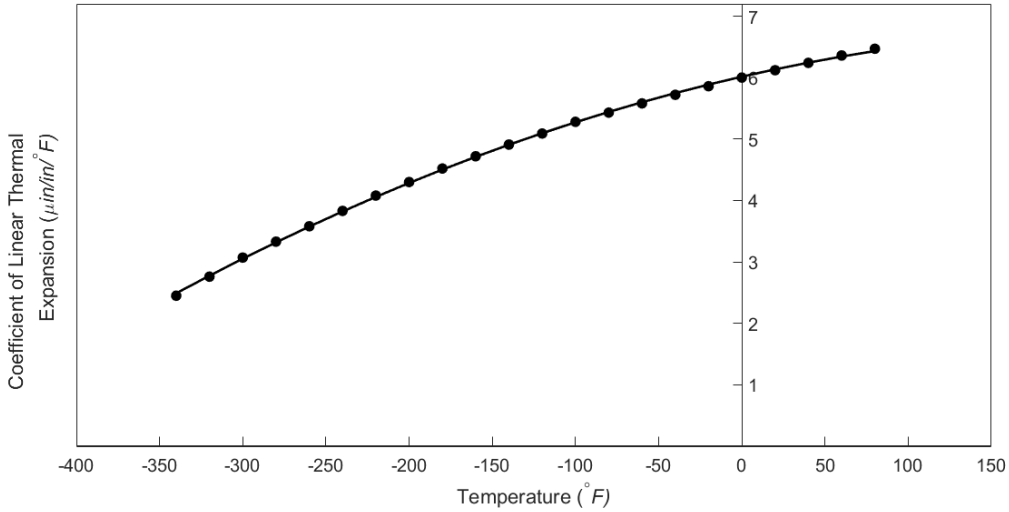



**Figure 5** Second-order polynomial regression model for the coefficient of thermal expansion as a function of temperature.

What do we find here? The contraction in the trunnion is not enough to meet the required specification of $0.015''$.

## Implementing the Solution

Although we were able to find out why the trunnion got stuck in the hub, we still need to find and implement a solution. What if the trunnion were immersed in a medium that was cooler than the dry-ice/alcohol mixture of $-108°F$ , say liquid nitrogen, which has a boiling temperature of $-321°F$? Will that be enough for the specified contraction in the trunnion?

As given in Equation (5)

$$\begin{aligned}\Delta D &= D\int_{T_{room}}^{T_{fluid}} (a_0 + a_1 T + a_2 T^2)dT \\ &= D\left[a_0 T + a_1\frac{T^2}{2} + a_2\frac{T^3}{3}\right]_{T_{room}}^{T_{fluid}} \\ &= D[a_0(T_{fluid} - T_{room}) + a_1\frac{({T_{fluid}}^2 - {T_{room}}^2)}{2} \\ &\quad + a_2\frac{({T_{fluid}}^3 - {T_{room}}^3)}{3}] \qquad (5 - repeated)\end{aligned}$$

which gives

$$\begin{aligned}\Delta D &= 12.363\begin{bmatrix} 6.0150\times10^{-6}\times(-321-80) \\ +6.1946\times10^{-9}\dfrac{\left((-321)^2-(80)^2\right)}{2} \\ -1.2278\times10^{-11}\dfrac{((-321)^3-(80)^3)}{3}\end{bmatrix} \\ &= -0.024420''\end{aligned}$$

The magnitude of this contraction is larger than the specified value of $0.015''$

**So here are some questions that you may want to ask yourself later in the course?**

1) What if the trunnion were immersed in liquid nitrogen (boiling temperature= $-321°$F)? Will that cause enough contraction in the trunnion?

2) Rather than regressing the thermal expansion coefficient data to a second-order polynomial so that one can find the contraction in the trunnion OD, how would you use the trapezoidal rule of integration for unequal segments?

3) What is the relative difference between the two results?

4) We chose a second-order polynomial for regression. Would a different order polynomial be a better choice for regression? Is there an optimum order of polynomial we could use?

# Chapter 01.01: Introduction to Numerical Methods

Lesson: Overview of Mathematical Processes Covered in This Course

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *enumerate the seven mathematical processes for which numerical methods are used.*

## Introduction

Numerical methods are techniques to approximate mathematical processes. This introductory numerical methods course will develop and apply numerical techniques for the following mathematical processes:

1) Roots of Nonlinear Equations

2) Simultaneous Linear Equations

3) Curve Fitting via Interpolation

4) Differentiation

5) Curve Fitting via Regression

6) Numerical Integration

7) Ordinary Differential Equations.

Some undergraduate courses in numerical methods may include topics of partial differential equations, optimization, and fast Fourier transforms as well.

## Roots of a Nonlinear Equation

The ubiquitous formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \qquad (1)$$

of finding the roots of a quadratic equation $ax^2 + bx + c = 0$ goes back to the ancient world. But in the real world, we get equations that are not just the quadratic ones. They can be polynomial equations of a higher order and transcendental equations.

Take an example of a floating ball shown in Figure 1, where you are asked to find the depth to which the ball will get submerged when floating in the water.

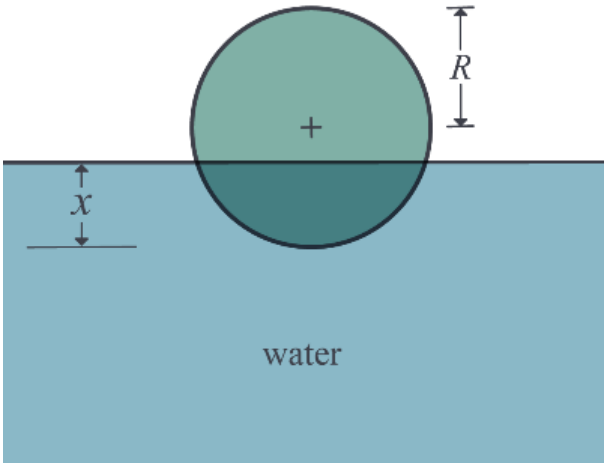



**Figure 1.** Depth to which a ball is floating

Assume that the ball has a density of 600 kg/m$^3$ and has a radius of 0.055 m. On applying the Newtons laws of motion and hence equating the weight of the ball to the buoyancy force, one finds that the depth, $x$ in meters, to which the ball is underwater and is given by

$$3.993 \times 10^{-4} - 0.165x^2 + x^3 = 0 \qquad (2)$$

Equation(2) is a cubic equation that you will need to solve. The equation will have three roots, and the root that is between 0 m (just touching the water surface) and 0.11 m (almost submerged) would be the depth to which the ball is submerged. The two other roots would be physically unacceptable. Note that a cubic equation with real coefficients can have a set of one real root and two complex roots or a set of three real roots. You may wonder why such an application could be important? Let's suppose you are filling in this tank with water, and you are using this ball as a control so that when the ball goes all the way to the top that the flow of the water stops – say in a fish tank that needs replenishing while the owner is away for a few weeks. So, we do need to figure out how much of the ball is submerged underwater.

A cubic equation can be solved exactly by radicals, but it is a tedious process. The same is true but even more complicated for a general fourth-order polynomial equation as well. However, there is no closed-form solution available for a general polynomial equation of fifth-order or more. So, one has to resort to numerical techniques to solve polynomial and other transcendental nonlinear equations (e.g., finding the nonzero roots of $\tan x = x$).

## Simultaneous Linear Equations

Ever since you were exposed to algebra, you have been solving simultaneous linear equations.

**Figure 2.** A rocket going upwards

Take this problem statement as an example. Suppose the upward velocity of a rocket (Figure 2) is given at three different times (Table 1).

**Table 1:** Velocity vs. time data

| Time, t (s) | Velocity, v (m/s) |
|---|---|
| 5 | 106.8 |
| 8 | 177.2 |
| 12 | 600.0 |

The velocity data is approximated by a polynomial as

$$v(t) = at^2 + bt + c \ , 5 \leq t \leq 12. \qquad (3)$$

To estimate the velocity at a time that is not given to us, we can set up the equations to find the coefficients $a, b, c$ of the velocity profile.

The polynomial in Equation (3) is going through three data points $(t_1, v_1), (t_2, v_2)$, and $(t_3, v_3)$ where from Table 1

$$t_1 = 5, v_1 = 106.8$$
$$t_2 = 8, v_2 = 177.2$$
$$t_3 = 12, v_3 = 600.0$$

Requiring that $v(t) = at^2 + bt + c$ passes through the three data points, gives

$$\begin{aligned} v(t_1) = v_1 &= at_1^2 + bt_1 + c \\ v(t_2) = v_2 &= at_2^2 + bt_2 + c \\ v(t_3) = v_3 &= at_3^2 + bt_3 + c \end{aligned}$$

Substituting the data $(t_1,\ v_1)$, $(t_2,\ v_2)$, and $(t_3,\ v_3)$ gives

$$\begin{aligned} a\left(5^2\right) + b(5) + c &= 106.8 \\ a\left(8^2\right) + b(8) + c &= 177.2 \\ a\left(12^2\right) + b(12) + c &= 600.0 \end{aligned}$$

Or

$$\begin{aligned} 25a + 5b + c &= 106.8 \\ 64a + 8b + c &= 177.2 \\ 144a + 12b + c &= 600.0 \end{aligned} \qquad (4)$$

Solving a few simultaneous linear equations such as the above set can be done without the knowledge of numerical techniques. However, imagine that instead of three given points, you were given 10 data points. Now the setting up as well as solving the set of 10 simultaneous linear equations without numerical techniques becomes laborious, if not impossible.

## Curve Fitting by Interpolation

Interpolation involves that given a function as a set of data points, how does one find the value of the function at points that are not given to us. For this, we choose a function, called an interpolant, and make it pass through all the points involved.

You may think that you have already used interpolation in courses such as Thermodynamics and Statistics. After all, it was just taking two points from a table at the back of the textbook or online and finding the value of the function at a point in between by using a straight line.

Take this problem statement as an example. Let's suppose the upward velocity of a rocket is given at three different times (Table 1).

If one asked you to estimate the velocity at 7 s, one might simply use the straight-line formula you are most accustomed to as given below.

Given $(t_1,\ v_1)$ and $(t_2,\ v_2)$, the value of the function $v$ at $t$ is given by

$$v = v_1 + \frac{v_2 - v_1}{t_2 - t_1}(t - t_1) \qquad (5)$$

Although this is possibly enough for courses such as Thermodynamics and Statistics, there are two questions to ask. Is the value calculated accurately, and how accurate is it? To know that, one needs

to calculate at least more than one value. In the above example of a rocket velocity vs. time, one can instead use a second-order polynomial interpolant and set up the three equations and three unknowns to find the unknown coefficients, $a$, $b$, and $c$ as given in the previous section.

$$v(t) = at^2 + bt + c, \ 5 \leq t \leq 12 \qquad (6)$$

Then, the resulting second-order polynomial can be used to find the velocity at $t = 7$ s.

The value obtained from the second-order polynomial (Equation (6)) can be considered to be a new measure of the value of $v(7)$, and the first-order polynomial (Equation (5)) result can be used to determine the accuracy of the results.

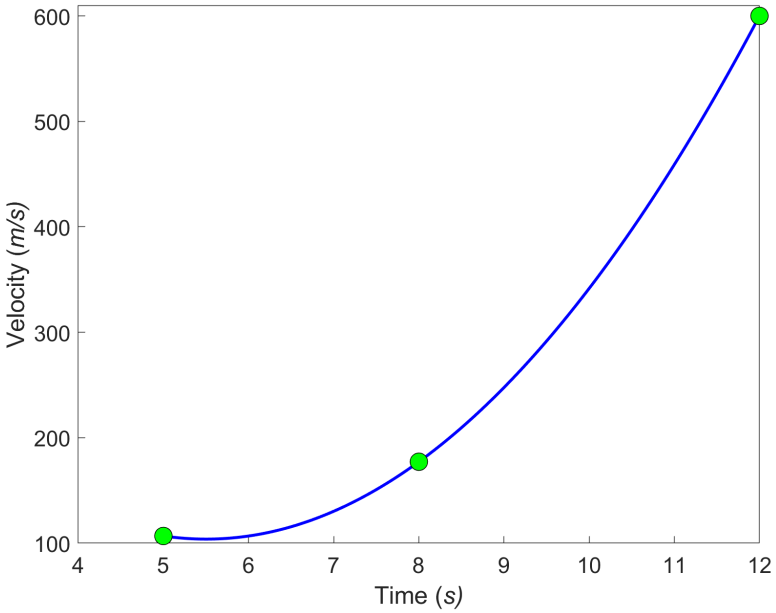



**Figure 3**. Second-order interpolant for velocity vs. time.

## Numerical Differentiation

You have taken a semester-long course in Differential Calculus, where you found derivatives of continuous functions. So let's suppose somebody gives you the velocity of a rocket as a continuous and at least once differentiable function of time and wants you to find acceleration. Indeed for this particular problem, you can use your differential calculus knowledge to differentiate the velocity function to get the acceleration and put in the value of time, $t = 7$ s. What if the velocity vs. time is not given as a continuous and at least once differentiable function? Instead, let's say the function is given at discrete data points (Table 1). How are you then going to find out what the acceleration at $t = 7$ s? Do we draw a straight line from $(5, 106.8)$ to $(8, 177.2)$ and use the straight-line slope as the estimate of acceleration? How do we know that this is adequate? We could incorporate all three points and find a second-order polynomial as given by Equation (6). This polynomial can now be differentiated to estimate the acceleration at $t = 7$ s. Now the two values can be used to evaluate the accuracy of the calculated acceleration.

## Curve Fitting by Regression

When we talked about curve fitting by interpolation, the chosen interpolant needs to go through all the points considered. What happens when we are given many data points, and we instead want a simplified formula to explain the relationship between two variables. See, for example, in Figure 4a, we are given the coefficient of linear thermal expansion data for cast steel as a function of temperature. Looking at the data, one may proclaim that a straight line could explain the data, and that is drawn in Figure 4b. How we draw this straight line is what is called regression. It would be based on minimizing some form of the residuals between what is observed (given data points) and what is predicted (straight line). It does not mean that every time you have data given to you, you draw a straight line. It is possible that a second-order polynomial or a transcendental function other than the first-order polynomial will be a better representation of this particular data. So these are the questions that we will answer when we discuss regression. We will also discuss the adequacy of linear regression models.

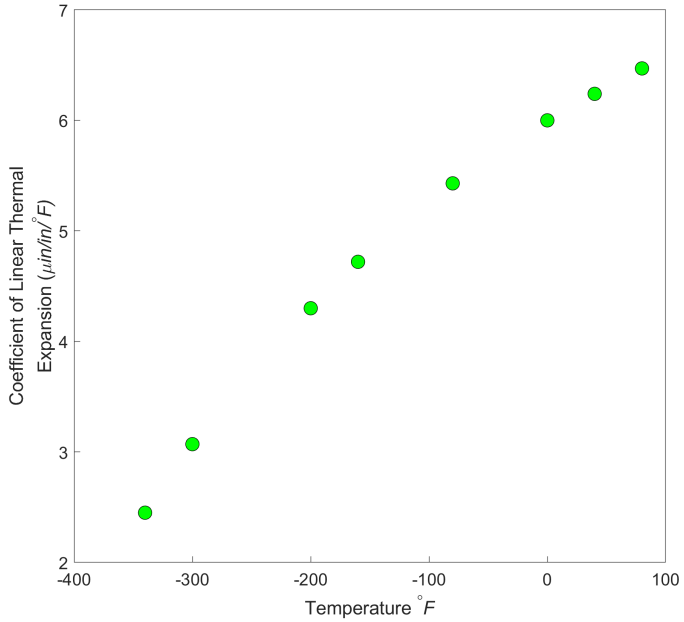



**Figure 4a.** Coefficient of linear thermal expansion for cast steel as a function of temperature

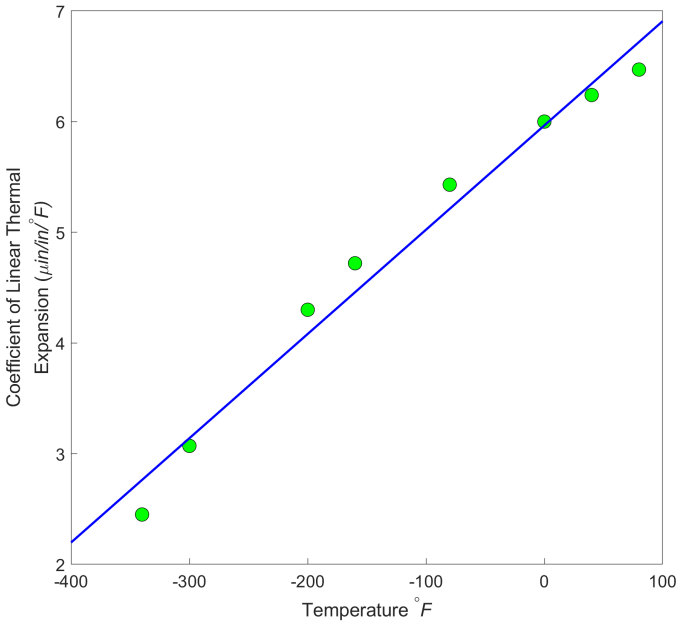



**Figure 4b.** Linear regression model of coefficient of linear thermal expansion for cast steel as a function of temperature

## Numerical Integration

You have taken a whole course on integral calculus. Now, why would we need to make numerical approximations of integrals? Just like the standard normal cumulative distribution function

$$\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} e^{-t^2/2} dt \qquad (7)$$

cannot be solved exactly, or when the integrand values are given at discrete data points, we need to use numerical methods of integration.

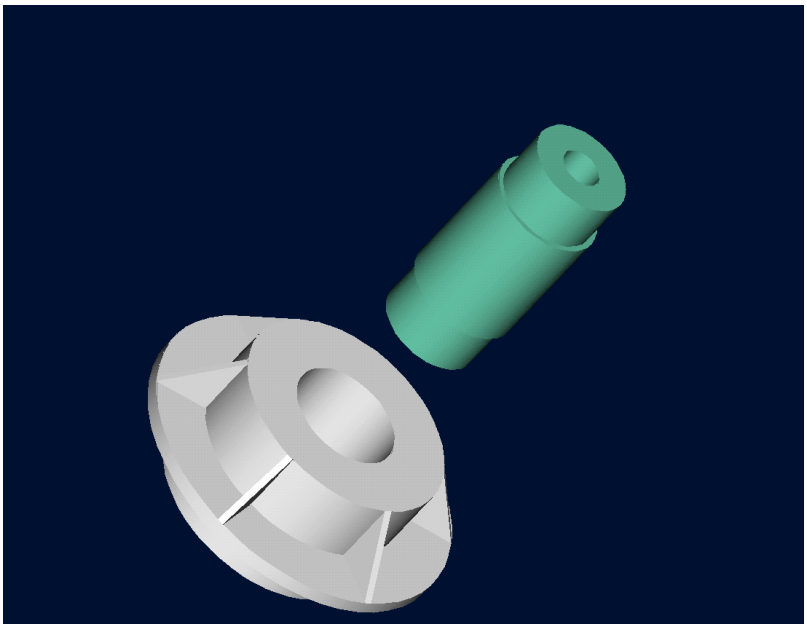


**Figure 5.** Trunnion of a fulcrum assembly of a bascule bridge

In the previous lesson, we looked at the example of contracting the diameter of a trunnion for a bascule bridge fulcrum assembly by dipping it in a mixture of dry ice and alcohol. The contraction is given by

$$\Delta D = D \int_{T_{room}}^{T_{fluid}} \alpha \, dT \qquad (8)$$

where

$D =$ outer diameter of the trunnion,

$\alpha =$ coefficient of linear thermal expansion that is varying with temperature

$T_{room} =$ room temperature

$T_{fluid} =$ temperature of dry-ice alcohol mixture.

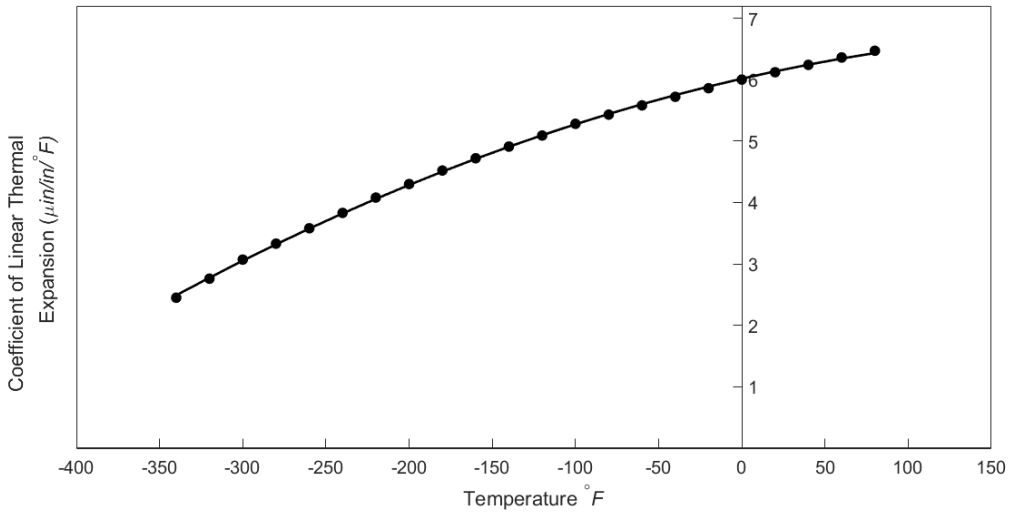



**Figure 6** Varying thermal expansion coefficient as a function of temperature for cast steel.

From Figure 4a, one can note that the coefficient of thermal expansion is only given at discrete temperatures and not as a known continuous function that could be integrated exactly. So we have to resort to numerical methods by approximating the data, say, by a second-order polynomial obtained via regression.

In Figure 6, the thermal expansion coefficient of typical cast steel is approximated by a second-order regression polynomial as given by Equation (9) (how we get this is a later lesson in regression) as

$$\alpha = -1.2278 \times 10^{-11} T^2 + 6.1946 \times 10^{-9} T + 6.0150 \times 10^{-6} \qquad (9)$$

The contraction of the diameter then is given by

$$\Delta D = D \int_{T_{room}}^{T_{fluid}} \left(-1.2278 \times 10^{-11} T^2 + 6.1946 \times 10^{-9} T + 6.015 \times 10^{-6}\right) dT \qquad (10)$$

and can now be calculated using integral calculus.

## Numerical Solution of Ordinary Differential Equations

Taking the same example of the trunnion being dipped in a dry-ice/alcohol mixture, one could ask the question - What would the temperature of the trunnion be after dipping it in the mixture for 30 minutes. The model is given by an ordinary differential equation for the temperature $\theta$ as a function of time, $t$.

$$-hA\left(\theta - \theta_a\right) = mC\frac{d\theta}{dt} \qquad (11)$$

where

$h$ = the convective cooling coefficient, W/m$^2$-K

$$A = \text{surface area, } \text{m}^2$$

$$\theta_a = \text{ambient temperature of dry-ice/alcohol mixture, K}$$

$$m = \text{mass of the trunnion, kg}$$

$$C = \text{specific heat of the trunnion, J/(kg-K)}$$

The differential Equation (11) can be solved exactly by using the classical solution, Laplace transform, or separation of variables techniques. So, where do numerical methods enter into the picture for this problem? For the temperature range of room temperatures to cold media such as dry-ice/alcohol, several of the variables in Equation (11) are not constant but change with the temperature. These include the convection coefficient $h$ as well as the specific heat $C$. Now, this differential equation has turned nonlinear as follows.

$$-h(\theta)A\,(\theta - \theta_a) = mC(\theta)\frac{d\theta}{dt} \qquad (12)$$

The ordinary differential Equation (12) cannot be solved by exact methods and would need to be solved by a numerical method.

In the above discussion, we have illustrated the need for numerical methods for each of the seven mathematical processes in the course. In the lessons to follow, we will be developing various numerical techniques to approximate the mathematical processes to calculate acceptable accurate values while calculating associated errors.

# Multiple Choice Test

## Chapter 01.01 Introduction to Numerical Methods

(1). Solving an engineering problem requires four steps. In order of sequence, the four steps are

(A) formulate, model, solve, implement

(B) formulate, solve, model, implement

(C) formulate, model, implement, solve

(D) model, formulate, implement, solve

(2). One of the roots of the equation $x^3 - 3x^2 + x - 3 = 0$ is

(A) $-1$

(B) $1$

(C) $\sqrt{3}$

(D) 3

(3). The solution to the set of equations

$$\begin{aligned} 25a + b + c &= 25 \\ 64a + 8b + c &= 71 \\ 144a + 12b + c &= 155 \end{aligned}$$

most nearly is $(a, b, c) =$

(A) $(1, 1, 1)$

(B) $(1, -1, 1)$

(C) $(1, 1, -1)$

(D) does not have a unique solution.

(4). The exact integral of $\int_0^{\frac{\pi}{4}} 2\cos 2x \; dx$ is most nearly

(A) $-1.000$

(B) $1.000$

(C) $0.000$

(D) $2.000$

(5). The value of $\frac{dy}{dx}(1.0)$, given $y = 2\sin(3x)$ most nearly is

(A) $-5.9399$

(B) $-1.980$

(C) $0.31402$

(D) $5.9918$

(6). The form of the exact solution of the ordinary differential equation

$2\frac{dy}{dx} + 3y = 5e^{-x},\ y(0) = 5$ is

(A) $Ae^{-1.5x} + Be^{x}$

(B) $Ae^{-1.5x} + Be^{-x}$

(C) $Ae^{1.5x} + Be^{-x}$

(D) $Ae^{-1.5x} + Bxe^{-x}$

For complete solution, go to

http://nm.mathforcollege.com/mcquizzes/01aae/quiz_01aae_introduction_answers.pdf

# Problem Set

## Chapter 01.01: Introduction to Numerical Methods

(1). Give one example of an engineering problem where each of the following mathematical procedures is used. If possible, draw from your experience in other classes or from any professional experience you have gathered to date.

a) Differentiation

b) Nonlinear equations

c) Simultaneous linear equations

d) Regression

e) Interpolation

f) Integration

g) Ordinary differential equations

(2). Only using your nonprogrammable calculator, find the root of

$$x^3 - 0.165x^2 + 3.993 \times 10^{-4} = 0$$

by any method. Hint: Find one root by hit and trial, and use long division for factoring the polynomial.

**Answer:** $0.06237758151,\ 0.1463595047,\ -0.04373708621$

(3). Solve the following system of simultaneous linear equations by any method

$$\begin{aligned} 25a + 5b + c &= 106.8 \\ 64a + 8b + c &= 177.2 \\ 144a + 12b + c &= 279.2 \end{aligned}$$

**Answer:** $a = 0.2904761905,\ b = 19.69047619,\ c = 1.085714286$

(4). You are given data for the upward velocity of a rocket as a function of time in the table below.

| $t$ | $v(t)$ |
|---|---|
| s | m/s |
| 10 | 305.31 |
| 20 | 701.53 |

Find the velocity at $t = 16s$.

**Answer:** 543.0420000 m/s

(5). Integrate exactly.

$$\int_0^{\pi/2} \sin 2x \; dx$$

**Answer:** 1

(6). Find

$$\frac{dy}{dx}(x = 1.4)$$

given

$$y = e^x + \sin(x)$$

**Answer:** 4.225167110

(7). Solve the following ordinary differential equation exactly.

$$\frac{dy}{dx} + y = e^{-x}, y(0) = 5$$

Also find $y(0),\ \frac{dy}{dx}\ (0),\ y(2.5),\ \frac{dy}{dx}\ (2.5)$

**Answer:** $y(0) = 5,\ \frac{dy}{dx}(0) = -4,\ y(2.5) = 0.61563,\ \frac{dy}{dx}(2.5) = -0.53355$

# Chapter 01.02: Quantifying Errors

Lesson: True Error

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *enumerate reasons why we need to measure errors*

2) *find the true and relative true error*

## Introduction

In any numerical analysis, errors will arise during the calculations. To be able to deal with the issue of errors, we

(A) identify where the error is coming from, followed by

(B) quantify the error, and lastly

(C) minimize the error as per our needs.

In this lesson, we concentrate on item (B) called quantifying the error, and specifically the true error.

## What is true error?

True error denoted by $E_t$ is the difference between the true value (also called the exact value) and the approximate value.

$$E_t = \text{True value} - \text{Approximate value} \qquad (1)$$

### Example 1

The derivative of a function $f(x)$ at a particular value of $x$ can be approximately calculated by

$$f'(x) \approx \frac{f(x+h) - f(x)}{h}$$

For $f(x) = 7e^{0.5x}$ and $h = 0.3$, find

a) the approximate value of $f'(2)$

b) the true value of $f'(2)$

c) the true error for part (a)

**Solution**

a)

$$f'(x) \approx \frac{f(x+h) - f(x)}{h}$$

For $x = 2$ and $h = 0.3$,

$$\begin{aligned} f'(2) &\approx \frac{f(2+0.3) - f(2)}{0.3} \\ &= \frac{f(2.3) - f(2)}{0.3} \\ &= \frac{7e^{0.5(2.3)} - 7e^{0.5(2)}}{0.3} \\ &= \frac{22.107 - 19.028}{0.3} \\ &= 10.263 \end{aligned}$$

b) The exact value of $f'(2)$ can be calculated by using our knowledge of differential calculus.

$$f(x) \ = 7e^{0.5x}$$

$$\begin{aligned} f'(x) &= 7 \times 0.5 \times e^{0.5x} \\ &= 3.5e^{0.5x} \end{aligned}$$

So the true value of $f'(2)$ is

$$\begin{aligned} f'(2) &= 3.5e^{0.5(2)} \\ &= 9.5140 \end{aligned}$$

c) True error is calculated as

$$\begin{aligned} E_t &= \text{True value} - \text{Approximate value} \\ &= 9.5140 - 10.263 \\ &= -0.749 \end{aligned}$$

The magnitude of true error does not show how bad the error is. A true error of $E_t = -0.749$ may seem to be small, but if the function given in Example 1 were $f(x) = 7 \times 10^{-6}e^{0.5x}$, the true error in calculating $f'(2)$ with $h = 0.3$, would be $E_t = -0.749 \times 10^{-6}$. This value of true error is smaller, even when the two problems are similar in that they use the same value of the function argument, $x = 2$, and the step size, $h = 0.3$. This recognition brings us to the definition of the relative true error.

## What is the relative true error?

Relative true error is denoted by $\epsilon_t$ and is defined as the ratio between the true error and the true value.

$$\epsilon_t = \frac{\text{True Error}}{\text{True Value}} \qquad (2)$$

**Example 2**

The derivative of a function $f(x)$ at a particular value of $x$ can be approximately calculated by

$$f'(x) \approx \frac{f(x+h) - f(x)}{h}$$

For $f(x) = 7e^{0.5x}$ and $h = 0.3$, find the relative true error in finding $f'(2)$.

**Solution**

From Example 1,

$$\begin{aligned} E_t &= \text{True value} - \text{Approximate value} \\ &= 9.5140 - 10.263 \\ &= -0.749 \end{aligned}$$

Relative true error is calculated as

$$\begin{aligned} \epsilon_t &= \frac{\text{True Error}}{\text{True Value}} \\ &= \frac{-0.749}{9.5140} \\ &= -0.078726 \end{aligned}$$

Relative true errors are also presented as percentages. For this example,

$$\begin{aligned} \epsilon_t &= -0.078726 \times 100\% \\ &= -7.8726\% \end{aligned}$$

Absolute relative true errors may also need to be calculated. In such cases,

$$\begin{aligned} |\epsilon_t| &= |-0.078726| \\ &= 0.078726 \\ &= 7.8726\% \end{aligned}$$

# Chapter 01.02: Quantifying Errors

Lesson: Approximate Errors

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *Calculate the approximate error,*

2) *Calculate relative approximate error.*

## Introduction

In any numerical analysis, errors will arise during the calculations. To be able to deal with the issue of errors, we need to

(A) identify where the error is coming from, followed by

(B) quantifying the error, and lastly

(C) minimize the error as per our needs.

In this section of the chapter, we concentrate on item (B) called quantifying the error, and specifically approximate error.

## What is approximate error?

In the previous lesson, we discussed how to calculate true errors. Such errors are calculated only if true values are known. An example where calculating true errors would be useful is when one is checking if a program is in working order, and you know some examples where the true error is known. But mostly, we will not have the luxury of knowing true values as to why would you want to find the approximate values when you already know the true values. So when we are solving a problem numerically, we will only have access to approximate values. We need to know how to quantify error for such cases, and such an error is appropriately called the approximate error.

Approximate error is denoted by $E_a$, and is defined as the difference between the present approximation and the previous approximation.

$$E_a = \text{Present Approximation} - \text{Previous Approximation} \qquad (1)$$

**Example 1**

The derivative of a function $f(x)$ at a particular value of $x$ can be approximately calculated by

$$f'(x) \approx \frac{f(x+h) - f(x)}{h}$$

For $f(x) = 7e^{0.5x}$ and at $x = 2$, find the following

a) $f'(2)$ using $h = 0.3$

b) $f'(2)$ using $h = 0.15$

c) approximate error for the value of $f'(2)$ for part (b)

**Solution**

The approximate expression for the derivative of a function is

$$f'(x) \approx \frac{f(x+h) - f(x)}{h}$$

a) For $x = 2$ and $h = 0.3$,

$$\begin{aligned} f'(2) &\approx \frac{f(2+0.3) - f(2)}{0.3} \\ &= \frac{f(2.3) - f(2)}{0.3} \\ &= \frac{7e^{0.5(2.3)} - 7e^{0.5(2)}}{0.3} \\ &= \frac{22.107 - 19.028}{0.3} \\ &= 10.263 \end{aligned}$$

b) Repeating the procedure of part (a) with $h = 0.15$

$$f'(x) \approx \frac{f(x+h) - f(x)}{h}$$

For $x = 2$ and $h = 0.15$,

$$\begin{aligned} f'(2) &\approx \frac{f(2+0.15) - f(2)}{0.15} \\ &= \frac{f(2.15) - f(2)}{0.15} \\ &= \frac{7e^{0.5(2.15)} - 7e^{0.5(2)}}{0.15} \\ &= \frac{20.510 - 19.028}{0.15} \\ &= 9.8800 \end{aligned}$$

c) So the approximate error, $E_a$ is

$$\begin{aligned} E_a &= \text{Present Approximation} - \text{Previous Approximation} \\ &= 9.8800 - 10.263 \\ &= -0.38300 \end{aligned}$$

The magnitude of approximate error does not reflect how bad the error is. An approximate error of $E_a = -0.38300$ in the above example may seem to be small, but repeating the above example for $f(x) = 7 \times 10^{-6} e^{0.5x}$, the approximate error in calculating $f'(2)$ with $h = 0.15$ would be $E_a = -0.38300 \times 10^{-6}$. This value of approximate error is much smaller, even when the two problems are similar in that they use the same value of the function argument, $x = 2$, and step sizes of $h = 0.15$ and $h = 0.3$. This recognition brings us to the definition of the relative approximate error in the next lesson.

## What is the relative approximate error?

As observed in the previous lesson, approximate errors generally do not reflect how bad or acceptable an error is. We hence define the relative approximate error, denoted by $\epsilon_a$ as the ratio between the approximate error and the present approximation.

$$\epsilon_a = \frac{\text{Approximate Error}}{\text{Present Approximation}} \qquad (2)$$

where

$$\text{Approximate Error, } E_a = \text{Current Approximation} - \text{Previous Approximation}$$

**Example 2**

The derivative of a function $f(x)$ at a particular value of $x$ can be approximately calculated by

$$f'(x) \approx \frac{f(x+h) - f(x)}{h}$$

For $f(x) = 7e^{0.5x}$, find the relative approximate error in calculating $f'(2)$ using values from $h = 0.3$ and $h = 0.15$.

**Solution**

From Example 1, the following values are already calculated

$$f'(2) \approx 10.263 \text{ using } h = 0.3 \text{ and}$$

$$f'(2) \approx 9.8800 \text{ using } h = 0.15.$$

Hence the approximate error

$$\begin{aligned} E_a &= \text{Present Approximation} - \text{Previous Approximation} \\ &= 9.8800 - 10.263 \\ &= -0.38300 \end{aligned}$$

The relative approximate error is calculated as

$$\begin{aligned} \epsilon_a &= \frac{\text{Approximate Error}}{\text{Present Approximation}} \\ &= \frac{-0.38300}{9.8800} \\ &= -0.038765 \end{aligned}$$

Relative approximate errors are also can be presented as percentages. For this example,

$$\begin{aligned} \epsilon_a &= -0.038765 \times 100\% \\ &= -3.8765\% \end{aligned}$$

Absolute relative approximate errors may also need to be calculated. In this example

$$\begin{aligned} |\epsilon_a| &= \ |-0.038765| \\ &= 0.038765 \end{aligned}$$

Absolute relative approximate errors also can be presented as percentages. For this example,

$$\begin{aligned} |\epsilon_a| &= 0.038765 \times 100\% \\ &= 3.8765\% \end{aligned}$$

# Chapter 01.02: Quantifying Errors

Lesson: Review of Significant Digits

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *identify how many significant digits a number has, whether given in decimal or scientific format.*

## What do we mean by significant digits?

Significant digits (also called significant figures) are the digits that are known reliably in a reported number.

Let us take an example. You may be asked to give your answer up to a certain number of decimal places. This fixed representation may bode well for items such as US currency, where one is generally interested in 2 decimal places to represent dollars as the integer part and cents as the fractional part. However, in engineering and science calculations, this can create issues. For example, the thermal expansion coefficient of steel at room temperature is 6.47 $\mu$in/in/$^\circ$F when given up to 2 decimal places. Changing the units of this property of steel to USCS of units, one would write it as 0.00000647 in/in/$^\circ$F, and if we were writing the value up to 2 decimal places, it would be written as 0.00 in/in/$^\circ$F. Hence, the emphasis in engineering is to write the numbers mostly in the scientific format as it represents the numbers with the proper number of significant digits. The number 0.00000647 in/in/$^\circ$F would be written as $6.47 \times 10^{-6}$ in/in/$^\circ$F with 3 significant digits, while 6.47 $\mu$in/in/$^\circ$F would be written as $6.47 \times 10^{0}$ $\mu$in/in/$^\circ$F with 3 significant digits.

Let's take another example. If someone asked me what the population of my county is, I would respond, "The population of the Hillsborough County in Florida, USA is $1,500,000$ (1.5 million)." But if I were going to give 100 dollars to every citizen of the county, I would have to find an exact count or at least a census estimate. That count was $1,471,968$ in the year 2019. So, you can see in my statement that the population is 1.5 million, and there are only two significant digits. In the statement that the population is $1,471,968$ there are seven significant digits. So, how do we differentiate the number of digits correct in $1,500,000$ and $1,471,968$ if no context was given? Well, for that, one uses scientific notation. For our data, we write as

$$1,500,000 = 1.5 \times 10^6$$

$$1,471,968 = 1.471968 \times 10^6$$

to signify the correct number of significant digits in the two numbers that represent the same variable.

**Example**

Give examples to illustrate all the rules of how many significant digits are in a given number.

**Solution**

2.789 has four significant digits as all nonzero digits are significant.

0.0439 has three significant digits as zeros to the left of the first nonzero number are not significant.

4.590 has four significant digits as all zeros to the right of a decimal point are significant.

4008 has four significant digits as any zero between nonzero digits is significant.

4208.07 has six significant digits as any zero between nonzero digits is significant.

4008.0 has five significant digits as all zeros to the right of a decimal point are significant.

4000.0 has five significant digits as all zeros to the right of a decimal point are significant.

15000.0 has six significant digits as all zeros to the right of a decimal point are significant.

15000 may represent 2, 3, 4, or 5 significant digits. Such vagueness can be addressed by using the scientific (also called floating point) format as given below.

$1.5 \times 10^3$ has two significant digits.

$1.50 \times 10^3$ has three significant digits.

$1.5000 \times 10^4$ has five significant digits.

An exact number has an infinite number of significant digits. If you have 5 books in your school bag, then the number can be written with an infinite number of trailing zeros as 5.000...

# Chapter 01.02: Quantifying Errors

Lesson: Relationship Between Significant Digits Correct and Relative Approximate Error

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *Relate the absolute relative approximate error to the number of significant digits at least correct in your answer.*

## While solving a mathematical model using numerical methods, how can we use the value of relative approximate error to figure out if we have achieved an acceptable answer?

In a numerical method that uses iterative methods, a user can calculate relative approximate error $\epsilon_a$ at the end of each iteration. The user may pre-specify a minimum acceptable tolerance called the pre-specified tolerance, $\epsilon_s$. If the absolute relative approximate error $|\epsilon_a|$ is less than or equal to the pre-specified tolerance $\epsilon_s$, that is, $|\epsilon_a| \leq \epsilon_s$, then the acceptable error has been reached and no more iterations would be required.

Alternatively, one may pre-specify how many significant digits they would like to be correct in their answer. In that case, if one wants at least $m$ significant digits to be correct in the answer, then you would need to have the absolute relative approximate error, $|\epsilon_a| \leq 0.5 \times 10^{2-m}\%$.

### Example 1

If one chooses 6 terms of the Maclaurin series for $e^x$ to calculate $e^{0.7}$, how many significant digits can you trust in the solution? Find your answer without knowing or using the exact answer.

### Solution

The Maclaurin series for $e^x$ is given by

$$e^x = 1 + x + \frac{x^2}{2!} + ..... \qquad (E1.1)$$

Using 6 terms in Equation (E1.1), we get the current approximation as

$$\begin{aligned} e^{0.7} &\cong 1 + 0.7 + \frac{0.7^2}{2!} + \frac{0.7^3}{3!} + \frac{0.7^4}{4!} + \frac{0.7^5}{5!} \\ &= 2.0136 \end{aligned}$$

Using 5 terms in Equation (E1.1), we get the previous approximation as

$$\begin{aligned} e^{0.7} &\cong 1 + 0.7 + \frac{0.7^2}{2!} + \frac{0.7^3}{3!} + \frac{0.7^4}{4!} \\ &= 2.0122 \end{aligned}$$

The percentage absolute relative approximate error is

$$\begin{aligned} |\epsilon_a| &= \left| \frac{\text{Current Approximation - Previous Approximation}}{\text{Present Approximation}} \right| \times 100 \\ &= \left| \frac{2.0136 - 2.0122}{2.0136} \right| \times 100 \\ &= 0.069527\% \end{aligned}$$

At least $m$ significant digits are expected to be correct when absolute relative approximate error, $|\epsilon_a| \leq 0.5 \times 10^{2-m}\%$. Since $|\epsilon_a| \leq 0.5 \times 10^{2-2}\% = 0.5\%$, but $|\epsilon_a| \geq 0.5 \times 10^{2-3}\% = 0.05\%$, at least 2 significant digits are correct.

The answer is 2 significant digits are at least correct in the estimated value

$$e^{0.7} \approx 2.0136$$

Alternatively, to find the least number of significant digits correct in the answer, $|\epsilon_a| \leq 0.5 \times 10^{2-2}\%$

$$0.069527 \ \leq \ 0.5 \times 10^{2-m}$$

$$\frac{0.069527}{0.5} \ \leq \ 10^{2-m}$$

$$0.1391 \ \leq \ 10^{2-m}$$

$$\log_{10}(0.1391) \ \leq \ \log_{10}(10^{2-m})$$

$$-0.8567 \leq 2 - m$$

$$m \leq 2.8567$$

Hence 2 significant digits are at least correct in the estimated value

$$e^{0.7} \approx 2.0136$$

# Multiple Choice Test

## Chapter 01.02 Quantifying Errors

(1). True error is defined as

(A) Present Approximation – Previous Approximation

(B) True Value – Approximate Value

(C) abs (True Value – Approximate Value)

(D) abs (Present Approximation – Previous Approximation)

(2). The expression for true error in calculating the derivative of $\sin(2x)$ at $x = \frac{\pi}{4}$ by using the approximate expression $f'(x) \approx \frac{f(x+h) - f(x)}{h}$ is

(A) $\frac{h - \cos(2h) - 1}{h}$

(B) $\frac{h - \cos(h) - 1}{h}$

(C) $\frac{1 - \cos(2h)}{h}$

(D) $\frac{\sin(2h)}{h}$

(3). The relative approximate error at the end of an iteration to find the root of an equation is 0.004%. The least number of significant digits we can trust in the solution is

(A) 2

(B) 3

(C) 4

(D) 5

(4). The number $0.01850 \times 10^3$ has _____ significant digits

(A) 3

(B) 4

(C) 5

(D) 6

(5). The following gas stations were cited for irregular dispensation by the Department of Agriculture. Which one cheated you the most relatively?

| **Station** | **Actual Gasoline dispensed** | **Gasoline Reading at pump** |
|---|---|---|
| Ser | 9.90 | 10.00 |
| Cit | 19.90 | 20.00 |
| Hus | 29.80 | 30.00 |
| She | 29.95 | 30.00 |

(A) Ser

(B) Cit

(C) Hus

(D) She

(6). The number of significant digits in the number 219900 is

(A) 4

(B) 5

(C) 6

(D) 4 or 5 or 6

For complete solution, go to

http://nm.mathforcollege.com/mcquizzes/01aae/quiz_01aae_measuringerror_answers.pdf

# Problem Set

## Chapter 01.02: Quantifying Errors

(1). The trigonometric function $\sin(x)$ can be calculated by using the following infinite series

$$\sin(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + .....$$

a) What is the value of $\sin(2.17)$ by using the first three terms in the given series?

b) What is the value of $\sin(2.17)$ by using the first four terms in the given series?

c) Use your calculator for the true value of $\sin(2.17)$?

d) What is the true error for the answer in part (a)?

e) What is the absolute true error for the answer in part (a)?

f) What is the relative true error for the answer in part (a)?

g) What is the absolute relative true error for the answer in part (a)?

h) What is the approximate error for the answer in part (b)?

i) What is the absolute approximate error for the answer in part (b)?

j) What is the relative approximate error for the answer in part (b)?

k) What is the absolute relative approximate error for the answer in part (b)?

l) Assume that you do not know the exact value of $\sin(2.17)$, how many significant digits are at least correct if you use four terms in the series?

m) What should be the pre-specified relative error tolerance if at least 4 significant digits are required to be correct in calculating $\sin(2.17)$?

**Answer:** *a*) 0.8679 *b*) 0.8229 *c*) 0.8257 *d*) $-0.04213$ *e*) 0.04213
*f*) $-5.102\%$ *g*) $5.102\%h$) $-0.04495$ *i*) 0.04495 *j*) $-5.462$ *k*) 5.462 *l*) 0 *m*) 0.005

(2). A Maclaurin series for a function is given by

$$f(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} + .....$$

What is the absolute relative approximate error if three terms are used for calculating $f(1.2)$?

**Answer:** 0.0221 or 2.21%

(3). A Maclaurin series for a function is given by

$$f(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} + .....$$

How many terms should be used in the series to consider that at least 2 significant digits are correct in your answer for $f(0.1)$?

**Answer:** 2

(4). A gas station owned by Valdez gives you 9.90 gallons of gasoline when you actually paid for 9.95 gallons. Another gas station owned by Hessup gives you 19.80 gallons of gasoline when you actually paid for 19.85 gallons of gasoline. If you only had these two gas stations available in your town, which one would you go to next time you had to fill up your car? Use the concepts learned in measuring errors to justify your answer.

**Answer:** Hessup

(5). The function $e^x$ can be calculated by using the following infinite Maclaurin series

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + .....$$

a) Use 5 terms to calculate the value of $e^{0.9}$?

b) How many significant digits in my calculation would be correct if I use 5 terms?

c) How do I know that I have used enough terms to calculate $e^{0.9}$, if I pre-specify a tolerance of 0.05%? What is the minimum number of terms I should use to achieve the pre-specified tolerance?

d) Where are the sources of error coming from in the above series?

**Answer:** *a*) 2.4538 *b*) 1 *c*) 7 *d*) mainly truncation, but round-off error is there also.

(6). How many significant digits are correct in the following numbers

a) 185000

b) 0.0185

c) 1.0185

d) $185 \times 10^3$

e) $1850 \times 10^2$

f) $0.01850 \times 10^5$

g) $0.0185 \times 10^5$

h) 100.00

i) 100.001

**Answer:** *a*) 3 or 4 or 5 or 6 *b*) 3 *c*) 5 *d*) 3 *e*) 4 *f*) 4 *g*) 3 *h*) 5 *i*) 6

(7). What is the correct normalized scientific notation for 0.029411765 with 4 significant digits?

**Answer:** $2.941 \times 10^{-2}$

# Chapter 01.03: Sources of Error

Lesson: Round-Off Errors

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *recognize that there are two inherent sources of error in numerical methods round-off and truncation error*

2) *define and exemplify round-off errors*

## Introduction

Error in solving an engineering or science problem can arise due to several factors. First, the error may be in the modeling technique. A mathematical model may be based on using assumptions that are not acceptable. For example, one may assume that the drag force on a car is proportional to the velocity of the car, but, actually it is proportional to the square of the velocity of the car. This itself can create huge errors in determining the performance of the car, no matter how accurate the numerical methods you may use. Second, errors may arise from mistakes in programs themselves or in the measurement of physical quantities. But, in applications of numerical methods themselves, the two errors we focus on are inherent to the subject, namely

1) round-off error

2) Truncation error.

## What is a round-off error?

A computer can only represent a number approximately. For example, a number like $\frac{1}{3}$ may be represented as 0.333333 on a PC. Then the round-off error, in this case, is

$\frac{1}{3} - 0.333333 = 0.0000003\bar{3}$. Then there are other numbers that cannot be represented exactly. For example, $\pi$ and $\sqrt{2}$ are numbers that need to be approximated in computer calculations.

## What problems can be created by round-off errors?

Round-off errors while storing numbers or while conducting arithmetic operations can propagate errors. Here is a real-life example of what happened when a small round-off error was responsible for a tragedy during the 1991 US-Iraq war.

Twenty-eight Americans were killed on February 25, 1991. An Iraqi Scud hit the Army barracks in Dhahran, Saudi Arabia. The Patriot defense system had failed to track and intercept the Scud. What was the cause of this failure?

The Patriot defense system consists of an electronic detection device called the range gate. It calculates the area in the air space where it should look for a Scud. To find out where it should aim next, it calculates the velocity of the Scud and the last time the radar detected the Scud. Time is saved in a register that has 24 bits in length. Since the internal clock of the system is measured for every one-tenth of a second, 1/10 is expressed in a 24 bit-register as 0.00011001100110011001100. However, this is not an exact representation. In fact, it would need infinite numbers of bits to represent 1/10 exactly. So, the error in the representation in decimal format is

**Figure 1** Patriot missile (Courtesy of the US Armed Forces,

(http://www.redstone.army.mil/history/archives/patriot/patriot.html)

$$\begin{aligned}\text{Error} &= \frac{1}{10} - (0\times 2^{-1} + 0\times 2^{-2} + 0\times 2^{-3} + 1\times 2^{-4} + ... + 1\times 2^{-22} + 0\times 2^{-23} + 0\times 2^{-24}) \\ &= \frac{1}{10} - \frac{209715}{2097152} \\ &= 9.537\times 10^{-8}\end{aligned}$$

The battery was on for 100 consecutive hours, hence causing an inaccuracy of

$$\begin{aligned}\text{Cumulative Error} &= 9.537\times 10^{-8}\frac{\text{s}}{0.1\text{ s}}\times 100\text{ hr}\times\frac{3600\text{ s}}{1\text{ hr}} \\ &= 0.3433\text{ s}\end{aligned}$$

The shift calculated in the range gate due to the error of 0.342 s was calculated as 687 m. For the Patriot missile defense system, the target is considered out of range if the shift is more than than

137 m. The shift of larger than 137 m resulted in the Scud not being targeted and hence killing 28 Americans in the barracks of Saudi Arabia.

## Show an example of how not carrying enough significant digits can affect the solution to a problem?

In Table 1, the instantaneous coefficient of thermal expansion of steel is given as a discrete function of temperature.

**Table 1:** Instantaneous Coefficient of Linear Thermal Expansion vs. Temperature for Steel

| Temperature, T | Instantaneous Coefficient of Linear Thermal Expansion, $\alpha$ |
|---|---|
| °F | $\mu$in/in/°F |
| 80 | 6.47 |
| 60 | 6.36 |
| 40 | 6.24 |
| 20 | 6.12 |
| 0 | 6.00 |
| −20 | 5.86 |
| −40 | 5.72 |
| −60 | 5.58 |
| −80 | 5.43 |
| −100 | 5.28 |
| −120 | 5.09 |
| −140 | 4.91 |
| −160 | 4.72 |
| −180 | 4.52 |
| −200 | 4.30 |
| −220 | 4.08 |
| −240 | 3.83 |
| −260 | 3.58 |
| −280 | 3.33 |
| −300 | 3.07 |
| −320 | 2.76 |
| −340 | 2.45 |

Regression is used to develop a simple formula in the form of a second-order polynomial to relate the coefficient of thermal expansion coefficient of steel as a function of temperature.

Using Microsoft Excel and using its trend line function, it gives the formula as shown in Figure 2.

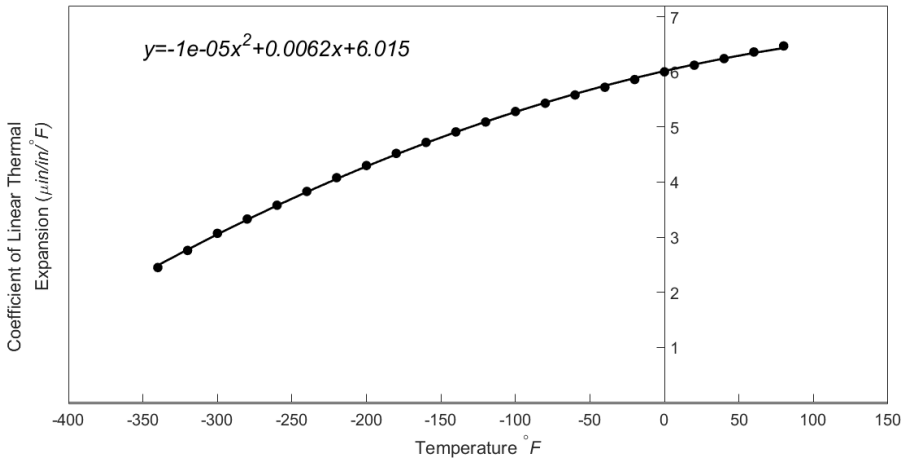



**Figure 2:** Regression Model with Default Format for Instantaneous Coefficient of Linear Thermal Expansion vs. Temperature for steel

The trend line given by Excel in its default format is

$$\alpha = -1 \times 10^{-5} T^2 + 0.0062T + 6.015 \qquad (1)$$

At

$$T = -300^\circ\text{F},$$

the predicted value of the instantaneous coefficient of the thermal expansion from Equation (1) is

$$\begin{aligned} \alpha &= -1 \times 10^{-5} \left(-300\right)^2 + 0.0062 \left(-300\right) + 6.015 \\ &= 3.26 \ \mu\text{in/in/}^\circ\text{F} \end{aligned}$$

From Table 1, the observed value

At

$$T = -300^\circ\text{F is}$$

$$\alpha = 3.07 \ \mu\text{in/in/}^\circ\text{F}$$

which gives a residual at $T = -300^\circ$F as

$$\begin{aligned} E_i &= 3.07 - 3.26 \\ &= -0.19 \ \mu\text{in/in/}^\circ\text{F} \end{aligned} \qquad (2)$$

When a graduate student showed me the above numbers for the observed and predicted values, they did not match what was getting depicted in Figure 2. In the figure, the predicted value seemed much closer to the observed value than what is calculated above.

So what was the cause of this discrepancy? Notice that the default format setting for the trend line (Equation (1)) in Excel is the general format. But, now change the format setting to scientific format with say five significant digits, and we get the same trend line as

$$\alpha = -1.2278 \times 10^{-5} T^2 + 6.1946 \times 10^{-3} T + 6.0150 \qquad (3)$$

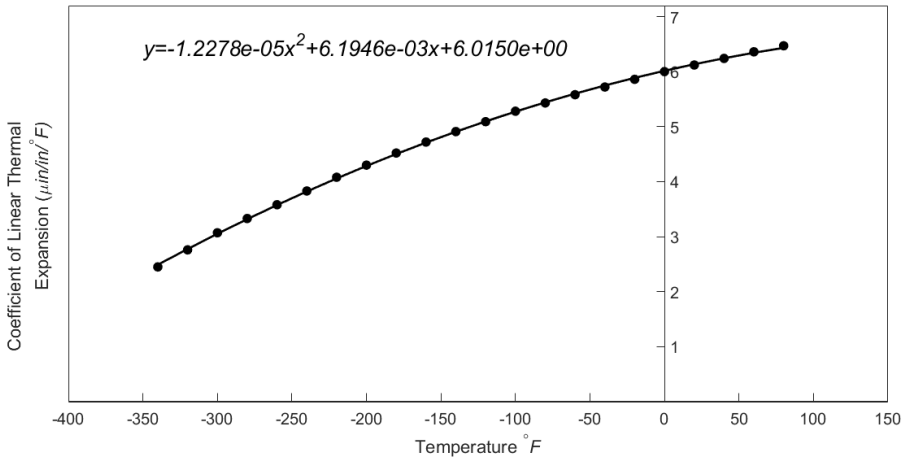



**Figure 3:** Regression Model with Scientific Format for Instantaneous Coefficient of Linear Thermal Expansion vs. Temperature for steel

At

$$T = -300°\text{F},$$

the predicted value of the instantaneous coefficient of the thermal expansion from Equation (3)

$$\begin{aligned} \alpha &= -1.2278 \times 10^{-5} \left(-300\right)^2 + 6.1946 \times 10^{-3} \left(-300\right) + 6.0150 \\ &= 3.05 \ \mu\text{in/in/}°\text{F} \end{aligned}$$

From Table 1, the observed value at

$$T = -300°\text{F}$$

is

$$\alpha = 3.07 \ \mu\text{in/in/}°\text{F}$$

which gives a residual of

$$\begin{aligned} E_i &= 3.07 - 3.05 \\ &= 0.02 \ \mu\text{in/in/}°\text{F} \end{aligned} \qquad (4)$$

The residual now looks reasonable.

The difference between the predicted value (Equation(2) when using the default format of Excel

$$\alpha = 3.255\ \mu\text{in/in/}^\circ\text{F}$$

and the predicted value when using scientific format with 5 significant digits (Equation(4))

$$\alpha = 3.051\ \mu\text{in/in/}^\circ\text{F}$$

is attributed fully to the round-off error.

## References


1. This reference is the full GAO report of the investigation that resulted after the accident. "Patriot Missile Defense – Software Problem Led to System Failure at Dhahran, Saudi Arabia", GAO Report, General Accounting Office, Washington DC, February 4, 1992.

2. It should be pointed out that the Patriot missile was originally designed to be a mobile system and not used as an anti-ballistic system. In mobile systems, the clocks are reset more often. As per the article *Operations: I Did Not Say You Could Do That!* by *Bill Barnes and Duke McMillin*, here are some important observations: "It turns out that the original use case for this system was to be mobile and to defend against aircraft that move much more slowly than ballistic missiles. Because the system was intended to be mobile, it was expected that the computer would be periodically rebooted. In this way, any clock-drift error would not be propagated over extended periods and would not cause significant errors in range calculation. Because the Patriot system was not intended to run for extended times, it was probably never tested under those conditions—explaining why the problem was not discovered until the war was in progress. The fact that the system was also designed as an antiaircraft system probably also enabled the inclusion of such a design flaw because slower-moving airplanes would be easier to track and, therefore, less dependent upon a highly accurate clock value."

# Chapter 01.03: Sources of Error

Lesson: Truncation error

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *define and exemplify truncation errors*

2) *recognize the difference between round-off and truncation error.*

## Introduction

Truncation error is defined as the error caused by approximating a mathematical procedure. Examples of mathematical procedure getting approximated include an infinite series where only the first few terms are used, integrals where the number of rectangles drawn to show the area under the curve is finite in number as opposed to infinite, and the derivative of a function is approximated by using the slope of a secant line as opposed to a tangent line.

## Examples of Truncation Error

For example, the Maclaurin series for $e^x$is given as

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + ... \qquad (1)$$

This series has an infinite number of terms, but when using this series to calculate $e^x$, only a finite number of terms can be used. For example, if one uses three terms to calculate $e^x$, then

$$e^x \approx 1 + x + \frac{x^2}{2!}$$

the truncation error for such an approximation for any value of $x$ is

$$\begin{aligned} \text{Truncation error} = \ & e^x - \left(1 + x + \frac{x^2}{2!}\right) \\ & = \left(1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + ....\right) - \left(1 + x + \frac{x^2}{2!}\right) \\ & = \frac{x^3}{3!} + \frac{x^4}{4!} + ..... \end{aligned}$$

**Example 1**

Given the following infinite series, find the truncation error for $x = 0.75$ if only the first three terms of the series are used.

$$S = 1 + x + x^2 + x^3 + \ldots, \ |x| < 1.$$

**Solution**

Using only the first three terms of the series for $x = 0.75$ gives

$$\begin{aligned} S_3 &= \left(1 + x + x^2\right)_{x=0.75} \\ &= 1 + 0.75 + (0.75)^2 \\ &= 2.3125 \end{aligned}$$

The sum of an infinite geometrical series

$$S = a + ar + ar^2 + ar^3 + \ldots, \ |r| < 1$$

is given by

$$S = \frac{a}{1 - r}$$

For our series, $a = 1$ and $r = 0.75$, to give

$$\begin{aligned} S &= \frac{1}{1 - 0.75} \\ &= 4 \end{aligned}$$

The truncation error ($TE$) hence is

$$\begin{aligned} TE &= 4 - 2.3125 \\ &= 1.6875 \end{aligned}$$

**Example 2**

Use the concept of the relative approximate error to illustrate through an example how truncation error can be controlled. Assume that you are not privy to the exact value.

**Solution**

Let us show through an example how many terms need to be considered in Equation (1) to calculate $e^{1.2}$ within 1% pre-specified tolerance. Of course, you are not privy to the exact answer.

The summation series for $e^{1.2}$ is given by

$$e^{1.2} = 1 + 1.2 + \frac{1.2^2}{2!} + \frac{1.2^3}{3!} + .....$$

In Table 1, we tabulate the value of $e^{1.2}$, approximate error and absolute relative approximate error as a function of the number of terms of the series used, $n$.

**Table 1**: Values and approximate errors for $e^{1.2}$ as a function of the number of terms of the Maclaurin series

| $n$ | $e^{1.2}$ | $E_a$ | $\lvert\epsilon_a\rvert\,\%$ |
|---|---|---|---|
| 1 | 1 | – | – |
| 2 | 2.2 | 1.2 | 54.546 |
| 3 | 2.92 | 0.72 | 24.658 |
| 4 | 3.208 | 0.288 | 8.9776 |
| 5 | 3.2944 | 0.0864 | 2.6226 |
| 6 | 3.315136 | 0.020736 | 0.6255 |

Using a minimum of 6 terms of the series yields an absolute relative approximate error $|\epsilon_a| < 1\%$. The reported value of $e^{1.2}$ is 3.3151 and is shown up to 5 significant digits. As a side note, the exact value, up to 5 significant digits, of $e^{1.2}$ is 3.3201.

# Chapter 01.03: Sources of Error

Lesson: Truncation error Extended Examples

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *exemplify truncation errors through numerical differentiation and integration*

## Can you give me other examples of truncation errors?

In many textbooks on numerical methods, the Maclaurin series is used as the only example to illustrate truncation error. This example may lead you to believe that truncation errors are just a result of using a finite number of terms of the series. However, truncation errors take place in other mathematical procedures as well. In this lesson, we take two such examples.

### Example 1

To find the exact derivative of a function, we define

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

But since we cannot use $h \to 0$, in numerical methods, we use a finite value of $h$, to give

$$f'(x) \approx \frac{f(x+h) - f(x)}{h}$$

So the truncation error is caused by choosing a finite value of $h$ as opposed to a $h \to 0$.

Find the truncation error for finding $f'(3)$ for $f(x) = x^2$ using a step size of $h = 0.2$

**Solution**

In finding $f'(3)$ for $f(x) = x^2$, we have the exact value of the derivative calculated as follows. Given

$$f(x) = x^2$$

from the definition of the derivative of a function,

$$
\begin{aligned}
f'(x) &= \lim_{h \to 0} \frac{f(x+h) - f(x)}{h} \\
&= \lim_{h \to 0} \frac{(x+h)^2 - (x)^2}{\Delta x} \\
&= \lim_{h \to 0} \frac{x^2 + 2xh + h^2 - x^2}{\Delta x} \\
&= \lim_{h \to 0} (2x + h) \\
&= 2x
\end{aligned}
$$

The above derivative is the same expression you would have obtained by directly using the formula from your differential calculus class

$$
\frac{d}{dx}(x^n) = nx^{n-1}
$$

By this formula for

$$
f(x) = x^2
$$

$$
f'(x) = 2x
$$

The exact value of $f'(3)$ is

$$
\begin{aligned}
f'(3) &= 2 \times 3 \\
&= 6
\end{aligned}
$$

Given the step size of $h = 0.2$, we get

$$
\begin{aligned}
f'(3) &= \frac{f(3 + 0.2) - f(3)}{0.2} \\
&= \frac{f(3.2) - f(3)}{0.2} \\
&= \frac{3.2^2 - 3^2}{0.2} \\
&= \frac{10.24 - 9}{0.2} \\
&= \frac{1.24}{0.2} \\
&= 6.2
\end{aligned}
$$

$$
\begin{aligned}
\text{Truncation Error} &= \text{Exact Value} - \text{Approximate Value} \\
&= 6 - 6.2 \\
&= -0.2
\end{aligned}
$$

Note: The example purposefully chose a simple function $f(x) = x^2$ with the value of $x = 2$ and $h = 0.2$ because we wanted to have no round-off error in our calculations so that the truncation error is the only error in the example.

Question: Can you reduce the truncation error by choosing a smaller $h$?

**Example 2**

Another example of truncation error is the numerical integration of a function,

$$I = \int_a^b f(x)dx$$

Exact calculations require us to calculate the area under the curve by adding the area of the rectangles, as shown in Figure 1. However, exact calculations would require an infinite number of such rectangles. Since we cannot choose an infinite number of rectangles, we will have a truncation error.

For the integral

$$\int_3^9 x^2 dx$$

find the truncation error if a two-segment left-hand Reimann sum with equal width of segments is used.

**Solution**

We have the exact value as

$$\begin{aligned}\int_3^9 x^2 dx &= \left[\frac{x^3}{3}\right]_3^9 \\ &= \left[\frac{9^3 - 3^3}{3}\right] \\ &= 234\end{aligned}$$

Using two rectangles of equal width to approximate the area (see Figure 1) under the curve, the approximate value of the integral

$$\begin{aligned}\int_3^9 x^2 dx \approx\ & \left.(x^2)\right|_{x=3}(6-3) + \left.(x^2)\right|_{x=6}(9-6) \\ &= (3^2)3 + (6^2)3 \\ &= 27 + 108 \\ &= 135\end{aligned}$$

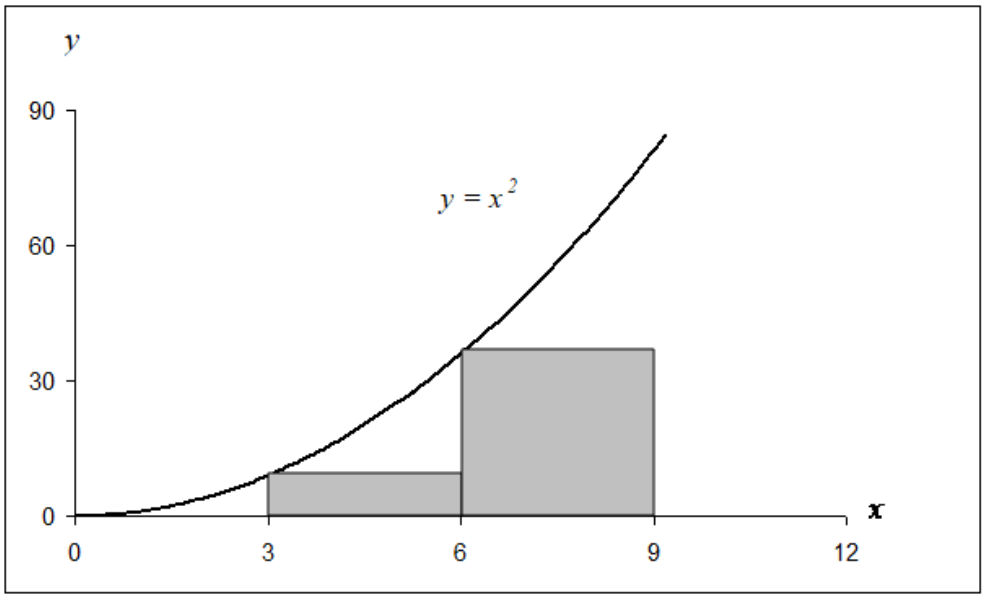



**Figure 1** Plot of $y = x^2$ showing the approximate area under the curve from $x = 3$ to $x = 9$ using two rectangles.

$$\begin{aligned} \text{Truncation Error} &= \text{Exact Value} - \text{Approximate Value} \\ &= 234 - 135 \\ &= 99. \end{aligned}$$

Note: Again, we purposefully chose a simple example with such numbers to produce numbers without round-off error in our calculations. This deliberate choice makes the obtained error purely due to truncation.

Question: Can you reduce the truncation error by choosing more rectangles as given in Figure 2? What is the truncation error when four rectangles are chosen?

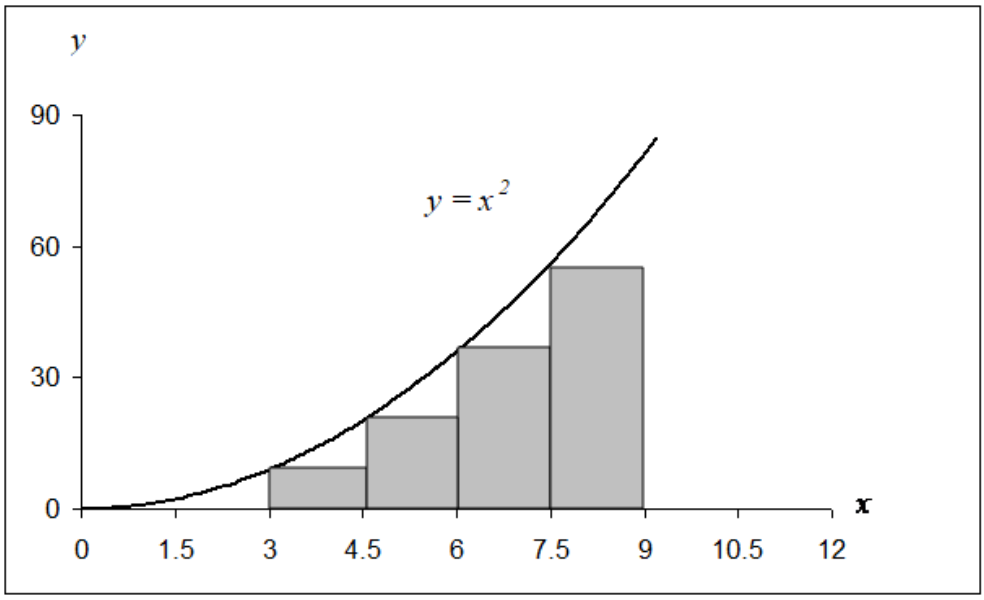



**Figure 2** Plot of $y = x^2$ showing the approximate area under the curve from $x = 3$ to $x = 9$ using four rectangles.

# Multiple Choice Test

## Chapter 01.03 Sources of Error

(1). Truncation error is caused by approximating

(A) irrational numbers

(B) fractions

(C) rational numbers

(D) exact mathematical procedures.

(2). A computer that represents only 4 significant digits with chopping would calculate $66.666 \times 33.333$ as

(A) 2220

(B) 2221

(C) 2221.17778

(D) 2222

(3). A computer that represents only 4 significant digits with rounding would calculate $66.666 \times 33.333$ as

(A) 2220

(B) 2221

(C) 2221.17778

(D) 2222

(4). The truncation error in calculating $f'(2)$ for $f(x) = x^2$ by $f'(x) \approx \dfrac{f(x+h) - f(x)}{h}$ with $h = 0.2$ is

(A) $-0.2$

(B) 0.2

(C) 4.0

(D) 4.2

(5). The truncation error in finding $\int_{-3}^{9} x^3 dx$ using LRAM (left end point Riemann approximation method) with equally portioned points $-3 < 0 < 3 < 6 < 9$ is

(A) 648

(B) 756

(C) 972

(D) 1620

(6). The number $\frac{1}{10}$ is registered in a fixed 6 bit-register with all bits used for the fractional part. The difference is accumulated every $\frac{1}{10}$th of a second for one day. The magnitude of the accumulated difference is

(A) 0.082

(B) 135

(C) 270

(D) 5400

For complete solution, go to

http://nm.mathforcollege.com/mcquizzes/01aae/quiz_01aae_sourceserror_answers.pdf

# Problem Set

## Chapter 01.03: Sources of Error

(1). What is the round-off error in representing 200/3 in a 6-significant digit computer that chops the last significant digit?

**Answer:** 0.00006666666....

(2). What is the round-off error in representing 200/3 in a 6-significant digit computer that rounds off the last significant digit?

**Answer:** $-0.00003333333...$

(3). What is the truncation error in the calculation of the $f'(x)$ that uses the approximation

$$f'(x) \approx \frac{f(x+\Delta x) - f(x)}{\Delta x}$$

for

$$f(x) = x^3,\ \Delta x = 0.4,\ \text{and}\ x = 5$$

**Answer:** $-6.16$

(4) What is the truncation error in the calculation of $\sin(\frac{\pi}{2})$ if only first five terms of the Maclaurin series are used for the calculation? Ignore the round-off error in your calculations.

$$\sin(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + .....$$

**Answer:** $-0.35427560 \times 10^{-5}$

(5). The integral $\int_3^9 x^2 dx$ can be calculated approximately by finding the area of the four rectangles as shown in Figure 1. What is the truncation error due to this approximation?

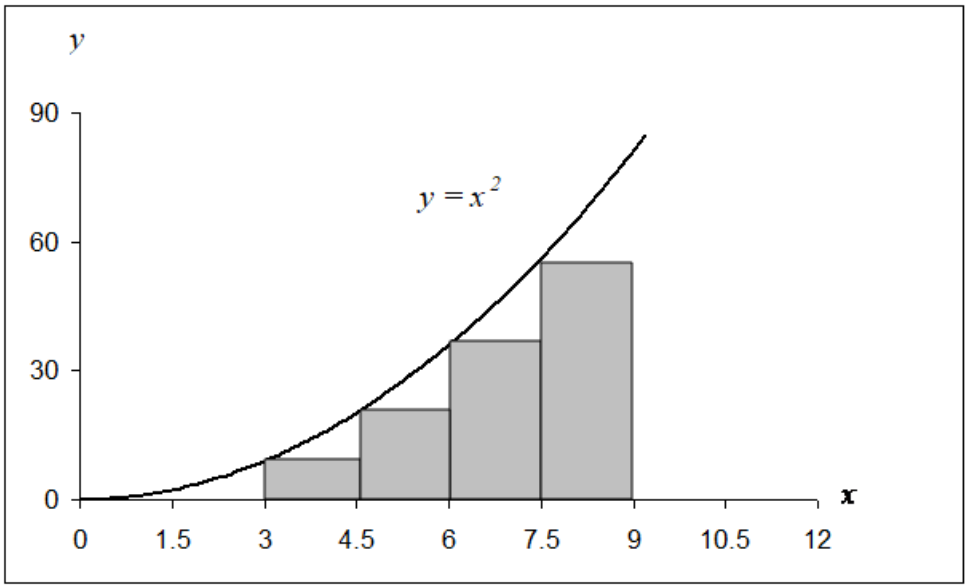



**Figure 1** Plot of $y = x^2$ showing the approximate area under the curve from $x = 3$ to $x = 9$ using four rectangles.

**Answer:** 51.75

(6). Below is the data given for linear coefficient of thermal expansion for steel as a function of temperature.

| Temperature | Instantaneous Thermal Expansion Coefficient |
|---|---|
| °F | $\mu$in/in/°F |
| 80 | 6.47 |
| 60 | 6.36 |
| 40 | 6.24 |
| 20 | 6.12 |
| 0 | 6.00 |
| −20 | 5.86 |
| −40 | 5.72 |
| −60 | 5.58 |
| −80 | 5.43 |
| −100 | 5.28 |
| −120 | 5.09 |
| −140 | 4.91 |
| −160 | 4.72 |
| −180 | 4.52 |
| −200 | 4.30 |
| −220 | 4.08 |
| −240 | 3.83 |
| −260 | 3.58 |
| −280 | 3.33 |
| −300 | 3.07 |
| −320 | 2.76 |

| Temperature | Instantaneous Thermal Expansion Coefficient |
|---|---|
| −340 | 2.45 |

Regression is used to come up with a simple formula – a second order polynomial to relate the linear coefficient of thermal expansion of steel as a function of temperature.

a) The formula given by MS Excel using default settings is shown in Figure 2. Using the formula, what is the value of the linear coefficient of thermal expansion at $T = -300$, $-200$, $-100$, 0 and 100°F? Compare these values with those given in the table above.

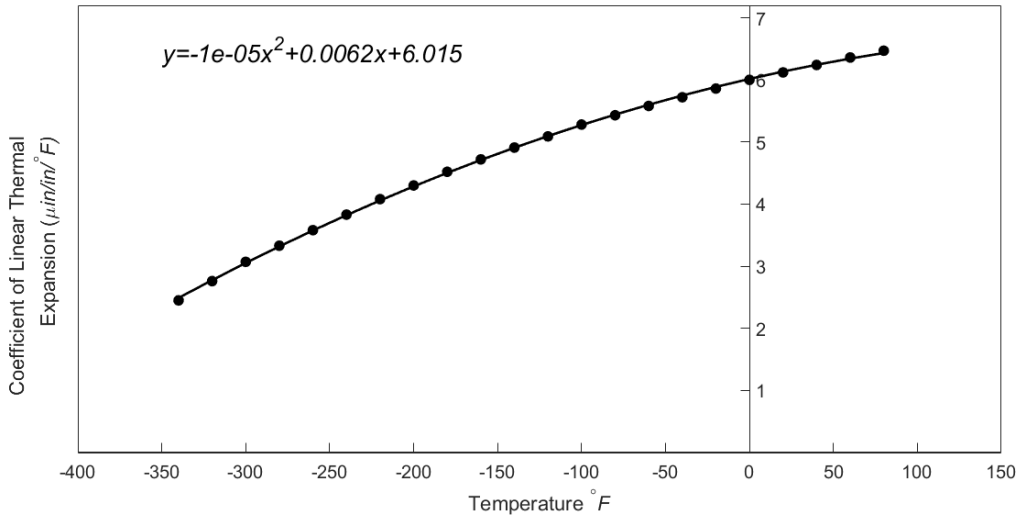



**Figure 2** Default trend line given by excel

(b) Now the default settings for the trend line (Figure 3) are changed to scientific format with five significant digits. Using the formula, what is the value of linear coefficient of thermal expansion at $T = -300$, $-200$,$-100$, 0 and 100°F? Compare these values with those given in the table above.

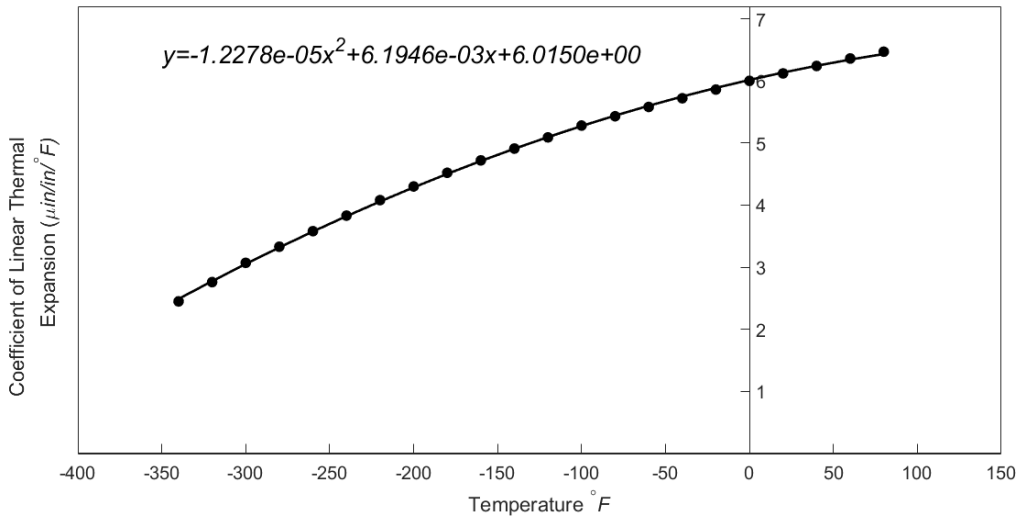


**Figure 3** Trend line given by Excel using scientific format.

c) Do you get different answers in part (a) and part (b) for the predicted value of the linear coefficient of thermal expansion at $T = -300°\text{F}$? Yes or No.

d) In part (c), what do you attribute the difference or lack of difference in answers to?

**Answer:** a) At $T = -300°\text{F}$, $\alpha = 3.255$ in/in/$^\circ$F. From table, $\alpha = 3.07$ in/in/$^\circ$F b) At $T = -300°\text{F}$, $\alpha = 3.0516$ in/in/$^\circ$F. From table, $\alpha = 3.07$ in/in/$^\circ$F c) No d) Answer not given intentionally.

# Chapter 01.04: Fixed-Point Binary Representation of Numbers

Lesson: Introduction to Binary Representation of Numbers

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *enumerate different bases of number systems and be able to convert base-*2 *numbers to base-*10 *numbers.*

## Introduction

In everyday life, we use a number system that has a base of 10 (decimal system). When a number is written in this system, we can quickly decipher it as we see such numbers throughout the day.

For example, look at the number 257.36. Each digit in 257.36 has a value of 0 through 9 and has a place value where the position of the number with respect to the decimal point determines its value. Each position in the decimal number corresponds to an integer exponent of 10. In the number 257.36, the 2 is in the place value of hundred, that is $10^2$, while the 3 is in the place of one-tenths, that is, $10^{-1}$. The number 257.36 can be written as

$$\begin{aligned} 257.36 &= 2 \times 10^2 + 5 \times 10^1 + 7 \times 10^0 + 3 \times 10^{-1} + 6 \times 10^{-2} \\ &= 200 + 50 + 7 + 0.3 + 0.06 \\ &= 257.36 \end{aligned}$$

In a binary system, we have a similar system as the base-10, but the only difference is that the base is made of only two digits 0 and 1. So it is a base-2 system. A number like (1011.0011) in base-2 represents an equivalent decimal number as

$$\begin{aligned} (1011.0011)_2 &= \begin{pmatrix} \left(1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0\right) \\ +(0 \times 2^{-1} + 0 \times 2^{-2} + 1 \times 2^{-3} + 1 \times 2^{-4}) \end{pmatrix}_{10} \\ &= \begin{pmatrix} (1 \times 8 + 0 \times 4 + 1 \times 2 + 1 \times 1) \\ +(0 \times 0.5 + 0 \times 0.25 + 1 \times 0.125 + 1 \times 0.0625) \end{pmatrix}_{10} \\ &= (8 + 0 + 2 + 1 + 0 + 0 + 0.125 + 0.0625)_{10} \\ &= (11.1875)_{10} \end{aligned}$$

Note the dot (.) that separates the integer and fractional part in the binary representation or any other representation is called the radix point.

**Example 1**

Find the equivalent of $(1101)_2$ in base-10.

**Solution**

$$\begin{aligned}(1101)_2 &= \left(1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0\right)_{10} \\ &= (1 \times 8 + 1 \times 4 + 0 \times 2 + 1 \times 1)_{10} \\ &= (8 + 4 + 0 + 1)_{10} \\ &= (13)_{10}\end{aligned}$$

**Example 2**

Find the equivalent of $(0.1101)_2$ in base-10.

**Solution**

$$\begin{aligned}(0.1101)_2 &= \left(1 \times 2^{-1} + 1 \times 2^{-2} + 0 \times 2^{-3} + 1 \times 2^{-4}\right)_{10} \\ &= (0.5 + 0.25 + 0 + 0.0625)_{10} \\ &= (0.8125)_{10}\end{aligned}$$

**Example 3**

Find the equivalent of $(1101.1101)_2$ in base-10.

**Solution**

From Example 1,

$$(1101)_2 = (13)_{10}$$

From Example 2,

$$(0.1101)_2 = (0.8125)_{10}$$

Hence

$$(1101.1101)_2 = (13.8125)_{10}$$

# Chapter 01.04: Fixed-Point Binary Representation of Numbers

Lesson: Fixed-Point Binary Representation of Numbers

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *convert a base-10 number to a base-2 number in a fixed format.*

## Introduction

To understand the binary system, we need to convert binary numbers to decimal numbers and vice-versa. We have already seen an example of how binary numbers can be converted to decimal numbers. Let us see how we can convert a decimal number to a binary number. For example, take the decimal number 11.1875.

First, look at the integer part: 11.

Divide 11 by 2. This division gives a quotient of 5 and a remainder of 1. Since the remainder is 1, $a_0 = 1$.

Divide the quotient 5 by 2. This division gives a quotient of 2 and a remainder of 1. Since the remainder is 1, $a_1 = 1$.

Divide the quotient 2 by 2. This division gives a quotient of 1 and a remainder of 0. Since the remainder is 0, $a_2 = 0$.

Divide the quotient 1 by 2. This division gives a quotient of 0 and a remainder of 1. Since the remainder is 1, $a_3 = 1$.

Since the quotient now is 0, the process is stopped. The above steps are summarized in Table 1.

**Table 1** Converting a base-10 integer to binary representation.

| | Quotient | Remainder |
|---|---|---|
| 11/2 | 5 | $1 = a_0$ |
| 5/2 | 2 | $1 = a_1$ |
| 2/2 | 1 | $0 = a_2$ |
| 1/2 | 0 | $1 = a_3$ |

Hence

$$\begin{aligned}(11)_{10} &= (a_3 a_2 a_1 a_0)_2 \\ &= (1011)_2\end{aligned}$$

For any integer, the algorithm for finding the binary equivalent is given in the flow chart in Figure 1.

Now let us look at the decimal part, that is, 0.1875.

Multiply 0.1875 by 2. This multiplication gives 0.375. The number before the decimal is 0, and the number after the decimal is 0.375. Since the number before the decimal is 0, $a_{-1} = 0$.

Multiply the number after the decimal, that is, 0.375 by 2. This multiplication gives 0.75. The number before the decimal is 0, and the number after the decimal is 0.75. Since the number before the decimal is 0, $a_{-2} = 0$.

Multiply the number after the decimal, that is, 0.75 by 2. This multiplication gives 1.5. The number before the decimal is 1, and the number after the decimal is 0.5. Since the number before the decimal is 1, $a_{-3} = 1$.

Multiply the number after the decimal, that is, 0.5 by 2. This multiplication gives 1.0. The number before the decimal is 1, and the number after the decimal is 0. Since the number before the decimal is 1, $a_{-4} = 1$.

Since the number after the decimal is 0, the conversion is complete. The above steps are summarized in Table 2.

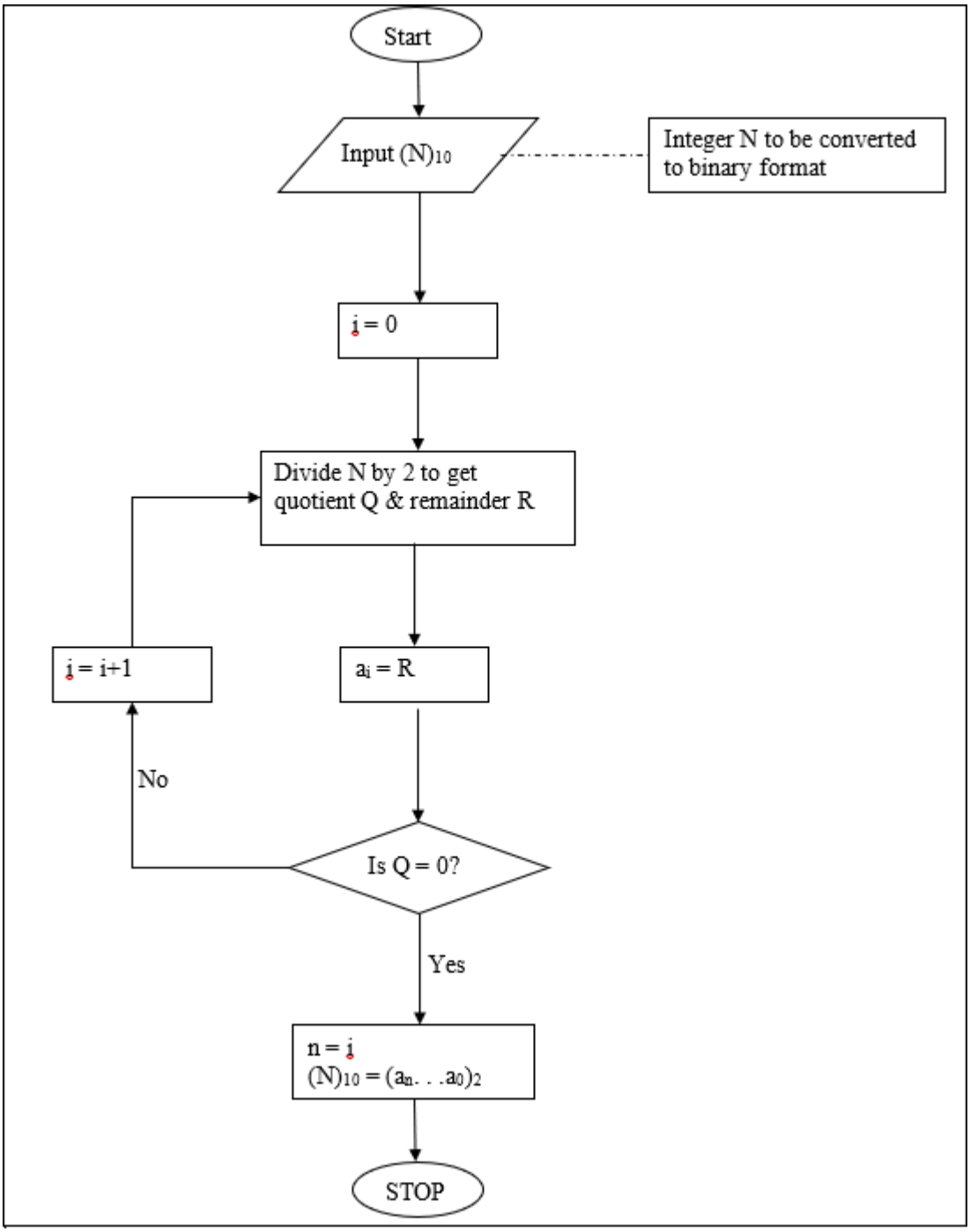



**Figure 1.** A flow chart to convert an integer from decimal to binary format

**Table 2**. Converting a base-10 fraction to binary representation.

| | Number | Number after decimal | Number before decimal |
|---|---|---|---|
| $0.1875 \times 2$ | 0.375 | 0.375 | $0 = a_{-1}$ |
| $0.375 \times 2$ | 0.75 | 0.75 | $0 = a_{-2}$ |

| | Number | Number after decimal | Number before decimal |
|---|---|---|---|
| $0.75 \times 2$ | 1.5 | 0.5 | $1 = a_{-3}$ |
| $0.5 \times 2$ | 1.0 | 0.0 | $1 = a_{-4}$ |

Hence

$$\begin{aligned}(0.1875)_{10} &= (a_{-1}a_{-2}a_{-3}a_{-4})_2 \\ &= (0.0011)_2\end{aligned}$$

The algorithm for any fraction is given in a flowchart in Figure 2.

Having calculated

$$(11)_{10} = (1011)_2$$

and

$$(0.1875)_{10} = (0.0011)_2,$$

we have

$$(11.1875)_{10} = (1011.0011)_2.$$

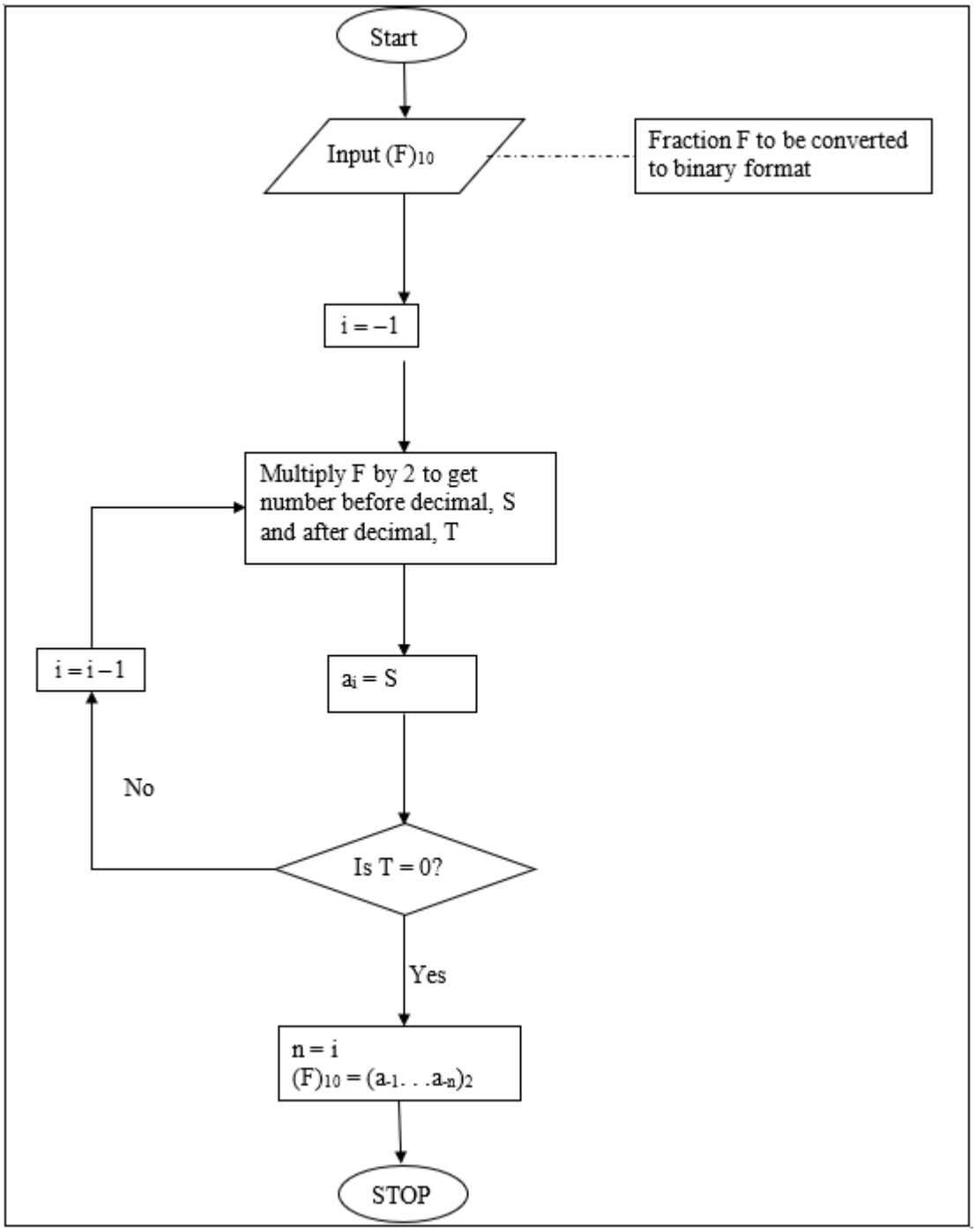



**Figure 2.** A flow chart to convert a fractional part from decimal to binary format

In the above example, when we were converting the fractional part of the number, we were left with 0 after the decimal number and used that as a place to stop. In many cases, we are never left with a 0 after the decimal number. For example, finding the binary equivalent of 0.3 is summarized in Table 3.

**Table 3**. Converting a base-10 fraction to approximate binary representation.

| | Number | Number after decimal | Number before decimal |
|---|---|---|---|
| $0.3 \times 2$ | 0.6 | 0.6 | $0 = a_{-1}$ |
| $0.6 \times 2$ | 1.2 | 0.2 | $1 = a_{-2}$ |
| $0.2 \times 2$ | 0.4 | 0.4 | $0 = a_{-3}$ |
| $0.4 \times 2$ | 0.8 | 0.8 | $0 = a_{-4}$ |
| $0.8 \times 2$ | 1.6 | 0.6 | $1 = a_{-5}$ |

As you can see, the process will never end. In this case, the number can only be approximated in binary format, that is,

$$\begin{aligned}(0.3)_{10} &\approx (a_{-1}a_{-2}a_{-3}a_{-4}a_{-5})_2 \\ &= (0.01001)_2\end{aligned}$$

**Example 1**

Convert $(13.875)_{10}$ to base 2.

**Solution**

For $(13)_{10}$, conversion to binary format is shown in Table 4.

**Table 4**. Conversion of base-10 integer to binary format.

| | Quotient | Remainder |
|---|---|---|
| 13/2 | 6 | $1 = a_0$ |
| 6/2 | 3 | $0 = a_1$ |
| 3/2 | 1 | $1 = a_2$ |
| 1/2 | 0 | $1 = a_3$ |

So

$$(13)_{10} = (1101)_2.$$

Conversion of $(0.875)_{10}$ to binary format is shown in Table 5.

**Table 5**. Converting a base-10 fraction to binary representation.

| | Number | Number after decimal | Number before decimal |
|---|---|---|---|
| $0.875 \times 2$ | 1.75 | 0.75 | $1 = a_{-1}$ |
| $0.75 \times 2$ | 1.5 | 0.5 | $1 = a_{-2}$ |
| $0.5 \times 2$ | 1.0 | 0.0 | $1 = a_{-3}$ |

So

$$(0.875)_{10} = (0.111)_2$$

Hence

$$(13.875)_{10} = (1101.111)_2$$

## But what is the mathematics behinds this process of converting a decimal number to binary format?

Let $z$ be the decimal number written as

$$z = x.y$$

where

$x$ is the integer part, and $y$ is the fractional part.

We want to find the binary equivalent of $x$. So, we can write

$$x = a_n 2^n + a_{n-1} 2^{n-1} + ... + a_0 2^0$$

If we can now find $a_0, ..., a_n$in the above equation, then

$$(x)_{10} = (a_n a_{n-1} ... a_0)_2$$

We now want to find the binary equivalent of $y$. So, we can write

$$y = b_{-1} 2^{-1} + b_{-2} 2^{-2} + ... + b_{-m} 2^{-m}$$

If we can now find $b_{-1}, ..., b_{-m}$in the above equation, then

$$(y)_{10} = (b_{-1} b_{-2} ... b_{-m})_2$$

Let us look at this using the same example as before.

**Example 2**

Convert $(11.1875)_{10}$ to base 2.

**Solution**

To convert $(11)_{10}$ to base 2, what is the highest power of 2 that is part of 11. That power is 3, as $2^3 = 8$ to give

$$11 = 2^3 + 3$$

What is the highest power of 2 that is part of 3? That power is 1, as $2^1 = 2$ to give

$$3 = 2^1 + 1$$

So

$$\begin{aligned} 11 &= 2^3 + 3 \\ &= 2^3 + 2^1 + 1 \end{aligned}$$

What is the highest power of 2 that is part of 1? That power is 0, as $2^0 = 1$ to give

$$1 = 2^0$$

Hence

$$\begin{aligned} (11)_{10} &= 2^3 + 2^1 + 1 \\ &= 2^3 + 2^1 + 2^0 \\ &= 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0 \\ &= (1011)_2 \end{aligned}$$

To convert $(0.1875)_{10}$ to base 2, we proceed as follows. What is the smallest negative power of 2 that is less than or equal to 0.1875? That power is $-3$ as $2^{-3} = 0.125$.

So

$$0.1875 = 2^{-3} + 0.0625$$

What is the next smallest negative power of 2 that is less than or equal to 0.0625? That power is $-4$ as $2^{-4} = 0.0625$.

So

$$0.1875 = 2^{-3} + 2^{-4}$$

Hence

$$\begin{aligned} (0.1875)_{10} &= 2^{-3} + 0.0625 = 2^{-3} + 2^{-4} \\ &= 0 \times 2^{-1} + 0 \times 2^{-2} + 1 \times 2^{-3} + 1 \times 2^{-4} \\ &= (0.0011)_2 \end{aligned}$$

Since

$$(11)_{10} = (1011)_2$$

and

$$(0.1875)_{10} = (0.0011)_2$$

we get

$$(11.1875)_{10} = (1011.0011)_2$$

One can show the above process algebraically for any general number, but it is out-of-scope for this course.

# Chapter 01.04: Fixed-Point Binary Representation of Numbers

Lesson: Writing a Fixed-Point Binary Number in Given Word Length

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *write a base-2 number in a given word length.*

## Introduction

A fixed-point binary number may need to be stored in a given number of bits called the word length. Some of the bits in the word would be used for the integer part and the rest of them for the fractional part. In addition, a bit would also be used to signify the sign of the number. This is best illustrated through an example.

### Example 1

Given that a fixed-point positive binary number is stored in eight bits word length, where the first six bits are used for the integer part and the next two for the fractional part. How would the $(13.875)_{10}$ be stored? Given that $(13.875)_{10} = (1101.111)_2$

### Solution

Since there are six bits for the integer part and two for the fractional part, we will first rewrite the number as

$$\begin{aligned}(13.875)_{10} &= (1101.111)_2 \\ &\approx (001101.11)_2\end{aligned}$$

This occupies the right number of bits for the integer part (6 bits) and the fractional part (2 bits). We write 1101 as 001101 to occupy the 6 places without changing it value. We approximate the fractional part .111 to .11 as we have only 2 bits for it.

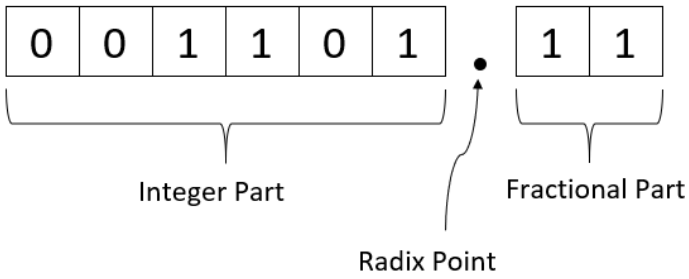



**Figure 1**. A fixed-point format.

# Multiple Choice Test

Chapter 01.04 Fixed-Point Binary Representation of Numbers

(1). $(25)_{10} = (?)_2$

(A) 100110

(B) 10011

(C) 11001

(D) 110010

(2). $(1101)_2 = (?)_{10}$

(A) 3

(B) 13

(C) 15

(D) 26

(3). $(25.375)_{10} = (?.?)_2$

(A) 100110.011

(B) 11001.011

(C) 10011.0011

(D) 10011.110

(4). Representing $\sqrt{2}$ in a fixed-point binary register with 2 bits for the integer part and 3 bits for the fractional part gives a round-off error of most nearly

(A) $-0.085709$

(B) 0.0392

(C) 0.1642

(D) 0.2892

(5). An engineer working for the Department of Defense is writing a program that transfers non-negative real numbers to integer format. To avoid overflow problems, the maximum non-negative integer that can be represented in a 5-bit binary integer word is

(A) 16

(B) 31

(C) 63

(D) 64

(6). For a numerically controlled machine, integers need to be stored in a memory location. The minimum number of bits needed for an integer binary word to represent all integers between 0 and 1024 is

(A) 8

(B) 9

(C) 10

(D) 11

For complete solution, go to

http://nm.mathforcollege.com/mcquizzes/01aae/quiz_01aae_binaryrepresentation_answers.pdf

# Problem Set

Chapter 01.04: Fixed-Point Binary Representation of Numbers

(1). Convert the following
a) $(19)_{10} = (?)_2$
b) $(75)_{10} = (?)_2$

**Answer:** *a*) 100111 *b*) 1001011

(2). Convert the following
a) $(110111)_2 = (?)_{10}$
b) $(11001)_2 = (?)_{10}$

**Answer:** *a*) 55 *b*) 25

(3). Convert the following
a) $(0.375)_{10} = (?)_2$
b) $(0.075)_{10} = (?)_2$

**Answer:** *a*) 0.011 *b*) 0.000100110011...

(4). Convert the following
a) $(0.110001)_2 = (0.?)_{10}$
b) $(0.0111)_2 = (0.?)_{10}$

**Answer:** *a*) 0.765625 *b*) 0.4375

(5). Convert the following
a) $(19.375)_{10} = (?.?)_2$
b) $(75.075)_{10} = (?.?)_2$

**Answer:** *a*) 10011.011 *b*) 1001011.00010011.....

(6). Convert the following
a) $(110111.110001)_2 = (?.?)_{10}$
b) $(11001.0111)_2 = (?.?)_{10}$

**Answer:** *a*) 55.765625 *b*) 25.4375

# Chapter 01.05: Floating-Point Binary Representation of Numbers

Lesson: Floating-Point Representation Background

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *revisit floating-point (scientific) representation for base-10 numbers*

2) *define mantissa, exponent, and base of a base-10 floating-point number*

3) *find the range of numbers that can be represented in fixed-point and floating-point format*

4) *identify that absolute true errors in representing numbers are bounded for the fixed format, while absolute relative true errors in representing numbers are bounded for floating-point format.*

## Introduction

To understand the floating-point representation advantages over fixed-point format in binary representation, we will discuss the base-10 format first.

## Fixed point representation

Fixed-point representation has a fixed number of places for the integer part and a fixed number of places for the fractional part. Take this as an example of fixed-point representation. Consider an old-time cash register that has three places for dollars (integer part) and two places for the cents (fractional part). One could ring purchases between 0 and 999.99 units of money. To keep it simple, we are not choosing negative numbers. Note that there are five (not six) working spaces in the cash register (the decimal number is shown just for clarification).

### How will the smallest number 0 be represented?

The number 0 will be represented as 000.00

### How will the largest number 999.99 be represented?

The number 999.99 will be represented as 999.99

**Now, look at any typical number between 0 and 999.99, say, 256.78. How would it be represented?**

The number 256.78 will be represented as 256.78

**What is the smallest difference between two exactly represented consecutive numbers?**

The difference is always 0.01. For example, the difference between the consecutive numbers that can be represented exactly, say 256.78 and 256.79 is 0.01. This difference is the same for every two consecutive numbers.

**What amount would one pay for an item if it costs 256.789?**

Depending on what method you use for rounding, the amount one would pay would be 256.79 if the number is rounded or 256.78 if the number is chopped.

**What true errors would occur in a transaction?**

No matter what transaction is completed on the above cash register, the maximum magnitude of the true error in the payment would be less than 0.01 if chopping is used and less than 0.05 if rounding is used.

**What relative errors would occur in a transaction?**

The magnitude of relative true errors for representing small numbers is going to be high, while for large numbers, the magnitude of the relative true error is going to be small. Let's illustrate that by choosing several different numbers. We will assume that we are using chopping.

For example, for 256.786, chopping would make it 256.78 and account for a round-off error of $256.786 - 256.78 = 0.006$. The relative true error in this case is

$$\begin{aligned} \varepsilon_t &= \frac{0.006}{256.786} \\ &= 0.000023366. \end{aligned}$$

For another number, 3.546, chopping would make it 3.54 and account for the round-off error of $3.546 - 3.54 = 0.006$. However, the relative true error in this case is

$$\begin{aligned} \varepsilon_t &= \frac{0.006}{3.546} \\ &= 0.0016920 \end{aligned}$$

For another number 0.016, chopping would make it 0.01 and account for the round-off error of $0.016 - 0.01 = 0.006$. However, the relative true error in this case is

$$\begin{aligned} \varepsilon_t &= \frac{0.006}{0.016} \\ &= 0.375. \end{aligned}$$

From the above examples, we see that when we used fixed point (also called decimal for base-10) format like in the cash register, we have control over the magnitude of the true error but not the magnitude of the relative true error.

The magnitude of the true error for any number that can be represented in the above configuration of three places for the integer part and two for the fractional part with chopping would be 0.01, that is, $10^{-p}$ where $p$ is the number of places used for the fractional part.

What would be the upper bound on the absolute true error be if, instead of chopping, rounding is used. It would be 0.005, that is, $0.5 \times 10^{-p}$, where $p$ is the number of places used for the fractional part.

Check this upper bound on the absolute true error out for yourself for a different configuration and different rounding-off choices. That is, choose a certain number of places for the integer and fractional parts, and use rounding or chopping.

**Example 1**

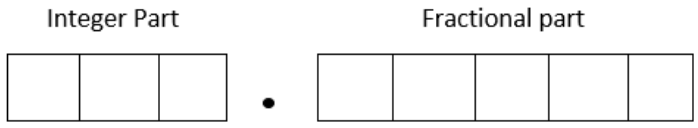
A base-10 non-negative number is represented using the fixed format with 8 places, where 3 places are used for the integer part, and 5 places are used for the fractional part. Chopping is used.

a) What is the upper bound on the absolute true error in representing a number?

b) What is the smallest value that can be represented?

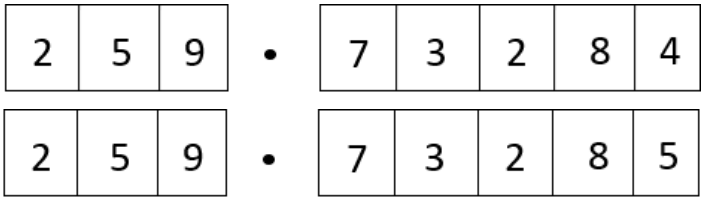
c) What is the largest number that can be represented?

**Solution**




a) The fixed-point format for base-10 numbers is the same as the decimal format. In the example, 3 places are used for the integer part, and 5 places are used for the fractional part. Also, only non-negative numbers (greater than or equal to zero) are represented.

To find the upper bound on the absolute true error in representing a number, take any two consecutive numbers that can be represented exactly. For example,




There are infinite numbers between the two. The number (recall that the line over 9 as $\bar{9}$ stands for 9 being a recurring number)

$$259.73284\bar{9}$$

would create the largest true error as it would get represented as

$$259.73284$$

because chopping is used to represent the numbers. The absolute true error is

$$\begin{aligned} |E_t| &= |259.732849\bar{9} - 259.73284| \\ &= |0.00000\bar{9}| \\ &= 0.00000\bar{9} \end{aligned}$$

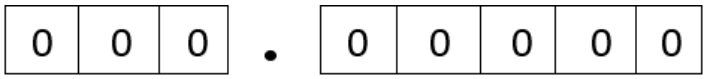
making

$$|E_t| < 0.00001 = 10^{-5}$$

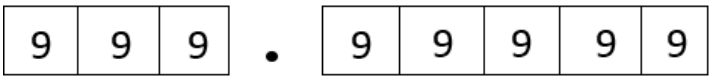
You can try any other two consecutive numbers, and you would find that $|E_t| < 10^{-5}$. Did you recognize that 5 is the number of places used for the fractional part?

Can you repeat the problem when rounding is used?

b) The smallest number that can be represented would have zeros in all the places, that is, 000.00000 and that would be simply zero.




c) The largest that can be represented would have 9's in all places, that is, 999.99999.




## Floating-point representation

**If I am interested in keeping relative true errors similar in magnitude for the range of numbers, what alternatives do I have?**

To keep the relative true error in the representation of numbers to be of a similar order for all numbers, one uses the floating-point representation of the number. You may know it better as the scientific format, but that terminology is only used for base-10 numbers, and that is the case here.

For example, in floating-point representation, a number

$$256.78 \text{ is written as } +2.5678 \times 10^2,$$

$$0.003678 \text{ is written as } +3.678 \times 10^{-3}, \text{ and}$$

$$-256.789 \text{ is written as } -2.56789 \times 10^2.$$

The general representation of a number in base-10 format is given as

$$\text{sign} \times \text{mantissa} \times 10^{\text{sign of exponent} \times \text{magnitude of exponent}}$$

or for a number $y$,

$$y = \sigma \times m \times 10^{\,p \times e}$$

where

$$\sigma = \text{sign of the number, } +1 \text{ or } -1$$

$$m \;=\; \text{mantissa, } 1 \leq \; m < 10$$

$$p = \text{sign of the exponent, } +1 \text{ or } -1$$

$$e \;= \text{integer exponent magnitude (also called ficand)}$$

Let us go back to the example where we had five spaces available for representing a number as in the cash register. Let us limit ourselves to positive numbers as we did previously but now also to positive exponents. Using the same number of five spaces, let us use the first four spaces for the mantissa and the last one for the exponent. The smallest number that can be represented is $1.000 \times 10^0$, and the largest number that can be represented is $9.999 \times 10^9$. The smallest number is not zero as the floating-point representation requires us to put a non-zero number before the decimal point, and the smallest non-zero number is 1. By using the floating-point representation, what we lose in accuracy, we gain in the range of numbers that can be represented. For our example, the range of numbers represented in the five spaces is [0, 999.99] for the fixed format and $[1,\ \ 9.999 \times 10^9]$ for the floating-point (scientific) format.

**How would one represent a number in the floating-point format?**

Take the previous example of 256.78. It would be represented as $2.567 \times 10^2$ if chopping is used and $2.568 \times 10^2$ if rounding is used.

Another example, the number 576329.78, would be represented as $5.763 \times 10^5$ if chopping is used and $5.763 \times 10^5$ if rounding is used.

**So, how much error is caused by the floating-point format representation using chopping?**

In representing, say 256.78, it would be represented as $2.567 \times 10^2$. The round-off error created is

$$E_t = 256.78 - 2.567 \times 10^2 = 0.08,$$

and the relative true error is

$$\varepsilon_t = \frac{0.08}{256.78} = 0.00031155,$$

In representing another number, 576329.78, it would be represented as $5.763 \times 10^5$. The round-off error created is

$$E_t = 576329.78 - 5.763 \times 10^5 = 29.78,$$

and the relative true error is

$$\varepsilon_t = \frac{29.78}{576329.78} = 0.000051672.$$

In representing, 576399.99, it would be represented as $5.763 \times 10^5$. The round-off error created is

$$E_t = 576399.99 - 5.763 \times 10^5 = 99.99,$$

and the relative true error is

$$\varepsilon_t = \frac{99.99}{576399.99} = 0.00017347.$$

In representing, 1.0009999, it would be represented as $1.000 \times 10^0$. The round-off error created is

$$E_t = 1.0009999 - 1.000 \times 10^0 = 0.0009999,$$

and the relative true error is

$$\varepsilon_t = \frac{0.0009999}{1.0009999} = 0.00099890.$$

What you see through these examples about the floating-point format is that although the true errors are large for large numbers and small for small numbers, it is the relative true errors that stay in the similar order for large as well as small numbers. In fact, the magnitude of the relative true error for any number that can be represented in the above configuration of four places for the mantissa and one for the exponent with chopping would be 0.001, that is, $10^{1-p}$ , where $p$ is the number of places used for the mantissa.

What would the upper bound on the absolute relative true error be if, instead of chopping, rounding is used. It would be 0.0005, that is, $0.5 \times 10^{1-p}$ , where $p$ is the number of places used for the mantissa.

Check this upper bound on the absolute relative true error out for yourself for a different configuration and different rounding-off choices. That is, choose a certain number of places for the mantissa and exponential parts, and use rounding or chopping.

**Example 2**

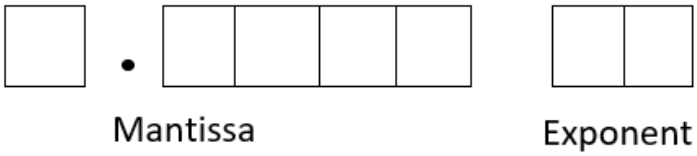
A base-10 non-negative number is represented using floating-point format with only non-negative exponents and 8 places, where 5 places are used for the mantissa and 2 places are used for the exponent part. Chopping is used.

a) What is the upper bound on the absolute relative true error in representing a number?

b) What is the smallest value that can be represented?

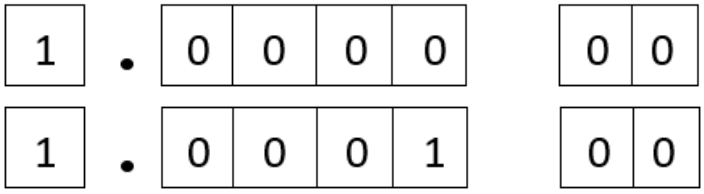
c) What is the largest number that can be represented?

**Solution**




a) The floating point format for base-10 numbers is the same as the scientific format. 5 places are used for the mantissa and 2 places are used for the exponent. The numbers are non-negative (greater than or equal to zero) and the exponents are non-negative (greater than or equal to zero).

To find the upper bound on the absolute relative true error in representing a number, take two consecutive numbers




There are infinite numbers between the two. The number

$$1.0000\bar{9} \times 10^0$$

would create the largest true error as it would be represented as

$$1.0000 \times 10^0$$

as chopping is used.

The absolute true error is

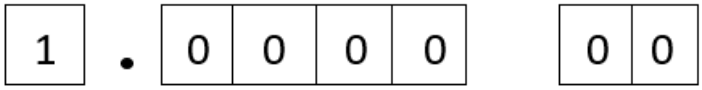
$$\begin{aligned} |E_t| &= |1.0000\bar{9} \times 10^0 - 1.0000 \times 10^0| \\ &= 0.0000\bar{9} \end{aligned}$$

The absolute relative true error is

$$\begin{aligned} |\epsilon_t| &= \left| \frac{1.0000\bar{9} \times 10^0 - 1.0000 \times 10^0}{1.0000\bar{9} \times 10^0} \right| \\ &= 0.000099998 \\ &< 0.0001 \\ &= 10^{-4} \end{aligned}$$

Any other number other than the one shown above would also yield an absolute relative true error which is less than $10^{-4}$, and hence $10^{-4}$ is an upper limit on the absolute relative true error in representing a number. Did you recognize that $|\epsilon_t| < 10^{-4}$ for all numbers where $-4 = 1 - 5$, and 5 is the number of places used for the magnitude of the mantissa?

b) The smallest number that can be represented is




which is simply

$$1.0000 \times 10^0$$
$$= 1$$

The scientific notation requires us to put a non-zero number before the decimal point, and the smallest non-zero number is 1.

c) The largest number that can be represented is

which is

$$9.9999 \times 10^{99}$$

Can you repeat the problem when rounding is used?

# Chapter 01.05: Floating-Point Binary Representation of Numbers

Lesson: Introduction to Binary Floating-Point Representation of Numbers

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *convert a base-*10 *number to a binary floating-point representation*

2) *convert a binary floating-point number to its equivalent base-*10 *number*

## How does the floating-point format work for binary representation?

A number $y$ would be written as

$$y = \sigma \times m \times 2^{p \times e}$$

where

$$\sigma = \text{sign of number (negative or positive)},$$

$$m = \text{magnitude of mantissa, } (1)_2 \leq m < (10)_2 \text{, that is, } (1)_{10} \leq m < (2)_{10},$$

$$p = \text{sign of the exponent (negative or positive)}$$

$$e = \text{magnitude of the exponent (an integer)}.$$

So, how will the number be stored in a binary floating-point format? Let us suppose a real number is represented in an 8-bit format (this is hypothetical and is chosen to keep the explanation simple) in a computer, and the architecture used by it is as follows – first bit is used for the sign of the number, second bit for the sign of the exponent, next four bits for the magnitude of the mantissa and the last two bits for the exponent.

In the first bit, you will put a 0 in it if the number is positive and 1 if it is negative.

In the second bit, you will put a 0 if the exponent is positive and 1 if it is negative.

In the next four bits, you will put the magnitude of the mantissa less one. Why? Since the magnitude of the mantissa is always greater than or equal to one and strictly less than 2, the leading 1 is not stored in the four bits of the mantissa, and only the digits after the radix point are stored.

In the next two bits, you will put the magnitude of the exponent.

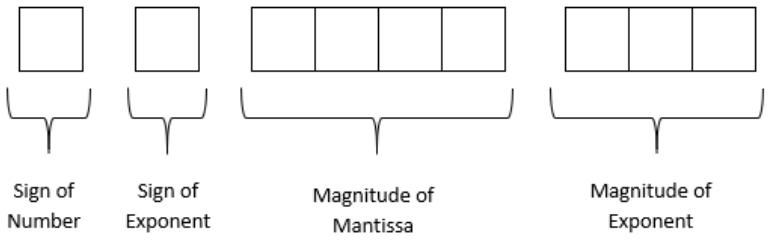
Looking at the example will make it clear how a number is represented in floating-point binary format.

**Example 1**

Represent $(54.75)_{10}$ in floating-point binary format. Assume that the number is written to a hypothetical format that is 9 bits long where the first bit is used for the sign of the number, the second bit for the sign of the exponent, the next four bits for the magnitude of the mantissa, and the next three bits for the magnitude of the exponent.

**Solution**

The format for the use of the 9-bits is as follows




Following the procedure in Chapter 01.04, the fixed-point binary format of the given base-10 number is

$$(54.75)_{10} = (110110.11)_2$$

Writing the number in floating-point format, we get

$$(54.75)_{10} = (1.1011011)_2 \times 2^{(5)_{10}}$$

The exponent 5 is equivalent in binary format as

$$(5)_{10} = (101)_2$$

Hence

$$(54.75)_{10} = (1.1011011)_2 \times 2^{(101)_2}$$

The sign of the number is positive, so the bit for the sign of the number will have zero in it.

The sign of the exponent is positive. So, the bit for the sign of the exponent will have zero in it.

The magnitude of mantissa will be

$$m = 1011$$

(There are only 4 places for the mantissa, and the leading 1 is not stored as it is always 1 by default).

The magnitude of the exponent is

$$e = 101,$$

We now have the representation as

| 0 | 0 | 1 | 0 | 1 | 1 | 1 | 0 | 1 |
|---|---|---|---|---|---|---|---|---|

**Example 2**

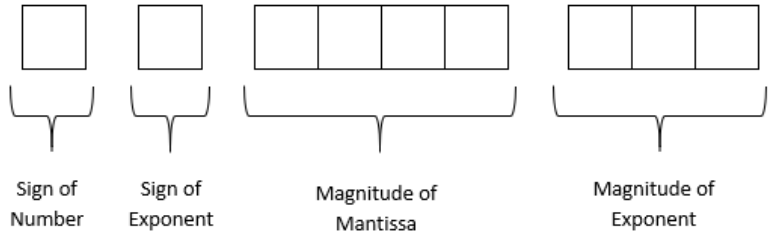
What number does the below given floating point format represent in base-10 format?

| 0 | 1 | 1 | 0 | 1 | 1 | 1 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|

Assume a hypothetical 9-bit word, where the first bit is used for the sign of the number, the second bit for the sign of the exponent, the next four bits for the magnitude of mantissa and the next three for the magnitude of the exponent.

**Solution**

The format for the use of the 9-bits is as follows




Given

| Bit Representation | Part of Floating-point number |
|---|---|
| 0 | Sign of number |
| 1 | Sign of exponent |
| 1011 | Magnitude of mantissa |
| 110 | Magnitude of exponent |

The first bit is 0, so the number is positive.

The second bit is 1, so the exponent is negative.

The next four bits, 1011, represent the magnitude of the mantissa.

The next three bits, 110, represent the magnitude of the exponent.

The number in binary format hence is

$$(1.1011)_2 \times 2^{-(110)_2}$$

The magnitude of the mantissa is

$$\begin{aligned} m &= (1.1011)_2 \\ &= \left(1 \times 2^0 + 1 \times 2^{-1} + 0 \times 2^{-2} + 1 \times 2^{-3} + 1 \times 2^{-4}\right)_{10} \\ &= (1.6875)_{10} \end{aligned}$$

The magnitude of the exponent is

$$\begin{aligned} e &= (110)_2 \\ &= \left(1 \times 2^2 + 1 \times 2^1 + 0 \times 2^0\right)_{10} \\ &= (6)_{10} \end{aligned}$$

Hence, the number $(1.1011)_2 \times 2^{-(110)_2}$ in base-10 format is

$$\begin{aligned} &= \left(1.6875 \times 2^{-6}\right)_{10} \\ &= (0.026367)_{10} \end{aligned}$$

**Example 3**

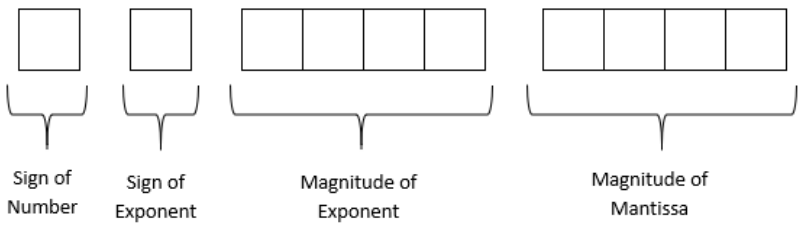
A machine stores floating-point numbers in a hypothetical 10-bit binary word. It employs the first bit for the sign of the number, the second one for the sign of the exponent, the next four bits for the magnitude of the exponent, and the last four for the magnitude of the mantissa.

a) Find how 0.02832 will be represented in the floating-point 10-bit word.

b) What is the decimal equivalent of the 10-bit word representation of part (a)?

**Solution**

The format for the use of the 10-bits is as follows




a) For the number, we have the integer part as 0, and the fractional part as 0.02832.

Let us first find the binary equivalent of the integer part

Integer part

$$(0)_{10} = (0)_2$$

Now we find the binary equivalent of the fractional part

| | Number | Number after decimal | Number before decimal |
|---|---|---|---|
| $0.02832 \times 2$ | 0.05664 | 0.05664 | $0 = a_{-1}$ |
| $0.05664 \times 2$ | 0.11328 | 0.11328 | $0 = a_{-2}$ |
| $0.11328 \times 2$ | 0.22656 | 0.22656 | $0 = a_{-3}$ |
| $0.22656 \times 2$ | 0.45312 | 0.45312 | $0 = a_{-4}$ |
| $0.45312 \times 2$ | 0.90624 | 0.90624 | $0 = a_{-5}$ |
| $0.90624 \times 2$ | 1.81248 | 0.81248 | $1 = a_{-6}$ |
| $0.81248 \times 2$ | 1.62496 | 0.62496 | $1 = a_{-7}$ |
| $0.62496 \times 2$ | 1.24992 | 0.24992 | $1 = a_{-8}$ |
| $0.24992 \times 2$ | 0.49984 | 0.49984 | $0 = a_{-9}$ |
| $0.49984 \times 2$ | 0.99968 | 0.99968 | $0 = a_{-10}$ |
| $0.99968 \times 2$ | 1.99936 | 0.99936 | $1 = a_{-11}$ |

Hence

$$\begin{aligned}(0.02832)_{10} &\cong (0.00000111001)_2 \\ &= (1.11001)_2 \times 2^{-6} \\ &\cong (1.1100)_2 \times 2^{-6}\end{aligned}$$

The binary equivalent of the magnitude of the exponent 6 is found as follows

| | Quotient | Remainder |
|---|---|---|
| 6/2 | 3 | $0 = a_0$ |
| 3/2 | 1 | $1 = a_1$ |
| 1/2 | 0 | $1 = a_2$ |

So

$$(6)_{10} = (110)_2$$

Hence

$$\begin{aligned}(0.02832)_{10} &= (1.1100)_2 \times 2^{-(110)_2} \\ &= (1.1100)_2 \times 2^{-(0110)_2}\end{aligned}$$

| Part of Floating-point number | Bit Representation |
|---|---|
| Sign of number is positive | 0 |
| Sign of exponent is negative | 1 |
| Magnitude of the exponent | 0110 |
| Magnitude of mantissa | 1100 |

The ten-bit representation bit by bit is

| 0 | 1 | 0 | 1 | 1 | 0 | 1 | 1 | 0 | 0 |
|---|---|---|---|---|---|---|---|---|---|

b) Converting the above floating-point representation from part (a) to base 10 by following Example 2 gives

$$
\begin{aligned}
&(1.1100)_2 \times 2^{-(0110)_2} \\
&= \left(1 \times 2^0 + 1 \times 2^{-1} + 1 \times 2^{-2} + 0 \times 2^{-3} + 0 \times 2^{-4}\right) \times 2^{-\left(0\times 2^3 + 1\times 2^2 + 1\times 2^1 + 0\times 2^0\right)} \\
&= (1.75)_{10} \times 2^{-(6)_{10}} \\
&= 0.02734375
\end{aligned}
$$

which is different from the original number which was 0.02832.

**Example 4**

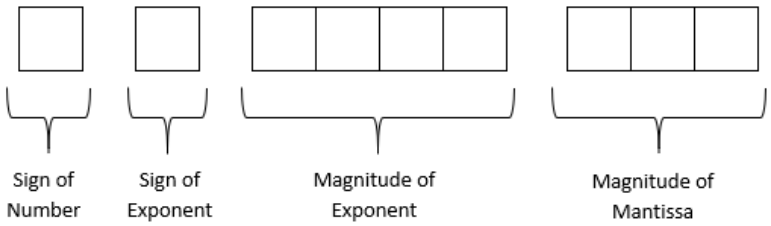
A machine stores a floating-point number in a hypothetical 9-bit word. It uses the first bit for the sign of the number, the second bit for the sign of the exponent, the next four bits for the magnitude of the exponent, and the last 3 bits for the magnitude of the mantissa.

a) What is the smallest base-10 positive number that can be represented in the given format?

b) What is the largest base-10 positive number that can be represented in the given format?

**Solution**

a) The format for the use of the 9-bits is as follows

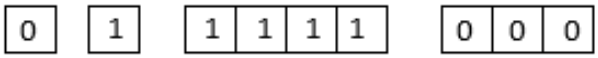



Since we are looking for the smallest positive number, the first bit for the sign of the number would be 0 as the number is positive. The second bit for the sign of the exponent would be 1 as that would give a negative exponent of 2, and hence result in a smaller number.

The next four bits would be filled with 1's because that will give us the largest exponent by the magnitude, and with the second bit being 1, it would make the exponent negative.

The next three bits of the magnitude of mantissa would be filled with 0's for the smallest number.

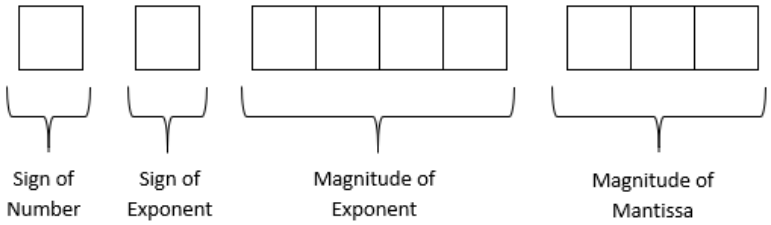
So the 9 bits would be




giving us

$$
\begin{aligned}
&+ (1.000)_2 \times 2^{-(1111)_2} \\
&= (1 \times 2^0) \times 2^{-(1\times2^3+1\times2^2+1\times2^1+1\times2^0)} \\
&= 1 \times 2^{-(8+4+2+1)} \\
&= 2^{-15} \\
&= 0.0000305176
\end{aligned}
$$

b) The format is as follows

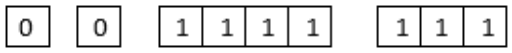



Since we are looking for the largest positive number, the first bit for the sign of the number would be 0 as the number is positive. The second bit for the sign of the exponent would be 0 as that would give a positive exponent of 2, and result in a larger number.

The next four bits would be filled with 1's because that will give us the largest exponent by the magnitude, and with the second bit being 0, it would be a positive exponent.

The next three bits for the magnitude of the mantissa would be filled with 1's for the largest number.

So the 9 bits would be




giving us

$$
\begin{aligned}
&+ (1.111)_2 \times 2^{+(1111)_2} \\
&= (1 \times 2^0 + 1 \times 2^{-1} + 1 \times 2^{-2} + 1 \times 2^{-3}) \times 2^{(1\times2^3+1\times2^2+1\times2^1+1\times2^0)} \\
&= 1.875 \times 2^{(15)} \\
&= 28672.0
\end{aligned}
$$

# Chapter 01.05: Floating-Point Binary Representation of Numbers

Lesson: Binary Floating-Point Representation of Numbers with Biased Exponent

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *convert a base-*10 *number to a binary floating-point representation with a biased exponent*

2) *convert a binary floating-point number with a biased exponent to its equivalent base-*10 *number*

## What is the floating-point format with a *biased* exponent for binary format? How is it different from the floating-point format with an *unbiased* exponent?

A number $y$ would be written as

$$y = \sigma \times m \times 2^e$$

where

$$\sigma = \text{sign of the number (negative or positive - use 0 for positive and 1 for negative),}$$

$$m = \text{magnitude of mantissa, } (1)_2 \leq m < (10)_2 \text{, that is, } (1)_{10} \leq m < (2)_{10} \text{, and}$$

$$e = \text{integer } \textit{biased} \text{ exponent.}$$

The only difference between the previous lesson of the binary floating-point format is that the exponent was unbiased in the previous lesson. In the unbiased exponent format, one bit was used for the sign of the exponent, and several bits were used for the magnitude of the exponent. In the biased exponent format, no bit is used for the sign of the exponent, and that bit is added to be used for the magnitude of the biased exponent. The reasons for using biased exponent are varied, but it is sufficient to say that it reduces computational effort and increases the range of representable numbers.

Let us suppose we are using 3 bits for the magnitude of the unbiased exponent and 1 bit for the sign of the exponent. For the unbiased exponent, we would then use 4 bits. But if there is no bit used for the sign of the number, how would we represent negative exponents. The following is how it is done.

The biased exponent is equally divided between negative and positive exponents. For example, we have four bits now for the biased exponent, and hence the minimum base-10 possible number that can be represented is

$$(0000)_2 = (0)_{10}$$

and the maximum base-10 possible number is

$$(1111)_2 = (15)_{10}.$$

Half of the maximum value is 7.5, and the integer part of it is 7. Hence, we will bias the unbiased exponent by 7 (add 7) and then store that value. When we have to unpack the exponent to its actual value, we will unbias it by 7 (subtract 7) from it. For example, if the following binary number

$$y = (1.011)_2 \times 2^{-(110)_2}$$

is represented with 4 bits for the biased exponent, we would proceed as follows. The exponent $-(110)_2$ is equivalent to $-(6)_{10}$. So, the biased exponent becomes

$$-6 + 7 = 1.$$

Then

$$(1)_{10} = (0001)_2.$$

Hence we would store the biased exponent in binary as 0001. If we had to unpack it, the biased exponent would be

$$(0001)_2 = (1)_{10}$$

and the unbiased exponent in base-10 would be

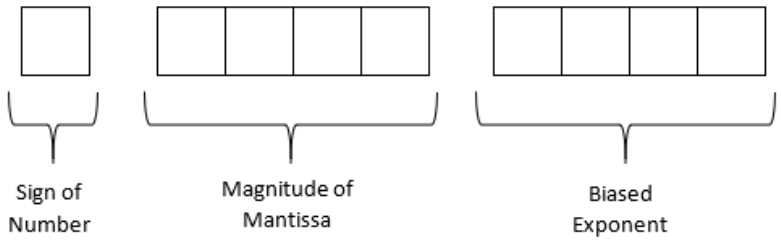
$$1 - 7 = -6.$$

**Example 1**

Represent $(54.75)_{10}$ in floating point binary format. Assume that the number is written to a hypothetical word that is 9 bits long where the first bit is used for the sign of the number, the next four bits for the magnitude of the mantissa, and the next four bits for the magnitude of the biased exponent.

**Solution**

The format for the use of 9 bits is as follows




Following procedures in Chapter 01.04, we get the fixed-point binary equivalent as

$$(54.75)_{10} = (110110.11)_2$$

Writing the number in floating-point format, we get

$$(54.75)_{10} = (1.1011011)_2 \times 2^{(5)_{10}}$$

Since the number of bits used for the biased exponent is 4, the bias would be by integer part of half of $(1111)_2$ to give $int(15/2) = (7)_{10}$. The biased exponent hence in base-10 is $5 + 7 = 12$

The biased exponent 12 is equivalent in binary format as

$$(12)_{10} = (1100)_2$$

Hence

$$(54.75)_{10} = (1.1011011)_2 \times 2^{(1100)_2} \text{ (with biased exponent)}$$

The sign of the number is positive, so the bit for the sign of the number will have zero in it.

$$\sigma = 0$$

There are only 4 places for the mantissa, and the leading 1 is not stored as it is always expected to be there. The magnitude of mantissa will be represented as

$$m = 1011$$

(There are only 4 places for the mantissa, and the leading 1 is not stored as it is always 1 by default).

The magnitude of the biased exponent

$$e = 1100,$$

and we have the representation as

| 0 | 1 | 0 | 1 | 1 | 1 | 1 | 0 | 0 |
|---|---|---|---|---|---|---|---|---|

**Example 2**

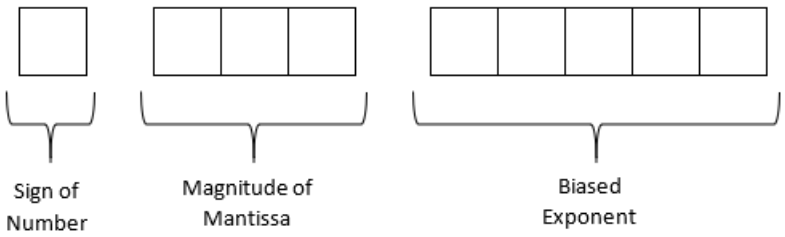
What number does the below-given floating-point format with a biased exponent represent in base-10 format?

| 1 | 1 | 1 | 0 | 1 | 1 | 0 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|

Assume a hypothetical 9-bit word, where the first bit is used for the sign of the number, the next three bits for the magnitude of mantissa, and the next five for the magnitude of the biased exponent.

**Solution**

The format for the use of 9 bits is as follows




Given

| Bit Representation | Part of Floating point number |
|---|---|
| 1 | The sign of the number |
| 110 | The magnitude of the mantissa |
| 11010 | The magnitude of the biased exponent |

The first bit is 1, so the number is negative.

The next three bits, 110, are the magnitude of the mantissa.

The next five bits, 11010, are for the biased exponent

Because there are five bits for the biased exponent, the bias would be by integer part of half of $(11111)_2$ to give

$$int(31/2) = (15)_{10}\,.$$

The biased exponent is

$$(11010)_2 = (26)_{10}\,,$$

and hence the unbiased (actual) exponent is

$$26 - 15 = 11 \text{ in base-10.}$$

Now

$$(11)_{10} = (1011)_2$$

The number in binary format hence is

$$-(1.110)_2 \times 2^{(1011)_2}$$

The magnitude of the mantissa is

$$\begin{aligned} m &= (1.110)_2 \\ &= (1 \times 2^0 + 1 \times 2^{-1} + 1 \times 2^{-2} + 0 \times 2^{-3})_{10} \\ &= (1.75)_{10} \end{aligned}$$

The magnitude of the exponent is

$$\begin{aligned} e &= (1011)_2 \\ &= 11 \end{aligned}$$

The number in the base-10 format hence is

$$\begin{aligned} &= -1.75 \times 2^{11} \\ &= -3584 \end{aligned}$$

## Appendix

**How are numbers represented in floating-point in a real computer?**

In an actual typical computer, a real number is stored as per the IEEE-754 (Institute of Electrical and Electronics Engineers) floating-point arithmetic format. To keep the discussion short and simple, let us point out the salient features of the single-precision format.

- A single-precision number uses 32 bits.
- A number $y$ is represented as

$$y = \sigma \times (1.a_1 a_2 \ldots a_{23}) \cdot 2^e$$

where

$$\sigma = \text{sign of the number (positive or negative)}$$

$$a_i = \text{entries of the mantissa can be only 0 or 1 for } i = 1, ..., 23$$

$$e = \text{the exponent}$$

Note the 1 before the radix point.

- The first bit represents the sign of the number (0 for a positive number and 1 for a negative number).

- The next eight bits represent the biased exponent. Note that there is no separate bit for the sign of the exponent. The sign of the exponent is taken care of by normalizing by adding 127 to the actual exponent. For example, in the previous example, the exponent was 6. It would be stored as the binary equivalent of $127 + 6 = 133$. Why is 127 and not some other number added to the actual exponent? Because in eight bits, the largest integer that can be represented is $(11111111)_2 = 255$, and halfway of 255 is 127 (integer part of 255/2). This allows negative and positive exponents to be represented equally. The normalized (also called biased) exponent has the range from 0 to 255, and hence the exponent e has the range of $-127 \leq e \leq 128$.

- If instead of using the biased exponent, let us suppose we still used eight bits for the exponent but used one bit for the sign of the exponent and seven bits for the exponent magnitude. In seven bits, the largest integer that can be represented is $(1111111)_2 = 127$, in which case the exponent $e$ range would have been smaller, that is,$-127 \leq e \leq 127$. By biasing the exponent, the unnecessary representation of a negative zero and positive zero exponents (which are the same) is also avoided.

- Actually, the biased exponent range used in the IEEE-754 format is not 0 to 255, but 1 to 254. Hence, exponent $e$ has a range of $-126 \leq e \leq 127$. So what are $e = -127$ and $e = 128$ used for? If $e = 128$ and all the mantissa entries are zeros, the number is $\pm\infty$ (the sign of infinity is governed by the sign bit), if $e = 128$ and the mantissa entries are not zero, the number being represented is Not a Number (NaN). Because of the leading 1 in the floating-point representation, the number zero cannot be represented exactly. That is why the number zero (0) is represented by $e = -127$ and all the mantissa entries being zero.

- The next twenty-three bits are used for the magnitude of the mantissa.

- The largest number by magnitude that is represented by this format is

$$\left(1 \times 2^0 + 1 \times 2^{-1} + 1 \times 2^{-2} + \ldots + 1 \times 2^{-22} + 1 \times 2^{-23}\right) \times 2^{127} = 3.40 \times 10^{38}$$

The smallest number by magnitude that is represented, other than zero, is

$$\left(1 \times 2^0 + 0 \times 2^{-1} + 0 \times 2^{-2} + \ldots + 0 \times 2^{-22} + 0 \times 2^{-23}\right) \times 2^{-126} = 1.18 \times 10^{-38}$$

Since 23 bits are used for the mantissa, the machine epsilon,

$$\begin{aligned} \epsilon_{mach} &= 2^{-23} \\ &= 1.19 \times 10^{-7}. \end{aligned}$$

## How are numbers represented in floating-point in double precision in a computer?

In double-precision of IEEE-754 format, a real number is stored in 64 bits.

- The first bit is used for the sign,

- the next 11 bits are used for the exponent, and

- the rest of the bits that are 52 are used for the mantissa.

So here are some questions that you may want to ask yourself? For double-precision format, can you find the following?

- the range of the biased exponent,
- the range of the corresponding unbiased exponent
- the smallest number that can be represented,
- the largest number that can be represented, and
- machine epsilon

# Chapter 01.05: Floating-Point Binary Representation of Numbers

Lesson: Accuracy of Binary Floating-Point Representation of Numbers

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *define machine epsilon*

2) *relate the round-off error in representing numbers to machine epsilon*

## How do you determine the accuracy of a floating-point binary format number?

The machine epsilon, $\epsilon_{mach}$ is a measure of the accuracy of a floating-point representation and is found by calculating the difference between 1 and the next number that can be represented as a machine number. That is machine epsilon, $\epsilon_{mach}$ is defined as the smallest number such that

$$1 + \epsilon_{mach} > 1$$

For example, assume a 10-bit hypothetical computer where the first bit is used for the sign of the number, the second bit for the sign of the exponent, the next four bits for the magnitude of the exponent and the next four for the magnitude of the mantissa.

We represent $(1)_{10}$ as $(1.0000)_2 \times 2^{(0000)_2}$ and hence in the 10-bits is represented as

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

The next higher number would be $(1.0001)_2 \times 2^{(0000)_2}$and that will be represented in the 10-bits as

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 |

The difference between the two numbers is

$$\begin{aligned}
&(1.0001)_2 \times 2^{(0000)_2} - (1.0000)_2 \times 2^{(0000)_2} \\
&= (0.0001)_2 \times 2^{(0000)_2} \\
&= (0 \times 2^{-1} + 0 \times 2^{-2} + 0 \times 2^{-3} + 1 \times 2^{-4})_{10} \times 2^{(0)_{10}} \\
&= (0.0625)_{10} \times (1)_{10} \\
&= (0.0625)_{10}.
\end{aligned}$$

The machine epsilon is

$$\epsilon_{mach} = 0.0625.$$

The machine epsilon, $\epsilon_{mach}$ can also simply be calculated as two to the negative power of the number of bits used for the mantissa.

$$\begin{aligned}\epsilon_{mach} &= 2^{-bits\ used\ for\ magnitude\ of\ mantissa} \\ &= 2^{-4} \\ &= 0.0625\end{aligned}$$

## What is the significance of machine epsilon for a student in an introductory course in numerical methods?

The machine epsilon is an upper bound on the absolute relative true error in representing a number. Let $y$ be the machine number representation of a number $x$; then we can show that the absolute relative true error in the representation is

$$\left|\frac{x-y}{x}\right| \leq \epsilon_{mach}$$

An example to illustrate follows.

### Example 1

A machine stores floating-point numbers in a hypothetical 10-bit binary word. It employs the first bit for the sign of the number, the second one for the sign of the exponent, the next four bits for the magnitude of the exponent, and the last four bits for the magnitude of the mantissa. Confirm that the magnitude of the relative true error that results from the approximate representation of 0.02832 in the 10-bit format is less than the machine epsilon.

**Solution**

From an example in an earlier lesson, the ten-bit representation of 0.02832 bit-by-bit for the above architecture is

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 0 | 1 | 0 | 1 | 1 | 0 | 1 | 1 | 0 | 0 |

Again from the same example, converting the above floating-point representation to base-10 gives

$$\begin{aligned}&(1.1100)_2 \times 2^{-(0110)_2} \\ &= (1.75)_{10} \times 2^{-(6)_{10}} \\ &= (0.02734375)_{10}\end{aligned}$$

The absolute relative true error between the base-10 numbers 0.02832 and its approximate representation 0.02734375 is

$$
\begin{aligned}
|\varepsilon_t| &= \left| \frac{0.02832 - 0.02734375}{0.02832} \right| \\
&= 0.034472
\end{aligned}
$$

which is less than the machine epsilon for a computer that uses 4 bits for the magnitude of mantissa, that is,

$$
\begin{aligned}
\varepsilon_{mach} &= 2^{-4} \\
&= 0.0625
\end{aligned}
$$

Take any other number that can be represented in the above format, and you will find that the absolute relative true error will always follow this bound, $0 \leq |\varepsilon_t| < \varepsilon_{mach}$ .

## Appendix

**What is the proof that the absolute relative true error in representing a number on a machine is always less than the machine epsilon?**

That is, prove that if $y$ is the machine number representation of a number $x$, then show that the absolute relative true error in the representation is

$$\left| \frac{x - y}{x} \right| \leq \epsilon_{mach}$$

**Proof:**

Let's limit the proof here to positive numbers and the use of chopping for machine representation.

If $x$ is a positive number we want to represent, it will be between a machine number $x_b$ below $x$ and a machine number $x_u$ above $x$.

If

$$x_b = (1.b_1 b_2 \ldots b_m)_2 \times 2^k$$

where

$$m = \text{number of bits used for the magnitude of the mantissa, then}$$

$$
\begin{aligned}
x_u &= [(1.b_1 b_2 \ldots b_m)_2 + (0.00 \ldots 1)_2] \times 2^k \\
&= \left[(1.b_1 b_2 \ldots b_m)_2 + 2^{-m}\right] \times 2^k \\
&= (1.b_1 b_2 \ldots b_m)_2 \times 2^k + 2^{-m} \times 2^k \\
&= (1.b_1 b_2 \ldots b_m)_2 \times 2^k + 2^{-m+k}
\end{aligned}
$$

Since $x$ is getting represented either as $x_b$ or $x_u$,

$$
\begin{aligned}
|x - y| &\leq |x_b - x_u| \\
&= 2^{-m+k}
\end{aligned}
$$

$$
\begin{aligned}
\left|\frac{x-y}{x}\right| &\leq \frac{2^{-m+k}}{x} \\
&\leq \frac{2^{-m+k}}{x_b} \\
&= \frac{2^{-m+k}}{(1 \cdot b_1 b_2 \ldots b_m)_2 2^k} \\
&= \frac{2^{-m}}{(1 \cdot b_1 b_2 \ldots b_m)_2} \\
&\leq 2^{-m} = \varepsilon_{mach}
\end{aligned}
$$

The above proof is for positive numbers that use chopping for machine representation. We can prove the same for negative numbers or when rounding is used for machine representation.

# Multiple Choice Test

Chapter 01.05 Floating-Point Binary Representation of Numbers

(1). A hypothetical computer stores real numbers in binary floating-point format in 8-bit words. The first bit is used for the sign of the number, the second bit for the sign of the exponent, the next two bits for the magnitude of the exponent, and the next four bits for the magnitude of the mantissa. Represent $e \approx 2.718$ in the 8-bit format.

(A) 00010101

(B) 00011010

(C) 00010011

(D) 00101010

(2). A hypothetical computer stores real numbers in binary floating-point format in 8-bit words. The first bit is used for the sign of the number, the second bit for the sign of the exponent, the next two bits for the magnitude of the exponent, and the next four bits for the magnitude of the mantissa. What is the base-10 number represented by $(10100111)_2$?

(A) $-5.75$

(B) $-2.875$

(C) $-1.75$

(D) $-0.359375$

(3). A hypothetical computer stores floating-point numbers in 8-bit binary words. The first bit is used for the sign of the number, the second bit for the sign of the exponent, the next two bits for the magnitude of the exponent, and the next four bits for the magnitude of the mantissa. The machine epsilon is most nearly

(A) $2^{-8}$

(B) $2^{-4}$

(C) $2^{-3}$

(D) $2^{-2}$

(4). A machine stores floating-point numbers in 7-bit binary word. The first bit is used for the sign of the number, the next three for the biased exponent, and the next three for the magnitude of the mantissa. The number $(0010110)_2$ represented in base-10 is

(A) 0.375

(B) 0.875

(C) 1.5

(D) 3.5

(5). A machine stores floating-point numbers in 7-bit binary word. The first bit is stored for the sign of the number, the next three for the biased exponent, and the next three for the magnitude of the mantissa. You are asked to represent 33.35 in the above word. The error you will get in this case would be

(A) underflow

(B) overflow

(C) NaN

(D) No error will be registered

(6). A hypothetical computer stores floating-point numbers in 9-bit binary words. The first bit is used for the sign of the number, the second bit for the sign of the exponent, the next three bits for the magnitude of the exponent, and the next four bits for the magnitude of the mantissa. Every second, the error between 0.1 and its binary representation in the 9-bit word is accumulated. The accumulated error in after one day most nearly is

(A) 0.002344

(B) 20.25

(C) 202.5

(D) 8640

For complete solution, go to

http://nm.mathforcollege.com/mcquizzes/01aae/quiz_01aae_floatingpoint_answers.pdf

# Problem Set

## Chapter 01.05: Floating-Point Binary Representation of Numbers

(1). A hypothetical computer stores real numbers in floating-point format in 8-bit binary words. The first bit is used for the sign of the number, the second bit for the sign of the exponent, the next two bits for the magnitude of the exponent, and the next four bits for the magnitude of the mantissa. Represent 3.1415 in the 8-bit format.

**Answer:** 0 0 0 1 1 0 0 1

(2). A hypothetical computer stores real numbers in floating-point format in 8-bit binary words. The first bit is used for the sign of the number, the second bit for the sign of the exponent, the next two bits for the magnitude of the exponent, and the next four bits for the magnitude of the mantissa. What number does 10101111 represent in the above given 8-bit format?

**Answer:** $-7.75$

(3). A hypothetical computer stores real numbers in floating-point format in 10-bit binary words. The first bit is used for the sign of the number, the second bit for the sign of the exponent, the next three bits for the magnitude of the exponent, and the next five bits for the magnitude of the mantissa. Represent $-0.0456$ in the 10-bit format.

**Answer:** 1 1 1 0 1 0 1 1 1 0

(4). A hypothetical computer stores real numbers in floating-point format in 10-bit binary words. The first bit is used for the sign of the number, the second bit for the sign of the exponent, the next three bits for the magnitude of the exponent, and the next five bits for the magnitude of the mantissa. What number does 1011010011 represent in the above given 10-bit format?

**Answer:** $-102$

(5). A machine stores floating-point numbers in 7-bit binary words. Employ first bit for the sign of the number, second bit for the sign of the exponent, next two bits for the magnitude of the exponent, and the last three bits for the magnitude of the mantissa.

a) By magnitude, what are the smallest negative and positive numbers in the system?

b) By magnitude, what are the largest negative and positive numbers in the system?

c) What is the machine epsilon?

d) Represent $e^1$ in the 7-bit format.

e) Represent 3.623 in the 7-bit format.

f) What is the next higher number, $x_2$ after $x_1$= 0110110 in the 7-bit format.

g) Find $\left|\dfrac{x_2 - x_1}{x_1}\right|$ from part (f) and compare with the machine epsilon.

**Answer:** *a*) $-0.125,\ 0.125$ *b*) $-15,\ 15$ *c*) 0.125 *d*) 0 0 0 1 0 1 0 *e*) 0 0 0 1 1 1 0 *f*) 0.46875
*g*) 0.07142. This value is less than the machine epsilon of 0.125. The absolute relative difference between consecutive numbers is always going to be less than the machine epsilon.

# Chapter 01.07: Taylor Theorem Revisited

Lesson: Introduction to Taylor Series

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *recall the formula for Taylor's theorem (pre-requisite to course), express Taylor's theorem in plain language, (pre-requisite to course)*

2) *apply Taylor's theorem to calculate a function at a point when the value of the function and its derivatives is given at another point.*

## Introduction

The use of the Taylor series exists in so many aspects of numerical methods that it is imperative to devote a separate chapter to reviewing it. You must have come across expressions such as

$$\cos(x) = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots \qquad (1)$$

$$\sin(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots \qquad (2)$$

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots \qquad (3)$$

All the above expressions are a special case of the Taylor series called the Maclaurin series.

## Why are applications of Taylor's theorem important for numerical methods?

Firstly, expressions such as given in Equations (1), (2), and (3) give you a way to find the approximate values of these functions by using the basic arithmetic operations of addition, subtraction, division, and multiplication.

Second, we will also see how numerical methods formulas can be derived from the Taylor series. Examples include finding the numerical formulas for derivatives of continuous and discrete functions and roots of nonlinear equations.

Third, the Taylor series also allows us to find the order of accuracy of numerical methods formulas, which in turn helps us to derive numerical methods that are even more computationally efficient. Examples in the course include the derivation of Richardson's extrapolation formulas for the first derivative approximation and numerical integration. The extrapolation formulas make the numerical methods even more efficient in these cases.

For this chapter, we will concentrate on a few introductory examples. Let's see how one can approximately find the value of $e^{0.25}$ using a Maclaurin series (a polynomial), and hence involves only the basic four arithmetic operations.

**Example 1**

Find the value of $e^{0.25}$ using the first five terms of the Maclaurin series.

**Solution**

The first five terms of the Maclaurin series for $e^x$ are

$$\begin{aligned} e^x &\approx 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} \\ e^{0.25} &\approx 1 + 0.25 + \frac{0.25^2}{2!} + \frac{0.25^3}{3!} + \frac{0.25^4}{4!} \\ &= 1.2840 \end{aligned}$$

The exact value of $e^{0.25}$ up to 5 significant digits is also 1.2840.

But the above discussion and example do not answer our question of what a Taylor series is. What is the Taylor series?

We can calculate the value of the function at $f(x)$ at $(x+h)$, provided the function $f(x)$ and all the derivatives of $f(x)$ are given at a single point, $x$

$$f(x+h) = f(x) + f'(x)\,h + \frac{f''(x)}{2!}h^2 + \frac{f'''(x)}{3!}h^3 + \ldots \quad (4)$$

Of course, the function $f(x)$ and all derivatives of $f(x)$ should exist and be continuous between $x$ and $x+h$.

## What does this mean in plain English?

As Archimedes would have said (without the fine print), "*Give me the value of the function at a single point, and the value of all (first, second, and so on) its derivatives, and I can give you the value of the function at any other point.*"

It is very important to note that the Taylor series is not asking for the expression of the function and its derivatives, just the value of the function and its derivatives at a single point.

*Now the fine print*: Yes, all the derivatives have to exist and be continuous between $x$ (the point where you are) to the point, $x+h$ where you want to calculate the function at. However, if you want to calculate the function approximately by using the $n^{\text{th}}$ order Taylor polynomial, then $1^{\text{st}}, 2^{\text{nd}}, ...., n^{\text{th}}$ derivatives need to exist and be continuous in the closed interval $[x, x+h]$, while the $(n+1)^{\text{th}}$ derivative needs to exist and be continuous in the open interval $(x, x+h)$.

## More Examples of Taylor Series

### Example 2

Find the value of $f(6)$ given that $\mathrm{f}(4) = 125$, $f'(4) = 74$, $f''(4) = 30$, $f'''(4) = 6$ and all other higher derivatives of $f(x)$ at $x = 4$ are zero.

**Solution**

$$f(x+h) = f(x) + f'(x)\,h + f''(x)\,\frac{h^2}{2!} + f'''(x)\,\frac{h^3}{3!} + \ldots \qquad (E2.1)$$

$$x = 4$$

$$\begin{aligned} h &= 6 - 4 \\ &= 2 \end{aligned}$$

Since fourth and higher derivatives of $f(x)$ are zero at $x = 4$, from Equation (E2.1)

$$f(4+2) = f(4) + f'(4)\,2 + f''(4)\,\frac{2^2}{2!} + f'''(4)\,\frac{2^3}{3!}$$

$$\begin{aligned} f(6) &= 125 + 74(2) + 30\left(\frac{2^2}{2!}\right) + 6\left(\frac{2^3}{3!}\right) \\ &= 125 + 148 + 60 + 8 \\ &= 341 \end{aligned}$$

Note that to find $f(6)$ exactly, we only needed the value of the function and all its derivatives at some other point, in this case, $x = 4$. We did not need the expression for the function or for its derivatives. Taylor series application would be redundant if we needed to know the expression for the function, as we could just substitute $x = 6$ in it to get the value of $f(6)$.

*Side Note*: Actually, the problem posed above was obtained from a known function $f(x) = x^3 + 3x^2 + 2x + 5$ and hence $f(4) = 125$, $f'(4) = 74$, $f''(4) = 30$, $f'''(4) = 6$, and all other higher derivatives are zero.

# Chapter 01.07: Taylor Theorem Revisited

Lesson: Application of Taylor Series

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *derive Taylor's polynomial for transcendental and trigonometric functions*

2) *calculate transcendental and trigonometric functions at a point when the value and its derivatives are given at some other point.*

## What You Have Learned So Far

In the previous lesson, we recalled some common Taylor series of a few transcendental functions. In this lesson, we show three examples. The first example illustrates how to derive the Maclaurin series for an exponential function. The second one finds the value of a trigonometric function at a specific point using the Taylor series. The third example derives the Maclaurin series for $sin(x)$.

### Example 1

Derive the Maclaurin series of

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots$$

### Solution

Maclaurin series is simply a Taylor series around the point $x = 0$.

$$f(x) = e^x,\ f(0) = 1$$

$$f'(x) = e^x,\ f'(0) = 1$$

$$f''(x) = e^x,\ f''(0) = 1$$

$$f'''(x) = e^x,\ f'''(0) = 1$$

$$f''''(x) = e^x, \ f''''(0) = 1$$

$$f'''''(x) = e^x, \ f'''''(0) = 1$$

Using the Taylor series now,

$$f(x+h) = f(x) + f'(x)\,h + f''(x)\,\frac{h^2}{2!} + f'''(x)\,\frac{h^3}{3!} + f''''(x)\,\frac{h^4}{4!} + f'''''(x)\,\frac{h^5}{5!} + ...$$

For $x = 0$

$$f(0+h) = f(0) + f'(0)\,h + f''(0)\,\frac{h^2}{2!} + f'''(0)\,\frac{h^3}{3!} + f''''(0)\,\frac{h^4}{4!} + f'''''(0)\,\frac{h^5}{5!} + ...$$

$$\begin{aligned} f(h) &= f(0) + f'(0)\,h + f''(0)\,\frac{h^2}{2!} + f'''(0)\,\frac{h^3}{3!} + f''''(0)\,\frac{h^4}{4!} + f'''''(0)\,\frac{h^5}{5!} + \dots \\ &= 1 + 1\,(h) + 1\frac{h^2}{2!} + 1\frac{h^3}{3!} + 1\frac{h^4}{4} + 1\frac{h^5}{5} + \dots \\ &= 1 + h + \frac{h^2}{2!} + \frac{h^3}{3!} + \frac{h^4}{4} + \frac{h^5}{5} + \dots \end{aligned}$$

and can be rewritten as

$$f(x) = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4} + \frac{x^5}{5} + \dots$$

and hence

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4} + \frac{x^5}{5} + \dots$$

**Example 2**

Take

$$f(x) = \sin(x).$$

We all know the value of

$$\sin\left(\frac{\pi}{2}\right) = 1.$$

We also know that

$$f'(x) = \cos(x) \ \text{ and } \ \cos\left(\frac{\pi}{2}\right) = 0.$$

Similarly

$$f''(x) = -\sin(x) \text{ and } \sin\left(\frac{\pi}{2}\right) = 1.$$

In a way, we know the value of $\sin(x)$ and all its derivatives at $x = \frac{\pi}{2}$ without any need for any major calculations. We do not need to use any calculators; just use our knowledge of differential calculus and trigonometry. Can you use the Taylor series and the information above to find the approximate value of $\sin(2)$?

**Solution**

$$x = \frac{\pi}{2}$$

$$x + h = 2$$

$$\begin{aligned} h &= 2 - x \\ &= 2 - \frac{\pi}{2} \\ &= 0.42920 \end{aligned}$$

Using the Taylor series

$$f(x+h) = f(x) + f'(x)\,h + f''(x)\,\frac{h^2}{2!} + f'''(x)\,\frac{h^3}{3!} + f''''(x)\,\frac{h^4}{4!} + \ldots$$

and

$$x = \frac{\pi}{2}$$

$$h = 0.42920$$

$$f(x) = \sin(x),\ f\left(\frac{\pi}{2}\right) = \sin\left(\frac{\pi}{2}\right) = 1$$

$$f'(x) = \cos(x),\ f'\left(\frac{\pi}{2}\right) = 0$$

$$f''(x) = -\sin(x),\ f''\left(\frac{\pi}{2}\right) = 1$$

$$f'''(x) = -\cos(x),\ f'''\left(\frac{\pi}{2}\right) = 0$$

$$f''''(x) = sin(x),\ f''''\left(\frac{\pi}{2}\right) = 1$$

Hence

$$f\left(\frac{\pi}{2}+h\right)=f\left(\frac{\pi}{2}\right)+f'\left(\frac{\pi}{2}\right)h+f''\left(\frac{\pi}{2}\right)\frac{h^2}{2!}+f'''\left(\frac{\pi}{2}\right)\frac{h^3}{3!}+\ f''''\left(\frac{\pi}{2}\right)\frac{h^4}{4!}+\ldots$$

$$\begin{aligned}f\left(\frac{\pi}{2}+0.42920\right)&=1+0\left(0.42920\right)-1\frac{\left(0.42920\right)^2}{2!}+0\frac{\left(0.42920\right)^3}{3!}\\&\quad+1\frac{\left(0.42920\right)^4}{4!}+\ldots\\0&=1+0-0.092106+0+0.00141393+\ldots\\&\cong 0.90931\end{aligned}$$

Within five significant digits, the value of $\sin(2)$ I get from my calculator is 0.90930, which is very close to the value I just obtained. Now you can get a more accurate value by using more terms of the series. In addition, you can now use the value calculated for $\sin(2)$ coupled with the value of $\cos(2)$ (which can be calculated by Taylor series just like this example or by using the $\sin^2 x+\cos^2 x\equiv 1$ identity) to find the value of $\sin(x)$ at some other point. In this way, we can find the value of $\sin(x)$ for any value from $x=0$ to $2\pi$ and then can use the periodicity of $\sin(x)$, that is $\sin(x)=\sin(x\pm 2n\pi),n=1,2,\ldots$ to calculate the value of $\sin(x)$ at any other point.

Let's now revisit how we obtain the Maclaurin series of a function.

**Example 3**

Derive the Maclaurin series of

$$\sin(x)=x-\frac{x^3}{3!}+\frac{x^5}{5!}-\frac{x^7}{7!}+\ldots$$

**Solution**

In Example 2, we wrote the Taylor series for $\sin(x)$ around the point $x=\frac{\pi}{2}$. The Maclaurin series is simply a Taylor series around the point $x=0$.

$$f(x)=\sin(x),\ f(0)=0$$

$$f'(x)=\cos(x),\ f'(0)=1$$

$$f''(x)=-\sin(x),\ f''(0)=0$$

$$f'''(x)=-\cos(x),\ f'''(0)=-1$$

$$f''''(x)=sin(x),\ f''''(0)=0$$

$$f'''''(x)=cos(x),\ f'''''(0)=1$$

Using the Taylor series now,

$$f(x+h) = f(x) + f'(x)\,h + f''(x)\,\frac{h^2}{2!} + f'''(x)\,\frac{h^3}{3!} + f''''(x)\,\frac{h^4}{4!} + f'''''(x)\,\frac{h^5}{5!}$$

Using $x = 0$

$$f(0+h) = f(0) + f'(0)\,h + f''(0)\,\frac{h^2}{2!} + f'''(0)\,\frac{h^3}{3!} + f''''(0)\,\frac{h^4}{4!} + f'''''(0)\,\frac{h^5}{5!}$$

$$\begin{aligned} f(h) &= f(0) + f'(0)\,h + f''(0)\,\frac{h^2}{2!} + f'''(0)\,\frac{h^3}{3!} + f''''(0)\,\frac{h^4}{4!} + f'''''(0)\,\frac{h^5}{5!} \\ &= 0 + 1\,(h) - 0\frac{h^2}{2!} - 1\frac{h^3}{3!} + 0\frac{h^4}{4} + 1\frac{h^5}{5} + \dots \\ &= h - \frac{h^3}{3!} + \frac{h^5}{5!} + \dots \end{aligned}$$

So

$$f(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$$

and hence

$$\sin(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$$

# Chapter 01.07: Taylor Theorem Revisited

Lesson: Remainder of Taylor Series

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *find the error in a Taylor series*

2) *find the upper bound in the error in a Taylor series.*

## Introduction

As you have noticed, the Taylor series has infinite terms. Only in special cases such as a polynomial of finite order does it have a finite number of terms. So, whenever you are using a Taylor series to calculate the value of a general function, it is being calculated approximately.

The Taylor polynomial of order $n$ of a function $f(x)$ with $(n+1)$ continuous derivatives in the domain $[x, x+h]$ is given by

$$f(x+h) = f(x) + f'(x)\,h + f''(x)\,\frac{h^2}{2!} + \ldots + f^{(n)}(x)\,\frac{h^n}{n!} + R_n(x+h)$$

where the remainder is given by

$$R_n(x+h) = \frac{(h)^{n+1}}{(n+1)!} f^{(n+1)}(c)\,.$$

where

$$x < c < x+h$$

that is,$c$ is some point in the domain $(x, x+h)$.

### Example 1

The Taylor series for $e^x$at point $x = 0$ is given by

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} + \ldots$$

a) What is the truncation (true) error in the representation of $e^1$ if only four terms of the series are used?

b) Use the remainder theorem to find the bounds of the truncation error.

**Solution**

a) If only four terms of the series are used, then

$$\begin{aligned} e^x &\approx 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} \\ e^1 &\approx 1 + 1 + \frac{1^2}{2!} + \frac{1^3}{3!} \\ &= 2.66667 \end{aligned}$$

The truncation (true) error would be the unused terms of the Taylor series, which then are

$$\begin{aligned} E_t &= \frac{x^4}{4!} + \frac{x^5}{5!} + \ldots \\ &= \frac{1^4}{4!} + \frac{1^5}{5!} + \ldots \\ &\cong 0.0516152 \end{aligned}$$

b) But is there any way to know the bounds of this error other than calculating it directly? Yes.

We know that

$$f(x+h) = f(x) + f'(x)\,h + \ldots + f^{(n)}(x)\,\frac{h^n}{n!} + R_n(x+h)$$

where

$$R_n(x+h) = \frac{(h)^{n+1}}{(n+1)!} f^{(n+1)}(c)\,,\ x < c < x+h, \text{ and}$$

$c$ is some point in the domain $(x, x+h)$. So, in this case, if we are using four terms of the Taylor series, the remainder is given by $(x = 0, n = 3)$

$$\begin{aligned} R_3(0+1) &= \frac{(1)^{3+1}}{(3+1)!} f^{(3+1)}(c) \\ &= \frac{1}{4!} f^{(4)}(c) \\ &= \frac{e^c}{24} \end{aligned}$$

Since

$$x < c < x + h$$

$$0 < c < 0 + 1$$

$$0 < c < 1$$

The error is bound between

$$\frac{e^0}{24} < R_3(1) < \frac{e^1}{24}$$

$$\frac{1}{24} < R_3(1) < \frac{e}{24}$$

$$0.041667 < R_3(1) < 0.113261$$

So, the upper bound of the error is 0.113261, which concurs with the calculated error of 0.0516152.

**Example 2**

The Taylor series for $e^x$at point $x = 0$ is given by

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} + \dots$$

As you can see in the previous example that by taking more terms, the error bounds decrease and hence you have a better estimate of $e^1$. How many terms it would require to get an approximation of $e^1$ within a magnitude of true error of less than $10^{-6}$?

**Solution**

Using $(n + 1)$ terms of the Taylor series gives an error bound of

$$R_n(x + h) = \frac{h^{n+1}}{(n+1)!} f^{(n+1)}(c)$$

$$x = 0, h = 1, f(x) = e^x$$

$$R_n(1) = \frac{(1)^{n+1}}{(n+1)!} f^{(n+1)}(c)$$
$$= \frac{(1)^{n+1}}{(n+1)!} e^c$$

Since

$$x < c < x + h$$

$$0 < c < 0 + 1$$

$$0 < c < 1$$

$$\frac{1}{(n+1)!} < |R_n(1)| < \frac{e}{(n+1)!}$$

So if we want to find out how many terms it would require to get an approximation of $e^1$ within a magnitude of true error of less than$10^{-6}$,

$$\frac{e}{(n+1)!} < 10^{-6}$$

$$(n+1)! > 10^6 e$$

$$(n+1)! > 10^6 \times 3$$

as we do not know the value of $e$, but it is less than 3.

$$n \geq 9$$

So, 9 terms or more will get $e^1$ within an error of $10^{-6}$ in its value.

We can do calculations such as the ones given above only for simple functions. To do a similar analysis of how many terms of the series are needed for a specified accuracy for any general function, we can do that based on the concept of absolute relative approximate errors discussed in Chapter 01.02 as follows.

We use the concept of absolute relative approximate error (see Chapter 01.02 for details), which is calculated after each term in the series is added. The maximum value of $m$, for which the absolute relative approximate error is less than $0.5 \times 10^{2-m}\%$ is the least number of significant digits correct in the answer. It establishes the accuracy of the approximate value of a function without the knowledge of the remainder of the Taylor series or the true error.

# Multiple Choice Test

## Chapter 01.07 Taylor Theorem Revisited

(1). The coefficient of the $x^5$ term in the Maclaurin polynomial for $\sin(2x)$ is

(A) 0

(B) 0.0083333

(C) 0.016667

(D) 0.26667

(2). Given $f(3) = 6$, $f'(3) = 8$, $f''(3) = 11$, and all other higher order derivatives of $f(x)$ are zero at $x = 3$, and assuming the function and all its derivatives exist and are continuous between $x = 3$ and $x = 7$, the value of $f(7)$ is

(A) 38.000

(B) 79.500

(C) 126.00

(D) 331.50

(3). Given that $y(x)$ is the solution to $\frac{dy}{dx} = y^3 + 2$, $y(0) = 3$, the value of $y(0.2)$ from a second order Taylor polynomial around $x = 0$ is

(A) 4.400

(B) 8.800

(C) 24.46

(D) 29.00

(4). The series $\sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!} 4^n$ is a Maclaurin series for the following function

(A) $\cos(x)$

(B) $\cos(2x)$

(C) $\sin(x)$

(D) $\sin(2x)$

(5). The function $\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt$ is called the error function. It is used in the field of probability and cannot be calculated exactly. However, one can expand the integrand as a Taylor polynomial and conduct integration. The approximate value of $\operatorname{erf}(2.0)$ using the first three terms of the Taylor series around $t = 0$ is

(A) $-0.75225$

(B) 0.99532

(C) 1.5330

(D) 2.8586

(6). Using the remainder of Maclaurin polynomial of $n^{th}$order for $f(x)$ defined as

$$R_n(x) = \frac{x^{n+1}}{(n+1)!} f^{(n+1)}(c), \ n \geq 0, \quad 0 \leq c \leq x$$

the order of the Maclaurin polynomial at least required to get an absolute true error of at most $10^{-6}$ in the calculation of $\sin(0.1)$ is (do not use the exact value of $\sin(0.1)$ or $\cos(0.1)$to find the answer, but the knowledge that $|\sin(x)| \leq 1$ and $|\cos(x)| \leq 1$).

(A) 3

(B) 5

(C) 7

(D) 9

For complete solution, go to

http://nm.mathforcollege.com/mcquizzes/01aae/quiz_01aae_taylorseries_answers.pdf

# Chapter 02.00: Physical Problem for Numerical Differentiation

Lesson: General Engineering

## Summary

To find the acceleration of a rocket requires differentiation of the velocity expression.

## Modeling

A rocket is traveling vertically and expels fuel at a velocity of 2000 m/s at a consumption rate of 2100 kg/s. The initial mass of the rocket is 140, 000 kg. If the rocket starts from rest, how can I calculate the acceleration of the rocket at 16 s?

**Figure 1** A rocket launched into space

Source of rocket picture: NASA Langley Research Center, Office of Education, **edu.larc.nasa.gov/pstp/**

If

$$m_0 = \text{initial mass of rocket at } t = 0 \text{ (kg)},$$

$$q = \text{rate at which fuel is expelled (kg/s)},$$

$$u = \text{velocity at which the fuel is being expelled (m/s)}$$

then since the fuel is expelled from the rocket, the mass of the rocket keeps decreasing with time. The mass of the rocket, $m$ at any time, $t$ is

$$m = m_o - qt \qquad (1)$$

The forces on the rocket at any time are found by applying Newton's second law of motion. Then

$$\sum F = ma \qquad (2)$$

The upward force is due to the fuel expulsion, while the downward force is due to gravity

$$uq - mg = ma \qquad (3)$$

Using Equations (1) and (3) in Equation (2) gives

$$uq - (m_o - qt)\, g = (m_o - qt)\, a \qquad (4)$$

where

$$g = \text{acceleration due to gravity } \left(\text{m/s}^2\right)$$

From Equation (4), we get

$$a = \frac{uq}{m_o - qt} - g \qquad (5a)$$

$$\frac{dv}{dt} = \frac{uq}{m_o - qt} - g \qquad (5b)$$

Integrating both sides of Equation (5b) gives

$$v = -u \log_e(m_0 - qt) - gt + C \qquad (6)$$

Since the rocket starts from rest

$$v = 0 \text{ at } t = 0$$

$$0 = -u \log_e(m_o) + C$$

$$C = u \log_e(m_o)$$

Hence Equation (6) reduces to

$$v = -u \log_e(m_o - qt) - gt + u \log_e(m_o)$$

$$\frac{dx}{dt} = u \; log_e(\frac{m_o}{m_o - qt}) - gt \qquad (7)$$

Using the following numerical values in Equation (7)

$$u = 2,000 \text{ m/s}$$

$$m_o = 140,000 \text{ kg}$$

$$q = 2100 \text{ kg/s}$$

$$g = 9.8 \text{ m/s}^2$$

$$t_0 = 0 \text{ s}$$

$$t_1 = 30 \text{ s}$$

gives

$$v = 2000 \log_e \left( \frac{14 \times 10^4}{14 \times 10^4 - 2100t} \right) - 9.8t. \qquad (8)$$

Can you numerically find the acceleration at $t = 16$ s? You may say that we do not need numerical or analytical differentiation to calculate the acceleration, as Equation (5a) directly gives us the acceleration of the rocket at any time. True! We are just doing this as an exercise to illustrate numerical differentiation, and we have a true expression of acceleration readily available for comparison with the numerical results.

## Questions

(1) Find the acceleration of the rocket at $t = 16$ s.

(2) Use different numerical differentiation techniques to find the acceleration at $t = 16$ s. Compare these results with the exact answer.

# Chapter 02.00: Physical Problem for Numerical Differentiation

Lesson: Chemical Engineering

## Summary

Estimating the rate at which area of containment is spreading in a lake requires numerical differentiation.

## Modeling

At about 11:56 am on June 22, 1969, near the city of Cleveland in Ohio, an oil slick on the Cuyahoga River caught fire that burned for 24 minutes. This incident on a navigable river acted as a catalyst for Congress to pass the Clean Water Act in 1972. The Federal Water Pollution Control Act prohibits the discharge of oil or oily waste substances or hazardous substances into or upon the navigable waters of the United States.

Interestingly, recreational boating is a growing activity in many waterways of the United States. Unfortunately, fuel leakages – however small – from so many boats can lead to the formation of large oil slicks. The ideal would be for the recreational boats to use fuels that can evaporate as quickly as the fuels leak onto the surface of the water.

**Figure 1** Waves generated on a pond by a pebble

A new fuel for recreational boats being developed at the local university was tested at an area pond by a team of engineers. The interest is to document the environmental impact of the fuel – how quickly does the slick spread? Table 1 shows the video camera record of the radius of the wave generated by a drop of the fuel that fell into the pond. To find the rate at which the contamination spreads requires numerical differentiation.

**Table 1** Radius of wave generated as a function of time.

| Time ($s$) | 0 | 0.5 | 1.0 | 1.5 | 2.0 | 2.5 | 3.0 | 3.5 | 4.0 | 4.5 | 5.0 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Radius ($m$) | 0 | 0.236 | 0.667 | 1.225 | 1.886 | 2.635 | 3.464 | 4.365 | 5.333 | 6.364 | 7.454 |

## Questions

(1) Compute the rate at which the radius of the drop was changing at $t = 2$s and $t = 5$s.

(2) Estimate the rate at which the area of the contaminant was spreading across the pond at $t = 2$s and $t = 5$s.

# Chapter 02.00: Physical Problem for Numerical Differentiation

Lesson: Civil Engineering

## Summary

Finding the stress concentration around a hole in a plate requires differentiation.

## Modeling

A rectangular plate under a uniform stress, $\sigma_0$, results in the stress state of

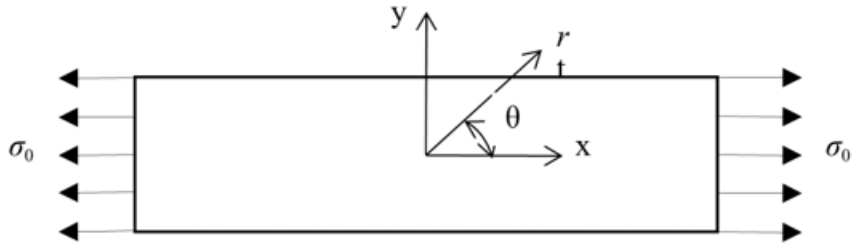



**Figure 1.** Rectangular plate under uniform stress.

where

$$\sigma_{xx} = \sigma_0$$

$$\sigma_{yy} = 0$$

$$\sigma_{zz} = 0$$

If the plate now has changed in the cross-section, the normal stresses will increase as a result of these changes. For example, if there is a circular hole in the plate (Figure 2) and it is subjected to the same uniform stress, $\sigma_0$, then stresses in the plate would increase. For example, if the radius of the hole in the plate was small compared to the length and width of the plate, the solution can be found analytically for $\sigma_{xx} = 3\sigma_0$ at $r = a$, $\theta = \pm\frac{\pi}{2}$.

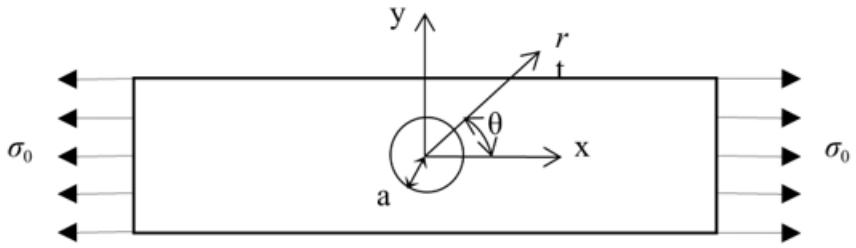



**Figure 2.** Rectangular plate with circular hole of radius $a$ under uniform stress.

This in fact is the highest normal stress that occurs in such a case. To specify such stresses we define a term called stress concentration, $k$, that is

$$k = \frac{Maximum\ Stress}{Nominal\ Stress}$$

So in this case

$$\begin{aligned} k &= \frac{(3\sigma_0)}{\sigma_0} \\ &= 3 \end{aligned}$$

Now what happens when the radius of the hole is not small as compared to the length and width of the plate? In most cases, the stresses cannot be found analytically, and we need to solve the problem using numerical techniques. One such method is called the finite difference method. Here is how it works for this problem. The partial differential equations for the plane stress problem are given by

$$\frac{\partial^2 u}{\partial x^2} + \frac{(1-\upsilon)}{2}\frac{\partial^2 v}{\partial y^2} + \frac{(1+\upsilon)}{2}\frac{\partial^2 v}{\partial x \partial y} = 0 \qquad (1)$$

$$\frac{\partial^2 v}{\partial y^2} + \frac{(1-\upsilon)}{2}\frac{\partial^2 u}{\partial x^2} + \frac{(1+\upsilon)}{2}\frac{\partial^2 u}{\partial x \partial y} = 0 \qquad (2)$$

Where

$$u = \text{displacement in x-direction}$$

$$v = \text{displacement in y-direction}$$

$$\upsilon = \text{Poisson's ratio}$$

To apply the finite difference method, the rectangular plate with the hole would be divided into $(m-1)$ segments along the x-axis and $(n-1)$ segments along the y-axis as shown in Figure 3. This results in nodes in the plate.

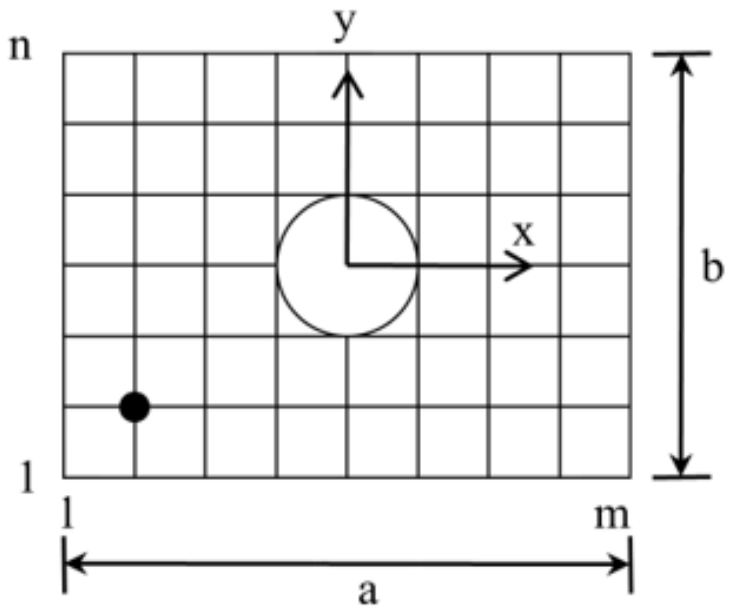



**Figure 3** Finite Difference Nodes of Rectangular Plate with Hole.

Now at each node such as a general node $(i,j)$ where the surrounding nodes do not have a node on the hole, the forward divided difference approximations can be applied as

$$\frac{\partial^2 u}{\partial x^2}|i,j \approx \frac{u_{i+1,j} - 2u_{i,j} + u_{i-1,j}}{(\Delta x)^2} \quad (3)$$

$$\frac{\partial^2 v}{\partial y^2}|i,j \approx \frac{v_{i,j+1} - 2v_{i,j} + v_{i,j-1}}{(\Delta y)^2} \quad (4)$$

$$\frac{\partial^2 v}{\partial x \partial y}|i,j \approx \frac{v_{i+1,j+1} - v_{i+1,j-1} - v_{i-1,j+1} - v_{i-1,j-1}}{(\Delta x)(\Delta y)} \quad (5)$$

$$\frac{\partial^2 u}{\partial x \partial y}|i,j \approx \frac{u_{i+1,j+1} - u_{i+1,j-1} - u_{i-1,j+1} - u_{i-1,j-1}}{(\Delta x)(\Delta y)} \quad (6)$$

Substituting these approximations (equations 3-6) in equation (1) and (2) gives

$$\begin{aligned} &\frac{u_{i+1,j} - 2u_{i,j} + u_{i-1,j}}{(\Delta x)^2} + \frac{(1-\upsilon)}{2}\left[\frac{v_{i,j+1} - 2v_{i,j} + v_{i,j-1}}{(\Delta y)^2}\right] \\ &+ \frac{(1+\upsilon)}{2}\left[\frac{v_{i+1,j+1} - v_{i+1,j-1} - v_{i-1,j+1} - v_{i-1,j-1}}{(\Delta x)(\Delta y)}\right] = 0 \end{aligned} \quad (7)$$

$$\frac{v_{i,j+1}-2v_{i,j}+v_{i,j-1}}{(\Delta y)^2}+\frac{(1-\upsilon)}{2}\left[\frac{u_{i+1,j}-2u_{i,j}+u_{i-1,j}}{(\Delta x)^2}\right]$$
$$+\frac{(1+\upsilon)}{2}\left[\frac{u_{i+1,j+1}-u_{i+1,j-1}-u_{i-1,j+1}-u_{i-1,j-1}}{(\Delta x)(\Delta y)}\right]=0 \quad (8)$$

Similarly one would write equations for the nodes that are around the hole and also for ones that are surrounded by the nodes on the hole. This is out of the scope of this topic but needs to be appreciated. We will also skip the application of boundary conditions at the outer four edges of the plate as well as the radius of the hole. However, one needs to realize that once all the equations are defined, we get as many unknowns (6: the displacement in the x- and y-directions, at each node) as we have equations by the approximations to equations (1) and (2), as well as the boundary conditions. This results in solving simultaneous linear equations. After the displacements are calculated, one can find stress and strain in the plate to consequently find the stress concentration. As an example, the problem description of one such case is given below.

To find the stresses and hence the stress concentration factor, one needs to find the derivatives of these displacements. In Table 1 the radial displacements $u$ are given along the $y$ -axis. The radius of the hole is 1.0 cm.

a) At $x = 0$, if the radial strain $\varepsilon_r$ is given by $\varepsilon_r = \dfrac{\partial u}{\partial r}$, find the radial strain at $r = 1.1cm$ using the forward divided difference method.

b) If the tangential strain at $r = 1.1cm,\ \theta = 90^\circ$ is given to you as $\varepsilon_\theta = 0.0029733$, find the hoop stress, $\sigma_\theta$, at $r = 1.1\ cm,\ \theta = 90^\circ$ if $\sigma_\theta = \dfrac{E}{1-\nu^2}(\varepsilon_r + \nu\varepsilon_\theta)$, where $E = 200$ GPa and $\nu\ = 0.3$.

**Table 1** Radial displacement as a function of location.

| $r\ (cm)$ | $u\ (cm)$ |
|---|---|
| 1.0 | ˇ0.0010000 |
| 1.1 | ˇ0.0010689 |
| 1.2 | ˇ0.0011088 |
| 1.3 | ˇ0.0011326 |
| 1.4 | ˇ0.0011474 |
| 1.5 | ˇ0.0011574 |
| 1.6 | ˇ0.0011650 |
| 1.7 | ˇ0.0011718 |
| 1.8 | ˇ0.0011785 |
| 1.9 | ˇ0.0011857 |

# Chapter 02.00: Physical Problem for Numerical Differentiation

Lesson: Computer Engineering

## Summary

Detect the edges (boundaries) in images requires differentiation.

## Modeling

Human vision is an intriguing ability. The fact that we can effortlessly recognize objects, people, and read this document belies the complexity of the task that a significant portion of our brain is devoted to solving. You might be surprised to learn that this almost photographic perception of the world that we have is not what our eyes send to our brain. What we perceive is a reconstruction from the noisy, shaky output of elementary "things" or "features" detected in the retina. There is strong evidence that the first level of processing done in the retina involves detecting something called "edges" or positions of transitions from dark to bright or bright to dark points in the images. These points usually coincide with boundaries of things.

**Figure 1** On left is an image, with the corresponding edge image shown on the right.

In Figure 1 above we see an example of an edge image. This process of "detecting" edges in an image is called edge detection and can be modeled as a differentiation process. In this module we will look into this differentiation process.

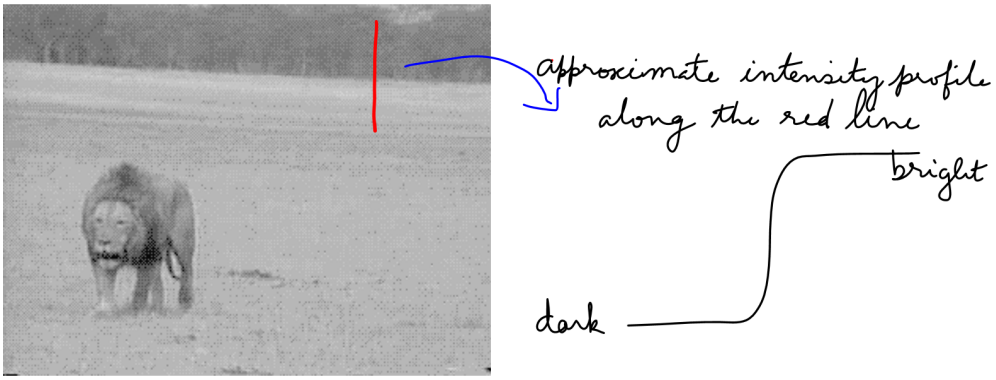



**Figure 2** Edges corresponds to points in the image where there are intensity changes.

In the above Figure we see the profile of the intensity value along a line (shown in red) cutting an edge in the image. We represent small values to represent dark points in the image and large values to detect bright regions in the image. The value increases and saturates to the value of the bright region. The edges can be marked at point of the maximum change in intensity, or where the derivative is a maximum. We will study the behavior of this derivative for a class of models of the edges.

**Example**

One way to model the edge, which is denote by $e^x$, where $x$ is the direction perpendicular to the edge, is using exponential functions.

$$e(x) = \begin{cases} 1 - e^{(-ax)} & \forall x \geq 0 \\ e^{(ax)} - 1 & \forall x \leq 0 \end{cases} \qquad (1)$$

Plots of this function for various values of $a$ are shown in the Figure 3 below. Ignore the fact that we have used negative values to represent intensity. We can always add a constant to shift the function "up", but the constant will not have any impact on the final derivative.

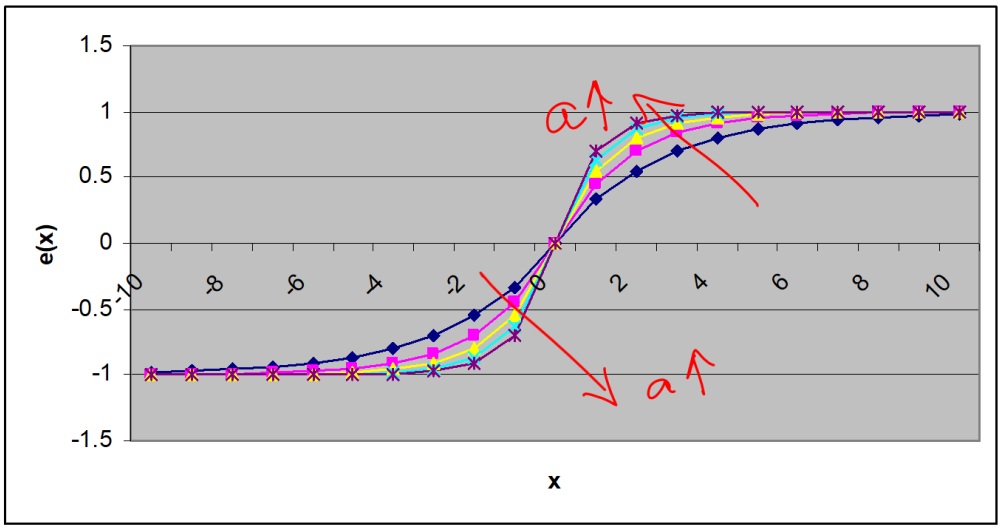



**Figure 3** Plot of the exponential edge profile model for various values of the modeling parameter $a$.

The derivative of this profile model is given by

$$e(x) = \begin{cases} 1 - e^{(-ax)} & \forall x \geq 0 \\ e^{(ax)} - 1 & \forall x \leq 0 \end{cases} \qquad (2)$$

(Note that at $x = 0$ both the derivatives have the same value of $a$; this shows that the function we used to model the edge profile is continuous in value and first derivative. This is just an aside fact that you might store in your brain for future use!)

What happens to the derivative as the underlying edge gets sharper? We see from Figure 4 that the value of the derivative gets larger and the peak gets sharper. Thus, strong edges, that is, edges with sharp intensity transitions will results in stronger response.

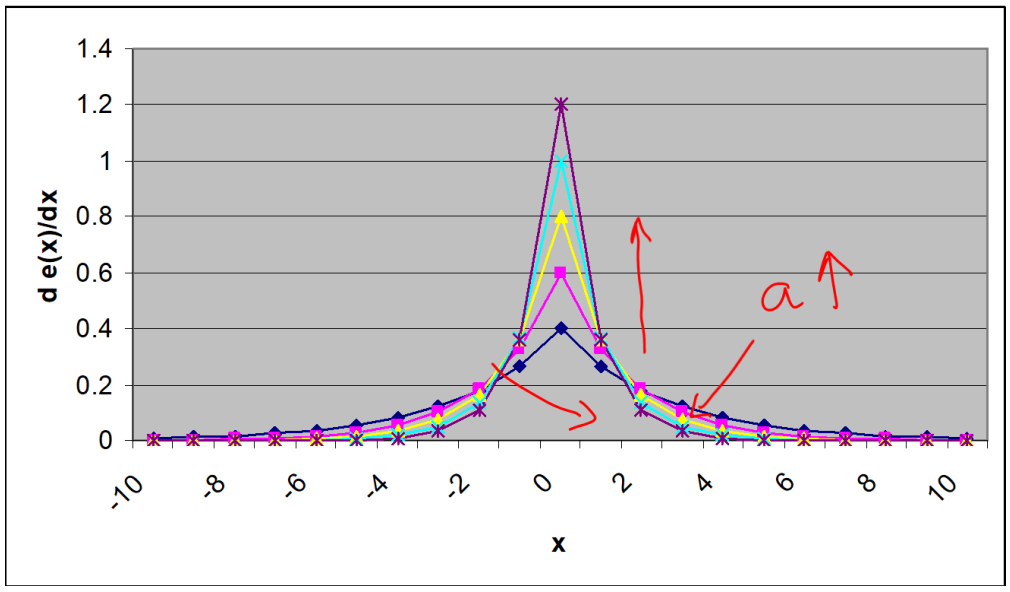



**Figure 4** Plot of the derivative of the edge profile as the parameter $a$ is increased.

## Questions

(1) What kind of "edge" is represented by the following edge profile model?

$$e(x) = e^{(-ax^2)}$$

Answer: Line edge: the edge will appear as a white line in a dark background. What happens to the width of this line as you change $a$.

(2) Compute the first derivative of the above function given in #1.

(3) Where, that is for what values of $x$, does this derivative reach a maximum or a minimum values? What happens to the distance between the maximum and minimum value locations as $a$ is increased?

(4) Where is the value of the derivative zero?

# Chapter 02.00: Physical Problem for Numerical Differentiation

Lesson: Electrical Engineering

## Summary

Zero-crossing requires calculation of the derivative of voltage.

## Modeling

The vast majority of commercial and industrial loads are inductive in character. Many electrical consumers find it advantageous to use capacitor banks to adjust their power factor toward 1.0 to avoid extensive rate penalties from their power utility. As load conditions change, it becomes necessary to add or remove capacitors from the bank. The most opportune time to do this is when the voltage across the capacitor is zero to avoid arcing and damages to the switches. Since the switches are slower than the AC waveforms, it is necessary to use a derivative of the voltage to anticipate the zero crossing.

Electrical systems are used for a wide-array of applications in the commercial and industrial world. Many of them perform physical work using motors, compressors, solenoids, and similar components. In almost all cases, these devices employ electro-magnetic principles and are thus inductive in nature. This presents an electrical load that result in a power factor lagging. Whenever the power factor of a load is not 1.0 (leading or lagging), it results in a discrepancy between the apparent power delivered to the load and the real power consumed by the load.

For most commercial and industrial power consumers, as opposed to residential, the electrical power utility bases its rate charge on the apparent power delivered. This often takes the form of a significant penalty for power factors much below 1.0 (e.g. 0.8). To help correct the power factor it is common to add capacitor banks in parallel with a load that can shift a lagging power factor toward 1.0. Under ideal circumstances, the capacitors could be connected and disconnected from the circuit at any time. Under practical conditions, however, this is not advisable.

The amount of energy in a capacitor is directly related to the voltage across the capacitor as shown in the familiar equation

$$E_c = \frac{1}{2} C V_c^2 \qquad (1)$$

When a switch tries to disconnect the capacitor, the energy resists this disconnection, and the result is often sparking and gapping as the switch opens. This not only is a significant source of noise in the circuit, but it also results in damage to the switch that greatly reduces its effective service life. The solution to this problem is quite simple; just open the switch when the stored energy is zero. This, of course, occurs when the voltage across the capacitor is zero.

Unfortunately, it is not just a matter of monitoring the voltage and activating the switch when the voltage is zero. The normal frequency for AC power systems is 60 Hz (50 Hz in Europe). This means that there are 120 zero crossings per second and the typical mechanical switch is simply not fast enough to react "instantaneously" at the zero crossing. To deal with this a smart capacitor bank switch is needed. It monitors the voltage and using numerical methods anticipates the point of zero crossing and initiates the switch action enough in advance to have it occur as close to the zero crossing as possible.

This can be done using a first-order approximation of the Taylor Series as shown by the following equations as well as Figure 1.

$$f(t+\Delta t) \cong f(t) + f^{'}(t)\,\Delta t = 0 \qquad (2)$$

$$\Delta t \cong -\frac{f(t)}{f^{'}(t)} \qquad (3)$$

If this formula looks a bit familiar, it is the same one used as the basis for Newton's Method as it is commonly used in root finding algorithms.

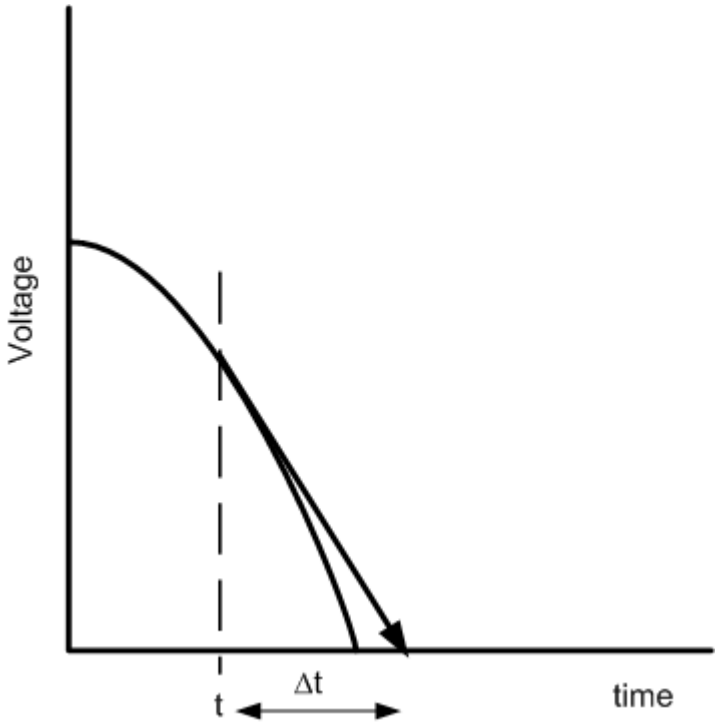



**Figure 1** Illustration of Zero-Crossing Prediction Using First-Order Taylor Series

Suppose a system containing a capacitor bank has been constructed and the data of Table 1 was sampled during operation. Using a numerical differentiating algorithm determine the derivative at each point and use the Taylor Series approximation to determine the value of $\Delta t$ which gives the future estimate of the zero crossing. Keep in mind that when $\Delta t$ is determined to be negative then the voltage has already crossed the zero point and it will be necessary to wait for the next half cycle of the waveform.

Most numerical methods for taking a derivative (e.g. the Central Difference Method) rely on samples taken both before and after the period in question. In this problem that is not practical since our desire

it to anticipate the future zero crossing at which the power should be turned off. Repeat your prior analysis using a differentiation algorithm that is one-sided, that is, one that only uses prior samples to determine the present value of the voltage's derivative.

**Table 1** Voltage Samples

| Time | Voltage | Time | Voltage |
|---|---|---|---|
| 1 | 0.62161 | 13 | −0.210796 |
| 2 | 0.362358 | 14 | 0.087499 |
| 3 | 0.070737 | 15 | 0.377978 |
| 4 | −0.227202 | 16 | 0.634693 |
| 5 | −0.504846 | 17 | 0.834713 |
| 6 | −0.737394 | 18 | 0.96017 |
| 7 | −0.904072 | 19 | 0.999859 |
| 8 | −0.989992 | 20 | 0.950233 |
| 9 | −0.98748 | 21 | 0.815725 |
| 10 | −0.896758 | 22 | 0.608351 |
| 11 | −0.725932 | 23 | 0.346635 |
| 12 | −0.490261 | 24 | 0.053955 |

## Questions

(1) What effect does the order of your differentiation algorithm have on the ability to predict correctly the point of zero crossing? Is there an order at which there are diminishing returns on the ability to identify the point of zero crossing?

(2) Using a higher-order approximation to the Taylor Series will also improve the prediction of the zero crossing, especially when it is a bit distant. Use a second-order model of the Taylor Series that requires both the first- and second-order derivative to determine $\Delta t$.

(3) In a real AC system, there could easily be quite a large amount of noise on the sampled voltages. Are there any reasonable algorithms for minimizing the effects this noise has on the results?

(4) The typical AC system in the United States uses a 60 Hz frequency (50 Hz in Europe). What is a reasonable sampling rate to use for the voltage sensor? How much does the required length for $\Delta t$ impact your answer. Did you account for the problem of question (c) in your answer?

(5) As in the data of Table 1, the samples have not been synchronized well with the cycles of the AC waveform (that is why you do not see 1 or -1 among the samples). Can you suggest a mechanism for bringing the samples in better synchronization with the cycles?

(6) Most industrial smart-sensors for this application use a prediction and correction method to detect the zero crossing, that is, they predict the crossing and then see how close they are for a few cycles prior to the actual disconnecting action. How might such an algorithm be structured?

(7) Would there be any advantage in using an interpolation or curve fitting algorithm (e.g. regression to a sinusoid) to better select the instant of true zero voltage?

(8) In a system with multiple loads (e.g. a three-phase system), the best instant of shut off for one phase of the capacitor banks may not be the best for the others. Is there any way to determine the point of best compromise or is it simply better to shut each of them off at the zero crossing and tolerate the temporary imbalance to the three-phase system?

# Chapter 02.00: Physical Problem for Numerical Differentiation

Lesson: Industrial Engineering

## Summary

Finding the reliability of manufactured parts over time requires differentiation.

## Modeling

The following is from a Firestone recall page (www.firestone-tire-recall.com/pages/overview.html -dead link).

"The Firestone tire recall is perhaps the most deadly auto safety crisis in American history. US regulators on 16 October, 2000 have raised the death count to 119 (the death count has steadily risen from 62, later to 88 and 101 deaths reported on 9/20/2000). Experts believe there may be as many as 250 deaths and more than 3000 catastrophic injuries associated with the defective tires. Most of the deaths occur in accidents involving the Ford Explorer which tends to rollover when one of the tires blows out.

In May 2000, the National Highway Transportation Safety Administration issued a letter to Ford and Firestone requesting information about the high incidence of tire failure on Ford Explorer vehicles. During July, Ford obtained and analyzed the data on tire failure. The data revealed that 15″ ATX and ATX II models and Wilderness AT tires had very high failure rates: the tread peels off. Many of the tires were made at a Decatur, Illinois plant. When the tires fail, the vehicle often rolls over and kills the occupants.

Ford Officials estimate the defect rate is 241 tires per million for 15 inch ATX and ATX II tires. By contrast Ford says there are no defects in 16 inch tires per million and only 2.3 incidents per million on other tires.

On August $9^{\text{th}}$, both companies decided on the recall. Ford and Firestone disagreed as to how to break the news. Bridgestone/Firestone officials wanted to read a statement at a joint briefing without answering any questions. Ford strongly disagreed with this strategy and warned of disaster if they refused questions. Ultimately questions were asked, many of them remain unanswered."

In our daily lives, we depend on a lot of different products. Their inability to perform may cause us inconveniences and sometimes more dire consequences as in the Firestone tire recall case. Therefore, companies are required to carefully study, document and control the reliability of their products before they are released to the market.

### Reliability Function

Reliability function is frequently used to describe the probability of an item operating under certain conditions for a certain amount of time without failure. The reliability function is a function of time,

in that every reliability value has an associated time value. In other words, one must specify a time value with the desired reliability value, that is, 95% reliability at 100 hours.

Mathematically, the reliability function $R(t)$ is the probability that a system will be successfully operating without failure in the interval from $t = 0$ to $t$,

$$\begin{aligned} R(t) &= P(T > t), t \geq 0 \\ &= \int_t^{\infty} f(s)ds \end{aligned} \tag{1}$$

where $T$ is a random variable representing the failure time or time-to-failure and $f(t)$ is the time to failure distribution function. More often than not, the time to failure distribution function is denoted by exponential, Weibull, normal or lognormal distribution based on the characteristics of the underlying product or the system. However, for complex practical systems the above distributions may not provide an accurate fit for the time to failure distribution function.

Also of interest is the failure distribution function $F(t)$ which describes the probability of an item failing before a certain period of time. As you may guess, the failure distribution function and the reliability function are related to each other by

$$F(t) + R(t) = 1 \tag{2}$$

where

$$F(t) = \int_{0,-\infty}^{t} f(s)ds \tag{3}$$

**Failure Rate Function**

The failure rate function, or hazard function, is also very important in reliability analysis because it specifies the rate of the system aging. The failure rate function $h(t)$ is defined as:

$$\begin{aligned} h\left(t\right) &= \lim_{\Delta\, t \to 0} \frac{R\left(t\right) - R\left(t + \Delta\, t\right)}{\Delta\, tR\left(t\right)} \\ &= \frac{f(t)}{R(t)} \end{aligned} \tag{4}$$

The importance of the failure rate function is that it indicates the changing rate in the aging behavior over the life of a population of components. For example, if the time to failure distribution function follows an exponential distribution with parameter $\lambda$, then the failure rate function is

$$\begin{aligned} h(t) &= \frac{f(t)}{R(t)} \\ &= \frac{\lambda e^{-\lambda t}}{e^{-\lambda t}} \\ &= \lambda \end{aligned} \tag{5}$$

This means that the failure rate function of the exponential distribution is a constant. In this case, the system does not have any aging property. This assumption is usually valid for software systems. However, for hardware systems, the failure rate could have other shapes. More information regarding reliability analysis can be found in [1].

A series systems, shown in Figure 1, is defined as $n$ independent components arranged in series. The reliability of the system at time $t$,$R_s(t)$, is the probability that all components survive to time $t$, or in other words,

$$R_s(t) = R_1(t)R_2(t)\dots R_n(t) = \prod_{i=1}^{n} R_i(t) \qquad (6)$$

**Figure 1** Block diagram of a series system

The following example relies on the research published in [2]. Low power portable direct methanol fuel cell (DMFC) systems provide higher energy density and longer operational life compared with Li-ion batteries widely used in portable electronic devices such as cellular phones, laptops, etc.

DMFC systems consist of three main subsystems, namely the fuel cell stack, fuel tank and the Balance of Plant which are assumed to be independently connected to each other in series as shown in Figure 2. These systems are still an immature technology and there is no publicly available reliable failure data for DMFC components.

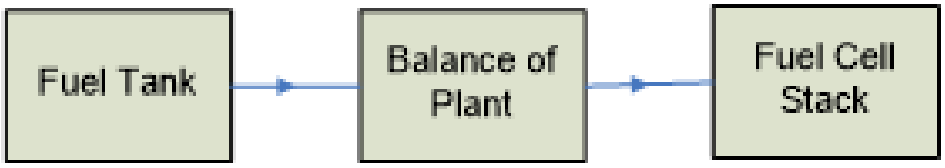



**Figure 2** Block diagram of a DFMC system

Consider the following reliability data in Table 1 for the DMFC system.

**Table 1**. Reliability of DFMC system

| $t$ $(hrs)$ | 0 | 1 | 10 | 100 | 1000 | 2000 | 3000 | 4000 | 5000 |
|---|---|---|---|---|---|---|---|---|---|
| $R(t)$ | 1 | 0.9999 | 0.9998 | 0.9980 | 0.9802 | 0.9609 | 0.9419 | 0.9233 | 0.9050 |

## Questions

(1) Calculate the value of the failure rate function, $h(t)$ at $t = 10$ hours.

(2) Calculate the value of the failure rate function, $h(t)$ at $t = 2000$ hours.

(3) Calculate the value of the failure rate function, $h(t)$ at $t = 4000$ hours.

(4) Compare the results you have in questions 1-3. What can be said about the failure rate function and the time to failure distribution function $F(t)$ of the DMFC system based on the data provided?

(5) You are told that the estimated failure rate for the fuel tank and the fuel cell stack are constant with values $5.42 \times 10^{-8}$ and $7.78 \times 10^{-6}$. Can you calculate an estimate the failure rate for the Balance of Plant?

(6) If you are told that the time to failure distribution function of the DFMC system is exponential with parameter $\lambda = 1.21 \times 10^{-5}$, calculate the accuracy of your answers to questions 1-3.

## References

[1] E. A. Elsayed (1996), Reliability Engineering, Addison-Wesley.

[2] N.S. Sisworahardjo, M.S.Alam and G. Aydinli, "Reliability and availability of low power portable direct methanol fuel cells", Journal of Power Sources, 177, 2008, p.412-418.

[3] Firestone and Ford tire controversy, Wikipedia page, https://en.wikipedia.org/wiki/Firestone_and_Ford_tire_controversy, last accessed May 26, 2021.

# Chapter 02.00: Physical Problem for Numerical Differentiation

Lesson: Mechanical Engineering

## Summary

Finding how the coefficient of linear thermal expansion varies with temperature requires numerical differentiation.

## Modeling

To make the fulcrum (Figure 1) of a bascule bridge, a long hollow steel shaft called the trunnion is shrink fit into a steel hub.

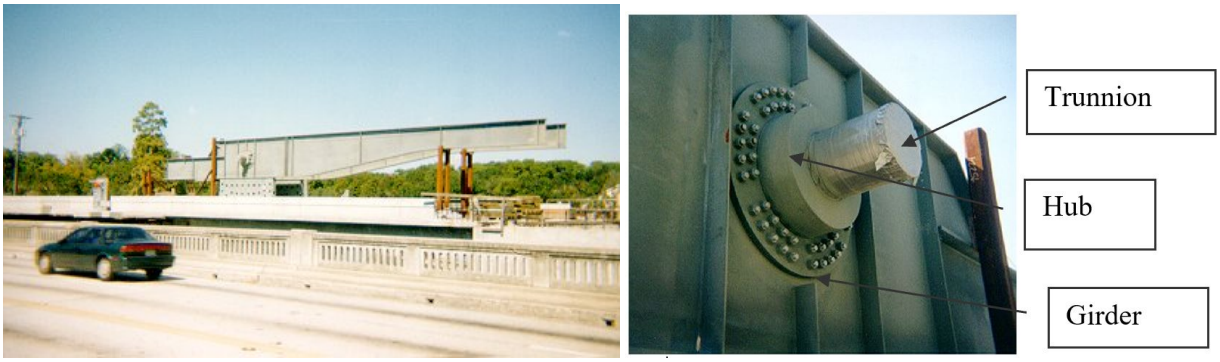



**Figure 1** Trunnion-Hub-Girder (THG) assembly.

This is done by first immersing the trunnion in a cold medium such as a dry-ice/alcohol mixture. After the trunnion reaches a steady-state temperature, that is the temperature of the cold medium, the trunnion outer diameter contracts, is taken out and slid through the hole of the hub (Figure 2).

When the trunnion heats up, it expands and creates an interference fit with the hub. In 1995, on one of the bridges in Florida, this assembly procedure did not work as designed. Before the trunnion could be inserted fully into the hub, the trunnion got stuck. So a new trunnion and hub had to be ordered worth $50,000. Coupled with construction delays, the total loss ran into more than a hundred thousand dollars.

Why did the trunnion get stuck - because the trunnion had not contracted enough to slide through the hole.

Now the same designer is working on making the fulcrum for another bascule bridge. Can you help him so that he does not make the same mistake?

For this new bridge, he needs to fit a hollow trunnion of outside diameter $12.363''$ in a hub of inner diameter $12.358''$. His plan is to put the trunnion in dry-ice/alcohol mixture (temperature of dry-ice/alcohol mixture is $-108°F$) to contract the trunnion so that it can be slid through the hole of the hub. To slide the trunnion without sticking, he has also specified a diametric clearance of at least $0.01''$. Assume the room temperature is $80°F$, is immersing it in a dry-ice/alcohol mixture a correct decision?

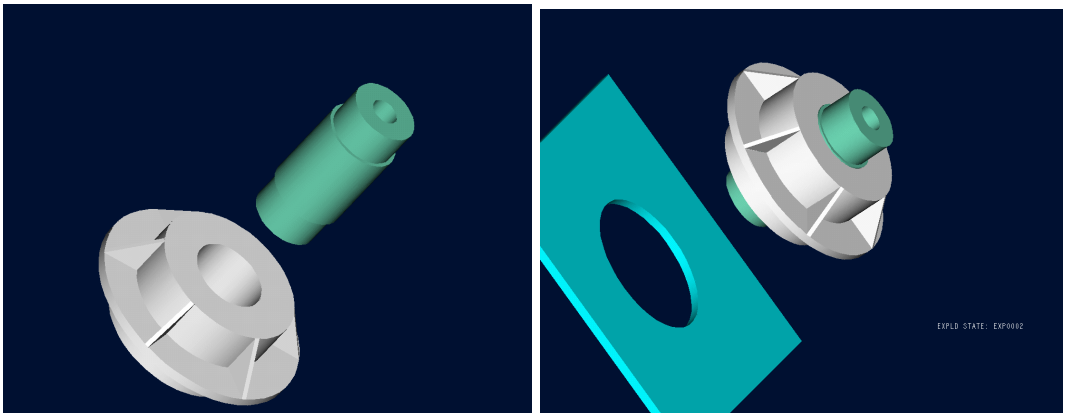



**Figure 2** Trunnion slid through the hub after contracting

Looking at the records of the designer for the previous bridge where the trunnion got stuck in the hub, it was found that he used the thermal expansion coefficient at room temperature to calculate the contraction in the trunnion diameter. In that case, the reduction, $\Delta D$ in the outer diameter of the trunnion is

$$\Delta D = D\alpha\Delta T \qquad (1)$$

where

$$D = \text{outer diameter of the trunnion,}$$

$$\alpha = \text{coefficient of thermal expansion coefficient at room temperature,}$$

$$\Delta T = \text{change in temperature,}$$

Given

$$D = 12.363''$$

$$\alpha = 6.817 \times 10^{-6} \frac{in}{in/°F} \text{ at } 80°F$$

$$\begin{aligned} \Delta T &= T_{fluid} - T_{room} \\ &= -108 - 80 \\ &= -188°F \end{aligned}$$

where

$$T_{fluid} = \text{temperature of dry-ice/alcohol mixture}$$

$$T_{room} = \text{room temperature}$$

The reduction in the trunnion outer diameter from Equation (1) is given by

$$\begin{aligned}\Delta D &= 12.363 \times \left(6.47 \times 10^{-6}\right)(-188) \\ &= -0.01504''\end{aligned}$$

So the trunnion is predicted to reduce in diameter by $0.01504''$. But is this enough reduction in diameter? As per the specifications, he needs the trunnion to contract by

$$\begin{aligned}&= \text{trunnion outside diameter} - \text{hub inner diameter} + \text{diametral clearance} \\ &= 12.363'' - 12.358'' + 0.01'' \\ &= 0.015''\end{aligned}$$

So, according to his calculations, it is enough to put the steel trunnion in dry-ice/alcohol mixture to get the desired contraction of $0.015''$ as he is predicting a contraction of $0.01504''$.

But as shown in Figure 3, the coefficient of linear thermal expansion of steel decreases with a decrease in temperature and is not constant over the range of temperature the trunnion goes through. Hence the above formula (Equation 1) would overestimate the thermal contraction. This assumption is the mistake he made in the calculations for the earlier bridge.

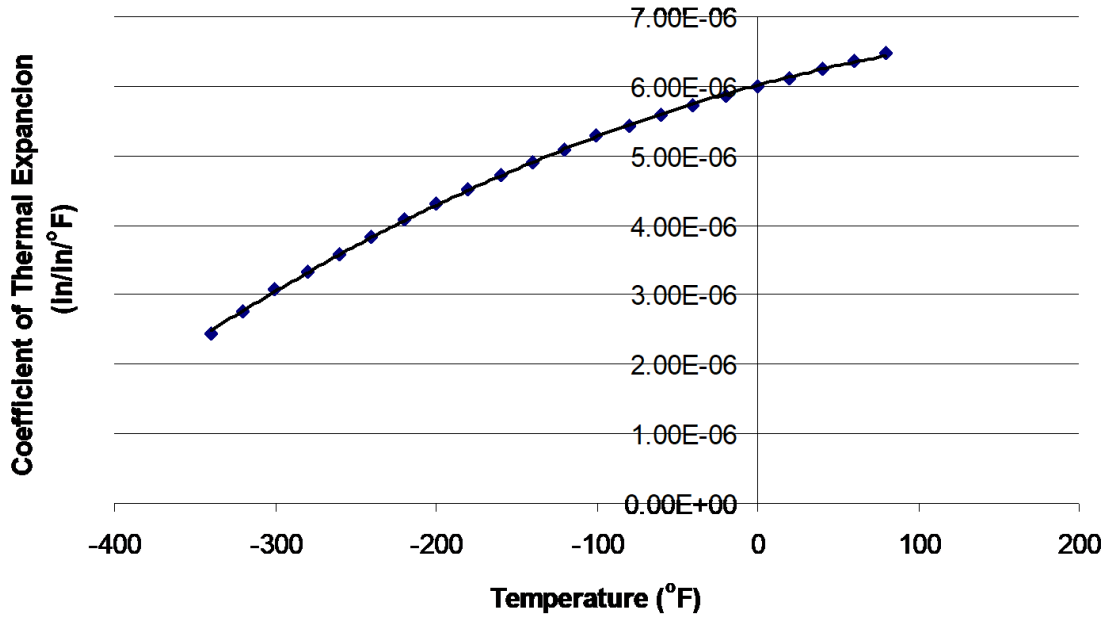


**Figure 3** Varying coefficient of linear thermal expansion as a function of temperature for cast steel.

To find the contraction of a steel cylinder immersed in a bath of liquid nitrogen, one needs to know the coefficient of linear thermal expansion data as a function of temperature. This data is given for steel in Table 1.

**Table 1** Coefficient of linear thermal expansion as a function of temperature

| Temperature, T ($^\circ F$) | Coefficient of linear thermal expansion, $\alpha$ ($\mu in/in/^\circ F$) |
|---|---|
| 80 | 6.47 |
| 40 | 6.24 |
| $-40$ | 5.72 |
| $-120$ | 5.09 |
| $-200$ | 4.30 |
| $-280$ | 3.33 |
| $-340$ | 2.45 |

An exercise to appreciate the way the thermal expansion coefficient changes with respect to the temperature is to look at the slope of the thermal expansion coefficient with respect to temperature at low and high temperatures.

## Questions

(1) Using the data from Table 1, is the rate of change of coefficient of thermal expansion with temperature more at $T = 80^\circ F$ than at $T = -340^\circ F$ ? Use any numerical differentiation technique.

(2) The data given in Table 1 can be regressed to $\alpha = a_0 + a_1 T + a_2 T^2$ to get $\alpha = 6.0217 \times 10^{-6} + 6.2782 \times 10^{-9} T - 1.2218 \times 10^{-11} T^2$. Compare the results from Problem 1 if you use the regression curve to find the rate of change of coefficient of thermal expansion with respect to temperature at $T = 80^\circ F$ and at $T = -340^\circ F$.

# Chapter 02.01: Prerequisites to Numerical Differentiation

Lesson: Prerequisites to Numerical Differentiation

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *recall the exact definition of differentiation*

2) *derive exact derivatives of a few functions*

## Introduction

In this lesson, we will review the necessary concepts of differential calculus to learn numerical differentiation. These include the concepts of finding exact derivatives of functions.

The derivative of a function represents the rate of change of a variable with respect to another variable. For example, the velocity of a body is defined as the rate of change of the location of the body with respect to time. The location is the *dependent* variable, while time is the *independent* variable. Now, if we measure the rate of change of velocity with respect to time, we get the acceleration of the body. In this case, velocity is the *dependent* variable, while time is the *independent* variable.

Whenever differentiation is introduced to a student, the two concepts of the secant line and tangent line (Figure 1) are revisited.

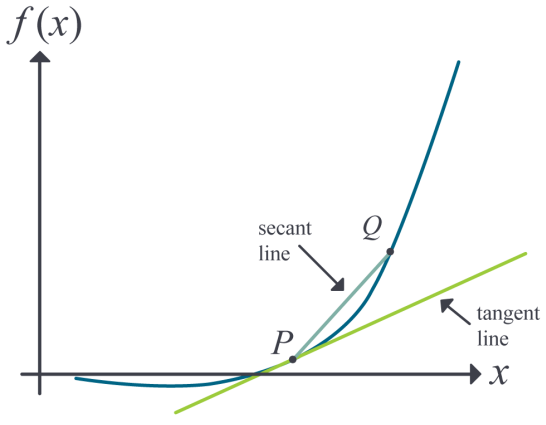


**Figure 1:** Function curve with tangent and secant lines

Let $P$ and $Q$ be two points on the curve, as shown in Figure 1. The secant line is the straight line drawn through $P$ and $Q$.

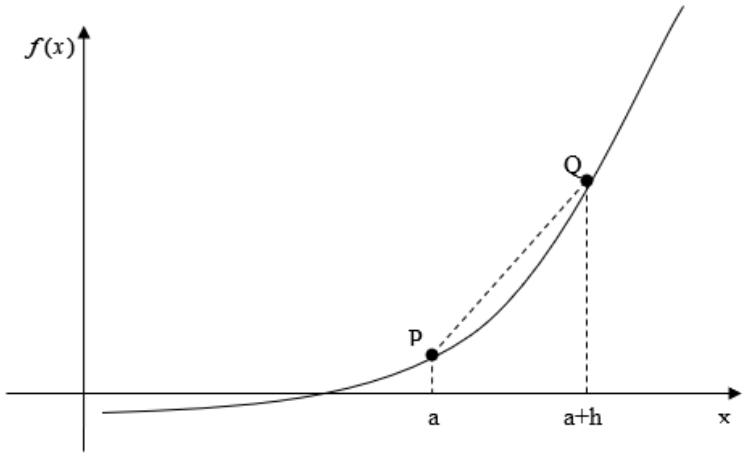



**Figure 2:** Calculation of the secant line

The slope of the secant line (Figure 2) is then given as

$$\begin{aligned} m_{PQ,secant} &= \frac{f(a+h)-f(a)}{(a+h)-a} \\ &= \frac{f(a+h)-f(a)}{h} \end{aligned} \qquad (1)$$

As $Q$ moves closer and closer to $P$, the limiting portion is called the tangent line. The slope of the tangent line $m_{PQ,tangent}$ then is the limiting value of $m_{PQ,secant}$ from Equation (1) as $h \to 0$.

$$m_{PQ,tangent} = \lim_{h \to 0} \frac{f(a+h)-f(a)}{h} \qquad (2)$$

**Example 1**

Find the slope of the secant line of the curve $y = 4x^2$ between points $(3, 36)$ and $(5, 100)$.

**Solution**

The slope of the secant line between $(3, 36)$ and $(5, 100)$ from Equation (1) is

$$\begin{aligned} m &= \frac{f(5)-f(3)}{5-3} \\ &= \frac{100-36}{5-3} \\ &= 32 \end{aligned}$$

**Example 2**

Find the slope of the tangent line of the curve $y = 4x^2$ at point $(3, 36)$.

**Solution**

The slope of the tangent line at $(3, 36)$ from Equation (2) is

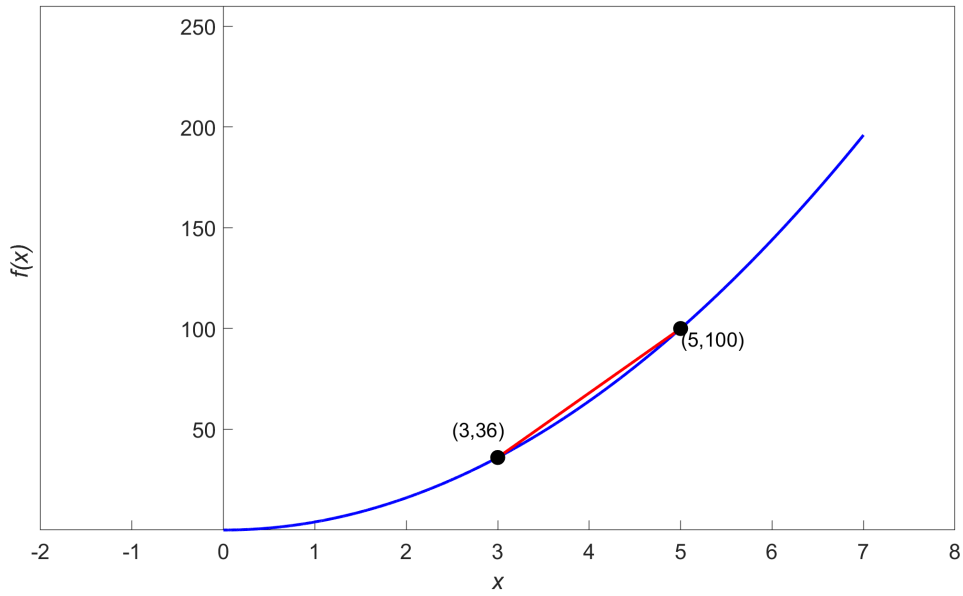



**Figure 3:** Calculation of the secant line for the function $y = 4x^2$

$$
\begin{aligned}
m &= \lim_{h \to 0} \frac{f(3+h) - f(3)}{h} \\
&= \lim_{h \to 0} \frac{4(3+h)^2 - 4(3)^2}{h} \\
&= \lim_{h \to 0} \frac{4(9 + h^2 + 6h) - 36}{h} \\
&= \lim_{h \to 0} \frac{36 + 4h^2 + 24h - 36}{h} \\
&= \lim_{h \to 0} \frac{h(4h + 24)}{h} \\
&= \lim_{h \to 0} (4h + 24) \\
&= 24
\end{aligned}
$$

## Derivative of a Function

Recall from calculus, the derivative of a function $f(x)$ at $x = a$ is defined as the slope of the tangent line at $x = a$

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h} \qquad (3)$$

**Example 3**

Find $f'(3)$ if $f(x) = 4x^2$.

**Solution**

$$\begin{aligned} f'(x) &= 4(2x) \\ &= 8x \\ &= 24 \end{aligned}$$

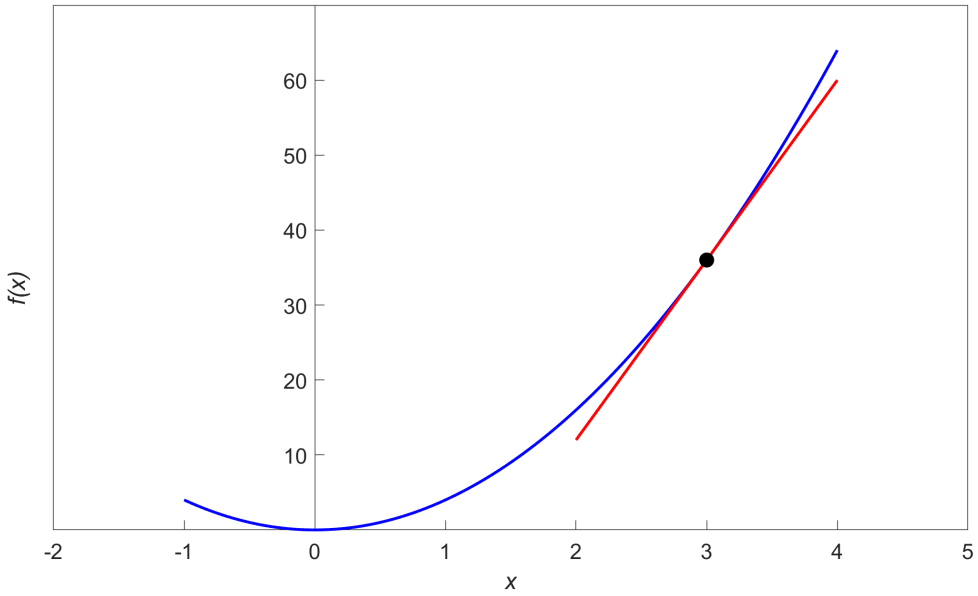



**Figure 4:** Calculation of the tangent line in the function $y = 4x^2$.

## Second Definition of Derivatives

There is another form of the definition of the derivative of a function. The derivative of the function $f(x)$ at $x = a$ is defined as

$$f'(a) = \lim_{x \to a} \frac{f(x) - f(a)}{x - a} \qquad (4)$$

As $x \to a$, the definition is nothing but the slope of the tangent line at $P$.

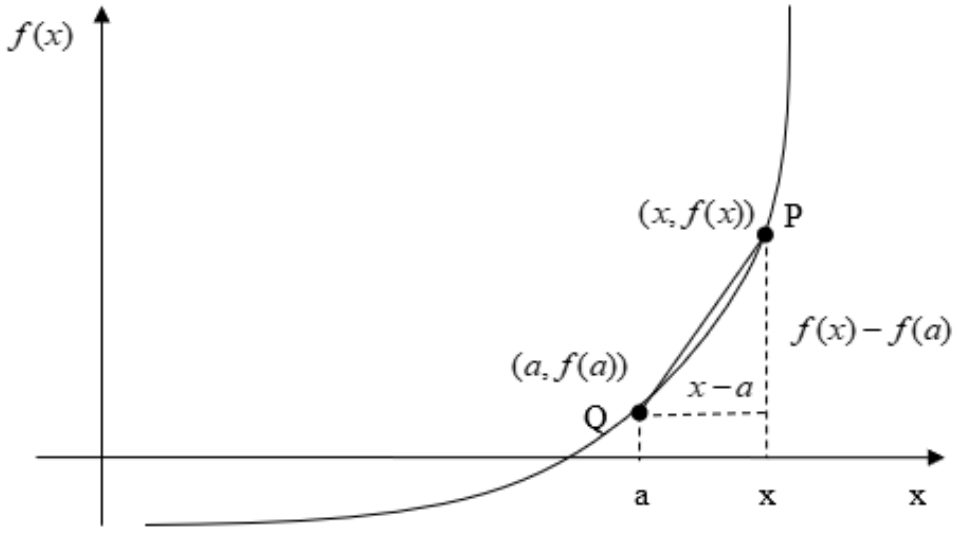



**Figure 5:** Graph showing the second definition of the derivative

**Example 4**

Find $f'(3)$ if $f(x) = 4x^2$ by using the form

$$f'(a) = \lim_{x \to a} \frac{f(x) - f(a)}{x - a}$$

of the definition of a derivative.

**Solution**

$$\begin{aligned}
f'(3) &= \lim_{x \to 3} \frac{f(x) - f(3)}{x - 3} \\
&= \lim_{x \to 3} \frac{4x^2 - 4(3)^2}{x - 3} \\
&= \lim_{x \to 3} \frac{4x^2 - 36}{x - 3} \\
&= \lim_{x \to 3} \frac{4(x^2 - 9)}{x - 3} \\
&= \lim_{x \to 3} \frac{4(x - 3)(x + 3)}{x - 3} \\
&= \lim_{x \to 3} 4(x + 3) \\
&= 4(3 + 3) \\
&= 24
\end{aligned}$$

## Higher-Order Derivatives

So far, we have limited our discussion to calculating the first derivative, $f'(x)$ of a function $f(x)$. What if we are asked to calculate higher-order derivatives of $f(x)$. A simple example of this is finding the acceleration of a body from a function that gives the location of the body as a function of time. The derivative of the location with respect to time is the velocity of the body, followed by the derivative of velocity with respect to time being the acceleration. Hence, the second derivative of the location function gives the acceleration function of the body.

**Example 5**

Given $f(x) = 3x^3 - 2x - 7$, find the second derivative, $f''(x)$ and the third derivative, $f'''(x)$. Also, find the value of $f''(2)$ and $f'''(2)$.

**Solution**

Given

$$f(x) = 3x^3 - 2x - 7$$

we have

$$\begin{aligned} f'(x) &= 3(3x^2) - 2 \\ &= 9x^2 - 2 \end{aligned}$$

$$\begin{aligned} f''(x) &= \frac{d}{dx}(f'(x)) \\ &= \frac{d}{dx}\left(9x^2 - 2\right) \\ &= 9\left(2x\right) \\ &= 18x \end{aligned}$$

$$\begin{aligned} f\left(2\right) &= 18 \times 2 \\ &= 36 \end{aligned}$$

$$\begin{aligned} f'''(x) &= \frac{d}{dx}(f''(x)) \\ &= \frac{d}{dx}(18x) \\ &= 18 \end{aligned}$$

$$f'''(2) = 18$$

## Other Notations of Derivatives

Derivatives can be denoted in several ways. For the first derivative, the notations include

$$f'(x),\ \frac{d}{dx}f(x),\ y', \dot{y},\ \text{and}\ \frac{dy}{dx}$$

For the second derivative, the notations include

$$f''(x),\ \frac{d^2}{dx^2}f(x),\ y'',\ \ddot{y},\ \text{and}\ \frac{d^2y}{dx^2}$$

For the $n^{th}$ derivative, the notations include

$$f^{(n)}(x),\ \frac{d^n}{dx^n}f(x),\ y^{(n)},\ \text{and}\ \frac{d^ny}{dx^n}$$

# Chapter 02.01: Prerequisites to Numerical Differentiation

Lesson: Equation of a Tangent Line

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *find the equation of the tangent line to a function.*

## Finding equations of a tangent line

One of the numerical methods used to solve a nonlinear equation is called the Newton-Raphson method. This method is based on the prerequisite knowledge of finding the tangent line to a curve at a point. Let us look at an example to illustrate finding the equation of the tangent line to a curve.

### Example 1

Find the equation of the line tangent to the function

$$f(x) = x^3 - 0.165x + 3.993 \times 10^{-4}$$

at $x = 0.05$.

### Solution

The line tangent is a straight line of the form

$$y = mx + c$$

To find the equation of the tangent line, let us first find the slope $m$ of the straight line.

$$f'(x) = 3x^2 - 0.165$$

At $x = 0.05$

$$\begin{aligned} f'(0.05) &= 3(0.05)^2 - 0.165 \\ &= -0.1575 \end{aligned}$$

Hence

$$m = -0.1575$$

To find the value of the $y$ -intercept $c$ of the straight line, we first find the value of the function at $x = 0.05$.

$$\begin{aligned} f(0.05) &= (0.05)^3 - 0.165(0.05) + 3.993 \times 10^{-4} \\ &= -0.0077257 \end{aligned}$$

The tangent line passes through the point $(0.05, -0.0077257)$, so

$$-0.0077257 = m(0.05) + c$$

$$-0.0077257 = -0.1575(0.05) + c$$

$$c = 0.0001493$$

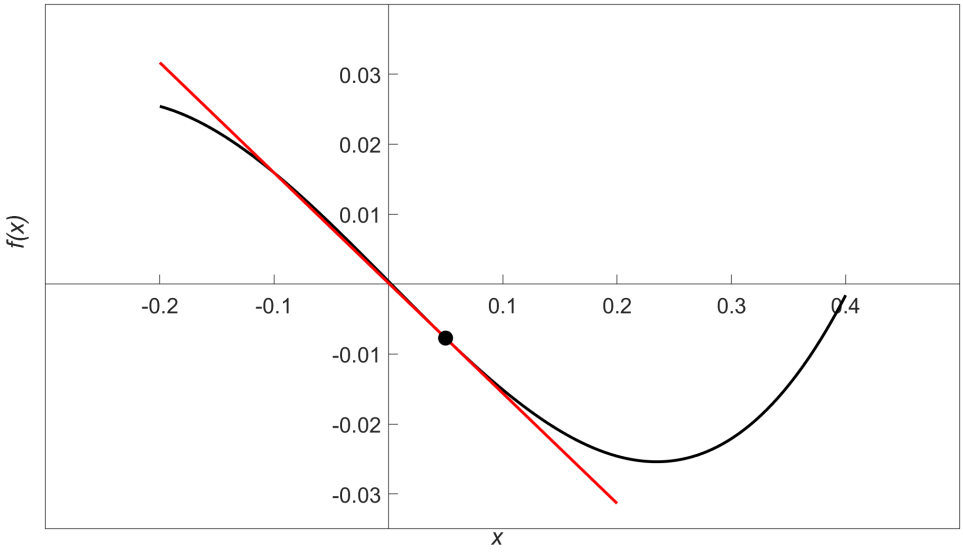



**Figure 1** Graph of function $f(x)$ and the tangent line at $x = 0.05$.

Hence,

$$\begin{aligned} y &= mx + c \\ &= -0.1575x + 0.0001493 \end{aligned}$$

is the equation of the tangent line.

# Chapter 02.01: Prerequisites to Numerical Differentiation

Lesson: Theorems Related to Differentiation

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *state several theorems of differentiation*

2) *apply the theorems of differentiation in finding the derivatives of complex functions.*

## Theorems of Differentiation

Several theorems of differentiation are revisited to show how one can find the exact derivative of different functions.

### Theorem 1

The derivative of a constant is zero. If $f(x) = k$, where $k$ is a constant, $f'(x) = 0$.

**Example 1**

Find the derivative of

$$f(x) = 6.$$

**Solution**

$$f(x) = 6$$

$$f'(x) = 0$$

### Theorem 2

The derivative of $f(x) = x^n$, where $n \neq 0$ is $f'(x) = nx^{n-1}$.

**Example 2**

Find the derivative of

$$f(x) = x^6.$$

**Solution**

$$f(x) = x^6$$

$$\begin{aligned} f'(x) &= 6x^{6-1} \\ &= 6x^5 \end{aligned}$$

**Example 3**

Find the derivative of

$$f(x) = x^{-6}.$$

**Solution**

$$f(x) = x^{-6}$$

$$\begin{aligned} f'(x) &= -6x^{-6-1} \\ &= -6x^{-7} \\ &= -\frac{6}{x^7} \end{aligned}$$

**Theorem 3**

The derivative of $f(x) = kg(x)$, where $k$ is a constant is $f'(x) = kg'(x)$.

**Example 4**

Find the derivative of

$$f(x) = 10x^6.$$

**Solution**

$$f(x) = 10x^6$$

$$\begin{aligned} f'(x) &= \frac{d}{dx}(10x^6) \\ &= 10\frac{d}{dx}x^6 \\ &= 10(6x^5) \\ &= 60x^5 \end{aligned}$$

**Theorem 4**

The derivative of $f(x) = u(x) \pm v(x)$ is $f'(x) = u'(x) \pm v'(x)$.

**Example 5**

Find the derivative of

$$f(x) = 3x^3 + 8.$$

**Solution**

$$f\,(x) \;= 3x^3 + 8.$$

$$f'\,(x) = \frac{d}{dx}(3x^3 + 8)$$

$$\begin{aligned} f'\,(x) &= \frac{d}{dx}(3x^3) + \frac{d}{dx}(8) \\ &= 3\frac{d}{dx}(x^3) + 0 \\ &= 3(3x^2) \\ &= 9x^2 \end{aligned}$$

**Theorem 5**

The derivative of

$$f(x) = u(x)v(x)$$

is

$$f'(x) = u(x)\frac{d}{dx}v(x) + v(x)\frac{d}{dx}u(x). \text{ (Product Rule)}$$

**Example 6**

Find the derivative of

$$f(x) = (2x^2 - 6)(3x^3 + 8)$$

**Solution**

Using the product rule as given by Theorem 5 where

$$f(x) = u(x)v(x)$$

$$f'(x) = u(x)\frac{d}{dx}v(x) + v(x)\frac{d}{dx}u(x)$$

$$f(x) = (2x^2 - 6)(3x^3 + 8)$$

$$u(x) = 2x^2 - 6$$

$$v(x) = 3x^3 + 8$$

Taking the derivative of $u(x)$,

$$\begin{aligned}
\frac{du}{dx} &= \frac{d}{dx}(2x^2 - 6) \\
&= \frac{d}{dx}(2x^2) - \frac{d}{dx}(6) \\
&= 2\frac{d}{dx}(x^2) - 0 \\
&= 2(2x) \\
&= 4x
\end{aligned}$$

Taking the derivative of $v(x)$,

$$\begin{aligned}
\frac{dv}{dx} &= \frac{d}{dx}(3x^3 + 8) \\
&= \frac{d}{dx}(3x^3) + \frac{d}{dx}(8) \\
&= 3\frac{d}{dx}(x^3) + 0 \\
&= 3(3x^2) \\
&= 9x^2
\end{aligned}$$

Using the formula for the product rule

$$\begin{aligned}
f'(x) &= u(x)\frac{d}{dx}v(x) + v(x)\frac{d}{dx}u(x) \\
&= (2x^2 - 6)(9x^2) + (3x^3 + 8)(4x) \\
&= 18x^4 - 54x^2 + 12x^4 + 32x \\
&= 30x^4 - 54x^2 + 32x
\end{aligned}$$

**Theorem 6**

The derivative of

$$f(x) = \frac{u(x)}{v(x)}$$

is

$$f'(x) = \frac{v(x)\frac{d}{dx}u(x) - u(x)\frac{d}{dx}v(x)}{(v(x))^2} \text{ (Quotient Rule)}$$

**Example 7**

Find the derivative of

$$f(x) = \frac{(2x^2 - 6)}{(3x^3 + 8)}.$$

**Solution**

Use the quotient rule of Theorem 6, if

$$f(x) = \frac{u(x)}{v(x)}$$

then

$$f'(x) = \frac{v(x)\frac{d}{dx}u(x) - u(x)\frac{d}{dx}v(x)}{(v(x))^2}$$

From

$$f(x) = \frac{(2x^2 - 6)}{(3x^3 + 8)}$$

we have

$$u(x) = 2x^2 - 6$$

$$v(x) = 3x^3 + 8$$

Taking the derivative of $u(x)$,

$$\begin{aligned}
\frac{du}{dx} &= \frac{d}{dx}(2x^2 - 6) \\
&= \frac{d}{dx}(2x^2) - \frac{d}{dx}(6) \\
&= 2\frac{d}{dx}(x^2) - 0 \\
&= 2(2x) \\
&= 4x
\end{aligned}$$

Taking the derivative of $v(x)$,

$$\begin{aligned}\frac{dv}{dx} &= \frac{d}{dx}(3x^3+8)\\ &= \frac{d}{dx}(3x^3)+\frac{d}{dx}(8)\\ &= 3\frac{d}{dx}\left(x^3\right)+0\\ &= 3(3x^2)\\ &= 9x^2\end{aligned}$$

Using the formula for the quotient rule,

$$\begin{aligned}f'(x) &= \frac{(3x^3+8)(4x)-(2x^2-6)(9x^2)}{(3x^3+8)^2}\\ &= \frac{12x^4+32x-18x^4+54x^2}{9x^6+48x^3+64}\\ &= \frac{-6x^4+54x^2+32x}{9x^6+48x^3+64}\end{aligned}$$

## Table of Derivatives

| $f(x)$ | $f'(x)$ |
|---|---|
| $x^n,\ n\neq 0$ | $nx^{n-1}$ |
| $kx^n,\ n\neq 0$ | $knx^{n-1}$ |
| $\sin(x)$ | $\cos(x)$ |
| $\cos(x)$ | $-\sin(x)$ |
| $\tan(x)$ | $\sec^2(x)$ |
| $\sinh(x)$ | $\cosh(x)$ |
| $\cosh(x)$ | $\sinh(x)$ |
| $\tanh(x)$ | $1-\tanh^2(x)$ |
| $\sin^{-1}(x)$ | $\frac{1}{\sqrt{1-x^2}}$ |
| $\cos^{-1}(x)$ | $\frac{-1}{\sqrt{1-x^2}}$ |
| $\tan^{-1}(x)$ | $\frac{1}{1+x^2}$ |
| $\csc(x)$ | $-\csc(x)\cot(x)$ |
| $\sec(x)$ | $\sec(x)\tan(x)$ |
| $\cot(x)$ | $-csc^2(x)$ |
| $csch(x)$ | $-\coth(x)csch(x)$ |
| $sech(x)$ | $-\tanh(x)sech(x)$ |
| $\coth(x)$ | $1-\coth^2(x)$ |
| $csc^{-1}(x)$ | $-\frac{\lvert x\rvert}{x^2\sqrt{x^2-1}}$ |
| $\sec^{-1}(x)$ | $\frac{\lvert x\rvert}{x^2\sqrt{x^2-1}}$ |
| $\cot^{-1}(x)$ | $\frac{-1}{1+x^2}$ |

| $f(x)$ | $f'(x)$ |
|---|---|
| $a^x$ | $\ln(a)a^x$ |
| $\ln(x)$ | $\frac{1}{x}$ |
| $\log_a(x)$ | $\frac{1}{x\ln(a)}$ |
| $e^x$ | $e^x$ |

## Chain Rule of Differentiation

Sometimes functions that need to be differentiated do not fall in the form of simple functions or the forms described previously. Such functions can be differentiated using the chain rule if they are of the form $f(g(x))$. The chain rule states

$$\frac{d}{dx}(f(g(x)) = f'(g(x))g'(x)$$

For example, to find $f'(x)$ of $f(x) = (3x^2 - 2x)^4$, one could use the chain rule.

$$g(x) = (3x^2 - 2x)$$

$$g'(x) = 6x - 2$$

$$f'(g(x)) = 4(g(x))^3$$

$$\frac{d}{dx}((3x^2 - 2x)^4) = 4(3x^2 - 2x)^3(6x - 2)$$

## Implicit Differentiation

Sometimes, the function to be differentiated is not given explicitly as an expression of the independent variable. In such cases, how do we find the derivatives? We will discuss this via examples.

**Example 8**

Find

$$\frac{dy}{dx} \text{ if } x^2 + y^2 = 2xy$$

**Solution**

$$x^2 + y^2 = 2xy$$

$$\frac{d}{dx}(x^2 + y^2) = \frac{d}{dx}(2xy)$$

$$\frac{d}{dx}(x^2) + \frac{d}{dx}(y^2) = \frac{d}{dx}(2xy)$$

$$2x + 2y\frac{dy}{dx} = 2x\frac{dy}{dx} + 2y$$

$$2y\frac{dy}{dx} - 2x\frac{dy}{dx} = 2y - 2x$$

$$(2y - 2x)\frac{dy}{dx} = 2y - 2x$$

$$\frac{dy}{dx} = \frac{2y - 2x}{2y - 2x}$$

$$\frac{dy}{dx} = 1$$

**Example 9**

If $x^2 - xy + y^2 = 5$, find the value of $y'$.

**Solution**

$$x^2 - xy + y^2 = 5$$

$$\frac{d}{dx}(x^2 - xy + y^2) = 5$$

$$\frac{d}{dx}(x^2) - \frac{d}{dx}(xy) + \frac{d}{dx}(y^2) = 0$$

$$2x - x\frac{dy}{dx} - y + 2y\frac{dy}{dx} = 0$$

$$(-x + 2y)\frac{dy}{dx} = -2x + y$$

$$\frac{dy}{dx} = \frac{y - 2x}{2y - x}$$

$$y' = \frac{y - 2x}{2y - x}$$

# Chapter 02.01: Prerequisites to Numerical Differentiation

Lesson: Minimum and Maximum of a Function of One Variable

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *find the minimum and maximum of a function of one variable.*

## Finding maximum and minimum of a function

The knowledge of the first derivative and second derivative of a function is used to find the minimum and maximum of a function. First, let us define what the maximum and minimum of a function are. Let $f(x)$ be a function in domain $D$, then

$f(a)$ is the maximum of the function if $f(a) \geq f(x)$ for all values of $x$ in the domain $D$.

$f(a)$ is the minimum of the function if $f(a) \leq f(x)$ for all values of $x$ in the domain $D$.

The minimum and maximum of a function are also the critical values of a function.

An extreme value can also occur in the interval $[c, d]$ at

1) endpoints $x = c, x = d$,

2) a point in $[c, d]$ where $f'(x) = 0$.

3) a point in $[c, d]$ where $f'(x)$ does not exist.

These critical points can be the local maxima and minima of the function (See Figure 2).

### Example 1

Find the minimum and maximum value of $f(x) = x^2 - 2x - 5$ in the interval $[0, 5]$.

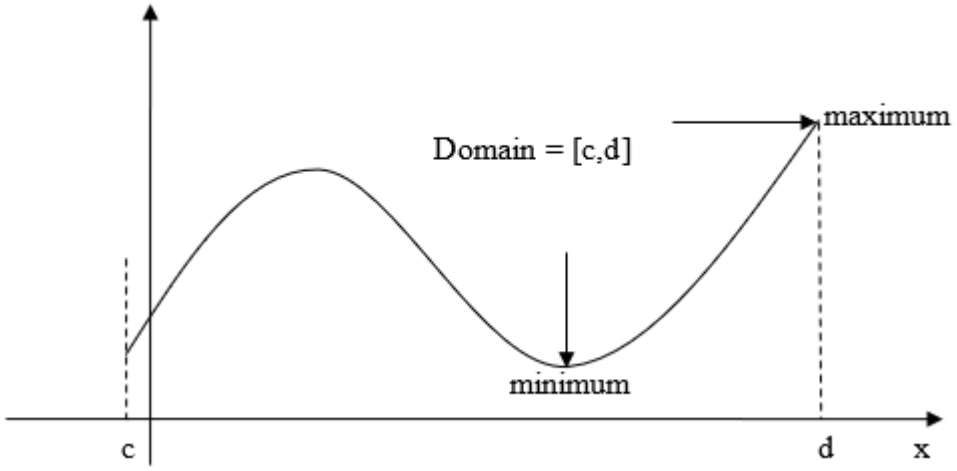



**Figure 1.** Graph illustrating the concepts of maximum and minimum

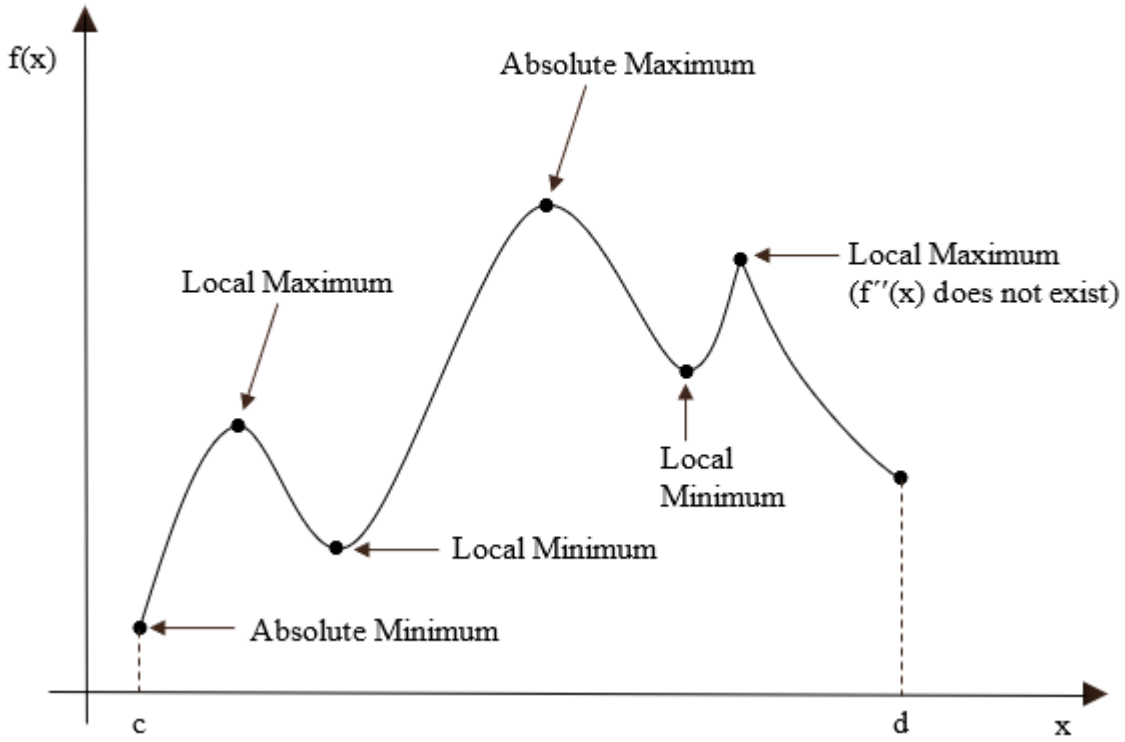



**Figure 2.** The plot showing critical points of $f(x)$ in $[c, d]$

**Solution**

$$f(x) = x^2 - 2x - 5$$

$$f'(x) = 2x - 2$$

$$f'(x) = 0 \text{ at } x = 1.$$

$$f'(x) \text{ exists everywhere in } [0, 5].$$

So, the critical points are $x = 0, x = 1, x = 5$.

$$f(0) = (0)^2 - 2(0) - 5 \ = -5$$

$$f(1) = (1)^2 - 2(1) - 5 \ = -6$$

$$f(5) = (5)^2 - 2(5) - 5 \ = 10$$

Hence, the minimum value of $f(x)$ occurs at $x = 1$, and the maximum value occurs at $x = 5$.

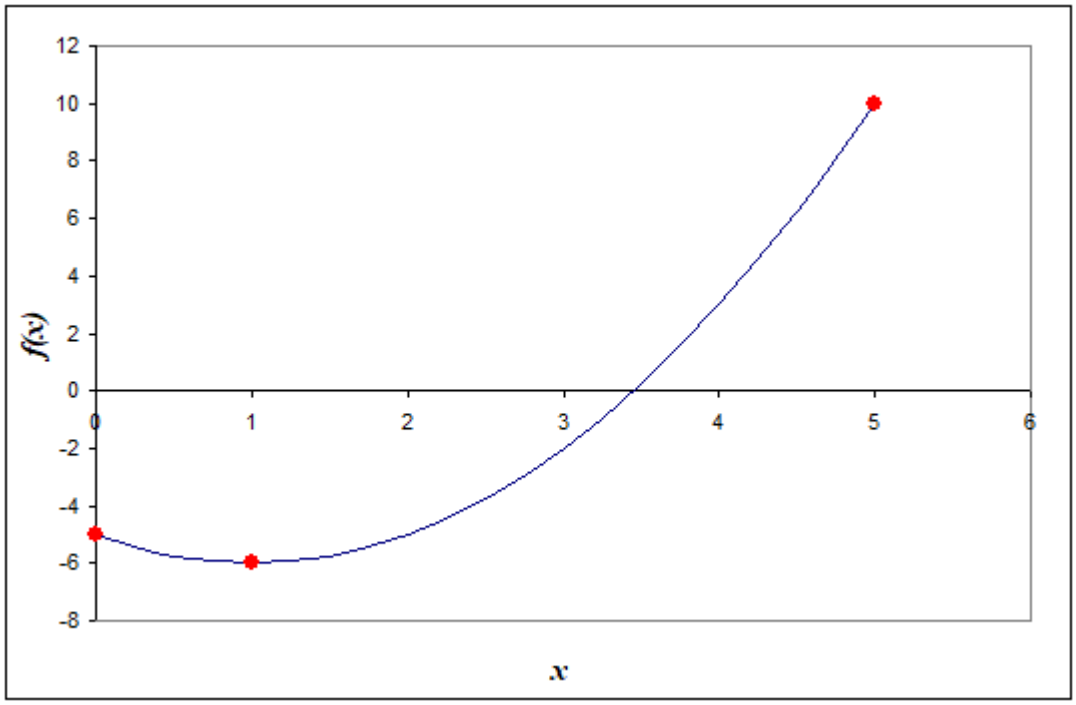



**Figure 3.** Maximum and minimum values of $f(x) = x^2 - 2x - 5$ over the interval $[0, 5]$

Figure 4 shows an example of a function that has no minimum or maximum value in the domain $(0, \infty)$.

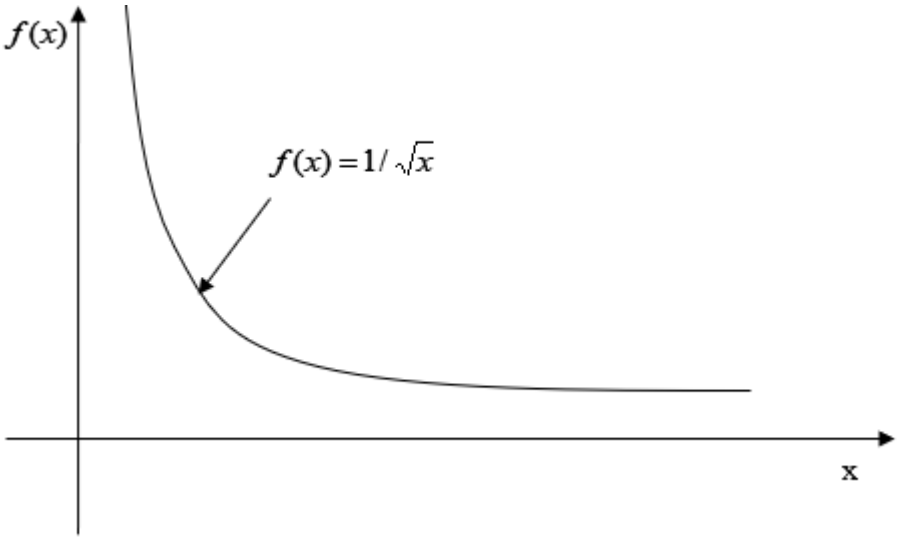



**Figure 4.** Function that has no maximum or minimum in an open interval.

Figure 5 shows the maximum of the function occurring at a singular point. The function $f(x)$ has a sharp corner at $x = a$, where $f'(x)$ does not exist.

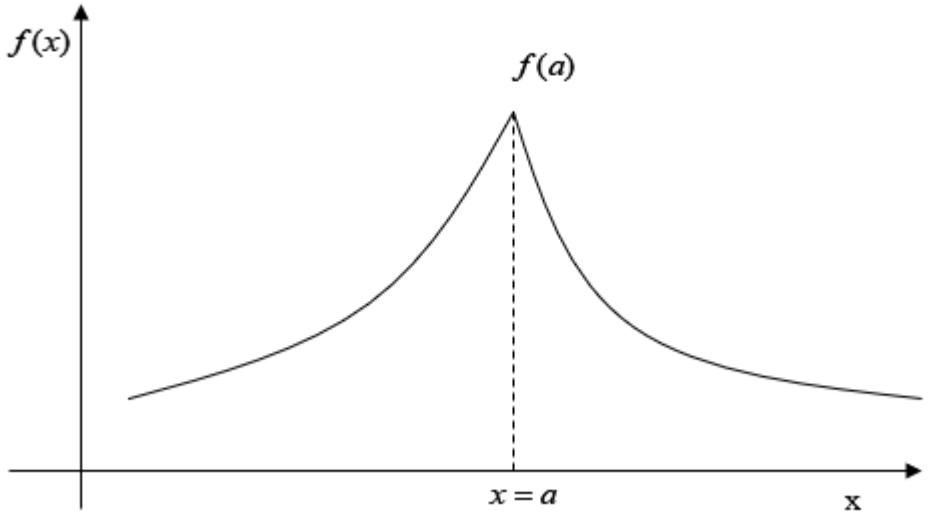



**Figure 5.** Graph demonstrates the concept of a singular point with discontinuous slope at $x = a$

**Example 2**

Find the maximum and minimum of $f(x) = 2x$ in the interval $[0, 5]$.

**Solution**

$$f(x) = 2x$$

$$f'(x) = 2$$

$$f'(x) \neq 0 \text{ on } [0, 5].$$

So the critical points are $x = 0$ and $x = 5$.

$$f(x) = 2x$$

$$f(0) = 2(0) = 0$$

$$f(5) = 2(5) = 10$$

So the minimum value of $f(x) = 2x$ is at $x = 0$, and the maximum value is at $x = 5$.

The point(s) where the second derivative of a function becomes zero is a way to know whether the critical point found in the first derivative test is a local minimum or maximum. Let $f(x)$ be a function in the interval $(c, d)$ and $f(a) = 0$.

$f(a)$ is a local maximum of the function if $f''(a) < 0$.

$f(a)$ is a local minimum of the function if $f''(a) > 0$.

If $f''(a) = 0$, then the second derivative does not offer any insight into the local maxima or minima.

**Example 3**

Remember Example 1 where we found $f'(x) = 0$ at $x = 1$ for $f(x) = x^2 - 2x - 5$ in the interval $[0, 5]$. Is $x = 1$ a local maxima or minima of the function?

**Solution**

$$f(x) = x^2 - 2x - 5$$

$$f'(x) = 2x - 2$$

$$f'(x) = 0 \text{ at } x = 1$$

$$f''(x) = 2$$
$$f''(1) = 2 > 0$$

So the $f(1)$ is the local minimum of the function.

## Applications of Derivatives

Below are some examples to show real-life applications of differentiation.

**Example 4**

A rain gutter cross-section is shown below.

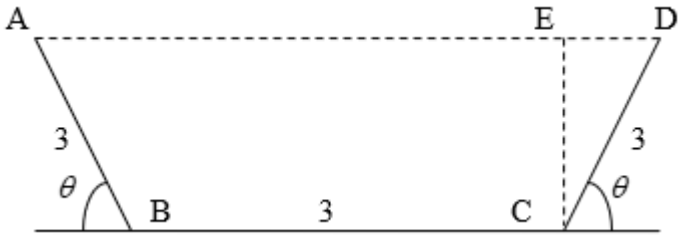



**Figure 6.** Gutter dimensions for example 4

What angle of $\theta$ would make the cross-sectional area of ABCD maximum? Note that common sense or intuition may lead us to believe that $\theta = \pi/4$ would maximize the cross-sectional area of ABCD. Question your intuition.

**Solution**

$$Area = \frac{1}{2}(\overline{BC}+\overline{AD})\times\overline{CE}$$

$$\begin{aligned}\overline{CE} &= \overline{CD}sin(\theta)\\ &= 3sin(\theta)\end{aligned}$$

$$\overline{BC} = 3$$

$$\overline{AD} = \overline{BC} + \overline{CD}\cos(\theta) + \overline{AB}cos(\theta)$$

$$\overline{AD} = 3 + 3cos(\theta) + 3cos(\theta)$$

$$\overline{AD} = 3 + 6cos(\theta)$$

$$\begin{aligned}Area &= \frac{1}{2}(3+3+6cos(\theta))(3sin(\theta))\\ &= 9sin(\theta) + 9sin(\theta)cos(\theta)\\ &= 9sin(\theta) + \frac{9}{2}sin(2\theta)\end{aligned}$$

$$\begin{aligned}\frac{dA}{d\theta} &= 9cos(\theta) + \frac{9}{2}\times 2cos(2\theta)\\ &= 9cos(\theta) + 9cos(2\theta)\end{aligned}$$

When is $\dfrac{dA}{d\theta} = 0$?

$$9cos(\theta) + 9cos(2\theta) = 0$$

$$\theta = \frac{\pi}{3}$$

The angle at which the area is maximum is $\theta = 60°$.

$$\begin{aligned} Area\left(\frac{\pi}{3}\right) &= 9sin\left(\frac{\pi}{3}\right) + \frac{9}{2}\sin\left(2\left(\frac{\pi}{3}\right)\right) \\ &= 9\left(\frac{\sqrt{3}}{2}\right) + \frac{9}{2}\left(\frac{\sqrt{3}}{2}\right) \\ &= \frac{27}{4}\sqrt{3} \end{aligned}$$

For the interval of $\theta = [0, \pi]$, the area at the endpoints is

$$Area(0) = 0$$

$$Area(\pi) = 0$$

**Example 5**

A classic example of the application of differentiation is to find the dimensions of a circular cylinder for a specific volume but which uses the least amount of material. Do this classic problem for a volume of $9m^3$.

**Solution**

The total surface area, $A$ of the cylinder is

$$\begin{aligned} A &= \text{top surface} + \text{side surface} + \text{bottom surface} \\ &= \pi r^2 + 2\pi rh + \pi r^2 \\ &= 2\pi r^2 + 2\pi rh \end{aligned}$$

The volume, $V$ of the cylinder is

$$V = \pi r^2 h$$

since

$$V = 9m^3.$$

We can write

$$9 = \pi r^2 h$$

$$h = \frac{9}{\pi r^2}$$

This gives the surface area just in terms of $r$ as

$$\begin{aligned} A &= 2\pi r^2 + 2\pi r \left(\frac{9}{\pi r^2}\right) \\ &= 2\pi r^2 + \frac{18}{r} \\ &= 2\pi r^2 + 18r^{-1} \end{aligned}$$

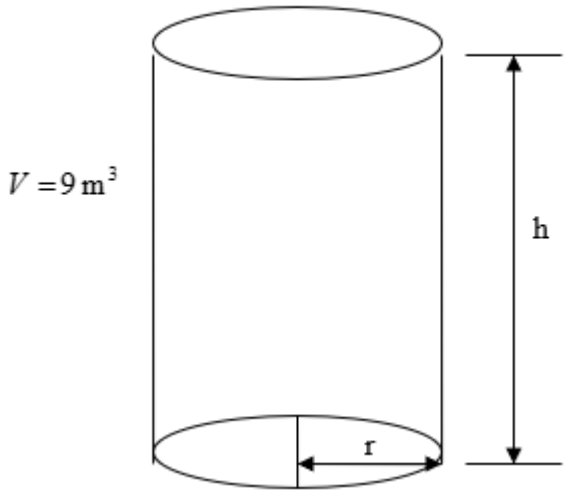



**Figure 7.** Cylinder Drawing for example 5

To find the minimum, take the first derivative of $A$ with respect to $r$ as

$$\begin{aligned} \frac{dA}{dr} &= 4\pi r + 18(-1)r^{-2} \\ &= 4\pi r - \frac{18}{r^2} \end{aligned}$$

Solving for

$$\frac{dA}{dr} = 0,$$

$$4\pi r - \frac{18}{r^2} = 0$$

$$4\pi r^3 - 18 = 0$$

$$r^3 = \frac{18}{4\pi}$$

$$\begin{aligned} r &= \left(\frac{18}{4\pi}\right)^{\frac{1}{3}} \\ &= 1.12725m \end{aligned}$$

Since

$$h = \frac{9}{\pi r^2},$$

$$\begin{aligned} h &= \frac{9}{\pi(1.12725)^2} \\ &= 2.25450m \end{aligned}$$

But does this value of $r$ correspond to a minimum?

$$\begin{aligned} \frac{d^2A}{dr^2} &= 4\pi - 18(-2)r^{-3} \\ &= 4\pi + \frac{36}{r^3} \\ &= 4\pi + \frac{36}{1.12725} \\ &= 44.5025 \end{aligned}$$

This value $\frac{d^2A}{dr^2} > 0$ for $r = 1.12725m$. As per the second derivative test, $r = 1.12725m$ corresponds to a minimum. The value of $r = 1.12725m$ also corresponds to the absolute minimum as $\frac{dA}{dr} = 0$ has only one real solution, and that $A(r)$ is a continuous function of $r$.

# Multiple Choice Test

## Chapter 02.01 Prerequisites to Numerical Differentiation

(1). The definition of the first derivative of a function $f(x)$ is

(A) $f'(x) = \dfrac{f(x+\Delta x) + f(x)}{\Delta x}$

(B) $f'(x) = \dfrac{f(x+\Delta x) - f(x)}{\Delta x}$

(C) $f'(x) = \lim\limits_{\Delta x \to 0} \dfrac{f(x+\Delta x) + f(x)}{\Delta x}$

(D) $f'(x) = \lim\limits_{\Delta x \to 0} \dfrac{f(x+\Delta x) - f(x)}{\Delta x}$

(2). Given $y = 5e^{3x} + \sin x$, $\dfrac{dy}{dx}$ is

(A) $5e^{3x} + \cos x$

(B) $15e^{3x} + \cos x$

(C) $15e^{3x} - \cos x$

(D) $2.666e^{3x} - \cos x$

(3). Given $y = \sin 2x$, $\dfrac{dy}{dx}$ at $x = 3$ is most nearly

(A) 0.9600

(B) 0.9945

(C) 1.920

(D) 1.989

(4). Given $y = x^3 \ln x$, $\dfrac{dy}{dx}$ is

(A) $3x^2 \ln x$

(B) $3x^2 \ln x + x^2$

(C) $x^2$

(D) $3x$

(5). The velocity of a body as a function of time is given as $v(t) = 5e^{-2t} + 4$, where $t$ is in seconds, and $v$ is in m/s. The acceleration in $m/s^2$ at $t = 0.6$ s is

(A) $-3.012$

(B) 5.506

(C) 4.147

(D) $-10.00$

(6). If $x^2 + 2xy = y^2$, then $\dfrac{dy}{dx}$ is

(A) $\dfrac{x+y}{y-x}$

(B) $2x + 2y$

(C) $\dfrac{x+1}{y}$

(D) $-x$

For complete solution, go to

http://nm.mathforcollege.com/mcquizzes/02dif/quiz_02dif_background_solution.pdf

# Problem Set

## Chapter 02.01: Prerequisites to Numerical Differentiation

(1). Find the slope of the secant line for

$$y = 4x^2 + 6x - 7$$

if the secant is drawn between points at $x = 3$ and $x = 5$.

**Answer:** 38

(2). Find the slope of the tangent line to

$$y = 4x^2 + 6x - 7$$

at $x = 3$.

**Answer:** 30

(3). Find the derivative of

$$y = 4x^2 + 6x - 7$$

at $x = 3$.

**Answer:** 30

(4). Find the derivative of

$$y = \frac{1}{5x - 3}$$

at $x = 2$.

**Answer:** $-\frac{5}{(5x-3)^2}$

(5). Find the derivative of

$$y = \frac{2x - 6}{5x - 9}$$

at $x = 4$.

**Answer:** 0.099174

(6). Find the derivative of

$$y = (2x - 6)(5x - 9)$$

at $x = 4$.

**Answer:** 32

(7). Find $y''$ at $x = 2$, given

$$y = 4x^2 + 6x - 7$$

**Answer:** 8

(8). Find $y''$ at $x = 4$, given

$$y = \frac{2x - 6}{5x - 9}$$

**Answer:** $-0.090158$

(9). Show that

$$\left| \frac{y''}{(1 + (y')^2)^{\frac{3}{2}}} \right| = \frac{1}{r}$$

if

$$x^2 + y^2 = r^2.$$

(10). Given

$$x^2y + xy^2 + y^2 = 5$$

find $y'$ and $y''$.

**Answer:** $y' = \dfrac{-y^2 - 2xy}{x^2 + 2yx + 2y}$

$y'' = \dfrac{2y\left(2x^4 + 6x^3y + 6x^2y^2 + 3xy^3 - 4y^2 + 3y^3\right)}{\left(x^2 + 2xy + 2y\right)^2}$

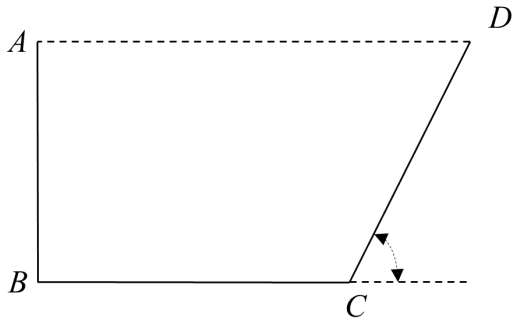
(11). For

$$f(x) = x^2 + 3x$$

find the critical points, and the minimum and maximum value in the domain $[-5, 1]$.

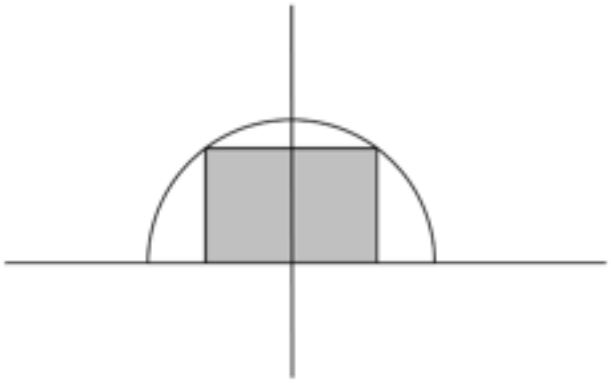
**Answer:** So the critical points are $x = -5,\ x = -1.5,\ x = 1$.
The maximum is at $x = -5$, minimum is at $x = -1.5$

(12). Find the optimum angle $\theta$ for maximizing the cross-section of the gutter given in the figure below. The total length of the sheet from which the gutter is made is $9''$, where $BC = 3''$.




**Answer:** 10.667

(13). A rectangle is inscribed in a semi-circle of radius 2 m. If the area of the rectangle is to be maximized, find the dimensions of the rectangle.



**Answer:** $2.828 \times 1.414$

# Chapter 02.02: Numerical Differentiation of Continuous Functions

Lesson: Numerical Differentiation of Continuous Functions - First Derivative

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *derive formulas for approximating the first derivative of a function,*

2) *use the developed formulas in examples to estimate derivatives of a function.*

## Introduction

The first derivative of a function $f(x)$ at $x$ is defined as

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

To be able to find a derivative numerically, one could make $h$ finite to give,

$$f'(x) \approx \frac{f(x+h) - f(x)}{h}.$$

Knowing the value of $x$ at which you want to find the derivative of $f(x)$, we choose a finite value of $h$ to find the value of $f'(x)$. To estimate the value of $f'(x)$, one such approximation is suggested as follows.

## Forward Difference Approximation of the First Derivative

For a finite $h$,

$$f'(x) \approx \frac{f(x+h) - f(x)}{h} \qquad (1)$$

The above is the forward divided difference approximation of the first derivative. It is called forward because you are taking a point ahead of $x$.

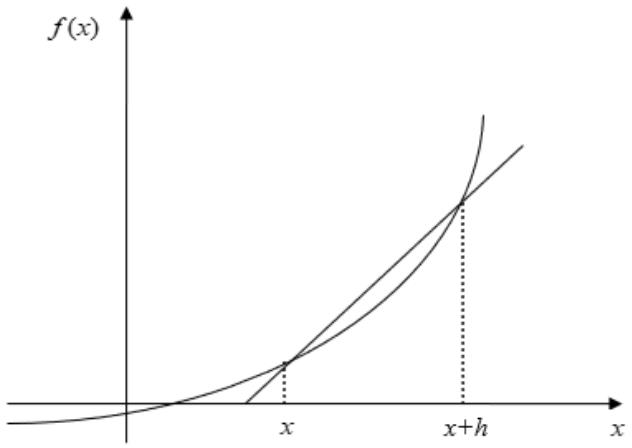



**Figure 1**: Graphical representation of forward difference approximation of first derivative.

**Example 1**

The velocity of a rocket is given by

$$v\left(t\right)=2000\ln\left[\frac{14\times10^{4}}{14\times10^{4}-2100t}\right]-9.8t,\ 0\leq t\leq30$$

where $v$ is given in m/s and $t$ is given in seconds. At $t=16$ s,

a) use the forward difference approximation of the first derivative of $v\left(t\right)$ to calculate the acceleration. Use a step size of $h=2$ s.

b) find the exact value of the acceleration of the rocket.

c) calculate the absolute relative true error for part (b).

**Solution**

(a)

$$a\left(t\right)\approx\frac{v\left(t+h\right)-v\left(t\right)}{h}$$

$$t=16$$

$$h=2$$

$$a\left(16\right)\approx\frac{v\left(18\right)-v\left(16\right)}{2}$$

$$\begin{aligned} v(18) &= 2000 \ln \left[ \frac{14 \times 10^4}{14 \times 10^4 - 2100(18)} \right] - 9.8(18) \\ &= 453.02 \text{ m/s} \end{aligned}$$

$$\begin{aligned} v(16) &= 2000 \ln \left[ \frac{14 \times 10^4}{14 \times 10^4 - 2100(16)} \right] - 9.8(16) \\ &= 392.07 \text{ m/s} \end{aligned}$$

Hence

$$\begin{aligned} a(16) &\approx \frac{v(18) - v(16)}{2} \\ &= \frac{453.02 - 392.07}{2} \\ &= 30.474 \text{ m/s}^2 \end{aligned}$$

(b) The exact value of $a(16)$ can be calculated by differentiating

$$v(t) = 2000 \ln \left[ \frac{14 \times 10^4}{14 \times 10^4 - 2100t} \right] - 9.8t$$

as

$$a(t) = \frac{d}{dt} [v(t)]$$

Knowing that

$$\frac{d}{dt} [\ln(at)] = \frac{1}{t}, \text{ and}$$

$$\frac{d}{dt} \left[ \frac{a}{b - ct} \right] = -\frac{ac}{(b - ct)^2}$$

$$\begin{aligned} a(t) &= 2000 \left( \frac{14 \times 10^4 - 2100t}{14 \times 10^4} \right) \frac{d}{dt} \left( \frac{14 \times 10^4}{14 \times 10^4 - 2100t} \right) - 9.8 \\ &= 2000 \left( \frac{14 \times 10^4 - 2100t}{14 \times 10^4} \right) (-1) \left( \frac{(14 \times 10^4)(-2100)}{(14 \times 10^4 - 2100t)^2} \right) - 9.8 \\ &= \frac{-4040 - 29.4t}{-200 + 3t} \end{aligned}$$

$$\begin{aligned} a(16) &= \frac{-4040 - 29.4(16)}{-200 + 3(16)} \\ &= 29.674 \text{ m/s}^2 \end{aligned}$$

(c) The absolute relative true error is

$$\begin{aligned}|\epsilon_t| &= \left|\frac{\text{True Value} - \text{Approximate Value}}{\text{True Value}}\right| \times 100 \\ &= \left|\frac{29.674 - 30.474}{29.674}\right| \times 100 \\ &= 2.6967\%\end{aligned}$$

## Backward Difference Approximation of the First Derivative

We know

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

For a finite $h$,

$$f'(x) \approx \frac{f(x+h) - f(x)}{h}$$

If $h$ is chosen as a negative number,

$$\begin{aligned}f'(x) &\approx \frac{f(x-h) - f(x)}{-h} \\ &= \frac{f(x) - f(x-h)}{h} \qquad (2)\end{aligned}$$

This is a backward difference approximation as you are taking a point backward from $x$.

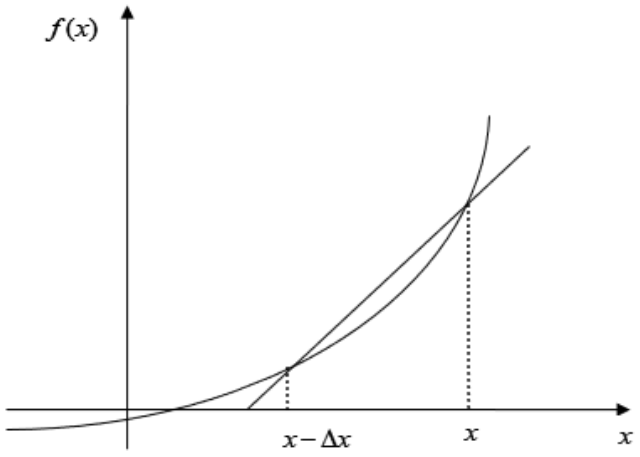



**Figure 2:** Graphical representation of backward difference approximation of the first derivative.

**Example 2**

The velocity of a rocket is given by

$$v(t) = 2000 \ln \left[ \frac{14 \times 10^4}{14 \times 10^4 - 2100t} \right] - 9.8t, \ 0 \leq t \leq 30$$

(a) Use the backward difference approximation of the first derivative of $v(t)$ to calculate the acceleration at $t = 16$ s. Use a step size of $h = 2$ s.

(b) Find the absolute relative true error for part (a).

**Solution**

(a)

$$a(t) \approx \frac{v(t+h) - v(t)}{h}$$

$$t = 16$$

$$h = 2$$

$$a(16) \approx \frac{v(16) - v(14)}{2}$$

$$\begin{aligned} v(16) &= 2000 \ln \left[ \frac{14 \times 10^4}{14 \times 10^4 - 2100(16)} \right] - 9.8(16) \\ &= 392.07 \text{ m/s} \end{aligned}$$

$$\begin{aligned} v(14) &= 2000 \ln \left[ \frac{14 \times 10^4}{14 \times 10^4 - 2100(14)} \right] - 9.8(14) \\ &= 334.24 \text{ m/s} \end{aligned}$$

$$\begin{aligned} a(16) &\approx \frac{v(16) - v(14)}{2} \\ &= \frac{392.07 - 334.24}{2} \\ &= 28.915 \text{ m/s}^2 \end{aligned}$$

(b) The exact value of the acceleration at $t = 16$ s from Example 1 is

$$a(16) = 29.674 \text{ m/s}^2$$

The absolute relative true error for the answer in part (a) is

$$\begin{aligned} |\epsilon_t| &= \left| \frac{29.674 - 28.915}{29.674} \right| \times 100 \\ &= 2.5584\% \end{aligned}$$

## Central Divided Difference Approximation of the First Derivative

If we took a point forward and backward of $x$ by a distance of $h$, then the slope of the secant line (Rise/Run) as shown in Figure 3, and approximating it as the first derivative of the function at $h$,

$$\begin{aligned} f'(x) &\approx \frac{f(x+h) - f(x-h)}{(x+h) - (x-h)} \\ &= \frac{f(x+h) - f(x-h)}{2h} \end{aligned} \tag{3}$$

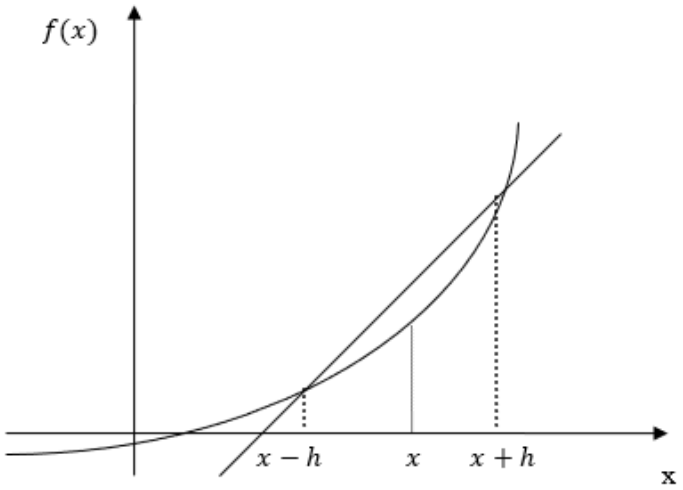



**Figure 3.** Graphical representation of central difference approximation of first derivative.

**Example 3**

The velocity of a rocket is given by

$$v(t) = 2000 \ln \left[ \frac{14 \times 10^4}{14 \times 10^4 - 2100t} \right] - 9.8t, 0 \leq t \leq 30.$$

(a) Use the central difference approximation of the first derivative of $v(t)$ to calculate the acceleration at $t = 16$ s. Use a step size of $h = 2$ s.

(b) Find the absolute relative true error for part (a).

**Solution**

(a)

$$a(t) \approx \frac{v(t+h) - v(t-h)}{2h}$$

$$t = 16$$

$$h = 2$$

$$\begin{aligned} a(16) &\approx \frac{v(16+2) - v(16-2)}{2(2)} \\ &= \frac{v(18) - v(14)}{4} \end{aligned}$$

$$\begin{aligned} v(18) &= 2000 \ln\left[\frac{14 \times 10^4}{14 \times 10^4 - 2100(18)}\right] - 9.8(18) \\ &= 453.02 \text{ m/s} \end{aligned}$$

$$\begin{aligned} v(14) &= 2000 \ln\left[\frac{14 \times 10^4}{14 \times 10^4 - 2100(14)}\right] - 9.8(14) \\ &= 334.24 \text{ m/s} \end{aligned}$$

$$\begin{aligned} a(16) &\approx \frac{v(18) - v(14)}{4} \\ &= \frac{453.02 - 334.24}{4} \\ &= 29.694 \text{ m/s}^2 \end{aligned}$$

(b) The exact value of the acceleration at $t = 16$ s from Example 1 was found as

$$a(16) = 29.674 \text{ m/s}^2$$

The absolute relative true error for the answer in part (a) is

$$\begin{aligned} |\epsilon_t| &= \left|\frac{\text{True Value} - \text{Approximate Value}}{\text{True Value}}\right| \times 100 \\ &= \left|\frac{29.674 - 29.694}{29.674}\right| \times 100 \\ &= 0.069157\% \end{aligned}$$

## Order of Accuracy of Divided Difference Schemes

When using the divided difference schemes, errors are introduced due to their approximate nature. However the order of the error is not the same.

The order of accuracy in the forward divided difference method is of $O(h)$.

$$f'(x) = \frac{f(x+h) - f(x)}{h} + O(h) \qquad (4)$$

The order of accuracy in the backward divided difference method is of $O(h)$.

$$f'(x) = \frac{f(x) - f(x-h)}{h} + O(h) \qquad (5)$$

The order of accuracy in the central divided difference method is of $O\left(h^2\right)$.

$$f'(x) = \frac{f(x+h) - f(x-h)}{2h} + O\left(h^2\right) \qquad (6)$$

The derivation and application of the order of the accuracy of the three divided difference schemes as given by Equations (4)-(6) is given in the next lesson in the textbook.

# Chapter 02.02: Numerical Differentiation of Continuous Functions

Numerical Differentiation of Continuous Functions - Second Derivative

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *derive formulas for approximating for higher-order derivatives,*

2) *use the developed formulas in examples to find higher derivatives of a function*

## Introduction

We used trigonometry and calculus to find the approximation of the first derivatives in the previous lessons. To find higher-order derivatives by such means is cumbersome and hinders a general approach to developing the numerical methods of approximating derivatives of any order.

One can use the Taylor series to approximate a higher-order derivative. As an illustration, here we derive the central difference approximation of the second derivative of a function $f(x)$ .

The Taylor series is given by

$$f(x+h) = f(x) + f'(x)\ h + \frac{f''(x)}{2!}(h)^2 + \frac{f'''(x)}{3!}(h)^3 + \frac{f''''(x)}{4!}(h)^4 + \ldots \qquad (1)$$

We can rewrite the Taylor series by substituting $-h$ for $h$,

$$f(x-h) = f(x) + f'(x)\ (-h) + \frac{f''(x)}{2!}(-h)^2 + \frac{f'''(x)}{3!}(-h)^3 + \frac{f''''(x)}{4!}(-h)^4 + \ldots$$

to get

$$f(x-h) = f(x) - f'(x)\ h + \frac{f''(x)}{2!}h^2 - \frac{f'''(x)}{3!}h^3 + \frac{f''''(x)}{4!}h^4 + \ldots \qquad (2)$$

Adding Equations (1) and (2) gives

$$f(x+h) + f(x-h) = 2f(x) + 2\frac{f''(x)}{2!}h^2 + 2\frac{f''''(x)}{4!}h^4 + \ldots$$

Moving all terms other than the $f''(x)$ to the left side

$$f(x+h)+f(x-h)-2f(x)-2\frac{f''''(x)}{4!}h^4-\ldots=f''(x)\,h^2$$

$$f''(x)=\frac{f(x+h)-2f(x)+f(x-h)}{h^2}-2\frac{f''''(x)}{4!}h^2-\ldots$$

Hence

$$f''(x)\approx\frac{f(x+h)-2f(x)+f(x-h)}{h^2}\qquad(3)$$

**Example 1**

The velocity of a rocket is given by

$$v(t)=2000\ln\left[\frac{14\times10^4}{14\times10^4-2100t}\right]-9.8t,\ 0\le t\le 30,$$

(a) Use the central difference approximation of the second derivative of $v(t)$ to calculate the jerk at $t=16$ s. Use a step size of $h=2$ s.

**Solution**

The second derivative of velocity with respect to time is called jerk. The second-order approximation of jerk then is

$$j(t)\approx\frac{v(t+h)-2v(t)+v(t-h)}{h^2}$$

$$t=16$$

$$h=2$$

$$\begin{aligned}t+h&=16+2\\&=18\end{aligned}$$

$$\begin{aligned}t-h&=16-2\\&=14\end{aligned}$$

$$j(16)\approx\frac{v(18)-2v(16)+v(14)}{(2)^2}$$

$$\begin{aligned}v(18)&=2000\ln\left[\frac{14\times10^4}{14\times10^4-2100(18)}\right]-9.8(18)\\&=453.02\text{ m/s}\end{aligned}$$

$$\begin{aligned} v(16) &= 2000 \ln\left[\frac{14\times10^4}{14\times10^4 - 2100(16)}\right] - 9.8(16) \\ &= 392.07 \text{ m/s} \end{aligned}$$

$$\begin{aligned} v(14) &= 2000 \ln\left[\frac{14\times10^4}{14\times10^4 - 2100(14)}\right] - 9.8(14) \\ &= 334.24 \text{ m/s} \end{aligned}$$

Hence

$$\begin{aligned} j(16) &\approx \frac{v(18) - 2v(16) + v(14)}{(2)^2} \\ &= \frac{453.02 - 2(392.07) + 334.24}{4} \\ &= 0.77969 \text{ m/s}^3 \end{aligned}$$

(b) The exact value of $j(16)$ can be calculated by differentiating

$$v(t) = 2000 \ln\left[\frac{14\times10^4}{14\times10^4 - 2100t}\right] - 9.8t$$

twice.

Now

$$a(t) = \frac{d}{dt}[v(t)]$$

Knowing that

$$\frac{d}{dt}[\ln(at)] = \frac{1}{t}, \text{ and}$$

$$\frac{d}{dt}\left[\frac{a}{b-ct}\right] = -\frac{ac}{(b-ct)^2}$$

$$\begin{aligned} a(t) &= 2000\left(\frac{14\times10^4 - 2100t}{14\times10^4}\right)\frac{d}{dt}\left(\frac{14\times10^4}{14\times10^4 - 2100t}\right) - 9.8 \\ &= 2000\left(\frac{14\times10^4 - 2100t}{14\times10^4}\right)(-1)\left(\frac{(14\times10^4)(-2100)}{(14\times10^4 - 2100t)^2}\right) - 9.8 \\ &= \frac{-4040 - 29.4t}{-200 + 3t} \end{aligned}$$

$$\begin{aligned} j\left(t\right) &= \frac{d}{dt}\left[a\left(t\right)\right] \\ &= \frac{d}{dt}\left[\frac{-4040-29.4t}{-200+3t}\right] \end{aligned}$$

Knowing that

$$\frac{d}{dt}\left[\frac{a+bt}{c+dt}\right] = -\frac{ad-bc}{\left(c+dt\right)^2}$$

we get

$$j\left(t\right) = \frac{18000}{\left(-200+3t\right)^2}$$

Hence

$$\begin{aligned} j\left(16\right) &= -\frac{(-4040)(3)-(-29.4)\left(-200\right)}{\left(-200+3\left(16\right)\right)^2} \\ &= 0.77908 \text{ m/s}^3 \end{aligned}$$

The absolute relative true error is

$$\begin{aligned} |\epsilon_t| &= \left|\frac{\text{True Value} - \text{Approximate Value}}{\text{True Value}}\right| \times 100 \\ &= \left|\frac{0.77908 - 0.77969}{0.77908}\right| \times 100 \\ &= 0.077992\% \end{aligned}$$

# Chapter 02.02: Numerical Differentiation of Continuous Functions

Error Analysis of Finite Difference Approximations

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *derive formulas from the Taylor series for approximating the first derivative of a function,*

2) *find the order of accuracy of divided difference formulas*

3) *use the order of accuracy results to derive more efficient numerical methods such as Richardson's extrapolation formula.*

## Introduction

The finite difference formulas we develop in previous lessons for the first and second derivatives of a function are approximate, as they have a truncation error.

Look at the example from the previous lessons, which asked to find the value of the acceleration (first derivative) at $t = 16s$ using a step size of $h = 2s$ by forward, backward, and central divided difference formulas. The velocity of a rocket was given by

$$v(t) = 2000 \ln \left[ \frac{14 \times 10^4}{14 \times 10^4 - 2100t} \right] - 9.8t, 0 \leq t \leq 30.$$

The results, along with the relative approximate true errors from the three difference approximations, are given in Table 1.

**Table 1.** Summary of $a(16)$ using different difference approximations

| Type of difference approximation | $a(16)$ $(m/s^2)$ | $\lvert\epsilon_t\rvert\,\%$ |
|---|---|---|
| Forward | 30.475 | 2.6967 |
| Backward | 28.915 | 2.5584 |
| Central | 29.695 | 0.069157 |

So the question arises that why is the central divided difference (CDD) approximation better than the forward divided difference (FDD) and backward divided difference (BDD) approximations. Is it an isolated case? The true errors in the forward and backward divided difference approximations are of

the same order. Is that also an isolated case? To illustrate this, we derive the order of accuracy of the three divided difference schemes.

**Example 1**

The forward divided difference scheme for finding the approximate first derivative of a function $f(x)$ is given by

$$f'(x) \approx \frac{f(x+h) - f(x)}{h}$$

Show that the order of the accuracy of the above formula is of the order of $h$.

**Solution**

Taylor's theorem says that if you know the value of a function $f(x)$ at a point $x$ and all its derivatives at that point, provided the derivatives are continuous between $x$ and $x+h$, then

$$f(x+h) = f(x) + f'(x)\,h + \frac{f''(x)}{2!}h^2 + \frac{f'''(x)}{3!}h^3 + \ldots \qquad (E1.1)$$

Taking the first term of the right-hand side of Equation (1) to the left-hand side

$$f(x+h) - f(x) = f'(x)\,h + \frac{f''(x)}{2!}h^2 + \frac{f'''(x)}{3!}h^3 + \ldots$$

Taking the first derivative term to the left-hand side and the function terms to the right-hand side

$$f'(x)\,h = f(x+h) - f(x) - \frac{f''(x)}{2!}h^2 - \frac{f'''(x)}{3!}h^3 - \ldots$$

Dividing both sides by $h$, to find $f'(x)$ gives

$$f'(x) = \frac{f(x+h) - f(x)}{h} - \frac{f''(x)}{2!}h - \frac{f'''(x)}{3!}h^2 - \ldots \qquad (E1.2)$$

The truncation error in the FDD formula for $f'(x)$ is hence given by

$$E_t = -\frac{f''(x)}{2!}h - \frac{f'''(x)}{3!}h^2 - \ldots \qquad (E1.3)$$

For small values of $h$, the predominant terms will be the first one in Equation (6), and hence the order of accuracy is determined to be of the $O(h)$.

$$f'(x) = \frac{f(x+h) - f(x)}{h} + O(h) \qquad (E1.4)$$

If you repeat the process for BDD, you will find the same order of accuracy

$$f'(x) = \frac{f(x) - f(x-h)}{h} + O(h) \text{ for BDD.} \qquad (E1.5)$$

**Example 2**

The central divided difference scheme for finding the approximate first derivative of a function $f(x)$ is given by

$$f'(x) \approx \frac{f(x+h) - f(x-h)}{2h}$$

Show that the order of the accuracy of the above formula is of the order of $h^2$.

**Solution**

Using the Taylor series, we write the value of the function $f(x)$ at $(x+h)$ in terms of the value of the function and its derivatives at $x$as

$$f(x+h) = f(x) + f'(x)h + \frac{f''(x)}{2!}h^2 + \frac{f'''(x)}{3!}h^3 + \frac{f''''(x)}{4!}h^4 + \frac{f'''''(x)}{5!}h^5 + \ldots \qquad (E2.1)$$

Using the Taylor series again, we write the value of the function $f(x)$at $(x-h)$ in terms of the value of the function and its derivatives at $x$as (simply done by substituting $(-h)$ for $h$ in Equation (1)

$$\begin{aligned} f(x-h) &= f(x) + f'(x)(-h) + \frac{f''(x)}{2!}(-h)^2 + \frac{f'''(x)}{3!}(-h)^3 + \frac{f''''(x)}{4!}(-h)^4 \\ &\quad + \frac{f'''''(x)}{5!}(-h)^5 + \ldots \\ &= f(x) - f'(x)h + \frac{f''(x))}{2!}h^2 - \frac{f'''(x))}{3!}h^3 + \frac{f''''(x))}{4!}h^4 - \frac{f'''''(x))}{5!}h^5 + \ldots \end{aligned} \qquad (E2.2)$$

Subtracting Equation (E2.2) from Equation (E2.1) gives

$$f(x+h) - f(x-h) = 2f'(x)h + \frac{2f'''(x)}{3!}h^3 + \frac{2f'''''(x)}{5!}h^5 + \ldots \qquad (E2.3)$$

Taking the first derivative term to the left-hand side and the function terms to the right-hand side

$$2f'(x)h = f(x+h) - f(x-h) - \frac{2f'''(x)}{3!}h^3 - \frac{2f'''''(x)}{5!}h^5 + \ldots \qquad (E2.4)$$

Dividing both sides by $2h$, to find $f'(x)$ gives

$$f'(x) = \frac{f(x+h) - f(x-h)}{2h} - \frac{f'''(x)}{3!}h^2 - \frac{f'''''(x)}{5!}h^4 - \ldots \qquad (E2.5)$$

The truncation error in the CDD formula for $f'(x)$ is hence given by

$$E_t = -\frac{f'''(x)}{3!}h^2 - \frac{f'''''(x)}{5!}h^4 - \ldots \qquad (E2.6)$$

For small values of $h$, the predominant terms will be the first one in Equation (E2.6), and hence the order of accuracy is determined to be of the $O\left(h^2\right)$.

$$f'(x) = \frac{f(x+h) - f(x-h)}{2h} + O\left(h^2\right) \qquad (E2.7)$$

Since the order of accuracy is higher for CDD than FDD and BDD, the true error being much smaller in CDD is not an isolated incident. Since the order of accuracy is of similar order in FDD and BDD, the true error being of a similar order for FDD and BDD is not an isolated incident either.

**Example 3**

To further investigate the issue of the true error for CDD being of the order of the square of the step size, that is, $h^2$, take the example of being asked to find the approximate value of the first derivative of $2e^{1.5x}$at $x = 3$ using CDD scheme. The exact value (up to 5 significant digits) using differential calculus is found to be = 270.05. The central divided difference approximation of the first derivative with $h = 0.1$ and then continuing to halve the step size are given in Table 2.

**Table 2.** Effect of step size on the error in CDD

| Step Size, $h$ | Central Divided Difference Approximation $\frac{f(x+h)-f(x-h)}{2h}$ | True Error = True Value − Approximate Value |
|---|---|---|
| 0.1 | 271.07 | −1.0138 |
| 0.05 | 270.30 | −0.25324 |
| 0.025 | 270.11 | −0.06330 |
| 0.0125 | 270.06 | −0.015824 |
| 0.00625 | 270.05 | −0.0039558 |

Comment on your observation of how the true error reduces with the step size.

**Solution**

In Table 2, we can observe that as the step size is halved, the true error is getting approximately quartered. This observation is in line with the derivation of the true error being of the order of $h^2$. Let's look at the contribution of the truncation terms as the step size decreases. That information is given in Table 3. The second column shows the true error.

**Table 3**. Contribution to true error as step size decreases in CDD.

| step size, $h$ | True Error $E_t$ | First Term of True Error $-\frac{f'''(x_i)}{3!}h^2$ | Rest of the True Error $-\frac{f'''''(x_i)}{5!}h^4 - ...$ | True Error divided by $h^2$ |
|---|---|---|---|---|
| 0.1 | −1.0138 | −1.0127 | $-1.1399 \times 10^{-3}$ | −101.38 |
| 0.05 | −0.25324 | −0.25317 | $-7.1216 \times 10^{-5}$ | −101.30 |
| 0.025 | −0.06330 | −0.063293 | $-4.4520 \times 10^{-6}$ | −101.27 |
| 0.0125 | −0.015824 | −0.015823 | $-2.7970 \times 10^{-7}$ | −101.27 |
| 0.00625 | −0.0039558 | −0.0039556 | $-1.8955 \times 10^{-8}$ | −101.27 |

In Table 3, you are able to see that the magnitude of the first term of the true error is much bigger than the magnitude of the rest of the true error terms and hence dominates the magnitude of the true error. Note that the value of the true error divided by $h^2$ also gets close to be a constant as

$h$becomes smaller. In fact, the value would converge to the coefficient term of $h^2$ in the true error term in Equation (6) which is 110.27 and is calculated as follows.

$$f(x) = 2e^{1.5x}$$

$$\begin{aligned} f'(x) &= 2(1.5)e^{1.5x} \\ &= 3e^{1.5x} \end{aligned}$$

$$\begin{aligned} f''(x) &= 3(1.5)e^{1.5x} \\ &= 4.5e^{1.5x} \end{aligned}$$

$$\begin{aligned} f'''(x) &= 4.5(1.5)e^{1.5x} \\ &= 6.750e^{1.5x} \end{aligned}$$

Hence,

$$\begin{aligned} \frac{f'''(x)}{3!} &= \frac{6.750e^{1.5x}}{3!} \\ &= 110.27 \end{aligned}$$

This value of 110.27 is close to the number we have in the last column of Table 3 as $h$ becomes small.

## Richardson's Extrapolation Formula for Central Divided Difference Approximation of First Derivative

Recognizing that the true error in the central divided difference formula for the first derivative is of the order the square of the step size, it allows us to develop more computationally efficient formulas for differentiation. These formulas are called Richardson's extrapolation formulas for numerical differentiation.

Since the true error

$$E_t = O(h^2)$$

$$E_t \approx Ch^2 \qquad (1)$$

Let $(AV)_h$ denote the approximate value of the first derivative as calculated by using the equation

$$f'(x) \approx \frac{f(x+h) - f(x-h)}{2h} \qquad (2)$$

and using the definition

$$E_t = \text{True Value - Approximate Value} \qquad (3)$$

we get,

$$Ch^2 \approx (TV) - (AV)_h \qquad (4)$$

where,

$$TV = \text{True value}$$

If we decrease the step size by half to $\frac{h}{2}$ , and $(AV)_{\frac{h}{2}}$ denotes the approximate value of the first derivative calculated using the central divided difference equation, then

$$C\left(\frac{h}{2}\right)^2 \approx (TV) - (AV)_{\frac{h}{2}} \qquad (5)$$

Multiplying Equation (5) by 4, gives

$$Ch^2 \approx 4\,(TV) - 4(AV)_{\frac{h}{2}} \qquad (6)$$

Subtracting Equation (6) from Equation (4) gives

$$Ch^2 - Ch^2 \approx ((TV) - (AV)_h) - \left(4\,(TV) - 4(AV)_{\frac{h}{2}}\right)$$

$$0 \approx -3\,(TV) - (AV)_h + 4(AV)_{\frac{h}{2}}$$

Taking the $(TV)$ term to the left side,

$$\begin{aligned} 3\,(TV) &\approx -(AV)_h + 4(AV)_{\frac{h}{2}} \\ (TV) &\approx \frac{-(AV)_h + 4(AV)_{\frac{h}{2}}}{3} \end{aligned} \qquad (7)$$

The above formula will allow us to find a better estimate of the first derivative just from the approximate values obtained using the step sizes $h$ and $\frac{h}{2}$ .

Equation (7) is generally rewritten as

$$\begin{aligned} (TV) &\approx \frac{-(AV)_h + 3\,(AV)_{\frac{h}{2}} + (AV)_{\frac{h}{2}}}{3} \\ &= (AV)_{\frac{h}{2}} + \frac{(AV)_{\frac{h}{2}} - (AV)_h}{3} \end{aligned} \qquad (8)$$

One of the reasons to write it this way is to show that adding a third of the difference between the approximate values obtained using step sizes of $h$ and $\frac{h}{2}$ to the most recent approximate value will give you a better estimate of the true value.

**Example 4**

Table 4 shows the approximate values of the first derivative of $f(x) = 2e^{1.5x}$ at $x = 3$ using different step sizes for the central divided difference scheme. Use the last two values of the scheme and Richardson's extrapolation formula to get a better estimate of $f'(3)$.

**Table 4**. CDD approximate values as a function of step size

| Step Size, $h$ | Central Divided Difference Approximation |
|---|---|
| 0.1 | 271.07 |
| 0.05 | 270.30 |
| 0.025 | 270.11 |

**Solution**

Using the last two values, we have

$$(AV)_{0.05} \approx 270.30$$

$$(AV)_{0.025} \approx 270.11$$

Using Equation (8)

$$\begin{aligned} TV \approx\ & (AV)_{\frac{h}{2}} + \frac{(AV)_{\frac{h}{2}} - (AV)_h}{3} \\ &= (AV)_{0.025} + \frac{(AV)_{0.025} - (AV)_{0.05}}{3} \\ &= 270.11 + \frac{270.11 - 270.30}{3} \\ &= 270.05 \end{aligned}$$

Coincidentally, this is the exact value of $f'(3)$ up to 5 significant digits.

**Example 5**

Using

$$f'(x) \approx \frac{f(x+h) - f(x-h)}{2h} \qquad (16)$$

two approximate values of $f'(10)$ are calculated using two different values of the step size $h$.

For $h = 2$,

$$f'(10) \cong 1315.$$

For $h = 0.5$,

$$f'(10) \cong 626.$$

Based on this information, can you get a better estimate of $f'(10)$? Find it.

**Solution**

The above approximate formula

$$f'(x) \approx \frac{f(x+h) - f(x-h)}{2h}$$

is the central divided difference formula for $f'(x)$.

The true error, $E_t$ associated with the formula is of the order of $h^2$. Hence the true error for a step size of $h_1$ is approximately $C{h_1}^2$, where $C$ is a constant of proportionality.

Since

$$\text{True Error} = \text{True Value} - \text{Approximate Value}$$

we have

$$C{h_1}^2 \approx TV - (AV)_{h_1} \qquad (E5.1)$$

where

$$TV = \text{True value of } f'(x)$$

$$(AV)_{h_1} = \text{ Approximate Value of } f'(x) \text{ using step size } h_1$$

Similarly, for a different step size $h_2$, the true error for a step size of $h_2$ is approximately $C{h_2}^2$.

Since

$$\text{True Error} = \text{True Value} - \text{Approximate Value}$$

$$C{h_2}^2 \approx TV - (AV)_{h_2} \qquad (E5.2)$$

where

$$(AV)_{h_2} = \text{ Approximate Value of } f'(x) \text{ using step size } h_2$$

Dividing Equation (E5.1) by Equation (E5.2), we get

$$\frac{Ch_1{}^2}{Ch_2{}^2} \approx \frac{TV - (AV)_{h_1}}{TV - (AV)_{h_2}}$$

which gives

$$TV \approx \frac{(AV)_{h_1} h_2{}^2 - (AV)_{h_2} h_1{}^2}{(h_2 - h_1)(h_2 + h_1)} \qquad (E5.3)$$

In our case,

$$h_1 = 2,\ (AV)_{h_1} = 1315,$$

$$h_2 = 0.5,\ (AV)_{h_2} = 626,$$

Then from Equation (E5.3), we get

$$\begin{aligned} TV &\approx \frac{(AV)_{h_1} h_2{}^2 - (AV)_{h_2} h_1{}^2}{(h_2 - h_1)(h_2 + h_1)} \\ &= \frac{1315 \times 0.5^2 - 626 \times 2^2}{(0.5 - 2)(0.5 + 2)} \\ &= 580.1 \end{aligned}$$

## Appendix

The formulas for the first derivative of a continuous function talked about in this and the previous lessons for forward and backward divided difference have the accuracy of the order of the step-size.

For FDD

$$f'(x) = \frac{f(x+h) - f(x)}{h} + O(h) \qquad (A.1)$$

For BDD

$$f'(x) = \frac{f(x) - f(x-h)}{h} + O(h) \qquad (A.2)$$

The question arises, are there more accurate formulas for the forward and backward divided difference for the first derivative of a function. Yes, there are and they are also derivable by manipulating the Taylor series. For example, note the following.

For FDD, we can derive the three-point formula

$$f'(x) = \frac{-3f(x) + 4f(x+h) - f(x+2h)}{2h} + O\left(h^2\right) \qquad (A.3)$$

For BDD, we can derive the three-point formula

$$f'(x) = \frac{f(x-2h) - 4f(x-h) + 3f(x)}{2h} + O\left(h^2\right) \qquad (A.4)$$

Both the above formulas have the same accuracy as the two-point CDD formula

$$f'(x) = \frac{f(x+h) - f(x-h)}{2h} + O\left(h^2\right) \qquad (A.5)$$

Getting matching accuracy becomes critical in the efficiency of solving for an application where one may be forced to use combinations of the FDD, BDD, and CDD formulas.

# Multiple Choice Test

## Chapter 02.02 Numerical Differentiation of Continuous Functions

(1). The definition of the first derivative of a function $f(x)$ is

(A) $f'(x) = \dfrac{f(x+\Delta x) + f(x)}{\Delta x}$

(B) $f'(x) = \dfrac{f(x+\Delta x) - f(x)}{\Delta x}$

(C) $f'(x) = \lim\limits_{\Delta x \to 0} \dfrac{f(x+\Delta x) + f(x)}{\Delta x}$

(D) $f'(x) = \lim\limits_{\Delta x \to 0} \dfrac{f(x+\Delta x) - f(x)}{\Delta x}$

(2). The exact derivative of $f(x) = x^3$ at $x = 5$ is most nearly

(A) 25.00

(B) 75.00

(C) 106.25

(D) 125.00

(3). Using the forwarded divided difference approximation with a step size of 0.2, the derivative of $f(x) = 5e^{2.3x}$ at $x = 1.25$ is

(A) 163.4

(B) 203.8

(C) 211.1

(D) 258.8

(4). A student finds the numerical value of $\dfrac{d}{dx}(e^x) = 20.220$ at $x = 3$ using a step size of 0.2. Which of the following methods did the student use to conduct the differentiation?

(A) Backward divided difference

(B) Calculus, that is, exact

(C) Central divided difference

(D) Forward divided difference

(5). Using the backward divided difference approximation, $\frac{d}{dx}(e^x) = 4.3715$ at $x = 1.5$ for a step size of 0.05. If you keep halving the step size to find $\frac{d}{dx}(e^x)$ at $x = 1.5$ before two significant digits can be considered to be at least correct in your answer, the step size would be (you cannot use the exact value to determine the answer)

(A) 0.05/2

(B) 0.05/4

(C) 0.05/8

(D) 0.05/16

(6). The heat transfer rate $q$ over a surface is given by

$$q = -kA\frac{dT}{dy}$$

where

$$k = \text{thermal conductivity } \left(\frac{\text{J}}{\text{s} \cdot \text{m} \cdot \text{K}}\right)$$

$$A = \text{surface area } \left(\text{m}^2\right)$$

$$T = \text{temperature (K)}$$

$$y = \text{distance normal to the surface (m)}$$

Given

$$k = 0.025\frac{\text{J}}{\text{s} \cdot \text{m} \cdot \text{K}}$$

$$A = 3 \text{ m}^2$$

the temperature $T$ over the surface varies as

$$T = -1493y^3 + 2200y^2 - 1076y + 500$$

The heat transfer rate $q$ at the surface most nearly is

(A) $-1076$ W

(B) 37.5 W

(C) 80.7 W

(D) 500 W

For complete solution, go to

http://nm.mathforcollege.com/mcquizzes/02dif/quiz_02dif_continuous_solution.pdf

# Problem Set

## Chapter 02.02: Numerical Differentiation of Continuous Functions

(1). Find the exact value of $\frac{dy}{dx}(x = 5)$, given

$$y = e^x + \sin(x)$$

**Answer:** 148.69

(2). Given the function $f(x) = \sin(2x)$ and using a step size of $h = 0.1$, find

a) $f'(\frac{\pi}{3})$ using the forward divided difference scheme. Also, find the absolute relative true error.

b) $f'(\frac{\pi}{3})$ using the backward divided difference scheme. Also, find the absolute relative true error.

c) $f'(\frac{\pi}{3})$ using the central divided difference scheme. Also, find the absolute relative true error.

**Answer:** *a)* $-1.1660,\ 0.1660$ *b)* $-0.8207,\ 0.1792$ *c)* $-0.99334,\ 0.0067$

(3). Using forward divided difference scheme, find the first derivative of the function $f(x) = \sin(2x)$ at $x = \frac{\pi}{3}$ correct within 2 significant digits. Start with a step size of $h = 0.01$ and keep halving it till you find the answer.

**Answer:** $-1.0043$, abs rel approx error $= 0.42\%, h = 0.0025$

(4). The velocity of a body is given by

$$v(t) = 20\ln(120 - 4t), 0 < t < 10$$

where $t$ is in seconds and $v$ is in m/s.

a) What is the exact acceleration at $t = 2$ s?

b) What is the acceleration at $t = 2$ s using any divided difference scheme of numerical differentiation with a step size of 0.25 s?

c) Use any numerical scheme to find jerk (rate of change of acceleration) at $t = 2$ s using a step size of 0.25 s.

**Answer:** *a)* $-0.7143$ m/s$^2$ *b)* $-0.7174$ m/s$^2$ (FDD),
$-0.7111$ m/s$^2$ (BDD), $-0.7143$ m/s$^2$ (CDD) *c)* $-0.0255$ m/s$^2$ using CDD

(5). Using forward divided difference method, the first derivative of a function $f(x)$ is found at $x = 0.8$. With a step size of 0.2, $f'(0.8) \approx 0.95375$, while with a step size of 0.1, $f'(0.8) \approx 0.96851$. Find a better estimate of $f'(0.8)$ with the above information knowing that the order of accuracy of the forward divided difference scheme is $O(\text{step size})$.

**Answer:** 0.98327

(6). The error in using central divided difference scheme to find the first derivative of the function, $f(x)$ is of $O(h^2)$, where $h$ is the step size. If the true error in finding $f'(\alpha)$ with a step size $h = 0.5$ is 1.6, estimate the true error in finding $f'(\alpha)$ with a step size of $h = 0.125$.

**Answer:** 0.1

(7). The voltage of a variable power source is given by

$$v(t) = 3\frac{dc}{dt} + 5c,$$

where $c$ is the current in amperes, time is in seconds, and $v$ is the voltage in volts. Find the voltage in volts at $t = 6$ s if $c = 5\cos(2t)$.

**Answer:** 37.194 V

# Chapter 02.03: Numerical Differentiation of Functions Given at Discrete Data Points

Lesson: Numerical Differentiation of Functions Given as Discrete Data Points - First Derivative

## Lesson Objectives

*After successful completion of this lesson, you should be able to:*

1) *choose the most proper scheme among FDD, BDD, and CDD based on order of accuracy and data availability,*

2) *find numerical values of the first derivatives of functions that are given at discrete data points.*

## Introduction

To find the derivatives of functions that are given at discrete points, several methods are available. Although these methods are mainly used when the data is spaced unequally, they can be used for data that is spaced equally as well.

We know

$$f'(x) = \lim_{\Delta x \to 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

We developed finite divided difference approximations for the first and second derivatives of continuous functions in a previous lesson. We can apply the same formulas for discrete functions as well. However, some limitations exist, and one should note them before applying any of the approximations.

So given $n+1$ data points in ascending order of $x$ values, $(x_0, y_0), (x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)$,

the value of $f'(x_i)$ by FDD for $x_i \leq x \leq x_{i+1}$, $i = 0, ..., n-1$, is given by

$$f'(x_i) \approx \frac{f(x_{i+1}) - f(x_i)}{x_{i+1} - x_i} \qquad (1)$$

the value of $f'(x_i)$ by BDD for $x_{i-1} \leq x \leq x_i$, $i = 1, ..., n$, is given by

$$f'(x_i) \approx \frac{f(x_i) - f(x_{i-1})}{x_i - x_{i-1}} \qquad (2)$$

the value of $f'(x_i)$ by CDD for $x_{i-1} \leq x \leq x_{i+1}$, $i = 1, ..., n-1$, is given by

$$f'(x_i) \approx \frac{f(x_{i+1}) - f(x_{i-1})}{x_{i+1} - x_{i-1}} \qquad (3)$$

provided that

$$x_{i+1} - x_i = x_i - x_{i-1}.$$

**Example 1**

The upward velocity of a rocket is given as a function of time in Table 1.

**Table 1** Velocity as a function of time.

| $t$ (s) | $v(t)$ (m/s) |
|---|---|
| 0 | 0 |
| 10 | 227.04 |
| 15 | 362.78 |
| 20 | 517.35 |
| 22.5 | 602.97 |
| 30 | 901.67 |

a) Use forward divided difference, if possible, to find the acceleration of the rocket at $t = 15$ s.

b) Use backward divided difference, if possible, to find the acceleration of the rocket at $t = 15$ s.

c) Use central divided difference, if possible, to find the acceleration of the rocket at $t = 15$ s.

**Solution**

a) To find the acceleration at $t = 15$ s with the forward divided difference method, a data point ahead of $t = 15$ s should be available. All these conditions are met, and we will use velocity values at $t = 15$ s and $t = 20$ s

$$a(t_i) \approx \frac{v(t_{i+1}) - v(t_i)}{t_{i+1} - t_i}$$

$$t_i = 15$$

$$t_{i+1} = 20$$

$$a(15) \approx \frac{v(20) - v(15)}{20 - 15}$$

$$= \frac{517.35 - 362.78}{5}$$

$$= 30.914 \text{ m/s}^2$$

b) To find the acceleration at $t = 15$ s with the backward divided difference method, a data point behind $t = 15$ s should be available. All these conditions are met, and we will use velocity values at $t = 10$ s and $t = 15$ s

$$a(t_i) \approx \frac{v(t_i) - v(t_{i-1})}{t_i - t_{i-1}}$$

$$t_{i-1} = 10$$

$$t_i = 15$$

$$a(15) \approx \frac{v(15) - v(10)}{15 - 10}$$

$$= \frac{362.78 - 227.04}{5}$$

$$= 27.148 \text{ m/s}^2$$

c) To find the acceleration at $t = 15$ s with the central divided difference method,

a data point behind $t = 15$ s should be available,

a data point ahead of $t = 15$ s should be available, and

the spacing should be the same on the time domain from $t = 15$ s.

All these conditions are met, and we will use velocity values at $t = 10$ s and $t = 20$ s

$$a(t_i) \approx \frac{v(t_{i+1}) - v(t_{i-1})}{t_{i+1} - t_{i-1}}$$

$$t_{i-1} = 10$$

$$t_{i+1} = 20$$

$$a(15) \approx \frac{v(20) - v(10)}{20 - 10}$$

$$= \frac{517.35 - 227.04}{10}$$

$$= 29.031 \text{ m/s}^2$$

The values of acceleration at $t = 15$ s obtained using the three divided difference schemes is given in Table 2.

**Table 2** Estimated acceleration for various divided difference schemes.

| Method | $a(15)$ (m/s$^2$) |
|---|---|
| Forward | 30.914 |
| Backward | 27.148 |
| Central | 29.031 |

# Chapter 02.03: Numerical Differentiation of Functions Given at Discrete Data Points

Lesson: Numerical Differentiation of Functions Given as Discrete Data Points - Second Derivative

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *find approximate values of the second derivatives of functions that are given at discrete data points*

## Introduction

To find the derivatives of functions that are given at discrete points, several methods are available. Although these methods are mainly used when the data is spaced unequally, they can be used for data spaced equally.

In a previous lesson, we developed finite divided difference approximations for the second derivatives of continuous functions. We can apply the same formulas for discrete functions as well. However, some limitations exist, and one should note them before applying any of the approximations.

So given $n+1$ data points in ascending order of $x$ values, $(x_0, y_0), (x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)$,

the value of $f''(x_i)$ by CDD for $x_{i-1} \leq x_i \leq x_{i+1}$, $i = 1, \ldots, n-1$, is given by

$$f''(x_i) \approx \frac{f(x_{i+1}) - 2f(x_i) + f(x_{i-1})}{(x_{i+1} - x_i)^2}$$

provided that

$$x_{i+1} - x_i = x_i - x_{i-1}.$$

### Example 1

The upward velocity of a rocket is given as a function of time in Table 1.

**Table 1** Velocity as a function of time.

| $t$ (s) | $v(t)$ m/s |
|---|---|
| 0 | 0 |
| 10 | 227.04 |
| 15 | 362.78 |
| 20 | 517.35 |

| $t$ (s) | $v(t)$ m/s |
|---|---|
| 22.5 | 602.97 |
| 30 | 901.67 |

Use central divided difference, if possible, to find the jerk of the rocket at $t = 15$ $s$.

**Solution**

To find the jerk at $t = 15$ s with the central divided difference method,

a data point behind $t = 15$ s should be available,

a data point ahead of $t = 15$ s should be available, and

the spacing should be the same on the time domain from $t = 15$ s.

All these conditions are met, and we will use velocity values at $t = 10$ s, $t = 15$ s, and $t = 20$ s

$$j\left(t_i\right) \approx \frac{v\left(t_{i+1}\right) - 2v\left(t_i\right) + v\left(t_{i-1}\right)}{\left(t_{i+1} - t_i\right)^2}$$

$$t_{i-1} = 10$$

$$t_i = 15$$

$$t_{i+1} = 20$$

$$\begin{aligned} j\left(15\right) &\approx \frac{v\left(20\right) - 2v\left(15\right) + v\left(10\right)}{\left(20 - 15\right)^2} \\ &= \frac{517.35 - 2 \times 362.78 + 227.04}{\left(20 - 15\right)^2} \\ &= 0.7532 \text{ m/s}^3 \end{aligned}$$

# Chapter 02.03: Numerical Differentiation of Functions Given at Discrete Data Points

Lesson: Direct Fit Polynomials to Find Derivatives of Functions Given as Discrete Data Points

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *find numerical derivatives of functions given at discrete points via the direct method of interpolation*

## Introduction

To find the derivatives of functions that are given at discrete points, several methods are available. Although the method of divided differences is quick to apply, the need for independent variable data to be equally spaced is a limitation for central divided difference scheme, and for higher-order derivatives. In such cases, polynomial interpolation is a method that can be used.

## Vandermonde Polynomial

In this method, given $n+1$ data points $(x_0, y_0), (x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)$, one can fit a $n^{th}$ order polynomial given by

$$P_n(x) = a_0 + a_1 x + \ldots\ldots + a_{n-1} x^{n-1} + a_n x^n \qquad (1)$$

To find the first derivative,

$$P_n'(x) = \frac{dP_n(x)}{dx} = a_1 + 2a_2 x + \ldots\ldots + (n-1)\, a_{n-1} x^{n-2} + n a_n x^{n-1} \qquad (2)$$

Similarly, other derivatives such as second, third, etc., can also be found.

**Example 1**

The upward velocity of a rocket is given as a function of time in Table 1.

**Table 1**. Velocity as a function of time

| $t$ (s) | $v(t)$ (m/s) |
|---|---|
| 0 | 0 |
| 10 | 227.04 |
| 15 | 362.78 |
| 20 | 517.35 |
| 22.5 | 602.97 |
| 30 | 901.67 |

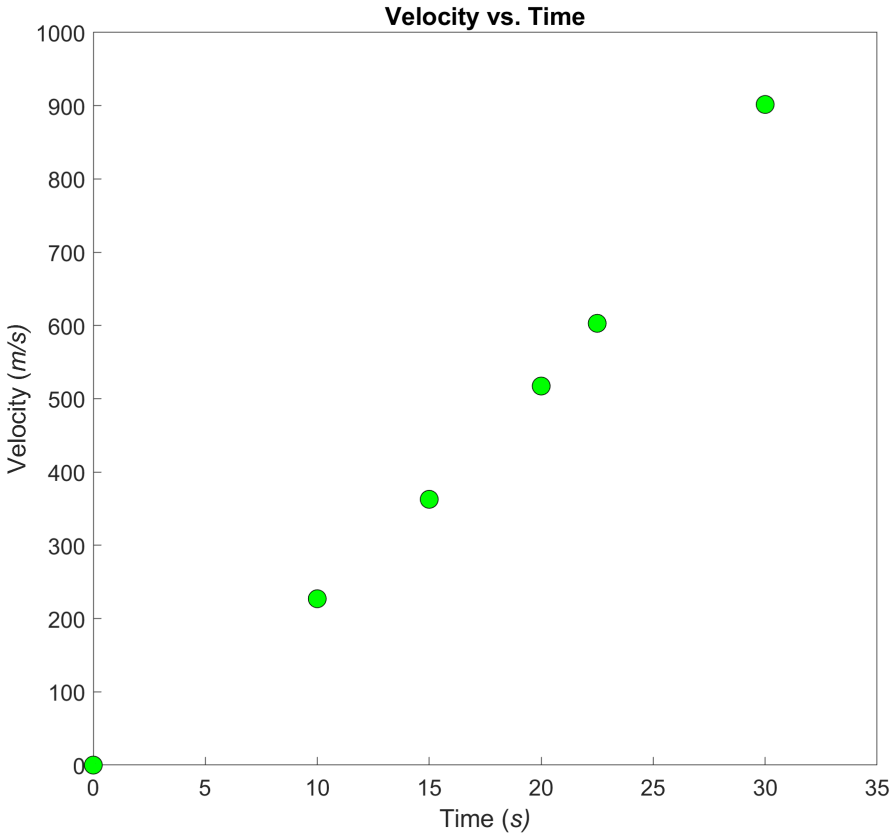



**Figure 1:** Velocity vs time data.

Using a second-order polynomial interpolant for velocity, find the acceleration of the rocket at $t = 16$ s.

**Solution**

For the second-order polynomial (also called quadratic interpolation), we choose the velocity given by

$$v\left(t\right) = a_0 + a_1 t + a_2 t^2$$

Since we want to find the velocity at $t = 16$ s, and we are using a second-order polynomial, we need to choose the three points closest to $t = 16$, and that also bracket $t = 16$ to evaluate it.

The three points are $t_0 = 10,\ t_1 = 15$, and $t_2 = 20$.

$$t_0 = 10,\ v(t_0) = 227.04$$

$$t_1 = 15,\ v(t_1) = 362.78$$

$$t_2 = 20,\ v(t_2) = 517.35$$

such that

$$\begin{aligned} v(10) &= 227.04 = a_0 + a_1(10) + a_2(10)^2 \\ v(15) &= 362.78 = a_0 + a_1(15) + a_2(15)^2 \\ v(20) &= 517.35 = a_0 + a_1(20) + a_2(20)^2 \end{aligned}$$

Writing the three equations in matrix form, we have

$$\begin{bmatrix} 1 & 10 & 100 \\ 1 & 15 & 225 \\ 1 & 20 & 400 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 227.04 \\ 362.78 \\ 517.35 \end{bmatrix}$$

Solving the above three equations gives

$$\begin{aligned} a_0 &= 12.050 \\ a_1 &= 17.733 \\ a_2 &= 0.37660 \end{aligned}$$

Hence

$$\begin{aligned} v(t) &= a_0 + a_1 t + a_2 t^2 \\ &= 12.050 + 17.733t + 0.37660t^2,\ 10 \le t \le 20 \end{aligned}$$

The acceleration at $t = 16$ is given by

$$a(16) = \left.\frac{d}{dt} v(t)\right|_{t=16}$$

We found

$$v(t) = 12.050 + 17.733t + 0.37660t^2,\ 10 \le t \le 20,$$

and hence

$$
\begin{aligned}
a(t) &= \frac{d}{dt} v(t) \\
&= \frac{d}{dt}\left(12.050 + 17.733t + 0.37660t^2\right) \\
&= 17.733 + 0.75320t,\ 10 \le t \le 20 \\
a(16) &= 17.733 + 0.75320(16) \\
&= 29.784 \text{ m/s}^2
\end{aligned}
$$

**Example 2**

The upward velocity of a rocket is given as a function of time in Table 2.

**Table 2.** Velocity as a function of time.

| $t$ (s) | $v(t)$ (m/s) |
|---|---|
| 0 | 0 |
| 10 | 227.04 |
| 15 | 362.78 |
| 20 | 517.35 |
| 22.5 | 602.97 |
| 30 | 901.67 |

Using a third-order polynomial interpolant for velocity, find the acceleration and jerk (it is a measure of discomfort caused to passengers in a vehicle) of the rocket at $t = 16$ s.

**Solution**

For the third-order polynomial (also called cubic interpolation), we choose the velocity given by

$$v(t) = a_0 + a_1 t + a_2 t^2 + a_3 t^3$$

Since we want to find the velocity at $t = 16$ s, and we are using a third-order polynomial, we need to choose the four points closest to $t = 16$, and that also bracket $t = 16$ to evaluate it.

The four points are $t_0 = 10,\ t_1 = 15, t_2 = 20$ and $t_3 = 22.5$.

$$t_0 = 10,\ v(t_0) = 227.04$$

$$t_1 = 15,\ v(t_1) = 362.78$$

$$t_2 = 20,\ v(t_2) = 517.35$$

$$t_3 = 22.5,\ v(t_3) = 602.97$$

such that

$$\begin{aligned}
v(10) = 227.04 &= a_0 + a_1(10) + a_2(10)^2 + a_3(10)^3 \\
v(15) = 362.78 &= a_0 + a_1(15) + a_2(15)^2 + a_3(15)^3 \\
v(20) = 517.35 &= a_0 + a_1(20) + a_2(20)^2 + a_3(20)^3 \\
v(22.5) = 602.97 &= a_0 + a_1(22.5) + a_2(22.5)^2 + a_3(22.5)^3
\end{aligned}$$

Writing the four equations in matrix form, we have

$$\begin{bmatrix} 1 & 10 & 100 & 1000 \\ 1 & 15 & 225 & 3375 \\ 1 & 20 & 400 & 8000 \\ 1 & 22.5 & 506.25 & 11391 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} 227.04 \\ 362.78 \\ 517.35 \\ 602.97 \end{bmatrix}$$

Solving the above four equations gives

$$\begin{aligned}
a_0 &= -4.3810 \\
a_1 &= 21.289 \\
a_2 &= 0.13065 \\
a_3 &= 0.0054606
\end{aligned}$$

Hence

$$\begin{aligned}
v(t) &= a_0 + a_1 t + a_2 t^2 + a_3 t^3 \\
&= -4.3810 + 21.289t + 0.13065t^2 + 0.0054606t^3, \ 10 \le t \le 22.5
\end{aligned}$$

The acceleration at $t = 16$ is given by

$$a(16) = \left.\frac{d}{dt} v(t)\right|_{t=16}$$

Given that

$$v(t) = -4.3810 + 21.289t + 0.13065t^2 + 0.0054606t^3, \ 10 \le t \le 22.5,$$

$$\begin{aligned}
a(t) &= \frac{d}{dt} v(t) \\
&= \frac{d}{dt}\left(-4.3810 + 21.289t + 0.13065t^2 + 0.0054606t^3\right) \\
&= 21.289 + 0.26130t + 0.016382t^2, \quad 10 \le t \le 22.5
\end{aligned}$$

$$\begin{aligned}
a(16) &= 21.289 + 0.26130(16) + 0.016382(16)^2 \\
&= 29.664 \text{ m/s}^2
\end{aligned}$$

The jerk at $t = 16$ s is given by

$$j\,(16) = \left.\frac{d}{dt}a\,(t)\right|_{t=16}$$

Given that

$$a\,(t) = 21.289 + 0.26130t + 0.016382t^2, \quad 10 \leq t \leq 22.5,$$

$$\begin{aligned} j\,(t) &= \frac{d}{dt}a\,(t) \\ &= \frac{d}{dt}\left(21.289 + 0.26130t + 0.016382t^2\right) \\ &= 0.26130 + 0.032764t, \quad 10 \leq t \leq 22.5 \end{aligned}$$

$$\begin{aligned} j\,(16) &= 0.26130 + 0.032764(16) \\ &= 0.785524 \text{ m/s}^3 \end{aligned}$$

# Chapter 02.03: Numerical Differentiation of Functions Given at Discrete Data Points

Lesson: Lagrange Interpolation to Find Derivatives of Functions Given as Discrete Data Points

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *find numerical derivatives of functions given at discrete points via the Lagrange interpolation*

## Introduction

To find the derivatives of functions that are given at discrete points, several methods are available. Although the method of divided differences is quick to apply, the need for independent variable data to be equally spaced is a limitation for central divided difference scheme, and for higher order derivatives. In such cases, polynomial interpolation is a method that can be used.

## Lagrange Polynomial

In this method, given $(x_0, y_0), \ldots, (x_n, y_n)$, one can fit a $n^{th}$ order Lagrangian polynomial given by

$$f_n(x) = \sum_{i=0}^{n} L_i(x) f(x_i)$$

where $n$ in $f_n(x)$ stands for the $n^{th}$ order polynomial that approximates the function $y = f(x)$ and

$$L_i(x) = \prod_{\substack{j=0 \\ j \neq i}}^{n} \frac{x - x_j}{x_i - x_j}$$

$L_i(x)$ is a weighting function that includes a product of $n-1$ terms with terms of $j = i$ omitted.

Then to find the first derivative, one can differentiate $f_n(x)$ once, and so on for other derivatives.

For example, the second order Lagrange polynomial passing through $(x_0, y_0), (x_1, y_1), and (x_2, y_2)$ is

$$f_2(x) = \frac{(x - x_1)(x - x_2)}{(x_0 - x_1)(x_0 - x_2)} f(x_0) + \frac{(x - x_0)(x - x_2)}{(x_1 - x_0)(x_1 - x_2)} f(x_1) + \frac{(x - x_0)(x - x_1)}{(x_2 - x_0)(x_2 - x_1)} f(x_2)$$

Differentiating the above equation gives

$$f_2{}'(x) = \frac{2x-(x_1+x_2)}{(x_0-x_1)(x_0-x_2)} f(x_0) + \frac{2x-(x_0+x_2)}{(x_1-x_0)(x_1-x_2)} f(x_1) + \frac{2x-(x_0+x_1)}{(x_2-x_0)(x_2-x_1)} f(x_2)$$

Differentiating again would give the second derivative as

$$f_2{}''(x) = \frac{2}{(x_0-x_1)(x_0-x_2)} f(x_0) + \frac{2}{(x_1-x_0)(x_1-x_2)} f(x_1) + \frac{2}{(x_2-x_0)(x_2-x_1)} f(x_2)$$

**Example 3**

The upward velocity of a rocket is given as a function of time in Table 3.

**Table 3** Velocity as a function of time.

| $t\ (s)$ | $v(t)\ (m/s)$ |
|---|---|
| 0 | 0 |
| 10 | 227.04 |
| 15 | 362.78 |
| 20 | 517.35 |
| 22.5 | 602.97 |
| 30 | 901.67 |

Determine the value of the acceleration at $t = 16s$ using second-order Lagrangian polynomial interpolation for velocity.

**Solution**

Since we want to find the velocity at $t = 16s$, and we are using a second-order polynomial, we need to choose the three points closest to $t = 16$ and that also bracket $t = 16$ to evaluate it.

The three points are $t_0 = 10,\ t_1 = 15$, and $t_2 = 20$.

$$t_0 = 10,\ v(t_0) = 227.04$$

$$t_1 = 15,\ v(t_1) = 362.78$$

$$t_2 = 20,\ v(t_2) = 517.35$$

For the second-order polynomial (also called quadratic interpolation), we choose the velocity given by

$$v(t) = \left(\frac{t-t_1}{t_0-t_1}\right)\left(\frac{t-t_2}{t_0-t_2}\right) v(t_0) + \left(\frac{t-t_0}{t_1-t_0}\right)\left(\frac{t-t_2}{t_1-t_2}\right) v(t_1) + \left(\frac{t-t_0}{t_2-t_0}\right)\left(\frac{t-t_1}{t_2-t_1}\right) v(t_2)$$

$$a(t) = \frac{2t-(t_1+t_2)}{(t_0-t_1)(t_0-t_2)} v(t_0) + \frac{2t-(t_0+t_2)}{(t_1-t_0)(t_1-t_2)} v(t_1) + \frac{2t-(t_0+t_1)}{(t_2-t_0)(t_2-t_1)} v(t_2)$$

$$
\begin{aligned}
a\left(16\right) &= \frac{2\left(16\right)-\left(15+20\right)}{\left(10-15\right)\left(10-20\right)}\left(227.04\right)+\frac{2\left(16\right)-\left(10+20\right)}{\left(15-10\right)\left(15-20\right)}\left(362.78\right) \\
&\quad+\frac{2\left(16\right)-\left(10+15\right)}{\left(20-10\right)\left(20-15\right)}\left(517.35\right) \\
&= -0.06\left(227.04\right)-0.08\left(362.78\right)+0.14\left(517.35\right) \\
&= 29.784 m/s^2
\end{aligned}
$$

# Multiple Choice Test

Chapter 02.03 Numerical Differentiation of Functions Given at Discrete Data Points

(1). The definition of the first derivative of a function $f(x)$ is

(A) $f'(x) = \dfrac{f(x+\Delta x)+f(x)}{\Delta x}$

(B) $f'(x) = \dfrac{f(x+\Delta x)-f(x)}{\Delta x}$

(C) $f'(x) = \lim\limits_{\Delta x \to 0} \dfrac{f(x+\Delta x)+f(x)}{\Delta x}$

(D) $f'(x) = \lim\limits_{\Delta x \to 0} \dfrac{f(x+\Delta x)-f(x)}{\Delta x}$

(2). Using the forward divided difference approximation with a step size of 0.2, the derivative of the function at $x = 2$ is given as

| $x$ | 1.8 | 2.0 | 2.2 | 2.4 | 2.6 |
|---|---|---|---|---|---|
| $f(x)$ | 6.0496 | 7.3890 | 9.0250 | 11.023 | 13.464 |

(A) 6.697

(B) 7.389

(C) 7.438

(D) 8.180

(3). A student finds the numerical value of $f'(x) = 20.220$ at $x = 3$ using a step size of 0.2. Which of the following methods did the student use to conduct the differentiation if $f(x)$ is given in the table below?

| $x$ | 2.6 | 2.8 | 3.0 | 3.2 | 3.4 | 3.6 |
|---|---|---|---|---|---|---|
| $f(x)$ | $e^{2.6}$ | $e^{2.8}$ | $e^{3}$ | $e^{3.2}$ | $e^{3.4}$ | $e^{3.6}$ |

(A) Backward divided difference

(B) Calculus, that is, exact

(C) Central divided difference

(D) Forward divided difference

(4). The upward velocity of a body is given as a function of time as

| $t$, s | 10 | 15 | 20 | 22 |
|---|---|---|---|---|
| $v$, m/s | 22 | 36 | 57 | 10 |

To find the acceleration at $t = 17$ s, a scientist finds a second-order polynomial approximation for the velocity, and then differentiates it to find the acceleration. The estimate of the acceleration in $\text{m/s}^2$ at $t = 17$ s is most nearly

(A) 4.060

(B) 4.200

(C) 8.157

(D) 8.498

(5). The velocity of a rocket is given as a function of time as

| $t$, s | 0 | 0.5 | 1.2 | 1.5 | 1.8 |
|---|---|---|---|---|---|
| $v$, m/s | 0 | 213 | 223 | 275 | 300 |

Allowed to use the forward divided difference, backward divided difference or central divided difference approximation of the first derivative, the best estimate for the acceleration $\left(a = \dfrac{dv}{dt}\right)$in $\text{m/s}^2$ of the rocket at $t = 1.5$ s is

(A) 83.33

(B) 128.33

(C) 173.33

(D) 183.33

(6). In a circuit with an inductor of inductance $L$, a resistor with resistance $R$, and a variable voltage source $E(t)$,

$E(t) = L\dfrac{di}{dt} + Ri$ The current, $i$, is measured at several values of time as

| | | | | |
|---|---|---|---|---|
| Time, $t$ (s) | 1.00 | 1.01 | 1.03 | 1.1 |
| Current, $i$ (A) | 3.10 | 3.12 | 3.18 | 3.24 |

If $L = 0.98$ Henries and $R = 0.142$ ohms, the most accurate expression for $E(1.00)$ is

(A) $0.98\left(\dfrac{3.24 - 3.10}{0.1}\right) + (0.142)(3.10)$

(B) $(0.142)(3.10)$

(C) $0.98\left(\dfrac{3.12 - 3.10}{0.01}\right) + (0.142)(3.10)$

(D) $0.98\left(\dfrac{3.12 - 3.10}{0.01}\right)$

For complete solution, go to

http://nm.mathforcollege.com/mcquizzes/02dif/quiz_02dif_discrete_solution.pdf

# Problem Set

Chapter 02.03: Numerical Differentiation of Functions Given at Discrete Data Points

(1). Given below is the value of a function at discrete values of $x$

| $x$ | 0 | 0.4 | 0.9 | 1.8 | 2.5 |
|---|---|---|---|---|---|
| $f(x)$ | 20 | 75 | 121 | 126 | 171 |

Estimate $f'(0.9)$ using

a) Forward divided difference scheme,

b) Backward divided difference scheme, and

c) Central divided difference scheme.

**Answer:** *a*) 5.5556 *b*) 92 *c*) 58.89

(2). The upward velocity of a rocket is given as a function of time in the table below

| $t$, s | 0 | 10 | 15 | 20 | 22.5 | 30 |
|---|---|---|---|---|---|---|
| $v(t)$, m/s | 0 | 227.04 | 362.78 | 517.35 | 602.97 | 901.67 |

Use the first, second and third-order polynomial direct method (Vandermonde) interpolation to find the acceleration at $t = 9.5$ s. What is the true error if I told you that the data given in the table was derived from the formula

$$v\left(t\right) = 2000 \ln\left[\frac{14 \times 10^4}{14 \times 10^4 - 2100t}\right] - 9.8t,\ 0 \leq t \leq 30$$

**Answer:** In $\text{m/s}^2$ : 22.704, 25.370, 25.152 True errors : 2.4814, −0.18498, 0.032927

(3). The upward velocity of a rocket is given as a function of time in the table below.

| $t$, s | 0 | 10 | 15 | 20 | 22.5 | 30 |
|---|---|---|---|---|---|---|
| $v(t)$, m/s | 0 | 227.04 | 362.78 | 517.35 | 602.97 | 901.67 |

From the data given, use the first, second and third-order Lagrangian method interpolation to find the acceleration at $t = 9.5$ s. What is the true error, if I told you that the data given in the table was derived from the formula

$$v\,(t) = 2000 \ln\left[\frac{14 \times 10^4}{14 \times 10^4 - 2100t}\right] - 9.8t, \quad 0 \le t \le 30$$

**Answer:** In m/s$^2$ : 22.704, 25.370, 25.152 True errors : 2.4814, −0.18498, 0.032927

(4). An aircraft position during an emergency landing exercise on a runway was timed as follows

| $t,\ s$ | 0 | 0.4 | 1.00 | 1.75 | 2.5 |
|---|---|---|---|---|---|
| $x,\ m$ | 20 | 71 | 110 | 161 | 178 |

What is your best estimate for the velocity of the aircraft at $t = 1.75$ s?

**Answer:** 45.33 m/s from CDD, 54.024 m/s if using third order polynomial interpolation

(5). An aircraft position during an emergency landing exercise on a runway was timed as follows

| $t,$ s | 0 | 0.4 | 1.00 | 1.75 | 2.5 |
|---|---|---|---|---|---|
| $x,$ m | 20 | 71 | 110 | 161 | 178 |

Estimate the acceleration of the jet fighter at $t = 1.75$ s by any method.

**Answer:** CDD gives −60.444 m/s$^2$; 3$^{\text{rd}}$ order polynomial for location gives −60.442 m/s$^2$

(6). The location of a futuristic car is given as a function of time

| $t,$ s | 0 | 0.5 | 1.2 | 1.5 | 1.8 |
|---|---|---|---|---|---|
| $x,$ m | 0 | 213 | 223 | 275 | 300 |

What is your best estimate for the momentum $M = mv$ ($m$ =mass, $v$ =velocity) of the car at $t = 1.5$ seconds? Assume the mass of the car is 2006 kg.

**Answer:** CDD gives 257437 kg-m/s; Using 3$^{\text{rd}}$ order polynomial gives 300354 kg-m/s

# Chapter 03.00: Physical Problem for Nonlinear Equations

Lesson: General Engineering

## Summary

Finding the depth to which a ball would float in water requires solving a nonlinear equation.

## Modeling

You are working for 'DOWN THE TOILET COMPANY' that makes floats for ABC commodes. The ball has a specific gravity of 0.6 and has a radius of 5.5 cm. You are asked to find the depth to which the ball will get submerged when floating in water (see Figure 1).

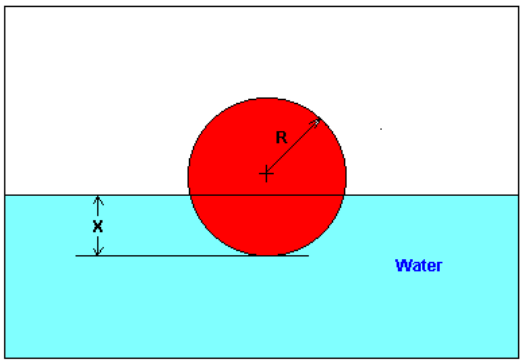



**Figure 1** Depth to which the ball is submerged in water

According to Newton's third law of motion, every action has an equal and opposite reaction. In this case, the weight of the ball is balanced by the buoyancy force (Figure 2).

$$\text{Weight of ball} = \text{Buoyancy force} \qquad (1)$$

The weight of the ball is given by

$$\begin{aligned}\text{Weight of ball} &= (\text{Volume of ball}) \times (\text{Density of ball}) \times (\text{Acceleration due to gravity}) \\ &= \left(\frac{4}{3}\pi R^3\right)(\rho_b)(g) \qquad (2)\end{aligned}$$

where

$$R = \text{radius of the ball (m)},$$

$$\rho = \text{density of the ball } \left(\text{kg/m}^3\right),$$

$$g = \text{acceleration due to gravity } \left(\text{m/s}^2\right).$$

The buoyancy force is given by

$$\begin{aligned}\text{Buoyancy force} &= \text{Weight of water displaced} \\ &= \text{(Volume of the ball under water) (Density of water)(Acceleration due to gravity)} \\ &= \pi x^2 \left(R - \frac{x}{3}\right) \rho_w g \qquad (3)\end{aligned}$$

where

$$x = \text{depth to which ball is submerged},$$

$$\rho_w = \text{density of water.}$$

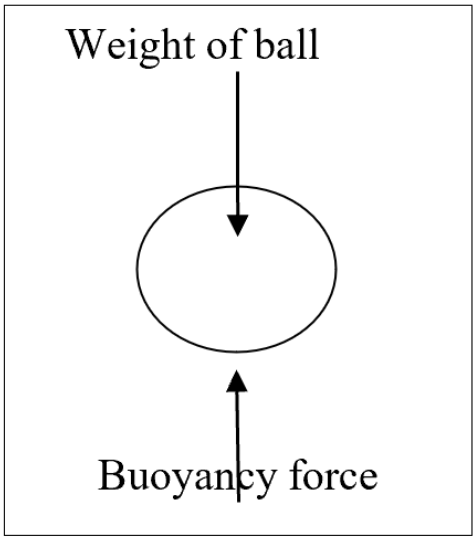



**Figure 2** Free Body Diagram showing the forces acting on the ball immersed in water

Now substituting Equations (2) and (3) in Equation (1),

$$\frac{4}{3}\pi R^3 \rho_b g = \pi x^2 \left(R - \frac{x}{3}\right) \rho_w g$$

$$4R^3 \rho_b = 3x^2 (R - \frac{x}{3}) \rho_w$$

$$4R^3\rho_b - 3x^2R\rho_w + x^3\rho_w = 0$$

$$4R^3\frac{\rho_b}{\rho_w} - 3x^2R + x^3 = 0$$

$$4R^3\gamma_b - 3x^2R + x^3 = 0 \qquad (4)$$

where

the specific gravity of the ball, $\gamma_b$ is given by

$$\gamma_b = \frac{\rho_b}{\rho_w} \qquad (5)$$

Given

$$R = 5.5 \text{ cm} = 0.055 \text{ m},$$

$$\gamma_b = 0.6, and$$

substituting in Equation (4), we get

$$4(0.055)^3(0.6) - 3x^2(0.055) + x^3 = 0$$

$$3.993 \times 10^{-4} - 0.165x^2 + x^3 = 0 \qquad (6)$$

Equation (6) is nonlinear. Solving it would give us the value of $x$, that is, the depth to which the ball is submerged underwater.

**Appendix: Derivation of the formula for the volume of a ball submerged underwater**

How do you find that the volume of the ball submerged underwater as given by

$$V = \frac{\pi h^2 (3r - h)}{3} \qquad (7)$$

where

$$r = \text{radius of the ball},$$

$$h = \text{height of the ball to which the ball is submerged.}$$

From calculus,

$$V = \int_{r-h}^{r} A dx \qquad (8)$$

where $A$ is the cross-sectioned area at a distance $x$ from the center of the sphere. The lower limit of integration is $x = r - h$ as that is where the water line is

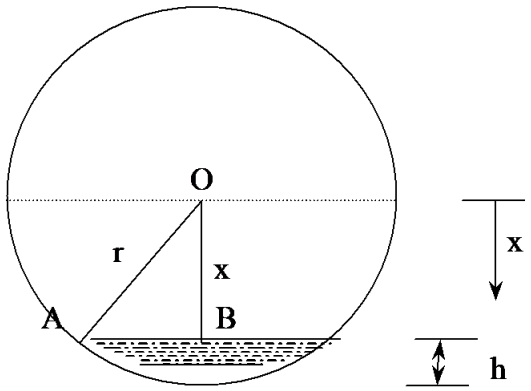



**Figure 3** Deriving the equation for the volume of ball underwater

and the upper limit is $r$ as that is the bottom of the sphere. So, what is the A at any location $x$. From Figure 3, for a location $x$,

$$OB = x$$

$$OA = r,$$

then

$$\begin{aligned} AB &= \sqrt{OA^2 - OB^2} \\ &= \sqrt{r^2 - x^2} \end{aligned} \qquad (9)$$

and AB is the radius of the area at $x$. So at location $B$ is

$$A = \pi(AB)^2 = \pi\left(r^2 - x^2\right) \qquad (10)$$

so

$$\begin{aligned} V &= \int_{r-h}^{r} \pi\left(r^2 - x^2\right) dx \\ &= \pi\left(r^2 x - \frac{x^3}{3}\right)_{r-h}^{r} \\ &= \pi\left[\left(r^2 r - \frac{r^3}{3}\right) - \left(r^2\left(r-h\right) - \frac{\left(r-h\right)^3}{3}\right)\right] \\ &= \frac{\pi h^2\left(3r - h\right)}{3}. \end{aligned} \qquad (11)$$

# Chapter 03.00: Physical Problem for Nonlinear Equations

Lesson: Chemical Engineering

## Summary

Designing a scale to find the volume of oil in a spherical tank requires solving a nonlinear equation.

## Modeling

Years ago, a businessperson called me and wanted to know how he could find how much oil was left in his storage tank. His tank was spherical and was 6 feet in diameter. Well, I suggested him to get a 8ft steel ruler and use it as a dipstick (Figure 1). Knowing the height

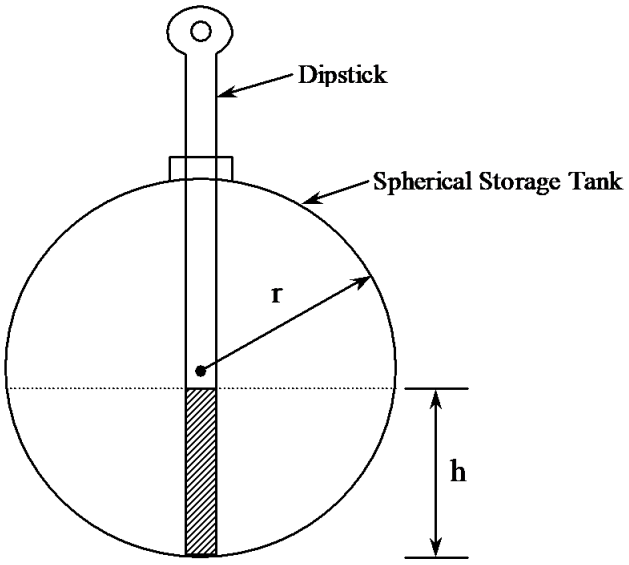



**Figure 1** Oil in a spherical storage tank.

to which the dip-stick would become wet with oil, one would know the height $h$ of the oil in the tank. The volume $V$ of oil left in the tank then is

$$V = \frac{\pi h^2 (3r - h)}{3} \quad (1)$$

where, $r$ is the radius of the tank. But, he did not stop there. He wanted me to design a steel ruler for him so that he would directly get the reading from the dipstick. How would I design such a dipstick?

The problem is inverse of what he wanted originally. To design a dipstick, I would have to mark the height corresponding to a volume. To do that I would need to solve the equation

$$V = \frac{\pi h^2 (3r - h)}{3}$$

for the height for a given volume and radius. For example, where would you mark the scale for $4ft^3$ of oil?

$$4 = \frac{\pi h^2 (3 \times 3 - h)}{3}$$

$$h^3 - 9h^2 + \frac{12}{\pi} = 0$$

$$f(h) = h^3 - 9h^2 + 3.8197 = 0$$

Therefore, this nonlinear equation needs to be solved. To mark the scale for other volumes, you will need to substitute the value for the volume and solve for $h$. Continue to do this for different preset volumes to develop the scale.

**Appendix A: Derivation of the formula for the volume of the oil based on radius of the tank and the height of oil in the tank**

How do you find that the volume of oil is given by

$$V = \frac{\pi h^2 (3r - h)}{3}$$

From calculus,

$$V = \int_{r-h}^{r} A dx$$

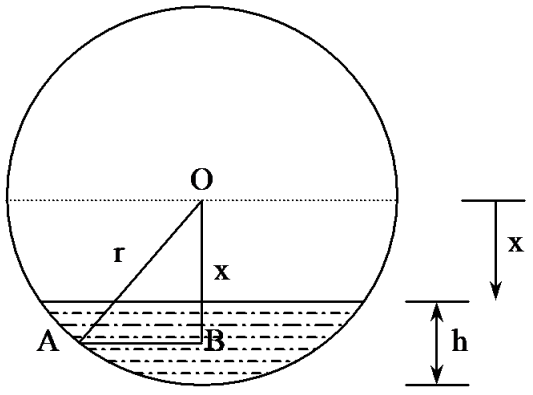


**Figure 2** Deriving the equation for volume of oil in the tank.

where $A$ is the cross-sectioned area at the location $x$, from the center of the sphere. The lower limit of integration is $x = r - h$as that is where the oil line is and the upper limit is $r$ as that is the bottom of the sphere. So, what is $A$ at any location $x$?
From Figure 2, for a location $x$

$$OB = x$$

$$OA = r$$

then

$$\begin{aligned} AB &= \sqrt{OA^2 - OB^2} \\ &= \sqrt{r^2 - x^2} \end{aligned}$$

and $AB$ is the radius of the area at $x$. So the area at location $x$ is

$$\begin{aligned} A &= \pi(AB)^2 \\ &= \pi\left(r^2 - x^2\right) \end{aligned}$$

So

$$\begin{aligned} V &= \int_{r-h}^{r} \pi\left(r^2 - x^2\right) dx \\ &= pi\left(r^2 x - \frac{x^3}{3}\right)_{r-h}^{r} \\ &= \pi\left[\left(r^2 r - \frac{r^3}{3}\right) - \left(r^2\left(r-h\right) - \frac{\left(r-h\right)^3}{3}\right)\right] \\ &= \frac{\pi h^2\left(3r - h\right)}{3}. \end{aligned}$$

# Chapter 03.00: Physical Problem for Nonlinear Equations

Lesson: Civil Engineering

## Summary

Finding the maximum sagging in a bookshelf requires solving a nonlinear equation.

## Modeling

My spouse asked me to build a bookshelf (Figure 1) for her books. Her books range from $8\frac{1}{2}''$ in to $11''$ in height, and would take $29''$ of space along the length. She asked me how much would the bookshelf sag, as she does not like sagging bookshelves we had during graduate student days. After all, our tolerance to sagging does sag after graduation! So, as a true engineer, before going to the local building store, I modeled the problem to find the maximum deflection of the bookshelf. I assumed a shelf of thickness $3/8''$ and width $12''$. The Young's modulus of a typical wood shelf material is 3.667 Msi.

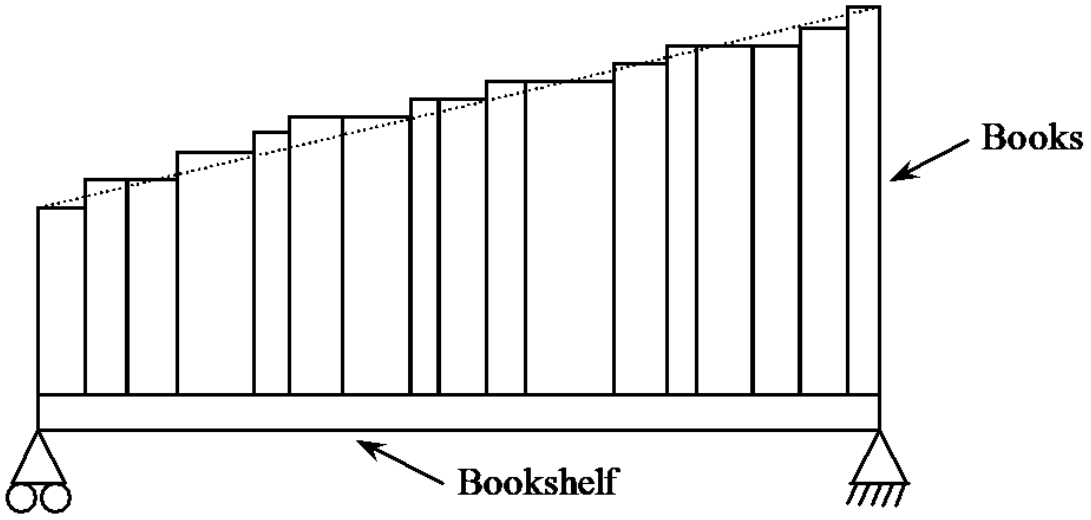



**Figure 1** Distributions of books on the bookshelf.

I assume the following

- 100 pages of $8\frac{1}{2}'' \times 11''$ book paper weighs 1 lb (on its website www.usps.gov, United States Post Office suggests 6 letter size pages weigh about 1 ounce, so 100 pages would be approximately a pound).
- 400 pages take up the space of $1''$ in thickness (I actually measured several textbooks and found an average.)
- the books decrease linearly in height from $11''$ from the right end to $8\frac{1}{2}''$ on the left, and the book width is $8\frac{1}{2}''$ for all books,

The total weight $W$ of the books is found as follows. If all the books were of dimensions $11'' \times 8\frac{1}{2}''$, the weight of the books would be

$$\begin{aligned} W &= 29in \times 400\frac{pages}{in} \times \frac{1lb}{100pages} \\ &= 116lb \end{aligned}$$

But since the books decrease in height from $11''$ to $8\frac{1}{2}''$, while the width of the books is assumed to be constant at $8\frac{1}{2}''$, the actual total weight of the books is calculated as

$$\begin{aligned} W &= \frac{\frac{1}{2}\left(11 + 8\frac{1}{2}\right)29}{(11)(29)} \times 116 \\ &\approx 103lb \end{aligned}$$

Now assuming that the weight '$W$' is distributed linearly from the left end to the right as a distribution $q$ (Figure 2),

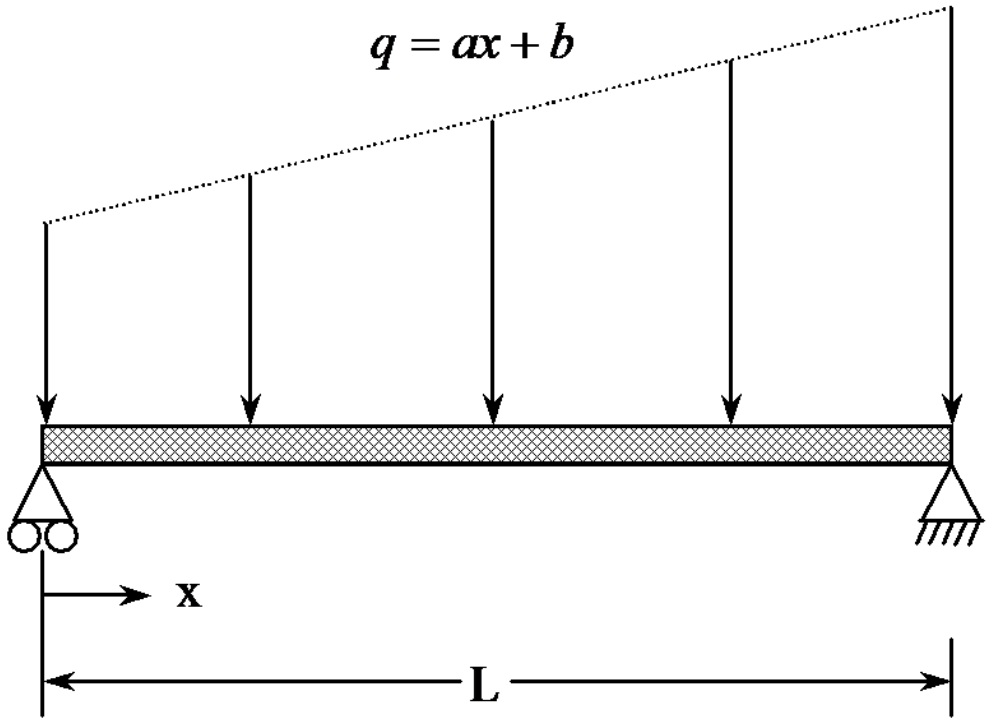



**Figure 2** Approximate weight distribution of books.

Then the total weight $W$ over the length $L$

$$\begin{aligned} W &= \int_0^L q dx \\ &= \int_0^L (ax+b)\, dx \\ &= a\frac{L^2}{2} + bL \quad (1) \end{aligned}$$

Also, the weight distribution can be assumed to be in the same ratio as the height of the books at the two ends, then

$$\frac{q\,(x=0)}{q\,(x=L)} = \frac{8.5}{11}$$

$$\frac{a\,(0)+b}{aL+b} = \frac{8.5}{11}$$

$$\frac{b}{aL+b} = \frac{8.5}{11} \quad (2)$$

Substituting

$$W = 103lb,$$

$$L = 29'',$$

in Equations (1) and (2), we get

$$a\frac{29^2}{2} + b\,(29) = 103$$

$$\frac{b}{a\,(29) + b} = \frac{8.5}{11}$$

or

$$420.5a + 29b = 103$$

$$-246.5a + 3b = 0$$

The above set of equations gives the solution as

$$a = 0.031403$$

$$b = 3.0964$$

So the weight distribution is given by

$$q = 0.031403x + 3.0964,\ 0 \leq x \leq 29$$

Now let us find the deflection in the beam. The deflection $y$ as a function of $x$ along the length of the beam is given by

$$\frac{d^2y}{dx^2} = \frac{M}{EI}$$

where

$$M = \text{Bending moment}\ \ (lb.in)$$

$$E = \text{Young's modulus}\ \ (psi)$$

$$I = \text{Second moment of area}\ \ \left(in^4\right)$$

To find the bending moment, we need to first find the reaction force at the support. Let $R_1$ and $R_2$ be the reactions at the left and right support, respectively (Figure 3). Then from the sum of forces in the vertical direction,

$$R_1 + R_2 = W$$

and the moment at the left support is zero as it is simply supported

$$R_2 L - \int_0^L (ax + b)\, x dx = 0$$

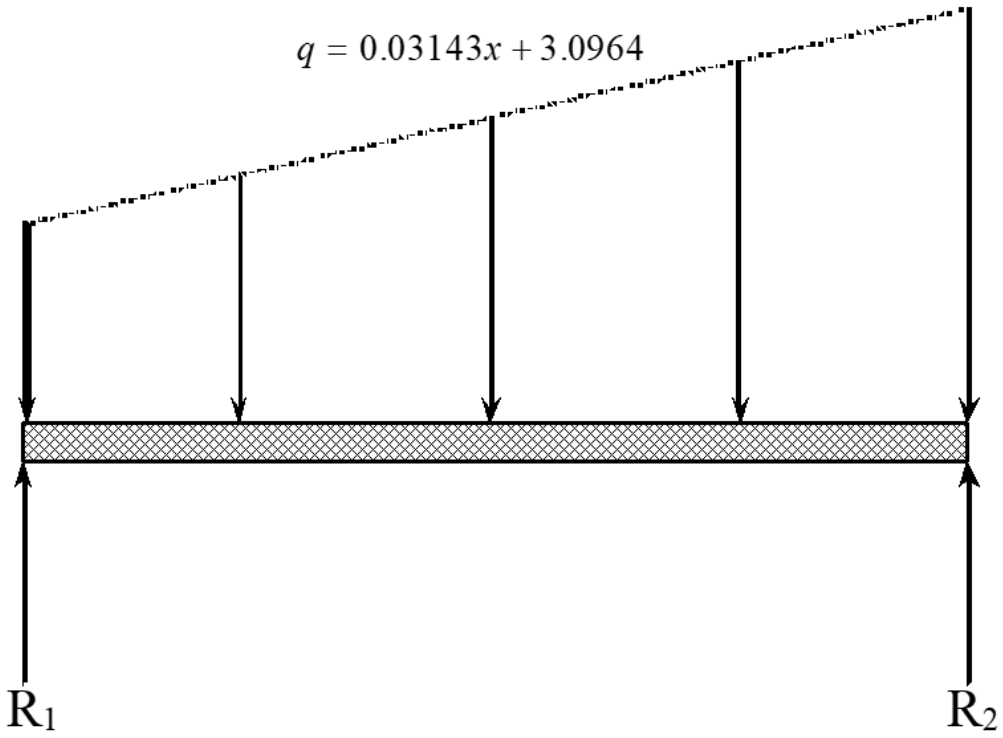



**Figure 3** Reaction at the end of shelf supports.

$$R_2 L - a\frac{L^3}{3} - \frac{bL^2}{2} = 0$$

Substituting

$$W = 103,$$

$$a = 0.031403,$$

$$b = 3.0964,$$

$$L = 29,$$

the equilibrium equations are

$$R_1 + R_2 = 103$$

$$R_2\,(29) - (0.031403)\,\frac{29^3}{3} - 3.0964\frac{29^2}{2} = 0$$

gives

$$R_1 = 49.300\ lb$$

$$R_2 = 53.701\ lb$$

The bending moment at any cross-section at a distance of $x$ from the left end is then given by summing the moments at a cross-section of distance $x$ from the left end as (Figure 4)

$$M\,(x) - R_1 x + \int_0^L \left(ax^{'} + b\right)\left(x - x^{'}\right) dx^{'} = 0$$

$$\begin{aligned} M\,(x) &= R_1 x \int_0^L \left(ax^{'} + b\right)(x - x')\,dx' \\ &= R_1 x - \frac{1}{6}ax^3 - \frac{1}{2}bx^2 \end{aligned}$$

Substituting value of$R_1, a$ and $b$,

$$R_1 = 49.299$$

$$a = 0.031403$$

$$b = 3.0964$$

the bending moment equation is given by

$$M\,(x) = 49.299x - \frac{1}{6}\,(0.031403)\,x^3 - \frac{1}{2}\,(3.0964)\,x^2$$

$$M\,(x) = 49.299x - 0.0052339x^3 - 1.5482x^2$$

So

$$EI\frac{d^2v}{dx^2} = M\left(x\right)$$

$$\frac{d^2v}{dx^2} = \frac{M\left(x\right)}{EI}$$

$$E = 3.667Msi$$

$$I = \frac{1}{12}ht^3$$

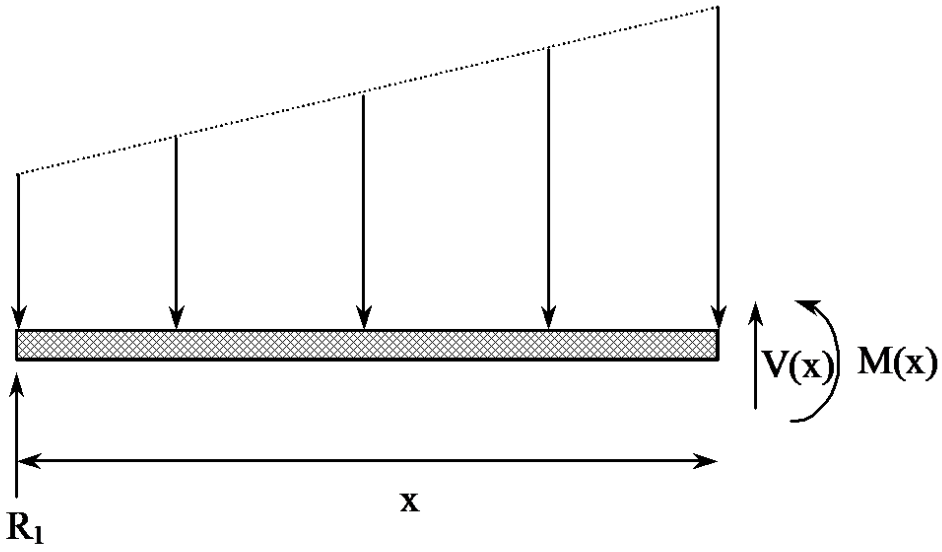



**Figure 4** Free body diagram to find bending moment at a cross-section.

where

$$h = \text{width of shelf}$$

$$t = \text{thickness of shelf}$$

$$I = \frac{1}{12} \times 12 \times \left(\frac{3}{8}\right)^3$$
$$= 0.052734in$$

Hence

$$\begin{aligned}\frac{d^2v}{dx^2} &= \frac{M(x)}{EI} \\ &= \frac{49.29x - 0.0052339x^3 - 1.5428x^2}{3.6667 \times 10^6 \times 0.052734} \\ &= 0.25496 \times 10^{-3}x - 0.27068 \times 10^{-7}x^3 - 0.80067 \times 10^{-5}x^2\end{aligned}$$

Integrating with respect to $x$ gives

$$\frac{dv}{dx} = 0.25496 \times 10^{-3}x^2 - 0.67670 \times 10^{-8}x^4 - 0.26689 \times 10^{-5}x^3 + c_1$$

Integrating once again with respect to $x$ gives

$$v(x) = 0.42493 \times 10^{-4}x^3 - 0.13534 \times 10^{-8}x^5 - 0.66723 \times 10^{-6}x^4 + c_1x + c_2$$

Now the boundary conditions at $x = 0$ and $x = L$ are that the displacement is zero at the ends,

$$v(0) = 0$$

$$v(L) = 0$$

So from

$$v(0) = 0$$

we get

$$c_2 = 0, \text{ and}$$

from

$$v(L) = 0$$

we get

$$0 = 0.42493 \times 10^{-3}L^3 - 0.13534 \times 10^{-8}L^5 - 0.66723 \times 10^{-6}L^4 + c_1L$$

Substituting $L = 29''$

$$0 = 0.42493 \times 10^{-3}(29)^3 - 0.13534 \times 10^{-8}(29)^5 - 0.66723 \times 10^{-6}(29)^4 + c_1(29)$$

$$c_1 = -0.018506$$

Hence the vertical deflection in the beam is given by

$$v\left(x\right)=0.42493\times10^{-4}x^{3}-0.13534\times10^{-8}x^{5}-0.66723\times10^{-6}x^{4}-0.018506x$$

But to find where the deflection would be maximum, we need to take the first derivative of the deflection to find where

$$\frac{dv}{dx}=0,$$

that is

$$\frac{d}{dx}(0.42493\times10^{-4}x^{3}-0.13534\times10^{-8}x^{5}-0.66723\times10^{-6}x^{4}-0.018506x)=0$$

$$f(x)=0.12748\times10^{-3}x^{2}-0.67670\times10^{-8}x^{4}-0.26689\times10^{-5}x^{3}-0.018506=0$$

Now here is a nonlinear equation that needs to solved to find where the maximum deflection occurs. Then substituting the obtained value of $x$ in $v\left(x\right)$ would give us the maximum sagging in the beam.

# Chapter 03.00: Physical Problem for Nonlinear Equations

Lesson: Computer Engineering

## Summary

For efficient design of the Cray supercomputer, it does not have a divide unit. It uses a numerical solution of a nonlinear equation to find the inverse of a number.

## Modeling

Many super computers do not have a unit to divide numbers. But why? Well, a divide operation in modern computers can take 20 to 25 clock cycles, and that is five times what it takes for multiplication [1]. Instead, a divide unit, based on numerically solving a nonlinear equation, is developed. This allows for a faster divide operation. This is how it works.

If you want to find the value of $b/a$ where $b$ and $a$ are real numbers, one can look at

$$\frac{b}{a} = b * \frac{1}{a} = b * c$$

where

$$c = \frac{1}{a}$$

So, if one is able to find $c$, we only need to multiply $b$ and $c$ to find $b/a$. So how do we find $c$ without the divide unit.

Equation

$$c = \frac{1}{a}$$

can be written as an equation

$$f(c) = ac - 1 = 0$$

If one is able to find the root of this equation without using a division, then we have the value of $1/a$.

Although we do not explain the numerical methods of solving nonlinear equations in this section of the notes, it becomes necessary to do so in this example. The Newton-Raphson method of solving

a nonlinear equation is used in finding $1/a$. The Newton Raphson method of solving an equation $f(c) = 0$ is given by the iterative formula

$$c_{i+1} = c_i - \frac{f(c_i)}{f'(c_i)},$$

where

$c_{i+1}$ is the new approximation of of the root of $f(c) = 0$ and

$c_i$ is the previous approximation of the root of $f(c) = 0$.

**What is the appropriate function to use to find the inverse of a?**

a) Using

$$f(c) = ac - 1 = 0$$

gives

$$f'(c) = a$$

and the Newton-Raphson method formula gives

$$c_{i+1} = c_i - \frac{ac_i - 1}{a}$$

$$c_{i+1} = \frac{1}{a}$$

This is of no use as it involves division.

b) Using

$$f(c) = a - \frac{1}{c} = 0$$

gives

$$f'(c) = \frac{1}{c^2}$$

$$\begin{aligned} c_{i+1} &= c_i - \frac{a - \frac{1}{c_i}}{\frac{1}{c_i^2}} \\ &= c_i - c_i^2 \left( a - \frac{1}{c_i} \right) \end{aligned}$$

$$c_{i+1} = c_i\,(2 - c_i a)$$

This one is the acceptable iterative formula to find the inverse of $a$ as it does not involve division.

Starting with an initial guess for the inverse of $a$, one can find newer approximations by using the above iterative formula. Each iteration requires two multiplications and one subtraction. However, the number of iterations required to find the inverse of $a$ very much depends on the initial approximation. More accurate is the starting approximation, less number of iterations are required to find the inverse of $a$. Since the convergence of Newton Raphson method is quadratic, it may take up to six iterations to get an accurate reciprocal in double precision. By using look-up tables for the initial approximation, the number of iterations required can be reduced to two [2]. Also, the operation of $(2 - c_i a)$ may be carried in a fused multiply-subtract unit to further reduce the clock cycles needed for the computation.

## References

1. Oberman, S.F., Flynn, M.J., "Division Algorithms and Implementations", IEEE Transactions on Computers, Vol. 46, No. 8, 1997.

2. Wong, D., Flynn, M.J., "Fat Division using accurate quotient approximations to reduce the number of iteratoimns", IEEE Transactions on Computers, Vol. 41, No. 8, 981-995, 1992.

## Appendix A

**Example of using Newton–Raphson method to find the inverse of a number**

Let us find $\dfrac{1}{2.5}$

The Newton –Raphson method formula is given by

$$c_{i+1} = c_i\,(2 - 2.5 c_i)$$

Starting with estimate of inverse as $c_0 = 0.5$,

$$\begin{aligned} c_1 &= 0.5\,[2 - 2.5\,(0.5)] \\ &= 0.375 \end{aligned}$$

$$\begin{aligned} c_2 &= 0.375\,[2 - 2.5\,(0.375)] \\ &= 0.3984 \end{aligned}$$

$$\begin{aligned} c_3 &= 0.3984\,[2 - 2.5\,(0.3984)] \\ &= 0.3999 \end{aligned}$$

$$\begin{aligned} c_4 &= 0.3999\,[2 - 2.5\,(0.3999)] \\ &= 0.4000 \end{aligned}$$

It took four iterations to find the inverse of 2.5 correct up to 4 significant digits.

# Chapter 03.00: Physical Problem for Nonlinear Equations

Lesson: Electrical Engineering

## Summary

Calibrating a thermistor requires solution of nonlinear equations.

## Modeling

Thermistors are temperature measuring devices based on that resistance of materials changes with temperature. To find whether the resistor is calibrated properly, one needs to solve a problem of a nonlinear equation.

Thermistors are temperature-measuring devices based on the principle that the thermistor material exhibits a change in electrical resistance with a change in temperature. By measuring the resistance of the thermistor material, one can then determine the temperature.

Thermistors are generally a piece of semiconductor (Figure 1) made from metal oxides such as those of manganese, nickel, cobalt, etc. These pieces may be made into a bead, disk, wafer, etc depending on the application.

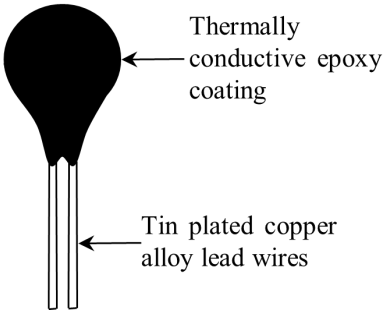



**Figure 1**. Sketch of a thermistor.

There are two types of thermistors – negative temperature coefficient (NTC) and positive temperature coefficient (PTC) thermistors. For NTCs, the resistance decreases with temperature, while for PTCs, the resistance increases with temperature. It is the NTCs that are generally used for temperature measurement.

Why would we want to use thermistors for measuring temperature as opposed to other choices such as thermocouples? It is because thermistors have

- high sensitivity giving more accuracy,

- a fast response to temperature changes for accuracy and quicker measurements, and
- relatively high resistance for decreasing the errors caused by the resistance of lead wires themselves.

But thermistors have a nonlinear output and are valued for a limited range. So, when a thermistor is manufactured, the manufacturer supplies a resistance vs. temperature curve. The curve generally used that gives an accurate representation is given by Steinhart and Hart equation

$$\frac{1}{T} = a_0 + a_1 \ln(R) + a_3 \{\ln(R)\}^3 \quad (1)$$

where

$T$ is temperature in Kelvin, and

$R$ is resistance in ohms.

$a_0, a_1, a_3$ are constants of the calibration curve.

As an example, for an actual thermistor – Part No 10K3A made by Betatherm sensors, the values of the three coefficients are given as

$$a_0 = 1.129241 \times 10^{-4}$$

$$a_1 = 2.341077 \times 10^{-4}$$

$$a_3 = 8.775468 \times 10^{-8}$$

and are found by measuring the resistance of the thermistor at three reference points (namely $0^0C, 25^0C$ and $70^0C$ in this case) and using equation (1) to set up three simultaneous linear equations to find the three constants $a_0, a_1, a_3$. The resulting Steinhart-Hart equation for the 10K3A Betatherm thermistor is

$$\frac{1}{T} = 1.129241 \times 10^{-3} + 2.341077 \times 10^{-4} \ln(R) + 8.775468 \times 10^{-8} \{\ln(R)\}^3 \quad (2)$$

where note that $T$ is in Kelvin and $R$ is in ohms.

Using a digital system to measure temperature, an analog system is used to measure the thermistor resistances and convert that to a temperature reading. You want to confirm that the Resistance vs. Temperature data compares well with the published $R/T$ data for the range for which the thermistor will be used. For example for the above thermistor, error of no more than $\pm 0.01^0C$ is acceptable. What is the range of the resistance then you can consider to be within this acceptable limit at $19^0C$? To find this we need to solve the equation for a temperature of $19 \pm 0.01 = 18.99^0C$ to $19.01^0C$ range. These equations are

$$\frac{1}{19.01 + 273.15} = 1.129241 \times 10^{-3} + 2.341077 \times 10^{-4} \ln(R) + 8.775468 \times 10^{-8} \{\ln(R)\}^3$$

$$3.42278 \times 10^{-3} = 1.129241 \times 10^{-3} + 2.341077 \times 10^{-4} \ln(R) + 8.775468 \times 10^{-8} \{\ln(R)\}^3 \quad (3)$$

and

$$\frac{1}{18.99 + 273.15} = 1.129241 \times 10^{-3} + 2.341077 \times 10^{-4} \ln(R) + 8.775468 \times 10^{-8} \{\ln(R)\}^3$$

$$3.42301 \times 10^{-3} = 1.129241 \times 10^{-3} + 2.341077 \times 10^{-4} \ln(R) + 8.775468 \times 10^{-8} \{\ln(R)\}^3 \quad (4)$$

Equations (3) and (4) are independent nonlinear equations that need to be solved for $R$.

## References


1. Betatherm sensors, http://www.betatherm.com

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3. Lavenuta, G., "Negative Temperature Coefficient Thermistors: Part 1: Characteristics, Materials, and Configurations", http://www.globalspec.com/cornerstone/ref/negtemp.html

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5. Steinhart, J.S. and Hart, S.R., 1968. "Calibration Curves for Thermistors," Deep Sea Research 15:497.

6. Sapoff, M. et al. 1982. "The Exactness of Fit of Resistance-Temperature Data of Thermistors with Third-Degree Polynomials," Temperature, Its Measurement and Control in Science and Industry, Vol. 5, James F. Schooley, ed., American Institute of Physics, New York, NY:875.

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## Questions

Answer the following questions

(1). Note that if we substitute $x = \ln(R)$, the equation becomes a cubic equation in $x$. The cubic equation will have three roots. Could some of these roots be complex? If so how many?

(2). Solving the cubic equation exactly would require major effort. However using numerical techniques, we can solve this equation and any other equation of the form $f(x) = 0$. Solve the above equation by all the methods you have learned assuming you want at least 3 significant digits to be correct in your answer.

(3). How can you use the knowledge of the physics of the problem to develop initial guess(es) for the numerical methods?

(4). If more than one root of the above equation is real, how do you choose the valid root? Or, are all the possible real roots physically acceptable?

# Chapter 03.00: Physical Problem for Nonlinear Equations

Lesson: Industrial Engineering

## Summary

Determine how many computers a business would have to sell to turn a profit. To find this number of computers to be sold, the physical model is a nonlinear equation.

## Modeling

You have been recently employed by a start-up computer assembly company called the "MOM AND POP COMPUTER SHOP". As a recent graduate with a bachelor's degree in industrial engineering, you have been asked by the president, to determine the minimum number of computers that the shop will have to sell to make a profit during the first year in business.

First, it is important to determine the first costs or capital costs, CC, associated with starting the business. The capital costs include such items as:

- computer assembly and diagnostic equipment,
- office furniture,
- workbenches, and
- initial inventory purchase, to name a few.

It is assumed that the capital costs, $CC$, associated with the business are

$$CC = \$20,000$$

Second, it is important to determine the repeated costs or operating costs, $OC$, associated with operating the business for the first year. Operating costs differ from capital costs mainly in that operating costs must be paid repeatedly (every year in this instance), in contrast to capital costs, which involve a one-time payment. Operating costs can be broken down into three major classifications:

(1) indirect costs or fixed costs, $FC$

(2) direct or variable costs, $VC$, and

(3) semivariable or regulated costs, $RC$,

where

$$OC = FC + VC + RC$$

Fixed costs include items that are not generally a function of the production level, such as:

- building lease,
- real estate taxes,
- utilities (DSL or cable modem, basic electric and local phone),
- insurance, and
- marketing, to name a few.

These fixed costs could, however, increase say if the production level reaches a point when a larger building is needed or when more phone lines needs to be installed. It is assumed that the fixed costs, $FC$, associated with the business are

$$FC = \$15,000$$

Variable costs include items that are a direct function, often linear, of the production level, such as:

- materials (computer components),
- utilities (production electric and long distance phone),
- labor (also includes supervision and payroll charges),
- maintenance, and
- distribution (packaging and shipping), to name a few.

It is assumed that the variable costs, $VC$, associated with the business are

$$VC = \$625n$$

where $n$ is the number of computers produced. Regulated costs include items that are also a direct function, though often non-linear, of the production level, such as:

- labor (also includes supervision and payroll charges),

For instance, if production levels reach a certain level, another employee may need to be hired. It is assumed that the regulated costs, $RC$, associated with the business are

$$RC = \$30n^{1.5}$$

Combining the fixed, variable and regulated costs gives the operating cost, that is,

$$CC = FC + VC + RC$$

$$CC = \$15,000 + \$500n + 30n^{1.5}$$

And the total cost, $TC$, to operate the business for the first year is the capital costs plus the operating costs, that is

$$TC = CC + OC$$

$$TC = \$20,000 + \$15,000 + \$625n + \$30n^{1.5}$$

$$TC = \$35,000 + \$625n + \$30n^{1.5}$$

All of the costs (capital, fixed, variable, regulated, operating, and total) are shown below in Figure 1.

Next, it is important to determine the total sales, $TS$, associated with operating the business for the first year. Total sales can be broken down into two major classifications:

(1) product sales, $PS$, and

(2) discounted sales, $DS$,

where

$$TS = PS - DS$$

If they sell $n$ computers, their product sales, PS, are

$$PS = \$15000n$$

This represents the \$1500 selling price for each computer. It is assumed that the discounted sales, $DS$, are

$$DS = \$10n^{1.5}$$

which represents a discount as the number of computers sold increases. The total sales is then given by

$$TS = PS - DS$$

$$TS = \$1500n - \$10n^{1.5}$$

All of the sales (product, discounted, and total) revenue are shown below in Figure 2.

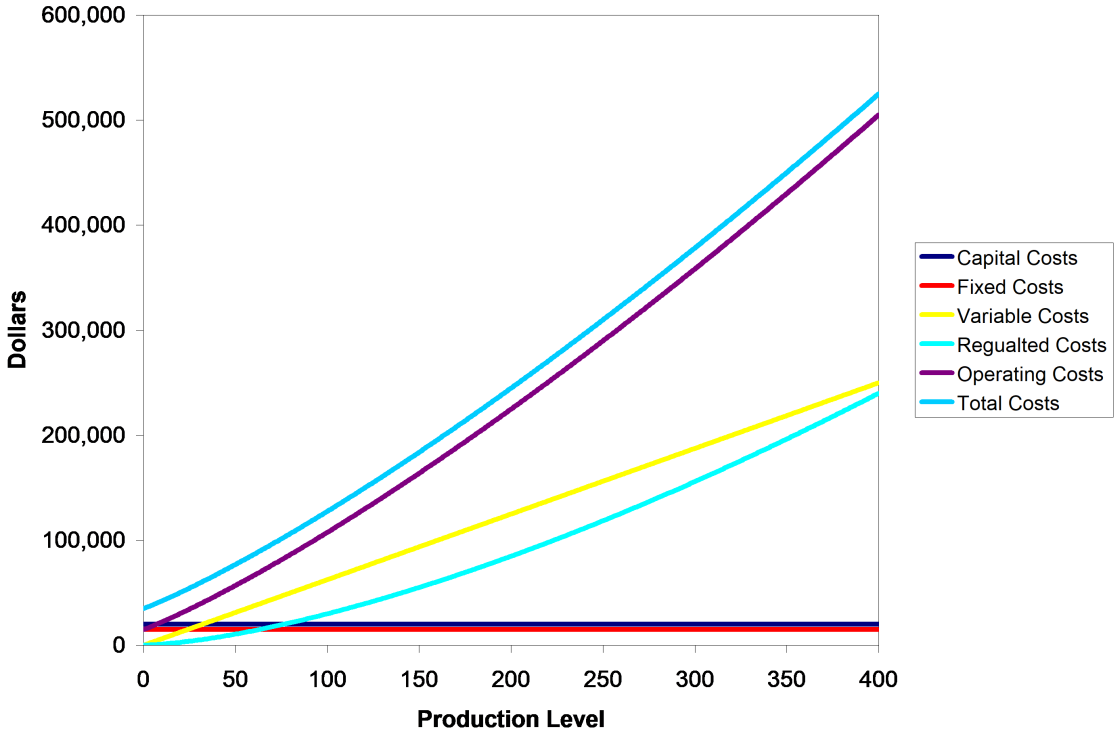



**Figure 1** Capital, Fixed, Variable, Regulated, Operating, And Total Costs

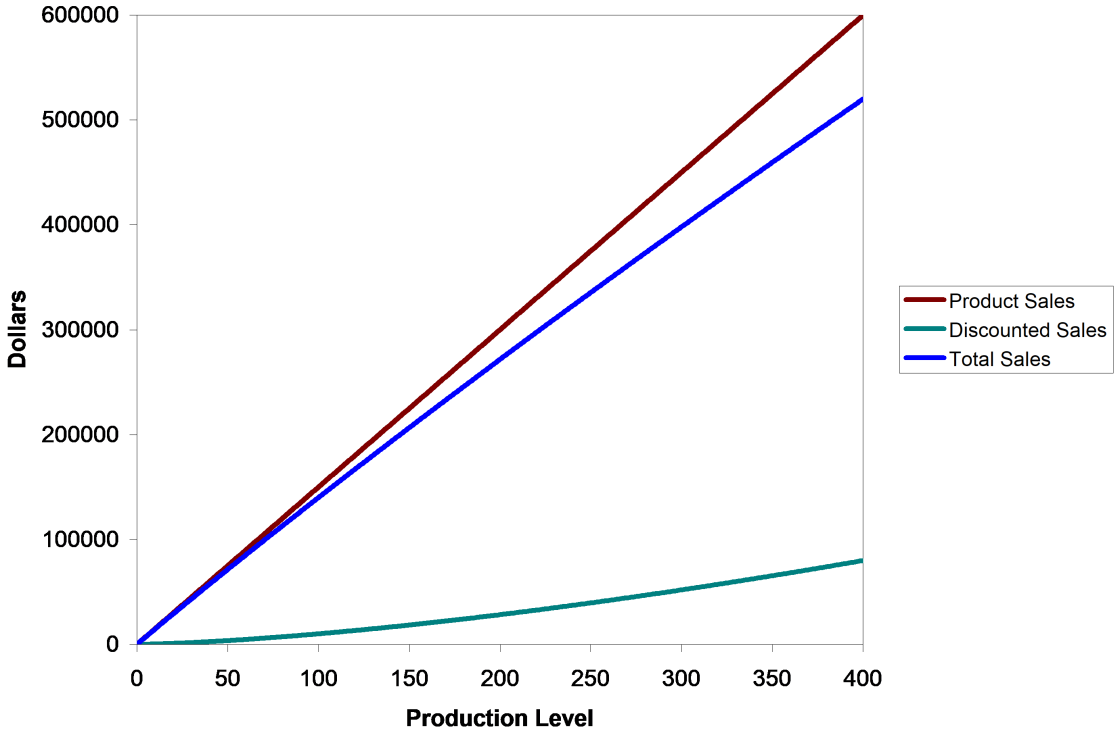


**Figure 2** Product, Discounted, And Total Sales

All of the costs and sales revenue are shown next in Figure 3.

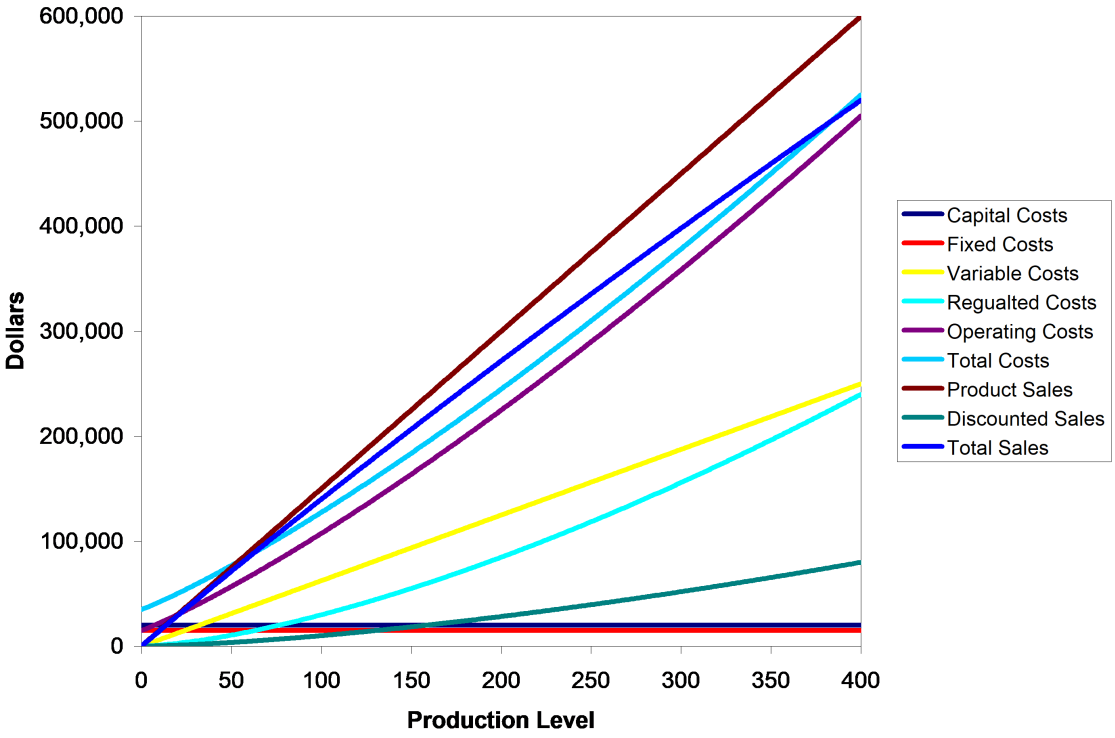



**Figure 3** All of the Costs and Sales Revenue

At the break-even point there is no profit, so the capital cost plus the product cost equals the product sales (see Figure 4).

$$TC = TS$$

$$\$35,000 + \$625n + \$30n^{1.5} = \$1500n - \$10n^{1.5}$$

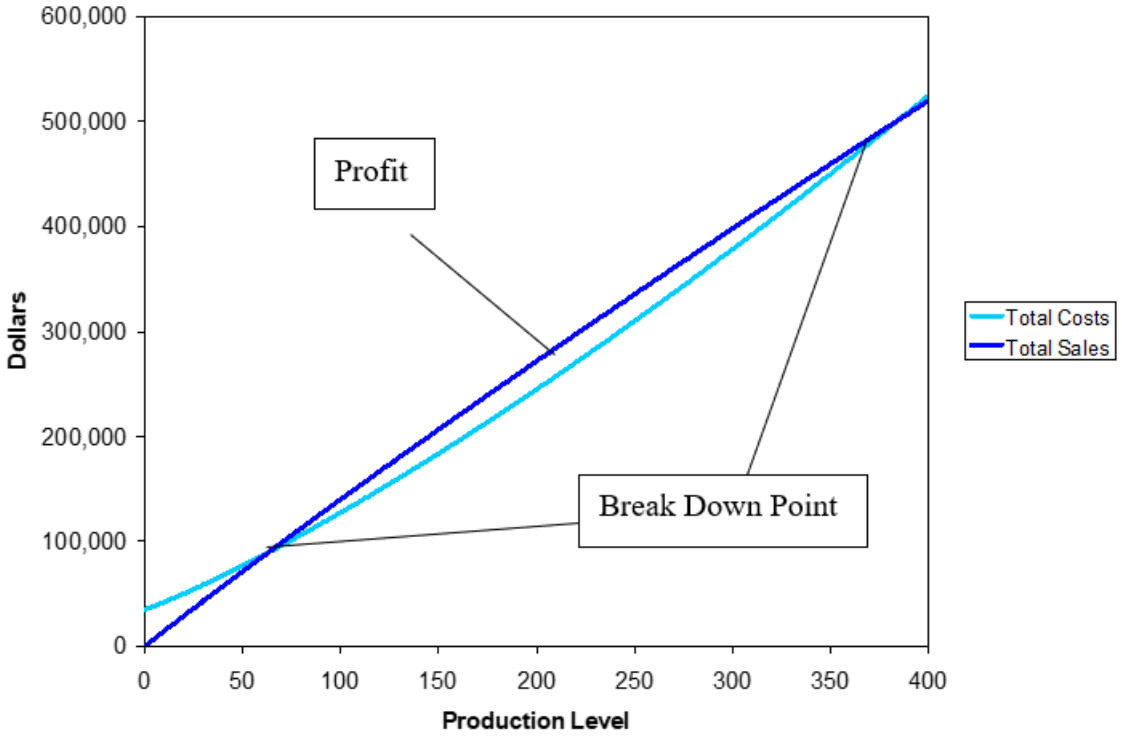



**Figure 4** Break-Even Point and Profit Margin

Simplifying the previous equation yields the following nonlinear equation that needs to solved.

$$\$35,000 - \$875n + \$40n^{1.5} = 0$$

The value of $n$ is the minimum number of computers that the shop will have to sell to make a profit. This is called the *break-even point*.

## Questions

(1) Will some of the roots be complex for the above nonlinear equation?

(2) Using numerical techniques, we can solve this equation and any other equation of the form $f(x) = 0$. Solve the above equation by all the methods you have learned assuming you want at least 3 significant digits to be correct in your answer.

(3) How can you use the knowledge of the problem to develop initial guess(es) for the numerical methods?

(4) Determine the production level that produces the most profit.

(5) Determine the second break-even point, after which a loss is realized.

(6) Reformulate the problem to determine the break-even point for the second year in business assuming that no capital costs are incurred.

# Chapter 03.00: Physical Problem for Nonlinear Equations

Lesson: Mechanical Engineering

## Summary

A physical problem of finding how much cooling a shaft needs to be shrink fit into a hollow hub. The temperature to which the cooling needs to be done is modeled as a nonlinear equation.

## Modeling

To make the fulcrum (Figure 1) of a bascule bridge, a long hollow steel shaft called the trunnion is shrink fit into a steel hub. The resulting steel trunnion-hub assembly is then shrink fit into the girder of the bridge.

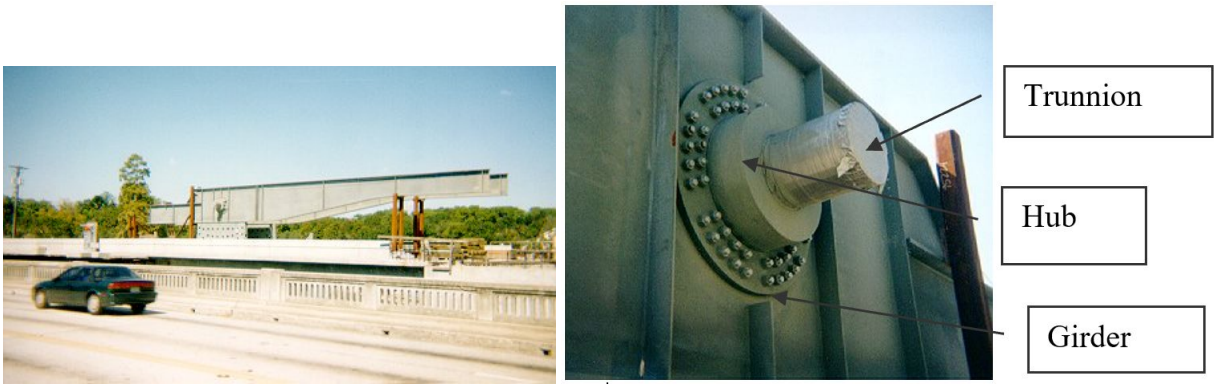



**Figure 1** Trunnion-Hub-Girder (THG) assembly.

This is done by first immersing the trunnion in a cold medium such as dry-ice/alcohol mixture. After the trunnion reaches the steady state temperature of the cold medium, the trunnion outer diameter contracts. The trunnion is taken out of the medium and slid through the hole of the hub (Figure 2).

When the trunnion heats up, it expands and creates an interference fit with the hub. In 1995, on one of the bridges in Florida, this assembly procedure did not work as designed. Before the trunnion could be inserted fully into the hub, the trunnion got stuck. So a new trunnion and hub had to be ordered at a cost of $50,000. Coupled with construction delays, the total loss was more than a hundred thousand dollars.

Why did the trunnion get stuck? This was because the trunnion had not contracted enough to slide through the hole.

Now the same designer is working on making the fulcrum for another bascule bridge. Can you help him/her so that he does not make the same mistake?

For this new bridge, he needs to fit a hollow trunnion of the outside diameter $12.363''$ in a hub of an inner diameter $12.358''$. He plans to put the trunnion in dry-ice/alcohol mixture (temperature of the fluid - dry ice/alcohol mixture is $-108°\text{F}$) to contract the trunnion so that it can be slided through the hole of the hub. To slide the trunnion without sticking, he has also specified a diametrical clearance of at least $0.01''$ between the trunnion and the hub. Assuming the room temperature is $80°\text{F}$, is immersing it in a dry-ice/alcohol mixture a correct decision? What temperature does he need to cool the trunnion to so that he gets the desired contraction?

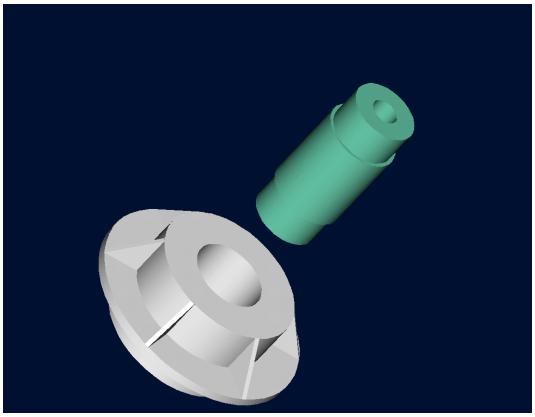


**Figure 2** Trunnion slided through the hub after contracting

Looking at the designer's record for the previous bridge (where the trunnion got stuck in the hub), it was found that they used the coefficient of linear thermal expansion at room temperature to calculate the contraction in the trunnion diameter. In that case, the reduction, $\Delta D$ in the outer diameter of the trunnion is

$$\Delta D = D\alpha\Delta T \qquad (1)$$

where

$$D = \text{outer diameter of the trunnion,}$$

$$\alpha = \text{coefficient of linear thermal expansion at room temperature, and}$$

$$\Delta T = \text{change in temperature,}$$

Given

$$D = 12.363''$$

$$\alpha = 6.817 \times 10^{-6}\text{in/in/°F at } 80°\text{F}$$

$$\begin{aligned}\Delta T &= T_{fluid} - T_{room} \\ &= -108 - 80 \\ &= -188^{\circ}\text{F}\end{aligned}$$

where

$$T_{fluid} = \text{temperature of dry-ice/alcohol mixture}$$

$$T_{room} = \text{room temperature}$$

the reduction in the trunnion outer diameter is given by

$$\begin{aligned}\Delta D &= (12.363)\left(6.47 \times 10^{-6}\right)(-188) \\ &= -0.01504''\end{aligned}$$

So the trunnion is predicted to reduce in diameter by $0.01504''$. But is this enough reduction in diameter? As per specifications, he needs the trunnion to contract by

$$\begin{aligned}&= \text{trunnion outside diameter} - \text{hub inner diameter} + \text{diametral clearance} \\ &= 12.363'' - 12.358'' + 0.01'' \\ &= 0.015''\end{aligned}$$

So, according to his calculations, immersing the steel trunnion in dry-ice/alcohol mixture gives the desired contraction of $0.015''$ as he is predicting a contraction of $0.01504''$.

But as shown in Figure 3, the coefficient of linear thermal expansion of steel decreases with temperature and is not constant over the range of temperature the trunnion goes through. Hence the above formula (Equation 1) would overestimate the thermal contraction. This assumption is the mistake he made in the calculations for the earlier bridge.

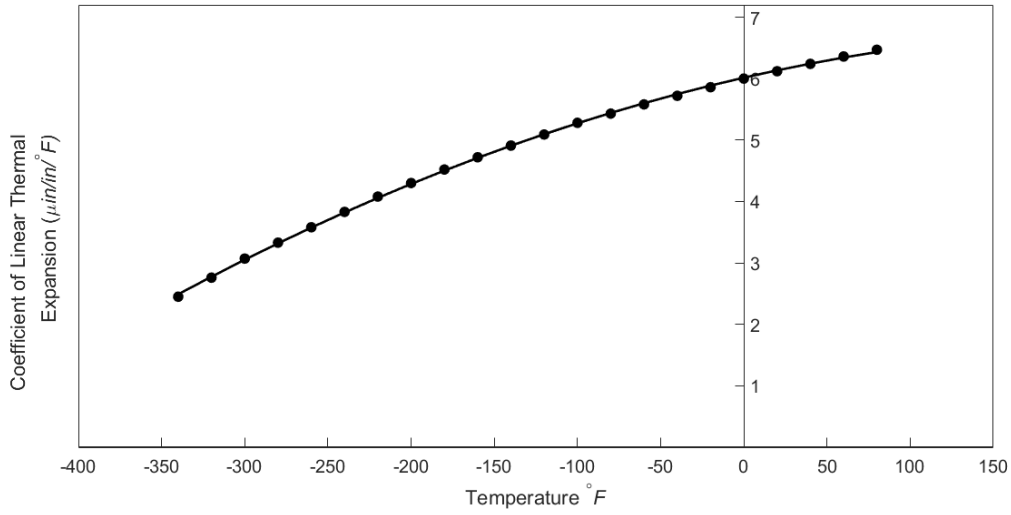


**Figure 3** Varying coefficient of linear thermal expansion as a function of temperature for cast steel.

The contraction in the diameter for the trunnion for which the coefficient of linear thermal expansion varies as a function of temperature is given by

$$\Delta D = D \int_{T_{room}}^{T_{fluid}} \alpha dT \qquad (2)$$

Note that Equation (2) reduces to Equation (1) if the coefficient of linear thermal expansion is assumed to be constant. In Figure 3, the coefficient of linear thermal expansion of typical cast steel is approximated by a second-order polynomial as

$$\alpha = -1.2278 \times 10^{-11} T^2 + 6.1946 \times 10^{-9} T + 6.0150 \times 10^{-6} \qquad (3)$$

Since the desired contraction is at least $0.015''$, that is,

$$\Delta D = -0.015'', \qquad (4)$$

From Equations (2), (3), and (4),

$$-0.015 = 12.363 \int_{80}^{T_{fluid}} \left(-1.2278 \times 10^{-11} T^2 + 6.1946 \times 10^{-9} T + 6.015 \times 10^{-6}\right) dT$$

$$-0.015 = 12.363 \left[-1.2278 \times 10^{-11} \frac{T^3}{3} + 6.1946 \times 10^{-9} \frac{T^2}{2} + 6.015 \times 10^{-6} T\right]_{80}^{T_{fluid}}$$

$$-0.015 = 12.363(-0.40927 \times 10^{-11} T_{fluid}^3 + 0.30973 \times 10^{-8} T_{fluid}^2 + 0.60150 \times 10^{-5} T_{fluid} - 0.49893 \times 10^{-3})$$

$$f(T_f) = -0.50598 \times 10^{-10} T_f^3 + 0.38292 \times 10^{-7} T_f^2 + 0.74363 \times 10^{-4} T_f + 0.88318 \times 10^{-2} = 0 \qquad (5)$$

One can solve this nonlinear equation to find the minimum fluid temperature needed to cool down the trunnion and get the desired contraction. Is cooling in dry-ice/alcohol mixture still your recommendation? You may be surprised that it would not be the correct decision to make.

# Chapter 03.01: Prerequisites to Numerical Methods for Solving Nonlinear Equations

Lesson: Review of Quadratic Equations

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *find the solutions of quadratic equations,*

2) *derive the formula for the solution of quadratic equations,*

3) *solve simple physical problems involving quadratic equations.*

## What are quadratic equations, and how do we solve them?

A quadratic equation has the form

$$ax^2 + bx + c = 0, \text{ where } a \neq 0$$

The solution to the above quadratic equation is given by

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

So the equation has two roots, and depending on the value of the discriminant, $b^2 - 4ac$, the equation may have real, complex, or repeated roots.

$$\text{If } b^2 - 4ac < 0, \text{ the roots are complex.}$$

$$\text{If } b^2 - 4ac > 0, \text{ the roots are real.}$$

$$\text{If } b^2 - 4ac = 0, \text{ the roots are real and repeated.}$$

**Example 1**

Derive the solution to $ax^2 + bx + c = 0$.

**Solution**

$$ax^2 + bx + c = 0 \qquad (E1.1)$$

Dividing both sides by $a$, $(a \neq 0)$, we get

$$x^2 + \frac{b}{a}x + \frac{c}{a} = 0$$

Note if $a = 0$, the solution to

$$ax^2 + bx + c = 0$$

is

$$x = -\frac{c}{b}$$

Rewrite Equation (E1.1) for $a \neq 0$

$$x^2 + \frac{b}{a}x + \frac{c}{a} = 0$$

as

$$\left(x + \frac{b}{2a}\right)^2 - \frac{b^2}{4a^2} + \frac{c}{a} = 0$$

$$\begin{aligned}\left(x + \frac{b}{2a}\right)^2 &= \frac{b^2}{4a^2} - \frac{c}{a} \\ &= \frac{b^2 - 4ac}{4a^2}\end{aligned}$$

$$\begin{aligned}x + \frac{b}{2a} &= \pm\sqrt{\frac{b^2 - 4ac}{4a^2}} \\ &= \pm\frac{\sqrt{b^2 - 4ac}}{2a}\end{aligned}$$

$$\begin{aligned}x &= -\frac{b}{2a} \pm \frac{\sqrt{b^2 - 4ac}}{2a} \\ &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}\end{aligned} \qquad (E1.2)$$

**Example 2**

A ball is thrown down at 50 mph from the top of a building. The building is 420 feet tall. Derive the equation that would let you find the time the ball takes to reach the ground.

**Solution**

The distance $s$ covered by the ball is given by

$$s = ut + \frac{1}{2}gt^2 \qquad (E2.1)$$

where

$$u = \text{initial velocity (ft/s)}$$

$$g = \text{acceleration due to gravity (ft/s}^2\text{)}$$

$$t = \text{time (s)}$$

Given

$$\begin{aligned} u &= 50\frac{\text{miles}}{\text{hour}} \times \frac{1 \text{ hour}}{3600 \text{ s}} \times \frac{5280 \text{ ft}}{1 \text{ mile}} \\ &= 73.33 \ \frac{\text{ft}}{\text{s}} \end{aligned}$$

$$g = 32.2\frac{\text{ft}}{\text{s}^2}$$

$$s = 420 \text{ ft}$$

we have from Equation (2.1)

$$420 = 73.33t + \frac{1}{2}\left(32.2\right) \ t^2$$

$$16.1t^2 + 73.33t - 420 = 0$$

The above equation is a quadratic equation, the solution of which would give the time it would take the ball to reach the ground. The solution of the quadratic equation is

$$\begin{aligned} t \ &= \frac{-73.33 \pm \sqrt{73.33^2 - 4 \times 16.1 \times (-420)}}{2(16.1)} \\ &= 3.315, -7.870 \end{aligned}$$

Since $t > 0$, the valid value of time $t$ is 3.315 s.

# Multiple Choice Test

Chapter 03.01: Prerequisites to Numerical Methods for Solving Nonlinear Equations

(1). The value of $x$ that satisfies $f(x) = 0$ is called most suitably the

(A) root of an equation $f(x) = 0$

(B) root of a function $f(x)$

(C) zero of an equation $f(x) = 0$

(D) none of the above

(2). A quadratic equation has _____ root(s).

(A) one

(B) two

(C) three

(D) four

(3). For a certain cubic equation, at least one of the roots is known to be a complex root. How many complex roots does the cubic equation have?

(A) one

(B) two

(C) three

(D) cannot be determined

(4). An equation such as $\tan x = x$ has _____ root(s).

(A) zero

(B) one

(C) two

(D) infinite

(5). A polynomial of order $n$ has _____ zeros.

(A) $n-1$

(B) $n$

(C) $n+1$

(D) $n+2$

(6). The velocity of a body is given by $v(t) = 5e^{-t} + 4$, where $t$ is in seconds and $v$ is in m/s. The velocity of the body is 6 m/s at $t =$ _____ seconds.

(A) 0.1823

(B) 0.3979

(C) 0.9163

(D) 1.609

For complete solution, go to

http://nm.mathforcollege.com/mcquizzes/03nle/quiz_03nle_background_solution.pdf

# Problem Set

Chapter 03.01: Prerequisites to Numerical Methods for Solving Nonlinear Equations

(1). Find the roots of the equation $2x^2 + 5x = -3$.

**Answer:** $-1.5,\ -1$

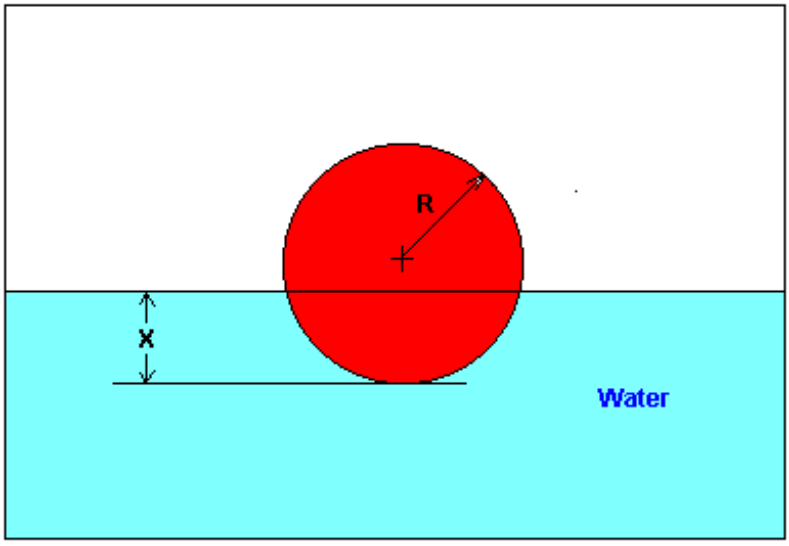
(2). Find the roots of the equation $2x^2 + 4x + 4 = 0$.

**Answer:** $-1 + i,\ -1 - i$

(3). Using any method, estimate the smallest non-negative root of the equation $\tan(x) = x$. Plot the function, $\tan(x)$ and $x$ to help you to find an approximate answer.

**Answer:** $x = 0$

(4). You are working for 'DOWN THE TOILET COMPANY' that makes floats for ABC commodes. The ball has a specific gravity of 0.6 and has a radius of 5.5 cm. You are asked to find the depth to which the ball will get submerged when floating in water.




The equation that gives the depth $x$ (unit of $x$ is m) to which the ball is submerged under water is given by

$$x^3 - 0.165x^2 + 3.993 \times 10^{-4} = 0$$

Note that the cubic equation will have three roots. Can some of these roots be complex. If so how many complex roots can it have? If more than one root of the above equation is real, how do you choose the acceptable root? Or are all the real roots physically acceptable?

**Answer:** The roots are $0.062378,\ 0.14636,\ -0.043737$. Only $x = 0.062378$ is acceptable as the root needs to lie between 0 and the diameter, $D = 0.11m$.

(5). A ball is thrown down at 50 mph from the top of a building. The building is 420 feet tall. The time in seconds the ball would take to reach the ground is given by

$$16.1t^2 + 73.33t - 420 = 0,\ t \geq 0$$

a) How many roots does the quadratic equation have?

b) Which of the two roots is a valid answer?

c) How much time would the ball take to reach the ground if it was just let go rather than thrown?

d) If the quadratic equation gave you complex roots, what would be your conclusion?

**Answer:** *a*) 2 *b*) 3.3149 *s* *c*) 5.1075 *s* *d*) Modeling of eqn must be wrong.

(6). A straight steel ruler that is $12''$ long at the initial temperature of 80°F contracts in length when dipped in a cold liquid. It is allowed to reach steady state. If the reduction in the length at steady state is found to be $0.001''$, find the nonlinear equation (you do not have to solve it) that will allow you to find the temperature of the liquid. The reduction in the length, $\Delta L$ is given by

$$\Delta L = L \int_{T_{\text{initial}}}^{T_{\text{liquid}}} \alpha dT$$

where

$$L = \text{initial length of the ruler,}$$

$$\alpha = -1.2 \times 10^{-11} T^2 + 6.2 \times 10^{-9} T + 6 \times 10^{-6}$$

$$(\text{units of } \alpha \text{ are } in/in/^\circ F, \text{ and units of temperature are } ^\circ F)$$

$$T_{\text{liquid}} = \text{temperature of the liquid,}$$

$$T_{\text{initial}} = \text{initial temperature of the steel ruler.}$$

**Answer:** $-4.8 \times 10^{-11}(T_{liquid})^3 + 3.72 \times 10^{-8}(T_{liquid})^2 + 7.2 \times 10^{-5}(T_{liquid}) - 4.9735 \times 10^{-3} = 0$

# Chapter 03.03: Bisection Method for Solving a Nonlinear Equation

Lesson: Background of Bisection Method

## Learning Objectives

*After successful completion of this lesson, you should be able to*

1) *articulate the background to the bisection method*

## Introduction

One of the first numerical methods developed to find the root of a nonlinear equation $f(x) = 0$ was the bisection method (also called the *binary-search* method). The procedure is based on the following theorem.

## Theorem

An equation$f(x) = 0$, where $f(x)$ is a real continuous function, has at least one root between $x_l$ and $x_u$ if $f(x_l)f(x_u) < 0$ (See Figure 1).

Note that if $f(x_l)f(x_u) > 0$, there may or may not be any root between $x_l$ and $x_u$ (Figures 2 and 3). If $f(x_l)f(x_u) < 0$, then there may be more than one root between $x_l$ and $x_u$ (Figure 4). So the theorem only guarantees one root between $x_l$ and $x_u$.

Since the method is based on finding the root between two points, the technique falls under bracketing methods.

Since the root is bracketed between two points, $x_l$ and $x_u$, one can find the mid-point, $x_m$ between $x_l$ and $x_u$. We hence get two new intervals $x_l$ and $x_m$, and $x_m$ and $x_u$.

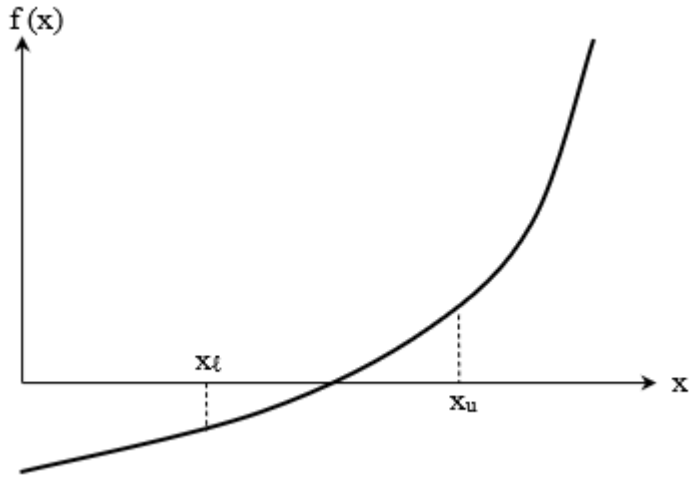



**Figure 1** At least one root exists between the two points if the function is real, continuous, and changes sign.

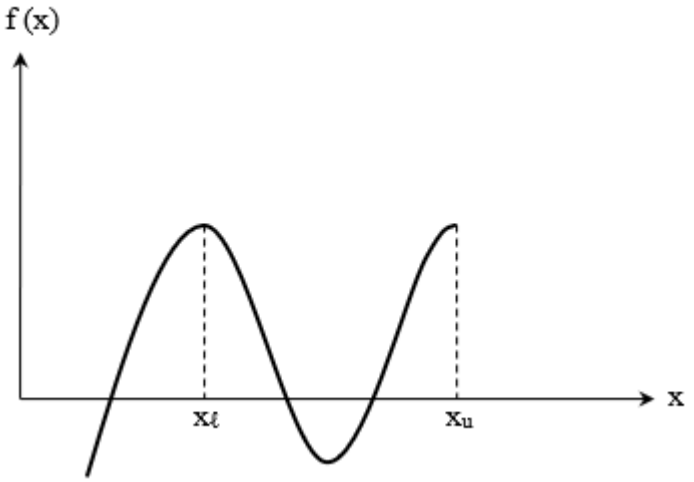



**Figure 2** If the function $f(x)$ does not change the sign between the two points, roots of the equation $f(x) = 0$ may still exist between the two points.

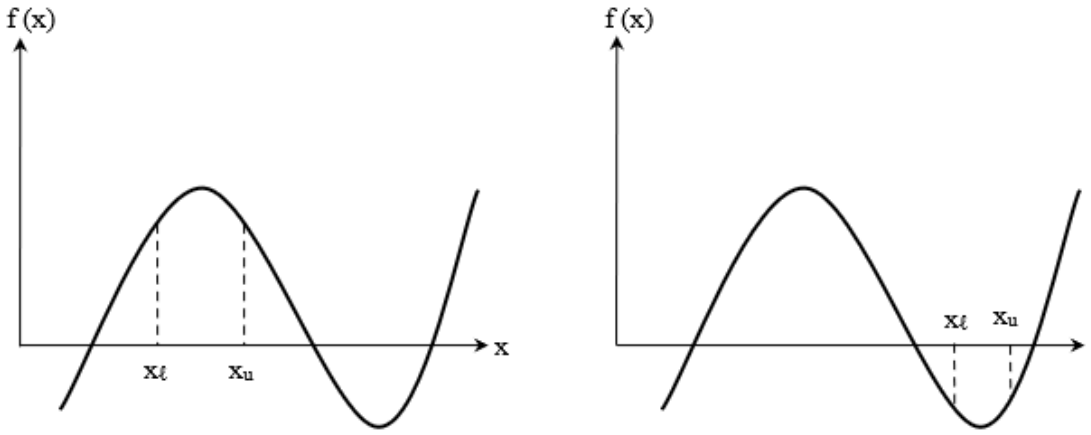



**Figure 3** If the function $f(x)$ does not change the sign between two points, there may not be any roots for the equation $f(x) = 0$ between the two points.

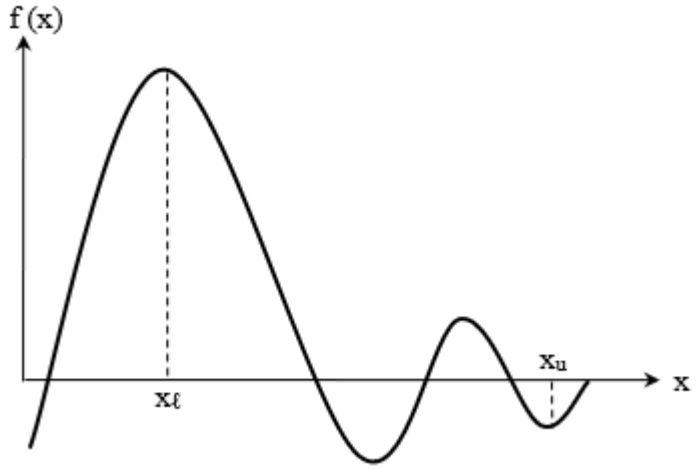



**Figure 4** If the function $f(x)$ changes the sign between the two points, more than one root for the equation $f(x) = 0$ may exist between the two points.

Is the root now between $x_l$ and $x_m$ or between $x_m$ and $x_u$? Well, one can find the sign of $f(x_l)f(x_m)$, and if $f(x_l)f(x_m) < 0$ then the new bracket is between $x_l$ and $x_m$, otherwise, it is between $x_m$ and $x_u$. So, you can see that you are halving the interval. As one repeats this process, the width of the interval $[x_l, x_u]$ becomes smaller and smaller, and you can zero on to the root of the equation $f(x) = 0$.

# Chapter 03.03: Bisection Method for Solving a Nonlinear Equation

Lesson: Bisection Method Algorithm

## Learning Objectives

*After successful completion of this lesson, you should be able to*

1) *write the algorithm for the bisection method of solving a nonlinear equation.*

## What is the bisection method, and what is it based on?

One of the first numerical methods developed to find the root of a nonlinear equation $f(x) = 0$ was the bisection method (also called the *binary-search* method). The procedure is based on the following theorem.

## Theorem

An equation$f(x) = 0$, where $f(x)$ is a real continuous function, has at least one root between $x_l$ and $x_u$ if $f(x_l)f(x_u) < 0$ (See Figure 1).

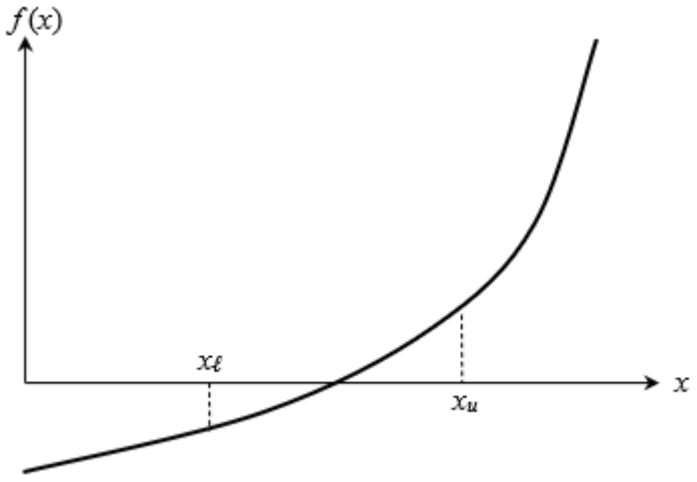


**Figure 1** At least one root exists between the two points if the function is real, continuous, and changes sign.

## Bisection method

Since the method is based on finding the root between two points, the technique falls under the category of bracketing methods.

Since the root is bracketed between two points, $x_l$ and $x_u$, one can find the mid-point, $x_m$ between $x_l$ and $x_u$. This gives us two new intervals $x_l$ and $x_m$, and $x_m$ and $x_u$.

Is the root now between $x_l$ and $x_m$ or between $x_m$ and $x_u$? Well, one can find the sign of $f(x_l)f(x_m)$, and if $f(x_l)f(x_m) < 0$ then the new bracket is between $x_l$ and $x_m$, otherwise, it is between $x_m$ and $x_u$. So, you can see that you are literally halving the interval. As one repeats this process, the width of the interval $[x_l, x_u]$ becomes smaller and smaller, and you can zero into the root of the equation $f(x) = 0$. The algorithm for the bisection method is given as follows.

## Algorithm for the bisection method

The steps to apply the bisection method to find the root of the equation $f(x) = 0$ are

Choose $x_l$ and $x_u$ as two guesses for the root such that$f(x_l)f(x_u) < 0$, or in other words, $f(x)$ changes sign between $x_l$ and $x_u$.

Estimate the root $x_m$ of the equation $f(x) = 0$ as the mid-point between $x_l$ and $x_u$ as

$$x_m = \frac{x_l + x_u}{2}$$

**Now check the following**

If $f(x_l)f(x_m) < 0$, then the root lies between $x_l$ and $x_m$; then $x_l = x_l$ and $x_u = x_m$.

If $f(x_l)f(x_m) > 0$, then the root lies between $x_m$ and $x_u$; then $x_l = x_m$ and $x_u = x_u$.

If $f(x_l)f(x_m) = 0$; then the root is $x_m$. Stop the algorithm if this is true.

Find the new estimate of the root

$$x_m = \frac{x_l + x_u}{2}$$

Find the absolute relative approximate error as

$$|\in_a| = \left| \frac{x_m^{new} - x_m^{old}}{x_m^{new}} \right| \times 100$$

where

$$x_m^{new} = \text{estimated root from the present iteration}$$

$x_m^{old}$ = estimated root from the previous iteration

Compare the absolute relative approximate error $|\in_a|$ with the pre-specified relative error tolerance $\in_s$. If $|\in_a| > \in_s$, then go to Step 3, else stop the algorithm. Note one should also check whether the number of iterations is more than the maximum number of iterations allowed. If so, one needs to terminate the algorithm and notify the user about it.

# Chapter 03.03: Bisection Method for Solving a Nonlinear Equation

Lesson: Application of Bisection Method

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *apply the bisection method to solve for roots of a nonlinear equation.*

## Applications

In the previous lesson, you learned the theory of the bisection method of solving a nonlinear equation. In this lesson, we apply the algorithm of the bisection method to solve a nonlinear equation.

### Example 1

Use the bisection method to find the root of the nonlinear equation

$$x^3 = 20$$

Use initial lower and upper guesses of 1 and 4, respectively.

Conduct three iterations to estimate the root of the equation.

Find the absolute relative approximate error at the end of each iteration.

Find the number of significant digits that are at least correct at the end of each iteration.

**Solution**

Rewrite the equation $x^3 = 20$ in the form $f(x) = 0$ that gives

$$f(x) = x^3 - 20 = 0$$

Check if the function changes the sign between the two initial guesses, $x_l$ and $x_u$. The initial guesses are given as $x_l = 1$ and $x_u = 4$

$$\begin{aligned} f(x_l) &= f(1) \\ &= 1^3 - 20 \\ &= -19 \end{aligned}$$

$$\begin{aligned} f(x_u) &= f(4) \\ &= 4^3 - 20 \\ &= 44 \end{aligned}$$

Hence

$$\begin{aligned} f(x_l)f(x_u) &= f(1)f(4) \\ &= (-19)(44) < 0 \end{aligned}$$

This change in sign tells us that the initial bracket of $[1, 4]$ given to us is valid.

Iteration 1

$$x_l = 1,\ x_u = 4$$

The estimate of the root is

$$\begin{aligned} x_m &= \frac{x_l + x_u}{2} \\ &= \frac{1+4}{2} \\ &= 2.5 \end{aligned}$$

Iteration 2

Find the value of the function at the midpoint from the previous iteration and use it to determine the new bracket.

$$\begin{aligned} f(x_m) &= f(2.5) \\ &= (2.5)^3 - 20 \\ &= -4.375 \end{aligned}$$

$$\begin{aligned} f(x_l)f(x_m) &= f(1)f(2.5) \\ &= (-19)(-4.375) > 0 \end{aligned}$$

Since $f(x_l)f(x_m) > 0$, the root does not lie between $x_l$ and $x_m$, but between $x_m$ and $x_u$, that is, 2.5 and 4.

$$x_l = 2.5,\ x_u = 4$$

The estimate of the root is

$$\begin{aligned} x_m &= \frac{x_l + x_u}{2} \\ &= \frac{2.5 + 4}{2} \\ &= 3.25 \end{aligned}$$

The absolute relative approximate error $|\varepsilon_a|$ at the end of Iteration 2 is

$$\begin{aligned} |\varepsilon_a| &= \left| \frac{x_m^{new} - x_m^{old}}{x_m^{new}} \right| \times 100 \\ &= \left| \frac{3.25 - 2.5}{3.25} \right| \times 100 \\ &= 23.1\% \end{aligned}$$

None of the significant digits are at least correct in the estimated root of the equation because the absolute relative approximate error is greater than 5%.

Iteration 3

Find the value of the function at the midpoint from the previous iteration and use it to determine the new bracket.

$$\begin{aligned} f(x_m) &= f(3.25) \\ &= (3.25)^3 - 20 \\ &= 14.3281 \end{aligned}$$

$$\begin{aligned} f(x_l)f(x_m) &= f(2.5)f(3.25) \\ &= (-4.375)(14.3281) < 0 \end{aligned}$$

Since $f(x_l)f(x_m) < 0$, the root does lie between $x_l$ and $x_m$, that is, 2.5 and 3.25.

$$x_l = 2.5, x_u = 3.25$$

The estimate of the root is

$$\begin{aligned} x_m &= \frac{x_l + x_u}{2} \\ &= \frac{2.5 + 3.25}{2} \\ &= 2.875 \end{aligned}$$

The absolute relative approximate error $|\varepsilon_a|$ at the end of Iteration 3 is

$$\begin{aligned}|\epsilon_a| &= \left|\frac{x_m^{new} - x_m^{old}}{x_m^{new}}\right| \times 100 \\ &= \left|\frac{2.875 - 2.5}{2.875}\right| \times 100 \\ &= 13.0\%\end{aligned}$$

Still, none of the significant digits are at least correct in the estimated root of the equation as the absolute relative approximate error is greater than 5%.

**Example 2**

You are working for 'DOWN THE TOILET COMPANY' that makes floats for ABC commodes. The floating ball has a specific gravity of 0.6 and has a radius of 5.5 cm. You are asked to find the depth to which the ball is submerged when floating in the water.

The equation that gives the depth $x$ to which the ball is submerged underwater is given by

$$x^3 - 0.165x^2 + 3.993 \times 10^{-4} = 0$$

Use the bisection method of finding roots of equations to find the
depth $x$ to which the ball is submerged underwater. Conduct three iterations to estimate the root of the above equation. Find the
absolute relative approximate error at the end of each iteration,
and the number of significant digits at least correct at the end of each iteration.

**Solution**

From the physics of the problem, the ball would be submerged somewhere between $x = 0$ and $x = 2R$, where

$$R = \text{radius of the ball,}$$

that is

$$0 \leq x \leq 2R$$

$$0 \leq x \leq 2(0.055)$$

$$0 \leq x \leq 0.11$$

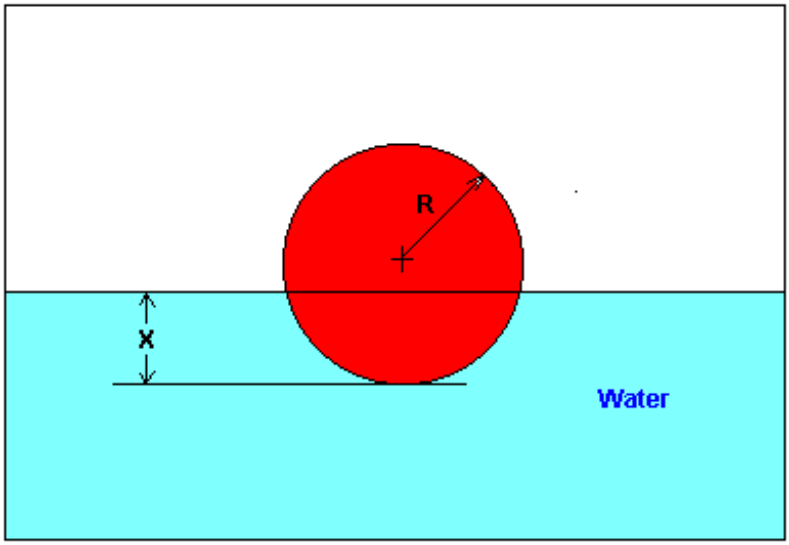



**Figure 5** Floating ball problem.

Let's us assume

$$x_l = 0,\ x_u = 0.11$$

Check if the function changes sign between $x_l$ and $x_u$.

$$\begin{aligned} f(x_l) &= f(0) \\ &= (0)^3 - 0.165(0)^2 + 3.993 \times 10^{-4} \\ &= 3.993 \times 10^{-4} \end{aligned}$$

$$\begin{aligned} f(x_u) &= f(0.11) \\ &= (0.11)^3 - 0.165(0.11)^2 + 3.993 \times 10^{-4} \\ &= -2.662 \times 10^{-4} \end{aligned}$$

Hence

$$\begin{aligned} f(x_l)f(x_u) &= f(0)f(0.11) \\ &= (3.993 \times 10^{-4})(-2.662 \times 10^{-4}) \\ &< 0 \end{aligned}$$

So, there is at least one root between $x_l$ and $x_u$, that is between 0 and 0.11.

**Iteration 1**

The estimate of the root is

$$\begin{aligned} x_m &= \frac{x_l + x_u}{2} \\ &= \frac{0 + 0.11}{2} \\ &= 0.055 \end{aligned}$$

$$\begin{aligned} f\left(x_m\right) &= f(0.055) \\ &= (0.055)^3 - 0.165(0.055)^2 + 3.993 \times 10^{-4} \\ &= 6.655 \times 10^{-5} \end{aligned}$$

$$\begin{aligned} f(x_l) f(x_m) &= f(0) f(0.055) \\ &= \left(3.993 \times 10^{-4}\right)\left(6.655 \times 10^{-4}\right) \\ &> 0 \end{aligned}$$

Hence the root is bracketed between $x_m$ and $x_u$, that is, between 0.055 and 0.11. So, the lower and upper limit of the new bracket is

$$x_l = 0.055,\ x_u = 0.11$$

At this point, the absolute relative approximate error $|\epsilon_a|$ cannot be calculated as we do not have a previous approximation.

**Iteration 2**

The estimate of the root is

$$\begin{aligned} x_m &= \frac{x_l + x_u}{2} \\ &= \frac{0.055 + 0.11}{2} \\ &= 0.0825 \end{aligned}$$

$$\begin{aligned} f(x_m) &= f(0.0825) \\ &= (0.0825)^3 - 0.165(0.0825)^2 + 3.993 \times 10^{-4} \\ &= -1.622 \times 10^{-4} \end{aligned}$$

$$\begin{aligned} f\left(x_l\right) f\left(x_m\right) &= f\left(0.055\right) f\left(0.0825\right) \\ &= \left(6.655 \times 10^{-5}\right) \times \left(-1.622 \times 10^{-4}\right) \\ &< 0 \end{aligned}$$

Hence, the root is bracketed between $x_l$ and $x_m$, that is, between 0.055 and 0.0825. So, the lower and upper limit of the new bracket is

$$x_l = 0.055,\ x_u = 0.0825$$

The absolute relative approximate error $|\epsilon_a|$ at the end of Iteration 2 is

$$\begin{aligned}
|\epsilon_a| &= \left|\frac{x_m^{\text{new}} - x_m^{\text{old}}}{x_m^{\text{new}}}\right| \times 100 \\
&= \left|\frac{0.0825 - 0.055}{0.0825}\right| \times 100 \\
&= 33.33\%
\end{aligned}$$

None of the significant digits are at least correct in the estimated root of $x_m = 0.0825$ because the absolute relative approximate error is greater than 5%.

**Iteration 3**

$$\begin{aligned}
x_m &= \frac{x_l + x_u}{2} \\
&= \frac{0.055 + 0.0825}{2} \\
&= 0.06875
\end{aligned}$$

$$\begin{aligned}
f\left(x_m\right) &= f(0.06875) \\
&= (0.06875)^3 - 0.165(0.06875)^2 + 3.993 \times 10^{-4} \\
&= -5.563 \times 10^{-5}
\end{aligned}$$

$$\begin{aligned}
f(x_l)f(x_m) &= f(0.055)f(0.06875) \\
&= (6.655 \times 10^5) \times (-5.563 \times 10^{-5}) \\
&< 0
\end{aligned}$$

Hence, the root is bracketed between $x_l$ and $x_m$, that is, between 0.055 and 0.06875. So the lower and upper limit of the new bracket is

$$x_l = 0.055,\ x_u = 0.06875$$

The absolute relative approximate error $|\epsilon_a|$ at the ends of Iteration 3 is

$$\begin{aligned}
|\epsilon_a| &= \left|\frac{x_m^{\text{new}} - x_m^{\text{old}}}{x_m^{\text{new}}}\right| \times 100 \\
&= \left|\frac{0.06875 - 0.0825}{0.06875}\right| \times 100 \\
&= 20\%
\end{aligned}$$

Still none of the significant digits are at least correct in the estimated root of the equation as the absolute relative approximate error is greater than 5%.

Seven more iterations were conducted, and these iterations are shown in Table 1.

**Table 1** Root of $f(x) = 0$ as function of number of iterations for bisection method.

| Iteration | $x_l$ | $x_u$ | $x_m$ | $\lvert\epsilon_a\rvert\,\%$ | $f(x_m)$ |
|---|---|---|---|---|---|
| 1 | 0.00000 | 0.11 | 0.055 | $---$ | $6.655 \times 10^{-5}$ |
| 2 | 0.055 | 0.11 | 0.0825 | 33.33 | $-1.622 \times 10^{-4}$ |
| 3 | 0.055 | 0.0825 | 0.06875 | 20.00 | $-5.563 \times 10^{-5}$ |
| 4 | 0.055 | 0.06875 | 0.06188 | 11.11 | $4.484 \times 10^{-6}$ |
| 5 | 0.06188 | 0.06875 | 0.06531 | 5.263 | $-2.593 \times 10^{-5}$ |
| 6 | 0.06188 | 0.06531 | 0.06359 | 2.702 | $-1.0804 \times 10^{-5}$ |
| 7 | 0.06188 | 0.06359 | 0.06273 | 1.370 | $-3.176 \times 10^{-6}$ |
| 8 | 0.06188 | 0.06273 | 0.0623 | 0.6897 | $6.497 \times 10^{-7}$ |
| 9 | 0.0623 | 0.06273 | 0.06252 | 0.3436 | $-1.265 \times 10^{-6}$ |
| 10 | 0.0623 | 0.06252 | 0.06241 | 0.1721 | $-3.0768 \times 10^{-7}$ |

At the end of $10^{\text{th}}$ iteration,

$$|\epsilon_a| = 0.1721\%$$

Hence the number of significant digits at least correct is given by the largest value of $m$ for which

$$|\epsilon_a| \leq 0.5 \times 10^{2-m}$$

$$0.1721 \leq 0.5 \times 10^{2-m}$$

$$0.3442 \leq 10^{2-m}$$

$$\log(0.3442) \leq 2 - m$$

$$m \leq 2 - \log(0.3442) = 2.463$$

So

$$m = 2$$

The number of significant digits at least correct in the estimated root of 0.06241 at the end of the $10^{\text{th}}$ iteration is 2.

# Chapter 03.03: Bisection Method for Solving a Nonlinear Equation

Lesson: Advantages and Pitfalls of Bisection Method

## Learning Objectives

*After successful completion of this lesson, you should be able to*

1) *enumerate the advantages of the bisection method and the reasoning behind them.*

2) *list the drawbacks of the bisection method and the reason behind them.*

## Advantages of the bisection method

a) The bisection method is always convergent. Since the technique brackets the root, the procedure is guaranteed to converge.

b) As iterations are conducted, the interval gets halved. So, one can guarantee the error in the solution of the equation.

## Drawbacks of bisection method

a) The convergence of the bisection method is slow as it is based on halving the interval.

b) If one of the initial guesses is closer to the root, it will take a larger number of iterations to reach the root.

c) If a function $f(x)$ is such that it just touches the x-axis (Figure 1) such as

$$f(x) = x^2 = 0$$

it will be unable to find the lower guess, $x_l$, and upper guess, $x_u$, such that

$$f(x_l) f(x_u) < 0$$

d) For functions $f(x)$ where there is a singularity, and it reverses the sign at the singularity, the bisection method may converge on the singularity (Figure 2). An example includes

$$f(x) = \frac{1}{x}$$

where $x_l = -2$, $x_u = 3$ are valid initial guesses which satisfy

$$f(x_l)f(x_u) < 0$$

However, the function is not continuous, and the theorem that a root exists is also not applicable.

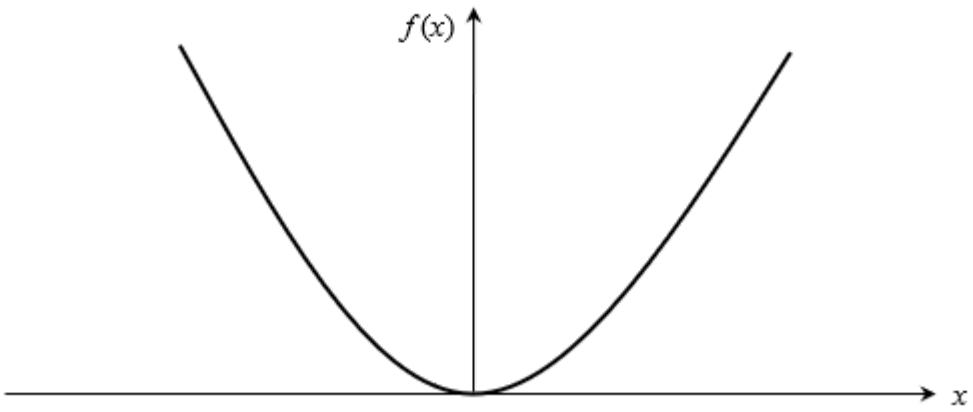



**Figure 1** The equation $f(x) = x^2 = 0$ has a single root at $x = 0$ that cannot be bracketed.

A singularity in a function is defined as a point where the function becomes infinite. For example, for a function such as $1/x$, the point of singularity is $x = 0$ as it becomes infinite.

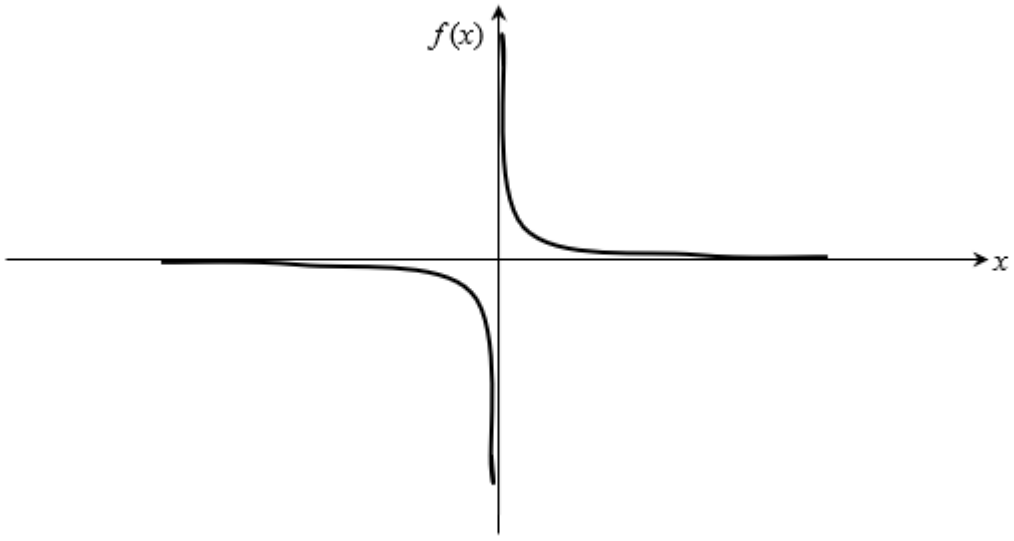



**Figure 2** The equation $f(x) = \dfrac{1}{x} = 0$ has no root but changes sign.

# Multiple Choice Test

## Chapter 03.03 Bisection Method for Solving a Nonlinear Equation

(1). The bisection method of finding roots of nonlinear equations falls under the category of a (an) _____ method.

(A) open

(B) bracketing

(C) random

(D) graphical

(2). If for a real continuous function $f(x)$, $f(a)f(b) < 0$, then in the domain of $[a, b]$ for $f(x) = 0$, there is (are)

(A) one root

(B) an undeterminable number of roots

(C) no root

(D) at least one root

(3). Assuming an initial bracket of $[1, 5]$, the second (at the end of 2 iterations) iterative value of the root of $te^{-t} - 0.3 = 0$ using the bisection method is

(A) 0

(B) 1.5

(C) 2

(D) 3

(4). To find the root of $f(x) = 0$, a scientist is using the bisection method. At the beginning of an iteration, the lower and upper guesses of the root are $x_l$ and $x_u$. At the end of the iteration, the absolute relative approximate error in the estimated value of the root would be

(A) $\left|\dfrac{x_u}{x_u + x_l}\right|$

(B) $\left|\dfrac{x_l}{x_u + x_l}\right|$

(C) $\left|\dfrac{x_u - x_l}{x_u + x_l}\right|$

(D) $\left|\dfrac{x_u + x_l}{x_u - x_l}\right|$

(5). For an equation like $x^2 = 0$, a root exists at $x = 0$. The bisection method cannot be adopted to solve this equation in spite of the root existing at $x = 0$ because the function $f(x) = x^2$

(A) is a polynomial

(B) has repeated zeros at $x = 0$

(C) is always non-negative

(D) has a slope equal to zero at $x = 0$

(6). The ideal gas law is given by

$$pv = RT$$

where $p$ is the pressure, $v$ is the specific volume, $R$ is the universal gas constant, and $T$ is the absolute temperature. This equation is only accurate for a limited range of pressure and temperature. Vander Waals came up with an equation that was accurate for larger ranges of pressure and temperature given by

$$\left(p + \frac{a}{v^2}\right)(v - b) = RT$$

where $a$ and $b$ are empirical constants dependent on a particular gas. Given the value of $R = 0.08$, $a = 3.592$, $b = 0.04267$, $p = 10$ and $T = 300$ (assume all units are consistent), one is going to find the specific volume, $v$, for the above values. Without finding the solution from the Vander Waals equation, what would be a good initial guess for $v$?

(A) 0

(B) 1.2

(C) 2.4

(D) 3.6

For complete solution, go to

http://nm.mathforcollege.com/mcquizzes/03nle/quiz_03nle_bisection_solution.pdf

# Problem Set

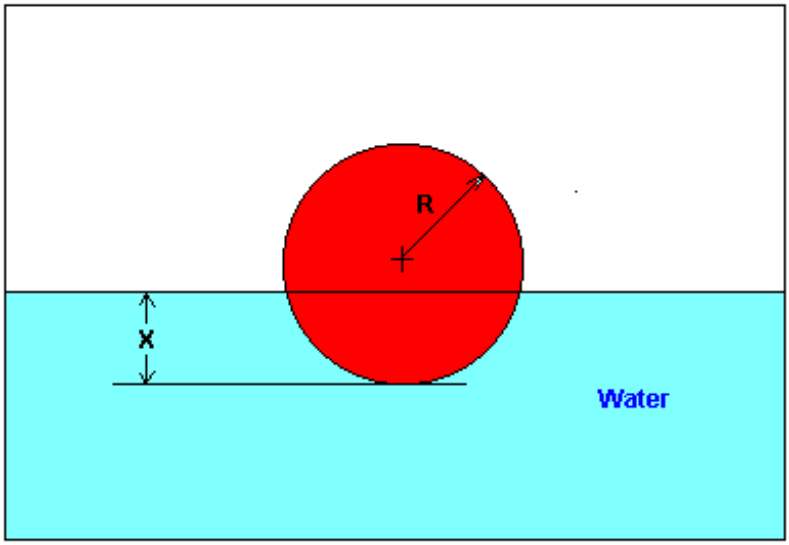
Chapter 03.03: Bisection Method for Solving a Nonlinear Equations

(1). Find the estimate of the root of $x^2 - 4 = 0$ by using the bisection method. Use initial guesses of 1.7 and 2.4. Conduct three iterations, and calculate the approximate error, true error, absolute relative approximate error, and absolute relative true error at the end of each iteration.

**Answer:**

| Iteration # | Root Estimate | Approx Error | True Error | Abs Rel Approx Error | Abs Rel True Error |
|---|---|---|---|---|---|
| 1 | 2.05 | − − − | −0.05 | − − − | 2.5% |
| 2 | 1.875 | −0.175 | 0.125 | 9.33% | 6.25% |
| 3 | 1.9625 | 0.0875 | 0.0375 | 4.45% | 1.875% |

(2). You are working for DOWN THE TOILET COMPANY that makes floats for ABC commodes. The ball has a specific gravity of 0.6 and has a radius of 5.5 cm. You are asked to find the depth to which the ball will get submerged when floating in the water.




The equation that gives the depth $x$ (unit of $x$ is m) to which the ball is submerged under water is given by

$$x^3 - 0.165x^2 + 3.993 \times 10^{-4} = 0$$

Solving it exactly would require some effort. However, using numerical techniques such as the bisection method, we can solve this equation and any other equation of the form $f(x) = 0$. Solve the above equation by the bisection method and do the following. Use initial guesses as $x = 0$ and $x = 0.11$.

a) Conduct three iterations.

b) Calculate the absolute relative approximate error at the end of each step.

c) Find the number of significant digits at least correct at the end of each iteration.

d) How can you use the knowledge of the physics of the problem to develop initial guesses for the bisection method?

**Answer:** *a*) $0.055, 0.0825, 0.06875$

*b*) $n/a, 33.333\%, 20.00\%$

*c*) $n/a, 0, 0$

*d*) Since the diameter is 0.11 m, initial guesses of 0 and 0.11 are reasonable starting values

(3). The velocity of a body is given by the following equation

$$v(t) = te^{-t} + \frac{1}{t}$$

where

$t$ is given in seconds and $v$ is given in m/s.

Find the time when the velocity of the body will be 0.35 m/s. Use bisection method and conduct three iterations. Use initial bracketing guess of $[1, 8]$. Show all steps in calculating the estimated root, absolute relative approximate error and the velocity of the body for each iteration. Also, tabulate your answers as iteration number, estimated root, absolute relative approximate error, and velocity of the body.

**Answer:**

| Iteration | Estimated Root | $\lvert\epsilon_a\rvert\%$ | $v(t)$ (m/s) |
|---|---|---|---|
| 1 | 4.500 | – | 0.27221 |
| 2 | 2.750 | 63.663 | 0.53944 |
| 3 | 3.6250 | 24.138 | 0.37247 |

(4). Enumerate the drawbacks of the bisection method of solving nonlinear equations.

**Answer:** Refer to the textbook

(5). The velocity of a body is given by

$$v = 5e^{-t} + 4 + \sin(t),$$

where

$v$ is given in m/s,

$t$ is given in s.

Derive the nonlinear equation that you will need to solve to find when the *acceleration* of the body would be 1.54 m/s$^2$. Find the solution of the equation by using three iterations of the bisection method.

**Answer:** The equation has no real roots. Note the question asks when acceleration is 1.54 m/s$^2$

(6). To solve the equation $f(x) = 0$, an engineer is using the bisection method. He starts with an initial valid bracket of $[2, 7]$. Find the maximum possible absolute true error in his estimate of the root at the end of two iterations? Show your reasoning clearly for your answer.

**Answer:** 1.25

(7). Estimate the next guess for the root of $x^2 - 16 = 0$ by using a modified bisection method as explained below. The initial bracket of $[1, 8]$ is found as a valid bracket. In the case of bisection method, the root estimated at the end of the first iteration is the midpoint between 1 and 8. Instead, in the modified bisection method, the root estimated at the end of the first iteration would be the point where the straight line drawn from the function at $x = 1$ to the function at $x = 8$ crosses the $x$ -axis. What is this estimate of the root?

**Answer:** 2.667

# Chapter 03.04: Newton-Raphson Method for Solving a Nonlinear Equation

Lesson: Tangent to a Curve

## Learning Objectives

*After successful completion of this lesson, you should be able to*

1) *find the first derivative of a function,*
2) *calculate the value of the first derivative of the function at a particular point,*
3) *relate the first derivative of the function at a specific point to the slope of the tangent line, and,*
4) *find where the tangent line crosses the x-axis.*

## Introduction

The formula for solving a nonlinear equation of the form $f(x) = 0$ by the Newton-Raphson method is given by

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$$

Some of the pre-requisites to learning the Newton-Raphson method are

a) finding the first derivative of a function,

b) calculating the value of the first derivative of the function at a particular point,

c) relating the first derivative of the function at a specific point to the slope of the tangent line, and

d) finding where the tangent line crosses the x-axis.

Let us illustrate the above concepts through an example.

### Example

For $f(x) = 0.13x^3$, find the following

a) Find $f'(x)$.

b) Find $f'(7)$.

c) Find the slope of the tangent line to the function at $x = 7$.

d) Find the angle the tangent line makes to the $x$-axis at $x = 7$. Write your answer both in degrees and radians.

e) Find where the tangent line to the function at $x = 7$ crosses the $x$-axis.

**Solution**

a)

$$f(x) = 0.13x^3$$

Using

$$\frac{d}{dx}(x^n) = nx^{n-1}, n \neq 0$$

and

$$\frac{d}{dx}(aq(x)) = a\frac{dq}{dx}$$

where $a$ is a constant, we get

$$\begin{aligned} f'(x) &= \frac{d}{dx}(0.13x^3) \\ &= 0.13\frac{d}{dx}(x^3) \\ &= 0.13(3)x^2 \\ &= 0.39x^2 \end{aligned}$$

b) From part (a)

$$f'(x) = 0.39x^2$$

$$\begin{aligned} f'(7) &= 0.39(7)^2 \\ &= 19.11 \end{aligned}$$

c) From part (b)

$$f'(7) = 19.11$$

The slope of the tangent line to the function at $x = 7$ is $f'(7)$ and that is 19.11.

d) From part (c), the slope of the tangent to the function at $x = 7$ is 19.11.

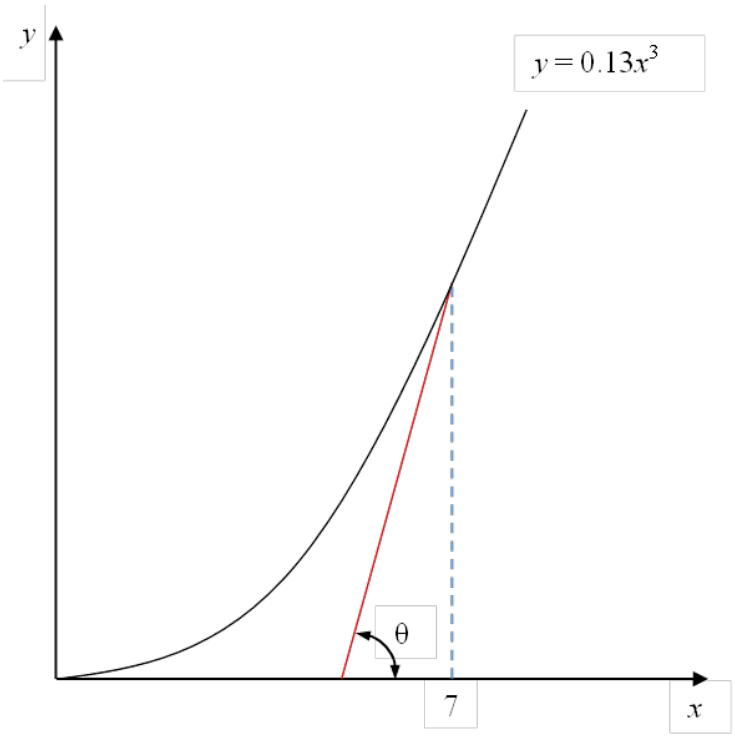



**Figure 1:** Tangent line crossing the x-axis

Hence

$$\tan\theta = 19.11$$

where

$$\theta = \text{angle made by the tangent line to the } x\text{-axis.}$$

Hence

$$\begin{aligned}\theta &= \tan^{-1}(19.11) \\ &= 87.004^\circ \text{ or } 1.5185 \text{ radians}\end{aligned}$$

e) The equation of a straight line is

$$y = mx + c$$

Since we know the slope of the tangent line, that is, 19.11, and we know one of the points on the tangent line is $(7, f(7)) = (7, 0.13 \times 7^3) = (7, 44.59)$, we can find the intercept of the straight line

$$44.59 = 19.11(7) + c$$

$$c = -89.18$$

Hence the equation of the tangent line is

$$y = 19.11x - 89.18$$

The tangent line crosses the $x$-axis when $y = 0$, that is,

$$0 = 19.11x - 89.18$$

$$\begin{aligned} x &= \frac{89.18}{19.11} \\ &= 4.667 \end{aligned}$$

# Chapter 03.04: Newton-Raphson Method for Solving a Nonlinear Equation

Lesson: Graphical Derivation of Newton-Raphson Method

## Learning Objectives

*After successful completion of this lesson, you should be able to*

1) *derive the Newton-Raphson method to solve a nonlinear equation*

2) *write the algorithm for the Newton-Raphson method to solve a nonlinear equation*

## Introduction

Methods such as the bisection method and the false position method of finding roots of a nonlinear equation $f(x) = 0$ require bracketing of the root by two guesses. Such methods are called *bracketing methods*. These methods are always convergent since they are based on reducing the interval between the two guesses so as to zero in on the root of the equation.

In the Newton-Raphson method, the root is not bracketed. In fact, only one initial guess of the root is needed to get the iterative process started to find the root of an equation. The method hence falls in the category of *open methods*. Convergence in open methods is not guaranteed, but it does so much faster than the bracketing methods if the method does converge.

## Derivation

The Newton-Raphson method is based on the principle that if the initial guess of the root of $f(x) = 0$ is at $x_i$, then if one draws the tangent to the curve at $(x_i, f(x_i)$, the point $x_{i+1}$ where the tangent crosses the $x$ -axis is an improved estimate of the root (Figure 1). One can then use $x_{i+1}$as the next point to draw the tangent line to the function $f(x)$ and find out where that tangent line crosses the $x$ -axis. Continuing this process brings us closer and closer to the root of the equation.

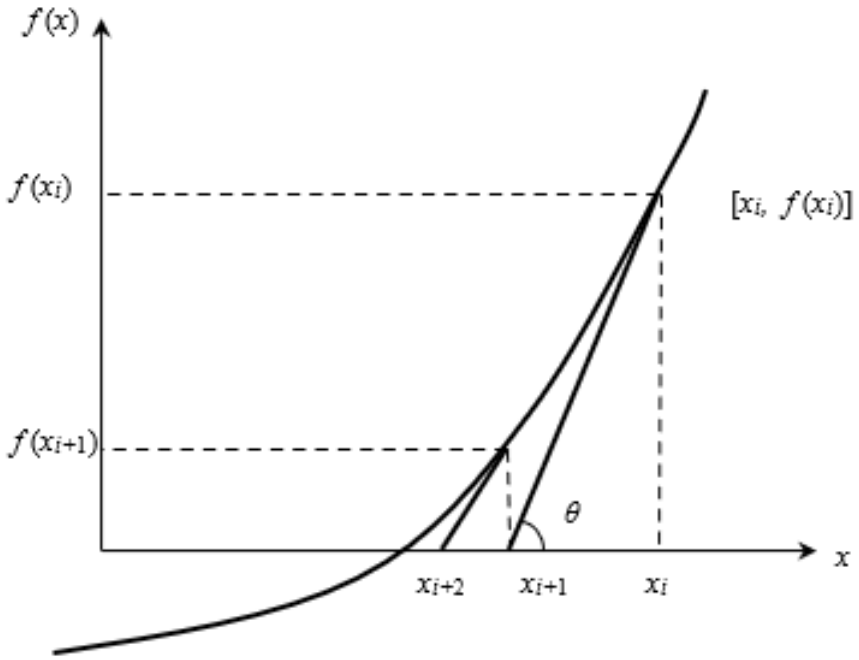



**Figure 1** Geometrical illustration of the Newton-Raphson method.

Using the definition of the slope of a function, at $x = x_i$

$$\begin{aligned} f'(x_i) &= \tan\theta \\ &= \frac{f(x_i) - 0}{x_i - x_{i+1}}, \end{aligned}$$

which gives

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)} \qquad (1)$$

Equation (1) is called the Newton-Raphson formula for solving nonlinear equations of the form $f(x) = 0$. So starting with an initial guess, $x_i$, one can find the next guess, $x_{i+1}$ by using Equation (1). One can repeat this process until one finds the root within a desirable tolerance.

## Algorithm

The steps of the Newton-Raphson method to find the root of an equation $f(x) = 0$ are

1) Evaluate $f'(x)$ symbolically

2) Use an initial guess of the root, $x_i$, to estimate the new value of the root, $x_{i+1}$, as

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$$

3) Find the absolute relative approximate error $|\epsilon_a|$ as

$$|\epsilon_a| = \left| \frac{x_{i+1} - x_i}{x_{i+1}} \right| \times 100 \qquad (2)$$

4) Compare the absolute relative approximate error with the pre-specified relative error tolerance, $\epsilon_s$. If $|\epsilon_a| > \epsilon_s$, then go to Step 2, else stop the algorithm. Also, check if the number of iterations has exceeded the maximum number of iterations allowed. If so, one needs to terminate the algorithm and notify the user.

# Chapter 03.04: Newton-Raphson Method for Solving a Nonlinear Equation

Lesson: Derivation of Newton-Raphson Method from Taylor Series

## Learning Objectives

*After successful completion of this lesson, you should be able to*

1) *derive the Newton-Raphson method from the Taylor series*

## Introduction

The Newton-Raphson method can also be derived from the Taylor series. For a general function $f(x)$, the Taylor series around the point $x_i$ is

$$f(x_{i+1}) = f(x_i) + f'(x_i)(x_{i+1} - x_i) + \frac{f''(x_i)}{2!}(x_{i+1} - x_i)^2 + \ldots$$

We are seeking a point where $f(x) = 0$, that is if we assume

$$f(x_{i+1}) = 0,$$

$$0 = f(x_i) + f'(x_i)(x_{i+1} - x_i) + \frac{f''(x_i)}{2!}(x_{i+1} - x_i)^2 + \ \ldots$$

Using only the first two terms of the series gives

$$0 \approx f(x_i) + f'(x_i)(x_{i+1} - x_i)$$

Solving for $x_{i+1}$ gives

$$x_{i+1} \approx x_i - \frac{f(x_i)}{f'(x_i)}$$

This is the same Newton-Raphson method formula as derived in a previous lesson using geometry and differential calculus.

# Chapter 03.04: Newton-Raphson Method for Solving a Nonlinear Equation

Lesson: Application of Newton-Raphson Method

## Learning Objectives

*After successful completion of this lesson, you should be able to*

1) *apply the Newton-Raphson method to solve for roots of a nonlinear equation.*

## Applications

In the previous lessons, we discussed the background to the Newton-Raphson Method of solving nonlinear equations. In this lesson, we take an example of how to apply the algorithm of the Newton-Raphson method to solve a nonlinear equation.

### Example 1

Solve the nonlinear equation $x^3 = 20$ by the Newton-Raphson method using an initial guess of $x_0 = 3$.

Conduct three iterations to estimate the root of the above equation.

Find the absolute relative approximate error at the end of each iteration.

Find the number of significant digits at least correct at the end of each iteration.

**Solution**

First, we rewrite the equation $x^3 = 20$ in the form $f(x) = 0$

$$f(x) = x^3 - 20$$

$$f'(x) = 3x^2$$

The initial guess is

$$x_0 = 3$$

Iteration 1

The estimate of the root is

$$\begin{aligned} x_1 &= x_0 - \frac{f(x_0)}{f'(x_0)} \\ &= 3 - \frac{3^3 - 20}{(3)^2} \\ &= 3 - \frac{7}{27} \\ &= 3 - 0.259259 \\ &= 2.74074 \end{aligned}$$

The absolute approximate relative error $|\epsilon_a|$ at the end of Iteration 1 is

$$\begin{aligned} |\epsilon_a| &= \left| \frac{x_1 - x_o}{x_1} \right| \times 100 \\ &= \left| \frac{2.74074 - 3}{2.74074} \right| \times 100 \\ &= 9.46\% \end{aligned}$$

The number of significant digits at least correct is 0, as you need an absolute relative approximate error of 5% or less for at least one significant digit to be correct in your result.

Iteration 2

The estimate of the root is

$$\begin{aligned} x_2 &= x_1 - \frac{f(x_1)}{f'(x_1)} \\ &= 2.74074 - \frac{(2.74074)^3 - 20}{3(2.74074)^2} \\ &= 2.74074 - \frac{0.587495}{22.5350} \\ &= 2.71467 \end{aligned}$$

The absolute relative approximate error $|\epsilon_a|$ at the end of Iteration 2 is

$$\begin{aligned} |\epsilon_a| &= \left| \frac{x_2 - x_1}{x_2} \right| \times 100 \\ &= \left| \frac{2.71467 - 2.74074}{2.71467} \right| \times 100 \\ &= 0.960\% \end{aligned}$$

The maximum value of $m$ for which $|\epsilon_a| \leq 0.5 \times 10^{2-m}\%$ is 1.71669. Hence, the number of significant digits at least correct is 1.

Iteration 3

The estimate of the root is

$$\begin{aligned} x_3 &= x_2 - \frac{f(x_2)}{f'(x_2)} \\ &= 2.71467 - \frac{(2.71467)^3 - 20}{(2.71476)^2} \\ &= 2.71467 - \frac{5.57925 \times 10^{-3}}{22.1083} \\ &= 2.71467 - 2.52360 \times 10^{-4} \\ &= 2.71442 \end{aligned}$$

The absolute relative approximate error $|\epsilon_a|$at the end of Iteration 3 is

$$\begin{aligned} |\epsilon_a| &= \left| \frac{2.71442 - 2.71467}{2.71442} \right| \times 100 \\ &= 0.00921\% \end{aligned}$$

The maximum value of $m$ for which $|\epsilon_a| \leq 0.5 \times 10^{2-m}\%$ is 3.73471. Hence, the number of significant digits at least correct is 3.

**Example 2**

You are working for 'DOWN THE TOILET COMPANY' that makes floats for ABC commodes. The floating ball has a specific gravity of 0.6 and has a radius of 5.5 cm. You are asked to find the depth to which the ball is submerged when floating in the water.

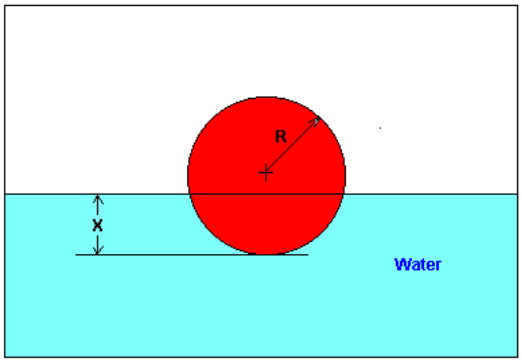



**Figure 2** Floating ball problem.

The equation that gives the depth $x$ in meters to which the ball is submerged underwater is given by

$$x^3 - 0.165x^2 + 3.993 \times 10^{-4} = 0$$

Use the Newton-Raphson method of finding roots of equations to find

a) the depth $x$ to which the ball is submerged underwater. Conduct three iterations to estimate the root of the above equation.

b) the absolute relative approximate error at the end of each iteration, and

c) the number of significant digits at least correct at the end of each iteration.

**Solution**

$$f(x) = x^3 - 0.165x^2 + 3.993 \times 10^{-4}$$

$$f'(x) = 3x^2 - 0.33x$$

Let us assume the initial guess of the root of $f(x) = 0$ is $x_0 = 0.05$ m. This is a reasonable guess (discuss why $x = 0$ and $x = 0.11$ m are not good choices) as the extreme values of the depth $x$ would be 0 and the diameter (0.11 m) of the ball.

**Iteration 1**

The estimate of the root is

$$\begin{aligned} x_1 &= x_0 - \frac{f(x_0)}{f'(x_0)} \\ &= 0.05 - \frac{(0.05)^3 - 0.165(0.05)^2 + 3.993 \times 10^{-4}}{3(0.05)^2 - 0.33(0.05)} \\ &= 0.05 - \frac{1.118 \times 10^{-4}}{-9 \times 10^{-3}} \\ &= 0.05 - (-0.01242) \\ &= 0.06242 \end{aligned}$$

The absolute relative approximate error $|\epsilon_a|$ at the end of Iteration 1 is

$$\begin{aligned} |\epsilon_a| &= \left| \frac{x_1 - x_0}{x_1} \right| \times 100 \\ &= \left| \frac{0.06242 - 0.05}{0.06242} \right| \times 100 \\ &= 19.90\% \end{aligned}$$

The number of significant digits at least correct is 0, as you need an absolute relative approximate error of 5% or less for at least one significant digit to be correct in your result.

**Iteration 2**

The estimate of the root is

$$
\begin{aligned}
x_2 &= x_1 - \frac{f(x_1)}{f'(x_1)} \\
&= 0.06242 - \frac{(0.06242)^3 - 0.165(0.06242)^2 + 3.993 \times 10^{-4}}{3(0.06242)^2 - 0.33(0.06242)} \\
&= 0.06242 - \frac{-3.97781 \times 10^{-7}}{-8.90973 \times 10^{-3}} \\
&= 0.06242 - \left(4.4646 \times 10^{-5}\right) \\
&= 0.06238
\end{aligned}
$$

The absolute relative approximate error $|\epsilon_a|$ at the end of Iteration 2 is

$$
\begin{aligned}
|\epsilon_a| &= \left|\frac{x_2 - x_1}{x_2}\right| \times 100 \\
&= \left|\frac{0.06238 - 0.06242}{0.06238}\right| \times 100 \\
&= 0.0716\%
\end{aligned}
$$

The maximum value of $m$ for which $|\epsilon_a| \leq 0.5 \times 10^{2-m}$ is 2.844. Hence, the number of significant digits at least correct in the answer is 2.

**Iteration 3**

The estimate of the root is

$$
\begin{aligned}
x_3 &= x_2 - \frac{f(x_2)}{f'(x_2)} \\
&= 0.06238 - \frac{(0.06238)^3 - 0.165(0.06238)^2 + 3.993 \times 10^{-4}}{3(0.06238)^2 - 0.33(0.06238)} \\
&= 0.06238 - \frac{4.44 \times 10^{-11}}{-8.91171 \times 10^{-3}} \\
&= 0.06238 - \left(-4.9822 \times 10^{-9}\right) \\
&= 0.06238
\end{aligned}
$$

The absolute relative approximate error $|\epsilon_a|$ at the end of Iteration 3 is

$$
\begin{aligned}
|\epsilon_a| &= \left|\frac{0.06238 - 0.06238}{0.06238}\right| \times 100 \\
&= 0
\end{aligned}
$$

The number of significant digits at least correct is 4, as only 4 significant digits are carried through in all the calculations.

# Chapter 03.04: Newton-Raphson Method for Solving a Nonlinear Equation

Lesson: Advantages and Pitfalls of Newton-Raphson Method

## Learning Objectives

*After successful completion of this lesson, you should be able to*

1) *enumerate the advantages of the Newton-Raphson method and the reasoning behind them.*

2) *list the drawbacks of the Newton-Raphson method and the reason behind them.*

## Advantages of the Newton-Raphson Method

1) Fast convergence: Newton-Raphson method converges fast when it converges. The method converges quadratically – one can interpret that the number of significant digits correct gets approximately squared at each iteration.

2) Only one initial estimate is needed: Newton-Raphson method is an open method that requires only one initial estimate of the root of the equation. However, since the method is an open method, convergence is not guaranteed.

## Drawbacks of the Newton-Raphson Method

1) Divergence at inflection points: If the selection of the initial guess or an iterated value of the root turns out to be close to the inflection point of the function $f(x)$ in the equation $f(x) = 0$, Newton-Raphson method may start diverging away from the root. It may then begin to converging back to the root. For example, to find the root of the equation

$$f(x) = (x-1)^3 + 0.512 = 0$$

then

$$f'(x) = 3(x-1)^2$$

and the Newton-Raphson method reduces to

$$x_{i+1} = x_i - \frac{({x_i}^3 - 1)^3 + 0.512}{3(x_i - 1)^2}$$

Starting with an initial guess of $x_0 = 5.0$, Table 1 shows the iterated values of the root of the equation. As you can observe, the root starts to diverge at Iteration 6 because the previous estimate of 0.92589 is close to the inflection point of $x = 1$ (the value of $f'(x)$ is zero at the inflection point). Eventually, after 12 more iterations, the root converges to the exact value of $x = 0.2$.

**Table 1** Divergence near an inflection point.

| Iteration Number | $x_i$ |
|---|---|
| 0 | 5.0000 |
| 1 | 3.6560 |
| 2 | 2.7465 |
| 3 | 2.1084 |
| 4 | 1.6000 |
| 5 | 0.92589 |
| 6 | −30.119 |
| 7 | −19.746 |
| 8 | −12.831 |
| 9 | −8.2217 |
| 10 | −5.1498 |
| 11 | −3.1044 |
| 12 | −1.7464 |
| 13 | −0.85356 |
| 14 | −0.28538 |
| 15 | 0.039784 |
| 16 | 0.17475 |
| 17 | 0.19924 |
| 18 | 0.2 |

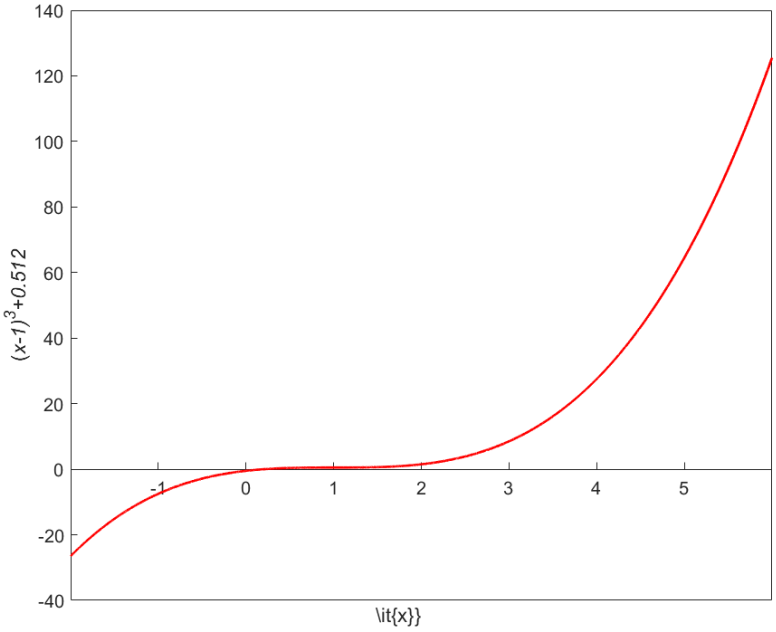



**Figure 1** Divergence at an inflection point for $f(x) = (x-1)^3 + 0.512 = 0$.

2) Division by zero: Division by zero may occur during the implementation of the Newton-Raphson method. For example, for the equation

$$f(x) = x^3 - 0.03x^2 + 2.4 \times 10^{-6} = 0$$

then

$$f'(x) = 3x^2 - 0.06x$$

and the Newton-Raphson method reduces to

$$x_{i+1} = x_i - \frac{{x_i}^3 - 0.03{x_i}^2 + 2.4 \times 10^{-6}}{3{x_i}^2 - 0.06x_i}$$

For $x_0 = 0$ or $x_0 = 0.02$, division by zero occurs (Figure 2). For an initial guess close to 0.02, such as $x_0 = 0.01999$, one may avoid division by zero, but then the denominator in the formula is a small number. For this case of initial guess of $x_0 = 0.01999$, as given in Table 2, even after 9 iterations, the Newton-Raphson method does not converge.

**Table 2** Division by near zero in Newton-Raphson method.

| Iteration Number | $x_i$ | $f(x_i)$ | $\lvert\in_a\rvert\,\%$ |
|---|---|---|---|
| 0 | 0.019990 | $-1.6000 \times 10^{-6}$ | $---$ |
| 1 | $-2.6480$ | 18.778 | 100.75 |
| 2 | $-1.7620$ | $-5.5638$ | 50.282 |
| 3 | $-1.1714$ | $-1.6485$ | 50.422 |
| 4 | $-0.77765$ | $-0.48842$ | 50.632 |
| 5 | $-0.51518$ | $-0.14470$ | 50.946 |
| 6 | $-0.34025$ | $-0.042862$ | 51.413 |
| 7 | $-0.22369$ | $-0.012692$ | 52.107 |
| 8 | $-0.14608$ | $-0.0037553$ | 53.127 |
| 9 | $-0.094490$ | $-0.0011091$ | 54.602 |

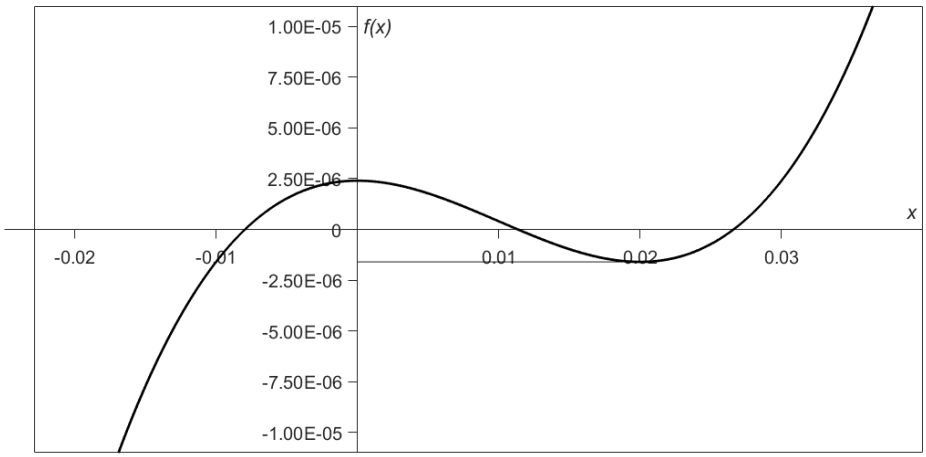



**Figure 2** Pitfall of division by zero or a near-zero number.

3) Oscillations near local maximum and minimum: Results obtained from the Newton-Raphson method may oscillate about the local maximum or minimum without converging on a root but converging on the local maximum or minimum. Eventually, it may lead to division by a number close to zero and may diverge.

For example, for

$$f(x) = x^2 + 2 = 0$$

the equation has no real roots (Figure 3 and Table 3).

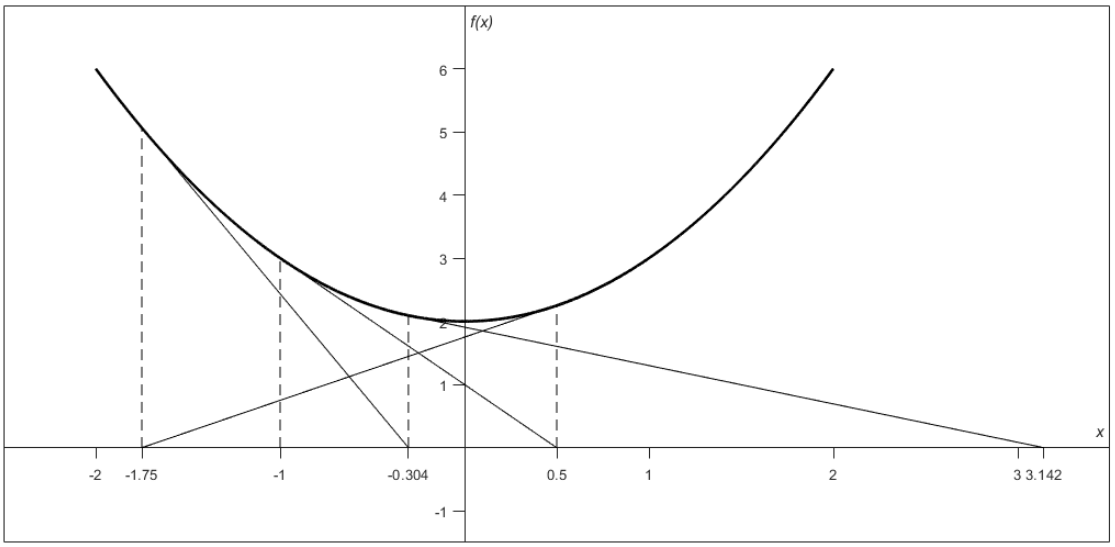



**Figure 3** Oscillations around local minima for $f(x) = x^2 + 2$.

**Table 3** Oscillations near local maxima and minima in Newton-Raphson method.

| Iteration Number | $x_i$ | $f(x_i)$ | $\lvert\in_a\rvert\,\%$ |
|---|---|---|---|
| 0 | $-1.0000$ | 3.00 | $---$ |
| 1 | 0.5 | 2.25 | 300.00 |
| 2 | $-1.75$ | 5.063 | 128.571 |
| 3 | $-0.30357$ | 2.092 | 476.47 |
| 4 | 3.1423 | 11.874 | 109.66 |
| 5 | 1.2529 | 3.570 | 150.80 |
| 6 | $-0.17166$ | 2.029 | 829.88 |
| 7 | 5.7395 | 34.942 | 102.99 |
| 8 | 2.6955 | 9.266 | 112.93 |
| 9 | 0.97678 | 2.954 | 175.96 |

4) Root jumping: In some cases where the function $f(x)$ is oscillating and has several roots, one may choose an initial guess close to a root. However, the guesses may jump and converge to some other root. For example, for solving the equation $\sin x = 0$ if you choose $x_0 = 2.4\pi = (7.539822)$ as an initial guess, it converges to the root of $x = 0$, as shown in Table 4 and Figure 4. However, one may have chosen this as an initial guess to converge to $x = 2\pi = 6.2831853$.

**Table 4** Root jumping in Newton-Raphson method.

| Iteration Number | $x_i$ | $f(x_i)$ | $\lvert\in_a\rvert\,\%$ |
|---|---|---|---|
| 0 | 7.539822 | 0.951 | $---$ |
| 1 | 4.462 | $-0.969$ | 68.973 |
| 2 | 0.5499 | 0.5226 | 711.44 |
| 3 | $-0.06307$ | $-0.06303$ | 971.91 |
| 4 | $8.376 \times 10^{-4}$ | $8.375 \times 10^{-5}$ | $7.54 \times 10^{4}$ |
| 5 | $-1.95861 \times 10^{-13}$ | $-1.95861 \times 10^{-13}$ | $4.28 \times 10^{10}$ |

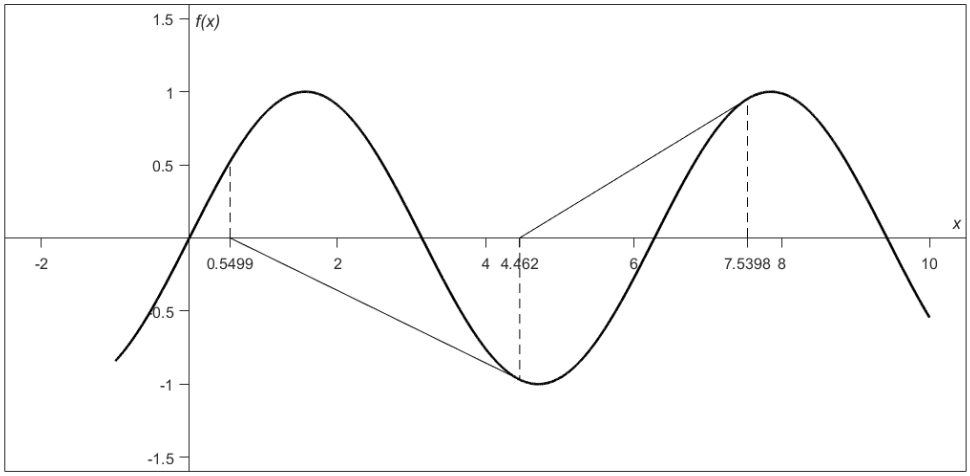



**Figure 4** Root jumping from intended location of root for $f(x) = \sin x = 0$

## Appendix A. What is an inflection point?

For a function $f(x)$, the point where the concavity changes from up-to-down or down-to-up is called its inflection point, that is, the sign of curvature changes sign. For example, for the function $f(x) = (x-1)^3$, the concavity changes at $x = 1$ (see Figure 3), and hence (1,0) is an inflection point.

For a function $f(x)$, for $(x, f(x))$ to be an inflection point, $f''(x) = 0$ is a necessary condition. An inflection point may or may not exist, though at a point where $f''(x) = 0$. For example, for $f(x) = x^4 - 16$, $f''(0) = 0$, but the concavity does not change at $x = 0$. Hence the point (0, –16) is not an inflection point of $f(x) = x^4 - 16$.

For $f(x) = (x-1)^3$, $f''(1) = 0$ and $f''(x)$ changes sign at $x = 1$ ($f''(x) < 0$ for $x = 1-$, and $f''(x) > 0$ for $x = 1+$), and thus brings up the *Inflection Point Theorem* for a function $f(x)$ that states the following.

"If $f'(c)$ exists and $f''(c)$ changes sign at $x = c$, then the point $(c, f(c))$ is an inflection point of the graph of $f$."

# Multiple Choice Test

Chapter 03.04 Newton-Raphson Method for Solving a Nonlinear Equation

(1). The Newton-Raphson method of finding roots of nonlinear equations falls under the category of _____ methods.

(A) bracketing

(B) open

(C) random

(D) graphical

(2). The Newton-Raphson method formula for finding the square root of a real number $R$ from the equation $x^2 - R = 0$ is,

(A) $x_{i+1} = \dfrac{x_i}{2}$

(B) $x_{i+1} = \dfrac{3x_i}{2}$

(C) $x_{i+1} = \dfrac{1}{2}\left(x_i + \dfrac{R}{x_i}\right)$

(D) $x_{i+1} = \dfrac{1}{2}\left(3x_i - \dfrac{R}{x_i}\right)$

(3). The next iterative value of the root of $x^2 - 4 = 0$ using the Newton-Raphson method, if the initial guess is 3, is

(A) 1.5

(B) 2.067

(C) 2.167

(D) 3.000

(4). The root of the equation $f(x) = 0$ is found by using the Newton-Raphson method. The initial estimate of the root is $x_0 = 3$, $f(3) = 5$. The angle made by the line tangent to the function $f(x)$ makes at $x = 3$ is $57^\circ$ with respect to the $x$-axis. The next estimate of the root, $x_1$ most nearly is

(A) $-3.2470$

(B) $-0.24704$

(C) 3.2470

(D) 6.2470

(5). The root of $x^3 = 4$ is found by using the Newton-Raphson method. The successive iterative values of the root are given in the table below.

| Iteration Number | Value of Root |
|---|---|
| 0 | 2.0000 |
| 1 | 1.6667 |
| 2 | 1.5911 |
| 3 | 1.5874 |
| 4 | 1.5874 |

The iteration number at which I would first trust at least two significant digits in the answer is

(A) 1

(B) 2

(C) 3

(D) 4

(6). The ideal gas law is given by

$pv = RT$

where $p$ is the pressure, $v$ is the specific volume, $R$ is the universal gas constant, and $T$ is the absolute temperature. This equation is only accurate for a limited range of pressure and temperature. Vander Waals came up with an equation that was accurate for a larger range of pressure and temperature given by

$$\left(p + \frac{a}{v^2}\right)(v - b) = RT$$

where $a$ and $b$ are empirical constants dependent on a particular gas. Given the value of $R = 0.08$, $a = 3.592$, $b = 0.04267$, $p = 10$ and $T = 300$ (assume all units are consistent), one is going to find the specific volume, $v$, for the above values. Without finding the solution from the Vander Waals equation, what would be a good initial guess for $v$?

(A) 0

(B) 1.2

(C) 2.4

(D) 3.6

For complete solution, go to

http://nm.mathforcollege.com/mcquizzes/03nle/quiz_03nle_newton_solution.pdf

# Problem Set

Chapter 03.04: Newton-Raphson Method for Solving a Nonlinear Equation

(1). Find the estimate of the root of $x^2 - 4 = 0$ by using the Newton-Raphson method. Assume that the initial guess of the root is 3. Conduct three iterations. Also, calculate the approximate error, true error, absolute relative approximate error, and absolute relative true error at the end of each iteration.

**Answer:**

| Iteration # | Root Estimate | Approx Error | True Error | Abs Rel Appr Err | Abs Rel True Err |
|---|---|---|---|---|---|
| 1 | 2.1667 | 0.8333 | −0.1667 | 27.77% | 8.335% |
| 2 | 2.0064 | −0.1603 | −0.0064 | 7.98% | 0.32% |
| 3 | 2.0002 | −0.0062 | −0.0002 | 0.31% | 0.001% |

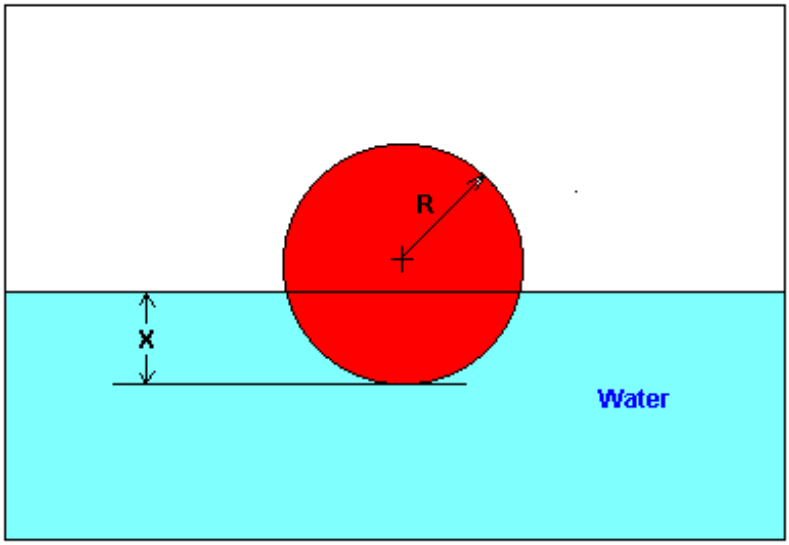
(2). The velocity of a body is given by $v(t) = 5e^{-t} + 6$, where $v$ is in m/s and $t$ is in seconds.

a) Use the Newton Raphson method to find the time when the velocity will be 7.0 m/s. Use two iterations and take $t = 2$ seconds as the initial guess.

b) What is the relative true error at the end of the second iteration for part (a)?

**Answer:** *a*) 1.5236, 1.6058 *b*) 0.2260%

(3). You are working for DOWN THE TOILET COMPANY that makes floats for ABC commodes. The ball has a specific gravity of 0.6 and has a radius of 5.5 cm. You are asked to find the depth to which the ball will get submerged when floating in the water.




The equation that gives the depth $x$ (unit of $x$ is m) to which the ball is submerged under water is given by

$$x^3 - 0.165x^2 + 3.993 \times 10^{-4} = 0$$

Solving it exactly would require some effort. However, using numerical techniques such as Newton-Raphson method, we can solve this equation and any other equation of the form $f(x) = 0$. Solve the above equation by the Newton-Raphson method and do the following.

a) Conduct three iterations. Use an initial guess $x = 0.055$ m.

b) Calculate the absolute relative approximate error at the end of each step.

c) Find the number of significant digits at least correct at the end of each iteration.

d) How could you have used the knowledge of the physics of the problem to develop initial guess for the Newton-Raphson method?

**Answer:** *a*) 0.06233, 0.06237, 0.06237

*b*) $n/a$, 0.06413%, 0%(0% if only 4 significant digits were carried in the calculations)

*c*) 0, 2, 4 (as only 4 significant digits were carried in the calculations)

*d*) Since the diameter is 0.11 m, a value of 0 or 0.11 seems to be a good initial estimate, but these guesses give division by zero. So, 0.055 would be a good initial estimate.

(4). The root of the equation $f(x) = 0$ is found by using the Newton-Raphson method. If the initial estimate of the root is assumed to be $x_0 = 3$, given $f(3) = 5$, and the angle the tangent makes to the function $f(x)$ at $x = 3$ is 57°, what is the next estimate of the root, $x_1$?

**Answer:** −0.24696

(5). The root of the equation $f(x) = 0$ is found by using the Newton-Raphson method. The initial estimate of the root is assumed to be $x_0 = 5.0$, and the angle the tangent makes with the $x$-axis to

the function $f(x)$ at $x_0 = 5.0$ is $89.236°$. If the next estimate of the root is, $x_1 = 3.693$, what is the value of the function $f(x)$ at $x = 5$?

**Answer:** 98.012

(6). Find the Newton-Raphson method formula for finding the square root of a real number $R$ from the equation $x^2 - R = 0$. Find the square root of 50.41 by using the formula. Assume 50.41 as the initial guess. How many iterations does it take to get at least 3 significant digits correct in your answer? Do not use the true value to answer the question.

**Answer:** Takes 6 iterations. Root at the end of $6^{\text{th}}$ iteration is 7.100, with the absolute relative approx error of 0.014%. Four significant digits were used in solving the problem.

# Chapter 03.05: Secant Method of Solving Nonlinear Equations

Lesson: Secant Method of Solving Nonlinear Equations

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *derive the secant method to solve for the roots of a nonlinear equation,*

2) *use the secant method to numerically solve a nonlinear equation.*

## What is the secant method and why would I want to use it instead of the Newton-Raphson method?

The Newton-Raphson method of solving a nonlinear equation $f(x) = 0$ is given by the iterative formula

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)} \quad (1)$$

One of the drawbacks of the Newton-Raphson method is that you have to evaluate the derivative of the function. With availability of symbolic manipulators such as Maple, MathCAD, MATHEMATICA and MATLAB, this process has become more convenient. However, it still can be a laborious process, and even intractable if the function is derived as part of a numerical scheme. To overcome these drawbacks, the derivative of the function, $f(x)$ is approximated as

$$f'(x_i) = \frac{f(x_i) - f(x_{i-1})}{x_i - x_{i-1}} \quad (2)$$

Substituting Equation (2) in Equation (1) gives

$$x_{i+1} = x_i - \frac{f(x_i)(x_i - x_{i-1})}{f(x_i) - f(x_{i-1})} \quad (3)$$

The above equation is called the secant method. This method now requires two initial guesses, but unlike the bisection method, the two initial guesses do not need to bracket the root of the equation. The secant method is an open method and may or may not converge. However, when secant method converges, it will typically converge faster than the bisection method. However, since the derivative is approximated as given by Equation (2), it typically converges slower than the Newton-Raphson method.

The secant method can also be derived from geometry, as shown in Figure 1. Taking two initial guesses, $x_{i-1}$ and $x_i$, one draws a straight line between $f(x_i)$ and $f(x_{i-1})$ passing through the $x$ -axis at $x_{i+1}$. $ABE$ and $DCE$ are similar triangles.

Hence

$$\frac{AB}{AE} = \frac{DC}{DE}$$

$$\frac{f(x_i)}{x_i - x_{i+1}} = \frac{f(x_{i-1})}{x_{i-1} - x_{i+1}}$$

On rearranging, the secant method is given as

$$x_{i+1} = x_i - \frac{f(x_i)(x_i - x_{i-1})}{f(x_i) - f(x_{i-1})}$$

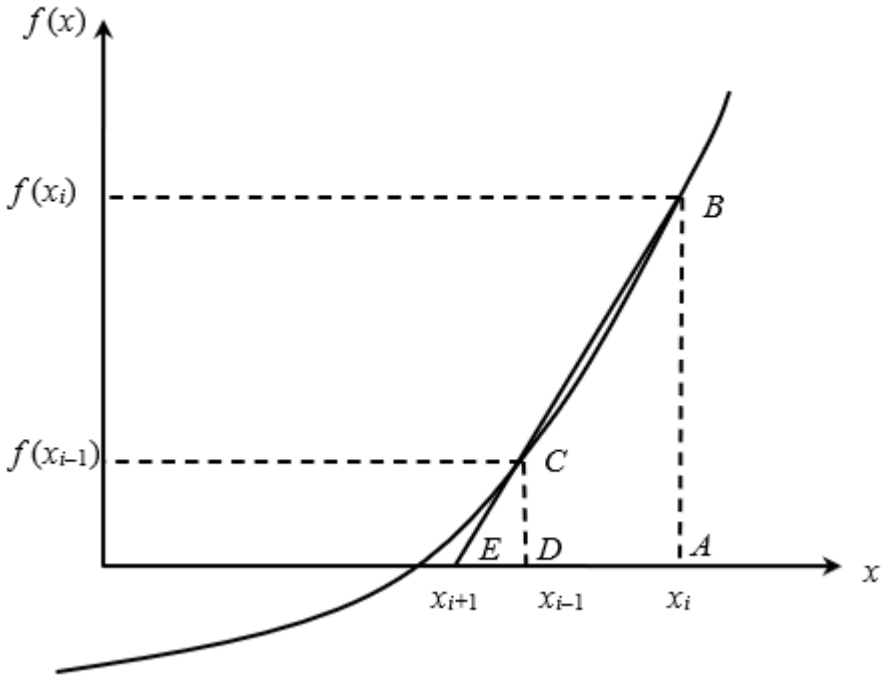



**Figure 1** Geometrical representation of the secant method.

## Application of Secant Method

Let's take an example to show the application of the secant method to solve a nonlinear equation.

### Example 1

You are working for 'DOWN THE TOILET COMPANY' that makes floats (Figure 2) for ABC commodes. The floating ball has a specific gravity of 0.6 and a radius of 5.5 cm. You are asked to find the depth to which the ball is submerged when floating in water.

The equation that gives the depth $x$ to which the ball is submerged under water is given by

$$x^3 - 0.165x^2 + 3.993 \times 10^{-4} = 0$$

Use the secant method of finding roots of equations to find the depth $x$ to which the ball is submerged under water. Conduct three iterations to estimate the root of the above equation. Find the absolute relative approximate error and the number of significant digits at least correct at the end of each iteration.

**Solution**

$f(x) = x^3 - 0.165x^2 + 3.993 \times 10^{-4}$

Let us assume the initial guesses of the root of $f(x) = 0$ as $x_{-1} = 0.02$ and $x_0 = 0.05$.

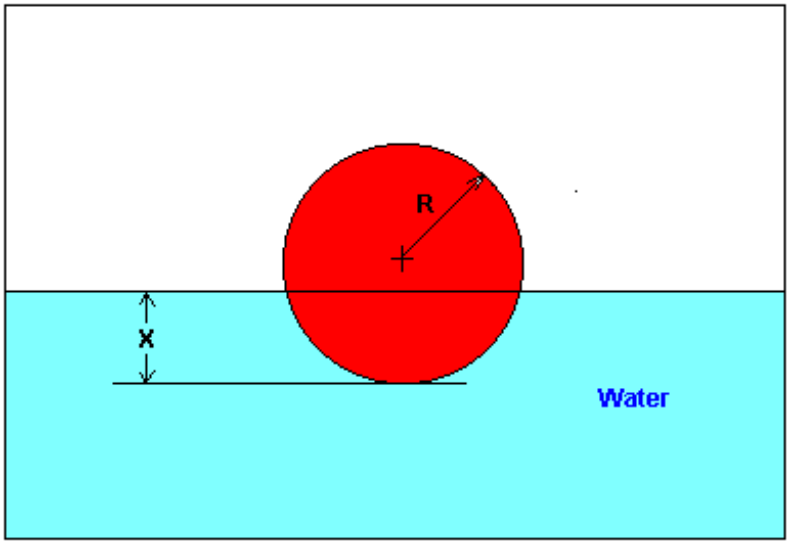



**Figure 2** Floating ball problem.

Iteration 1

The estimate of the root is

$$\begin{aligned}
x_1 &= x_0 - \frac{f(x_0)(x_0 - x_{-1})}{f(x_0) - f(x_{-1})} \\
&= x_0 - \frac{\left(x_0^3 - 0.165x_0^2 + 3.993 \times 10^{-4}\right) \times (x_0 - x_{-1})}{\left(x_0^3 - 0.165x_0^2 + 3.993 \times 10^{-4}\right) - \left(x_{-1}^3 - 0.165x_{-1}^2 + 3.993 \times 10^{-4}\right)} \\
&= 0.05 - \frac{\left[0.05^3 - 0.165(0.05)^2 + 3.993 \times 10^{-4}\right] \times [0.05 - 0.02]}{\left[0.05^3 - 0.165(0.05)^2 + 3.993 \times 10^{-4}\right] - \left[0.02^3 - 0.165(0.02)^2 + 3.993 \times 10^{-4}\right]} \\
&= 0.06461
\end{aligned}$$

The absolute relative approximate error $|\epsilon_a|$ at the end of Iteration 1 is

$$
\begin{aligned}
|\epsilon_a| &= \left|\frac{x_1 - x_0}{x_1}\right| \times 100 \\
&= \left|\frac{0.06461 - 0.05}{0.06461}\right| \times 100 \\
&= 22.62\%
\end{aligned}
$$

The number of significant digits at least correct is 0, as you need an absolute relative approximate error of 5% or less for one significant digit to be correct in your result.

Iteration 2

$$
\begin{aligned}
x_2 &= x_1 - \frac{f(x_1)(x_1 - x_0)}{f(x_1) - f(x_0)} \\
&= x_1 - \frac{\left(x_1^3 - 0.165x_1^2 + 3.993 \times 10^{-4}\right) \times (x_1 - x_0)}{(x_1^3 - 0.165x_1^2 + 3.993 \times 10^{-4}) - (x_0^3 - 0.165x_0^2 + 3.993 \times 10^{-4})} \\
&= 0.06461 - \frac{\left[0.06461^3 - 0.165\left(0.06461\right)^2 + 3.993 \times 10^{-4}\right] \times (0.06461 - 0.05)}{\left[0.06461^3 - 0.165\left(0.06461\right)^2 + 3.993 \times 10^{-4}\right] - \left[0.05^3 - 0.165\left(0.05\right)^2 + 3.993 \times 10^{-4}\right]} \\
&= 0.06241
\end{aligned}
$$

The absolute relative approximate error $|\epsilon_a|$ at the end of Iteration 2 is

$$
\begin{aligned}
|\epsilon_a| &= \left|\frac{x_2 - x_1}{x_2}\right| \times 100 \\
&= \left|\frac{0.06241 - 0.06461}{0.06241}\right| \times 100 \\
&= 3.525\%
\end{aligned}
$$

The number of significant digits at least correct is 1, as you need an absolute relative approximate error of 5% or less.

Iteration 3

$$
\begin{aligned}
x_3 &= x_2 - \frac{f(x_2)(x_2 - x_1)}{f(x_2) - f(x_1)} \\
&= x_2 - \frac{\left(x_2^3 - 0.165x_2^2 + 3.993 \times 10^{-4}\right) \times (x_2 - x_1)}{(x_2^3 - 0.165x_2^2 + 3.993 \times 10^{-4}) - (x_1^3 - 0.165x_1^2 + 3.993 \times 10^{-4})} \\
&= 0.06241 - \frac{\left[0.06241^3 - 0.165\left(0.06241\right)^2 + 3.993 \times 10^{-4}\right] \times (0.06241 - 0.06461)}{\left[0.06241^3 - 0.165\left(0.06241\right)^2 + 3.993 \times 10^{-4}\right] - \left[0.06461^3 - 0.165\left(0.06461\right)^2 + 3.993 \times 10^{-4}\right]} \\
&= 0.06238
\end{aligned}
$$

The absolute relative approximate error $|\epsilon_a|$ at the end of Iteration 3 is

$$
\begin{aligned}
|\epsilon_a| &= \left|\frac{x_3 - x_2}{x_3}\right| \times 100 \\
&= \left|\frac{0.06238 - 0.06241}{0.06238}\right| \times 100 \\
&= 0.0595\%
\end{aligned}
$$

The number of significant digits at least correct is 2, as you need an absolute relative approximate error of 0.5% or less. Table 1 shows the secant method calculations for the results from the above problem.

**Table 1** Secant method results as a function of iterations.

| Iteration Number,n | $x_{i-1}$ | $x_i$ | $x_{i+1}$ | $\lvert\epsilon_a\rvert$ | $f(x_{i+1})$ |
|---|---|---|---|---|---|
| 1 | 0.02 | 0.05 | 0.06461 | 22.62 | $-1.9812 \times 10^{-5}$ |
| 2 | 0.05 | 0.06461 | 0.06241 | 3.525 | $-3.285 \times 10^{-7}$ |
| 3 | 0.06461 | 0.06241 | 0.06238 | 0.0595 | $2.0252 \times 10^{-9}$ |
| 4 | 0.06241 | 0.06238 | 0.06238 | $3.64 \times 10^{-4}$ | $-1.8576 \times 10^{-13}$ |

# Multiple Choice Test

## Chapter 03.0?: Secant Method

(1). The secant method of finding roots of nonlinear equations falls under the category of _____ methods.

(A) bracketing

(B) graphical

(C) open

(D) random

(2). The secant method formula for finding the square root of a real number $R$ from the equation $x^2 - R = 0$is

(A) $\dfrac{x_i x_{i-1} + R}{x_i + x_{i-1}}$

(B) $\dfrac{x_i x_{i-1}}{x_i + x_{i-1}}$

(C) $\dfrac{1}{2}\left(x_i + \dfrac{R}{x_i}\right)$

(D) $\dfrac{2x_i^2 + x_i x_{i-1} - R}{x_i + x_{i-1}}$

(3). The next iterative value of the root of $x^2 - 4 = 0$ using secant method, if the initial guesses are 3 and 4, is

(A) 2.2857

(B) 2.5000

(C) 5.5000

(D) 5.7143

(4). The root of the equation $f(x) = 0$ is found by using the secant method. Given one of the initial estimates is $x_0 = 3$, $f(3) = 5$, and the angle the secant line makes with the $x$-axis is $57°$, the next estimate of the root, $x_1$, is

(A) $-3.2470$

(B) $-0.24704$

(C) 3.247

(D) 6.2470

(5). For finding the root of $\sin\ x = 0$ by the secant method, the following choice of initial guesses would not be appropriate.

(A) $\frac{\pi}{4}$ and $\frac{\pi}{2}$

(B) $\frac{\pi}{4}$ and $\frac{3\pi}{4}$

(C) $-\frac{\pi}{2}$ and $\frac{\pi}{2}$

(D) $\frac{\pi}{3}$ and $\frac{\pi}{2}$

(6). When drugs are given orally to a patient, the drug concentration $c$ in the blood stream at time $t$ is given by a formula

$$c = Kte^{-at}$$

where $K$ is dependent on parameters such as the dose administered while $a$ is dependent on the absorption and elimination rates of the drug. If $K = 2$ and $a = 0.25$, and $t$ is in seconds and $c$ is in mg/ml, the time at which the maximum concentration is reached is given by the solution of the equation,

(A) $2te^{-0.25t} = 0$

(B) $2e^{-0.25t} - 2te^{-0.25t} = 0$

(C) $2e^{-0.25t} - 0.5te^{-0.25t} = 0$

(D) $2te^{-0.25t} = 2$

For complete solution, go to

http://nm.mathforcollege.com/mcquizzes/03nle/quiz_03nle_secant_solution.pdf

# Problem Set

## Chapter 03.05: Secant Method for Solving a Nonlinear Equations

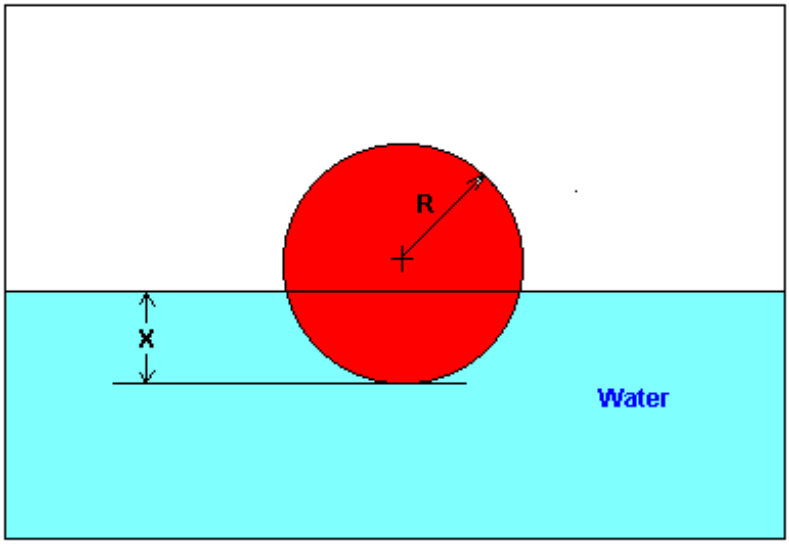
(1). Find the estimate of the root of $x^2 - 4 = 0$ by using secant method, if initial guesses of the roots are 3 and 5. Conduct three iterations. Also, calculate the approximate error, true error, absolute relative approximate error, and absolute relative true error at the end of each iteration.

(2). The velocity of a body is given by $v(t) = 5e^{-t} + 6$, where $v$ is in m/s and $t$ is in seconds.

a) Use secant method to find when the velocity will be 7.0 m/s. Use only two iterations and take $t = 2$ and $t = 3.5$ seconds as the initial guesses.

b) What is the relative true error at the end of the second iteration for part (a)?

(3). You are working for DOWN THE TOILET COMPANY that makes floats for ABC commodes. The ball has a specific gravity of 0.6 and has a radius of 5.5 cm. You are asked to find the depth to which the ball will get submerged when floating in water.




The equation that gives the depth $x$ (unit of $x$ is in m) to which the ball is submerged under water is given by

$$x^3 - 0.165x^2 + 3.993 \times 10^{-4} = 0$$

Solving it exactly would require some effort. However using numerical techniques such as secant method, we can solve this equation and any other equation of the form $f(x) = 0$. Solve the above equation by secant method and do the following.

a) Conduct three iterations. Use initial guesses of 0.055 and 0.0825.

b) Calculate the absolute relative approximate error at the end of each step.

c) Find the number of significant digits that are at least correct at the end of each iteration.

d) How could you have used the knowledge of the physics of the problem to develop initial guesses for the secant method?

(4). The root of the equation $f(x) = 0$ is found by using secant method. The initial guesses for the root are assumed to be $x_0 = 3$ and $x_1 = 2$. Given $f(3) = 28$ and $f(2) = 4$, what is the

a) angle the secant makes with the $x$ -axis,

b) next estimate of the root, $x_2$.

(5). The root of the equation $f(x) = 0$ is found by using secant method. If the initial estimates of the root are $x_0 = 5.0$ and $x_1 = 2.0$, and the angle the secant makes with the $x$ -axis between $x_0 = 5.0$ and $x_1 = 2.0$ is $88.75^\circ$. If the next estimate of the root is $x_2 = 1.9130$, what is the value of the function $f(x)$ at $x = 2$?

(6). Find the secant method formula for finding the square root of a real number $R$ from the equation $x^2 - R = 0$. Use the formula to find the square root of 50.41. Assume 50.41 and 25.205 as the initial guesses. How many iterations does it take to get at least 3 significant digits correct in your answer? Do not calculate the true value to answer the question.

# Chapter 03.07: Newton-Raphson Method For Solving Simultaneous Nonlinear Equations

Lesson: Newton-Raphson Method For Solving Simultaneous Nonlinear Equations

## Learning Objectives

*After reading this chapter, you should be able to:*

1) *derive the Newton-Raphson method formula for simultaneous nonlinear equations,*

2) *develop the algorithm of the Newton-Raphson method for solving simultaneous nonlinear equations,*

3) *use the Newton-Raphson method to solve a set of simultaneous nonlinear equations,*

4) *model a real-life problem that results in a set of simultaneous nonlinear equations.*

## Introduction

Several physical systems result in a mathematical model in terms of simultaneous nonlinear equations. A set of such equations can be written as

$$\begin{aligned} &f_1(x_1, x_2, ..., x_n) = 0 \\ &... \\ &... \\ &f_n(x_1, x_2, ..., x_n) = 0 \end{aligned} \qquad (1)$$

The solution to these simultaneous nonlinear equations are values of $x_1, x_2, ..., x_n$ which satisfy all the above $n$ equations. The number of set of solutions to these equations could be none, unique, more than one but finite, or infinite. In this chapter, we use the Newton-Raphson method to solve these equations.

The Newton-Raphson method of solving a single nonlinear equation, $f(x) = 0$, can be derived using first-order Taylor series (first two terms of Taylor series) and are given by

$$f(x_{i+1}) = f(x_i) + f'(x_i)(x_{i+1} - x_i) \qquad (2)$$

where

$$x_i = \text{previous estimate of root}$$

$$x_{i+1} = \text{present estimate of root}$$

Since we are looking for $x_{i+1}$where $f(x_{i+1})$ becomes zero, Equation (2) can be re-written as

$$0 = f(x_i) + f'(x_i)(x_{i+1} - x_i)$$

and then as a recursive formula as

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)} \qquad (3)$$

## Derivation

Now how do we extend the same to simultaneous nonlinear equations? For sake of simplicity, let us limit the number of nonlinear equations to two as

$$u(x, y) = 0 \qquad (4a)$$

$$v(x, y) = 0 \qquad (4b)$$

The first order Taylor-series for nonlinear equation is (4a) & (4b) are

$$u(x_{i+1}, y_{i+1}) = u(x_i, y_i) + \left.\frac{\partial u}{\partial x}\right|_{x_i, y_i} (x_{i+1} - x_i) + \left.\frac{\partial u}{\partial y}\right|_{x_i, y_i} (y_{i+1} - y_i) \qquad (5a)$$

$$v(x_{i+1}, y_{i+1}) = v(x_i, y_i) + \left.\frac{\partial v}{\partial x}\right|_{x_i, y_i} (x_{i+1} - x_i) + \left.\frac{\partial v}{\partial y}\right|_{x_i, y_i} (y_{i+1} - y_i) \qquad (5b)$$

We are looking for $(x_{i+1}, y_{i+1})$ where $u(x_{i+1}, y_{i+1})$ and $v(x_{i+1}, y_{i+1})$ are zero. Hence

$$0 = u(x_i, y_i) + \left.\frac{\partial u}{\partial x}\right|_{x_i, y_i} (x_{i+1} - x_i) + \left.\frac{\partial u}{\partial y}\right|_{x_i, y_i} (y_{i+1} - y_i) \qquad (6a)$$

$$0 = v(x_i, y_i) + \left.\frac{\partial v}{\partial x}\right|_{x_i, y_i} (x_{i+1} - x_i) + \left.\frac{\partial v}{\partial y}\right|_{x_i, y_i} (y_{i+1} - y_i) \qquad (6b)$$

Writing

$$x_{i+1} - x_i = \Delta x \qquad (7a)$$

$$y_{i+1} - y_i = \Delta y \qquad (7b)$$

we get

$$0 = u(x_i, y_i) + \left.\frac{\partial u}{\partial x}\right|_{x_i,y_i} \Delta x + \left.\frac{\partial u}{\partial y}\right|_{x_i,y_i} \Delta y \qquad (8a)$$

$$0 = v(x_i, y_i) + \left.\frac{\partial v}{\partial x}\right|_{x_i,y_i} \Delta x + \left.\frac{\partial v}{\partial y}\right|_{x_i,y_i} \Delta y \qquad (8b)$$

Rewriting Equations (8a) and (8b)

$$\left.\frac{\partial u}{\partial x}\right| x_i, y_i \Delta x + \left.\frac{\partial u}{\partial y}\right|_{x_i,y_i} \Delta y = -u(x_i, y_i) \qquad (9a)$$

$$\left.\frac{\partial v}{\partial x}\right| x_i, y_i \Delta x + \left.\frac{\partial v}{\partial y}\right|_{x_i,y_i} \Delta y = -v(x_i, y_i) \qquad (9b)$$

and then in the matrix form

$$\begin{bmatrix} \left.\frac{\partial u}{\partial x}\right|_{x_i,y_i} & \left.\frac{\partial u}{\partial y}\right|_{x_i,y_i} \\ \left.\frac{\partial v}{\partial x}\right|_{x_i,y_i} & \left.\frac{\partial v}{\partial y}\right|_{x_i,y_i} \end{bmatrix} \begin{bmatrix} \Delta x \\ \Delta y \end{bmatrix} = \begin{bmatrix} -u(x_i, y_i) \\ -v(x_i, y_i) \end{bmatrix} \qquad (10)$$

Solving Equation (10) would give us $\Delta x$ and $\Delta y$. Since the previous estimate of the root is $(x_i, y_i)$, one can find from Equation (7)

$$x_{i+1} = x_i + \Delta x \qquad (11a)$$

$$y_{i+1} = y_i + \Delta y \qquad (11b)$$

This process is repeated till one obtains the root of the equation within a prespecified tolerance,$\varepsilon_s$ such that

$$|\varepsilon_a|_x = \left|\frac{x_{i+1} - x_i}{x_{i+1}}\right| \times 100 < \varepsilon_s$$

$$|\varepsilon_a|_y = \left|\frac{y_{i+1} - y_i}{y_{i+1}}\right| \times 100 < \varepsilon_s$$

**Example 1**

Find the roots of the simultaneous nonlinear equations

$$x^2 + y^2 = 50$$
$$x - y = 10$$

Use an initial guess of $(x, y) = (2, -4)$. Conduct two iterations.

**Solution**

First put the equations in the form

$$u(x, y) = 0$$

$$v(x, y) = 0$$

to give

$$u(x, y) = x^2 + y^2 - 50 = 0$$
$$v(x, y) = x - y - 10 = 0$$

Now

$$\frac{\partial u}{\partial x} = 2x$$

$$\frac{\partial u}{\partial y} = 2y$$

$$\frac{\partial v}{\partial x} = 1$$

$$\frac{\partial v}{\partial y} = -1$$

Hence from Equation (10)

$$\begin{bmatrix} \left.\frac{\partial u}{\partial x}\right|_{x_i,y_i} & \left.\frac{\partial u}{\partial y}\right|_{x_i,y_i} \\ \left.\frac{\partial v}{\partial x}\right|_{x_i,y_i} & \left.\frac{\partial v}{\partial y}\right|_{x_i,y_i} \end{bmatrix} \begin{bmatrix} \Delta x \\ \vdots \\ \Delta y \end{bmatrix} = \begin{bmatrix} -u(x_i, y_i) \\ \vdots \\ -v(x_i, y_i) \end{bmatrix}$$

$$\begin{bmatrix} 2x_i & 2y_i \\ 1 & -1 \end{bmatrix} \begin{bmatrix} \Delta x \\ \Delta y \end{bmatrix} = \begin{bmatrix} -{x_i}^2 - y_i^2 + 50 \\ -x_i + y_i + 10 \end{bmatrix}$$

Iteration 1

The initial guess

$$(x_i, y_i) = (2, -4)$$

Hence

$$\begin{bmatrix} 2(2) & 2(-4) \\ 1 & -1 \end{bmatrix} \begin{bmatrix} \Delta x \\ \Delta y \end{bmatrix} = \begin{bmatrix} -(2)^2 - (-4)^2 + 50 \\ -(2) + (-4) + 10 \end{bmatrix}$$

$$\begin{bmatrix} 4 & -8 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} \Delta x \\ \Delta y \end{bmatrix} = \begin{bmatrix} 30 \\ 4 \end{bmatrix}$$

Solving the equations by any method of your choice, we get

$$\begin{aligned} \Delta x &= 0.5000 \\ \Delta y &= -3.500 \end{aligned}$$

Since

$$\begin{aligned} \Delta x &= x_2 - x_1 = 0.5000 \\ \Delta y &= y_2 - y_1 = -3.500 \end{aligned}$$

we get

$$\begin{aligned} x_2 &= x_1 + 0.5000 \\ &= 2 + 0.5000 \\ &= 2.5000 \end{aligned}$$

$$\begin{aligned} y_2 &= y_1 + (-3.500) \\ &= -4 + (-3.500) \\ &= -7.500 \end{aligned}$$

The absolute relative approximate errors at the end of the first iteration are

$$\begin{aligned} |\varepsilon_a|_x &= \left| \frac{x_2 - x_1}{x_2} \right| \times 100 \\ &= \left| \frac{2.500 - 2.000}{2.500} \right| \times 100 \\ &= 20.00\% \end{aligned}$$

$$\begin{aligned} |\varepsilon_a|_y &= \left| \frac{y_2 - y_1}{y_2} \right| \times 100 \\ &= \left| \frac{-7.500 - (-4.000)}{-7.500} \right| \times 100 \end{aligned}$$

Iteration 2

The estimate of the root at the end of iteration#1 is

$$(x_2, y_2) = (2.500 - 7.500)$$

Hence

$$\begin{bmatrix} 2(2.500) & 2(-7.500) \\ 1 & -1 \end{bmatrix} \begin{bmatrix} \Delta x \\ \Delta y \end{bmatrix} = \begin{bmatrix} -(2.500)^2 - (-7.500)^2 + 50 \\ -(2.500) + (-7.500) + 10 \end{bmatrix}$$

$$\begin{bmatrix} 5 & -15 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} \Delta x \\ \Delta y \end{bmatrix} = \begin{bmatrix} -12.50 \\ 0 \end{bmatrix}$$

Solving the above equations by any method of your choice gives

$$\Delta x = 1.250$$

$$\Delta y = 1.250$$

Since

$$\Delta x = x_3 - x_2 = 1.250$$

$$\Delta y = y_3 - y_2 = 1.250$$

giving

$$\begin{aligned} x_3 &= x_2 + 1.250 \\ &= 2.500 + 1.250 \\ &= 3.750 \end{aligned}$$

$$\begin{aligned} y_3 &= y_2 + 1.250 \\ &= -7.500 + 1.250 \\ &= -6.250 \end{aligned}$$

The absolute relative approximate error at the end of the second iteration is

$$\begin{aligned} |\epsilon_a|_x &= \left| \frac{x_3 - x_2}{x_3} \right| \times 100 \\ &= \left| \frac{3.750 - 2.5}{3.750} \right| \times 100 \\ &= 33.33\% \end{aligned}$$

$$\begin{aligned} |\epsilon_a|_y &= \left| \frac{y_3 - y_2}{y_3} \right| \times 100 \\ &= \left| \frac{-6.250 - (-7.500)}{-6.250} \right| \times 100 \\ &= 20.00\% \end{aligned}$$

Although not asked in the example problem statement, the estimated values of the root and the absolute relative approximate errors are given below in Table 1.

**Table 1:** Estimate of the root and absolute relative approximate error.

| Iteration number, $i$ | $x_i$ | $y_i$ | $\lvert\epsilon_a\rvert_x\,\%$ | $\lvert\epsilon_a\rvert_y\,\%$ |
|---|---|---|---|---|
| 1 | 2.500 | −7.500 | 20.00 | 46.67 |
| 2 | 3.750 | −6.250 | 33.33 | 20.00 |
| 3 | 4.375 | −5.625 | 14.29 | 11.11 |
| 4 | 4.688 | −5.312 | 6.667 | 5.882 |
| 5 | 4.844 | −5.156 | 3.226 | 3.030 |
| 6 | 4.922 | −5.078 | 1.587 | 1.538 |

The exact solution to which the above scheme is converging to is $(x, y) = (5, -5)$

**Example 2**

A 5 m long gutter is made from a flat sheet of aluminum which is 5 m × 0.21 m. The shape of the gutter cross-section is shown in Figure 1 and is made by bending the sheet at two locations at an angle $\theta$ (Figure 2). What are the values of $s$ and $\theta$, that will maximize the volume capacity of the gutter so that it drains water quickly during a heavy rainfall?

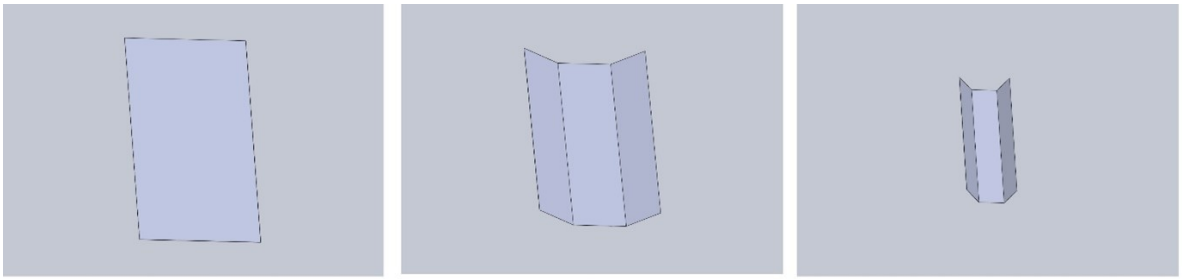


**Figure 1** Sheet metal bent to form a gutter

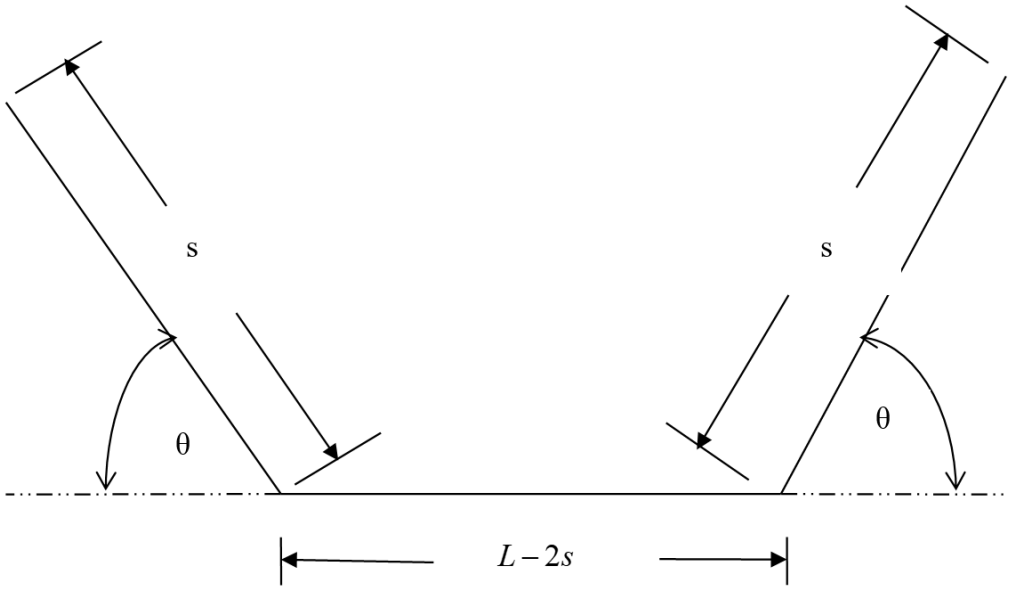



**Figure 2:** Labeling the parameters and points of the gutter

**Solution**

One needs to maximize the cross-sectional area of the gutter. The cross-sectional area $G$ of the gutter is given by

$$\begin{aligned} G &= \text{Area of trapezoid ABCD} \\ &= \frac{1}{2}(\text{AB} + \text{CD})(\text{FC}) \qquad (E2.1) \end{aligned}$$

where

$$\text{AB} \;=\; L - 2\;s \qquad (E2.2)$$

$$\begin{aligned} CD &= AB + EA + BF \\ &= (L - 2s) + s\cos(\theta) + s\cos(\theta) \\ &= (L - 2s) \;+\; 2s\cos(\theta) \qquad (E2.3) \end{aligned}$$

$$\begin{aligned} FC &= BC\sin(\theta) \\ &= s\sin(\theta) \qquad (E2.4) \end{aligned}$$

hence

$$G = \frac{1}{2}(AB \;+ CD)(DE)$$

$$G(s,\theta) = \frac{1}{2}\left[(L-2s) + (L-2s) + 2s\cos(\theta)\right] s\sin(\theta)$$
$$= \left[L - 2s + s\cos(\theta)\right]\ s\sin(\theta) \qquad (E2.5)$$

For example, for $\theta = 0°$, the cross-sectional area of the gutter becomes

$$G = 0$$

as it represents a flat sheet, for $\theta = 180°$, the cross-sectional area of the gutter becomes

$$G = 0$$

as it represents an overlapped sheet, and for $\theta = 90°$, it represents a rectangular cross-sectional area with

$$G = (L-2s)s$$

For the given $L = 0.21$ m,

$$G(s,\theta) = (0.21 - 2s + s\cos\theta)s\sin\theta$$

Then

$$\frac{\partial G}{\partial s} = -s^2\sin^2\theta + (0.21 - 2s + s\cos\theta)s\cos\theta$$

$$\frac{\partial G}{\partial \theta} = (-2 + \cos\theta)s\sin\theta + (0.21 + 2s + s\cos\theta)\sin\theta$$

By solving the equations

$$f_1(s,\theta) = -s\sin^2\theta + (0.21 - 2s + s\cos\theta)s\cos\theta = 0$$

$$f_2(s,\theta) = (-2 + \cos\theta)s\sin\theta + (0.21 - 2s + s\cos\theta)\sin\theta = 0$$

we can find the local minimas and maximas of $G(s,\theta)$. One of the local maximas may also be the absolute maximum. The values of $s$ and $\theta$ that correspond to the maximum of $G(s,\theta)$ are what we are looking for. Can you solve the equations to find the corresponding values of $s$ and $\theta$ for maximum value of $G$? Intuitively, what do you think would be these values of $s$ and $\theta$?

## Appendix A: General matrix form of solving simultaneous nonlinear equations.

The general system of equations is given by

$$\begin{aligned} &f_1(x_1, x_2, ..., x_n) = 0 \\ &f_2(x_1, x_2, ..., x_n) = 0 \\ &... \\ &... \\ &... \\ &f_n(x_1, x_2, ..., x_n) = 0 \end{aligned} \qquad (A.1)$$

We can rewrite in a form as

$$F(\hat{x}) = \hat{0} \qquad (A.2)$$

where

$$F(\hat{x}) = \begin{bmatrix} f_1(\hat{x}) \\ f_2(\hat{x}) \\ . \\ . \\ f_n(\hat{x}) \end{bmatrix} \qquad (A.3)$$

$$\hat{0} = \begin{bmatrix} 0 \\ 0 \\ . \\ . \\ 0 \end{bmatrix} \qquad (A.4)$$

$$\hat{x} = \begin{bmatrix} x_1 \\ x_2 \\ . \\ . \\ x_n \end{bmatrix}^T \qquad (A.5)$$

The Jacobian of the system of equations by using Newton-Raphson method then is

$$J(x_1, x_2, ..., x_n) = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} \cdots \frac{\partial f_1}{\partial x_n} \\ ..... \\ ..... \\ ..... \\ \frac{\partial f_n}{\partial x_1} \cdots \frac{\partial f_n}{\partial x_n} \end{bmatrix} \qquad (A.6)$$

Using the Jacobian for the Newton-Raphson method guess

$$[J]\Delta\hat{x} = -F(\hat{x}) \qquad (A.7)$$

where

$$\Delta\hat{x} = \hat{x}_{\text{new}} - \hat{x}_{\text{old}} \qquad (A.8)$$

Hence

$$\Delta\hat{x} = -[J]^{-1}F(\hat{x})$$

$$\hat{x}_{\text{new}} - \hat{x}_{\text{old}} = -[J]^{-1}F(\hat{x})$$

$$\hat{x}_{\text{new}} = \hat{x}_{\text{old}} - [J]^{-1}F(\hat{x}) \qquad (A.10)$$

Evaluating the inverse of $[J]$ in Equation (A.10) is computationally more intensive than solving Equation (A.7) for $\Delta\hat{x}$and calculating $\hat{x}_{\text{new}}$ as

$$\hat{x}_{\text{new}} = \hat{x}_{\text{old}} + \Delta\hat{x} \qquad (A.11)$$

# Problem Set

## Chapter 03.07:Newton Raphson Method for Solving Simultaneous Nonlinear Equations

(1). Solve the following set of nonlinear equations using Newton-Raphson method.

$$xy = 2$$

$$x^2 + y = 5$$

Start with an initial guess of $(3, 4)$.

(2). Write the drawbacks of Newton-Raphson method and how you would address them.

(3). The following simultaneous nonlinear equations are found while modeling two continuous non-alcoholic stirred tank reactions

$$f_1 = (1 - R)\left[\frac{D}{10(1+\beta_1)} - \varphi_1\right]\exp\left(\frac{10\varphi_1}{1 + 10\varphi_1/\gamma}\right) - \varphi_1$$

$$f_2 = \varphi_1 - (1+\beta_2)\varphi_2 + (1 - R) \times \left[\frac{D}{10} - \beta_1\varphi_1 - (1+\beta_2)\varphi_2\right]\exp\left(\frac{10\varphi_2}{1 + 10\varphi_2/\gamma}\right)$$

Given $\gamma = 1000,\ D = 22,\ \beta_1 = 2,\ \beta_2 = 2,\ R = 0.935$, find a suitable solution for $(\varphi_1, \varphi_2)$, given that $\varphi_i \in [0, 1], i = 1, 2$

Source: Michael J. Hirsch, Panos M. Pardalos, Mauricio G.C Resende, Solving systems of nonlinear equations with continuous GRASP, Nonlinear Analysis: Real World Applications, Volume 10, Issue 4, August 2009, Pages 2000-2006, ISSN 1468-1218, http://dx.doi.org/10.1016/j.nonrwa.2008.03.006

(4). The Lorenz equations are given by

$$\frac{dx}{dt} = \sigma(y - x)$$

$$\frac{dy}{dt} = x(p - z) - y$$

$$\frac{dz}{dt} = xy - \beta z$$

Assuming $\sigma = 3,\ p = 26,\ \beta = 1$, the steady state equations are written as

$$0 = 3(y - x)$$

$$0 = x(26 - z) - y$$

$$0 = xy - z$$

Write the Jacobian for the above system of equations.

(5). Use MATLAB or equivalent program to find the solution to the problem stated in Example 2.

# Chapter 04.00: Physical Problem for Simultaneous Linear Equations

Lesson: General Engineering

## Summary

To find the velocity profile of a rocket requires a solution of simultaneous linear equations.

## Modeling

The upward velocity of a rocket is given at three different times in the following table

| Time, t | Velocity, v |
|---|---|
| s | m/s |
| 5 | 106.8 |
| 8 | 177.2 |
| 12 | 279.2 |

The velocity data is approximated by a polynomial as

**Figure 1** A rocket launched into space

$$v(t) = at^2 + bt + c,\ 5 \leq t \leq 12. \qquad (1)$$

Set up the equations in matrix form to find the coefficients $a,\ b,\ c$ of the velocity profile.

The polynomial is going through three data points $(t_1, v_1), (t_2, v_2)$, and $(t_3, v_3)$ where from the above table

$$t_1 = 5, v_1 = 106.8$$

$$t_2 = 8, v_2 = 177.2$$

$$t_3 = 12, v_3 = 279.2$$

Requiring that $v(t) = at^2 + bt + c$ passes through the three data points gives

$$
\begin{aligned}
v\left(t_1\right) &= v_1 = at_1^2 + bt_1 + c \\
v\left(t_2\right) &= v_2 = at_2^2 + bt_2 + c \\
v\left(t_3\right) &= v_3 = at_3^2 + bt_3 + c
\end{aligned}
$$

Substituting the data $\left(t_1,\ v_1\right),\ \left(t_2,\ v_2\right),\ \left(t_3,\ v_3\right)$ gives

$$
\begin{aligned}
a\left(5^2\right) + b\left(5\right) + c &= 106.8 \\
a\left(8^2\right) + b\left(8\right) + c &= 177.2 \\
a\left(12^2\right) + b\left(12\right) + c &= 279.2
\end{aligned}
$$

or

$$
\begin{aligned}
25a + 5b + c &= 106.8 \\
64a + 8b + c &= 177.2 \\
144a + 12b + c &= 279.2
\end{aligned}
$$

This set of equations can be rewritten in the matrix form as

$$
\begin{bmatrix} 25a+ & 5b+ & c \\ 64a+ & 8b+ & c \\ 144a+ & 12b+ & c \end{bmatrix} = \begin{bmatrix} 106.8 \\ 177.2 \\ 279.2 \end{bmatrix}
$$

The above equation can be written as a linear combination as follows

$$
a\begin{bmatrix} 25 \\ 64 \\ 144 \end{bmatrix} + b\begin{bmatrix} 5 \\ 8 \\ 12 \end{bmatrix} + c\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 106.8 \\ 177.2 \\ 279.2 \end{bmatrix}
$$

and further using matrix multiplications gives

$$
\begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 106.8 \\ 177.2 \\ 279.2 \end{bmatrix}
$$

The solution of the above three simultaneous linear equations will give the value of $a, b, c$.

## Questions

(1) Solve for the values of $a, b, c$.

(2) Verify if you get back the value of the velocity data at $t = 5$ s.

(3) Estimate the velocity of the rocket at $t = 7.5$ s?

(4) Estimate the acceleration of the rocket at $t = 7.5$ s?

(5) Estimate the distance covered by the rocket between $t = 5.5$ s and 8.9 s.

(6) If the following data is given for the velocity of the rocket as a function of time, and you are asked to use a quadratic polynomial to approximate the velocity profile to find the velocity at $t = 16$ s, what data points would you choose and why?

| t | v(t) |
|---|---|
| s | m/s |
| 0 | 0 |
| 10 | 227.04 |
| 15 | 362.78 |
| 20 | 517.35 |
| 22.5 | 602.97 |
| 30 | 901.67 |

# Chapter 04.00: Physical Problem for Simultaneous Linear Equations

Lesson: Chemical Engineering

## Summary

Estimating nickel in organic phase requires solution of simultaneous linear equations

## Modeling

Liquid-liquid extraction depends on the ability of some metal ions to form metal complexes with organic acids. The method is used to separate, concentrate, and purify metals and organic compounds. Liquid-liquid extraction was the technique used to produce weapon grade uranium during the arms race (cold war) era. The technique is also used to recover noble metals used in catalytic processes such as oil refining etc.

In liquid-liquid extraction, the metal ion in the aqueous phase is recovered by mixing the aqueous phase with an organic phase. The metal ion forms a complex with the organic phase and floats on top of the aqueous phase. The organic phase can be decanted and separated from the aqueous phase and the complexed metal ion recovered in a useful form using an acid (nitric acid for nitrates, sulfuric acid for sulfates etc).

A liquid-liquid extraction process conducted in the Electrochemical Materials Laboratory involved the extraction of nickel from the aqueous phase into an organic phase. A typical experimental data from the laboratory is given in Table 1.

**Table 1** Aqueous and organic phase concentration of nickel.

| **Ni aqueous phase (g/l)** | 2 | 2.5 | 3 | 3.5 | 4 |
|---|---|---|---|---|---|
| **Ni organic phase (g/l)** | 8.57 | 10 | 12 | 14 | 15.66 |

Estimate the amount of nickel in organic phase when 2.3 g/l is in the aqueous phase. Use quadratic interpolation.

The polynomial is going through three data points

$$(a_1, g_1)\,,\ (a_2, g_2)\,, \text{ and } (a_3, g_3)$$

where from the above table

$$a_1 = 2, g_1 = 8.57$$

$$a_2 = 2.5, g_2 = 10$$

$$a_3 = 3, g_3 = 12$$

Requiring that $g = x_1 a^2 + x_2 a + x_3$ passes through the three data points gives

$$g(a_1) = g_1 = x_1 a_1^2 + x_2 a_1 + x_3$$

$$g(a_2) = g_2 = x_1 a_2^2 + x_2 a_2 + x_3$$

$$g(a_3) = g_3 = x_1 a_3^2 + x_2 a_3 + x_3$$

Substituting the data

$$(a_1, g_1),\ (a_2,\ g_2),\ (a_3,\ g_3)$$

$$x_1 (2)^2 + x_2 (2) + x_3 = 8.57$$

$$x_1 (2.5)^2 + x_2 (2.5) + x_3 = 10$$

$$x_1 (3)^2 + x_2 (3) + x_3 = 12$$

gives

$$4x_1 + 2x_2 + x_3 = 8.57$$

$$6.25x_1 + 2.5x_2 + x_3 = 10$$

$$9x_1 + 3x_2 + x_3 = 12$$

This set of equations can be rewritten in the matrix form as

$$\begin{bmatrix} 4x_1 + & 2x_2 + & x_3 \\ 6.25x_1 + & 2.5x_2 + & x_3 \\ 9x_1 + & 3x_2 + & x_3 \end{bmatrix} = \begin{bmatrix} 8.57 \\ 10 \\ 12 \end{bmatrix}$$

The above equations can be written as a linear combination as follows

$$x_1 \begin{bmatrix} 4 \\ 6.25 \\ 9 \end{bmatrix} + x_2 \begin{bmatrix} 2 \\ 2.5 \\ 3 \end{bmatrix} + x_3 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 8.57 \\ 10 \\ 12 \end{bmatrix}$$

and further using matrix multiplication gives

$$\begin{bmatrix} 4 & 2 & 1 \\ 6.25 & 25 & 1 \\ 9 & 3 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 8.57 \\ 10 \\ 12 \end{bmatrix}$$

The solution of the above simultaneous linear equations will give the value of $x_1, x_2, x_3$.

## Questions

(1) Verify if you get back the value of the Ni organic phase when the Ni aqueous phase is 2.5 g/l.

(2) Estimate the value of the Ni organic phase when the Ni aqueous phase is 2.78 g/l

(3) Estimate the error between linear interpolation and quadratic interpolation values obtained for nickel in organic phase when 2.78 g/l is in the aqueous phase.

# Chapter 04.00: Physical Problem for Simultaneous Linear Equations

Lesson: Civil Engineering

## Summary

A pressure vessel made of a compounded cylinder has a higher capability to handle internal pressure over a single cylinder. To find this capability, one has to solve a set of simultaneous linear equations.

## Modeling

A pressure vessel can only be subjected to an amount of internal pressure that is limited by the strength of the material used. For example, take a pressure vessel of an internal radius of $a = 5''$ and outer radius, $b = 8''$, made of ASTM 36 steel (yield strength of ASTM 36 steel is 36 ksi). How much internal pressure can this pressure vessel take before it is considered to have failed?

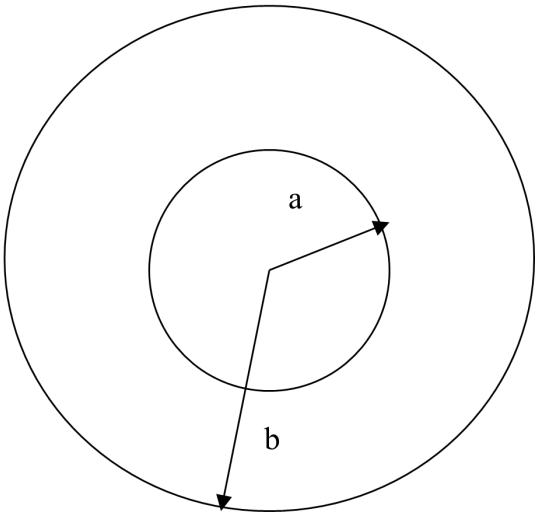



**Figure 1** A single-cylinder pressure vessel with internal radius, $a$ and outer radius, $b$.

The hoop and radial stress in a cylindrical pressure vessel is given by [1]

$$\sigma_r = \frac{a^2 p_i}{b^2 - a^2}\left(1 - \frac{b^2}{r^2}\right) \qquad (1)$$

$$\sigma_\theta = \frac{a^2 p_i}{b^2 - a^2}\left(1 + \frac{b^2}{r^2}\right) \qquad (2)$$

The maximum normal stress anywhere in the cylinder is the hoop stress at the inner radius, $a$

$$\sigma_\theta|\, i\left(\frac{b^2 + a^2}{b^2 - a^2}\right)_{\max} \qquad (3)$$

Assuming a factor of safety of 2, while the yield strength is given as 36 ksi,

$$\frac{36 \times 10^3}{2} = p_i\left(\frac{8^2 + 5^2}{8^2 - 5^2}\right)$$

$$p_i = 7.887 ksi$$

You can see from Equation (3) that even for $b >> a$, the maximum internal pressure one can apply is only $p_i = 18 ksi$. Therefore, what can an engineer do to maximize the internal pressure while keeping the material and radial dimensions the same? They can use a compounded cylinder. They can create a compounded cylinder by shrink fitting one cylinder into another, and hence creating pre-existing favorable stresses to allow more internal pressure. Let us see how that would work?

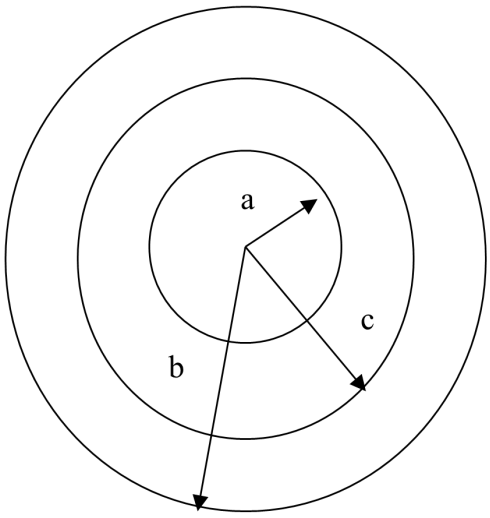



**Figure 2** A compounded cylinder pressure vessel with internal radius, $a$, outer radius, $b$, and interface at $r = c$.

Let us make the compounded cylinder of two cylinders (Figure 2). Cylinder 1 has an internal radius of $a = 5''$, and outer radius $c = 6.5''$, while Cylinder 2 has an internal radius of $c = 6.5''$ and outer radius,

$b = 8''$. Assume that that radial interference, $\delta = 0.007''$ occurs at the interface of a compounded cylinder at $r = c = 6.5''$. How does then one find the pressure that can be applied to the compounded cylinder of internal radius, $a = 5''$ and outer radius, $b = 8''$?

For cylinder 1, the radial displacement, $u_1$ is given by

$$u_1 = c_1 r + \frac{c_2}{r} \qquad (4)$$

the radial stress, $\sigma_r^1$and hoop stress, $\sigma_\theta^1$ by

$$\sigma_r^1 = \frac{E}{1-\nu^2}\left[c_1(1+\nu) - c_2\left(\frac{1-\nu}{r^2}\right)\right] \qquad (5)$$

$$\sigma_\theta^1 = \frac{E}{1-\nu^2}\left[c_1(1+\nu) + c_2\left(\frac{1-\nu}{r^2}\right)\right] \qquad (6)$$

where

$$E = \text{Young's modulus of steel,}$$

$$\nu = \text{Poisson's ratio of steel.}$$

For cylinder 2, the radial displacements, $u_2$is given by

$$u_2 = c_3 r + \frac{c_4}{r} \qquad (7)$$

the radial stress, $\sigma_r^2$ and hoop stress, $\sigma_\theta^2$ by

$$\sigma_r^2 = \frac{E}{1-\nu^2}\left[c_3(1+\nu) - c_4\left(\frac{1-\nu}{r^2}\right)\right] \qquad (8)$$

$$\sigma_\theta^2 = \frac{E}{1-\nu^2}\left[c_3(1+\nu) + c_4\left(\frac{1-\nu}{r^2}\right)\right] \qquad (9)$$

So if one is able to find the four constants, $c_1, c_2, c_3$ and $c_4$, one can find the stresses in the compounded cylinder to be able to find what internal pressure can be applied. So how do we find the four unknown constants?

The boundary and interface conditions are the following.

The radial stress at the inner radius, $r = a$ is the applied internal pressure

$$\sigma_r^1(r = a) = -p_i \qquad (10)$$

The radial stress is continuous at the interface, $r = c$

$$\sigma_r^1(r = c) = \sigma_r^2(r = c) \qquad (11)$$

The radial displacement at the interface, $r = c$ has a jump of the radial interference, $\delta$

$$u_2\left(r=c\right)-u_1\left(r=c\right)=\delta \qquad (12)$$

The radial stress at the outer radius, $r=c$ is

$$\sigma_r^2\left(r=b\right)=0 \qquad (13)$$

This will set up four equations and four unknowns, if we know what internal pressure we are applying. Assume we are applying the same pressure as the single cylinder can take, that is,$p_i = 7.887$ksi and let us see later what stresses it creates in the compounded cylinder.

Assuming $E=30\times10^6$psi, $\nu=0.3$, Equations (10) through (13) become

$$\frac{30\times10^6}{1-0.3^2}\left[c_1\left(1+0.3\right)-c_2\left(\frac{1-0.3}{5^2}\right)\right]=-7.887\times10^3$$

$$\frac{30\times10^6}{1-0.3^2}\left[c_1\left(1+0.3\right)-c_2\left(\frac{1-0.3}{6.5^2}\right)\right]=\frac{30\times10^6}{1-0.3^2}\left[c_3\left(1+0.3\right)-c_4\left(\frac{1-0.3}{6.5^2}\right)\right]$$

$$c_3\left(6.5\right)+\frac{c_4}{6.5}-c_1\left(6.5\right)-\frac{c_2}{6.5}=0.007$$

$$\frac{30\times10^6}{1-0.3^2}\left[c_3\left(1+0.3\right)-c_4\left(\frac{1-0.3}{8^2}\right)\right]=0 \qquad (14a-d)$$

Writing the Equations (14a-d) in matrix form, we get

$$\begin{bmatrix} 4.2857\times10^7 & -9.2307\times10^5 & 0 & 0 \\ 4.2857\times10^7 & -5.4619\times10^5 & -4.2857\times10^7 & 5.4619\times10^5 \\ -6.5 & -0.15384 & 6.5 & 0.15384 \\ 0 & 0 & 4.2857\times10^7 & -3.6057\times10^5 \end{bmatrix}\begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{bmatrix}=\begin{bmatrix} -7.887\times10^3 \\ 0 \\ 0.007 \\ 0 \end{bmatrix} \qquad (15)$$

Solving these four simultaneous linear equations, we can find the four constants.

**References**

(1) A.C. Ugural, S.K. Fenster, Advanced strength and applied elasticity, Third Edition, Prentice Hall, New York, 1995.

(2) J.E. Shigley, C.R. Mischke, Chapter 19 - Limits and fits, Standard handbook of machine design, McGraw-Hill, New York, 1986.

## Questions

(1) Find the unknown constants of Equation (14) using different numerical methods.

(2) Knowing that the critical points in the compounded cylinder are $r = a, c-, c+,$ and $b$, find the maximum hoop stress in the compounded cylinder. What is its value compared to the maximum hoop stress allowable of 18 ksi?

(3) Find the maximum internal pressure you can apply to the compounded cylinder? Compare it with the maximum possible internal pressure for a single cylinder of the same dimensions.

(4) The radial interference at the interface is created by making the inner cylinder 1 to have a larger outer radius than the inner radius of cylinder 2. Standard interference fits dictate the limits of these dimensions. If cylinder 2 is fit into cylinder 1, there is an upper and lower limit by which the nominal diameter of each cylinder varies at the interface. This limit L in thousands of an inch, is given by [2]

$$L = CD^{1/3}$$

where $D$ (nominal diameter) is in inches, and the coefficient $C$,based on the type of fit, is given in Table 1 below.

| Cylinder | Limit | Class of fit | |
|---|---|---|---|
| | | FN2 | FN3 |
| 1 | Lower | 0.000 | 0.000 |
| | Upper | 0.907 | 0.907 |
| 2 | Lower | 2.717 | 3.739 |
| | Upper | 3.288 | 4.310 |

Assuming FN2 fit at the interface, find the maximum internal pressure you would recommend.

# Chapter 04.00: Physical Problem for Simultaneous Linear Equations

Lesson: Computer Engineering

## Summary

To infer the surface shape from images requires solution of simultaneous linear equations.

## Modeling

Human vision has the remarkable ability to infer 3D shapes from 2D images. When we look at 2D photographs or TV we do not see them as 2D shapes, rather as 3D entities with surfaces and volumes. Perception research has unraveled many of the cues that are used by us. The intriguing question is can we replicate some of these abilities on a computer? To this end, in this assignment we are going to look at one way to engineer a solution to the 3D shape from 2D images problem. Apart from the pure fun of solving a new problem, there are many practical applications of this technology such as in automated inspection of machine parts, inference of obstructing surfaces for robot navigation, and even in robot assisted surgery.

Image is a collection of gray level values at set of predetermined sites known as pixels, arranged in an array. These gray level values are also known as image intensities. The registered intensity at an image pixel is dependent on a number of factors such as the lighting conditions, surface orientation and surface material properties. The nature of lighting and its placement can drastically affect the appearance of a scene. Can we infer shape of the underlying surface given images as in the images below in Figure 1.

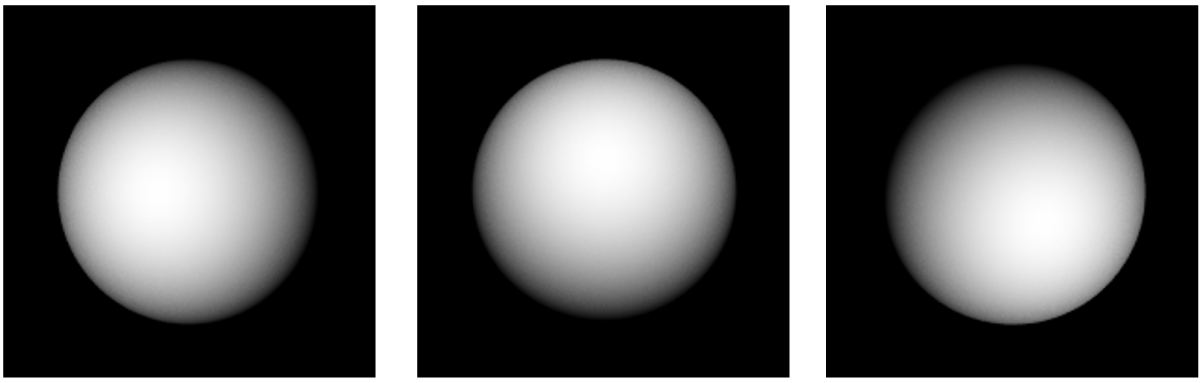


**Figure 1** Images of a surface taken with three different light directions. Can you guess the shape of the underlying surface?

**Physics of the Problem**

To be able to reconstruct the shape of the underlying surface, we have to first understand the image formation process. The simplest image formation model assumes that the camera is far away from the scene so that we have assume that the image is a scaled version of the world. The simplest light model consists of a point light that is far away. This is not an unrealistic assumption. A good example of such a light source is the sun. We also assume that the surface is essentially matte that reflects lights uniformly in all directions, unlike specular (or mirror-like surfaces). These kinds of surfaces are called Lambertian surfaces; examples include walls, and carpet.

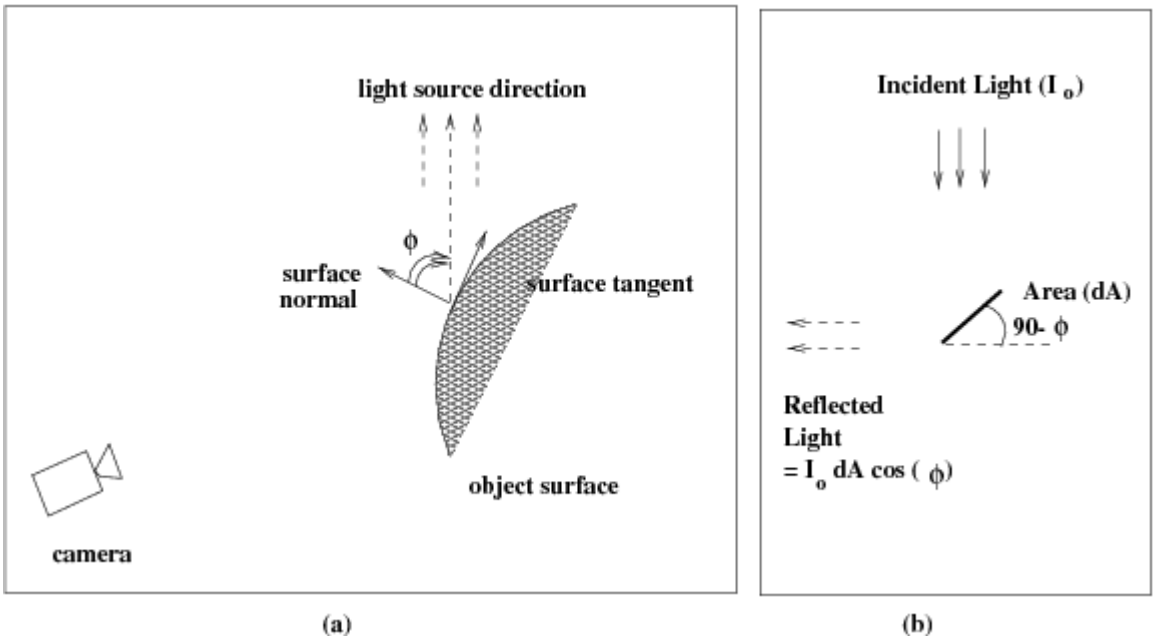



**Figure 2** Relationship between the surface and the light source. (b) Amount of light reflected by an elemental area (dA) is proportional to the cosine of the angle between the light source direction and the surface normal

The image brightness of a Lambertian surface is dependent on the local surface orientation with respect to the light source. Since the point light source is far away we will assume that the incident light is locally uniform and comes from a single direction, i.e, the light rays are parallel to each other. This situation is illustrated in Figure 2. Let the incident light intensity be denoted by $I_0$. Also let the angle between the light source and local surface orientation be denoted by $\varphi$. Then the registered image intensity, $I(u, v)$, of that point is given by

$$I(u, v) = I_0 \rho \cos(\varphi) = I_0 \rho (n_x s_x + n_y s_y + n_z s_z)$$

where the surface normal, $n$, and the light source direction, $s$, are given by:

$$n = \begin{bmatrix} n_x \\ n_y \\ n_z \end{bmatrix}, \; s = \begin{bmatrix} s_x \\ s_y \\ s_z \end{bmatrix}$$

and $\rho$ is a number capturing the surface reflection property at the location $(x, y)$, and It is referred to as the surface albedo. Black surfaces tend to have low albedo and white surfaces tend to high albedo.

Note that the registered intensity in the image does not depend on the camera location because a Lambertian surface reflects light equally in all directions. This would not be true for specular surfaces and the corresponding image formation equation would involve the viewing direction.

The variation in image intensity is essentially dependent on the local surface orientation. If the surface normal and the light source directions are aligned then we have the maximum intensity and the observed intensity of the lowest when the angle between the light source and the local surface orientation is $90^0$. Thus, given the knowledge of light source and the local surface albedo, it should be possible to infer the local surface orientation from the local image intensity variations. This is what we explore next.

The mapping of surface orientation to image intensity is many to one. Thus, it is not possible to infer the surface orientation from just one intensity image in the absence of any other knowledge. We need multiple samples per point in the scene. How many samples do we need? The vector specifying the surface normal has three components, which implies that we need three. So, we engineer a setup to infer surface orientation from image intensities. Instead of just one image of a scene, let us take three images of the same scene, without moving either the camera or the scene, but with three different light sources, turned on one at a time. These three different light sources are placed at different locations in space. Let the three light source directions, relative to the camera, be specified by the vectors

$$s^1 = \begin{bmatrix} s_x^1 \\ s_y^1 \\ s_z^1 \end{bmatrix}, \; s^2 = \begin{bmatrix} s_x^2 \\ s_y^2 \\ s_z^2 \end{bmatrix}, \; s^3 = \begin{bmatrix} s_x^3 \\ s_y^3 \\ s_z^3 \end{bmatrix}$$

Corresponding pixels in the three images would have three different intensities $I_1$, $I_2$, and $I_3$ for three light source directions. Let the surface normal corresponding to the pixel under consideration be denoted by

$$n = \begin{bmatrix} n_x \\ n_y \\ n_z \end{bmatrix}$$

Assuming Lambertian surfaces, the three intensities can be related to the surface normal and the light source directions

$$I_i = I_0 \rho (n_x s_x^i + n_y s_y^i + n_z s_z^i), \; \forall i = 1, 2, 3$$

In these three equations, the known variables are the intensities, $I_1$, $I_2$, $I_3$, and the light source directions, $s_1$, $s_2$, $s_3$. The unknowns are the incident intensity, $I_3$, surface albedo, $\rho$ and the surface normal, $n$. These unknowns can be bundled into three unknown variables, $m_x = I_0 \rho n_x$, $m_y = I_0 \rho n_y$, and $m_z = I_0 \rho n_z$. We will recover the surface normal by normalizing the recovered $m$ vector, using the fact that the magnitude of the normal is one. The normalizing constant will give us the product $I_0 \rho$. Thus, for each point in the image, we have three simultaneous equations in three unknowns.

$$\begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} s_x^1 & s_y^1 & s_z^1 \\ s_x^2 & s_y^2 & s_z^2 \\ s_x^3 & s_y^3 & s_z^3 \end{bmatrix} \begin{bmatrix} m_x \\ m_y \\ m_z \end{bmatrix}$$

**Worked Out Example**

Consider the middle of the sphere in Figure 1. We know that the surface normal points towards the camera (i.e. towards the viewer). Assume a 3D coordinate system centered at the camera with $x$

-direction along the horizontal direction, $y$ -direction along the vertical direction, and $z$ -direction is away from the camera into the scene. Then the actual surface normal of the middle of the sphere is given by [0, 0, -1] – the negative denotes that it point in the direction opposite the z-axis. Let us see how close our estimate is to the actual value.

The intensity of the middle of the sphere in the three views, $I_1 = 247$, $I_2 = 248$, and $I_3 = 239$, respectively. The light directions for the three images are along [5, 0, -20], [0, 5, -20], and [-5, -5, -20], respectively. Normalizing the three vectors we get the normal directions towards the lights and construct the 3 by 3 matrix

$$\begin{bmatrix} s_x^1 & s_y^1 & s_z^1 \\ s_x^2 & s_y^2 & s_z^2 \\ s_x^3 & s_y^3 & s_z^3 \end{bmatrix} = \begin{bmatrix} 0.2425 & 0 & -0.9701 \\ 0 & 0.2425 & -0.9701 \\ -0.2357 & -0.2357 & -0.9428 \end{bmatrix}$$

Solving the corresponding 3 simultaneous equations, we arrive the following solution for the $m$ -vector:

$$\begin{bmatrix} m_x \\ m_y \\ m_z \end{bmatrix} = \begin{bmatrix} 0.0976 \\ 4.2207 \\ -254.5774 \end{bmatrix}$$

Normalizing this vector we get the surface normal

$$\begin{bmatrix} n_x \\ n_y \\ n_z \end{bmatrix} = \begin{bmatrix} 0.0004 \\ 0.0166 \\ -0.9999 \end{bmatrix}$$

The corresponding normalizing constant is 254.6124, which is the product of the intensity of the illumination and the surface (albedo) reflectance factor ($I_0\rho$). Compare the estimate of the surface normal to the actual one. The difference is quantization effects – in images we can represent intensities as a finite sized integer – 8-bit integers in our case.

We can repeat the above computations for each point in the scene to arrive at estimates of the corresponding surface normals. Figure 3(a) is a visualization of the surface normals thus estimated as a vector field. In Figure 3(b), we see the product $I_0\rho$ visualized as image intensities. As expected, it is the same at all points on the sphere. In another problem module, how do we recover the underlying surface from these surface normals?

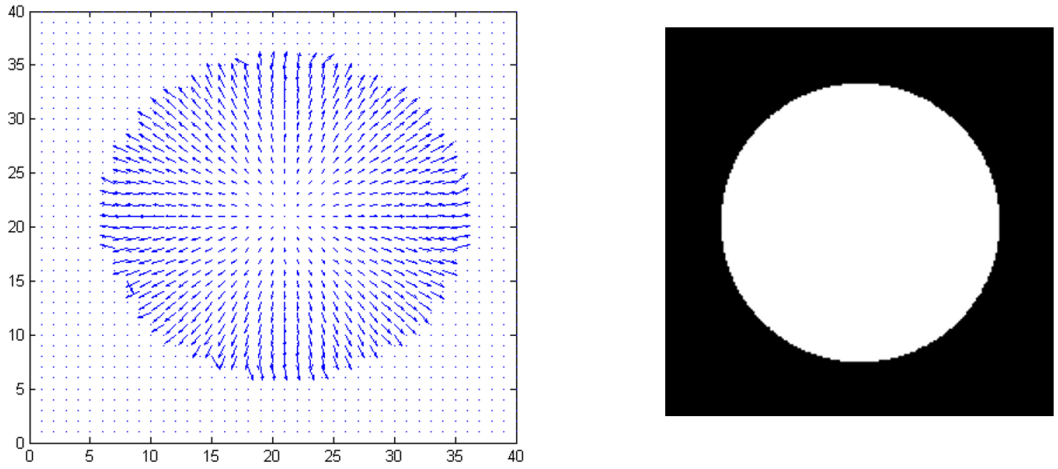



**Figure 3** (a) Recovered surface normal at each point on the sphere. Just the first two components of the vectors are shown as a arrows. (b) Recovered product $I_0\rho$ for all points

## Questions

(1) What can you infer about the surface normal for the brightest point in the image? What about the darkest point in the scene?

(2) What assumptions do you have to make to make the above inferences?

# Chapter 04.00: Physical Problem for Simultaneous Linear Equations

Lesson: Electrical Engineering

## Summary

Designing three-phase loads in AC systems require the solution of simultaneous linear equations.

## Modeling

Three-phase AC systems are the norm for most industrial applications. AC power in the form of voltage and current it delivered from the power company using three-phase distribution systems and many larger loads are three-phase loads in the form of motors, compressors, or similar. Sources and loads can be configured in either wye (where sources or loads are connected from line to neutral/ground) or delta (where sources or loads are connected from line to line) configurations and mixing between the types is common. Figure 1 shows the general wiring of a wye-wye three-phase system modeling all of the impedances typically found in such a system.

During the typical analysis undertaken in most circuits textbooks, it is assumed that the system is entirely balanced. This means that all the source, line, and load impedances are equivalent, that is,

$$Z_a = Z_b = Z_c$$

$$Z_{aA} = Z_{bB} = Z_{cC}$$

$$Z_{AN} = Z_{bN} = Z_{CN}$$

Under this assumption, the circuit is then typically reduced to a single-phase equivalent circuit model and the resultant circuit is solved with a single loop equation. What happens, however, when the system is unbalanced? Typically because the three load impedances $Z_{AN}, Z_{BN}$ and $Z_{CN}$ are not equal, which results in different currents through each load, is often measured in terms of the percentage difference between the load currents.

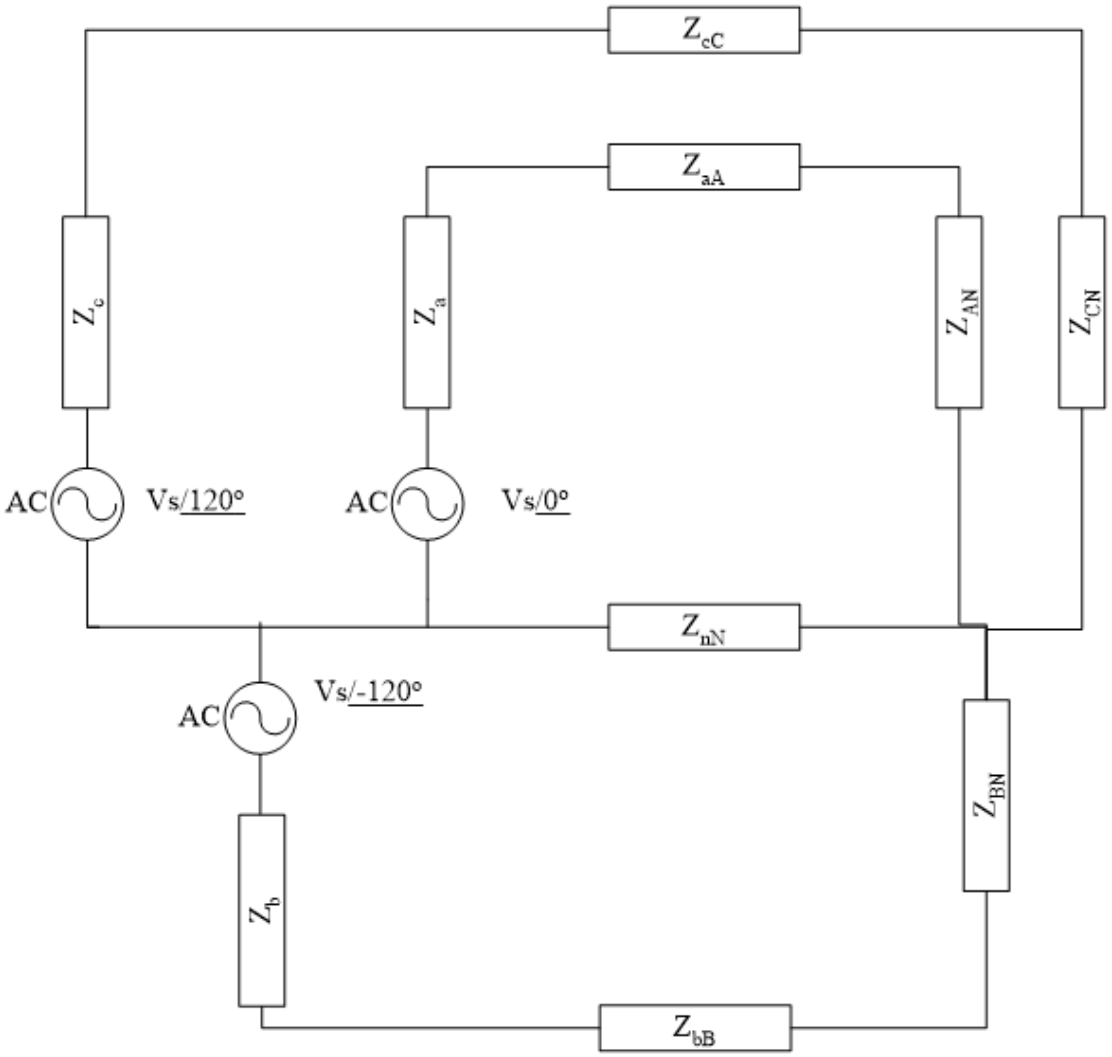



**Figure 1** A Three-Phase Wye-Wye System with Positive Phase Sequence

Creating an imbalance in a three-phase system is not all that difficult. Consider a small business operating in an isolated leg of the power grid so that localized aspects of a load are not "balanced" by other neighboring loads. Let's assume that the primary load for this system is a 45 kVA set of three-phase motors at 0.8 power factor lagging and, further, that the electrician that did the wiring for the lighting mistakenly connected two banks of lights to the A phase, one to the B phase and none to the C phase creating an imbalance in the system. Each of these lighting loads is 1500 W. The load for this system is shown in Figure 2.

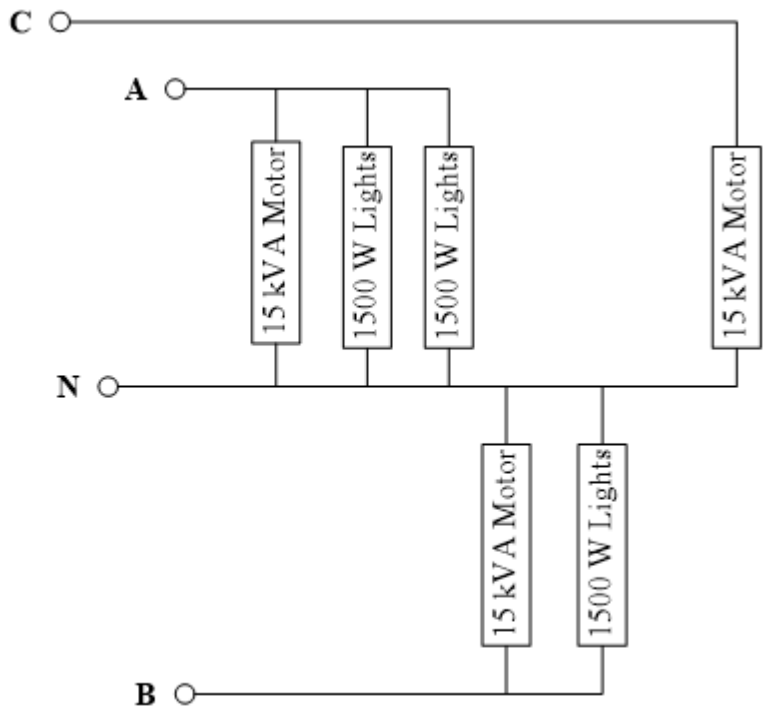



**Figure 2** Model of the System Load

The impedance of each of the loads can be determined by examining the power consumed in each phase of the system

$$A : 3000 + 15000/\underline{-36.87^\circ} = 3000 + 12000 - j9000 = 15000 - j9000$$
$$= 17.49/\underline{-30.96^\circ} kVA$$

$$B : 1500 + 15000/\underline{-36.87^\circ} = 1500 + 12000 - j9000 = 13500 - j9000$$
$$= 16.22/\underline{-33.69^\circ} kVA$$

$$C : 15.00/\underline{-36.87^\circ} kVA$$

Converting these to impedances using the formula

$$S = \frac{|V|^2}{Z} \text{ with } V = 120V \text{ yields :}$$

$$Z_{AN} = 0.8233/\underline{30.96^\circ}\Omega = R_A + jX_A = 0.7060 + j0.4236\Omega$$

$$Z_{BN} = 0.8878/\underline{33.39^\circ}\Omega = R_B + jX_B = 0.7387 + j0.4925\Omega$$

$$Z_{CN} = 0.9600/\underline{36.87^\circ}\Omega = R_C + jX_C = 0.7680 + j0.5760\Omega$$

For the rest of this analysis we will assume that each phase of the system has an equivalent source and line impedance of $R_s + jX_s = 0.0300 + j0.0200\Omega$ and that the ground return wire has an impedance of$R_n + jZ_n = 0.0100 + j0.0080\Omega$. This yields the equivalent circuit of Figure 3.

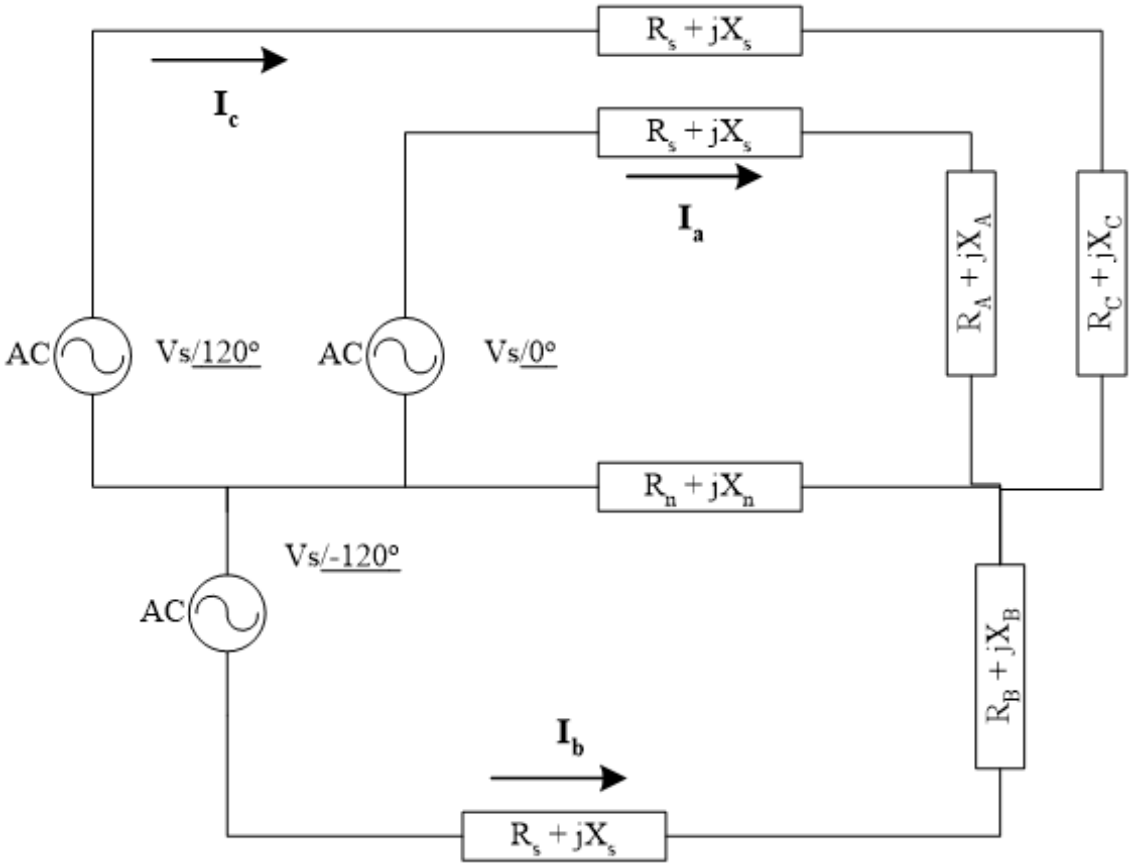



**Figure 3** Equivalent Circuit Model for the Working Problem

The circuit can be analyzed using three loop equations using the currents $I_a, I_b$, and $I_c$ shown in Figure 3. For loop A this yields the complex equation:

Loop A:

$$-V_s/\underline{0^\circ} + I_a\left(R_s + jX_s + R_A + jX_A\right) + \left(I_a + I_b + I_c\right)\left(R_n + jX_n\right) = 0$$

with loops B and C yielding similar results. Assuming that our simultaneous equation solver is not capable of handling complex numbers we can turn the loop A equation into two separate non-complex equations addressing both the real and imaginary parts. Using $I_a = I_{ar} + jI_{ai}$ and collecting terms yields:

Real A:

$$I_{ar}\left(R_s + R_A + R_n\right) - I_{ai}\left(X_s + X_A + X_n\right) + I_{br}R_n - I_{bi}X_n + I_{cr}R_n - I_{ci}X_n = 120 \quad (1)$$

Imaginary A:

$$I_{ar}\left(X_s + X_A + X_n\right) + I_{ai}\left(R_s + R_A + R_n\right) + I_{br}X_n + I_{bi}R_n + I_{cr}X_n + I_{ci}R_n = 0 \quad (2)$$

Applying the same analysis to the B and C loops yields the remaining equations for the system.

Real B:

$$I_{ar}R_n - I_{ai}X_n + I_{br}\left(R_s + R_B + R_n\right) - I_{bi}\left(X_s + X_B + X_n\right) + I_{cr}R_n - I_{ci}X_n = -60 \quad (3)$$

Imaginary B:

$$I_{ar}X_n + I_{ai}R_n + I_{br}\left(X_s + X_B + X_n\right) + I_{bi}\left(R_s + R_B + R_n\right) + I_{cr}X_n + I_{ci}R_n = -103.9 \quad (4)$$

Real C:

$$I_{ar}R_n - I_{ai}X_n + I_{br}R_n - I_{bi}X_n + I_{cr}\left(R_s + R_C + R_n\right) - I_{ci}\left(X_s + X_C + X_n\right) = -60 \quad (5)$$

Imaginary C:

$$I_{ar}X_n + I_{ai}R_n + I_{br}X_n + I_{bi}R_n + I_{cr}\left(X_s + X_C + X_n\right) + I_{ci}\left(R_s + R_C + R_n\right) = 103.9 \quad (6)$$

This yields a system of six linear equations and six unknowns ($I_{ar}, I_{ai}, I_{br}, I_{bi}, I_{cr},\ and\ I_{ci}$) that can be solved by any conventional means. This is shown in matrix form in Figure 4.

$$\begin{bmatrix} 0.7460 & -0.4516 & 0.0100 & -0.0080 & 0.0100 & -0.0080 \\ 0.4516 & 0.7460 & 0.0080 & 0.0100 & 0.0080 & 0.0100 \\ 0.0100 & -0.0080 & 0.7787 & -0.5205 & 0.0100 & -0.0080 \\ 0.0080 & 0.0100 & 0.5205 & 0.7787 & 0.0080 & 0.0100 \\ 0.0100 & -0.0080 & 0.0100 & -0.0080 & 0.8080 & -0.6040 \\ 0.0080 & 0.0100 & 0.0080 & 0.0100 & 0.6040 & 0.8080 \end{bmatrix} \begin{bmatrix} I_{ar} \\ I_{ai} \\ I_{br} \\ I_{bi} \\ I_{cr} \\ I_{ci} \end{bmatrix} = \begin{bmatrix} 120.0 \\ 0.000 \\ -60.00 \\ -103.9 \\ -60.00 \\ 103.9 \end{bmatrix} \quad (7)$$

**Figure 4** Complete System of Equations

Once the currents are known it is a simple procedure to determine the voltages across the three motor terminals ($V_{AN}, V_{BN},\ and\ V_{CN}$) using Ohm's Law.

$$V_{AN} = \left(I_{ar} + jI_{ai}\right)\left(R_A + jX_A\right)$$

$$V_{BN} = \left(I_{br} + jI_{bi}\right)\left(R_B + jX_B\right)$$

$$V_{CN} = \left(I_{cr} + jI_{ci}\right)\left(R_C + jX_C\right)$$

To better evaluate the imbalance, the percentage difference in currents through the actual load elements is more often considered. The best way to think of this as three people pulling and pushing together. If they do not pull and push in balance, things can become unstable. In the case of a three-phase motor, this can result in significant wobble with corresponding wear in the bearings and other parts. To determine the current through each load is again computed using Ohm's Law.

$$I_{Aload} = \frac{V_{AN}}{Z_{3\varphi}},\ I_{Bload} = \frac{V_{BN}}{Z_{3\varphi}},\ I_{Cload} = \frac{V_{CN}}{Z_{3\varphi}}$$

$$Z_{3\varphi} = 0.7680 + j0.5760\Omega$$

where $Z_{3\varphi}$ was computed earlier as $Z_C$. Do not forget that the lighting loads are separate.

## Questions

(1) What would be the ramifications of solving the problem directly using the three complex linear equations? Could we do it using an approach like Gauss-Jordan Elimination? What about some of the other numerical methods used to solve simultaneous linear equations?

(2) This problem is only interesting if the ground return leg $Z_{nN}$is non-zero. Otherwise, we have three loop equations that are completely independent of each other and can be solved directly. Why is that the case?

(3) A much more interesting and practical problem occurs when the motor load is a Delta configuration. Since it does not have the ground return line in the middle it results in additional loop equations. Sketch the equivalent circuit for a system with a Wye source and a mix of Delta and Wye loads. Write the set of equations that result from this system. Solve them.

# Chapter 04.00: Physical Problem for Simultaneous Linear Equations

Lesson: Industrial Engineering

## Summary

Maximizing assembly line yield.

## Modeling

A company that manufactures small toys recently received a contract from a fast-food company, to manufacture three toys, at a low cost, to be added to kid's lunches. The company has to manufacture toys for boys (toy B), girls (toy G) and a generic version (toy U). Furthermore, based on the demand and demographics, the fast food company has specified that 5% more girl's toys than boy's toys should be produced, and that there is no constraint is specified on the number of generic toys. The components of each toy (B, G, and U) must be injection molded out of plastic (Process 1) and then assembled (Process 2). After the toys have been designed, it is determined that the following production times will be needed on each toy:

- Toy B will require 2 minutes for injection molding 6 toys and 1 minute for assembling all 6 toys.
- Toy G will require 2 minutes for injection molding 12 toys and 8 minutes for assembling all 12 toys.
- Toy U will require 4 minutes for injection molding 6 toys and 2 minutes for assembling all 6 toys.

Note that because of daily scheduled maintenance of the injection molding machine, it can only run for a maximum of 756 out of 1440 minutes a day, whereas the assembly line works 3 shifts a day with scheduled breaks for a maximum of 1260 out of 1440 minutes per day. An industrial engineer working for the toy company is asked to determine the production schedule that maximizes, on a daily basis, the use of both the injection molding machine and the assembly line.

### Background

The variables need to solve the problem are listed in Table 1

**Table 1**. Variables for these different toys.

| Variable | Toy B | Toy G | Toy U |
|---|---|---|---|
| Time (minutes) required in Process 1 per toy | $B_1$ | $G_1$ | $U_1$ |
| Time (minutes) required in Process 2 per toy | $B_2$ | $G_2$ | $U_2$ |
| Total manufactured per day | $X_B$ | $X_G$ | $X_U$ |

The total time required to produce toys in process 1 (injection molding) has to be equal to the maximum minutes per day that process 1 can run, that is,

$$B_1X_B + G_1X_G + U_1X_U = M_1$$

where $M_1$ is the maximum minutes that process 1 can run per day. Similarly, for process 2 (assembly),

$$B_2X_B + G_2X_G + U_2X_U = M_2$$

where $M_2$ is the maximum minutes that process 2 can run per day. Finally, the constraint of 5% more girl's toys than boy's toys is expressed as

$$1.05X_B = X_G \text{ or}$$

$$1.05X_B - X_G = 0$$

The previous three simultaneous linear equations can be expressed in matrix form as follows

$$\begin{bmatrix} B_1 & G_1 & U_1 \\ B_2 & G_2 & U_2 \\ 1.05 & -1 & 0 \end{bmatrix} \begin{Bmatrix} X_B \\ X_G \\ X_U \end{Bmatrix} = \begin{Bmatrix} M_1 \\ M_2 \\ 0 \end{Bmatrix}$$

The input variables to the preceding simultaneous linear equations are

$$B_1 = \frac{2}{6} = \frac{1}{3} \text{ toy per minute}$$

$$B_2 = \frac{1}{6} \text{ toy per minute}$$

$$G_1 = \frac{2}{12} = \frac{1}{6} \text{ toy per minute}$$

$$G_2 = \frac{8}{12} = \frac{2}{3} \text{ toy per minute}$$

$$U_1 = \frac{4}{6} = \frac{2}{3} \text{ toy per minute}$$

$$U_2 = \frac{2}{6} = \frac{1}{3} \text{ toy per minute}$$

$$M_1 = 756 \text{ minutes per day}$$

$$M_2 = 1260 \text{ minutes per day}$$

Substituting onto the matrix representation of the simultaneous linear equations yields

$$\frac{1}{6}\begin{bmatrix} 2 & 1 & 4 \\ 1 & 4 & 2 \\ 6.3 & -6 & 0 \end{bmatrix}\begin{Bmatrix} X_B \\ X_G \\ X_U \end{Bmatrix} = \begin{Bmatrix} 756 \\ 1260 \\ 0 \end{Bmatrix}$$

One needs to solve these simultaneous linear equations to find the number of boys, girl and unisex toys for maximizing the manufacturing facility yield.

# Chapter 04.00: Physical Problem for Simultaneous Linear Equations

Lesson: Mechanical Engineering


## Summary

To find the diametrical contraction during shrink-fitting in a trunnion requires one to first find how the coefficient of linear thermal expansion of steel is related to temperature. The relation is given through a second-order polynomial, and calculating the coefficients of the polynomial requires solving a set of simultaneous linear equations.


## Modeling

To make the fulcrum (Figure 1) of a bascule bridge, a long hollow steel shaft called the trunnion is shrink fit into a steel hub. The resulting steel trunnion-hub assembly is then shrunk-fit into the girder of the bridge.

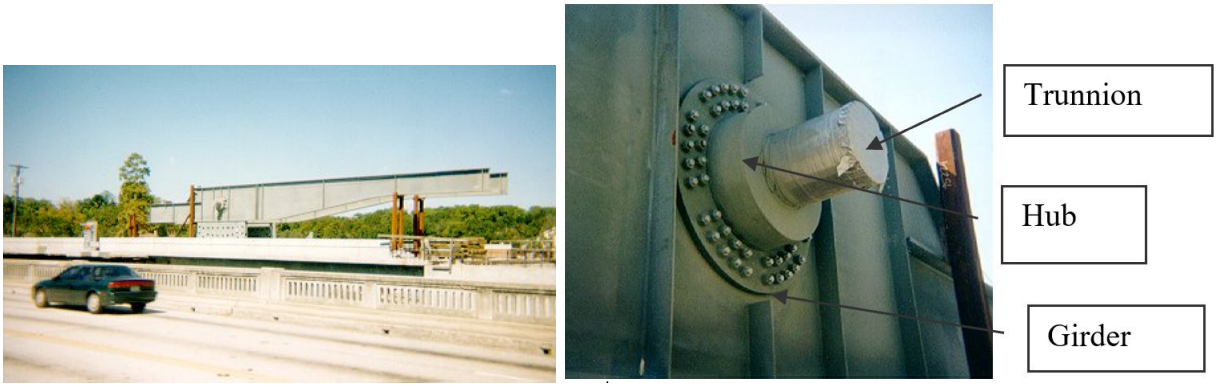



**Figure 1** Trunnion-Hub-Girder (THG) assembly.

This is done by first immersing the trunnion in a cold medium such as dry-ice/alcohol mixture. After the trunnion reaches the steady-state temperature, that is, the temperature of the cold medium, the trunnion outer diameter contracts. The trunnion is taken out of the medium and slid through the hole of the hub (Figure 2).

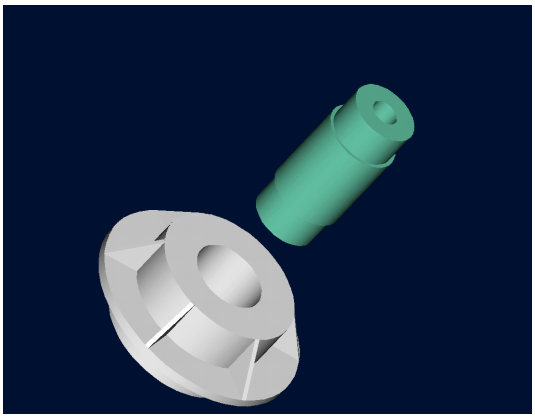


**Figure 2** Trunnion slid through the hub after contracting

When the trunnion heats up, it expands and creates an interference fit with the hub. In 1995, on one of the bridges in Florida, this assembly procedure did not work as designed. Before the trunnion could be inserted fully into the hub, the trunnion got stuck. Luckily the trunnion was taken out before it got stuck permanently. Otherwise, a new trunnion and hub would be needed to be ordered at a cost of $50,000. Coupled with construction delays, the total loss could have been more than a hundred thousand dollars.

Why did the trunnion get stuck? This was because the trunnion had not contracted enough to slide through the hole. Can you find out why?

A hollow trunnion of an outside diameter $12.363''$ is to be fitted in a hub of an inner diameter $12.358''$. The trunnion was put in dry-ice/alcohol mixture (temperature of the fluid - dry ice/alcohol mixture is $-108°$F) to contract the trunnion so that it can be slided through the hole of the hub. To slide the trunnion without sticking, a diametrical clearance of at least $0.01''$ is required between the trunnion and the hub. Assuming the room temperature is $80°$F, is immersing it in dry-ice/alcohol mixture a correct decision?

**Solution**

To calculate the contraction in the diameter of the trunnion, the coefficient of linear thermal expansion at room temperature is used. In that case, the reduction, $\Delta D$ in the outer diameter of the trunnion is

$$\Delta D = D\alpha\Delta T \quad (1)$$

where

$$D = \text{outer diameter of the trunnion,}$$

$$\alpha = \text{coefficient of linear thermal expansion at room temperature, and}$$

$$\Delta T = \text{change in temperature,}$$

Given

$$D = 12.363''$$

$$\alpha = 6.817 \times 10^{-6} in/in/^{\circ}\text{F at } 80^{\circ}\text{F}$$

$$\begin{aligned}\Delta T &= T_{fluid} - T_{room} \\ &= -108 - 80 \\ &= -188^{\circ}\text{F}\end{aligned}$$

where

$$T_{fluid} = \text{temperature of dry-ice/alcohol mixture}$$

$$T_{room} = \text{room temperature}$$

the reduction in the trunnion outer diameter is given by

$$\begin{aligned}\Delta D &= (12.363)\left(6.47 \times 10^{-6}\right)(-188) \\ &= -0.01504''\end{aligned}$$

So, the trunnion is predicted to reduce in diameter by $0.01504''$. But is this enough reduction in diameter? As per specifications, he needs the trunnion to contract by

$$\begin{aligned}&= \text{trunnion outside diameter} - \text{hub inner diameter} + \text{diametral clearance} \\ &= 12.363'' - 12.358'' + 0.01'' \\ &= 0.015''\end{aligned}$$

So, according to his calculations, immersing the steel trunnion in dry-ice/alcohol mixture gives the desired contraction of $0.015''$ as we predict a contraction of $0.01504''$.

But as shown in Figure 3, the coefficient of linear thermal expansion of steel decreases with temperature and is not constant over the range of temperature the trunnion goes through. Hence, Equation 1 would overestimate the thermal contraction.

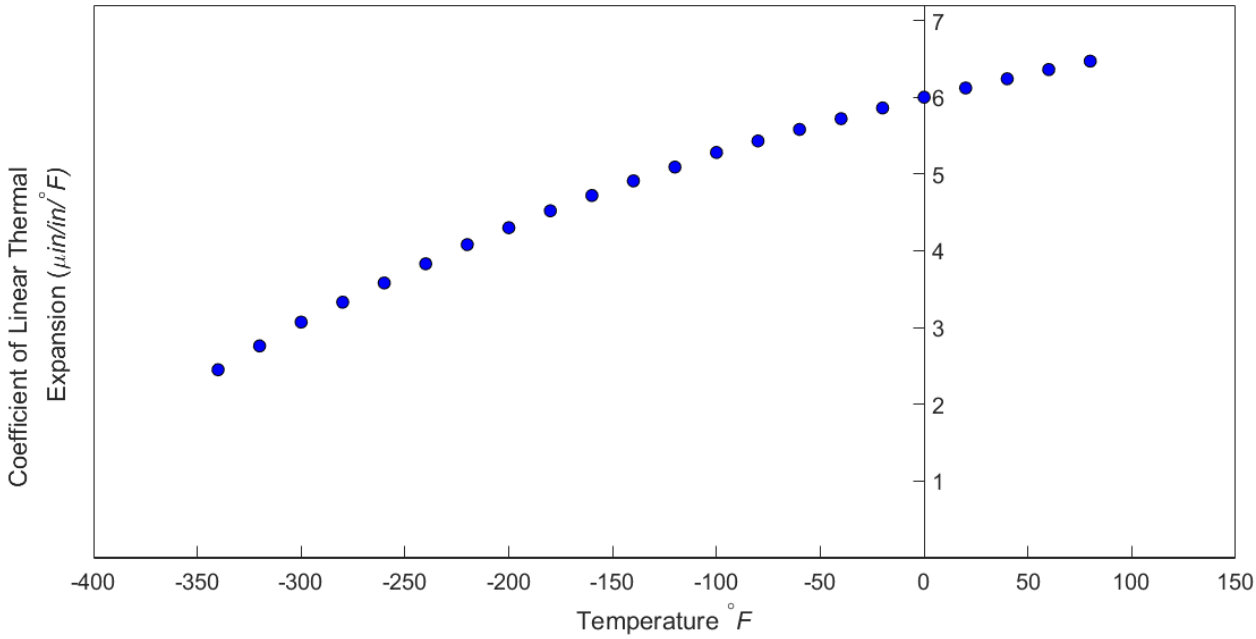



**Figure 3** Varying coefficient of linear thermal expansion as a function of temperature for cast steel.

The contraction in the diameter for the trunnion for which the coefficient of linear thermal expansion varies as a function of temperature is given by

$$\Delta D = D \int_{T_{room}}^{T_{fluid}} \alpha dT \quad (2)$$

So one needs to find the curve to find the coefficient of linear thermal expansion as a function of temperature. This curve is found by regression, where we best fit a polynomial through the data given in Table 1.

**Table 1** Coefficient of linear thermal expansion as a function of temperature.

| Temperature | Coefficient of linear thermal expansion |
|---|---|
| °F | $\mu$in/in/°F |
| 80 | 6.47 |
| 60 | 6.36 |
| 40 | 6.24 |
| 20 | 6.12 |
| 0 | 6.00 |
| −20 | 5.86 |
| −40 | 5.72 |
| −60 | 5.58 |
| −100 | 5.28 |
| −120 | 5.09 |
| −140 | 4.91 |
| −160 | 4.72 |
| −180 | 4.52 |
| −200 | 4.30 |

| Temperature | Coefficient of linear thermal expansion |
|---|---|
| −220 | 4.08 |
| −240 | 3.83 |
| −260 | 3.58 |
| −280 | 3.33 |
| −300 | 3.07 |
| −320 | 2.76 |
| −340 | 2.45 |

Assuming that the coefficient of linear thermal expansion is related to temperature by a second-order polynomial,

$$\alpha = a_0 + a_1 T + a_2 T^2 \quad (3)$$

Given the data points $(\alpha_1, T_1)$, $(\alpha_2, T_2)$, ….,$(\alpha_n, T_n)$ as in Figure 3 and Table 1, the sum of the square of the residuals (sum of the square of the differences between the observed and predicted values) is

$$\begin{aligned} S_r &= \sum_{i=1}^{n} \left(\alpha_i - \{a_0 + a_1 T_i + a_2 T_i^2\}\right)^2 \\ &= \sum_{i=1}^{n} \left(\alpha_i - a_0 - a_1 T_i - a_2 T_i^2\right)^2 \quad (4) \end{aligned}$$

To minimize the value of the sum of the square of the residuals, we take the derivative with respect to each of the three unknown coefficients, $a_0$, $a_1$, and $a_2$ to give

$$\begin{aligned} \frac{\partial S_r}{\partial a_0} &= \sum_{i=1}^{n} 2\left(\alpha_i - a_0 - a_1 T_i - a_2 T_i^2\right)(-1) \\ &= 2\left[-\sum_{i=1}^{n} \alpha_i + n a_0 + a_1 \sum_{i=1}^{n} T_i + a_2 \sum_{i=1}^{n} T_i^2\right] \end{aligned}$$

$$\begin{aligned} \frac{\partial S_r}{\partial a_1} &= \sum_{i=1}^{n} 2\left(\alpha_i - a_0 - a_1 T_i - a_2 T_i^2\right)(-T_i) \\ &= 2\left[-\sum_{i=1}^{n} \alpha_i T_i + a_0 \sum_{i=1}^{n} T_i + a_1 \sum_{i=1}^{n} {T_i}^2 + a_2 \sum_{i=1}^{n} T_i^3\right] \end{aligned}$$

$$\begin{aligned} \frac{\partial S_r}{\partial a_2} &= \sum_{i=1}^{n} 2\left(\alpha_i - a_0 - a_1 T_i - a_2 T_i^2\right)\left(-T_i^2\right) \\ &= 2\left[-\sum_{i=1}^{n} \alpha_i {T_i}^2 + a_0 \sum_{i=1}^{n} {T_i}^2 + a_1 \sum_{i=1}^{n} T_i^3 + a_2 \sum_{i=1}^{n} T_i^4\right] \quad (5) \end{aligned}$$

Setting three partial derivatives in Equation (5) equal to zero gives,

$$na_0 + a_1 \sum_{i=1}^{n} T_i + a_2 \sum_{i=1}^{n} T_i^2 = \sum_{i=1}^{n} \alpha_i$$

$$a_0 \sum_{i=1}^{n} T_i + a_1 \sum_{i=1}^{n} T_i^2 + a_2 \sum_{i=1}^{n} T_i^3 = \sum_{i=1}^{n} \alpha_i T_i$$

$$a_0 \sum_{i=1}^{n} T_i^2 + a_1 \sum_{i=1}^{n} T_i^3 + a_2 \sum_{i=1}^{n} T_i^4 = \sum_{i=1}^{n} \alpha_i T_i^2 \quad (6)$$

The set of equations given by equations(6a), (6b), and (6c) are simultaneous linear equations. The number of data points in Figure (3) is 24, as given in Table 1. Hence

$$n = 24$$

$$\sum_{i=1}^{24} T_i = -2860$$

$$\sum_{i=1}^{24} T_i^2 = 7.26 \times 10^5$$

$$\sum_{i=1}^{24} T_i^3 = -1.86472 \times 10^8$$

$$\sum_{i=1}^{24} T_i^4 = 5.24357 \times 10^{10}$$

$$\sum_{i=1}^{24} \alpha_i = 1.057 \times 10^{-4}$$

$$\sum_{i=1}^{24} \alpha_i T_i = -1.04162 \times 10^{-2}$$

$$\sum_{i=1}^{24} \alpha_i T_i^2 = 2.56799$$

$$\begin{aligned} &24a_0 - 2860a_1 + 7.26 \times 10^5 a_2 = 1.057 \times 10^{-4} \\ &-2860a_0 + 7.26 \times 10^5 a_1 - 1.8647 \times 10^8 a_2 = -1.04162 \times 10^{-2} \\ &7.26 \times 10^5 a_0 - 1.86472 \times 10^8 a_1 + 5.24357 \times 10^{10} a_2 = 2.56799 \end{aligned} \quad (7)$$

In matrix form, the three simultaneous linear equations can be written as

$$\begin{bmatrix} 24 & -2860 & 7.26 \times 10^5 \\ -2860 & 7.26 \times 10^5 & -1.86472 \times 10^8 \\ 7.26 \times 10^5 & -1.86472 \times 10^8 & 5.24357 \times 10^{10} \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 1.057 \times 10^{-4} \\ -1.04162 \times 10^{-2} \\ 2.56799 \end{bmatrix} \quad (8)$$

## Questions

(1) Can you now find the contraction in the trunnion outer diameter?

(2) Is the magnitude of contraction more than $0.015''$ as required?

(3) If that is not the case, what if the trunnion were immersed in liquid nitrogen (boiling temperature$=-321°$F)? Will that give enough contraction in the trunnion?

(4) Redo problem #1 using a third-order polynomial as the regression model. How much different is the estimate of contraction using the third-order polynomial?

(5) Find the optimum polynomial order to use for the regression model.

(6) Find the effect of the number of significant digits used in solving the set of the equations for problem #4 as you must have noticed that the order of the numbers in the coefficient matrix varies quite a bit.

# Chapter 04.01: Prerequisites to Simultaneous Linear Equations

Lesson: Definition of Matrices and Special Matrices

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *Define what a matrix is.*

2) *Identify special types of matrices.*

## What does a matrix look like?

Matrices are everywhere. If you have used a spreadsheet such as Excel or wrote numbers in a table, you have used a matrix. Matrices make the presentation of numbers clearer and make calculations easier to program. Look at the matrix below about the sale of tires in a Blowoutr'us store – given by quarter and make of tires.

$$\begin{array}{c} \\ Tirestone \\ Michigan \\ Copper \end{array} \begin{array}{c} \begin{array}{cccc} Q1. & Q2. & Q3. & Q4. \end{array} \\ \begin{bmatrix} 25 & 20 & 3 & 2 \\ 5 & 10 & 15 & 25 \\ 6 & 16 & 7 & 27 \end{bmatrix} \end{array}$$

If one wants to know how many *Copper* tires were sold in *Quarter* 4, we go along the row *Copper* and column *Q4* and find that it is 27.

## So, what is a matrix?

A *matrix* is a rectangular array of elements. The elements can be symbolic expressions or/and numbers. Matrix $[A]$ is denoted by

$$[A] = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{\mathrm{mn}} \end{bmatrix}$$

Row $i$ of $[A]$ has $n$ elements and is

$$[a_{i1}a_{i2}\dots a_{in}]$$

and column $j$ of $[A]$ has $m$ elements and is

$$\begin{bmatrix} a_{1j} \\ a_{2j} \\ \vdots \\ a_{mj} \end{bmatrix}$$

Each matrix has rows and columns, and this defines the size of the matrix. If a matrix $[A]$ has $m$ rows and $n$ columns, the size of the matrix is denoted by $m \times n$. The matrix $[A]$ may also be denoted by $[A]_{m\times n}$ to show that $[A]$ is a matrix with $m$ rows and $n$ columns.

Each entry in the matrix is called the entry or element of the matrix and is denoted by $a_{ij}$ where $i$ is the row number, and $j$ is the column number of the element.

The matrix for the tire sales example could be denoted by the matrix $A$ as

$$[A] = \begin{bmatrix} 25 & 20 & 3 & 2 \\ 5 & 10 & 15 & 25 \\ 6 & 16 & 7 & 27 \end{bmatrix}$$

There are 3 rows and 4 columns, so the size of the matrix is $3 \times 4$. In the above $[A]$ matrix, $a_{34} = 27$.

## What are the special types of matrices?

**Vector:** A vector is a matrix that has only one row or one column. There are two types of vectors – row vectors and column vectors.

## Row Vector

If a matrix $[B]$ has one row, it is called a row vector $[B] = [b_1\ b_2 \dots\dots b_n]$ and $n$ is the dimension of the row vector.

### Example 1

Give an example of a row vector.

**Solution**

$$[B] = [25\ \ 20\ \ 3\ \ 2\ \ 0]$$

is an example of a row vector of dimension 5.

## Column vector

If a matrix $[C]$ has one column, it is called a column vector

$$[C] = \begin{bmatrix} c_1 \\ \vdots \\ \vdots \\ c_m \end{bmatrix}$$

and $m$ is the dimension of the vector.

**Example 2**

Give an example of a column vector.

**Solution**

$$[C] = \begin{bmatrix} 25 \\ 5 \\ 6 \end{bmatrix}$$

is an example of a column vector of dimension 3.

## Submatrix

If some row(s) or/and column(s) of a matrix $[A]$ are deleted (no rows or columns may be deleted), the remaining matrix is called a submatrix of $[A]$.

## Square matrix

If the number of rows $m$ of a matrix is equal to the number of columns $n$ of a matrix $[A]$, that is, $m = n$, then $[A]$ is called a square matrix. The entries $a_{11}, a_{22}, ..., a_{nn}$ are called the *diagonal elements* of a square matrix. Sometimes the diagonal of the matrix is also called the *principal* or *main of the matrix*.

**Example 3**

Give an example of a square matrix.

**Solution**

$$[A] = \begin{bmatrix} 25 & 20 & 3 \\ 5 & 10 & 15 \\ 6 & 15 & 7 \end{bmatrix}$$

is a square matrix as it has the same number of rows and columns, that is, 3. The diagonal elements of $[A]$ are

$$a_{11} = 25, \quad a_{22} = 10, \quad a_{33} = 7.$$

## Upper triangular matrix

A $n \times n$ matrix for which $a_{ij} = 0, \quad i > j$ for all $i, j$ is called an upper triangular matrix. That is, all the elements below the diagonal entries are zero.

**Example 4**

Give an example of an upper triangular matrix.

**Solution**

$$[A] = \begin{bmatrix} 10 & -7 & 0 \\ 0 & -0.001 & 6 \\ 0 & 0 & 15005 \end{bmatrix}$$

is an upper triangular matrix.

## Lower triangular matrix

A $n \times n$ matrix for which $a_{ij} = 0, \quad j > i$ for all $i, j$ is called a lower triangular matrix. That is, all the elements above the diagonal entries are zero.

**Example 5**

Give an example of a lower triangular matrix.

**Solution**

$$[A] = \begin{bmatrix} 1 & 0 & 0 \\ 0.3 & 1 & 0 \\ 0.6 & 2.5 & 1 \end{bmatrix}$$

is a lower triangular matrix.

## Diagonal matrix

A square matrix with all non-diagonal elements equal to zero is called a diagonal matrix, that is, only the diagonal entries of the square matrix can be non-zero ($a_{ij} = 0, \quad i \neq j$).

**Example 6**

Give examples of a diagonal matrix.

**Solution**

$$[A] = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 2.1 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$

is a diagonal matrix.

Any or all the diagonal entries of a diagonal matrix can be zero. For example

$$[A] = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 2.1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

is also a diagonal matrix.

## Identity matrix

A diagonal matrix with all diagonal elements equal to 1 is called an identity matrix, ($a_{ij} = 0, \quad i \neq j$) for all $i, j$ and $a_{ii} = 1$ for all $i$).

**Example 7**

Give an example of an identity matrix.

**Solution**

$$[A] = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

is an identity matrix.

# Chapter 04.01: Prerequisites to Simultaneous Linear Equations

Lesson: Binary Matrix Operations

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *Add one matrix to another*
2) *Subtract one matrix from another*
3) *Multiply one matrix by another*

## How do you add two matrices?

Two matrices $[A]$ and $[B]$ can be added if they are of the same size. The addition is then shown as

$$[C] = [A] + [B]$$

where

$$c_{ij} = a_{ij} + b_{ij}$$

### Example 1

Add the following two matrices.

$$[A] = \begin{bmatrix} 5 & 2 & 3 \\ 1 & 2 & 7 \end{bmatrix}$$

$$[B] = \begin{bmatrix} 6 & 7 & -2 \\ 3 & 5 & 19 \end{bmatrix}$$

### Solution

$$
\begin{aligned}
[C] &= [A] + [B] \\
&= \begin{bmatrix} 5 & 2 & 3 \\ 1 & 2 & 7 \end{bmatrix} + \begin{bmatrix} 6 & 7 & -2 \\ 3 & 5 & 19 \end{bmatrix} \\
&= \begin{bmatrix} 5+6 & 2+7 & 3-2 \\ 1+3 & 2+5 & 7+19 \end{bmatrix} \\
&= \begin{bmatrix} 11 & 9 & 1 \\ 4 & 7 & 26 \end{bmatrix}
\end{aligned}
$$

## How do you subtract two matrices?

Two matrices $[A]$ and $[B]$ can be subtracted if they are the same size. The subtraction is then shown as

$$[D] = [A] - [B]$$

where

$$d_{ij} = a_{ij} - b_{ij}$$

**Example 2**

Subtract matrix $[B]$ from matrix $[A]$.

$$[A] = \begin{bmatrix} 5 & 2 & 3 \\ 1 & 2 & 7 \end{bmatrix}$$

$$[B] = \begin{bmatrix} 6 & 7 & -2 \\ 3 & 5 & 19 \end{bmatrix}$$

**Solution**

$$
\begin{aligned}
[D] &= [A] - [B] \\
&= \begin{bmatrix} 5 & 2 & 3 \\ 1 & 2 & 7 \end{bmatrix} - \begin{bmatrix} 6 & 7 & -2 \\ 3 & 5 & 19 \end{bmatrix} \\
&= \begin{bmatrix} (5-6) & (2-7) & (3-(-2)) \\ (1-3) & (2-5) & (7-19) \end{bmatrix} \\
&= \begin{bmatrix} -1 & -5 & 5 \\ -2 & -3 & -12 \end{bmatrix}
\end{aligned}
$$

## How do I multiply two matrices?

Two matrices $[A]$ and $[B]$ can be multiplied only if the number of columns of $[A]$ is equal to the number of rows of $[B]$ to give

$$[C]_{m\times n} = [A]_{m\times p}\,[B]_{p\times n}$$

If $[A]$ is a $m \times p$ matrix and $[B]$ is a $p \times n$ matrix, the resulting matrix $[C]$ is a $m \times n$ matrix.

So how does one calculate the elements of $[C]$ matrix?

$$\begin{aligned} c_{ij} &= \sum_{k=1}^{p} a_{ik}b_{kj} \\ &= a_{i1}b_{1j} + a_{i2}b_{2j} + \ldots + a_{ip}b_{pj} \end{aligned}$$

for each $i = 1,\ 2,\ \ldots\ ,\ m$ and $j = 1,\ 2,\ \ldots\ ,\ n$.

To put it in simpler terms, the $i^{th}$ row and $j^{th}$ column of the $[C]$ matrix in $[C] = [A]\,[B]$ is calculated by multiplying the $i^{th}$ row of $[A]$ by the $j^{th}$ column of $[B]$. That is

$$\begin{aligned} c_{ij} &= \lceil a_{i1} \quad a_{i2} \quad \ldots \quad a_{ip} \rceil \begin{bmatrix} b_{1j} \\ b_{2j} \\ \vdots \\ b_{pj} \end{bmatrix} \\ &= a_{i1}b_{1j} + a_{i2}b_{2j} + \ldots + a_{ip}b_{pj} \\ &= \sum_{k=1}^{p} a_{ik}b_{kj} \end{aligned}$$

**Example 3**

Given

$$[A] = \begin{bmatrix} 5 & 2 & 3 \\ 1 & 2 & 7 \end{bmatrix}$$

$$[B] = \begin{bmatrix} 3 & -2 \\ 5 & -8 \\ 9 & -10 \end{bmatrix}$$

Find

$$[C] = [A]\,[B]$$

**Solution**

$c_{12}$ can be found by multiplying the first row of $[A]$ by the second column of $[B]$,

$$
\begin{aligned}
c_{12} &= \begin{bmatrix} 5 & 2 & 3 \end{bmatrix} \begin{bmatrix} -2 \\ -8 \\ -10 \end{bmatrix} \\
&= (5)(-2) + (2)(-8) + (3)(-10) \\
&= -56
\end{aligned}
$$

Similarly, one can find the other elements of $[C]$ to give

$$
[C] = \begin{bmatrix} 52 & -56 \\ 76 & -88 \end{bmatrix}
$$

# Chapter 04.01: Prerequisites to Simultaneous Linear Equations

Lesson: Setting Up Problems in Matrix Form

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *Develop simultaneous linear equations model from a physical problem*

2) *Set up simultaneous linear equations in matrix form*

## Matrix algebra is used for solving systems of equations. Can you illustrate this concept?

Matrix algebra is used to solve a system of simultaneous linear equations. In fact, for many mathematical procedures such as the solution to a set of nonlinear equations, interpolation, integration, and differential equations, the solutions reduce to a set of simultaneous linear equations. Let us illustrate with an example for interpolation.

### Example 1

The upward velocity of a rocket is given at three different times on the following table.

**Table 1.** Velocity vs. time data for a rocket

| Time, $t$ | Velocity, $v$ |
|---|---|
| s | (m/s) |
| 5 | 106.8 |
| 8 | 177.2 |
| 12 | 279.2 |

The velocity data is approximated by a polynomial as

$$v(t) = at^2 + bt + c,\ 5 \le t \le 12 \qquad (E1.1)$$

Set up the equations in matrix form to find the coefficients $a, b, c$ of the velocity profile.

**Solution**

The polynomial is going through three data points $(t_1, v_1), (t_2, v_2),$ and $(t_3, v_3)$ where from Table 1

$$t_1 = 5,\ v_1 = 106.8$$

$$t_2 = 8,\ v_2 = 177.2$$

$$t_3 = 12,\ v_3 = 279.2$$

Requiring that $v(t) = at^2 + bt + c$ passes through the three data points gives

$$\begin{aligned} v(t_1) = v_1 &= at_1^2 + bt_1 + c \\ v(t_2) = v_2 &= at_2^2 + bt_2 + c \\ v(t_3) = v_3 &= at_3^2 + bt_3 + c \end{aligned}$$

Substituting the data $(t_1, v_1), (t_2, v_2),$ *and* $(t_3, v_3)$ gives

$$\begin{aligned} a\left(5^2\right) + b(5) + c &= 106.8 \\ a\left(8^2\right) + b(8) + c &= 177.2 \\ a\left(12^2\right) + b(12) + c &= 279.2 \end{aligned}$$

or

$$\begin{aligned} 25a + 5b + c &= 106.8 \\ 64a + 8b + c &= 177.2 \\ 144a + 12b + c &= 279.2 \end{aligned}$$

This set of equations can be rewritten in the matrix form as

$$\begin{bmatrix} 25a+ & 5b+ & c \\ 64a+ & 8b+ & c \\ 144a+ & 12b+ & c \end{bmatrix} = \begin{bmatrix} 106.8 \\ 177.2 \\ 279.2 \end{bmatrix}$$

The above equation can be written as a linear combination as follows

$$a\begin{bmatrix} 25 \\ 64 \\ 144 \end{bmatrix} + b\begin{bmatrix} 5 \\ 8 \\ 12 \end{bmatrix} + c\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 106.8 \\ 177.2 \\ 279.2 \end{bmatrix}$$

and further using matrix multiplication gives

$$\begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{bmatrix}\begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 106.8 \\ 177.2 \\ 279.2 \end{bmatrix}$$

The above is an illustration of why matrix algebra is needed. The complete solution to the set of equations is given later in this chapter.

A general set of $m$ linear equations and $n$ unknowns,

$$
\begin{aligned}
&a_{11}x_1 + a_{12}x_2 + ..... + a_{1n}x_n = c_1 \\
&a_{21}x_1 + a_{22}x_2 + ..... + a_{2n}x_n = c_2 \\
&..... \\
&..... \\
&a_{m1}x_1 + a_{m2}x_2 + ..... + a_{\mathrm{mn}}x_n = c_m
\end{aligned}
$$

can be rewritten in the matrix form as

$$
\begin{bmatrix}
a_{11} & a_{12} & . & . & a_{1n} \\
a_{21} & a_{22} & . & . & a_{2n} \\
\vdots & & & & \vdots \\
\vdots & & & & \vdots \\
a_{m1} & a_{m2} & . & . & a_{\mathrm{mn}}
\end{bmatrix}
\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ \vdots \\ x_n \end{bmatrix}
=
\begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ \vdots \\ c_m \end{bmatrix}
$$

Denoting the matrices by $[A]$, $[X]$, and $[C]$, the system of equation is $[A]\ [X] = [C]$, where $[A]$ is called the coefficient matrix, $[C]$ is called the right-hand side vector and $[X]$ is called the solution vector.

Sometimes $[A]\ [X] = [C]$ system of equations is written in the augmented form, that is,

$$
[A \vdots C] =
\left[\begin{array}{cccc:c}
a_{11} & a_{12} & \cdots & a_{1n} & c_1 \\
a_{21} & a_{22} & \cdots & a_{2n} & c_2 \\
\vdots & \vdots & \ddots & & \vdots \\
a_{m1} & a_{m2} & \cdots & a_{mn} & c_n
\end{array}\right]
$$

As an example, for the set of equations

$$
\begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{bmatrix}
\begin{bmatrix} a \\ b \\ c \end{bmatrix}
=
\begin{bmatrix} 106.8 \\ 177.2 \\ 279.2 \end{bmatrix}
$$

the augmented matrix form is

$$
\left[\begin{array}{ccc|c}
25 & 5 & 1 & 106.8 \\
64 & 8 & 1 & 177.2 \\
144 & 12 & 1 & 279.2
\end{array}\right]
$$

# Chapter 04.01: Prerequisites to Simultaneous Linear Equations

Lesson: Inverse of a Square Matrix

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *define the inverse of a matrix*

2) *know important statements about the inverse of a matrix*

3) *solve a set of equations where the inverse of the coefficient matrix is given*

## Can you divide two matrices?

If $[A]\ [B] = [C]$ is defined, it might seem intuitive that $[A] = \dfrac{[C]}{[B]}$, but matrix division is not defined like that. However, an inverse of a matrix can be defined for certain types of square matrices. The inverse of a square matrix $[A]$, if existing, is denoted by $[A]^{-1}$ such that

$$[A]\ [A]^{-1} = [I] = [A]^{-1}[A]$$

where $[I]$ is the identity matrix.

In other words, let $A$ be a square matrix. If $[B]$ is another square matrix of the same size such that $[B]\ [A] = [I]$, then $[B]$ is the inverse of $[A]$. $[A]$ is then called to be invertible or nonsingular. If $[A]^{-1}$ does not exist, $[A]$ is called noninvertible or singular.

If $[A]$ and $[B]$ are two $n \times n$ matrices such that $[B]\ [A] = [I]$, then these statements are also true

a) $[B]$ is the inverse of $[A]$
b) $[A]$ is the inverse of $[B]$
c) $[A]$ and $[B]$ are both invertible
d) $[A][B] = [I]$.
e) $[A]$ and $[B]$ are both nonsingular
f) all columns of $[A]$ and $[B]$ are linearly independent
g) all rows of $[A]$ and $[B]$ are linearly independent.

**Example 1**

Determine if

$$[B] = \begin{bmatrix} 3 & 2 \\ 5 & 3 \end{bmatrix}$$

is the inverse of

$$[A] = \begin{bmatrix} -3 & 2 \\ 5 & -3 \end{bmatrix}$$

***Solution***

$$\begin{aligned} [B][A] &= \begin{bmatrix} 3 & 2 \\ 5 & 3 \end{bmatrix} \begin{bmatrix} -3 & 2 \\ 5 & -3 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= [I] \end{aligned}$$

Since

$$[B]\,[A] = [I]\,,$$

$[B]$ is the inverse of $[A]$, and $[A]$ is the inverse of $[B]$.

But, we can also show that

$$\begin{aligned} [A][B] &= \begin{bmatrix} -3 & 2 \\ 5 & -3 \end{bmatrix} \begin{bmatrix} 3 & 2 \\ 5 & 3 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= [I] \end{aligned}$$

to prove that $[A]$ is the inverse of $[B]$.

## Can I use the concept of the inverse of a matrix to find the solution of a set of equations [A][X] = [C]?

Yes, if the number of equations is the same as the number of unknowns, the coefficient matrix $[A]$ is a square matrix.

Given

$$[A]\ [X] = [C]$$

Then, if $[A]^{-1}$ exists, multiplying both sides by $[A]^{-1}$.

$$[A]^{-1}[A][X] = [A]^{-1}[C]$$

$$[I]\ [X] = [A]^{-1}[C]$$

$$[X] = [A]^{-1}[C]$$

This implies that if we are able to find $[A]^{-1}$, the solution vector of $[A]\ [X] = [C]$ is simply a multiplication of $[A]^{-1}$ and the right-hand side vector, $[C]$.

## How do I find the inverse of a matrix?

If $[A]$ is a $n \times n$ matrix, then $[A]^{-1}$ is a $n \times n$ matrix, and according to the definition of inverse of a matrix

$$[A]\ [A]^{-1} = [I]$$

Denoting

$$[A] = \begin{bmatrix} a_{11} & a_{12} & \cdot & \cdot & a_{1n} \\ a_{21} & a_{22} & \cdot & \cdot & a_{2n} \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ a_{n1} & a_{n2} & \cdot & \cdot & a_{nn} \end{bmatrix}$$

$$[A]^{-1} = \begin{bmatrix} a'_{11} & a'_{12} & \cdot & \cdot & a'_{1n} \\ a'_{21} & a'_{22} & \cdot & \cdot & a'_{2n} \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ a'_{n1} & a'_{n2} & \cdot & \cdot & a'_{\mathrm{nn}} \end{bmatrix}$$

$$[I] = \begin{bmatrix} 1 & 0 & \cdot & \cdot & \cdot & 0 \\ 0 & 1 & & & & 0 \\ 0 & & \cdot & & & \cdot \\ \cdot & & & 1 & & \cdot \\ \cdot & & & & \cdot & \cdot \\ 0 & \cdot & \cdot & \cdot & \cdot & 1 \end{bmatrix}$$

Using the definition of matrix multiplication, the first column of the $[A]^{-1}$ matrix can then be found by solving

$$\begin{bmatrix} a_{11} & a_{12} & \cdot & \cdot & a_{1n} \\ a_{21} & a_{22} & \cdot & \cdot & a_{2n} \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ a_{n1} & a_{n2} & \cdot & \cdot & a_{\mathrm{nn}} \end{bmatrix} \begin{bmatrix} a'_{11} \\ a'_{21} \\ \cdot \\ \cdot \\ a'_{n1} \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ \cdot \\ \cdot \\ 0 \end{bmatrix}$$

Similarly, one can find the other columns of the $[A]^{-1}$ matrix by changing the right-hand side accordingly.

**Example 2**

The upward velocity of the rocket is given by

**Table 1.** Velocity vs. time data for a rocket

| Time, t (s) | Velocity, v (m/s) |
|---|---|
| 5 | 106.8 |
| 8 | 177.2 |
| 12 | 279.2 |

In an earlier example, we wanted to approximate the velocity profile by

$$v(t) = at^2 + bt + c,\ 5 \le t \le 12$$

We found that the coefficients $a,\ b,$ and $c$ in $v(t)$ are given by solving

$$\begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 106.8 \\ 177.2 \\ 279.2 \end{bmatrix}$$

First, find the inverse of

$$[A] = \begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{bmatrix}$$

and then use the definition of inverse to find the coefficients $a,\ b,$ and $c$, and the velocity profile

**Solution**

If

$$[A]^{-1} = \begin{bmatrix} a'_{11} & a'_{12} & a'_{13} \\ a'_{21} & a'_{22} & a'_{23} \\ a'_{31} & a'_{32} & a'_{33} \end{bmatrix}$$

is the inverse of $[A]$, then

$$\begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{bmatrix} \begin{bmatrix} a'_{11} & a'_{12} & a'_{13} \\ a'_{21} & a'_{22} & a'_{23} \\ a'_{31} & a'_{32} & a'_{33} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

gives three sets of equations

$$\begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{bmatrix} \begin{bmatrix} a'_{11} \\ a'_{21} \\ a'_{31} \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{bmatrix} \begin{bmatrix} a'_{12} \\ a'_{22} \\ a'_{32} \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{bmatrix} \begin{bmatrix} a'_{13} \\ a'_{23} \\ a'_{33} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Solving the above three sets of equations separately gives

$$\begin{bmatrix} a'_{11} \\ a'_{21} \\ a'_{31} \end{bmatrix} = \begin{bmatrix} 0.04762 \\ -0.9524 \\ 4.571 \end{bmatrix}$$

$$\begin{bmatrix} a'_{12} \\ a'_{22} \\ a'_{32} \end{bmatrix} = \begin{bmatrix} -0.08333 \\ 1.417 \\ -5.000 \end{bmatrix}$$

$$\begin{bmatrix} a'_{13} \\ a'_{23} \\ a'_{33} \end{bmatrix} = \begin{bmatrix} 0.03571 \\ -0.4643 \\ 1.429 \end{bmatrix}$$

Hence

$$[A]^{-1} = \begin{bmatrix} 0.04762 & -0.08333 & 0.03571 \\ -0.9524 & 1.417 & -0.4643 \\ 4.571 & -5.000 & 1.429 \end{bmatrix}$$

Now

$$[A]\,[X] = [C]$$

where

$$[X] = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$$

$$[C] = \begin{bmatrix} 106.8 \\ 177.2 \\ 279.2 \end{bmatrix}$$

Using the definition of $[A]^{-1}$,

$$[A]^{-1}\,[A]\,[X] = [A]^{-1}\,[C]$$

$$[X] = [A]^{-1}[C]$$

$$\begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 0.04762 & -0.08333 & 0.03571 \\ -0.9524 & 1.417 & -0.4643 \\ 4.571 & -5.000 & 1.429 \end{bmatrix} \begin{bmatrix} 106.8 \\ 177.2 \\ 279.2 \end{bmatrix}$$

Conducting matrix multiplication of the right hand side gives

$$\begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 0.2905 \\ 19.69 \\ 1.086 \end{bmatrix}$$

So

$$v(t) = 0.2905t^2 + 19.69t + 1.086,\ 5 \leq t \leq 12$$

# Chapter 04.01: Introduction to Matrix Algebra

Lesson: Introduction to Matrix Algebra

## Learning Objectives

*After reading this chapter, you should be able to*

1) *define what a matrix is.*

2) *identify special types of matrices, and*

3) *identify when two matrices are equal.*

## What does a matrix look like?

Matrices are everywhere. If you have used a spreadsheet such as Excel or written numbers in a table, you have used a matrix. Matrices make presentation of numbers clearer and make calculations easier to program. Look at the matrix below about the sale of tires in a Blowoutr'us store – given by quarter and make of tires.

$$\begin{array}{c} \\ Tirestone \\ Michigan \\ Copper \end{array} \begin{array}{c} \begin{array}{cccc} Q1. & Q2. & Q3. & Q4 \end{array} \\ \begin{bmatrix} 25 & 20 & 3 & 2 \\ 5 & 10 & 15 & 25 \\ 6 & 16 & 7 & 27 \end{bmatrix} \end{array}$$

If one wants to know how many *Copper* tires were sold in *Quarter* 4, we go along the row *Copper* and column $Q4$ and find that it is 27.

## So, what is a matrix?

A *matrix* is a rectangular array of elements. The elements can be symbolic expressions or/and numbers. Matrix $[A]$ is denoted by

$$[A] = \begin{bmatrix} a_{11} & a_{12} & ..... & a_{1n} \\ a_{21} & a_{22} & ..... & a_{2n} \\ \vdots & & & \vdots \\ a_{m1} & a_{m2} & ..... & a_{mn} \end{bmatrix}$$

Row $i$ of $[A]$ has $n$ elements and is

$$[a_{i1}a_{i2}....a_{in}]$$

and column $j$ of $[A]$ has $m$ elements and is

$$\begin{bmatrix} a_{1j} \\ a_{2j} \\ \vdots \\ a_{mj} \end{bmatrix}$$

Each matrix has rows and columns and this defines the size of the matrix. If a matrix $[A]$ has $m$ rows and $n$ columns, the size of the matrix is denoted by $m \times n$. The matrix $[A]$ may also be denoted by $[A]_{m\times n}$ to show that $[A]$ is a matrix with $m$ rows and $n$ columns.

Each entry in the matrix is called the entry or element of the matrix and is denoted by $a_{ij}$ where $i$ is the row number and $j$ is the column number of the element.

The matrix for the tire sales example could be denoted by the matrix $[A]$ as

$$[A] = \begin{bmatrix} 25 & 20 & 3 & 2 \\ 5 & 10 & 15 & 25 \\ 6 & 16 & 7 & 27 \end{bmatrix}$$

There are 3 rows and 4 columns, so the size of the matrix is $3 \times 4$. In the above $[A]$ matrix, $a_{34} = 27$.

## What are the special types of matrices?

Vector: A vector is a matrix that has only one row or one column. There are two types of vectors – row vectors and column vectors.

## Row Vector:

If a matrix $[B]$ has one row, it is called a row vector $[B] = [b_1\ b_2\ ...\ b_n]$

and $n$ is the dimension of the row vector.

**Example 1**

Give an example of a row vector.

**Solution**

$$[B] = [25 \quad 20 \quad 3 \quad 2 \quad 0]$$

is an example of a row vector of dimension 5.

## Column vector:

If a matrix $[C]$ has one column, it is called a column vector

$$[C] = \begin{bmatrix} c_1 \\ \vdots \\ \vdots \\ c_m \end{bmatrix}$$

and $m$ is the dimension of the vector.

**Example 2**

Give an example of a column vector.

**Solution**

$$[C] = \begin{bmatrix} 25 \\ 5 \\ 6 \end{bmatrix}$$

is an example of a column vector of dimension 3.

## Submatrix:

If some row(s) or/and column(s) of a matrix $[A]$ are deleted (no rows or columns may be deleted), the remaining matrix is called a submatrix of $[A]$.

**Example 3**

Find some of the submatrices of the matrix

$$[A] = \begin{bmatrix} 4 & 6 & 2 \\ 3 & -1 & 2 \end{bmatrix}$$

**Solution**

$$\begin{bmatrix} 4 & 6 & 2 \\ 3 & -1 & 2 \end{bmatrix}, \quad \begin{bmatrix} 4 & 6 \\ 3 & -1 \end{bmatrix}, \quad \begin{bmatrix} 4 & 6 & 2 \end{bmatrix}, [4], \begin{bmatrix} 2 \\ 2 \end{bmatrix}$$

are some of the submatrices of $[A]$. Can you find other submatrices of $[A]$?

## Square matrix:

If the number of rows $m$ of a matrix is equal to the number of columns $n$ of a matrix $[A]$, that is, $m = n$, then $[A]$ is called a square matrix. The entries $a_{11}, a_{22}, ..., a_{nn}$ are called the *diagonal elements* of a square matrix. Sometimes the diagonal of the matrix is also called the *principal* or *main of the matrix.*

**Example 4**

Give an example of a square matrix.

**Solution**

$$[A] = \begin{bmatrix} 25 & 20 & 3 \\ 5 & 10 & 15 \\ 6 & 15 & 7 \end{bmatrix}$$

is a square matrix as it has the same number of rows and columns, that is, 3. The diagonal elements of $[A]$ are $a_{11} = 25, \quad a_{22} = 10, \quad a_{33} = 7$.

## Upper triangular matrix:

A $n \times n$ matrix for which $a_{ij} = 0, \quad i > j$ for all $i, j$ is called an upper triangular matrix. That is, all the elements below the diagonal entries are zero.

**Example 5**

Give an example of an upper triangular matrix.

**Solution**

$$[A] = \begin{bmatrix} 10 & -7 & 0 \\ 0 & -0.001 & 6 \\ 0 & 0 & 15005 \end{bmatrix}$$

is an upper triangular matrix.

## Lower triangular matrix:

A $n \times n$ matrix for which $a_{ij} = 0, \quad j > i$ for all $i, j$ is called a lower triangular matrix. That is, all the elements above the diagonal entries are zero.

**Example 6**

Give an example of a lower triangular matrix.

**Solution**

$$[A] = \begin{bmatrix} 1 & 0 & 0 \\ 0.3 & 1 & 0 \\ 0.6 & 2.5 & 1 \end{bmatrix}$$

is a lower triangular matrix.

## Diagonal matrix:

A square matrix with all non-diagonal elements equal to zero is called a diagonal matrix, that is, only the diagonal entries of the square matrix can be non-zero, ($a_{ij} = 0, \quad i \neq j$).

**Example 7**

Give examples of a diagonal matrix.

**Solution**

$$[A] = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 2.1 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$

is a diagonal matrix.

Any or all the diagonal entries of a diagonal matrix can be zero. For example

$$[A] = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 2.1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

is also a diagonal matrix.

## Identity matrix:

A diagonal matrix with all diagonal elements equal to 1 is called an identity matrix, ($a_{ij} = 0, \quad i \neq j$ for all $i, j$ and $a_{ii} = 1$ for all $i$).

**Example 8**

Give an example of an identity matrix.

**Solution**

$$[A] = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

is an identity matrix.

## Zero matrix:

A matrix whose all entries are zero is called a zero matrix, ($a_{ij} = 0$ for all $i$ and $j$).

**Example 9**

Give examples of a zero matrix.

**Solution**

$$[A] = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$[B] = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$[C] = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$[D] = \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$$

are all examples of a zero matrix.

## Tridiagonal matrices:

A tridiagonal matrix is a square matrix in which all elements not on the following are zero - the major diagonal, the diagonal above the major diagonal, and the diagonal below the major diagonal.

**Example 10**

Give an example of a tridiagonal matrix.

**Solution**

$$[A] = \begin{bmatrix} 2 & 4 & 0 & 0 \\ 2 & 3 & 9 & 0 \\ 0 & 0 & 5 & 2 \\ 0 & 0 & 3 & 6 \end{bmatrix}$$

is a tridiagonal matrix.

## Do non-square matrices have diagonal entries?

Yes, for a $m \times n$ matrix $[A]$ , the diagonal entries are $a_{11}, a_{22}..., a_{k-1,k-1}, a_{kk}$ where $k = min\{m,\ n\}$.

**Example 11**

What are the diagonal entries of

$$[A] = \begin{bmatrix} 3.2 & 5 \\ 6 & 7 \\ 2.9 & 3.2 \\ 5.6 & 7.8 \end{bmatrix}$$

**Solution**

The diagonal elements of $[A]$ are $a_{11} = 3.2$ *and* $a_{22} = 7$.

## Diagonally Dominant Matrix:

A $n \times n$ square matrix $[A]$ is a diagonally dominant matrix (also called weak diagonally dominant) if

$$|a_{ii}| \geq \sum_{\substack{j=1 \\ i \neq j}}^{n} |a_{ij}|$$

$$$$ for $i = 1, 2, ....., n$

that is, for each row, the absolute value of the diagonal element is greater than or equal to the sum of the absolute values of the rest of the elements of that row.

**Example 12**

Give examples of diagonally dominant matrices and not diagonally dominant matrices.

**Solution**

The matrix

$$[A] = \begin{bmatrix} 15 & 6 & 7 \\ 2 & -4 & -2 \\ 3 & 2 & 6 \end{bmatrix}$$

is a diagonally dominant matrix as

$$|a_{11}| = |15| = 15 \geq |a_{12}| + |a_{13}| = |6| + |7| = 13$$

$$|a_{22}| = |-4| = 4 \geq |a_{21}| + |a_{23}| = |2| + |-2| = 4$$

$$|a_{33}| = |6| = 6 \geq |a_{31}| + |a_{32}| = |3| + |2| = 5$$

The matrix

$$[B] = \begin{bmatrix} -15 & 6 & 9 \\ 2 & -4 & 2 \\ 3 & -2 & 5 \end{bmatrix}$$

is a diagonally dominant matrix as

$$|b_{11}| = |-15| = 15 \geq |b_{12}| + |b_{13}| = |6| + |9| = 15$$

$$|b_{22}| = |-4| = 4 \geq |b_{21}| + |b_{23}| = |2| + |2| = 4$$

$$|b_{33}| = |5| = 5 \geq |b_{31}| + |b_{32}| = |3| + |-2| = 5$$

The inequalities are satisfied for all rows.

The matrix

$$[C] = \begin{bmatrix} -15.01 & 6 & 9 \\ 2 & -4 & 2 \\ 3 & -2 & 5 \end{bmatrix}$$

is a diagonally dominant matrix as

$$|c_{11}| = |-15.01| = 15.01 \geq |c_{12}| + |c_{13}| = |6| + |9| = 15$$

$$|c_{22}| = |-4| = 4 \geq |c_{21}| + |c_{23}| = |2| + |2| = 4$$

$$|c_{33}| = |5| = 5 \geq |c_{31}| + |c_{32}| = |3| + |-2| = 5$$

The inequalities are satisfied for all rows.

The matrix

$$[D] = \begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{bmatrix}$$

is not diagonally dominant as

$$|d_{22}| = |8| = 8 \leq |d_{21}| + |d_{23}| = |64| + |1| = 65$$

The inequalities are not satisfied for all rows.

## When are two matrices considered to be equal?

Two matrices $[A]$ and $[B]$ are equal if the size of $[A]$ and $[B]$ is the same (number of rows and columns of $[A]$ are same as that of $[B]$) and $a_{ij} = b_{ij}$ for all $i$ and $j$.

**Example 13**

What would make

$$[A] = \begin{bmatrix} 2 & 3 \\ 6 & 7 \end{bmatrix}$$

to be equal to

$$[B] = \begin{bmatrix} b_{11} & 3 \\ 6 & b_{22} \end{bmatrix}$$

**Solution**

The two matrices $[A]$and $[B]$ would be equal if $b_{11} = 2$ and $b_{22} = 7$.

## Appendix A

In this appendix, we go beyond the discussion of a diagonally dominant matrix, and discuss weak diagonally dominant matrix, strictly diagonally dominant matrix, irreducible diagonally dominant matrix and reducible matrices. We also discuss the implications of such matrices.

## What is a diagonally dominant matrix?

A $n \times n$ square matrix $[A]$ is a diagonally dominant matrix if

$$|a_{ii}| \geq \sum_{\substack{j=1 \\ i \neq j}}^{n} |a_{ij}| \text{ for } i = 1, 2, ...., n$$

that is, for each row, the absolute value (also called magnitude) of the diagonal element is greater than or equal to the sum of the absolute values of the rest of the elements of that row.

**Example A.1**

Give examples of matrices that are diagonally dominant and those that are not diagonally dominant.

**Solution**

The matrix

$$[A] = \begin{bmatrix} 15 & 6 & 7 \\ 2 & -4.1 & -2 \\ 3 & 2 & 6 \end{bmatrix}$$

is a diagonally dominant matrix.

Why? Because for each and every row, the answer to the question below is Yes.

*Row* 1: Is $|a_{11}| \geq |a_{12}| + |a_{13}|$ ? Yes, because

$$|a_{11}| = |15| = 15, |a_{12}| + |a_{13}| = |6| + |7| = 13, 15 \geq 13$$

*Row* 2: Is $|a_{22}| \geq |a_{21}| + |a_{23}|$ ? Yes, because

$$|a_{22}| = |-4.1| = 4.1, |a_{21}| + |a_{23}| = |2| + |-2| = 4, 4.1 \geq 4$$

*Row* 3: Is $|a_{33}| \geq |a_{31}| + |a_{32}|$ ? Yes, because

$$|a_{33}| = |6| = 6, |a_{31}| + |a_{32}| = |3| + |2| = 5, 6 \geq 5$$

---

The matrix

$$[B] = \begin{bmatrix} -15 & 6 & 9 \\ 2 & -4 & -2 \\ 3 & -2 & 5 \end{bmatrix}$$

is a diagonally dominant matrix.

Why? Because for each and every row, the answer to the question below is Yes.

*Row* 1: Is $|b_{11}| \geq |b_{12}| + |b_{13}|$ ? Yes, because

$$|b_{11}| = |15| = 15, |b_{12}| + |b_{13}| = |6| + |9| = 15, 15 \geq 15$$

*Row* 2: Is $|b_{22}| \geq |b_{21}| + |b_{23}|$ ? Yes, because

$$|b_{22}| = |-4| = 4, \quad |b_{21}| + |b_{23}| = |2| + |-2| = 4, 4 \geq 4$$

*Row* 3: Is $|b_{33}| \geq |b_{31}| + |b_{32}|$? Yes, because

$$|b_{33}| = |5| = 5, |b_{31}| + |b_{32}| = |3| + |2| = 5, 5 \geq 5$$

---

The matrix

$$[C] = \begin{bmatrix} -15 & 6 & 9 \\ 2 & -4.1 & -2 \\ 3 & -2 & 5 \end{bmatrix}$$

is a diagonally dominant matrix.

Why? Because for each and every row, the answer to the question below is Yes.

*Row* 1: Is $|c_{11}| \geq |c_{12}| + |c_{13}|$? Yes, because

$$|c_{11}| = |15| = 15, |c_{12}| + |c_{13}| = |6| + |9| = 15, 15 \geq 15$$

*Row* 2: Is $|c_{22}| \geq |c_{21}| + |c_{23}|$? Yes, because

$$|c_{22}| = |-4.1| = 4, |c_{21}| + |c_{23}| = |2| + |-2| = 4, 4.1 \geq 4$$

*Row* 3: Is $|c_{33}| \geq |c_{31}| + |c_{32}|$? Yes, because

$$|c_{33}| = |5|, |c_{31}| + |c_{32}| = |3| + |2| = 5, 5 \geq 5$$

---

The matrix

$$[D] = \begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{bmatrix}$$

is *not* a diagonally dominant matrix.

Why? Because for each and every row, the answer to the question below is not a Yes.

*Row* 1: Is $|d_{11}| \geq |d_{12}| + |d_{13}|$? Yes, because

$$|d_{11}| = |25| = 25, |d_{12}| + |d_{13}| = |5| + |1| = 6, 25 \geq 6$$

*Row* 2: Is $|d_{22}| \geq |d_{21}| + |d_{23}|$? No, because

$$|d_{22}| = |8| = 8, |d_{21}| + |d_{23}| = |64| + |1| = 65, 8 < 65$$

*Row* 3: Is $|d_{33}| \geq |d_{31}| + |d_{32}|$? No, because

$$|d_{33}| = |1| = 1, |d_{31}| + |d_{32}| = |144| + |12| = 156, 1 < 156$$

**What is a weak diagonally dominant matrix?**

The answer is simple – the definition of a weak(ly) diagonally dominant matrix is identical to that of a diagonally dominant matrix as the inequality used for the check is a weak inequality of greater than or equal to ($\geq$).

**What is a strictly diagonally dominant matrix?**

A $n \times n$ square matrix is a strictly diagonally dominant matrix if

$$|a_{ii}| > \sum_{\substack{j=1 \\ i \neq j}}^{n} |a_{ij}| \text{ for } i = 1, 2, ...., n$$

that is, for each row, the absolute value of the diagonal element is strictly greater than the sum of the absolute values of the rest of the elements of that row.

**Example A.2**

Give examples of strictly diagonally dominant matrices and not strictly diagonally dominant matrices.

**Solution**

The matrix

$$[A] = \begin{bmatrix} 15 & 6 & 7 \\ 2 & -4.1 & -2 \\ 3 & 2 & 6 \end{bmatrix}$$

is a strictly diagonally dominant matrix

Why? Because for each and every row, the answer to the question below is Yes.

*Row* 1: Is $|a_{11}| > |a_{12}| + |a_{13}|$? Yes, because

$$|a_{11}| = |15| = 15, |a_{12}| + |a_{13}| = |6| + |7| = 13, 15 > 13.$$

*Row* 2: Is $|a_{22}| > |a_{21}| + |a_{23}|$? Yes, because

$$|a_{22}| = |-4.1| = 4.1, |a_{21}| + |a_{23}| = |2| + |-2| = 4, 4.1 > 4$$

*Row* 3: Is $|a_{33}| > |a_{31}| + |a_{32}|$? Yes, because

$$|a_{33}| = |6| = 6, |a_{31}| + |a_{32}| = |3| + |2| = 5, 6 > 5$$

---

The matrix

$$[B] = \begin{bmatrix} 13 & 6 & 7 \\ 2 & -4.1 & -2 \\ 3 & 2 & 6 \end{bmatrix}$$

is a *not* a strictly diagonally dominant matrix

Why? Because for each and every row, the answer to the question below is *not* a Yes.

*Row* 1: Is $|b_{11}| > |b_{12}| + |b_{13}|$? No, because

$$|b_{11}| = |13| = 13, |b_{12}| + |b_{13}| = |6| + |7| = 13, 13 = 13$$

*Row* 2: Is $|b_{22}| > |b_{21}| + |b_{23}|$? Yes, because

$$|b_{22}| = |-4.1| = 4.1, |b_{21}| + |b_{23}| = |2| + |-2| = 4, 4.1 > 4$$

*Row* 3: Is $|b_{33}| > |b_{31}| + |b_{32}|$? Yes, because

$$|b_{33}| = |6| = 6, |b_{31}| + |b_{32}| = |3| + |2| = 5, 6 > 5$$

---

The matrix

$$[C] = \begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{bmatrix}$$

is *not* a strictly diagonally dominant matrix.

Why? Because for each and every row, the answer to the question below is *not* a Yes.

*Row* 1: Is $|c_{11}| \geq |c_{12}| + |c_{13}|$? Yes, because

$$|c_{11}| = |25| = 25, |c_{12}| + |c_{13}| = |5| + |1| = 6, 25 > 6$$

*Row* 2: Is $|c_{22}| \geq |c_{21}| + |c_{23}|$? No, because

$$|c_{22}| = |8| = 8, |c_{21}| + |c_{23}| = |64| + |1| = 65, 8 < 65$$

*Row* 3: Is $|c_{33}| \geq |c_{31}| + |c_{32}|$? No, because

$$|c_{33}| = |1| = 1, c_{31}| + |c_{32}| = |144| + |12 \,| = 156, 1 < 156$$

**What is an irreducible diagonally dominant matrix?**

A $n \times n$ square matrix is an irreducible diagonally dominant matrix if

$$[A] \text{ is irreducible,}$$

$$|a_{ii}| \geq \sum_{\substack{j=1 \\ i \neq j}}^{n} |a_{ij}| \text{ for } i = 1, 2, ...., n \text{ and}$$

$$|a_{ii}| > \sum_{\substack{j=1 \\ i \neq j}}^{n} |a_{ij}| \text{ for at least one row, } i = 1, 2, ...., n$$

The second condition means that for each row, the absolute value (also called magnitude) of the diagonal element is greater than or equal to the sum of the absolute values of the rest of the elements of that row. The third condition means that for at least one row, the absolute value (also called magnitude) of the diagonal element is greater than the sum of the absolute values of the rest of the elements of that row.

**Example**

Give examples of matrices that are irreducibly diagonally dominant and those that are not irreducibly diagonally dominant.

**Solution**

The matrix

$$[A] = \begin{bmatrix} 15 & 6 & 7 \\ 2 & -4.1 & -2 \\ 3 & 2 & 6 \end{bmatrix}$$

is an irreducible diagonally dominant matrix.

Because $[A]$ irreducible and the answer to every question below is Yes.

*Row* 1: Is $|a_{11}| \geq |a_{12}| + |a_{13}|$? Yes, because

$$|a_{11}| = |15|, |a_{12}| + |a_{13}| = |6| + |7| = 13, 15 > 13.$$

*Row* 2: Is $|a_{22}| \geq |a_{21}| + |a_{23}|$? Yes, because

$$|a_{22}| = |-4.1| = 4.1, |a_{21}| + |a_{23}| = |2| + |-2| = 4, 4.1 > 4$$

*Row* 3: Is $|a_{33}| \geq |a_{31}| + |a_{32}|$? Yes, because

$$|a_{33}| = |6| \, |a_{31}| + |a_{32}| = |3| + |2| = 5, 6 > 5$$

Is the inequality satisfied strictly for at least one row? Yes, it is satisfied for Row 1 (Row 2 and Row 3 do too satisfy the inequality strictly but is not necessary)

---

The matrix

$$[B] = \begin{bmatrix} -15 & 6 & 9 \\ 2 & -4 & -2 \\ 3 & -2 & 5 \end{bmatrix}$$

is a not an irreducible diagonally dominant matrix.

Because $[B]$ irreducible but the answer to every question below is not a Yes.

*Row* 1: Is $|b_{11}| \geq |b_{12}| + |b_{13}|$? Yes, because

$$|b_{11}| = |15| = 15, |b_{12}| + |b_{13}| = |6| + |9| = 15, 15 \geq 15$$

*Row* 2: Is $|b_{22}| \geq |b_{21}| + |b_{23}|$? Yes, because

$$|b_{22}| = |-4| = 4, |b_{21}| + |b_{23}| = |2| + |-2| = 4, 4 \geq 4$$

*Row* 3: Is $|b_{33}| \geq |b_{31}| + |b_{32}|$? Yes, because

$$|b_{33}| = |5| = 5, |b_{31}| + |b_{32}| = |3| + |2| = 5, 5 \geq 5$$

Is the inequality satisfied strictly for at least one row? No.

---

The matrix

$$[C] = \begin{bmatrix} -15 & 6 & 9 \\ 2 & -4.1 & -2 \\ 3 & -2 & 5 \end{bmatrix}$$

is an irreducible diagonally dominant matrix.

Because $[C]$ irreducible and the answer to every question below is Yes.

*Row* 1: Is $|c_{11}| \geq |c_{12}| + |c_{13}|$? Yes, because

$$|c_{11}| = |15|, |c_{12}| + |c_{13}| = |6| + |9| = 15, 15 \geq 15$$

*Row* 2: Is $|c_{22}| \geq |c_{21}| + |c_{23}|$? Yes, because

$$|c_{22}| = |-4.1| = 4.1, |c_{21}| + |c_{23}| = |2| + |-2| = 4, 4.1 \geq 4$$

*Row* 3: Is $|c_{33}| \geq |c_{31}| + |c_{32}|$? Yes, because

$$|c_{33}| = |5|, |c_{31}| + |c_{32}| = |3| + |2| = 5, 5 \geq 5$$

Is the inequality satisfied strictly for at least one row? Yes, it is satisfied for Row 2.

---

The matrix

$$[D] = \begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{bmatrix}$$

is *not* an irreducible diagonally dominant matrix.

Because $[D]$ is irreducible but the answer to every question below is not a Yes.

*Row* 1: Is $|d_{11}| \geq |d_{12}| + |d_{13}|$? Yes, because

$$|d_{11}| = |25| = 25, |d_{12}| + |d_{13}| = |5| + |1| = 6, 25 > 6$$

*Row* 2: Is $|d_{22}| \geq |d_{21}| + |d_{23}|$? No, because

$$|d_{22}| = |8| = 8, |d_{21}| + |d_{23}| = |64| + |1| = 65, 8 < 65$$

*Row* 3: Is $|d_{33}| \geq |d_{31}| + |d_{32}|$? No, because

$$|d_{33}| = |1| = 1, |d_{31}| + |d_{32}| = |144| + |12 \,| = 156, 1 < 156$$

There is no need to check for strict inequality condition.

**What is an irreducible matrix?**

A square matrix is called *reducible* matrix if the following is true. Take the indices $i = 1, 2, ...., n$ and see if they can be divided into two disjoint nonempty sets $i_1, i_2, ...., i_\alpha$ and $j_1, j_2, ...., j_\beta$ such that

$$n = \alpha + \beta, \text{ and}$$

and

$$a_{i_k j_l} = 0, \ k = 1, 2, ...., \alpha \text{ and } l = 1, 2, ...., \beta$$

If the square matrix is not reducible, it is called an *irreducible* matrix.

A square matrix $[A]$ is called *reducible* matrix if and only if for any perturbation matrix $[P]$, the matrix multiplication $[P]^T[A][P]$ results in a block upper triangular matrix.

**Give examples of irreducible and reducible matrices.**

The matrix

$$[A] = \begin{bmatrix} 0 & 5 & 7 \\ 8 & 0 & 0 \\ 10 & 0 & 0 \end{bmatrix}$$

is an irreducible matrix.

The matrix

$$[B] = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 4 & 6 \\ 10 & 0 & 0 \end{bmatrix}$$

is a reducible matrix. Why? Take the indices $i = 1, 2, 3$ and see that they can be divided into two disjoint nonempty sets 1 and 2,3 such that,

$$\alpha = 1, \beta = 2, \text{ giving } \alpha + \beta = 1 + 2 = 3, \text{ and}$$

$$b_{i_k j_l} = 0, \ k = 1 \text{ and } l = 1, 2$$

## Implications of diagonally dominant matrices?

If a square matrix is strictly diagonally dominant

- then the matrix is non-singular [1].
- then if the matrix is symmetric with non-negative diagonal entries, the matrix is positive semi-definite [1].
- then if the matrix is the coefficient matrix for a set of simultaneous linear equations, the iterative Gauss-Seidel numerical method will always converge [2].
- then if the matrix is the coefficient matrix for a set of simultaneous linear equations, the iterative Jordan numerical method will always converge [2].
- then if the diagonal entries of the matrix are positive, the real parts of the matrix eigenvalues are positive [1].
- then if the diagonal entries of the matrix are negative, the real parts of the matrix eigenvalues are negative [1].
- then if the matrix is column dominant, no pivoting is needed for Gaussian elimination [2].
- then if the matrix is column dominant, no pivoting is needed for LU factorization [2].

If a square matrix is irreducible diagonally dominant

- then if the matrix is the coefficient matrix for a set of simultaneous linear equations, the iterative Gauss-Seidel numerical method will always converge.
- then if the matrix is the coefficient matrix for a set of simultaneous linear equations, the iterative Jordan numerical method will always converge.
- the matrix is non-singular [2].

If a square matrix is diagonally dominant (also called weakly diagonally dominant)

- then if the matrix is column dominant, no pivoting is needed for Gaussian elimination [3].
- then if the matrix is column dominant, no pivoting is needed for LU factorization [3].

**References**

[1] Briggs, Keith. "Diagonally Dominant Matrix." FromMathWorld–A Wolfram
Web Resource, created by Eric W. Weisstein.
http://mathworld.wolfram.com/DiagonallyDominantMatrix.html
[2] Diagonally Dominant Matrix, see
https://en.wikipedia.org/wiki/Diagonally_dominant_matrix, last accessed
on November 4, 2016.
[3] "Lecture 4: A Gaussian Elimination Example", see
http://www.cs.yale.edu/homes/spielman/BAP/lect4.pdf, last accessed on
November 4, 2016.

## Key Terms:

Matrix

Vector

Submatrix

Square matrix

Equal matrices

Zero matrix

Identity matrix

Diagonal matrix

Upper triangular matrix

Lower triangular matrix

Tri-diagonal matrix

# Multiple Choice Test

## Chapter 04.01 Introduction to Matrices

(1). For an $n \times n$ upper triangular matrix $[A]$,

(A) $a_{ij} = 0, i > j$

(B) $a_{ij} = 0, j > i$

(C) $a_{ij} \neq 0, i > j$

(D) $a_{ij} \neq 0, j > i$

(2). Which one of these square matrices is strictly diagonally dominant?

(A) $\begin{bmatrix} 5 & 7 & 0 \\ 3 & -6 & 2 \\ 2 & 2 & 9 \end{bmatrix}$

(B) $\begin{bmatrix} 7 & -5 & -2 \\ 6 & -13 & -7 \\ 6 & -7 & -13 \end{bmatrix}$

(C) $\begin{bmatrix} 8 & -5 & -2 \\ 6 & -14 & -7 \\ 6 & -7 & -13 \end{bmatrix}$

(D) $\begin{bmatrix} 8 & 5 & 2 \\ 6 & 14 & 7 \\ 6 & 7.5 & 14 \end{bmatrix}$

(3). The order of the following matrix is

$$\begin{bmatrix} 4 & -6 & -7 & 2 \\ 3 & 2 & -5 & 6 \end{bmatrix}$$

(A) $4 \times 2$

(B) $2 \times 4$

(C) $8 \times 1$

(D) not defined

(4). To make the following two matrices equal

$$[A] = \begin{bmatrix} 5 & -6 & 7 \\ 3 & 2 & 5 \end{bmatrix}$$

$$[B] = \begin{bmatrix} 5 & p & 7 \\ 3 & 2 & 5 \end{bmatrix}$$

the value of $p$ is

(A) $-6$

(B) 6

(C) 0

(D) 7

(5). For a square $n \times n$ matrix $[A]$ to be an identity matrix,

(A) $a_{ij} \neq 0, i = j; a_{ij} = 0, i = j$

(B) $a_{ij} = 0, i \neq j; a_{ij} = 1, i = j$

(C) $a_{ij} = 0, i \neq j; a_{ij} = i, i = j$

(D) $a_{ij} = 0, i \neq j; a_{ij} > 0, i = j$

(6). To make the following square matrix to be diagonally dominant, the value of $p$ needs to be

$$\begin{bmatrix} 6 & -2 & -4 \\ 7 & 9 & 1 \\ 8 & -5 & p \end{bmatrix}$$

(A) greater than or equal to 13

(B) greater than 3

(C) greater than or equal to 3

(D) greater than 13

For complete solution, go to

https://ma.mathforcollege.com/mcquizzes/04sle/quiz_04sle_introduction_solution.pdf

# Multiple Choice Test

## Chapter 04.01 Prerequisites to Simultaneous Linear Equations

(1). Given

$$[A] = \begin{bmatrix} 6 & 2 & 3 & 9 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 4 & 5 \\ 0 & 0 & 0 & 6 \end{bmatrix}$$

then $[A]$ is a _____ matrix.

(A) diagonal

(B) identity

(C) lower triangular

(D) upper triangular

(2). A square matrix $[A]$ is lower triangular if

(A) $a_{ij} = 0, j > i$

(B) $a_{ij} = 0, i > j$

(C) $a_{ij} \neq 0, i > j$

(D) $a_{ij} \neq 0, j > i$

(3). Given

$$[A] = \begin{bmatrix} 12.3 & -12.3 & 20.3 \\ 11.3 & -10.3 & -11.3 \\ 10.3 & -11.3 & -12.3 \end{bmatrix}, \ [B] = \begin{bmatrix} 2 & 4 \\ -5 & 6 \\ 11 & -20 \end{bmatrix}$$

then if

$[C] = [A][B]$, then

$c_{31} =$ _____

(A) $-58.2$

(B) $-37.6$

(C) 219.4

(D) 259.4

(4). The following system of equations has _____ solution(s).

$$x + y = 2$$

$$6x + 6y = 12$$

(A) infinite

(B) no

(C) two

(D) unique

(5). Consider there are only two computer companies in a country. The companies are named *Dude* and *Imac*. Each year, company *Dude* keeps $1/5^{th}$ of its customers, while the rest switch to *Imac*. Each year, *Imac* keeps $1/3^{rd}$ of its customers, while the rest switch to *Dude*. If in 2003, *Dude* had $1/6^{th}$ of the market and *Imac* had $5/6^{th}$ of the market, what will be the share of *Dude* computers when the market becomes stable?

(A) 37/90

(B) 5/11

(C) 6/11

(D) 53/90

(6). Three kids - Jim, Corey, and David receive an inheritance of $\$2,253,453$. The money is put in three trusts but is not divided equally, to begin with. Corey's trust is three times that of David's because Corey made an $A$ in Dr. Kaw's class. Each trust is put in an interest-generating investment. The three trusts of Jim, Corey, and David pays an interest of 6%, 8%, 11%, respectively. The total interest of all the three trusts combined at the end of the first year is $\$190,740.57$. The equations to find the trust money of Jim ($J$), Corey ($C$), and David ($D$) in a matrix form is

(A) $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 3 & -1 \\ 0.06 & 0.08 & 0.11 \end{bmatrix} \begin{bmatrix} J \\ C \\ D \end{bmatrix} = \begin{bmatrix} 2,253,453 \\ 0 \\ 190,740.57 \end{bmatrix}$

(B) $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & -3 \\ 0.06 & 0.08 & 0.11 \end{bmatrix} \begin{bmatrix} J \\ C \\ D \end{bmatrix} = \begin{bmatrix} 2,253,453 \\ 0 \\ 190,740.57 \end{bmatrix}$

(C) $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & -3 \\ 6 & 8 & 11 \end{bmatrix} \begin{bmatrix} J \\ C \\ D \end{bmatrix} = \begin{bmatrix} 2,253,453 \\ 0 \\ 190,740.57 \end{bmatrix}$

(D) $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 3 & -1 \\ 6 & 8 & 11 \end{bmatrix} \begin{bmatrix} J \\ C \\ D \end{bmatrix} = \begin{bmatrix} 2,253,453 \\ 0 \\ 19,074,057 \end{bmatrix}$

For complete solution, go to

http://nm.mathforcollege.com/mcquizzes/04sle/quiz_04sle_background_solution.pdf

# Problem Set

Chapter 04.01: Prerequisites to Simultaneous Linear Equations

## Introduction Exercise

(1). Write an example of a row vector of dimension 4.

**Answer:** $\begin{bmatrix} 5 & 6 & 2 & 3 \end{bmatrix}$

(2). Write an example of a column vector of dimension 4.

**Answer:** $\begin{bmatrix} 5 \\ -7 \\ 3 \\ 2.5 \end{bmatrix}$

(3). Write an example of a square matrix of order $4 \times 4$.

**Answer:** $\begin{bmatrix} 9 & 0 & -2 & 3 \\ -2 & 3 & 5 & 1 \\ 1.5 & 6 & 7 & 8 \\ 1.1 & 2 & 3 & 4 \end{bmatrix}$

(4). Write an example of a tri-diagonal matrix of order $4 \times 4$.

**Answer:** $\begin{bmatrix} 6 & 3 & 0 & 0 \\ 2.1 & 2 & 2.2 & 0 \\ 0 & 6.2 & -3 & 3.5 \\ 0 & 0 & 2.1 & 4.1 \end{bmatrix}$

(5). Write an example of a identity matrix of order $5 \times 5$.

**Answer:** $\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$

(6). Write an example of a upper triangular matrix of order $4 \times 4$.

**Answer:** $\begin{bmatrix} 6 & 2 & 3 & 9 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 4 & 5 \\ 0 & 0 & 0 & 6 \end{bmatrix}$

(7). Write an example of a lower triangular matrix of order $4 \times 4$.

**Answer:** $\begin{bmatrix} 2 & 0 & 0 & 0 \\ 3 & 1 & 0 & 0 \\ 4 & 2 & 4 & 0 \\ 5 & 3 & 5 & 6 \end{bmatrix}$

(8). Which of these matrices are strictly diagonally dominant?

(A) $[A] = \begin{bmatrix} 15 & 6 & 7 \\ 2 & -4 & 2 \\ 3 & 2 & 6 \end{bmatrix}$

(B) $[A] = \begin{bmatrix} 5 & 6 & 7 \\ 2 & -4 & 2 \\ 3 & 2 & -5 \end{bmatrix}$

(C) $[A] = \begin{bmatrix} 5 & 3 & 2 \\ 6 & -8 & 2 \\ 7 & -5 & 12 \end{bmatrix}$

**Answer:** (A) Yes (B) No (C) No

(9). Find all the submatrices of

$$[A] = \begin{bmatrix} 10 & -7 & 0 \\ 0 & -0.001 & 6 \end{bmatrix}$$

**Answer:** $[10]$ $[-7]$ , $[0]$, $[-0.001]$, $[6]$ $\begin{bmatrix} 10 \\ 0 \end{bmatrix}$, $\begin{bmatrix} -7 \\ -.001 \end{bmatrix}$, $\begin{bmatrix} 0 \\ 6 \end{bmatrix}$, $\begin{bmatrix} 10 & -7 & 0 \end{bmatrix}$, $\begin{bmatrix} 0 & -0.001 & 6 \end{bmatrix}$, $\begin{bmatrix} 10 & -7 \\ 0 & -0.001 \end{bmatrix}$,$\begin{bmatrix} 10 & 0 \\ 0 & 6 \end{bmatrix}$, $\begin{bmatrix} -7 & 0 \\ -0.001 & 6 \end{bmatrix}$, $[10, -7]$ , $[10, 0]$, $[-7, 0]$ , $[0, 6]$ , $[0, -0.001]$, $[-0.001, 6]$.

(10). If

$$[A] = \begin{bmatrix} 4 & -1 \\ 0 & 2 \end{bmatrix},$$

what are $b_{11}$ and $b_{12}$ in

$$[B] = \begin{bmatrix} b_{11} & b_{12} \\ 0 & 4 \end{bmatrix}$$

if $[B] = 2[A]$.

**Answer:** $8, -2$

(11). Are matrix

$$[A] = \begin{bmatrix} 10 & -7 & 0 \\ 0 & -0.001 & 6 \end{bmatrix}$$

and matrix

$$[B] = \begin{bmatrix} 10 & 0 \\ -7 & -0.001 \\ 0 & 6 \end{bmatrix}$$

equal?

**Answer:** No

(12). A square matrix $[A]$ is lower triangular if

(A) $a_{ij} = 0$ for $i > j$

(B) $a_{ij} = 0$ for $j > i$

(C) $a_{ij} = 0$ for $i = j$

(D) $a_{ij} = 0$ for $i + j = odd\ integer$

**Answer:** B

(13). A square matrix $[A]$ is upper triangular if

(A) $a_{ij} = 0$ for $i > j$

(B) $a_{ij} = 0$ for $j > i$

(C) $a_{ij} = 0$ for $i = j$

$a_{ij} = 0$ for $i + j = odd\ integer$

**Answer:** A

# Problem Set

Chapter 04.01: Prerequisites to Simultaneous Linear Equations

(1). Write an example of a row vector of dimension 4.

**Answer:** $[B] = \begin{bmatrix} 2 & 1 & 5 & 7 \end{bmatrix}$

(2). Write an example of a column vector of dimension 4.

**Answer:** $[A] = \begin{bmatrix} 1 \\ 8 \\ 9 \\ 5 \end{bmatrix}$

(3). Write an example of a square matrix of order $4 \times 4$.

**Answer:** $[A] = \begin{bmatrix} 1 & 8 & 6 & 2 \\ 9 & 3 & 4 & 1 \\ 7 & 6 & 2 & 5 \\ 8 & 3 & 6 & 4 \end{bmatrix}$

(4). Write an example of a tri-diagonal matrix of order $4 \times 4$.

**Answer:** $[A] = \begin{bmatrix} 2 & 4 & 0 & 0 \\ 2 & 3 & 9 & 0 \\ 0 & 0 & 5 & 2 \\ 0 & 0 & 3 & 6 \end{bmatrix}$

(5). Write an example of a identity matrix of order $5 \times 5$.

**Answer:** $[A] = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$

(6). Write an example of a upper triangular matrix of order $4 \times 4$.

**Answer:** $[A] = \begin{bmatrix} 1 & 3 & 6 & 0 \\ 0 & 5 & 8 & 2 \\ 0 & 0 & 4 & 9 \\ 0 & 0 & 0 & 2 \end{bmatrix}$

(7). Write an example of a lower triangular matrix of order $4 \times 4$.

**Answer:** $[A] = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 3 & 5 & 0 & 0 \\ 9 & 6 & 4 & 0 \\ 7 & 8 & 1 & 2 \end{bmatrix}$

(8). Which of these matrices are strictly diagonally dominant?

a)

$$[A] = \begin{bmatrix} 15 & 6 & 7 \\ 2 & -4 & 2 \\ 3 & 2 & 6 \end{bmatrix}$$

b)

$$[A] = \begin{bmatrix} 5 & 6 & 7 \\ 2 & -4 & 2 \\ 3 & 2 & -5 \end{bmatrix}$$

c)

$$[A] = \begin{bmatrix} 5 & 3 & 2 \\ 6 & -8 & 2 \\ 7 & -5 & 12 \end{bmatrix}$$

**Answer:** None of them

(9). For the following matrices

$$[A] = \begin{bmatrix} 3 & 0 \\ -1 & 2 \\ 1 & 1 \end{bmatrix}, \; [B] = \begin{bmatrix} 4 & -1 \\ 0 & 2 \end{bmatrix}, \; [C] = \begin{bmatrix} 5 & 2 \\ 3 & 5 \\ 6 & 7 \end{bmatrix},$$

find where possible

a) 4[A]+5[C]

b) [A][B]

c) [A]-2[C]

**Answer:** a) $[C] = \begin{bmatrix} 37 & 10 \\ 11 & 33 \\ 34 & 39 \end{bmatrix}$

b) $[C] = \begin{bmatrix} 12 & -3 \\ -4 & 5 \\ 4 & 1 \end{bmatrix}$

c) $[C] = \begin{bmatrix} -7 & -4 \\ -7 & -8 \\ -11 & -13 \end{bmatrix}$

(10). Food orders are taken from two engineering departments for a takeout. The order is tabulated below.

Food order:

$$\begin{array}{c} \\ \text{Mechanical} \\ \text{Civil} \end{array} \begin{array}{c} \begin{array}{ccc} \text{Chicken Sandwich} & \text{Fries} & \text{Drink} \end{array} \\ \begin{bmatrix} 25 & 35 & 25 \\ 21 & 20 & 21 \end{bmatrix} \end{array}$$

However they have a choice of buying this food from three different restaurants. Their prices for the three food items are tabulated below

Price Matrix:

$$\begin{array}{c} \\ \text{ChickenSandwich} \\ \text{Fries} \\ \text{Drink} \end{array} \begin{array}{c} \begin{array}{ccc} \text{McFat} & \text{Burcholestrol} & \text{Kentucky Sodium} \end{array} \\ \begin{bmatrix} 2.42 & 2.38 & 2.46 \\ 0.93 & 0.90 & 0.89 \\ 0.95 & 1.03 & 1.13 \end{bmatrix} \end{array}$$

Show how much each department will pay for their order at each restaurant. Which restaurant would be more economical to order from for each department?

**Answer:** Burcholesterol is the cheapest for the mechanical department, and McFat is the cheapest for the civil department.

(11). Given

$$[A] = \begin{bmatrix} 12.3 & -12.3 & 10.3 \\ 11.3 & -10.3 & -11.3 \\ 10.3 & -11.3 & -12.3 \end{bmatrix},$$

$$[B] = \begin{bmatrix} 2 & 4 \\ -5 & 6 \\ 11 & -20 \end{bmatrix}$$

Find $[C]$ if $[C] = [A][B]$ .

**Answer:** $[C] = \begin{bmatrix} 199.4 & -230.6 \\ -50.2 & 209.4 \\ -58.2 & 219.4 \end{bmatrix}$

(12). The set of equations

$$\begin{bmatrix} 1 & 2 & 5 \\ 7 & 3 & 9 \\ 8 & 5 & 14 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 8 \\ 19 \\ 27 \end{bmatrix}$$

has unique, no, or infinite solutions. Which is it?

**Answer:** infinite solutions

(13). For what values of $a$ will the following simultaneous linear equations

$$x_1 + x_2 + x_3 = 4$$
$$x_3 = 2$$
$$\left(a^2 - 4\right) x_1 + x_3 = a - 2$$

have

a) Unique solution

b) No solution

c) Infinite solutions

**Answer:** Intentionally not provided

(14). Find if

$$[A] = \begin{bmatrix} 5 & -2.5 \\ -2 & 3 \end{bmatrix}$$

and

$$[B] = \begin{bmatrix} 0.3 & 0.25 \\ 0.2 & 0.5 \end{bmatrix}$$

are inverse of each other.

**Answer:** Yes

(15). Find if

$$[A] = \begin{bmatrix} 5 & 2.5 \\ 2 & 3 \end{bmatrix}$$

and

$$[B] = \begin{bmatrix} 0.3 & -0.25 \\ 0.2 & 0.5 \end{bmatrix}$$

are inverse of each other.

**Answer:** No

(16). Three kids - Jim, Corey and David receive an inheritance of $\$2,253,453$. The money is put in three trusts but is not divided equally to begin with. Corey gets three times what David gets because Corey made an "A" in Dr. Kaw's class. Each trust is put in an interest generating investment. The three trusts of Jim, Corey and David pays an interest of 6%, 8%, 11%, respectively. The combined interest gained in all the three trusts at the end of the first year is $\$190,740.57$. How much money was invested in each trust? Set the following as equations in a matrix form. Identify the unknowns. Do not solve for the unknowns.

**Answer:** $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & -3 \\ 0.06 & 0.08 & 0.11 \end{bmatrix} \begin{bmatrix} J \\ C \\ D \end{bmatrix} = \begin{bmatrix} 2,253,453 \\ 0 \\ 190,740.57 \end{bmatrix}$

(17). By any scientific method, find the second column of the inverse of

$$\begin{bmatrix} 1 & 2 & 0 \\ 4 & 5 & 0 \\ 0 & 0 & 13 \end{bmatrix}$$

**Answer:** $\begin{bmatrix} 0.6666 \\ -0.3333 \\ 0 \end{bmatrix}$

(18). Just write out the inverse of (no need to show any work)

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 5 \end{bmatrix}$$

**Answer:** $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0.5 & 0 & 0 \\ 0 & 0 & 0.25 & 0 \\ 0 & 0 & 0 & 0.20 \end{bmatrix}$

(19). Solve $[A][X] = [B]$ for $[X]$ if

$$[A]^{-1} = \begin{bmatrix} 10 & -7 & 0 \\ 2 & 2 & 5 \\ 2 & 0 & 6 \end{bmatrix},$$

and

$$[B] = \begin{bmatrix} 7 \\ 2.5 \\ 6.012 \end{bmatrix}.$$

**Answer:** $[X] = \begin{bmatrix} 52.5 \\ 49.06 \\ 50.072 \end{bmatrix}$

(20). Find the missing coefficients in the following matrices.

$$\begin{bmatrix} 1 & 2 & 0 \\ 4 & 5 & 0 \\ 0 & 0 & 13 \end{bmatrix} \begin{bmatrix} X & a'_{12} & X \\ X & a'_{22} & X \\ X & a'_{32} & X \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

**Answer:** $[A]^{-1} = \begin{bmatrix} -1.666 & 0.6666 & 0 \\ 1.333 & -0.3333 & 0 \\ 0 & 0 & 0.07692 \end{bmatrix}$

# Chapter 04.02: Vectors

Lesson: Vectors

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *define a vector,*
2) *add and subtract vectors,*
3) *find linear combinations of vectors and their relationship to a set of equations,*
4) *explain what it means to have a linearly independent set of vectors, and*
5) *find the rank of a set of vectors.*

## What is a vector?

A vector is a collection of numbers in a definite order. If it is a collection of $n$ numbers, it is called a $n$-dimensional vector. So, the vector $\overrightarrow{A}$ given by

$$\overrightarrow{A} = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}$$

is a $n$-dimensional column vector with $n$ components, $a_1, a_2, ....., a_n$. The above is a column vector. A row vector $[B]$ is of the form $\overrightarrow{B} = [b_1, b_2, ...., b_n]$ where $\overrightarrow{B}$ is a $n$-dimensional row vector with $n$ components $b_1, b_2, ...., b_n$.

### Example 1

Give an example of a 3-dimensional column vector.

**Solution**

Assume a point in space is given by its $(x, y, z)$ coordinates. Then if the value of $x = 3,\ y = 2,\ z = 5$, the column vector corresponding to the location of the points is

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \\ 5 \end{bmatrix}$$

## When are two vectors equal?

Two vectors $\overrightarrow{A}$ and $\overrightarrow{B}$ are equal if they are of the same dimension and if their corresponding components are equal.

Given

$$\overrightarrow{A} = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}$$

and

$$\overrightarrow{B} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$$

then $\overrightarrow{A} = \overrightarrow{B}$ if $a_i = b_i, \quad i = 1, 2, ....., n.$

**Example 2**

What are the values of the unknown components in $\overrightarrow{B}$ if

$$\overrightarrow{A} = \begin{bmatrix} 2 \\ 3 \\ 4 \\ 1 \end{bmatrix}$$

and

$$\overrightarrow{B} = \begin{bmatrix} b_1 \\ 3 \\ 4 \\ b_4 \end{bmatrix}$$

and $\overrightarrow{A} = \overrightarrow{B}$.

**Solution**

$$b_1 = 2, b_4 = 1$$

## How do you add two vectors?

Two vectors can be added only if they are of the same dimension and the addition is given by

$$\begin{aligned}[A]+[B] &= \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} + \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix} \\ &= \begin{bmatrix} a_1+b_1 \\ a_2+b_2 \\ \vdots \\ a_n+b_n \end{bmatrix}\end{aligned}$$

**Example 3**

Add the two vectors

$$\overrightarrow{A} = \begin{bmatrix} 2 \\ 3 \\ 4 \\ 1 \end{bmatrix}$$

and

$$\overrightarrow{B} = \begin{bmatrix} 5 \\ -2 \\ 3 \\ 7 \end{bmatrix}$$

**Solution**

$$\begin{aligned}\overrightarrow{A}+\overrightarrow{B} &= \begin{bmatrix} 2 \\ 3 \\ 4 \\ 1 \end{bmatrix} + \begin{bmatrix} 5 \\ -2 \\ 3 \\ 7 \end{bmatrix} \\ &= \begin{bmatrix} 2+5 \\ 3-2 \\ 4+3 \\ 1+7 \end{bmatrix} \\ &= \begin{bmatrix} 7 \\ 1 \\ 7 \\ 8 \end{bmatrix}\end{aligned}$$

**Example 4**

A store sells three brands of tires: Tirestone, Michigan and Copper. In quarter 1, the sales are given by the column vector

$$\overrightarrow{A}_1 = \begin{bmatrix} 25 \\ 5 \\ 6 \end{bmatrix}$$

where the rows represent the three brands of tires sold – Tirestone, Michigan and Copper respectively. In quarter 2, the sales are given by

$$\overrightarrow{A}_2 = \begin{bmatrix} 20 \\ 10 \\ 6 \end{bmatrix}$$

What is the total sale of each brand of tire in the first half of the year?

**Solution**

The total sales would be given by

$$\begin{aligned} \overrightarrow{C} &= \overrightarrow{A}_1 + \overrightarrow{A}_2 \\ &= \begin{bmatrix} 25 \\ 5 \\ 6 \end{bmatrix} + \begin{bmatrix} 20 \\ 10 \\ 6 \end{bmatrix} \\ &= \begin{bmatrix} 25+20 \\ 5+10 \\ 6+6 \end{bmatrix} \\ &= \begin{bmatrix} 45 \\ 15 \\ 12 \end{bmatrix} \end{aligned}$$

So, the number of Tirestone tires sold is 45, Michigan is 15 and Copper is 12 in the first half of the year.

## What is a null vector?

A null vector (also called zero vector) is where all the components of the vector are zero.

**Example 5**

Give an example of a null vector or zero vector.

**Solution**

The vector

$$\begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

is an example of a zero or null vector.

## What is a unit vector?

A unit vector $\overrightarrow{U}$ is defined as

$$\overrightarrow{U} = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix}$$

where

$$\sqrt{u_1^2 + u_2^2 + u_3^2 + \ldots + u_n^2} = 1$$

**Example 6**

Give examples of 3-dimensional unit column vectors.

**Solution**

Examples include

$$\begin{bmatrix} \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \text{ etc.}$$

## How do you multiply a vector by a scalar?

If $k$ is a scalar and $\overrightarrow{A}$ is a $n$-dimensional vector, then

$$\begin{aligned} k\overrightarrow{A} &= k \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} \\ &= \begin{bmatrix} ka_1 \\ ka_2 \\ \vdots \\ ka_n \end{bmatrix} \end{aligned}$$

**Example 7**

What is $2\overrightarrow{A}$ if

$$\overrightarrow{A} = \begin{bmatrix} 25 \\ 20 \\ 5 \end{bmatrix}$$

**Solution**

$$\begin{aligned} 2\overrightarrow{A} &= 2 \begin{bmatrix} 25 \\ 20 \\ 5 \end{bmatrix} \\ &= \begin{bmatrix} 2 \times 25 \\ 2 \times 20 \\ 2 \times 5 \end{bmatrix} \\ &= \begin{bmatrix} 50 \\ 40 \\ 10 \end{bmatrix} \end{aligned}$$

**Example 8**

A store sells three brands of tires: Tirestone, Michigan and Copper. In quarter 1, the sales are given by the column vector

$$\overrightarrow{A} = \begin{bmatrix} 25 \\ 25 \\ 6 \end{bmatrix}$$

If the goal is to increase the sales of all tires by at least 25% in the next quarter, how many of each brand should be sold?

**Solution**

Since the goal is to increase the sales by 25%, one would multiply the $\overrightarrow{A}$ vector by 1.25,

$$\begin{aligned} \overrightarrow{B} &= 1.25 \begin{bmatrix} 25 \\ 25 \\ 6 \end{bmatrix} \\ &= \begin{bmatrix} 31.25 \\ 31.25 \\ 7.5 \end{bmatrix} \end{aligned}$$

Since the number of tires must be an integer, we can say that the goal of sales is

$$\overrightarrow{B} = \begin{bmatrix} 32 \\ 32 \\ 8 \end{bmatrix}$$

## What do you mean by a linear combination of vectors?

Given

$$\overrightarrow{A}_1, \overrightarrow{A}_2, ....., \overrightarrow{A}_m$$

as $m$ vectors of same dimension $n$, and if $k_1, k_2, ..., k_m$ are scalars, then

$$k_1\overrightarrow{A}_1 + k_2\overrightarrow{A}_2 + ..... + k_m\overrightarrow{A}_m$$

is a linear combination of the $m$ vectors.

**Example 9**

Find the linear combinations
(a) $\overrightarrow{A} - \overrightarrow{B}$ and
(b) $\overrightarrow{A} + \overrightarrow{B} - 3\overrightarrow{C}$
where

$$\overrightarrow{A} = \begin{bmatrix} 2 \\ 3 \\ 6 \end{bmatrix}, \overrightarrow{B} = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}, \overrightarrow{C} = \begin{bmatrix} 10 \\ 1 \\ 2 \end{bmatrix}$$

**Solution**

(a)

$$\begin{aligned} \overrightarrow{A} - \overrightarrow{B} &= \begin{bmatrix} 2 \\ 3 \\ 6 \end{bmatrix} - \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} \\ &= \begin{bmatrix} 2-1 \\ 3-1 \\ 6-2 \end{bmatrix} \\ &= \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix} \end{aligned}$$

(b)

$$\begin{aligned} \overrightarrow{A} + \overrightarrow{B} - 3\overrightarrow{C} &= \begin{bmatrix} 2 \\ 3 \\ 6 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} - 3\begin{bmatrix} 10 \\ 1 \\ 2 \end{bmatrix} \\ &= \begin{bmatrix} 2+1-30 \\ 3+1-3 \\ 6+2-6 \end{bmatrix} \\ &= \begin{bmatrix} -27 \\ 1 \\ 2 \end{bmatrix} \end{aligned}$$

## What do you mean by vectors being linearly independent?

A set of vectors $\overrightarrow{A}_1, \overrightarrow{A}_2, \ldots, \overrightarrow{A}_m$ are considered to be linearly independent if

$$k_1\overrightarrow{A}_1 + k_2\overrightarrow{A}_2 + ..... + k_m\overrightarrow{A}_m = \overrightarrow{0}$$

has only one solution of

$$k_1 = k_2 = ..... = k_m = 0$$

**Example 10**

Are the three vectors

$$\overrightarrow{A}_1 = \begin{bmatrix} 25 \\ 64 \\ 144 \end{bmatrix}, \quad \overrightarrow{A}_2 = \begin{bmatrix} 5 \\ 8 \\ 12 \end{bmatrix}, \quad \overrightarrow{A}_3 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

linearly independent?

**Solution**

Writing the linear combination of the three vectors

$$k_1 \begin{bmatrix} 25 \\ 64 \\ 144 \end{bmatrix} + k_2 \begin{bmatrix} 5 \\ 8 \\ 12 \end{bmatrix} + k_3 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

gives

$$\begin{bmatrix} 25k_1 + 5k_2 + k_3 \\ 64k_1 + 8k_2 + k_3 \\ 144k_1 + 12k_2 + k_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

The above equations have only one solution, $k_1 = k_2 = k_3 = 0$. However, how do we show that this is the only solution? This is shown below.

The above equations are

$$25k_1 + 5k_2 + k_3 = 0 \qquad (1)$$

$$64k_1 + 8k_2 + k_3 = 0 \qquad (2)$$

$$144k_1 + 12k_2 + k_3 = 0 \qquad (3)$$

Subtracting Eqn (1) from Eqn (2) gives

$$39k_1 + 3k_2 = 0$$

$$k_2 = -13k_1 \qquad (4)$$

Multiplying Eqn (1) by 8 and subtracting it from Eqn (2) that is first multiplied by 5 gives

$$120k_1 - 3k_3 = 0$$

$$k_3 = 40k_1 \qquad (5)$$

Remember we found Eqn (4) and Eqn (5) just from Eqns (1) and (2).

Substitution of Eqns (4) and (5) in Eqn (3) for $k_1$ and $k_2$ gives

$$144k_1 + 12(-13k_1) + 40k_1 = 0$$

$$28k_1 = 0$$

$$k_1 = 0$$

This means that $k_1$ has to be zero, and coupled with (4) and (5), $k_2$ and $k_3$ are also zero. So the only solution is $k_1 = k_2 = k_3 = 0$. The three vectors hence are linearly independent.

**Example 11**

Are the three vectors

$$\overrightarrow{A}_1 = \begin{bmatrix} 1 \\ 2 \\ 5 \end{bmatrix}, \ \overrightarrow{A}_2 = \begin{bmatrix} 2 \\ 5 \\ 7 \end{bmatrix}, \ \overrightarrow{A}_3 = \begin{bmatrix} 6 \\ 14 \\ 24 \end{bmatrix}$$

linearly independent?

**Solution**

By inspection,

$$\overrightarrow{A}_3 = 2\overrightarrow{A}_1 + 2\overrightarrow{A}_2$$

or

$$-2\overrightarrow{A}_1 - 2\overrightarrow{A}_2 + \overrightarrow{A}_3 = \overrightarrow{0}$$

So the linear combination

$$k_1\overrightarrow{A}_1 + k_2\overrightarrow{A}_2 + k_3\overrightarrow{A}_3 = \overrightarrow{0}$$

has a non-zero solution

$$k_1 = -2,\ k_2 = -2,\ k_3 = 1$$

Hence, the set of vectors is linearly dependent.

What if I cannot prove by inspection, what do I do? Put the linear combination of three vectors equal to the zero vector,

$$k_1 \begin{bmatrix} 1 \\ 2 \\ 5 \end{bmatrix} + k_2 \begin{bmatrix} 2 \\ 5 \\ 7 \end{bmatrix} + k_3 \begin{bmatrix} 6 \\ 14 \\ 24 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

to give

$$k_1 + 2k_2 + 6k_3 = 0 \qquad (1)$$

$$2k_1 + 5k_2 + 14k_3 = 0 \qquad (2)$$

$$5k_1 + 7k_2 + 24k_3 = 0 \qquad (3)$$

Multiplying Eqn (1) by 2 and subtracting from Eqn (2) gives

$$k_2 + 2k_3 = 0$$

$$k_2 = -2k_3 \qquad (4)$$

Multiplying Eqn (1) by 2.5 and subtracting from Eqn (2) gives

$$-0.5k_1 - k_3 = 0$$

$$k_1 = -2k_3 \qquad (5)$$

Remember we found Eqn (4) and Eqn (5) just from Eqns (1) and (2).

Substitute Eqn (4) and (5) in Eqn (3) for $k_1$ and $k_2$ gives

$$5\left(-2k_3\right) + 7\left(-2k_3\right) + 24k_3 = 0$$

$$-10k_3 - 14k_3 + 24k_3 = 0$$

$$0 = 0$$

This means any values satisfying Eqns (4) and (5) will satisfy Eqns (1), (2) and (3) simultaneously.

For example, chose

$k_3 = 6$, then

$k_2 = -12$ from Eqn (4), and

$k_1 = -12$ from Eqn (5).

Hence we have a nontrivial solution of $\begin{bmatrix} k_1 & k_2 & k_3 \end{bmatrix} = \begin{bmatrix} -12 & -12 & 6 \end{bmatrix}$. This implies the three given vectors are linearly dependent. Can you find another nontrivial solution?

What about the following three vectors?

$$\begin{bmatrix} 1 \\ 2 \\ 5 \end{bmatrix}, \begin{bmatrix} 2 \\ 5 \\ 7 \end{bmatrix}, \begin{bmatrix} 6 \\ 14 \\ 25 \end{bmatrix}$$

Are they linearly dependent or linearly independent?

Note that the only difference between this set of vectors and the previous one is the third entry in the third vector. Hence, equations (4) and (5) are still valid. What conclusion do you draw when you plug in equations (4) and (5) in the third equation: $5k_1 + 7k_2 + 25k_3 = 0$? What has changed?

**Example 12**

Are the three vectors

$$\overrightarrow{A}_1 = \begin{bmatrix} 25 \\ 64 \\ 89 \end{bmatrix}, \ \overrightarrow{A}_2 = \begin{bmatrix} 5 \\ 8 \\ 13 \end{bmatrix}, \ \overrightarrow{A}_3 = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$$

linearly independent?

**Solution**

Writing the linear combination of the three vectors and equating to zero vector

$$k_1 \begin{bmatrix} 25 \\ 64 \\ 89 \end{bmatrix} + k_2 \begin{bmatrix} 5 \\ 8 \\ 13 \end{bmatrix} + k_3 \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

gives

$$\begin{bmatrix} 25k_1 + 5k_2 + k_3 \\ 64k_1 + 8k_2 + k_3 \\ 89k_1 + 13k_2 + 2k_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

In addition to $k_1 = k_2 = k_3 = 0$, one can find other solutions for which $k_1, \ k_2, \ k_3$are not equal to zero. For example, $k_1 = 1, \ k_2 = -13, \ k_3 = 40$ is also a solution as

$$1 \begin{bmatrix} 25 \\ 64 \\ 89 \end{bmatrix} - 13 \begin{bmatrix} 5 \\ 8 \\ 13 \end{bmatrix} + 40 \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Hence $\overrightarrow{A}_1, \quad \overrightarrow{A}_2, \quad \overrightarrow{A}_3$ are linearly dependent.

## What do you mean by the rank of a set of vectors?

From a set of $n$-dimensional vectors, the maximum number of linearly independent vectors in the set is called the rank of the set of vectors. *Note that the rank of the vectors can never be greater than the vectors dimension.*

**Example 13**

What is the rank of

$$\overrightarrow{A}_1 = \begin{bmatrix} 25 \\ 64 \\ 144 \end{bmatrix}, \ \overrightarrow{A}_2 = \begin{bmatrix} 5 \\ 8 \\ 12 \end{bmatrix}, \ \overrightarrow{A}_3 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}?$$

**Solution**

Since we found in Example 2.10 that $\overrightarrow{A}_1, \ \overrightarrow{A}_2, \ \overrightarrow{A}_3$ are linearly independent, the rank of the set of vectors $\overrightarrow{A}_1, \ \overrightarrow{A}_2, \ \overrightarrow{A}_3$ is 3. If we were given another vector $\overrightarrow{A}_4$, the rank of the set of the vectors $\overrightarrow{A}_1, \ \overrightarrow{A}_2, \ \overrightarrow{A}_3, \overrightarrow{A}_4$ would still be 3 as the rank of a set of vectors is always less than or equal to the dimension of the vectors and that at least $\overrightarrow{A}_1, \ \overrightarrow{A}_2, \ \overrightarrow{A}_3$ are linearly independent.

**Example 14**

What is the rank of

$$\overrightarrow{A}_1 = \begin{bmatrix} 25 \\ 64 \\ 89 \end{bmatrix}, \ \overrightarrow{A}_2 = \begin{bmatrix} 5 \\ 8 \\ 13 \end{bmatrix}, \ \overrightarrow{A}_3 = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}?$$

**Solution**

In Example 2.12, we found that $\overrightarrow{A}_1, \ \overrightarrow{A}_2, \ \overrightarrow{A}_3$ are linearly dependent, the rank of $\overrightarrow{A}_1, \ \overrightarrow{A}_2, \ \overrightarrow{A}_3$ is hence not 3, and is less than 3. Is it 2? Let us choose two of the three vectors

$$\overrightarrow{A}_1 = \begin{bmatrix} 25 \\ 64 \\ 89 \end{bmatrix}, \ \overrightarrow{A}_2 = \begin{bmatrix} 5 \\ 8 \\ 13 \end{bmatrix}$$

Linear combination of $\overrightarrow{A}_1$ and $\overrightarrow{A}_2$ equal to zero has only one solution – the trivial solution. Therefore, the rank is 2.

**Example 15**

What is the rank of

$$\overrightarrow{A}_1 = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}, \ \overrightarrow{A}_2 = \begin{bmatrix} 2 \\ 2 \\ 4 \end{bmatrix}, \ \overrightarrow{A}_3 = \begin{bmatrix} 3 \\ 3 \\ 5 \end{bmatrix}?$$

**Solution**

From inspection,

$$\overrightarrow{A}_2 = 2\overrightarrow{A}_1,$$

that implies

$$2\overrightarrow{A}_1 - \overrightarrow{A}_2 + 0\overrightarrow{A}_3 = \overrightarrow{0}$$

Hence

$$k_1\overrightarrow{A}_1 + k_2\overrightarrow{A}_2 + k_3\overrightarrow{A}_3 = \overrightarrow{0}$$

has a nontrivial solution.
So $\overrightarrow{A}_1$, $\overrightarrow{A}_2$, $\overrightarrow{A}_3$ are linearly dependent, and hence the rank of the three vectors is not 3. Since

$$\overrightarrow{A}_2 = 2\overrightarrow{A}_1,$$

$\overrightarrow{A}_1$ and $\overrightarrow{A}_2$ are linearly dependent, but

$$k_1\overrightarrow{A}_1 + k_3\overrightarrow{A}_3 = \overrightarrow{0}.$$

has trivial solution as the only solution. So $\overrightarrow{A}_1$ and $\overrightarrow{A}_3$ are linearly independent. The rank of the above three vectors is 2.

## Prove that if a set of vectors contains the null vector, the set of vectors is linearly dependent.

Let $\overrightarrow{A}_1, \overrightarrow{A}_2, ....., \overrightarrow{A}_m$ be a set of $n$-dimensional vectors, then

$$k_1\overrightarrow{A}_1 + k_2\overrightarrow{A}_2 + \ \ldots\ + k_m\overrightarrow{A}_m = \overrightarrow{0}$$

is a linear combination of the $m$ vectors. Then assuming if $\overrightarrow{A}_1$ is the zero or null vector, any value of $k_1$ coupled with $k_2 = k_3 = \ .. \ . \ = k_m = 0$ will satisfy the above equation. Hence, the set of vectors is linearly dependent as more than one solution exists.

## Prove that if a set of $m$ vectors is linearly independent, then a subset of the $m$ vectors also has to be linearly independent.

Let this subset of vectors be

$$\overrightarrow{A}_{a1}, \overrightarrow{A}_{a2}, \ldots, \overrightarrow{A}_{\mathrm{ap}}$$

where $p < m$.

Then if this subset of vectors is linearly dependent, the linear combination

$$k_1 \overrightarrow{A}_{a1} + k_2 \overrightarrow{A}_{a2} + \ldots + k_p \overrightarrow{A}_{\mathrm{ap}} = \overrightarrow{0}$$

has a non-trivial solution.

So

$$k_1 \overrightarrow{A}_{a1} + k_2 \overrightarrow{A}_{a2} + \ldots + k_p \overrightarrow{A}_{\mathrm{ap}} + 0\,\overrightarrow{A}_{a(p+1)} + \ldots\ldots + 0\,\overrightarrow{A}_{\mathrm{am}} = \overrightarrow{0}$$

also has a non-trivial solution too, where $\overrightarrow{A}_{a(p+1)}, \ldots, \overrightarrow{A}_{\mathrm{am}}$are the rest of the $(m-p)$ vectors. However, this is a contradiction. Therefore, a subset of linearly independent vectors cannot be linearly dependent.

## Prove that if a set of vectors is linearly dependent, then at least one vector can be written as a linear combination of others.

Let $\overrightarrow{A}_1, \overrightarrow{A}_2, \ldots, \overrightarrow{A}_m$ be linearly dependent set of vectors, then there exists a set of scalars $k_1, \ldots, k_m$ not all of which are zero for the linear combination equation

$$k_1 \overrightarrow{A}_1 + k_2 \overrightarrow{A}_2 + \ldots + k_m \overrightarrow{A}_m = \overrightarrow{0}.$$

Let $k_p$ be one of the non-zero values of $k_i,\ i = 1, \ldots, m$, that is, $k_p \neq 0$, then

$$A_p = -\frac{k_2}{k_p}\overrightarrow{A}_2 - \ \ldots\ - \frac{k_{p-1}}{k_p}\overrightarrow{A}_{p-1} - \frac{k_{p+1}}{k_p}\overrightarrow{A}_{p+1} - \ \ldots\ - \frac{k_m}{k_p}\overrightarrow{A}_m$$

and that proves the theorem.

## Prove that if the dimension of a set of vectors is less than the number of vectors in the set, then the set of vectors is linearly dependent.

Can you prove it?

## How can vectors be used to write simultaneous linear equations?

If a set of $m$ simultaneous linear equations with $n$ unknowns is written as

$$a_{11}x_1 + \ \ldots\ + a_{1n}x_n = c_1$$

$$a_{21}x_1 + \ \ldots\ + a_{2n}x_n = c_2$$

$$\begin{array}{cc} \vdots & \vdots \\ \vdots & \vdots \end{array}$$

$$a_{m1}x_1 + \ \ldots \ + a_{\mathrm{mn}}x_n = c_n$$

where

$x_1, x_2, \ldots, x_n$are the unknowns, then in the vector notation they can be written as

$$x_1\overrightarrow{A}_1 + x_2\overrightarrow{A}_2 + \ldots + x_n\overrightarrow{A}_n = \overrightarrow{C}$$

where

$$\overrightarrow{A}_1 = \begin{bmatrix} a_{11} \\ \vdots \\ a_{m1} \end{bmatrix}$$

where

$$\overrightarrow{A}_1 = \begin{bmatrix} a_{11} \\ \vdots \\ a_{m1} \end{bmatrix}$$

$$\overrightarrow{A}_2 = \begin{bmatrix} a_{12} \\ \vdots \\ a_{m2} \end{bmatrix}$$

$$\overrightarrow{A}_n = \begin{bmatrix} a_{1n} \\ \vdots \\ a_{\mathrm{mn}} \end{bmatrix}$$

$$\overrightarrow{C}_1 = \begin{bmatrix} c_1 \\ \vdots \\ c_m \end{bmatrix}$$

The problem now becomes whether you can find the scalars $x_1, x_2, ....., x_n$ such that the linear combination

$$x_1\overrightarrow{A}_1 + ..... + x_n\overrightarrow{A}_n$$

is equal to the $\overrightarrow{C}$, that is

$$x_1\overrightarrow{A}_1 + ..... + x_n\overrightarrow{A}_n = \overrightarrow{C}$$

**Example 16**

Write

$$25x_1 + 5x_2 + x_3 = 106.8$$

$$64x_1 + 8x_2 + x_3 = 177.2$$

$$144x_1 + 12x_2 + x_3 = 279.2$$

as a linear combination of set of vectors equal to another vector.

**Solution**

$$\begin{bmatrix} 25x_1 & +5x_2 & +x_3 \\ 64x_1 & +8x_2 & +x_3 \\ 144x_1 & +12x_2 & +x_3 \end{bmatrix} = \begin{bmatrix} 106.8 \\ 177.2 \\ 279.2 \end{bmatrix}$$

$$x_1 \begin{bmatrix} 25 \\ 64 \\ 144 \end{bmatrix} + x_2 \begin{bmatrix} 5 \\ 8 \\ 12 \end{bmatrix} + x_3 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 106.8 \\ 177.2 \\ 279.2 \end{bmatrix}$$

## What is the definition of the dot product of two vectors?

Let $\overrightarrow{A} = [a_1, a_2, \ldots, a_n]$ and $\overrightarrow{B} = [b_1, b_2, \ldots, b_n]$ be two $n$-dimensional vectors. Then the dot product of the two vectors $\overrightarrow{A}$ and $\overrightarrow{B}$ is defined as

$$\overrightarrow{A} \cdot \overrightarrow{B} = a_1 b_1 + a_2 b_2 + \ldots + a_n b_n = \sum_{i=1}^{n} a_i b_i$$

A dot product is also called an inner product.

**Example 17**

Find the dot product of the two vectors $\overrightarrow{A} = [4, 1, 2, 3]$ and $\overrightarrow{B} = [3, 1, 7, 2]$.

**Solution**

$$\begin{aligned} \overrightarrow{A} \cdot \overrightarrow{B} &= [4, 1, 2, 3] \ . \ [3, 1, 7, 2] \\ &= (4)(3) + (1)(1) + (2)(7) + (3)(2) \\ &= 33 \end{aligned}$$

**Example 18**

A product line needs three types of rubber as given in the table below.

| Rubber Type | Weight (lbs) | Cost per pound ($) |
|---|---|---|
| A | 200 | 20.23 |
| B | 250 | 30.56 |
| C | 310 | 29.12 |

Use the definition of a dot product to find the total price of the rubber needed.

**Solution**

The weight vector is given by

$$\overrightarrow{W} = [200, 250, 310]$$

and the cost vector is given by

$$\overrightarrow{C} = [20.23, 30.56, 29.12]$$

The total cost of the rubber would be the dot product of $\overrightarrow{W}$ and $\overrightarrow{C}$.

$$\begin{aligned} \overrightarrow{W} \cdot \overrightarrow{C} &= [200, 250, 310] \cdot [20.23, 30.56, 29.12] \\ &= (200)(20.23) + (250)(30.56) + (310)(29.12) \\ &= 4046 + 7640 + 9027.2 \\ &= \$20713.20 \end{aligned}$$

## Key Terms:

*Vector*

*Addition of vectors*

*Rank*

*Dot Product*

*Subtraction of vectors*

*Unit vector*

*Scalar multiplication of vectors*

*Null vector*

*Linear combination of vectors*

*Linearly independent vectors*

# Multiple Choice Test

Chapter 04.02 Vectors

(1). A set of equations

$$4x_1 + 7x_2 + 11x_3 = 13$$

$$17x_1 + 39x_2 + 23x_3 = 31$$

$$13x_1 + 67x_2 + 59x_3 = 37$$

can also be written as

(A) $x_1 \begin{bmatrix} 4 \\ 17 \\ 13 \end{bmatrix} + x_2 \begin{bmatrix} 7 \\ 39 \\ 23 \end{bmatrix} + x_3 \begin{bmatrix} 11 \\ 23 \\ 59 \end{bmatrix} = \begin{bmatrix} 13 \\ 31 \\ 37 \end{bmatrix}$

(B) $4 \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + 39 \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + 59 \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 13 \\ 31 \\ 37 \end{bmatrix}$

(C) $x_1 \begin{bmatrix} 4 \\ 7 \\ 11 \end{bmatrix} + x_2 \begin{bmatrix} 17 \\ 39 \\ 23 \end{bmatrix} + x_3 \begin{bmatrix} 13 \\ 67 \\ 59 \end{bmatrix} = \begin{bmatrix} 13 \\ 31 \\ 37 \end{bmatrix}$

(D) $x_1 \begin{bmatrix} 13 \\ 17 \\ 4 \end{bmatrix} + x_2 \begin{bmatrix} 67 \\ 39 \\ 7 \end{bmatrix} + x_3 \begin{bmatrix} 59 \\ 23 \\ 11 \end{bmatrix} = \begin{bmatrix} 57 \\ 13 \\ 31 \end{bmatrix}$

(2). The magnitude of the vector, $V = (5, -3, 2)$ is

(A) 4

(B) 10

(C) $\sqrt{38}$

(D) $\sqrt{20}$

(3). The rank of the vector

$$\vec{A} \begin{bmatrix} 2 \\ 3 \\ 7 \end{bmatrix}, \begin{bmatrix} 6 \\ 9 \\ 21 \end{bmatrix}, \begin{bmatrix} 3 \\ 2 \\ 7 \end{bmatrix}$$

is

(A) 1

(B) 2

(C) 3

(D) 4

(4). If $\overrightarrow{A} = (5, 2, 3)$and $\overrightarrow{B} = (6, -7, 3)$, then $4\overrightarrow{A} + 5\overrightarrow{B}$is

(A) $(50, -5, 6)$

(B) $(50, -27, 27)$

(C) $(11, -5, 6)$

(D) $(20, 8, 12)$

(5). The dot product of two vectors $\overrightarrow{A}$ and $\overrightarrow{B}$

$$\overrightarrow{A} = 3i + 5j + 7k$$

$$\overrightarrow{B} = 11i + 13j + 17k$$

most nearly is

(A) 14.80

(B) 33.00

(C) 56.00

(D) 217.0

(6). The angle in degrees between two vectors $\overrightarrow{u}$ and $\overrightarrow{v}$

$$\overrightarrow{u} = 3i + 5j + 7k$$

$$\overrightarrow{v} = 11i + 13j + 17k$$

most nearly is

(A) 8.124

(B) 11.47

(C) 78.52

(D) 81.88

For complete solution, go to

https://ma.mathforcollege.com/mcquizzes/04sle/quiz_04sle_vectors.pdf

# Problem Set

Chapter 04.02: Vectors

(1). For

$$\overrightarrow{A} = \begin{bmatrix} 2 \\ 9 \\ -7 \end{bmatrix}, \overrightarrow{B} = \begin{bmatrix} 3 \\ 2 \\ 5 \end{bmatrix}, \overrightarrow{C} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

find $\overrightarrow{A} + \overrightarrow{B}$ and $2\overrightarrow{A} - 3\overrightarrow{B} + \overrightarrow{C}$.

**Answer:** $\begin{bmatrix} 5 \\ 11 \\ -2 \end{bmatrix}$; $\begin{bmatrix} -4 \\ 13 \\ -28 \end{bmatrix}$

(2). Are

$$\overrightarrow{A} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \overrightarrow{B} = \begin{bmatrix} 1 \\ 2 \\ 5 \end{bmatrix}, \overrightarrow{C} = \begin{bmatrix} 1 \\ 4 \\ 25 \end{bmatrix}$$

linearly independent?.

What is the rank of the above set of vectors?

**Answer:** 3

(3). Are

$$\overrightarrow{A} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \overrightarrow{B} = \begin{bmatrix} 1 \\ 2 \\ 5 \end{bmatrix}, \overrightarrow{C} = \begin{bmatrix} 3 \\ 5 \\ 7 \end{bmatrix}$$

linearly independent?.

What is the rank of the above set of vectors?

**Answer:** 3

(4). Are

$$\overrightarrow{A} = \begin{bmatrix} 1 \\ 2 \\ 5 \end{bmatrix}, \overrightarrow{B} = \begin{bmatrix} 2 \\ 4 \\ 10 \end{bmatrix}, \overrightarrow{C} = \begin{bmatrix} 1.1 \\ 2.2 \\ 5.5 \end{bmatrix}$$

linearly independent?

What is the rank of the above set of vectors?

**Answer:** No;1

(5). If a set of vectors contains the null vector, the set of vectors is linearly

(A) Independent

(B) Dependent?

**Answer:** B

(6). If a set of vectors is linearly independent, a subset of the vectors is linearly

(A) Independent.

(B) Dependent.

**Answer:** A

(7). If a set of vectors is linearly dependent, then

(A) At least one vector can be written as a linear combination of others.

(B) At least one vector is a null vector.

**Answer:** A

(8). If the dimension of a set of vectors is less than the number of vectors in the set, then the set of vectors is linearly

(A) Dependent.

(B) Independent.

**Answer:** A

(9). Find the dot product of $\overrightarrow{A} = (2, 1, 2.5, 3)$and $\overrightarrow{B} = (-3, 2, 1, 2.5)$

**Answer:** 6

(10). If $\overrightarrow{u}, \overrightarrow{v}, \overrightarrow{w}$ are three nonzero vector of 2-dimensions, then

(A) $\overrightarrow{u}, \overrightarrow{v}, \overrightarrow{w}$ are linearly independent

(B) $\overrightarrow{u}, \overrightarrow{v}, \overrightarrow{w}$ are linearly dependent

(C) $\overrightarrow{u}, \overrightarrow{v}, \overrightarrow{w}$ are unit vectors

(D) $k_1\overrightarrow{u} + k_2\overrightarrow{v} + k_3\overrightarrow{v} = \overrightarrow{0}$ has a unique solution.

**Answer:** B

(11). $\overrightarrow{u}$and $\overrightarrow{v}$are two non-zero vectors of dimension $n$. Prove that if $\overrightarrow{u}$and $\overrightarrow{v}$are linearly dependent, there is a scalar $q$ such that $\overrightarrow{v} = q\overrightarrow{u}$.

**Answer:** Hint :

Start with $k_1\overrightarrow{u} + k_2\overrightarrow{v} = \overrightarrow{0}$

Show that $k_1 \neq 0$and $k_2 \neq 0$ because $\overrightarrow{u}$and $\overrightarrow{v}$ are both nonzero.

Hence

$$\begin{aligned}\overrightarrow{v} &= -\frac{k_1}{k_2}\overrightarrow{u} \\ &= q\overrightarrow{u} \qquad q = -\frac{k_1}{k_2}\end{aligned}$$

(12). $\overrightarrow{u}$and $\overrightarrow{v}$are two non-zero vectors of dimension $n$. Prove that if there is a scalar $q$ such that $\overrightarrow{v} = q\overrightarrow{u}$, then $\overrightarrow{u}$and $\overrightarrow{v}$are linearly dependent.

**Answer:** Hint:

Since

$\overrightarrow{v} = q\overrightarrow{u}$

$\overrightarrow{v} - q\overrightarrow{u} = \overrightarrow{0}$

$q \neq 0$, otherwise $\overrightarrow{v} = \overrightarrow{0}$

So the equation

$$k_1\overrightarrow{v} + k_2\overrightarrow{u} = \overrightarrow{0}$$

has a non trivial solution of

$$k_1 = 1, k_2 = q \neq 0.$$

# Chapter 04.03: Binary Matrix Operations

Lesson: Binary Matrix Operations

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *add, subtract, and multiply matrices, and*

2) *apply rules of binary operations on matrices.*

## *How do you add two matrices?*

Two matrices $[A]$ and $[B]$ can be added only if they are the same size. The addition is then shown as

$$[C] = [A] + [B]$$

where

$$c_{ij} = a_{ij} + b_{ij}$$

**Example 1**

Add the following two matrices.

$[A] = \begin{bmatrix} 5 & 2 & 3 \\ 1 & 2 & 7 \end{bmatrix}$ $[B] = \begin{bmatrix} 6 & 7 & -2 \\ 3 & 5 & 19 \end{bmatrix}$

**Solution**

$$\begin{aligned} [C] &= [A] + [B] \\ &= \begin{bmatrix} 5 & 2 & 3 \\ 1 & 2 & 7 \end{bmatrix} + \begin{bmatrix} 6 & 7 & -2 \\ 3 & 5 & 19 \end{bmatrix} \\ &= \begin{bmatrix} 5+6 & 2+7 & 3-2 \\ 1+3 & 2+5 & 7+19 \end{bmatrix} \\ &= \begin{bmatrix} 11 & 9 & 1 \\ 4 & 7 & 26 \end{bmatrix} \end{aligned}$$

**Example 2**

Blowout r'us store has two store locations $A$ and $B$, and their sales of tires are given by make (in rows) and quarters (in columns) as shown below.

$$[A] = \begin{bmatrix} 25 & 20 & 3 & 2 \\ 5 & 10 & 15 & 25 \\ 6 & 16 & 7 & 27 \end{bmatrix}$$

$$[B] = \begin{bmatrix} 20 & 5 & 4 & 0 \\ 3 & 6 & 15 & 21 \\ 4 & 1 & 7 & 20 \end{bmatrix}$$

where the rows represent the sale of Tirestone, Michigan and Copper tires respectively and the columns represent the quarter number: 1, 2, 3 and 4. What are the total tire sales for the two locations by make and quarter?

**Solution**

$$\begin{aligned} [C] &= [A] + [B] \\ &= \begin{bmatrix} 25 & 20 & 3 & 2 \\ 5 & 10 & 15 & 25 \\ 6 & 16 & 7 & 27 \end{bmatrix} + \begin{bmatrix} 20 & 5 & 4 & 0 \\ 3 & 6 & 15 & 21 \\ 4 & 1 & 7 & 20 \end{bmatrix} \\ &= \begin{bmatrix} (25+20) & (20+5) & (3+4) & (2+0) \\ (5+3) & (10+6) & (15+15) & (25+21) \\ (6+4) & (16+1) & (7+7) & (27+20) \end{bmatrix} \\ &= \begin{bmatrix} 45 & 25 & 7 & 2 \\ 8 & 16 & 30 & 46 \\ 10 & 17 & 14 & 47 \end{bmatrix} \end{aligned}$$

So, if one wants to know the total number of Copper tires sold in quarter 4 at the two locations, we would look at Row 3 – Column 4 to give $c_{34} = 47$.

## How do you subtract two matrices?

Two matrices $[A]$ and $[B]$ can be subtracted only if they are the same size. The subtraction is then shown as

$$[D] = [A] - [B]$$

where

$$d_{ij} = a_{ij} - b_{ij}$$

**Example 3**

Subtract matrix $[B]$ from matrix $[A]$.

$$[A] = \begin{bmatrix} 5 & 2 & 3 \\ 1 & 2 & 7 \end{bmatrix}$$

$$[B] = \begin{bmatrix} 6 & 7 & -2 \\ 3 & 5 & 19 \end{bmatrix}$$

**Solution**

$$\begin{aligned} [D] &= [A] - [B] \\ &= \begin{bmatrix} 5 & 2 & 3 \\ 1 & 2 & 7 \end{bmatrix} - \begin{bmatrix} 6 & 7 & -2 \\ 3 & 5 & 19 \end{bmatrix} \\ &= \begin{bmatrix} (5-6) & (2-7) & (3-(-2)) \\ (1-3) & (2-5) & (7-19) \end{bmatrix} \\ &= \begin{bmatrix} -1 & -5 & 5 \\ -2 & -3 & -12 \end{bmatrix} \end{aligned}$$

**Example 4**

Blowout r'us has two store locations $A$ and $B$ and their sales of tires are given by make (in rows) and quarters (in columns) as shown below.

$$[A] = \begin{bmatrix} 2520\,3 & 2 \\ 5\,101525 \\ 6\,16\,7\,27 \end{bmatrix}$$

$$[B] = \begin{bmatrix} 205\,4 & 0 \\ 3\,61521 \\ 4\,1\,7\,20 \end{bmatrix}$$

where the rows represent the sale of Tirestone, Michigan and Copper tires respectively and the columns represent the quarter number: 1, 2, 3, and 4. How many more tires did store $A$ sell than store $B$ of each brand in each quarter?

**Solution**

$$\begin{aligned} [D] &= [A] - [B] \\ &= \begin{bmatrix} 2520\,3 & 2 \\ 5\,101525 \\ 6\,16\,7\,27 \end{bmatrix} - \begin{bmatrix} 205\,4 & 0 \\ 3\,61521 \\ 4\,1\,7\,20 \end{bmatrix} \\ &= \begin{bmatrix} (25-20)(20-5) & (3-4) & (2-0) \\ (5-3) & (10-6)(15-15)(25-21) \\ (6-4) & (16-1) & (7-7) & (27-20) \end{bmatrix} \\ &= \begin{bmatrix} 515{-}12 \\ 2\,4 & 0\,4 \\ 215\,0\,7 \end{bmatrix} \end{aligned}$$

So, if you want to know how many more Copper tires were sold in quarter 4 in store $A$ than store $B$, $d_{34} = 7$. Note that $d_{13} = -1$ implies that store $A$ sold 1 less Michigan tire than store $B$ in quarter 3.

## How do I multiply two matrices?

Two matrices $[A]$ and $[B]$ can be multiplied only if the number of columns of $[A]$ is equal to the number of rows of $[B]$ to give

$$[C]_{m\times n} = [A]_{m\times p}\,[B]_{p\times n}$$

If $[A]$ is a $m \times p$ matrix and $[B]$ is a $p \times n$ matrix, the resulting matrix $[C]$ is a $m \times n$ matrix.

So how does one calculate the elements of $[C]$ matrix?

$$\begin{aligned} c_{ij} &= \sum_{k=1}^{p} a_{ik} b_{kj} \\ &= a_{i1}b_{1j} + a_{i2}b_{2j} + \ldots + a_{ip}b_{pj} \end{aligned}$$

for each $i = 1,\ 2,\ \ldots\ ,\ m$ and $j = 1,\ 2,\ \ldots\ ,\ n$.

To put it in simpler terms, the $i^{th}$ row and $j^{th}$ column of the $[C]$ matrix in $[C] = [A]\,[B]$ is calculated by multiplying the $i^{th}$ row of $[A]$ by the $j^{th}$ column of $[B]$. That is

$$\begin{aligned} c_{ij} &= \lceil a_{i1}a_{i2} \quad \ldots \quad a_{\mathrm{ip}} \rceil \begin{bmatrix} b_{1j} \\ b_{2j} \\ \vdots \\ b_{\mathrm{pj}} \end{bmatrix} \\ &= a_{i1}b_{1j} + a_{i2}b_{2j} + \ldots + a_{\mathrm{ip}}b_{\mathrm{pj}} \\ &= \sum_{k=1}^{p} a_{\mathrm{ik}} b_{\mathrm{kj}} \end{aligned}$$

**Example 5**

Given

$$[A] = \begin{bmatrix} 5 & 2 & 3 \\ 1 & 2 & 7 \end{bmatrix}$$

$$[B] = \begin{bmatrix} 3 & -2 \\ 5 & -8 \\ 9 & -10 \end{bmatrix}$$

Find

$$[C] = [A]\,[B]$$

**Solution**

$c_{12}$ can be found by multiplying the first row of $[A]$ by the second column of $[B]$,

$$\begin{aligned} c_{12} &= \begin{bmatrix} 5 & 2 & 3 \end{bmatrix} \begin{bmatrix} -2 \\ -8 \\ -10 \end{bmatrix} \\ &= (5)(-2) + (2)(-8) + (3)(-10) \\ &= -56 \end{aligned}$$

Similarly, one can find the other elements of $[C]$ to give

$$[C] = \begin{bmatrix} 52 & -56 \\ 76 & -88 \end{bmatrix}$$

**Example 6**

Blowout r'us store location $A$ and the sales of tires are given by make (in rows) and quarters (in columns) as shown below

$$[A] = \begin{bmatrix} 2520\ 3 & 2 \\ 5\ 101525 \\ 6\ 16\ 7\ 27 \end{bmatrix}$$

where the rows represent the sale of Tirestone, Michigan and Copper tires respectively and the columns represent the quarter number: 1, 2, 3, and 4. Find the per quarter sales of store $A$ if the following are the prices of each tire:

Tirestone = \$33.25

Michigan = \$40.19

Copper = \$25.03

**Solution**

The answer is given by multiplying the price matrix by the quantity of sales of store $A$. The price matrix is $\begin{bmatrix} 33.25 & 40.19 & 25.03 \end{bmatrix}$, so the per-quarter sales of store $A$ would be given by:

$$[C] = \begin{bmatrix} 33.25 & 40.19 & 25.03 \end{bmatrix} \begin{bmatrix} 2520\ 3 & 2 \\ 5\ 101525 \\ 6\ 16\ 7\ 27 \end{bmatrix}$$

$$c_{ij} = \sum_{k=1}^{3} a_{ik} b_{kj}$$

$$
\begin{aligned}
c_{11} &= \sum_{k=1}^{3} a_{1k} b_{k1} \\
&= a_{11} b_{11} + a_{12} b_{21} + a_{13} b_{31} \\
&= (33.25)(25) + (40.19)(5) + (25.03)(6) \\
&= \$1182.38
\end{aligned}
$$

Similarly

$$c_{12} = \$1467.38$$

$$c_{13} = \$877.81$$

$$c_{14} = \$1747.06$$

Therefore, each quarter sales of store $A$ in dollars is given by the four columns of the row vector

$$[C] = [1182.38 \quad 1467.38 \quad 877.81 \quad 1747.06]$$

Remember, since we are multiplying a $1 \times 3$ matrix by a $3 \times 4$ matrix, the resulting matrix is a $1 \times 4$ matrix.

## What is the scalar multiplication of a matrix?

If $[A]$ is a $m \times n$ matrix and $k$ is a real number, then the multiplication $[A]$ by a scalar $k$ is another $m \times n$ matrix $[B]$, where

$b_{ij} = k\ a_{ij}$ for all $i,\ j$.

**Example 7**

Let

$$[A] = \begin{bmatrix} 2.1 & 3 & 2 \\ 5 & 1 & 6 \end{bmatrix}$$

Find $2\,[A]$

**Solution**

$$
\begin{aligned}
2\,[A] &= 2 \begin{bmatrix} 2.1 & 3 & 2 \\ 5 & 1 & 6 \end{bmatrix} \\
&= \begin{bmatrix} 2 \times 2.1 & 2 \times 3 & 2 \times 2 \\ 2 \times 5 & 2 \times 1 & 2 \times 6 \end{bmatrix} \\
&= \begin{bmatrix} 4.2 & 6 & 4 \\ 10 & 2 & 12 \end{bmatrix}
\end{aligned}
$$

## What is a linear combination of matrices?

If $[A_1], [A_2], \ \ldots \ , \ [A_p]$ are matrices of the same size and $k_1, \ k_2, \ \ldots \ , \ k_p$ are scalars, then

$$k_1 [A_1] + k_2 [A_2] + \ \ldots \ + k_p [A_p]$$

is called a linear combination of $[A_1], [A_2], \ \ldots \ , \ [A_p]$.

**Example 8**

If

$$[A_1] = \begin{bmatrix} 5 & 6 & 2 \\ 3 & 2 & 1 \end{bmatrix}, \ [A_2] = \begin{bmatrix} 2.1 & 3 & 2 \\ 5 & 1 & 6 \end{bmatrix}, \ [A_3] = \begin{bmatrix} 0 & 2.2 & 2 \\ 3 & 3.5 & 6 \end{bmatrix}$$

then find

$$[A_1] + 2 [A_2] - 0.5 [A_3]$$

**Solution**

$$[A_1] + 2 [A_2] - 0.5 [A_3]$$

$$\begin{aligned} &= \begin{bmatrix} 5 & 6 & 2 \\ 3 & 2 & 1 \end{bmatrix} + 2 \begin{bmatrix} 2.1 & 3 & 2 \\ 5 & 1 & 6 \end{bmatrix} - 0.5 \begin{bmatrix} 0 & 2.2 & 2 \\ 3 & 3.5 & 6 \end{bmatrix} \\ &= \begin{bmatrix} 5 & 6 & 2 \\ 3 & 2 & 1 \end{bmatrix} + \begin{bmatrix} 4.2 & 6 & 4 \\ 10 & 2 & 12 \end{bmatrix} - 0.5 \begin{bmatrix} 0 & 1.1 & 1 \\ 1.5 & 1.75 & 3 \end{bmatrix} \\ &= \begin{bmatrix} 9.2 & 10.9 & 5 \\ 11.5 & 2.25 & 10 \end{bmatrix} \end{aligned}$$

## What are some of the rules of binary matrix operations?

**Commutative law of addition**

If $[A]$ and $[B]$ are $m \times n$ matrices, then

$$[A] + [B] = [B] + [A]$$

**Associative law of addition**

If $[A]$, $[B]$, and $[C]$ are all $m \times n$ matrices, then

$$[A] + ([B] + [C]) = ([A] + [B]) + [C]$$

**Associative law of multiplication**

If $[A]$, $[B]$, and $[C]$ are $m \times n$, $n \times p$, and $p \times r$ size matrices, respectively, then

$$[A]\,([B]\,[C]) = ([A]\,[B])\,[C]$$

and the resulting matrix size on both sides of the equation is $m \times p$.

**Distributive law**

If $[A]$ and $[B]$ are $m \times n$ size matrices, and $[C]$ and $[D]$ are $n \times p$ size matrices

$$[A]\,([C] + [D]) = [A]\,[C] + [A]\,[D]$$

$$([A] + [B])\,[C] = [A]\,[C] + [B]\,[C]$$

And the resulting matrix size on both sides of the equation is $m \times p$.

**Example 9**

Illustrate the associative law of multiplication of matrices using

$$[A] = \begin{bmatrix} 1 & 2 \\ 3 & 5 \\ 0 & 2 \end{bmatrix}, \quad [B] = \begin{bmatrix} 2 & 5 \\ 9 & 6 \end{bmatrix}, \quad [C] = \begin{bmatrix} 2 & 1 \\ 3 & 5 \end{bmatrix}$$

**Solution**

$$\begin{aligned} [B]\,[C] &= \begin{bmatrix} 2 & 5 \\ 9 & 6 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 3 & 5 \end{bmatrix} \\ &= \begin{bmatrix} 19 & 27 \\ 36 & 39 \end{bmatrix} \end{aligned}$$

$$[A]([B][C]) = \begin{bmatrix} 1 & 2 \\ 3 & 5 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 19 & 27 \\ 36 & 39 \end{bmatrix}$$
$$= \begin{bmatrix} 91 & 105 \\ 237 & 276 \\ 72 & 78 \end{bmatrix}$$

$$[A][B] = \begin{bmatrix} 1 & 2 \\ 3 & 5 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 2 & 5 \\ 9 & 6 \end{bmatrix}$$
$$= \begin{bmatrix} 20 & 17 \\ 51 & 45 \\ 18 & 12 \end{bmatrix}$$

$$([A][B])[C] = \begin{bmatrix} 20 & 17 \\ 51 & 45 \\ 18 & 12 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 3 & 5 \end{bmatrix}$$
$$= \begin{bmatrix} 91 & 105 \\ 237 & 276 \\ 72 & 78 \end{bmatrix}$$

The above illustrates the associative law of multiplication of matrices.

## Is [A][B] = [B][A]?

If $[A][B]$ exists, number of columns of $[A]$ has to be same as the number of rows of $[B]$ and if $[B][A]$ exists, number of columns of $[B]$ has to be same as the number of rows of $[A]$. Now for $[A][B] = [B][A]$, the resulting matrix from $[A][B]$ and $[B][A]$ has to be of the same size. This is only possible if $[A]$ and $[B]$ are square and are of the same size. Even then in general $[A][B] \neq [B][A]$

**Example 10**

Determine if

$$[A][B] = [B][A]$$

For the following matrices

$$[A] = \begin{bmatrix} 6 & 3 \\ 2 & 5 \end{bmatrix}, \quad [B] = \begin{bmatrix} -3 & 2 \\ 1 & 5 \end{bmatrix}$$

**Solution**

$$[A][B] = \begin{bmatrix} 6 & 3 \\ 2 & 5 \end{bmatrix} \begin{bmatrix} -3 & 2 \\ 1 & 5 \end{bmatrix}$$
$$= \begin{bmatrix} -15 & 27 \\ -1 & 29 \end{bmatrix}$$

$$[B][A] = \begin{bmatrix} -3 & 2 \\ 1 & 5 \end{bmatrix} \begin{bmatrix} 6 & 3 \\ 2 & 5 \end{bmatrix}$$
$$= \begin{bmatrix} -14 & 1 \\ 16 & 28 \end{bmatrix}$$

$$[A][B] \neq [B][A]$$

## Key Terms:

*Addition of matrices*

*Subtraction of matrices*

*Multiplication of matrices*

*Scalar Product of matrices*

*Linear Combination of Matrices*

*Rules of Binary Matrix Operation*

# Multiple Choice Test

## Chapter 04.03 Binary Matrix Operations

(1). If $[A] = \begin{bmatrix} 5 & 6 \\ 7 & -3 \end{bmatrix}$ and $[B] = \begin{bmatrix} 2 \\ -3 \end{bmatrix}$ then $[A][B] =$

(A) $\begin{bmatrix} -8 \\ 23 \end{bmatrix}$

(B) $\begin{bmatrix} 10 & 12 \\ 14 & 9 \end{bmatrix}$

(C) $\begin{bmatrix} -2 & 5 \end{bmatrix}$

(D) not possible

(2). For the product $[A][B]$to be possible

(A) the number of rows of $[A]$ needs to be the same as the number of columns of $[B]$

(B) the number of columns of $[A]$ needs to be the same as the number of rows of $[B]$

(C) the number of rows of $[A]$ and$[B]$ needs to be the same

(D) the number of columns of $[A]$ and$[B]$ needs to be the same

(3). If $[A] = \begin{bmatrix} 50 & 60 \\ 20 & -30 \end{bmatrix}$ then $6[A]$is equal to

(A) $\begin{bmatrix} 50 & 360 \\ 120 & -180 \end{bmatrix}$

(B) $\begin{bmatrix} 300 & 60 \\ 20 & -30 \end{bmatrix}$

(C) $\begin{bmatrix} 300 & 360 \\ 120 & -180 \end{bmatrix}$

(D) $\begin{bmatrix} 56 & 66 \\ 26 & -24 \end{bmatrix}$

(4). $[A]$ and $[B]$ are square matrices of $n \times n$ order. Then $([A] - [B])([A] - [B])$ is equal to

(A) $[A]^2 + [B]^2 - 2[A][B]$

(B) $[A]^2 + [B]^2$

(C) $[A]^2 - [B]^2$

(D) $[A]^2 + [B]^2 - [A][B] - [B][A]$

(5). Given $[A]$ is a rectangular matrix and $c[A] = [0]$, then choose the most appropriate answer.

(A) $C = 0$

(B) $C = 0$ and $[A] = [0]$

(C) $C = 0$ or $[A] = [0]$

(D) $C = 0$ and $[A]$ is a non-zero matrix

(6). You sell Jupiter and Fickers Candy bars. The sales in January are 25 and 30 of Jupiter and Fickers, respectively. In February, the sales are 75 and 35 of Jupiter and Fickers, respectively. If a Jupiter bar costs \$2 and a Fickers bar costs \$7, then if

$$[A] = \begin{bmatrix} 25 & 30 \\ 75 & 35 \end{bmatrix}, \ \textit{and}$$

$$[B] = \begin{bmatrix} 2 \\ 7 \end{bmatrix},$$

the total sales amount in each month is given by

(A) $[B][A]$

(B) $[A][B]$

(C) $2[A]$

(D) $7[A]$

For complete solution, go to

https://ma.mathforcollege.com/mcquizzes/04sle/quiz_04sle_binarymatrixoperations_solution.pdf

# Problem Set

Chapter 04.03: Binary Matrix Operations

(1). For the following matrices

$$[A] = \begin{bmatrix} 3 & 0 \\ -1 & 2 \\ 1 & 1 \end{bmatrix}, [B] = \begin{bmatrix} 4 & -1 \\ 0 & 2 \end{bmatrix}, [C] = \begin{bmatrix} 5 & 2 \\ 3 & 5 \\ 6 & 7 \end{bmatrix}$$

Find where possible

(A) $4[A] + 5[C]$

(B) $[A][B]$

(C) $[A] = 2[C]$

**Answer:**

$$\text{(A)} = \begin{bmatrix} 37 & 10 \\ 11 & 33 \\ 34 & 39 \end{bmatrix}$$

$$\text{(B)} = \begin{bmatrix} 12 & -3 \\ -4 & 5 \\ 4 & 1 \end{bmatrix}$$

$$\text{(C)} = \begin{bmatrix} -7 & -4 \\ -7 & -8 \\ -11 & -13 \end{bmatrix}$$

(2). Food orders are taken from two engineering departments for a takeout. The order is tabulated below.

Food order:

| | Chicken Sandwich | Fries | Drink |
|---|---|---|---|
| Mechanical | 25 | 35 | 25 |
| Civil | 21 | 20 | 21 |

However they have a choice of buying this food from three different restaurants. Their prices for the three food items are tabulated below

Price Matrix:

$$\begin{array}{cc} & \begin{matrix}\text{McFat} & \text{Burcholestrol} & \text{Kentucky Sodium}\end{matrix} \\ \begin{matrix}\text{Chicken Sandwich} \\ \text{Fries} \\ \text{Drink}\end{matrix} & \begin{bmatrix}2.42 & 2.38 & 2.46 \\ 0.93 & 0.90 & 0.89 \\ 0.95 & 1.03 & 1.13\end{bmatrix} \end{array}$$

Show how much each department will pay for their order at each restaurant. Which restaurant would be more economical to order from for each department?

**Answer:** The cost in dollars is 116.80, 116.75, 120.90 for the Mechanical Department at three fast food joints. So BurCholestrol is the cheapest for the Mechanical Department. The cost in dollars is 89.37, 89.61, 93.19 for the Civil Department at three fast food joints. McFat is the cheapest for the Civil Department.

(3). Given

$$[A] = \begin{bmatrix}2 & 3 & 5 \\ 6 & 7 & 9 \\ 2 & 1 & 3\end{bmatrix}$$

$$[B] = \begin{bmatrix}3 & 5 \\ 2 & 9 \\ 1 & 6\end{bmatrix}$$

$$[C] = \begin{bmatrix}5 & 2 \\ 3 & 9 \\ 7 & 6\end{bmatrix}$$

Illustrate the distributive law of binary matrix operations

$$[A]([B] + [C]) = [A][B] + [A][C]$$

**Answer:**

$$[B] + [C] = \begin{bmatrix}3 & 5 \\ 2 & 9 \\ 1 & 6\end{bmatrix} + \begin{bmatrix}5 & 2 \\ 3 & 9 \\ 7 & 6\end{bmatrix}$$

$$= \begin{bmatrix}8 & 7 \\ 5 & 18 \\ 8 & 12\end{bmatrix}$$

$$[A]([B] + [C]) = \begin{bmatrix}71 & 128 \\ 155 & 276 \\ 45 & 68\end{bmatrix}$$

$$[A][B] = \begin{bmatrix}17 & 67 \\ 41 & 147 \\ 11 & 37\end{bmatrix}$$

$$[A][C] = \begin{bmatrix}54 & 61 \\ 114 & 129 \\ 34 & 31\end{bmatrix}$$

$$[A]\,[B] + [A]\,[C] = \begin{bmatrix} 71 & 128 \\ 155 & 276 \\ 45 & 68 \end{bmatrix}$$

(4). Let $[I]$ be a $n \times n$ identity matrix. Show that $[A][[I] = [I][A] = [A]$ for every $n \times n$ matrix $[A]$.

Let $[C]_{n\times n} = [A]_{n\times n}\,[I]_{n\times n}$

**Answer:** Hint: $c_{ij} = \sum_{p=1}^{n} a_{ip} i_{pj}$

$= a_{i1} i_{1j} + \ldots\ldots + a_{i,j-1} i_{j-1,j} + a_{ij} i_{jj} + a_{i(j+1)} i_{(j+1)j} + \ldots\ldots + a_{in} i_{nj}$

Since

$i_{ij} = 0$ for $i \neq j$

$= 1$ for $i = j$

$c_{ij} = a_{ij}$

So $[A] = [A]\,[I]$

Similarly do the other case

$[I]\,[A] = [A]$. Just do it!

(5). Consider there are only two computer companies in a country. The companies are named *Dude* and *Imac*. Each year, company *Dude* keeps $1/5^{\text{th}}$ of its customers, while the rest switch to *Imac*. Each year, *Imac* keeps $1/3^{\text{rd}}$ of its customers, while the rest switch to *Dude*. If in 2002, *Dude* has $1/6^t h$ of the market and *Imac* has $5/6^t h$ of the market.

(A) What is the distribution of the customers between the two companies in 2003? Write the answer first as multiplication of two matrices.

(B) What would be distribution when the market becomes stable?

**Answer:**

(A) At the end of 2002, Dude has
$\frac{1}{5} \times \frac{1}{6} + \frac{2}{3} \times \frac{5}{6} = 0.589$.
Imac has
$\frac{4}{5} \times \frac{1}{6} + \frac{1}{3} \times \frac{5}{6} = 0.411$
In matrix form $\begin{bmatrix} \frac{1}{5} & \frac{2}{3} \\ \frac{4}{5} & \frac{1}{3} \end{bmatrix} \begin{bmatrix} \frac{1}{6} \\ \frac{5}{6} \end{bmatrix} = \begin{bmatrix} 0.589 \\ 0.411 \end{bmatrix}$

(B) Stable distribution is [10/22 12/22] (Try to do this part of the problem first by finding the distribution five years from now).

(6). Given

$$[A] = \begin{bmatrix} 12.3 & -12.3 & 10.3 \\ 11.3 & -10.3 & -11.3 \\ 10.3 & -11.3 & -12.3 \end{bmatrix},$$

$$[B] = \begin{bmatrix} 2 & 4 \\ -5 & 6 \\ 11 & -20 \end{bmatrix}$$

$[A][B]$ matrix size is _____

**Answer:** $3 \times 2$

(7). Given

$$[A] = \begin{bmatrix} 12.3 & -12.3 & 10.3 \\ 11.3 & -10.3 & -11.3 \\ 10.3 & -11.3 & -12.3 \end{bmatrix},$$

$$[B] = \begin{bmatrix} 2 & 4 \\ -5 & 6 \\ 11 & -20 \end{bmatrix}$$

if $[C] = [A]\,[B]$, then $c_{31}$= _____

**Answer:** $(10.3 \times 2) + ((-5) \times (-11.3)) + (11 \times (-12.3)) = -58.2$

# Chapter 04.04: Unary Matrix Operations

Lesson: Unary Matrix Operations

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *know what unary operations are,*
2) *find the transpose of a square matrix and its relationship to symmetric matrices,*
3) *find the trace of a matrix, and*
4) *find the determinant of a matrix by the cofactor method.*

## *What is the transpose of a matrix?*

Let $[A]$ be a $m \times n$ matrix. Then $[B]$ is the transpose of $[A]$ if $b_{ji} = a_{ij}$ for all $i$ and $j$. That is, the $i^{th}$ row and the $j^{th}$ column element of $[A]$ is the $j^{th}$ row and $i^{th}$ column element of $[B]$. Note, $[B]$ would be a $n \times m$ matrix. The transpose of $[A]$ is denoted by $[A]^T$.

### Example 1

Find the transpose of

$$[A] = \begin{bmatrix} 25 & 20 & 3 & 2 \\ 5 & 10 & 15 & 25 \\ 6 & 16 & 7 & 27 \end{bmatrix}$$

### Solution

The transpose of $[A]$ is

$$[A]^T = \begin{bmatrix} 25 & 5 & 6 \\ 20 & 10 & 16 \\ 3 & 15 & 7 \\ 2 & 25 & 27 \end{bmatrix}$$

Note, the transpose of a row vector is a column vector and the transpose of a column vector is a row vector.

Also, note that the transpose of a transpose of a matrix is the matrix itself. That is,

$$\left([A]^T\right)^T = [A]$$

Also,

$$([A] + [B])^T = [A]^T + [B]^T \; ; \; (cA)^T = cA^T$$

## What is a symmetric matrix?

A square matrix $[A]$ with real elements were $a_{ij} = a_{ji}$ for $i = 1, \; 2, \; \ldots \; , \; n$ and $j = 1, \; 2, \; \ldots \; , \; n$ is called a symmetric matrix. This is the same as saying that if $[A] = [A]^T$, then $[A]^T$ is a symmetric matrix.

**Example 2**

Give an example of a symmetric matrix

**Solution**

$$[A] = \begin{bmatrix} 21.2 & 3.2 & 6 \\ 3.2 & 21.5 & 8 \\ 6 & 8 & 9.3 \end{bmatrix}$$

Is a symmetric matrix as $a_{12} = a_{21} = 3.2, \;\; a_{13} = a_{31} = 6$ and $a_{13} = a_{31} = 8$.

## What is a skew-symmetric matrix?

A $n \times n$ matrix is skew-symmetric if $a_{ij} = -a_{ji}$, for $i = 1, \; \ldots \; , \; n$ and $j = 1, \; \ldots \; , \; n$. This the same as

$$[A] = -\,[A]^T$$

**Example 3**

Give an example of a skew-symmetric matrix

**Solution**

$$\begin{bmatrix} 0 & 1 & 2 \\ -1 & 0 & -5 \\ -2 & 5 & 0 \end{bmatrix}$$

Is a skew symmetric matrix as $a_{12} = -a_{21} = 1; \;\; a_{13} = -a_{31} = 2; \; a_{23} = -a_{32} = -5$. Since $a_{ii} = -a_{ii}$ only if $a_{ii} = 0$, all the diagonal elements of a skew-symmetric matrix have to be zero.

## What is the trace of a matrix?

The trace of a $n \times n$ matrix $[A]$ is the sum of the diagonal entries of $[A]$. That is

$$tr\,[A] = \sum_{i=1}^{n} a_{ii}$$

**Example 4**

Find the trace of:

$$[A] = \begin{bmatrix} 15 & 6 & 7 \\ 2 & -4 & 2 \\ 3 & 2 & 6 \end{bmatrix}$$

**Solution**

$$\begin{aligned} tr\,[A] &= \sum_{i=1}^{n} a_{ii} \\ &= 15 + (-4) + 6 \\ &= 17 \end{aligned}$$

**Example 5**

The sales of tires are given by make (rows) and quarters (columns) for Blowout r'us store location $A$, as shown below

$$[A] = \begin{bmatrix} 2520\ 3\ \ 2 \\ 5\ 101525 \\ 6\ 16\ 7\ 27 \end{bmatrix}$$

Where the rows represent the sales of Tirestone, Michigan, and Copper tires, and the columns represent the quarter numbers 1, 2, 3, 4.

Find the total yearly revenue of store $A$ if the prices of tires vary by quarters as follows

$$[B] = \begin{bmatrix} 33.2530.0135.0230.05 \\ 40.1938.0241.0338.23 \\ 25.0322.0227.0322.95 \end{bmatrix}$$

Where the rows represent the cost of each tire made by Tirestone, Michigan, and Copper, and the columns represent the quarter numbers.

**Solution**

$$[C] = [B]^T$$

**Example 6**

Blowout r'us store location $A$ and the sales of tires are given by make (in rows) and quarters (in columns) as shown below

$$
\begin{aligned}
[A] &= \begin{bmatrix} 2520\ 3\ \ 2 \\ 5\ 101525 \\ 6\ 16\ 7\ 27 \end{bmatrix} \\
&= \begin{bmatrix} 33.2530.0135.0230.05 \\ 40.1938.0241.0338.23 \\ 25.0322.0227.0322.95 \end{bmatrix} \\
&= \begin{bmatrix} 33.2540.1925.03 \\ 30.0138.0222.02 \\ 35.0241.0327.03 \\ 30.0538.2322.95 \end{bmatrix}
\end{aligned}
$$

Recognize now that if we find $[A]\,[C]$, we get

$$
\begin{aligned}
[D] &= [A]\,[C] \\
&= \begin{bmatrix} 2520\ 3\ \ 2 \\ 5\ 101525 \\ 6\ 16\ 7\ 27 \end{bmatrix} \begin{bmatrix} 33.2540.1925.03 \\ 30.0138.0222.02 \\ 35.0241.0327.03 \\ 30.0538.2322.95 \end{bmatrix} \\
&= \begin{bmatrix} 1597 & 1965 & 1193 \\ 1743 & 2152 & 1325 \\ 1736 & 2169 & 1311 \end{bmatrix}
\end{aligned}
$$

The diagonal elements give the sales of each brand of tire for the whole year. That is

$d_{11} = \$1597$ (Tirestone sales)

$d_{22} = \$2152$ (Michigan sales)

$d_{33} = \$1597$ (Copper sales)

The total yearly sales of all three brans of tires are

$$
\begin{aligned}
\sum_{i=1}^{3} d_{ii} &= 1597 + 2152 + 1311 \\
&= \$5060
\end{aligned}
$$

And this is the trace of matrix $[D]$.

## Define the determinant of a matrix.

The determinant of a square matrix is a single unique real number corresponding to a matrix. For a matrix $[A]$, determinant is denoted by $|A|$ or $\det(A)$. So do not use $[A]$ and $|A|$ interchangeably.

For a $2 \times 2$ matrix

$$[A] = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

$$\det(A) = a_{11}a_{22} - a_{12}a_{21}$$

## How does one calculate the determinant of any square matrix?

Let $[A]$ be a $n \times n$ matrix. The minor of entry $a_{ij}$ is denoted by $M_{ij}$ and is defined as the determinant of the $(n-1) \times (n-1)$ submatrix of $[A]$, where the submatrix is obtained by deleting the $i^{th}$ row and $j^{th}$ column of the matrix $[A]$. The determinant is then given by

$$\det(A) = \sum_{j=1}^{n} (-1)^{i+j} a_{ij} M_{ij} \ for\ any\ i = 1,\ 2,\ \ldots\ ,\ n$$

or

$$\det(A) = \sum_{i=1}^{n} (-1)^{i+j} a_{ij} M_{ij} \ for\ any\ j = 1,\ 2,\ \ldots\ ,\ n$$

Couple that with $\det(A) = a_{11}$ for a $1 \times 1$ matrix $[A]$, we can always reduce the determinant of a matrix to determinants of $1 \times 1$ matrices. The number $(-1)^{i+j} M_{ij}$ is called the cofactor of $a_{ij}$ and is denoted by $c_{ij}$. The formula for the determinant can then be written as

$$\det(A) = \sum_{j=1}^{n} a_{ij} C_{ij} \ for\ any\ i = 1,\ 2,\ \ldots\ ,\ n$$

or

$$\det(A) = \sum_{i=1}^{n} a_{ij} C_{ij} \ for\ any\ j = 1,\ 2,\ \ldots\ ,\ n$$

Determinants are not generally calculated using this method as it becomes computationally intensive for large matrices. For a $n \times n$ matrix, it requires arithmetic operations proportional to $n!$.

**Example 6**

Find the determinant of

$$[A] = \begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{bmatrix}$$

**Solution**

Method 1:

$$\det(A) = \sum_{j=1}^{3} (-1)^{i+j} a_{ij} M_{ij} \quad for \quad any \quad i = 1, \quad 2, \quad 3$$

Let us choose $i = 1$ in the formula

$$\begin{aligned} \det(A) &= \sum_{j=1}^{3} (-1)^{1+j} a_{1j} M_{1j} \\ &= (-1)^{1+1} a_{11} M_{11} + (-1)^{1+2} a_{12} M_{12} + (-1)^{1+3} a_{13}\ M_{13} \\ &= a_{11} M_{11} - a_{12} M_{12} + a_{13} M_{13} \end{aligned}$$

$$\begin{aligned} M_{11} &= \begin{vmatrix} 8 & 1 \\ 12 & 1 \end{vmatrix} \\ &= -4 \end{aligned}$$

$$\begin{aligned} M_{12} &= \begin{vmatrix} 64 & 1 \\ 144 & 1 \end{vmatrix} \\ &= -80 \end{aligned}$$

$$\begin{aligned} M_{13} &= \begin{vmatrix} 64 & 8 \\ 144 & 12 \end{vmatrix} \\ &= -384 \end{aligned}$$

$$\begin{aligned} det(A) &= a_{11} M_{11} - a_{12} M_{12} + a_{13} M_{13} \\ &= 25(-4) - 5(-80) + 1(-384) \\ &= -100 + 400 - 384 \\ &= -84 \end{aligned}$$

Also for $i = 1$,

$$\det(A) = \sum_{j=1}^{3} a_{1j} C_{1j}$$

$$\begin{aligned} C_{11} &= (-1)^{1+1} M_{11} \\ &= M_{11} \\ &= -4 \end{aligned}$$

$$\begin{aligned} C_{12} &= (-1)^{1+2} M_{12} \\ &= -M_{12} \\ &= 80 \end{aligned}$$

$$\begin{aligned} C_{13} &= (-1)^{1+3} M_{13} \\ &= M_{13} \\ &= -384 \end{aligned}$$

$$\begin{aligned} \det(A) &= a_{11}C_{11} + a_{21}C_{21} + a_{31}C_{31} \\ &= (25)(-4) + (5)(80) + (1)(-384) \\ &= -100 + 400 - 384 \\ &= -84 \end{aligned}$$

Method 2:

$$\det(A) = \sum_{i=1}^{3} (-1)^{i+j} a_{ij} M_{ij} \ \textit{for any } j = 1,\ 2,\ 3$$

Let us choose $j = 2$ in the formula

$$\begin{aligned} \det(A) &= \sum_{i=1}^{3} (-1)^{i+2} a_{i2} M_{i2} \\ &= (-1)^{1+2} a_{12} M_{12} + (-1)^{2+2} a_{22} M_{22} + (-1)^{3+2} a_{32} M_{32} \\ &= -a_{12} M_{12} + a_{22} M_{22} - a_{32} M_{32} \end{aligned}$$

$$\begin{aligned} M_{12} &= \begin{vmatrix} 64 & 1 \\ 144 & 1 \end{vmatrix} \\ &= -80 \end{aligned}$$

$$\begin{aligned} M_{22} &= \begin{vmatrix} 25 & 1 \\ 144 & 1 \end{vmatrix} \\ &= -119 \end{aligned}$$

$$M_{32} = \begin{vmatrix} 25 & 1 \\ 64 & 1 \end{vmatrix}$$
$$= -39$$

$$\begin{aligned} det(A) &= -a_{12}M_{12} + a_{22}M_{22} - a_{32}M_{32} \\ &= -5(-80) + 8(-119) - 12(-39) \\ &= 400 - 952 + 468 \\ &= -84 \end{aligned}$$

In terms of cofactors for $j = 2$,

$$\det(A) = \sum_{i=1}^{3} a_{i2}C_{i2}$$

$$\begin{aligned} C_{12} &= (-1)^{1+2} M_{12} \\ &= -M_{12} \\ &= 80 \end{aligned}$$

$$\begin{aligned} C_{22} &= (-1)^{2+2} M_{22} \\ &= M_{22} \\ &= -119 \end{aligned}$$

$$\begin{aligned} C_{32} &= (-1)^{3+2} M_{32} \\ &= -M_{32} \\ &= 39 \end{aligned}$$

$$\begin{aligned} \det(A) &= a_{12}C_{12} + a_{22}C_{22} + a_{32}C_{32} \\ &= (5)(80) + (8)(-119) + (12)(39) \\ &= 400 - 952 + 468 \\ &= -84 \end{aligned}$$

## Is there a relationship between det($AB$), and det($A$) and det($B$)?

Yes, if $[A]$ and $[B]$ are square matrices of same size, then

$$det(AB) = det(A)det(B)$$

## Are there some other theorems that are important in finding the determinant of a square matrix?

**Theorem 1:** If a row or a column in a $n \times n$ matrix $[A]$ is zero, then $det(A) = 0$.

**Theorem 2:** Let $[A]$ be a $n \times n$ matrix. If a row is proportional to another row, then $det(A) = 0$.

**Theorem 3:** Let $[A]$ be a $n \times n$ matrix. If a column is proportional to another column, then $det(A) = 0$.

**Theorem 4:** Let $[A]$ be a $n \times n$matrix. If a column or row is multiplied by $k$ to result in matrix $k$, then $det(B) = kdet(A)$.

**Theorem 5:** Let $[A]$ be a $n \times n$ upper or lower triangular matrix, then $det(A) = \prod_{i=1}^{n} a_{ii}$.

### Example 7

What is the determinant of

$$[A] = \begin{bmatrix} 0 & 2 & 6 & 3 \\ 0 & 3 & 7 & 4 \\ 0 & 4 & 9 & 5 \\ 0 & 5 & 2 & 1 \end{bmatrix}$$

### Solution

Since one of the columns (first column in the above example) of $[A]$ is a zero, $det(A) = 0$.

### Example 8

What is the determinant of

$$[A] = \begin{bmatrix} 2 & 1 & 6 & 4 \\ 3 & 2 & 7 & 6 \\ 5 & 4 & 2 & 10 \\ 9 & 5 & 3 & 18 \end{bmatrix}$$

### Solution

$det(A)$ is zero because the fourth column

$$\begin{bmatrix} 4 \\ 6 \\ 10 \\ 18 \end{bmatrix}$$

is 2 times the first column

$$\begin{bmatrix} 2 \\ 3 \\ 5 \\ 9 \end{bmatrix}$$

**Example 9**

If the determinant of

$$[A] = \begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{bmatrix}$$

is $-84$, then what is the determinant of

$$[B] = \begin{bmatrix} 25 & 10.5 & 1 \\ 64 & 16.8 & 1 \\ 144 & 25.2 & 1 \end{bmatrix}$$

**Solution**

Since the second column of $[B]$ is 2.1 times the second column of $[A]$

$$\begin{aligned} \det(B) &= 2.1 \det(A) \\ &= (2.1)(-84) \\ &= -176.4 \end{aligned}$$

**Example 10**

Given the determinant of

$$[A] = \begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{bmatrix}$$

is $-84$, what is the determinant of

$$[B] = \begin{bmatrix} 25 & 5 & 1 \\ 0 & -4.8 & -1.56 \\ 144 & 12 & 1 \end{bmatrix}$$

**Solution**

Since $[B]$ is simply obtained by subtracting the second row of $[A]$ by 2.56 times the first row of $[A]$,

$$\begin{aligned} det(B) &= det(A) \\ &= -84 \end{aligned}$$

**Example 11**

What is the determinant of

$$[A] = \begin{bmatrix} 25 & 5 & 1 \\ 0 & -4.8 & -1.56 \\ 0 & 0 & 0.7 \end{bmatrix}$$

**Solution**

Since $[A]$ is an upper triangular matrix

$$\begin{aligned} \det(A) &= \prod_{i=1}^{3} a_{ii} \\ &= a_{11} \times a_{22} \times a_{33} \\ &= 25 \times (-4.8) \times 0.7 \\ &= -84 \end{aligned}$$

## Key Terms:

*Transpose*

*Symmetric Matrix*

*Skew-Symmetric Matrix*

*Trace of Matrix*

*Determinant*

# Multiple Choice Test

## Chapter 04.04 Unary Matrix Operations

(1). If the determinant of a $4 \times 4$ matrix is given as 20, then the determinant of $5\,[A]$ is

(A) 100

(B) 12500

(C) 25

(D) 62500

(2). If the matrix product $[A]\ [B]\ [C]$ is defined, then $([A]\,[B]\,[C])^T$ is

(A) $[C]^T\ [B]^T\ [A]^T$

(B) $[A]^T\ [B]^T\ [C]^T$

(C) $[A]\ [B]\ [C]^T$

(D) $[A]^T\ [B]\ [C]$

(3). The trace of a matrix

$$\begin{bmatrix} 5 & 6 & -7 \\ 9 & -11 & 13 \\ -17 & 19 & 23 \end{bmatrix}$$

is

(A) 17

(B) 39

(C) 40

(D) 110

(4). A square $n \times n$ matrix $[A]$ is symmetric if

(A) $a_{ij} = a_{ji},\ i = j$ for all $i, j$

(B) $a_{ij} = a_{ji},\ i \neq j$ for all $i, j$

(C) $a_{ij} = -a_{ji},\ i = j$ for all $i, j$

(D) $a_{ij} = -a_{ji},\ i \neq j$ for all $i, j$

(5). The determinant of the matrix

$$\begin{bmatrix} 25 & 5 & 1 \\ 0 & 3 & 8 \\ 0 & 9 & a \end{bmatrix}$$

is 50. The value of $a$ is then

(A) 0.6667

(B) 24.67

(C) $-23.33$

(D) 5.556

(6). $[A]$ is a $5 \times 5$ matrix and a matrix $[B]$ is obtained by the row operations of replacing $Row\ 1$ with $Row\ 3$, and then $Row\ 3$ is replaced by a linear combination of $2 \times Row\ 3 + 4 \times Row\ 2$. If $\det(A) = 17$, then $\det(B)$ is equal to

(A) 12

(B) $-34$

(C) $-112$

(D) 112

For complete solution, go to

https://ma.mathforcollege.com/mcquizzes/04sle/quiz_04sle_unarymatrixoperations_solution.pdf

# Problem Set

## Chapter 04.04: Unary Matrix Operations

(1). Let

$$[A] = \begin{bmatrix} 25 & 3 & 6 \\ 7 & 9 & 2 \end{bmatrix}.$$

Find $[A]^T$

**Answer:** $\begin{bmatrix} 25 & 7 \\ 3 & 9 \\ 6 & 2 \end{bmatrix}$

(2). If $[A]$ and $[B]$ are two $n \times n$ symmetric matrices, show that $[A] + [B]$ is also symmetric. Hint: Let $[C] = [A] + [B]$

**Answer:** $c_{ij} = a_{ij} + b_{ij}$ for all *i, j.*
and

$c_{ji} = a_{ji} + b_{ji}$ for all *i, j.*
$c_{ji} = a_{ij} + b_{ij}$ as $[A]$ and $[B]$ are symmetric
Hence $c_{ji} = c_{ij}$.

(3). Give an example of a $4 \times 4$ symmetric matrix.

(4). Give an example of a $4 \times 4$ skew-symmetric matrix.

(5). What is the trace of

(A) $[A] = \begin{bmatrix} 7 & 2 & 3 & 4 \\ -5 & -5 & -5 & -5 \\ 6 & 6 & 7 & 9 \\ -5 & 2 & 3 & 10 \end{bmatrix}$

(B) For

$$[A] = \begin{bmatrix} 10 & -7 & 0 \\ -3 & 2.099 & 6 \\ 5 & -1 & 5 \end{bmatrix}$$

Find the determinant of $[A]$ using the cofactor method.

**Answer:** a)19
b) $-150.05$

(6). $\det(3[A])$ of a $n \times n$ matrix is

(A) $3\det(A)$

(B) $3\det(A)$

(C) $3^n \det(A)$

(D) $9\det(A)$

**Answer:** C

(7). For a $5 \times 5$ matrix $[A]$, the first row is interchanged with the fifth row, the determinant of the resulting matrix $[B]$is

(A) $\det(A)$

(B) $-\det(A)$

(C) $5\det(A)$

(D) $2\det(A)$

**Answer:** A

(8). $\det \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix}$ is

(A) 0

(B) 1

(C) $-1$

(D) $\infty$

**Answer:** C

(9). Without using the cofactor method of finding determinants, find the determinant of

$$[A] = \begin{bmatrix} 0 & 0 & 0 \\ 2 & 3 & 5 \\ 6 & 9 & 2 \end{bmatrix}$$

**Answer:** 0: Can you answer why?

(10). Without using the cofactor method of finding determinants, find the determinant of

$$[A] = \begin{bmatrix} 0 & 0 & 2 & 3 \\ 0 & 2 & 3 & 5 \\ 6 & 7 & 2 & 3 \\ 6.6 & 7.7 & 2.2 & 3.3 \end{bmatrix}$$

**Answer:** 0: Can you answer why?

(11). Without using the cofactor method of finding determinants, find the determinant of

$$[A] = \begin{bmatrix} 5 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 2 & 5 & 6 & 0 \\ 1 & 2 & 3 & 9 \end{bmatrix}$$

**Answer:** $5 \times 3 \times 6 \times 9 = 810$: Can you answer why?

(12). Given the matrix

$$[A] = \begin{bmatrix} 125 & 25 & 5 & 1 \\ 512 & 64 & 8 & 1 \\ 1157 & 89 & 13 & 1 \\ 8 & 4 & 2 & 1 \end{bmatrix}$$

and

$det(A) = -32400$

find the determinant of

(A) $[A] = \begin{bmatrix} 125 & 25 & 5 & 1 \\ 512 & 64 & 8 & 1 \\ 1141 & 81 & 9 & -1 \\ 8 & 4 & 2 & 1 \end{bmatrix}$

(B) $[A] = \begin{bmatrix} 125 & 25 & 1 & 5 \\ 512 & 64 & 1 & 8 \\ 1157 & 89 & 1 & 13 \\ 8 & 4 & 1 & 2 \end{bmatrix}$

(C) $[B] = \begin{bmatrix} 125 & 25 & 5 & 1 \\ 1157 & 89 & 13 & 1 \\ 512 & 64 & 8 & 1 \\ 8 & 4 & 2 & 1 \end{bmatrix}$

(D) $[C] = \begin{bmatrix} 125 & 25 & 5 & 1 \\ 1157 & 89 & 13 & 1 \\ 8 & 4 & 2 & 1 \\ 512 & 64 & 8 & 1 \end{bmatrix}$

(E) $[D] = \begin{bmatrix} 125 & 25 & 5 & 1 \\ 512 & 64 & 8 & 1 \\ 1157 & 89 & 13 & 1 \\ 16 & 8 & 4 & 2 \end{bmatrix}$

**Answer:** A) $-32400$ B) 32400 C) 32400 D) $-32400$ E) $-64800$

(13). What is the transpose of

$$[A] = \begin{bmatrix} 25 & 20 & 3 & 2 \\ 5 & 10 & 15 & 25 \\ 6 & 16 & 7 & 27 \end{bmatrix}$$

**Answer:** $[A]^T = \begin{bmatrix} 25 & 5 & 6 \\ 20 & 10 & 16 \\ 3 & 15 & 7 \\ 2 & 25 & 27 \end{bmatrix}$

(14). What values of the missing numbers will make this a skew-symmetric matrix?

$$[A] = \begin{bmatrix} 0 & 3 & ? \\ ? & 0 & ? \\ 21 & ? & 0 \end{bmatrix}$$

**Answer:** $\begin{bmatrix} 0 & 3 & -21 \\ -3 & 0 & 4 \\ 21 & -4 & 0 \end{bmatrix}$

(15). What values of the missing number will make this a symmetric matrix?

$$[A] = \begin{bmatrix} 2 & 3 & ? \\ ? & 6 & 7 \\ 21 & ? & 5 \end{bmatrix}$$

**Answer:** $\begin{bmatrix} 2 & 3 & 21 \\ 3 & 6 & 7 \\ 21 & 7 & 5 \end{bmatrix}$

(16). Find the determinant of

$$[A] = \begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 5 \end{bmatrix}$$

**Answer:** The determinant of $[A]$ is,

$$\begin{aligned} & 25 \begin{bmatrix} 8 & 1 \\ 12 & 5 \end{bmatrix} - 5 \begin{bmatrix} 64 & 1 \\ 144 & 5 \end{bmatrix} + 1 \begin{bmatrix} 64 & 8 \\ 144 & 12 \end{bmatrix} \\ & = 25(28) - 5(176) + 1(-384) \\ & = -564 \end{aligned}$$

(17). What is the determinant of an upper triangular matrix $[A]$ that is of order $n \times n$?

**Answer:** The determinant of an upper triangular matrix is the product of its diagonal elements,$\prod_{i=1}^{n} a_{ii}$

(18). Given the determinant of

$$[A] = \begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & a \end{bmatrix}$$

is$-564$, find $a$.

**Answer:** $det(A) = -120a + 36$

$120a + 36 = 564$

$a = 5$

(19). Why is the determinant of the following matrix zero?

$$[A] = \begin{bmatrix} 0 & 0 & 0 \\ 2 & 3 & 5 \\ 6 & 9 & 2 \end{bmatrix}$$

**Answer:** The first row of the matrix is zero, hence, the determinant of the matrix is zero.

(20). Why is the determinant of the following matrix zero?

$$[A] = \begin{bmatrix} 0 & 0 & 2 & 3 \\ 0 & 2 & 3 & 5 \\ 6 & 7 & 2 & 3 \\ 6.6 & 7.7 & 2.2 & 3.3 \end{bmatrix}$$

**Answer:** Row 4 of the matrix is 1.1 times Row 3. Hence, its determinant is zero.

(21). Show that if $[A]\ [B] = [I]$, where $[A]$ , $[B]$ and $[I]$ are matrices of $n \times n$ size and $[I]$ is an identity matrix, then $det(A) \neq 0$ and $det(B) \neq 0$.

**Answer:** We know that $det(AB) = det(A)det(B)$.

$$[A][B] = [I]$$
$$det(AB) = det(I)$$

$$det(I) = \prod_{i=1}^{n} a_{ii} = \prod_{i=1}^{n} 1 = 1$$
$$det(A)det(B) = 1$$

Therefore,

$det(A) \neq 0$ and

$det(B) \neq 0$.

# Chapter 04.05: System of Equations

Lesson: System of Equations

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *setup simultaneous linear equations in matrix form and vice-versa,*

2) *understand the concept of the inverse of a matrix,*

3) *know the difference between a consistent and inconsistent system of linear equations, and*

4) *learn that a system of linear equations can have a unique solution, no solution or infinite solutions.*

## Matrix algebra is used for solving systems of equations. Can you illustrate this concept?

Matrix algebra is used to solve a system of simultaneous linear equations. In fact, for many mathematical procedures such as the solution to a set of nonlinear equations, interpolation, integration, and differential equations, the solutions reduce to a set of simultaneous linear equations. Let us illustrate with an example for interpolation.

### Example 1

The upward velocity of a rocket is given at three different times on the following table.

**Table 5.1.** Velocity vs. time data for a rocket

| Time, $t$ | Velocity, $v$ |
|---|---|
| (s ) | (m/s) |
| 5 | 106.8 |
| 8 | 177.2 |
| 12 | 279.2 |

The velocity data is approximated by a polynomial as

$$v(t) = at^2 + bt + c, 5 \leq t \leq 12$$

Set up the equations in matrix form to find the coefficients $a, b, c$ of the velocity profile.

**Solution**

The polynomial is going through three data points $(t_1, v_1), (t_2, v_2), and (t_3, v_3)$ where from table 5.1.

$$t_1 = 5,\ v_1 = 106.8$$

$$t_2 = 8,\ v_2 = 177.2$$

$$t_3 = 12,\ v_3 = 279.2$$

Requiring that $v(t) = at^2 + bt + c$ passes through the three data points gives

$$v(t_1) = v_1 = at_1^2 + bt_1 + c$$

$$v(t_2) = v_2 = at_2^2 + bt_2 + c$$

$$v(t_3) = v_3 = at_3^2 + bt_3 + c$$

Substituting the data $(t_1, v_1), (t_2, v_2),\ and\ (t_3, v_3)$ gives

$$a\left(5^2\right) + b(5) + c = 106.8$$

$$a\left(8^2\right) + b(8) + c = 177.2$$

$$a\left(12^2\right) + b(12) + c = 279.2$$

or

$$25a + 5b + c = 106.8$$

$$64a + 8b + c = 177.2$$

$$144a + 12b + c = 279.2$$

This set of equations can be rewritten in the matrix form as

$$\begin{bmatrix} 25a+ & 5b+ & c \\ 64a+ & 8b+ & c \\ 144a+ & 12b+ & c \end{bmatrix} = \begin{bmatrix} 106.8 \\ 177.2 \\ 279.2 \end{bmatrix}$$

The above equation can be written as a linear combination as follows

$$a\begin{bmatrix}25\\64\\144\end{bmatrix}+b\begin{bmatrix}5\\8\\12\end{bmatrix}+c\begin{bmatrix}1\\1\\1\end{bmatrix}=\begin{bmatrix}106.8\\177.2\\279.2\end{bmatrix}$$

and further using matrix multiplication gives

$$\begin{bmatrix}25 & 5 & 1\\64 & 8 & 1\\144 & 12 & 1\end{bmatrix}\begin{bmatrix}a\\b\\c\end{bmatrix}=\begin{bmatrix}106.8\\177.2\\279.2\end{bmatrix}$$

The above is an illustration of why matrix algebra is needed. The complete solution to the set of equations is given later in this chapter.

A general set of $m$ linear equations and $n$ unknowns,

$$a_{11}x_1+a_{12}x_2+\cdots\cdots+a_{1n}x_n=c_1$$

$$a_{21}x_1+a_{22}x_2+\cdots\cdots+a_{2n}x_n=c_2$$

.....

.....

$$a_{m1}x_1+a_{m2}x_2+.....+a_{mn}x_n=c_m$$

can be rewritten in the matrix form as

$$\begin{bmatrix}a_{11} & a_{12} & . & . & a_{1n}\\a_{21} & a_{22} & . & . & a_{2n}\\\vdots & & & & \vdots\\\vdots & & & & \vdots\\a_{m1} & a_{m2} & . & . & a_{mn}\end{bmatrix}\begin{bmatrix}x_1\\x_2\\.\\.\\x_n\end{bmatrix}=\begin{bmatrix}c_1\\c_2\\.\\.\\c_m\end{bmatrix}$$

Denoting the matrices by $[A]$, $[X]$, and $[C]$, the system of equation is

$[A]\ [X]=[C]$, where $[A]$ is called the coefficient matrix, $[C]$ is called the right hand side vector and $[X]$ is called the solution vector.

Sometimes $[A]\ [X]=[C]$ systems of equations are written in the augmented form. That is

$$\left[A\vdots C\right]=\left[\begin{array}{cccc:c}a_{11} & a_{12} & ..... & a_{1n} & c_1\\a_{21} & a_{22} & ..... & a_{2n} & c_2\\\vdots & & & & \vdots\\\vdots & & & & c_n\\a_{m1} & a_{m2} & ..... & a_{mn} & \end{array}\right]$$

## A system of equations can be consistent or inconsistent. What does that mean?

A system of equations $[A]\ [X] = [C]$ is consistent if there is a solution, and it is inconsistent if there is no solution. However, a consistent system of equations does not mean a unique solution, that is, a consistent system of equations may have a unique solution or infinite solutions (Figure 1).

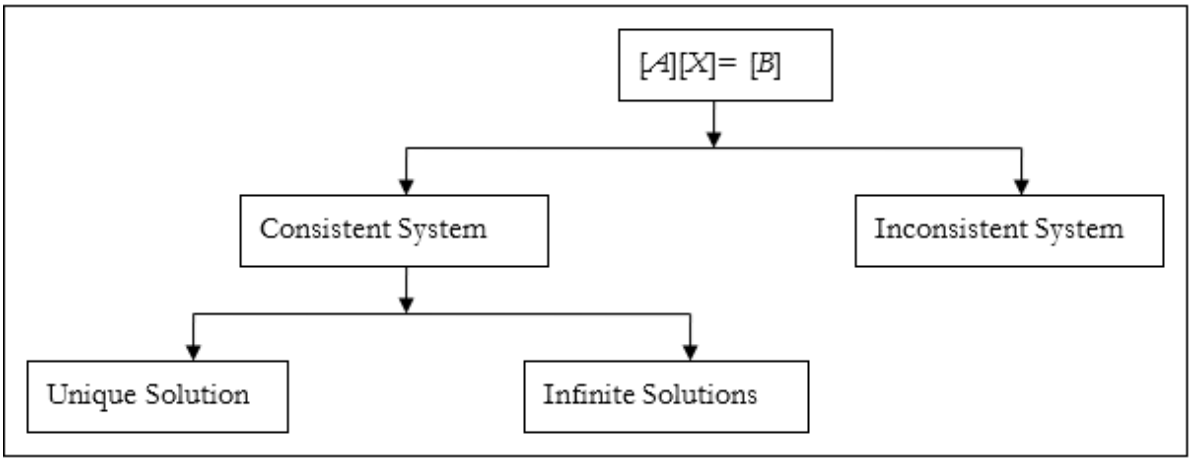



**Figure 5.1.** Consistent and inconsistent system of equations flow chart.

**Example 2**

Give examples of consistent and inconsistent system of equations.

**Solution**

(a) The system of equations

$$\begin{bmatrix} 2 & 4 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 6 \\ 4 \end{bmatrix}$$

is a consistent system of equations as it has a unique solution, that is,

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

(b) The system of equations

$$\begin{bmatrix} 2 & 4 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 6 \\ 3 \end{bmatrix}$$

is also a consistent system of equations but it has infinite solutions as given as follows.

Expanding the above set of equations,

$$2x + 4y = 6$$

$$x + 2y = 3$$

you can see that they are the same equation. Hence, any combination of $(x, y)$ that satisfies

$$2x + 4y = 6$$

is a solution. For example $(x, y) = (1, 1)$ is a solution. Other solutions include $(x, y) = (0.5, 1.25)$, $(x, y) = (0, 1.5)$, and so on.

(c) The system of equations

$$\begin{bmatrix} 2 & 4 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 6 \\ 4 \end{bmatrix}$$

is inconsistent as no solution exists.

## How can one distinguish between a consistent and inconsistent system of equations?

A system of equations $[A]\ [X] = [C]$ is *consistent* if the rank of $A$ is equal to the rank of the augmented matrix $\left[A \vdots C\right]$

A system of equations $[A]\ [X] = [C]$ is *inconsistent* if the rank of $A$ is less than the rank of the augmented matrix $\left[A \vdots C\right]$.

But, what do you mean by rank of a matrix?

The rank of a matrix is defined as the order of the largest square submatrix whose determinant is not zero.

**Example 3**

What is the rank of

$[A] = \begin{bmatrix} 3 & 1 & 2 \\ 2 & 0 & 5 \\ 1 & 2 & 3 \end{bmatrix}$?

**Solution**

The largest square submatrix possible is of order 3 and that is $[A]$ itself. Since $det(A) = -23 \neq 0$, the rank of $[A] = 3$.

**Example 4**

What is the rank of

$$[A] = \begin{bmatrix} 3 & 1 & 2 \\ 2 & 0 & 5 \\ 5 & 1 & 7 \end{bmatrix} ?$$

**Solution**

The largest square submatrix of $[A]$ is of order 3 and that is $[A]$ itself. Since $det(A) = 0$, the rank of $[A]$ is less than 3. The next largest square submatrix would be a $2 \times 2$ matrix. One of the square submatrices of $[A]$ is

$$[B] = \begin{bmatrix} 3 & 1 \\ 2 & 0 \end{bmatrix}$$

and $det(B) = -2 \neq 0$. Hence the rank of $[A]$ is 2. There is no need to look at other $2 \times 2$ submatrices to establish that the rank of $[A]$ is 2.

**Example 5**

How do I now use the concept of rank to find if

$$\begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 106.8 \\ 177.2 \\ 279.2 \end{bmatrix}$$

is a consistent or inconsistent system of equations?

**Solution**

The coefficient matrix is

$$[A] = \begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{bmatrix}$$

and the right-hand side vector is

$$[C] = \begin{bmatrix} 106.8 \\ 177.2 \\ 279.2 \end{bmatrix}$$

The augmented matrix is

$$[B] = \left[\begin{array}{ccc:c} 25 & 5 & 1 & 106.8 \\ 64 & 8 & 1 & 177.2 \\ 144 & 12 & 1 & 279.2 \end{array}\right]$$

Since there are no square submatrices of order 4 as $[B]$ is a $3 \times 3$ matrix, the rank of $[B]$ is at most 3. So let us look at the square submatrices of $[B]$ of order 3; if any of these square submatrices have determinant not equal to zero, then the rank is 3. For example, a submatrix of the augmented matrix $[B]$ is

$$[D] = \begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{bmatrix}$$

has$det(D) = -84 \neq 0$.

Hence the rank of the augmented matrix $[B]$ is 3. Since $[A] = [D]$, the rank of $[A]$ is 3. Since the rank of the augmented matrix $[B]$ equals the rank of the coefficient matrix $[A]$, the system of equations is consistent.

**Example 6**

Use the concept of rank of matrix to find if

$$\begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 89 & 13 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 106.8 \\ 177.2 \\ 284.0 \end{bmatrix}$$

is consistent or inconsistent?

**Solution**

The coefficient matrix is given by

$$[A] = \begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 89 & 13 & 2 \end{bmatrix}$$

and the right-hand side

$$[C] = \begin{bmatrix} 106.8 \\ 177.2 \\ 284.0 \end{bmatrix}$$

The augmented matrix is

$$[B] = \begin{bmatrix} 25 & 5 & 1 & : 106.8 \\ 64 & 8 & 1 & : 177.2 \\ 89 & 13 & 2 & : 284.0 \end{bmatrix}$$

Since there are no square submatrices of order 4 as $[B]$ is a $4 \times 3$ matrix, the rank of the augmented $[B]$ is at most 3. So let us look at square submatrices of the augmented matrix $[B]$ of order 3 and see if any of these have determinants not equal to zero. For example, a square submatrix of the augmented matrix $[B]$ is

$$[D] = \begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 89 & 13 & 2 \end{bmatrix}$$

has $det(D) = 0$. This means, we need to explore other square submatrices of order 3 of the augmented matrix $[B]$ and find their determinants.

That is,

$$[E] = \begin{bmatrix} 5 & 1 & 106.8 \\ 8 & 1 & 177.2 \\ 13 & 2 & 284.0 \end{bmatrix}$$

$$det(E) = 0$$

$$[F] = \begin{bmatrix} 25 & 5 & 106.8 \\ 64 & 8 & 177.2 \\ 89 & 13 & 284.0 \end{bmatrix}$$

$$det(F) = 0$$

$$[G] = \begin{bmatrix} 25 & 1 & 106.8 \\ 64 & 1 & 177.2 \\ 89 & 2 & 284.0 \end{bmatrix}$$

$$det(G) = 0$$

All the square submatrices of order $3 \times 3$ of the augmented matrix $[B]$ have a zero determinant. So the rank of the augmented matrix $[B]$ is less than 3. Is the rank of augmented matrix $[B]$ equal to 2?. One of the $2 \times 2$ submatrices of the augmented matrix $[B]$ is

$$[H] = \begin{bmatrix} 25 & 5 \\ 64 & 8 \end{bmatrix}$$

and

$$det(H) = -120 \neq 0$$

So, the rank of the augmented matrix $[B]$ is 2.

Now we need to find the rank of the coefficient matrix $[B]$.

$$[A] = \begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 89 & 13 & 2 \end{bmatrix}$$

and

$$det(A) = 0$$

So, the rank of the coefficient matrix $[A]$ is less than 3. A square submatrix of the coefficient matrix $[A]$ is

$$[J] = \begin{bmatrix} 5 & 1 \\ 8 & 1 \end{bmatrix}$$

$$det(J) = -3 \neq 0$$

So, the rank of the coefficient matrix $[A]$ is 2.

Hence, rank of the coefficient matrix $[A]$ equals the rank of the augmented matrix $[B]$. So the system of equations $[A]\ [X] = [C]$ is consistent.

**Example 7**

Use the concept of rank to find if

$$\begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 89 & 13 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 106.8 \\ 177.2 \\ 280.0 \end{bmatrix}$$

is consistent or inconsistent.

**Solution**

The augmented matrix is

$$[B] = \begin{bmatrix} 25 & 5 & 1 & : 106.8 \\ 64 & 8 & 1 & : 177.2 \\ 89 & 13 & 2 & : 280.0 \end{bmatrix}$$

Since there are no square submatrices of order $4 \times 4$ as the augmented matrix $[B]$ is a $4 \times 3$ matrix, the rank of the augmented matrix $[B]$ is at most 3. So let us look at square submatrices of the augmented matrix $(B)$ of order 3 and see if any of the $3 \times 3$ submatrices have a determinant not equal to zero. For example, a square submatrix of order $3 \times 3$ of $[B]$

$$[D] = \begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 89 & 13 & 2 \end{bmatrix}$$

$$\det(D) \ = \ 0$$

So, it means, we need to explore other square submatrices of the augmented matrix $[B]$

$$[E] = \begin{bmatrix} 5 & 1 & 106.8 \\ 8 & 1 & 177.2 \\ 13 & 2 & 280.0 \end{bmatrix}$$

$det(E) = 12.0 \neq 0$.

So, the rank of the augmented matrix $[B]$ is 3.

The rank of the coefficient matrix $[A]$ is 2 from the previous example.

Since the rank of the coefficient matrix $[A]$ is less than the rank of the augmented matrix $[B]$, the system of equations is inconsistent. Hence, no solution exists for $[A]\ [X] = [C]$.

## If a solution exists, how do we know whether it is unique?

In a system of equations $[A]\ [X] = [C]$ that is consistent, the rank of the coefficient matrix $[A]$ is the same as the augmented matrix $[A\,|C\ ]$. If in addition, the rank of the coefficient matrix $[A]$ is same as the number of unknowns, then the solution is unique; if the rank of the coefficient matrix $[A]$ is less than the number of unknowns, then infinite solutions exist.

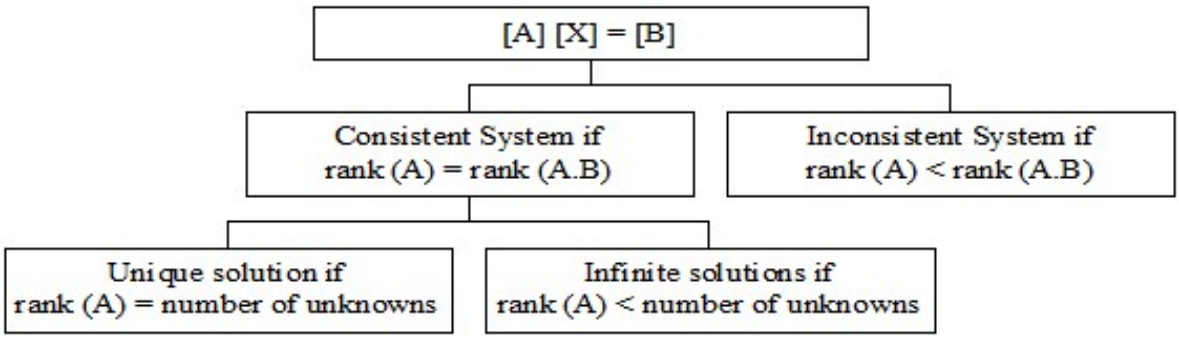



**Figure 5.2.** Flow chart of conditions for consistent and inconsistent system of equations.

**Example 8**

We found that the following system of equations

$$\begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 106.8 \\ 177.2 \\ 279.2 \end{bmatrix}$$

is a consistent system of equations. Does the system of equations have a unique solution or does it have infinite solutions?

**Solution**

The coefficient matrix is

$$[A] = \begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{bmatrix}$$

and the right-hand side is

$$[C] = \begin{bmatrix} 106.8 \\ 177.2 \\ 279.2 \end{bmatrix}$$

While finding out whether the above equations were consistent in an earlier example, we found that the rank of the coefficient matrix $(A)$ equals rank of augmented matrix $\left[A \vdots C\right]$ equals 3.

The solution is unique as the number of unknowns = 3 = rank of $(A)$.

**Example 9**

We found that the following system of equations

$$\begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 89 & 13 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 106.8 \\ 177.2 \\ 284.0 \end{bmatrix}$$

is a consistent system of equations. Is the solution unique or does it have infinite solutions.

**Solution**

While finding out whether the above equations were consistent, we found that the rank of the coefficient matrix $[A]$equals the rank of augmented matrix $\left(A \vdots C\right)$ equals 2

Since the rank of $[A] = 2 <$ number of unknowns = 3, infinite solutions exist.

## If we have more equations than unknowns in [A] [X] = [C], does it mean the system is inconsistent?

No, it depends on the rank of the augmented matrix $\left[A \vdots C\right]$ and the rank of $[A]$.

(a) For example

$$\begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \\ 89 & 13 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 106.8 \\ 177.2 \\ 279.2 \\ 284.0 \end{bmatrix}$$

is consistent, since

rank of augmented matrix = 3

rank of coefficient matrix = 3

Now since the rank of $(A)$ = 3 = number of unknowns, the solution is not only consistent but also unique.

(b) For example

$$\begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \\ 89 & 13 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 106.8 \\ 177.2 \\ 279.2 \\ 280.0 \end{bmatrix}$$

is inconsistent, since

rank of augmented matrix = 4

rank of coefficient matrix = 3

(c) For example

$$\begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 50 & 10 & 2 \\ 89 & 13 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 106.8 \\ 177.2 \\ 213.6 \\ 280.0 \end{bmatrix}$$

is consistent, since

rank of augmented matrix = 2

rank of coefficient matrix = 2

But since the rank of $[A] = 2 <$ the number of unknowns = 3, infinite solutions exist.

## Consistent systems of equations can only have a unique solution or infinite solutions. Can a system of equations have more than one but not infinite number of solutions?

No, you can only have either a unique solution or infinite solutions. Let us suppose $[A][X] = [C]$has two solutions $[Y]$ and $[Z]$ so that

$$[A][Y] = [C]$$

$$[A][Z] = [C]$$

If $r$ is a constant, then from the two equations

$$r\,[A]\,[Y] = r\,[C]$$

$$(1-r)\,[A]\,[Z] = (1-r)\,[C]$$

Adding the above two equations gives

$$r\,[A]\,[Y] + (1-r)\,[A]\,[Z] = r\,[C] + (1-r)\,[C]$$

$$[A]\,(r\,[Y] + (1-r)\,[Z]) = [C]$$

Hence

$$r\,[Y] + (1-r)\,[Z]$$

is a solution to

$$[A]\,[X] = [C]$$

Since $r$ is any scalar, there are infinite solutions for $[A]\ [X] = [C]$ of the form

$$r\,[Y] + (1-r)\,[Z]$$

## Can you divide two matrices?

If $[A]\ [B] = [C]$ is defined, it might seem intuitive that $[A] = \dfrac{[C]}{[B]}$, but matrix division is not defined like that. However an inverse of a matrix can be defined for certain types of square matrices. The inverse of a square matrix $[A]$, if existing, is denoted by $[A]^{-1}$ such that

$$[A]\ [A]^{-1} = [I] = [A]^{-1}[A]$$

Where $[I]$ is the identity matrix.

In other words, let $[A]$ be a square matrix. If $[B]$ is another square matrix of the same size such that $[B]\ [A] = [I]$, then $[B]$ is the inverse of $[A]$. $[A]$ is then called to be invertible or nonsingular. If $[A]^{-1}$ does not exist, $[A]$ is called noninvertible or singular.

If $[A]$ and $[B]$ are two $n \times n$ matrices such that $[B]\ [A] = [I]$, then these statements are also true

$[B]$ is the inverse of $[A]$

$[A]$ is the inverse of $[B]$

$[A]$ and $[B]$ are both invertible

$[A]\ [B]=[I]$.

$[A]$ and $[B]$ are both nonsingular

all columns of $[A]$ and $[B]$ are linearly independent

all rows of $[A]$ and $[B]$ are linearly independent.

**Example 10**

Determine if

$$[B] = \begin{bmatrix} 3 & 2 \\ 5 & 3 \end{bmatrix}$$

is the inverse of

$$[A] = \begin{bmatrix} -3 & 2 \\ 5 & -3 \end{bmatrix}$$

**Solution**

$$\begin{aligned} [B][A] &= \begin{bmatrix} 3 & 2 \\ 5 & 3 \end{bmatrix} \begin{bmatrix} -3 & 2 \\ 5 & -3 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= [I] \end{aligned}$$

Since

$$[B]\,[A] = [I]\,,$$

$[B]$ is the inverse of $[A]$ and $[A]$ is the inverse of $[B]$.

But, we can also show that

$$\begin{aligned} [A][B] &= \begin{bmatrix} -3 & 2 \\ 5 & -3 \end{bmatrix} \begin{bmatrix} 3 & 2 \\ 5 & 3 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= [I] \end{aligned}$$

to show that $[A]$ is the inverse of $[B]$.

## Can I use the concept of the inverse of a matrix to find the solution of a set of equations [A] [X] = [C]?

Yes, if the number of equations is the same as the number of unknowns, the coefficient matrix $[A]$ is a square matrix.

Given

$$[A]\ [X] = [C]$$

Then, if $[A]^{-1}$ exists, multiplying both sides by $[A]^{-1}$.

$$[A]^{-1}[A][X] = [A]^{-1}[C]$$

$$[I]\ [X] = [A]^{-1}[C]$$

$$[X] = [A]^{-1}[C]$$

This implies that if we are able to find $[A]^{-1}$, the solution vector of $[A]\ [X] = [C]$ is simply a multiplication of $[A]^{-1}$ and the right hand side vector, $[C]$.

## How do I find the inverse of a matrix?

If $[A]$ is a $n \times n$ matrix, then $[A]^{-1}$ is a $n \times n$ matrix and according to the definition of inverse of a matrix

$$[A]\ [A]^{-1} = [I]$$

Denoting

$$[A] = \begin{bmatrix} a_{11} & a_{12} & \cdot & \cdot & a_{1n} \\ a_{21} & a_{22} & \cdot & \cdot & a_{2n} \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ a_{n1} & a_{n2} & \cdot & \cdot & a_{nn} \end{bmatrix}$$

$$[A]^{-1} = \begin{bmatrix} a'_{11} & a'_{12} & \cdot & \cdot & a'_{1n} \\ a'_{21} & a'_{22} & \cdot & \cdot & a'_{2n} \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ a'_{n1} & a'_{n2} & \cdot & \cdot & a'_{nn} \end{bmatrix}$$

$$[I] = \begin{bmatrix} 1 & 0 & \cdot & \cdot & \cdot & 0 \\ 0 & 1 & & & & 0 \\ 0 & & \cdot & & & \cdot \\ \cdot & & & 1 & & \cdot \\ \cdot & & & & \cdot & \cdot \\ 0 & \cdot & \cdot & \cdot & \cdot & 1 \end{bmatrix}$$

Using the definition of matrix multiplication, the first column of the $[A]^{-1}$ matrix can then be found by solving

$$\begin{bmatrix} a_{11} & a_{12} & \cdot & \cdot & a_{1n} \\ a_{21} & a_{22} & \cdot & \cdot & a_{2n} \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ a_{n1} & a_{n2} & \cdot & \cdot & a_{nn} \end{bmatrix} \begin{bmatrix} a'_{11} \\ a'_{21} \\ \cdot \\ \cdot \\ a'_{n1} \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ \cdot \\ \cdot \\ 0 \end{bmatrix}$$

Similarly, one can find the other columns of the $[A]^{-1}$ matrix by changing the right hand side accordingly.

**Example 11**

The upward velocity of the rocket is given by

**Table 5.2.** Velocity vs time data for a rocket

| Time, $t$ (s) | Velocity, $v$ (m/s) |
|---|---|
| 5 | 106.8 |
| 8 | 177.2 |
| 12 | 279.2 |

In an earlier example, we wanted to approximate the velocity profile by

$$v(t) = at^2 + bt + c, 5 \leq t \leq 12$$

We found that the coefficients $a,\ b,\ and\ c$ in $v(t)$ are given by

$$\begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 106.8 \\ 177.2 \\ 279.2 \end{bmatrix}$$

First, find the inverse of

$$[A] = \begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{bmatrix}$$

and then use the definition of inverse to find the coefficients $a,\ b,\ and\ c$.

**Solution**

If

$$[A]^{-1} = \begin{bmatrix} a'_{11} & a'_{12} & a'_{13} \\ a'_{21} & a'_{22} & a'_{23} \\ a'_{31} & a'_{32} & a'_{33} \end{bmatrix}$$

is the inverse of $[A]$, then

$$\begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{bmatrix} \begin{bmatrix} a'_{11} & a'_{12} & a'_{13} \\ a'_{21} & a'_{22} & a'_{23} \\ a'_{31} & a'_{32} & a'_{33} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

gives three sets of equations

$$\begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{bmatrix} \begin{bmatrix} a'_{11} \\ a'_{21} \\ a'_{31} \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{bmatrix} \begin{bmatrix} a'_{12} \\ a'_{22} \\ a'_{32} \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{bmatrix} \begin{bmatrix} a'_{13} \\ a'_{23} \\ a'_{33} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Solving the above three sets of equations separately gives

$$\begin{bmatrix} a'_{11} \\ a'_{21} \\ a'_{31} \end{bmatrix} = \begin{bmatrix} 0.04762 \\ -0.9524 \\ 4.571 \end{bmatrix}$$

$$\begin{bmatrix} a'_{12} \\ a'_{22} \\ a'_{32} \end{bmatrix} = \begin{bmatrix} -0.08333 \\ 1.417 \\ -5.000 \end{bmatrix}$$

$$\begin{bmatrix} a'_{13} \\ a'_{23} \\ a'_{33} \end{bmatrix} = \begin{bmatrix} 0.03571 \\ -0.4643 \\ 1.429 \end{bmatrix}$$

Hence

$$[A]^{-1} = \begin{bmatrix} 0.04762 & -0.08333 & 0.03571 \\ -0.9524 & 1.417 & -0.4643 \\ 4.571 & -5.000 & 1.429 \end{bmatrix}$$

Now

$$[A]\,[X] = [C]$$

where

$$[X] = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$$

$$[C] = \begin{bmatrix} 106.8 \\ 177.2 \\ 279.2 \end{bmatrix}$$

Using the definition of $[A]^{-1}$,

$$[A]^{-1}\,[A]\,[X] = [A]^{-1}\,[C]$$

$$[X] = [A]^{-1}\,[C]$$

$$\begin{bmatrix} 0.04762 & -0.08333 & 0.03571 \\ -0.9524 & 1.417 & -0.4643 \\ 4.571 & -5.000 & 1.429 \end{bmatrix} \begin{bmatrix} 106.8 \\ 177.2 \\ 279.2 \end{bmatrix}$$

Hence

$$\begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 0.2905 \\ 19.69 \\ 1.086 \end{bmatrix}$$

So

$$v(t) = 0.2905t^2 + 19.69t + 1.086, 5 \leq t \leq 12$$

## Is there another way to find the inverse of a matrix?

For finding the inverse of small matrices, the inverse of an invertible matrix can be found by

$$[A]^{-1} = \frac{1}{\det(A)} adj(A)$$

where

$$adj(A) = \begin{bmatrix} C_{11} & C_{12} & \cdots & C_{1n} \\ C_{21} & C_{22} & & C_{2n} \\ \vdots & & & \\ C_{n1} & C_{n2} & \cdots & C_{nn} \end{bmatrix}^T$$

where $C_{ij}$ are the cofactors of $a_{ij}$. The matrix

$$\begin{bmatrix} C_{11} & C_{12} & \cdots & C_{1n} \\ C_{21} & C_{22} & \cdots & C_{2n} \\ \vdots & & & \vdots \\ C_{n1} & \cdots & \cdots & C_{nn} \end{bmatrix}$$

itself is called the matrix of cofactors from $[A]$. Cofactors are defined in Chapter 4.

**Example 12**

Find the inverse of

$$[A] = \begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{bmatrix}$$

**Solution**

From Example 4.6 in Chapter 04.06, we found

$$\det(A) = -84$$

Next we need to find the adjoint of $[A]$. The cofactors of $A$ are found as follows.

The minor of entry $a_{11}$ is

$$\begin{aligned} M_{11} &= \begin{vmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{vmatrix} \\ &= \begin{vmatrix} 8 & 1 \\ 12 & 1 \end{vmatrix} \\ &= -4 \end{aligned}$$

The cofactors of entry $a_{11}$ is

$$\begin{aligned} C_{11} &= (-1)^{1+1} M_{11} \\ &= M_{11} \\ &= -4 \end{aligned}$$

The minor of entry $a_{12}$ is

$$\begin{aligned} M_{12} &= \begin{vmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{vmatrix} \\ &= \begin{vmatrix} 64 & 1 \\ 144 & 1 \end{vmatrix} \\ &= -80 \end{aligned}$$

The cofactor of entry $a_{12}$ is

$$\begin{aligned} C_{12} &= (-1)^{1+2} M_{12} \\ &= -M_{12} \\ &= -(-80) \\ &= 80 \end{aligned}$$

Similarly

$$C_{13} = -384$$

$$C_{21} = 7$$

$$C_{22} = -119$$

$$C_{23} = 420$$

$$C_{31} = -3$$

$$C_{32} = 39$$

$$C_{33} = -120$$

Hence the matrix of cofactors of $[A]$ is

$$[C] = \begin{bmatrix} -4 & 80 & -384 \\ 7 & -119 & 420 \\ -3 & 39 & -120 \end{bmatrix}$$

The adjoint of matrix $[A]$ is $[C]^T$,

$$\begin{aligned} adj\,(A) &= [C]^T \\ &= \begin{bmatrix} -4 & 7 & -3 \\ 80 & -119 & 39 \\ -384 & 420 & -120 \end{bmatrix} \end{aligned}$$

Hence

$$\begin{aligned} [A]^{-1} &= \frac{1}{\det(A)} adj\,(A) \\ &= \frac{1}{-84} \begin{bmatrix} -4 & 7 & -3 \\ 80 & -119 & 39 \\ -384 & 420 & -120 \end{bmatrix} \\ &= \begin{bmatrix} 0.04762 & -0.08333 & 0.03571 \\ -0.9524 & 1.417 & -0.4643 \\ 4.571 & -5.000 & 1.429 \end{bmatrix} \end{aligned}$$

## If the inverse of a square matrix [A] exists, is it unique?

Yes, the inverse of a square matrix is unique, if it exists. The proof is as follows. Assume that the inverse of $[A]$ is $[B]$ and if this inverse is not unique, then let another inverse of $[A]$ exist called $[C]$.

If $[B]$ is the inverse of $[A]$, then

$$[B]\ [A] = [I]$$

Multiply both sides by $[C]$,

$$[B]\ [A]\ [C] = [I]\ [C]$$

$$[B]\ [A]\ [C] = [C]$$

Since $[C]$ is inverse of $[A]$,

$$[A]\ [C] = [I]$$

Multiply both sides by $[B]$,

$$[B]\ [I]\ = [C]$$

$$[B]\ = [C]$$

This shows that $[B]$ and $[C]$ are the same. So the inverse of $[A]$ is unique.

## Key Terms:

*Consistent system*

*Inconsistent system*

*Infinite solutions*

*Unique solution*

*Rank*

*Inverse*

# Multiple Choice Test

## Chapter 04.05 System of Equations

(1). A $3 \times 4$ matrix can have a rank of at most

(A) 3

(B) 4

(C) 5

(D) 12

(2). Three kids – Jim, Corey and David receive an inheritance of $\$2,253,453$. The money is put in three trusts but is not divided equally to begin with. Corey gets three times what David gets because Corey made an "A" in Dr. Kaw's class. Each trust is put in an interest generating investment. The three trusts of Jim, Corey and David pay an interest of 6%, 8%, 11%, respectively. The total interest of all the three trusts combined at the end of the first year is $\$190,740.57$. How much money was invested in each trust? The equations in a matrix form are

(A) $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 3 \\ .06 & .08 & .11 \end{bmatrix} \begin{bmatrix} J \\ C \\ D \end{bmatrix} = \begin{bmatrix} 2253543 \\ 0 \\ 190740.57 \end{bmatrix}$

(B) $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & -3 \\ .06 & .08 & .11 \end{bmatrix} \begin{bmatrix} J \\ C \\ D \end{bmatrix} = \begin{bmatrix} 2253543 \\ 0 \\ 190740.57 \end{bmatrix}$

(C) $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & -3 \\ 6 & 8 & 11 \end{bmatrix} \begin{bmatrix} J \\ C \\ D \end{bmatrix} = \begin{bmatrix} 2253543 \\ 0 \\ 190740.57 \end{bmatrix}$

(D) $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & -3 \\ .06 & .08 & .11 \end{bmatrix} \begin{bmatrix} J \\ C \\ D \end{bmatrix} = \begin{bmatrix} 2253543 \\ 0 \\ 190740.57 \end{bmatrix}$

(3). Which of the following matrices does not have an inverse?

(A) $\begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$

(B) $\begin{bmatrix} 6 & 7 \\ 12 & 14 \end{bmatrix}$

(C) $\begin{bmatrix} 6 & 0 \\ 0 & 7 \end{bmatrix}$

(D) $\begin{bmatrix} 0 & 6 \\ 7 & 0 \end{bmatrix}$

(4). The set of equations

$$\begin{bmatrix} 1 & 2 & 5 \\ 2 & 3 & 7 \\ 5 & 8 & 19 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 18 \\ 26 \\ 70 \end{bmatrix}$$

has

(A) no solution

(B) finite number of solutions

(C) a unique solution

(D) infinite solutions

(5). Given a system of $[A]\,[X] = [C]$where $[A]$is $n \times n$ matrix and $[X]$and $[C]$are $n \times 1$ matrices, a unique solution $[X]$exists if

(A) rank of $[A]$ = rank of $\left[A \vdots C\right]$

(B) rank of $[A]$ = rank of $\left[A \vdots C\right] = n$

(C) rank of $[A]$ < rank of $\left[A \vdots C\right]$

(D) rank of $[A]$ = rank of $\left[A \vdots C\right] < n$

(6). If $[A]\,[X] = \begin{bmatrix} -13 \\ 76 \\ 38 \end{bmatrix}$ and $[A]^{-1} = \begin{bmatrix} 1 & 2 & -4 \\ -8 & 2 & 16 \\ 2 & 4 & 8 \end{bmatrix}$ then

(A) $[X] = \begin{bmatrix} -13.000 \\ 864.00 \\ 582.00 \end{bmatrix}$

(B) one cannot find a unique $[X]$.

(C) $[X] = \begin{bmatrix} -1.0000 \\ 2.0000 \\ 4.0000 \end{bmatrix}$

(D) no solutions of $[X]$ are possible

For complete solution, go to

https://ma.mathforcollege.com/mcquizzes/04sle/quiz_04sle_systemofequations_solution.pdf

# Problem Set

## Chapter 04.05: System of Equations

(1). For a set of equations $[A]\ [X] = [B]$, a unique solution exists if

(A) rank (A) = rank $\left(A \vdots B\right)$

(B) rank (A) = rank $\left(A \vdots B\right)$ and rank (A) = number of unknowns

(C) rank (A) = rank $\left(A \vdots B\right)$ and rank (A) = number of rows of (A).

**Answer:** B

(2). The rank of matrix

$A = \begin{bmatrix} 4 & 4 & 4 & 4 \\ 4 & 4 & 4 & 4 \\ 4 & 4 & 4 & 4 \\ 4 & 4 & 4 & 4 \end{bmatrix}$ is

(A) 1

(B) 2

(C) 3

(D) 4

**Answer:** A

(3). A $3 \times 4$ matrix can have a rank of at most

(A) 3

(B) 4

(C) 5

(D) 12

**Answer:** A

(4). If $[A]\ [X] = [C]$ has a unique solution, where the order of $[A]$ is $3 \times 3$, $[X]$ is $3 \times 1$, then the rank of $[A]$ is

(A) 2

(B) 3

(C) 4

(D) 5

**Answer:** B

(5). Show if the following system of equations is consistent or inconsistent. If they are consistent, determine if the solution would be unique or infinite ones exist.

$$\begin{bmatrix} 1 & 2 & 5 \\ 7 & 3 & 9 \\ 8 & 5 & 14 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 8 \\ 19 \\ 27 \end{bmatrix}$$

**Answer:** Consistent; Infinite solutions

(6). Show if the following system of equations is consistent or inconsistent. If they are consistent, determine if the solution would be unique or infinite ones exist.

$$\begin{bmatrix} 1 & 2 & 5 \\ 7 & 3 & 9 \\ 8 & 5 & 14 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 8 \\ 19 \\ 28 \end{bmatrix}$$

**Answer:** Inconsistent

(7). Show if the following system of equations is consistent or inconsistent. If they are consistent, determine if the solution would be unique or infinite ones exist.

$$\begin{bmatrix} 1 & 2 & 5 \\ 7 & 3 & 9 \\ 8 & 5 & 13 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 8 \\ 19 \\ 28 \end{bmatrix}$$

**Answer:** Consistent; Unique

(8). The set of equations

$$\begin{bmatrix} 1 & 2 & 5 \\ 7 & 3 & 9 \\ 8 & 5 & 14 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 8 \\ 19 \\ 27 \end{bmatrix}$$

has

(A) Unique solution

(B) No solution

(C) Infinite solutions

**Answer:** C

(9). For what values of $a$ will the following equation have

$x_1 + x_2 + x_3 = 4$

$x_3 = 2$

$\left(a^2 - 4\right) x_1 + x_3 = a - 2$

(A) Unique solution

(B) No solution

(C) Infinite solutions

**Answer:** If $a \neq +2$ or $-2$, then there will be a unique solution If $a = +2$ *or* $-2$, then there will be no solution.
Possibility of infinite solutions does not exist.

(10). Find if

$$[A] = \begin{bmatrix} 5 & -2.5 \\ -2 & 3 \end{bmatrix}$$

and

$$[B] = \begin{bmatrix} 0.3 & 0.25 \\ 0.2 & 0.5 \end{bmatrix}$$

are inverse of each other.

**Answer:** Yes

(11). Find if

$$[A] = \begin{bmatrix} 5 & 2.5 \\ 2 & 3 \end{bmatrix}$$

and

$$lbrackB] = \begin{bmatrix} 0.3 & -0.25 \\ 0.2 & 0.5 \end{bmatrix}$$

are inverse of each other.

**Answer:** No

(12). Find the

(A) cofactor matrix

(B) adjoint matrix

of

$$[A] = \begin{bmatrix} 3 & 4 & 1 \\ 2 & -7 & -1 \\ 8 & 1 & 5 \end{bmatrix}$$

**Answer:** $\begin{bmatrix} -34 & -18 & 58 \\ -19 & 7 & 29 \\ 3 & 5 & -29 \end{bmatrix} \begin{bmatrix} -34 & -19 & 3 \\ -18 & 7 & 5 \\ 58 & 29 & -29 \end{bmatrix}$

(13). Find $[A]^{-1}$ using any method for
$[A] = \begin{bmatrix} 3 & 4 & 1 \\ 2 & -7 & -1 \\ 8 & 1 & 5 \end{bmatrix}$

**Answer:** $[A]^{-1} = \begin{bmatrix} 2.931 \times 10^{-1} & 1.638 \times 10^{-1} & -2.586 \times 10^{-2} \\ 1.552 \times 10^{-1} & -6.034 \times 10^{-2} & -4.310 \times 10^{-2} \\ -5.000 \times 10^{-1} & -2.500 \times 10^{-1} & 2.500 \times 10^{-1} \end{bmatrix}$

(14). Prove that if $[A]$ and $[B]$ are both invertible and are square matrices of same order, then

$$([A][B])^{-1} = [B]^{-1}[A]^{-1}$$

**Answer:**

$$([A][B])^{-1} = [B]^{-1}[A]^{-1}$$

Let

$$[C] = [A][B]$$

$$\begin{aligned} [C][B]^{-1} &= [A][B][B]^{-1} \\ &= [A][I] \\ &= [A] \end{aligned}$$

Again

$[C] = [A][B]$

$$\begin{aligned} [A]^{-1}[C] &= [A]^{-1}[A][B] \\ &= [I][B] \\ &= [B] \end{aligned}$$

So

$$[C]\,[B]^{-1} = [A]\,; \qquad (1)$$

$$[A]^{-1}\,[C] = [B]\,; \qquad (2)$$

From (1) and (2)

$$[C]\,[B]^{-1}\,[A]^{-1}\,[C] = [A]\,[B]$$

$$[A]\,[B]\,[B]^{-1}\,[A]^{-1}\,[A]\,[B] = [A]\,[B]$$

$$[A]^{-1}\,[A]\,[B]\,[B]^{-1}\,[A]^{-1}\,[A]\,[B] = \left[A^{-1}\right]\,[A]\,[B]$$

$$[B]\,[B]^{-1}\left[A^{-1}\right]\,[A]\,[B] = [B]$$

$$\left[B^{-1}\right]\,[B]\,[B]^{-1}\left[A^{-1}\right]\,[A]\,[B] = [B]^{-1}\,[B]$$
$$[B]^{-1}\left[A^{-1}\right]\,[A]\,[B] = [I]\,.$$

(15). What is the inverse of a square diagonal matrix? Does it always exist?

**Answer: Hint:** Inverse of a square n×n diagonal matrix $[A]$ is $[A]^{-1} = \begin{bmatrix} \frac{1}{a_{11}} & 0 & \cdots & 0 \\ 0 & \frac{1}{a_{22}} & \cdots & 0 \\ 0 & & & \vdots \\ \vdots & \cdots & \cdots & \frac{1}{a_{nn}} \end{bmatrix}$

So inverse exists only if $a_{ii} \neq 0$ for all $i$.

(16). $[A]$ and $[B]$ are square matrices. If $[A]\,[B] = [0]$ and $[A]$ is invertible, show$[B] = 0$.

**Answer:**

$$[A]\,[B] = [0]$$
$$\left[A^{-1}\right]\,[A]\,[B] = [A]^{-1}\,[0]$$

(17). If $[A]\,[B]\,[C] = [I]$, where $[A]$, $[B]$ and $[C]$ are of the same size, show that $[B]$ is invertible.

**Answer:** Hint: $det(AB) = det(A)det(B)$

(18). Prove if $[B]$ is invertible, $[A]\,[B]^{-1} = [B]^{-1}[A]$ if and only if $[A]\,[B] = [B][A]$

**Answer:** Hint: Multiply by $[B]^{-1}$ on both sides, $[A]\,[B]\,[B]^{-1} = [B]^{-1}\,[A]\,[B]^{-1}$

(19). For

$$[A] = \begin{bmatrix} 10 & -7 & 0 \\ -3 & 2.099 & 6 \\ 5 & -1 & 5 \end{bmatrix}$$

$$[A]^{-1} = \begin{bmatrix} -0.1099 & -0.2333 & 0.2799 \\ -0.2999 & -0.3332 & 0.3999 \\ 0.04995 & 0.1666 & 6.664 \times 10^{-5} \end{bmatrix}$$

Show

$$det\,(A) = \frac{1}{det\,(A^{-1})}.$$

**Answer:** Not provided on purpose

(20). For what values of $a$ does the linear system have

$$\begin{aligned} x + y &= 2 \\ 6x + 6y &= a \end{aligned}$$

(A) infinite solutions

(B) unique solution

**Answer:**

(A) 12

(B) not possible

(21). Three kids - Jim, Corey and David receive an inheritance of $\$2, \$253, \$453$. The money is put in three trusts but is not divided equally to begin with. Corey gets three times more than David because Corey made an "A" in Dr. Kaw's class. Each trust is put in an interest generating investment. The three trusts of Jim, Corey and David pays an interest of $6\%, 8\%, 11\%$, respectively. The total interest of all the three trusts combined at the end of the first year is $\$190, \$740.57$. How much money was invested in each trust? Set the following as equations in a matrix form. Identify the unknowns. Do not solve for the unknowns.

**Answer:** $J + C + D = \$2, \$253, \$453$

$$C = 3D$$

$$0.06J + 0.08C + 0.11D = \$190, 740.57$$

In matrix form

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & -3 \\ 0.06 & 0.08 & 0.11 \end{bmatrix} \begin{bmatrix} J \\ C \\ D \end{bmatrix} = \begin{bmatrix} 2,253,453 \\ 0 \\ 190,740.57 \end{bmatrix}$$

(22). What is the rank of

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 6 & 7 \\ 6 & 10 & 13 \end{bmatrix}?$$

Justify your answer.

**Answer:** In the above matrix, 2(Row 1) + Row 2 = Row 3. Hence, rank is less than 3. Row 1 and Row 2 are linearly independent. Hence, the rank of the matrix is 2.

(23). What is the rank of

$$\begin{bmatrix} 1 & 2 & 3 & 6 \\ 4 & 6 & 7 & 17 \\ 6 & 10 & 13 & 29 \end{bmatrix}?$$

Justify your answer.

**Answer:** The determinant of all the $3 \times 3$ sub-matrices is zero. Hence, the rank is less than 3. Determinant of

$$\begin{bmatrix} 2 & 3 \\ 6 & 7 \end{bmatrix} = -4 \neq 0.$$

(24). What is the rank of

$$\begin{bmatrix} 1 & 2 & 3 & 6 \\ 4 & 6 & 7 & 18 \\ 6 & 10 & 13 & 30 \end{bmatrix}?$$

Justify your answer.

**Answer:** In the above matrix, 2(Row 1) + Row 2 = Row 3. Hence, rank is less than 3 as the 3 rows are linearly dependant. Determinant of

$$\begin{bmatrix} 2 & 3 \\ 6 & 7 \end{bmatrix} = -4 \neq 0.$$

Hence, the rank is 2.

(25). How many solutions does the following system of equations have

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 6 & 7 \\ 6 & 10 & 13 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 6 \\ 17 \\ 29 \end{bmatrix} ?$$

Justify your answer.

**Answer:** Rank of \$ A = 2\$\ Rank of $A|C = 2$\ Number of unknowns = 3.\ There are infinite solutions since rank of A is less than the number of unknowns.

(26). How many solutions does the following system of equations have

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 6 & 7 \\ 6 & 10 & 13 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 6 \\ 18 \\ 30 \end{bmatrix} ?$$

Justify your answer.

**Answer:** Rank of $A = 2$\ Rank of $A|C = 2$\ Number of unknowns = 3.\ There are infinite solutions since rank of A is less than the number of unknowns.

(27). By any scientific method, find the second column of the inverse of

$$\begin{bmatrix} 1 & 2 & 0 \\ 4 & 5 & 0 \\ 0 & 0 & 13 \end{bmatrix}.$$

**Answer:** $$\begin{bmatrix} 1 & 2 & 0 \\ 4 & 5 & 0 \\ 0 & 0 & 13 \end{bmatrix} \begin{bmatrix} X & a'_{12} & X \\ X & a'_{22} & X \\ X & a'_{32} & X \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$a'_{12} + 2a'_{22} = 0$$
$$4a'_{12} + 5a'_{22} = 1$$
$$13a'_{32} = 0$$

Simplifying,

$$\begin{bmatrix} a'_{12} \\ a'_{22} \\ a'_{32} \end{bmatrix} = \begin{bmatrix} 0.667 \\ -0.333 \\ 0 \end{bmatrix}$$

(28). Just write out the inverse of (no need to show any work)

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 5 \end{bmatrix}$$

**Answer:** $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \frac{1}{2} & 0 & 0 \\ 0 & 0 & \frac{1}{4} & 0 \\ 0 & 0 & 0 & \frac{1}{5} \end{bmatrix}$

(29). Solve $[A]\ [X] = [B]$ for $[X]$ if

$$[A]^{-1} = \begin{bmatrix} 10 & -7 & 0 \\ 2 & 2 & 5 \\ 2 & 0 & 6 \end{bmatrix}$$

and

$$[B] = \begin{bmatrix} 7 \\ 2.5 \\ 6.012 \end{bmatrix}$$

**Answer:**

$$[X] = [A] - 1[B] = \begin{bmatrix} 10 & -7 & 0 \\ 2 & 2 & 5 \\ 2 & 0 & 6 \end{bmatrix} \begin{bmatrix} 7 \\ 2.5 \\ 6.012 \end{bmatrix}$$
$$= \begin{bmatrix} 52.5 \\ 49.06 \\ 50.072 \end{bmatrix}$$

(30). Let $[A]$ be a $3 \times 3$ matrix. Suppose

$$[X] = \begin{bmatrix} 7 \\ 2.5 \\ 6.012 \end{bmatrix}$$

is a solution to the homogeneous set of equations $[A]\ [X] = [0]$ (the right hand side is a zero vector of order $3 \times 1$). Does $[A]$ have an inverse? Justify your answer.

**Answer:** Given

$[A]\ [X] = [0]$
If $[A]^{-1}$exists, then
$[A]^{-1}\ [A]\ [X] = [A]^{-1}[0]$
$[I]\ [X] = [0]$
$[X] = [0]$
This contradicts the given value of $[X]$. Hence, $[A]^{-1}$ does not exist.

(31). Is the set of vectors

$$\overrightarrow{A} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \overrightarrow{B} = \begin{bmatrix} 1 \\ 2 \\ 5 \end{bmatrix}, \overrightarrow{C} = \begin{bmatrix} 1 \\ 4 \\ 25 \end{bmatrix}$$

linearly independent? Justify your answer.

**Answer:** The set of vectors are linearly independent.

(32). What is the rank of the set of vectors

$$\overrightarrow{A} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \overrightarrow{B} = \begin{bmatrix} 1 \\ 2 \\ 5 \end{bmatrix}, \overrightarrow{C} = \begin{bmatrix} 1 \\ 3 \\ 6 \end{bmatrix} ?$$

Justify your answer.

**Answer:** Since, the 3 vectors are linearly independent as proved above, the rank of the 3 vectors is 3.

(33). What is the rank of

$$\overrightarrow{A} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \overrightarrow{B} = \begin{bmatrix} 2 \\ 2 \\ 4 \end{bmatrix}, \overrightarrow{C} = \begin{bmatrix} 3 \\ 3 \\ 5 \end{bmatrix} ?$$

Justify your answer.

**Answer:** By inspection, $\overrightarrow{C} = \overrightarrow{A} + \overrightarrow{B}$. Hence, the 3 vectors are linearly dependent, and the rank is less than 3. Linear combination of $\overrightarrow{A}$ and $\overrightarrow{B}$, that is, $K_1 \overrightarrow{A} + K_2 \overrightarrow{B} = 0$ has only one solution $K_1 = K_2 = 0$. Therefore, the rank is 2.

# Chapter 04.06: Gaussian Elimination Method for Solving Simultaneous Linear Equations

Lesson: Theory of Naive Gauss Elimination Method

## Learning Objectives

*After successful completion of this lesson, you should be able to*

1) *write the algorithm to solve a set of simultaneous linear equations using Naive Gauss elimination method*

## How is a set of equations solved numerically by Gaussian elimination method?

One of the most popular techniques for solving simultaneous linear equations is the Gaussian elimination method. The approach is designed to solve a general set of $n$ equations and $n$ unknowns

$$\begin{aligned}
&a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \ldots + a_{1n}x_n = b_1\\
&a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + \ldots + a_{2n}x_n = b_2\\
&\vdots \qquad\qquad\qquad \vdots\\
&a_{n1}x_1 + a_{n2}x_2 + a_{n3}x_3 + \ldots + a_{nn}x_n = b_n
\end{aligned}$$

Gaussian elimination consists of two parts

1) Forward Elimination of Unknowns: In this part, an unknown is eliminated in the second through the last equation. This way, the equations are *reduced* to one equation and one unknown in each equation.

2) Back Substitution: In this part, starting from the last equation, each of the unknowns is found.

## Forward Elimination of Unknowns:

In the first step of the forward elimination part, the first unknown, $x_1$ is eliminated from all rows below the first row. The first equation is selected as the pivot equation to eliminate $x_1$. So, to eliminate $x_1$ in the second equation, one divides the first equation by $a_{11}$ (hence called the pivot element) and then multiplies it by $a_{21}$. This operation is the same as multiplying the first equation by $a_{21}/a_{11}$ to give

$$a_{21}x_1 + \frac{a_{21}}{a_{11}}a_{12}x_2 + \ldots + \frac{a_{21}}{a_{11}}a_{1n}x_n = \frac{a_{21}}{a_{11}}b_1$$

Now, this equation can be subtracted from the second equation to give

$$\left(a_{22} - \frac{a_{21}}{a_{11}}a_{12}\right)x_2 + \ldots + \left(a_{2n} - \frac{a_{21}}{a_{11}}a_{1n}\right)x_n = b_2 - \frac{a_{21}}{a_{11}}b_1$$

or

$$a'_{22}x_2 + \ldots + a'_{2n}x_n = b'_2$$

where

$$\begin{aligned} a'_{22} &= a_{22} - \frac{a_{21}}{a_{11}}a_{12} \\ &\vdots \\ a'_{2n} &= a_{2n} - \frac{a_{21}}{a_{11}}a_{1n} \end{aligned}$$

This procedure of eliminating $x_1$, is now repeated for the third equation to the $n^{th}$ equation to reduce the set of equations as

$$\begin{aligned} &a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \ldots + a_{1n}x_n = b_1 \\ &a'_{22}x_2 + a'_{23}x_3 + \ldots + a'_{2n}x_n = b'_2 \\ &a'_{32}x_2 + a'_{33}x_3 + \ldots + a'_{3n}x_n = b'_3 \\ &\vdots \qquad\qquad \vdots \qquad\qquad \vdots \\ &a'_{n2}x_2 + a'_{n3}x_3 + \ldots + a'_{\mathrm{nn}}x_n = b'_n \end{aligned}$$

This marks the end of the first step of the forward elimination part. Now for the second step of the forward elimination part, we start with the second equation as the pivot equation and $a'_{22}$ as the pivot element. So, to eliminate $x_2$ in the third equation, one divides the second equation by $a'_{22}$ (the pivot element) and then multiply it by$a'_{32}$. This operation is the same as multiplying the second equation by $a'_{32}/a'_{22}$ and subtracting it from the third equation. This subtraction makes the coefficient of $x_2$ zero in the third equation. The same procedure is now repeated for the fourth equation till the $n^{\mathrm{th}}$equation to give

$$\begin{aligned} &a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \ldots + a_{1n}x_n = b_1 \\ &a'_{22}x_2 + a'_{23}x_3 + \ldots + a'_{2n}x_n = b'_2 \\ &a''_{33}x_3 + \ldots + a''_{3n}x_n = b''_3 \\ &\vdots \qquad\qquad \vdots \\ &a''_{n3}x_3 + \ldots + a''_{\mathrm{nn}}x_n = b''_n \end{aligned}$$

The next steps of the forward elimination part are conducted by using the third equation as a pivot equation. That is, there will be a total of $n-1$ steps in the forward elimination part of the method. At the end of $n-1$ steps within the forward elimination part, we get a set of equations that look like

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \ldots + a_{1n}x_n &= b_1 \\ a'_{22}x_2 + a'_{23}x_3 + \ldots + a'_{2n}x_n &= b'_2 \\ a''_{33}x_3 + \ldots + a''_{3n}x_n &= b''_3 \\ \vdots \qquad\qquad\qquad\qquad & \vdots \\ a_{nn}^{(n-1)}x_n &= b_n^{(n-1)} \end{aligned}$$

## Back Substitution:

In this part, the equations are solved starting from the last equation as it has only one unknown.

$$x_n = \frac{b_n^{(n-1)}}{a_{nn}^{(n-1)}}$$

Then the second last equation, that is the $(n-1)^{th}$ equation, has two unknowns: $x_n$ and $x_{n-1}$, but $x_n$ is already known. This recognition reduces the $(n-1)^{th}$ equation also to one unknown. Back substitution hence can be represented for all equations by the formula

$$x_i = \frac{b_i^{(i-1)} - \sum_{j=i+1}^{n} a_{ij}^{(i-1)} x_j}{a_{ii}^{(i-1)}} \text{ for } i = n-1, \quad n-2, \ldots, \ 1$$

and

$$x_n = \frac{b_n^{(n-1)}}{a_{nn}^{(n-1)}}$$

# Chapter 04.06: Gaussian Elimination Method for Solving Simultaneous Linear Equations

Lesson: Application of Naive Gauss Elimination Method

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *solve a set of simultaneous linear equations using Naïve Gauss elimination.*

## Applications

In the previous lesson, you learned the theory behind Naive Gauss Elimination. In this lesson, we will apply the theory to solve a set of simultaneous linear equations.

### Example 1

The upward velocity of a rocket is given at three different times in Table 1.

**Table 1**: Velocity vs. time data.

| Time, t (s) | Velocity, v (m/s) |
|---|---|
| 5 | 106.8 |
| 8 | 177.2 |
| 12 | 279.2 |

The velocity data is approximated by a polynomial as

$$v(t) = a_1 t^2 + a_2 t + a_3,\ 5 \le t \le 12$$

The coefficients $a_1, a_2$, and $a_3$ for the above expression are given by

$$\begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} 106.8 \\ 177.2 \\ 279.2 \end{bmatrix}$$

Find the values of $a_1, a_2$, and $a_3$ using the Naïve Gauss elimination method. Find the velocity at $t = 6,\ 7.5,\ 9,\ 11$ seconds.

**Solution**

The augmented matrix is

$$\begin{bmatrix} 25 & 5 & 1 & | & 106.8 \\ 64 & 8 & 1 & | & 177.2 \\ 144 & 12 & 1 & | & 279.2 \end{bmatrix}$$

**Forward Elimination of Unknowns**

Since there are three equations, there will be two steps of the forward elimination part.

First step

Divide Row 1 by 25 and then multiply it by 64, that is, multiply Row 1 by $64/25 = 2.56$.

$$\big([25 \quad 5 \quad 1 \quad | \quad 106.8]\big) \times 2.56 \text{ gives Row 1 as}$$

$$[64 \quad 12.8 \quad 2.56 \quad | \quad 273.408]$$

Subtract the result from Row 2

$$\begin{array}{rrrrr} & [64 & 8 & 1 & | \quad 177.2] \\ -[ & 64 & 12.8 & 2.56 & | \quad 273.408] \\ \hline & 0 & -4.8 & -1.56 & -96.208 \end{array}$$

to get the resulting equations as

$$\begin{bmatrix} 25 & 5 & 1 & | & 106.8 \\ 0 & -4.8 & -1.56 & | & -96.208 \\ 144 & 12 & 1 & | & 279.2 \end{bmatrix}$$

Divide Row 1 by 25 and then multiply it by 144, that is, multiply Row 1 by $144/25 = 5.76$.

$$\big([25 \quad 5 \quad 1 \quad | \quad 106.8]\big) \times 5.76 \text{ gives Row 1 as}$$

$$[144 \quad 28.8 \quad 5.76 \quad | \quad 615.168]$$

Subtract the result from Row 3

$$\begin{array}{rrrrr} & [144 & 12 & 1 & | \quad 279.2] \\ -[ & 144 & 28.8 & 5.76 & | \quad 615.168] \\ \hline & 0 & -16.8 & -4.76 & -335.968 \end{array}$$

to get the resulting equations as

$$\begin{bmatrix} 25 & 5 & 1 & | & 106.8 \\ 0 & -4.8 & -1.56 & | & -96.208 \\ 0 & -16.8 & -4.76 & | & -335.968 \end{bmatrix}$$

Second step

We now divide Row 2 by $-4.8$ and then multiply by $-16.8$, that is, multiply Row 2 by $-16.8/-4.8 = 3.5$.

$$([0 \quad -4.8 \quad -1.56 \quad | \quad -96.208]) \times 3.5 \text{ gives Row 2 as}$$

$$[0 \quad -16.8 \quad -5.46 \quad | -336.728]$$

Subtract the result from Row 3

$$\begin{array}{rrrrr} & [0 & -16.8 & -4.76 & | \quad -335.968] \\ -\ [0 & -16.8 & -5.46 & | \quad -336.728] \\ \hline 0 & 0 & 0.7 & & 0.76 \end{array}$$

to get the resulting equations as

$$\left[\begin{array}{ccc|c} 25 & 5 & 1 & 106.8 \\ 0 & -4.8 & -1.56 & -96.208 \\ 0 & 0 & 0.7 & 0.76 \end{array}\right]$$

$$\begin{bmatrix} 25 & 5 & 1 \\ 0 & -4.8 & -1.56 \\ 0 & 0 & 0.7 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} 106.8 \\ -96.208 \\ 0.76 \end{bmatrix}$$

**Back substitution**

From the third equation

$$0.7a_3 = 0.76$$

$$\begin{aligned} a_3 &= \frac{0.76}{0.7} \\ &= 1.08571 \end{aligned}$$

Substituting the value of $a_3$ in the second equation,

$$-4.8a_2 - 1.56a_3 = -96.208$$

$$\begin{aligned} a_2 &= \frac{-96.208 + 1.56a_3}{-4.8} \\ &= \frac{-96.208 + 1.56 \times 1.08571}{-4.8} \\ &= 19.6905 \end{aligned}$$

Substituting the value of $a_2$ and $a_3$ in the first equation,

$$25a_1 + 5a_2 + a_3 = 106.8$$

$$\begin{aligned} a_1 &= \frac{106.8 - 5a_2 - a_3}{25} \\ &= \frac{106.8 - 5 \times 19.6905 - 1.08571}{25} \\ &= 0.290472 \end{aligned}$$

Hence the solution vector is

$$\begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} 0.290472 \\ 19.6905 \\ 1.08571 \end{bmatrix}$$

The polynomial that passes through the three data points is then

$$\begin{aligned} v(t) &= a_1 t^2 + a_2 t + a_3 \\ &= 0.290472t^2 + 19.6905t + 1.08571, \quad 5 \leq t \leq 12 \end{aligned} \qquad (E1.1)$$

Since we want to find the velocity at $t = 6, 7.5, 9$ and 11 seconds, we could simply substitute each value of $t$ in $v(t) = 0.290472t^2 + 19.6905t + 1.08571$ and find the corresponding velocity. For example, at $t = 6$

$$\begin{aligned} v(6) &= 0.290472(6)^2 + 19.6905(6) + 1.08571 \\ &= 129.686 \ \text{m/s} \end{aligned}$$

However, we could also find all the needed values of velocity at $t = 6, 7.5, 9, 11$ seconds using the matrix multiplication form of Equation (E1.1).

$$v(t) = \begin{bmatrix} 0.290472 & 19.6905 & 1.08571 \end{bmatrix} \begin{bmatrix} t^2 \\ t \\ 1 \end{bmatrix}$$

So, if we want to find $v(6), v(7.5), v(9), v(11)$, it is given by

$$\begin{aligned} [v(6)\, v(7.5)\, v(9)\, v(11)] &= \begin{bmatrix} 0.290472 & 19.6905 & 1.08571 \end{bmatrix} \begin{bmatrix} 6^2 & 7.5^2 & 9^2 & 11^2 \\ 6 & 7.5 & 9 & 11 \\ 1 & 1 & 1 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 0.290472 & 19.6905 & 1.08571 \end{bmatrix} \begin{bmatrix} 36 & 56.25 & 81 & 121 \\ 6 & 7.5 & 9 & 11 \\ 1 & 1 & 1 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 129.686 & 165.104 & 201.828 & 252.828 \end{bmatrix} \end{aligned}$$

$$v(6) = 129.686 \ \text{m/s}$$

$$v(7.5) = 165.104 \text{ m/s}$$

$$v(9) = 201.828 \text{ m/s}$$

$$v(11) = 252.828 \text{ m/s}$$

# Chapter 04.06: Gaussian Elimination Method for Solving Simultaneous Linear Equations

Lesson: Finding Determinant of a Square Matrix Using Gaussian Elimination

## Learning Objectives

*After successful completion of this lesson, you should be able to*

1) *use the forward elimination part of Naive Gauss elimination method to find the determinant of a square matrix,*

2) *enumerate theorems related to the determinant of matrices,*

3) *relate the zero and non-zero value of the determinant of a square matrix to the existence or non-existence of the matrix inverse.*

## Can we use Naive Gauss elimination methods to find the determinant of a square matrix?

One of the more efficient ways to find the determinant of a square matrix is by taking advantage of the following two theorems on a determinant of matrices coupled with Naive Gauss elimination.

**Theorem 1:**

Let $[A]$ be a $n \times n$ matrix. Then, if $[B]$ is a $n \times n$ matrix that results from adding or subtracting a multiple of one row to another row, then $det(A) = det(B)$ (The same is true for column operations also).

**Theorem 2:**

Let $[A]$ be a $n \times n$ matrix that is upper triangular, lower triangular, or diagonal, then

$$\begin{aligned}\det(A) &= a_{11} \times a_{22} \times ... \times a_{ii} \times ... \times a_{nn} \\ &= \prod_{i=1}^{n} a_{ii}\end{aligned}$$

This formula implies that if we apply the forward elimination part of the Naive Gauss elimination method, the determinant of the matrix stays the same according to Theorem 1. Then since at the end of the forward elimination part, the resulting matrix is upper triangular, the determinant will be given by Theorem 2.

**Example 1**

Find the determinant of

$$[A] = \begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{bmatrix}$$

**Solution**

If we conduct all the steps of the forward elimination part using the Naive Gauss elimination method on $[A]$, it will give us the following upper triangular matrix (refer to the example in the previous lesson of Naive Gauss elimination for the process)

$$[B] = \begin{bmatrix} 25 & 5 & 1 \\ 0 & -4.8 & -1.56 \\ 0 & 0 & 0.7 \end{bmatrix}$$

According to Theorem 2

$$\begin{aligned} \det(B) &= 25 \times (-4.8) \times 0.7 \\ &= -84.00 \end{aligned}$$

and since forward elimination part involve subtracting multiple of one row from another, using the Theorem 1

$$\begin{aligned} \det(A) &= \det(B) \\ &= -84.00 \end{aligned}$$

## What if I cannot find the determinant of the matrix using the Naive Gauss elimination method, for example, if I get division by zero problems during the Naive Gauss elimination method?

Well, in that case, you can apply Gaussian elimination with partial pivoting. However, the determinant of the resulting upper triangular matrix may differ by sign. The following theorem applies in addition to the previous two to find the determinant of a square matrix.

**Theorem 3:**

Let $[A]$ be a $n \times n$ matrix. Then, if $[B]$ is a matrix that results from switching one row with another row, then $\det(B) = -\det(A)$.

**Example 2**

Find the determinant of

$$[A] = \begin{bmatrix} 10 & -7 & 0 \\ -3 & 2.099 & 6 \\ 5 & -1 & 5 \end{bmatrix}$$

**Solution**

The end of the forward elimination part of Gaussian elimination with partial pivoting, we would obtain

$$[B] = \begin{bmatrix} 10 & -7 & 0 \\ 0 & 2.5 & 5 \\ 0 & 0 & 6.002 \end{bmatrix}$$

$$\begin{aligned} \det(B) &= 10 \times 2.5 \times 6.002 \\ &= 150.05 \end{aligned}$$

Since rows were switched once during the forward elimination part of Gaussian elimination with partial pivoting,

$$\begin{aligned} \det(A) &= -\det(B) \\ &= -150.05 \end{aligned}$$

**Example 3**

Prove

$$\det(A) = \frac{1}{\det(A^{-1})}$$

**Solution**

$$[A][A]^{-1} = [I]$$

$$\det(AA^{-1}) = det(I)$$

$$\det(A)\det(A^{-1}) = 1$$

$$\det(A) = \frac{1}{\det(A^{-1})}$$

If $[A]$ is a $n \times n$ matrix and $\det(A) \neq 0$, what other statements are equivalent to it?

(1) $[A]$ is invertible.

(2) $[A]^{-1}$ exists.

(3) $[A]\ [X] = [C]$ has a unique solution.

(4) $[A]\ [X] = [0]$ only solution is $[X] = [0]$.

(5) $[A]\ [A]^{-1} = [I] = [A]^{-1}\ [A]$.

# Chapter 04.06: Gaussian Elimination Method for Solving Simultaneous Linear Equations

Section: Pitfalls of Naive Gauss Elimination Method

## Learning Objectives

*After successful completion of this lesson, you should be able to*

1) *enumerate the pitfalls of the Naive Gauss elimination method*

2) *show the pitfalls of the Naive Gauss elimination method through examples*

## Are there any pitfalls of the Naive Gauss elimination method?

Yes, there are two pitfalls of the Naive Gauss elimination method.

**Division by zero:** First, it is possible for division by zero to occur during the beginning of any of the $n-1$ steps of the forward elimination part.

For example

$$\begin{aligned} 5x_2 + 6x_3 &= 11 \\ 4x_1 + 5x_2 + 7x_3 &= 16 \\ 9x_1 + 2x_2 + 3x_3 &= 15 \end{aligned}$$

will result in division by zero in the first step of the forward elimination part as the coefficient of $x_1$ in the first equation is zero, as is evident when we write the equations in matrix form.

$$\begin{bmatrix} 0 & 5 & 6 \\ 4 & 5 & 7 \\ 9 & 2 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 11 \\ 16 \\ 15 \end{bmatrix}$$

But what about the equations below: Is division by zero a problem?

$$\begin{aligned} 5x_1 + 6x_2 + 7x_3 &= 18 \\ 10x_1 + 12x_2 + 3x_3 &= 25 \\ 20x_1 + 17x_2 + 19x_3 &= 56 \end{aligned}$$

Written in matrix form,

$$\begin{bmatrix} 5 & 6 & 7 \\ 10 & 12 & 3 \\ 20 & 17 & 19 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 18 \\ 25 \\ 56 \end{bmatrix}$$

there is no issue of division by zero in the the first step of forward elimination part. The pivot element is the coefficient of $x_1$ in the first equation, that is, 5, and that is a non-zero number. However, at the end of the first step of the forward elimination part, we get the equations in the following matrix form

$$\begin{bmatrix} 5 & 6 & 7 \\ 0 & 0 & -11 \\ 0 & -7 & -9 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 18 \\ -11 \\ -16 \end{bmatrix}$$

Now at the beginning of the $2^{nd}$ step of the forward elimination part, the coefficient of $x_2$ in Equation 2 would be used as the pivot element. That element is zero and hence would create the division by zero problem.

So it is important to consider that the possibility of division by zero can occur at the beginning of any of the $n-1$ steps of the forward elimination part in the Naive Gauss elimination method.

**Round-off error:** Second, the Naive Gauss elimination method is prone to round-off errors. These errors are higher when there are many equations as errors propagate more than when there are a small number of equations. Also, if there is a large variation in the order of the magnitude of the numbers, and subtraction of close numbers takes place, it may also create large errors. See the examples below.

**Example 1**

Use Naive Gauss elimination to solve

$$\begin{aligned} 20x_1 + 15x_2 + 10x_3 &= 45 \\ -3x_1 - 2.249x_2 + 7x_3 &= 1.751 \\ 5x_1 + x_2 + 3x_3 &= 9 \end{aligned}$$

Use six significant digits with chopping in your calculations.

**Solution**

Putting the equations in the matrix form, we get

$$\begin{bmatrix} 20 & 15 & 10 \\ -3 & -2.249 & 7 \\ 5 & 1 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 45 \\ 1.751 \\ 9 \end{bmatrix}$$

The augmented matrix is

$$\left[\begin{array}{ccc|c} 20 & 15 & 10 & 45 \\ -3 & -2.249 & 7 & 1.751 \\ 5 & 1 & 3 & 9 \end{array}\right]$$

**Forward Elimination**

First step

Divide Row 1 by 20 and then multiply it by $-3$, that is, multiply Row 1 by $-3/20 = -0.15$.

$$([20 \;\; 15 \;\; 10 \;\; | \;\; 45]) \times -0.15 \text{ gives Row 1 as}$$

$$[-3 \;\; -2.25 \;\; -1.5 \;\; | \;\; -6.75]$$

Subtract the result from Row 2

$$\begin{array}{rrrcr} [-3 & -2.249 & 7 & | & 1.751] \\ -[-3 & -2.25 & -1.5 & | & -6.75] \\ \hline 0 & 0.001 & 8.5 & & 8.501 \end{array}$$

to get the resulting equations as

$$\begin{bmatrix} 20 & 15 & 10 & | & 45 \\ 0 & 0.001 & 8.5 & | & 8.501 \\ 5 & 1 & 3 & | & 9 \end{bmatrix}$$

Divide Row 1 by 20 and then multiply it by 5, that is, multiply Row 1 by $5/20 = 0.25$

$$[20 \;\; 15 \;\; 10 \;\; | \;\; 45] \times 0.25 \text{ gives Row 1 as}$$

$$[5 \;\; 3.75 \;\; 2.5 \;\; | \;\; 11.25]$$

Subtract the result from Row 3

$$\begin{array}{rrrcr} [5 & 1 & 3 & | & 9] \\ -[5 & 3.75 & 2.5 & | & 11.25] \\ \hline 0 & -2.75 & 0.5 & & -2.25 \end{array}$$

to get the resulting equations as

$$\begin{bmatrix} 20 & 15 & 10 & | & 45 \\ 0 & 0.001 & 8.5 & | & 8.501 \\ 0 & -2.75 & 0.5 & | & -2.25 \end{bmatrix}$$

Second step

Now for the second step of the forward elimination part, we will use Row 2 as the pivot equation and eliminate Row 3: Column 2.

Divide Row 2 by 0.001 and then multiply it by $-2.75$, that is, multiply Row 2 by $-2.75/0.001 = -2750$.

$$[0 \;\; 0.001 \;\; 8.5 \;\; | \;\; 8.501] \times -2750 \text{ gives Row 2 as}$$

$$[0 \quad -2.75 \quad -23375 \quad | \quad -23377.75]$$

Rewriting within 6 significant digits with chopping

$$[0 \quad -2.75 \quad -23375 \quad | \quad -23377.7]$$

Subtract the result from Row 3

$$\begin{array}{rrrrr} [0 & -2.75 & 0.5 & | & -2.25] \\ -[0 & -2.75 & -23375 & | & -23377.7] \\ \hline 0 & 0 & 23375.5 & & 23375.45 \end{array}$$

Rewriting within 6 significant digits with chopping

$$[0 \quad 0 \quad 23375.5 \quad | \quad 23375.4]$$

to get the resulting equations as

$$\begin{bmatrix} 20 & 15 & 10 & | & 45 \\ 0 & 0.001 & 8.5 & | & 8.501 \\ 0 & 0 & 23375.5 & | & 23375.4 \end{bmatrix}$$

This is the end of the forward elimination part.

**Back substitution**

We can now solve the above equations by back substitution. From the third equation,

$$23375.5x_3 = 23375.4$$

$$\begin{aligned} x_3 &= \frac{23375.4}{23375.5} \\ &= 0.999995 \end{aligned}$$

Substituting the value of $x_3$ in the second equation

$$0.001x_2 + 8.5x_3 = 8.501$$

$$\begin{aligned} x_2 &= \frac{8.501 - 8.5x_3}{0.001} \\ &= \frac{8.501 - 8.5 \times 0.999995}{0.001} \\ &= \frac{8.501 - 8.49995}{0.001} \\ &= \frac{0.00105}{0.001} \\ &= 1.05 \end{aligned}$$

Substituting the value of $x_3$ and $x_2$ in the first equation,

$$20x_1 + 15x_2 + 10x_3 = 45$$

$$\begin{aligned} x_1 &= \frac{45 - 15x_2 - 10x_3}{20} \\ &= \frac{45 - 15 \times 1.05 - 10 \times 0.999995}{20} \\ &= \frac{45 - 15.75 - 9.99995}{20} \\ &= \frac{29.25 - 9.99995}{20} \\ &= \frac{19.2500}{20} \\ &= 0.9625 \end{aligned}$$

This is the end of the back substitution part.

Hence the solution is

$$\begin{aligned} [X] &= \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \\ &= \begin{bmatrix} 0.9625 \\ 1.05 \\ 0.999995 \end{bmatrix} \end{aligned}$$

Compare this with the exact solution of

$$\begin{aligned} [X] &= \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \\ &= \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \end{aligned}$$

**Example 2**

In Example 1, we used Naive Gauss elimination to solve

$$\begin{aligned} 20x_1 + 15x_2 + 10x_3 &= 45 \\ -3x_1 - 2.249x_2 + 7x_3 &= 1.751 \\ 5x_1 + x_2 + 3x_3 &= 9 \end{aligned}$$

using six significant digits with chopping in your calculations? Repeat the problem, but now use five significant digits with chopping in your calculations.

**Solution**

Putting the equations in the matrix form, we get

$$\begin{bmatrix} 20 & 15 & 10 \\ -3 & -2.249 & 7 \\ 5 & 1 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 45 \\ 1.751 \\ 9 \end{bmatrix}$$

The augmented matrix is

$$\left[\begin{array}{ccc|c} 20 & 15 & 10 & 45 \\ -3 & -2.249 & 7 & 1.751 \\ 5 & 1 & 3 & 9 \end{array}\right]$$

**Forward Elimination of Unknowns**

First step

Divide Row 1 by 20 and then multiply it by $-3$, that is, multiply Row 1 by $-3/20 = -0.15$.

$$([20 \;\; 15 \;\; 10 \;\; | \;\; 45]) \times -0.15 \text{ gives Row 1 as}$$

$$[-3 \;\; -2.25 \;\; -1.5 \;\; | \;\; -6.75]$$

Subtract the result from Row 2

$$\begin{array}{rrrcr} [-3 & -2.249 & 7 & | & 1.751] \\ -[-3 & -2.25 & -1.5 & | & -6.75] \\ \hline 0 & 0.001 & 8.5 & & 8.501 \end{array}$$

to get the resulting equations as

$$\left[\begin{array}{ccc|c} 20 & 15 & 10 & 45 \\ 0 & 0.001 & 8.5 & 8.501 \\ 5 & 1 & 3 & 9 \end{array}\right]$$

Divide Row 1 by 20 and then multiply it by 5, that is, multiply Row 1 by $5/20 = 0.25$.

$$[20 \;\; 15 \;\; 10 \;\; | \;\; 45] \times 0.25 \text{ gives Row 1 as}$$

$$[5 \;\; 3.75 \;\; 2.5 \;\; | \;\; 11.25]$$

Subtract the result from Row 3

$$\begin{array}{rrrcr} [5 & 1 & 3 & | & 9] \\ -[5 & 3.75 & 2.5 & | & 11.25] \\ \hline 0 & -2.75 & 0.5 & & -2.25 \end{array}$$

to get the resulting equations as

$$\begin{bmatrix} 20 & 15 & 10 & | & 45 \\ 0 & 0.001 & 8.5 & | & 8.501 \\ 0 & -2.75 & 0.5 & | & -2.25 \end{bmatrix}$$

Second step

Now for the second step of the forward elimination part, we will use Row 2 as the pivot equation and eliminate Row 3: Column 2.

Divide Row 2 by 0.001 and then multiply it by $-2.75$, that is, multiply Row 2 by $-2.75/0.001 = -2750$.

$$[0 \quad 0.001 \quad 8.5 \quad | \quad 8.501] \times -2750 \text{ gives Row 2 as}$$

$$[0 \quad -2.75 \quad -23375 \quad | \quad -23377.75]$$

Rewriting within 5 significant digits with chopping

$$[0 \quad -2.75 \quad -23375 \quad | \quad -23377]$$

Subtract the result from Row 3

$$\begin{array}{cccc} [0 & -2.75 & 0.5 & | \quad -2.25] \\ -[0 & -2.75 & -23375 & | \quad -23377] \\ \hline 0 & 0 & 23375.5 & 23374.75 \end{array}$$

Rewriting with 5 significant digits with chopping

$$[0 \quad 0 \quad 23375 \quad | \quad 23374]$$

to get the resulting equations as

$$\begin{bmatrix} 20 & 15 & 10 & | & 45 \\ 0 & 0.001 & 8.5 & | & 8.501 \\ 0 & 0 & 23375 & | & 23374 \end{bmatrix}$$

This marks the end of the forward elimination part.

**Back substitution**

We can now solve the above equations by back substitution. From the third equation,

$$\begin{aligned} 23375x_3 &= 23374 \\ x_3 &= \frac{23374}{23375} \\ &= 0.99995 \end{aligned}$$

Substituting the value of $x_3$ in the second equation

$$0.001x_2 + 8.5x_3 = 8.501$$

$$\begin{aligned}
x_2 &= \frac{8.501 - 8.5x_3}{0.001} \\
&= \frac{8.501 - 8.5 \times 0.99995}{0.001} \\
&= \frac{8.501 - 8.499575}{0.001} \\
&= \frac{8.501 - 8.4995}{0.001} \\
&= \frac{0.0015}{0.001} \\
&= 1.5
\end{aligned}$$

Substituting the value of $x_3$ and $x_2$ in the first equation,

$$20x_1 + 15x_2 + 10x_3 = 45$$

$$\begin{aligned}
x_1 &= \frac{45 - 15x_2 - 10x_3}{20} \\
&= \frac{45 - 15 \times 1.5 - 10 \times 0.99995}{20} \\
&= \frac{45 - 22.5 - 9.9995}{20} \\
&= \frac{22.5 - 9.9995}{20} \\
&= \frac{12.5005}{20} \\
&= \frac{12.500}{20} \\
&= 0.625
\end{aligned}$$

This marks the end of the back substitution part.

Hence the solution is

$$\begin{aligned}
[X] &= \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \\
&= \begin{bmatrix} 0.625 \\ 1.5 \\ 0.99995 \end{bmatrix}
\end{aligned}$$

Compare this with the exact solution of

$$[X] = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

Reducing the number of significant digits carried from 6 to 5 results in a substantial round-off error. In the next lesson, we will discuss how we can alleviate the two pitfalls of the Naive Gauss elimination method.

# Chapter 04.06: Gaussian Elimination Method for Solving Simultaneous Linear Equations

Lesson: Theory of Gaussian Elimination with Partial Pivoting Method

## Learning Objectives

*After successful completion of this lesson, you should be able to*

1) *write the algorithm to solve a set of simultaneous linear equations using Gaussian elimination with Partial Pivoting.*

## What are some techniques for improving the Naïve Gauss elimination method?

As seen in the examples in the previous lesson, round-off errors were large when five significant digits were used as opposed to six significant digits. One method of decreasing the round-off error would be to use more significant digits, that is, use double or quad precision for representing the numbers. However, this would not avoid possible division by zero errors in the Naïve Gauss elimination method. To avoid division by zero as well as reduce (not eliminate) round-off error, Gaussian elimination with partial pivoting is the method of choice.

## How does Gaussian elimination with partial pivoting differ from Naïve Gauss elimination?

The two methods are the same, except at the beginning of each step of the forward elimination part, a row switching is done based on the following criterion. If there are $n$ equations, then there are $n-1$ steps of the forward elimination part. At the beginning of the $k^{th}$ step of the forward elimination part, one finds the maximum of

$$|a_{kk}|, |a_{k+1,k}|, \ldots\ldots, |a_{nk}|$$

Then if the maximum of these values is $|a_{pk}|$ , $k \leq p \leq n$, then switch rows $p$ and $k$.

The other steps of the forward elimination part are the same as the Naïve Gauss elimination method. The back substitution part stays exactly the same as the Naive Gauss elimination method.

# Chapter 04.06: Gaussian Elimination Method for Solving Simultaneous Linear Equations

Lesson: Application of Gaussian Elimination with Partial Pivoting Method

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *solve a set of simultaneous linear equations using the Gauss elimination method with partial pivoting*

## Applications

In the previous lessons, you learned why we modify the Naive Gauss elimination method to the Gaussian elimination with partial pivoting. Then we discussed, the theory behind the method. In this lesson, we will apply the theory to solve a set of simultaneous linear equations.

### Example 1

In the examples in the previous lesson, we used Naive Gauss elimination to solve

$$\begin{aligned} 20x_1 + 15x_2 + 10x_3 &= 45 \\ -3x_1 - 2.249x_2 + 7x_3 &= 1.751 \\ 5x_1 + x_2 + 3x_3 &= 9 \end{aligned}$$

using five and six significant digits with chopping in the calculations. When we used five significant digits with chopping, the solution found was

$$\begin{aligned} [X] &= \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \\ &= \begin{bmatrix} 0.625 \\ 1.5 \\ 0.99995 \end{bmatrix} \end{aligned}$$

This is quite different from the exact solution of

$$[X] = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$
$$= \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

Redo using Gaussian elimination with partial pivoting. Use five significant digits with chopping in all your calculations.

**Solution**

The augmented matrix is

$$\left[\begin{array}{ccc|c} 20 & 15 & 10 & 45 \\ -3 & -2.249 & 7 & 1.751 \\ 5 & 1 & 3 & 9 \end{array}\right]$$

Forward Elimination of Unknowns

Now for the first step of the forward elimination part, the absolute value of the first column elements below Row 1 is

$$|20|, |-3|, |5|$$

or

$$20,\ 3,\ 5$$

So, the largest absolute value is in Row 1. So as per Gaussian elimination with partial pivoting, the switch is between Row 1 and Row 1 to give

$$\left[\begin{array}{ccc|c} 20 & 15 & 10 & 45 \\ -3 & -2.249 & 7 & 1.751 \\ 5 & 1 & 3 & 9 \end{array}\right]$$

Divide Row 1 by 20 and then multiply it by $-3$, that is, multiply Row 1 by $-3/20 = -0.15$.

$$([20 \quad 15 \quad 10 \quad | \quad 45]\ ) \times -0.15 \text{ gives Row 1 as}$$

$$[-3 \quad -2.25 \quad -1.5 \quad | \quad -6.75]$$

Subtract the result from Row 2

$$\begin{array}{rrrcr} [-3 & -2.249 & 7 & | & 1.751] \\ -[-3 & -2.25 & -1.5 & | & -6.75] \\ \hline 0 & 0.001 & 8.5 & & 8.501 \end{array}$$

to get the resulting equations as

$$\begin{bmatrix} 20 & 15 & 10 & | & 45 \\ 0 & 0.001 & 8.5 & | & 8.501 \\ 5 & 1 & 3 & | & 9 \end{bmatrix}$$

Divide Row 1 by 20 and then multiply it by 5, that is, multiply Row 1 by $5/20 = 0.25$.

$$[20 \quad 15 \quad 10 \quad | \quad 45] \times 0.25 \text{ gives Row 1 as}$$

$$[5 \quad 3.75 \quad 2.5 \quad | \quad 11.25]$$

Subtract the result from Row 3

$$\begin{array}{rrrrr} [5 & 1 & 3 & | & 9] \\ -[5 & 3.75 & 2.5 & | & 11.25] \\ \hline 0 & -2.75 & 0.5 & & -2.25 \end{array}$$

to get the resulting equations as

$$\begin{bmatrix} 20 & 15 & 10 & | & 45 \\ 0 & 0.001 & 8.5 & | & 8.501 \\ 0 & -2.75 & 0.5 & | & -2.25 \end{bmatrix}$$

This marks the end of the first step of the forward elimination part.

Now for the second step of the forward elimination part, the absolute value of the second column elements below Row 1 is

$$|0.001|, |-2.75|$$

or

$$0.001, \ 2.75$$

So, the largest absolute value is in Row 3. So, Row 2 is switched with Row 3 to give

$$\begin{bmatrix} 20 & 15 & 10 & | & 45 \\ 0 & -2.75 & 0.5 & | & -2.25 \\ 0 & 0.001 & 8.5 & | & 8.501 \end{bmatrix}$$

Divide Row 2 by $-2.75$ and then multiply it by 0.001, that is, multiply Row 2 by $0.001/-2.75 = -0.00036363$.

$$[0 \quad -2.75 \quad 0.5 \quad | \quad -2.25] \times -0.00036363 \text{ gives Row 2 as}$$

$$[0 \quad 0.00099998 \quad -0.00018182 \quad | \quad 0.00081816]$$

Subtract the result from Row 3

$$\begin{array}{llll} [0 & 0.001 & 8.5 & | \quad 8.501] \\ -[0 & 0.00099998 & -0.00018182 & | \quad 0.00081816] \\ \hline 0 & 0 & 8.50018182 & \quad 8.50018184 \end{array}$$

Rewriting within 5 significant digits with chopping

$$[0 \quad 0 \quad 8.5001 \quad | \quad 8.5001]$$

to get the resulting equations as

$$\begin{bmatrix} 20 & 15 & 10 & | & 45 \\ 0 & -2.75 & 0.5 & | & -2.25 \\ 0 & 0 & 8.5001 & | & 8.5001 \end{bmatrix}$$

Back substitution

$$8.5001x_3 = 8.5001$$

$$\begin{aligned} x_3 &= \frac{8.5001}{8.5001} \\ &= 1 \end{aligned}$$

Substituting the value of $x_3$ in Row 2

$$-2.75x_2 + 0.5x_3 = -2.25$$

$$\begin{aligned} x_2 &= \frac{-2.25 - 0.5x_3}{-2.75} \\ &= \frac{-2.25 - 0.5 \times 1}{-2.75} \\ &= \frac{-2.25 - 0.5}{-2.75} \\ &= \frac{-2.75}{-2.75} \\ &= 1 \end{aligned}$$

Substituting the value of $x_3$ and $x_2$ in Row 1

$$20x_1 + 15x_2 + 10x_3 = 45$$

$$
\begin{aligned}
x_1 &= \frac{45 - 15x_2 - 10x_3}{20} \\
&= \frac{45 - 15 \times 1 - 10 \times 1}{20} \\
&= \frac{45 - 15 - 10}{20} \\
&= \frac{30 - 10}{20} \\
&= \frac{20}{20} \\
&= 1
\end{aligned}
$$

So, the solution is

$$
\begin{aligned}
[X] &= \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \\
&= \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}
\end{aligned}
$$

This, in fact, is the exact solution. By coincidence only, in this case, the round-off error is not only reduced but fully removed.

**Example 2**

The upward velocity of a rocket is given at three different times in Table 1.

**Table 1**: Velocity vs. time data.

| Time, $t$ (s) | Velocity, $v$ (m/s) |
|---|---|
| 5 | 106.8 |
| 8 | 177.2 |
| 12 | 279.2 |

The velocity data is approximated by a polynomial as

$$
v(t) = a_1 t^2 + a_2 t + a_3, 5 \leq t \leq 12
$$

The coefficients $a_1, a_2$, and $a_3$ for the above expression are given by

$$
\begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} 106.8 \\ 177.2 \\ 279.2 \end{bmatrix}
$$

Find the values of $a_1, a_2$, and $a_3$ using the Gaussian elimination method with partial pivoting showing 6 significant digits. Find the velocity at $t = 6,\ 7.5,\ 9,\ 11$ seconds.

**Solution**

The augmented matrix is

$$\left[\begin{array}{ccc|c} 25 & 5 & 1 & 106.8 \\ 64 & 8 & 1 & 177.2 \\ 144 & 12 & 1 & 279.2 \end{array}\right]$$

Forward Elimination of unknowns

For the first step of the forward elimination part, the absolute value of the first column elements is

$$|25|,\ |64|,\ |144|$$

Or

$$25,\ 64,\ 144$$

So, the largest absolute value is in the Row 3. As per the Gaussian elimination with partial pivoting, we switch Row 3 by Row 1 to give

$$\left[\begin{array}{ccc|c} 144 & 12 & 1 & 279.2 \\ 64 & 8 & 1 & 177.2 \\ 25 & 5 & 1 & 106.8 \end{array}\right]$$

Divide Row 1 by 144 and then multiply it be 64, that is, multiply Row 1 by $64/144 = 0.444444$

$([144 \quad 12 \quad 1 \quad | \quad 279.2]) \times 0.444444$ gives Row 1 as

$$[64 \quad 5.33333 \quad 0.444444 \quad | \quad 124.089]$$

Subtract the result from Row 2

$$\begin{array}{rrrcr} [64 & 8 & 1 & | & 177.2] \\ -[64 & 5.33333 & 0.444444 & | & 124.089] \\ \hline 0 & 2.66667 & 0.555556 & & 53.1111 \end{array}$$

To get the resulting equation as

$$\left[\begin{array}{ccc|c} 144 & 12 & 1 & 279.2 \\ 0 & 2.66667 & 0.555556 & 53.1111 \\ 25 & 5 & 1 & 106.8 \end{array}\right]$$

Divide Row 1 by 144 and then multiply it by 25, that is, multiply Row 1 by $25/144 = 0.173611$.

$([144 \quad 12 \quad 1 \quad | \quad 279.2]) \times 0.173611$ gives Row 1 as

$$[25 \quad 2.08333 \quad 0.172611 \quad | \quad 48.4722]$$

Subtract the result from Row 3

$$\begin{array}{r} [25 \quad 5 \quad\quad 1 \quad\quad | \quad 106.8] \\ -[25 \quad 2.08333 \quad 0.173611 \quad | \quad 48.4722] \\ \hline 0 \quad 2.91667 \quad 0.826389 \quad\quad 58.3278 \end{array}$$

to get the resulting equations as

$$\begin{bmatrix} 144 & 12 & 1 & | & 279.2 \\ 0 & 2.66667 & 0.555556 & | & 53.1111 \\ 0 & 2.91667 & 0.826389 & | & 58.3278 \end{bmatrix}$$

This marks the end of the first step of forward elimination part.

Now for the second step of forward elimination part, the absolute value of the second column elements Row 2 and below are

$$|2.66667|, \ |2.91667|$$

Or

$$2.66667, \ 2.91667$$

The largest value is in Row 3. So, Row 2 is switched with Row 3 to give

$$\begin{bmatrix} 144 & 12 & 1 & | & 279.2 \\ 0 & 2.91667 & 0.826389 & | & 58.3278 \\ 0 & 2.66667 & 0.555556 & | & 53.1111 \end{bmatrix}$$

Divide Row 2 by 2.91667, then multiply it by 2.66667, that is, multiply Row 2 by $2.66667/2.91667 = 0.914286$.

$([0 \quad 2.91667 \quad 0.826389 \quad | \quad 58.3278]) \times 0.914286$ gives Row 2 as

$$[0 \quad 2.66667 \quad 0.755556 \quad | \quad 53.3283]$$

Subtract the result from Row 3

$$\begin{array}{r} [0 \quad\quad 2.66667 \quad\quad 0.555556 \quad\quad | \quad 53.1111] \\ -[0 \quad\quad 2.66667 \quad\quad 0.755556 \quad\quad | \quad 53.3283] \\ \hline 0 \quad\quad 0 \quad\quad -0.2 \quad\quad -0.217189 \end{array}$$

To get the resulting equations as

$$\left[\begin{array}{ccc|c} 144 & 12 & 1 & 279.2 \\ 0 & 2.91667 & 0.826389 & 58.3278 \\ 0 & 0 & -0.2 & -0.217189 \end{array}\right]$$

Back Substitution

$$-0.2a_3 = -0.217189$$

$$\begin{aligned} a_3 &= \frac{-0.217189}{-0.2} \\ a_3 &= 1.08595 \end{aligned}$$

Substituting the value of $a_3$ in Row 2

$$2.91667a_2 + 0.826389a_3 = 58.3278$$

$$\begin{aligned} a_2 &= \frac{58.3278 - 0.826389 \times a_3}{2.91667} \\ &= \frac{58.3278 - 0.826389 \times 1.08595}{2.91667} \\ &= \frac{57.4304}{2.91667} \\ &= 19.6903 \end{aligned}$$

Substituting the value of $a_3$ and $a_2$ in Row 1

$$144a_1 + 12a_2 + 1a_3 = 279.2$$

$$\begin{aligned} a_1 &= \frac{279.2 - 12a_2 - 1a_3}{144} \\ &= \frac{279.2 - 12 \times 19.6903 - 1.08595}{144} \\ &= \frac{41.8305}{144} \\ &= 0.290489 \end{aligned}$$

Hence, the solution vector is

$$\begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} 0.290489 \\ 19.6903 \\ 1.08595 \end{bmatrix}$$

The polynomial that passes through the three data points is then

$$\begin{aligned} v(t) &= a_1 t^2 + a_2 t + a_3 \\ &= 0.290489t^2 + 19.6903t + 1.08595, \quad 5 \le t \le 12 \end{aligned}$$

Since we want to find the velocity at $t = 6,\ 7.5,\ 9$ *and* 11 seconds, we could simply substitute each value of $t$ in $v(t) = 0.290489t^2 + 19.6903t + 1.08595$ and find the corresponding velocity. For example, at $t = 6$

$$\begin{aligned} v(6) &= 0.290489(6)^2 + 19.6903(6) + 1.08595 \\ &= 129.685 \text{ m/s} \end{aligned}$$

However, we could also find all the needed values of velocity at $t = 6,\ 7.5,\ 9,\ 11$ seconds using matrix multiplication.

$$v(t) = \begin{bmatrix} 0.290489 & 19.6903 & 1.08595 \end{bmatrix} \begin{bmatrix} t^2 \\ t \\ 1 \end{bmatrix}$$

So, if we want to find $v(6),\ v(7.5),\ v(9),\ v(11)$, it is given by

$$\begin{aligned} [v(6)\, v(7.5)\, v(9)\, v(11)] &= \begin{bmatrix} 0.290489 & 19.6903 & 1.08595 \end{bmatrix} \begin{bmatrix} 6^2 & 7.5^2 & 9^2 & 11^2 \\ 6 & 7.5 & 9 & 11 \\ 1 & 1 & 1 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 0.290489 & 19.6903 & 1.08595 \end{bmatrix} \begin{bmatrix} 36 & 56.25 & 81 & 121 \\ 6 & 7.5 & 9 & 11 \\ 1 & 1 & 1 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 129.685 & 165.103 & 201.828 & 252.828 \end{bmatrix} \end{aligned}$$

$$v(6) = 129.685 \text{ m/s}$$

$$v(7.5) = 165.103 \text{ m/s}$$

$$v(9) = 201.828 \text{ m/s}$$

$$v(11) = 252.828 \text{ m/s}$$

# Multiple Choice Test

## Chapter 04.06 Gaussian Elimination Method for Solving Simultaneous Linear Equations

(1). The goal of the forward elimination part in the Naive Gauss elimination method is to reduce the coefficient matrix to a (an) _____ matrix.

(A) diagonal

(B) identity

(C) lower triangular

(D) upper triangular

(2). Division by zero during the forward elimination part in Naive Gaussian elimination of the set of equations $[A][X] = [C]$ implies the coefficient matrix $[A]$

(A) is invertible

(B) is nonsingular

(C) may be singular or nonsingular

(D) is singular

(3). Using a computer with four significant digits with chopping, the Naive Gauss elimination solution to

$$0.0030x_1 + 55.23x_2 = 58.12$$
$$6.239x_1 - 7.123x_2 = 47.23$$

is

(A) $x_1 = 26.66;\ x_2 = 1.051$

(B) $x_1 = 8.769;\ x_2 = 1.051$

(C) $x_1 = 8.800;\ x_2 = 1.000$

(D) $x_1 = 8.771;\ x_2 = 1.052$

(4). Using a computer with four significant digits with chopping, the Gaussian elimination with partial pivoting method solution to

$$0.0030x_1 + 55.23x_2 = 58.12$$
$$6.239x_1 - 7.123x_2 = 47.23$$

is

(A) $x_1 = 26.66;\ x_2 = 1.051$

(B) $x_1 = 8.769;\ x_2 = 1.051$

(C) $x_1 = 8.800;\ x_2 = 1.000$

(D) $x_1 = 8.771;\ x_2 = 1.052$

(5). At the end of the forward elimination part of the Naive Gauss elimination method on the following equations

$$\begin{bmatrix} 4.2857 \times 10^7 & -9.2307 \times 10^5 & 0 & 0 \\ 4.2857 \times 10^7 & -5.4619 \times 10^5 & -4.2857 \times 10^7 & 5.4619 \times 10^5 \\ -6.5 & -0.15384 & 6.5 & 0.15384 \\ 0 & 0 & 4.2857 \times 10^7 & -3.6057 \times 10^5 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{bmatrix} = \begin{bmatrix} -7.887 \times 10^3 \\ 0 \\ 0.007 \\ 0 \end{bmatrix}$$

the resulting equations in matrix form are given by

$$\begin{bmatrix} 4.2857 \times 10^7 & -9.2307 \times 10^5 & 0 & 0 \\ 0 & 3.7688 \times 10^5 & -4.2857 \times 10^7 & 5.4619 \times 10^5 \\ 0 & 0 & -26.9140 & 0.579684 \\ 0 & 0 & 0 & 5.62500 \times 10^5 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{bmatrix} = \begin{bmatrix} -7.887 \times 10^3 \\ 7.887 \times 10^3 \\ 1.19530 \times 10^{-2} \\ 1.90336 \times 10^4 \end{bmatrix}$$

The determinant of the original coefficient matrix is

(A) 0.00

(B) $4.2857 \times 10^7$

(C) $5.486 \times 10^{19}$

(D) $-2.445 \times 10^{20}$

(6). The following data is given for the velocity of the rocket as a function of time. To find the velocity at $t = 21s$, you are asked to use a quadratic polynomial, $v(t) = at^2 + bt + c$, to approximate the velocity profile.

| $t$ | $(s)$ | 0 | 14 | 15 | 20 | 30 | 35 |
|---|---|---|---|---|---|---|---|
| $v(t)$ | $(m/s)$ | 0 | 227.04 | 362.78 | 517.35 | 602.97 | 901.67 |

The correct set of equations that will find $a$, $b$ and $c$ are

(A) $\begin{bmatrix} 176 & 14 & 1 \\ 225 & 15 & 1 \\ 400 & 20 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 227.04 \\ 362.78 \\ 517.35 \end{bmatrix}$

(B) $\begin{bmatrix} 225 & 15 & 1 \\ 400 & 20 & 1 \\ 900 & 30 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 362.78 \\ 517.35 \\ 602.97 \end{bmatrix}$

(C) $\begin{bmatrix} 0 & 0 & 1 \\ 225 & 15 & 1 \\ 400 & 20 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 0 \\ 362.78 \\ 517.35 \end{bmatrix}$

(D) $\begin{bmatrix} 400 & 20 & 1 \\ 900 & 30 & 1 \\ 1225 & 35 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 517.35 \\ 602.97 \\ 901.67 \end{bmatrix}$

For complete solution, go to

http://nm.mathforcollege.com/mcquizzes/04sle/quiz_04sle_gaussianelimination_solution.pdf

# Problem Set

## Chapter 04.06: Gaussian Elimination Method for Solving Simultaneous Linear Equations

(1). Use Naive Gauss elimination to solve

$$4x_1 + x_2 - x_3 = -2$$
$$5x_1 + x_2 + 2x_3 = 4$$
$$6x_1 + x_2 + x_3 = 6$$

**Answer:** $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 3 \\ -13 \\ 1 \end{bmatrix}$

(2). Assume that you are using a computer with four significant digits with chopping. Use the Naive Gauss elimination method to solve

$$4x_1 + x_2 - x_3 = -2$$
$$5x_1 + x_2 + 2x_3 = 4$$
$$6x_1 + x_2 + x_3 = 6$$

**Answer:** $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 3.000 \\ -13.00 \\ 1.000 \end{bmatrix}$

(3). For

$$[A] = \begin{bmatrix} 10 & -7 & 0 \\ -3 & 2.099 & 6 \\ 5 & -1 & 5 \end{bmatrix}$$

Find the determinant of $[A]$ using the forward elimination part of Naive Gauss elimination method.

**Answer:** $-150.05$

(4). At the end of the forward elimination part using the Naive Gauss elimination method on the coefficient matrix

$$[A] = \begin{bmatrix} 25 & c & 1 \\ 64 & a & 1 \\ 144 & b & 1 \end{bmatrix}$$

$[A]$ reduces to

$$[B] = \begin{bmatrix} 25 & 5 & 1 \\ 0 & -4.8 & -1.56 \\ 0 & 0 & 0.7 \end{bmatrix}$$

What is the determinant of $[A]$?

**Answer:** $-84$

(5). Use Gaussian elimination with partial pivoting method to solve

$$4x_1 + x_2 - x_3 = -2$$
$$5x_1 + x_2 + 2x_3 = 4$$
$$6x_1 + x_2 + x_3 = 6$$

**Answer:** $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 3 \\ -13 \\ 1 \end{bmatrix}$

(6). Assume that you are using a computer with four significant digits with chopping, use Gaussian elimination with partial pivoting method to solve

$$4x_1 + x_2 - x_3 = -2$$
$$5x_1 + x_2 + 2x_3 = 4$$
$$6x_1 + x_2 + x_3 = 6$$

**Answer:** $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2.998 \\ -12.99 \\ 1.000 \end{bmatrix}$

# Chapter 04.07: LU Decomposition Method for Solving Simultaneous Linear Equations

Lesson: Theory of LU Decomposition Method

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *solve a set of simultaneous linear equations using the LU decomposition method*

2) *decompose a nonsingular square matrix into the LU form.*

## I hear about LU decomposition used as a method to solve a set of simultaneous linear equations. What is it?

We already studied two numerical methods of finding the solution to simultaneous linear equations – Naive Gauss elimination and Gaussian elimination with partial pivoting. Then, why do we need to learn yet another method? To appreciate why LU decomposition could be a better choice than the Gauss elimination techniques in some cases, let us first discuss what LU decomposition is about.

For a nonsingular square matrix $[A]$ on which one can successfully conduct the Naive Gauss elimination forward elimination part, one can write it as

$$[A] = [L]\,[U]$$

where

$$[L] = \text{Lower triangular matrix}$$

$$[U] = \text{Upper triangular matrix}$$

Then if one is solving a set of equations

$$[A]\,[X] = [C]\,,$$

then

$$[L]\,[U]\,[X] = [C] \text{ as } ([A] = [L]\,[U])$$

Multiplying both sides by $[L]^{-1}$,

$$[L]^{-1}[L][U][X] = [L]^{-1}[C]$$

$$[I][U][X] = [L]^{-1}[C] \text{ as } \left([L]^{-1}[L] = [I]\right)$$

$$[U][X] = [L]^{-1}[C] \text{ as } ([I]\ [U] = [U])$$

Let

$$[L]^{-1}[C] = [Z]$$

then

$$[L][Z] = [C] \qquad (1)$$

and

$$[U][X] = [Z] \qquad (2)$$

So we can solve Equation (1) first for $[Z]$ by using forward substitution and then use Equation (2) to calculate the solution vector $[X]$ by back substitution.

## So how do I decompose a nonsingular matrix [A], that is, how do I find [A] = [L][U]?

If the forward elimination part of the Naive Gauss elimination methods can be applied on a nonsingular matrix $[A]$, then $[A]$ can be decomposed into $LU$ form as

$$[A] = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \cdots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & \dots & 0 \\ l_{21} & 1 & \cdots & 0 \\ \vdots & \vdots & \cdots & \vdots \\ l_{n1} & l_{n2} & \cdots & 1 \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} & \dots & u_{1n} \\ 0 & u_{22} & \cdots & u_{2n} \\ \vdots & \vdots & \cdots & \vdots \\ 0 & 0 & \cdots & u_{nn} \end{bmatrix}$$

The elements of the $[U]$ matrix are exactly the same as the coefficient matrix one obtains at the end of the forward elimination part in Naive Gauss elimination.

The lower triangular matrix $[L]$ has 1 in its diagonal entries. The non-zero elements on the non-diagonal elements in $[L]$ are multipliers that made the corresponding entries zero in the upper triangular matrix $[U]$ during the forward elimination.

# Chapter 04.07: LU Decomposition Method for Solving Simultaneous Linear Equations

Lesson: Application of LU Decomposition Method

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *decompose a nonsingular matrix into the LU form.*

*2) solve a set of simultaneous linear equations using the LU decomposition method*

## Applications

So far, you now know how the LU decomposition method is used to solve simultaneous linear equations. We also learned how to decompose the coefficient matrix to use forward substitution and back substitution parts to solve a set of simultaneous linear equations..

In this lesson, we apply what we learned in the previous lesson.

### Example 1

Find the LU decomposition of the matrix

$$[A] = \begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{bmatrix}$$

### Solution

$$\begin{aligned} [A] &= [L]\,[U] \\ &= \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix} \end{aligned}$$

The $[U]$ matrix is the same as found at the end of the forward elimination part of the Naive Gauss elimination method, that is

$$[U] = \begin{bmatrix} 25 & 5 & 1 \\ 0 & -4.8 & -1.56 \\ 0 & 0 & 0.7 \end{bmatrix}$$

To find $l_{21}$ and $l_{31}$, find the multipliers that were used to make the $a_{21}$ and $a_{31}$ elements zero in the first step of the forward elimination part of the Naive Gauss elimination method. They were

$$\begin{aligned} l_{21} &= \frac{64}{25} \\ &= 2.56 \end{aligned}$$

$$\begin{aligned} l_{31} &= \frac{144}{25} \\ &= 5.76 \end{aligned}$$

To find $l_{32}$, what multiplier was used to make $a_{32}$ element zero? Remember $a_{32}$ element was made zero in the second step of the forward elimination part. The $[A]$ matrix at the beginning of the second step of the forward elimination part was

$$\begin{bmatrix} 25 & 5 & 1 \\ 0 & -4.8 & -1.56 \\ 0 & -16.8 & -4.76 \end{bmatrix}$$

So

$$\begin{aligned} l_{32} &= \frac{-16.8}{-4.8} \\ &= 3.5 \end{aligned}$$

Hence

$$[L] = \begin{bmatrix} 1 & 0 & 0 \\ 2.56 & 1 & 0 \\ 5.76 & 3.5 & 1 \end{bmatrix}$$

Confirm $[L][U] = [A]$.

$$\begin{aligned} [L][U] &= \begin{bmatrix} 1 & 0 & 0 \\ 2.56 & 1 & 0 \\ 5.76 & 3.5 & 1 \end{bmatrix} \begin{bmatrix} 25 & 5 & 1 \\ 0 & -4.8 & -1.56 \\ 0 & 0 & 0.7 \end{bmatrix} \\ &= \begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{bmatrix} \end{aligned}$$

**Example 2**

Use the LU decomposition method to solve the following simultaneous linear equations.

$$\begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} 106.8 \\ 177.2 \\ 279.2 \end{bmatrix}$$

**Solution**

Recall that

$$[A]\,[X] = [C]$$

and if

$$[A] = [L]\,[U]$$

then first solving

$$[L]\,[Z] = [C]$$

and then

$$[U]\,[X] = [Z]$$

gives the solution vector $[X]$.

Now in Example 1, we showed

$$\begin{aligned} [A] &= [L]\,[U] \\ &= \begin{bmatrix} 1 & 0 & 0 \\ 2.56 & 1 & 0 \\ 5.76 & 3.5 & 1 \end{bmatrix} \begin{bmatrix} 25 & 5 & 1 \\ 0 & -4.8 & -1.56 \\ 0 & 0 & 0.7 \end{bmatrix} \end{aligned}$$

First, solve

$$[L]\ \ [Z] = [C]$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 2.56 & 1 & 0 \\ 5.76 & 3.5 & 1 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} = \begin{bmatrix} 106.8 \\ 177.2 \\ 279.2 \end{bmatrix}$$

to give

$$\begin{aligned} z_1 &= 106.8 \\ 2.56z_1 + z_2 &= 177.2 \\ 5.76z_1 + 3.5z_2 + z_3 &= 279.2 \end{aligned}$$

Forward substitution starting from the first equation gives

$$z_1 = 106.8$$

$$\begin{aligned} z_2 &= 177.2 - 2.56 z_1 \\ &= 177.2 - 2.56 \times 106.8 \\ &= -96.208 \end{aligned}$$

$$\begin{aligned} z_3 &= 279.2 - 5.76 z_1 - 3.5 z_2 \\ &= 279.2 - 5.76 \times 106.8 - 3.5 \times (-96.208) \\ &= 0.76 \end{aligned}$$

Hence

$$\begin{aligned} [Z] &= \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} \\ &= \begin{bmatrix} 106.8 \\ -96.208 \\ 0.76 \end{bmatrix} \end{aligned}$$

This matrix is the same as the right-hand side obtained at the end of the forward elimination part of the Naive Gauss elimination method. Is this a coincidence?

Now solve

$$[U]\,[X] = [Z]$$

$$\begin{bmatrix} 25 & 5 & 1 \\ 0 & -4.8 & -1.56 \\ 0 & 0 & 0.7 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} 106.8 \\ -96.208 \\ 0.76 \end{bmatrix}$$

$$\begin{aligned} 25a_1 + 5a_2 + a_3 &= 106.8 \\ -4.8a_2 - 1.56a_3 &= -96.208 \\ 0.7a_3 &= 0.76 \end{aligned}$$

From the third equation

$$0.7a_3 = 0.76$$

$$\begin{aligned} a_3 &= \frac{0.76}{0.7} \\ &= 1.0857 \end{aligned}$$

Substituting the value of $a_3$ in the second equation,

$$-4.8a_2 - 1.56a_3 = -96.208$$

$$\begin{aligned} a_2 &= \frac{-96.208 + 1.56a_3}{-4.8} \\ &= \frac{-96.208 + 1.56 \times 1.0857}{-4.8} \\ &= 19.691 \end{aligned}$$

Substituting the value of $a_2$ and $a_3$ in the first equation,

$$25a_1 + 5a_2 + a_3 = 106.8$$

$$\begin{aligned} a_1 &= \frac{106.8 - 5a_2 - a_3}{25} \\ &= \frac{106.8 - 5 \times 19.691 - 1.0857}{25} \\ &= 0.29048 \end{aligned}$$

Hence the solution vector is

$$\begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} 0.29048 \\ 19.691 \\ 1.0857 \end{bmatrix}$$

# Chapter 04.07: LU Decomposition Method for Solving Simultaneous Linear Equations

Lesson: Finding Inverse of a Matrix Using LU Decomposition Method

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *find the inverse of a matrix using the LU decomposition method.*

## How do I find the inverse of a square matrix using LU decomposition?

A matrix $[B]$ is the inverse of $[A]$ if

$$[A][B] = [I] = [B][A]$$

How can we use LU decomposition to find the inverse of the matrix? Assume the first column of $[B]$ (the inverse of $[A]$) is

$$[b_{11}\ b_{12} \dots\ \dots b_{n1}]^T$$

Then from the above definition of an inverse and the definition of matrix multiplication

$$[A]\begin{bmatrix} b_{11} \\ b_{21} \\ \vdots \\ b_{n1} \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

Similarly, the second column of $[B]$ is given by

$$[A]\begin{bmatrix} b_{12} \\ b_{22} \\ \vdots \\ b_{n2} \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix}$$

Similarly, all columns of $[B]$ can be found by solving $n$ different sets of equations with the column of the right-hand side being the $n$ columns of the identity matrix.

In the previous lessons, we learned how to solve simultaneous linear equations using the LU decomposition method.

**Example 1**

Use LU decomposition to find the inverse of

$$[A] = \begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{bmatrix}$$

**Solution**

Knowing that from the previous lesson,

$$\begin{aligned} [A] &= [L]\,[U] \\ &= \begin{bmatrix} 1 & 0 & 0 \\ 2.56 & 1 & 0 \\ 5.76 & 3.5 & 1 \end{bmatrix} \begin{bmatrix} 25 & 5 & 1 \\ 0 & -4.8 & -1.56 \\ 0 & 0 & 0.7 \end{bmatrix} \end{aligned}$$

We can solve for the first column of $[B] = [A]^{-1}$ by solving for

$$\begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{bmatrix} \begin{bmatrix} b_{11} \\ b_{21} \\ b_{31} \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

First, solve

$$[L]\,[Z] = [C]\,,$$

that is

$$\begin{bmatrix} 1 & 0 & 0 \\ 2.56 & 1 & 0 \\ 5.76 & 3.5 & 1 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

to give

$$\begin{aligned} z_1 &= 1 \\ 2.56z_1 + z_2 &= 0 \\ 5.76z_1 + 3.5z_2 + z_3 &= 0 \end{aligned}$$

Forward substitution starting from the first equation gives

$$z_1 = 1$$

$$\begin{aligned} z_2 &= 0 - 2.56 z_1 \\ &= 0 - 2.56\,(1) \\ &= -2.56 \end{aligned}$$

$$\begin{aligned} z_3 &= 0 - 5.76 z_1 - 3.5 z_2 \\ &= 0 - 5.76\,(1) - 3.5\,(-2.56) \\ &= 3.2 \end{aligned}$$

Hence

$$\begin{aligned} [Z] &= \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} \\ &= \begin{bmatrix} 1 \\ -2.56 \\ 3.2 \end{bmatrix} \end{aligned}$$

Now solve

$$[U]\,[X] = [Z]$$

that is

$$\begin{bmatrix} 25 & 5 & 1 \\ 0 & -4.8 & -1.56 \\ 0 & 0 & 0.7 \end{bmatrix} \begin{bmatrix} b_{11} \\ b_{21} \\ b_{31} \end{bmatrix} = \begin{bmatrix} 1 \\ -2.56 \\ 3.2 \end{bmatrix}$$

$$\begin{aligned} 25 b_{11} + 5 b_{21} + b_{31} &= 1 \\ -4.8 b_{21} - 1.56 b_{31} &= -2.56 \\ 0.7 b_{31} &= 3.2 \end{aligned}$$

Backward substitution starting from the third equation gives

$$\begin{aligned} b_{31} &= \frac{3.2}{0.7} \\ &= 4.571 \end{aligned}$$

$$\begin{aligned} b_{21} &= \frac{-2.56 + 1.56 b_{31}}{-4.8} \\ &= \frac{-2.56 + 1.56(4.571)}{-4.8} \\ &= -0.9524 \end{aligned}$$

$$
\begin{aligned}
b_{11} &= \frac{1 - 5b_{21} - b_{31}}{25} \\
&= \frac{1 - 5(-0.9524) - 4.571}{25} \\
&= 0.04762
\end{aligned}
$$

Hence the first column of the inverse of $[A]$ is

$$
\begin{bmatrix} b_{11} \\ b_{21} \\ b_{31} \end{bmatrix} = \begin{bmatrix} 0.04762 \\ -0.9524 \\ 4.571 \end{bmatrix}
$$

Similarly, solving

$$
\begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{bmatrix} \begin{bmatrix} b_{12} \\ b_{22} \\ b_{32} \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \text{ gives } \begin{bmatrix} b_{12} \\ b_{22} \\ b_{32} \end{bmatrix} = \begin{bmatrix} -0.08333 \\ 1.417 \\ -5.000 \end{bmatrix}
$$

and solving

$$
\begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{bmatrix} \begin{bmatrix} b_{13} \\ b_{23} \\ b_{33} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \text{ gives } \begin{bmatrix} b_{13} \\ b_{23} \\ b_{33} \end{bmatrix} = \begin{bmatrix} 0.03571 \\ -0.4643 \\ 1.429 \end{bmatrix}
$$

Hence

$$
[A]^{-1} = \begin{bmatrix} 0.04762 & -0.08333 & 0.03571 \\ -0.9524 & 1.417 & -0.4643 \\ 4.571 & -5.000 & 1.429 \end{bmatrix}
$$

Can you confirm the following for the above example?

$$
[A]\ [A]^{-1} = [I] = [A]^{-1}\,[A]
$$

Did you also note that the decomposition of $[A]$ to $[L][U]$ needed to be done only once?

# Chapter 04.07: LU Decomposition Method for Solving Simultaneous Linear Equations

Lesson: Computational Efficiency

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *justify why using the LU decomposition method is more efficient than Gaussian elimination in some cases.*

## LU decomposition looks more complicated than Gaussian elimination. Do we use LU decomposition because it is computationally more efficient than Gaussian elimination to solve a set of n equations given by $[A][X] = [C]$?

Note: The formulas for the computational time in this lesson assume it takes 4 clock cycles each for an add, subtract, or multiply operation, and 16 clock cycles for a divide operation as is the case for a typical AMD®-K7 chip. <http://www.isi.edu/~draper/papers/mwscas07_kwon.pdf>. These clock cycle times taken by an arithmetic operation and the number of such operations allow us to find the computational time for each of the processes.

For a square matrix $[A]$ of $n \times n$ size, the computational time $CT|_{DE}$ to decompose the $[A]$ matrix to $[L][U]$ form is given by

$$CT|_{DE} = T\left(\frac{8n^3}{3} + 4n^2 - \frac{20n}{3}\right), \qquad (1)$$

where

$$T = \text{clock cycle time}$$

The computational time $CT|_{FS}$ for forward substitution: $[L]\,[Z] = [C]$ is given by

$$CT|_{FS} = T\left(4n^2 - 4n\right) \qquad (2)$$

The computational time $CT|_{BS}$ for back substitution: $[U]\,[X] = [Z]$ is given by

$$CT|_{BS} = T\left(4n^2 + 12n\right) \qquad (3)$$

So, the total computational time to solve a set of simultaneous linear equations by LU decomposition is

$$\begin{aligned} CT|_{LU} &= CT|_{DE} + CT|_{FS} + CT|_{BS} \\ &= T\left(\frac{8n^3}{3} + 4n^2 - \frac{20n}{3}\right) + T\left(4n^2 - 4n\right) + T\left(4n^2 + 12n\right) \\ &= T\left(\frac{8n^3}{3} + 12n^2 + \frac{4n}{3}\right) \qquad (4) \end{aligned}$$

Now let us look at the computational time taken by Naive Gaussian elimination. The computational time $CT|_{FE}$ for the forward elimination part

$$CT|_{FE} = T\left(\frac{8n^3}{3} + 8n^2 - \frac{32n}{3}\right), \qquad (5)$$

and the computational time $CT|_{BS}$ for the back substitution part is

$$CT|_{BS} = T\left(4n^2 + 12n\right) \qquad (6)$$

So, the total computational time $CT|_{GE}$ to solve a set of simultaneous linear equations by Gaussian Elimination is

$$\begin{aligned} CT|_{GE} &= CT|_{FE} + CT|_{BS} \\ &= T\left(\frac{8n^3}{3} + 8n^2 - \frac{32n}{3}\right) + T\left(4n^2 + 12n\right) \\ &= T\left(\frac{8n^3}{3} + 12n^2 + \frac{4n}{3}\right) \qquad (7) \end{aligned}$$

The computational times for Gaussian elimination and LU decomposition are identical.

## This has confused me further! Why should I learn LU decomposition method when it takes the same computational time as Gaussian elimination, and that too when the two methods are closely related in the procedure? Please convince me that LU decomposition has its place in solving linear equations!

We now know to convince you that the LU decomposition method has its place in the solution of simultaneous linear equations. Let us examine an example where the LU decomposition method would be computationally more efficient than Gaussian elimination.

Remember, in trying to find the inverse of the matrix $[A]$ in a previous lesson, the problem reduces to solving $n$ sets of equations with the $n$ columns of the identity matrix as the RHS vector. For calculations of each column of the inverse of the $[A]$ matrix, the coefficient matrix $[A]$ matrix in the set of equation $[A][X] = [C]$ does not change. So, if we use the LU decomposition method, the

decomposition $[A] = [L][U]$ needs to be done only once, the forward substitution (Equation 1) $n$ times, and the back substitution (Equation 2) $n$ times.

Therefore, the total computational time $CT|_{inverse\ LU}$ required to find the inverse of a matrix using LU decomposition is

$$\begin{aligned} CT|_{inverse\ LU} &= 1 \times CT|_{DE} + n \times CT|_{FS} + n \times CT|_{BS} \\ &= 1 \times T\left(\frac{8n^3}{3} + 4n^2 - \frac{20n}{3}\right) + n \times T\left(4n^2 - 4n\right) + n \times T\left(4n^2 + 12n\right) \\ &= T\left(\frac{32n^3}{3} + 12n^2 - \frac{20n}{3}\right) \qquad (8) \end{aligned}$$

In comparison, if the Naive Gaussian elimination method were used to find the inverse of a matrix, the forward elimination part, as well as the back substitution part will have to be done $n$ times. The total computational time $CT|_{inverse\ GE}$ required to find the inverse of a matrix by using Naive Gaussian elimination then is

$$\begin{aligned} CT|_{inverse\ GE} &= n \times CT|_{FE} + n \times CT|_{BS} \\ &= n \times T\left(\frac{8n^3}{3} + 8n^2 - \frac{32n}{3}\right) + n \times T\left(4n^2 + 12n\right) \\ &= T\left(\frac{8n^4}{3} + 12n^3 + \frac{4n^2}{3}\right) \qquad (9) \end{aligned}$$

Clearly, for large $n$, $CT|_{inverse\ GE} >> CT|_{inverse\ LU}$ as $CT|_{inverse\ GE}$ has the dominating terms of $n^4$ and $CT|_{inverse\ LU}$has the dominating terms of $n^3$. For large values of $n$, the Gaussian elimination method would take more computational time (approximately $n/4$ times – prove it) than the LU decomposition method. Typical values of the ratio of the computational time for different values of $n$ are given in Table 1.

**Table 1.** Comparing computational times of finding the inverse of a matrix using the LU decomposition and Naive Gaussian elimination methods.

| $n$ | 10 | 100 | 1000 | 10000 |
|---|---|---|---|---|
| $CT\|_{inverseGE}/CT\|_{inverseLU}$ | 3.28 | 25.83 | 250.8 | 2501 |

Are you convinced now that LU decomposition has its place in solving systems of equations when we have to solve for multiple right-hand side vectors while the coefficient matrix stays unchanged?

## Appendix

How much computational time does it take to conduct back substitution in the LU Decomposition method?

**Solution**

At the beginning of back substitution, the set of equations is of the form

$$[U]\,[X] = [B]$$

where

$$[U] = \text{upper triangular matrix, } n \times n,$$
$$[X] = \text{unknown vector, } n \times 1, \text{ and}$$

$$[B] = \text{right-hand side vector, } n \times 1.$$

The algorithm for finding the unknown vector is

$$x_n = \frac{b_n}{u_{nn}} \qquad (A.1)$$

$$x_i = \frac{b_i - \sum_{j=i+1}^{n} u_{ij} x_j}{u_{ii}},\ i = n-1,\ n-2,\ ...,\ 2,\ 1 \qquad (A.2)$$

**Hint 1**

For an arithmetic progression series summation

$$S = a_1 + a_2 + ... + a_{m-1} + a_m$$

$$\frac{m}{2}(a_1 + a_m)$$

or

$$\frac{m}{2}(2a_1 + (m-1)d)$$

where

$$d = a_{i+1} - a_i, i = 1, 2.., m-1$$

**Hint 2**

When you add $n$ terms together, there are $n-1$ additions.

Now let us see how many arithmetic operations are needed to complete the algorithm.

**Number of Divisions**

See equation (1). We have one operation of division for the calculation of $x_n$.

See equation (2). For each $x_i$, $i = n-1, n-2, ..., 2, 1$, the result is divided once by $u_{ii}$.

So, the total number of divisions is $1 + (n-1) = n$

**Number of Subtractions**

See equation (2). For each $x_i$, $i = n-1, n-2, ..., 2, 1$, we have 1 subtraction in the numerator.

So, the total number of subtractions is $n-1$.

**Number of Multiplications**

See equation (2). For a particular $i$, there are $(n-i)$ multiplications – just count the $u_{ij}x_j$ terms within the summation in the numerator. Since $i$ takes the values of $i = n-1, n-2, ..., 2, 1$, you can see that

when $i = n-1$, there is $(n-(n-1)) = 1$ multiplication,

when $i = n-2$, there are $(n-(n-2)) = 2$ multiplications,

$\vdots$

when $i = 1$, there are $(n-1) = n-1$ multiplications.

So, the total number of multiplications is

$$\begin{aligned} & 1 + 2 + ... + (n-1) \\ &= \frac{n-1}{2}(1 + (n-1)) \\ &= \frac{n(n-1)}{2} \end{aligned}$$

**Number of Additions**

For a particular $i$, there are $(n-i-1)$ additions – just see the number of $u_{ij}x_j$ terms within the summation in the numerator of equation (2).

Since $i$ takes the values of $i = n-1, n-2, ..., 2, 1$, you can see that

when $i = n-1$, there is $(n-(n-1)-1)) = 0$ addition,

when $i = n-2$, there is $(n-(n-2)-1)) = 1$ addition,

$\vdots$

when $i = 1$, there are $(n-(1)-1)) = n-2$ additions.

So, the total number of additions is

$$\begin{aligned} & 0 + 1 + 2 + ... + (n-2) \\ &= \frac{n-1}{2}(0 + (n-2)) \\ &= \frac{(n-1)(n-2)}{2} \end{aligned}$$

**Computational Time**

Assuming it takes 4 clock cycles for each addition, subtraction, and multiplication, and 16 clock cycles for each division, then if $C$ is the clock cycle time,

computational time spent on divisions= $n \times (16C) = 16Cn$

computational time spent on subtractions= $(n-1) \times (4C) = 4C(n-1)$

computational time spent on multiplications= $\frac{n(n-1)}{2} \times (4C) = 2Cn(n-1)$

computational time spent on additions= $\frac{(n-1)(n-2)}{2} \times (4C) = 2C(n-1)(n-2)$

The total computational time spent on back substitution then is

$$
\begin{aligned}
&= \text{Total time for divisions} + \text{Total time for subtractions} \\
&+ \text{Total time for multiplications} + \text{Total time for additions} \\
&= (16Cn) + (4Cn - 4C) + (2Cn^2 - 2Cn) + (2Cn^2 - 6Cn + 4C) \\
&= 4Cn^2 + 12Cn \\
&= C(4n^2 + 12n)
\end{aligned}
$$

**Questions Related to the Appendix**

1. How much computational time does it take to conduct forward substitution?

2. How much computational time does it take to conduct forward elimination?

3. How much computational time does it take to conduct back substitutions?

4. How much computational time does it take to conduct Naive Gaussian elimination?

5. How much computational time does it take to conduct LU decomposition?

6. How much computational time does it take to find the inverse of a square matrix using LU decomposition?

7. How much computational time does it take to find the inverse of a square matrix using Gaussian elimination?

# Multiple Choice Test

## Chapter 04.07 LU Decomposition Method for Solving Simultaneous Linear Equations

(1). The $[L][U]$ decomposition method is computationally more efficient than Naive Gauss elimination for solving

(A). a single set of simultaneous linear equations.

(B). multiple sets of simultaneous linear equations with different coefficient matrices and the same right-hand side vectors.

(C). multiple sets of simultaneous linear equations with the same coefficient matrix and different right-hand side vectors.

(D). less than ten simultaneous linear equations.

(2). The lower triangular matrix $[L]$ in the $[L][U]$ decomposition of the matrix given below

$$\begin{bmatrix} 25 & 5 & 4 \\ 10 & 8 & 16 \\ 8 & 12 & 22 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ \updownarrow_{21} & 1 & 0 \\ \updownarrow_{31} & \updownarrow_{32} & 1 \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$$

is

(A) $\begin{bmatrix} 1 & 0 & 0 \\ 0.40000 & 1 & 0 \\ 0.32000 & 1.7333 & 1 \end{bmatrix}$

(B) $\begin{bmatrix} 25 & 5 & 4 \\ 0 & 6 & 14.400 \\ 0 & 0 & -4.2400 \end{bmatrix}$

(C) $\begin{bmatrix} 1 & 0 & 0 \\ 10 & 1 & 0 \\ 8 & 12 & 0 \end{bmatrix}$

(D) $\begin{bmatrix} 1 & 0 & 0 \\ 0.40000 & 1 & 0 \\ 0.32000 & 1.5000 & 1 \end{bmatrix}$

(3). The upper triangular matrix $[U]$ in the $[L][U]$ decomposition of the matrix given below

$$\begin{bmatrix} 25 & 5 & 4 \\ 0 & 8 & 16 \\ 0 & 12 & 22 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ \updownarrow_{21} & 1 & 0 \\ \updownarrow_{31} & \updownarrow_{32} & 1 \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$$

is

(A) $\begin{bmatrix} 1 & 0 & 0 \\ 0.40000 & 1 & 0 \\ 0.32000 & 1.7333 & 1 \end{bmatrix}$

(B) $\begin{bmatrix} 25 & 5 & 4 \\ 0 & 6 & 14.400 \\ 0 & 0 & -4.2400 \end{bmatrix}$

(C) $\begin{bmatrix} 25 & 5 & 4 \\ 0 & 8 & 16 \\ 0 & 0 & -2 \end{bmatrix}$

(D) $\begin{bmatrix} 1 & 0.2000 & 0.16000 \\ 0 & 1 & 2.4000 \\ 0 & 0 & -4.240 \end{bmatrix}$

(4). For a given $2000 \times 2000$ matrix $[A]$, assume that it takes about 15 seconds to find the inverse of $[A]$ by the use of the $[L]\,[U]$ decomposition method, that is, finding the $[L]\,[U]$ once, and then doing forward substitution and back substitution 2000 times using the 2000 columns of the identity matrix as the right-hand side vector. The approximate time, in seconds, that it will take to find the inverse if found by repeated use of the Naive Gauss elimination method, that is, doing forward elimination and back substitution 2000 times by using the 2000 columns of the identity matrix as the right-hand side vector is most nearly

(A) 300

(B) 1500

(C) 7500

(D) 30000

(5). The algorithm for solving a set of $n$ equations $[A]\,[X] = [C]$, where $[A] = [L]\,[U]$ involves solving$[L]\,[Z] = [C]$ by forward substitution. The algorithm to solve $[L]\,[Z] = [C]$ is given by

(A) $z_1 = c_1/l_{11}$

for $i$ from 2 to $n$ do

sum = 0

for $j$ from 1 to $i$ do

sum = sum + $l_{\text{ij}} * z_j$

end do

$z_i = (c_i - \text{sum})/l_{\text{ii}}$

end do

(B) $z_1 = c_1/l_{11}$

for $i$ from 2 to $n$ do

sum = 0

for $j$ from 1 to $(i-1)$ do

sum = sum + $l_{ij} * z_j$

end do

$z_i = (c_i - \text{sum})/l_{ii}$

end do

(C) $z_1 = c_1/l_{11}$

for $i$ from 2 to $n$ do

for $j$ from 1 to $(i-1)$do

sum = sum + $l_{ij} * z_j$

end do

$z_i = (c_i - \text{sum})/l_{ii}$

end do

(D) for $i$ from 2 to $n$ do

sum = 0

for $j$ from 1 to $(i-1)$ do

sum = sum + $l_{ij} * z_j$

end do

$z_i = (c_i - \text{sum})/l_{ii}$

end do

(6). To solve boundary value problems, a numerical method based on the finite difference method is used. This results in simultaneous linear equations with tridiagonal coefficient matrices. These are solved using a specialized $[L]\,[U]$ decomposition method.

Choose the set of equations that approximately solves the boundary value problem

$$\frac{d^2y}{dx^2} = 6x - 0.5x^2, \quad y\,(0) = 0, \quad y\,(12) = 0, \quad 0 \le x \le 12$$

The second derivative in the above equation is approximated by the second-order accurate central divided difference approximation as learned in the numerical differentiation lessons (Chapter 02.02). A step size of $h = 4$ is used, and hence the value of $y$ can be found approximately at equidistantly placed 4 nodes between $x = 0$ and $x = 12$.

(A) $$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0.0625 & 0.125 & 0.0625 & 0 \\ 0 & 0.0625 & 0.125 & 0.0625 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 16.0 \\ 16.0 \\ 0 \end{bmatrix}$$

(B) $$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0.0625 & -0.125 & 0.0625 & 0 \\ 0 & 0.0625 & -0.125 & 0.0625 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 16.0 \\ 16.0 \\ 0 \end{bmatrix}$$

(C) $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0.0625 & -0.125 & 0.0625 & 0 \\ 0 & 0.0625 & -0.125 & 0.0625 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 16.0 \\ 16.0 \\ 0 \end{bmatrix}$

(D) $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0.0625 & 0.125 & 0.0625 & 0 \\ 0 & 0.0625 & 0.125 & 0.0625 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 16.0 \\ 16.0 \end{bmatrix}$

For complete solution, go to

http://nm.mathforcollege.com/mcquizzes/04sle/quiz_04sle_ludecomposition_solution.pdf

# Problem Set

Chapter 04.07: LU Decomposition Method for Solving Simultaneous Linear Equations

(1). Show that LU decomposition method is computationally a more efficient way of finding the inverse of a square matrix than using Gaussian elimination.

**Answer:** See Book

(2). Use LU decomposition to find [L] and [U]

$$4x_1 + x_2 - x_3 = -2$$
$$5x_1 + x_2 + 2x_3 = 4$$
$$6x_1 + x_2 + x_3 = 6$$

**Answer:** $[L][U] = \begin{bmatrix} 1 & 0 & 0 \\ 1.25 & 1 & 0 \\ 1.5 & 2 & 1 \end{bmatrix} \begin{bmatrix} 4 & 1 & -1 \\ 0 & -0.25 & 3.25 \\ 0 & 0 & -4 \end{bmatrix}$

(3). Find the inverse

$$[A] = \begin{bmatrix} 3 & 4 & 1 \\ 2 & -7 & -1 \\ 8 & 1 & 5 \end{bmatrix}$$

using the LU decomposition method.

**Answer:** $[A]^{-1} = \begin{bmatrix} 0.29310 & 0.169379 & -0.025862 \\ 0.1551 & -0.060345 & -0.043103 \\ -0.5 & -0.25 & 0.25 \end{bmatrix}$

(4). Fill in the blanks for the unknowns in the LU decomposition of the matrix given below

$$\begin{bmatrix} 25 & 5 & 4 \\ 75 & 7 & 16 \\ 12.5 & 12 & 22 \end{bmatrix} = \begin{bmatrix} \updownarrow_{11} & 0 & 0 \\ \updownarrow_{21} & \updownarrow_{22} & 0 \\ \updownarrow_{31} & \updownarrow_{32} & \updownarrow_{33} \end{bmatrix} \begin{bmatrix} 25 & 5 & 4 \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$$

**Answer:** $[L] = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 0.5 & -1.1875 & 1 \end{bmatrix}$ ; $[U] = \begin{bmatrix} 25 & 5 & 4 \\ 0 & -8 & 4 \\ 0 & 0 & 24.75 \end{bmatrix}$

(5). Show that the nonsingular matrix

$$[A] = \begin{bmatrix} 0 & 2 \\ 2 & 0 \end{bmatrix}$$

cannot be decomposed into the LU form.

**Answer:** Since the $a_{11} = 0$, the first step of the forward elimination part of Gaussian elimination will involve a division by zero.

(6). The LU decomposition of

$$[A] = \begin{bmatrix} 4 & 1 & -1 \\ 5 & 1 & 2 \\ 6 & 1 & 1 \end{bmatrix}$$

is given by

$$\begin{bmatrix} 4 & 1 & -1 \\ 5 & 1 & 2 \\ 6 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 1.25 & 1 & 0 \\ 1.5 & 2 & 1 \end{bmatrix} \begin{bmatrix} ?? & ?? & ?? \\ 0 & ?? & ?? \\ 0 & 0 & ?? \end{bmatrix}$$

Find the upper triangular matrix in the above decomposition?

**Answer:** $[U] = \begin{bmatrix} 4 & 1 & -1 \\ 0 & -0.25 & 3.25 \\ 0 & 0 & -4 \end{bmatrix}$

# Chapter 04.08: Gauss-Seidel Method of Solving Simultaneous Linear Equations

Gauss-Seidel Method

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *solve a set of equations using the Gauss-Seidel method,*

2) *recognize the advantages and pitfalls of the Gauss-Seidel method, and*

3) *determine under what conditions the Gauss-Seidel method always converges.*

## Why do we need another method to solve a set of simultaneous linear equations?

In certain cases, such as when a system of equations is large, iterative methods of solving equations are more advantageous. Elimination methods, such as Gaussian elimination, are prone to large round-off errors for a large set of equations. Iterative methods, such as the Gauss-Seidel method, give the user control of the round-off error. Also, if the physics of the problem are well known, initial guesses needed in iterative methods can be made more judiciously leading to faster convergence.

What is the algorithm for the Gauss-Seidel method? Given a general set of $n$ equations and $n$ unknowns, we have

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + ... + a_{1n}x_n = c_1$$

$$a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + ... + a_{2n}x_n = c_2$$

$$\vdots \qquad\qquad \vdots$$

$$a_{n1}x_1 + a_{n2}x_2 + a_{n3}x_3 + ... + a_{nn}x_n = c_n$$

If the diagonal elements are non-zero, each equation is rewritten for the corresponding unknown, that is, the first equation is rewritten with $x_1$ on the left hand side, the second equation is rewritten with $x_2$ on the left hand side and so on as follows

$$x_1 = \frac{c_1 - a_{12}x_2 - a_{13}x_3 \ldots\ldots - a_{1n}x_n}{a_{11}}$$

$$x_2 = \frac{c_2 - a_{21}x_1 - a_{23}x_3 \ldots\ldots - a_{2n}x_n}{a_{22}}$$

$$\vdots$$

$$\vdots$$

$$x_{n-1} = \frac{c_{n-1} - a_{n-1,1}x_1 - a_{n-1,2}x_2 \ldots\ldots - a_{n-1,n-2}x_{n-2} - a_{n-1,n}x_n}{a_{n-1,n-1}}$$

$$x_n = \frac{c_n - a_{n1}x_1 - a_{n2}x_2 - \ldots\ldots - a_{n,n-1}x_{n-1}}{a_{nn}}$$

These equations can be rewritten in a summation form as

$$x_1 = \frac{c_1 - \sum_{\substack{j=1 \\ j \neq 1}}^{n} a_{1j}x_j}{a_{11}}$$

$$x_2 = \frac{c_2 - \sum_{\substack{j=1 \\ j \neq 2}}^{n} a_{2j}x_j}{a_{22}}$$

$$\vdots$$

$$x_{n-1} = \frac{c_{n-1} - \sum_{\substack{j=1 \\ j \neq n-1}}^{n} a_{n-1,j}x_j}{a_{n-1,n-1}}$$

$$x_n = \frac{c_n - \sum_{\substack{j=1 \\ j \neq n}}^{n} a_{nj}x_j}{a_{nn}}$$

Hence for any row $i$,

$$x_i = \frac{c_i - \sum_{\substack{j=1 \\ j \neq i}}^{n} a_{ij}x_j}{a_{ii}}, i = 1, 2, \ldots, n.$$

Now to find $x_i$'s, one assumes an initial guess for the $x_i$'s and then uses the rewritten equations to calculate the new estimates. Remember, one always uses the most recent estimates to calculate the

next estimates, $x_i$. At the end of each iteration, one calculates the absolute relative approximate error for each $x_i$ as

$$|\in_a|_i = \left| \frac{x_i^{\text{new}} - x_i^{\text{old}}}{x_i^{\text{new}}} \right| \times 100$$

where $x_i^{\text{new}}$is the recently obtained value of $x_i$, and $x_i^{\text{old}}$ is the previous value of $x_i$.

When the absolute relative approximate error for each $x_i$ is less than the pre-specified tolerance, the iterations are stopped.

**Example 1**

The upward velocity of a rocket is given at three different times in the following table

**Table 1:** Velocity vs. time data.

| Time, $t$ $(s)$ | Velocity, $v$ $(m/s)$ |
|---|---|
| 5 | 106.8 |
| 8 | 177.2 |
| 12 | 279.2 |

The velocity data is approximated by a polynomial as

$$v(t) = a_1 t^2 + a_2 t + a_3, 5 \leq t \leq 12$$

Find the values of $a_1,\ a_2, and\ a_3$ using the Gauss-Seidel method. Assume an initial guess of the solution as

$$\begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 5 \end{bmatrix}$$

and conduct two iterations.

**Solution**

The polynomial is going through three data points $(t_1, v_1), (t_2, v_2), and\, (t_3, v_3)$ where from the above table

$$t_1 = 5, v_1 = 106.8$$

$$t_2 = 8, v_2 = 177.2$$

$$t_3 = 12, v_3 = 279.2$$

Requiring that $v(t) = a_1 t^2 + a_2 t + a_3$passes through the three data points gives

$$v(t_1) = v_1 = a_1 t_1^2 + a_2 t_1 + a_3$$

$$v(t_2) = v_2 = a_1 t_2^2 + a_2 t_2 + a_3$$

$$v(t_3) = v_3 = a_1 t_3^2 + a_2 t_3 + a_3$$

Substituting the data $(t_1, v_1), (t_2, v_2), and (t_3, v_3)$ gives

$$a_1\left(5^2\right) + a_2(5) + a_3 = 106.8$$

$$a_1\left(8^2\right) + a_2(8) + a_3 = 177.2$$

$$a_1\left(12^2\right) + a_2(12) + a_3 = 279.2$$

or

$$25a_1 + 5a_2 + a_3 = 106.8$$

$$64a_1 + 8a_2 + a_3 = 177.2$$

$$144a_1 + 12a_2 + a_3 = 279.2$$

The coefficients $a_1, a_2, and a_3$ for the above expression are given by

$$\begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} 106.8 \\ 177.2 \\ 279.2 \end{bmatrix}$$

Rewriting the equations gives

$$a_1 = \frac{106.8 - 5a_2 - a_3}{25}$$

$$a_2 = \frac{177.2 - 64a_1 - a_3}{8}$$

$$a_3 = \frac{279.2 - 144a_1 - 12a_2}{1}$$

Iteration 1

Given the initial guess of the solution vector as

$$\begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 5 \end{bmatrix}$$

we get

$$\begin{aligned} a_1 &= \frac{106.8 - 5(2) - (5)}{25} \\ &= 3.6720 \end{aligned}$$

$$\begin{aligned} a_2 &= \frac{177.2 - 64\,(3.6720) - (5)}{8} \\ &= -7.8150 \end{aligned}$$

$$\begin{aligned} a_3 &= \frac{279.2 - 144\,(3.6720) - 12\,(-7.8510)}{1} \\ &= -155.36 \end{aligned}$$

The absolute relative approximate error for each $x_i$ then is

$$\begin{aligned} |\in_a|_1 &= \left| \frac{3.6720 - 1}{3.6720} \right| \times 100 \\ &= 72.76\% \end{aligned}$$

$$\begin{aligned} |\in_a|_2 &= \left| \frac{-7.8510 - 2}{-7.8510} \right| \times 100 \\ &= 125.47\% \end{aligned}$$

$$\begin{aligned} |\in_a|_3 &= \left| \frac{-155.36 - 5}{-155.36} \right| \times 100 \\ &= 103.22\% \end{aligned}$$

At the end of the first iteration, the estimate of the solution vector is

$$\begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} 3.6720 \\ -7.8510 \\ -155.36 \end{bmatrix}$$

and the maximum absolute relative approximate error is 125.47.

Iteration 2

The estimate of the solution vector at the end of Iteration #1 is

$$\begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} 3.6720 \\ -7.8510 \\ -155.36 \end{bmatrix}$$

Now we get

$$\begin{aligned} a_1 &= \frac{106.8 - 5\,(-7.8510) - (-155.36)}{25} \\ &= 12.056 \end{aligned}$$

$$\begin{aligned} a_2 &= \frac{177.2 - 64\,(12.056) - (-155.36)}{8} \\ &= -54.882 \end{aligned}$$

$$\begin{aligned} a_3 &= \frac{279.2 - 144\,(12.056) - 12\,(-54.882)}{1} \\ &= -798.34 \end{aligned}$$

The absolute relative approximate error for each $x_i$ then is

$$\begin{aligned} |\in_a|_1 &= \left| \frac{12.056 - 3.6720}{12.056} \right| \times 100 \\ &= 69.543\% \end{aligned}$$

$$\begin{aligned} |\in_a|_2 &= \left| \frac{-54.882 - (-7.8510)}{-54.882} \right| \times 100 \\ &= 85.695\% \end{aligned}$$

$$\begin{aligned} |\in_a|_3 &= \left| \frac{-798.34 - (-155.36)}{-798.34} \right| \times 100 \\ &= 80.540\% \end{aligned}$$

At the end of the second iteration the estimate of the solution vector is

$$\begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} 12.056 \\ -54.882 \\ -798.54 \end{bmatrix}$$

and the maximum absolute relative approximate error is 85.695%.

Conducting more iterations gives the following values for the solution vector and the corresponding absolute relative approximate errors.

| Iteration | $a_1$ | $\lvert\epsilon_a\rvert_1\%$ | $a_2$ | $\lvert\epsilon_a\rvert_2\%$ | $a_3$ | $\lvert\epsilon_a\rvert_3\%$ |
|---|---|---|---|---|---|---|
| 1 | 3.6720 | 72.767 | −7.8510 | 125.47 | −155.36 | 103.22 |
| 2 | 12.056 | 69.543 | −54.882 | 85.695 | −798.34 | 80.540 |
| 3 | 47.182 | 74.447 | −255.51 | 78.521 | −3448.9 | 76.852 |
| 4 | 193.33 | 75.595 | −1093.4 | 76.632 | −14440 | 76.116 |
| 5 | 800.53 | 75.850 | −4577.2 | 76.112 | −60072 | 75.963 |
| 6 | 3322.6 | 75.906 | −19049 | 75.972 | −249580 | 75.931 |

As seen in the above table, the solution estimates are not converging to the true solution of

$$a_1 = 0.29048$$

$$a_2 = 19.690$$

$$a_3 = 1.0857$$

## The above system of equations does not seem to converge. Why?

Well, a pitfall of most iterative methods is that they may or may not converge. However, the solution to a certain class of system of simultaneous equations does always converge using the Gauss-Seidel method. This class of system of equations is where the coefficient matrix $[A]$ in $[A][X] = [C]$ is diagonally dominant, that is

$$|a_{ii}| \geq \sum_{\substack{j=1 \\ j \neq i}}^{n} |a_{ij}| \text{ for all } i$$

$$|a_{ii}| > \sum_{\substack{j=1 \\ j \neq i}}^{n} |a_{ij}| \text{ for at least one } i$$

If a system of equations has a coefficient matrix that is not diagonally dominant, it may or may not converge. Fortunately, many physical systems that result in simultaneous linear equations have a diagonally dominant coefficient matrix, which then assures convergence for iterative methods such as the Gauss-Seidel method of solving simultaneous linear equations.

### Example 2

Find the solution to the following system of equations using the Gauss-Seidel method.

$$12x_1 + 3x_2 - 5x_3 = 1$$

$$x_1 + 5x_2 + 3x_3 = 28$$

$$3x_1 + 7x_2 + 13x_3 = 76$$

Use

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

as the initial guess and conduct two iterations.

**Solution1**

The coefficient matrix

$$[A] = \begin{bmatrix} 12 & 3 & -5 \\ 1 & 5 & 3 \\ 3 & 7 & 13 \end{bmatrix}$$

is diagonally dominant as

$$|a_{11}| = |12| = 12 \geq |a_{12}| + |a_{13}| = |3| + |-5| = 8$$

$$|a_{22}| = |5| = 5 \geq |a_{21}| + |a_{23}| = |1| + |3| = 4$$

$$|a_{33}| = |13| = 13 \geq |a_{31}| + |a_{32}| = |3| + |7| = 10$$

and the inequality is strictly greater than for at least one row. Hence, the solution should converge using the Gauss-Seidel method.

Rewriting the equations, we get

$$x_1 = \frac{1 - 3x_2 + 5x_3}{12}$$

$$x_2 = \frac{28 - x_1 - 3x_3}{5}$$

$$x_3 = \frac{76 - 3x_1 - 7x_2}{13}$$

Assuming an initial guess of

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

Iteration 1

$$x_1 = \frac{1 - 3\,(0) + 5\,(1)}{12}$$
$$= 0.50000$$

$$x_2 = \frac{28 - (0.50000) - 3\,(1)}{5}$$
$$= 4.9000$$

$$x_3 = \frac{76 - 3\,(0.50000) - 7\,(4.9000)}{13}$$
$$= 3.0923$$

The absolute relative approximate error at the end of the first iteration is

$$|\in_a|_1 = \left|\frac{0.50000 - 1}{0.50000}\right| \times 100$$
$$= 100.00\%$$

$$|\in_a|_2 = \left|\frac{4.9000 - 0}{4.9000}\right| \times 100$$
$$= 100.00\%$$

$$|\in_a|_3 = \left|\frac{3.0923 - 1}{3.0923}\right| \times 100$$
$$= 67.662\%$$

The maximum absolute relative approximate error is 100.00%

Iteration 2

$$x_1 = \frac{1 - 3\,(4.9000) + 5\,(3.0923)}{12}$$
$$= 0.14679$$

$$x_2 = \frac{28 - (0.14679) - 3\,(3.0923)}{5}$$
$$= 3.7153$$

$$x_3 = \frac{76 - 3\,(0.14679) - 7\,(3.7153)}{13}$$
$$= 3.8118$$

At the end of second iteration, the absolute relative approximate error is

$$|\in_a|_1 = \left|\frac{0.14679 - 0.50000}{0.14679}\right| \times 100$$
$$= 240.61\%$$

$$|\in_a|_2 = \left|\frac{3.7153 - 4.9000}{3.7153}\right| \times 100$$
$$= 31.889\%$$

$$|\in_a|_3 = \left|\frac{3.8118 - 3.0923}{3.8118}\right| \times 100$$
$$= 18.874\%$$

The maximum absolute relative approximate error is 240.61%. This is greater than the value of 100.00% we obtained in the first iteration. Is the solution diverging? No, as you conduct more iterations, the solution converges as follows.

| Iteration | $x_1$ | $\lvert\epsilon_a\rvert_1\%$ | $x_2$ | $\lvert\epsilon_a\rvert_2\%$ | $x_3$ | $\lvert\epsilon_a\rvert_3\%$ |
|---|---|---|---|---|---|---|
| 1 | 0.50000 | 100.00 | 4.9000 | 100.00 | 3.0923 | 67.662 |
| 2 | 0.14679 | 240.61 | 3.7153 | 31.889 | 3.8118 | 18.874 |
| 3 | 0.74275 | 80.236 | 3.1644 | 17.408 | 3.9708 | 4.0064 |
| 4 | 0.94675 | 21.546 | 3.0281 | 4.4996 | 3.9971 | 0.65772 |
| 5 | 0.99177 | 4.5391 | 3.0034 | 0.82499 | 4.0001 | 0.074383 |
| 6 | 0.99919 | 0.74307 | 3.0001 | 0.10856 | 4.0001 | 0.00101 |

This is close to the exact solution vector of

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ 4 \end{bmatrix}$$

**Example 3**

Given the system of equations

$$3x_1 + 7x_2 + 13x_3 = 76$$

$$x_1 + 5x_2 + 3x_3 = 28$$

$$12x_1 + 3x_2 - 5x_3 = 1$$

find the solution using the Gauss-Seidel method. Use

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

as the initial guess.

**Solution1**

Rewriting the equations, we get

$$x_1 = \frac{76 - 7x_2 - 13x_3}{3}$$

$$x_2 = \frac{28 - x_1 - 3x_3}{5}$$

$$x_3 = \frac{1 - 12x_1 - 3x_2}{-5}$$

Assuming an initial guess of

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

the next six iterative values are given in the table below.

| Iteration | $a_1$ | $\lvert\epsilon_a\rvert_1\%$ | $a_2$ | $\lvert\epsilon_a\rvert_2\%$ | $a_3$ | $\lvert\epsilon_a\rvert_3\%$ |
|---|---|---|---|---|---|---|
| 1 | 21.000 | 95.238 | 0.80000 | 100.00 | 50.680 | 98.027 |
| 2 | $-196.15$ | 110.71 | 14.421 | 94.453 | $-462.30$ | 110.96 |
| 3 | 1995.0 | 109.83 | $-116.02$ | 112.43 | 4718.1 | 109.80 |
| 4 | $-20149$ | 109.90 | 1204.6 | 109.63 | $-47636$ | 109.90 |
| 5 | $2.0364 \times 10^5$ | 109.89 | $-12140$ | 109.92 | $4.8144 \times 10^5$ | 109.89 |
| 6 | $3322.6 \times 10^6$ | 109.89 | $1.2272 \times 10^5$ | 109.89 | $-4.8653 \times 10^6$ | 109.89 |

You can see that this solution is not converging and the coefficient matrix is not diagonally dominant. The coefficient matrix

$$[A] = \begin{bmatrix} 3 & 7 & 13 \\ 1 & 5 & 3 \\ 12 & 3 & -5 \end{bmatrix}$$

is not diagonally dominant as

$$|a_{11}| = |3| = 3 \leq |a_{12}| + |a_{13}| = |7| + |13| = 20$$

Hence, the Gauss-Seidel method may or may not converge.

However, it is the same set of equations as the previous example and that converged. The only difference is that we exchanged first and the third equation with each other and that made the coefficient matrix not diagonally dominant.

Therefore, it is possible that a system of equations can be made diagonally dominant if one exchanges the equations with each other. However, it is not possible for all cases. For example, the following set of equations

$$x_1 + x_2 + x_3 = 3$$

$$2x_1 + 3x_2 + 4x_3 = 9$$

$$x_1 + 7x_2 + x_3 = 9$$

cannot be rewritten to make the coefficient matrix diagonally dominant.

## Key Terms:

Gauss-Seidel method

Convergence of Gauss-Seidel method

Diagonally dominant matrix

# Multiple Choice Test

## Chapter 04.08 Gauss-Seidel Method of Solving

(1). A square matrix $[A]_{n\times n}$ is diagonally dominant if

(A) $|a_{ii}| \geq \sum\limits_{\substack{j=1 \\ i\neq j}}^{n} |a_{ij}|$, $i = 1, 2, ..., n$

(B) $|a_{ii}| \geq \sum\limits_{\substack{j=1 \\ i\neq j}}^{n} |a_{ij}|$, $i = 1, 2, ..., n$ and $|a_{ii}| > \sum_{\substack{j=1 \\ i\neq j}}^{n} |a_{ij}|$, for any $i = 1, 2, ..., n$

(C) $|a_{ii}| \geq \sum\limits_{j=1}^{n} |a_{ij}|$, $i = 1, 2, ..., n$ and $|a_{ii}| > \sum_{j=1}^{n} |a_{ij}|$, for any $i = 1, 2, ..., n$

(D) $|a_{ii}| \geq \sum\limits_{j=1}^{n} |a_{ij}|$, $i = 1, 2, ..., n$

(2). Using $[x_1, x_2, x_3] = [1, 3, 5]$ as the initial guess, the values of $[x_1, x_2, x_3]$ after three iterations in the Gauss-Seidel method for

$$\begin{bmatrix} 12 & 7 & 3 \\ 1 & 5 & 1 \\ 2 & 7 & -11 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ -5 \\ 6 \end{bmatrix}$$

are

(A) $[-2.8333 - 1.4333 - 1.9727]$

(B) $[1.4959 - 0.90464 - 0.84914]$

(C) $[0.90666 - 1.0115 - 1.0243]$

(D) $[1.2148 - 0.72060 - 0.82451]$

(3). To ensure that the following system of equations,

$$\begin{array}{rrrr} 2x_1+ & 7x_2- & 11x_3 = & 6 \\ x_1+ & 2x_2+ & x_3 = & -5 \\ 7x_1+ & 5x_2+ & 2x_3 = & 17 \end{array}$$

converges using the Gauss-Seidel method, one can rewrite the above equations as follows:

(A) $\begin{bmatrix} 2 & 7 & -11 \\ 1 & 2 & 1 \\ 7 & 5 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 6 \\ -5 \\ 17 \end{bmatrix}$

(B) $\begin{bmatrix} 7 & 5 & 2 \\ 1 & 2 & 1 \\ 2 & 7 & -11 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 17 \\ -5 \\ 6 \end{bmatrix}$

(C) $\begin{bmatrix} 7 & 5 & 2 \\ 1 & 2 & 1 \\ 2 & 7 & -11 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 6 \\ -5 \\ 17 \end{bmatrix}$

(D) The equations cannot be rewritten in a form to ensure convergence.

(4). For $\begin{bmatrix} 12 & 7 & 3 \\ 1 & 5 & 1 \\ 2 & 7 & -11 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 22 \\ 7 \\ -2 \end{bmatrix}$ and using $\begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 1 \end{bmatrix}$ as the initial guess, the values of $\begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix}$ are found at the end of each iteration as

| Iteration # | $x_1$ | $x_2$ | $x_3$ |
|---|---|---|---|
| 1 | 0.41667 | 1.1167 | 0.96818 |
| 2 | 0.93990 | 1.0184 | 1.0008 |
| 3 | 0.98908 | 1.0020 | 0.99931 |
| 4 | 0.99899 | 1.0003 | 1.0000 |

At what first iteration number would you trust at least 1 significant digit in your solution?

(A) 1

(B) 2

(C) 3

(D) 4

(5). The algorithm for the Gauss-Seidel method to solve $[A][X] = [C]$ is given as follows when using $n$ max iterations. The initial value of $[X]$ is stored in $[X]$.

| (A) Sub | (B) Sub |
|---|---|
| Seidel$(n, a, x, rhs, nmax)$ | Seidel$(n, a, x, rhs, nmax)$ |
| For $k = 1$ To $nmax$ | For $k = 1$ To $nmax$ |
| For $i = 1$ To $n$ | For $i = 1$ To $n$ |
| For $j = 1$ To $n$ | $Sum = 0$ |
| If $(i <> j)$ Then | For $j = 1$ To $n$ |
| $Sum = Sum + a(i, j) * x(j)$ | If $(i <> j)$ Then |
| endif | $Sum = Sum + a(i, j) * x(j)$ |
| Next $j$ | endif |
| $x(i) = (rhs(i) - \text{Sum})/a(i, i)$ | Next $j$ |
| Next $i$ | $x(i) = (rhs(i) - \text{Sum})/a(i, i)$ |
| Next $j$ | Next $i$ |
| End Sub | Next $k$ |
| | End Sub |

| (C) Sub | (D) Sub |
|---|---|
| Seidel$(n, a, x, rhs, nmax)$ | Seidel$(n, a, x, rhs, nmax)$ |
| For $k = 1$To $nmax$ | For $k = 1$To{nmax}$ |
| For $i = 1$To $n$ | For $i = 1$ To $n$ |
| $Sum = 0$ | $Sum = 0$ |
| For $j = 1$To $n$ | For $j = 1$ To $n$ |
| $Sum = Sum + a(i, j) * x(j)$ | If $(i <> j)$ Then |
| Next $j$ | $Sum = Sum + a(i, j) * x(j)$ |
| $x(i) = (rhs(i) - \text{Sum})/a(i, i)$ | endif |
| Next $i$ | Next $j$ |
| Next $k$ | $x(i) = (rhs(i) - \text{Sum})/a(i, i)$ |
| End Sub | Next $i$ |
| | Next $k$ |
| | End Sub |

(6). Thermistors measure temperature, have a nonlinear output and are valued for a limited range. So when a thermistor is manufactured, the manufacturer supplies a resistance vs. temperature curve. An accurate representation of the curve is generally given by

$$\frac{1}{T} = a_0 + a_1 ln(R) + a_2 \{\ln(R)\}^2 + a_3 \{\ln(R)\}^3$$

where $T$ is temperature in Kelvin, $R$ is resistance in ohms, and $a_0, a_1, a_2, a_3$ are constants of the calibration curve. Given the following for a thermistor

| $R$ | $T$ |
|---|---|
| $ohm$ | $^\circ C$ |
| 1101.0 | 25.113 |
| 911.3 | 30.131 |
| 636.0 | 40.120 |
| 451.1 | 50.128 |

the value of temperature in $^{\circ}C$ for a measured resistance of 900 ohms most nearly is

(A) 30.002

(B) 30.473

(C) 31.272

(D) 31.445

For complete solution, go to

http://nm.mathforcollege.com/mcquizzes/04sle/quiz_04sle_gaussseidal_solution.pdf

# Multiple Choice Test

Chapter 04.08 Gauss-Seidel Method for Solving Simultaneous Linear Equations

## Gauss-Seidel Method Exercise

(1). In a system of equation $[A][X] = [C]$, if $[A]$ is diagonally dominant, then GaussSeidal-Seidel method

(A) always converges

(B) may or may not converge

(C) always diverges

**Answer:** A

(2). In a system of equations $[A][X] = [C]$, if $[A]$ is not diagonally dominant, then Gauss-Seidel method

(A) Always converges

(B) May or may not converge

(C) Always diverges.

**Answer:** B

(3). In a system of equations $[A][X] = [C]$, if $[A]$ is not diagonally dominant, the system of equations can always be rewritten to make it diagonally dominant.

(A) True

(B) False

**Answer:** B

(4). Solve the following system of equations using Gauss-Seidel method

$$\begin{gathered} 12x_1 + 7x_2 + 3x_3 = 2 \\ x_1 + 5x_2 + x_3 = -5 \\ 2x_1 + 7x_2 - 11x_3 = 6 \end{gathered}$$

Conduct 3 iterations, calculate the maximum absolute relative approximate error at the end of each iteration and choose $\begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix} = \begin{bmatrix} 1 & 3 & 5 \end{bmatrix}$ as your initial guess.

**Answer:** $\begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix} = \begin{bmatrix} 0.90666 & -1.0115 & -1.0243 \end{bmatrix}$ $\begin{bmatrix} |\in_a|_1 & |\in_a|_2 & |\in_a|_3 \end{bmatrix} = \begin{bmatrix} 65.001\% & 10.564\% & 17.099\% \end{bmatrix}$
$\begin{bmatrix} |\in_a|_1 & |\in_a|_2 & |\in_a|_3 \end{bmatrix} = \begin{bmatrix} 65.001\% & 10.564\% & 17.099\% \end{bmatrix}$

(5). Solve the following system of equations using Gauss-Seidel method

$12x_1 + 7x_2 + 3x_3 = 2$
$x_1 + 5x_2 + x_3 = -5$
$2x_1 + 7x_2 - 11x_3 = 6$
Conduct 3 iterations, calculate the maximum absolute relative approximate error at the end of each iteration, and choose $\begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix} = \begin{bmatrix} 1 & 3 & 5 \end{bmatrix}$ as your initial guess.

**Answer:** $\begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix} = \begin{bmatrix} 0.90666 & -1.0115 & -1.0243 \end{bmatrix}$

(6). Solve the following system of equations using Gauss-Seidel Method
$x_1 + 5x_2 + x_3 = 5$
$12x_1 + 7x_2 + 3x_3 = 2$
$2x_1 + 7x_2 - 11x_3 = 6$
Conduct 3 iterations, calculate the maximum absolute relative approximate error at the end of each iteration, and choose $\begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix} = \begin{bmatrix} 1 & 3 & 5 \end{bmatrix}$ as your initial guess.

**Answer:** $\begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix} = \begin{bmatrix} -1163.7 & 1947.6 & 1027.2 \end{bmatrix}$

$\begin{bmatrix} |\in_a|_1 & |\in_a|_2 & |\in_a|_3 \end{bmatrix} = \begin{bmatrix} 89.156\% & 89.139\% & 89.183\% \end{bmatrix}$

# Chapter 04.09: Adequacy of Solutions

Lesson: Adequacy of Solutions

## Learning Objectives

*After reading this chapter, you should be able to:*

(1). *know the difference between ill-conditioned and well-conditioned systems of equations,*

(2). *define and find the norm of a matrix*

(3). *define and evaluate the condition number of an invertible square matrix*

(4). *relate the condition number of a coefficient matrix to the ill or well conditioning of the system of simultaneous linear equations, that is, how much can you trust the solution of the simultaneous linear equations.*

## What do you mean by ill-conditioned and well-conditioned system of equations?

A system of equations is considered to be **well-conditioned** if a small change in the coefficient matrix or a small change in the right hand side results in a small change in the solution vector.

A system of equations is considered to be **ill-conditioned** if a small change in the coefficient matrix or a small change in the right hand side results in a large change in the solution vector.

### Example 1

Is this system of equations well-conditioned?

$$\begin{bmatrix} 1 & 2 \\ 2 & 3.999 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 4 \\ 7.999 \end{bmatrix}$$

### Solution

The solution to the above set of equations is

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

Make a small change in the right-hand side vector of the equations

$$\begin{bmatrix} 1 & 2 \\ 2 & 3.999 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 4.001 \\ 7.998 \end{bmatrix}$$

gives

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -3.999 \\ 4.000 \end{bmatrix}$$

Make a small change in the coefficient matrix of the equations

$$\begin{bmatrix} 1.001 & 2.001 \\ 2.001 & 3.998 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 4 \\ 7.999 \end{bmatrix}$$

gives

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 3.994 \\ 0.001388 \end{bmatrix}$$

This last systems of equation "looks" ill-conditioned because a small change in the coefficient matrix or the right hand side resulted in a large change in the solution vector.

**Example 2**

Is this system of equations well-conditioned?

$$\begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 4 \\ 7 \end{bmatrix}$$

**Solution**

The solution to the above equations is

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

Make a small change in the right-hand side vector of the equations.

$$\begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 4.001 \\ 7.001 \end{bmatrix}$$

gives

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1.999 \\ 1.001 \end{bmatrix}$$

Make a small change in the coefficient matrix of the equations.

$$\begin{bmatrix} 1.001 & 2.001 \\ 2.001 & 3.001 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 4 \\ 7 \end{bmatrix}$$

gives

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2.003 \\ 0.997 \end{bmatrix}$$

This system of equation "looks" well-conditioned because small changes in the coefficient matrix or the right-hand side resulted in small changes in the solution vector.

## So, what if the system of equations is ill conditioned or well-conditioned?

Well, if a system of equations is ill-conditioned, we cannot trust the solution as much. Revisit the velocity problem, Example 5.1 in Chapter 5. The values in the coefficient matrix $[A]$ are squares of time, etc. For example, if instead of $a_{11} = 25$, you used $a_{11} = 24.99$, would you want this small change to make a huge difference in the solution vector. If it did, would you trust the solution?

Later we will see how much (quantifiable terms) we can trust the solution in a system of equations. Every invertible square matrix has a **condition number** and coupled with the **machine epsilon**, we can quantify how many significant digits one can trust in the solution.

## To calculate the condition number of an invertible square matrix, I need to know what the norm of a matrix means. How is the norm of a matrix defined?

Just like the determinant, the norm of a matrix is a simple unique scalar number. However, the norm is always positive and is defined for all matrices – square or rectangular, and invertible or noninvertible square matrices.

One of the popular definitions of a norm is the row sum norm (also called the uniform-matrix norm). For a $m \times n$ matrix $[A]$, the row sum norm of $[A]$ is defined as

$$\|A\|_\infty = \max_{1 \leq i \leq m} \sum_{j=1}^{n} |a_{ij}|$$

that is, find the sum of the absolute value of the elements of each row of the matrix $[A]$. The maximum out of the $m$ such values is the row sum norm of the matrix $[A]$.

**Example 3**

Find the row sum norm of the following matrix [A].

$$A = \begin{bmatrix} 10 & -7 & 0 \\ -3 & 2.099 & 6 \\ 5 & -1 & 5 \end{bmatrix}$$

**Solution**

$$\begin{aligned} \|A\|_\infty &= \max_{1 \leq i \leq 3} \sum_{j=1}^{3} |a_{ij}| \\ &= max\left[(|10| + |-7| + |0|), (|-3| + |2.099| + |6|), (|5| + |-1| + |5|)\right] \\ &= max\left[(10 + 7 + 0), (3 + 2.099 + 6), (5 + 1 + 5)\right] \\ &= max\left[17,\ 11.099,\ 11\right] \\ &= 17 \end{aligned}$$

## How is the norm related to the conditioning of the matrix?

Let us start answering this question using an example. Go back to the *ill-conditioned* system of equations,

$$\begin{bmatrix} 1 & 2 \\ 2 & 3.999 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 4 \\ 7.999 \end{bmatrix}$$

that gives the solution as

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

Denoting the above set of equations as

$$[A]\,[X] = [C]$$

$$\|X\|_\infty = 2$$

$$\|C\|_\infty = 7.999$$

Making a small change in the right-hand side,

$$\begin{bmatrix} 1 & 2 \\ 2 & 3.999 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 4.001 \\ 7.998 \end{bmatrix}$$

gives

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -3.999 \\ 4.000 \end{bmatrix}$$

Denoting the above set of equations by

$$[A]\,[X'] = [C']$$

right hand side vector is found by

$$[\Delta C] = [C'] - [C]$$

and the change in the solution vector is found by

$$[\Delta X] = [X'] - [X]$$

then

$$\begin{aligned} [\Delta C] &= \begin{bmatrix} 4.001 \\ 7.998 \end{bmatrix} - \begin{bmatrix} 4 \\ 7.999 \end{bmatrix} \\ &= \begin{bmatrix} 0.001 \\ -0.001 \end{bmatrix} \end{aligned}$$

and

$$[\Delta X] = \begin{bmatrix} -3.999 \\ 4.000 \end{bmatrix} - \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$
$$= \begin{bmatrix} -5.999 \\ 3.000 \end{bmatrix}$$

then

$$\|\Delta C\|_\infty = 0.001$$

$$\|\Delta X\|_\infty = 5.999$$

The relative change in the norm of the solution vector is

$$\frac{\|\Delta X\|_\infty}{\|X\|_\infty} = \frac{5.999}{2}$$

$$= 2.9995$$

The relative change in the norm of the right-hand side vector is

$$\frac{\|\Delta C\|_\infty}{\|C\|_\infty} = \frac{0.001}{7.999}$$

$$= 1.250 \times 10^{-4}$$

See the small relative change of $1.250 \times 10^{-4}$ in the right-hand side vector norm results in a large relative change in the solution vector norm of 2.9995.

In fact, the ratio between the relative change in the norm of the solution vector and the relative change in the norm of the right-hand side vector is

$$\frac{\|\Delta X\|_\infty / \|X\|_\infty}{\|\Delta C\|_\infty / \|C\|_\infty} = \frac{2.9995}{1.250 \times 10^{-4}}$$
$$= 23993$$

Let us now go back to the *well-conditioned* system of equations.

$$\begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 4 \\ 7 \end{bmatrix}$$

gives

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

Denoting the system of equations by

$$[A]\,[X] = [C]$$

$$\|X\|_\infty = 2$$

$$\|C\|_\infty = 7$$

Making a small change in the right-hand side vector

$$\begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 4.001 \\ 7.001 \end{bmatrix}$$

gives

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1.999 \\ 1.001 \end{bmatrix}$$

Denoting the above set of equations by

$$[A]\,[X'] = [C']$$

the change in the right-hand side vector is then found by

$$[\Delta C] = [C'] - [C]$$

and the change in the solution vector is

$$[\Delta X] = [X'] - [X]$$

then

$$\begin{aligned} [\Delta C] &= \begin{bmatrix} 4.001 \\ 7.001 \end{bmatrix} - \begin{bmatrix} 4 \\ 7 \end{bmatrix} \\ &= \begin{bmatrix} 0.001 \\ 0.001 \end{bmatrix} \end{aligned}$$

and

$$[\Delta X] = \begin{bmatrix} 1.999 \\ 1.001 \end{bmatrix} - \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$
$$= \begin{bmatrix} -0.001 \\ 0.001 \end{bmatrix}$$

then

$$\|\Delta C\|_\infty = 0.001$$

$$\|\Delta X\|_\infty = 0.001$$

The relative change in the norm of solution vector is

$$\frac{\|\Delta X\|_\infty}{\|X\|_\infty} = \frac{0.001}{2}$$
$$= 5 \times 10^{-4}$$

The relative change in the norm of the right-hand side vector is

$$\frac{\|\Delta C\|_\infty}{\|C\|_\infty} = \frac{0.001}{7}$$
$$= 1.429 \times 10^{-4}$$

See the small relative change in the right-hand side vector norm of $1.429 \times 10^{-4}$ results in the small relative change in the solution vector norm of $5 \times 10^{-4}$.

In fact, the ratio between the relative change in the norm of the solution vector and the relative change in the norm of the right-hand side vector is

$$\frac{\|\Delta X\|_\infty / \|X\|_\infty}{\|\Delta C\|_\infty / \|C\|_\infty} = \frac{5 \times 10^{-4}}{1.429 \times 10^{-4}}$$
$$= 3.5$$

## What are some of the properties of norms?

1. For a matrix $[A]$, $\|A\| \geq 0$
2. For a matrix $[A]$ and a scalar k, $\|kA\| = |k| \, \|A\|$
3. For two matrices $[A]$ and $[B]$ of same order, $\|A + B\| \leq \|A\| + \|B\|$
4. For two matrices $[A]$ and $[B]$ that can be multiplied as $[A]$ $[B]$, $\|AB\| \leq \|A\| \, \|B\|$

**Is there a general relationship that exists between $\|\mathbf{\Delta X}\| / \|\mathbf{X}\|$ and $\|\mathbf{\Delta C}\| / \|\mathbf{C}\|$ or between $\|\mathbf{\Delta X}\| / \|\mathbf{X}\|$ and $\|\mathbf{\Delta A}\| / \|\mathbf{A}\|$? If so, it could help us identify well-conditioned and ill conditioned system of equations.**

**If there is such a relationship, will it help us quantify the conditioning of the matrix? That is, will it tell us how many significant digits we could trust in the solution of a system of simultaneous linear equations?**

There is a relationship that exists between

$$\frac{\|\Delta X\|}{\|X\|} \text{and} \frac{\|\Delta C\|}{\|C\|}$$

and between

$$\frac{\|\Delta X\|}{\|X\|} \text{and} \frac{\|\Delta A\|}{\|A\|}$$

These relationships are

$$\frac{\|\Delta X\|}{\|X\|} \leq \|A\| \quad \left\|A^{-1}\right\| \frac{\|\Delta C\|}{\|C\|}$$

and

$$\frac{\|\Delta X\|}{\|X + \Delta X\|} \leq \|A\| \quad \left\|A^{-1}\right\| \frac{\|\Delta A\|}{\|A\|}$$

The above two inequalities show that the relative change in the norm of the right-hand side vector or the coefficient matrix can be amplified by as much as $\|A\| \left\|A^{-1}\right\|$.

This number $\|A\| \left\|A^{-1}\right\|$ is called the **condition number** of the matrix and coupled with the machine epsilon, we can quantify the accuracy of the solution of $[A]$ $[X] = [C]$.

**Prove for** $[A]\ [X] = [C]$

that

$$\frac{\|\Delta X\|}{\|X + \Delta X\|} \leq \quad \|A\| \quad \left\|A^{-1}\right\| \frac{\|\Delta A\|}{\|A\|}$$

**Proof** Let

$$[A] \quad [X] = [C] \qquad (1)$$

Then if $[A]$ is changed to $[A']$, the $[X]$ will change to $[X']$, such

that

$$[A'] \quad [X'] = [C] \qquad (2)$$

From Equations (1) and (2),

$$[A] \quad [X] = [A'] [X']$$

Denoting change in $[A]$ and $[X]$ matrices as $[\Delta A]$ and $[\Delta X]$, respectively

$$[\Delta A] = [A'] - [A]$$

$$[\Delta X] = [X'] - [X]$$

then

$$[A] [X] = \quad ([A] + [\Delta A])\ ([X] + [\Delta X])$$

Expanding the above expression

$$[A] [X] = \ [A] [X] + [A] [\Delta X] + [\Delta A] [X] + [\Delta A] [\Delta X]$$

$$[0] = [A] [\Delta X] + [\Delta A] ([X] + [\Delta X])$$

$$-[A] [\Delta X] = [\Delta A] ([X] + [\Delta X])$$

$$[\Delta X] = -[A]^{-1} [\Delta A] ([X] + [\Delta X])$$

Applying the theorem of norms, that the norm of multiplied matrices is less than the multiplication of the individual norms of the matrices,

$$\|\Delta X\| \leq \left\|A^{-1}\right\| \|\Delta A\| \|X + \Delta X\|$$

Multiplying both sides by $\|A\|$

$$\|A\| \ \|\Delta X\| \leq \|A\| \left\|A^{-1}\right\| \|\Delta A\| \|X + \Delta X\|$$

$$\frac{\|\Delta X\|}{\|X + \Delta X\|} \leq \|A\| \ \left\|A^{-1}\right\| \ \frac{\|\Delta A\|}{\|A\|}$$

## How do I use the above theorems to find how many significant digits are correct in my solution vector?

The relative error in a solution vector norm is $\leq$ Cond (A) $\times$relative error in right hand side vector norm.

The possible relative error in the solution vector norm is $\leq Cond(A) \times \in_{mach}$

Hence $Cond(A) \times \in_{mach}$ should give us the number of significant digits, $m$ that are at least correct in our solution by finding out the largest value of $m$ for which $Cond(A) \times \in_{mach}$ is less than $0.5 \times 10^{-m}$.

### Example 4

How many significant digits can I trust in the solution of the following system of equations?

$$\begin{bmatrix} 1 & 2 \\ 2 & 3.999 \end{bmatrix} \ \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$$

### Solution

For

$$[A] = \begin{bmatrix} 1 & 2 \\ 2 & 3.999 \end{bmatrix}$$

it can be shown

$$[A]^{-1} = \begin{bmatrix} -3999 & 2000 \\ 2000 & -1000 \end{bmatrix}$$

$$\|A\|_\infty = 5.999$$

$$\left\|A^{-1}\right\|_\infty = 5999$$

$$\begin{aligned} Cond\,(A) &= \|A\| \left\|A^{-1}\right\| \\ &= 5.999 \times 5999.4 \\ &= 35990 \end{aligned}$$

Assuming single precision with 23 bits used in the mantissa for real numbers, the machine epsilon is

$$\begin{aligned} \in_{mach} &= 2^{-23} \\ &= 0.119209 \times 10^{-6} \end{aligned}$$

$$\begin{aligned} Cond(A) \times \in_{mach} &= 35990 \times 0.119209 \times 10^{-6} \\ &= 0.4290 \times 10^{-2} \end{aligned}$$

For what maximum positive value of $m$, would $Cond(A) \times \in_{mach}$ be less than or equal to $0.5 \times 10^{-m}$

$$0.4290 \times 10^{-2} \leq 0.5 \times 10^{-m}$$

$$0.8580 \times 10^{-2} \leq 10^{-m}$$

$$log(0.8580 \times 10^{-2}) \leq log(10^{-m})$$

$$-2.067 \leq -m$$

$$m \leq 2.067$$

$$m \leq 2$$

So, two significant digits are at least correct in the solution vector.

**Example 5**

How many significant digits can I trust in the solution of the following system of equations?

$$\begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 4 \\ 7 \end{bmatrix}$$

**Solution**

For

$$[A] = \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix}$$

it can be shown

$$[A]^{-1} = \begin{bmatrix} -3 & 2 \\ 2 & -1 \end{bmatrix}$$

Then

$$\|A\|_\infty = 5,$$

$$\left\|A^{-1}\right\|_\infty = 5.$$

$$\begin{aligned} Cond\ (A) &= \|A\| \quad \left\|A^{-1}\right\| \\ &= 5 \times 5 \\ &= 25 \end{aligned}$$

Assuming single precision with 23 bits used in the mantissa for real numbers, the machine epsilon

$$\begin{aligned} \in_{mach} &= 2^{-23} \\ &= 0.119209 \times 10^{-6} \end{aligned}$$

$$\begin{aligned} Cond(A) \times \in_{mach} &= 25 \times 0.119209 \times 10^{-6} \\ &= 0.2980 \times 10^{-5} \end{aligned}$$

For what maximum positive value of $m$, would $Cond(A)\times \in_{mach}$ be less than or equal to $0.5 \times 10^{-m}$

$$0.2980 \times 10^{-5} \leq 0.5 \times 10^{-m}$$

$$m \leq 5$$

So, five significant digits are at least correct in the solution vector.

## Key Terms:

*Ill-Conditioned matrix*

*Well-Conditioned matrix*

*Norm*

*Condition Number*

*Machine Epsilon*

*Significant Digits*

# Multiple Choice Test

## Chapter 04.09 Adequacy of Solutions

(1). The row sum norm of the matrix

$$[A] = \begin{bmatrix} 6 & -7 & 3 & 13 \\ 19 & -21 & 23 & -29 \\ 41 & 47 & -51 & 61 \end{bmatrix}$$

is

(A) 29

(B) 61

(C) 98

(D) 200

(2). The adequacy of the solution of simultaneous linear equations

$$[A]\,[X] = [C]$$

depends on

(A) the condition number of coefficient matrix $[A]$

(B) the machine epsilon

(C) the condition number for matrix $[A]$and the machine epsilon

(D) norm of the coefficient matrix $[A]$

(3). Given a set of equations in matrix form

$$[A]\,[X] = [C]\,,$$

$$\|A\| = 250, \left\|A^{-1}\right\| = 40 \;\; and$$

$$\varepsilon_{mach} = 0.119 \times 10^{-6},$$

then the number of significant digits you can at least trust in the solution are

(A) 1

(B) 2

(C) 3

(D) 4

(4). The solution to a set of simultaneous linear equations

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 44 \\ 94 \\ 138 \end{bmatrix}$$

is given as

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \\ 7 \end{bmatrix}$$

The solution to another set of simultaneous linear equations is given by (note the coefficient matrix is the same as above)

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 43.99 \\ 93.98 \\ 138.03 \end{bmatrix}$$

is given as

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 214.01 \\ -208.01 \\ 60 \end{bmatrix}$$

Based on the row sum norm, the condition number of the coefficient matrix is greater than (choose the largest possible value)

(A) 1

(B) 138

(C) 4500

(D) 139320

(5). The condition number of the $n \times n$ identity matrix based on the row sum norm is

(A) 0

(B) 1

(C) $n$

(D) $n^2$

(6). Let $[A] = \begin{bmatrix} 1 & 2+\delta \\ 2-\delta & 1 \end{bmatrix}$. Based on the row sum norm and given that $\delta \to 0, \delta > 0$, the condition number of the matrix is

(A) $\dfrac{3-\delta}{3+\delta}$

(B) $\dfrac{9-\delta^2}{3-\delta^2}$

(C) $\dfrac{(3+\delta)^2}{3-\delta^2}$

(D) $\dfrac{3-2\delta-\delta^2}{3-\delta^2}$

# Multiple Choice Test

Chapter 04.09 Adequacy of Solutions

## Adequacy of Solutions Exercise

(1). The adequacy of the solution of simultaneous linear equations depends on

(A) Condition number

(B) Machine epsilon

(C) Product of condition number and machine epsilon

(D) Norm of the matrix.

Answer: C

(2). If a system of equations $[A]\ [X] = [C]$ is ill conditioned, then

(A) $det(A) = 0$

(B) $Cond(A) = 1$

(C) $Cond(A)$ is large.

(D) $\|A\|$ is large.

Answer: C

(3). If $Cond(A) = 10^4$ and $\in_{mach}= 0.119 \times 10^{-6}$, then in $[A]\ [X] = [C]$, at least these many significant digits are correct in your solution,

(A) 0

(B) 1

(C) 2

(D) 3

Answer: C

(4). Make a small change in the coefficient matrix of $\begin{bmatrix} 1 & 2 \\ 2 & 3.999 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 4 \\ 7.999 \end{bmatrix}$ and find

$$\frac{\|\Delta X\|_\infty / \|X\|_\infty}{\|\Delta A\|_\infty / \|A\|_\infty}$$

Is it a large or small number? How is this number related to the condition number of the coefficient matrix?

Answer: Changing $[A]$ to

$$\begin{bmatrix} 1.001 & 2.001 \\ 2.001 & 4.000 \end{bmatrix}$$

Results in solution of

$$\begin{bmatrix} 5999 \\ -2999 \end{bmatrix}$$

$$\frac{\frac{\|\Delta X\|_\infty}{\|X\|_\infty}}{\frac{\|\Delta A\|_\infty}{\|A\|_\infty}} = \frac{\frac{5999.7}{2}}{\frac{0.002}{5.999}}$$

$4= 8.994 \times 10^6\$$

(5). Make a small change in the coefficient matrix of
$\begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = [4\ 7]$
and find

$$\frac{\|\Delta X\|_\infty / \|X\|_\infty}{\|\Delta A\|_\infty / \|A\|_\infty}.$$

Is it a large or a small number? Compare your results with the previous problem. How is this number related to the condition number of the coefficient matrix?

Answer: Changing $[A]$ to

$$\begin{bmatrix} 1.001 & 2.001 \\ 2.001 & 3.000 \end{bmatrix}$$

Results in solution of

$$\begin{bmatrix} 2.003 \\ 0.9970 \end{bmatrix}$$

$$\frac{\frac{\|\Delta X\|_\infty}{\|X\|_\infty}}{\frac{\|\Delta A\|_\infty}{\|A\|_\infty}} = \frac{\frac{0.003}{2.000}}{\frac{0.002}{5}}$$

$$= 3.75$$

(6). Prove

$$\frac{\|\Delta X\|}{\|X\|} \quad \|A\| \ \left\|A^{-1}\right\| \ \frac{\|\Delta C\|}{\|C\|}$$

Answer: Use theorem that if $[A][B] = [C]$then $\|A\|\,\|B\| \geq \|C\|$

(7). For

$$[A] = \begin{bmatrix} 10 & -7 & 0 \\ -3 & 2.099 & 6 \\ 5 & -1 & 5 \end{bmatrix}$$

gives

$$[A]^{-1} = \begin{bmatrix} -0.1099 & -0.2333 & 0.2799 \\ -0.2999 & -0.3332 & 0.3999 \\ 0.04995 & 0.1666 & 6.664 \times 10^{-5} \end{bmatrix}$$

(A) What is the condition number of $[A]$?

(B) How many significant digits can we at least trust in the solution of $[A]\ [X] = [C]$ if $\in_{mach}= 0.1192 \times 10^{-6}$?

(C) Without calculating the inverse of the matrix $[A]$, can you estimate the condition number of $[A]$ using the theorem in problem#6?

Answer:

(A) $\|A\| = 17$
$\left\|A^{-1}\right\| = 1.033$
$Cond(A) = 17.56$

(B) 5

(C) Try different values of right hand side of $C = [\pm 1 \quad \pm 1 \quad \pm 1]^T$ with signs chosen randomly. Then $\left\|A^{-1}\right\| \leq \|X\|$ obtained from solving equation set $[A]\ [X] = [C]$ as $\|C\| = 1$.

(8). Prove that the $Cond(A) \geq 1$.

Answer: We know that

$$\|A\ B\| \leq \|A\|\,\|B\|$$

then if

$$[B] = [A]^{-1},$$

$$\left\|A\ A^{-1}\right\| \leq \|A\|\,\left\|A^{-1}\right\|$$

$$\|I\| \leq \|A\|\,\left\|A^{-1}\right\|$$

$$1 \leq \|A\|\,\left\|A^{-1}\right\|$$

$$\|A\|\,\left\|A^{-1}\right\| \geq 1$$

$$Cond(A) \geq 1$$

# Chapter 04.10: Eigenvalues and Eigenvectors

Lesson: Eigenvalues and Eigenvectors

## Learning Objectives

*After successful completion, you should be able to:*

1) *define eigenvalues and eigenvectors of a square matrix,*

2) *find eigenvalues and eigenvectors of a square matrix,*

3) *relate eigenvalues to the singularity of a square matrix, and*

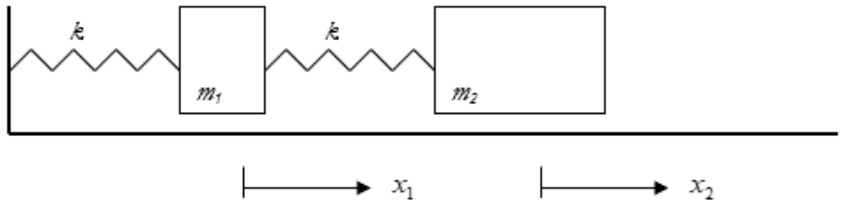
4) *use the power method to numerically find the largest eigenvalue in magnitude of a square matrix and the corresponding eigenvector.*

## What does eigenvalue mean?

The word eigenvalue comes from the German word *Eigenwert* where Eigen means *characteristic* and Wert means *value*. However, what the word means is not on your mind! You want to know why I need to learn about eigenvalues and eigenvectors. Once I give you an example of an application of eigenvalues and eigenvectors, you will want to know how to find these eigenvalues and eigenvectors.

## Can you give me a physical example application of eigenvalues and eigenvectors?

Look at the spring-mass system as shown in the picture below.




Assume each of the two mass-displacements to be denoted by $x_1$ and $x_2$, and let us assume each spring has the same spring constant $k$. Then by applying Newton's 2nd and 3rd law of motion to develop a force-balance for each mass we have

$$m_1 \frac{d^2x_1}{dt^2} = -kx_1 + k(x_2 - x_1)$$

$$m_2 \frac{d^2x_2}{dt^2} = -k(x_2 - x_1)$$

Rewriting the equations, we have

$$m_1 \frac{d^2x_1}{dt^2} - k(-2x_1 + x_2) = 0$$

$$m_2 \frac{d^2x_2}{dt^2} - k(x_1 - x_2) = 0$$

Let $m_1 \ = 10,\ m_2 = 20, k = 15$

$$10 \frac{d^2x_1}{dt^2} - 15(-2x_1 + x_2) = 0$$

$$20 \frac{d^2x_2}{dt^2} - 15(x_1 - x_2) = 0$$

From vibration theory, the solutions can be of the form

$$x_i = A_i \sin(\omega t - \Theta)$$

where

$A_i$= amplitude of the vibration of mass $i$,

$\omega$ = frequency of vibration,

$\Theta$ = phase shift.

then

$$\frac{d^2x_i}{dt^2} = -A_i w^2 Sin(\omega t - \Theta)$$

Substituting $x_i$and $\frac{d^2x_i}{dt^2}$in equations,

$$-10A_1\omega^2 - 15(-2A_1 + A_2) = 0$$

$$-20A_2\omega^2 - 15(A_1 - A_2) = 0$$

gives

$$(-10\omega^2 + 30)A_1 - 15A_2 = 0$$

$$-15A_1 + (-20\omega^2 + 15)A_2 = 0$$

or

$$(-\omega^2 + 3)A_1 - 1.5A_2 = 0$$

$$-0.75A_1 + (-\omega^2 + 0.75)A_2 = 0$$

In matrix form, these equations can be rewritten as

$$\begin{bmatrix} -\omega^2 + 3 & -1.5 \\ -0.75 & -\omega^2 + 0.75 \end{bmatrix} \begin{bmatrix} A_1 \\ A_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 3 & -1.5 \\ -0.75 & 0.75 \end{bmatrix} \begin{bmatrix} A_1 \\ A_2 \end{bmatrix} - \omega^2 \begin{bmatrix} A_1 \\ A_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Let $\omega^2 = \lambda$

$$[A] = \begin{bmatrix} 3 & -1.5 \\ -0.75 & 0.75 \end{bmatrix}$$

$$[X] = \begin{bmatrix} A_1 \\ A_2 \end{bmatrix}$$

$$[A]\ [X] - \lambda[X] = 0$$

$$[A]\ [X] = \lambda[X]$$

In the above equation, $\lambda$ is the eigenvalue and $[X]$ is the eigenvector corresponding to $\lambda$. As you can see, if we know $\lambda$ for the above example we can calculate the natural frequency of the vibration

$\omega = \sqrt{\lambda}$

Why are the natural frequencies of vibration important? Because you do not want to have a forcing force on the spring-mass system close to this frequency as it would make the amplitude $A_i$ very large and make the system unstable.

## What is the general definition of eigenvalues and eigenvectors of a square matrix?

If $[A]$ is a $n \times n$ matrix, then $[X] \neq \overrightarrow{0}$ is an eigenvector of $[A]$ if

$$[A]\ [X] = \lambda[X]$$

where $\lambda$ is a scalar and $[X] \neq 0$. The scalar $\lambda$ is called the eigenvalue of $[A]$and $[X]$ is called the eigenvector corresponding to the eigenvalue$\lambda$.

## How do I find eigenvalues of a square matrix?

To find the eigenvalues of a $n \times n$ matrix $[A]$ , we have

$$[A]\ [X] = \lambda [X]$$

$$[A]\ [X] - \lambda [X] = 0$$

$$[A]\ [X] - \lambda\ [I][X] = 0$$

$$([A]\ \ - [\lambda][I])[X] = 0$$

Now for the above set of equations to have a nonzero solution,

$$det([A] - \lambda [I]) = 0$$

This left-hand side can be expanded to give a polynomial in $\lambda$ and solving the above equation would give us values of the eigenvalues. The above equation is called the characteristic equation of $[A]$.

For a $[A]$ $n \times n$ matrix, the characteristic polynomial of $A$ is of degree $n$ as follows

$$det([A] - \lambda [I]) = 0$$

giving

$$\lambda^n + c_1\lambda^{n-1} + c_2\lambda^{n-2} + - - + c_n = 0$$

Hence. this polynomial has $n$ roots.

### Example 1

Find the eigenvalues of the physical problem discussed in the beginning of this chapter, that is, find the eigenvalues of the matrix

$$[A] = \begin{bmatrix} 3 & -1.5 \\ -0.75 & 0.75 \end{bmatrix}$$

### Solution

$$[A] - \lambda [I] = \begin{bmatrix} 3-\lambda & -1.5 \\ -0.75 & 0.75-\lambda \end{bmatrix}$$

$$det([A] - \lambda\,[I]) = (3-\lambda)(0.75-\lambda) - (-0.75)(-1.5) = 0$$

$$2.25 - 0.75\lambda - 3\lambda + \lambda^2 - 1.125 = 0$$

$$\lambda^2 - 3.75\lambda + 1.125 = 0$$

$$\begin{aligned}\lambda &= \frac{-(-3.75) \pm \sqrt{(-3.75)^2 - 4(1)(1.125)}}{2(1)} \\ &= \frac{3.75 \pm 3.092}{2} \\ &= 3.421,\ 0.3288\end{aligned}$$

So the eigenvalues are 3.421 and 0.3288.

**Example 2**

Find the eigenvectors of

$$A = \begin{bmatrix} 3 & -1.5 \\ -0.75 & 0.75 \end{bmatrix}$$

**Solution**

The eigenvalues have already been found in Example 1 as

$$\lambda_1 = 3.421, \lambda_2 = 0.3288$$

Let

$$[X] = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

be the eigenvector corresponding to

$$\lambda_1 = 3.421$$

Hence

$$([A] - \lambda_1[I])[X] = 0$$

$$\left\{ \begin{bmatrix} 3 & -1.5 \\ -0.75 & 0.75 \end{bmatrix} - 3.421 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right\} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$\begin{bmatrix} -0.421 & -1.5 \\ -0.75 & -2.671 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

If

$$x_1 = s$$

then

$$-0.421s - 1.5x_2 = 0$$
$$x_2 = -0.2808s$$

The eigenvector corresponding to $\lambda_1 = 3.421$ then is

$$[X] = \begin{bmatrix} s \\ -0.2808s \end{bmatrix}$$
$$= s \begin{bmatrix} 1 \\ -0.2808 \end{bmatrix}$$

The eigenvector corresponding to

$$\lambda_1 = 3.421$$

is

$$\begin{bmatrix} 1 \\ -0.2808 \end{bmatrix}$$

Similarly, the eigenvector corresponding to

$$\lambda_2 = 0.3288$$

is

$$\begin{bmatrix} 1 \\ 1.781 \end{bmatrix}$$

**Example 3**

Find the eigenvalues and eigenvectors of

$$[A] = \begin{bmatrix} 1.5 & 0 & 1 \\ -0.5 & 0.5 & -0.5 \\ -0.5 & 0 & 0 \end{bmatrix}$$

**Solution**

The characteristic equation is given by

$$det([A] - \lambda[I]) = 0$$

$$\det \begin{bmatrix} 1.5 - \lambda & 0 & 1 \\ -0.5 & 0.5 - \lambda & -0.5 \\ -0.5 & 0 & -\lambda \end{bmatrix} = 0$$

$$(1.5-\lambda)[(0.5-\lambda)(-\lambda)-(-0.5)(0)]+(1)[(-0.5)(0)-(-0.5)(0.5-\lambda)]=0$$

$$-\lambda^3+2\lambda^2-1.25\lambda+0.25=0$$

To find the roots of the characteristic polynomial equation

$$-\lambda^3+2\lambda^2-1.25\lambda+0.25=0$$

we find that the first root by observation is

$$\lambda=1$$

as substitution of $\lambda=1$gives

$$(-1)^3+2(1)^2-1.25(1)+0.25=0$$
$$0=0$$

So

$$(\lambda-1)$$

is a factor of

$$-\lambda^3+2\lambda^2-1.25\lambda+0.25$$

To find the other factors of the characteristic polynomial, we first conduct long division

$$-\lambda^2+\lambda+0.25$$

$$(\ \lambda-1)\overline{-\lambda^3+2\lambda^2-1.25\lambda+0.25}$$

$$-\lambda^3+\lambda^2$$

$$\overline{\lambda^2-1.25\lambda+0.25}$$

$$\lambda^2-\lambda$$

$$\overline{-0.25\lambda+0.25}$$

$$\overline{-0.25\lambda+0.25}$$

Hence

$$-\lambda^3 + 2\lambda^2 - 1.25\lambda + 0.25 = (\lambda - 1)(-\lambda^2 + \lambda + 0.25)$$

To find zeroes of$-\lambda^2 + \lambda + 0.25$, we solve the quadratic equation,

$$-\lambda^2 + \lambda + 0.25 = 0$$

to give

$$\begin{aligned}\lambda &= \frac{-(1) \pm \sqrt{(1)^2 - (4)(-1)(0.25)}}{2(-1)} \\ &= \frac{-1 \pm \sqrt{0}}{-2} \\ &= 0.5, 0.5\end{aligned}$$

So $\setminus$ $\lambda = 0.5$ and $\lambda = 0.5$ are the zeroes of

$$-\lambda^2 + \lambda + 0.5$$

giving

$$-\lambda^2 + \lambda + 0.25 = -(\lambda - 0.5)(\lambda - 0.5)$$

Hence

$$-\lambda^3 + 2\lambda^2 - 1.25\lambda + 0.25 = 0$$

can be rewritten as

$$-(\lambda - 1)(\lambda - 0.5)(\lambda - 0.5) = 0$$

to give the roots as

$$\lambda = 1, \quad 0.5, \quad 0.5$$

These are the three roots of the characteristic polynomial equation and hence the eigenvalues of matrix [A].

Note that there are eigenvalues that are repeated. Since there are only two distinct eigenvalues, there are only two eigenspaces. But, corresponding to $\lambda = 0.5$ there should be two eigenvectors that form a basis for the eigenspace corresponding to $\lambda = 0.5$.

Given

$$[(A - \lambda I)]\ [X] = 0$$

then

$$\begin{bmatrix} 1.5-\lambda & 0 & 1 \\ -0.5 & 0.5-\lambda & -0.5 \\ -0.5 & 0 & -\lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

For $\lambda = 0.5$,

$$\begin{bmatrix} 1 & 0 & 1 \\ -0.5 & 0 & -0.5 \\ -0.5 & 0 & -0.5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Solving this system gives

$$x_1 = -a,\ x_2 = b,\ x_3 = a$$

So

$$\begin{aligned} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} &= \begin{bmatrix} -a \\ b \\ a \end{bmatrix} \\ &= \begin{bmatrix} -a \\ 0 \\ a \end{bmatrix} + \begin{bmatrix} 0 \\ b \\ 0 \end{bmatrix} \\ &= a \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} + b \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \end{aligned}$$

So, the vectors $\begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$ form a basis for the eigenspace for the eigenvalue $\lambda = 0.5$ and are the two eigenvectors corresponding to $\lambda = 0.5$.

For $\lambda = 1$,

$$\begin{bmatrix} 0.5 & 0 & 1 \\ -0.5 & -0.5 & -0.5 \\ -0.5 & 0 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Solving this system gives

$$x_1 = a,\ x_2 = -0.5a,\ x_3 = -0.5a$$

The eigenvector corresponding to $\lambda = 1$ is

$$\begin{bmatrix} a \\ -0.5a \\ -0.5a \end{bmatrix} = a \begin{bmatrix} 1 \\ -0.5 \\ -0.5 \end{bmatrix}$$

Hence the vector

$$\begin{bmatrix} 1 \\ -0.5 \\ -0.5 \end{bmatrix}$$

is a basis for the eigenspace for the eigenvalue of$\lambda = 1$, and is the eigenvector corresponding to $\lambda = 1$.

## What are some of the theorems of eigenvalues and eigenvectors?

**Theorem 1:** If $[A]$ is a $n \times n$ triangular matrix – upper triangular, lower triangular or diagonal, the eigenvalues of $[A]$ are the diagonal entries of$[A]$.

**Theorem 2:** $\lambda = 0$ is an eigenvalue of $[A]$ if $[A]$ is a singular (noninvertible) matrix.

**Theorem 3:** $[A]$ and $[A]^T$ have the same eigenvalues.

**Theorem 4:** Eigenvalues of a symmetric matrix are real.

**Theorem 5:** Eigenvectors of a symmetric matrix are orthogonal, but only for distinct eigenvalues.

**Theorem 6:** $|det(A)|$ is the product of the absolute values of the eigenvalues of$[A]$.

### Example 4

What are the eigenvalues of

$$[A] = \begin{bmatrix} 6 & 0 & 0 & 0 \\ 7 & 3 & 0 & 0 \\ 9 & 5 & 7.5 & 0 \\ 2 & 6 & 0 & -7.2 \end{bmatrix}$$

### Solution

Since the matrix $[A]$ is a lower triangular matrix, the eigenvalues of $[A]$ are the diagonal elements of$[A]$. The eigenvalues are

$$\lambda_1 = 6,\ \lambda_2 = 3,\ \lambda_3 = 7.5,\ \lambda_4 = -7.2$$

### Example 5

One of the eigenvalues of

$$[A] = \begin{bmatrix} 5 & 6 & 2 \\ 3 & 5 & 9 \\ 2 & 1 & -7 \end{bmatrix}$$

is zero. Is $[A]$ invertible?

### Solution

$\lambda = 0$ is an eigenvalue of $[A]$, that implies $[A]$ is singular and is not invertible.

**Example 6**

Given the eigenvalues of

$$[A] = \begin{bmatrix} 2 & -3.5 & 6 \\ 3.5 & 5 & 2 \\ 8 & 1 & 8.5 \end{bmatrix}$$

are

$$\lambda_1 = -1.547,\ \lambda_2 = 12.33,\ \lambda_3 = 4.711$$

What are the eigenvalues of $[B]$ if

$$[B] = \begin{bmatrix} 2 & 3.5 & 8 \\ -3.5 & 5 & 1 \\ 6 & 2 & 8.5 \end{bmatrix}$$

**Solution**

Since $[B] = [A]^T$, the eigenvalues of $[A]$ and $[B]$ are the same. Hence eigenvalues of $[B]$ also are

$$\lambda_1 = -1.547,\ \lambda_2 = 12.33,\ \lambda_3 = 4.711$$

**Example 7**

Given the eigenvalues of

$$[A] = \begin{bmatrix} 2 & -3.5 & 6 \\ 3.5 & 5 & 2 \\ 8 & 1 & 8.5 \end{bmatrix}$$

are

$$\lambda_1 = -1.547,\ \lambda_2 = 12.33,\ \lambda_3 = 4.711$$

Calculate the magnitude of the determinant of the matrix.

**Solution**

Since

$$\begin{aligned} |det[A]| &= |\lambda_1|\ |\lambda_2|\ |\lambda_3| \\ &= |-1.547|\ |12.33|\ |4.711| \\ &= 89.88 \end{aligned}$$

## How does one find eigenvalues and eigenvectors numerically?

One of the most common methods used for finding eigenvalues and eigenvectors is the power method. It is used to find the largest eigenvalue in an absolute sense. Note that if this largest eigenvalues is repeated, this method will not work. Also this eigenvalue needs to be distinct. The method is as follows:

(1). Assume a guess $[X^{(0)}]$ for the eigenvector in

$$[A]\ [X] = \lambda[X]$$

equation. One of the entries of $[X^{(0)}]$ needs to be unity.

(2). Find

$$[Y^{(1)}] = [A]\ [X^{(0)}]$$

(3). Scale $[Y^{(1)}]$ so that the chosen unity component remains unity.

$$[Y^{(1)}] = \lambda^{(1)}[X^{(1)}]$$

(4). Repeat steps (2) and (3) with

$[X] = [X^{(1)}]$ to get $[X^{(2)}]$.

(5). Repeat the steps 2 and 3 until the value of the eigenvalue converges.

If $E_s$is the pre-specified percentage relative error tolerance to which you would like the answer to converge to, keep iterating until

$$\left|\frac{\lambda^{(i+1)} - \lambda^{(i)}}{\lambda^{(i+1)}}\right| \times 100 \leq E_s$$

where the left-hand side of the above inequality is the definition of absolute percentage relative approximate error, denoted generally by$E_s$ A pre-specified percentage relative tolerance of $0.5 \times 10^{2-m}$ implies at least $m$ significant digits are current in your answer. When the system converges, the value of $\lambda$ is the largest (in absolute value) eigenvalue of $[A]$.

### Example 8

Using the power method, find the largest eigenvalue and the corresponding eigenvector of

$$[A] = \begin{bmatrix} 1.5 & 0 & 1 \\ -0.5 & 0.5 & -0.5 \\ -0.5 & 0 & 0 \end{bmatrix}$$

### Solution

Assume

$$[X^{(0)}] = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$[A]\ [X^{(0)}] = \begin{bmatrix} 1.5 & 0 & 1 \\ -0.5 & 0.5 & -0.5 \\ -0.5 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$
$$= \begin{bmatrix} 2.5 \\ -0.5 \\ -0.5 \end{bmatrix}$$

$$Y^{(1)} = 2.5 \begin{bmatrix} 1 \\ -0.2 \\ -0.2 \end{bmatrix}$$

$$\lambda^{(1)} = 2.5$$

We will choose the first element of $[X^{(0)}]$ to be unity.

$$[X^{(1)}] = \begin{bmatrix} 1 \\ -0.2 \\ -0.2 \end{bmatrix}$$

$$[A]\ [X^{(1)}] = \begin{bmatrix} 1.5 & 0 & 1 \\ -0.5 & 0.5 & -0.5 \\ -0.5 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ -0.2 \\ -0.2 \end{bmatrix}$$
$$= \begin{bmatrix} 1.3 \\ -0.5 \\ -0.5 \end{bmatrix}$$

$$[X^{(2)}] = 1.3 \begin{bmatrix} 1 \\ -0.3846 \\ -0.3846 \end{bmatrix}$$

$$\lambda^{(2)} = 1.3$$

$$[X^{(2)}] = \begin{bmatrix} 1 \\ -0.3846 \\ -0.3846 \end{bmatrix}$$

The absolute relative approximate error in the eigenvalues is

$$|\varepsilon_a| = \left| \frac{\lambda^{(2)} - \lambda^{(1)}}{\lambda^{(2)}} \right| \times 100$$
$$= \left| \frac{1.3 - 1.5}{1.5} \right| \times 100$$
$$= 92.307\%$$

Conducting further iterations, the values of $\lambda^{(i)}$and the corresponding eigenvectors is given in the table below

| $i$ | $\lambda^{(i)}$ | $[X^{(i)}]$ | $\lvert\varepsilon_a\rvert$ (%) |
|---|---|---|---|
| 1 | 2.5 | $\begin{bmatrix} 1 \\ -0.2 \\ -0.2 \end{bmatrix}$ | ——— |
| 2 | 1.3 | $\begin{bmatrix} 1 \\ -0.38462 \\ -0.38462 \end{bmatrix}$ | 92.307 |
| 3 | 1.1154 | $\begin{bmatrix} 1 \\ -0.44827 \\ -0.44827 \end{bmatrix}$ | 16.552 |
| 4 | 1.0517 | $\begin{bmatrix} 1 \\ -0.47541 \\ -0.47541 \end{bmatrix}$ | 6.0529 |
| 5 | 1.02459 | $\begin{bmatrix} 1 \\ -0.48800 \\ -0.48800 \end{bmatrix}$ | 1.2441 |

The exact value of the eigenvalue is $\lambda = 1$

and the corresponding eigenvector is

$$[X] = \begin{bmatrix} 1 \\ -0.5 \\ -0.5 \end{bmatrix}$$

## Key Terms:

*Eigenvalue*

*Eigenvectors*

*Power method*

# Multiple Choice Test

## Chapter 04.10 Eigenvalues and Eigenvectors

(1). The eigenvalues of

$$\begin{bmatrix} 5 & 6 & 17 \\ 0 & -19 & 23 \\ 0 & 0 & 37 \end{bmatrix}$$

are

(A) $-19,\ \ 5,\ \ 37$

(B) $19,\ \ -5,\ \ -37$

(C) $2,\ \ -3,\ \ 7$

(D) $3,\ \ -5,\ \ 37$

(2). If $\begin{bmatrix} -4.5 \\ -4 \\ 1 \end{bmatrix}$ is an eigenvector of $\begin{bmatrix} 8 & -4 & 2 \\ 4 & 0 & 2 \\ 0 & -2 & -4 \end{bmatrix}$, the eigenvalue corresponding to the eigenvector is

(A) 1

(B) 4

(C) $-4.5$

(D) 6

(3). The eigenvalues of the following matrix

$$\begin{bmatrix} 3 & 2 & 9 \\ 7 & 5 & 13 \\ 6 & 17 & 19 \end{bmatrix}$$

are given by solving the cubic equation

(A) $\lambda^3 - 27\lambda^2 + 167\lambda - 285$

(B) $\lambda^3 - 27\lambda^2 - 122\lambda - 313$

(C) $\lambda^3 + 27\lambda^2 + 167\lambda + 285$

(D) $\lambda^3 + 23.23\lambda^2 - 158.3\lambda + 313$

(4). The eigenvalues of a $4 \times 4$ matrix $[A]$ are given as $2, -3, 13$, and $7$. The $|\det(A)|$ then is

(A) 546

(B) 19

(C) 25

(D) cannot be determined

(5). If one of the eigenvalues of $[A]_{n \times n}$ is zero, it implies

(A) The solution to $[A][X] = [C]$ system of equations is unique

(B) The determinant of $[A]$ is zero

(C) The solution to $[A][X] = [0]$ system of equations is trivial

(D) The determinant of $[A]$ is nonzero

(6).Given that matrix $[A] = \begin{bmatrix} 8 & -4 & 2 \\ 4 & 0 & 2 \\ 0 & -2 & -3 \end{bmatrix}$ has an eigenvalue value of 4 with the corresponding Eigenvectors of $[x] = \begin{bmatrix} -4.5 \\ -4 \\ 1 \end{bmatrix}$, then $[A]^5[X]$ is

(A) $\begin{bmatrix} -18 \\ -16 \\ 4 \end{bmatrix}$

(B) $\begin{bmatrix} -4.5 \\ -4 \\ 1 \end{bmatrix}$

(C) $\begin{bmatrix} -4608 \\ -4096 \\ 1024 \end{bmatrix}$

(D) $\begin{bmatrix} -0.004395 \\ -0.003906 \\ 0.0009766 \end{bmatrix}$

For complete solution, go to

https://ma.mathforcollege.com/mcquizzes/04sle/quiz_04sle_eigenvaluesandeigenvectors_solution.pdf

# Problem Set

Chapter 04.10 Eigenvalues and Eigenvectors

## Eigenvalues and Eigenvectors Exercise

(1). The eigenvalues $\lambda$ of matrix $[A]$ are found by solving the equation(s)?

(A) $[A]\ [X] = [I]$

(B) $[A]\ [X] - \lambda[I] = 0$

(C) $|A = 0|$

(D) $|A - \lambda I| = 0$

**Answer:** D

(2). Find the eigenvalues and eigenvectors of

$$[A] = \begin{bmatrix} 10 & 9 \\ 2 & 3 \end{bmatrix}$$

using the determinant method

**Answer:** $(12, 1)$, $\begin{bmatrix} 0.9762 \\ 0.2169 \end{bmatrix}$, $\begin{bmatrix} 0.8381 \\ -0.8381 \end{bmatrix}$

(3). Find the eigenvalues and eigenvectors of

$$[A] = \begin{bmatrix} 4 & 0 & 1 \\ -2 & 0 & 1 \\ 2 & 0 & 1 \end{bmatrix}$$

using the determinant method

**Answer:** $(0, 4, 5615, 0.43845)$, $\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0.87193 \\ -0.27496 \\ 0.48963 \end{bmatrix}, \begin{bmatrix} -0.27816 \\ 3.5284 \\ 0.99068 \end{bmatrix}$

(4). Find the eigenvalues of these matrices by inspection

(A) $\begin{bmatrix} 2 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & 6 \end{bmatrix}$

(B) $\begin{bmatrix} 3 & 5 & 7 \\ 0 & -2 & 1 \\ 0 & 0 & 0 \end{bmatrix}$

(C) $\begin{bmatrix} 2 & 0 & 0 \\ 3 & 5 & 0 \\ 2 & 1 & 6 \end{bmatrix}$

**Answer:**

(A) $2, -3, 6$

(B) $3, -2, 0$

(C) $2, 5, 6$

(5). Find the largest eigenvalue in magnitude and its corresponding vector by using the power method

$$[A] = \begin{bmatrix} 4 & 0 & 1 \\ -2 & 0 & 1 \\ 2 & 0 & 1 \end{bmatrix}$$

Start with an initial guess of the eigenvector as

$$\begin{bmatrix} 1 \\ -0.5 \\ 0.5 \end{bmatrix}$$

**Answer:** 4.5615, $\begin{bmatrix} 1 \\ -0.31534 \\ 0.56154 \end{bmatrix}$ after 4 iterations

(6). Prove if $\lambda$ is an eigenvalue of $[A]$, then $\frac{1}{\lambda}$ is an eigenvalue of $[A]^{-1}$.

(7). Prove that square matrices $[A]$ and $[A]^T$ have the same eigenvalues.

(8). Show that $|\det(A)|$ is the product of the absolute values of the eigenvalues of $[A]$.

# Chapter 05.00: Physical Problem for Interpolation

Lesson: General Engineering

## Summary

The upward velocity of a rocket is given as a function of time at discrete data points. To find the velocity at a particular time requires interpolation.

## Problem Statement

To find the altitude, velocity, and acceleration profile of a rocket, a velocity probe in the rocket (Figure 1) measures its velocity. Table 1 shows some typical values of the rocket velocity profile as a function of time.

**Figure 1** A rocket launched into space: Source: NASA Langley Research Center, Office of Education, **edu.larc.nasa.gov/pstp/**

To determine the velocity at a particular time, one needs to interpolate the velocity vs. time data. Although you may be familiar with linear interpolation, where you draw a straight line between two data points, you also want to know how accurate your estimate is. This forces you to use other interpolation functions such as quadratic and cubic polynomials.

**Table 1**. Velocity as a function of time

| $t$ (s) | $v(t)$ (m/s) |
|---|---|
| 0 | 0 |
| 10 | 227.04 |
| 15 | 362.78 |
| 20 | 517.35 |
| 22.5 | 602.97 |
| 30 | 901.67 |

Can you also find the distance covered by the rocket from one point of time to another? Can you find the acceleration of the rocket at a particular time?

# Chapter 05.00: Physical Problem for Interpolation

Lesson: Chemical Engineering

## Summary

Textbook notes of a real world problem using interpolation.

## Problem Statement

Well, I am from India and we are in the habit of drinking "afternoon tea." The other day, my wife asked me to heat up some water in our kettle. I put 4 cups (you cannot have just 1 cup) of water in the kettle and put it on over our new flat-top burner.

"You are an engineer", quipped my wife teasingly. "Can you estimate how long it would take for the water to boil and the kettle to make that whistling sound?" Yes, she clearly knows that whistling sound reminds me of all the horror movies that keep me awake at night.

**Figure 1.** A kettle of water on a flat burner.

## Solution

A cup of water is about 200 ml in volume. So the total volume of water is about 800 ml. The burner for our flat top is rated at 1200 W. From first law of thermodynamics [1],

$$\Delta H + \Delta E_p + \Delta E_k = Q - W_{sh}$$

where

$$\Delta H = \text{change in internal energy},$$

$$\Delta E_p = \text{change in potential energy},$$

$$\Delta E_k = \text{change in kinetic energy},$$

$$Q = \text{heat added to the system},$$

$$W_{sh} = \text{work that is called shaft work.}$$

In this example

$$\Delta E_p = 0,$$

$$\Delta E_k = 0,$$

$$W_{sh} = 0,$$

giving

$$\Delta H = Q.$$

Assuming no heat is lost as the kettle is assumed to be thermally insulated, the amount of heat needed is

$$\begin{aligned} Q &= \Delta H \\ &= mC_p\Delta T \end{aligned}$$

where

$$m = \text{mass of water (kg)},$$

$$C_p = \text{specific heat} \left( \frac{J}{kg -^\circ C} \right),$$

$$\Delta T = \text{change in temperature},$$

and the values are given as

$$m = 800ml \times \frac{1kg}{1lit}$$
$$= 0.8kg$$

$$C_p = 4814 \frac{T}{Kg -^\circ K} \text{ from (Reference [1] - Table 8.1)}$$

$$\Delta T = 100^\circ C - 22^\circ C$$
$$= 78^\circ C$$

Assuming that the room temperature is $22^\circ C$ and the boiling temperature of water is $100^\circ C$.

$$Q = mC_p \Delta T$$
$$= (0.8)\,(4814)\,(78)$$
$$= 300394J.$$

Since the wattage of the heater is 1200 W, the time it would take to boil is

$$= \frac{300394}{1200}$$
$$\cong 250\ s$$
$$\cong 4\text{min } 10\text{s}$$

But, I do not see any interpolation here. One of the approximations made in the above formula is that the specific heat is constant over the temperature range of $22^\circ C$ to $100^\circ C$. But it is not a constant given in the Table 1.

**Table 1.** Specific heat of water as a function of temperature [2].

| Temperature | Specific heat |
|---|---|
| $^\circ C$ | $\frac{J}{kg.^\circ C}$ |
| 22 | 4181 |
| 42 | 4179 |
| 52 | 4186 |
| 82 | 4199 |
| 100 | 4217 |

One assumption one may make is to use the specific heat at the average temperature. In this case it is $\frac{22+100}{2} = 61^\circ C$ .

So how do we find $C_p\,(61^\circ C)$? We use interpolation to do that, that is, finding the value of a discrete function at a point that is not given to us. Using $C_p\,(61^\circ C)$ will give us a better estimate of how much time it would take to boil the water.

## References

(1) Levenspiel, Octave, Understanding Engineering Thermo, Prentice Hall, New Jersey, 1996.

(2) Incropera, F.P. and DeWitt, D.P., Introduction to Heat Transfer, Wiley, 4th edition, 2001.

## QUESTIONS

(1) Using the specific heat at the average temperature, how much is the difference in the estimated time for boiling the water.

(2) Use first, second and third order polynomial interpolation to estimate $C_p\,(61°C)$ by all the methods (except spline) you learned in class. What is the absolute relative approximate error for each order of polynomial approximation? How many significant digits are at least correct in your solution.

(3) Just by looking at the data in Table 1, it may be clear that the calculated time using interpolation will not be very different from that found using the approximate specific heat. But in case of solids, it can be quite a different story. For example, to calculate heat required to raise the temperature of graphite from room temperature to $800°C$ for pyrolization, one needs to use proper specific heat data. Check yourself to see the difference between using specific heat at room temperature and specific heat at average temperature for the following problem. Find the heat required to raise the temperature of 1 kg of graphite from room temperature of $22°C$ to $800°C$, given the table of specific heat vs. temperature below.

**Table 2** Specific heat of graphite as a function of temperature.

| Temperature | Specific heat |
|---|---|
| $°C$ | $\frac{J}{kg.°C}$ |
| $-73$ | 420 |
| 127 | 1070 |
| 327 | 1370 |
| 527 | 1620 |
| 727 | 1820 |

# Chapter 05.00: Physical Problem for Interpolation

Lesson: Civil Engineering

## Summary

Bass fishing never got so technical. Find how interpolation can help you to have a great catch and let you tell others a true fish story.

## Problem Statement

This is a conversation between a former instructor (Autar) and civil engineering alumni named John Q. This is just what we heard!

**Autar:** "That is interesting! You are now a professional civil engineer and love to go bass fishing."

**John Q:** "Actually, my civil engineering education is helping me in being a good bass fisherman!"

**Autar:** "How is that?"

**John Q:** "Well, if you know where the thermocline is in the lake, you can find a lot of bass there!"

**Autar:** "Glad I asked? Educate me."

**John Q:** "Well, the water in a lake generally has three layers – epilimonon, thermocline and hypolimnion as shown in Figure 1 below. The thermocline layer is sandwiched between the epilimonon and the hypolimnion. The sun heats the water at the surface and this layer (epilimonon) of warm water floats over a layer (hypolimnion) of colder water, as the warm water is less dense then cold water. As days become hotter during the summer the layers become very distinct. Between these two layers of warm and cold water, you have a thin layer called the thermocline. Bass love the thermocline. So if you know where this thin layer of thermocline is, you can have a great catch."

**Autar:** "But why do bass fish like the thermocline?"

**John Q:** "Well, the upper layer (epilimonon) has too much light for the bass to be calm, while the lower layer (hypolimnion) has too little oxygen. The thermocline is also ideal for algae growth. So, it is the place of choice for the bass. You will be wasting your time if you fish below the thermocline."

**Autar:** "So how does one find where the thermocline is?"

**John Q:** "Well, you got electronic gadgets, such as depth finders, LCRs, and thermometers to do that. What some of these gadgets measure is the depth at which drastic change in temperature takes place? Just look at the data in Figure 2 taken in a lake in Pennsylvania. It shows the temperature data as a function of depth. You can see where it changes temperature suddenly. That is where the thermocline is.

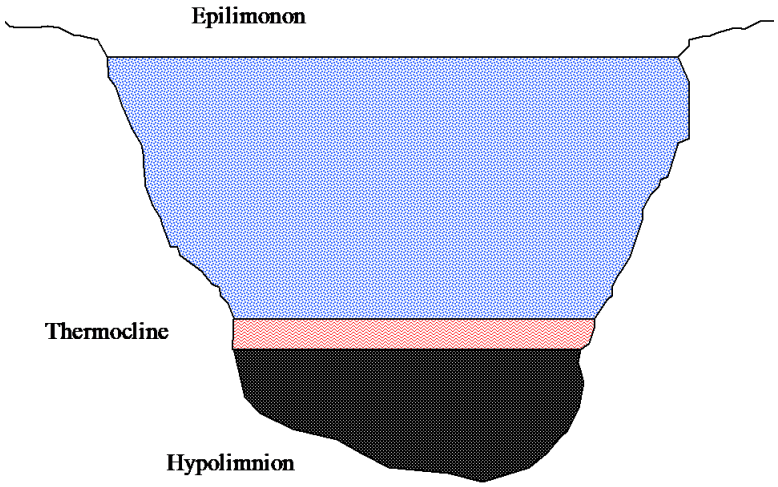
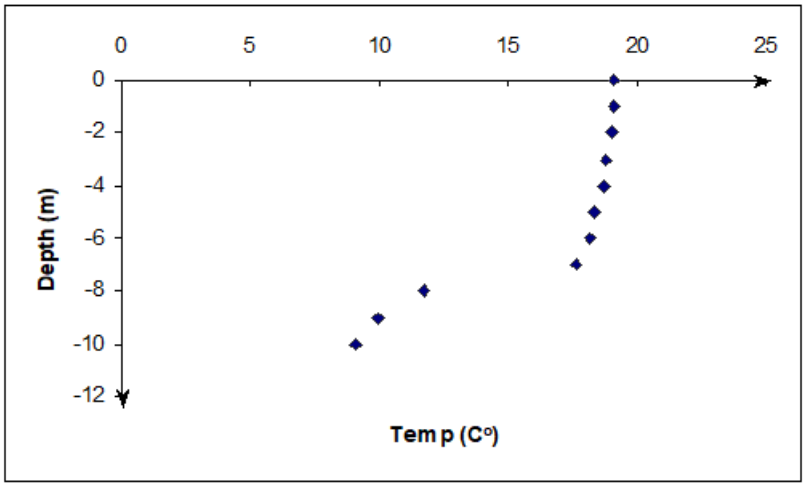



**Figure 1** Three layers of lake stratification.




| Temperature | Depth |
|---|---|
| °C | m |
| 19.1 | 0 |
| 19.1 | -1 |
| 19 | -2 |
| 18.8 | -3 |
| 18.7 | -4 |
| 18.3 | -5 |
| | |
| 18.2 | -6 |
| 17.6 | -7 |
| 11.7 | -8 |
| 9.9 | -9 |
| 9.1 | -10 |

**Figure 2** Temperature as a function of depth (Data courtesy of Ms. Bartlett - http://www.lehigh.edu/~infolios/becky/lakegraph.htm (last accessed January 2012).

**Autar:** "So what one can do is interpolate the temperature vs. depth data. The depth at which the thermocline occurs is the inflection point of the temperature-depth curve."

**John Q:**"You are still in the habit of using jargon like inflection points. It still gives me beautiful dreams of Calculus II!"

**Autar:** "Yes, simply said, the inflection point is where the second derivative of temperature with respect to the depth becomes zero. That is $\frac{d^2T}{dz^2} = 0$, where $T$ is the temperature at depth $z$. I think I am going to taking this problem to the classroom to teach my students about interpolation. Thanks to you."

# Chapter 05.00: Physical Problem for Interpolation

Lesson: Computer Engineering

## Summary

A robot arm path needs to be developed over several points on a flat plate. The path needs to be smooth to avoid sudden jerky motion and at the same time needs to be short.

**Peter:** "Dr. Kaw, I am taking a course in Manufacturing. We are solving the following problem. A robot arm with a rapid laser is used to do a quick quality check, such as hole radius, on six holes on a rectangular plate $15'' \times 10''$ at several points as shown in this table.

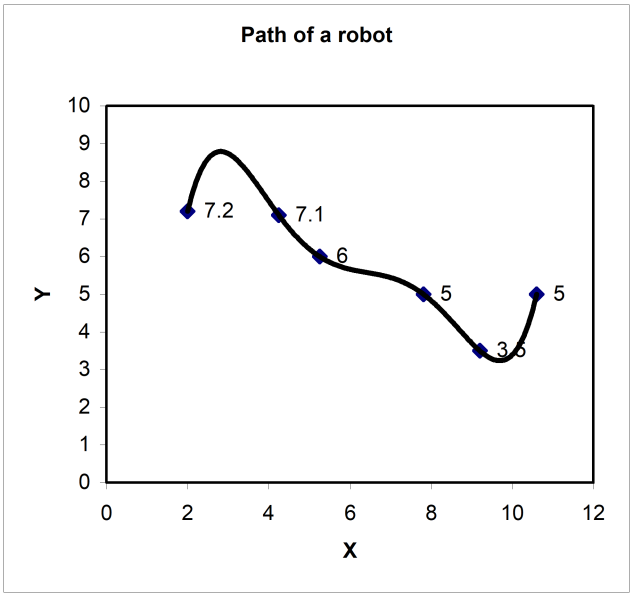



**Figure 1.** Path of a robot arm

**Table 1.** Coordinates of the points on the table

| X | Y |
|---|---|
| **inches** | **inches** |
| 2.00 | 7.2 |
| 4.25 | 7.1 |
| 5.25 | 6.0 |
| 7.81 | 5.0 |

| X | Y |
|---|---|
| 9.20 | 3.5 |
| 10.60 | 5.0 |

I am using Excel to fit a fifth order polynomial through the 6 points. But, when I plot the polynomial, it is giving a long path!"

**Kaw**: "Why do you not just join the consecutive points by a straight line; just like the kids do at Pizza Hut with those 'Connect the dots' activities?"

**Peter**: "You are making me hungry and I wish it were that easy. The path of the robot going from one point to another needs to be smooth so as to avoid sharp jerks in the arm that can otherwise create premature wear and tear of the robot arm."

**Kaw**: "As I recall, you took my course in Numerical Methods. What was that – one year ago?"

**Peter**: "Yes, your memory is sharp but my retention from that course – can we not talk about that!"

**Kaw**: "Come into my office. I wrote this program using Maple as you did. See this function, $f(x) = 1/(1 + 25x^2)$. I am choosing 7 points equidistantly between –1 and 1. Now look at the sixth order interpolating polynomial and the original function (See Figure 1). See the oscillations in the interpolating polynomial. In 1901, Runge [5] used this example function to show that higher order interpolation is a bad idea. A better alternative to getting a better representation of the curve is using splines. This is the solution to your problem as well. It will give you a smooth curve with less oscillations, and a shorter path. Try it!"

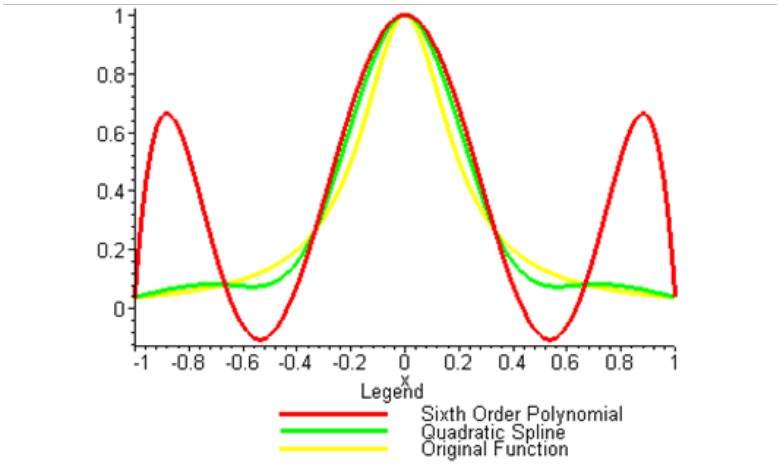



**Figure 2** Runge's function interpolated

# Chapter 05.00: Physical Problem for Interpolation

Lesson: Electrical Engineering


## Summary

Thermistors measure temperature based on the principle that resistance of thermistor material changes with temperature. Hence, a manufacturer supplies a resistance vs. temperature calibration curve. This curve is developed using interpolation.


Thermistors are temperature-measuring devices based on the principle that the thermistor material exhibits a change in electrical resistance with a change in temperature. By measuring the resistance of the thermistor material, one can then determine the temperature.

Thermistors are generally a piece of semiconductor made from metal oxides such as those of manganese, nickel, cobalt, etc. These pieces may be made into a bead, disk, wafer, etc depending on the application.

There are two types of thermistors – negative temperature coefficient (NTC) and positive temperature coefficient (PTC) thermistors. For NTCs, the resistance decreases with temperature, while for PTCs, the resistance increases with temperature. It is the NTCs that are generally used for temperature measurement.

Why would we want to use thermistors for measuring temperature as opposed to other choices such as thermocouples? It is because thermistors have high sensitivity giving more accuracy, a fast response to temperature changes for accuracy and quicker measurements, and relatively high resistance for decreasing the errors caused by the resistance of lead wires themselves.

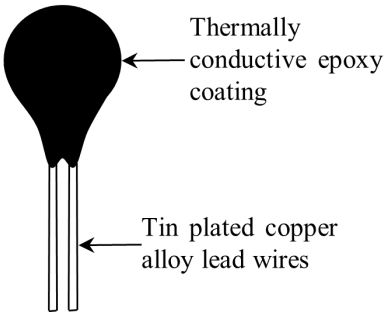



**Figure 1**. A typical thermistor

But thermistors have a nonlinear output and are valued for a limited range. So, when a thermistor is manufactured, the manufacturer supplies a resistance vs. temperature curve. The curve generally used that gives an accurate representation is given by

$$\frac{1}{T} = a_0 + a_1 \ln(R) + a_2 \{\ln(R)\}^2 + a_3 \{\ln(R)\}^3 \quad (1)$$

where

$T$ is temperature in Kelvin, and

$R$ is resistance in ohms.

$a_0, a_1, a_2, a_3$ are constants of the calibration curve.

Making change of variables

$$y = \frac{1}{T}, \text{ and}$$

$$x = \ln R,$$

we can change the calibration curve to a polynomial

$$y = a_0 + a_1 x + a_2 x^2 + a_3 x^3.$$

So if one is able to find the constants of the above formula, one can then use the calibration curve to find the temperature.

Given below is the data of resistance vs. temperature for a thermistor. Can you find the calibration curve?

**Table 1** Resistance vs. temperature data for calibration of a thermistor

| R | T |
|---|---|
| Ohm | $^\circ C$ |
| 1101.0 | 25.113 |
| 911.3 | 30.131 |
| 636.0 | 40.120 |
| 451.1 | 50.128 |

## References


[1] Betatherm sensors, http://www.betatherm.com

[2] Valvanao, J., "Measuring Temeparture Using Thermistors", Curcuit Cellar Online, August 2000, http://www.circuitcellar.com/online

[3] Lavenuta, G., "Negative Temperature Coefficient Thermistors: Part 1: Characteristics, Materials, and Configurations", http://www.globalspec.com/cornerstone/ref/negtemp.html

[4] Potter, D., "Measuring Temperature with Thermistors – a Tutorial", National Instruments Application Note 065, http://www.seas.upenn.edu/courses/belab/ReferenceFiles/Thermisters/an065.pdf

[5] Steinhart, J.S. and Hart, S.R., 1968. "Calibration Curves for Thermistors," Deep Sea Research 15:497.

[6] Sapoff, M. et al. 1982. "The Exactness of Fit of Resistance-Temperature Data of Thermistors with Third-Degree Polynomials," Temperature, Its Measurement and Control in Science and Industry, Vol. 5, James F. Schooley, ed., American Institute of Physics, New York, NY:875.

[7] Siwek, W.R., et al. 1992. "A Precision Temperature Standard Based on the Exactness of Fit of Thermistor Resistance-Temperature Data Using Third Degree Polynomials," Temperature, Its Measurement and Control in Science and Industry, Vol. 6, James F. Schooley, ed., American Institute of Physics, New York, NY:491-496.

# Chapter 05.00: Physical Problem for Interpolation

Lesson: Industrial Engineering

## Summary

To program a milling machine to make a cam profile, one needs to use interpolation to develop the path of the profile. The interpolation then requires a solution of simultaneous linear equations.

## Problem Statement

An Industrial Engineer needs to program a Computer Numerical Control (CNC) milling machine to fabricate a cam profile that was designed by a Industrial Engineer to operate the intake valves in an internal combustion engine. The Mechanical Engineer's task was to design a disk cam (rotating counterclockwise) to move a radial roller follower (in the vertical y-direction) as shown in Figure 1 below.

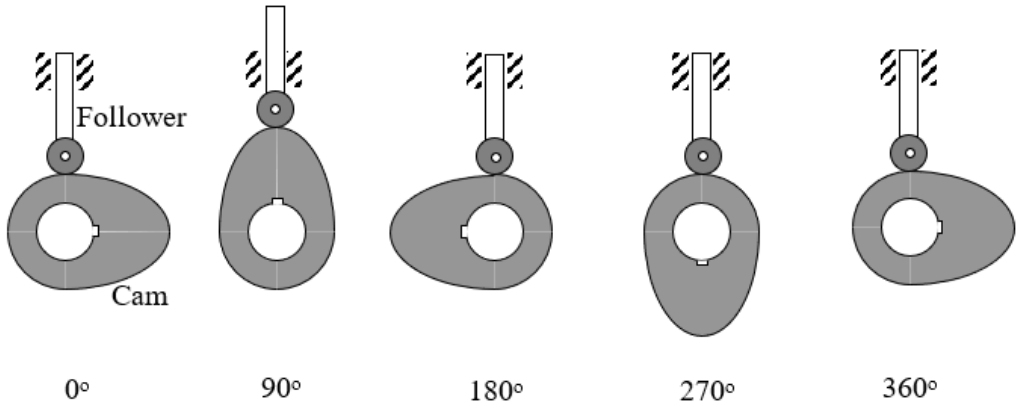



**Figure 1** Motion of cam and follower.

Specifically, the cam is to move the follower as described in Table 1 below.

**Table 1** Cam follower movement as a function of cam rotation.

| Cam rotation from X-axis | Follower movement in Y-direction |
|---|---|
| 0° | 0.0 |
| 90° | 1.0 |
| 180° | 0.0 |
| 270° | 0.0 |

| Cam rotation from X-axis | Follower movement in Y-direction |
|---|---|
| 360° | 0.0 |

## Solution

The Mechanical Engineer has specified seven points along the profile of the cam (see Figure below) at 30° increments as shown in Figure 2 below.

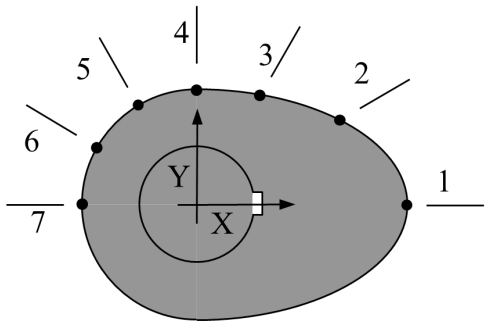



**Figure 2** Schematic of cam profile

The geometry of the cam (i.e., coordinates of the seven points on the cam surface) are given in Table 2 below.

**Table 2** Geometry of the cam.

| Point | Angle from X-axis | $X$ | $Y$ |
|---|---|---|---|
| 1 | −90° | 2.20 | 0.00 |
| 2 | −60° | 1.28 | 0.88 |
| 3 | −30° | 0.66 | 1.14 |
| 4 | −0° | 0.00 | 1.20 |
| 5 | 30° | −0.60 | 1.04 |
| 6 | 60° | −1.04 | 0.60 |
| 7 | 90° | −1.20 | 0.00 |

The Industrial Engineer is responsible for fitting a *smooth* curve through the 7 points keeping in mind that the final curve must have a infinite slope at points 1 and 7, and zero slope at point 4.

# Chapter 05.00: Physical Problem for Interpolation

Lesson: Mechanical Engineering

## Summary

Find the coefficient of linear thermal expansion of steel at a specific temperature to estimate whether a steel shaft will cool down enough to shrink fit into a hollow hub. The coefficient of linear thermal expansion is to be found by using interpolation from a given table of coefficient of linear thermal expansion of steel as a function of temperature.

## Problem Statement

To make the fulcrum (Figure 1) of a bascule bridge, a long hollow steel shaft called the trunnion is shrink fit into a steel hub.

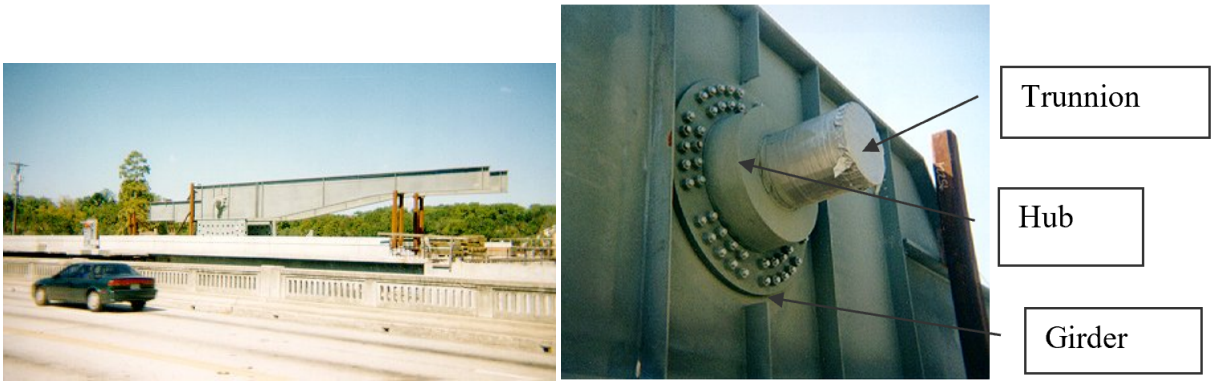



**Figure 1** Trunnion-Hub-Girder (THG) assembly.

This is done by first immersing the trunnion in a cold medium such as a dry-ice/alcohol mixture. After the trunnion reaches a steady-state temperature, that is, the temperature of the cold medium, the trunnion outer diameter contracts, is taken out and slid through the hole of the hub (Figure 2).

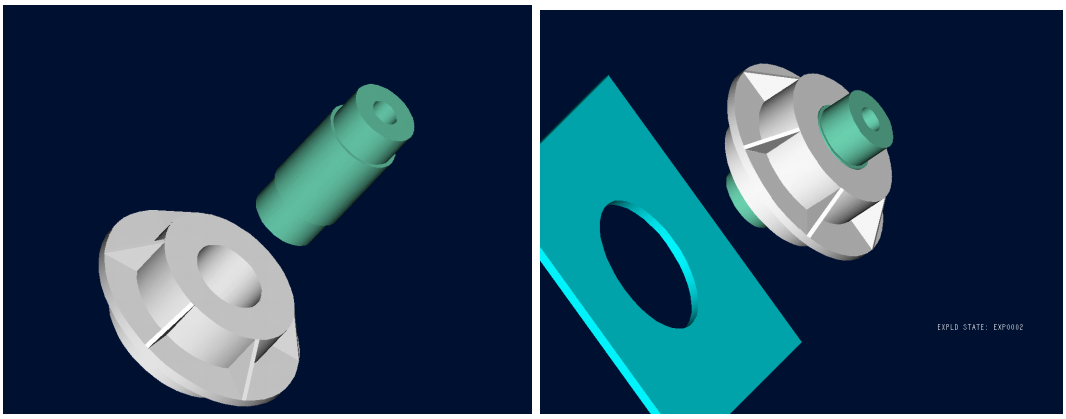



**Figure 2** Trunnion slid through the hub after contracting

When the trunnion heats up, it expands and creates an interference fit with the hub. In 1995, on one of the bridges in Florida, this assembly procedure did not work as designed. Before the trunnion could be inserted fully into the hub, the trunnion got stuck. So a new trunnion and hub had to be ordered worth $50,000. Coupled with construction delays, the total loss ran into more than a hundred thousand dollars.

Why did the trunnion get stuck? This was because the trunnion had not contracted enough to slide through the hole.

Now the same designer is working on making the fulcrum for another bascule bridge. Can you help him so that he does not make the same mistake?

For this new bridge, he needs to fit a hollow trunnion of an outside diameter $12.363''$ in a hub of an inner diameter $12.358''$. His plan is to put the trunnion in dry-ice/alcohol mixture (the temperature of dry ice/alcohol mixture is $-108°$F) to contract the trunnion so that it can be slid through the hole of the hub. To slide the trunnion without sticking, he has also specified a diametral clearance of at least $0.01''$. Assume the room temperature is $80°$F, is immersing it in a dry-ice/alcohol mixture a correct decision?

## Solution

Looking at the records of the designer for the previous bridge where the trunnion got stuck in the hub, it was found that he used the coefficient of linear thermal expansion at room temperature to calculate the contraction in the trunnion diameter. In that case, the reduction, $\Delta D$ in the outer diameter of the trunnion is

$$\Delta D = D\alpha\Delta T \qquad (1)$$

where

$$D = \text{outer diameter of the trunnion,}$$

$$\alpha = \text{coefficient of linear thermal expansion at room temperature,}$$

$$\Delta T = \text{change in temperature.}$$

Given

$$D = 12.363''$$

$$\alpha = 6.817 \times 10^{-6} \text{ in/in/°F at } 80°\text{F}$$

$$\begin{aligned} \Delta T &= T_{\text{fluid}} - T_{\text{room}} \\ &= -108 - 80 \\ &= -188°\text{F} \end{aligned}$$

where

$$T_{fluid} = \text{temperature of dry-ice/alcohol mixture}$$

$$T_{room} = \text{room temperature}$$

the reduction in the trunnion outer diameter is given by

$$\begin{aligned} \Delta D &= 12.363 \times \left(6.47 \times 10^{-6}\right)(-188) \\ &= -0.01504'' \end{aligned}$$

So the trunnion is predicted to reduce in diameter by $0.01504''$. But is this enough reduction in diameter? As per the specifications, he needs the trunnion to contract by

$$\begin{aligned} &= \text{trunnion outside diameter} - \text{hub inner diameter} + \text{diametric clearance} \\ &= 12.363'' - 12.358'' + 0.01'' \\ &= 0.015'' \end{aligned}$$

So, according to his calculations, it is enough to put the steel trunnion in a dry-ice/alcohol mixture to get the desired contraction of $0.015''$ as he is predicting a contraction of $0.01504''$.

But as shown in the graph below, the coefficient of linear thermal expansion of steel decreases with temperature and is not constant over the range of temperature the trunnion goes through. Hence the above formula (Equation 1) would overestimate the thermal contraction. This assumption is the mistake he made in the calculations for the earlier bridge.

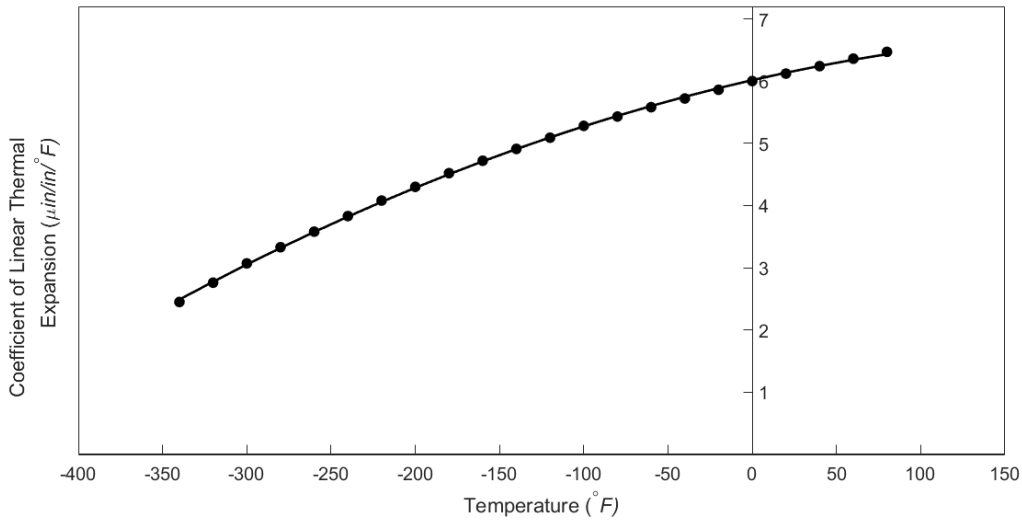



**Figure 3** Varying coefficient of linear thermal expansion as a function of temperature for cast steel.

To get a better estimate of the contraction in the diameter, we can use the coefficient of linear thermal expansion value at the average temperature. The average temperature would be

$$\begin{aligned} T_{avg} &= \frac{-108 + 80}{2} \\ &= -14^{\circ}\text{F} \end{aligned} \qquad (2)$$

Now given the table of coefficient of linear thermal expansion as a function of temperature as given below, we can use polynomial interpolation to find the coefficient of linear thermal expansion at the average temperature of $-14^{\circ}F$ and find the contraction using Equation (1).

**Table 1** Coefficient of linear thermal expansion vs. temperature

| Temperature (°F) | coefficient of linear thermal expansion ($\mu$in/in/°F) |
|---|---|
| 80 | 6.47 |
| 60 | 6.36 |
| 40 | 6.24 |
| 20 | 6.12 |
| 0 | 6.00 |
| −20 | 5.86 |
| −40 | 5.72 |
| −60 | 5.58 |
| −80 | 5.43 |
| −100 | 5.28 |
| −120 | 5.09 |
| −140 | 4.91 |
| −160 | 4.72 |
| −180 | 4.52 |
| −200 | 4.30 |
| −220 | 4.08 |

| Temperature (°F) | coefficient of linear thermal expansion ($\mu$in/in/°F) |
|---|---|
| −240 | 3.83 |
| −260 | 3.58 |
| −280 | 3.33 |
| −300 | 3.07 |
| −320 | 2.76 |
| −340 | 2.45 |

## Questions

1. Is cooling in dry-ice/alcohol mixture still your recommendation to get the required contraction in the trunnion?

# Chapter 05.01: Prerequisites to Interpolation

Lesson: Prerequisites to Interpolation

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *define interpolation*

2) *interpolate using a straight line.*

## What is interpolation?

Many times, as given in Figure 1, data is given only at discrete points such as $(x_0, y_0)$, $(x_1, y_1)$, ....., $(x_{n-1}, y_{n-1})$, $(x_n, y_n)$. So, how then does one find the value of $y$ at any other value of $x$? Well, a continuous function $f(x)$ may be used to represent the $n+1$ data values with $f(x)$ passing through the $n+1$ points (Figure 1). Then one can find the value of $y$ at any other value of $x$. This curve fitting technique is called *interpolation*.

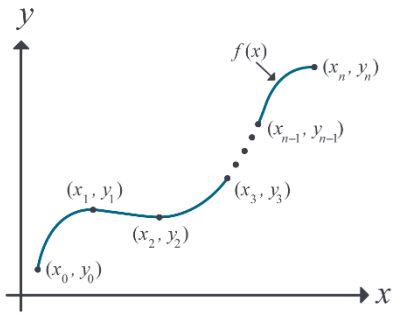



**Figure 1.** Interpolation of a function given at discrete points

The function $f(x)$ chosen for interpolation is called the *interpolant*. The value of $y$ at any other value of $x$ that is not at the discrete data points is called the *interpolated value*. Of course, if $x$ falls outside the domain of $x$ for which the data is given, it is no longer interpolation and is called *extrapolation*.

So, what kind of function $f(x)$ should one choose? A polynomial is a common choice for an interpolating function because polynomials are easy to

(A) evaluate,

(B) differentiate, and

(C) integrate

relative to other choices such as a trigonometric and exponential series. For example, given data points $(2,4),(3,10),(5,16)$, one may interpolate the data to

$$y(x) = a_0 + a_1 x + a_2 x^2,\ 2 \leq x \leq 5$$

Other examples of usable interpolants are

$$y(x) = a_0 + a_1 e^x + a_2 e^{2x},\ 2 \leq x \leq 5$$

$$y(x) = a_0 + a_1 \sin x + a_2 \sin 2x,\ 2 \leq x \leq 5$$

The problem hence reduces to finding the unknown coefficients, $a_0$, $a_1$, and $a_2$. Once these are found, one can estimate the value of the dependent variable value $y$ at any value of $x$ within the bounds of the $x$ values.

# Chapter 05.01: Prerequisites to Interpolation

Lesson: Uniqueness of Interpolating Polynomials

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *derive why a polynomial of degree n or less that passes through n + 1 data points is unique.*

## Proof for a unique polynomial of degree $n$ or less passes through $n+1$ data points.

Let us use proof by contradiction. If the polynomial is not unique, then at least two polynomials of order $n$ or less pass through the $n+1$ data points.

Assume two polynomials $P_n(x)$ and $Q_n(x)$ go through $n+1$ data points,

$$(x_0, y_0), (x_1, y_1), \ldots, (x_n, y_n)$$

Then

$$R_n(x) = P_n(x) - Q_n(x) \qquad (1)$$

Since $P_n(x)$and $Q_n(x)$ pass through all the $n+1$ data points,

$$P_n(x_i) = Q_n(x_i), i = 0, \ldots, n \qquad (2)$$

Hence

$$R_n(x_i) = P_n(x_i) - Q_n(x_i) = 0, i = 0, \ldots, n \qquad (3)$$

The $n^{th}$ order polynomial $R_n(x)$ has $n+1$ zeros. A polynomial of order $n$ can have $n+1$ zeros only if it is identical to a zero polynomial, that is,

$$R_n(x) \equiv 0 \qquad (4)$$

Hence from Equation (1)

$$P_n(x) \equiv Q_n(x)$$

**Example**

Show that a unique polynomial of degree 2 or less passes through 3 data points

**Solution**

Let us use proof by contradiction. If the polynomial is not unique, then at least two polynomials of order 3 or less pass through the 3 data points.

Assume two polynomials $P_2(x)$ and $Q_2(x)$ go through 3 data points (Figure 1),

$$(x_0, y_0), (x_1, y_1), (x_2, y_2)$$

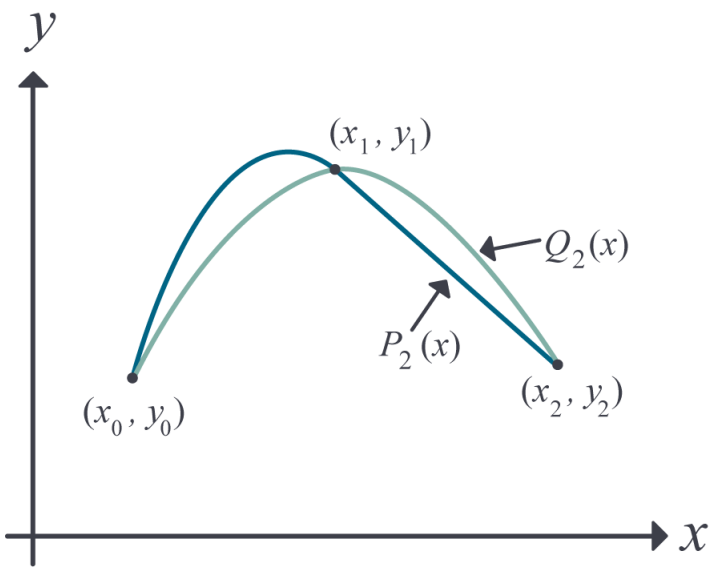



**Figure 1**. Interpolating polynomials through three points.

Then

$$R_2(x) = P_2(x) - Q_2(x) \qquad (E1.1)$$

Since $P_2(x)$and $Q_2(x)$ pass through all the 3 data points,

$$P_2(x_i) = Q_2(x_i), i = 0,\ 1,\ 2 \qquad (E1.2)$$

Hence

$$R_2(x_i) = P_2(x_i) - Q_2(x_i) = 0, i = 0,\ 1,\ 2 \qquad (E1.3)$$

Let $R_2(x)$ be written as

$$R_2(x) = a_0 + a_1 x + a_2 x^2. \qquad (E1.4)$$

The $2^{nd}$ order polynomial $R_2(x)$ has 3 zeros and these zeros are at $x_1$, $x_2$, and $x_3$. Then from Equation (E1.4)

$$\begin{aligned} R_2(x_1) &= a_0 + a_1 x_1 + a_2 x_1^2 = 0 \\ R_2(x_2) &= a_0 + a_1 x_2 + a_2 x_2^2 = 0 \\ R_2(x_3) &= a_0 + a_1 x_3 + a_2 x_3^2 = 0 \end{aligned} \qquad (E1.5)$$

which in the matrix form gives

$$\begin{bmatrix} 1 & x_1 & x_1^2 \\ 1 & x_2 & x_2^2 \\ 1 & x_3 & x_3^2 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \qquad (E1.6)$$

The above set of equations has a trivial solution, that is, $a_1 = a_2 = a_3 = 0$. But is this the only solution? That is true if the coefficient matrix is invertible.

The determinant of the coefficient matrix can be found symbolically with the forward elimination steps of Naïve Gauss elimination to give

$$\begin{aligned} \det \begin{bmatrix} 1 & x_1 & x_1^2 \\ 1 & x_2 & x_2^2 \\ 1 & x_3 & x_3^2 \end{bmatrix} &= x_2 x_3^2 - x_2^2 x_3 - x_1 x_3^2 + x_1^2 x_3 + x_1 x_2^2 - x_1^2 x_2 \\ &= (x_1 - x_2)(x_2 - x_3)(x_3 - x_1) \end{aligned} \qquad (E1.7)$$

Since

$$x_1 \neq x_2 \neq x_3$$

the determinant is non-zero. Hence the coefficient matrix is invertible. $a_1 = a_2 = a_3 = 0$ is the only solution, that is, $R_2(x) \equiv 0$ giving $P_2(x) \equiv Q_2(x)$ showing that the polynomial has to be unique.

# Multiple Choice Test

## Chapter 05.01 Prerequisites to Interpolation

(1). The number of different polynomials that can go through two fixed data points $(x_1, y_1)$ and $(x_2, y_2)$ is

(A) 0

(B) 1

(C) 2

(D) infinite

(2). Given $n + 1$ data pairs, a unique polynomial of degree _____ passes through $n + 1$ data points.

(A) $n + 1$

(B) $n + 1$ or less

(C) $n$

(D) $n$ or less

(3). The following function(s) can be used for interpolation:

(A) polynomial

(B) exponential

(C) trigonometric

(D) all of the above

(4). Polynomials are the most commonly used functions for interpolation because they are easy to

(A) evaluate

(B) differentiate

(C) integrate

(D) evaluate, differentiate and integrate

(5). Given $n+1$ data points $(x_0, y_0), (x_1, y_1), ....., (x_{n-1}, y_{n-1}), (x_n, y_n)$, assume you pass a function $f(x)$ through all the data points. If now the value of the function $f(x)$ is required to be found outside the range of the given $x$-data, the procedure is called

(A) extrapolation

(B) interpolation

(C) guessing

(D) regression

(6). Given three data points $(1, 6)$, $(3, 28)$, and $(10, 231)$, it is found that the function $y = 2x^2 + 3x + 1$ passes through the three data points. Your estimate of $y$ at $x = 2$ is most nearly

(A) 6

(B) 15

(C) 17

(D) 28

For complete solution, go to

http://nm.mathforcollege.com/mcquizzes/05inp/quiz_05inp_background_solution.pdf

# Chapter 05.02: Direct Method of Interpolation

Lesson: Direct Method of Interpolation

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *find the interpolant through the direct method,*

2) *choose the correct data points for interpolation,*

3) *solve problems using the direct method of interpolation,*

4) *use the direct method interpolants to find derivatives of discrete functions,*

5) *use the direct method interpolants to find integrals of discrete functions.*

## Direct Method

The direct method (also called the Vandermonde polynomial method) of interpolation is based on the following premise. Given $n+1$ data points, fit a polynomial of order $n$ as given below

$$y = a_0 + a_1 x + \ldots\ldots\ldots\ldots\ldots + a_n x^n \qquad (1)$$

through the data, where $a_0, a_1, \ldots\ldots\ldots, a_n$ are $n+1$ real constants. Since $n+1$ values of $y$ are given at $n+1$ values of $x$, one can write $n+1$ equations. Then the $n+1$ constants, $a_0, a_1, \ldots\ldots\ldots, a_n$ can be found by solving the $n+1$ simultaneous linear equations. To find the value of $y$ at a given value of $x$, simply substitute the value of $x$ in Equation 1.

But, it is not necessary to use all the data points. How does one then choose the order of the polynomial and what data points to use? This concept of choosing points and the direct method of interpolation are best illustrated using examples.

### Example 1

The upward velocity of a rocket is given as a function of time in Table 1.

**Table 1.** Velocity as a function of time.

| $t$ (s) | $v(t)$ (m/s) |
|---|---|
| 0 | 0 |
| 10 | 227.04 |
| 15 | 362.78 |
| 20 | 517.35 |
| 22.5 | 602.97 |
| 30 | 901.67 |

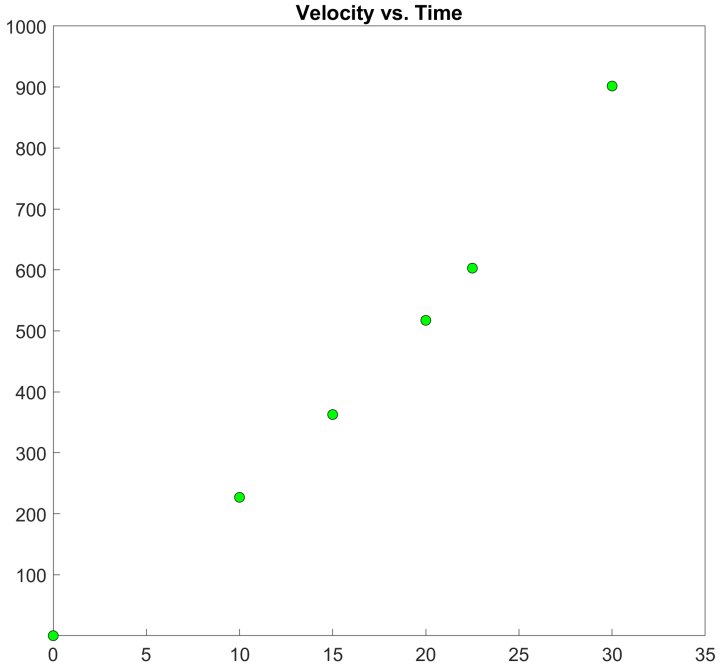



**Figure 1.** Graph of velocity vs. time data for the rocket example.

Estimate the velocity at $t = 16$ seconds using the direct method of interpolation with a first-order polynomial.

**Solution**

For first-order polynomial interpolation (also called linear interpolation), the velocity given by

$$v\left(t\right) = a_0 + a_1 t \qquad (E1.1)$$

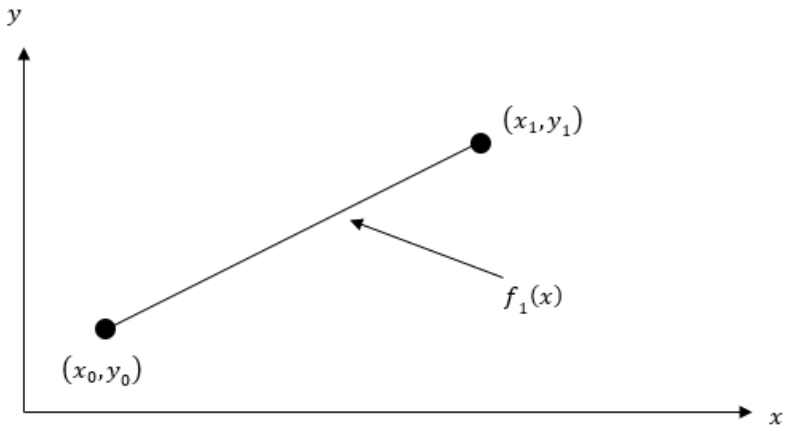



**Figure 2.** Linear interpolation.

Since we want to find the velocity at $t = 16$, and we are using a first-order polynomial, we need to choose the two data points that are closest to $t = 16$ that also bracket $t = 16$ to evaluate it. The two points are $t_0 = 15$ and $t_1 = 20$.

Then

$$t_0 = 15,\ v(t_0) = 362.78$$
$$t_1 = 20,\ v(t_1) = 517.35$$

Equation (E1.1) gives

$$v(15) = a_0 + a_1(15) = 362.78$$
$$v(20) = a_0 + a_1(20) = 517.35$$

Writing the equations in matrix form, we have

$$\begin{bmatrix} 1 & 15 \\ 1 & 20 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} 362.78 \\ 517.35 \end{bmatrix}$$

Solving the above two equations gives

$$a_0 = -100.93$$
$$a_1 = 30.914$$

Hence from Equation (E1.1)

$$\begin{aligned} v(t) &= a_0 + a_1 t \\ &= -100.93 + 30.914t,\ 15 \le t \le 20 \end{aligned}$$

At $t = 16$,

$$v(16) = -100.92 + 30.914 \times 16$$
$$= 393.70 \text{ m/s}$$

**Example 2**

The upward velocity of a rocket is given as a function of time in Table 2.

**Table 2.** Velocity as a function of time.

| $t$ (s) | $v(t)$ (m/s) |
|---|---|
| 0 | 0 |
| 10 | 227.04 |
| 15 | 362.78 |
| 20 | 517.35 |
| 22.5 | 602.97 |
| 30 | 901.67 |

a) Estimate the velocity at $t = 16$ seconds using the direct method of interpolation with a second-order polynomial.

b) Find the absolute relative approximate error for the second-order polynomial approximation.

c) Using the second-order polynomial interpolant for velocity from part (a), find the distance covered by the rocket from $t = 11$ $s$ to $t = 16$ $s$.

d) Using the second-order polynomial interpolant for velocity from part (a), find the acceleration of the rocket at $t = 16$ $s$.

**Solution**

For second-order polynomial interpolation (also called quadratic interpolation), the velocity is given by

$$v(t) = a_0 + a_1 t + a_2 t^2 \qquad (E2.1)$$

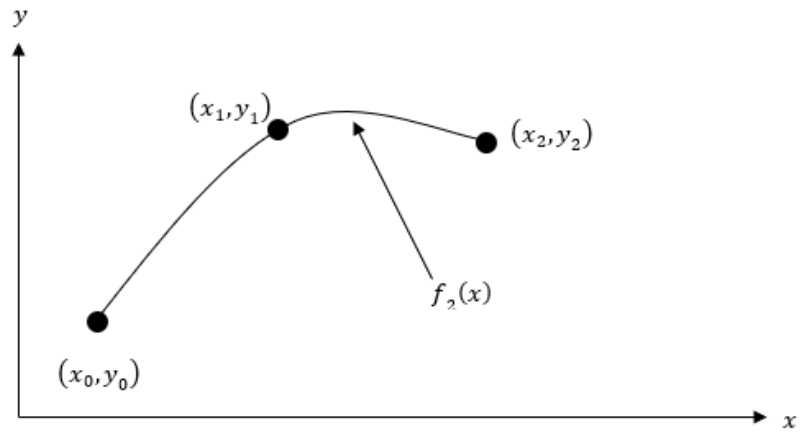


**Figure 3.** Quadratic interpolation.

a) Since we want to find the velocity at $t = 16$, and we are using a second-order polynomial, we need to choose the three data points that are closest to $t = 16$ that also bracket $t = 16$ to evaluate it. The three points are $t_0 = 10, \ \ t_1 = 15, \text{ and } t_2 = 20.$

Then

$$\begin{aligned} t_0 &= 10, \ v(t_0) = 227.04 \\ t_1 &= 15, \ v(t_1) = 362.78 \\ t_2 &= 20, \ v(t_2) = 517.35 \end{aligned}$$

Equation (E2.1) gives

$$\begin{aligned} v(10) &= a_0 + a_1(10) + a_2(10)^2 = 227.04 \\ v(15) &= a_0 + a_1(15) + a_2(15)^2 = 362.78 \\ v(20) &= a_0 + a_1(20) + a_2(20)^2 = 517.35 \end{aligned}$$

Writing the three equations in matrix form, we have

$$\begin{bmatrix} 1 & 10 & 100 \\ 1 & 15 & 225 \\ 1 & 20 & 400 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 227.04 \\ 362.78 \\ 517.35 \end{bmatrix}$$

Solving the above three equations gives

$$\begin{aligned} a_0 &= 12.050 \\ a_1 &= 17.733 \\ a_2 &= 0.37660 \end{aligned}$$

Hence from Equation (E2.1)

$$v(t) = 12.050 + 17.733t + 0.37660t^2, \ 10 \le t \le 20 \qquad (E2.2)$$

At $t = 16$,

$$\begin{aligned} v(16) &= 12.050 + 17.7333(16) + 0.37660(16)^2 \\ &= 392.19 \text{ m/s} \end{aligned}$$

b) The absolute relative approximate error $|\epsilon_a|$ obtained between the results from the first- and second-order polynomial is

$$\begin{aligned} |\epsilon_a| &= \left| \frac{392.19 - 393.70}{392.19} \right| \times 100 \\ &= 0.38410\% \end{aligned}$$

c) The distance covered by the rocket between $t = 11\ s$ and $t = 16\ s$ can be calculated from the interpolating polynomial (Equation E2.2)

$$v(t) = 12.050 + 17.733t + 0.37660t^2,\ 10 \leq t \leq 20$$

Note that the polynomial is valid between $t = 10$ s and $t = 22.5$ s and hence includes the limits of integration of $t = 11$ s and $t = 16$ s can

So

$$\begin{aligned} s(16) - s(11) &= \int_{11}^{16} v(t)\, dt \\ &= \int_{11}^{16} \left(12.050 + 17.733t + 0.37660t^2\right)\, dt \\ &= \left[12.050t + 17.733\frac{t^2}{2} + 0.37660\frac{t^3}{3}\right]_{11}^{16} \\ &= 1604.3 \text{ m} \end{aligned}$$

d) The acceleration at $t = 16$ s is given by

$$a(16) = \left.\frac{d}{dt}v(t)\right|_{t=16}$$

Given that from Equation (E2.2)

$$v(t) = 12.050 + 17.733t + 0.37660t^2,\ 10 \leq t \leq 20$$

we get

$$\begin{aligned} a(t) &= \frac{d}{\mathrm{dt}}v(t) \\ &= \frac{d}{\mathrm{dt}}\left(12.050 + 17.733t + 0.37660t^2\right) \\ &= 17.733 + 0.75320t,\ 10 \leq t \leq 20 \end{aligned}$$

Hence

$$\begin{aligned} a(16) &= 17.733 + 0.75320(16) \\ &= 29.784 \text{ m/s}^2 \end{aligned}$$

**Example 3**

The upward velocity of a rocket is given as a function of time in Table 3.

**Table 3.** Velocity as a function of time.

| $t$ (s) | $v(t)$ (m/s) |
|---|---|
| 0 | 0 |
| 10 | 227.04 |
| 15 | 362.78 |
| 20 | 517.35 |
| 22.5 | 602.97 |
| 30 | 901.67 |

a) Determine the value of the velocity at $t = 16$ seconds using the direct method of interpolation with a third-order polynomial.

b) Find the absolute relative approximate error for the third-order polynomial approximation.

c) Using the third-order polynomial interpolant for velocity from part (a), find the distance covered by the rocket from $t = 11\ s$ to $t = 16\ s$.

d) Using the third-order polynomial interpolant for velocity from part (a), find the acceleration of the rocket at $t = 16\ s$.

**Solution**

a) For third-order polynomial interpolation (also called cubic interpolation), we choose the velocity given by

$$v\,(t) = a_0 + a_1 t + a_2 t^2 + a_3 t^3 \qquad (E3.1)$$

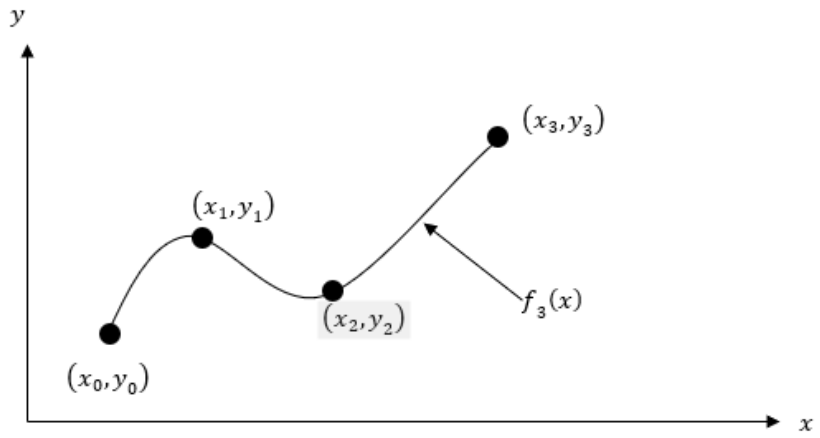



**Figure 4.** Cubic interpolation.

Since we want to find the velocity at $t = 16$, and we are using a third-order polynomial, we need to choose the four data points closest to $t = 16$ that also brackets $t = 16$ to evaluate it.

The four points are $t_0 = 10, t_1 = 15,\ t_2 = 20$ and $t_3 = 22.5$.

Then

$$
\begin{aligned}
t_0 &= 10,\ v(t_0) = 227.04 \\
t_1 &= 15,\ v(t_1) = 362.78 \\
t_2 &= 20,\ v(t_2) = 517.35 \\
t_3 &= 22.5,\ v(t_3) = 602.97
\end{aligned}
$$

Equation (E3.1)

$$
\begin{aligned}
v(10) &= a_0 + a_1(10) + a_2(10)^2 + a_3(10)^3 = 227.04 \\
v(15) &= a_0 + a_1(15) + a_2(15)^2 + a_3(15)^3 = 362.78 \\
v(20) &= a_0 + a_1(20) + a_2(20)^2 + a_3(20)^3 = 517.35 \\
v(22.5) &= a_0 + a_1(22.5) + a_2(22.5)^2 + a_3(22.5)^3 = 602.97
\end{aligned}
$$

Writing the four equations in matrix form, we have

$$
\begin{bmatrix}
1 & 10 & 100 & 1000 \\
1 & 15 & 225 & 3375 \\
1 & 20 & 400 & 8000 \\
1 & 22.5 & 506.25 & 11391
\end{bmatrix}
\begin{bmatrix}
a_0 \\ a_1 \\ a_2 \\ a_3
\end{bmatrix}
=
\begin{bmatrix}
227.04 \\ 362.78 \\ 517.35 \\ 602.97
\end{bmatrix}
$$

Solving the above four equations gives

$$
\begin{aligned}
a_0 &= -4.2540 \\
a_1 &= 21.266 \\
a_2 &= 0.13204 \\
a_3 &= 0.0054347
\end{aligned}
$$

Hence

$$
\begin{aligned}
v(t) &= a_0 + a_1 t + a_2 t^2 + a_3 t^3 \\
&= -4.2540 + 21.266t + 0.13204t^2 + 0.0054347t^3,\ 10 \le t \le 22.5
\end{aligned}
\qquad (E3.2)
$$

From Equation (E3.2)

$$
\begin{aligned}
v(16) &= -4.2540 + 21.266(16) + 0.13204(16)^2 + 0.0054347(16)^3 \\
&= 392.06 \text{ m/s}
\end{aligned}
$$

b) The absolute percentage relative approximate error $|\epsilon_a|$ for the value obtained for $v(16)$ between second and third-order polynomial is

$$
\begin{aligned}
|\epsilon_a| &= \left| \frac{392.06 - 392.19}{392.06} \right| \times 100 \\
&= 0.033269\%
\end{aligned}
$$

c) The distance covered by the rocket between $t = 11\ s$ and $t = 16\ s$ can be calculated from the interpolating polynomial (Equation E3.2)

$$v(t) = -4.2540 + 21.266t + 0.13204t^2 + 0.0054347t^3,\ 10 \leq t \leq 22.5$$

Note that the polynomial is valid between $t = 10$ s and $t = 22.5$ s and hence includes the limits of integration of $t = 11$ s and $t = 16$ s.

So

$$\begin{aligned} s(16) - s(11) &= \int_{11}^{16} v(t)\,\{dt \\ &= \int_{11}^{16} \left(-4.2540 + 21.266t + 0.13204t^2 + 0.0054347t^3\right) dt \\ &= \left[-4.2540t + 21.266\frac{t^2}{2} + 0.13204\frac{t^3}{3} + 0.0054347\frac{t^4}{4}\right]_{11}^{16} \\ &= 1605.0 \text{ m} \end{aligned}$$

d) The acceleration at $t = 16$ s is given by

$$a(16) = \left.\frac{d}{dt}v(t)\right|_{t=16}$$

Given that from Equation (E3.2)

$$v(t) = -4.2540 + 21.266t + 0.13204t^2 + 0.0054347t^3,\ 10 \leq t \leq 22.5$$

we get

$$\begin{aligned} a(t) &= \frac{d}{dt}v(t) \\ &= \frac{d}{dt}\left(-4.2540 + 21.266t + 0.13204t^2 + 0.0054347t^3\right) \\ &= 21.266 + 0.26408t + 0.016304t^2,\ 10 \leq t \leq 22.5 \end{aligned}$$

Hence

$$\begin{aligned} a(16) &= 21.266 + 0.26408(16) + 0.016304(16)^2 \\ &= 29.665 \text{ m/s}^2 \end{aligned}$$

# Multiple Choice Test

## Chapter 05.02 Direct Method of Interpolation

(1). Given $n+1$ data pairs, a unique polynomial of degree _____ passes through $n+1$ data points.

(A) $n+1$

(B) $n+1$ or less

(C) $n$

(D) $n$ or less

(2). The data of the velocity of a body as a function of time is given as follows.

| | | | | | |
|---|---|---|---|---|---|
| Time $(s)$ | 0 | 15 | 18 | 22 | 24 |
| Velocity $(m/s)$ | 22 | 24 | 37 | 25 | 123 |

The velocity in m/s at $16s$ using linear polynomial interpolation is most nearly

(A) 27.867

(B) 28.333

(C) 30.429

(D) 43.000

(3). The following data of the velocity of a body as a function of time is given as follows.

| | | | | | |
|---|---|---|---|---|---|
| Time $(s)$ | 0 | 15 | 18 | 22 | 24 |
| Velocity $(m/s)$ | 22 | 24 | 37 | 25 | 123 |

The velocity in m/s at $16s$ using quadratic polynomial interpolation is most nearly

(A) 27.867

(B) 28.333

(C) 30.429

(D) 43.000

(4). The following data of the velocity of a body is given as a function of time

| | | | | | |
|---|---|---|---|---|---|
| Time, $t$ $(s)$ | 0 | 15 | 18 | 22 | 24 |
| Velocity, $v$ $(m/s)$ | 22 | 24 | 37 | 25 | 123 |

Using quadratic interpolation, the interpolant

$v(t) = 8.667t^2 - 349.67t + 3523,\ 18 \leq t \leq 24$

approximates the velocity of the body. From this information, one of the times in seconds at which the velocity of the body is 35 m/s during the above time interval of $t = 18s$ to $t = 24s$ is

(A) 18.667

(B) 20.850

(C) 22.200

(D) 22.294

(5). The following data of the velocity of a body is given as a function of time

| | | | | | |
|---|---|---|---|---|---|
| Time, $t$ $(s)$ | 0 | 15 | 18 | 22 | 24 |
| Velocity, $v$ $(m/s)$ | 22 | 24 | 37 | 25 | 123 |

One of the interpolant approximations for the velocity from the above data is given as

$$v(t) = 8.6667t^2 - 349.67t + 3523,\ 18 \leq t \leq 24$$

Using the above interpolant, the distance in meters covered by the body between $t = 19s$ and $t = 22s$ is most nearly

(A) 10.337

(B) 88.500

(C) 93.000

(D) 168.00

(6). The following data of the velocity of a body is given as a function of time.

| | | | | | |
|---|---|---|---|---|---|
| Time $(s)$ | 0 | 15 | 18 | 22 | 24 |
| Velocity $(m/s)$ | 22 | 24 | 37 | 25 | 123 |

If you were going to use quadratic interpolation to find the value of the velocity at $t = 14.9s$ , what three data points of time would you choose for interpolation?

(A) $0, 15, 18$

(B) $15, 18, 22$

(C) $0, 15, 22$

(D) $0, 18, 24$

For complete solution, go to

http://nm.mathforcollege.com/mcquizzes/05inp/quiz_05inp_direct_solution.pdf

# Problem Set

## Chapter 05.02: Direct Method of Interpolation

(1). The following data of the velocity of a body as a function of time is given

| Time $(s)$ | 0 | 15 | 18 | 22 | 24 |
|---|---|---|---|---|---|
| Velocity $(m/s)$ | 22 | 24 | 37 | 25 | 123 |

What is the velocity at $t = 14$ seconds using linear polynomial interpolation?

**Answer:** 23.86 $m/s$

(2). The following data of the velocity of a body as a function of time is given

| Time $(s)$ | 0 | 15 | 18 | 22 | 24 |
|---|---|---|---|---|---|
| Velocity $(m/s)$ | 22 | 24 | 37 | 25 | 123 |

What is the velocity at $t = 14$ seconds using quadratic polynomial interpolation?

**Answer:** 20.60 $m/s$

(3). The following data of the velocity of a body as a function of time is given

| Time $(s)$ | 0 | 15 | 18 | 22 | 24 |
|---|---|---|---|---|---|
| Velocity $(m/s)$ | 22 | 24 | 37 | 25 | 123 |

What is the velocity at $t = 14$ seconds using cubic polynomial interpolation?

**Answer:** 17.30 $m/s$

(4). The following data of the velocity of a body as a function of time is given as follows.

| Time $(s)$ | 0 | 15 | 18 | 22 | 24 |
|---|---|---|---|---|---|
| Velocity $(m/s)$ | 22 | 24 | 37 | 25 | 123 |

Using quadratic interpolation, the interpolant

$$v(t) = 9.500t^2 - 383.0t + 3853$$

approximates the velocity of the body between 18 and 24 seconds. From this information, can you find out when the velocity of the body is 26 m/s during the above time interval?

**Answer:** 18.287 $s$ or 22.028 $s$

(5). The following data of the velocity of a body as a function of time is given as follows.

| Time $(s)$ | 0 | 15 | 18 | 22 | 24 |
|---|---|---|---|---|---|
| Velocity $(m/s)$ | 22 | 24 | 37 | 25 | 123 |

Find the value of velocity at $t = 16$ seconds using the following interpolation function for the velocity of the body

$$v(t) = a_0 + a_1 \sin(t) + a_2 \sin(2t)$$

where $a_0,\ a_1$, and $a_2$ are the unknown constants.

**Answer:** 24.875 $m/s$ (Intermediate answers $a_0 = 25.008$, $a_1 = -9.2636$; $a_2 = -5.0770$ (arguments of sin are radians )

(6). You are given data for the upward velocity of a rocket as a function of time in the table below.

| $t\ (s)$ | 0 | 10 | 15 | 20 | 22.5 | 30 |
|---|---|---|---|---|---|---|
| $v(t)\ (m/s)$ | 0 | 227.04 | 362.78 | 517.35 | 602.97 | 901.67 |

a) Determine the value of the velocity at $t = 16$ seconds using first order polynomial direct method.

b) Determine the value of the velocity at $t = 16$ seconds using second order polynomial direct method. Find the absolute relative approximate error for approximation from the second order polynomial. How many significant digits would you at least trust in the solution?

c) Determine the value of the velocity at $t = 16$ seconds using third order polynomial direct method. Find the absolute relative approximate error for approximation from the third order polynomial. How many significant digits would you at least trust in the solution?

d) What is the true error for part (a), (b) and (c), if I told you that the data given in the table above was derived from the formula

$$v(t) = 2000 \ln\left[\frac{14 \times 10^4}{14 \times 10^4 - 2100t}\right] - 9.8t, \quad 0 \le t \le 30$$

**Answer:** *a*) 393.69 $m/s$

*b*) 392.19 $m/s,\ 0.38247\%,$ 2sig digits

*c*) 392.05 $m/s,\ 0.035561\%,$ 3 sig digits

*d*) Exact 392.08 $m/s$

| Interpolant | True Error ($E_t$) |
|---|---|
| $1^{st}$ order | $-1.61$ |
| $2^{nd}$ order | $-0.11$ |
| $3^{rd}$ order | 0.03 |

(7). You are given data for the upward velocity of a rocket as a function of time in the table below.

| $t$ $(s)$ | 0 | 10 | 15 | 20 | 22.5 | 30 |
|---|---|---|---|---|---|---|
| $v(t)$ $(m/s)$ | 0 | 227.04 | 362.78 | 517.35 | 602.97 | 901.67 |

a) Use linear interpolant approximation of velocity to find the acceleration at t = 16s.

b) Use quadratic interpolant approximation of velocity to find the acceleration at t = 16s.

c) Find the distance covered by the rocket from t =5s to 16s? Use any method.

**Answer:** *a*) 30.914 $m/s^2$

*b*) 29.784 $m/s^2$

*c*) Method 1: 2672.3 $m$ by integrating 3rd order polynomial interpolant
Method 2: 2704.2 $m$ by Trapezoidal rule with unequal segments
Other methods are possible too – average method is unacceptable.

# Chapter 05.03: Newton's Divided Difference Method of Interpolation

Lesson: Newton's Divided Difference Method of Interpolation

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *find the interpolant through the Newton's divided difference interpolation,*

2) *choose the correct data points for interpolation,*

3) *solve problems using the Newton's divided difference interpolation,*

4) *use Newton's divided difference interpolants to find derivatives of discrete functions,*

5) *use the Newton's divided difference interpolants to find integrals of discrete functions.*

## Introduction

Polynomial interpolation involves finding a polynomial of order $n$ that passes through the $n+1$ points. One of the methods of interpolation is called Newton's divided difference polynomial method. Other methods include the direct method and the Lagrangian interpolation method. We will discuss Newton's divided difference polynomial method in this chapter.

## Newton's Divided Difference Polynomial Method

To illustrate this method, linear and quadratic interpolation is presented first. Then, the general form of Newton's divided difference polynomial method is presented. To illustrate the general form, cubic interpolation is shown in Figure 1.

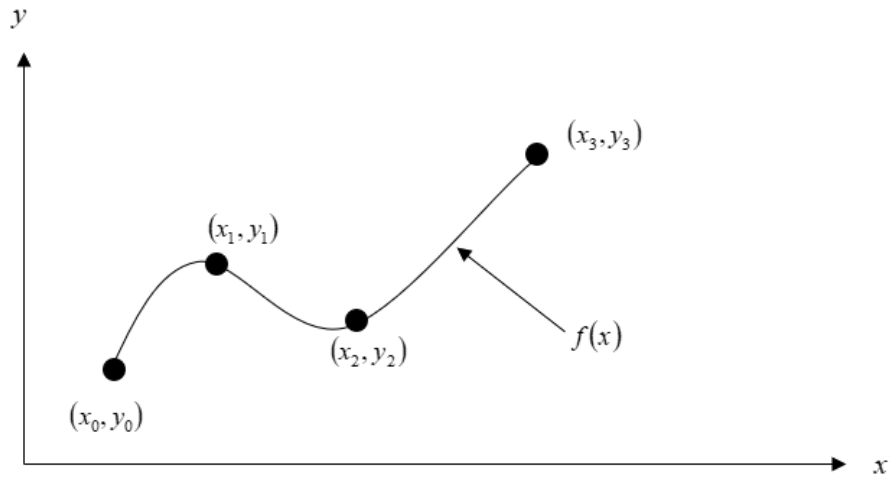



**Figure 1** Interpolation of discrete data.

## Linear Interpolation

Given $(x_0, y_0)$ and $(x_1, y_1)$, fit a linear interpolant through the data. Noting $y = f(x)$ and $y_1 = f(x_1)$, assume the linear interpolant $f_1(x)$ is given by (Figure 2)

$$f_1(x) = b_0 + b_1(x - x_0)$$

Since at $x = x_0$,

$$f_1(x_0) = f(x_0) = b_0 + b_1(x_0 - x_0) = b_0$$

and at $x = x_1$,

$$\begin{aligned} f_1(x_1) = f(x_1) &= b_0 + b_1(x_1 - x_0) \\ &= f(x_0) + b_1(x_1 - x_0) \end{aligned}$$

giving

$$b_1 = \frac{f(x_1) - f(x_0)}{x_1 - x_0}$$

So

$$\begin{aligned} b_0 &= f(x_0) \\ b_1 &= \frac{f(x_1) - f(x_0)}{x_1 - x_0} \end{aligned}$$

giving the linear interpolant as

$$f_1(x) = b_0 + b_1(x - x_0)$$

$$f_1(x) = f(x_0) + \frac{f(x_1) - f(x_0)}{x_1 - x_0}(x - x_0)$$

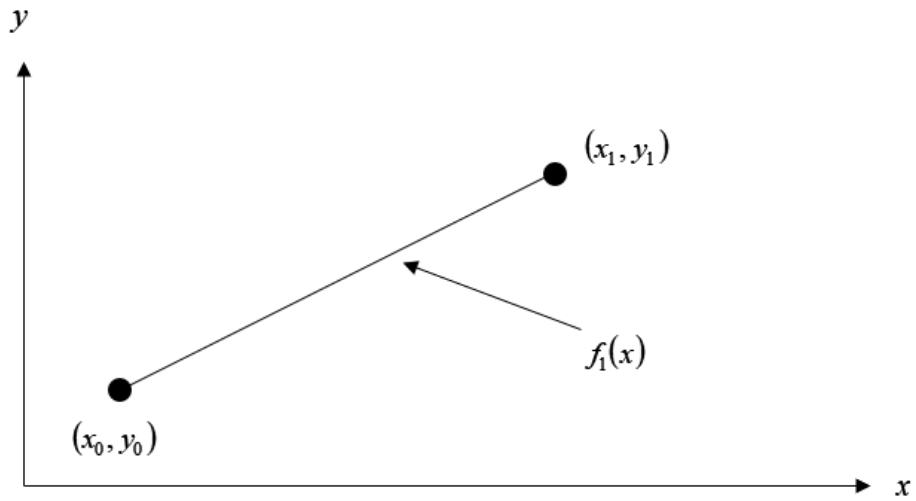



**Figure 2** Linear interpolation.

**Example 1**

The upward velocity of a rocket is given as a function of time in Table 1 (Figure 3).

**Table 1** Velocity as a function of time.

| $t$ $(s)$ | $v(t)$ (m/s) |
|---|---|
| 0 | 0 |
| 10 | 227.04 |
| 15 | 362.78 |
| 20 | 517.35 |
| 22.5 | 602.97 |
| 30 | 901.67 |

Determine the value of the velocity at $t = 16$ seconds using first order polynomial interpolation by Newton's divided difference polynomial method.

**Solution**

For linear interpolation, the velocity is given by

$$v(t) = b_0 + b_1(t - t_0)$$

Since we want to find the velocity at $t = 16$, and we are using a first order polynomial, we need to choose the two data points that are closest to $t = 16$ that also bracket $t = 16$ to evaluate it. The two points are $t = 15$ and $t = 20$.

Then

$$t_0 = 15,\ v(t_0) = 362.78$$

$$t_1 = 20,\ v(t_1) = 517.35$$

gives

$$\begin{aligned} b_0 &= v(t_0) \\ &= 362.78 \end{aligned}$$

$$\begin{aligned} b_1 &= \frac{v(t_1) - v(t_0)}{t_1 - t_0} \\ &= \frac{517.35 - 362.78}{20 - 15} \\ &= 30.914 \end{aligned}$$

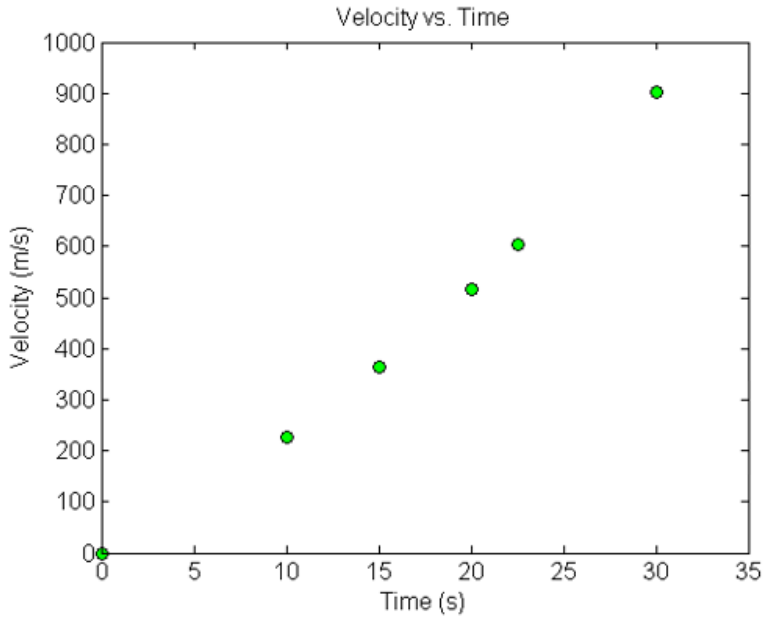



**Figure 3** Graph of velocity vs. time data for the rocket example.

Hence

$$\begin{aligned} v(t) &= b_0 + b_1(t - t_0) \\ &= 362.78 + 30.914(t - 15),\ 15 \le t \le 20 \end{aligned}$$

At $t = 16$,

$$\begin{aligned} v(16) &= 362.78 + 30.914(16 - 15) \\ &= 393.69 \text{ m/s} \end{aligned}$$

If we expand

$$v(t) = 362.78 + 30.914(t - 15),\ 15 \leq t \leq 20$$

we get

$$v(t) = -100.93 + 30.914t,\ 15 \leq t \leq 20$$

and this is the same expression as obtained in the direct method.

## Quadratic Interpolation

Given $(x_0, y_0)$, $(x_1, y_1)$, and $(x_2, y_2)$, fit a quadratic interpolant through the data. Noting $y = f(x)$, $y_0 = f(x_0)$, $y_1 = f(x_1)$, and $y_2 = f(x_2)$, assume the quadratic interpolant $f_2(x)$ is given by

$$f_2(x) = b_0 + b_1(x - x_0) + b_2(x - x_0)(x - x_1)$$

At $x = x_0$,

$$\begin{aligned} f_2(x_0) = f(x_0) &= b_0 + b_1(x_0 - x_0) + b_2(x_0 - x_0)(x_0 - x_1) \\ &= b_0 \end{aligned}$$

$$b_0 = f(x_0)$$

At $x = x_1$

$$f_2(x_1) = f(x_1) = b_0 + b_1(x_1 - x_0) + b_2(x_1 - x_0)(x_1 - x_1)$$

$$f(x_1) = f(x_0) + b_1(x_1 - x_0)$$

giving

$$b_1 = \frac{f(x_1) - f(x_0)}{x_1 - x_0}$$

At $x = x_2$

$$f_2(x_2) = f(x_2) = b_0 + b_1(x_2 - x_0) + b_2(x_2 - x_0)(x_2 - x_1)$$

$$f(x_2) = f(x_0) + \frac{f(x_1) - f(x_0)}{x_1 - x_0}(x_2 - x_0) + b_2(x_2 - x_0)(x_2 - x_1)$$

Giving

$$b_2 = \frac{\dfrac{f(x_2) - f(x_1)}{x_2 - x_1} - \dfrac{f(x_1) - f(x_0)}{x_1 - x_0}}{x_2 - x_0}$$

Hence the quadratic interpolant is given by

$$\begin{aligned} f_2(x) &= b_0 + b_1(x - x_0) + b_2(x - x_0)(x - x_1) \\ &= f(x_0) + \frac{f(x_1) - f(x_0)}{x_1 - x_0}(x - x_0) \\ &\quad + \frac{\dfrac{f(x_2) - f(x_1)}{x_2 - x_1} - \dfrac{f(x_1) - f(x_0)}{x_1 - x_0}}{x_2 - x_0}(x - x_0)(x - x_1) \end{aligned}$$

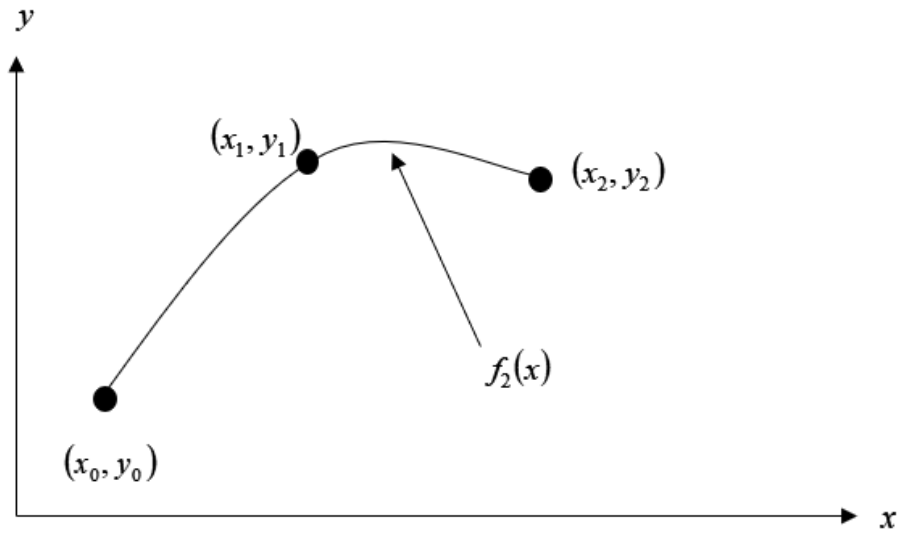



**Figure 4** Quadratic interpolation.

**Example 2**

The upward velocity of a rocket is given as a function of time in Table 2.

**Table 2** Velocity as a function of time.

| $t$ $(s)$ | $v(t)$ (m/s) |
|---|---|
| 0 | 0 |
| 10 | 227.04 |
| 15 | 362.78 |
| 20 | 517.35 |
| 22.5 | 602.97 |

| $t$ $(s)$ | $v(t)$ (m/s) |
|---|---|
| 30 | 901.67 |

Determine the value of the velocity at $t = 16$ seconds using second order polynomial interpolation using Newton's divided difference polynomial method.

**Solution**

For quadratic interpolation, the velocity is given by

$$v(t) = b_0 + b_1(t - t_0) + b_2(t - t_0)(t - t_1)$$

Since we want to find the velocity at $t = 16$, and we are using a second order polynomial, we need to choose the three data points that are closest to $t = 16$ that also bracket $t = 16$ to evaluate it. The three points are $t_0 = 10$, $t_1 = 15$, and $t_2 = 20$.

Then

$$t_0 = 10,\ v(t_0) = 227.04$$

$$t_1 = 15,\ v(t_1) = 362.78$$

$$t_2 = 20,\ v(t_2) = 517.35$$

gives

$$\begin{aligned} b_0 &= v(t_0) \\ &= 227.04 \end{aligned}$$

$$\begin{aligned} b_1 &= \frac{v(t_1) - v(t_0)}{t_1 - t_0} \\ &= \frac{362.78 - 227.04}{15 - 10} \\ &= 27.148 \end{aligned}$$

$$\begin{aligned} b_2 &= \frac{\dfrac{v(t_2) - v(t_1)}{t_2 - t_1} - \dfrac{v(t_1) - v(t_0)}{t_1 - t_0}}{t_2 - t_0} \\ &= \frac{\dfrac{517.35 - 362.78}{20 - 15} - \dfrac{362.78 - 227.04}{15 - 10}}{20 - 10} \\ &= \frac{30.914 - 27.148}{10} \\ &= 0.37660 \end{aligned}$$

Hence

$$\begin{aligned} v(t) &= b_0 + b_1(t - t_0) + b_2(t - t_0)(t - t_1) \\ &= 227.04 + 27.148(t - 10) + 0.37660(t - 10)(t - 15),\ 10 \leq t \leq 20 \end{aligned}$$

At $t = 16$,

$$\begin{aligned} v(16) &= 227.04 + 27.148(16 - 10) + 0.37660(16 - 10)(16 - 15) \\ &= 392.19 \text{ m/s} \end{aligned}$$

If we expand

$$v(t) = 227.04 + 27.148(t - 10) + 0.37660(t - 10)(t - 15),\ 10 \leq t \leq 20$$

we get

$$v(t) = 12.05 + 17.733t + 0.37660t^2,\ 10 \leq t \leq 20$$

This is the same expression obtained by the direct method.

## General Form of Newton's Divided Difference Polynomial

In the two previous cases, we found linear and quadratic interpolants for Newton's divided difference method. Let us revisit the quadratic polynomial interpolant formula

$$f_2(x) = b_0 + b_1(x - x_0) + b_2(x - x_0)(x - x_1)$$

where

$$b_0 = f(x_0)$$

$$b_1 = \frac{f(x_1) - f(x_0)}{x_1 - x_0}$$

$$b_2 = \frac{\dfrac{f(x_2) - f(x_1)}{x_2 - x_1} - \dfrac{f(x_1) - f(x_0)}{x_1 - x_0}}{x_2 - x_0}$$

Note that $b_0$, $b_1$, and $b_2$ are finite divided differences. $b_0$, $b_1$, and $b_2$ are the first, second, and third finite divided differences, respectively. We denote the first divided difference by

$$f[x_0] = f(x_0)$$

the second divided difference by

$$f[x_1, x_0] = \frac{f(x_1) - f(x_0)}{x_1 - x_0}$$

and the third divided difference by

$$\begin{aligned} f[x_2, x_1, x_0] &= \frac{f[x_2, x_1] - f[x_1, x_0]}{x_2 - x_0} \\ &= \frac{\dfrac{f(x_2) - f(x_1)}{x_2 - x_1} - \dfrac{f(x_1) - f(x_0)}{x_1 - x_0}}{x_2 - x_0} \end{aligned}$$

where $f[x_0], f[x_1, x_0]$, and $f[x_2, x_1, x_0]$ are called bracketed functions of their variables enclosed in square brackets.

Rewriting,

$$f_2(x) = f[x_0] + f[x_1, x_0](x - x_0) + f[x_2, x_1, x_0](x - x_0)(x - x_1)$$

This leads us to writing the general form of the Newton's divided difference polynomial for $n + 1$ data points, $(x_0, y_0), (x_1, y_1), \ldots\ldots, (x_{n-1}, y_{n-1}), (x_n, y_n)$, as

$f_n(x) = b_0 + b_1(x - x_0) + \ldots + b_n(x - x_0)(x - x_1)...(x - x_{n-1})$

where

$$b_0 = f[x_0]$$

$$b_1 = f[x_1, x_0]$$

$$b_2 = f[x_2, x_1, x_0]$$

$$\vdots$$

$$b_{n-1} = f[x_{n-1}, x_{n-2}, \ldots, x_0]$$

$$b_n = f[x_n, x_{n-1}, \ldots, x_0]$$

where the definition of the $m^{\text{th}}$ divided difference is

$$\begin{aligned} b_m &= f[x_m, \ldots\ldots, x_0] \\ &= \frac{f[x_m, \ldots\ldots, x_1] - f[x_{m-1}, \ldots\ldots, x_0]}{x_m - x_0} \end{aligned}$$

From the above definition, it can be seen that the divided differences are calculated recursively.

For an example of a third order polynomial, given $(x_0, y_0)$, $(x_1, y_1)$, $(x_2, y_2)$, and $(x_3, y_3)$,

$$f_3(x) = f[x_0] + f[x_1, x_0](x - x_0) + f[x_2, x_1, x_0](x - x_0)(x - x_1) + f[x_3, x_2, x_1, x_0](x - x_0)(x - x_1)(x - x_2)$$

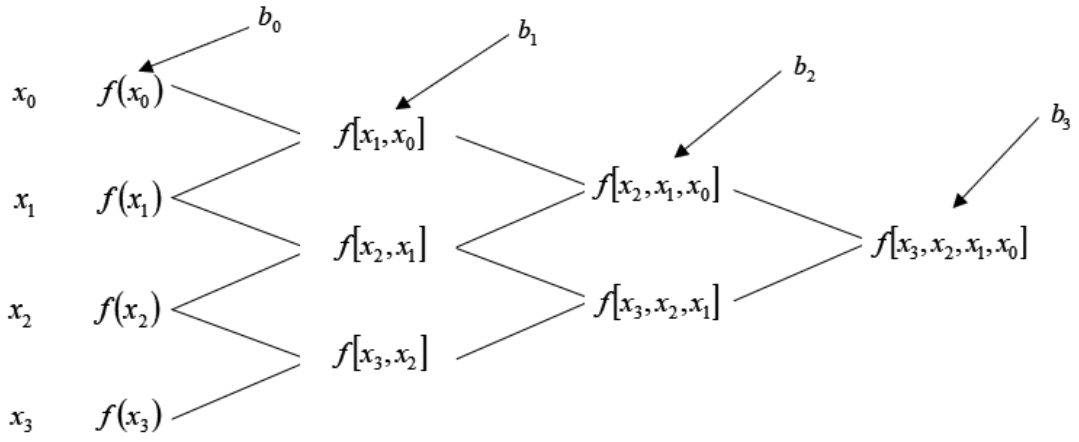



**Figure 5** Table of divided differences for a cubic polynomial.

**Example 3**

The upward velocity of a rocket is given as a function of time in Table 3.

**Table 3** Velocity as a function of time.

| $t$ $(s)$ | $v(t)$ $(m/s)$ |
|---|---|
| 0 | 0 |
| 10 | 227.04 |
| 15 | 362.78 |
| 20 | 517.35 |
| 22.5 | 602.97 |
| 30 | 901.67 |

a) Determine the value of the velocity at $t = 16$ seconds with third order polynomial interpolation using Newton's divided difference polynomial method.

b) Using the third order polynomial interpolant for velocity, find the distance covered by the rocket from $t = 11$ s to $t = 16$ s.

c) Using the third order polynomial interpolant for velocity, find the acceleration of the rocket at $t = 16$ s.

**Solution**

a) For a third order polynomial, the velocity is given by

$$v(t) = b_0 + b_1(t - t_0) + b_2(t - t_0)(t - t_1) + b_3(t - t_0)(t - t_1)(t - t_2)$$

Since we want to find the velocity at $t = 16$, and we are using a third order polynomial, we need to choose the four data points that are closest to $t = 16$ that also bracket $t = 16$ to evaluate it. The four data points are $t_0 = 10$, $t_1 = 15$, $t_2 = 20$, and $t_3 = 22.5$.

Then

$$t_0 = 10,\ v(t_0) = 227.04$$

$$t_1 = 15,\ v(t_1) = 362.78$$

$$t_2 = 20,\ v(t_2) = 517.35$$

$$t_3 = 22.5,\ v(t_3) = 602.97$$

gives

$$\begin{aligned} b_0 &= v[t_0] \\ &= v(t_0) \\ &= 227.04 \end{aligned}$$

$$\begin{aligned} b_1 &= v[t_1, t_0] \\ &= \frac{v(t_1) - v(t_0)}{t_1 - t_0} \\ &= \frac{362.78 - 227.04}{15 - 10} \\ &= 27.148 \end{aligned}$$

$$\begin{aligned} b_2 &= v[t_2, t_1, t_0] \\ &= \frac{v[t_2, t_1] - v[t_1, t_0]}{t_2 - t_0} \end{aligned}$$

$$\begin{aligned} v[t_2, t_1] &= \frac{v(t_2) - v(t_1)}{t_2 - t_1} \\ &= \frac{517.35 - 362.78}{20 - 15} \\ &= 30.914 \end{aligned}$$

$$v[t_1, t_0] = 27.148$$

$$\begin{aligned} b_2 &= \frac{v[t_2, t_1] - v[t_1, t_0]}{t_2 - t_0} \\ &= \frac{30.914 - 27.148}{20 - 10} \\ &= 0.37660 \end{aligned}$$

$$
\begin{aligned}
b_3 &= v[t_3, t_2, t_1, t_0] \\
&= \frac{v[t_3, t_2, t_1] - v[t_2, t_1, t_0]}{t_3 - t_0}
\end{aligned}
$$

$$
v[t_3, t_2, t_1] = \frac{v[t_3, t_2] - v[t_2, t_1]}{t_3 - t_1}
$$

$$
\begin{aligned}
v[t_3, t_2] &= \frac{v(t_3) - v(t_2)}{t_3 - t_2} \\
&= \frac{602.97 - 517.35}{22.5 - 20} \\
&= 34.248
\end{aligned}
$$

$$
\begin{aligned}
v[t_2, t_1] &= \frac{v(t_2) - v(t_1)}{t_2 - t_1} \\
&= \frac{517.35 - 362.78}{20 - 15} \\
&= 30.914
\end{aligned}
$$

$$
\begin{aligned}
v[t_3, t_2, t_1] &= \frac{v[t_3, t_2] - v[t_2, t_1]}{t_3 - t_1} \\
&= \frac{34.248 - 30.914}{22.5 - 15} \\
&= 0.44453
\end{aligned}
$$

$$
v[t_2, t_1, t_0] = 0.37660
$$

$$
\begin{aligned}
b_3 &= \frac{v[t_3, t_2, t_1] - v[t_2, t_1, t_0]}{t_3 - t_0} \\
&= \frac{0.44453 - 0.37660}{22.5 - 10} \\
&= 5.4347 \times 10^{-3}
\end{aligned}
$$

Hence

$$
\begin{aligned}
v(t) &= b_0 + b_1(t - t_0) + b_2(t - t_0)(t - t_1) + b_3(t - t_0)(t - t_1)(t - t_2) \\
&= 227.04 + 27.148(t - 10) + 0.37660(t - 10)(t - 15) \\
&\quad + 5.5347 \times 10^{-3}(t - 10)(t - 15)(t - 20)
\end{aligned}
$$

At $t = 16$,

$$
\begin{aligned}
v(16) &= 227.04 + 27.148(16 - 10) + 0.37660(16 - 10)(16 - 15) \\
&\quad + 5.5347 \times 10^{-3}(16 - 10)(16 - 15)(16 - 20) \\
&= 392.06 \ m/s
\end{aligned}
$$

b) The distance covered by the rocket between $t = 11$ s and $t = 16$ s can be calculated from the interpolating polynomial

$$\begin{aligned} v(t) &= 227.04 + 27.148(t-10) + 0.37660(t-10)(t-15) \\ &\quad + 5.5347 \times 10^{-3}(t-10)(t-15)(t-20) \\ &= -4.2541 + 21.265t + 0.13204t^2 + 0.0054347t^3, \ 10 \le t \le 22.5 \end{aligned}$$

Note that the polynomial is valid between $t = 10$ and $t = 22.5$ and hence includes the limits of $t = 11$ and $t = 16$.

So

$$\begin{aligned} s(16) - s(11) &= \int_{11}^{16} v(t)dt \\ &= \int_{11}^{16} (-4.2541 + 21.265t + 0.13204t^2 + 0.0054347t^3)dt \\ &= \left[ -4.2541t + 21.265\frac{t^2}{2} + 0.13204\frac{t^3}{3} + 0.0054347\frac{t^4}{4} \right]_{11}^{16} \\ &= 1605 \ m \end{aligned}$$

c) The acceleration at $t = 16$ is given by

$$a(16) = \left. \frac{d}{dt} v(t) \right|_{t=16}$$

$$\begin{aligned} a(t) &= \frac{d}{dt} v(t) \\ &= \frac{d}{dt} \left( -4.2541 + 21.265t + 0.13204t^2 + 0.0054347t^3 \right) \\ &= 21.265 + 0.26408t + 0.016304t^2 \end{aligned}$$

$$\begin{aligned} a(16) &= 21.265 + 0.26408(16) + 0.016304(16)^2 \\ &= 29.664 \ m/s^2 \end{aligned}$$

# Multiple Choice Test

## Chapter 05.03 Newton's Divided Difference Method of Interpolation

(1). If a polynomial of degree $n$ has more than $n$ zeros, then the polynomial is

(A) oscillatory

(B) zero everywhere

(C) quadratic

(D) not defined

(2). The following $x$ -$y$ data is given.

| | | | |
|---|---|---|---|
| $x$ | 15 | 18 | 22 |
| $y$ | 24 | 37 | 25 |

The Newton's divided difference second order polynomial for the above data is given by

$$f_2(x) = b_0 + b_1 (x - 15) + b_2 (x - 15)(x - 18)$$

The value of $b_1$ is most nearly

(A) $-1.0480$

(B) 0.14333

(C) 4.3333

(D) 24.000

(3). The polynomial that passes through the following $x$ -$y$ data

| | | | |
|---|---|---|---|
| $x$ | 18 | 22 | 24 |
| $y$ | ? | 25 | 123 |

is given by

$$8.125x^2 - 324.75x + 3237, \ 18 \leq x \leq 24$$

The corresponding polynomial using Newton's divided difference polynomial is given by

$$f_2(x) = b_0 + b_1 (x - 18) + b_2 (x - 18)(x - 22)$$

The value of $b_2$ is

(A) 0.25000

(B) 8.1250

(C) 24.000

(D) not obtainable with the information given

(4). Velocity vs. time data for a body is approximated by a second order Newton's divided difference polynomial as

$$v(t) = b_0 + 39.622 (t - 20) + 0.5540 (t - 20)(t - 15), \quad 10 \le t \le 20$$

The acceleration in $m/s^2$ at $t = 15$ is

(A) 0.55400

(B) 39.622

(C) 36.852

(D) not obtainable with the given information

(5). The path that a robot is following on a $x$-$y$ plane is found by interpolating four data points as

| | | | | |
|---|---|---|---|---|
| $x$ | 2 | 4.5 | 5.5 | 7 |
| $y$ | 7.5 | 7.5 | 6 | 5 |

$$y(x) = 0.15238x^3 - 2.2571x^2 + 9.6048x - 3.9000$$

The length of the path from $x = 2$ to $x = 7$ is

(A) $\sqrt{(7.5 - 7.5)^2 + (4.5 - 2)^2} + \sqrt{(6 - 7.5)^2 + (5.5 - 4.5)^2} + \sqrt{(5 - 6)^2 + (7 - 5.5)^2}$

(B) $\int_2^7 \sqrt{1 + (0.15238x^3 - 2.2571x^2 + 9.6048x - 3.9000)^2}dx$

(C) $\int_2^7 \sqrt{1 + (0.45714x^2 - 4.5142x + 9.6048)^2}dx$

(D) $\int_2^7 (0.15238x^3 - 2.2571x^2 + 9.6048x - 3.9000)dx$

(6). The following data of the velocity of a body is given as a function of time.

| Time ($s$) | 0 | 15 | 18 | 22 | 24 |
|---|---|---|---|---|---|
| Velocity ($m/s$) | 22 | 24 | 37 | 25 | 123 |

If you were going to use quadratic interpolation to find the value of the velocity at $t = 14.9$ seconds, the three data points of time you would choose for interpolation are

(A) $0, 15, 18$

(B) $15, 18, 22$

(C) $0, 15, 22$

(D) $0, 18, 24$

For complete solution, go to

http://nm.mathforcollege.com/mcquizzes/05inp/quiz_05inp_newton_solution.pdf

# Problem Set

## Chapter 05.03: Newton's Divided Difference Method of Interpolation

(1). The following data of the velocity of a body as a function of time is given

| Time $(s)$ | 0 | 15 | 18 | 22 | 24 |
|---|---|---|---|---|---|
| Velocity $(m/s)$ | 22 | 24 | 37 | 25 | 123 |

Find the coefficients of the Newton's divided difference interpolating first order polynomial to find the velocity at $t = 14$ s.

**Answer:** velocity at 14 seconds = 23.867 m/s

(2). The following data of the velocity of a body as a function of time is given

| Time $(s)$ | 0 | 15 | 18 | 22 | 24 |
|---|---|---|---|---|---|
| Velocity $(m/s)$ | 22 | 24 | 37 | 25 | 123 |

Find the coefficients of the Newton's divided difference interpolating second order polynomial to find the velocity at $t = 14$ s.

**Answer:** velocity at 14 seconds = 20.6 m/s

(3). The following data of the velocity of a body as a function of time is given

| Time $(s)$ | 0 | 15 | 18 | 22 | 24 |
|---|---|---|---|---|---|
| Velocity $(m/s)$ | 22 | 24 | 37 | 25 | 123 |

Find the coefficients of the Newton's divided difference interpolating third order polynomial to find the velocity at $t = 14$ s.

**Answer:** velocity at 14 seconds = 17.339 m/s

(4). You are given data for the upward velocity of a rocket as a function of time in the table below.

| $t,\ (s)$ | 0 | 10 | 15 | 20 | 22.5 | 30 |
|---|---|---|---|---|---|---|
| $v(t),\ (m/s)$ | 0 | 227.04 | 362.78 | 517.35 | 602.97 | 901.67 |

a) Use Newton's divided difference linear interpolant approximation of velocity to find the acceleration at $t = 16$ s.

b) Use Newton's divided difference quadratic interpolant approximation of velocity to find the acceleration at $t = 16$ s.

c) Find the distance covered by the rocket from $t = 5$ s to 16 s? Use any method.

**Answer:** *a)* $a(16) = 30.914$ m/s$^2$ *b)* $a(16) = 29.784$ m/s$^2$

(5). The acceleration-time data for a small rocket is given in tabular form below.

| Time $(s)$ | 10 | 12 | 14 | 16 | 18 | 20 | 22 | 24 |
|---|---|---|---|---|---|---|---|---|
| Acceleration $m/s^2$ | 106.6 | 94.1 | 80.9 | 68.0 | 56.2 | 45.8 | 37.1 | 30.1 |

a) Use Newton's divided difference quadratic polynomial interpolation to find the acceleration at $t = 15.5$ seconds. Be sure to choose your base points for good accuracy.

b) Use the quadratic interpolant of part (a) to find the change in the velocity of the rocket between $t = 14.1$ and $t = 15.8$ seconds.

**Answer:** *a)* $a(15.5) = 71.122$ m/s$^2$ *b)* Change in velocity = 126.94 m/s

(6). A robot follows a path generated by a quadratic interpolant from x = 2 to x = 4. The interpolant passes through three consecutive data points $(2, 4)$, $(3, 9)$ and $(4, 16)$ and is given by $y = x^2$. Find the length of the interpolant path from $x = 2$ to $x = 4$. You can approximate a general integral by

$$\int_a^b f(x)dx \approx \frac{(b-a)}{6}\left[f(a) + 4f\left(\frac{a+b}{2}\right) + f(b)\right]$$

**Answer:** Length = 12.172

# Chapter 05.05: Spline Method of Interpolation

Lesson: Why Do We Need Spline Interpolation?

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *justify why higher-order interpolation is a bad idea,*

2) *how spline interpolation can avoid the pitfalls of higher-order interpolation.*

## Introduction

Polynomial interpolation involves finding a polynomial of order $n$ or less that passes through the $n+1$ points. Several methods to obtain such a polynomial include the direct method (also called the Vandermonde polynomial method), Newton's divided difference polynomial method, and the Lagrangian interpolation method.

So is the spline method yet another method of obtaining this $n^{\text{th}}$ order polynomial. ..... NO!

Actually, when $n$ becomes large, in many cases, one may get oscillatory behavior in the interpolating polynomial. This phenomenon was illustrated by Runge when he interpolated data based on a simple function of

$$y = \frac{1}{1+25x^2} \qquad (1)$$

on an interval of $[-1, 1]$.

For example, take six equidistantly spaced points in $[-1, 1]$ and find $y$ at these points from Equation (1). These values are given in Table 1.

**Table 1.** Six equidistantly spaced points in $[-1, 1]$.

| $x$ | $y = \dfrac{1}{1+25x^2}$ |
|---|---|
| $-1.0$ | 0.038461 |
| $-0.6$ | 0.1 |
| $-0.2$ | 0.5 |
| 0.2 | 0.5 |
| 0.6 | 0.1 |
| 1.0 | 0.038461 |

Now through these six data points, one can pass a fifth-order interpolating polynomial.

$$f_5(x) = 1.2019x^4 - 1.7308x^2 + 0.56731, \ -1 \leq x \leq 1 \qquad (2)$$

On plotting (Figure 1) the fifth-order polynomial (Equation (2)) and the original function (Equation (1)), you can see that they do not match well. The polynomial does go through the six data pairs of Table 1, but it does not even come close to the original function at many other points. Just look at $x = 0.85$, the value of the original function is 0.052459, but the fifth-order polynomial gives you a value of $-0.055762$. That is a whopping relative true error of 206.30%.

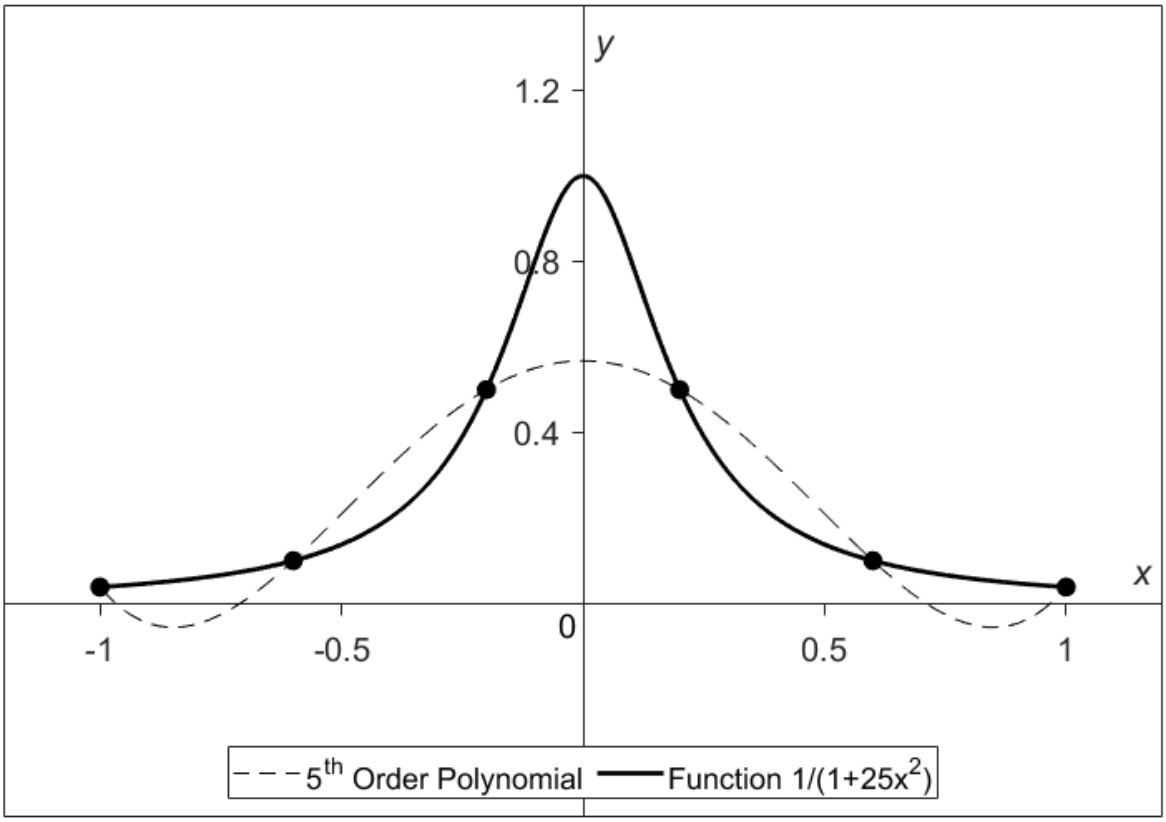



**Figure 1.** Fifth order polynomial interpolation with six equidistant points.

One may think that choosing more points in the interval $[-1, 1]$ will give us a better match between the original function and the interpolant, but it diverges even more (Figure 2). Here 20 equidistant points were chosen in the interval $[-1, 1]$ to draw a $19^{th}$ order interpolating polynomial. It, however, did do a better job of approximating the data but except near the ends where the approximation is worse than before. At our chosen point, $x = 0.85$, the value of the original function is 0.052459, while we get $-0.62944$ from the $19^{th}$ order polynomial. That gives a big whopper relative true error of 1299.9%.

In fact, Runge found that as the order of the polynomial approaches infinity, the polynomial diverges even more in the interval of $-1 < x < -0.726$ and $0.726 < x < 1$.

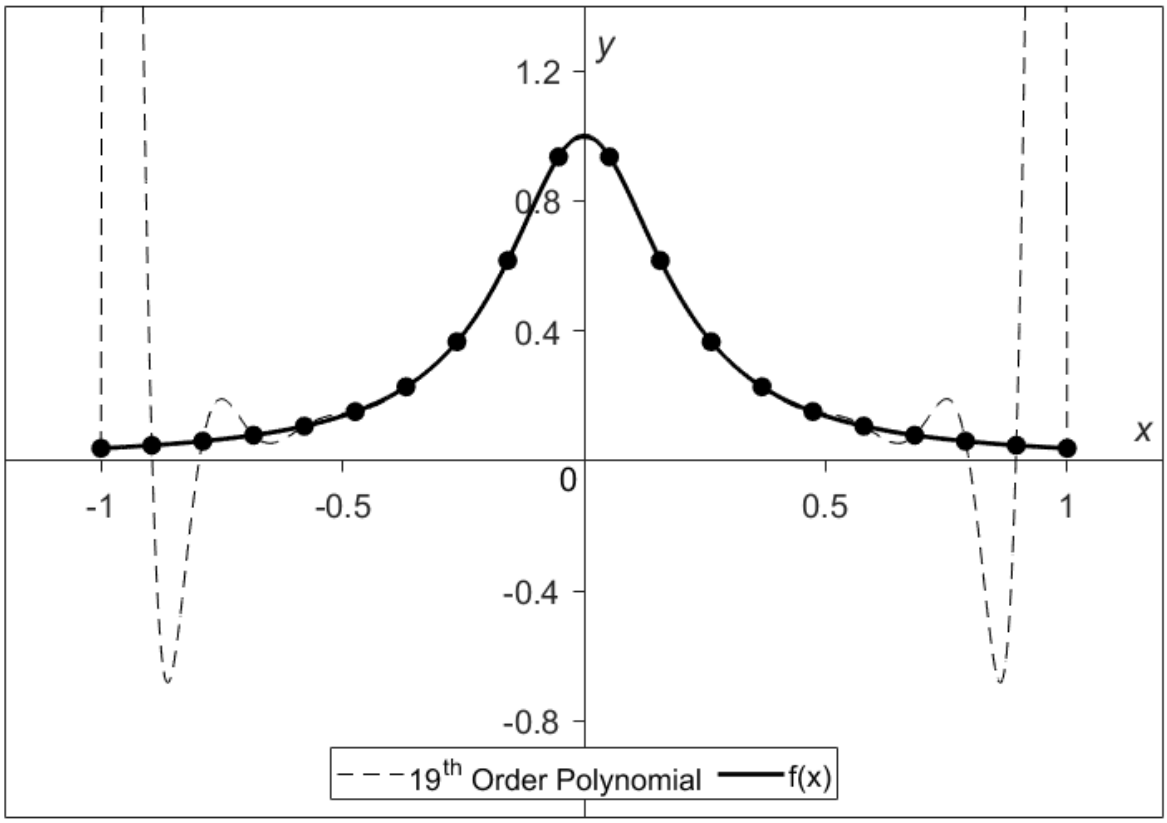



**Figure 2.** Nineteenth order polynomial interpolation with twenty equidistant points

So, what is the solution to using information from more data points, but at the same time keeping the function reasonably true to the data behavior? The answer is in spline interpolation and will be discussed in the following lessons. The most common types of spline interpolation used are linear, quadratic, and cubic.

# Chapter 05.05: Spline Method of Interpolation

Lesson: Linear Spline Interpolation

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *develop linear spline interpolant to given data points,*

2) *estimate unknown data points from the linear spline interpolant.*

## Linear Spline Interpolation

Given $(x_0, y_0), (x_1, y_1), \ldots\ldots, (x_{n-1}, y_{n-1}), (x_n, y_n)$, find the interpolating linear spline (Figure 1) to the data. This procedure simply involves drawing straight lines between consecutive points.

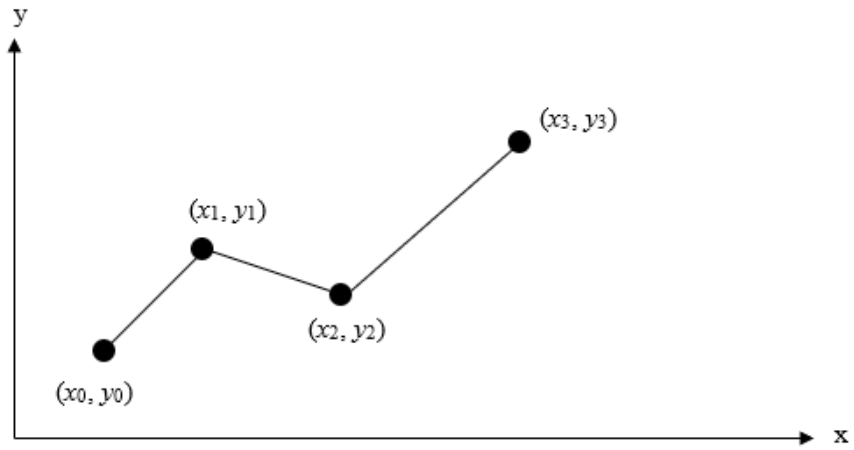



**Figure 1.** Linear splines.

Assuming that the above data is given in ascending order, the interpolating linear spline, also called spline of degree 1, $f(x)$ is given by

$$\begin{aligned} f(x) &= f(x_0) + \frac{f(x_1) - f(x_0)}{x_1 - x_0}(x - x_0), \ x_0 \le x \le x_1, \\ &= f(x_1) + \frac{f(x_2) - f(x_1)}{x_2 - x_1}(x - x_1), \ x_1 \le x \le x_2, \\ &\vdots \\ &= f(x_{n-1}) + \frac{f(x_n) - f(x_{n-1})}{x_n - x_{n-1}}(x - x_{n-1}), \ x_{n-1} \le x \le x_n. \end{aligned} \tag{1}$$

Note that $y_i = f(x_i)$ in the above expression and that the terms of

$$\frac{f(x_i) - f(x_{i-1})}{x_i - x_{i-1}} \qquad (2)$$

are simply slopes between $x_{i-1}$ and $x_i$.

### Example 1

The upward velocity of a rocket is given as a function of time in Table 1 (Figure 2).

**Table 1.** Velocity as a function of time.

| $t$ (s) | $v(t)$ (m/s) |
|---|---|
| 0 | 0 |
| 10 | 227.04 |
| 15 | 362.78 |
| 20 | 517.35 |
| 22.5 | 602.97 |
| 30 | 901.67 |

Determine the value of the velocity at $t = 16$ seconds using an interpolating linear spline.

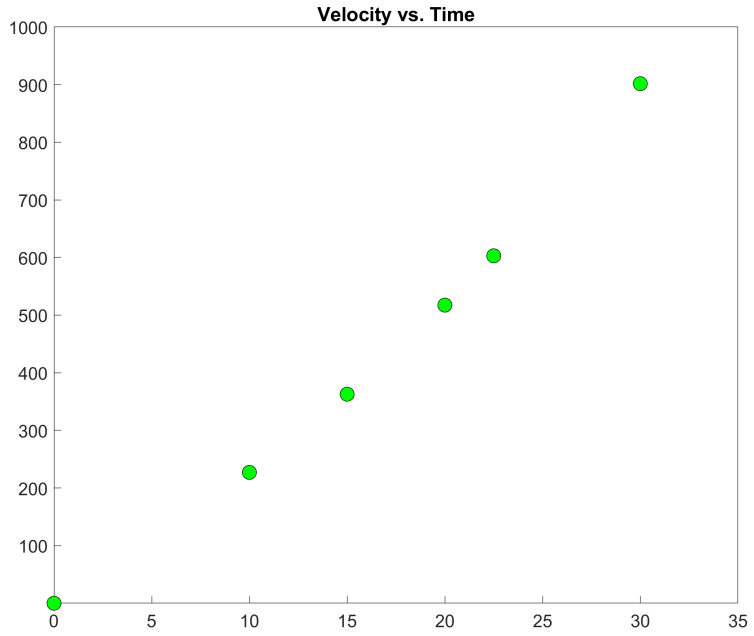



**Figure 2.** Graph of velocity vs. time data for the rocket example.

### Solution

Since we want to evaluate the velocity at $t = 16$ and use linear spline interpolation, we need to choose the two data points closest to $t = 16$ that also bracket $t = 16$ to evaluate it. The two points are $t_0 = 15$ and $t_1 = 20$.

Then

$$t_0 = 15,\ v(t_0) = 362.78$$

$$t_1 = 20,\ v(t_1) = 517.35$$

gives

$$\begin{aligned} v(t) &= v(t_0) + \frac{v(t_1) - v(t_0)}{t_1 - t_0}(t - t_0) \\ &= 362.78 + \frac{517.35 - 362.78}{20 - 15}(t - 15) \\ &= 362.78 + 30.913(t - 15),\ 15 \leq t \leq 20 \end{aligned}$$

At $t = 16$,

$$\begin{aligned} v(16) &= 362.78 + 30.913(16 - 15) \\ &= 393.7 \text{ m/s} \end{aligned}$$

Linear spline interpolation is no different from linear polynomial interpolation. Linear spline interpolation still uses data only from the two consecutive data points, and data from other points is not used at all. Also, at the interior points of the data, the slope of the spline changes abruptly, which implies that the first derivative is "artificially" not continuous at these points. So how do we overcome these two drawbacks? We can do so by using quadratic or cubic spline interpolation. We discuss these types of spline interpolation in upcoming lessons.

# Chapter 05.05: Spline Method of Interpolation

Lesson: Quadratic Spline Interpolation

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *derive interpolating quadratic spline for discrete data.*

## Interpolating Quadratic Spline

Quadratic spline interpolation is a method to curve fit data. For quadratic spline interpolation, piecewise quadratics approximates the data between two consecutive data points (Figure 1). Given $(x_0, y_0), (x_1, y_1), ....., (x_{n-1}, y_{n-1}), (x_n, y_n)$, fit an interpolating quadratic spline through the data. The quadratics of the spline are given by

$$\begin{aligned} f(x) &= a_1x^2 + b_1x + c_1,\ x_0 \le x \le x_1 \\ &= a_2x^2 + b_2x + c_2,\ x_1 \le x \le x_2 \\ &\vdots \\ &= a_nx^2 + b_nx + c_n,\ x_{n-1} \le x \le x_n \end{aligned}$$

So how does one find the coefficients of these quadratics? There are $3n$ such coefficients

$$a_i,\ i = 1, 2, ....., n$$

$$b_i,\ i = 1, 2, ....., n$$

$$c_i,\ i = 1, 2, ....., n$$

To find $3n$ unknowns, one needs to set up $3n$ equations and then simultaneously solve them. These $3n$ equations are found as follows.

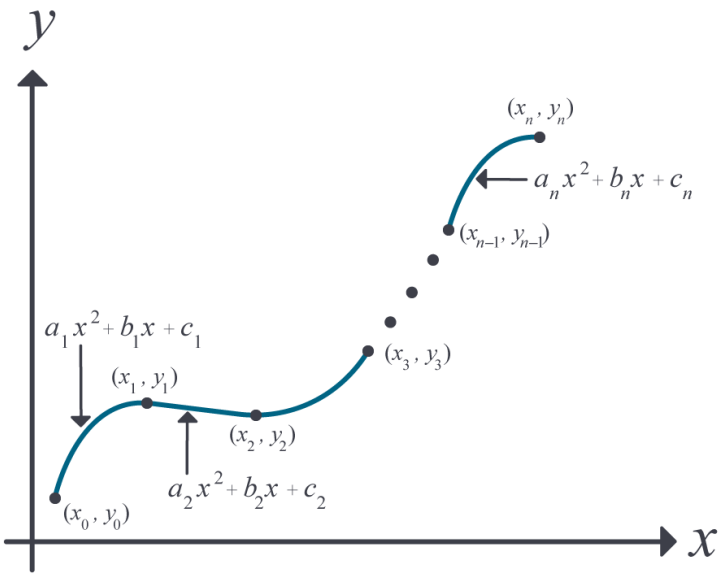



**Figure 1**. Quadratic spline interpolation

1. Each quadratic goes through two consecutive data points

$$
\begin{aligned}
&a_1 {x_0}^2 + b_1 x_0 + c_1 = f(x_0)\\
&a_1 {x_1}^2 + b_1 x_1 + c_1 = f(x_1)\\
&\vdots\\
&a_i {x_{i-1}}^2 + b_i x_{i-1} + c_i = f(x_{i-1})\\
&a_i {x_i}^2 + b_i x_i + c_i = f(x_i)\\
&\vdots\\
&a_n {x_{n-1}}^2 + b_n x_{n-1} + c_n = f(x_{n-1})\\
&a_n {x_n}^2 + b_n x_n + c_n = f(x_n)
\end{aligned}
$$

This condition gives $2n$ equations as there are $n$ quadratics going through two consecutive data points.

2. The first derivatives of two consecutive quadratics are continuous at the common interior points. For example, the derivative of the first quadratic

$$a_1 x^2 + b_1 x + c_1$$

is

$$2a_1x + b_1$$

The derivative of the second quadratic

$$a_2x^2 + b_2x + c_2$$

is

$$2a_2x + b_2$$

and the two are equal at the common interior point $x = x_1$ giving

$$2a_1x_1 + b_1 = 2a_2x_1 + b_2$$

$$2a_1x_1 + b_1 - 2a_2x_1 - b_2 = 0$$

Similarly, at the other interior points, $x_2, \ldots, x_{n-1}$,

$$\begin{aligned}
&2a_2x_2 + b_2 - 2a_3x_2 - b_3 = 0\\
&\vdots\\
&2a_ix_i + b_i - 2a_{i+1}x_i - b_{i+1} = 0\\
&\vdots\\
&2a_{n-1}x_{n-1} + b_{n-1} - 2a_nx_{n-1} - b_n = 0
\end{aligned}$$

Since there are $(n-1)$ interior points, we have $(n-1)$ such equations. So far, the total number of equations obtained is $(2n) + (n-1) = (3n-1)$ equations. We still then need one more equation.

We can assume the first quadratic is linear, that is,

$$a_1 = 0$$

Some assume the last quadratic is linear, that is,

$$a_n = 0$$

Others rightly base it on which interval is smaller, $[x_0, x_1]$ or $[x_{n-1}, x_n]$. If $|x_1 - x_0| \leq |x_n - x_{n-1}|$, then one would choose $a_1 = 0$, else choose $a_n = 0$.

This gives us $3n$ simultaneous linear equations and $3n$ unknowns. These can be solved by several techniques used to solve a general set of simultaneous linear equations.

# Chapter 05.05: Spline Method of Interpolation

Lesson: Application of Quadratic Spline Interpolation

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *conduct quadratic spline interpolation on discrete data,*

2) *differentiate an interpolating quadratic spline as needed,*

3) *integrate an interpolating quadratic spline as needed.*

## Applications

In the previous lesson, you learned the theory of the quadratic spline interpolation method. In this lesson, we apply the theory to given interpolate discrete data to an interpolating quadratic spline.

### Example 1

The upward velocity of a rocket is given as a function of time as

**Table 1.** Velocity as a function of time.

| $t$ (s) | $v(t)$ (m/s) |
|---|---|
| 0 | 0 |
| 10 | 227.04 |
| 15 | 362.78 |
| 20 | 517.35 |
| 22.5 | 602.97 |
| 30 | 901.67 |

a) Determine the value of the velocity at $t = 16$ seconds using quadratic spline interpolation.

b) Using the interpolating quadratic spline, find the distance covered by the rocket from $t = 11s$ to $t = 16s$.

c) Using the interpolating quadratic spline, find the acceleration of the rocket at $t = 16s$.

**Solution**

a) Since there are six data points, five quadratics pass through them.

$$\begin{aligned} v(t) &= a_1 t^2 + b_1 t + c_1,\ 0 \le t \le 10 \\ &= a_2 t^2 + b_2 t + c_2,\ 10 \le t \le 15 \\ &= a_3 t^2 + b_3 t + c_3,\ 15 \le t \le 20 \\ &= a_4 t^2 + b_4 t + c_4,\ 20 \le t \le 22.5 \\ &= a_5 t^2 + b_5 t + c_5,\ 22.5 \le t \le 30 \end{aligned}$$

The equations are found as follows.

1. Each quadratic passes through two consecutive data points.

Quadratic $a_1 t^2 + b_1 t + c_1$ passes through $t = 0$ and $t = 10$.

$$a_1(0)^2 + b_1(0) + c_1 = 0 \qquad (E1.1)$$

$$a_1(10)^2 + b_1(10) + c_1 = 227.04 \qquad (E1.2)$$

Quadratic $a_2 t^2 + b_2 t + c_2$ passes through $t = 10$ and $t = 15$.

$$a_2(10)^2 + b_2(10) + c_2 = 227.04 \qquad (E1.3)$$

$$a_2(15)^2 + b_2(15) + c_2 = 362.78 \qquad (E1.4)$$

Quadratic $a_3 t^2 + b_3 t + c_3$ passes through $t = 15$ and $t = 20$.

$$a_3(15)^2 + b_3(15) + c_3 = 362.78 \qquad (E1.5)$$

$$a_3(20)^2 + b_3(20) + c_3 = 517.35 \qquad (E1.6)$$

Quadratic $a_4 t^2 + b_4 t + c_4$passes through $t = 20$ and $t = 22.5$.

$$a_4(20)^2 + b_4(20) + c_4 = 517.35 \qquad (E1.7)$$

$$a_4(22.5)^2 + b_4(22.5) + c_4 = 602.97 \qquad (E1.8)$$

Quadratic $a_5 t^2 + b_5 t + c_5$ passes through $t = 22.5$ and $t = 30$.

$$a_5(22.5)^2 + b_5(22.5) + c_5 = 602.97 \qquad (E1.9)$$

$$a_5(30)^2 + b_5(30) + c_5 = 901.67 \qquad (E1.10)$$

2. The quadratics have continuous derivatives at the common interior data points.

At $t = 10$

$$2a_1(10) + b_1 - 2a_2(10) - b_2 = 0 \qquad ((E1.11)$$

At $t = 15$

$$2a_2(15) + b_2 - 2a_3(15) - b_3 = 0 \qquad (E1.12)$$

At $t = 20$

$$2a_3(20) + b_3 - 2a_4(20) - b_4 = 0 \qquad (E1.13)$$

At $t = 22.5$

$$2a_4(22.5) + b_4 - 2a_5(22.5) - b_5 = 0 \qquad (E1.14)$$

3. Assuming the last quadratic $a_5t^2 + b_5t + c_5$ is linear (why did we choose the last spline to be linear over the first one?),

$$a_5 = 0 \qquad (E1.15)$$

Combining Equations (E1.1) – (E1.15) in matrix form gives

$$\begin{bmatrix}
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
100 & 10 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 100 & 10 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 225 & 15 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 225 & 15 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 400 & 20 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 400 & 20 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 506.25 & 22.5 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 506.25 & 22.5 & 1 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 900 & 30 & 1 \\
20 & 1 & 0 & -20 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 30 & 1 & 0 & -30 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 40 & 1 & 0 & -40 & -1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 45 & 1 & 0 & -45 & -1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0
\end{bmatrix}
\begin{bmatrix}
a_1 \\ b_1 \\ c_1 \\ a_2 \\ b_2 \\ c_2 \\ a_3 \\ b_3 \\ c_3 \\ a_4 \\ b_4 \\ c_4 \\ a_5 \\ b_5 \\ c_5
\end{bmatrix}
=
\begin{bmatrix}
0 \\ 227.04 \\ 227.04 \\ 362.78 \\ 362.78 \\ 517.35 \\ 517.35 \\ 602.97 \\ 602.97 \\ 901.67 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0
\end{bmatrix}$$

Solving the above 15 simultaneous linear equations for the 15 unknowns gives

| $i$ | $a_i$ | $b_i$ | $c_i$ |
|---|---|---|---|
| 1 | −0.15667 | 24.271 | 0 |
| 2 | 1.2021 | −2.9053 | 135.88 |
| 3 | −0.44893 | 46.627 | −235.61 |
| 4 | 2.2315 | −60.589 | 836.55 |

| $i$ | $a_i$ | $b_i$ | $c_i$ |
|---|---|---|---|
| 5 | 0 | 39.827 | −293.13 |

Therefore, the interpolating quadratic spline is given by

$$\begin{aligned} v(t) &= -0.15667t^2 + 24.271t,\ 0 \le t \le 10 \\ &= 1.2021t^2 - 2.9053t + 135.88,\ 10 \le t \le 15 \\ &= -0.44893t^2 + 46.627t - 235.61,\ 15 \le t \le 20 \\ &= 2.2315t^2 - 60.589t + 836.55,\ 20 \le t \le 22.5 \\ &= 39.827t - 293.13,\ 22.5 \le t \le 30 \end{aligned} \qquad (E1..16)$$

At $t = 16$ s

$$\begin{aligned} v(16) &= -0.44893(16)^2 + 46.627(16) - 235.61 \\ &= 395.50 \text{ m/s} \end{aligned}$$

b) The distance covered by the rocket between 11 s and 16 s seconds can be calculated as

$$s(16) - s(11) = \int_{11}^{16} v(t)\,dt$$

But since several quadratics are valid over the limits of integration, we need to break the integral accordingly. Using Equation (E1.16),

$$\begin{aligned} v(t) &= 1.2021t^2 - 2.9053t + 135.88,\ 10 \le t \le 15 \\ &= -0.44893t^2 + 46.627t - 235.61,\ 15 \le t \le 20 \end{aligned}$$

Then

$$\begin{aligned} s(16) - s(11) &= \int_{11}^{16} v(t)\,dt \\ &= \int_{11}^{15} v(t)\,dt + \int_{15}^{16} v(t)\,dt \end{aligned}$$

Substituting proper quadratics gives

$$
\begin{aligned}
s(16) - s(11) &= \int_{11}^{15} 1.2021t^2 - 2.9053t + 135.88 \; dt \\
&\quad + \int_{15}^{16} -0.44893t^2 + 46.627t - 235.61dt \\
&= \left[ 1.2021\frac{t^3}{3} - 2.9053\frac{t^2}{2} + 135.88t \right]_{11}^{15} \\
&\quad + \left[ -0.44893\frac{t^3}{3} + 46.627\frac{t^2}{2} - 235.61t \right]_{15}^{16} \\
&= 1211.49 + 379.22 \\
&= 1590.7 \text{ m}
\end{aligned}
$$

c) What is the acceleration at $t = 16$ s?

$$
a\,(16) = \left. \frac{dv}{dt} \right|_{t=16}
$$

Here we use the third quadratic from Equation (E1.15)

$$
-0.44893t^2 + 46.627t - 235.61, \;\; 15 \leq t \leq 20
$$

as $t = 16$ is in the domain of $15 \leq t \leq 20$.

So

$$
\begin{aligned}
a\,(t) &= \frac{d}{dt} v\,(t) \\
&= \frac{d}{dt} \left( -0.44893t^2 + 46.627t - 235.61 \right) \\
&= -0.89786t + 46.627, \;\; 15 \leq t \leq 20
\end{aligned}
$$

Hence

$$
\begin{aligned}
a\,(16) &= -0.89786\,(16) + 46.627 \\
&= 32.261 \text{ m/s}^2
\end{aligned}
$$

# Chapter 05.05: Spline Method of Interpolation

Lesson: Outline of Cubic Spline Interpolation

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *outline the derivation of cubic spline interpolation.*

## Introduction

Just like quadratic spline interpolation, cubic spline interpolation is a method to curve fit data. The cubic spline interpolation is the most common degree of spline used.

## Interpolating Cubic Spline

Given $n+1$ data points, $(x_0, y_0),\ (x_1, y_1), \ldots,\ (x_n, y_n)$, fit an interpolating cubic spline through the data. For cubic spline interpolation, piecewise cubic polynomials approximate the data between two consecutive points. The interpolating cubic spline is given by the cubics as

$$\begin{aligned} f(x) &= a_1x^3 + b_1x^2 + c_1x + d_1,\ x_0 \le x \le x_1 \\ &= a_2x^3 + b_2x^2 + c_2x + d_2,\ x_1 \le x \le x_2 \\ &\vdots \\ &= a_nx^3 + b_nx^2 + c_nx + d_n,\ x_{n-1} \le x \le x_n \end{aligned}$$

So, how does one find the coefficients of these cubics? There are $4n$ such coefficients as given below.

$$\begin{aligned} a_i,\ i &= 1,\ 2, \ldots,\ n \\ b_i,\ i &= 1,\ 2, \ldots,\ n \\ c_i,\ i &= 1,\ 2, \ldots,\ n \\ d_i,\ i &= 1,\ 2, \ldots,\ n \end{aligned}$$

To find the $4n$ unknowns, we need to set up $4n$ simultaneous linear equations. The outline of setting up these equations is as follows. We have $n+1$ data points. Out of these $n+1$ points, two are the exterior (end) data points, while there are $((n+1)-2 = n-1)$ interior data points.

1. Each cubic must go through two consecutive points. Since we have $n$ splines, we will get $2n$ equations.

2. The first derivative of two consecutive cubics is continuous at the common interior data point. Since we have $(n-1)$ interior points, we will get $2(n-1)$ equations

3. The second derivative of two consecutive cubics is continuous at the common interior data point. Since we have $(n-1)$ interior points, we will get $2(n-1)$ equations

So far, we have outlined that

$$\begin{aligned}&(2n) + (n-1) + (n-1)\\ &= 4n - 2\end{aligned}$$

equations can be set up. We need two more equations, and several types of conditions can be used to get them. A few common conditions used are as follows.

1. Assume that the first and last cubic splines are quadratics ($a_1 = 0$ and $a_n = 0$) to get the two more equations. This assumption is called the quadratic end condition.

2. Assume the second derivative of the first and last cubic spline is zero ($6a_1x_0 + 2b_1 = 0$ and $6a_nx_n + 2b_n = 0$) at the respective exterior points to get the two more equations. This assumption is called the natural end condition.

# Multiple Choice Test

## Chapter 05.05 Spline Method of Interpolation

(1). The following $n$ data points, $(x_1, y_1)$, $(x_2, y_2)$, ..... $(x_n, y_n)$, are given. For conducting quadratic spline interpolation, the $x$-data needs to be

(A) equally spaced

(B) placed in ascending or descending order of $x$ -values

(C) integers

(D) positive

(2). In cubic spline interpolation,

(A) the first derivatives of the cubics are continuous at the interior data points

(B) the second derivatives of the cubics are continuous at the interior data points

(C) the first and the second derivatives of the cubics are continuous at the interior data points

(D) the third derivatives of the cubics are continuous at the interior data points

(3). The following incomplete $y$ vs. $x$ data is given.

| $x$ | 1 | 2 | 4 | 6 | 7 |
|---|---|---|---|---|---|
| $y$ | 5 | 11 | ???? | ???? | 32 |

The data is fit by interpolating quadratic splines given by

$$\begin{aligned} f(x) &= ax - 1,\ 1 \le x \le 2 \\ &= -2x^2 + 14x - 9,\ 2 \le x \le 4 \\ &= bx^2 + cx + d,\ 4 \le x \le 6 \end{aligned}$$

$$= 25x^2 - 303x + 928, \quad 6 \le x \le 7$$

where $a, b, c,$ and $d$ are constants. The value of $c$ is most nearly

(A) $-303.00$

(B) $-144.50$

(C) 0.0000

(D) 14.000

(4). The following incomplete $y$ vs. $x$ data is given.

| $x$ | 1 | 2 | 4 | 6 | 7 |
|---|---|---|---|---|---|
| $y$ | 5 | 11 | ???? | ???? | 32 |

The data is fit by interpolating quadratic spline given by

$$\begin{aligned} f(x) &= ax - 1,\ 1 \le x \le 2, \\ &= -2x^2 + 14x - 9,\ 2 \le x \le 4 \\ &= bx^2 + cx + d,\ 4 \le x \le 6 \\ &= ex^2 + fx + g,\ 6 \le x \le 7 \end{aligned}$$

where $a, b, c, d, e, f,$ and $g$ are constants. The value of $\dfrac{df}{dx}$ at $x = 2.6$ most nearly is

(A) $-144.50$

(B) $-4.0000$

(C) 3.6000

(D) 12.200

(5). The following incomplete $y$ vs. $x$ data is given.

| $x$ | 1 | 2 | 4 | 6 | 7 |
|---|---|---|---|---|---|
| $y$ | 5 | 11 | ???? | ???? | 32 |

The data is fit by interpolating quadratic spline given by

$$\begin{aligned} f(x) &= ax - 1,\ 1 \le x \le 2, \\ &= -2x^2 + 14x - 9,\ 2 \le x \le 4 \\ &= bx^2 + cx + d,\ 4 \le x \le 6 \\ &= 25x^2 - 303x + 928,\ 6 \le x \le 7 \end{aligned}$$

where $a, b, c,$ and $d$ are constants. What is the estimated value of $\displaystyle\int_{1.5}^{3.5} f(x)\,dx$?

(A) 23.500

(B) 25.667

(C) 25.750

(D) 28.000

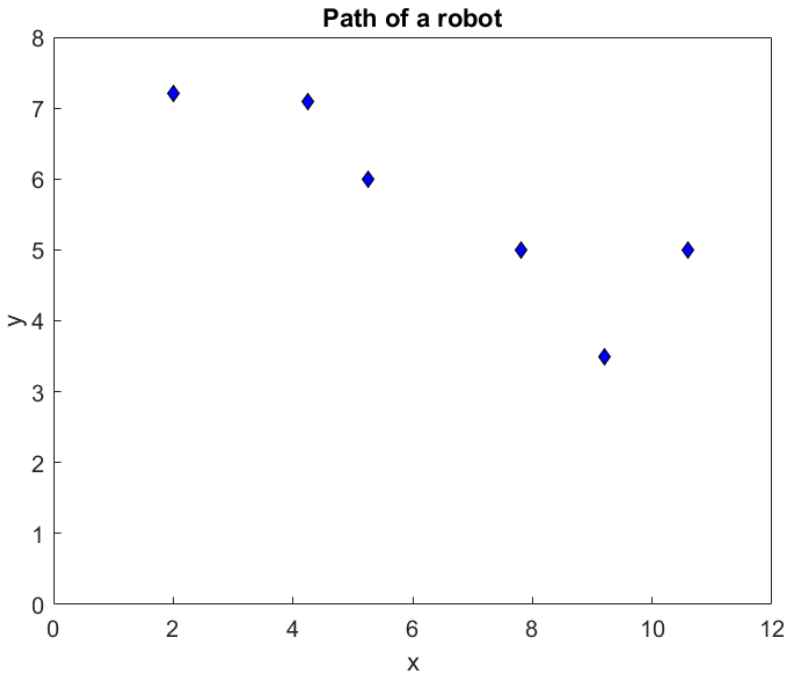
(6). A robot needs to follow a path that passes consecutively through six points, as shown in the figure. To find the shortest path that is also smooth, you would recommend which of the following?

(A) Pass a fifth-order polynomial through the data

(B) Pass interpolating linear spline through the data

(C) Pass interpolating quadratic spline through the data

(D) Regress the data to a second-order polynomial




For complete solution, go to

http://nm.mathforcollege.com/mcquizzes/05inp/quiz_05inp_spline_solution.pdf

# Problem Set

## Chapter 05.05: Spline Method of Interpolation

(1). The following $y$ vs $x$ data is given.

| $x$ | 2 | 3 | 6 |
|---|---|---|---|
| $y$ | 4.75 | 5.25 | 45 |

a) Set up the equations to solve for the interpolating quadratic spline that goes through the data.

b) Use a program such as MATLAB to solve the equations and then write down the interpolating quadratic spline.

c) Estimate the value of $y(3.6)$.

**Answer:** *a)*

$$\begin{bmatrix} 4 & 2 & 1 & 0 & 0 & 0 \\ 9 & 3 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 9 & 3 & 1 \\ 0 & 0 & 0 & 36 & 6 & 1 \\ 6 & 1 & 0 & -6 & -1 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} a_1 \\ b_1 \\ c_1 \\ a_2 \\ b_2 \\ c_2 \end{bmatrix} = \begin{bmatrix} 4.75 \\ 5.25 \\ 5.25 \\ 45 \\ 0 \\ 0 \end{bmatrix}$$

*b)*

| $i$ | $a_i$ | $b_i$ | $c_i$ |
|---|---|---|---|
| 1 | 0 | 0.5 | 3.75 |
| 2 | 4.25 | −25 | 42 |

*c)* 7.08

(2). The following $y$ vs $x$ data is given.

| $x$ | 1 | 2 | 3 | 5 | 6 |
|---|---|---|---|---|---|
| $y$ | 4.75 | 4 | 5.25 | 15 | 45 |

The data is fit by interpolating quadratic spline given by

$$
\begin{aligned}
f(x) &= -0.75x + 5.5, 1 \leq x \leq 2 \\
&= 2x^2 - 8.75x + 13.5, 2 \leq x \leq 3 \\
&= cx^2 + gx + h, 3 \leq x \leq 5 \\
&= jx^2 + kx + l, 5 \leq x \leq 6
\end{aligned}
$$

where $c,\ g,\ h,\ j,\ k,$ and $l$ are constants.

a) Find the value of $c,\ g,\ h,\ j,\ k,$ and $l$.

b) Compare the value of the function at $x = 2.3$ using interpolating linear spline and interpolating quadratic spline.

**Answer:**

$a)$ $c = 0.8125$
$g = -1.625$
$h = 2.8125$
$j = 23.5$
$k = -228.5$
$l = 570$

$b)$ linear $= 4.375;$ quadratic $= 3.955$

(3). The following incomplete $y$ vs. $x$ data is given.

| $x$ | 1 | 2.2 | 3.7 | 5.1 | 6 |
|---|---|---|---|---|---|
| $y$ | 4.25 | 6 | 5.25 | 15.1 | ????? |

The data is fit by interpolating quadratic spline given by

$$
\begin{aligned}
f(x) &= 1.4583x + 2.7917, 1 \leq x \leq 2.2 \\
&= -1.3056x^2 + 7.2028x - 3.5278, 2.2 \leq x \leq 3.7 \\
&= cx^2 + gx + h, 3.7 \leq x \leq 5.1 \\
&= jx^2 + kx + 1, 5.1 \leq x \leq 6
\end{aligned}
$$

where $c,\ g,\ h,\ j,\ k,$ and $l$ are constants. What is the value of $g$? Show all your steps clearly.

**Answer:** $g = -52.6391$

(4). The following incomplete $y$ vs. $x$ data is given.

| $x$ | 1 | 2.2 | 3.7 | 5.1 | 6 |
|---|---|---|---|---|---|
| $y$ | 4.25 | 6 | 5.25 | ???? | ????? |

The data is fit by interpolating quadratic spline given by

$$
\begin{aligned}
f(x) &= 1.4583x + 2.7917, 1 \leq x \leq 2.2 \\
&= -1.3056x^2 + 7.2028x - 3.5272, 2.2 \leq x \leq 3.7 \\
&= cx^2 + gx + h, 3.7 \leq x \leq 5.1 \\
&= jx^2 + kx + l, 5.1 \leq x \leq 6
\end{aligned}
$$

where $c,\ g,\ h,\ j,\ k,$ and $l$ are constants. What is the value of $\dfrac{df}{dx}$ at $x = 2.67$?

**Answer:** 0.23133

(5). The following incomplete $y$ vs. $x$ data is given.

| $x$ | 1 | 2.2 | 3.7 | 5.1 | 6 |
|---|---|---|---|---|---|
| $y$ | 4.25 | 6 | 5.25 | ???? | ????? |

The data is fit by interpolating quadratic spline given by

$$
\begin{aligned}
f(x) &= 1.4583x + 2.7917, 1 \leq x \leq 2.2 \\
&= -1.3056x^2 + 7.2028x - 3.5272, 2.2 \leq x \leq 3.7 \\
&= cx^2 + gx + h, 3.7 \leq x \leq 5.1 \\
&= jx^2 + kx + l, 5.1 \leq x \leq 6
\end{aligned}
$$

where $c,\ g,\ h,\ j,\ k,$ and $l$ are constants. What is the value of $\displaystyle\int_{1.5}^{2.5} f(x)dx$?

**Answer:** 5.6965

(6). Given three data points $(1, 6),\ (3, 28)$, and $(10, 231)$, it is found that the function $y = 2x^2 + 3x + 1$ passes through the three data points. To find the length of the polynomial curve from $x = 5$ to $x = 8$, a student uses the formula, $S = \int_a^b \sqrt{1 + (dy/dx)^2}dx$. However, this formula gives the student an integral that cannot be solved exactly. Instead, the student approximates the polynomial curve by drawing interpolating linear spline that consists of piecewise functions from $x = 5$ to $x = 6$, from $x = 6$ to $x = 7$ and from $x = 7$ to $x = 8$. The student then finds the length of the interpolating linear spline from $x = 5$ to $x = 8$. What is the student's estimate of the length of the function from $x = 5$ to $x = 8$?

**Answer:** 87.052 (getting 87 as the answer is wrong)

# Chapter 05.06: Extrapolation is a Bad Idea

Lesson: Extrapolation is a Bad Idea

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *enumerate why using extrapolation can be a bad idea.*

2) *show through an example why extrapolation can be a bad idea.*

## Description

This conversation illustrates the pitfall of using extrapolation.

(Due to certain reasons, this student wishes to remain anonymous.)

This takes place in Summer Session B – July 2001

**Student:** "Hey, Dr. Kaw! Look at this cool new cell phone I just got!"

**Kaw:** "That's nice. It better not ring in my class or it's mine."

**Student:** "What would you think about getting stock in this company?"

**Kaw:** "What company is that?"

**Student:** "WorldCom! They're the world's leading global data and internet company."

**Kaw:** "So?"

**Student:** "They've just closed the deal today to merge with Intermedia Communications, based right here in Tampa!"

**Kaw:** "Yeah, and . . . ?"

**Student:** "The stock's booming! It's at $14.11 per share and promised to go only one way—up! We'll be millionaires if we invest now!"

**Kaw:** "You might not want to assume their stock will keep rising . . . besides, I'm skeptical of their success. I don't want you putting yourself in financial 'jeopardy!' over some silly extrapolation. Take a look at these NASDAQ composite numbers (Table 1)."

**Student:** "That's only up to two years ago . . . "

**Kaw:** "That's right. Looking at this data, don't you think you should've invested back then?"

**Student:** "Well, didn't the composite drop after that?"

**Kaw:** "Right again, but look what you would've hoped for if you had depended on that trend continuing (Figure 1)."

**Student:** "So you're saying that . . . ?"

**Kaw:** "You should seldom depend on extrapolation as a source of approximation! Just take a look at how wrong you would have been (Table 2)."

**Table 1**. End of year NASDAQ composite data

| End of year | NASDAQ |
|---|---|
| 1 | 751.96 |
| 2 | 1052.13 |
| 3 | 1291.03 |
| 4 | 1570.35 |
| 5 | 2192.69 |
| 6 | 4069.31 |

Note: The range of years in Table 1 are actually between 1994 (Year 1) and 1999 (Year 9). Numbers start from 1 to avoid round-off errors and near singularities in matrix calculations.

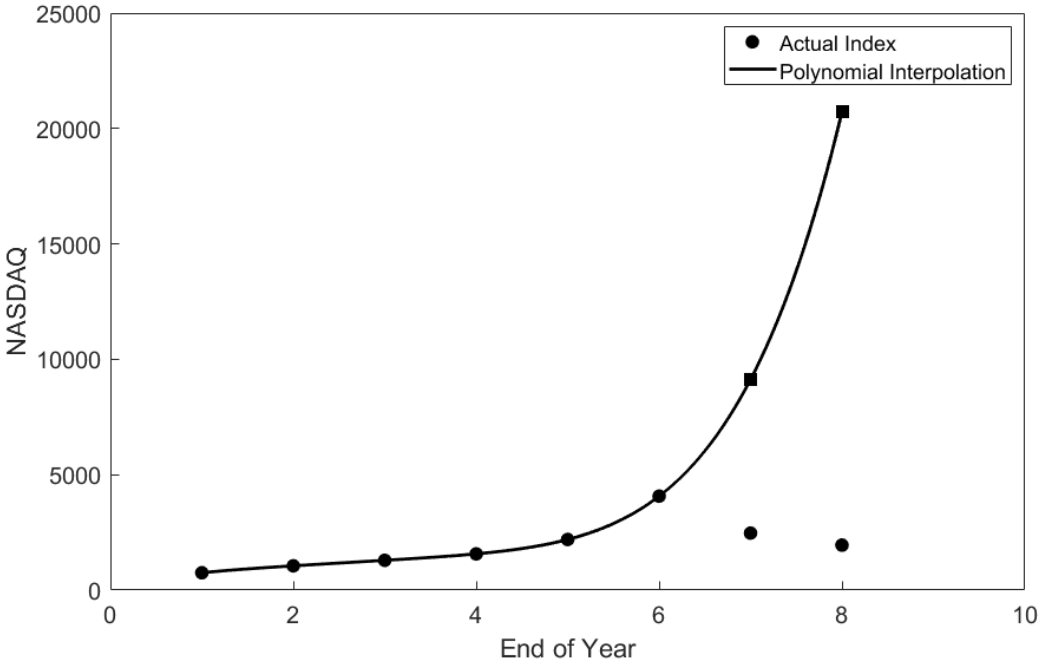



**Figure 1** Data from 1994 to 1999 extrapolated to yield results for 2000 and 2001 using polynomial extrapolation.

**Table 2** Absolute relative true error of polynomial interpolation.

| End of Year | Actual | Fifth order polynomial interpolation | Absolute relative true error |
|---|---|---|---|
| 2000 | 2471 | 9128 | 269.47% |
| 2001 | 1950 | 20720 | 962.36% |

**Student:** "Now wait a sec! I wouldn't have been quite that wrong. What if I had used cubic splines instead of a fifth-order interpolant?"

**Kaw:** "Let's find out."

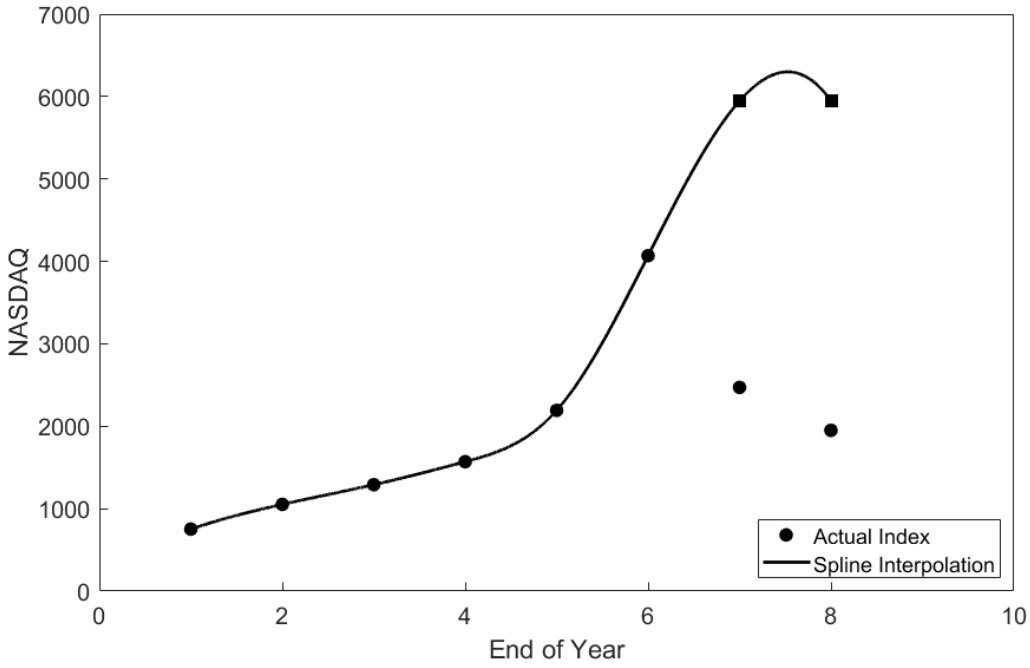



**Figure 2** Data from 1994 to 1999 extrapolated to yield results for 2000 and 2001 using cubic spline interpolation.

**Table 3** Absolute relative true error of cubic spline interpolation

| End of Year | Actual | Cubic spline interpolation | Absolute relative true error |
|---|---|---|---|
| 2000 | 2471 | 5945.9 | 140.63% |
| 2001 | 1950 | 5947.4 | 204.99% |

**Student:** "There you go. That didn't take so long (Figure 2 and Table 3)."

**Kaw:** "Well, let's think about what this data means. If you had gone ahead and invested, thinking your projected yield would follow the spline, you would have only been 205% (Table 3) wrong, as opposed to being 962% (Table 2) wrong by following the polynomial. That's not so bad, is it?"

**Student:** "Okay, you've got a point. Maybe I'll hold off on being an investor and just use the cell phone."

**Kaw:** "You've got a point, too—you're brighter than you look ... that is if you turn off the phone before coming to class."

* * * * *

<One year later ... July 2002>

**Student:** "Hey, Dr. Kaw! Whatcha got for me today?"

**Kaw:** "The Computational Methods students just took their interpolation test today, so here you go. <hands stack of tests to student> Time to grade them!"

**Student:** <Grunt!> "That's a lot of paper! Boy, interpolation ... learned that a while ago."

**Kaw:** "You haven't forgotten my lesson to you about not extrapolating, have you?"

**Student:** "Of course not! Haven't you seen the news? WorldCom just closed down 93% from 83¢ on June 25 to 6¢ per share! They've had to recalculate their earnings, so your skepticism really must've spread. Did you have an"in" on what was going on?"

**Kaw:** "Oh, of course not. I'm just an ignorant numerical methods professor."

# Chapter 05.10: Length of a Curve

Lesson: Length of a Curve

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *estimate the length of a curve given by a continuous function,*

2) *estimate the length of a curve found by interpolation of discrete data points.*

## Introduction

The length, $L$ of a smooth function $y(x)$ from $(a, f(a))$ *to* $(b, f(b))$is given analytically by (Figure 1)

$$L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx \qquad (1)$$

Over the interval $[a, b]$, the integrand is required to be integrable, $y(x)$ needs to be differentiable, and $y'(x)$ has to be continuous. The length can be thought of as the distance one would travel if traversing along the path of the curve. An application would be to calculate the length of the path traversed by a robot that moves from one point to another on a continuous curve.

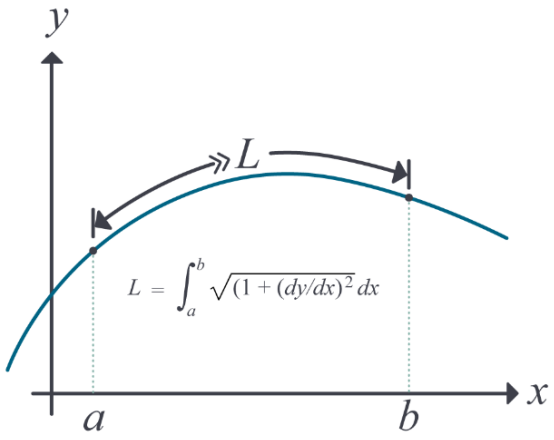


**Figure 1.** Length of a curve.

**Example 1**

Plot the function $y = \frac{1}{3}(x^2+2)^{\frac{3}{2}}$ and find the length of the curve from $x = 0$ to $x = 3$.

**Solution**

We use the equation

$$L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx \qquad (E1.1)$$

We have

$$y = \frac{1}{3}(x^2+2)^{\frac{3}{2}}$$

So,

$$\begin{aligned} \frac{dy}{dx} &= \left(\frac{1}{3}\right) \times \left(\frac{3}{2}\right) \times \left(x^2+2\right)^{\frac{3}{2}\ -1} \times (2x) \\ &= x\sqrt{x^2+2} \end{aligned} \qquad (E1.2)$$

Substituting Equation (E1.2) in Equation (E1.1), the length of the curve from $x = 0$ to $x = 3$ is

$$\begin{aligned} L &= \int_0^3 \sqrt{1 + \left(x\sqrt{x^2+2}\right)^2}\, dx \\ &= \int_0^3 \sqrt{1 + x^2\,(x^2+2)}dx \\ &= \int_0^3 \sqrt{1 + x^4 + 2x^2}dx \\ &= \int_0^3 \sqrt{(x^2+1)^2}dx \\ &= \int_0^3 (x^2+1)dx \\ &= \left[\frac{x^3}{3} + x\right]_0^3 \\ &= 12 \end{aligned}$$

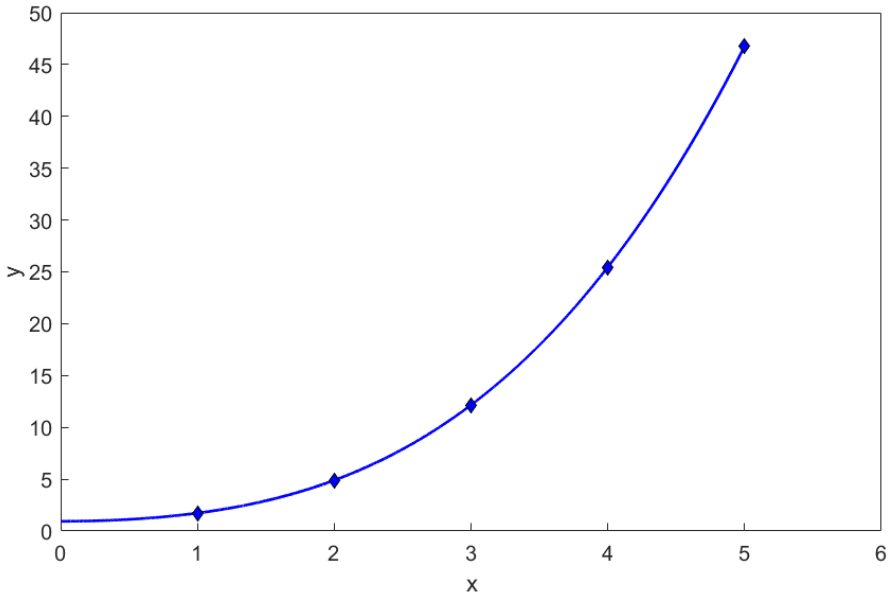



**Figure 2.** Plot of the function $y = \frac{1}{3}(x^2+2)^{\frac{3}{2}}$

**Example 2**

A function passes consecutively through the three points $(2,8)$, $(4,18)$, $(8,5)$ and is found by as a $2^{nd}$ order polynomial interpolant.

a) Find the function.

b) Find the approximate length of the function from $x = 2$ to $x = 8$. You can use the Simpson's $1/3^{rd}$ rule of integration rule as given by

$$\int_a^b f(x)dx \approx \frac{(b-a)}{6}\left[f(a) + 4f\left(\frac{a+b}{2}\right) + f(b)\right]$$

c) What is the absolute relative true error for part (b) if the exact length from $(2,8)$ to $(8,5)$ of the second-order polynomial is 26.042?

***Solution***

a) The second-order interpolant is given by

$$y = a_0 + a_1 x + a_2 x^2$$

Since it passes through $(2,8),(4,18),(8,5)$, we can set up three equations as

$$\begin{aligned} a_0 + a_1 2 + a_2 2^2 &= 8 \\ a_0 + a_1 4 + a_2 4^2 &= 18 \qquad (E2.1 - E2.3) \\ a_0 + a_1 8 + a_2 8^2 &= 5 \end{aligned}$$

Solving Equations (E2.1), (E2.2), (E2.3) simultaneously gives

$$a_0 = -13,\ a_1 = 13.25,\ a_2 = -1.375$$

The resulting interpolant is

$$y = -13 + 13.25x - 1.375x^2 \qquad (E2.4)$$

b) The length of the curve $y(x)$ from $a$ to $b$ is given by

$$L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx \qquad (E2.5)$$

Since

$$y = -13 + 13.25x - 1.375x^2$$

$$\begin{aligned} \frac{dy}{dx} &= \frac{d}{dx}(-13 + 13.25x - 1.375x^2) \\ &= 13.25 - 1.375(2x) \\ &= 13.25 - 2.75x \end{aligned} \qquad (E2.6)$$

Substituting Equation (E2.6) in Equation (E2.5), the length of the curve from $x = 2$ to $x = 8$ is given by

$$L = \int_2^8 \sqrt{1 + (13.25 - 2.75x)^2}\, dx \qquad (E2.7)$$

Recall from the problem statement of part (b), integral given by Equation (E2.7) is asked to be found approximately by using Simpson's $1/3^{\text{rd}}$ rule, that is,

$$\int_a^b f(x)dx \approx \frac{(b-a)}{6}\left[f(a) + 4f\left(\frac{a+b}{2}\right) + f(b)\right]$$

In this case

$$f(x) = \sqrt{1 + (13.25 - 2.75x)^2}\ ,\ a = 2,\ b = 8$$

Hence Equation (E2.7) gives

$$\begin{aligned} L &= \int_2^8 f(x)dx \\ &\approx \frac{(8-2)}{6}\left[f(2) + 4f\left(\frac{2+8}{2}\right) + f(8)\right] \\ &= f(2) + 4f(5) + f(8) \\ &= 7.814 + 4(1.118) + 8.807 \\ &= 21.093 \end{aligned}$$

c)

$$\text{True value } = 26.042$$

$$\text{Approximate value } = 21.093$$

So the absolute relative true error is

$$\begin{aligned} |\epsilon_t| &= \left| \frac{\text{True value} - \text{Approximate value}}{\text{True value}} \right| \times 100 \\ &= \left| \frac{26.042 - 21.093}{26.042} \right| \times 100 \\ &= 19.003\% \end{aligned}$$

# Chapter 05.10: Length of a Curve

Lesson: Comparing Lengths of Curves

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *find the shortest smooth path through consecutive points, and*

2) *compare the lengths of different paths.*

## Introduction

In the previous lesson, you learned how to find the length of a curve numerically. In this lesson, via an example, we apply what we learned to find the shortest but smoothest path to go through consecutive points.

### Example 1

**Peter:** "Dr. Kaw, I am taking a course in manufacturing. We are solving the following problem. A robot arm with a rapid laser is used to do a quick quality check, such as the radius of hole, on six holes on a rectangular plate $15'' \times 10''$ at several points as shown in Table 1 and Figure 1.

**Table 1** The coordinate values of six holes on a rectangle plate.

| $x$ | $y$ |
|---|---|
| 2.00 | 7.2 |
| 4.5 | 7.1 |
| 5.25 | 6.0 |
| 7.81 | 5.0 |
| 9.20 | 3.5 |
| 10.60 | 5.0 |

I am using Excel to fit a fifth-order polynomial through the 6 points. But, when I plot the polynomial, it is taking a long path! (Figure 2)"

**Kaw:** "Why do you not just join the consecutive points by a straight line, just like the kids do at Pizza Hut™ with those 'Connect the dots' activities?"

**Peter:** "You are making me hungry, and I wish it were that easy. The path of the robot going from one point to another point needs to be smooth so as to avoid sharp jerks in the arm that can otherwise create premature wear and tear of the robot arm."

**Kaw:** "As I recall, you took my course in Numerical Methods. What was that ..... one year ago?"

**Peter:** "Yes, your memory is sharp, but my retention from that course – can we not talk about that?!?"

**Kaw:** "Come into my office. I wrote this program using Maple. See this function, $f(x) = 1/(1+25x^2)$. I am choosing 7 points equidistantly (Table 2) between $-1$ and 1.

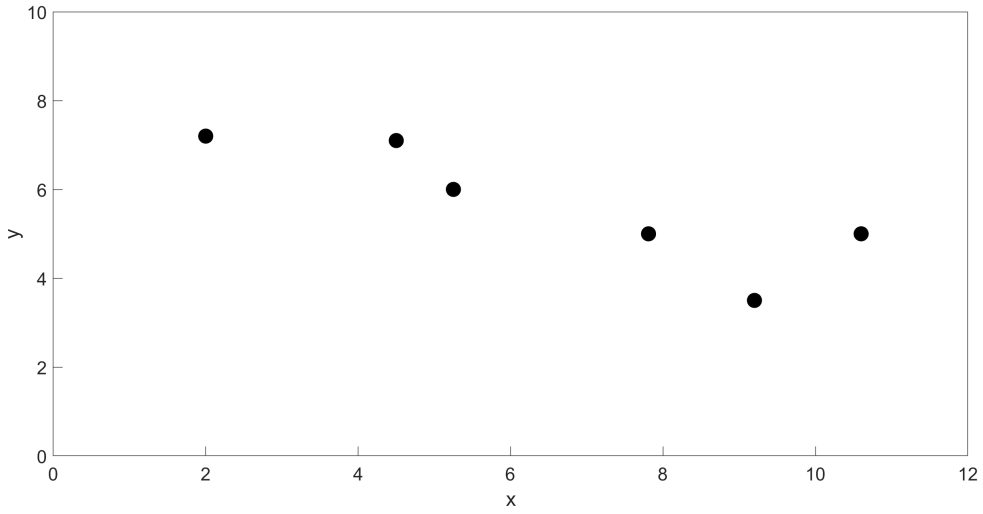



**Figure 1** Locations of holes on the rectangular plate.

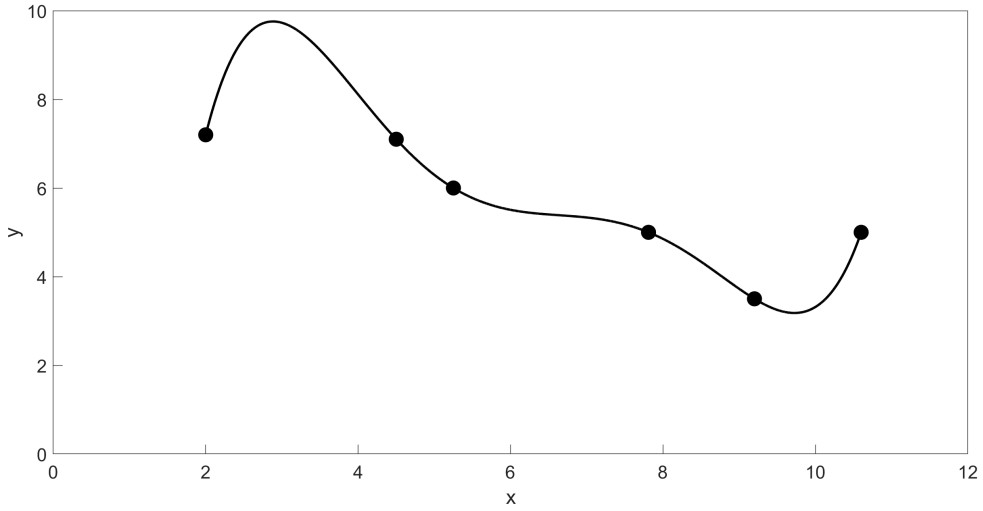



**Figure 2** Approximating the path of the robot using 5th order polynomial.

Now, look at the sixth order interpolating polynomial and the original function (Figure 3). See the oscillations in the interpolating polynomial. In 1901, Runge used this example function to show that higher-order interpolation is a bad idea. One of the solutions to your robot path problem is to use

quadratic or cubic spline interpolation. That gives a smooth curve with fewer oscillations and a smoother and shorter path."

**Table 2** The coordinate values of 7 equidistantly spaced points.

| $x$ | $y = \dfrac{1}{1+25x^2}$ |
|---|---|
| $-1$ | 0.038462 |
| $-0.66667$ | 0.0826 |
| $-0.33333$ | 0.264706 |
| 0 | 1 |
| 0.333333 | 0.264706 |
| 0.666667 | 0.082569 |
| 1 | 0.0385 |

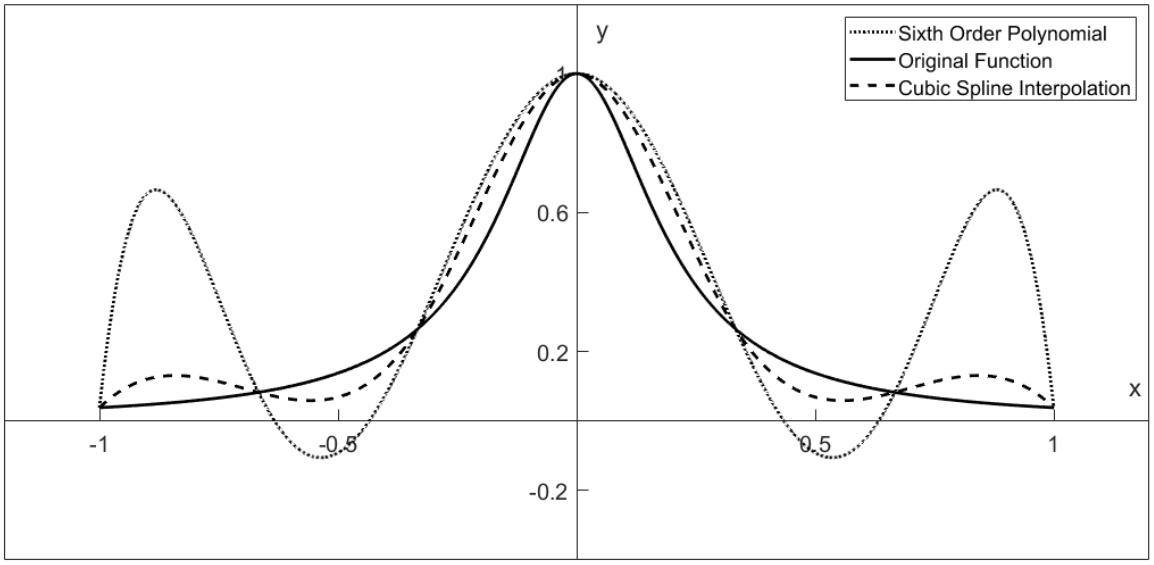



**Figure 3** Runge's function interpolated.

**Peter:** "Okay. Let's give that a try."

**Kaw:** "Now, let's try generating a set of cubic splines to go through the data:"

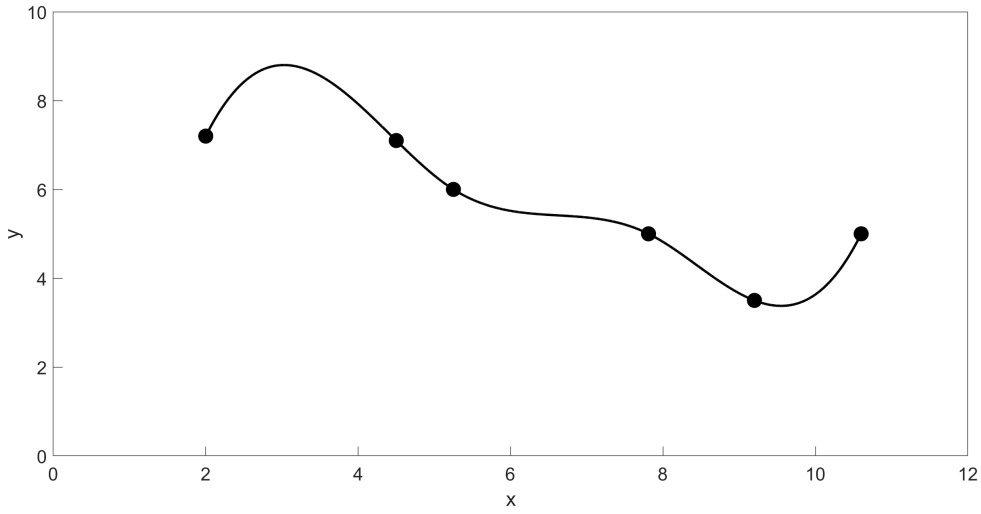



**Figure 4** Path of the robot arm using cubic spline interpolation.

**Peter:** "Wow! That (Figure 4) looks much better!"

**Kaw:** "It may look better, but let's find out for sure. See if you can combine the two plots (Figure 5) and compare the lengths of each path."

**Peter:** "The length of a path if from to is given by

$$S = \int_a^b \sqrt{1 + \left(\frac{df}{dx}\right)^2}\, dx \qquad (1)$$

Right?"

**Table 3** Comparison of the length of curves.

| **Type of interpolation** | **5th order polynomial** | **Cubic Spline** |
|---|---|---|
| Length of Curve | $14.919''$ | $11.248''$ |

**Kaw:** "Yes! You solved the problem. See Table 3 for answers."

**Peter:** "I guess your class was good for something, after all, Dr. Kaw."

**Kaw:** "Are you sure? You could have always fallen back on the connecting-the-dots method. Besides, you don't want to grow up ... you're a Pizza Hut™ kid, right?"

**Peter:** "No, that's a Toys R' Us™ kid."

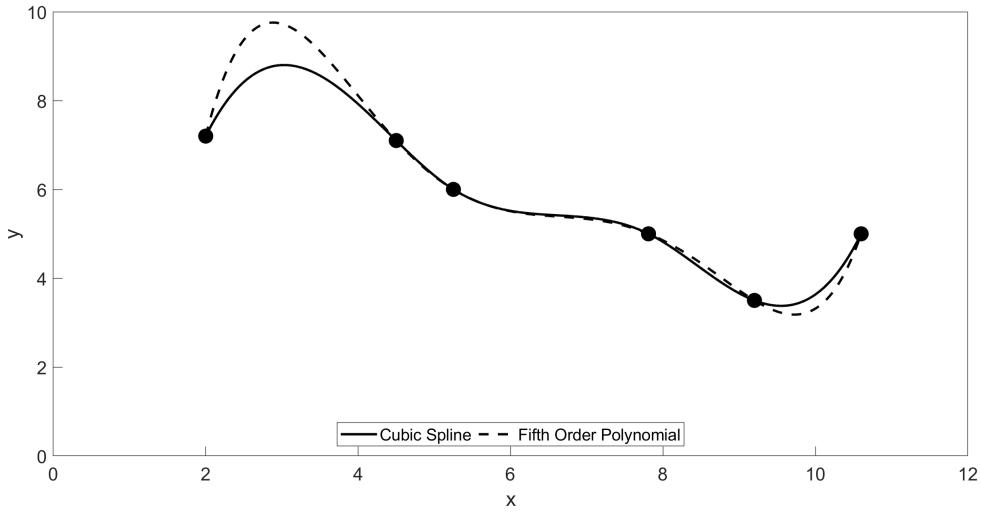



**Figure 5** Path of robot arm compared using polynomial interpolation and cubic spline interpolation.

# Multiple Choice Test

## Chapter 05.10 Length of a Curve

(1). The arc length of a smooth cartesian curve $f(x)$ from $a$ to $b$ is given by

(A) $\int_a^b f(x)\,dx$

(B) $(b-a)^2 + (f(b) - f(a))^2$

(C) $\int_a^b \sqrt{\left(1 + \frac{dy}{dx}\right)}dx$

(D) $\int_a^b \sqrt{\left(1 + \left(\frac{dy}{dx}\right)^2\right)}dx$

(2). Which of these definite integrals gives the length of the arc of the function $f(x) = 3x + \sin x$ from $x = 2$ to $x = 5$

(A) $\int_2^5 (3x + \sin x)dx$

(B) $\int_2^5 \sqrt{1 + (3 + \cos x)}dx$

(C) $\int_2^5 \sqrt{1 + (3 + \cos x)^2}dx$

(D) $\int_2^5 \sqrt{1 + (3x + \sin x)^2}dx$

(3). The length of the curve $y = \sqrt{1 - x^2}$ from $x = 0$ to $x = 1$ is most nearly equal to

(A) 0.7854

(B) 1.414

(C) 1.518

(D) 1.571

(4). A robotic drawing pencil is using linear spline interpolation to trace a path consecutively through three data points $(4, 20)$, $(6, 10)$, and $(9, 25)$. What is the length of the path that the pencil traces if it begins at the first data point and ends at the last data point?

(A) 15.297

(B) 10.198

(C) 7.071

(D) 25.495

(5). A path is traversed consecutively through three points $(2, 4)$, $(5, 11)$, $(8, 3)$ using a quadratic polynomial. Estimate the exact length of the path up to at least 4 significant digits. Use only MATLAB or some other program to do this problem - do not do it manually as it will take time.

(A) 16.701

(B) 6.0827

(C) 16.160

(D) 51.387

(6). A cable hangs between two poles that are 50 feet apart. The shape of the cable is given by

$$f(x) = a \cosh\left(\frac{x}{a}\right), \ -25 \leq x \leq 25,$$

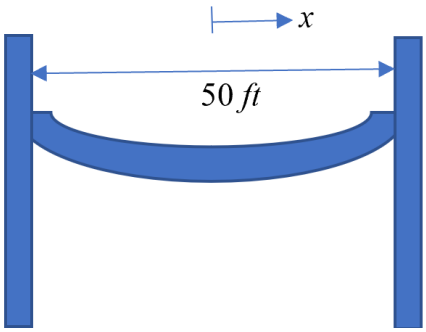
where $a$ is dependent on the tension in the cable and the weight per unit length of the cable, and $x$ is measured between the poles but from the center of the cable. If the length of the cable is 60 feet, the formula that will allow us to find the value of $a$ is

(A) $2\ a^2 \sinh\left(\frac{25}{a}\right) = 60$

(B) $2\ a\ sinh\left(\frac{25}{a}\right) = 60$

(C) $2\ a\ sinh\left(\frac{25}{a}\right) = 50$

(D) $2\ a^2 \sinh\left(\frac{25}{a}\right) = 50$



# Chapter 06.00: Physical Problem for Regression

Lesson: General Engineering

## Summary

A student wants to pick the best mousetrap out of many for use in a mousetrap car. To do so, one requires to regress the torsion in the mousetrap spring to the angle through which it is opened.

## Modeling

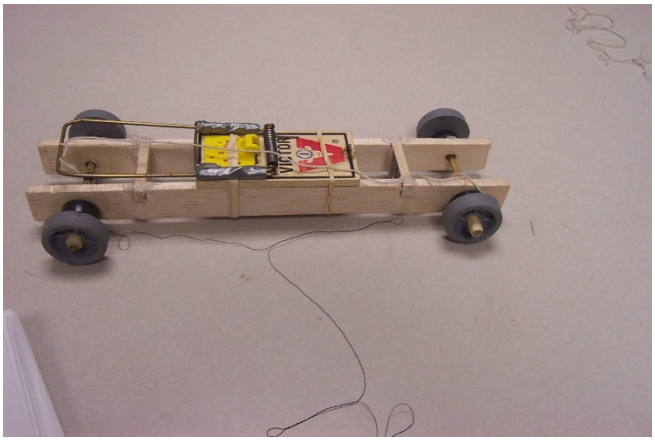


**Figure 1** Mousetrap car.

In the Summer of 2003, a high school student worked with me to study the physics of a mousetrap car to build the car that would go the fastest and/or the farthest.

### So what is a mousetrap car?

A mousetrap car (Figure 1) is a vehicle that is powered by the energy stored in the spring of a mousetrap. One of the basic ways most mousetrap vehicles are set into motion is by connecting the lever of the mousetrap bar through a string to the car's axle. As the mousetrap lever is released, the tension that was built up in the spring is released, and the car sets into motion.

So to get the car to go the farthest and fastest, we need to store as much potential energy in the mousetrap spring. Since there is no specification on the mousetrap other than using a certain brand, we could use a mousetrap with the largest spring constant as that would translate to the largest potential energy stored.

Therefore, we bought several mousetrap springs and conducted a simple experiment to determine the spring constant of the mousetrap spring. These values of the spring constant would then allow us to find the potential energy stored in the mousetraps.

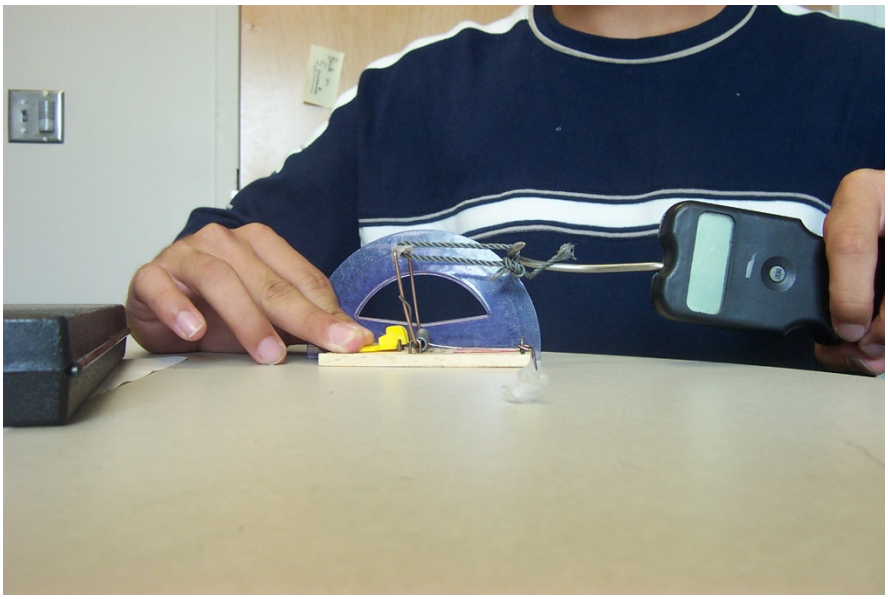


**Figure 2** Experimental setup to measure the spring constant.

To find the spring constant of the mousetrap spring, we conducted the following experiment. We pulled the end of the mousetrap lever by a force probe (a fishhook) and measured the angle through which it is pulled by a protractor (Figure 2). We pulled on the end of the spring via a string so that the force on the lever is applied at a right angle to the lever. This way, the torque applied is simply the product of the force, as measured by the fishhook (Figure 3) and the length of the lever arm.

**Figure 3** Force probe designed to measure the amount of force an object contains.

We measured the force needed to pull the lever for several different angles. The data for one of the mousetraps is given in Table 1.

The relationship between the torque applied and the angle of the rotation of the spring rotation is assumed to be a straight line

$$T = k_0 + k_1\theta \qquad (1)$$

where

$T$ = Torque (N-m)

$\theta$ = Angle through which the spring is rotated

**Table 1** Force vs. angle of lever rotation (Mousetrap #1).

| Angle (degrees) | Force (lbs) |
|---|---|
| 40 | 0.9 |
| 55 | 1.0 |
| 65 | 1.1 |
| 90 | 1.2 |
| 110 | 1.5 |

The data in Table 1 is converted to the SI system of units, and the torque is calculated using the measured lever moment arm of 47 mm, that is

$$T = FL \qquad (2)$$

where,

$T =$ Torque (N-m)

$F$ = Force applied (N)

$L =$ Moment arm (m)

| Angle | Torque |
|---|---|
| Radians | N-m |
| 0.698132 | 0.188224 |
| 0.959931 | 0.209138 |
| 1.134464 | 0.230052 |
| 1.570796 | 0.250965 |
| 1.919862 | 0.313707 |

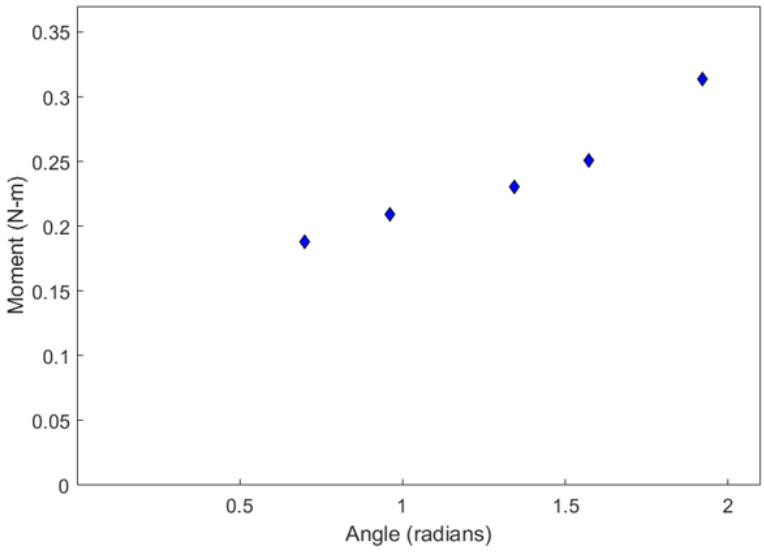



**Figure 4** Data and plot of torque vs angle of rotation.

Other mousetrap springs were tested accordingly, and the data is given in Tables 2 and 3.

From these tables, we can find the $T = k_0 + k_1\theta$ relationship for each of three mousetraps. The potential energy, $U$ stored in the spring would be given by

$$\begin{aligned} U &= \int_{\theta_{low}}^{\theta_{high}} T d\theta \\ &= \int_{\theta_{low}}^{\theta_{high}} (k_0 + k_1\ \theta) d\theta \end{aligned}$$

Knowing that in our case, $\theta_{low} = 0$ and $\theta_{high} = \pi$, the maximum potential energy stored

$$\begin{aligned} U_{\max} &= \int_0^{\pi} (k_0 + k_1\ \theta) d\theta \\ &= k_0\pi + k_1\frac{\pi^2}{2} \end{aligned} \tag{3}$$

**Table 2** Force vs. angle of lever rotation (Mousetrap #2).

| Angle (degrees) | Force (lbs) |
|---|---|
| 38 | 0.9 |
| 58 | 1.0 |
| 65 | 1.1 |
| 90 | 1.2 |
| 120 | 1.6 |

**Table 3** Force vs. angle of lever rotation (Mousetrap #3).

| Angle (degrees) | Force (lbs) |
|---|---|
| 40 | 0.9 |
| 57 | 1.0 |
| 65 | 1.1 |
| 90 | 1.2 |
| 135 | 1.7 |

## Questions

(1) Find the constants of the regression model for the spring for the three cases (Tables 1, 2, and 3).

(2) Based on maximum potential energy stored, which of the three springs would you choose?

# Chapter 06.00: Physical Problem for Regression

Lesson: Chemical Engineering

## Summary

Infrared spectroscopy requires regression.

## Modeling

Chemical bonds have specific frequencies at which they will vibrate. These frequencies depend on the length of the bonds and the mass of the atoms at either end of bonds. Hence, vibrations of a molecule can be studied by shining infrared light onto the surface of the molecule. If one end of the molecular bond has a different charge from the other end (dipole moment), then the molecule can absorb infrared spectrum. However, this absorption occurs only at a certain fixed frequencies characteristic of the molecule and its component parts. Thus, an infrared light reflected from the surface of the bond will show absorption peaks characteristic of the molecule. This forms the basis of Infrared Spectroscopy (IRS).

To measure a sample, infrared light at a specific frequency is beamed onto the sample and the amount of energy absorbed is recorded. A chart is built up when this is repeated for several other frequencies. By examining the chart, one experienced in the art can identify the substance.

Fourier Transform Infrared Spectroscopy is a measurement technique for collecting infrared spectra and analyzing it. The wavenumber is the inverse of frequency while absorbance is proportional to the energy of the infrared light absorbed.

Table 1gives the FT-IR (Fourier Transform Infra Red) data of a 1:1 (by weight) mixture of ethylene carbonate (EC) and dimethyl carbonate (DMC). One would like to develop an equation which relates the absorbance as a function of wave number.

**Table 1** Absorbance as a function of wavenumber

| Wavenumber | Absorbance |
|---|---|
| $cm^{-1}$ | (arbitrary unit) |
| 804.184 | 0.1591 |
| 808.041 | 0.1447 |
| 815.755 | 0.1045 |
| 821.540 | 0.0731 |
| 827.326 | 0.0439 |
| 829.254 | 0.0357 |
| 831.183 | 0.0285 |
| 833.111 | 0.0226 |

| Wavenumber | Absorbance |
|---|---|
| 835.040 | 0.0178 |
| 836.968 | 0.0140 |
| 838.897 | 0.0109 |
| 840.825 | 0.0087 |
| 846.611 | 0.0050 |
| 852.396 | 0.0039 |
| 860.110 | 0.0045 |
| 869.753 | 0.0073 |
| 877.467 | 0.0142 |
| 881.324 | 0.0206 |
| 883.252 | 0.0250 |
| 885.181 | 0.0304 |
| 889.038 | 0.0448 |
| 892.895 | 0.0649 |
| 896.752 | 0.0910 |
| 900.609 | 0.1204 |

## Questions

(1) Find a best-fit equation (polynomial) to the data trend.

(2) A student had used only the first 8 data points in his best-fit equation in (a), What conclusion would he/she reach about the data? Would the conclusion reached from the use of the first 8 data points be valid for the whole data set?

# Chapter 06.00: Physical Problem for Regression

Lesson: Civil Engineering

## Summary

Finding the Young's modulus of a unidirectional graphite/epoxy composite material requires regression.

## Modeling

A composite is a structural material that consists of combining two or more constituents. The constituents are combined at a macroscopic level and are not soluble in each other. One constituent is called the *reinforcing phase* and the one in which it is embedded is called the *matrix*. The reinforcing phase material may be in the form of fibers, particles, or flakes. The matrix phase materials are generally continuous. Examples of composite systems include concrete reinforced with steel and epoxy reinforced with graphite fibers, etc.

A special case of composite materials are advanced composites, and are defined as composite materials which are traditionally used in the aerospace industries. These composites have high performance reinforcements of a thin diameter in a matrix material such as epoxy and aluminum. Examples are Graphite/Epoxy, Kevlar/Epoxy, and Boron/Aluminum composites. These materials have now found applications in commercial industries as well.

Monolithic metals and their alloys cannot always meet the demands of today's advanced technologies. Only by combining several materials can one meet the performance requirements. For example, trusses and benches used in satellites need to be dimensionally stable in space during temperature changes between$-256^\circ F\,(-160^\circ C)$ and $200^\circ F\,(93.3^\circ C)$. Limitations on coefficient of thermal expansion hence are low and may be of the order of $1 \times 10^{-7} in/in/^\circ F\left(1.8 \times 10^{-7} m/m/^\circ C\right)$. Monolithic materials cannot meet these requirements, which leave composites, such as Graphite/Epoxy, as the only materials to satisfy this requirement.

In many cases, using composites is more efficient. For example, in the highly competitive airline market, one is continuously looking for ways to lower the overall mass of the aircraft without decreasing the stiffness and strength of its components. This is possible by replacing conventional metal alloys with composite materials. Even if the composite material costs may be higher, the reduction in the number of parts in an assembly and the savings in fuel costs make them more profitable. Reducing one pound (0.453 kg) of mass in a commercial aircraft can save up to 360 gallons (1360 liters) of fuel per year; and fuel expenses are 25% of the total operating costs of a commercial airline.

The general test method to find the Young's modulus of a composite material is the ASTM Test Method for Tensile Properties of Fiber-Resin Composites (D3039).

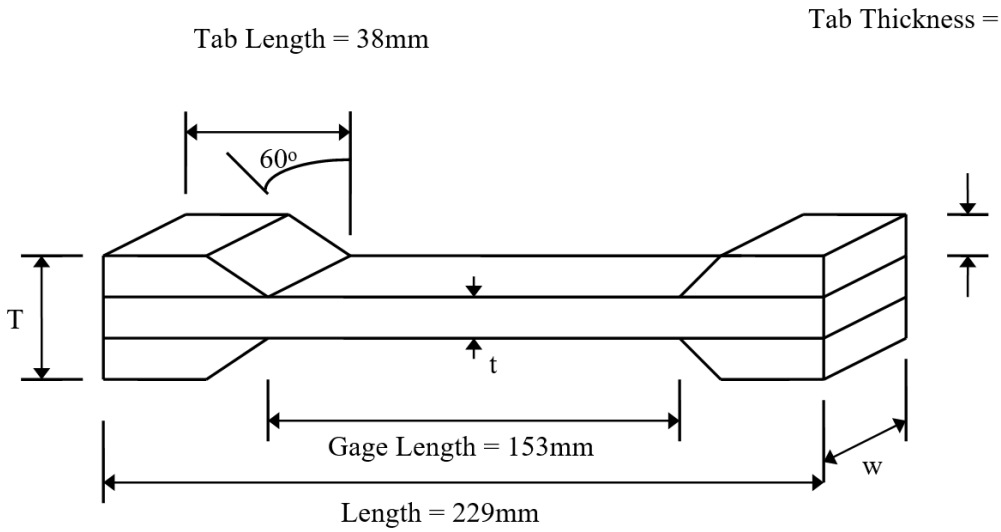



**Figure 1** Schematic of Test Specimen used to find the Young's modulus of a composite.

A tensile test geometry (Figure 1) to find the longitudinal tensile strength consists of 6-8 plies of $0^0$ plies which are 12.5 mm (2 inches) wide and 229 mm (10 inches) long. The specimen is mounted with strain gages in the longitudinal and transverse direction. Tensile stresses are applied on the specimen at a rate of about 0.5 - 1 mm/min (0.02 to 0.04 in/min). A total of 40-50 data points for stress and strain are taken until a specimen fails. The stress in the longitudinal direction is plotted as a function of longitudinal strain as shown in Figure 2. The data is reduced using linear regression. The longitudinal Young's modulus is the initial slope of the longitudinal stress, $\sigma_1$ vs strain, longitudinal strain, $\varepsilon_1$ curve.

| **Strain** (%) | **Stress** (MPa) |
|---|---|
| 0 | 0 |
| 0.183 | 306 |
| 0.36 | 612 |
| 0.5324 | 917 |
| 0.702 | 1223 |
| 0.867 | 1529 |
| 1.0244 | 1835 |
| 1.1774 | 2140 |
| 1.329 | 2446 |
| 1.479 | 2752 |
| 1.5 | 2767 |
| 1.56 | 2896 |

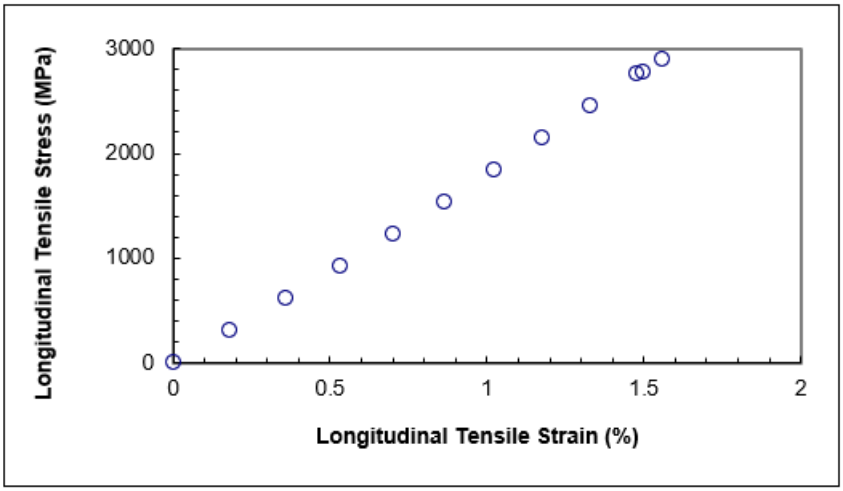



**Figure 2** Longitudinal stress as a function of longitudinal strain for a graphite/epoxy composite.

The first order polynomial that one may fit to the data is

$$\sigma = E_1 \varepsilon \quad (1)$$

## Questions

(1) Find the Young's modulus, $E_1$?

(2) What is the longitudinal ultimate tensile strength?

(3) What is the longitudinal ultimate strain?

(4) What value of $E_1$ do you get if you use the general linear regression model?

# Chapter 06.00: Physical Problem for Regression

Lesson: Computer Engineering

## Summary

Extraction of road networks in images by function fitting requires regression

## Modeling

Automated cartography deals with building or updating maps automatically from aerial or satellite images. Gone are the days when map updating was a laborious process that took a long period to accomplish. There are now many computer vision methods that one can employ to assist and ease the job of the cartographer. For instance, consider the aerial image shown, which is of a park with a lake. One of the tasks is to extract the road network that is visible. The road network appears as bright linear strips. On the right of that figure is shown the road network

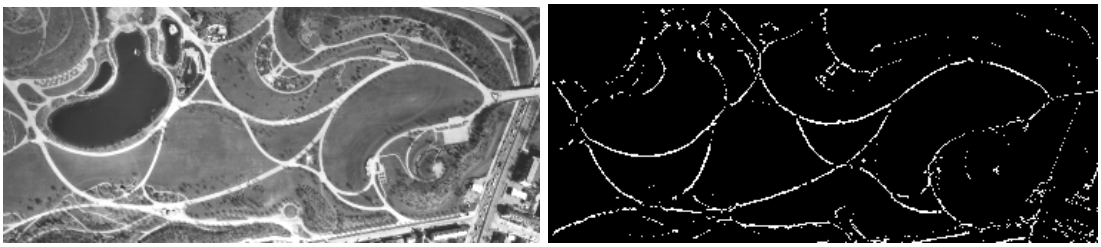


**Figure 1.** On left is an aerial image of park in Munich, Germany. We see the roads and a lake. On the right is the road network extracted from the image.

that is automatically extracted from the image. In this module, we will see how to achieve this by fitting models to the local intensities in the image.

Image is a collection of gray level values at set of predetermined sites known as pixels, arranged in an array. These gray level values are also known as image intensities. In the above figure we see the profile of the intensity value along a line (shown in red) cutting a road in an image. We represent small values to represent dark points in the image and large values to detect bright regions in the image. The value increases and then decreases. This intensity profile can be modeled as an inverted parabola. The road can be detected by identifying pixels in the image that exhibit this intensity profile around it. To detect all the road pixels, we would have to repeat this process for intensity profile along the vertical as well along the horizontal directions at each pixel.

Let the intensities profile at a pixel be denoted by the values:$y_{-K}, \ldots, y_0, \ldots, y_K$, the index 0 is used to denote the pixel location under consideration and we look at K pixels on either sides of it. We will use $x = -K, \ldots, 0, \ldots K$ to denote these pixel locations that are the independent variables.

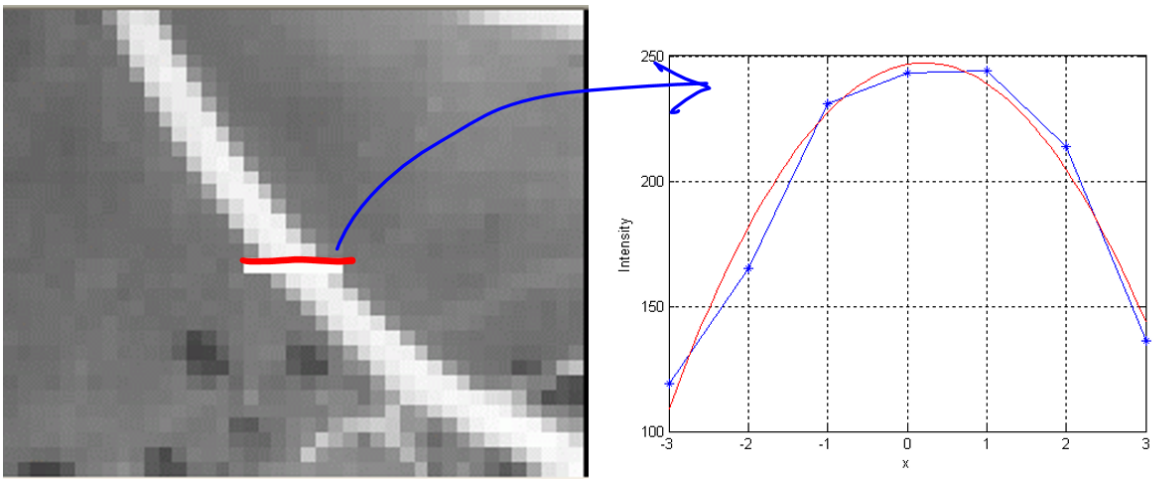



**Figure 2.** On left is a magnified portion of an image with a piece of the road. Note we can observe the individual pixels. On the right, we plot the intensities of the pixels marked by a red line. The raw intensities are plotted in blue, with fitted function in red.

Consider a polynomial model of the intensities.

$$y(x) = ax^2 + bx + c$$

We fit this second order polynomial model to the observed intensities to minimize the fit error denoted by

$$\begin{aligned} e &= \sum_{k=-K}^{k=K} (y_k - y(k))^2 \\ &= \sum_{k=-K}^{k=K} (y_k - ak^2 - bk - c)^2 \end{aligned}$$

The unknown parameters,$a$, $b$, and $c$ can be estimated to be the values that minimize the above error. The necessary condition is that the derivative of the error with respect to these parameters be zero.

$$\frac{\partial e}{\partial a} = -\sum 2k^2(y_k - ak^2 - bk - c) = 0$$

$$\frac{\partial e}{\partial b} = -\sum 2k(y_k - ak^2 - bk - c) = 0$$

$$\frac{\partial e}{\partial c} = -\sum 2(y_k - ak^2 - bk - c) = 0$$

The three equations can be expressed in matrix form as

$$\begin{bmatrix} \sum_k k^4 & \sum_k k^3 & \sum_k k^2 \\ \sum_k k^3 & \sum_k k^2 & \sum_k k^1 \\ \sum_k k^2 & \sum_k k^1 & \sum_k 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} \sum_k k^2 y_k \\ \sum_k k \ y_k \\ \sum_k y_k \end{bmatrix}$$

We thus have three linear equations with three unknown, which can be solved using a variety of methods. The conditions corresponding to the road intensity profile, which is an inverted parabola, can be found by using the observation that the maximum of the fitted polynomial should happen at or near $k = 0$. The condition for a maximum value is given by $a < 0$ and the location of the maximum is given by $-b/2a$, which should be between $-0.5$ and $0.5$ pixels. Note that 0.5 denotes the halfway location between the pixel at location 0 and the next pixel, which is at 1. The value of the maximum is given by $c$, which should be high because we are looking of bright road edges.

**Worked Out Example**

Consider the following intensity profile: $yk = 119,\ 165,\ 231,\ 243,\ 244,\ 214,\ 136$ corresponding to $k = -3,\ -2,\ -1,\ 0,\ 1,\ 2,\ 3$. This intensity profile corresponds to the one shown in Figure 2. The corresponding simultaneous equation for best least square fit is given by

$$\begin{bmatrix} 196 & 0 & 28 \\ 0 & 28 & 0 \\ 28 & 0 & 0 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 4286 \\ 162 \\ 1352 \end{bmatrix}$$

The solution is given by $a = -13.3571, b = 5.7857$, and $c = 246.5714$. Since $a$ is negative, we have a maximum value, which is located at $-b/2a = -0.2166$. This location is near 0 and the value at zero is 246.57, which denotes a bright value. All of these quantities suggest that we have a road at the pixel we are considering.

## QUESTIONS

Instead of processing the an image, which is laid out in a 2D plane, along just the horizontal and vertical directions, it is more natural to consider fitting a 2D surface model, whose general form is given by

$$y(x, y) = ax^2 + by^2 + cxy + dx + ey + f$$

(1) Work out the regression equation for fitting this surface model to intensities in a $K$ by $K$ neighborhood around a pixel.

(2) How would you determine, based on the values of the estimated parameters, whether the pixel is on a road edge or not, that is, the fitted surface has a *roof-like* shape?

# Chapter 06.00: Physical Problem for Regression

Lesson: Electrical Engineering

## Summary

Many electrical devices have non-linear behavior. It is often advantageous to substitute a model for the device using simpler linear components. In the case of a diode, this is often done with a DC supply and a series resistor. To determine an appropriate value for the resistor, regression is used.

## Modeling

All electrical devices exhibit non-linear behavior to some extent. This is caused by many factors including material and thermal properties. For most of the basic components, a simple linear model is used that is sufficient for most operating conditions. This is the case for resistors, capacitors, and inductors. Semiconductor devices, however, are specifically chosen for their non-linear behaviors. This allows them to perform gating, switching, and amplifying operations. Consider the case of a simple diode, like the 1N4001, whose forward-bias VI-characteristic is shown in Figure 1.

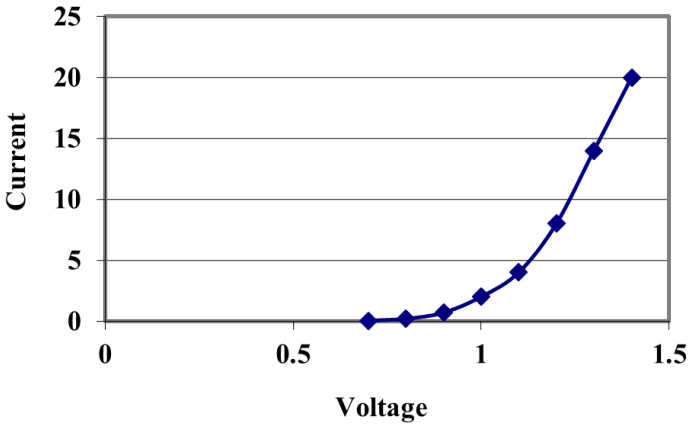



**Figure 1** VI Forward-Bias Characteristic of a 1N4001 Diode [1]

In many applications it is suitable to model the forward-bias behavior of the diode as a simple voltage drop of about 0.7 V [2, pp. 70,1]. Other types of diodes, such as LEDs, have voltage drops on the order of 2.0 to 2.5 V. This modeling is not very realistic because it fails to account for the increasing voltage

drop as current flow through the diode increases as is shown in Figure 1. If our analysis requires us to know how much current a diode can supply, then the ideal model is clearly inadequate.

It would be, of course, possible to model the diode using some form of interpolated curve, but that would almost certainly force any kind of analysis to be able to only use numerical tools. A much better solution would be to develop a model using simpler linear components that could be used in place of the diode. This is typically done using a DC supply and a series resistor as shown in Figure 2 [2, pp. 72-5].

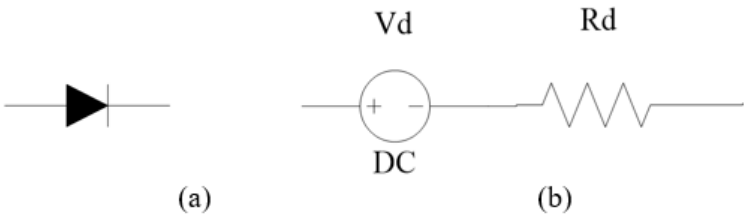



**Figure 2.** (a) Ideal Diode and (b) Forward Bias Model

The question then becomes, what are the appropriate values for $V_d$ and $R_d$? The answer will vary from diode to diode, but good values can be determined using the numerical method of linear regression. The model of Figure 2b in effect replaces the characteristic of Figure 1 with that of Figure 3. Remember that for this to work we are still assuming the diode is sufficiently forward-biased to turn it on which means that the voltage across the diode must exceed $V_d$.

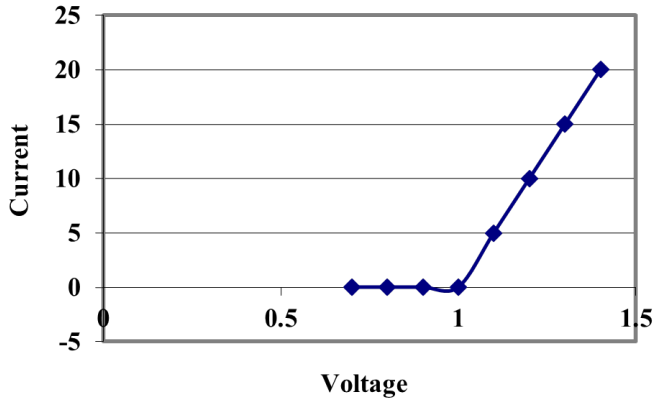



**Figure 3.** VI Characteristic of Forward Bias Model

Choosing suitable values for $V_d$ and $R_d$ will depend on the potential use for the diode. In a small signal application (low current) it would be more appropriate to choose current values from the original characteristic curve that are well-below 5 A. and for large-signal applications it would be a better fit to use current values of 5 A and above. To study this problem, let us take a closer look at the actual data used to generate the characteristic in Figure 1. This is shown in Table 1.

The simplest way to determine the $V_d$ and $R_d$ from the selected data is to generate a trend line for the region in question. This can be done using linear regression since a linear regression trend line has the effect of minimizing the mean-square error between the modeled line and the actual data. Using

the standard linear regression model using $\beta_0$ and $\beta_1$ parameters yields the following equation for the trend line.

$$I \cong \beta_1 V + \beta_0$$

Once $\beta_0$ and $\beta_1$are known $V_d$and $R_d$ can be computed as follows:

$$V_d = -\frac{\beta_0}{\beta_1}$$

$$R_d = \frac{1}{\beta_1}$$

**Table 1.** VI Characteristic of a Forward-Biased Diode

| Voltage, V | Current, A | Small Signal | Large Signal |
|---|---|---|---|
| 0.0 | 0.00 | *N* | *N* |
| 0.6 | 0.01 | *Y* | *N* |
| 0.7 | 0.05 | *Y* | *N* |
| 0.8 | 0.20 | *Y* | *N* |
| 0.9 | 0.70 | *Y* | *Y* |
| 1.0 | 2.00 | *Y* | *Y* |
| 1.1 | 4.00 | *Y* | *Y* |
| 1.2 | 8.00 | *N* | *Y* |
| 1.3 | 14.00 | *N* | *Y* |
| 1.4 | 20.00 | *N* | *Y* |

The effects of this choice are shown in Figure 4. When the data from the small-signal column of Table 1 is used to generate the regression line, it results in the model of Figure 4a and when the data from the large-signal column of Table 2 is used, the result is the model of Figure 4b.

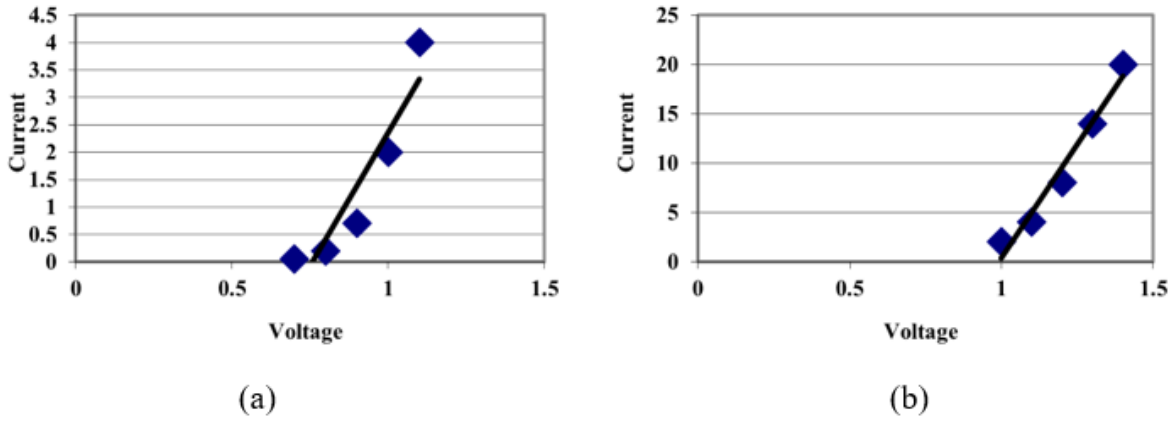



**Figure 4** (a) Small-Signal Model and (b) Large-Signal Model for a Diode

The small-signal model agrees with the typical diode modeling where $V_d$ is approximately 0.7V and will model the diode fairly well for currents below about 4 A. For a higher-current circuit the large-signal model with $V_d$ approximately 1 V is a better choice, but will not model the data as well when the current is 2 A or below.

Complete the analysis by determining more exact values for $V_d$ and $R_d$ for both of these models. Use the small signal and large signal columns of Table 1 to select the data to use in your analysis.

## Questions

(1) This analysis can be extended to handle other diodes. Find the data sheet for a 1N4150 or an LED, which shows the VI-Characteristic and derive suitable small- and large-signal forward-bias models.

(2) A similar analysis can be used to model the reverse-bias behavior of a diode. Use the datasheet from (1) to develop an appropriate model.

(3) Model the reverse-bias characteristics for a zener diode (e.g. the 1N4678 to 1N4717 family).

(4) The use of linear regression only reduces the modeling/approximating error for the sample data points provided. In this problem, we actually have a two-piece linear model (the flat portion up to $V_d$ when the diode turns on and the forward-bias model involving $R_d$ after the diode is on). This may not adequately model the entire forward-bias characteristic. Using your model values for $V_d$ and $R_d$ as a start, examine the overall characteristic and pick values for $V_d$ and $R_d$ that minimize the mean-square error over the entire forward bias region. This will have to be a combination of the error caused by having $V_d$ to the right of the actual turn-on point as well as the error in the turned-on region. How close is this model to our simpler approach that modeled each piece separately?

## References

[1] 1N4001 Data Sheet, Fairchild Semiconductor, http://www.fairchildsemi.com/ds/1N/1N4007.pdf, accessed May 12, 2005.

[2] Malvino, A., Electronic Principles, Sixth Edition, Glencoe McGraw-Hill, 1999.

# Chapter 06.00: Physical Problem for Regression

Lesson: Industrial Engineering

## Summary

Finding the number of toys a company should manufacture per day to maximize their injection molding and assembly line requires regression.

## Modeling

An industrial engineer working for a manufacturing company has noticed a deviation in the accuracy of a machine after it runs for long periods without a cool down cycle. This is especially concerning because the company wants to increase production (longer machine operating times without a cool down) because of a large contract the company will start in 3-4 months. The industrial engineer decides to monitor the machining process to determine the point (hours of operation) when the machine is producing parts that could be out of tolerance. Over the course of several months, the industrial engineer monitored the machining process to determine a relationship between hours of machine use and millimeters off target the machine was. The data collected is shown in tabular form (Table 1) and scatter plot (Figure 1).

**Table 1.** Off target measured as a fubtion of machine use.

| Hours of Machine Use | Millimeters off Target |
|---|---|
| 30 | 1.10 |
| 31 | 1.21 |
| 32 | 1.00 |
| 33 | 1.21 |
| 34 | 1.25 |
| 35 | 1.23 |
| 36 | 1.24 |
| 37 | 1.28 |
| 38 | 1.30 |
| 39 | 1.30 |
| 40 | 1.38 |
| 41 | 1.35 |
| 42 | 1.38 |
| 43 | 1.38 |
| 44 | 1.40 |
| 45 | 1.42 |

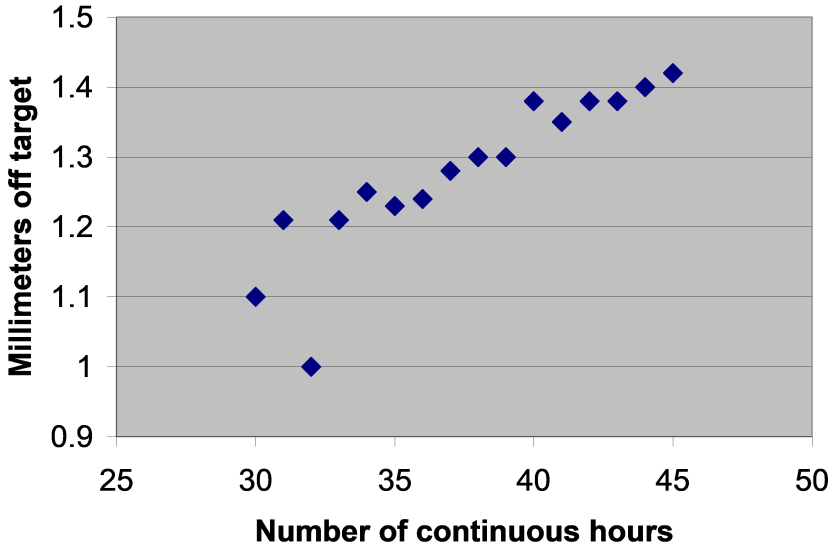



**Figure 1.** Off target measurement as a function of machine use.

Based on the above data, the industrial engineer would like to determine the number of hours of machine use that would produce a 2 millimeters off target because many parts would fail quality check at that point. Determine the number of hours of operation that produces 2 millimeter off target based on a least squares fit for the data.

### Background

In order to determine a relationship between millimeters off target and hours of machine use, a curve (for example, a linear polynomial) needs to be fit to the data. This is done by regression where we best fit a curve through the data given in the previous table. In this case, we may best fit the data to a first order polynomial, that is

$$h = a_0 + a_1 t$$

Where $h$ is the hours of machine use and $t$ is the millimeters off target. The values of the coefficients in the above equations will be found by linear regression. Knowing the values of $a_0$ and $a_1$, we can determine the millimeters off target as a function of hours of machine use. For example, if we want to find the time when the machine will be 2 mm off target, then

$$2 = a_0 + a_1 t$$

giving

$$t = \frac{2 - a_0}{a_1}$$

## Questions

(1) Fit the data in the previous table using a linear regression, and determine the hours of machine use that will make the machine 2 mm off target.

(2) Fit the data in the previous table using a second order polynomial regression, and determine the hours of machine use that will make the machine 2 millimeters off target.

(3) Find whether a linear or second order polynomial regression is a better fit to the data.

# Chapter 06.00: Physical Problem for Regression

Lesson: Mechanical Engineering

## Summary

To find the diametrical contraction during shrink-fitting in a trunnion requires one to first find how the thermal expansion coefficient of steel is related to temperature. The relation is given through a second-order polynomial regression model.

## Modeling

To make the fulcrum (Figure 1) of a bascule bridge, a long hollow steel shaft called the trunnion is shrink fit into a steel hub. The resulting steel trunnion-hub assembly is then shrink fit into the girder of the bridge.

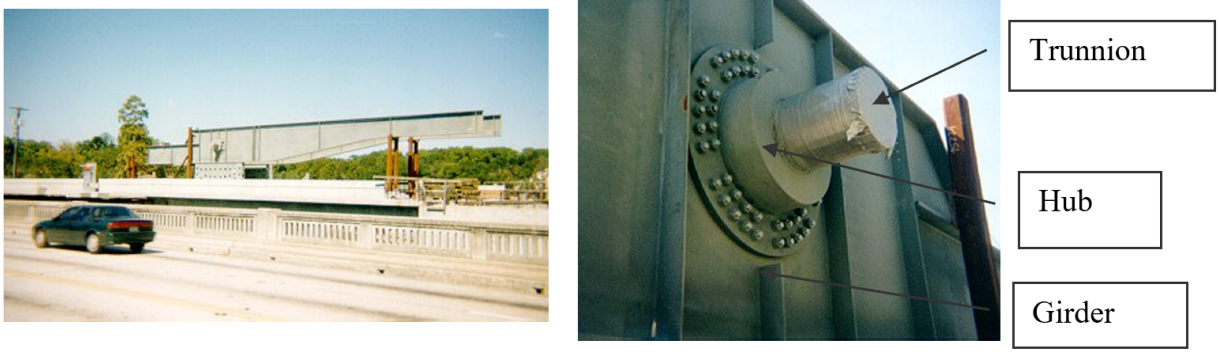



**Figure 1** Trunnion-Hub-Girder (THG) assembly.

This shrink-fitting is done by first immersing the trunnion in a cold medium such as a dry-ice/alcohol mixture. After the trunnion reaches the steady-state temperature, that is, the temperature of the cold medium, the trunnion outer diameter contracts. The trunnion is taken out of the medium and slid through the hole of the hub (Figure 2).

When the trunnion heats up, it expands and creates an interference fit with the hub. In 1995, on one of the bridges in Florida, this assembly procedure did not work as designed. Before the trunnion could be inserted fully into the hub, the trunnion got stuck. Luckily, the trunnion was taken out before it got stuck permanently. Otherwise, a new trunnion and hub would be needed to be ordered at the cost of $50,000. Coupled with construction delays, the total loss could have been more than a hundred thousand dollars.

Why did the trunnion get stuck? This failure was because the trunnion had not contracted enough to slide through the hole. Can you find out why?

A hollow trunnion of outside diameter $12.363''$ is to be fitted in a hub of inner diameter $12.358''$. The trunnion was put in a dry ice/alcohol mixture (temperature of the fluid - dry ice/alcohol mixture is $-108°$F) to contract the trunnion so that it can be slid through the hole of the hub. To slide the trunnion without sticking, a diametrical clearance of at least $0.01''$ is required between the trunnion and the hub. Assuming the room temperature is 80°F, is immersing it in a dry-ice/alcohol mixture a correct decision?

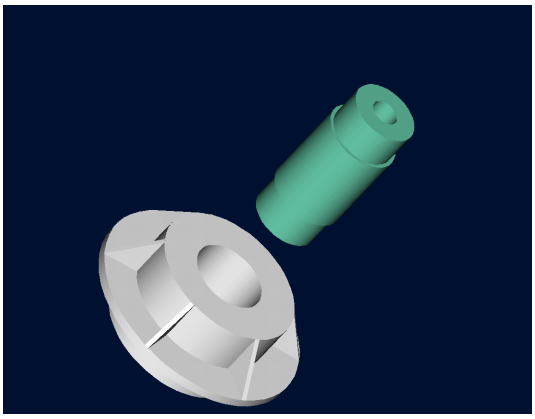


**Figure 2** Trunnion slid through the hub after contracting.

## Solution

To calculate the contraction in the diameter of the trunnion, coefficient of linear thermal expansion at room temperature is used. In that case the reduction, $\Delta D$ in the outer diameter of the trunnion is

$$\Delta D = D\alpha\Delta T \qquad (1)$$

where

$$D = \text{outer diameter of the trunnion,}$$

$$\alpha = \text{coefficient of linear thermal expansion at room temperature, and}$$

$$\Delta T = \text{change in temperature.}$$

Given

$$D = 12.363''$$

$$\alpha = 6.817 \times 10^{-6} \text{ in/in/°F at } 80°\text{F}$$

$$\begin{aligned}\Delta T &= T_{fluid} - T_{room} \\ &= -188^{\circ}\text{F}\end{aligned}$$

where

$$T_{fluid} = \text{temperature of dry-ice/alcohol mixture,}$$

$$T_{room} = \text{room temperature,}$$

the reduction in the trunnion outer diameter (Equation 1) is given by

$$\begin{aligned}\Delta D &= (12.363)\left(6.47 \times 10^{-6}\right)(-188) \\ &= -0.01504''\end{aligned}$$

So the trunnion is predicted to reduce in diameter by $0.01504''$. But is this enough reduction in diameter? As per specifications, he needs the trunnion to contract by

$$\begin{aligned}&= \text{trunnion outside diameter - hub inner diameter + diametric clearance} \\ &= 12.363'' - 12.358'' + 0.01'' \\ &= 0.015''\end{aligned}$$

So, according to his calculations, immersing the steel trunnion in a dry-ice/alcohol mixture gives the desired contraction of $0.015''$ as we predict a contraction of $0.01504''$.

But as shown in Figure 3 and Table 1, the coefficient of linear thermal expansion of steel decreases with temperature and is not constant over the range of temperature the trunnion goes through. Hence, the above formula (Equation 1) would overestimate the thermal contraction.

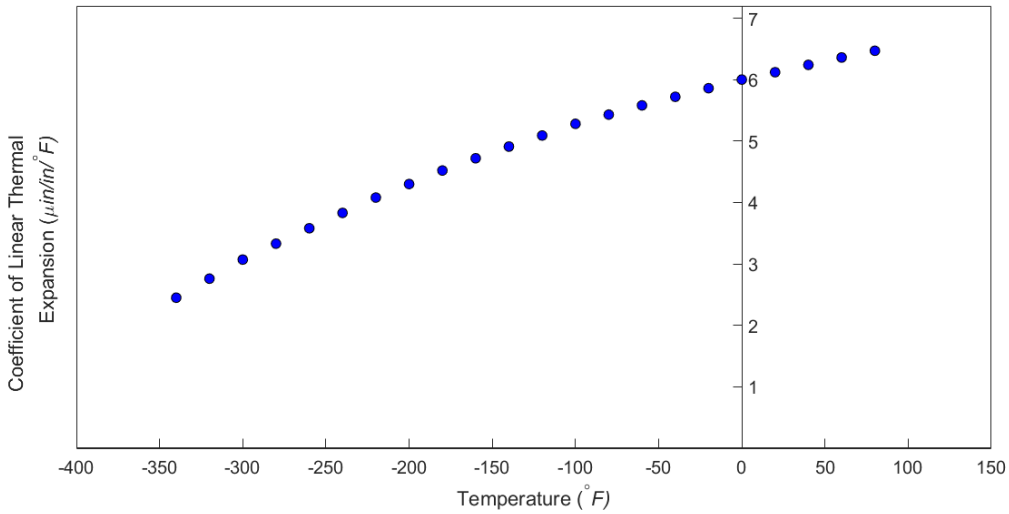


**Figure 3** Varying coefficient of linear thermal expansion as a function of temperature for cast steel.

The contraction in the diameter for the trunnion for which the coefficient of linear thermal expansion varies as a function of temperature is given by

$$\Delta D = D \int_{T_{room}}^{T_{fluid}} \alpha dT \qquad (2)$$

So one needs to find the curve to find the coefficient of linear thermal expansion as a function of temperature. This curve is found by regression where we best fit a function through the data given in Table 1. In this case, we may fit a second-order polynomial

$$\alpha = a_0 + a_1 \times T + a_2 \times T^2. \qquad (3)$$

The values of the coefficients in the above equations will be found by polynomial regression. Knowing the values of $a_0$, $a_1$,and $a_2$, we can then find the contraction in the trunnion diameter from Equation (2) and (3) as

$$\begin{aligned} \Delta D &= D \int_{T_{room}}^{T_{fluid}} (a_0 + a_1 \times T + a_2 \times T^2) dT \\ &= D[a_0 \times (T_{fluid} - T_{room}) + a_1 \times \frac{({T_{fluid}}^2 - {T_{room}}^2)}{2} + a_2 \times \frac{({T_{fluid}}^3 - {T_{room}}^3)}{3}] \end{aligned} \qquad (4)$$

## Questions

(1) Find the contraction in the outer diameter of the trunnion.

(2) As required, is the magnitude of contraction more than $0.015''$?

(3) If the magnitude of the contraction is less than $0.015''$, what if the trunnion were immersed in liquid nitrogen (boiling temperature= $-321^\circ F$)? Will that give enough contraction in the trunnion?

(4) Rather than regressing the data to a second-order polynomial so that one can find the contraction in the outer diameter of the trunnion, how would you use the trapezoidal rule of integration for unequal segments to find the contraction? What is the relative difference between the two results? The data for the coefficient of linear thermal expansion as a function of temperature is given in Table 1.

(5) We chose a second-order polynomial for regression. What would make a different order polynomial a better choice for regression? Is there an optimum order of polynomial you can find?

**Table 1** Coefficient of linear thermal expansion as a function of temperature.

| Temperature | Coefficient of Thermal Expansion |
|---|---|
| $^\circ F$ | $\mu in/in/^\circ F$ |
| 80 | 6.47 |
| 60 | 6.36 |
| 40 | 6.24 |
| 20 | 6.12 |
| 0 | 6.00 |

| Temperature | Coefficient of Thermal Expansion |
|---|---|
| −20 | 5.86 |
| −40 | 5.72 |
| −60 | 5.58 |
| −80 | 5.43 |
| −100 | 5.28 |
| −120 | 5.09 |
| −140 | 4.91 |
| −160 | 4.72 |
| −180 | 4.52 |
| −200 | 4.30 |
| −220 | 4.08 |
| −240 | 3.83 |
| −260 | 3.58 |
| −280 | 3.33 |
| −300 | 3.07 |
| −320 | 2.76 |
| −340 | 2.45 |

# Chapter 06.01: Prerequisites to Regression

Lesson: Simple Statistics

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *compute the average of a set of numbers,*

2) *compute the total sum of squares of a set of numbers,*

3) *compute the variance of a set of numbers,*

4) *compute the standard deviation of a set of numbers.*

## Introduction

In regression analysis, we are required to find the average, variance, and standard deviation of a set of numbers. In this lesson, we cover some simple descriptive statistics.

For a given set of $n$ numbers $(y_1, y_2, \ldots, y_n)$, the average or arithmetic mean $\bar{y}$ is defined by

$$\overline{y} = \frac{\sum_{i=1}^{n} y_i}{n} \qquad (1)$$

The total sum of the squares of the difference between the numbers and the mean (sometimes, just called total sum of squares) $S_t$ is defined as

$$S_t = \sum_{i=1}^{n} (y_i - \bar{y})^2 \qquad (2)$$

The variance of the numbers $\sigma^2$ is defined by

$$\sigma^2 = \frac{\sum_{i=1}^{n} (y_i - \bar{y})^2}{n-1} \qquad (3)$$

The standard deviation $\sigma$of the numbers is defined by

$$\sigma = \sqrt{\frac{\sum_{i=1}^{n} (y_i - \bar{y})^2}{n-1}} \qquad (4)$$

Standard deviation is the measure of the dispersion of the set of data from its mean. If the data points are further from the mean, the standard deviation is high; if the data points are close to the mean, the standard deviation is low.

**Example 1.**

Given the numbers (5, 8, 50, 3, 7), calculate the average, total sum of the squares, variance, and standard deviation of the numbers.

**Solution**

The average of a set of numbers is given by

$$\overline{y} = \frac{\sum_{i=1}^{n} y_i}{n}$$

In the problem,

$$n = 5,\ y_1 = 5,\ y_2 = 8,\ y_3 = 50,\ y_4 = 3,\ y_5 = 7$$

The average of the numbers is

$$\begin{aligned} \bar{y} &= \frac{5 + 8 + 50 + 3 + 7}{5} \\ &= 14.6 \end{aligned}$$

The total sum of the squares is

$$\begin{aligned} S_t &= \sum_{i=1}^{n} (y_i - \bar{y})^2 \\ &= \sum_{i=1}^{n} (y_i - 14.6)^2 \\ &= (5 - 14.6)^2 + (8 - 14.6)^2 + (50 - 14.6)^2 + (3 - 14.6)^2 + (7 - 14.6)^2 \\ &= 1581.2 \end{aligned}$$

The variance is

$$\begin{aligned}\sigma^2 &= \frac{\sum_{i=1}^{n}(y_i - \bar{y})^2}{n-1} \\ &= \frac{S_t}{n-1} \\ &= \frac{181.2}{5-1} \\ &= 395.3\end{aligned}$$

The standard deviation is

$$\begin{aligned}\sigma &= \sqrt{\sigma^2} \\ &= \sqrt{395.3} \\ &= 19.88\end{aligned}$$

# Chapter 06.01: Prerequisites to Regression

Lesson: Absolute Minimum of a Function of One Variable

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *Find the local minimum of a twice-differentiable continuous single-variant function. This knowledge will assist you in deriving the parameters of regression models in the lessons to follow.*

## Minimum of a twice differentiable continuous function

In regression, we are asked to minimize a differentiable, continuous function of one or more variables. In this primer, we cover the basics of finding the minimum of a continuous function that is twice differentiable with domain D.

Absolute Minimum Value: Given a function $f(x)$ with domain D, then $f(c)$ is the absolute minimum on D if and only if $f(x) \geq f(c)$ for all $x$ in D.

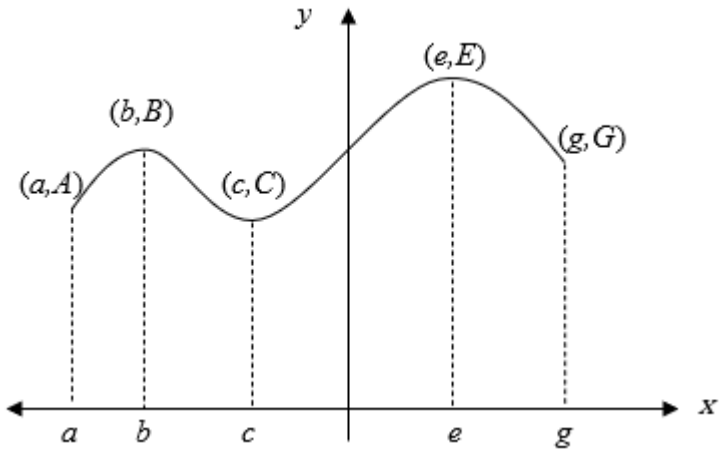



**Figure 1.** Sketch showing local and absolute minimums and maximums for a function.

Look at Figure 1, where if the domain D is given by the interval $[a, g]$, then C= absolute minimum of the function, as it is the smallest value of the function in the domain D. To find the absolute minimum of a continuous function with domain D, we look at the value of the function at the endpoints of D and also check where $f'(x) = 0$.

These points where$f'(x) = 0$ are critical points and could be local extreme (local minimum or local maximum) values. If $f''(x) > 0$ at any of these points where local extremes occur, then it corresponds

to a local minimum. Out of all the local minimums and the values at the domain ends, one can find the minimum of all such values. The point where this minimum exists is then the location of the absolute minimum value, and the value of the function at that point is the absolute minimum of the function.

**Example 1**

Find the location of the minimum of a polynomial $25 - 20x + 4x^2$.

**Solution**

$$\begin{aligned} f'(x) &= \frac{d}{dx}(25 - 20x + 4x^2) \\ &= \frac{d}{dx}(25) + \frac{d}{dx}(-20x) + \frac{d}{dx}(4x^2) \\ &= 0 - 20 + 4\frac{d}{dx}(x^2) \\ &= -20x + 4(2x) \\ &= -20 + 8x \end{aligned}$$

Check where $f'(x) = 0$

$$-20 + 8x = 0$$

gives

$$\begin{aligned} x &= \frac{20}{8} \\ &= 2.5 \end{aligned}$$

Now check for $f''(x)$

$$\begin{aligned} f'(x) &= -20 + 8x \\ f''(x) &= \frac{d}{dx}(-20 + 8x) \\ &= 8 \\ f''(2.5) &= 8 \end{aligned}$$

Since $f'(2.5) = 0$ and$f''(2.5) > 0$, the function has a local minimum at $x = 2.5$. Is this point also the location of the absolute minimum? Yes, because of two reasons – firstly, the value of the function approaches infinity (maximum value) at the endpoints of the domain $(-\infty, \infty)$. This check is made to look for extremes at the endpoints of the domain. Secondly, there is only one critical point at which $f'(x) = 0$ and $f''(x) > 0$.

**Example 2**

Given $(x, y)$ data points $(5, 10),\ (6, 15),\ (10, 20)$, and

$$S = \sum_{i=1}^{3} (y_i - ax_i)^2$$

Find the value of $a$ where the minimum of the summation series occurs.

**Solution**

The $(x, y)$ data pairs are given as follows

$$x_1 = 5,\ x_2 = 6,\ x_3 = 10,\ y_1 = 10,\ y_2 = 15,\ y_3 = 20$$

Calculating $S$

$$\begin{aligned} S &= \sum_{i=1}^{3} (y_i - ax_i)^2 \\ &= (y_1 - ax_1)^2 + (y_2 - ax_2)^2 + (y_3 - ax_3)^2 \\ &= (10 - 5a)^2 + (15 - 6a)^2 + (20 - 10a)^2 \\ &= 161a^2 - 680a + 725 \end{aligned}$$

Finding the first derivative

$$\begin{aligned} \frac{dS}{da} &= \frac{d}{da}(161a^2 - 680a + 725) \\ &= 161(2a) - 680 \\ &= 322a - 680 \end{aligned}$$

Using

$$\frac{dS}{da} = 0$$

gives

$$322a - 680 = 0$$

$$\begin{aligned} a &= \frac{680}{322} \\ &= 2.111 \end{aligned}$$

For the second derivative test

$$\begin{aligned}\frac{d^2S}{da^2} &= \frac{d}{da}\left(\frac{dS}{da}\right)\\ &= \frac{d}{da}(322a - 680)\\ &= 322\end{aligned}$$

$$\frac{d^2S}{da^2}(2.11) = 322$$

Since,

$$\frac{dS}{da} = 0 \text{ at } a = 2.11 \text{ and}$$

$$\frac{d^2S}{da^2}(2.11) > 0.$$

a local minimum exists at $a = 2.11$. Also, since $S$ is a continuous function of $a$ and it has only one point where $\frac{dS}{da} = 0$ and $\frac{d^2S}{da^2} > 0$, this local minimum is the absolute minimum.

**Alternative Solution**

Look at the solution if we had not expanded the summation.

$$S = \sum_{i=1}^{n} (y_i - ax_i)^2$$

Using the chain rule, if $u$ is a function of variable $a$

$$\frac{d}{da}(u^2) = 2u\frac{du}{da}$$

then for

$$\frac{dS}{da} = 0$$

we get

$$\sum_{i=1}^{n} 2(y_i - ax_i)(-x_i) = 0$$

$$\sum_{i=1}^{n} -2y_ix_i + 2a{x_i}^2 = 0$$

$$\sum_{i=1}^{3} -2y_ix_i + \sum_{i=1}^{3} 2a{x_i}^2 = 0$$

$$-2\sum_{i=1}^{3} y_i x_i + 2a\sum_{i=1}^{3} {x_i}^2 = 0$$

$$2a\sum_{i=1}^{3} {x_i}^2 = 2\sum_{i=1}^{3} y_i x_i$$

$$a = \frac{\sum_{i=1}^{3} y_i x_i}{\sum_{i=1}^{3} {x_i}^2}$$

Substituting given values of $(x_i, y_i)$ gives

$$\begin{aligned} a &= \frac{(10 \times 5) + (15 \times 6) + (20 \times 10)}{5^2 + 6^2 + 10^2} \\ &= \frac{340}{161} \\ &= 2.11 \end{aligned}$$

We found

$$\frac{dS}{da} = \sum_{i=1}^{3} (-2y_i x_i + 2a{x_i}^2)$$

Then

$$\begin{aligned} \frac{d^2 S}{da^2} &= \sum_{i=1}^{3} 2{x_i}^2 \\ &= 2(5)^2 + 2(6)^2 + 2(10)^2 \\ &= 322 \end{aligned}$$

A local minimum hence exists at $a = 2.11$. Also, since $S$ is a continuous function of $a$ and it has only one point where $\frac{dS}{da} = 0$ and $\frac{d^2 S}{da^2} > 0$, this local minimum is the absolute minimum.

# Chapter 06.01: Prerequisites to Regression

Lesson: Partial Derivatives

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *know the definition of a partial derivative,*

2) *find partial derivatives of a function.*

## Introduction

In regression, we are asked to find partial derivatives of functions. In this lesson, we cover the definition of a partial derivative and find partial derivatives of a simple function.

If someone gives a function $f(x)$ of one variable $x$, then we already know how to find the derivative $f'(x)$. However, functions can be of more than one independent variable. How can we then calculate the rate of change of a function with respect to each variable? This is simply done by defining partial derivatives, where one can find the derivative with respect to one variable while treating other variables as constants. For example, for a function $f(x, y)$, the partial derivative with respect to $x$ is defined as

$$\frac{\partial f}{\partial x} = \lim_{\Delta x \to 0} \frac{f(x + \Delta x, y) - f(x, y)}{\Delta x} \qquad (1)$$

and the partial derivative with respect to $y$ is defined as

$$\frac{\partial f}{\partial y} = \lim_{\Delta y \to 0} \frac{f(x, y + \Delta y) - f(x, y)}{\Delta y} \qquad (2)$$

**Example 1**

Given

$$f(x, y) = 2x^3y^2 + 7x^2y^2,$$

find

$$\frac{\partial f}{\partial x} \text{ and } \frac{\partial f}{\partial y}.$$

**Solution**

$$f(x,y) = 2x^3y^2 + 7x^2y^2$$

$$\frac{\partial f}{\partial x} = \frac{\partial}{\partial x}(2x^3y^2 + 7x^2y^2)$$

Since all variables other than $x$ are considered to be constant,

$$\begin{aligned}\frac{\partial f}{\partial x} &= 2y^2\frac{\partial}{\partial x}(x^3) + 7y^2\frac{\partial}{\partial x}(x^2)\\ &= 2y^2(3x^2) + 7y^2(2x)\\ &= 6x^2y^2 + 14xy^2\end{aligned}$$

Now

$$\frac{\partial f}{\partial y} = \frac{\partial}{\partial y}(2x^3y^2 + 7x^2y^2)$$

Since all the variables other than $y$ are considered to be constant,

$$\begin{aligned}\frac{\partial f}{\partial y} &= 2x^3\frac{\partial}{\partial y}(y^2) + 7x^2\frac{\partial}{\partial y}(y^2)\\ &= 2x^3(2y) + 7x^2(2y)\\ &= 4x^3y + 14x^2y\end{aligned}$$

# Chapter 06.01: Prerequisites to Regression

Lesson: Absolute Minimum of a Function of Multiple Variables

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *explain the first-order optimality condition,*

2) *apply the first-order optimality condition to find potential local minimums of a multivariant, continuous, and twice-differentiable function.*

## Recap: Local Minimum of a Single-Variant Function

In a previous lesson, we learned about the optimality conditions for finding extreme points and how to apply them to find a local minimum of a single-variant function. The optimality conditions are:

- First-order optimality condition: $f'(x) = 0$ (zero slope)
- Second-order optimality condition: $f''(x) > 0$ (bowl-shaped, also called concave up)

## Local Minimum of a Multivariant Function

The ideas for finding a local minimum of a multivariant function $y = f(x_1, x_2, x_3, \ldots, x_n)$ are very similar to those for a single-variant function. The first-and second-order derivatives of the function can help us find local minimums.

## First-Order Optimality Condition

The condition remains the same: $f'(x) = 0$, where $x$ is a vector of $n$ independent variables $(x_1, x_2, x_3, \ldots, x_n)$. However, since $f(x)$ is now a multivariant function $f(x) = f(x_1, x_2, x_3, \ldots, x_n)$, $f'(x)$ is now a vector, called **gradient**. Each element of the gradient vector is the corresponding partial derivative. The **first-order optimality condition** can now be more explicitly written as:

$$f'(x) = \begin{pmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \\ \vdots \\ \frac{\partial f}{\partial x_n} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix} \qquad (1)$$

In other words, we have

$$\frac{\partial f}{\partial x_i} = 0, \ \forall i = 1, \ 2, \ldots, n \qquad (2)$$

where the symbol $\forall$ means "for all."

## Second-Order Optimality Condition:

Similarly, the second-order optimality condition has to do with the second-order derivative of the function $f''(x)$. For a multivariant function, its second-order derivative is defined by the Hessian matrix:

$$H(x) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} & \cdots & \frac{\partial^2 f}{\partial x_2 \partial x_n} \\ \vdots & \vdots & \vdots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \frac{\partial^2 f}{\partial x_n \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_n^2} \end{bmatrix} \qquad (3)$$

The shape of the function is determined by the definiteness of $H(x)$. The second-order optimality discussion is beyond the scope of this course.

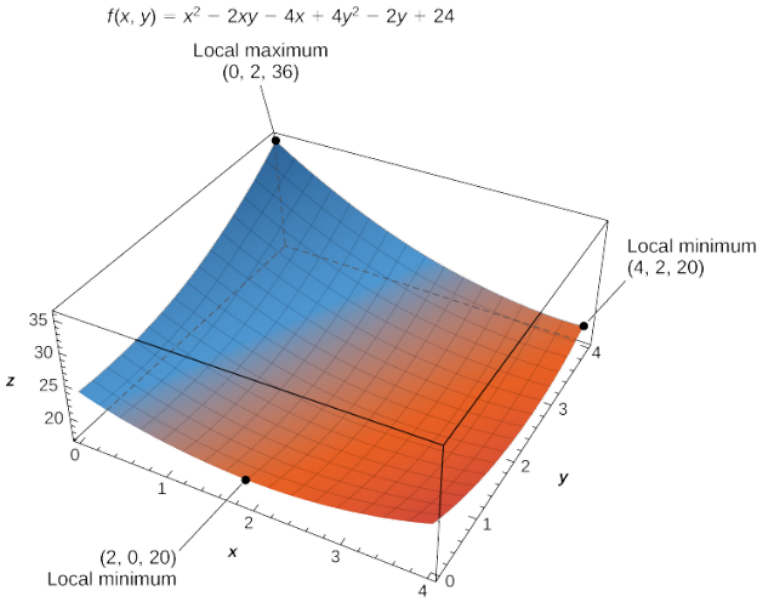



**Figure 1.** The function $f(x, y)$ has two global minima and one global maximum over its domain (Source: University Calculus by Hahn et al. Permission via Creative Commons License).

## Critical Points of a Function of Two Variables

Let $z = f(x, y)$ be a function of two variables $x$ and $y$. Then $(x_0,\ y_0)$ is considered to be a critical point of $f$ if $f$ is differentiable on an open set that contains $(x_0, y_0)$ and if one of two conditions is true.

$$\begin{array}{ll} (1) & \left.\dfrac{\partial f}{\partial x}\right|_{x_0,y_0} = 0; \quad \left.\dfrac{\partial f}{\partial y}\right|_{x_0,y_0} = 0 \\ (2) & \left.\dfrac{\partial f}{\partial x}\right|_{x_0,y_0} \text{ or } \left.\dfrac{\partial f}{\partial y}\right|_{x_0,y_0} \text{ does not exist} \end{array} \tag{4}$$

**Example 1**

Find the critical points of the function

$$f\,(x, y) = y^4 + 8xy - 4y - 4x$$

**Solution**

Now

$$\begin{aligned} \frac{\partial f}{\partial x} &= \frac{\partial}{\partial x}\left(y^4 + 8xy - 4y - 4x\right) \\ &= 8y - 4 \\ \frac{\partial f}{\partial y} &= \frac{\partial}{\partial y}\left(y^4 + 8xy - 4y - 4x\right) \\ &= 4y^3 + 8x - 4 \end{aligned}$$

Putting

$$\frac{\partial f}{\partial x} = 0$$

gives

$$8y - 4 = 0$$

$$\begin{aligned} y &= \frac{4}{8} \\ &= 0.5 \end{aligned}$$

Putting

$$\frac{\partial f}{\partial y} = 0$$

gives

$$4y^3 + 8x - 4 = 0$$

$$x = \frac{4 \ - \ 4y^3}{8}$$

Then at

$$y = 0.5$$

since

$$\frac{\partial f}{\partial y} = 0$$

we get

$$x = \frac{4 \ - \ 4(0.5)^3}{8}$$
$$= 0.4375$$

Hence $(x, y) = (0.4375,\ 0.5)$ is a critical point of the function. This critical point could be corresponding to a local minimum, a local maximum, or a saddle point.

The following appendix is not used for the course but is included here for completion. It is the second derivative test that would establish if a critical point correspondence to a local minimum. And following that, one can find the absolute minimum by looking at all the local minima and the values of the function at the endpoints.

## Appendix

Let $z = f\,(x, y)$ be a function of two variables $x$ and $y$. Let the first and second-order partial derivatives be continuous on a domain containing $(x_0, y_0)$. Then $f(x, y)$ has a local minimum at $(x_0, y_0)$ if all the conditions below are met.

$$\left.\frac{\partial f}{\partial x}\right|_{x_0,y_0} = 0, \text{ and } \left.\frac{\partial f}{\partial y}\right|_{x_0,y_0} = 0,$$

$$\left( \left.\frac{\partial^2 f}{\partial x^2}\right|_{x_0,y_0} \right) \left( \left.\frac{\partial^2 f}{\partial y^2}\right|_{x_0,y_0} \right) - \left( \left.\frac{\partial^2 f}{\partial x \partial y}\right|_{x_0,y_0} \right)^2 > 0, \text{ and}$$

$$\left.\frac{\partial^2 f}{\partial x^2}\right|_{x_0,y_0} > 0 \qquad (A.1)$$

To find the absolute minimum, one needs to choose the smallest value amongst all the local minimums, and check the value of the function at the boundary of the domain.

# Multiple Choice Test

## Chapter 06.01 Prerequisites to Regression

(1). The average of 7 numbers is given 12.6. If 6 of the numbers are 5, 7, 9, 12, 17, and 10, the remaining number is

(A) $-47.9$

(B) $-47.4$

(C) 15.6

(D) 28.2

(2). The average and standard deviation of 7 numbers are given a 8.142 and 5.005, respectively. If 5 numbers are 5, 7, 9, 12, and 17, the other two numbers are

(A) $-0.1738,\ 7.175$

(B) $3.396,\ 12.890$

(C) $3.500,\ 3.500$

(D) $4.488,\ 2.512$

(3). A local minimum of a continuous function in the interval $(-\infty, \infty)$ exists at $x = a$ if

(A) $f'(a) = 0, f''(a) = 0$

(B) $f'(a) = 0, f''(a) < 0$

(C) $f'(a) = 0, f''(a) > 0$

(D) $f'(a) = 0, f''(a)$ does not exist

(4). The absolute minimum of a function $f(x) = x^2 + 2x - 15$ in the interval $(-\infty, \infty)$ exists at $x =$_____ and is _____.

(A) $x = -1, f(-1) = -16$

(B) $x = -1, f(-1) = 0$

(C) $x = 3, f(3) = 0$

(D) $x = 5, f(5) = 0$

(5). The first order partial derivative with respect to $x$ of $u(x, y) = x^2y^3 + 6x^3e^{2y}$

(A) $y^3 + 6e^{2y}$

(B) $3x^2y^2 + 18x^3e^{2y}$

(C) $2xy^3 + 18x^2e^{2y}$

(D) $2xy^3 + 24x^2e^{2y}$

(6). The critical point(s) $(x, y)$ of the function $f(x, y) = y^3 + 4xy - 16y - 4x^2$ is (are)

(A) $(-4/3, 1), (-8/3, 2)$

(B) $(4/3, 8/3)$ and $(-1, -2)$

(C) $(-4/3, -8/3)$ and $(1, 2)$

(D) $(0, 0)$

For the complete solution, go to

http://nm.mathforcollege.com/mcquizzes/06reg/quiz_06reg_background_solution_new.pdf

# Problem Set

## Chapter 06.01: Prerequisites to Regression

(1). Enumerate three items that a scientist should do for effective use of regression analysis.

(2). Enumerate three uses of regression analysis.

(3). Enumerate three common abuses of regression analysis.

(4). Does regression analysis prove causality? If not, what does it do.

(5). Give an example of each of abuses of regression – extrapolation, generalization, and misidentification.

(6). What are the differences between regression and interpolation?

# Chapter 06.03: Linear Regression

Lesson: Introduction to Linear Regression

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *Define a residual for a linear regression model,*

2) *Explain the concept of the least-squares method as an optimization approach,*

3) *Explain why other criteria of finding the regression model do not work.*

## Introduction

The problem statement for a regression model is as follows. Given $n$ data pairs $(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)$, best fit $y = f(x)$ to the data (Figure 1).

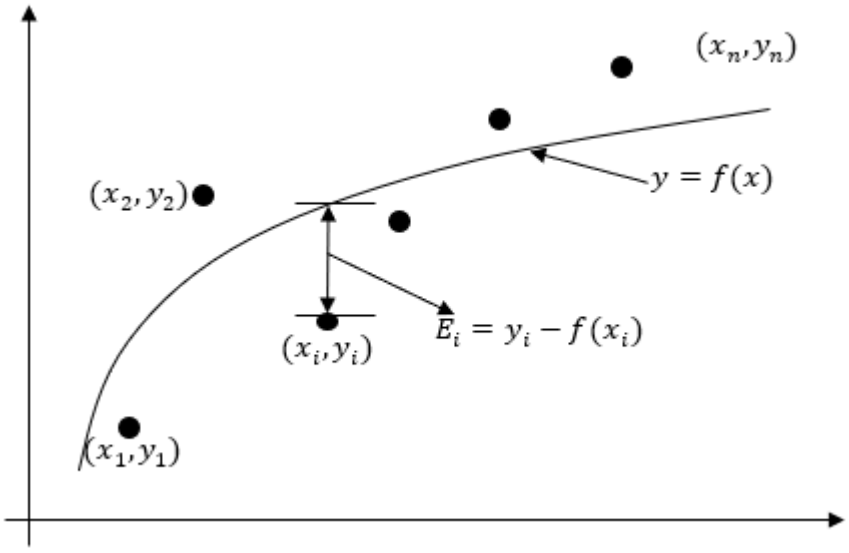



**Figure 1**. A general regression model for discrete $y$ vs. $x$ data

Linear regression is the most popular regression model. In this model, we wish to predict response to $n$ data points $(x_1, y_1), (x_2, y_2), \ldots\ldots, (x_n, y_n)$ by a regression model given by

$$y = a_0 + a_1 x \qquad (1)$$

where $a_0$ and $a_1$ are the constants of the regression model.

A measure of goodness of fit, that is, how well $a_0+a_1x$ predicts the response variable $y$ is the magnitude of the residual $E_i$ at each of the $n$ data points.

$$E_i = y_i - (a_0 + a_1 x_i) \qquad (2)$$

Ideally, if all the residuals $E_i$ are zero, one has found an equation in which all the points lie on the model. Thus, minimization of the residuals is an objective of obtaining regression coefficients.

The most popular method to minimize the residual is the least-squares method, where the estimates of the constants of the models are chosen such that the sum of the squared residuals is minimized, that is, minimize

$$S_r = \sum_{i=1}^{n} {E_i}^2 \qquad (3)$$

Why minimize the sum of the square of the residuals, $S_r$?

Why not, for instance, minimize the sum of the residual errors or the sum of the absolute values of the residuals? Alternatively, constants of the model can be chosen such that the average residual is zero without making individual residuals small. Would any of these criteria yield unbiased parameters with the smallest variance? All of these questions will be answered. Look at the example data in Table 1.

**Table 1** Data points.

| $x$ | $y$ |
|---|---|
| 2.0 | 4.0 |
| 3.0 | 6.0 |
| 2.0 | 6.0 |
| 3.0 | 8.0 |

To explain this data by a straight line regression model,

$$y = a_0 + a_1 x \qquad (4)$$

Let us use minimizing $\sum_{i=1}^{n} E_i$ as a criterion to find $a_0$ and $a_1$. Assume randomly that

$$y = 4x - 4 \qquad (5)$$

as the resulting regression model (Figure 2).

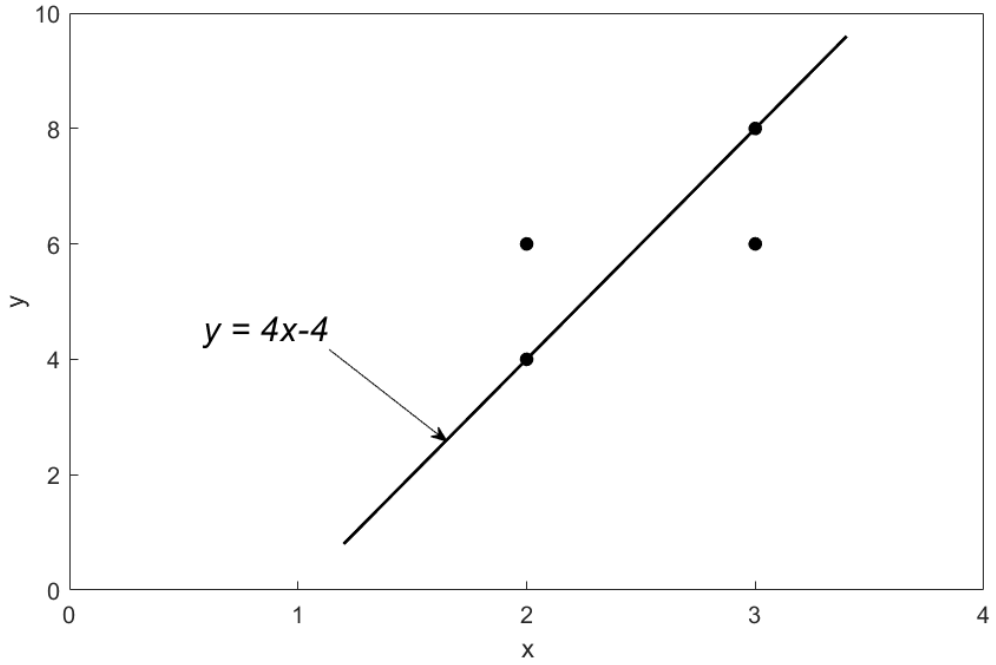



**Figure 2.** Regression curve $y = 4x - 4$ for $y$ vs. $x$ data.

The sum of the residuals $\sum_{i=1}^{4} E_i = 0$ as shown in Table 2.

**Table 2.** The residuals at each data point for regression model $y = 4x - 4$.

| $x$ | $y$ | $y_{predicted}$ | $E = y - y_{predicted}$ |
|---|---|---|---|
| 2.0 | 4.0 | 4.0 | 0.0 |
| 3.0 | 6.0 | 8.0 | $-2.0$ |
| 2.0 | 6.0 | 4.0 | 2.0 |
| 3.0 | 8.0 | 8.0 | 0.0 |
| | | | $\sum_{i=1}^{4} E_i = 0$ |

So does this give us the smallest possible sum of residuals? For this data, it does as $\sum_{i=1}^{4} E_i = 0$, and it cannot be made any smaller. But does it give unique values for the parameters of the regression model? No, because, for example, a straight-line model (Figure 3)

$$y = 6 \qquad (6)$$

also gives $\sum_{i=1}^{4} E_i = 0$ as shown in Table 3.

In fact, there are many other straight lines for this data for which the sum of the residuals $\sum_{i=1}^{4} E_i = 0$. We hence find the regression models are not unique, and therefore this criterion of minimizing the sum of the residuals is a bad one.

**Table 3.** The residuals at each data point for regression model $y = 6$

| $x$ | $y$ | $y_{\text{predicted}}$ | $E = y - y_{predicted}$ |
|---|---|---|---|
| 2.0 | 4.0 | 6.0 | $-2.0$ |
| 3.0 | 6.0 | 6.0 | 0.0 |
| 2.0 | 6.0 | 6.0 | 0.0 |
| 3.0 | 8.0 | 6.0 | 2.0 |
| | | | $\sum_{i=1}^{4} E_i = 0$ |

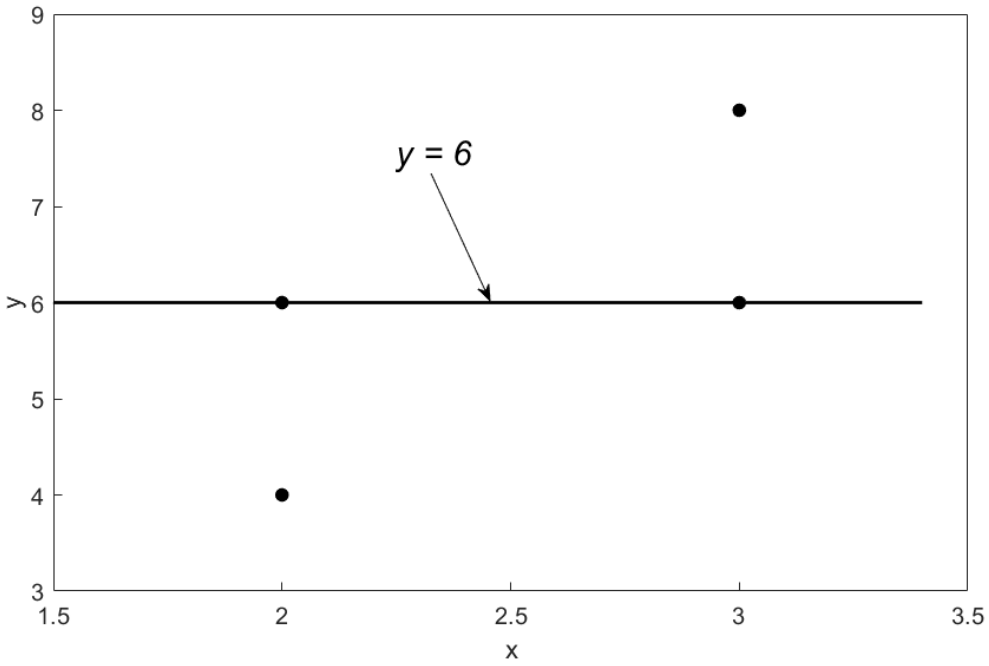



**Figure 3.** Regression curve $y = 6$ for $y$ vs. $x$ data.

You may think that the reason the criterion of minimizing $\sum_{i=1}^{n} E_i$ does not work is because negative residuals cancel with positive residuals. So, is minimizing the sum of absolute values of the residuals, that is, $\sum_{i=1}^{n} |E_i|$ better? Let us look at the same example data given in Table 1. For the regression model $y = 4x - 4$, the sum of the absolute value of residuals $\sum_{i=1}^{4} |E_i| = 4$ as shown in Table 4.

**Table 4.** The absolute residuals at each data point when employing $y = 4x - 4$.

| $x$ | $y$ | $y_{predicted}$ | $E = y - y_{predicted}$ |
|---|---|---|---|
| 2.0 | 4.0 | 4.0 | 0.0 |
| 3.0 | 6.0 | 8.0 | 2.0 |
| 2.0 | 6.0 | 4.0 | 2.0 |
| 3.0 | 8.0 | 8.0 | 0.0 |
| | | | $\sum_{i=1}^{4} \lvert E_i \rvert = 4$ |

The value of $\sum_{i=1}^{4} |E_i| = 4$ also exists for the straight-line model $y = 6$. (see Table 5).

**Table 5.** The absolute residuals at each data point for regression model $y = 6$

| $x$ | $y$ | $y_{predicted}$ | $E = y - y_{predicted}$ |
|---|---|---|---|
| 2.0 | 4.0 | 6.0 | $-2.0$ |
| 3.0 | 6.0 | 6.0 | 0.0 |
| 2.0 | 6.0 | 6.0 | 0.0 |
| 3.0 | 8.0 | 6.0 | 2.0 |
| | | | $\sum_{i=1}^{4} \lvert E_i \rvert = 4$ |

No other straight-line model that you may choose for this data has $\sum_{i=1}^{4} |E_i| < 4$. And there are many other straight lines for which the sum of absolute values of the residuals $\sum_{i=1}^{4} |E_i| = 4$. We hence find that the regression models are not unique, and hence the criterion of minimizing the sum of the absolute values of the residuals is also a bad one.

To get a unique regression model, the least-squares criterion where we minimize the sum of the square of the residuals

$$\begin{aligned} S_r &= \sum_{i=1}^{n} {E_i}^2 \\ &= \sum_{i=1}^{n} (y_i - a_0 - a_1 x_i)^2 \end{aligned} \qquad (7)$$

is recommended. The formulas obtained for the regression constants $a_0$ and $a_1$ are given below and will be derived in the next lesson.

$$a_0 = \frac{\sum_{i=1}^{n} y_i \sum_{i=1}^{n} x_i^2 - \sum_{i=1}^{n} x_i \sum_{i=1}^{n} x_i y_i}{n \sum_{i=1}^{n} x_i^2 - \left( \sum_{i=1}^{n} x_i \right)^2} \qquad (8)$$

$$a_1 = \frac{n\sum_{i=1}^{n} x_i y_i - \sum_{i=1}^{n} x_i \sum_{i=1}^{n} y_i}{n\sum_{i=1}^{n} x_i^2 - \left(\sum_{i=1}^{n} x_i\right)^2} \tag{9}$$

The formula for $a_0$ can also be written as

$$\begin{aligned} a_0 &= \frac{\sum_{i=1}^{n} y_i}{n} - a_1 \frac{\sum_{i=1}^{n} x_i}{n} \\ &= \overline{y} - a_1 \overline{x} \end{aligned} \tag{10}$$

# Chapter 06.03: Linear Regression

Lesson: Straight-Line Regression Model without an Intercept

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *derive constants of linear regression model without an intercept,*

2) *use the derived formula to find the constants of the nonlinear regression model from given data.*

## Introduction

In this model, we wish to predict response to $n$ data points $(x_1, y_1), (x_2, y_2), \ldots\ldots, (x_n, y_n)$ by a regression model given by

$$y = a_1 x \qquad (1)$$

where $a_1$ is the only constant of the regression model.

A measure of goodness of fit, that is, how well $a_1 x$ predicts the response variable $y$ is the sum of the square of the residuals, $S_r$

$$\begin{aligned} S_r &= \sum_{i=1}^{n} {E_i}^2 \\ &= \sum_{i=1}^{n} (y_i - a_1 x_i)^2 \qquad (2) \end{aligned}$$

To find $a_1$, we look for the value of $a_1$ for which $S_r$ is the absolute minimum.

We will begin by conducting the first derivative test. Take the derivative of Equation (2)

$$\frac{dS_r}{da_1} = 2\sum_{i=1}^{n} (y_i - a_1 x_i)(-x_i) = 0 \qquad (3)$$

Now putting

$$\frac{dS_r}{da_1} = 0$$

gives

$$2\sum_{i=1}^{n}\left(y_i - a_1 x_i\right)\left(-x_i\right) = 0$$

giving

$$-2\sum_{i=1}^{n} y_i x_i + 2\sum_{i=1}^{n} a_1 x_i^2 = 0$$

$$-2\sum_{i=1}^{n} y_i x_i + 2a_1\sum_{i=1}^{n} x_i^2 = 0$$

Solving the above equation for $a_1$ gives

$$a_1 = \frac{\sum_{i=1}^{n} y_i x_i}{\sum_{i=1}^{n} x_i^2} \qquad (4)$$

Let's conduct the second derivative test.

$$\begin{aligned}\frac{d^2 S_r}{da_1{}^2} &= \frac{d}{da_1}\left(2\sum_{i=1}^{n}\left(y_i - a_1 x_i\right)\left(-x_i\right)\right)\\ &= \frac{d}{da_1}\sum_{i=1}^{n}(-2x_i y_i + 2a_1 x_i{}^2)\\ &= \sum_{i=1}^{n} 2x_i{}^2 > 0 \qquad (5)\end{aligned}$$

for at most one $x_i \neq 0$, which is a pragmatic assumption that all the $x$-values are not zero.

This inequality shows that the Equation (2) value of $a_1$ corresponds to a location of local minimum. Since the sum of the squares of the residuals, $S_r$ is a continuous function of $a_1$, that $S_r$ has only one point where $\frac{dS_r}{da_1} = 0$, and at that point, we have $\frac{d^2 S_r}{da_1{}^2} > 0$, it corresponds not only to a local minimum but an absolute minimum as well. Hence, Equation (4) gives us the value of the constant, $a_1$ of the regression model, $y = a_1 x$.

### Example 1

To find the longitudinal modulus of a composite material, the following data, as given in Table 1, is collected.

**Table 1** Stress vs. strain data for a composite material.

| Strain (%) | Stress (MPa) |
|---|---|
| 0 | 0 |
| 0.183 | 306 |
| 0.36 | 612 |
| 0.5324 | 917 |
| 0.702 | 1223 |
| 0.867 | 1529 |
| 1.0244 | 1835 |
| 1.1774 | 2140 |
| 1.329 | 2446 |
| 1.479 | 2752 |
| 1.5 | 2767 |
| 1.56 | 2896 |

Find the longitudinal modulus $E$ using the regression model.

$$\sigma = E\varepsilon$$

**Solution**

Rewriting data from Table 1 in the base SI system of units is given in Table 2.

**Table 2** Stress vs. strain data for a composite in SI system of units

| Strain (m/m) | Stress (Pa) |
|---|---|
| 0.0000 | 0.0000 |
| $1.8300 \times 10^{-3}$ | $3.0600 \times 10^{8}$ |
| $3.6000 \times 10^{-3}$ | $6.1200 \times 10^{8}$ |
| $5.3240 \times 10^{-3}$ | $9.1700 \times 10^{8}$ |
| $7.0200 \times 10^{-3}$ | $1.2230 \times 10^{9}$ |
| $8.6700 \times 10^{-3}$ | $1.5290 \times 10^{9}$ |
| $1.0244 \times 10^{-2}$ | $1.8350 \times 10^{9}$ |
| $1.1774 \times 10^{-2}$ | $2.1400 \times 10^{9}$ |
| $1.3290 \times 10^{-2}$ | $2.4460 \times 10^{9}$ |
| $1.4790 \times 10^{-2}$ | $2.7520 \times 10^{9}$ |
| $1.5000 \times 10^{-2}$ | $2.7670 \times 10^{9}$ |
| $1.5600 \times 10^{-2}$ | $2.8960 \times 10^{9}$ |

Using Equation (4) gives

$$E = \frac{\sum_{i=1}^{n} \sigma_i \varepsilon_i}{\sum_{i=1}^{n} {\varepsilon_i}^2} \qquad (E1.1)$$

The summations used in Equation (E1.1) are given in Table 3.

**Table 3** Tabulation for Example 2 for needed summations

| $i$ | $\varepsilon$ | $\sigma$ | $\varepsilon^2$ | $\varepsilon\sigma$ |
|---|---|---|---|---|
| 1 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| 2 | $1.8300 \times 10^{-3}$ | $3.0600 \times 10^{8}$ | $3.3489 \times 10^{-6}$ | $5.5998 \times 10^{5}$ |
| 3 | $3.6000 \times 10^{-3}$ | $6.1200 \times 10^{8}$ | $1.2960 \times 10^{-5}$ | $2.2032 \times 10^{6}$ |
| 4 | $5.3240 \times 10^{-3}$ | $9.1700 \times 10^{8}$ | $2.8345 \times 10^{-5}$ | $4.8821 \times 10^{6}$ |
| 5 | $7.0200 \times 10^{-3}$ | $1.2230 \times 10^{9}$ | $4.9280 \times 10^{-5}$ | $8.5855 \times 10^{6}$ |
| 6 | $8.6700 \times 10^{-3}$ | $1.5290 \times 10^{9}$ | $7.5169 \times 10^{-5}$ | $1.3256 \times 10^{7}$ |
| 7 | $1.0244 \times 10^{-2}$ | $1.8350 \times 10^{9}$ | $1.0494 \times 10^{-4}$ | $1.8798 \times 10^{7}$ |
| 8 | $1.1774 \times 10^{-2}$ | $2.1400 \times 10^{9}$ | $1.3863 \times 10^{-4}$ | $2.5196 \times 10^{7}$ |
| 9 | $1.3290 \times 10^{-2}$ | $2.4460 \times 10^{9}$ | $1.7662 \times 10^{-4}$ | $3.2507 \times 10^{7}$ |
| 10 | $1.4790 \times 10^{-2}$ | $2.7520 \times 10^{9}$ | $2.1874 \times 10^{-4}$ | $4.0702 \times 10^{7}$ |
| 11 | $1.5000 \times 10^{-2}$ | $2.7670 \times 10^{9}$ | $2.2500 \times 10^{-4}$ | $4.1505 \times 10^{7}$ |
| 12 | $1.5600 \times 10^{-2}$ | $2.8960 \times 10^{9}$ | $2.4336 \times 10^{-4}$ | $4.5178 \times 10^{7}$ |
| $\sum_{i=1}^{12}$ | | | $1.2764 \times 10^{-3}$ | $2.3337 \times 10^{8}$ |

$$n = 12$$

$$\sum_{i=1}^{12} \varepsilon_i^2 = 1.2764 \times 10^{-3}$$

$$\sum_{i=1}^{12} \sigma_i \varepsilon_i = 2.3337 \times 10^{8}$$

From Equation (E1.1)

$$\begin{aligned} E &= \frac{\sum_{i=1}^{12} \sigma_i \varepsilon_i}{\sum_{i=1}^{12} {\varepsilon_i}^2} \\ &= \frac{2.3337 \times 10^{8}}{1.2764 \times 10^{-3}} \\ &= 1.8284 \times 10^{11} \text{ Pa} \\ &= 182.84 \text{ GPa} \end{aligned}$$

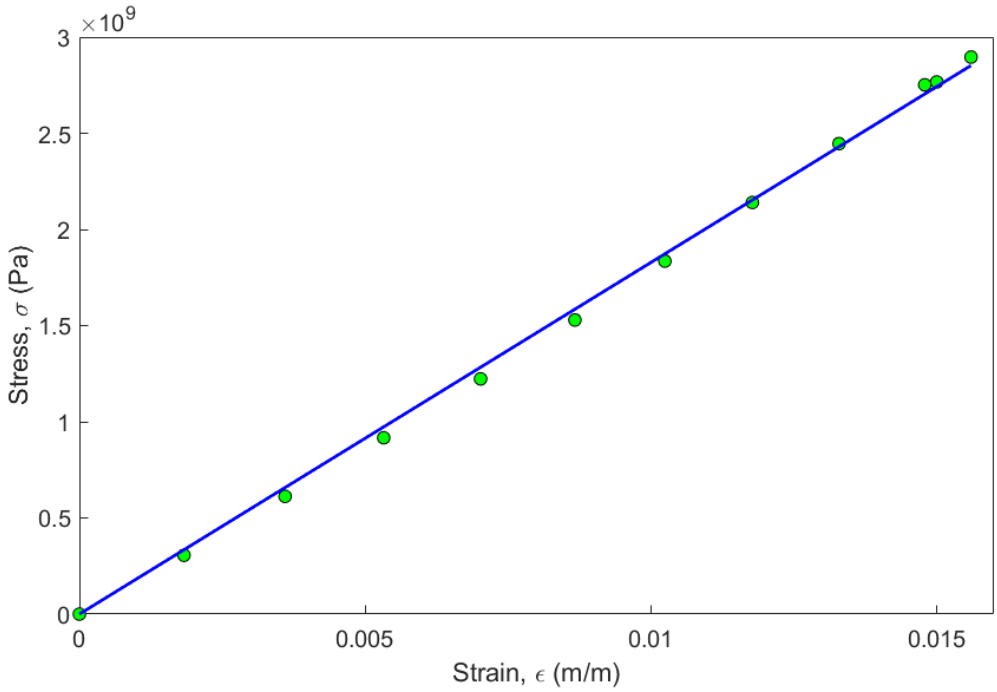



**Figure 1.** Stress vs strain data and regression model for a composite material uniaxial test

# Chapter 06.03: Linear Regression

Lesson: Theory of General Straight-Line Regression Model

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *derive the constants of a linear regression model based on the least-squares method criterion.*

## Introduction

In this model, we best fit a general straight line $y = a_0 + a_1 x$ to the $n$ data points $(x_1, y_1)$, $(x_2, y_2), \ldots,$ $(x_n, y_n)$ Let us use the least-squares criterion where we minimize the sum of the square of the residuals, $S_r$

$$S_r = \sum_{i=1}^{n} {E_i}^2$$
$$= \sum_{i=1}^{n} (y_i - a_0 - a_1 x_i)^2 \qquad (1)$$

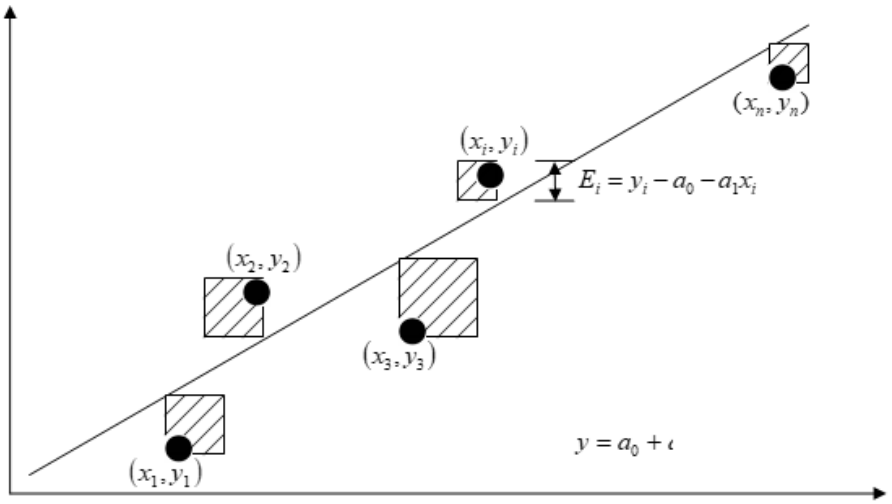


**Figure 1.** Linear regression of $y$ vs. $x$ data showing residuals and square of residual at a typical point, $x_i$

To find $a_0$ and $a_1$, we need to calculate where the sum of the square of the residuals, $S_r$ is the absolute minimum. We start this process of finding the absolute minimum first by

1) taking the partial derivative of $S_r$ with respect to $a_0$ and $a_1$ and set them equal to zero, and

2) conducting the second derivative test.

Taking the partial derivative of $S_r$ with respect to $a_0$ and $a_1$ and set them equal to zero

$$\frac{\partial S_r}{\partial a_0} = 2\sum_{i=1}^{n}\left(y_i - a_0 - a_1 x_i\right)(-1) = 0 \qquad (2)$$

$$\frac{\partial S_r}{\partial a_1} = 2\sum_{i=1}^{n}\left(y_i - a_0 - a_1 x_i\right)(-x_i) = 0 \qquad (3)$$

Dividing both sides by 2 and expanding the summations in Equations (2) and (3) gives,

$$-\sum_{i=1}^{n} y_i + \sum_{i=1}^{n} a_0 + \sum_{i=1}^{n} a_1 x_i = 0$$

$$-\sum_{i=1}^{n} y_i x_i + \sum_{i=1}^{n} a_0 x_i + \sum_{i=1}^{n} a_1 x_i^2 = 0$$

Noting that

$$\sum_{i=1}^{n} a_0 = a_0 + a_0 + \ldots + a_0 = n a_0$$

we get

$$n a_0 + a_1 \sum_{i=1}^{n} x_i = \sum_{i=1}^{n} y_i \qquad (4)$$

$$a_0 \sum_{i=1}^{n} x_i + a_1 \sum_{i=1}^{n} x_i^2 = \sum_{i=1}^{n} x_i y_i \qquad (5)$$

Solving the above two simultaneous linear equations (4) and (5) gives

$$a_1 = \frac{n\sum_{i=1}^{n} x_i y_i - \sum_{i=1}^{n} x_i \sum_{i=1}^{n} y_i}{n\sum_{i=1}^{n} x_i^2 - \left(\sum_{i=1}^{n} x_i\right)^2} \qquad (6)$$

$$a_0 = \frac{\sum_{i=1}^{n} x_i^2 \sum_{i=1}^{n} y_i - \sum_{i=1}^{n} x_i \sum_{i=1}^{n} x_i y_i}{n \sum_{i=1}^{n} x_i^2 - \left( \sum_{i=1}^{n} x_i \right)^2} \qquad (7)$$

Redefining

$$S_{xy} = \sum_{i=1}^{n} x_i y_i - n \overline{x} \overline{y}$$

$$S_{xx} = \sum_{i=1}^{n} x_i^2 - n \overline{x}^2$$

$$\overline{x} = \frac{\sum_{i=1}^{n} x_i}{n}$$

$$\overline{y} = \frac{\sum_{i=1}^{n} y_i}{n}$$

we can also rewrite the constants $a_0$ and $a_1$ from Equations (6) and (7) as

$$a_1 = \frac{S_{xy}}{S_{xx}} \qquad (8)$$

$$a_0 = \overline{y} - a_1 \overline{x} \qquad (9)$$

Putting the first derivative equations equal to zero only gives us a critical point. For a general function, it could be a local minimum, a local maximum, a saddle point, or none of the previous. The second derivative test, though, given in the "optional" appendix below, shows that it is a local minimum. Now, is this local minimum also the absolute minimum? Yes, because the first derivative test gave us only one solution, and that $S_r$ is a continuous function of $a_0$ and $a_1$.

## Appendix

### Question

Given $n$ data pairs, $(x_1, y_1), \ldots, (x_n, y_n)$, do the values of the two constants $a_0$ and $a_1$in the least-squares straight-line regression model $y = a_0 + a_1 x$ correspond to the absolute minimum of the sum of the squares of the residuals? Are these constants of regression unique?

### Solution

Given $n$ data pairs$(x_1, y_1), \ldots, (x_n, y_n)$, the best fit for the straight-line regression model

$$y = a_0 + a_1 x \qquad (A.1)$$

is found by the method of least squares. Starting with the sum of the squares of the residuals $S_r$

$$S_r = \sum_{i=1}^{n} (y_i - a_0 - a_1 x_i)^2 \qquad (A.2)$$

and using

$$\frac{\partial S_r}{\partial a_0} = 0 \qquad (A.3)$$

$$\frac{\partial S_r}{\partial a_1} = 0 \qquad (A.4)$$

gives two simultaneous linear equations whose solution is

$$a_1 = \frac{n\sum_{i=1}^{n} x_i y_i - \sum_{i=1}^{n} x_i \sum_{i=1}^{n} y_i}{n\sum_{i=1}^{n} x_i^2 - \left(\sum_{i=1}^{n} x_i\right)^2} \qquad (A.5a)$$

$$a_0 = \frac{\sum_{i=1}^{n} x_i^2 \sum_{i=1}^{n} y_i - \sum_{i=1}^{n} x_i \sum_{i=1}^{n} x_i y_i}{n\sum_{i=1}^{n} x_i^2 - \left(\sum_{i=1}^{n} x_i\right)^2} \qquad (A.5b)$$

But do these values of $a_0$ and $a_1$ give the absolute minimum value of $S_r$ (Equation (A.2))? The first derivative analysis only tells us that these values give local minima or maxima of $S_r$, and not whether they give an absolute minimum or maximum. So, we still need to figure out if they correspond to an absolute minimum.

We need first to conduct a second derivative test to find out whether the point $(a_0, a_1)$ from Equation (A.5) gives a local minimum of $S_r$. Only then can we show if this local minimum also corresponds to the absolute minimum (or maximum).

What is the second derivative test for a local minimum of a function of two variables?

If you have a function $f(x, y)$ and we found a critical point $(a, b)$ from the first derivative test, then $(a, b)$ is a minimum point if

$$\frac{\partial^2 f}{\partial x^2}\frac{\partial^2 f}{\partial y^2} - \left(\frac{\partial^2 f}{\partial x \partial y}\right)^2 > 0, \text{ and} \qquad (A.6)$$

$$\frac{\partial^2 f}{\partial x^2} > 0 \text{ or } \frac{\partial^2 f}{\partial y^2} > 0 \qquad (A.7)$$

From Equation (A.2)

$$\frac{\partial S_r}{\partial a_0} = \sum_{i=1}^{n} 2\left(y_i - a_0 - a_1 x_i\right)(-1)$$
$$= -2\sum_{i=1}^{n}\left(y_i - a_0 - a_1 x_i\right) \qquad (A.8)$$

$$\frac{\partial S_r}{\partial a_1} = \sum_{i=1}^{n} 2\left(y_i - a_0 - a_1 x_i\right)(-x_i)$$
$$= -2\sum_{i=1}^{n}\left(x_i y_i - a_0 x_i - a_1 x_i^2\right) \qquad (A.9)$$

then

$$\frac{\partial^2 S_r}{\partial a_0^2} = -2\sum_{i=1}^{n} -1$$
$$= 2n \qquad (A.10)$$

$$\frac{\partial^2 S_r}{\partial a_1^2} = 2\sum_{i=1}^{n} x_i^2 \qquad (A.11)$$

$$\frac{\partial^2 S_r}{\partial a_0 \partial a_1} = 2\sum_{i=1}^{n} x_i \qquad (A.12)$$

So, we satisfy condition (A.7), because from Equation (A.10), we see that $2n$ is a positive number. Although not required, from Equation (A.11), we see that $2\sum_{i=1}^{n} x_i^2$ is also a positive number as assuming that all $x$ data points are NOT zero is reasonable.

Is the other condition (Equation (A.6)) for $S_r$ being a minimum met? Yes, we can show (*proof not given that the term is positive*)

$$\frac{\partial^2 S_r}{\partial a_0^2}\frac{\partial^2 S_r}{\partial a_1^2} - \left(\frac{\partial^2 S_r}{\partial a_0 \partial a_1}\right)^2 = (2n)\left(2\sum_{i=1}^{n} x_i^2\right) - \left(2\sum_{i=1}^{n} x_i\right)^2$$
$$= 4\left[n\sum_{i=1}^{n} x_i^2 - \left(\sum_{i=1}^{n} x_i\right)^2\right]$$
$$= 4\sum_{\substack{i=1 \\ i<j}}^{n} (x_i - x_j)^2 > 0 \qquad (A.13)$$

So, the values of $a_0$ and $a_1$ that we have in Equation (A.5) do correspond to a local minimum of $S_r$. Now, is this local minimum also the absolute minimum? Yes, because the first derivative test gave us only one solution, and that $S_r$ is a continuous function of $a_0$ and $a_1$.

As a side note, the denominator in Equations (A.5a) and (A.5b) is nonzero, as shown by Equation (A.13). This nonzero value proves that $a_0$ and $a_1$ are finite numbers.

# Chapter 06.03: Linear Regression

Lesson: Application of General Straight-Line Regression Model

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *calculate the constants of a linear regression model.*

## Recap

In the previous lesson, we derived the formulas for the linear regression model. In this lesson, we show the application of the formulas to an applied engineering problem.

### Example 1

The torque $T$ needed to turn the torsional spring of a mousetrap through an angle, $\theta$ is given below

**Table 1.** Torque versus angle for a torsion spring.

| $Angle$, $\theta$<br>Radians | $Torque$, $T$<br>$\mathrm{N \cdot m}$ |
|---|---|
| 0.698132 | 0.188224 |
| 0.959931 | 0.209138 |
| 1.134464 | 0.230052 |
| 1.570796 | 0.250965 |
| 1.919862 | 0.313707 |

Find the constants $k_1$ and $k_2$ of the regression model

$$T = k_1 + k_2\theta \qquad (E1.1)$$

### Solution

For the linear regression model,

$$T = k_1 + k_2\theta$$

the constants of the regression model are given by

$$k_2 = \frac{n\sum_{i=1}^{5}\theta_i T_i - \sum_{i=1}^{5}\theta_i \sum_{i=1}^{5} T_i}{n\sum_{i=1}^{5}\theta_i^2 - \left(\sum_{i=1}^{5}\theta_i\right)^2} \qquad (E1.2)$$

$$k_1 = \overline{T} - k_2\overline{\theta} \qquad (E1.3)$$

Table 2 shows the summations needed for the calculation of the above two constants $k_1$ and $k_2$ of the regression model.

**Table 2.** Tabulation of data for calculation of needed summations.

| $i$ | $\theta$ | $T$ | $\theta^2$ | $T\theta$ |
|---|---|---|---|---|
| | $Radians$ | $N \cdot m$ | $Radians^2$ | $N \cdot m$ |
| 1 | 0.698132 | 0.188224 | $4.87388 \times 10^{-1}$ | $1.31405 \times 10^{-1}$ |
| 2 | 0.959931 | 0.209138 | $9.21468 \times 10^{-1}$ | $2.00758 \times 10^{-1}$ |
| 3 | 1.134464 | 0.230052 | 1.2870 | $2.60986 \times 10^{-1}$ |
| 4 | 1.570796 | 0.250965 | 2.4674 | $3.94215 \times 10^{-1}$ |
| 5 | 1.919862 | 0.313707 | 3.6859 | $6.02274 \times 10^{-1}$ |
| $\sum_{i=1}^{5}$ | 6.2831 | 1.1921 | 8.8491 | 1.5896 |

Using the summations from the last row of Table 2, we get

$$n = 5$$

From Equation (E1.2)

$$\begin{aligned} k_2 &= \frac{n\sum_{i=1}^{5}\theta_i T_i - \sum_{i=1}^{5}\theta_i \sum_{i=1}^{5} T_i}{n\sum_{i=1}^{5}\theta_i^2 - \left(\sum_{i=1}^{5}\theta_i\right)^2} \\ &= \frac{5(1.5896) - (6.2831)(1.1921)}{5(8.8491) - (6.2831)^2} \\ &= 9.6091 \times 10^{-2}\text{N-m/rad} \end{aligned}$$

To find $k_1$

$$\begin{aligned}\overline{T} &= \frac{\sum_{i=1}^{5} T_i}{n} \\ &= \frac{1.1921}{5} \\ &= 2.3842 \times 10^{-1} N - m\end{aligned}$$

$$\begin{aligned}\overline{\theta} &= \frac{\sum_{i=1}^{5} \theta_i}{n} \\ &= \frac{6.2831}{5} \\ &= 1.2566 \; radians\end{aligned}$$

From Equation (E1.3),

$$\begin{aligned}k_1 &= \overline{T} - k_2 \overline{\theta} \\ &= 2.3842 \times 10^{-1} - (9.6091 \times 10^{-2})(1.2566) \\ &= 1.1767 \times 10^{-1} \text{N-m}\end{aligned}$$

# Multiple Choice Test

Chapter 06.03 Linear Regression

(1). Given $(x_1, y_1), (x_2, y_2), ....., (x_n, y_n)$, best fitting data to $y = f(x)$ by least squares requires minimization of

(A) $\sum_{i=1}^{n} [y_i - f(x_i)]$

(B) $\sum_{i=1}^{n} |y_i - f(x_i)|$

(C) $\sum_{i=1}^{n} [y_i - f(x_i)]^2$

(D) $\sum_{i=1}^{n} (y_i - \bar{y})^2, \ \bar{y} = \dfrac{\sum_{i=1}^{n} y_i}{n}$

(2). The following data

| $x$ | 1 | 20 | 30 | 40 |
|---|---|---|---|---|
| $y$ | 1 | 400 | 800 | 1300 |

is regressed with least squares regression to $y = a_0 + a_1 x$. The value of $a_1$ most nearly is

(A) 27.480

(B) 28.956

(C) 32.625

(D) 40.000

(3). The following data

| $x$ | 1 | 20 | 30 | 40 |
|---|---|---|---|---|
| $y$ | 1 | 400 | 800 | 1300 |

is regressed with least squares regression to $y = a_1 x$. The value of $a_1$ most nearly is

(A) 27.480

(B) 28.956

(C) 32.625

(D) 40.000

(4). An instructor gives the same $y$ vs. $x$ data as given below to four students and asks them to regress the data with least squares regression to $y = a_0 + a_1 x$.

| $x$ | 1 | 10 | 20 | 30 | 40 |
|---|---|---|---|---|---|
| $y$ | 1 | 100 | 400 | 600 | 1200 |

They each come up with four different answers for the straight-line regression model. Only one is correct. Which one is the correct model? (additional exercise - without using the regression formulas for $a_0$ and $a_1$, can you find the correct model)

(A) $y = 60x - 1200$

(B) $y = 30x - 200$

(C) $y = -139.43 + 29.684x$

(D) $y = 1 + 22.782x$

(5). A torsion spring of a mousetrap is twisted through an angle of 180°. The torque vs. angle data is given below.

| Torsion, $T(N-m)$ | 0.110 | 0.189 | 0.230 | 0.250 |
|---|---|---|---|---|
| Angle, $\theta(rad)$ | 0.10 | 0.50 | 1.1 | 1.5 |

The relationship between the torque and the angle is $T = a_0 + a_1\theta$.

The amount of strain energy stored in the mousetrap spring in Joules is

(A) 0.29872

(B) 0.41740

(C) 0.84208

(D) 1561.8

(6). A scientist finds that regressing the $y$ vs. $x$ data given below to $y = a_0 + a_1 x$ results in the coefficient of determination for the straight-line model, $r^2$ to being zero.

| $x$ | 1 | 3 | 11 | 17 |
|---|---|---|---|---|
| $y$ | 2 | 6 | 22 | ? |

The missing value for $y$ at $x = 17$ most nearly is

(A) $-2.4444$

(B) 2.0000

(C) 6.8889

(D) 34.000

For complete solution, go to

http://nm.mathforcollege.com/mcquizzes/06reg/quiz_06reg_linear_solution.pdf

# Problem Set

## Chapter 06.03: Linear Regression

(1). Given the following data of $y$ vs. $x$

| $x$ | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| $y$ | 1 | 4 | 9 | 16 | 25 |

The data is regressed to a straight line $y = -7 + 6x$. What is the residual at $x = 4$?

**Answer:** $-1$

(2). The force vs. displacement data for a linear spring is given below. $F$ is the force in Newtons and $x$ is the displacement in meters. Assume displacement data is known more accurately.

| Displacement, $x$ $(m)$ | 10 | 15 | 20 |
|---|---|---|---|
| Force, $F$ $(N)$ | 100 | 200 | 400 |

If the $F$ vs $x$ data is regressed to $F = a + kx$, what is the value of $k$ by minimizing the sum of the square of the residuals?

**Answer:** 30 $N/m$

(3). A torsion spring of a mousetrap is twisted through an angle of 180°. The torque vs. angle data is given below.

| $\Theta$ $(rad)$ | 0.12 | 0.50 | 1.1 |
|---|---|---|---|
| $T$ $(Nm)$ | 0.25 | 1.00 | 2.0 |

Assuming that the torque and the angle are related via a general straight line as

$T = k_0 + k_1\ \theta$, regress the above data to the straight-line model.

**Answer:** $0.06567 + 1.7750\theta$

(4). The force vs. displacement data for a linear spring is given below. $F$ is the force in Newtons and $x$ is the displacement in meters. Assume displacement data is known more accurately.

| Displacement, $x$ $(m)$ | 10 | 15 | 20 |
|---|---|---|---|
| Force, $F$ $(N)$ | 100 | 200 | 400 |

If the $F$ vs. $x$ data is regressed to $F = kx$, what is the value of $k$ by minimizing the sum of the square of the residuals?

**Answer:** 16.55 $N/m$

(5). Given the following data of $y$ vs. $x$

| $x$ | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| $y$ | 1 | 1.1 | 0.9 | 0.96 | 1.01 |

If the $y$ vs. $x$ data is regressed to a constant line given by $y = a$, where $a$ is a constant, what is the value of $a$ by minimizing the sum of the square of the residuals.

**Answer:** 0.994

(6). To find the longitudinal modulus of composite, the following data as given in Table 1.

| Strain (%) | Stress (MPa) |
|---|---|
| 0 | 0 |
| 0.183 | 306 |
| 0.36 | 612 |
| 0.5324 | 917 |
| 0.702 | 1223 |
| 0.867 | 1529 |
| 1.0244 | 1835 |
| 1.1774 | 2140 |
| 1.329 | 2446 |
| 1.479 | 2752 |
| 1.5 | 2767 |
| 1.56 | 2896 |

**Table 1** Stress versus strain data for a composite material.

Find the longitudinal modulus, $E$ using the regression model. (Hint: $\sigma = E\varepsilon$)

**Answer:** 182.8 $GPa$

# Chapter 06.04: Nonlinear Regression

Lesson: Introduction to Nonlinear Regression

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *identify different nonlinear regression models,*

2) *recall the formula for the sum of the squares of the residuals for a general regression model.*

## Introduction

Many physical phenomena have a nonlinear relationship between variables. For example, unlike the linear spring you see in a weighing machine at your local grocery store in the produce section, a spring in the car's suspension system follows a nonlinear relationship between force and its displacement. However, the fundamentals of finding the constants of nonlinear regression models are similar to what we use for linear regression modes.

The problem statement for a nonlinear regression model is still the same, that is, given $n$ data pairs $(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)$, best fit $y = f(x)$ to the data.

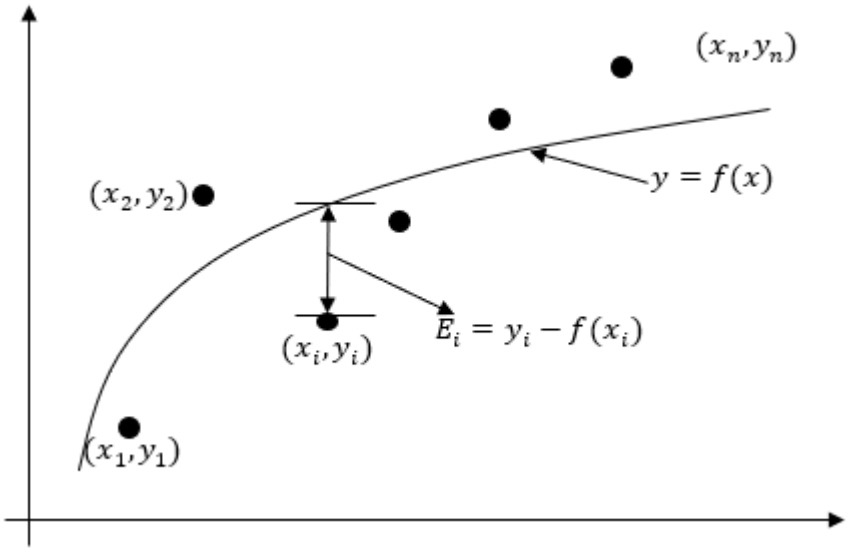



**Figure 1**. Nonlinear regression model for discrete $y$ vs. $x$ data

The residual at each data point $x_i$ is found

$$E_i = y_i - f(x_i) \qquad (1)$$

to get the sum of the square of the residuals as

$$\begin{aligned} S_r &= \sum_{i=1}^{n} E_i^2 \\ &= \sum_{i=1}^{n} \left(y_i - f(x_i)\right)^2 \end{aligned} \qquad (2)$$

Now, one minimizes the square of the residuals $S_r$ with respect to the constants of the regression model $y = f(x)$.

Common nonlinear models in science and engineering include the following.

## Exponential model

Given $(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)$, best fit $y = ae^{bx}$ to the data. In this model, the constants of the regression model are $a$ and $b$. An example of the model is the case of a radioactive dye such as Technitium-99 given to patients who are going through a CT scan of their body to diagnose an illness such as excess fat on a liver. The radiation intensity of the radioactive dye decreases with time and is governed by an exponentially decaying model (Figure 2). The interest in such a model would be to find out how long substantial radiation stays in the body.

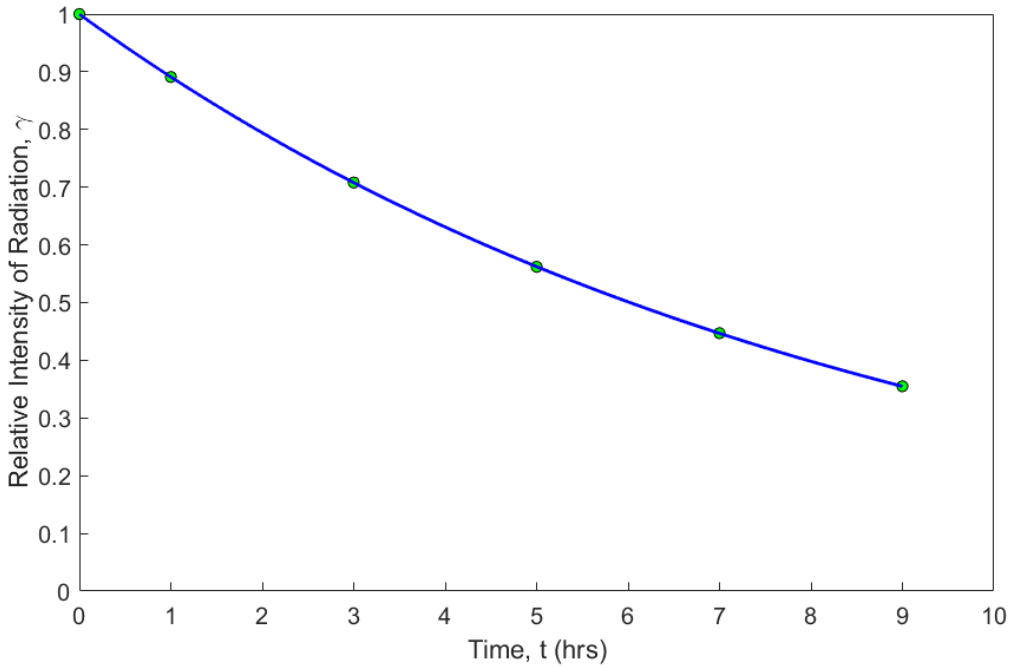


**Figure 2** Relative intensity of radiation as a function of time using an exponential regression model.

## Power model

Given $(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)$, best fit $y = ax^b$ to the data. In this model, the constants of the regression model are $a$ and $b$. An example of the model is a falling parachutist, where the drag force on the parachute is related through the power of the velocity with which they are falling. An interest in such a model would arise from designing the parachute such that the drag is low enough to maintain control of fall but high enough for the fall to be safe.

## Saturation growth model

Given $(x_1, y_1)$, $(x_2, y_2), \ldots, (x_n, y_n)$, best fit $y = \dfrac{ax}{b+x}$ to the data. In this model, the constants of the regression model are $a$ and $b$. An example of the model is the case of how good animation looks. How good an animation looks is measured by a variable called performance and is a function of the frame rate. Frame rate is the frequency at which consecutive frames (images) appear on display. The higher the frame rate, the more natural animation looks to the human eye, but the human eye cannot distinguish the increased performance after a certain frame rate. This model is an example of a saturation growth model where at zero frame rate, the performance is zero as one is looking at a static image, and at high frame rates, the performance level saturates.

## Harmonic decline model

Given $(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)$, best fit $y = \dfrac{b}{1+ax}$ to the data. In this model, the constants of the regression model are $a$ and $b$. An example of the model is how much oil would be produced from a well as a function of time. An interest in such a model would arise to estimate the life of the oil well to find out when it should be abandoned.

In all the above examples, one would find the sum of the square of the residuals and minimize it with respect to the constants of the model.

For example, take the case of the exponential model. Given $(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)$, best fit $y = ae^{\mathrm{bx}}$ to the data. The variables $a$ and $b$ are the constants of the exponential model. The residual at each data point $x_i$ is

$$E_i = y_i - ae^{bx_i} \qquad (3)$$

The sum of the square of the residuals is

$$\begin{aligned} S_r &= \sum_{i=1}^{n} E_i^2 \\ &= \sum_{i=1}^{n} \left(y_i - ae^{bx_i}\right)^2 \end{aligned} \qquad (4)$$

All one must do is to minimize the sum of the square of the residuals with respect to $a$ and $b$. The challenge lies as the resulting equations, unlike in linear regression, turn out to be simultaneous nonlinear equations. We cover this issue in the upcoming lessons.

# Chapter 06.04: Nonlinear Regression

Lesson: Exponential Regression Model with Transformation

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *derive constants of nonlinear regression models using transformation of data,*

2) *use the derived formula to find the constants of the nonlinear regression model from given data.*

## Derivation of nonlinear regression models

In a previous lesson, we have discussed linear regression models. But when data is following a nonlinear trend, we need to develop nonlinear regression models. A simple example of such models is the drag force on a parachute, which is related to the square of the velocity of the parachutist. Other examples are the decay of radioactivity intensity of nuclear material, which is related exponentially to passage of time, or the height of a human that saturates as the person ages.

The derivation of the nonlinear models follows the same concept of finding the sum of the squares of the residuals and minimizing with respect to the constants of the regression model. For brevity, we limit our discussion to the exponential model in this lesson.

## Exponential model

Given $(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)$, best fit $y = ae^{bx}$ to the data (Figure 1).

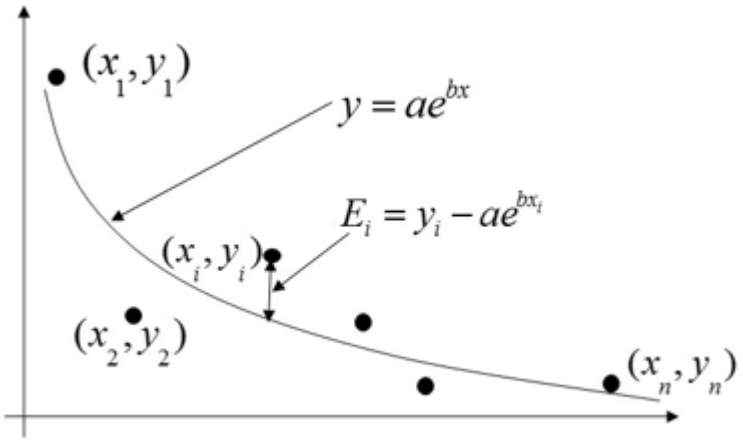


**Figure 1.** Exponential regression model for $y$ vs. $x$ data

The variables $a$ and $b$ are the constants of the exponential model. The residual $E_i$ at each data point $x_i$ is

$$E_i = y_i - ae^{bx_i} \qquad (1)$$

The sum of the square of the residuals is

$$\begin{aligned} S_r &= \sum_{i=1}^{n} E_i^2 \\ &= \sum_{i=1}^{n} \left(y_i - ae^{bx_i}\right)^2 \end{aligned} \qquad (2)$$

To find the constants $a$ and $b$ of the exponential model, we minimize $S_r$ by differentiating with respect to $a$ and $b$ and equating the resulting equations to zero

$$\frac{\partial S_r}{\partial a} = \sum_{i=1}^{n} 2\left(y_i - ae^{bx_i}\right)\left(-e^{bx_i}\right) = 0$$

$$\frac{\partial S_r}{\partial b} = \sum_{i=1}^{n} 2\left(y_i - ae^{bx_i}\right)\left(-ax_i e^{bx_i}\right) = 0 \qquad (3a, b)$$

Expanding Equations (3a,b) gives

$$-2\sum_{i=1}^{n} y_i e^{bx_i} + 2a\sum_{i=1}^{n} e^{2bx_i} = 0$$

$$-2a\sum_{i=1}^{n} y_i x_i e^{bx_i} + 2a^2\sum_{i=1}^{n} x_i e^{2bx_i} = 0 \qquad (4a, b)$$

Examining the Equations (4a, b) above shows that we need to solve simultaneous nonlinear equations for the regression constants $a$ and $b$ and need numerical methods. Is there an alternative?

It is sometimes possible to use simple linear regression formulas to estimate the parameters of a nonlinear model. This procedure involves first transforming the given data, followed by using formulas derived for linear regression. Data for several nonlinear models such as exponential, power, and growth can be transformed, but not so for all nonlinear models.

## Exponential Model through Transformation of Data

Given $(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)$, best fit $y = ae^{bx}$ to the data by using the transformation of data. The variables $a$ and $b$ are the constants of the exponential model

$$y = ae^{bx} \qquad (5)$$

Taking the natural log of both sides of Equation (5) gives

$$\ln y = \ln a + bx \qquad (6)$$

Let

$$z = \ln y$$

$$a_0 = \ln a \text{ implying } a = e^{a_o}$$

$$a_1 = b \qquad (7)$$

then

$$z = a_0 + a_1 x \qquad (8)$$

For the transformed data of $z$ versus $x$, we can use the linear regression formulas. Hence, the constants $a_0$ and $a_1$ can be found as

$$a_1 = \frac{n\sum_{i=1}^{n} x_i z_i - \sum_{i=1}^{n} x_i \sum_{i=1}^{n} z_i}{n\sum_{i=1}^{n} x_i^2 - \left(\sum_{i=1}^{n} x_i\right)^2}$$

$$a_0 = \overline{z} - a_1 \overline{x} \qquad (9a, b)$$

When the constants $a_0$ and $a_1$ are found, the original constants of the exponential model are found as given in Equation (7)

$$b = a_1$$
$$a = e^{a_0}$$

**Example 1**

Many patients get concerned when a medical test involves an injection of radioactive material. For example, for scanning a gallbladder, a few drops of Technetium-99m isotope are used. Half of the technetium-99m would be gone in about 6 hours. However, it takes about 24 hours for the radiation levels to reach what we are exposed to in day-to-day activities. Below is given the relative intensity of radiation as a function of time.

**Table 1**. Relative intensity of radiation as a function of time

| $t(hrs)$ | 0 | 1 | 3 | 5 | 7 | 9 |
|---|---|---|---|---|---|---|
| $\gamma$ | 1.000 | 0.891 | 0.708 | 0.562 | 0.447 | 0.355 |

If the level of the relative intensity of radiation is related to time via an exponential formula $\gamma = Ae^{\lambda t}$, using the transformation of data, find

a) the value of the regression constants $A$ and $\lambda$,

b) the half-life of Technium-99m, and

c) the radiation intensity after 24 hours.

**Solution**

a)

$$\gamma = Ae^{\lambda t} \qquad (E1.1)$$

Taking the natural logarithm on both sides,

$$\ln(\gamma) = \ln(A) + \lambda t \qquad (E1.2)$$

Assuming

$$y = \ln \gamma$$

$$a_0 = \ln(A) \qquad (E1.3)$$

$$a_1 = \lambda \qquad (E1.4)$$

we get

$$y = a_0 + a_1 t$$

This is a linear relationship between $y$ and $t$.Then

$$a_1 = \frac{n\sum_{i=1}^{n} t_i y_i - \sum_{i=1}^{n} t_i \sum_{i=1}^{n} y_i}{n\sum_{i=1}^{n} t_i^2 - \left(\sum_{i=1}^{n} t_i\right)^2}$$

$$a_0 = \bar{y} - a_1 \bar{t} \qquad (1.5a, b)$$

Table 2 shows the summations one would need for calculating $a_0$ and $a_1$.

**Table 2** Summations of data to calculate constants of the model.

| $i$ | $t_i$ | $y_i$ | $y_i = lny_i$ | $t_i y_i$ | $t_i^2$ |
|---|---|---|---|---|---|
| 1 | 0 | 1 | 0.00000 | 0.0000 | 0.0000 |
| 2 | 1 | 0.891 | −0.11541 | −0.11541 | 1.0000 |
| 3 | 3 | 0.708 | −0.34531 | −1.0359 | 9.0000 |
| 4 | 5 | 0.562 | −0.57625 | −2.8813 | 25.0000 |
| 5 | 7 | 0.447 | −0.80520 | −5.6364 | 49.0000 |
| 6 | 9 | 0.355 | −1.0356 | −9.3207 | 81.0000 |
| $\sum_{i=1}^{6}$ | 25.0000 | | −2.8778 | −18.990 | 165.00 |

$$n = 6$$

$$\sum_{i=1}^{6} t_i = 25.000$$

$$\sum_{i=1}^{6} y_i = -2.8778$$

$$\sum_{i=1}^{6} t_i y_i = -18.990$$

$$\sum_{i=1}^{6} t_i^2 = 165.00$$

From Equation (1.5a, b) we have

$$\begin{aligned} a_1 &= \frac{6\left(-18.990\right) - \left(25\right)\left(-2.8778\right)}{6\left(165.00\right) - \left(25\right)^2} \\ &= -0.11505 \end{aligned}$$

$$\begin{aligned} a_0 &= \frac{-2.8778}{6} - \left(-0.11505\right)\frac{25}{6} \\ &= -2.6150 \times 10^{-4} \end{aligned}$$

Since

$$a_0 = \ln\left(A\right)$$

$$\begin{aligned} A &= e^{a_0} \\ &= e^{-2.6150\times10^{-4}} \\ &= 0.99974 \end{aligned}$$

$$\lambda = a_1 = -0.11505$$

The regression formula then is

$$\gamma = 0.99974 \times e^{-0.11505t}$$

b) Half-life is when

$$\gamma = \left. \frac{1}{2}\gamma \right|_{t=0}$$

$$0.99974 \times e^{-0.11505t} = \frac{1}{2}\left(0.99974\right) e^{-0.11505(0)}$$

$$e^{-0.11508t} = 0.5$$

$$-0.11505\ t = \ln(0.5)$$

$$t = 6.0248 \text{ hours}$$

c) The relative intensity of radiation after 24 hours is

$$\begin{aligned} \gamma &= 0.99974 e^{-0.11505(24)} \\ &= 0.063200 \end{aligned}$$

This value of relative intensity implies that only

$$\frac{6.3200 \times 10^{-2}}{0.99974} \times 100 = 6.3216\%$$

of the initial radioactivity is left after 24 hours.

# Chapter 06.04: Nonlinear Regression

Lesson: Nonlinear Regression Model Without Transformation

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *derive constants of a nonlinear exponential regression model,*

2) *use the derived formula to find the constants of the exponential regression model from given data.*

## Nonlinear models using least squares

The development of the least-squares estimation for nonlinear models does not generally yield linear equations and hence is not easy to solve. Solving for the constants of the nonlinear models mostly results in simultaneous nonlinear equations. We illustrate this challenge through an exponential model.

## Exponential model

Given $(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)$, best fit $y = ae^{bx}$ to the data. The variables $a$ and $b$ are the constants of the exponential model. The residual at each data point $x_i$, is

$$E_i = y_i - ae^{bx_i} \qquad (1)$$

The sum of the square of the residuals is

$$\begin{aligned} S_r &= \sum_{i=1}^{n} E_i^2 \\ &= \sum_{i=1}^{n} \left(y_i - ae^{bx_i}\right)^2 \end{aligned} \qquad (2)$$

To find the constants $a$ and $b$ of the exponential model, we minimize $S_r$ by differentiating $S_r$ with respect to $a$ and $b$ and equating the resulting equations to zero

$$\frac{\partial S_r}{\partial a} = \sum_{i=1}^{n} 2\left(y_i - ae^{bx_i}\right)\left(-e^{bx_i}\right) = 0$$

$$\frac{\partial S_r}{\partial b} = \sum_{i=1}^{n} 2\left(y_i - ae^{bx_i}\right)\left(-ax_i e^{bx_i}\right) = 0 \qquad (3a, b)$$

Expanding Equation (3a,b) gives

$$-2\sum_{i=1}^{n} y_i e^{bx_i} + 2a\sum_{i=1}^{n} e^{2bx_i} = 0$$

$$-2a\sum_{i=1}^{n} y_i x_i e^{bx_i} + 2a^2\sum_{i=1}^{n} x_i e^{2bx_i} = 0 \qquad (4a, b)$$

Simplifying Equation (4a,b) gives

$$-\sum_{i=1}^{n} y_i e^{bx_i} + a\sum_{i=1}^{n} e^{2bx_i} = 0$$

$$-\sum_{i=1}^{n} y_i x_i e^{bx_i} + a\sum_{i=1}^{n} x_i e^{2bx_i} = 0 \qquad (5a, b)$$

Equations (5a) and (5b) are nonlinear in $a$ and $b$ and thus not in a closed-form to be solved as was the case for linear regression. In general, iterative methods (such as Gauss-Newton iteration method, method of steepest descent, Marquardt's method, direct search, etc) must be used to find values of $a$ and $b$.

However, in this case, from Equation (5a), $a$ can be written explicitly in terms of $b$ as

$$a = \frac{\sum_{i=1}^{n} y_i e^{bx_i}}{\sum_{i=1}^{n} e^{2bx_i}} \qquad (6)$$

Substituting Equation (6) in (5b) gives

$$\sum_{i=1}^{n} y_i x_i e^{bx_i} - \frac{\sum_{i=1}^{n} y_i e^{bx_i}}{\sum_{i=1}^{n} e^{2bx_i}} \sum_{i=1}^{n} x_i e^{2bx_i} = 0 \qquad (7)$$

This equation is still nonlinear in $b$ and can be solved best by numerical methods such as the bisection method or the secant method.

**Example 1**

Many patients get concerned when a test involves an injection of radioactive material. For example, for scanning a gallbladder, a few drops of Technetium-99m isotope are used. Half of the technetium-99m would be gone in about 6 hours. It, however, takes about 24 hours for the radiation levels to reach what we are exposed to in day-to-day activities. Below is given the relative intensity of radiation as a function of time.

**Table 1** Relative intensity of radiation as a function of time

| $t$ (hrs) | 0 | 1 | 3 | 5 | 7 | 9 |
|---|---|---|---|---|---|---|
| $\gamma$ | 1.000 | 0.891 | 0.708 | 0.562 | 0.447 | 0.355 |

If the level of the relative intensity of radiation is related to time via an exponential formula $\gamma = Ae^{\lambda t}$, find

a). the value of the regression constants $A$ and $\lambda$,

b). the half-life of Technium-99m, and

c). the radiation intensity after 24 hours.

**Solution**

a) The value of $\lambda$ is given by solving the nonlinear Equation (7),

$$f(\lambda) = \sum_{i=1}^{n} \gamma_i t_i e^{\lambda t_i} - \frac{\sum_{i=1}^{n} \gamma_i e^{\lambda t_i}}{\sum_{i=1}^{n} e^{2\lambda t_i}} \sum_{i=1}^{n} t_i e^{2\lambda t_i} = 0 \qquad (E1.1)$$

Then the value of $A$ from Equation (6) takes the form,

$$A = \frac{\sum_{i=1}^{n} \gamma_i e^{\lambda t_i}}{\sum_{i=1}^{n} e^{2\lambda t_i}} \qquad (E1.2)$$

Equation (E1.1) can be solved for $\lambda$ using the bisection method. To estimate the initial guesses, we assume $\lambda = -0.120$ and $\lambda = -0.110$. We need to check whether these values first bracket the root of $f(\lambda) = 0$. At $\lambda = -0.120$, the table below shows the evaluation of $f(-0.120)$.

**Table 2** Summation value for calculation of constants of the model

| $i$ | $t_i$ | $\gamma_i$ | $\gamma_i t_i e^{\lambda t_i}$ | $\gamma_i e^{\lambda t_i}$ | $e^{2\lambda t_i}$ | $t_i e^{2\lambda t_i}$ |
|---|---|---|---|---|---|---|
| 1 | 0 | 1 | 0.00000 | 1.00000 | 1.00000 | 0.00000 |
| 2 | 1 | 0.891 | 0.79205 | 0.79205 | 0.78663 | 0.78663 |
| 3 | 3 | 0.708 | 1.4819 | 0.49395 | 0.48675 | 1.4603 |
| 4 | 5 | 0.562 | 1.5422 | 0.30843 | 0.30119 | 1.5060 |
| 5 | 7 | 0.447 | 1.3508 | 0.19297 | 0.18637 | 1.3046 |
| 6 | 9 | 0.355 | 1.0850 | 0.12056 | 0.11533 | 1.0379 |
| $\sum_{i=1}^{6}$ | | | 6.2501 | 2.9062 | 2.8763 | 6.0954 |

From Table 2

$$n = 6$$

$$\sum_{i=1}^{6} \gamma_i t_i e^{-0.120 t_i} = 6.2501$$

$$\sum_{i=1}^{6} \gamma_i e^{-0.120 t_i} = 2.9062$$

$$\sum_{i=1}^{6} e^{2(-0.120) t_i} = 2.8763$$

$$\sum_{i=1}^{6} t_i e^{2(-0.120) t_i} = 6.0954$$

$$\begin{aligned} f(-0.120) &= (6.2501) - \frac{2.9062}{2.8763}(6.0954) \\ &= 0.091357 \end{aligned}$$

Similarly

$$f(-0.110) = -0.10099$$

Since

$$f(-0.120) \times f(-0.110) < 0,$$

the value of $\lambda$ falls in the bracket of $[-0.120, -0.110]$. The next guess of the root then is

$$\begin{aligned} \lambda &= \frac{-0.120 + (-0.110)}{2} \\ &= -0.115 \end{aligned}$$

Continuing with the bisection method, the root of $f(\lambda) = 0$ is found as $\lambda = -0.11508$. This value of the root was obtained after 20 iterations with an absolute relative approximate error of less than 0.000008%.

From Equation (E1.2), $A$ can be calculated as

$$\begin{aligned} A &= \frac{\sum_{i=1}^{6} \gamma_i e^{\lambda t_i}}{\sum_{i=1}^{6} e^{2\lambda t_i}} \\ &= \frac{\begin{array}{c} 1 \times e^{-0.11508(0)} + 0.891 \times e^{-0.11508(1)} + 0.708 \times e^{-0.11508(3)} + \\ 0.562 \times e^{-0.11508(5)} + 0.447 \times e^{-0.11508(7)} + 0.355 \times e^{-0.11508(9)} \end{array}}{\begin{array}{c} e^{2(-0.11508)(0)} + e^{2(-0.11508)(1)} + e^{2(-0.11508)(3)} + \\ e^{2(-0.11508)(5)} + e^{2(-0.11508)(7)} + e^{2(-0.11508)(9)} \end{array}} \\ &= \frac{2.9373}{2.9378} \\ &= 0.99983 \end{aligned}$$

The regression formula is hence given by

$$\gamma = 0.99983\ e^{-0.11508t}$$

b) Half-life of Technetium-99m is when

$$\gamma = \left. \frac{1}{2}\gamma \right|_{t=0}$$

$$0.99983 \times e^{-0.11508t} = \frac{1}{2}(0.99983)e^{-0.11508(0)}$$

$$e^{-0.11508t} = 0.5$$

$$-0.11508t = \ln(0.5)$$

$$t = 6.0232 \text{ hours}$$

c) The relative intensity of the radiation after 24 hours is

$$\begin{aligned} \gamma &= 0.99983 \times e^{-0.11508(24)} \\ &= 6.3160 \times 10^{-2} \end{aligned}$$

This value of relative intensity implies that only

$$\frac{6.3160 \times 10^{-2}}{0.99983} \times 100 = 6.3171\%$$

of the initial radioactivity is left after 24 hours.

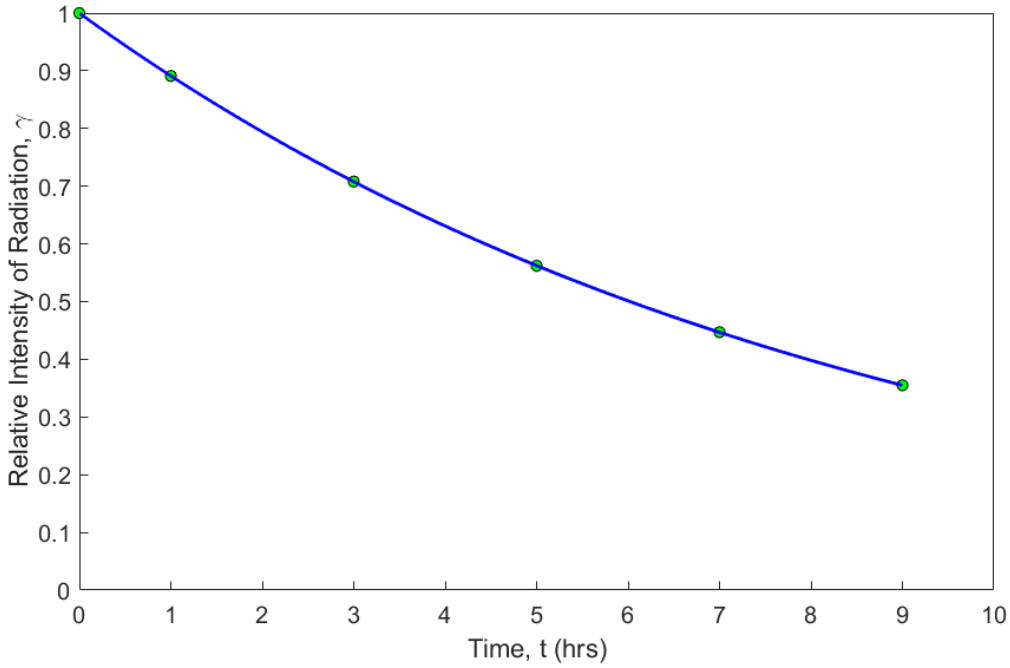



**Figure 1** Relative intensity of radiation as a function of time using an exponential regression model.

How different are the constants of the model when compared to when the data is transformed?

The regression formula obtained without transforming the data is

$$\gamma = 0.99983 \; e^{-0.11508t}$$

and the regression formula obtained with transforming the data is

$$\gamma = 0.99974e^{-0.11505t}$$

Such proximity of the constants of the model for this example may lead us to believe that it does not matter much whether we transform the data or not. Far from it, as we will see in the next example.

**Example 2**

Given the data below, regress the data to $y = e^{bx}$ with and without data transformation.

| $x$ | $y$ |
|---|---|
| 0 | 1.0000 |
| 5 | 0.8326 |
| 10 | 0.6738 |
| 15 | 0.5837 |
| 20 | 0.5150 |
| 25 | 0.4163 |
| 40 | 0.3219 |
| 60 | 0.2466 |
| 90 | 0.1803 |

**Solution**

Regress $(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)$ data to

$$y = e^{bx} \qquad (E2.1)$$

regression model.

Transforming the data

The value of $b$ can be found by transforming the data by taking the natural log of both sides of the model equation as

$$\ln(y) = \ln(e^{bx}) \qquad (E2.2)$$

$$\ln(y) = bx \qquad (E2.3)$$

Assuming

$$z = \ln y \qquad (E2.4)$$

We get a special linear model (intercept is zero) relating the $z$ data to $x$,

$$z = bx \qquad (E2.5)$$

and this linear model on minimizing the sum of the squares of the residuals gives

$$b = \frac{\sum_{i=1}^{n} x_i \ln(y_i)}{\sum_{i=1}^{n} x_i^2} \qquad (E2.6)$$

Substituting the values of the data points gives

$$\sum_{i=1}^{9} x_i \ln(y_i) = -331.64$$

$$\sum_{i=1}^{n} x_i^2 = 14675$$

$$\begin{aligned} b &= \frac{-331.64}{14675} \\ &= -0.02260 \end{aligned}$$

The regression model obtained with transforming the data is hence given by

$$y = e^{-0.02260x}$$

Without transforming the data

Here we need to start from the sum of the square of the residuals of the original model (Equation E2.1), and minimize the sum with respect to $b$. The residual is given by

$$E_i = y_i - ae^{bx_i} \qquad (E2.7)$$

The sum of the square of the residuals is

$$\begin{aligned} S_r &= \sum_{i=1}^{n} E_i^2 \\ &= \sum_{i=1}^{n} \left(y_i - e^{bx_i}\right)^2 \qquad (E2.8) \end{aligned}$$

To find the constant $b$ of the exponential model, we minimize $S_r$ by differentiating with respect to $b$ and equating the resulting expression to zero

$$\frac{dS_r}{db} = \sum_{i=1}^{n} 2\left(y_i - e^{bx_i}\right)\left(-x_i e^{bx_i}\right) = 0 \qquad (E2.9)$$

Expanding and simplifying Equation (E2.9) gives

$$\sum_{i=1}^{n} (-y_i x_i e^{bx_i} + x_i e^{2bx_i}) = 0$$

This is a nonlinear equation in terms of $b$, and can be solved by numerical methods such as bisection method. The value of $b$ obtained is

$$b = -0.03071$$

From the above solution, the regression formula obtained without transforming the data is

$$y = e^{-0.03071x}$$

The regression formula obtained with transforming the data is

$$y = e^{-0.02260x}$$

Clearly, the two models are not close, and you can see this in Figure 2 as well.

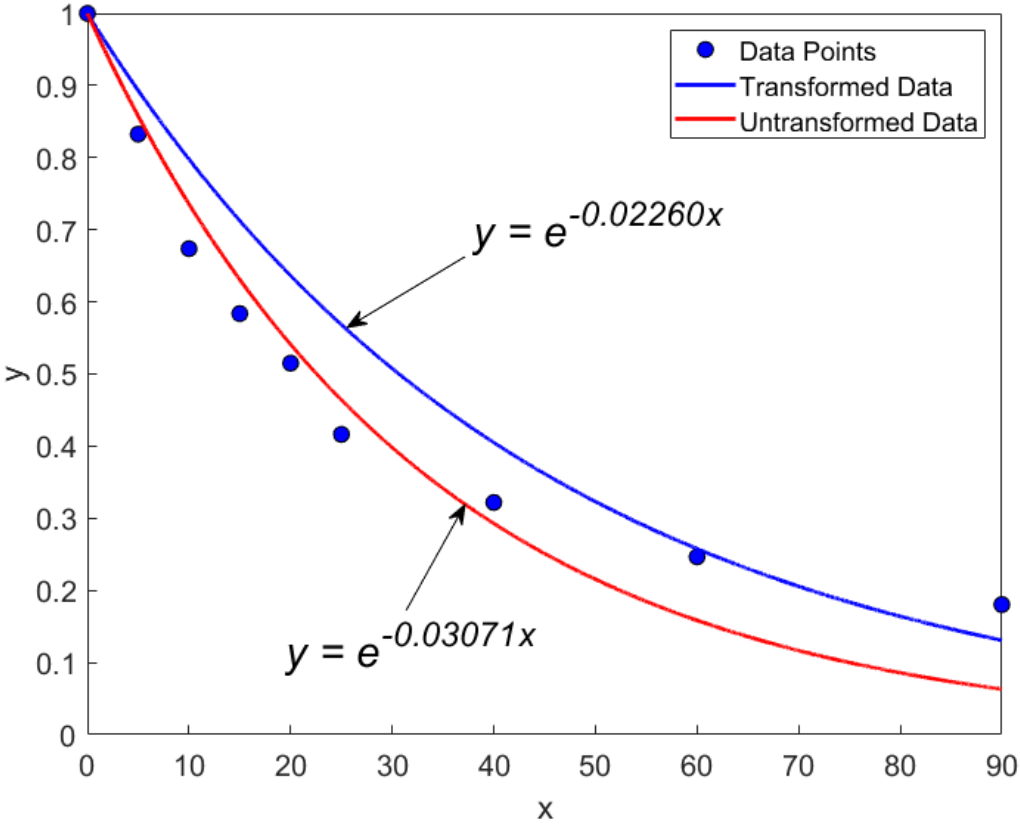



**Figure 2**. Comparing an exponential regression model with and without data transformation

# Chapter 06.04: Nonlinear Regression

Lesson: Polynomial Regression Model

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *derive formulas for the regression constants of a polynomial regression model*

2) *apply the polynomial regression model to given data.*

## Polynomial Models

Given $n$ data points $(x_1, y_1), (x_2, y_2), ....., (x_n, y_n)$, use the least-squares method to regress the data to an $m^{th}$ order polynomial.

$$y = a_0 + a_1 x + a_2 x^2 + \cdots\cdots + a_m x^m, \ m < n \qquad (1)$$

The residual at each data point is given by

$$E_i = y_i - a_0 - a_1 x_i - ... - a_m x_i^m \qquad (2)$$

The sum of the square of the residuals is given by

$$\begin{aligned} S_r &= \sum_{i=1}^{n} E_i^2 \\ &= \sum_{i=1}^{n} (y_i - a_0 - a_1 x_i - ... - a_m x_i^m)^2 \end{aligned} \qquad (3)$$

To find the constants of the polynomial regression model, we put the derivatives with respect to $a_i, \ i = 1, 2, \ldots, m$ to zero, that is,

$$\frac{\partial S_r}{\partial a_0} = \sum_{i=1}^{n} 2\left(y_i - a_0 - a_1 x_i - ... - a_m x_i^m\right)(-1) = 0$$
$$\frac{\partial S_r}{\partial a_1} = \sum_{i=1}^{n} 2\left(y_i - a_0 - a_1 x_i - ... - a_m x_i^m\right)(-x_i) = 0$$
$$\vdots \quad = \quad \vdots$$
$$\frac{\partial S_r}{\partial a_m} = \sum_{i=1}^{n} 2\left(y_i - a_0 - a_1 x_i - ... - a_m x_i^m\right)(-x_i^m) = 0$$

Setting these equations in matrix form gives

$$\begin{bmatrix} n & \left(\sum_{i=1}^{n} x_i\right) & ... \left(\sum_{i=1}^{n} x_i^m\right) \\ \left(\sum_{i=1}^{n} x_i\right) & \left(\sum_{i=1}^{n} x_i^2\right) & ... \left(\sum_{i=1}^{n} x_i^{m+1}\right) \\ ... & ... & .... \\ \left(\sum_{i=1}^{n} x_i^m\right) & \left(\sum_{i=1}^{n} x_i^{m+1}\right) & ... \left(\sum_{i=1}^{n} x_i^{2m}\right) \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ ... \\ a_m \end{bmatrix} = \begin{bmatrix} \sum_{i=1}^{n} y_i \\ \sum_{i=1}^{n} x_i y_i \\ ... \\ \sum_{i=1}^{n} x_i^m y_i \end{bmatrix} \tag{4}$$

The above equations are solved for $a_0, a_1, ..., a_m$.

**Example 1**

To find the contraction of a steel cylinder, one wishes to regress the coefficient of linear thermal expansion data to temperature.

**Table 1** Coefficient of linear thermal expansion at given different temperatures

| Temperature, $T$ (°F) | Coefficient of thermal expansion, $\alpha$(in/in/°F) |
|---|---|
| 80 | $6.47 \times 10^{-6}$ |
| 40 | $6.24 \times 10^{-6}$ |
| $-40$ | $5.72 \times 10^{-6}$ |
| $-120$ | $5.09 \times 10^{-6}$ |
| $-200$ | $4.30 \times 10^{-6}$ |
| $-280$ | $3.33 \times 10^{-6}$ |
| $-340$ | $2.45 \times 10^{-6}$ |

Regress the above data to $\alpha = a_0 + a_1 T + a_2 T^2$

**Solution**

Since $\alpha = a_0 + a_1 T + a_2 T^2$ is the quadratic relationship between the coefficient of linear thermal expansion and the temperature, the coefficients $a_0,\ a_1,\ a_2$ are found as follows

$$\begin{bmatrix} n & \left(\sum_{i=1}^{n} T_i\right) & \left(\sum_{i=1}^{n} T_i^2\right) \\ \left(\sum_{i=1}^{n} T_i\right) & \left(\sum_{i=1}^{n} T_i^2\right) & \left(\sum_{i=1}^{n} T_i^3\right) \\ \left(\sum_{i=1}^{n} T_i^2\right) & \left(\sum_{i=1}^{n} T_i^3\right) & \left(\sum_{i=1}^{n} T_i^4\right) \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} \sum_{i=1}^{n} \alpha_i \\ \sum_{i=1}^{n} T_i \alpha_i \\ \sum_{i=1}^{n} T_i^2 \alpha_i \end{bmatrix} \qquad (E1.1)$$

**Table 2** Summations for calculating constants of the model

| $i$ | $T(^\circ\mathrm{F})$ | $\alpha(\mathrm{in/in/^\circ F})$ | $T^2$ | $T^3$ |
|---|---|---|---|---|
| 1 | 80 | $6.4700 \times 10^{-6}$ | $6.4000 \times 10^{3}$ | $5.1200 \times 10^{5}$ |
| 2 | 40 | $6.2400 \times 10^{-6}$ | $1.6000 \times 10^{3}$ | $6.4000 \times 10^{4}$ |
| 3 | $-40$ | $5.7200 \times 10^{-6}$ | $1.6000 \times 10^{3}$ | $-6.4000 \times 10^{4}$ |
| 4 | $-120$ | $5.0900 \times 10^{-6}$ | $1.4400 \times 10^{4}$ | $-1.7280 \times 10^{6}$ |
| 5 | $-200$ | $4.3000 \times 10^{-6}$ | $4.0000 \times 10^{4}$ | $-8.0000 \times 10^{6}$ |
| 6 | $-280$ | $3.3300 \times 10^{-6}$ | $7.8400 \times 10^{4}$ | $-2.1952 \times 10^{7}$ |
| 7 | $-340$ | $2.4500 \times 10^{-6}$ | $1.1560 \times 10^{5}$ | $-3.9304 \times 10^{7}$ |
| $\sum_{i=1}^{7}$ | $-8.6000 \times 10^{2}$ | $3.3600 \times 10^{-5}$ | $2.5800 \times 10^{5}$ | $-7.0472 \times 10^{7}$ |

**Table 2** (cont)

| $i$ | $T^4$ | $T \times \alpha$ | $T^2 \times \alpha$ |
|---|---|---|---|
| 1 | $4.0960 \times 10^{7}$ | $5.1760 \times 10^{-4}$ | $4.1408 \times 10^{-2}$ |
| 2 | $2.5600 \times 10^{6}$ | $2.4960 \times 10^{-4}$ | $9.9840 \times 10^{-3}$ |
| 3 | $2.5600 \times 10^{6}$ | $-2.2880 \times 10^{-4}$ | $9.1520 \times 10^{-3}$ |
| 4 | $2.0736 \times 10^{8}$ | $-6.1080 \times 10^{-4}$ | $7.3296 \times 10^{-2}$ |
| 5 | $1.6000 \times 10^{9}$ | $-8.6000 \times 10^{-4}$ | $1.7200 \times 10^{-1}$ |
| 6 | $6.1466 \times 10^{9}$ | $-9.3240 \times 10^{-4}$ | $2.6107 \times 10^{-1}$ |
| 7 | $1.3363 \times 10^{10}$ | $-8.3300 \times 10^{-4}$ | $2.8322 \times 10^{-1}$ |
| $\sum_{i=1}^{7}$ | $2.1363 \times 10^{10}$ | $-2.6978 \times 10^{-3}$ | $8.5013 \times 10^{-1}$ |

$$n = 7$$

$$\sum_{i=1}^{7} T_i = -8.6000 \times 10^{-2}$$

$$\sum_{i=1}^{7} T_i^2 = 2.5580 \times 10^{5}$$

$$\sum_{i=1}^{7} T_i^3 = -7.0472 \times 10^{7}$$

$$\sum_{i=1}^{7} T_i^4 = 2.1363 \times 10^{10}$$

$$\sum_{i=1}^{7} \alpha_i = 3.3600 \times 10^{-5}$$

$$\sum_{i=1}^{7} T_i \alpha_i = -2.6978 \times 10^{-3}$$

$$\sum_{i=1}^{7} T_i^2 \alpha_i = \ 8.5013 \times 10^{-1}$$

From Equation (E1.1), we have

$$\begin{bmatrix} 7.0000 & -8.6000 \times 10^2 & 2.5800 \times 10^5 \\ -8.600 \times 10^2 & 2.5800 \times 10^5 & -7.0472 \times 10^7 \\ 2.5800 \times 10^5 & -7.0472 \times 10^7 & 2.1363 \times 10^{10} \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 3.3600 \times 10^{-5} \\ -2.6978 \times 10^{-3} \\ 8.5013 \times 10^{-1} \end{bmatrix}$$

Solving the above system of simultaneous linear equations, we get

$$\begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 6.0217 \times 10^{-6} \\ 6.2782 \times 10^{-9} \\ -1.2218 \times 10^{-11} \end{bmatrix}$$

The polynomial regression model hence is

$$\begin{aligned} \alpha &= a_0 + a_1 T + a_2 T^2 \\ &= 6.0217 \times 10^{-6} + 6.2782 \times 10^{-9} T - 1.2218 \times 10^{-11} T^2 \end{aligned}$$

# Chapter 06.04: Nonlinear Regression

Lesson: Optimum Order of a Polynomial Regression Model

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *Choose an optimum order of polynomial for polynomial regression models.*

## Introduction

Often, you may not have the privilege or enough knowledge of the problem to choose the type of regression model. Some may suggest fitting the data to a polynomial model. But the question that you would begin to ask then is − What order of polynomial should I use?

Say, if you are given $n$ data points, $(x_1, y_1), (x_2, y_2), ....., (x_n, y_n)$, you could use the least-squares method to regress the data to an $m^{th}$ order polynomial, where the order, $m$ can be as low as 0 and as high as $n-1$.

$$y = a_0 + a_1 x + a_2 x^2 + \cdots\cdots + a_m x^m,\ 0 \le m \le n-1 \qquad (1)$$

Look at Figures 1 and 2, where the data from Table 1, which has 22 data points, has been regressed to a first-order and second-order polynomial.

**Table 1** Instantaneous thermal expansion coefficient as a function of temperature for steel

| Temperature | Instantaneous Thermal Expansion |
|---|---|
| °F | $\mu$in/in/°F |
| 80 | 6.47 |
| 60 | 6.36 |
| 40 | 6.24 |
| 20 | 6.12 |
| 0 | 6.00 |
| −20 | 5.86 |
| −40 | 5.72 |
| −60 | 5.58 |
| −80 | 5.43 |
| −100 | 5.28 |
| −120 | 5.09 |
| −140 | 4.91 |

| Temperature | Instantaneous Thermal Expansion |
|---|---|
| −160 | 4.72 |
| −180 | 4.52 |
| −200 | 4.30 |
| −220 | 4.08 |
| −240 | 3.83 |
| −260 | 3.58 |
| −280 | 3.33 |
| −300 | 3.07 |
| −320 | 2.76 |
| −340 | 2.45 |

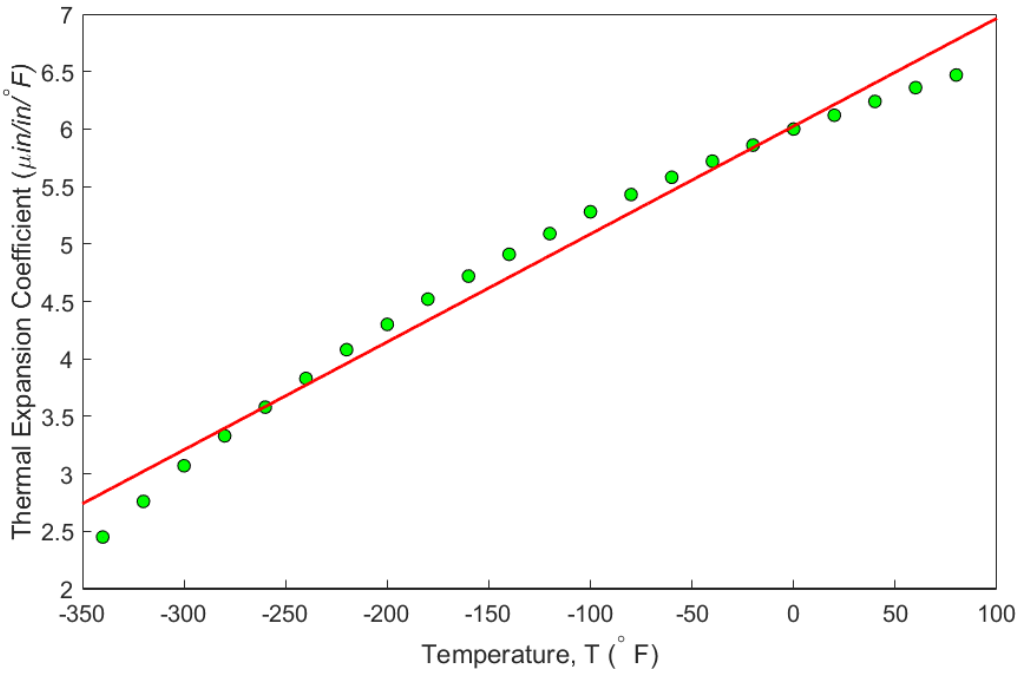



**Figure 1(a)** Polynomial regression model $\alpha = 0.009387T + 6.025$ for the linear coefficient of thermal expansion vs. temperature data for steel.

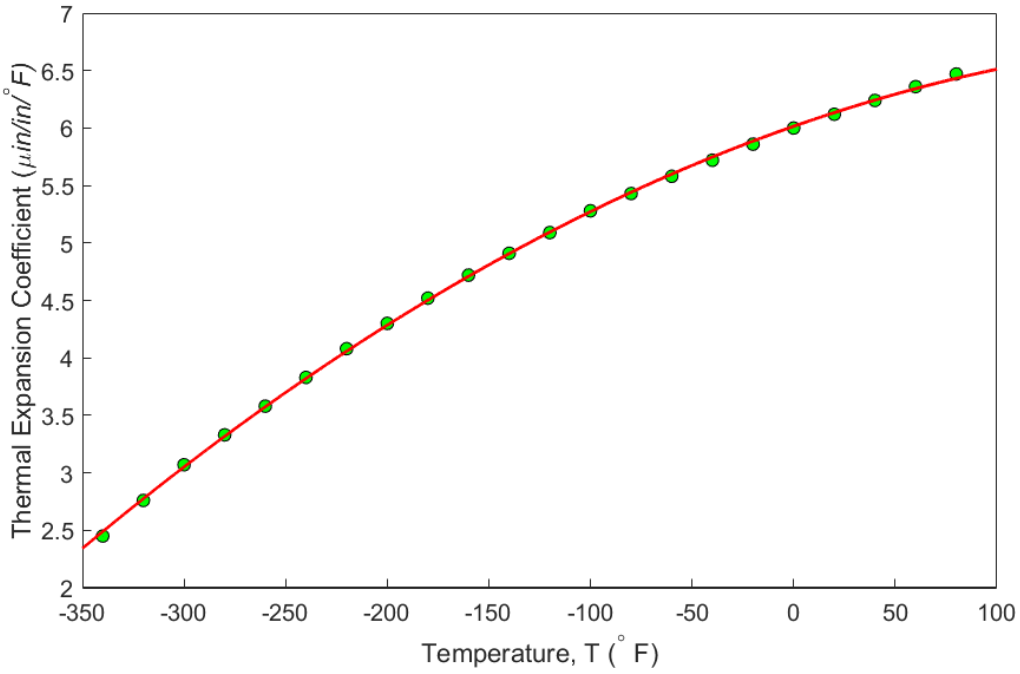



**Figure 1(b).** Polynomial regression model $\alpha = -0.00001228T^2 + 0.006196T + 6.015$ for the linear coefficient of thermal expansion vs. temperature data for steel.

Visual observation of Figure 1(b) may make us believe that the second-order polynomial looks acceptable. However, we should be basing such decisions on quantitative measurements as we may not have the luxury of a visual inspection. Even if we did, the visual interpretation is speculative.

One may hence suggest using a polynomial order for which the sum of the squares of the residuals, $S_r$ is a minimum. For the data in Table 1, the sum of the square of the residuals, $S_{rm}$ as a function of the order of the polynomial, $m$ is shown in Table 2, Figure (2a), and Figure (2b).

**Table 2.**

| Order of polynomial, $m$ | $S_{rm}$ |
|---|---|
| 1 | 0.6912 |
| 2 | 0.007732 |
| 3 | 0.001063 |
| 4 | 0.001013 |
| 5 | 0.0009019 |
| 6 | 0.0006700 |

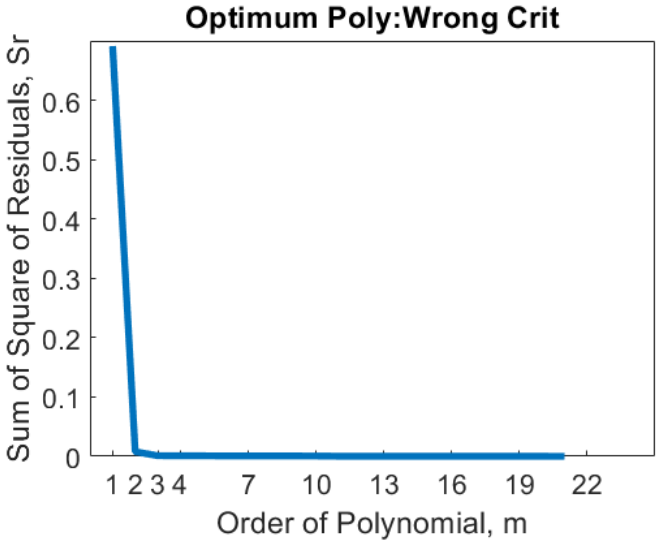



**Figure 2(a):** Sum of square of residuals as a function of the order of the polynomial. This graph starts at $m = 1$

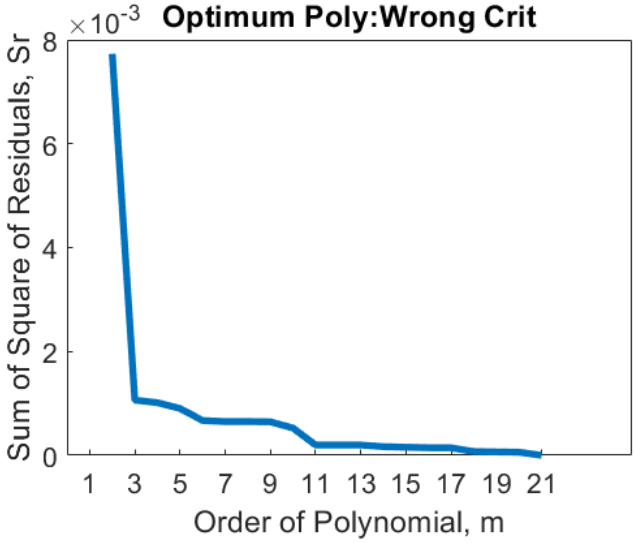



**Figure 2(b):** Sum of square of residuals as a function of the order of the polynomial. This graph is the same as Figure 2(a) but starts at $m = 2$

The value of $S_{rm}$ continues to decrease with the increase in the order of the polynomial (Figures 2a and 2b), and in fact, we will get $S_{rm} = 0$ when the chosen polynomial order is one less than the number of data points (of course, we assume there is only one $y$-value for each $x$-value). In this case, the regression curve would go through all the points and be a perfect fit. In fact, it is polynomial interpolation for this extreme case. Using the criterion of minimizing $S_{rm}$ with respect to the order of the polynomial is hence not acceptable.

Instead, the criterion suggested to use is as follows. Given $(x_1, y_1), (x_2, y_2), \ldots\ldots, (x_n, y_n)$ data points, choose the degree of the polynomial, $m$ for which the variance as computed by

$$\text{Variance} = \frac{S_{rm}}{n-(m+1)} \qquad (2)$$

is a minimum or when there is no statistically significant decrease in its value as the degree of the polynomial is increased, where

$$S_{rm} = \text{sum of the square of the residuals for the polynomial of order, } m.$$

Note the reason $m+1$ is in the formula, as there are as many constants in the polynomial regression model of order, $m$. Also, $S_{rm} = 0$, when $n = m+1$. As the order of the polynomial, $m$ increases, both the numerator, $S_{rm}$, and the denominator, $n-(m+1)$, decrease. These competing decreases would allow us to optimize the order of the polynomial.

For the data in Table 1, the variance is shown as a function of the order of the polynomial in Table 3 (shown only up to $m$=6), and Figures (3a) and (3b).

**Table 3** Variance as a function of the order of the polynomial.

| Order of polynomial, $m$ | $\frac{S_{rm}}{n-(m+1)}$ |
|---|---|
| 1 | 0.03455 |
| 2 | 0.0004069 |
| 3 | 0.00005906 |
| 4 | 0.00005961 |
| 5 | 0.00005637 |
| 6 | 0.00004467 |

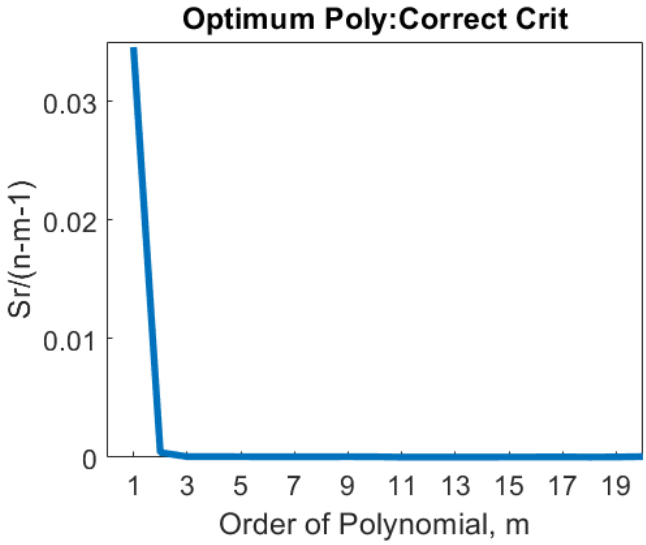



**Figure 3(a):** Variance as a function of the order of the polynomial. This graph starts at $m = 1$

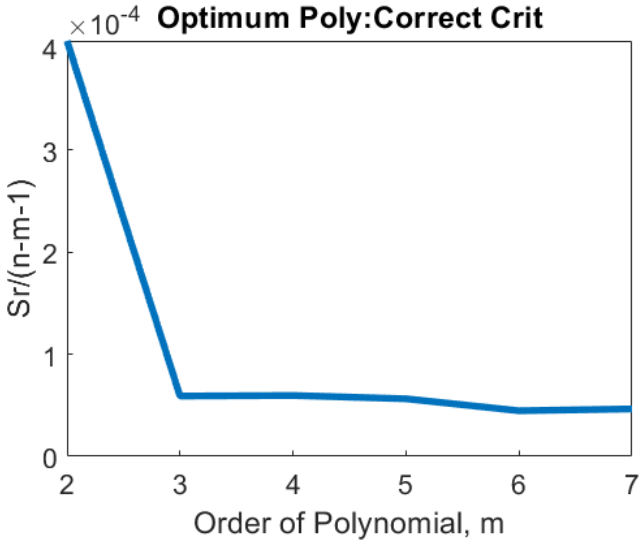



**Figure 3(b):** Sum of square of residuals as a function of the order of the polynomial. This graph is the same as Figure 3(a) but starts at $m = 2$

So, what order of polynomial would you choose? A second- or third-order polynomial would be the right choice as little statistical change occurs in the variance value after $m = 2$.

# Chapter 06.04: Nonlinear Regression

Lesson: Other Nonlinear Regression Models

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *derive constants of other popular nonlinear regression models*

2) *use the derived formula to find the constants of nonlinear regression models from given data with and without transformation.*

## Introduction

In the previous lessons, we talked about several nonlinear regression models, such as exponential. In this lesson, we continue our discussion on some growth and power regression models.

## Growth model

Common growth models standard in scientific fields have been developed and are used successfully for specific situations. The growth models are used to describe how something grows with changes in the regressor variable (often the time). Examples in this category include the growth of thin films or population with time and are of the form

$$y = \frac{a}{1 + be^{-cx}} \qquad (1)$$

where $a,\ b$ and $c$ are the constants of the model.

At $x = 0$, $y = \dfrac{a}{1+b}$ and
as $x \to \infty$, $y \to a$.

The residuals at each data point $x_i$, are

$$E_i = y_i - \frac{a}{1 + be^{-cx_i}} \qquad (2)$$

The sum of the square of the residuals is

$$S_r = \sum_{i=1}^{n} E_i^2$$
$$= \sum_{i=1}^{n} \left( y_i - \frac{a}{1 + be^{-cx_i}} \right)^2 \quad (3)$$

To find the constants $a$, $b$, and $c$, we minimize $S_r$ by differentiating $S_r$ with respect to $a$, $b$ and $c$, and equating the resulting equations to zero.

$$\frac{\partial S_r}{\partial a} = \sum_{i=1}^{n} \left( \frac{2e^{cx_i} \left[ ae^{cx_i} - y_i \left( e^{cx_i} + b \right) \right]}{\left( e^{cx_i} + b \right)^2} \right) = 0,$$
$$\frac{\partial S_r}{\partial b} = \sum_{i=1}^{n} \left( \frac{2ae^{cx_i} \left[ by_i + e^{cx_i} \left( y_i - a \right) \right]}{\left( e^{cx_i} + b \right)^3} \right) = 0,$$
$$\frac{\partial S_r}{\partial c} = \sum_{i=1}^{n} \left( \frac{-2abx_i e^{cx_i} \left[ by_i + e^{cx_i} \left( y_i - a \right) \right]}{\left( e^{cx_i} + b \right)^3} \right) = 0. \qquad (4a, b, c)$$

One can use the Newton-Raphson method to solve the above set of simultaneous nonlinear equations for the constants of the regression model, $a$, $b$, and $c$.

**Example 1**

The height of a child is measured at different ages as follows.

**Table 1** Height of the child at different ages.

| $t$ (yrs) | 0 | 5 | 8 | 12 | 16 | 18 |
|---|---|---|---|---|---|---|
| $H$(in) | 20 | 36.2 | 52 | 60 | 69.2 | 70 |

Predict the height of the child as an adult of 30 years of age using the growth model,

$$H = \frac{a}{1 + be^{-ct}}$$

**Solution**

The saturation growth model of height, $H$ vs. age, $t$ is given as

$$H = \frac{a}{1 + be^{-ct}} \qquad (E1.1)$$

where the constants $a$, $b$, and $c$ are the roots of the simultaneous nonlinear equation system

$$\sum_{i=1}^{6} \left( \frac{2e^{ct_i} \left[ ae^{ct_i} - H_i \left( e^{ct_i} + b \right) \right]}{\left( e^{ct_i} + b \right)^2} \right) = 0$$

$$\sum_{i=1}^{6}\left(\frac{2ae^{ct_i}\left[bH_i+e^{ct_i}\left(H_i-a\right)\right]}{\left(e^{ct_i}+b\right)^3}\right)=0$$

$$\sum_{i=1}^{6}\left(\frac{-2abt_ie^{ct_i}\left[bH_i+e^{ct_i}\left(H_i-a\right)\right]}{\left(e^{ct_i}+b\right)^3}\right)=0 \qquad (E1.2a,b,c)$$

We need initial guesses of the roots to get the iterative process started to find the root of those equations. Suppose we use three of the given data points such as $(0,20),(12,60)$, and $(18,70)$ to find the initial guesses of roots; we have

$$20=\frac{a}{1+be^{-c(0)}}$$

$$60=\frac{a}{1+be^{-c(12)}}$$

$$70=\frac{a}{1+be^{-c(18)}}$$

One can solve three unknowns $a$, $b$, and $c$ for the initial guesses from the three equations as

$$a=7.5534\times10^{1}$$

$$b=2.7767$$

$$c=1.9772\times10^{-1}$$

Applying the Newton-Raphson method for simultaneous nonlinear equations with the above initial guesses, one can get the roots of Equations (1.2a,b,c) as

$$a=7.4321\times10^{1}$$

$$b=2.8233$$

$$c=2.1715\times10^{-1}$$

The saturation growth model of the height of the child then is

$$H=\frac{7.4321\times10^{1}}{1+2.8233e^{-2.1715\times10^{-1}t}}$$

The predicted height of the child as an adult of 30 years of age is

$$\begin{aligned}H&=\frac{7.4321\times10^{1}}{1+2.8233e^{-2.1715\times10^{-1}\times(30)}}\\&=74''\end{aligned}$$

## Logistic Growth Model

In the logistic growth model, an example of a growth model in which a measurable quantity $y$ varies with some quantity $x$ is

$$y = \frac{ax}{b+x} \qquad (5)$$

For $x = 0$, $y = 0$ while as $x \to \infty$, $y \to a$. As noticed in the previous growth model, we had to solve three simultaneous nonlinear equations. Many times, one can transform the data and then use formulas derived for linear regression.

To transform the data for this model, we rewrite Equation (5) as

$$\begin{aligned} \frac{1}{y} &= \frac{b+x}{ax} \\ &= \frac{b}{a}\frac{1}{x} + \frac{1}{a} \end{aligned} \qquad (6)$$

Let

$$z = \frac{1}{y}$$

$$w = \frac{1}{x}$$

$$a_0 = \frac{1}{a} \text{ implying that } a = \frac{1}{a_0}$$

$$a_1 = \frac{b}{a} \text{ implying that } b = \ a_1 \times a = \frac{a_1}{a_0}$$

Then

$$z = a_0 + a_1 w \qquad (7)$$

The relationship between $z$ and $w$ is linear with the coefficients $a_0$ and $a_1$ found as follows.

$$a_1 = \frac{n\sum_{i=1}^{n} w_i z_i - \sum_{i=1}^{n} w_i \sum_{i=1}^{n} z_i}{n\sum_{i=1}^{n} w_i^2 - \left(\sum_{i=1}^{n} w_i\right)^2}$$

$$a_0 = \frac{\sum_{i=1}^{n} z_i}{n} - a_1 \frac{\sum_{i=1}^{n} w_i}{n} \qquad (8a, b)$$

Finding $a_0$ and $a_1$, then gives the constants of the original growth model as

$$a = \frac{1}{a_0}$$
$$b = \frac{a_1}{a_0} \qquad (9a, b)$$

## Logarithmic Functions

The form for the log regression models is

$$y = \beta_0 + \beta_1 \ln(x) \qquad (10)$$

Equation (10) is a linear function between $y$ and $\ln(x)$ and the usual least-squares method applies in which $y$ is the response variable and $\ln(x)$ is the regressor.

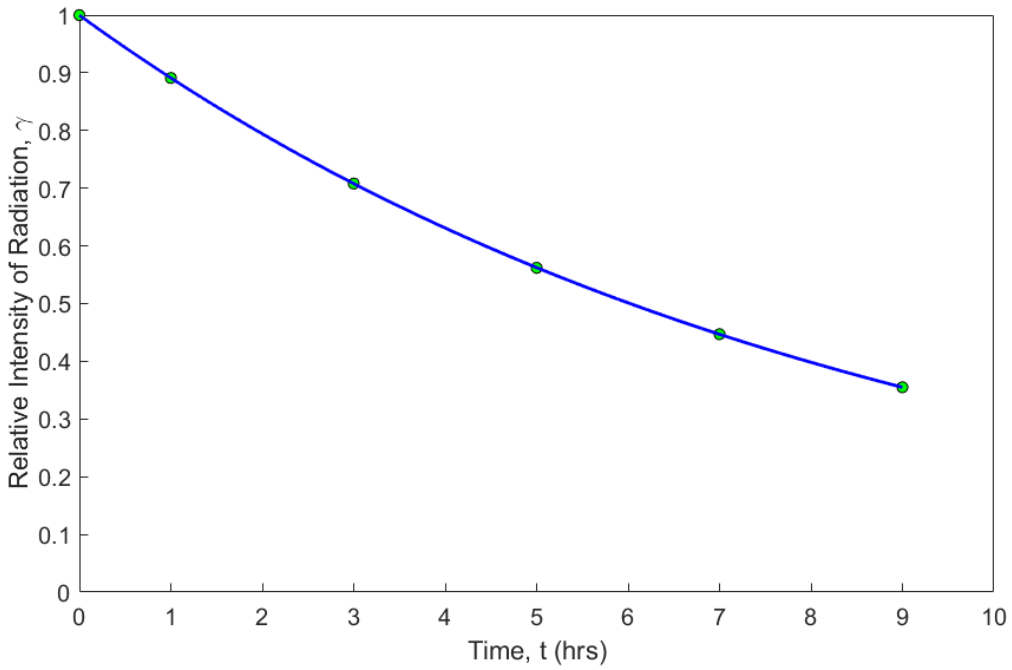



**Figure 1** Exponential regression model with transformed data for relative intensity of radiation as a function of time.

### Example 2

Sodium borohydride is a potential fuel for fuel cell. The following overpotential ($\eta$) vs. current ($i$) data was obtained in a study conducted to evaluate its electrochemical kinetics.

**Table 2** Electrochemical Kinetics of borohydride data.

| $\eta$ $(V)$ | $-0.29563$ | $-0.24346$ | $-0.19012$ | $-0.18772$ | $-0.13407$ | $-0.0861$ |
|---|---|---|---|---|---|---|
| $i$ $(A)$ | 0.00226 | 0.00212 | 0.00206 | 0.00202 | 0.00199 | 0.00195 |

At the conditions of the study, it is known that the relationship that exists between the overpotential ($\eta$) and current ($i$) can be expressed as

$$\eta = a + b \ln i$$

where $a$ is an electrochemical kinetics parameter of borohydride on the electrode. Use the data in Table 2 to evaluate the values of $a$ and $b$.

**Solution**

Following the least-squares method, Table 3 tabulates the summations where

$$x = \ln i$$

$$y = \eta$$

We obtain

$$y = a + bx \qquad (E2.1)$$

This is a linear relationship between $y$ and $x$, and the coefficients $b$ and $a$ are found as follow

$$b = \frac{n\sum_{i=1}^{n} x_i y_i - \sum_{i=1}^{n} x_i \sum_{i=1}^{n} y_i}{n\sum_{i=1}^{n} x_1^2 - \left(\sum_{i=1}^{n} x_i\right)^2}$$

$$a = \bar{y} - b\bar{x} \qquad (E2.2a, b)$$

**Table 3** Summation values for calculating constants of model

| # | $i$ | $y = \eta$ | $x = \ln(i)$ | $x^2$ | $xy$ |
|---|---|---|---|---|---|
| 1 | 0.00226 | $-0.29563$ | $-6.0924$ | 37.117 | 1.8011 |
| 2 | 0.00212 | $-0.24346$ | $-6.1563$ | 37.901 | 1.4988 |
| 3 | 0.00206 | $-0.19012$ | $-6.1850$ | 38.255 | 1.1759 |
| 4 | 0.00202 | $-0.18772$ | $-6.2047$ | 38.498 | 1.1647 |
| 5 | 0.00199 | $-0.13407$ | $-6.2196$ | 38.684 | 0.83386 |
| 6 | 0.00195 | $-0.08610$ | $-6.2399$ | 38.937 | 0.53726 |
| $\sum_{i=1}^{6}$ | 0.012400 | $-1.1371$ | $-37.098$ | 229.39 | 7.0117 |

$$n = 6$$

$$\sum_{i=1}^{6} x_i = -37.098$$

$$\sum_{i=1}^{6} y_i = -1.1371$$

$$\sum_{i=1}^{6} x_i y_i = 7.0117$$

$$\sum_{i=1}^{6} x_i^2 = 229.39$$

$$\begin{aligned} b &= \frac{6\,(7.0117) - (-37.098)\,(-1.1371)}{6\,(229.39) - (-37.098)^2} \\ &= -1.3601 \end{aligned}$$

$$\begin{aligned} a &= \frac{-1.1371}{6} - (-1.3601)\,\frac{-37.098}{6} \\ &= -8.5990 \end{aligned}$$

Hence

$$\eta = -8.5990 - 1.3601 \times \ln i$$

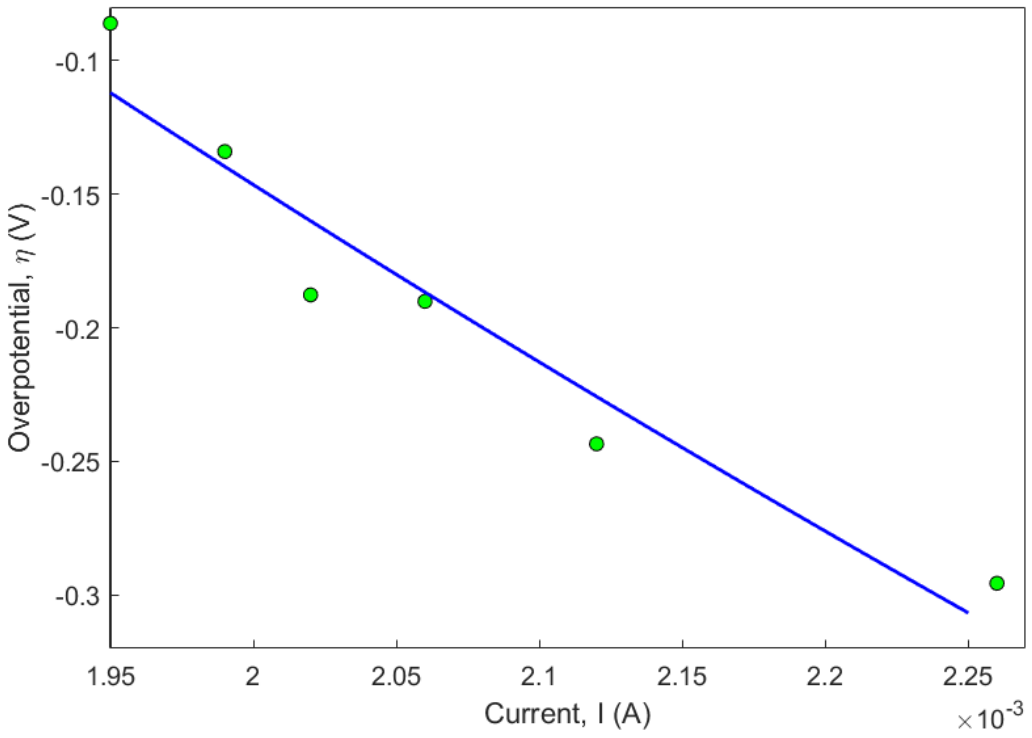



**Figure 2** Overpotential as a function of current.$\eta(V)$

## Power Functions

The power function equation describes many scientific and engineering phenomena. In chemical engineering, the rate of a chemical reaction is often written in power function form as

$$y = ax^b \qquad (11)$$

The least-squares method is applied to the power function by first transforming the data (the assumption is that $b$ is not known). If the only unknown is $a$, then a linear relation exists between $x^b$ and $y$.

When both $a$ and $b$ are unknowns, the transformation of the data is as follows.

$$\ln(y) = \ln(a) + b\ln(x) \qquad (12)$$

The resulting equation shows a linear relation between $\ln y$ and $\ln(x)$.

Let

$$z = \ln y$$

$$w = \ln(x)$$

$$a_0 = \ln a \text{ implying } a = e^{a_0}$$

$$a_1 = b$$

we get

$$z = a_0 + a_1 w$$

$$a_1 = \frac{n\sum_{i=1}^{n} w_i z_i - \sum_{i=1}^{n} w_i \sum_{i=1}^{n} z_i}{n\sum_{i=1}^{n} w_i^2 - \left(\sum_{i=1}^{n} w_i\right)^2}$$

$$a_0 = \frac{\sum_{i=1}^{n} z_i}{n} - a_1 \frac{\sum_{i=1}^{n} w_i}{n} \qquad (13)$$

Since $a_0$ and $a_1$ can be found, the original constants of the model are

$$b = a_1$$
$$a = e^{a_0} \qquad (14a, b)$$

**Example 3**

The progress of a homogeneous chemical reaction is followed, and it is desired to evaluate the rate constant and the order of the reaction. The rate law expression for the reaction is known to follow the power function form

$$-r = kC^n$$

Use the data provided in the table to obtain $n$ and $k$.

**Table 4** Chemical kinetics.

| $C_A$(gmol/l) | 4 | 2.25 | 1.45 | 1.0 | 0.65 | 0.25 | 0.006 |
|---|---|---|---|---|---|---|---|
| $-r_A$(gmol/l · s) | 0.398 | 0.298 | 0.238 | 0.198 | 0.158 | 0.098 | 0.048 |

**Solution**

Taking the natural log of both sides of Equation (35), we obtain

$$\ln(-r) = \ln(k) + n\ln(C) \qquad (E3.1)$$

Let

$$z = \ln(-r)$$

$$w = \ln(C)$$

$$a_0 = \ln(k)$$

implying that

$$\begin{aligned} k &= e^{a_0} \\ a_1 &= n \end{aligned} \qquad (3.2a, b)$$

We get

$$z = a_0 + a_1 w$$

This is a linear relation between $z$ and $w$, where

$$a_1 = \frac{m\sum_{i=1}^{m} w_i z_i - \sum_{i=1}^{m} w_i \sum_{i=1}^{m} z_i}{m\sum_{i=1}^{m} w_i^2 - \left(\sum_{i=1}^{m} w_i\right)^2}$$

$$a_0 = \frac{\sum_{i=1}^{m} z_i}{m} - a_1 \frac{\sum_{i=1}^{m} w_i}{m} \qquad (E3.3)$$

**Table 5** Kinetics rate law using power function

| $i$ | $C$ | $-r$ | $w$ | $z$ | $wz$ | $w^2$ |
|---|---|---|---|---|---|---|
| 1 | 4 | 0.398 | 1.3863 | −0.92130 | −1.2772 | 1.9218 |
| 2 | 2.25 | 0.298 | 0.8109 | −1.2107 | −0.9818 | 0.65761 |
| 3 | 1.45 | 0.238 | 0.3716 | −1.4355 | −0.5334 | 0.13806 |
| 4 | 1 | 0.198 | 0.0000 | −1.6195 | 0.0000 | 0.00000 |
| 5 | 0.65 | 0.158 | −0.4308 | −1.8452 | 0.7949 | 0.18557 |
| 6 | 0.25 | 0.098 | −1.3863 | −2.3228 | 3.2201 | 1.9218 |
| 7 | 0.006 | 0.048 | −5.1160 | −3.0366 | 15.535 | 26.173 |
| $\sum_{i=1}^{7}$ | | | −4.3643 | −12.391 | 16.758 | 30.998 |

$$m = 7$$

$$\sum_{i=1}^{7} w_i = -4.3643$$

$$\sum_{i=1}^{7} z_i = -12.391$$

$$\sum_{i=1}^{7} w_i z_i = 16.758$$

$$\sum_{i=1}^{7} w_i^2 = 30.998$$

From Equation (E3.3)

$$\begin{aligned} a_1 &= \frac{7 \times (16.758) - (-4.3643) \times (-12.391)}{7 \times (30.998) - (-4.3643)^2} \\ &= 0.31943 \end{aligned}$$

$$\begin{aligned} a_0 &= \frac{-12.391}{7} - (0.31943)\,\frac{-4.3643}{7} \\ &= -1.5711 \end{aligned}$$

From Equation (E3.2a) and (E3.2b), we obtain

$$k = e^{-1.5711} = 0.20782$$

$$n = a_1 = 0.31941$$

Finally, the model of progress of that chemical reaction is

$$-r = 0.20782 \times C^{0.31941}$$

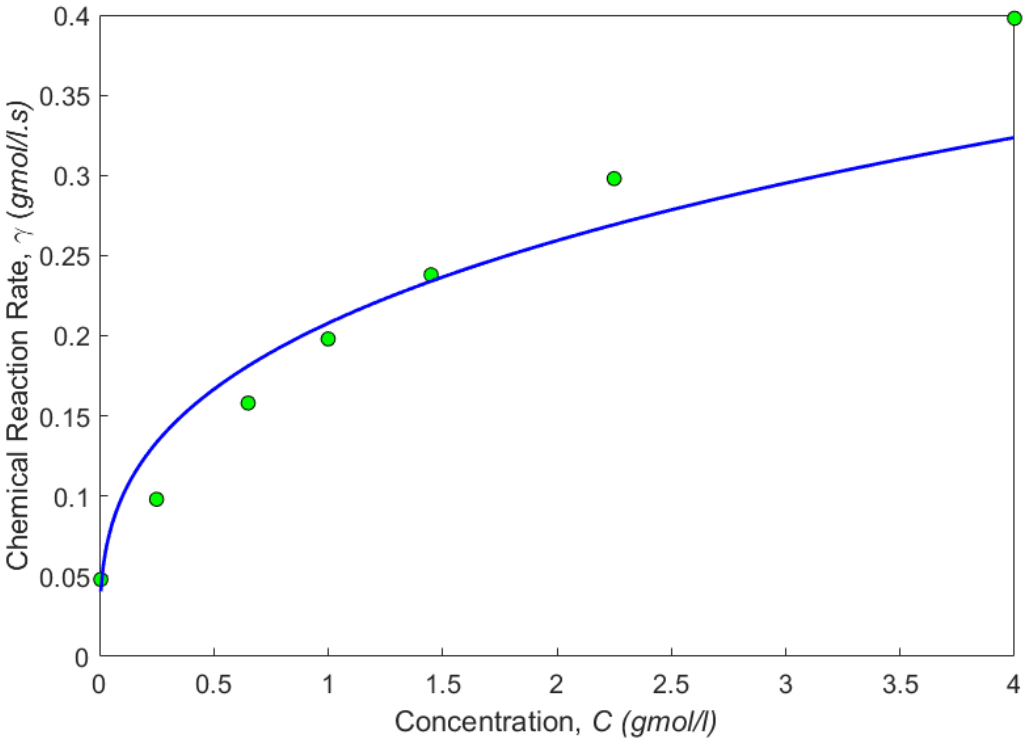



**Figure 3** Kinetic chemical reaction rate as a function of concentration.

# Multiple Choice Test

Chapter 06.04: Nonlinear Regression

(1). When using transformed data to find the constants of the regression model $y = ae^{bx}$ to best fit $(x_1, y_1), (x_2, y_2), \ldots\ldots (x_n, y_n)$, the sum of the square of the residuals that is minimized is

(A) $\sum_{i=1}^{n} \left(y_i - ae^{bx_i}\right)^2$

(B) $\sum_{i=1}^{n} \left(\ln(y_i) - \ln(a) - bx_i\right)^2$

(C) $\sum_{i=1}^{n} \left(y_i - \ln(a) - bx_i\right)^2$

(D) $\sum_{i=1}^{n} \left(\ln(y_i) - \ln(a) - b\ln(x_i)\right)^2$

(2). It is suspected from theoretical considerations that the rate of water flow from a firehouse is proportional to some power of the nozzle pressure. Assume pressure data is more accurate. You are transforming the data.

| Flow rate, $F$ (gallons/min) | 96 | 129 | 135 | 145 | 168 | 235 |
|---|---|---|---|---|---|---|
| Pressure, $p$ (psi) | 11 | 17 | 20 | 25 | 40 | 55 |

The exponent of the nozzle pressure in the regression model $F = ap^b$ most nearly is

(A) 0.49721

(B) 0.55625

(C) 0.57821

(D) 0.67876

(3). The stress-strain curve is given as $\sigma = k_1 \varepsilon e^{-k_2 \varepsilon}$ for concrete in compression, where $\sigma$ is the stress and $\varepsilon$ is the strain. To transform the data, one would use the following form

(A) $\ln(\sigma) = \ln(k_1) + \ln(\varepsilon) - k_2\varepsilon$

(B) $\ln\left(\frac{\sigma}{\varepsilon}\right) = \ln(k_1) - k_2\varepsilon$

(C) $\ln\left(\frac{\sigma}{\varepsilon}\right) = \ln(k_1) + k_2\varepsilon$

(D) $\ln(\sigma) = \ln(k_1\varepsilon) - k_2\varepsilon$

(4). In nonlinear regression, finding the constants of the model requires solving simultaneous nonlinear equations. However, in the exponential model $y = ae^{bx}$ that is best fit to $(x_1, y_1), (x_2, y_2), ....., (x_n, y_n)$, the value of $b$ can be found as a solution of a single nonlinear equation. That nonlinear equation is given by

(A) $\sum_{i=1}^{n} y_i x_i e^{bx_i} - \sum_{i=1}^{n} y_i e^{bx_i} \sum_{i=1}^{n} x_i = 0$

(B) $\sum_{i=1}^{n} y_i x_i e^{bx_i} - \frac{\sum_{i=1}^{n} y_i e^{bx_i}}{\sum_{i=1}^{n} e^{2bx_i}} \sum_{i=1}^{n} x_i e^{2bx_i} = 0$

(C) $\sum_{i=1}^{n} y_i x_i e^{bx_i} - \frac{\sum_{i=1}^{n} y_i e^{bx_i}}{\sum_{i=1}^{n} e^{2bx_i}} \sum_{i=1}^{n} e^{bx_i} = 0$

(D) $\sum_{i=1}^{n} y_i e^{bx_i} - \frac{\sum_{i=1}^{n} y_i e^{bx_i}}{\sum_{i=1}^{n} e^{2bx_i}} \sum_{i=1}^{n} x_i e^{2bx_i} = 0$

(5). There is a functional relationship between the mass density $\rho$ of air and the altitude $h$ above sea level.

| Altitude above sea level, $h$ (km) | 0.32 | 0.64 | 1.28 | 1.60 |
|---|---|---|---|---|
| Mass Density, $\rho$ (kg/m$^3$) | 1.15 | 1.10 | 1.05 | 0.95 |

In the regression model $\rho = k_1 e^{-k_2 h}$, the constant $k_2$ is found as $k_2 = 0.1315$. Assume the mass density of air at the top of the atmosphere is $1/1000^{th}$ of the mass density of air at sea level. The altitude in kilometers of the top of the atmosphere most nearly is

(A) 46.2

(B) 46.6

(C) 49.7

(D) 52.5

(6). A steel cylinder at 80°F of length 12″ is placed in a commercially available liquid nitrogen bath(−315°F). If the thermal expansion coefficient of steel behaves as a second-order polynomial function of temperature and the polynomial is found by regressing the data below,

| Temperature, $T$ (°F) | Thermal expansion Coefficient, $\alpha$ ($\mu$in/in/$^\circ$F) |
|---|---|
| −320 | 2.76 |
| −240 | 3.83 |
| −160 | 4.72 |
| −80 | 5.43 |
| 0 | 6.00 |
| 80 | 6.47 |

the reduction in the length of the cylinder in inches most nearly is

(A) 0.0219

(B) 0.0231

(C) 0.0235

(D) 0.0307

For complete solution, go to

http://nm.mathforcollege.com/mcquizzes/06reg/quiz_06reg_nonlinear_solution.pdf

# Problem Set

## Chapter 06.04: Nonlinear Regression

(1). It is suspected from theoretical considerations that the rate of flow from a fire hose is proportional to some power of the nozzle pressure. Determine whether the speculation is true. What is the exponent of the data? Assume pressure data is more accurate. You are allowed to transform the data.

| Flow rate (gallons/min), $F$ | 94 | 118 | 147 | 180 | 230 |
|---|---|---|---|---|---|
| Pressure (psi), $p$ | 10 | 16 | 25 | 40 | 60 |

**Answer:** $30.213p^{0.49101}$

(2). The following force vs. displacement data is given for a nonlinear spring.

| Displacement, $x$ (m) | 10 | 15 | 20 |
|---|---|---|---|
| Force, $F$ (N) | 100 | 200 | 400 |

Starting by minimizing the sum of the square of the residuals and without data transformation, the above $F$ vs. $x$ data is regressed to $F = kx^2$. Find the value of $k$.

**Answer:** 0.97450 N/m$^2$

(3). Data points $(x_1, y_1), (x_2, y_2), ....., (x_n, y_n)$ is regressed to $y = \dfrac{1}{(ax+b)^2}$. The constants of the model, $a$ and $b$ are solved by data transformation. Find the formulas for calculating $a$ and $b$?

Hint: Take the inverse and square-root of both sides to transform the data

**Answer:** Answer not shown intentionally - follow the hint.

(4). Fit the regression model $\sigma = K_1 \varepsilon e^{-K_2 \varepsilon}$ to the following stress-strain data of concrete. You are allowed to transform the data. Units of stress, $\sigma$ are psi, and units of strain, $\varepsilon$ are $\mu$in/in.

| Stress, $\sigma$ (psi) | 2250 | 3575 | 4250 | 4400 | 4200 |
|---|---|---|---|---|---|
| Strain, $\varepsilon$ ($\mu$in/in) | 500 | 1000 | 1500 | 2000 | 2375 |

**Answer:** $K_1 = 5.8406$, $K_2 = 4.9439 \times 10^{-4}$ (note the units for strain for which this $K_2$ is given is $\mu$in/in). In the final answer, strain is in $\mu$in/in, and stress in psi. The units of $K_2$ are in/$\mu$in. The units of $K_1$ are psi/($\mu$in/in) which is the same as Msi.

(5). It is desired to obtain a functional relationship between the mass density $\rho$ of air and the altitude $h$ above the sea level for the dynamic analysis of bodies moving within the earth's atmosphere. Use the regression model $\rho = k_1 e^{-k_2 h}$ to fit the data given below. Find the constants $k_1$ and $k_2$. You are allowed to transform the data.

| Altitude (kilometers) | Mass density ($kg/m^3$) |
|---|---|
| 0.32 | 1.15 |
| 0.64 | 1.10 |
| 1.28 | 1.05 |
| 1.60 | 0.95 |

**Answer:** $k_1 = 1.2053,\ k_2 = 0.13395$, where in the model, $h$ is in kms, and $\rho$ is in $kg/m^3$

(6). You are working for Valdez SpillProof Oil Company as a petroleum engineer. Your boss is asking you to estimate the life of an oil well. The analysis used in the industry is called the *decline curve analysis*, where the barrels of oil produced per unit time are plotted against time, and the curve is extrapolated. One of the standard curves used is the harmonic decline model, that is

$$q = \frac{b}{1 + at}$$

where $q$ is the rate of production and $t$ is the time, $b$ and $a$ are the constants of the regression model.

| Time $(t)$, month | 2 | 6 | 10 | 14 | 18 |
|---|---|---|---|---|---|
| Rate of production $(q)$, barrels per day | 260 | 189 | 120 | 87 | 75 |

a) Find the constants of the regression model. Hint: You are allowed to transform the data, if possible.

b) Find the total life of an oil field if 5 barrels per day is considered the production at which the field needs to be abandoned for further production.

c) What does $b$ stand for physically?

**Answer:** *a*) $461.85/(1 + 0.29069t)$

*b*) 314.30 months

*c*)Answer not given intentionally

(7). From the physical understanding of a phenomenon, the velocity of a body is suspected to follow the relationship $v = at^2$, where $v$ is the velocity and $t$ is the time. Given $n$ data points $(t_1, v_1), (t_2, v_2), ....., (t_n, v_n)$, start from the fundamentals of minimizing the sum of the squares of the residuals to derive a single equation in terms of the time and velocity data to find the value of $a$. Simplify the equation. Show fill derivation that includes that the derived value of $a$ corresponds to an absolute minimum of the sum of the squares of the residuals. You are not allowed to transform the data.

**Answer:** $a = \dfrac{\sum_{i=1}^{n} v_i t_i^2}{\sum_{i=1}^{n} t_i^4}$

(8). The population for a small community is given as a function of time.

| time, $t$ (years) | 0 | 5 | 10 |
|---|---|---|---|
| Population, $p$ | 100 | 165 | 314 |

By minimizing the sum of the square of the residuals and without data transformation, the model is regressed to $p = ae^{bt}$, where $a$ and $b$ are constants of the model. The value of $b$ was found to be 0.1199. Find the value of $a$.

**Answer:** $p = 94.136e^{0.1199t}$

(9). The data given above is actually the weight (lbs) of a baby as a function of the age (months) of the baby

| $t$ | 0 | 2 | 4 | 6 | 18 |
|---|---|---|---|---|---|
| $W$ | 7.5 | 11.25 | 14.30 | 16.00 | 26.00 |

a) Fit $W = at^2 + bt + c$ to the data to find $W(360)$

b) We expect that the weight of the baby to saturate as the baby reaches adulthood. Suggest a different model and estimate $W(360)$. Compare your results with part(a).

**Answer:** *a*) $W = -0.03642t^2 + 1.6659t + 7.7810;\ W(360) = -4110.2$

*b*) Answer intentionally not given

(10). The temperature of a copper sphere cooling in air is measured as a function of time to yield the following data.

| Time, $t$ (s) | 0.2 | 0.6 | 1.0 | 2.0 |
|---|---|---|---|---|
| Temperature, $T$ (C) | 146.0 | 129.5 | 114.8 | 85.1 |

From theoretical considerations, an exponential decrease of temperature is expected as a function of time. One can assume a regression model given by $T = Ae^{-0.3t}$ . Starting from minimizing the sum of the squares of the residuals, find the constant of the model, $A$.

**Answer:** $A = 155.02$

(11). There is a functional relationship between the mass density $\rho$ of air and altitude $h$ above sea level, and this is found from the data below.

| | | | | |
|---|---|---|---|---|
| Altitude above sea level, $h$ (km) | 0.32 | 0.64 | 1.28 | 1.60 |
| Mass Density, $\rho$ (kg/m$^3$) | 1.15 | 1.10 | 1.05 | 0.95 |

This functional relationship is found by regression using the model $\rho = k_1 e^{-k_2 h}$. The constant $k_2$ is found as $k_2 = 0.1315$. Assuming the mass density of air at the top of the atmosphere is $1/1000^{th}$ of the mass density of air at sea level, find the altitude above the sea level in kilometers of the top of the atmosphere.

**Answer:** 52.53 km

# Chapter 06.05: Adequacy of Linear Regression Models

Lesson: Introduction to Adequacy of Linear Regression Models

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *find why adequacy of a linear regression model is critical,*

2) *answer the two questions that need to be considered for a linear regression model to be adequate.*

## Introduction

In the application of regression models, one objective is to obtain an equation $y = f(x)$ that best describes the $n$ response data points $(x_1, y_1), (x_2, y_2), \ldots\ldots, (x_n, y_n)$. Consequently, we are faced with answering two basic questions.

1) Does the model $y = f(x)$ describe the data adequately, that is, is there an adequate fit?

2) How well does the model predict the response variable (predictability)?

To answer these questions, we limit our discussion to straight-line models as nonlinear models require a different approach. Some authors claim that nonlinear model parameters are not unbiased.

For our discussion, we take example data to go through the process of model evaluation. Given below is the data for the coefficient of thermal expansion vs. temperature for steel.

**Table 1**. Values of the coefficient of thermal expansion vs. temperature.

| $T$ (°F) | $\alpha$ ($\mu$in/in/°F) |
|---|---|
| −340 | 2.45 |
| −260 | 3.58 |
| −180 | 4.52 |
| −100 | 5.28 |
| −20 | 5.86 |
| 60 | 6.36 |

We assume a linear relationship between the coefficient of thermal expansion and temperature data as

$$\alpha(T) = a_0 + a_1 T \qquad (1)$$

Following the procedure for conducting linear regression as given in a previous lesson, we get

$$\alpha(T) = 6.0325 + 0.0096964T \qquad (2)$$

The adequacy of a linear regression model can be determined through four checks.

1) Check if the data and corresponding regression line look visually acceptable.

2) Check how many scaled residuals are in the $[-2, 2]$ range.

3) Check the coefficient of determination.

4) Check the assumption of the inherent randomness of the residuals.

The procedure to make these checks and the criteria for acceptability are discussed in the next lesson.

# Chapter 06.05: Adequacy of Linear Regression Models

Lesson: Criteria for Adequacy of Linear Regression Models

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *Make the first check of the adequacy of the regression model by plotting the data and the linear regression model.*

2) *Calculate a standard estimate of the error.*

3) *Calculate scaled residuals.*

4) *Make the second check of the adequacy of the linear regression model by calculating how many scaled residuals are in the* [-2,2] *bracket.*

5) *Calculate the coefficient of determination.*

6) *Calculate correlation coefficient.*

7) *Make the third check of the adequacy of the linear regression model by calculating how close the coefficient of determination is to* 1.

8) *Make the fourth check of the adequacy of the linear regression model by calculating if it meets the assumptions of random errors.*

## Check 1. Plot the data and the regression model.

In this check, we visually check the plot of the data and the regression model. As an example, given below is the data for the coefficient of thermal expansion vs. temperature for steel.

**Table 1**. Values of the coefficient of thermal expansion vs. temperature.

| $T(^\circ\text{F})$ | $\alpha$ ($\mu$in/in/$^\circ$F) |
|---|---|
| −340 | 2.45 |
| −260 | 3.58 |
| −180 | 4.52 |
| −100 | 5.28 |
| −20 | 5.86 |
| 60 | 6.36 |

We assume a linear relationship between the coefficient of thermal expansion and temperature data as

$$\alpha(T) = a_0 + a_1 T \qquad (1)$$

Following the procedure for conducting linear regression as given in the earlier lessons, we get

$$\alpha(T) = 6.0325 + 0.0096964T \qquad (2)$$

Figure 1 shows the data and the regression model.

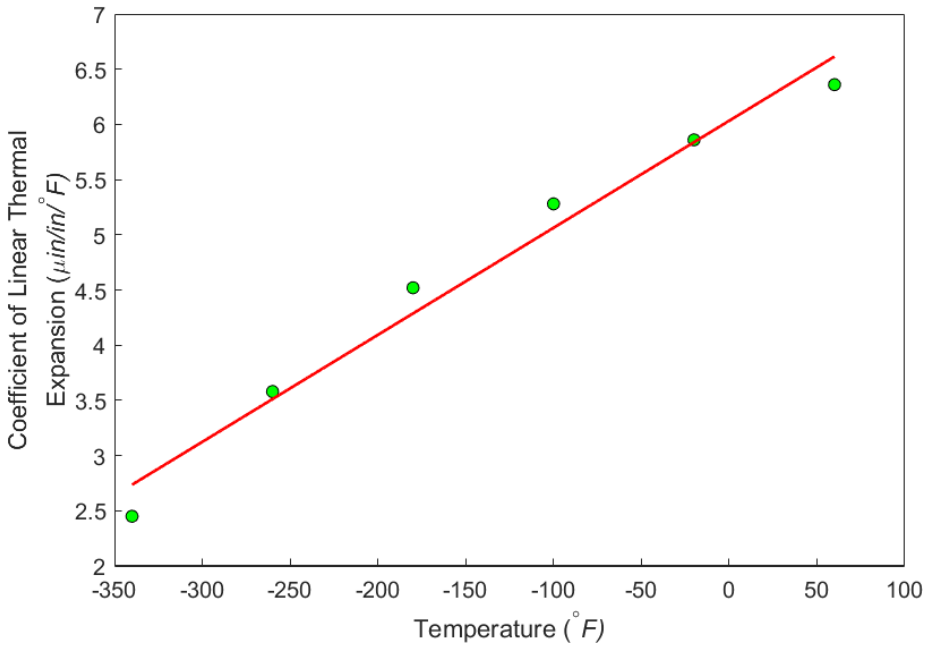



**Figure 1**. Plot of the coefficient of thermal expansion vs. temperature data points and regression line.

So what is the check? From a visual check, the regression line and the data are close to each other, and it looks like the model explains the data adequately.

## Check 2. Is the standard estimate of error within bounds?

Denoted by $s_{y/x}$ for $y$ vs. $x$ data, the standard error of estimate is used in the criterion to check the adequacy of a linear regression model.

For $y$ vs. $x$ data with $n$ data points, the standard error of estimate $s_{y/x}$ is defined as

$$s_{y/x} = \sqrt{\frac{S_r}{n-2}} \qquad (3)$$

where

$$S_r = \sum_{i=1}^{n} (y_i - a_0 - a_1 x_i)^2 \qquad (4)$$

Take the same example for data given in Table 1. The residuals for the data are given in Table 2.

**Table 2** Residuals for data.

| $T_i$ | $\alpha_i$ | $a_0 + a_1 T_i$ | $\alpha_i - a_0 - a_1 T_i$ |
|---|---|---|---|
| $-340$ | 2.45 | 2.7357 | $-0.28571$ |
| $-260$ | 3.58 | 3.5114 | 0.068571 |
| $-180$ | 4.52 | 4.2871 | 0.23286 |
| $-100$ | 5.28 | 5.0629 | 0.21714 |
| $-20$ | 5.86 | 5.8386 | 0.021429 |
| 60 | 6.36 | 6.6143 | $-0.25429$ |

Calculate the sum of the square of residuals as

$$\begin{aligned} S_r &= (-0.28571)^2 + (0.068571)^2 + (0.23286)^2 \\ &\quad + (0.21714)^2 + (0.021429)^2 + (-0.25429)^2 \\ &= 0.25283 \end{aligned}$$

The standard error of estimate as given by Equation (3) is

$$\begin{aligned} s_{\alpha/T} &= \sqrt{\frac{S_r}{n-2}} \\ &= \sqrt{\frac{0.25283}{6-2}} \\ &= 0.25141 \end{aligned}$$

The units of $s_{\alpha/T}$ are the same as the units of $\alpha$.

How is the value of the standard error of estimate interpreted? We may say that, on average, the absolute difference between the observed and predicted values is $s_{\alpha/T}$, that is, 0.25141 $\mu$in/in/°F. We can also look at the value of $s_{\alpha/T}$ as follows. The observed $\alpha$ values are between $\pm 2 s_{\alpha/T}$ of the predicted value (see Figure 1). This would lead us to believe that the value of $\alpha$ in the example is expected to be accurate within $\pm 2 s_{\alpha/T} = \pm 2 \times 0.25141 = \pm 0.50282$ $\mu$in/in/°F.

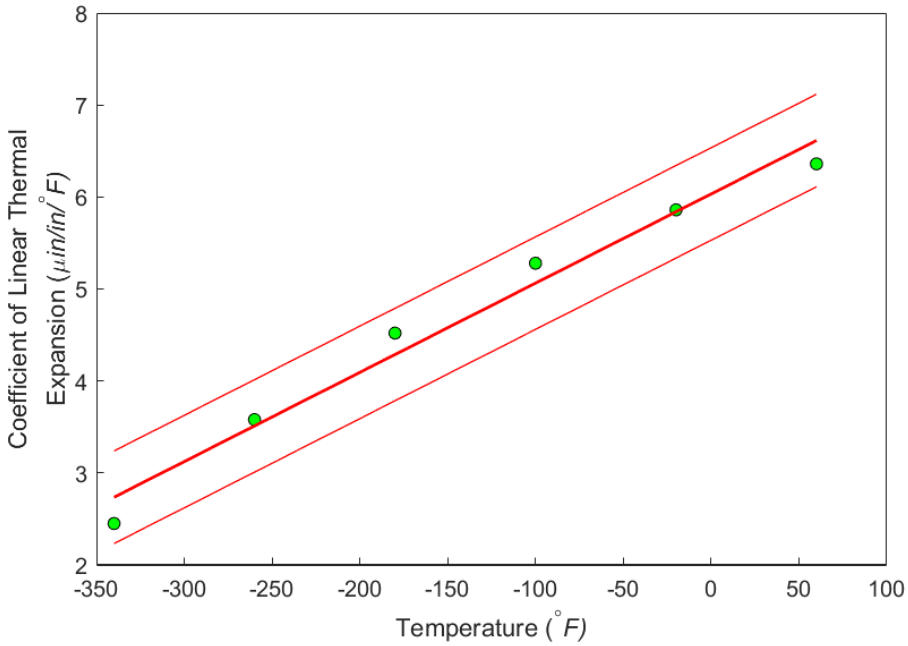



**Figure 2**. Plotting the linear regression line and showing the regression standard error.

So what is the check? Check if 95% of the observed $\alpha$ values are between $\pm 2s_{\alpha/T}$ of the predicted value. Since it would be impractical to visually make this observation for a large data set, one needs to do so programmatically. Simply calculate the scaled residual for each data pair as given by dividing the residual by the standard error of estimate.

$$\text{Scaled residual} = \frac{\alpha_i - a_0 - a_1 T_i}{s_{\alpha/T}} \qquad (5)$$

and check if 95% of the scaled residuals are in $[-2, 2]$ domain.

For this example,

$$s_{\alpha/T} = 0.25141$$

and the scaled residuals are shown in Table 3.

**Table 3**. Residuals and scaled residuals for data.

| $T_i$ | $\alpha_i$ | $\alpha_i - a_0 - a_1 T_i$ | Scaled Residuals |
|---|---|---|---|
| $-340$ | 2.45 | $-0.28571$ | $-1.1364$ |
| $-260$ | 3.58 | 0.068571 | 0.27275 |
| $-180$ | 4.52 | 0.23286 | 0.92622 |
| $-100$ | 5.28 | 0.21714 | 0.86369 |
| $-20$ | 5.86 | 0.021429 | 0.085235 |
| 60 | 6.36 | $-0.25429$ | $-1.0115$ |

All the scaled residuals are in the $[-2, 2]$ domain, and hence 95% or more of the scaled residuals are in the $[-2, 2]$ domain. It looks like the model explains the data adequately.

## Check 3. How close is the coefficient of determination to one?

Denoted by $r^2$, the coefficient of determination is used in another criterion for checking the adequacy of the linear regression model.

To find the formula for the coefficient of determination, let us start examining some measures of discrepancies between the whole data and some key central tendency. Look at the two equations given below.

$$S_r = \sum_{i=1}^{n} (\alpha_i - a_0 - a_1 T_i)^2 \qquad (6)$$

$$S_t = \sum_{i=1}^{n} (\alpha_i - \bar{\alpha})^2 \qquad (7)$$

where

$$\bar{\alpha} = \frac{\sum_{i=1}^{n} \alpha_i}{n}$$

Equation (6) is the spread of the data from the regression model (called the sum of the squares of the residuals, $S_r$) and Equation (7) is the spread of the data from the average (called the total sum of squares, $S_t$).

For the example data

$$\begin{aligned} \bar{\alpha} &= \frac{\sum_{i=1}^{6} \alpha_i}{6} \\ &= \frac{2.45 + 3.58 + 4.52 + 5.28 + 5.86 + 6.36}{6} \\ &= 4.6750 \ \mu in/in/^\circ F \end{aligned}$$

Table 4 shows the difference between observed values and the average.

**Table 4**. Difference between observed values and the average.

| $T_i$ | $\alpha_i$ | $\alpha_i - \bar{\alpha}$ |
|---|---|---|
| −340 | 2.45 | −2.2250 |
| −260 | 3.58 | −1.0950 |
| −180 | 4.52 | −0.15500 |
| −100 | 5.28 | 0.60500 |
| −20 | 5.86 | 1.1850 |
| 60 | 6.36 | 1.6850 |

Using Equation (7), the total sum of squares, $S_t$ (see column 3 of Table 5), is

$$
\begin{aligned}
S_t &= \sum_{i=1}^{n} (\alpha_i - \bar{\alpha})^2 \\
&= (-2.2250)^2 + (-1.0950)^2 + (-0.15500)^2 + (0.60500)^2 + (1.1850)^2 + (1.6850)^2 \\
&= 10.783
\end{aligned}
$$

Table 5 shows the residuals of the data.

**Table 5**. Residuals for data.

| $T_i$ | $\alpha_i$ | $a_0 + a_1 T_i$ | $\alpha_i - a_0 - a_1 T_i$ |
|---|---|---|---|
| $-340$ | 2.45 | 2.7357 | $-0.28571$ |
| $-260$ | 3.58 | 3.5114 | 0.068571 |
| $-180$ | 4.52 | 4.2871 | 0.23286 |
| $-100$ | 5.28 | 5.0629 | 0.21714 |
| $-20$ | 5.86 | 5.8386 | 0.021429 |
| 60 | 6.36 | 6.6143 | $-0.25429$ |

Using Equation (6), the sum of the square of residuals (see Column 4 of Table 6) is

$$
\begin{aligned}
S_r &= (-0.28571)^2 + (0.068571)^2 + (0.23286)^2 + (0.21714)^2 \\
&\quad + (0.021429)^2 + (-0.25429)^2 \\
&= 0.25283
\end{aligned}
$$

What inferences can we make from Equations (6) and (7)?

Equation (7) measures the discrepancy between the data and the average. Recall that the average of the data is a single value that measures the central tendency of the whole data. Equation (7) contrasts with Equation (6) as Equation (6) uses the vertical distance of the point from the regression line (another measure of central tendency) to measure the tendency from the straight line.

The objective of least-squares method is to obtain a compact equation that best describes all the data points. The mean can also be used to describe all the data points. The magnitude of the sum of squares of deviation from the mean or from the least-squares line *is, therefore, a good indicator of how well the mean or least-squares characterizes the whole data, respectively*. We can liken the sum of squares deviation around the mean $S_t$ as the error or variability in the dependent variable $y$ without considering the independent variable $x$, while $S_r$ as the sum of squares deviation around the least square regression line or variability in $y$ remaining after the variability with $x$ has been considered.

The difference between these two parameters $(S_t - S_r)$ measures the error due to describing or characterizing the data. However, $(S_t - S_r)$ value by itself is not a good criterion to use. Just see for yourself in the example how this difference would be perceived differently for the example data if we changed the units of the predicted variable $\alpha$ from ($\mu$in/in/°F) to (in/in/°F). The value of $(S_t - S_r)$ is 10.53017 when units of ($\mu i$in/in/°F) are used, and $10.53017 \times 10^{-12}$ when units of (in/in/°F) are used.

A relative comparison of this difference $(S_t - S_r)$ to $S_t$ is a better-normalized indicator and is defined as the *coefficient of determination* $r^2$, where

$$r^2 = \frac{S_t - S_r}{S_t} \qquad (8)$$

For the example

$$\begin{aligned} r^2 &= \frac{10.783 - 0.25283}{10.783} \\ &= 0.97655 \end{aligned}$$

The limits of the values of $r^2$ are between 0 and 1. What do these limiting values of $r^2$ mean? If $r^2 = 0$, then $S_t = S_r$, which means that regressing the data to a straight line does nothing to explain the data any further. If $r^2 = 1$, then $S_r = 0$, which means that the straight line passes through all the data points and is a perfect fit.

So what is the check? Based on the value obtained above for $r^2 = 0.97655$, we can claim that 97.7% of the original uncertainty in the value of $\alpha$ can be explained by the straight-line regression model

$$\alpha(T) = 6.0325 + 0.0096964T.$$

This high value of $r^2$, close to 1, means that the linear regression model is adequate based on this criterion.

Correlation Coefficient: The square root of the coefficient of determination $r^2$ is called the correlation coefficient, $r$. It can vary from $-1$ to 1 and represents the strength of the association between the predicted (dependent) and predictor (independent) variables. The sign of the correlation coefficient is the same as the sign of the slope of the linear regression line. As a simple rule of thumb, the strength of the correlation (Table 6) is interpreted by the absolute value of the correlation coefficient.

**Table 6**. Strength of correlation

| $\|r\|$ | Correlation Interpretation |
|---|---|
| 0.80 to 1.00 | Very Strong |
| 0.60 to 0.79 | Strong |
| 0.40 to 0.59 | Moderate |
| 0.20 to 0.39 | Low |
| 0.00 to 0.19 | Little or none |

For the example

$$\begin{aligned} r &= \sqrt{0.97655} \\ &= +0.98820 \end{aligned}$$

The value is positive because the slope of the linear regression curve is positive. The high value of the correlation coefficient can be interpreted as a very strong correlation between the predicted variable $\alpha$ and the predictor variable $T$.

## Check 4. Find if the model meets the assumptions of random errors.

These assumptions include 1) that the residuals are negative as well as positive to give a mean of zero, 2) the variation of the residuals as a function of the independent variable is random, 3) the residuals follow a normal distribution, and 4) that there is no auto-correlation between the data points.

To illustrate this better, we have an extended data set for the example that we took. Instead of 6 data points, this set has 22 data points (Table 7). Drawing conclusions from small or large data sets for checking the assumption of random error is not recommended.

**Table 7**. Instantaneous thermal expansion coefficient as a function of temperature.

| Temperature | Instantaneous Thermal Expansion |
|---|---|
| °F | $\mu$in/in/°F |
| 80 | 6.47 |
| 60 | 6.36 |
| 40 | 6.24 |
| 20 | 6.12 |
| 0 | 6.00 |
| −20 | 5.86 |
| −40 | 5.72 |
| −60 | 5.58 |
| −80 | 5.43 |
| −100 | 5.28 |
| −120 | 5.09 |
| −140 | 4.91 |
| −160 | 4.72 |
| −180 | 4.52 |
| −200 | 4.30 |
| −220 | 4.08 |
| −240 | 3.83 |
| −260 | 3.58 |
| −280 | 3.33 |
| −300 | 3.07 |
| −320 | 2.76 |
| −340 | 2.45 |

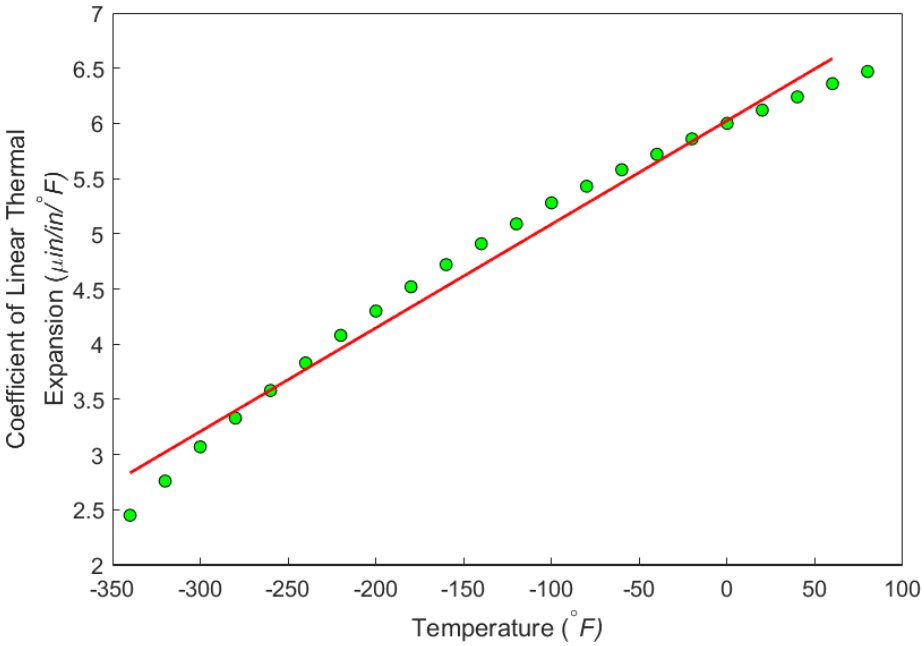



**Figure 3**. Plot of thermal expansion coefficient vs. temperature data points and regression line for more data points.

Regressing the data from Table 7 to the straight-line regression line

$$\alpha(T) = a_0 + a_1 T$$

and following the procedure for conducting linear regression as given in Chapter 06.03, we get (Figure 3)

$$\alpha = 6.0248 + 0.0093868T \qquad (9)$$

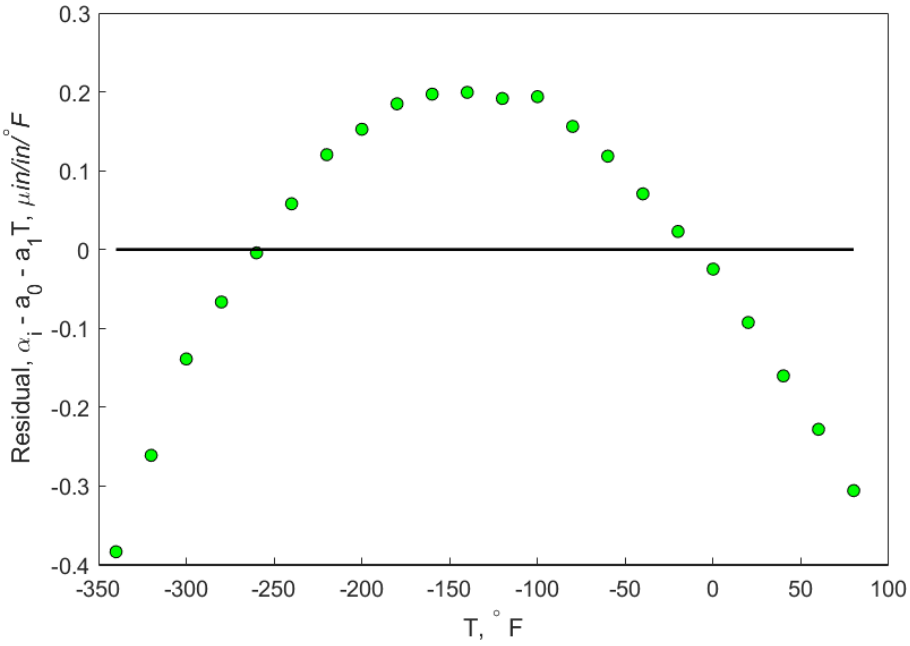



**Figure 4**. Plot of residuals.

Figure 4 shows the residuals for the example as a function of temperature. Although the residuals seem to average to zero, but within a range, they do not exhibit this zero mean. For an initial value of $T$, the averages are below zero. For the middle values of $T$, the averages are above zero, and again for the final values of $T$, the averages are below zero. This non-random variation may be considered a violation of the model assumption.

Figure 4 also shows that the residuals for the example are following a nonlinear variance which is a clear violation of the model assumption of constant variance.

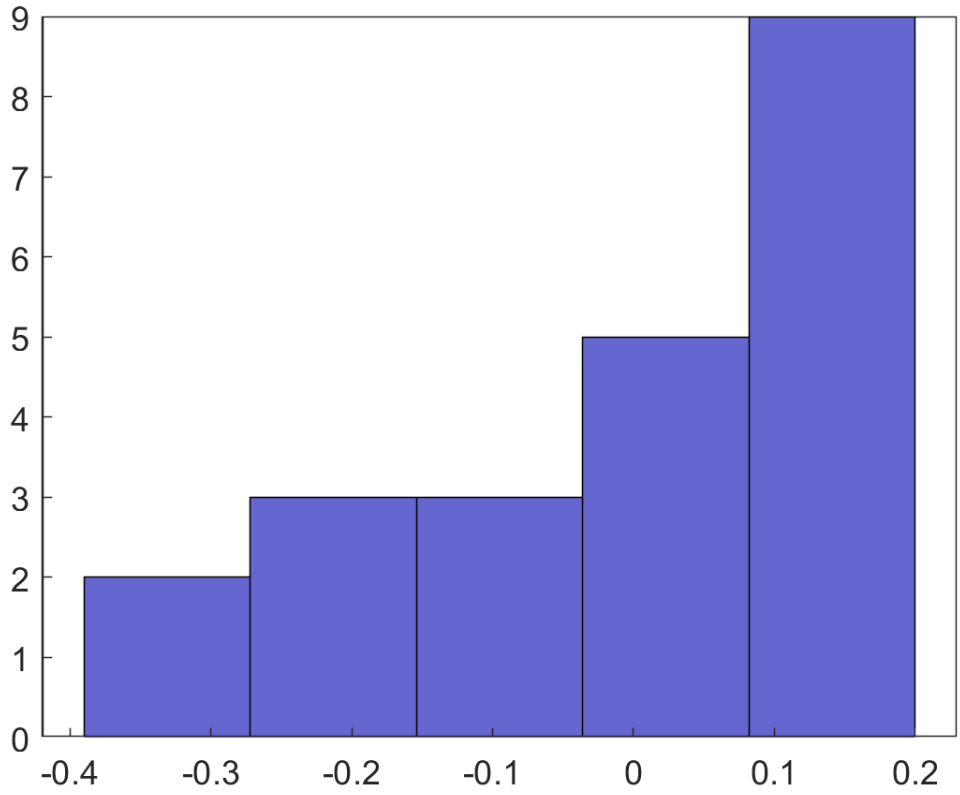



**Figure 5**. Histogram of residuals.

Figure 5 shows the histogram of the residuals. Clearly, the histogram is not showing a normal distribution and hence violates the model assumption of normality.

To check that there is no autocorrelation between observed values, the following rule of thumb can be used. If $n$ is the number of data points, and $q$ is the number of times the sign of the residual changes, then if

$$\frac{(n-1)}{2} - \sqrt{n-1} \leq \ q \leq \frac{n-1}{2} + \sqrt{n-1} \qquad (10)$$

you most likely do not have an autocorrelation. For the example, $n = 22$, then

$$\frac{(22-1)}{2} - \sqrt{22-1} \leq \ q \leq \frac{22-1}{2} + \sqrt{22-1}$$

$$5.9174 \leq \ q \leq 15.083$$

is not satisfied as $q = 2$. So this model assumption is violated.

# Chapter 06.05: Adequacy of Linear Regression Models

Lesson: Abuses of Regression

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *enumerate abuses of regression*

2) *give examples related to abuses of regression.*

## Introduction

There are three common cases of abuse of regression analysis. They are the following.

1) Extrapolation

2) Generalization

3) Misidentification

## Extrapolation

If you were dealing in the stock market or even interested in it, then you might remember the stock market crash of March 2000. During 1997-1999, investors thought they would double their money every year. They started buying fancy cars and houses on credit and living a high life. Little did they know that the whole market was hyped on speculation and little economic sense. The Enron and MCI financial fiascos soon followed.

Let us look if we could have safely extrapolated the NASDAQ index from past years. Below is the table of NASDAQ index, $S$, as a function of the end of year number, $t$ (Year 1 is the end of the year 1994, and Year 6 is the end of the year 1999).

**Table 1** NASDAQ index as a function of year number.

| Year Number, t | NASDAQ Index {S} |
|---|---|
| 1 (1994) | 752 |
| 2 (1995) | 1052 |
| 3 (1996) | 1291 |

| Year Number, t | NASDAQ Index {S} |
|---|---|
| 4 (1997) | 1570 |
| 5 (1998) | 2193 |
| 6 (1999) | 4069 |

Note: NASDAQ (National Association of Securities Dealers Automated Quotations) index is a composite index based on the stock market value of 3,000 companies. The NASDAQ index began on February 5, 1971, with a base value of 100. Twenty years later, in 1995, the NASDAQ index crossed the 1000 mark. It rose as high as 5132 on March 10, 2000, and currently is at a value of 13829 (April 8, 2021).

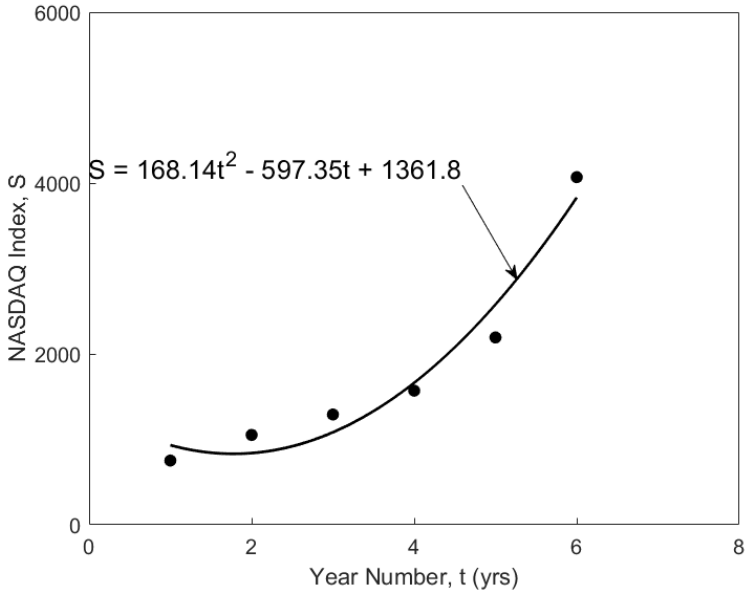



**Figure 1** The regression line of the NASDAQ Index as a function of year number.

A relationship $S = a_0 + a_1 t + a_2 t^2$ between the NASDAQ index, $S$, and the year number, $t$, is developed using least square regression and is found to be $S = 168.14t^2 - 597.37t + 1361.8$. The data and the regression line are shown in Figure 1. The data is given only for Years 1 through 6, and it is desired to calculate the value for $t > 6$, and that is extrapolation outside the model data. The error inherent in this model is shown in Table 2 and Figure 2. Look at the Year 7 and 8 that were not included in the data – the error between the predicted and actual values is 119% and 277%, respectively.

**Table 2** NASDAQ index as a function of year number.

| Year Number (**t**) | NASDAQ Index (**S**) | Predicted Index | Absolute Relative True Error (%) |
|---|---|---|---|
| 1 (1994) | 752 | 933 | 24 |
| 2 (1995) | 1052 | 840 | 20 |
| 3 (1996) | 1291 | 1083 | 16 |
| 4 (1997) | 1570 | 1663 | 6 |

| Year Number (**t**) | NASDAQ Index (**S**) | Predicted Index | Absolute Relative True Error (%) |
|---|---|---|---|
| 5 (1998) | 2193 | 2579 | 18 |
| 6 (1999) | 4069 | 3831 | 6 |
| 7 (2000) | 2471 | 5419 | 119 |
| 8 (2001) | 1951 | 7344 | 276 |

This illustration is not exaggerated, and it is important that careful use of any given model equation is always employed. At all times, it is imperative to infer the domain of independent variables for which a given equation is valid.

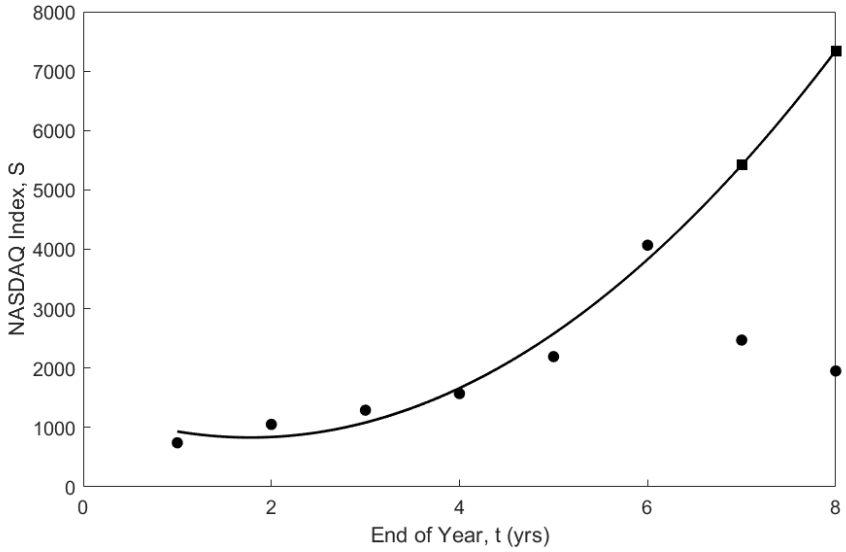



**Figure 2** Extrapolated curve and actual data for Years 7 and 8.

## Generalization

Generalization could arise when unsupported or exaggerated claims are made. It is not often possible to measure all predictor variables relevant in a study. For example, a study carried out about the efficacy of a drug might have inadvertently been restricted to men only. Shall we then generalize the result to be applicable to all? Such use of regression is an abuse since the limitations imposed by the data restrict the use of the prediction equations to men.

## Misidentification

Finally, misidentification of causation is a classic abuse of regression analysis equations. Regression analysis can only aid in the confirmation or refutation of a causal model - the model must, however, have a theoretical basis. In a chemical reacting system in which two species react to form a product,

the amount of product formed or amount of reacting species vary with time. Although a regression equation of species concentration and time can be obtained, one cannot attribute time as the causal agent for the varying species concentration. Regression analysis cannot prove causality. Instead, it can only substantiate or contradict causal assumptions. Anything outside this is an abuse of the regression analysis method.

# Multiple Choice Test

## Chapter 06.05 Adequacy of Linear Regression Models

(1). For a linear regression model to be considered adequate, the percentage of scaled residuals that need to be in the range $[-2, 2]$ is greater than or equal to

(A) 5%

(B) 50%

(C) 90%

(D) 95%

(2). Given 20 data pairs of $y$ vs. $x$ data are regressed to a straight line. The straight line regression model is given by $y = 9 - 5x$, and the coefficient of determination is found to be 0.59. The correlation coefficient is

(A) $-0.7681$

(B) $-0.3481$

(C) 0.7681

(D) 0.3481

(3). The following $y$ vs. $x$ data is regressed to a straight line.

| $x$ | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|
| $y$ | 0.3 | 0.4 | 0.57 | 0.6 | 1.77 |

The linear regression model is found to be $y = -1.4700 + 0.3140x$. The coefficient of determination is

(A) 0.3046

(B) 0.4319

(C) 0.6956

(D) 0.8339

(4). Many times, you may not know what regression model to use for given discrete data. In such cases, a suggestion may be to use a polynomial model. But the question remains – what is the order of polynomial shoud I use? For example, if you are given 10 data points, you can regress the data to a polynomial order 0, 1, 2, 3, 4, 5, 6, 7, 8, or 9. Below is the question you are asked to answer.

If $S_p$ is the sum of the squares of the residuals for the regression polynomial model of order $p$, the criterion you would use to find the optimum order of the polynomial would be to find the minimum of $\frac{S_p}{m-p}$ for all possible polynomial orders. If you have 30 data points, then the value of $m$ in the formula is

(A) 10

(B) 29

(C) 30

(D) 50

(5). On regressing $n$ data pairs $(x_1, y_1), ..., (x_n, y_n)$ to a linear regression model $y = a_0 + a_1 x$, a scientist finds the regression model to have zero slope. The regression model then is given by

(A) $y = \dfrac{\sum_{i=1}^{n} x_i}{n}$

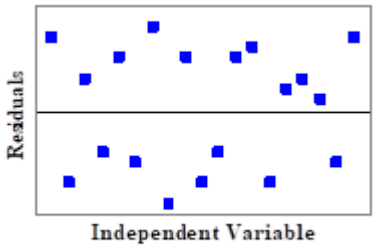
(B) $y = \dfrac{\sum_{i=1}^{n} y_i}{n}$

(C) $y = 0$

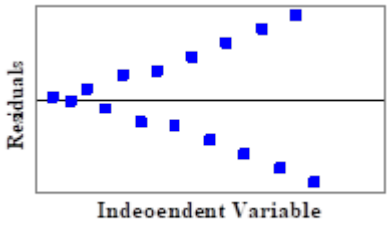
(D) $y = \sqrt{\dfrac{\sum_{i=1}^{n} (y_i - \bar{y})^2}{n-1}}$

(6). Out of the following patterns of residuals, which one is the most acceptable for a linear regression model?

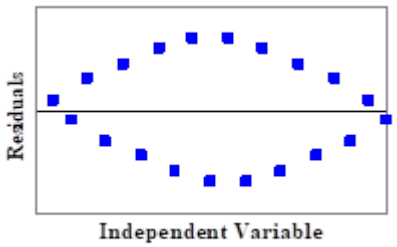
(A)



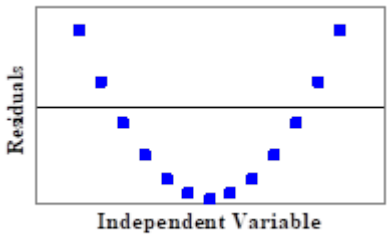
(B)




(C)




(D)




For complete solution, go to

http://nm.mathforcollege.com/mcquizzes/06reg/quiz_06reg_adequacy.pdf

# Problem Set

## Chapter 06.05: Adequacy of Linear Regression Models

(1). Water is flowing through a circular pipe of 0.5 ft radius, and flow velocity (ft/s) measurements are made from the center to the wall of the pipe as follows.

| Radial location, $r$ (ft) | 0 | 0.083 | 0.17 | 0.25 | 0.33 | 0.42 | 0.50 |
|---|---|---|---|---|---|---|---|
| Velocity, $v$(ft/s) | 10 | 9.72 | 8.88 | 7.5 | 5.6 | 3.1 | 0 |

A scientist regresses the data to a straight line and is given by $v(r) = -19.92r + 11.39$.

a) How much is the absolute relative true error in the flow rate using the above velocity expression if the actual flow rate is $3.9\text{ft}^3/\text{s}$?

b) Regress the data to a second-order polynomial of $r$. What is the absolute relative true error for this case?

c) Regress the data to a specified second-order polynomial, $v = a_0(1 - 4r^2)$. Find the value of $a_0$. What is the absolute relative true error in this case?

**Answer:** *a*) $3.7306\ \text{ft}^3/\text{s},\ 4.34\%$
*b*) $3.9496\ \text{ft}^3/\text{s},\ 1.27\%$
*c*) $3.9322\ \text{ft}^3/\text{s},\ 0.83\%$

(2). The following $y$ vs. $x$ data is regressed to a straight line and is given by $y = -1.398 + 0.3x$.

| $x$ | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|
| $y$ | 0.3 | 0.4 | 0.51 | 0.6 | 1.7 |

Find the coefficient of determination.

**Answer:** 0.69441

(3). Many times, you may not know what regression model to use for the given data. In such cases, a suggestion may be to use a polynomial. But the question remains, what is the order of the polynomial to use? For example, if you are given 10 data points, you can regress the data to a polynomial of order 1, 2, 3, 4, 5, 6, 7, 8 or 9. What order of polynomial would you use? An instructor suggests four different criteria. Only one is correct. Which one would you choose?

A) Find the sum of the square of residuals, $S_r$ for all possible polynomials, and choose the order for which $S_r$ is minimum.

B) Find $\dfrac{\text{sum of the square of residuals}}{(\text{number of data points-order of polynomial})}$ for all possible polynomials and choose the order for which it is minimum.

C) Find $\dfrac{\text{sum of the square of residuals}}{(\text{number of data points-order of polynomial - 1})}$ for all possible polynomials and choose the order for which it is minimum.

D) Find $\dfrac{\text{sum of the square of residuals}}{(\text{number of data points-order of polynomial} + 1)}$ for all possible polynomials and choose the order for which it is minimum.

**Answer:** $(C)$

(4). The following $y$ vs. $x$ data is regressed to a straight line and is given by $y = -1.398 + 0.3x$.

| $x$ | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|
| $y$ | 0.3 | 0.4 | 0.51 | 0.6 | 1.7 |

Find the scaled residuals and check how many of them are between in the range $[-2, 2]$.

**Answer:** All scaled residuals are in the range $[-2, 2]$.

(5). Water is flowing through a circular pipe of 0.5 ft radius, and flow velocity (ft/s) measurements are made from the center to the wall of the pipe as follows

| Radial Location, $r$ (ft) | 0 | 0.083 | 0.17 | 0.25 | 0.33 | 0.42 | 0.50 |
|---|---|---|---|---|---|---|---|
| Velocity, $v$ ( ft/s) | 10 | 9.72 | 8.88 | 7.5 | 5.6 | 3.1 | 0 |

A scientist regresses the data to a straight line and is given by $v(r) = -19.92r + 11.39$. Verify if the model is adequate based on what you learned in class.

**Answer:** Make all four checks - residual plot characteristics may be a cause for rejection

(6). Water is flowing through a circular pipe of 0.5 ft radius, and flow velocity (ft/s) measurements are made from the center to the wall of the pipe as follows

| Radial Location, $r$ (ft) | 0 | 0.083 | 0.17 | 0.25 | 0.33 | 0.42 | 0.50 |
|---|---|---|---|---|---|---|---|
| Velocity, $v$ (ft/s) | 10 | 9.72 | 8.88 | 7.5 | 5.6 | 3.1 | 0 |

Use an appropriate regression model to find the following.

a) Velocity at $r = 0.3$ ft.

b) Rate of change of velocity with respect to the radial location at $r = 0.3$ ft.

c) An engineer, who thinks they know everything, calculates the flow rate out of the pipe by averaging the given velocities and multiplying it by the cross-sectional area of the pipe. How will you convince

them that they need to use a different (better) approach. Hint: The flow rate $Q$ out of the pipe is given by the formula $Q = \int_0^{0.5} vdA = \int_0^{0.5} 2\pi r v dr$.

**Answer:** Use $v = a(1 - r^2/0.5^2)$

*a*) 6.4082 ft/s,
*b*) $-24.032$ ft$^2$/s
*c*) get 5.0266 $ft^3/s$ by average method, get 3.9322 ft$^3$/s by using the integration formula

(7). For the purpose of shrinking a trunnion into a hub, the reduction of diameter, $\Delta D$ of a trunnion shaft by cooling it through a temperature change of $\Delta T$, is given by

$$\Delta D = D \int_{T_{\text{room}}}^{T_{\text{fluid}}} \alpha dT$$

where

$$D = \text{original diameter (in)}$$

$$\alpha = \text{linear coefficient of thermal expansion (in/in/}^\circ\text{F)}$$

$$T_{\text{fluid}} = \text{temperature of dry-ice/alcohol mixture, (}^\circ\text{F)}$$

$$T_{\text{room}} = \text{room temperature, (}^\circ\text{F)}$$

Given below is the table of the linear coefficient of thermal expansion vs. temperature

| Temperature (°F) | Linear coefficient of thermal expansion coefficient ($\mu in/in/^\circ$F) |
|---|---|
| 80 | 6.47 |
| 0 | 6.00 |
| $-60$ | 5.58 |
| $-160$ | 4.72 |
| $-260$ | 3.58 |
| $-340$ | 2.45 |

A diametric contraction of $0.015''$ is needed in the trunnion.

$$\text{Nominal outer diameter of the trunnion} = 12.363''$$

$$\text{Temperature of dry-ice/alcohol mixture} = -108^\circ\text{F}$$

$$\text{Temperature of liquid nitrogen} = -321^\circ\text{F}$$

Room temperature = 80°F

a) Find the regression curve (a straight line is not acceptable) for the coefficient of linear thermal expansion as a function of temperature.

b) Find the diametric contraction of the trunnion if it is cooled in dry- ice/alcohol mixture? Can this cooling method be recommended to get the needed contraction?

c) If the answer to part (b) is no, would you recommend cooling it in liquid nitrogen? Why?

**Answer:** *a*) $-0.1219 \times 10^{-10}T^2 + 0.6313 \times 10^{-8}T + 0.6023 \times 10^{-5}$
*b*) $0.0137''$, not enough magnitude of contraction - cannot be recommended
*c*) $0.0244''$, enough magnitude of contraction - can be recommended

# Chapter 07.00: Physical Problem for Integration

Lesson: General Engineering

## Summary

The upward velocity of a rocket is given. To find the distance covered by the rocket requires integration.

## Modeling

A rocket is going vertically up and expels fuel at a velocity 2000 m/s at a consumption rate of 2100 kg/s. The initial mass of the rocket is $140,000$ kg. If the rocket starts from rest at $t = 0$ seconds, how can I calculate the vertical distance covered by the rocket from $t = 8$ to $t = 30$ seconds?

**Figure 1** A rocket launched into space

If

$$m_0 = \text{initial mass of rocket at } t = 0 \text{ (kg)}$$

$$q = \text{rate at which fuel is expelled (kg/sec)}$$

$$u = \text{velocity at which the fuel is being expelled (m/s)}$$

Then since fuel is expelled from the rocket, the mass of the rocket keeps decreasing with time. The mass of the rocket, $m$ at any time $t$

$$m = m_0 - qt \qquad (1)$$

The forces on the rocket at any time are found by applying Newton's second law of motion. Then

$$\sum F = ma \qquad (2)$$

gives

$$uq - mg = ma \qquad (3)$$

Substituting Equation (1) in Equation (3) gives

$$uq - (m_0 - qt)g = (m_0 - qt)a \qquad (4)$$

where

$$g = \text{acceleration due to gravity (m/s}^2\text{)}$$

Equation (4) is simplified to

$$a = \frac{uq}{m_0 - qt} - g \qquad (5)$$

$$\frac{d^2x}{dt^2} = \frac{uq}{m_0 - qt} - g \qquad (6)$$

$$\frac{dx}{dt} = -u\ln(m_0 - qt) - gt + C \qquad (7)$$

Since the rocket starts from rest

$$\frac{dx}{dt} = 0 \text{ at } t = 0$$

$$0 = u\ln(m_0) + C$$

$$C = u\ln(m_0)$$

Hence

$$\begin{aligned}\frac{dx}{dt} &= -u\ln(m_0 - qt) - gt + u\ln(m_0) \\ &= u\ln\left(\frac{m_0}{m_0 - qt}\right) - gt \qquad (8)\end{aligned}$$

Then the distance covered by the rocket from $t = t_0$ to $t = t_1$ is,

$$x = \int_{t_0}^{t_1} \left[u\ln\left(\frac{m_0}{m_0 - qt}\right) - gt\right] dt \qquad (9)$$

Let us substitute the values into the above equation. A rocket is going vertically up and expels fuel at a velocity 2000 m/s at a consumption rate of 2100 kg/s. The initial mass of the rocket is $140,000$ kg. If the rocket starts from rest at $t = 0$ seconds, how can I calculate the vertical distance covered by the rocket from $t = 8$ to $t = 30$ seconds?

Substituting

$$u = 2000 \text{ m/s}$$

$$m_0 = 140000 \text{ kg}$$

$$q = 2100 \text{ kg/s}$$

$$g = 9.8 \text{ m/s}^2$$

$$t_0 = 8 \text{ s}$$

$$t_1 = 30 \text{ s}$$

$$x = \int_8^{30} \left(2000\ln\left[\frac{140000}{140000 - 2100t}\right] - 9.8t\right) dt$$

# Chapter 07.00: Physical Problem for Integration

Lesson: Chemical Engineering

## Summary

These are textbook notes of a physical problem for integration

## Modeling

Although a fuel cell provides DC electricity like an ordinary battery, it differs from an ordinary battery from the fact that its fuel source is externally supplied and as long as the fuel is being supplied, the fuel cell can in theory run continuously without needing a recharge. The two electrochemical reactions in a fuel cell occur at the electrodes (or poles) to which reactants are continuously fed. The reaction at the negative electrode (anode) is maintained by supplying a fuel such as hydrogen or methanol, whereas the positive electrode (cathode) reaction is maintained by the supply of oxygen or air. The two electrodes (anode and cathode) are separated by a proton conducting membrane – polymer electrolyte membrane (PEM).

A schematic diagram showing the operating principles of fuel cell that utilizes methanol as fuel, i.e., a Direct Methanol Fuel Cell (DMFC) is shown below.

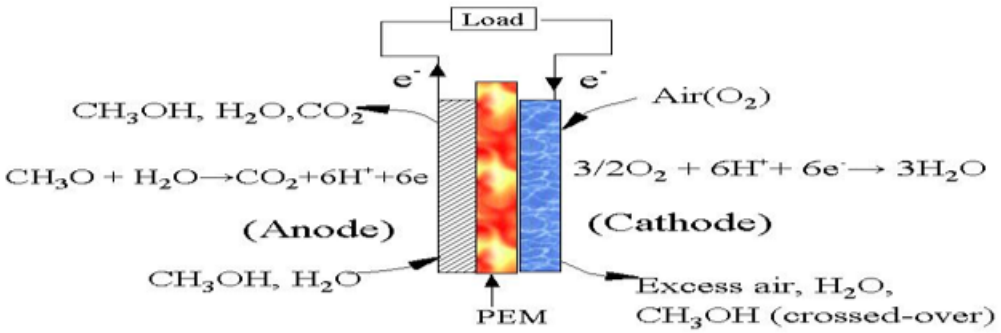



**Figure 1**. Direct methanol fuel cell

DC current is produced in the DMFC when methanol is electrochemically oxidized at the anode electrocatalyst. The electrons produced from the oxidation reaction leave the anode and travel through the external circuit to the cathode electrocatalyst where they are consumed together with oxygen in a reduction reaction. The circuit is maintained within the cell by the conduction of protons in the electrolyte (PEM).

Because of ease of design and its ability to withstand high temperature and pressure operation, most fuel cells today use PEM such as Nafion™.

The overall reaction occurring in the DMFC is equivalent to the direct combustion of methanol:

$$2CH_3OH + 3O_2 \rightarrow 4H_2O + 2CO_2$$

One of the problems affecting the performance and hence commercialization of the direct methanol fuel cell (DMFC) is the depolarization of the oxygen reduction electrode caused by methanol cross-over. Depolarization as used here means that the presence of methanol limits the oxygen reduction and hence the amount of electricity produced. The exact role of methanol in limiting oxygen reduction can be inferred by understanding the physics and chemistry of the reaction. In an attempt to understand the mechanism of the depolarization process, an electro-kinetic model for mixed oxygen-methanol current on platinum was developed in our laboratory. A very simplified model [1] of the reaction developed suggests a functional relation of the form

$$T = -\int_{x_1}^{x_2} \left( \frac{6.73x + 6.725 \times 10^{-8} + 7.26 \times 10^{-4} C_{me}}{3.62 \times 10^{-12} x + 3.908 \times 10^{-8} x C_{me}} \right) dx$$

where

$$T = \text{time},\ s$$

$$x = \text{Concentration of oxygen},\ moles/cm^3,$$

$$C_{me} = \text{Concentration of methanol},\ moles/cm^3,$$

The parameters [1] in the above equation are

$$C_{me} = 5 \times 10^{-4}\ moles/cm^3$$

The initial concentration is

$$x(t = 0) = 1.22 \times 10^{-6}\ moles/cm^3$$

## Questions

(1) Evaluate the time required for 50% of the initial oxygen concentration to be consumed in the fuel cell in the presence of methanol.

(2) Repeat (a) in the absence of methanol (that is, $C_{me} = 0$). What can you infer from the result in (a) and (b)?

(3) Plot time vs. oxygen concentration for the following concentrations of oxygen - $x = [1.22, 1.20, 1.0, 0.8, 0.6, 0.4, 0.2] \times 10^{-6} moles/cm^3$ both in the presence and absence of methanol.

## References

[1] Itoe, R.N., Analysis of simultaneous oxygen reduction and methanol oxidation process in a direct methanol fuel cell, MS Thesis, Department of Chemical and Biomedical Engineering, Florida A&M University (1999).

# Chapter 07.00: Physical Problem for Integration

Lesson: Civil Engineering

## Summary

Find if the concentration of benzene is above or below the toxicity limit at a critical distance from its source.

## Modeling

In a class X528 well in New York State, you are asked to find if the concentration of benzene is below the toxicity level of $0.5mg/l$ at a distance of $36m$ from the point of contamination. The contamination at the source is constant at $3.5mg/l$ for a whole year.

The equations governing transport of groundwater contaminant are quite complex. They depend on several parameters such as "the molecular diffusion of the contaminant, physical and chemical isotropy of the medium, and actual direction of the groundwater flow" [1].

Here we consider that the groundwater is flowing in one-direction, $x$ and the aquifer has isotropic and homogeneous physical and chemical characteristics. In that case, the governing equation [2] for the concentration $c(x,t)$ of the contaminant is given by

$$\frac{\partial c}{\partial t} + u\frac{\partial c}{\partial x} = D\frac{\partial^2 c}{\partial x^2} \quad (1)$$

where

$$u = \text{velocity of ground water flow in the } x \text{ -direction } (m/s)$$

$$t = \text{time } (s)$$

$$x = \text{distance from source } (m)$$

$$D = \text{dispersion coefficient } \left(m^2\right).$$

To solve this partial differential equation (1) [3], we use Laplace transforms. Applying Laplace transform to both sides on the variable $t$,

$$L\left(\frac{\partial c}{\partial t}+u\frac{\partial c}{\partial x}\right)=L\left(D\frac{\partial^2 c}{\partial x^2}\right)$$

$$sC-c\left(x,0\right)+u\frac{dC}{dx}=D\frac{d^2C}{dx^2}\quad(2)$$

where

$$L[c(x,t)]=C\left(x,s\right)$$

Since the initial concentration is zero,

$$c\left(x,0\right)=0,$$

Equation (2) becomes

$$sC+u\frac{dc}{dx}=D\frac{d^2C}{dx^2}$$

$$D\frac{d^2C}{dx^2}-u\frac{dC}{dx}-sC=0\quad(3)$$

This is a homogenous ordinary differential equation. It is linear with fixed coefficients. The characteristics equation of the above ordinary differential equation (3) is

$$Dm^2-um-s=0$$

$$m=\frac{u\pm\sqrt{u^2+4Ds}}{2D}\quad(4)$$

The homogeneous as well as the complete solution of Equation (3) is

$$C\left(x,s\right)=A\left(s\right)e^{\frac{u+\sqrt{u^2+4Ds}}{2D}x}+B\left(s\right)e^{\frac{u-\sqrt{u^2+4Ds}}{2D}x}\quad(5)$$

At $x=\infty$, that is far away from the source, the concentration of the pollutant is zero, that is $c\left(\infty,s\right)=0$, that is, $C\left(\infty,s\right)=0$. This forces $A\left(s\right)=0$as the exponent of the exponential term is always positive as $u,D,s>0$. The exponent of the exponential term on the second term will be negative as $\sqrt{u^2+uDs}>u$for all values of $u,D,s$as $u,D,s>0$. Equation (5) hence reduces to

$$C\left(x,s\right)=B\left(s\right)e^{\frac{u-\sqrt{u^2+4Ds}}{2D}x}\quad(6)$$

At $x=0$, $c=c_0$ (initial concentration, $c_0$), then $C\left(0,s\right)=\frac{c_0}{s}$. Substituting this in equation (6) gives

$$\frac{c_o}{s}=B\left(s\right)$$

then equation (6) is

$$C(x,s) = \frac{c_0}{s} e^{\frac{u-\sqrt{u^2+4Ds}}{2D}x}$$
$$= \left(c_0 e^{\frac{ux}{2D}}\right)\left(\frac{e^{-\frac{\sqrt{u^2+4Ds}}{2D}x}}{s}\right)$$

$$L^{-1}[C(x,s)] = L^{-1}\left[\left(c_0 e^{\frac{ux}{2D}}\right)\left(\frac{e^{-\frac{\sqrt{u^2+4Ds}}{2D}x}}{s}\right)\right]$$

$$c(x,t) = c_0 e^{\frac{ux}{2D}} L^{-1}\left(\frac{e^{-\frac{\sqrt{u^2+4Ds}}{2D}x}}{s}\right)$$
$$= C_0 e^{\frac{ux}{2D}} L^{-1}\left[\frac{e^{-\frac{x}{\sqrt{D}}\sqrt{\frac{u^2}{4D}+s}}}{S}\right] \quad (7)$$

Let

$$a = \frac{x}{\sqrt{D}}$$

$$b = \frac{u}{2\sqrt{D}}$$

Then Equation (7) becomes

$$c(x,t) = c_0 e^{ab} L^{-1}\left(\frac{e^{-a\sqrt{b^2+s}}}{s}\right) \quad (8)$$

Using the formulas

$$L^{-1}(e^{-a\sqrt{s}}) = \left(\frac{a}{2\sqrt{\pi t^3}} e^{-\frac{a^2}{4t}}\right), a > 0,$$

and

$$L^{-1}(F(s+a)) = e^{-at} f(t)$$

then

$$L^{-1}\left(e^{-a\sqrt{b^2+s}}\right) = \frac{a}{2\sqrt{\pi t^3}} e^{-\frac{a^2}{4t}-b^2 t} \quad (9)$$

Then from

$$\int_0^t f(\tau)\, d\tau = \frac{F(s)}{s}$$

we get

$$
\begin{aligned}
L^{-1}\left(\frac{e^{-a\sqrt{b^2+s}}}{s}\right) &= \int_0^t \frac{a}{2\sqrt{\pi\tau^3}} e^{-\frac{a^2}{4\tau}} e^{-b^2\tau} d\tau \\
&= e^{-ab}\int_0^t \frac{a}{2\sqrt{\pi\tau^3}} e^{-\frac{(a-2b\tau)^2}{4\tau}} d\tau \\
&= e^{-ab}\int_0^t \left(\frac{a+2b\tau}{4\sqrt{\pi\tau^3}} + \frac{a-2b\tau}{4\sqrt{\pi\tau^3}}\right) e^{-\frac{(a-2b\tau)^2}{4\tau}} d\tau \\
&= e^{-ab}\int_0^t \frac{a+2b\tau}{4\sqrt{\pi\tau^3}} e^{-\frac{(a-2b\tau)^2}{4\tau}} d\tau + e^{-ab}\int_0^t \frac{a-2b\tau}{4\sqrt{\pi\tau^3}} e^{-\frac{(a-2b\tau)^2}{4\tau}} \\
&= e^{-ab}\int_0^t \frac{a+2b\tau}{4\sqrt{\pi\tau^3}} e^{-\frac{(a-2b\tau)^2}{4\tau}} d\tau + e^{-ab}\int_0^t \frac{a-2b\tau}{4\sqrt{\pi\tau^3}} e^{-\frac{(a-2b\tau)^2}{4\tau} - \frac{8ab\tau}{4\tau} + \frac{8ab\tau}{4\tau}} \\
&= e^{-ab}\int_0^t \frac{a+2b\tau}{4\sqrt{\pi\tau^3}} e^{-\frac{(a-2b\tau)^2}{4\tau}} d\tau + e^{ab}\int_0^t \frac{a-2b\tau}{4\sqrt{\pi\tau^3}} e^{-\frac{(a+2b\tau)^2}{4\tau}} d\tau
\end{aligned}
$$

Let

$$p = \frac{a-2b\tau}{\sqrt{4\tau}},$$

$$q = \frac{a+2b\tau}{\sqrt{4\tau}}$$

then

$$
\begin{aligned}
L^{-1}\left(\frac{e^{-a\sqrt{b^2+s}}}{s}\right) &= \frac{e^{-ab}}{\sqrt{\pi}}\int_\infty^{\frac{a-2bt}{\sqrt{4t}}} e^{-p^2} dp - \frac{e^{ab}}{\sqrt{\pi}}\int_\infty^{\frac{a+2bt}{\sqrt{4t}}} e^{-q^2} dq \\
&= \frac{e^{-ab}}{2} erfc\left(\frac{a-2bt}{2\sqrt{t}}\right) + \frac{e^{ab}}{2} erfc\left(\frac{a+2bt}{2\sqrt{t}}\right) \qquad (10)
\end{aligned}
$$

Using Equation (10), Equation (8) becomes

$$
\begin{aligned}
c(x,t) &= c_0 e^{ab}\left[\frac{e^{-ab}}{2} erfc\left(\frac{a-2bt}{2\sqrt{t}}\right) + \frac{e^{ab}}{2} erfc\left(\frac{a+2bt}{2\sqrt{t}}\right)\right] \\
&= \frac{c_0}{2}\left[erfc\left(\frac{a-2bt}{2\sqrt{t}}\right) + \frac{e^{2ab}}{2} erfc\left(\frac{a+2bt}{2\sqrt{t}}\right)\right] \qquad (11)
\end{aligned}
$$

Substituting back

$$a = -\frac{x}{\sqrt{D}},$$

$$b = \frac{u}{2\sqrt{D}}$$

$$c(x,t) = \frac{c_0}{2}\left[erfc\left(\frac{x-ut}{2\sqrt{Dt}}\right) + e^{\frac{ux}{D}} erfc\left(\frac{x+ut}{2\sqrt{Dt}}\right)\right] \quad (12)$$

The velocity of the ground flow, $u$ is given by

$$u = \frac{k}{n_{ed}}\frac{dh}{dl} \quad (13)$$

where

$$n_{ed} = \text{effective Darcian porosity,}$$

$$K = \text{permeability,}$$

$$\frac{dh}{dl} = \text{ground water gradient.}$$

Assuming

$$n_{ed} = 22\% = 0.22$$

$$K = 0.002\frac{cm}{s}$$

$$\frac{dh}{dl} = 0.01\frac{cm}{cm}.$$

Then from Equation (13),

$$\begin{aligned} u &= \frac{0.002}{0.22}(0.01) \\ &= 9.091 \times 10^{-5}\frac{cm}{s}. \end{aligned}$$

So in the formula for concentration given by Equation (12), substituting

$$D = 0.01\frac{cm^2}{s}$$

$$c_0 = 3.5\frac{mg}{L}$$

$$x = 36m$$

$$t = 1 \text{ year} = 3.15 \times 10^7 s$$

the concentration of benzene after 1 year at a distance of $36m$is

$$c = \frac{3.5}{2} erfc\left(\frac{36 - \left(9.091 \times 10^{-5}\right)\left(3.15 \times 10^{7}\right)}{2\sqrt{(0.01)\left(3.15 \times 10^{7}\right)}}\right) +$$
$$\frac{3.5}{2} e^{\frac{\left(9.091 \times 10^{-5}\right)(36)}{0.01}} erfc\left(\frac{36 + \left(9.091 \times 10^{-5}\right)\left(3.15 \times 10^{7}\right)}{2\sqrt{(0.01)\left(3.15 \times 10^{7}\right)}}\right)$$
$$= 1.75[erfc\,(0.6560) + e^{32.73} erfc\,(5.758)] \quad (14)$$

As you can see to calculate the concentration of benzene at $x = 36m$, we need to calculate $erfc\,(0.6560)$ and $erfc\,(5.758)$. How is $erfc\,(x)$ defined?

$$erfc\,(x) = \int_{\infty}^{x} e^{-z^2} dz$$

So

$$erfc\,(0.6560) = \int_{\infty}^{0.6560} e^{-z^2} dz$$

Since $e^{-z^2}$decays rapidly as $z \to \infty$, we will approximate

$$erfc\,(0.6560) = \int_{5}^{0.6560} e^{-z^2} dz$$

So to find whether the concentration of benzene is below toxicity level, we need to find the integral numerically.

## Questions

(1) What is the concentration of benzene after 2 years?

(2) After 1 year, at what distance does the concentration go below the toxicity level of 0.5 mg/L?

## References

(1) EPA Region 5 Water Draft UIC Class IV and V Site Assessment Guidelines, http://www.epa.gov/R5water/uic/r5guid/siteasst.htm, last accessed July 2004.

(2) Fetter, C.W., Jr., Applied Hydrology, Merill Publishing Co, Columbus, 1980.

(3) Li, W., Differential Equations of Hydraulic Transients, Dispersion, and Groundwater Flow, Prentice Hall, Englewood Cliffs, NJ, 1972

# Chapter 07.00: Physical Problem for Integration

Lesson: Computer Engineering

## Summary

To infer the surfaces from vector fields

## Modeling

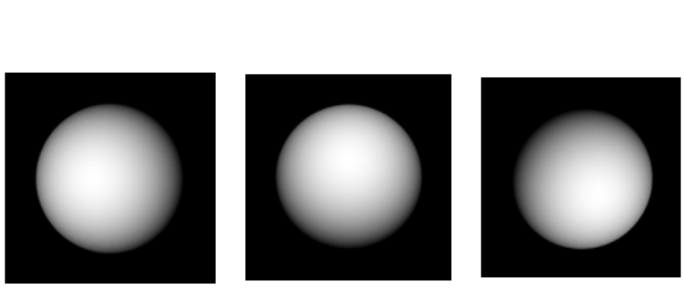
Human vision has the remarkable ability to infer 3D shapes from 2D images. When we look at 2D photographs or TV we do not see them as 2D shapes, rather as 3D entities with surfaces and volumes. Perception research has unraveled many of the cues that are used by us. The intriguing question is: can we replicate some of these abilities on a computer? To this end, in this assignment we are going to look at one way to engineer a solution to the 3D shape from 2D images problem. Apart from the pure fun of solving a new problem, there are many practical applications of this technology such as in automated inspection of machine parts, inference of obstructing surfaces for robot navigation, and even in robot-assisted surgery.

Image is a collection of gray level values at a set of predetermined sites known as pixels, arranged in an array. These gray level values are also known as image intensities. The registered intensity at an image pixel is dependent on a number of factors such as the lighting conditions, surface orientation, and surface material properties. The nature of lighting and its placement can drastically affect the appearance of a scene. In another module on simultaneous linear equations, we saw how to infer the surface normal vectors for each point in the scene, given three images of the scene taken with three different light sources. In Figure 1 we see vector field that we have inferred form the three images. In this module, we will see how we can *integrate this vector field to arrive at a surface.*

To be able to reconstruct the shape of the underlying surface, we have to first understand the how the surface normal is related to the underlying surface. Figure 2 shows the schematic of the camera centered coordinate axis that we can use to formulate the problem. The $z$ -direction is away from the camera towards the scene. Let the scene surface be parameterized by the function.

$$z = g(x, y)$$

The equation of the local surface normal can be related to this function as follows. If we find two tangents on the surface then the cross product of these tangents will give us the surface normal. Two tangents along the $x$ and $y$ -directions can easily be specified in terms of the derivatives along these directions. Figure 2 shows the underlying geometry that can be used to arrive at these equations.



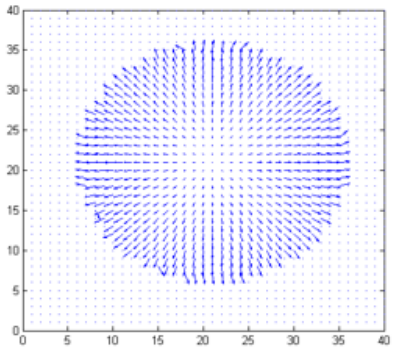
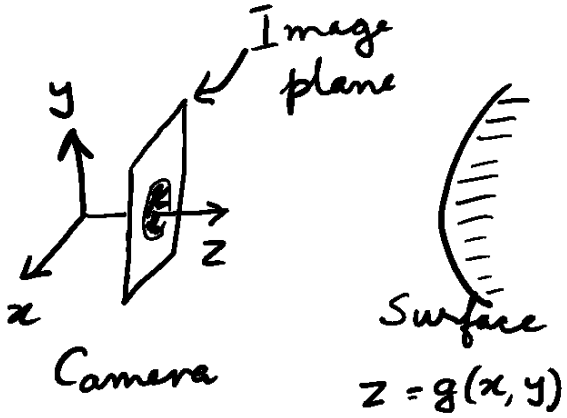



**Figure 1.** The first three images are of a sphere taken with three different light source positions. The right image is a vector field representation of the surface normal vectors estimated from these three images. Can you compute the underlying surface representation from this vector field?

$$\vec{t}_x = \begin{bmatrix} 1 \\ 0 \\ \frac{\partial g(x,y)}{\partial x} \end{bmatrix} \text{ and } \vec{t}_y = \begin{bmatrix} 0 \\ 1 \\ \frac{\partial g(x,y)}{\partial y} \end{bmatrix}$$



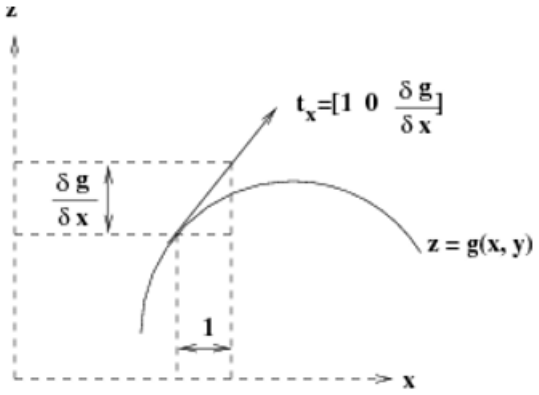




**Figure 2.** Relationship between surface tangent at a point to underlying surface equation. On the

top is a simplified representation of the imaging geometry. The surface is assumed to be far away from the camera. On the bottom is a schematic of the local geometry on the scene surface.

The local surface normal, $n$, will be along the direction of the cross product of $t_x$ and $t_y$.

$$c\overrightarrow{n} = -\overrightarrow{t}_x \times \overrightarrow{t}_y = \begin{bmatrix} \frac{\partial g(x,y)}{\partial x} \\ \frac{\partial g(x,y)}{\partial x} \\ -1 \end{bmatrix}$$

Note that there is 1800 ambiguity in the specification of the surface normal. We need the negative sign because we want the surface normal to be oriented towards the camera and the $z$ -axis is pointed away from the camera. Also, note that the cross product gives as a vector along the surface normal. We have to normalize the vector to arrive at the surface normal. For notational ease let

$$p = \frac{\partial g(x,y)}{\partial x} \text{ and}$$
$$q = \frac{\partial g(x,y)}{\partial y}$$

Then,

$$\overrightarrow{n} = \begin{bmatrix} n_x \\ n_y \\ n_z \end{bmatrix} = \frac{1}{\sqrt{p^2+q^2+1}} \begin{bmatrix} p \\ q \\ -1 \end{bmatrix}$$

We see that the surface normal can be expressed in terms of the derivatives of the underlying surface. From this equation, we can express the local surface derivatives in terms of ratios of the surface normal components.

$$\frac{\partial g(x,y)}{\partial x} = \frac{n_x}{n_y} \text{ and}$$
$$\frac{\partial g(x,y)}{\partial y} = \frac{n_y}{n_z}$$

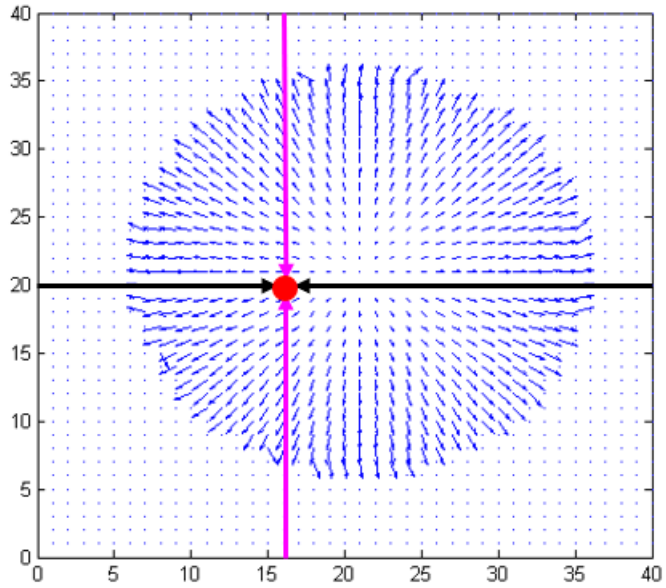



**Figure 3.** Integration paths to arrive at the depth surface value for the point marked by the red dot.

The black lines denote possible paths over which the x-partial derivative can be integrated. And, the purple lines denote possible paths to integrate the y-partial derivative. The paths are shown overlaid on the input vector field representing the partial derivatives of the surface along x and y directions.

We arrive at the final surface equation by integration these partial derivatives. Since we have a 2D field, we can perform the integration along many paths. The simplest paths are along the $x$ (or $y$) axis.

$$g(u,v) = \int_o^u \frac{\partial g(x,v)}{\partial x} dx \text{ or}$$

$$g(u,v) = \int_o^v \frac{\partial g(u,y)}{\partial y} dy$$

One could also start from the right edge of the image and integrate the partial derivative with respect to $x$ towards the left edge, or start from the bottom of the image and integrate the $y$ -partial derivative towards the top. These directions are depicted in Figure 3. The equivalent integrals are as follows. The image is assumed to N pixels by M pixels in size and the negative sign arises because of the direction of the integration.

$$g(u,v) = -\int_N^u \frac{\partial g(x,v)}{\partial x} dx \text{ or}$$

$$g(u,v) = -\int_M^v \frac{\partial g(u,y)}{\partial y} dy$$

Ideally, all the four integrals should result in the same value, however, for noisy, real world data they will never be the same. One solution is to take the average of these four integrals as the final value.

Note that these integrals have to be evaluated for each location $(u, v)$ in the image to arrive at the full surface.

**Worked Out Example**

Consider the problem of estimating the surface height along the line passing through the center of the sphere in Figure 1. Figure 4 (a) shows the input estimates for the partial derivatives along the x direction, $p = \frac{\partial g(x,y)}{\partial x}$ for the sphere along this line. The raw data for this plot is given below.

| x | g |
|---|---|
| 1.00000 | 0.00000 |
| 2.00000 | 0.00000 |
| 3.00000 | 0.00000 |
| 4.00000 | 0.00000 |
| 5.00000 | 0.00000 |
| 6.00000 | 0.00000 |
| 7.00000 | 0.00000 |
| 8.00000 | 0.00000 |
| 9.00000 | 0.00000 |
| 10.00000 | 0.00000 |
| 11.00000 | 0.00000 |
| 12.00000 | 0.00000 |
| 13.00000 | 0.00000 |
| 14.00000 | 0.00000 |
| 15.00000 | 0.00000 |
| 16.00000 | 0.00000 |
| 17.00000 | 0.00000 |
| 18.00000 | 0.00000 |
| 19.00000 | 0.00000 |
| 20.00000 | 0.00000 |
| 21.00000 | 0.00000 |
| 22.00000 | 0.00000 |
| 23.00000 | 0.00000 |
| 24.00000 | 0.00000 |
| 25.00000 | 0.00000 |
| 26.00000 | 0.00000 |
| 27.00000 | 0.00000 |
| 28.00000 | 0.00000 |
| 29.00000 | 0.00000 |
| 30.00000 | 5.42857 |
| 31.00000 | 4.40000 |
| 32.00000 | 3.82362 |
| 33.00000 | 3.23097 |
| 34.00000 | 2.74844 |
| 35.00000 | 2.43579 |
| 36.00000 | 2.19293 |
| 37.00000 | 2.04677 |
| 38.00000 | 1.91851 |
| 39.00000 | 1.88649 |
| 40.00000 | 1.62019 |

| x | g |
|---|---|
| 41.00000 | 1.55007 |
| 42.00000 | 1.47428 |
| 43.00000 | 1.44755 |
| 44.00000 | 1.36134 |
| 45.00000 | 1.28187 |
| 46.00000 | 1.14868 |
| 47.00000 | 1.15089 |
| 48.00000 | 1.14027 |
| 49.00000 | 1.05379 |
| 50.00000 | 0.98459 |
| 51.00000 | 1.00329 |
| 52.00000 | 0.96249 |
| 53.00000 | 0.90161 |
| 54.00000 | 0.88597 |
| 55.00000 | 0.83243 |
| 56.00000 | 0.78901 |
| 57.00000 | 0.80149 |
| 58.00000 | 0.74730 |
| 59.00000 | 0.73934 |
| 60.00000 | 0.70364 |
| 61.00000 | 0.67288 |
| 62.00000 | 0.66161 |
| 63.00000 | 0.62773 |
| 64.00000 | 0.61402 |
| 65.00000 | 0.60783 |
| 66.00000 | 0.58271 |
| 67.00000 | 0.55106 |
| 68.00000 | 0.52999 |
| 69.00000 | 0.50641 |
| 70.00000 | 0.50146 |
| 71.00000 | 0.46053 |
| 72.00000 | 0.45829 |
| 73.00000 | 0.43148 |
| 74.00000 | 0.41627 |
| 75.00000 | 0.39025 |
| 76.00000 | 0.38186 |
| 77.00000 | 0.35635 |
| 78.00000 | 0.34187 |
| 79.00000 | 0.32304 |
| 80.00000 | 0.31064 |
| 81.00000 | 0.27967 |
| 82.00000 | 0.27367 |
| 83.00000 | 0.26137 |
| 84.00000 | 0.24340 |
| 85.00000 | 0.23128 |
| 86.00000 | 0.20758 |
| 87.00000 | 0.19566 |
| 88.00000 | 0.19566 |
| 89.00000 | 0.17807 |
| 90.00000 | 0.16632 |

| x | g |
|---|---|
| 91.00000 | 0.14409 |
| 92.00000 | 0.12686 |
| 93.00000 | 0.11039 |
| 94.00000 | 0.10481 |
| 95.00000 | 0.08266 |
| 96.00000 | 0.06623 |
| 97.00000 | 0.04945 |
| 98.00000 | 0.04383 |
| 99.00000 | 0.03309 |
| 100.00000 | 0.00038 |
| 101.00000 | −0.00516 |
| 102.00000 | −0.02149 |
| 103.00000 | −0.03232 |
| 104.00000 | −0.04317 |
| 105.00000 | −0.05421 |
| 106.00000 | −0.06513 |
| 107.00000 | −0.09250 |
| 108.00000 | −0.09820 |
| 109.00000 | −0.11468 |
| 110.00000 | −0.12011 |
| 111.00000 | −0.13697 |
| 112.00000 | −0.14808 |
| 113.00000 | −0.18176 |
| 114.00000 | −0.18776 |
| 115.00000 | −0.19904 |
| 116.00000 | −0.20511 |
| 117.00000 | −0.22784 |
| 118.00000 | −0.25084 |
| 119.00000 | −0.26862 |
| 120.00000 | −0.26960 |
| 121.00000 | −0.30463 |
| 122.00000 | −0.30577 |
| 123.00000 | −0.33591 |
| 124.00000 | −0.33720 |
| 125.00000 | −0.35061 |
| 126.00000 | −0.37626 |
| 127.00000 | −0.38823 |
| 128.00000 | −0.40213 |
| 129.00000 | −0.42330 |
| 130.00000 | −0.44994 |
| 131.00000 | −0.47173 |
| 132.00000 | −0.47897 |
| 133.00000 | −0.50129 |
| 134.00000 | −0.54231 |
| 135.00000 | −0.54480 |
| 136.00000 | −0.57621 |
| 137.00000 | −0.56790 |
| 138.00000 | −0.61925 |
| 139.00000 | −0.63884 |
| 140.00000 | −0.65569 |

| x | g |
|---|---|
| 141.00000 | −0.68145 |
| 142.00000 | −0.73088 |
| 143.00000 | −0.75263 |
| 144.00000 | −0.76928 |
| 145.00000 | −0.79030 |
| 146.00000 | −0.80970 |
| 147.00000 | −0.81065 |
| 148.00000 | −0.88367 |
| 149.00000 | −0.89782 |
| 150.00000 | −0.92400 |
| 151.00000 | −1.00067 |
| 152.00000 | −1.03536 |
| 153.00000 | −1.08947 |
| 154.00000 | −1.09584 |
| 155.00000 | −1.20465 |
| 156.00000 | −1.22123 |
| 157.00000 | −1.26090 |
| 158.00000 | −1.30056 |
| 159.00000 | −1.33557 |
| 160.00000 | −1.49718 |
| 161.00000 | −1.53483 |
| 162.00000 | −1.70123 |
| 163.00000 | −1.69167 |
| 164.00000 | −1.89507 |
| 165.00000 | −2.06525 |
| 166.00000 | −2.24788 |
| 167.00000 | −2.45874 |
| 168.00000 | −2.80478 |
| 169.00000 | −3.22446 |
| 170.00000 | −3.86839 |
| 171.00000 | −4.00000 |
| 172.00000 | −4.00000 |
| 173.00000 | 0.00000 |
| 174.00000 | 0.00000 |
| 175.00000 | 0.00000 |
| 176.00000 | 0.00000 |
| 177.00000 | 0.00000 |
| 178.00000 | 0.00000 |
| 179.00000 | 0.00000 |
| 180.00000 | 0.00000 |
| 181.00000 | 0.00000 |
| 182.00000 | 0.00000 |
| 183.00000 | 0.00000 |
| 184.00000 | 0.00000 |
| 185.00000 | 0.00000 |
| 186.00000 | 0.00000 |
| 187.00000 | 0.00000 |
| 188.00000 | 0.00000 |
| 189.00000 | 0.00000 |
| 190.00000 | 0.00000 |

| x | g |
|---|---|
| 191.00000 | 0.00000 |
| 192.00000 | 0.00000 |
| 193.00000 | 0.00000 |
| 194.00000 | 0.00000 |
| 195.00000 | 0.00000 |
| 196.00000 | 0.00000 |
| 197.00000 | 0.00000 |
| 198.00000 | 0.00000 |
| 199.00000 | 0.00000 |
| 200.00000 | 0.00000 |

This data needs to be numerically integrated to arrive at height values. Figure 4 (b) shows the plot of the surface height, as computed by integrating starting from left and from right. We used the trapezoidal integration method. Notice the small discrepancy, which is due to real world data noise. The overall shape does look circular, which should serve as a sanity check on the calculations.

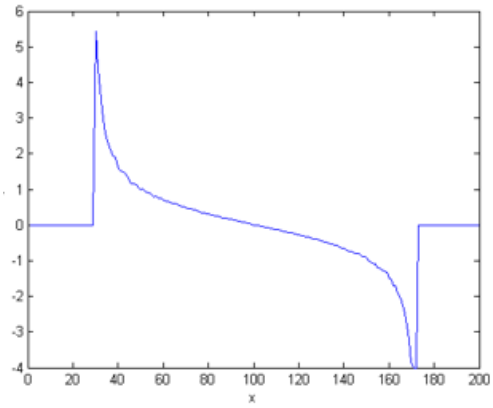



**Figure 4(a)**

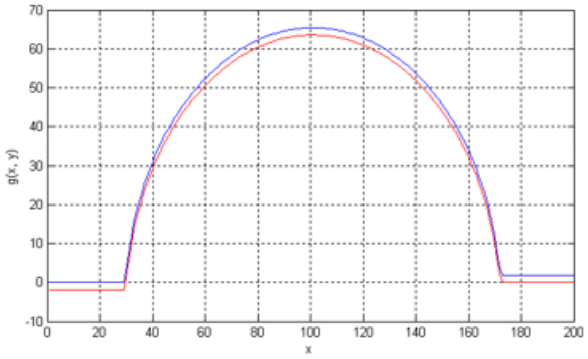



**Figure 4(b)**

**Figure 4.** (a) The plot of the estimates for the partial derivatives along the x direction, $p = \frac{\partial g(x,y)}{\partial x}$ for the sphere along a horizontal line passing through the middle of the sphere. (b) Plot of the integrated value at each point along the horizontal. The blue plot corresponds to integrating from left to right and the red plot corresponds to integrating from right to left.

If we repeat the above process along the vertical direction (along image columns), then we will arrive at two more estimates for the middle point of the sphere, which can be averaged to arrive at one estimate. If we repeat this for each point on the sphere, not just the middle, then we will arrive at the surface representation for the full sphere. Figure 5 shows the averaged estimate of the sphere as estimated from the input vector field. Note the spherical nature of the final estimate.

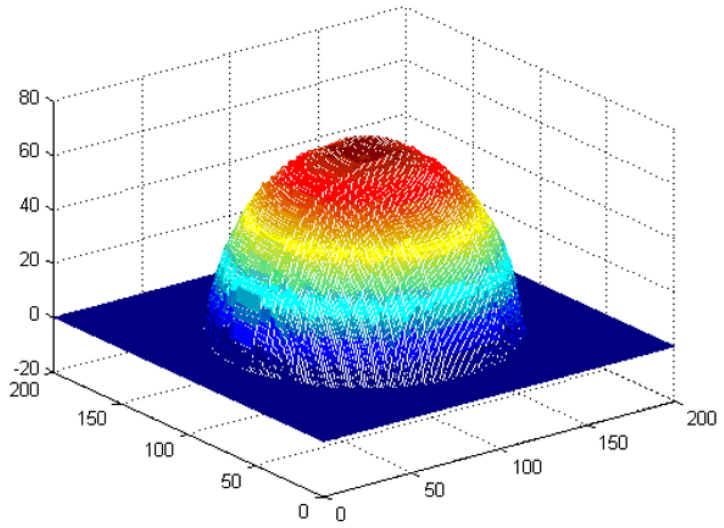



**Figure 5.** Estimated surface height for the vector field in Figure 1(c) as computed by averaging the estimates of the integrals along four directions, for each point.

## Questions

(1) Write code to recover the surface height at each point in the image given the vector field.

(2) For any point on the sphere, specify 25 different paths along which you can integrate.

(3) Study how the roughness of the estimated surface changes with number of path integrals that are averaged.

# Chapter 07.00: Physical Problem for Integration

Lesson: Electrical Engineering

## Summary

Not all electrical components, especially off-the-shelf components match their nominal value. Variations in materials and manufacturing as well as operating conditions can affect their value. Suppose a circuit is designed such that it requires a specific component value, how confident can we be that the variation in the component value will result in acceptable circuit behavior? To solve this problem a probability density function will need to be integrated to determine the confidence interval.

## Modeling

Today's consumer products rely heavily on small electronic circuits to provide much of their functionality. Each of these circuits is built around electrical components that, like all devices, vary in value due to variations in materials, manufacturing, and operating conditions. For example, capacitors are notoriously variable often operating as much as $+/-20\%$ from their nominal value. Clearly, a variation in capacitor value can have an affect on the operation of an electronic circuit. If the statistical properties of the capacitor and resistor value distributions are known, stochastic methods can be used to compute this effect. Unfortunately, this is often not the case and we are frequently left to use empirical methods to determine this effect. A common technique is to use a Monte Carlo method where a circuit will be built (or simulated) many times with different components from a lot and the resulting behavior analyzed. The goal will be to determine a confidence value that a randomly selected circuit built with these components will behave within a certain interval.

Consider the case of an op-amp based square wave generator as shown in Figure 1. Its operative behavior in terms of period is described by the equation [1, pp. 625-627]

$$T = 2RC \ln\left(1 + 2\frac{R_2}{R_1}\right)$$

Clearly, the period/frequency of this circuit is quite susceptible to variations in all the components 1 except the op-amp, resistor, $R_3$ and the zener diodes. How then can the likely behavior to all the possible combination of variations in the components $R, C, R_1$ and $R_2$ be determined? The answer is to apply a Monte Carlo method and some basic statistics.

If we assume that the statistical distribution of the circuit behavior is normal[1] then a two-tailed confidence interval test can be used to determine the likelihood that the circuit will behave acceptably [2, pp. 113-27]. An examination of the literature will show that a confidence interval can be established through the use of the z-test. For a sample of size n with known standard deviation $\sigma$ it can be determined with confidence $(1-\alpha)$ that the measured value will stay in the following range

$$\bar{x} - z\,(\alpha/2)\,\sigma < value < \bar{x} + z\,(\alpha/2)\,\sigma,$$

where

$\overline{x}$ is the sample mean of the measured value.

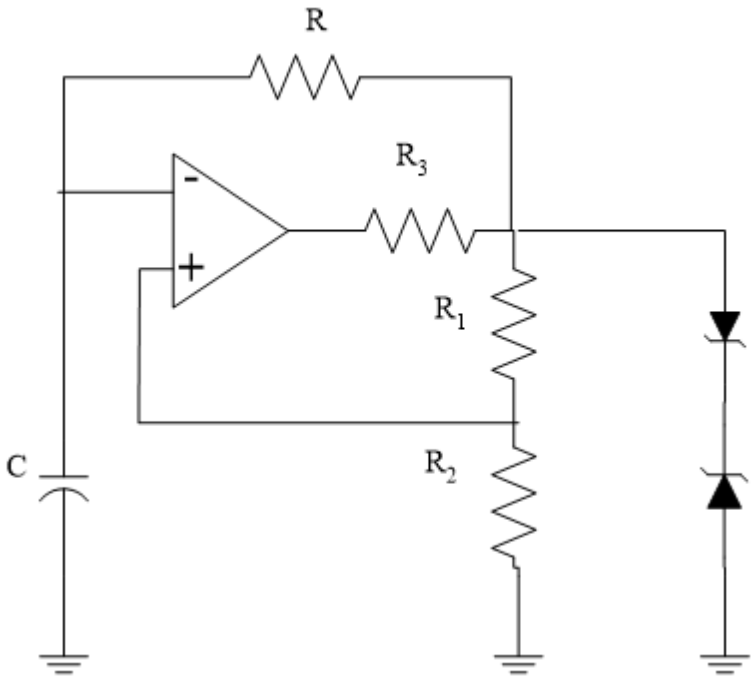



**Figure 1** Square-Wave Oscillator

If $\alpha$ is 0.05 then there is 90% confidence that the value of all possible circuits will be within the indicated range about the sample mean. When the sample size $n$ is smaller than 30 or the population standard deviation,$\sigma$ is not known the sample standard deviation,$s$ can be substituted and the normal distribution must be replaced with the Student-T distribution [2, pp. 174-7].

Interpretation of the term $z\,(\alpha/2)$ and $t\,(\alpha/2, n-2)$ often leads to a lot of confusion. It is the $z-$ or $t-$ score value that is necessary so that $(1-\alpha)$ of the area under the probability distribution is inside the desired range. This is often referred to, in part, as a two-tailed test. Figure 2 shows this graphically.

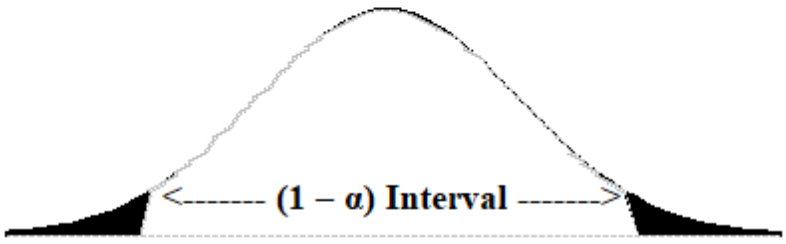


**Figure 2** $(1-\alpha)$ Confidence Interval

$z(\alpha/2)$ is the value so that when the normal distribution is integrated from $-z(\alpha/2)$ to $+z(\alpha/2)$ that the resulting area is$(1-\alpha)$. This integration is summarized by the following equation.

$$(1-\alpha) = \int_{-z(\alpha/2)}^{z(\alpha/2)} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx$$

The analysis is similar for the Student-T distribution and $t(\alpha/2, n-2)$.

The problem is that there is no closed-form solution to the integration of either the normal distribution or the Student-T distribution. These are typically handled via look-up tables of the cumulative distributions from $-\infty$ to $z$ [2, pp. 513-14].

In the case of a confidence interval computation generally everything is known except for $z(\alpha/2)$. However, in the case of a circuit, that analysis can be reversed to determine with what confidence the average circuit behavior will be in a given interval. In other words, suppose that the goal is to determine how likely it is that the square-wave generator for Figure 1 will behave with a frequency of 11 kHz. For example, suppose that suitable components have been chosen for $R, C, R_1$ and $R_2$ for a lot and that 25circuits were built with the following results being measured:

Number of Samples: $n = 25$

Average Period: $\bar{x} = 0.995ms$

Standard Deviation: $\sigma = 0.02ms$

What is the likelihood, with this batch of components, that any given oscillator will have its frequency within 5% of the target of $1kHz$? Rearranging the confidence interval formula and solving for $z(\alpha/2)$ yields the following two equations. One for the upper end for $\mu < 1.053ms$ and the other for the upper end for$\mu > 0.952ms$[3].

$$z(\alpha_1/2) = \frac{(\mu_1 - \bar{x})}{\sigma} = \frac{(1.053m - 0.995m)}{0.02m} = 2.9$$

$$z(\alpha_2/2) = \frac{(\mu_2 - \bar{x})}{\sigma} = \frac{(0.952m - 0.995m)}{0.02m} = -2.15$$

The likelihood of this happening can then be determined by finding the total area under the normal distribution for the range in question

$$(1-\alpha) = \int_{-2.15}^{2.9} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx$$

## Questions

(1) There is no real way to know the population standard deviation, $\sigma$, so the more complex formula using the sample standard deviation, $s$ , and the Student-T distribution would have to be used, especially when $n$ is less than 30 30 [2, pp. 174-19]. The entire analysis is the same except the integral is now the more complicated.

$$(1-\alpha) = \int_{-t(\alpha/2,m)}^{t(\alpha/2,m)} \left\{ \frac{\Gamma\left(\frac{m+1}{2}\right)}{\sqrt{m\pi}\Gamma\left(\frac{m}{2}\right)} \left(1+\frac{x^2}{m}\right)^{-\left(\frac{m+1}{2}\right)} \right\} dx$$

where

$$\Gamma(1) = 1$$

$$\Gamma(1/2) = \sqrt{\pi}$$

$$\Gamma(m) = (m-1)\,\Gamma(m-1)$$

Perform the computations using the Student-T distribution.

(2) The more common way of using these statistics is to compute the actual confidence interval centered about the sample data. This means that $\alpha$ is known and $z(\alpha/2)$ or $t(\alpha/2, n-2)$ have to be computed. This can be done by repeatedly trying possible values for $z$ or $t$ and evaluating the integral. This is an excellent application for a root-finding algorithm. If an algorithm like bisection method of solving nonlinear equations were used, how would you pick the initial bounds for the $x$ or $t$ value?

(3) Because there are actual formulas for the functional inside the integral and not just sampled points on the curve in question, it is possible to apply an interval halving approach and a Romberg extrapolation to solve these integrals. Set up the algorithm for doing this.

(4) Normal and Student-T distribution tables like those in [2, pp. 513-4] are limited in their choices. Integrating these distributions from $-\infty$ can be used to generate a more complete table. To do this an algorithm must take advantage of the symmetry in the distributions since it is not possible to integrate numerically with an infinite bound. When the $z$ or $t$ value is less than zero simply report 0.5 minus the integral from 0 to $z$or $t$, and when the $z$or $t$ value is greater than zero simply report 0.5 plus the integral from 0 to $z$ or $t$. Set-up the algorithm for doing this.

(5) In the analysis above it was assumed that the population of circuits was roughly normal, this should generally be tested before applying the assumption. There exist goodness-of-fit tests to see if the population is sufficiently normal. For more information, see [2, pp. 265-8].

## References

[1] Millman, J., Micro-Electronics: Digital and Analog Circuits and Systems, McGraw Hill, 1979.

[2] Walpole, R. and Myers, R., Probability and Statistics for Engineers and Scientists, 2nd Ed., MacMillan, 1978.

# Chapter 07.00: Physical Problem for Integration

Lesson: Industrial Engineering


## Summary

The number of sheets in a roll of toilet paper are sampled at a factory. Even if the number of the sheets in every chosen sample turns out to be what is advertised or more, the probability of every roll having the advertised number or more sheets is not 100%. To find out this probability requires numerical integration.


## Modeling

An industrial engineer works as a quality control engineer for a company making toilet paper. The company advertises that every roll of toilet paper has at least 250 sheets. It is the industrial (quality) engineer's responsibility to validate this claim by sampling rolls off of the assembly line and determining the probability (that is, confidence) that the company can make a claim. Note that validating the claim is important to avoid frivolous lawsuits associated with false advertising.

Let us assume that the number of sheets in a roll of toilet paper, $y$, is governed by the normal probability distribution (see figure on next page), that is,

$$y \sim N(\mu, \sigma^2) \qquad (1)$$

where N is a normal random variable described by its mean, $\mu$ , and standard deviation, $\sigma$. The probability distribution function of the normal random variable $y$ is

$$f(y) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(1/2)[y-\mu)/\sigma]^2} \qquad (2)$$

and is shown in Figure 1.

Note that the previously described normal probability density function was first derived by Abraham de Moivre in 1733 in his book called *The Doctrine of Chance*. Initially, the discovery went unnoticed until it was re-derived by Laplace and Gauss about 50 years later. Hence, it is often called the Gaussian distribution. The normal probability distribution, which is a continuous distribution, is derived from the binomial probability distribution, which is a discrete distribution. The discrete binomial distribution is given below

$$f(y) = \frac{n!}{k!\,(n-k)!} p^k\,(1-p)^{n-k} \qquad (3)$$

where $n$ is the number of samples and $k$ is the number of successes with a probability of $p$. When $n$ is large, the binomial distribution approaches the normal distribution. In fact, when $n$ is greater than or equal to 6, the normal distribution is an excellent approximation to the binomial distribution (Figure 2).

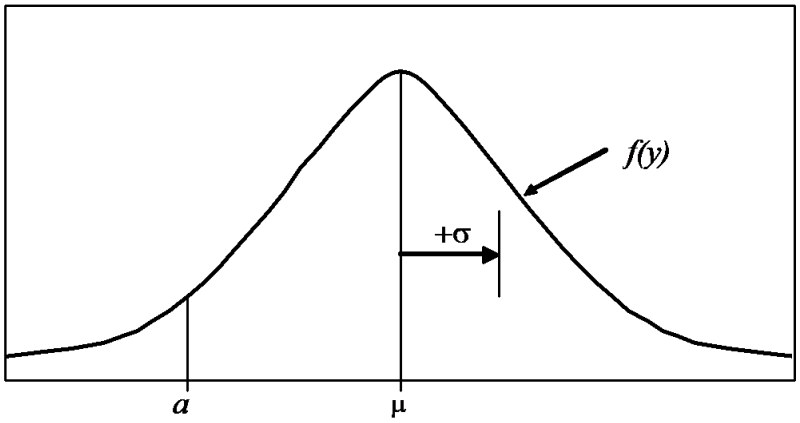



**Figure 1** Probability distribution function, $f(y) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(1/2)[y-\mu)/\sigma]^2}$

The actual proof that the binomial distribution approaches the normal distribution involves a special case of the central limit theorem and is beyond the scope of this problem. Furthermore, the derivation also involves the inclusion of a continuity correction factor.

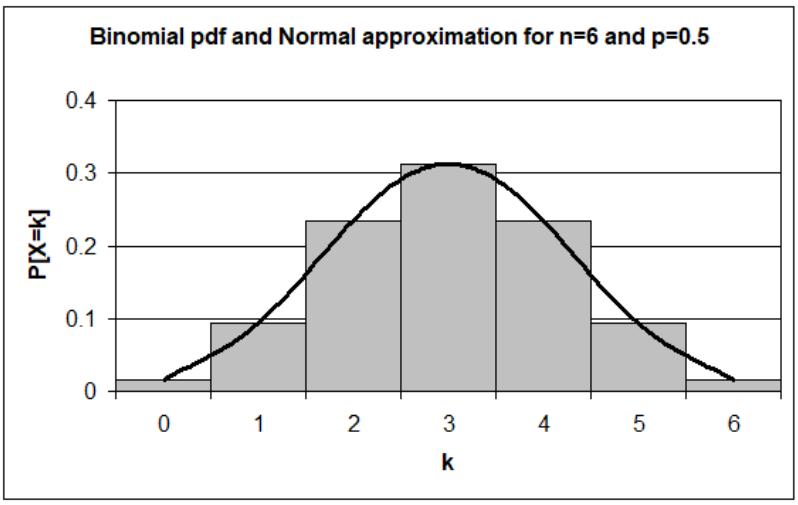



**Figure 2** Binomial pdf and normal approximation for $n = 6$ and $p = 0.5$

Since the quality engineer cannot sample every roll of toilet paper and count the number of sheets, the mean and standard deviation are estimated based on a sample size of $n$ rolls of toilet paper. The

mean, $\bar{y}$, and standard deviation, $s$, of the sample (that is, the sample mean and sample standard deviation) are estimates of the actual mean, $\mu$, and standard deviation, $\sigma$, and are given by

$$\bar{y} = \frac{\sum_{i=1}^{n} y_i}{n} \approx \mu \qquad (4)$$

$$s = \sqrt{\frac{\sum_{i=1}^{n} (y_i - \bar{y})^2}{n-1}} \approx \sigma \qquad (5)$$

where $n$ is the number of samples (that is, rolls of toilet paper). The probability that the number of sheets in a roll is greater than or equal to some specified number, $a$, is the area under the probability density function (that is, integral) from $a$ to $\infty$, and is given by

$$\begin{aligned} P(y \geq a) &= \int_a^{\infty} f(y)dy \\ &= \int_a^{\infty} \frac{1}{\sigma\sqrt{2\pi}} e^{-(1/2)[(y-\mu)/\sigma]^2} dy \end{aligned} \qquad (6)$$

Note that the result of Equation (6) is a number from 0 to 1 that correlates to a 0% to 100% probability, respectively.

A co-operative education student working for an industrial (quality) engineer samples 10 rolls of toilet paper every week and determines that the number of sheets in each roll is

| Roll | Number of sheets |
|---|---|
| 1 | 253 |
| 2 | 250 |
| 3 | 251 |
| 4 | 252 |
| 5 | 253 |
| 6 | 253 |
| 7 | 252 |
| 8 | 254 |
| 9 | 252 |
| 10 | 252 |

The industrial (quality) engineer calculates the following mean and standard deviation of the sample of 10 rolls of toilet paper.

$$\bar{y} = \frac{253 + 250 + 251 + 252 + 253 + 253 + 252 + 254 + 252 + 252}{10} = 252.2 \approx \mu$$

$$s = \sqrt{\frac{\begin{array}{c}(253-252.2)^2+(250-252.2)^2+(251-252.2)^2+\\(252-252.2)^2+(253-252.2)^2+(253-252.2)^2+\\(252-252.2)^2+(254-252.2)^2+(252-252.2)^2+(252-252.2)^2\end{array}}{10-1}} = 1.135 \approx \sigma$$

Using Equation (6), the probability that the number of sheets in a roll is greater than or equal to 250 sheets is

$$P(y \geq 250) = \int_{250}^{\infty} \frac{1}{1.135\sqrt{2\pi}} e^{-(1/2)[(y-252.2)/1.135]^2} dy$$

$$P(y \geq 250) = \int_{250}^{\infty} 0.3515 e^{-0.3881(y-252.2)^2} dy$$

$$P(y \geq 250) = ??$$

## Questions

(1) Although all the rolls have more than 250 sheets, why do we not get a 100% probability that all the rolls have 250 sheets or more?

(2) Change one of the data points to 248 sheets and rework the problem.

(3) To find the integral numerically in Equation (6), one may substitute infinity by 5. How will you justify this substitution quantitatively?

# Chapter 07.00: Physical Problem for Integration

Lesson: Mechanical Engineering

## Summary

To find the diametrical contraction during shrink-fitting in a trunnion requires one to first find how the coefficient of linear thermal expansion of steel is related to temperature. The relation may be given through a second-order polynomial. This model is then used in integration to find the diametrical contraction.

## Modeling

To make the fulcrum (Figure 1) of a bascule bridge, a long hollow steel shaft called the trunnion is shrink fit into a steel hub. The resulting steel trunnion-hub assembly is then shrink fit into the girder of the bridge.

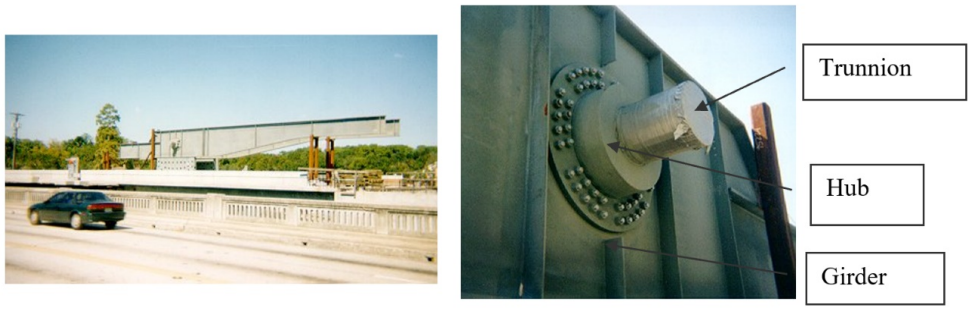



**Figure 1** Trunnion-Hub-Girder (THG) assembly.

This is done by first immersing the trunnion in a cold medium such as a dry-ice/alcohol mixture. After the trunnion reaches the steady-state temperature, that is, the temperature of the cold medium, the trunnion outer diameter contracts. The trunnion is taken out of the medium and slid though the hole of the hub (Figure 2).

When the trunnion heats up, it expands and creates an interference fit with the hub. In 1995, on one of the bridges in Florida, this assembly procedure did not work as designed. Before the trunnion could be inserted fully into the hub, the trunnion got stuck. Luckily the trunnion was taken out before it got stuck permanently. Otherwise, a new trunnion and hub would be needed to be ordered at a cost of $50,000. Coupled with construction delays, the total loss could have been more than a hundred thousand dollars.

Why did the trunnion get stuck? This failure took place because the trunnion had not contracted enough to slide through the hole. Can you find out why?

A hollow trunnion of outside diameter $12.363''$ is to be fitted in a hub of inner diameter $12.358''$. The trunnion was put in a dry ice/alcohol mixture (temperature of the fluid - dry ice/alcohol mixture is $-108°$F) to contract the trunnion so that it can be slid through the hole of the hub. To slide the trunnion without sticking, a diametrical clearance of at least $0.01''$ is required between the trunnion and the hub. Assuming the room temperature is 80°F, is immersing it in a dry-ice/alcohol mixture a correct decision?

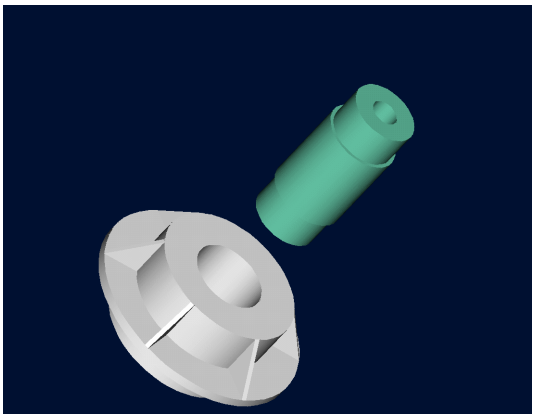


**Figure 2** Trunnion slided through the hub after contracting

To calculate the contraction in the diameter of the trunnion, the coefficient of linear thermal expansion at room temperature is used. In that case, the reduction, $\Delta D$ in the outer diameter of the trunnion is

$$\Delta D = D\alpha\Delta T \qquad (1)$$

where

$$D = \text{outer diameter of the trunnion,}$$

$$\alpha = \text{coefficient of linear thermal expansion at room temperature, and}$$

$$\Delta T = \text{change in temperature,}$$

Given

$$D = 12.363''$$

$$\alpha = 6.817 \times 10^{-6} \text{ in/in/°F at } 80°\text{F}$$

$$\begin{aligned} \Delta T &= T_{fluid} - T_{room} \\ &= -188°\text{F} \end{aligned}$$

where

$$T_{fluid} = \text{temperature of dry-ice/alcohol mixture,}$$

$$T_{room} = \text{room temperature,}$$

the reduction in the trunnion outer diameter is given by

$$\begin{aligned}\Delta D &= (12.363)\left(6.47 \times 10^{-6}\right)(-188) \\ &= -0.01504''\end{aligned}$$

So the trunnion is predicted to reduce in diameter by $0.01504''$. But is this enough reduction in diameter? As per specifications, he needs the trunnion to contract by

$$\begin{aligned}&= \text{trunnion outside diameter} - \text{hub inner diameter} + \text{diametral clearance} \\ &= 12.363'' - 12.358'' + 0.01'' \\ &= 0.015''\end{aligned}$$

So, according to his calculations, immersing the steel trunnion in a dry-ice/alcohol mixture gives the desired contraction of $0.015''$ as we predict a contraction of $0.01504''$.

But as shown in Figure 3, the coefficient of linear thermal expansion of steel decreases with temperature and is not constant over the range of temperature the trunnion goes through. Hence the above formula (Equation (1)) would overestimate the thermal contraction.

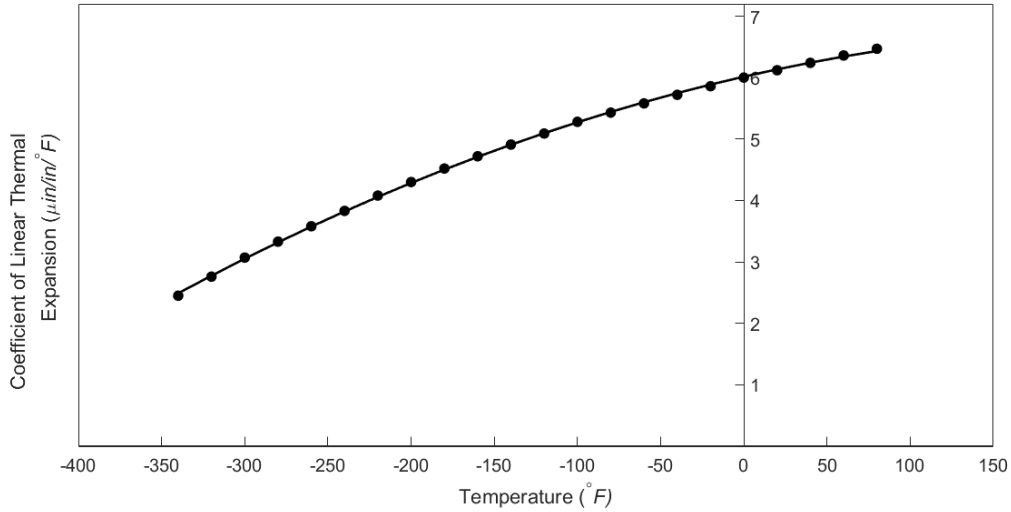



**Figure 3** Varying coefficient of linear thermal expansion as a function of temperature for cast steel.

The contraction in the diameter for the trunnion for which the thermal expansion coefficient varies as a function of temperature is given by

$$\Delta D = D \int_{T_{room}}^{T_{fluid}} \alpha dT \qquad (2)$$

Note that Equation (2) reduces to Equation (1) if the coefficient of thermal expansion is assumed to be constant. In Figure 3, the coefficient of linear thermal expansion of typical cast steel is approximated by a second-order polynomial as

$$\alpha = -1.2278 \times 10^{-11} T^2 + 6.1946 \times 10^{-9} T + 6.0150 \times 10^{-6} \qquad (3)$$

From Equations (2) and (3)

$$\Delta D = 12.363 \int_{80}^{-108} \left(-1.2278 \times 10^{-11} T^2 + 6.1946 \times 10^{-9} T + 6.015 \times 10^{-6}\right) dT \qquad (4)$$

## Questions

(1) Can you now find the contraction of the trunnion outer diameter?

(2) Is the magnitude of contraction more than $0.015''$ as required?

(3) If that is not the case, what if the trunnion were immersed in liquid nitrogen (boiling temperature=$-321°$F)? Will that give enough contraction in the trunnion?

(4) Rather than regressing the data to a second-order polynomial so that one can find the contraction in the trunnion outer diameter, how would you use the trapezoidal rule of integration for unequal segments? What is the relative difference between the two results? The data for the coefficient of linear thermal expansion as a function of temperature is given below.

**Table 1** coefficient of linear thermal expansion as a function of temperature.

| Temperature | Coefficient of Thermal Expansion |
|---|---|
| °F | $\mu$in/in/°F |
| 80 | 6.47 |
| 60 | 6.36 |
| 40 | 6.24 |
| 20 | 6.12 |
| 0 | 6.00 |
| −20 | 5.86 |
| −40 | 5.72 |
| −60 | 5.58 |
| −80 | 5.43 |
| −100 | 5.28 |
| −120 | 5.09 |
| −140 | 4.91 |
| −160 | 4.72 |
| −180 | 4.52 |
| −200 | 4.30 |
| −220 | 4.08 |
| −240 | 3.83 |
| −260 | 3.58 |
| −280 | 3.33 |
| −300 | 3.07 |
| −320 | 2.76 |

| Temperature | Coefficient of Thermal Expansion |
|---|---|
| −340 | 2.45 |

The second-order polynomial is derived using regression analysis which is another mathematical process where numerical methods are employed. Regression analysis approximates discrete data such as the coefficient of linear thermal expansion vs. temperature data as a continuous function. This is an excellent example of where one has to use numerical methods for more than one process to solve a real-life problem.

# Chapter 07.01: Prerequisites to Numerical Integration

Lesson: Prerequisites to Numerical Integration

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *define an integral,*

2) *use Riemann's sum to approximately calculate integrals,*

3) *use Riemann's sum and its limit to find the exact expression of integrals, and*

4) *find exact integrals of a few functions.*

## What is integration?

The dictionary definition of *integration* is combining parts so that they work together or form a whole. Mathematically, integration stands for finding the area under a curve from one point to another. It is represented by

$$\int_a^b f(x)dx \qquad (1)$$

where the symbol $\int$ is the integral sign, and $a$ and $b$ are the lower and upper limits of integration, respectively, the function $f(x)$ is the integrand of the integral, and $x$ is the variable of integration. Figure 1 represents a graphical demonstration of the concept.

## Riemann Sum

Let $f$ be defined on the closed interval $[a, b]$, and let $\Delta$ be an arbitrary partition of $[a, b]$ such as: $a = x_0 < x_1 < x_2 < \ldots < x_{n-1} < x_n = b$, where $\Delta x_i$ is the length of the $i^{th}$ subinterval (Figure 1).

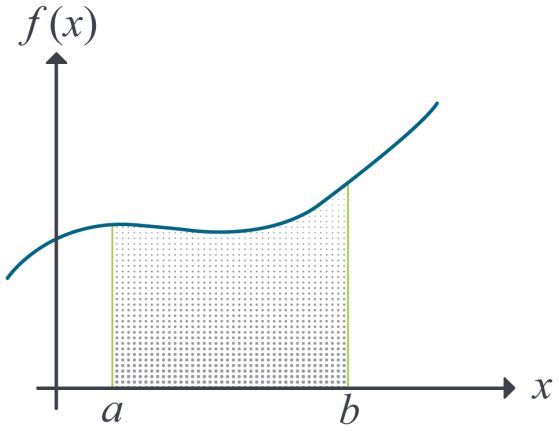



**Figure 1** The definite integral as the area of a region under the curve, Area $= \int_a^b f(x)dx$.

If $c_i$ is any point in the $i^{th}$ subinterval, then the sum

$$\sum_{i=1}^{n} f(c_i)\Delta x_i, \;\; x_{i-1} \leq c_i \leq x_i \qquad (2)$$

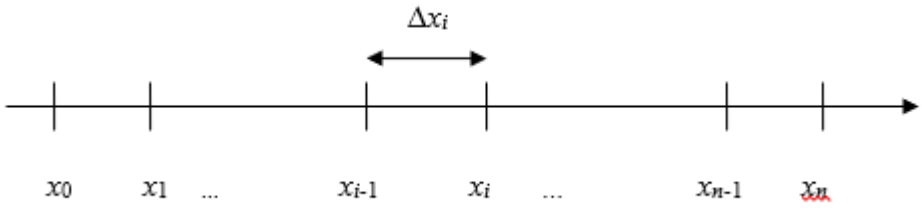



**Figure 2** Division of interval into $n$ segments.

is called a Riemann sum of the function $f$ for the partition $\Delta$ on the interval $[a, b]$. For a given partition $\Delta$, the length of the longest subinterval is called the norm of the partition. It is denoted by $\|\Delta\|$ (the norm of $\Delta$). The following limit is used to define the definite integral.

$$\lim_{\|\Delta\| \to 0} \sum_{i=1}^{n} f(c_i)\Delta x_i = I \qquad (3)$$

This limit exists if and only if for any positive number $\varepsilon$, there exists a positive number $\delta$ such that for every partition $\Delta$ of $[a, b]$ with $\|\Delta\| < \delta$, it follows that

$$|I - \sum_{i=1}^{n} f(c_i)\Delta x_i| < \varepsilon \qquad (4)$$

for any choice of $c_i$ in the $i^{th}$ subinterval of $\Delta$.

If the limit of a Riemann sum of $f$ exists, then the function $f$ is said to be integrable over $[a, b]$ and the Riemann sum of $f$ on $[a, b]$ approaches the number $I$.

$$\lim_{\|\Delta\| \to 0} \sum_{i=1}^{n} f\left(c_i\right) \Delta x_i = I \qquad (5)$$

where

$$I = \int_a^b f(x) dx \qquad (6)$$

**Example 1**

Find the area of the region between the parabola $y = x^2$ and the $x$-axis on the interval $[0, 4.5]$. Use right Riemann's sum with four partitions of equal segment width. Compare the value with the exact value.

**Solution**

We evaluate the integral for the area as a limit of Riemann sums. We sketch the region (Figure 3), and partition $[0, 4.5]$ into four subintervals of length

$$\begin{aligned} \Delta x &= \frac{4.5 - 0}{4} \\ &= 1.125 \end{aligned}.$$

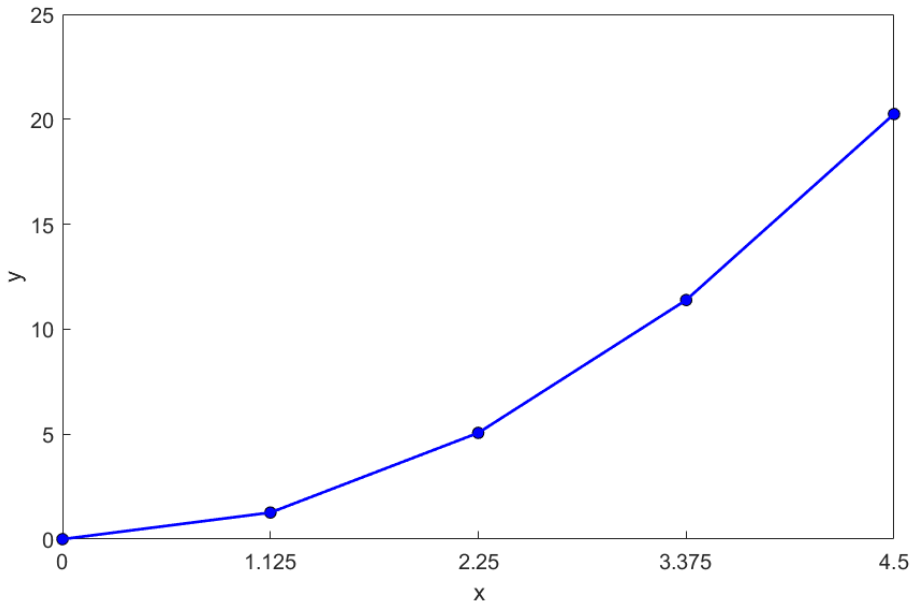



**Figure 3** Graph of the function $y = x^2$.

The points of partition are

$$x_0 = 0,\ x_1 = 1.125,\ x_2 = 2.25,\ x_3 = 3.375,\ x_4 = 4.5$$

Let us choose $c_i$ to be right-hand endpoint of its subinterval. Thus,

$$c_1 = x_1 = 1.125,\ c_2 = x_2 = 2.25,\ c_3 = x_3 = 3.375,\ c_4 = x_4 = 4.5$$

The rectangles defined by these choices have the following areas:

$$\begin{aligned} f(c_1)\,\Delta x &= f(1.125) \times (1.125) \\ &= (1.125)^2(1.125) \\ &= 1.4238 \end{aligned}$$

$$\begin{aligned} f(c_2)\,\Delta x &= f(2.25) \times (1.125) \\ &= (2.25)^2(1.125) \\ &= 5.6953 \end{aligned}$$

$$\begin{aligned} f(c_3)\,\Delta x &= f(3.375) \times (1.125) \\ &= (3.375)^2(1.125) \\ &= 12.814 \end{aligned}$$

$$\begin{aligned} f(c_4)\,\Delta x &= f(4.5) \times (1.125) \\ &= (4.5)^2(1.125) \\ &= 22.781 \end{aligned}$$

The sum of the areas then is the approximate value of the integral

$$\begin{aligned} \int_0^{4.5} x^2 dx &\approx \sum_{i=1}^{4} f(c_i)\Delta x, \\ &= 1.4238 + 5.6953 + 12.814 + 22.781 \\ &= 42.715 \end{aligned}$$

How does this compare with the exact value of the integral $\int_0^{4.5} x^2 dx$?

$$\begin{aligned} \int_0^{4.5} x^2 dx &= \left[\frac{x^3}{3}\right]_0^{4.5} \\ &= \frac{4.5^3}{3} - \frac{0^3}{3} \\ &= 30.375 \end{aligned}$$

**Example 2**

Find the exact area of the region between the parabola $y = x^2$ and the $x-$ axis on the interval $[0, b]$. Use the right Riemann's sum with equal segment widths.

**Solution**

Note that in Example 1 for $y = x^2$ that

$$\begin{aligned} f(c_i)\,\Delta x &= (i\Delta x)^2\,(\Delta x) \\ &= i^2\,(\Delta x)^3 \end{aligned} \qquad (E2.1)$$

Thus, the sum of these areas, if the interval is divided into $n$ equal segments, is

$$\begin{aligned} S_n &= \sum_{i=1}^{n} f(c_i)\Delta x \\ &= \sum_{i=1}^{n} i^2(\Delta x)^3 \\ &= (\Delta x)^3 \sum_{i=1}^{n} i^2 \end{aligned} \qquad (E2.2)$$

Since

$$\Delta x = \frac{b}{n}, \text{ and} \qquad (E2.3)$$

$$\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6} \qquad (E2.4)$$

Substituting Equations (E2.3) and (E2.4) in Equation (E2.1)

$$\begin{aligned} S_n &= \frac{b^3}{n^3}\frac{n(n+1)(2n+1)}{6} \\ &= \frac{b^3}{6}\frac{2n^2+n+2n+1}{n^2} \\ &= \frac{b^3}{6}\left(2+\frac{3}{n}+\frac{1}{n^2}\right) \end{aligned}$$

The definition of a definite integral can now be used

$$\int_a^b f(x)dx = \lim_{|\Delta x|\to 0} \sum_{i=1}^{n} f(c_i)\Delta x$$

To find the area under the parabola from $x = 0$ to $x = b$, we have

$$\begin{aligned}\int_0^b x^2 dx &= \lim_{|\Delta|\to 0} \sum_{i=1}^{n} f(c_i)\Delta x \\ &= \lim_{n\to\infty} S_n \\ &= \lim_{n\to\infty} \frac{b^3}{6}\left(2 + \frac{3}{n} + \frac{1}{n^2}\right) \\ &= \frac{b^3}{6}(2 + 0 + 0) \\ &= \frac{b^3}{3}\end{aligned}$$

## Mean Value of a Function

The mean value $\overline{f}$ of a function $f$ in an interval $[a, b]$ is given by

$$\overline{f} = \frac{1}{b-a}\int_a^b f(x)dx \qquad (7)$$

**Example 3**

Find the average value of $f(x) = (x-1)^2$ over the interval $[0, 3]$. At what point in the given interval does the function assume this average value?

**Solution**

$$\begin{aligned}\overline{f} &= \frac{1}{b-a}\int_a^b f(x)\,dx \\ &= \frac{1}{3-0}\int_a^b (x-1)^2\,dx \\ &= \frac{1}{3-0}\left[\frac{(x-1)^3}{3}\right]_0^3 \\ &= \frac{1}{3-0}\left[\frac{(3-1)^3}{3} - \frac{(0-1)^3}{3}\right] \\ &= 1\end{aligned}$$

The average value of the function $f$ over the interval $[0, 3]$ is 1.

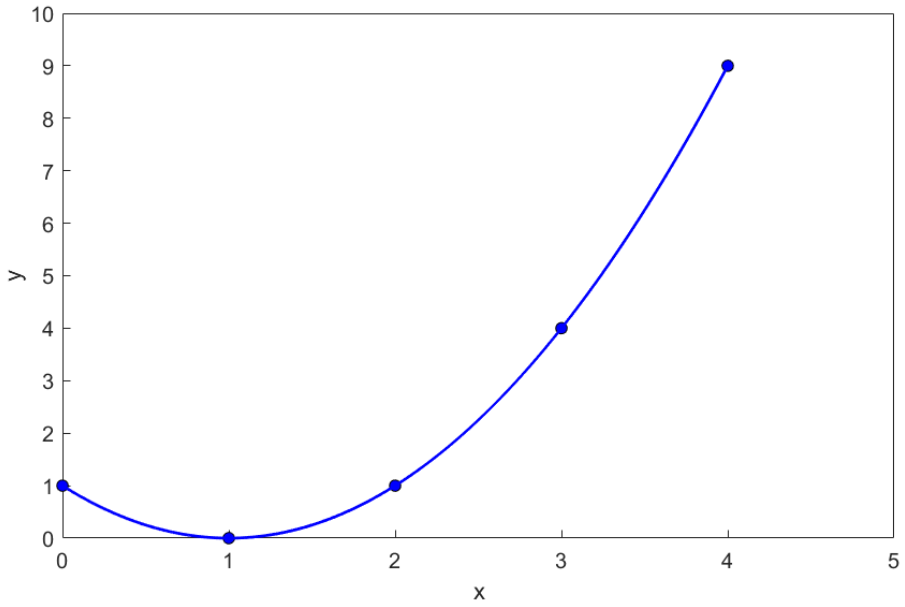



**Figure 4** The function $f(x) = (x-1)^2$.

The function assumes the average value at the points where

$$(x-1)^2 = 1$$

$$x^2 - 2x + 1 = 1$$

$$x^2 - 2x = 0$$

$$x(x-2) = 0$$

$$x = 0 \text{ and } x = 2$$

As evident from Figure 4, the function value is 1 at $x = 0$ and $x = 2$

Table 1 lists a number of standard indefinite integral forms.

**Table 1** A brief table of integrals

| Integrals |
|---|
| $\int dx = x + C$ |
| $\int af(x)dx = a\int f(x)dx + C$ |

Integrals

$$\int [u(x) \pm v(x)dx] = \int u(x)dx \pm \int v(x)dx + C$$

$$\int x^n dx = \frac{x^n + 1}{n + 1} + C$$

$$\int udv = uv - \int vdu + C$$

$$\int \frac{dx}{ax + b} = \frac{1}{a} \ln |ax + b| + C$$

$$\int a^x dx = \frac{a^x}{\ln a} + C$$

$$\int e^{ax} dx = \frac{e^{ax}}{a} + C$$

$$\int \cos x dx = \sin x + C$$

$$\int \sin x dx = -\cos x + C$$

$$\int \tan x dx = -ln|\cos x| + C = ln|\sec x| + C$$

$$\int \sec(ax)dx = \frac{1}{a} \ln |\sec(ax) + tan(ax)| + C$$

$$\int \cot x dx = -ln|\csc x| + C = ln|\sin x| + C$$

$$\int \sec^2 axdx = \frac{1}{a} \tan(ax) + C$$

$$\int \sec(x) \tan(x)dx = \sec(x) + C$$

$$\int \csc(x) \cot(x)dx = -\csc(x) + C$$

# Multiple Choice Test

## Chapter 07.01 Prerequisites to Numerical Integration

(1). Physically, integrating $\int_a^b f(x)dx$ means finding the

(A) area under the curve from $a$ to $b$

(D) area to the left of point $a$

(C) area to the right of point $b$

(D) area above the curve from $a$ to $b$

(2). The mean value of a function $f(x)$ from $a$ to $b$ is given by

(A) $\dfrac{f(a)+f(b)}{2}$

(D) $\dfrac{f(a)+2f\left(\dfrac{a+b}{2}\right)+f(b)}{4}$

(C) $\displaystyle\int_a^b f(x)dx$

(D) $\dfrac{\displaystyle\int_a^b f(x)dx}{b-a}$

(3). The exact value of $\displaystyle\int_{0.2}^{2.2} xe^x dx$ is most nearly

(A) 7.8036

(D) 11.807

(C) 14.034

(D) 19.611

(4). $\displaystyle\int_{0.2}^{2} f(x)dx$ for

$$f(x) = x, 0 \leq x \leq 1.2$$
$$= x^2, 1.2 < x \leq 2.4$$

is most nearly

(A) 1.9800

(D) 2.6640

(C) 2.7907

(D) 4.7520

(5). The area of a circle of radius $a$ can be found by the following integral

(A) $\int_0^a \left(a^2 - x^2\right) dx$

(D) $\int_0^{2\pi} \sqrt{a^2 - x^2} dx$

(C) $4\int_0^a \sqrt{a^2 - x^2} dx$

(D) $\int_0^a \sqrt{a^2 - x^2} dx$

(6). Velocity distribution of a fluid flow through a pipe varies along the radius and is given by $v(r)$. The flow rate through the pipe of radius $a$ is given by

(A) $\pi v(a) a^2$

(D) $\pi \frac{v(0) + v(a)}{2} a^2$

(C) $\int_0^a v(r) dr$

(D) $2\pi \int_0^a v(r) r dr$

For complete solution, go to

http://nm.mathforcollege.com/mcquizzes/07int/quiz_07int_background_solution.pdf

# Chapter 07.02: Trapezoidal Rule of Integration

Lesson: Single-Application Trapezoidal Rule

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *derive the trapezoidal rule of integration,*

2) *use the trapezoidal rule of integration to solve problems,*

## Introduction

The trapezoidal rule is based on the Newton-Cotes formula that if one approximates the integrand by an $n^{th}$ order polynomial, then the integral of the function is approximated by the integral of that $n^{th}$ order polynomial. Integrating polynomials is simple and is based on the calculus formula.

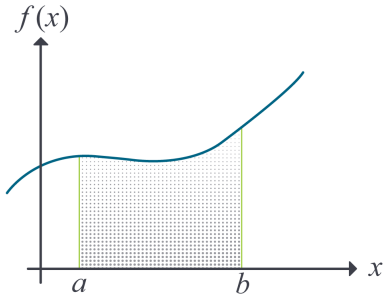



**Figure 1** Integration of a function

$$\int_a^b x^n dx = \left(\frac{b^{n+1} - a^{n+1}}{n+1}\right), n \neq -1 \qquad (1)$$

So if we want to approximate the integral

$$I = \int_a^b f(x)dx \qquad (2)$$

to find the value of the above integral, one assumes

$$f(x) \approx f_n(x)$$

where

$$f_n(x) = a_0 + a_1 x + \ldots + a_{n-1} x^{n-1} + a_n x^n. \qquad (3)$$

where $f_n(x)$ is a $n^{th}$ order polynomial. The trapezoidal rule assumes $n = 1$, that is, approximating the integral by a linear polynomial (straight line),

$$\int_a^b f(x)dx \approx \int_a^b f_1(x)dx \qquad (4)$$

**Method 1: Derived from Calculus**

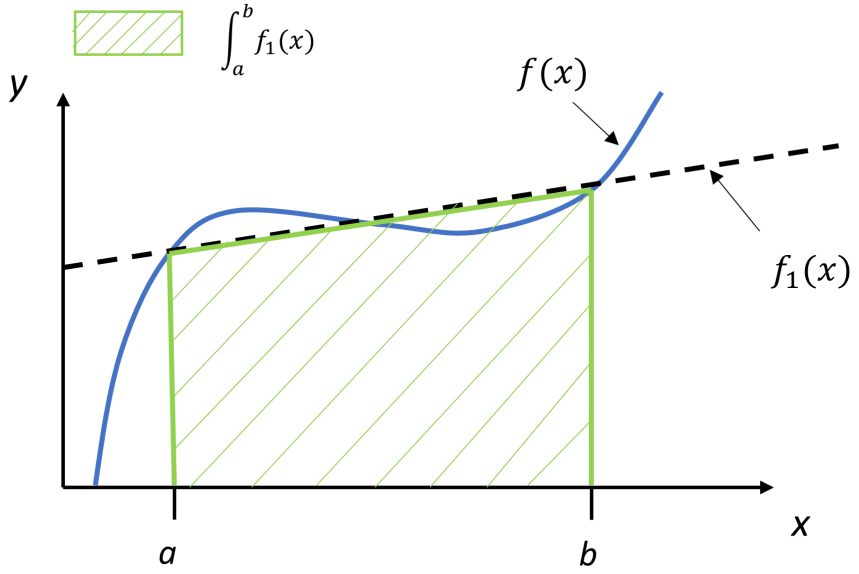



**Figure 2**. Approximating the function by a first-order polynomial to derive the trapezoidal rule

Approximating the integrand $f(x)$ by a first-order polynomial $f_1(x)$, that is, $f_1(x) = a_0 + a_1 x$,

$$\begin{aligned} \int_a^b f(x)dx &\approx \int_a^b f_1(x)dx \\ &= \int_a^b (a_0 + a_1 x)dx \\ &= a_0(b-a) + a_1 \left( \frac{b^2 - a^2}{2} \right) \end{aligned} \qquad (5)$$

But what are $a_0$ and $a_1$? Now if one chooses, $(a, f(a))$ and $(b, f(b))$ as the two points to approximate $f(x)$ by a straight line from $a$ to $b$,

$$f(a) = f_1(a) = a_0 + a_1 a$$
$$f(b) = f_1(b) = a_0 + a_1 b \qquad (6a, b)$$

Solving Equations (6a) and (6b) for $a_0$ and $a_1$, we get

$$a_1 = \frac{f(b) - f(a)}{b - a}$$

$$a_0 = \frac{f(a)b - f(b)a}{b - a} \qquad (7a, b)$$

Substituting values of $a_0$ and $a_1$ from Equations (7a) and (7b) in Equation (5) gives,

$$\int_a^b f(x)dx \approx \frac{f(a)b - f(b)a}{b - a}(b - a) + \frac{f(b) - f(a)}{b - a}\frac{b^2 - a^2}{2}$$
$$= (b - a)\left[\frac{f(a) + f(b)}{2}\right] \qquad (8)$$

Note that the trapezoidal rule can be interpreted as the width of the interval $[a, b]$ multiplied to an approximate average value of the function. Recall the definition of the mean of the function $\overline{f}$ of a function $f$ in an interval $[a, b]$ is given by

$$\overline{f} = \frac{1}{b - a}\int_a^b f(x)dx \qquad (9)$$

**Example 1**

The following integral is given

$$\int_{0.1}^{1.3} 5xe^{-2x}dx$$

a) Use the trapezoidal rule to estimate the value of the integral.

b) Find the true error, $E_t$ for part (a).

c) Find the absolute relative true error, $|\varepsilon_t|$ for part (a).

**Solution**

a)

$$\int_b^a f(x)dx \approx \frac{(b - a)}{2}\left[f(a) + f(b)\right], \qquad (8 - \text{repeated})$$

where

$$a = 0.1$$

$$b = 1.3$$

Then

$$\begin{aligned}\int_{0.1}^{1.3} f(x)dx &\approx \frac{(1.3-0.1)}{2}\left[f(0.1)+f(1.3)\right] \\ &= 0.6\left[f(0.1)+f(1.3)\right] \\ &= 0.6[5(0.1)e^{-2(0.1)}+5(1.3)e^{-2(1.3)}] \\ &= 0.6(0.4094+0.4828) \\ &= 0.5353\end{aligned}$$

b) The true error is

$$\begin{aligned}E_t = &\text{ True Value} - \text{Approximate Value} \\ &= 0.8939 - 0.5353 \\ &= 0.3586\end{aligned}$$

c) The absolute relative true error, $|\varepsilon_t|$ would then be

$$\begin{aligned}|\varepsilon_t| &= \left|\frac{\text{True Error}}{\text{True Value}}\right| \times 100 \\ &= \left|\frac{0.3586}{0.8939}\right| \times 100 \\ &= 40.12\%\end{aligned}$$

**Example 2**

The vertical distance covered by a rocket from $t = 8$ to $t = 30$ seconds is given by

$$x = \int_8^{30} \left(2000 \ln\left[\frac{140000}{140000 - 2100t}\right] - 9.8t\right) \text{dt}$$

a) Use the trapezoidal rule to find the distance covered for $t = 8$ to $t = 30$ seconds.

b) Find the true error, $E_t$ for part (a).

c) Find the absolute relative true error for part (a).

**Solution**

a)

$$I \approx (b-a)\left[\frac{f(a)+f(b)}{2}\right],$$

where

$$a = 8$$

$$b = 30$$

$$f(t) = 2000 \ln \left[ \frac{140000}{140000 - 2100t} \right] - 9.8t$$

$$\begin{aligned} f(8) &= 2000 \ln \left[ \frac{140000}{140000 - 2100(8)} \right] - 9.8(8) \\ &= 177.27 \text{ m/s} \end{aligned}$$

$$\begin{aligned} f(30) &= 2000 \ln \left[ \frac{140000}{140000 - 2100(30)} \right] - 9.8(30) \\ &= 901.67 \text{ m/s} \end{aligned}$$

$$\begin{aligned} I &\approx (30 - 8) \left[ \frac{177.27 + 901.67}{2} \right] \\ &= 11868 \text{ m} \end{aligned}$$

b) The exact value of the above integral is

$$\begin{aligned} x &= \int_8^{30} \left( 2000 \ln \left[ \frac{140000}{140000 - 2100t} \right] - 9.8t \right) dt \\ &= 11061 \text{ m} \end{aligned}$$

so the true error is

$$\begin{aligned} E_t &= \text{True Value} - \text{Approximate Value} \\ &= 11061 - 11868 \\ &= -807 \text{ m} \end{aligned}$$

c) The absolute relative true error, $|\epsilon_t|$, would then be

$$\begin{aligned} |\epsilon_t| &= \left| \frac{\text{True Error}}{\text{True Value}} \right| \times 100 \\ &= \left| \frac{11061 - 11868}{11061} \right| \times 100 \\ &= 7.2958\% \end{aligned}$$

## Appendix

This appendix shows other methods of deriving the trapezoidal rule of integration. The method of undetermined coefficients is significant to learn as it forms the basis of the Gauss quadrature rule.

### Method 2: Derived from Calculus Using Newton's divided difference polynomial

$f_1(x)$ can also be approximated by using Newton's divided difference polynomial as

$$f_1(x) = f(a) + \frac{f(b) - f(a)}{b - a}(x - a) \qquad (A.1)$$

Hence

$$\begin{aligned}
\int_a^b f(x)dx &\approx \int_a^b f_1(x)dx \\
&= \int_a^b \left[ f(a) + \frac{f(b) - f(a)}{b - a}(x - a) \right] dx \\
&= \left[ f(a)x + \frac{f(b) - f(a)}{b - a} \left( \frac{x^2}{2} - ax \right) \right]_a^b \\
&= f(a)b - f(a)a + \left( \frac{f(b) - f(a)}{b - a} \right) \left( \frac{b^2}{2} - ab - \frac{a^2}{2} + a^2 \right) \\
&= f(a)b - f(a)a + \left( \frac{f(b) - f(a)}{b - a} \right) \left( \frac{b^2}{2} - ab + \frac{a^2}{2} \right) \\
&= f(a)b - f(a)a + \left( \frac{f(b) - f(a)}{b - a} \right) \frac{1}{2} (b - a)^2 \\
&= f(a)b - f(a)a + \frac{1}{2} (f(b) - f(a)) (b - a) \\
&= f(a)b - f(a)a + \frac{1}{2} f(b)b - \frac{1}{2} f(b)a - \frac{1}{2} f(a)b + \frac{1}{2} f(a)a \\
&= \frac{1}{2} f(a)b - \frac{1}{2} f(a)a + \frac{1}{2} f(b)b - \frac{1}{2} f(b)a \\
&= (b - a) \left[ \frac{f(a) + f(b)}{2} \right] \qquad (A.2)
\end{aligned}$$

This gives the same result as Equation (8) because $f_1(x)$ is just a different form of writing the same first-order polynomial.

### Method 3: Derived from Geometry

The trapezoidal rule can also be derived from geometry. Look at Figure 2. The area under the curve $f_1(x)$ is the area of a trapezoid. The integral

$$\int_a^b f(x)dx \approx \text{Area of trapezoid}$$

$$= \frac{1}{2}(\text{Sum of the length of parallel sides}) \times (\text{Perpendicular distance between the parallel sides})$$

$$= \frac{1}{2}\left(f(b) + f(a)\right)(b-a)$$

$$= (b-a)\left[\frac{f(a)+f(b)}{2}\right] \qquad (A.3)$$

**Method 4: Derived from Method of Coefficients**

Choose the integral $\int_a^b f(x)dx$ approximated as follows.

$$\int_a^b f(x)dx \approx c_1 f(a) + c_2 f(b) \qquad (A.4)$$

The coefficients $c_1$ and $c_2$ are undetermined. We will find these coefficients such that the right-hand side is exact for integrals of a straight line $a_0 + a_1 x$.

So from exact integration

$$\int_a^b (a_0 + a_1 x)\, dx = \left[a_0 x + a_1 \frac{x^2}{2}\right]_a^b$$

$$= a_0 (b-a) + a_1 \left(\frac{b^2 - a^2}{2}\right) \qquad (A.5)$$

But we want the right-hand side formula to give the same result as Equation (A.5) for $f(x) = a_0 + a_1 x$ which is

$$c_1 f(a) + c_2 f(b) = c_1(a_0 + a_1 b) + c_2(a_0 + a_1 b)$$

$$= a_0 (c_1 + c_2) + a_1 (c_1 a + c_2 b) \qquad (A.6)$$

Hence from Equations (A.5) and (A.6),

$$a_0 (b-a) + a_1 \left(\frac{b^2 - a^2}{2}\right) = a_0 (c_1 + c_2) + a_1 (c_1 a + c_2 b) \qquad (A.7)$$

Since $a_0$ and $a_1$ are arbitrary constants for the chosen general straight line, the coefficients of $a_0$ and $a_1$ need to be equal. That gives

$$c_1 + c_2 = b - a \qquad (A.8a)$$

$$c_1 a + c_2 b = \frac{b^2 - a^2}{2} \qquad (A.8b)$$

Multiplying Equation (A.8a) by $a$ and subtracting from Equation (A.8b) gives

$$c_2 = \frac{b-a}{2} \qquad (A.9a)$$

Substituting the value of $c_2$ from Equation (A.9a) in Equation (A.8a) gives

$$c_1 = \frac{b-a}{2} \qquad (A.9b)$$

Therefore, from Equation (A.4), (A.9a), and (A.9b),

$$\begin{aligned} \int_a^b f(x)dx &\approx c_1 f(a) + c_2 f(b) \\ &= \frac{b-a}{2} f(a) + \frac{b-a}{2} f(b) \end{aligned} \qquad (A.10)$$

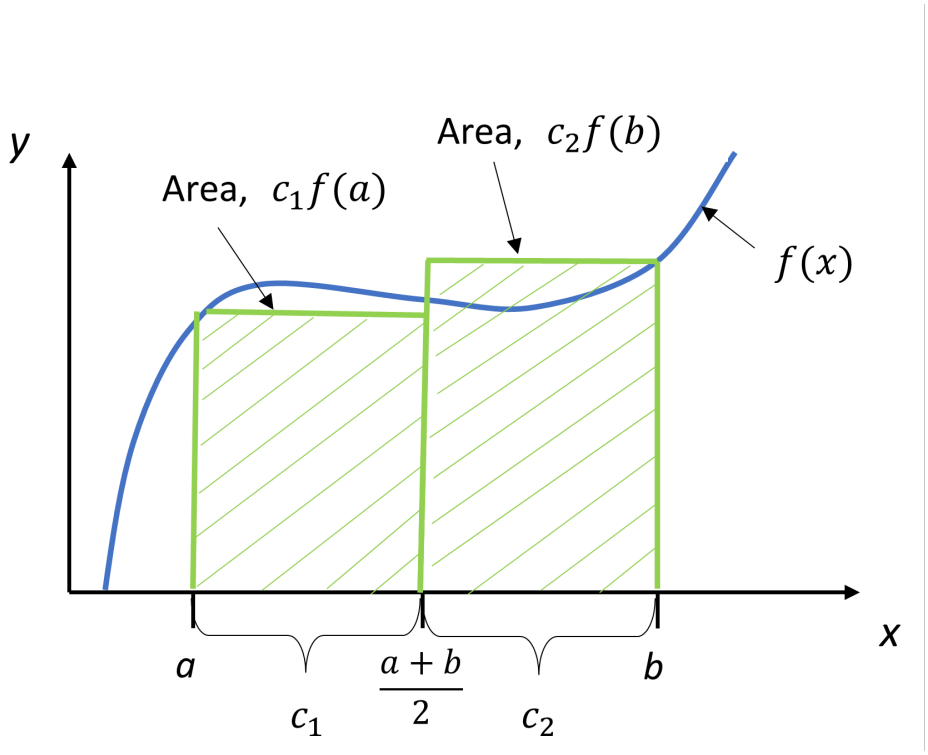



**Figure 3** Area by the method of coefficients.

The interpretation is that $f(x)$ is evaluated at points $a$ and $b$, and each function evaluation is given a weight of $\frac{b-a}{2}$. Geometrically, Equation (8) is looked at as the area of a trapezoid, while Equation (A.10) is viewed as the sum of the area of two rectangles, as shown in Figure 3.

# Chapter 07.02: Trapezoidal Rule of Integration

Lesson: Composite Trapezoidal Rule

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *derive the composite (also called multiple-segment) trapezoidal rule of integration,*

2) *use the composite (also called multiple-segment) trapezoidal rule of integration to solve problems*

## Introduction

A single segment trapezoidal rule seldom gives you acceptable results for an integral. Instead for higher accuracy and its control, we can use the composite (also called multiple-segment) trapezoidal rule where the integral is broken into segments, and the single-segment trapezoidal rule is applied over each segment.

## Derivation

Divide $(b-a)$ into $n$ equal segments, as shown in Figure 1. Then the width of each segment is

$$h = \frac{b-a}{n} \qquad (1)$$

The integral $I$ can be broken into $n$ integrals as

$$\begin{aligned} I &= \int_a^b f(x)dx \\ &= \int_a^{a+h} f(x)dx + \int_{a+h}^{a+2h} f(x)dx + ... + \int_{a+(n-2)h}^{a+(n-1)h} f(x)dx + \int_{a+(n-1)h}^{b} f(x)dx \end{aligned} \qquad (2)$$

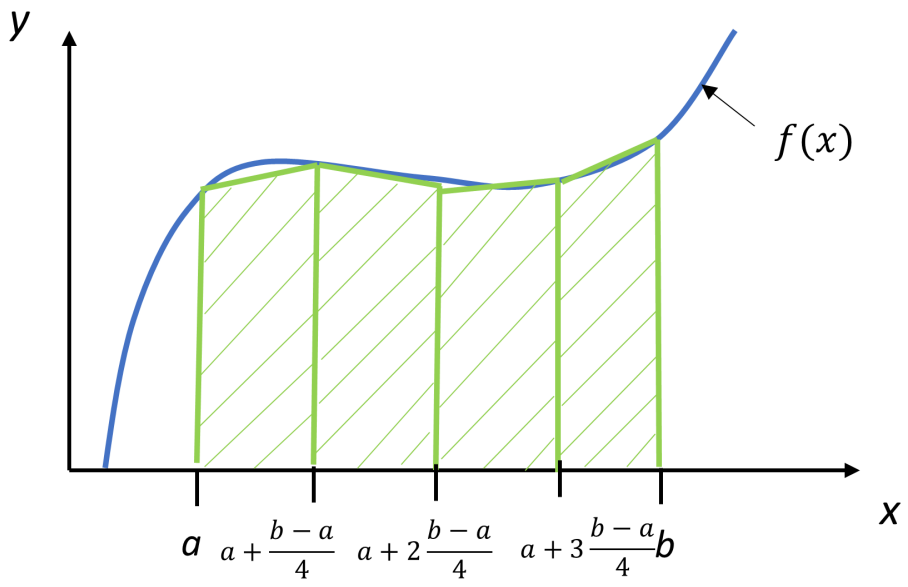



**Figure 1** Composite ($n = 4$) trapezoidal rule

Applying single-segment trapezoidal rule on Equation (2) on each segment gives

$$\int_a^b f(x)dx \approx [(a+h)-a]\left[\frac{f(a)+f(a+h)}{2}\right]$$
$$+[(a+2h)-(a+h)]\left[\frac{f(a+h)+f(a+2h)}{2}\right]$$
$$+ \ldots\ldots\ldots\ldots\ldots$$
$$+[(a+(n-1)h)-(a+(n-2)h)]\left[\frac{f(a+(n-2)h)+f(a+(n-1)h)}{2}\right]$$
$$+[b-(a+(n-1)h)]\left[\frac{f(a+(n-1)h)+f(b)}{2}\right]$$

$$=h\left[\frac{f(a)+f(a+h)}{2}\right]+h\left[\frac{f(a+h)+f(a+2h)}{2}\right]$$
$$+ \ldots\ldots\ldots\ldots\ldots$$
$$+h\left[\frac{f(a+(n-2)h)+f(a+(n-1)h)}{2}\right]$$
$$+h\left[\frac{f(a+(n-1)h)+f(b)}{2}\right]$$

$$=h\left[\frac{f(a)+2f(a+h)+2f(a+2h)+\ldots+2f(a+(n-1)h)+f(b)}{2}\right]$$
$$=\frac{h}{2}\left[f(a)+2\left\{\sum_{i=1}^{n-1}f(a+ih)\right\}+f(b)\right]$$
$$=\frac{b-a}{2n}\left[f(a)+2\left\{\sum_{i=1}^{n-1}f(a+ih)\right\}+f(b)\right] \qquad (3)$$

**Example 1**

The following integral is given:

$$\int_{0.1}^{1.3} 5xe^{-2x}dx$$

a) Use the composite trapezoidal rule to estimate the value of this integral. Use three segments.

b) Find the true error $E_t$ for part (a).

c) Find the absolute relative true error $|\varepsilon_t|$ for part (a).

**Solution**

a) The solution using the composite trapezoidal rule with 3 segments is applied as follows.

$$I \approx \frac{b-a}{2n}\left[f(a)+2\sum_{i=1}^{n-1}f(a+ih)+f(b)\right]$$

$$n = 3$$

$$a = 0.1$$

$$b = 1.3$$

$$\begin{aligned} h &= \frac{b-a}{n} \\ &= \frac{1.3-0.1}{3} \\ &= 0.4 \end{aligned}$$

From Equation (3),

$$I \approx \frac{1.3-0.1}{6}\left[f(0.1) + 2\sum_{i=1}^{3-1} f(0.1+0.4i) + f(1.3)\right]$$

$$I \approx \frac{1.3-0.1}{6}\left[f(0.1) + 2\sum_{i=1}^{2} f(0.1+0.4i) + f(1.3)\right]$$

$$\begin{aligned} &= 0.2[f(0.1) + 2f(0.5) + 2f(0.9) + f(1.3)] \\ &= 0.2[5 \times 0.1 \times e^{-2(0.1)} + 2(5 \times 0.5 \times e^{-2(0.5)}) + 2(5 \times 0.9 \times e^{-2(0.9)}) + 5 \times 1.3 \times e^{-2(1.3)}] \\ &= 0.84385 \end{aligned}$$

b) The exact value of the above integral can be found by integration by parts and is

$$\int_{0.1}^{1.3} 5xe^{-2x}dx = 0.89387$$

So the true error is

$$\begin{aligned} E_t &= \text{True Value} - \text{Approximate Value} \\ &= 0.89387 - 0.84385 \\ &= 0.05002 \end{aligned}$$

c) The absolute relative true error is

$$\begin{aligned} |\varepsilon_t| &= \left|\frac{\text{True Error}}{\text{True Value}}\right| \times 100 \\ &= \left|\frac{0.05002}{0.89387}\right| \times 100 \\ &= 5.5959\% \end{aligned}$$

**Example 2**

The vertical distance covered by a rocket from $t = 8$ to $t = 30$ seconds is given by

$$x = \int_8^{30} \left( 2000 \ln \left[ \frac{140000}{140000 - 2100t} \right] - 9.8t \right) \mathrm{dt}$$

a) Use the composite trapezoidal rule to find the distance covered from $t = 8$ to $t = 30$ seconds. Use two segments.

b) Find the true error, $E_t$ for part (a).

c) Find the absolute relative true error for part (a).

**Solution**

a) The solution using the composite trapezoidal rule with 2-segments is

$$I \approx \frac{b-a}{2n} \left[ f(a) + 2 \left\{ \sum_{i=1}^{n-1} f(a + ih) \right\} + f(b) \right]$$

$$n = 2$$

$$a = 8$$

$$b = 30$$

$$\begin{aligned} h &= \frac{b-a}{n} \\ &= \frac{30-8}{2} \\ &= 11 \end{aligned}$$

$$\begin{aligned} I &\approx \frac{30-8}{2(2)} \left[ f(8) + 2 \left\{ \sum_{i=1}^{2-1} f(8 + 11i) \right\} + f(30) \right] \\ &= \frac{22}{4} \left[ f(8) + 2f(19) + f(30) \right] \\ &= \frac{22}{4} \left[ 177.27 + 2(484.75) + 901.67 \right] \\ &= 11266 \text{ m} \end{aligned}$$

b) The exact value of the above integral is

$$\begin{aligned} x &= \int_8^{30} \left( 2000 \ln \left[ \frac{140000}{140000 - 2100t} \right] - 9.8t \right) dt \\ &= 11061 \text{ m} \end{aligned}$$

So the true error is

$$\begin{aligned} E_t &= \text{True Value} - \text{Approximate Value} \\ &= 11061 - 11266 \\ &= -205 \text{ m} \end{aligned}$$

c) The absolute relative true error,$|\epsilon_t|$, would then be

$$\begin{aligned} |\epsilon_t| &= \left| \frac{\text{True Error}}{\text{True Value}} \right| \times 100 \\ &= \left| \frac{11061 - 11266}{11061} \right| \times 100 \\ &= 1.8537\% \end{aligned}$$

**Table 1** Values obtained using composite trapezoidal rule for

$$x = \int_8^{30} \left( 2000 \ln \left[ \frac{140000}{140000 - 2100t} \right] - 9.8t \right) \text{dt}$$

| $n$ | Approximate Value | $E_t$ | $\lvert\epsilon_t\rvert\,\%$ | $\lvert\epsilon_a\rvert\,\%$ |
|---|---|---|---|---|
| 1 | 11868 | $-807$ | 7.296 | $---$ |
| 2 | 11266 | $-205$ | 1.853 | 5.343 |
| 3 | 11153 | $-91.4$ | 0.8265 | 1.019 |
| 4 | 11113 | $-51.5$ | 0.4655 | 0.3594 |
| 5 | 11094 | $-33.0$ | 0.2981 | 0.1669 |
| 6 | 11084 | $-22.9$ | 0.2070 | 0.09082 |
| 7 | 11078 | $-16.8$ | 0.1521 | 0.05482 |
| 8 | 11074 | $-12.9$ | 0.1165 | 0.03560 |

## Error in Composite Trapezoidal Rule

Without proof, the true error for a single-segment Trapezoidal rule for $\int_a^b f(x)dx$ is given by

$$E_t = -\frac{(b-a)^3}{12} f''(\zeta), \ a < \zeta < b \qquad (4)$$

where

$$\zeta \ \text{ is some point in } \ [a, b] \, .$$

What is the error then in the composite trapezoidal rule for $\int_a^b f(x)dx$?

It is given by

$$E_t \approx -\frac{(b-a)^3}{12n^2}\overline{f''} \qquad (5)$$

where

$$\overline{f''} = \text{some approximate average value of the second derivative of the function in } [a,b]\,.$$

$$n = \text{number of segments.}$$

Recall that the width of each segment is

$$h = \frac{b-a}{n}$$

## Appendix

The appendix gives you two more examples for showing the application of the composite trapezoidal rule.

**Example A.1**

Use the composite trapezoidal rule to find the area under the curve

$$f(x) = \frac{300x}{1+e^x}$$

from $x = 0$ to $x = 10$. Use 2 segments.

**Solution**

Using two segments, we get

$$h = \frac{10-0}{2} = 5$$

$$f(0) = \frac{300(0)}{1+e^0} = 0$$

$$f(5) = \frac{300(5)}{1+e^5} = 10.039$$

$$f(10) = \frac{300(10)}{1+e^{10}} = 0.136$$

From Equation (3)

$$
\begin{aligned}
I &\approx \frac{b-a}{2n}\left[f(a)+2\left\{\sum_{i=1}^{n-1} f(a+ih)\right\}+f(b)\right] \\
&= \frac{10-0}{2(2)}\left[f(0)+2\left\{\sum_{i=1}^{2-1} f(0+5)\right\}+f(10)\right] \\
&= \frac{10}{4}\left[f(0)+2f(5)+f(10)\right] \\
&= \frac{10}{4}\left[0+2(10.039)+0.136\right] \\
&= 50.537
\end{aligned}
$$

So what is the true value of this integral?

$$\int_0^{10} \frac{300x}{1+e^x}dx = 246.59$$

The absolute relative true error is

$$
\begin{aligned}
|\epsilon_t| &= \left|\frac{246.59-50.535}{246.59}\right| \times 100 \\
&= 79.506\%
\end{aligned}
$$

Why is the true value different from the approximate values? Just take a look at Figure 2. As you can see, the area under the "trapezoids" (yeah, they really look like triangles now) covers a small portion of the area under the curve. As we add more segments, the approximated value quickly approaches the true value.

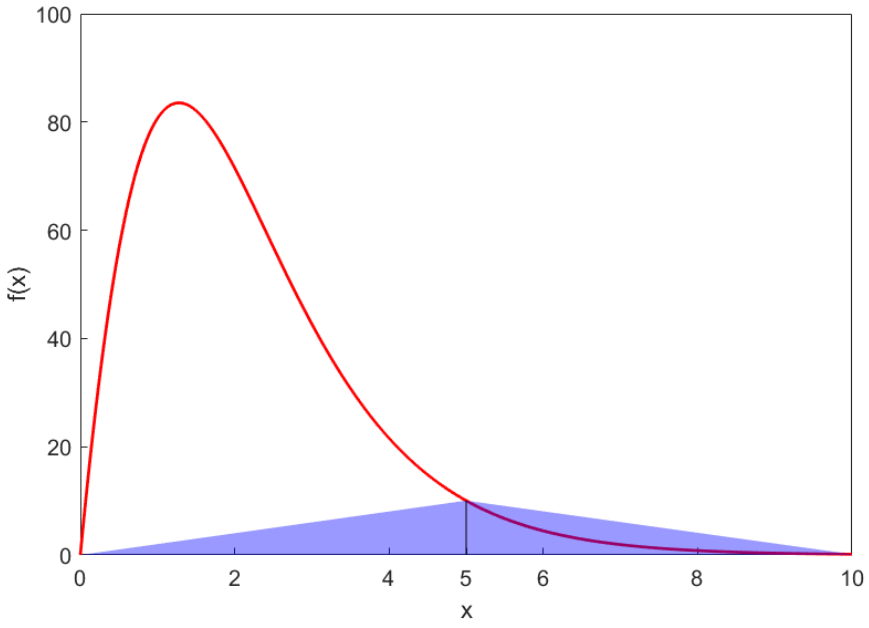


**Figure 2.** Composite trapezoidal rule approximation.

**Table 1** Values obtained using the composite trapezoidal rule for $\int_0^{10} \frac{300x}{1+e^x} dx$.

| $n$ | *Approximate* $Value$ | $E_t$ | $\lvert\epsilon_t\rvert$ |
|---|---|---|---|
| 1 | 0.681 | 245.91 | 99.724% |
| 2 | 50.535 | 196.05 | 79.505% |
| 4 | 170.61 | 75.978 | 30.812% |
| 8 | 227.04 | 19.546 | 7.927% |
| 16 | 241.70 | 4.887 | 1.982% |
| 32 | 245.37 | 1.222 | 0.495% |
| 64 | 246.28 | 0.305 | 0.124% |

**Example A.2**

Use composite trapezoidal rule to find

$$I = \int_0^2 \frac{1}{\sqrt{x}} dx$$

**Solution**

We cannot use the trapezoidal rule for this integral, as the value of the integrand at $x = 0$ is infinite. However, it is known that discontinuities in a curve do not change the area under it. We can assume any value for the function at $x = 0$. The algorithm to define the function so that we can use the composite trapezoidal rule is given below.

Function $f(x)$

If $x = 0$ Then $f = 0$

If $x \neq 0$ Then $f = x^{(-0.5)}$

End Function

Basically, we are just assigning the function a value of zero at $x = 0$. Everywhere else, the function is continuous. This assumption means the true value of our integral is just that—true. Let's see what happens using the composite trapezoidal rule.

Using two segments, we get

$$h = \frac{2-0}{2} = 1$$

$$f(0) = 0$$

$$f(1) = \frac{1}{\sqrt{1}} = 1$$

$$f(2) = \frac{1}{\sqrt{2}} = 0.70711$$

$$\begin{aligned} I &\approx \frac{b-a}{2n}\left[f(a) + 2\left\{\sum_{i=1}^{n-1} f(a+ih)\right\} + f(b)\right] \\ &= \frac{2-0}{2(2)}\left[f(0) + 2\left\{\sum_{i=1}^{2-1} f(0+1)\right\} + f(2)\right] \\ &= \frac{2}{4}\left[f(0) + 2f(1) + f(2)\right] \\ &= \frac{2}{4}\left[0 + 2(1) + 0.70711\right] \\ &= 1.3536 \end{aligned}$$

So what is the true value of this integral?

$$\int_0^2 \frac{1}{\sqrt{x}} dx = 2.8284$$

thus making the absolute relative true error

$$\begin{aligned} |\epsilon_t| &= \left|\frac{2.8284 - 1.3536}{2.8284}\right| \times 100 \\ &= 52.145\% \end{aligned}$$

Table 2 shows how the approximate value converges to the true value as the number of segments is increased.

**Table 2** Values obtained using the composite trapezoidal rule for $\int_0^2 \frac{1}{\sqrt{x}} dx$.

| $n$ | Approximate Value | $E_t$ | $\lvert\epsilon_t\rvert$ |
|---|---|---|---|
| 2 | 1.354 | 1.474 | 52.14% |
| 4 | 1.792 | 1.036 | 36.64% |
| 8 | 2.097 | 0.731 | 25.85% |
| 16 | 2.312 | 0.516 | 18.26% |
| 32 | 2.463 | 0.365 | 12.91% |
| 64 | 2.570 | 0.258 | 9.128% |
| 128 | 2.646 | 0.182 | 6.454% |
| 256 | 2.699 | 0.129 | 4.564% |
| 512 | 2.737 | 0.091 | 3.227% |

| $n$ | Approximate Value | $E_t$ | $\lvert\epsilon_t\rvert$ |
|---|---|---|---|
| 1024 | 2.764 | 0.064 | 2.282% |
| 2048 | 2.783 | 0.045 | 1.613% |
| 4096 | 2.796 | 0.032 | 1.141% |

# Chapter 07.02: Trapezoidal Rule of Integration

Lesson: Error Analysis of Trapezoidal Rule

## Learning Objective

*After successful completion of this lesson, you should be able to:*

1) *derive the formula for the true error in the composite (also called multiple-segment) trapezoidal rule of integration.*

## Error in Composite Trapezoidal Rule

The true error for a single segment trapezoidal rule is given by

$$E_t = -\frac{(b-a)^3}{12} f''(\zeta),\ a < \zeta < b \qquad (1)$$

where

$$\zeta \text{ is some point in } [a,b].$$

What is the error then in the composite trapezoidal rule? It is the sum of the errors from each of the $n$ segments. The error in each segment is that of the single segment trapezoidal rule. The error in each segment is

$$\begin{aligned} E_1 &= -\frac{[(a+h)-a]^3}{12} f''(\zeta_1), a < \zeta_1 < a+h \\ &= -\frac{h^3}{12} f''(\zeta_1) \end{aligned}$$

$$\begin{aligned} E_2 &= -\frac{[(a+2h)-(a+h)]^3}{12} f''(\zeta_2), a+h < \zeta_2 < a+2h \\ &= -\frac{h^3}{12} f''(\zeta_2) \end{aligned}$$

$$\vdots$$

$$\begin{aligned} E_i &= -\frac{[(a+ih)-(a+(i-1)h)]^3}{12} f''(\zeta_i), a+(i-1)h < \zeta_i < a+ih \\ &= -\frac{h^3}{12} f''(\zeta_i) \end{aligned}$$

$$\vdots$$

$$E_{n-1} = -\frac{[\{a+(n-1)h\} - \{a+(n-2)h\}]^3}{12} f''(\zeta_{n-1}), a+(n-2)h < \zeta_{n-1} < a+(n-1)h$$
$$= -\frac{h^3}{12} f''(\zeta_{n-1})$$

$$E_n = -\frac{[b - \{a+(n-1)h\}]^3}{12} f''(\zeta_n), a+(n-1)h < \zeta_n < b$$
$$= -\frac{h^3}{12} f''(\zeta_n)$$

Hence the total error in the composite trapezoidal rule is

$$E_t = \sum_{i=1}^{n} E_i$$
$$= -\frac{h^3}{12} f''(\zeta_1) - \frac{h^3}{12} f''(\zeta_2) - \ldots - \frac{h^3}{12} f''(\zeta_n)$$
$$= -\frac{h^3}{12} \sum_{i=1}^{n} f''(\zeta_i)$$

Since $h = \dfrac{b-a}{n}$,

$$E_t = -\frac{(b-a)^3}{12n^3} \sum_{i=1}^{n} f''(\zeta_i)$$
$$= -\frac{(b-a)^3}{12n^2} \frac{\sum_{i=1}^{n} f''(\zeta_i)}{n}$$

The term $\dfrac{\sum_{i=1}^{n} f''(\zeta_i)}{n}$ is an approximate average value of the second derivative $f''(x), a < x < b$.

Hence

$$E_t \approx -\frac{(b-a)^3}{12n^2} \overline{f''} \qquad (2)$$

In Table 1, the approximate value of the integral

$$\int_8^{30} \left( 2000 \ln \left[ \frac{140000}{140000 - 2100t} \right] - 9.8t \right) dt$$

is given as a function of the number of segments in a composite trapezoidal rule. You can visualize that as the number of segments is doubled, the true error gets approximately quartered.

**Table 1** Values obtained using the composite trapezoidal rule for

$$x = \int_8^{30} \left( 2000 \ln \left[ \frac{140000}{140000 - 2100t} \right] - 9.8t \right) dt.$$

| n | Approximate Value | $E_t$ | $\lvert\epsilon_t\rvert\,\%$ | $\lvert\epsilon_a\rvert\,\%$ |
|---|---|---|---|---|
| 2 | 11266 | $-205$ | 1.853 | 5.343 |
| 4 | 11113 | $-52$ | 0.4701 | 0.3594 |
| 8 | 11074 | $-13$ | 0.1175 | 0.03560 |
| 16 | 11065 | $-4$ | 0.03616 | 0.00401 |

For example, for the composite trapezoidal rule with 2 segments, the true error is $-205$, and a quarter of that error is $-51.25$. That is close to the true error of $-52$ for the composite trapezoidal rule with 4 segments.

Can you answer the question, *why is the true error not exactly* $-51.25$*?* How does this information help us in numerical integration? You will find out that this forms the basis of Richardson's extrapolation formula for integration based on the trapezoidal rule, where we use the argument that true error gets approximately quartered when the number of segments is doubled. Richardson's extrapolation formula based on the trapezoidal rule is computationally more efficient than using the composite trapezoidal rule by itself.

## Richardson's Extrapolation Formula for Trapezoidal Rule

From Equation (2). the true error, $E_t$, in the composite trapezoidal rule with $n$ segments is estimated as

$$E_t \approx \alpha \frac{1}{n^2} \qquad (3)$$

Assuming C to be the constant of proportionality

$$E_t \approx \frac{C}{n^2} \qquad (4)$$

Since

$$E_t = \text{True Value} - \text{Approximate Value}$$
$$E_t = (TV) - I_n$$

where

$$(TV) = \text{true value}$$

$$I_n = \text{approximate value using } n\text{-segments}$$

Then from Equations (4) and (5),

$$\frac{C}{n^2} \approx (TV) - I_n \qquad (6)$$

If the number of segments is doubled from $n$ to $2n$ in the composite trapezoidal rule,

$$\frac{C}{(2n)^2} \approx (TV) - I_{2n} \qquad (7)$$

Equations (6) and (7) can be solved simultaneously for $(TV)$ to get

$$(TV) \approx I_{2n} + \frac{I_{2n} - I_n}{3}$$

**Example 1**

The vertical distance in meters covered by a rocket from $t = 8$ to $t = 30$ seconds is given by

$$x = \int_8^{30} \left( 2000 \ln \left[ \frac{140000}{140000 - 2100t} \right] - 9.8t \right) dt$$

a) Use Richardson's extrapolation formula to find the distance covered. Use the composite trapezoidal rule results with 2 and 4 segments given in Table 1.

b) Find the true error for part (a).

c) Find the absolute relative true error for part (a).

**Solution**

a)

$$I_2 = 11266 \text{ m}$$

$$I_4 = 11113 \text{ m}$$

Using Richardson's extrapolation formula for the trapezoidal rule, the true value is approximated by

$$TV \approx I_{2n} + \frac{I_{2n} - I_n}{3}$$

and choosing $n = 2$,

$$
\begin{aligned}
TV &\approx I_4 + \frac{I_4 - I_2}{3} \\
&= 11113 + \frac{11113 - 11266}{3} \\
&= 11062 \text{ m}
\end{aligned}
$$

b) The exact value of the above integral is

$$
\begin{aligned}
x &= \int_8^{30} \left( 2000 \ln \left[ \frac{140000}{140000 - 2100t} \right] - 9.8t \right) dt \\
&= 11061 \text{ m}
\end{aligned}
$$

So the true error

$$
\begin{aligned}
E_t &= \text{True Value} - \text{Approximate Value} \\
&= 11061 - 11062 \\
&= -1 \text{ m}
\end{aligned}
$$

c) The absolute relative true error, $|\epsilon_t|$, would then be

$$
\begin{aligned}
|\epsilon_t| &= \frac{\text{True Error}}{\text{True Value}} \times 100 \\
&= \left| \frac{11061 - 11062}{11061} \right| \times 100 \\
&= 0.00904\%
\end{aligned}
$$

Table 2 shows Richardson's extrapolation results using 1, 2, 4, and 8 segments. Results are compared with those with the composite trapezoidal rule.

**Table 2** Values obtained using Richardson's extrapolation formula for the trapezoidal rule for $x = \int_8^{30} \left( 2000 \ln \left[ \frac{140000}{140000 - 2100t} \right] - 9.8t \right) dt$.

| n | Trapezoidal rule | $E_t\%$ for trapezoidal rule | Richardson's Extrapolation | $E_t\%$ for Richardson's extrapolation |
|---|---|---|---|---|
| 1 | 11868 | 7.296 | – | – |
| 2 | 11266 | 1.854 | 11065 | 0.03616 |
| 4 | 11113 | 0.4655 | 11062 | 0.009041 |
| 8 | 11074 | 0.1165 | 11061 | 0.0000 |

# Multiple Choice Test

## Chapter 07.02 Trapezoidal Rule of Integration

(1). What is the highest order of polynomial for which the composite trapezoidal rule of integration with two segments is exact?

(A) first

(B) second

(C) third

(D) fourth

(2). The estimated value of $\int_{0.2}^{2.2} xe^x dx$ by the using single application trapezoidal rule is most nearly

(A) 11.672

(B) 11.807

(C) 20.099

(D) 24.119

(3). The estimated value of $\int_{0.2}^{2.2} xe^x dx$ by using the composite trapezoidal rule with three segments is most nearly

(A) 11.672

(B) 11.807

(C) 12.811

(D) 14.633

(4). The velocity of a body is given by

$$\begin{aligned} v(t) &= 2t,\ 1 \leq t \leq 5 \\ &= 5t^2 + 3,\ 5 < t \leq 14 \end{aligned}$$

where $t$ is given in seconds, and $v$ is given in m/s. Use the composite trapezoidal rule with two segments to find the distance covered by the body from $t = 2$ to $t = 9$ seconds.

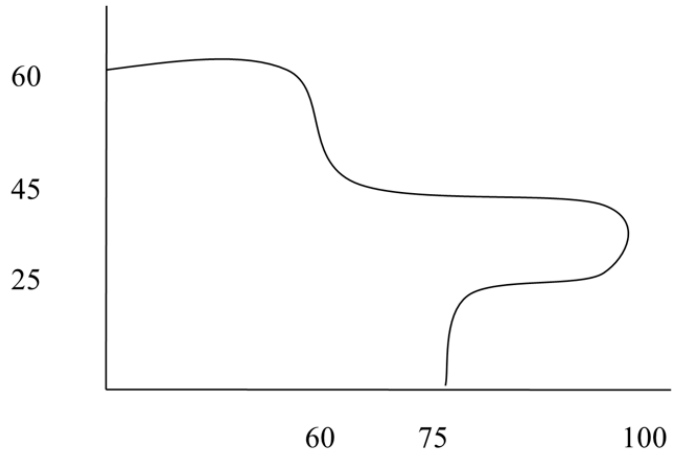
(A) $935.0m$

(B) $1039.7m$

(C) $1260.9m$

(D) $5048.9m$

(5). The shaded area shows a plot of land available for sale. Your best estimate of the area of the land is most nearly




(A) $2500m^2$

(B) $4775m^2$

(C) $5250m^2$

(D) $6000m^2$

(6). The following data of the velocity of a body is given as a function of time.

| Time, $t$ (s) | 0 | 15 | 18 | 22 | 24 |
|---|---|---|---|---|---|
| Velocity, $v$ (m/s) | 22 | 24 | 37 | 25 | 123 |

Using the trapezoidal rule with unequal segments, the distance covered by the body from $t = 12$ s to $t = 18$ s is

(A) 162.90 m

(B) 166.00 m

(C) 181.70 m

(D) 436.50 m

For complete solution, go to

http://nm.mathforcollege.com/mcquizzes/07int/quiz_07int_trapcontinous_solution.pdf

# Problem Set

## Chapter 07.02: Trapezoidal Rule of Integration

(1). Find the value of $\int_{0.2}^{2.2} xe^x dx$ by using composite trapezoidal rule with two segments.

**Answer:** 14.033

(2). Find the value of $\int_{-3.4}^{2.2} x^2 e^x dx$ by using composite trapezoidal rule with four segments.

**Answer:** 33.875

(3). The upward velocity of a rocket is given by

$$v(t) = 200\ln(t+1) - 10t, t > 0$$

where $t$ is given in s and $v$ is given in m/s.

a) Use composite trapezoidal rule with two segments to calculate the distance covered by the rocket from $t = 0$ s to $t = 5$ s.

b) What is the true value of the distance covered by the rocket from $t = 0$ s to $t = 5$ s?

c) What is the true error in part (a)?

d) What is the relative true error in part (a)?

e) What is the absolute relative true error in percentage for part (a)?

f) Use composite trapezoidal rule with four segments to calculate the distance covered by the rocket from $t = 0$ s to $t = 5$ s.

g) What is the absolute relative approximate error in percentage for part (e), assuming the previous approximation is from part (a)?

h) Based on the answer from part (g), how many significant digits are correct in the answer in part (g).

**Answer:** *a*) 949.32 m
*b*) 1025.1 m
*c*) 75.790 m
*d*) 0.073933
*e*) 7.3933%
*f*) 1004.4 m
*g*) 5.4865%
*h*) 0

(4). Find the value of the integral $\int_1^5 3e^{-2t}dt$ using composite trapezoidal rule with two segments.

**Answer:** 0.42101

(5). For the integral $\int_1^4 xe^{-2x}dx$, find the following

a) Exact integral using your calculus knowledge (Hint: $\int udv = uv - \int vdu + C$)

b) Value of integral using composite trapezoidal rule with two segments.

c) Value of integral using composite trapezoidal rule with four segments.

d) Calculate the true error for part (b) and part (c). Is the true error for part (c) approximately a quarter of the true error for part (b)? Explain.

**Answer:** *a*) 0.10075
*b*) 0.12778
*c*) 0.10719
*d*) $-0.027028,\ -0.00644$

(6). In composite trapezoidal rule, if $E_n$ is the true error using $n$ ($n \geq 1$) segments, then the true error using composite trapezoidal rule with $2n$ sgements approximately is _____ of $E_n$.

**Answer:** 1/4

(7). The upward velocity of a body is given by

$$v(t) = \ln\left(\frac{160}{160 - 9t}\right), t > 0$$

where $t$ is given in seconds, and $v$ in m/s.

What is the distance covered by the body from $t = 5$ to $t = 9$ seconds? Use composite trapezoidal rule with four segments.

**Answer:** 2.0280 m

(8). The true error for the single application trapezoidal rule used to calculate the approximate value of the integral $\int_a^b f(x)dx$ is given by

$$E_t = -\frac{(b-a)^3}{12}f'(\zeta), a \leq \zeta \leq b$$

For the integral $\int_3^6 3e^{1.1x}dx$, what is the value of $\zeta$?

**Answer:** 4.7386

(9). The approximate value of the integral of $\int_1^5 f(x)dx$ is found by using single application trapezoidal rule as 0.35958. Given $f(2) = 0.7447$, $f(3) = 0.10454$ and $f(4) = -0.50730$, what is the approximate value of the integral of $\int_1^5 f(x)dx$ using the composite trapezoidal rule with two segments.

**Answer:** 0.38887

(10). The true error for trapezoidal rule used to calculate the approximate value of the integral $\int_a^b f(x)dx$ is given by

$$E_t = -\frac{(b-a)^3}{12} f''(\zeta), a \leq \zeta \leq b$$

For the integral $\int_2^5 (3x^2 + dx + e)dx$, where constants $d$ and $e$ are not given to you, the single application trapezoidal rule gives the value of the integral as 201.0. Find the exact value of the integral.

$$E_t = -\frac{(b-a)^3}{12} f'(\zeta), a \leq \zeta \leq b$$

**Answer:** 187.5

(11). A quadrature rule is developed by a scientist

$$\int_a^b f(x)dx \approx c_1 f(a).$$

The value of $c_1$ is found by assuming that the formula is exact for integrals of $f(x) = a_0 x^2$, where $a_0$ is an arbitrary constant. What is the value of $c_1$ in terms of $a$ and $b$.

**Answer:** $(b^3 - a^3)/(3a^2)$

# Chapter 07.03: Simpson's 1/3rd Rule of Integration

Lesson: Single Application Simpson's 1/3rd Rule

## Learning Objective

*After successful completion of this lesson, you should be able to:*

1) *derive the single application of Simpson's 1/3rd rule formula,*

2) *use the single application of Simpson's 1/3rd rule of integration to solve problems.*

## What is integration?

Integration is the process of measuring the area under a function plotted on a graph. Why would we want to integrate a function? Among the most common examples are finding the velocity of a body from an acceleration function, and displacement of a body from a velocity function. Throughout many engineering fields, there are (what sometimes seems like) countless applications for integral calculus. You can read about some of these applications in Chapters 07.00A-07.00G.

Sometimes, the evaluation of expressions involving these integrals can become daunting, if not indeterminate. For this reason, a wide variety of numerical methods has been developed to simplify the integral. Here, we will discuss Simpson's 1/3 rule of integral approximation, which improves upon the accuracy of the trapezoidal rule.

Here, we will discuss the Simpson's 1/3 rule of approximating integrals of the form

$$I = \int_a^b f(x)\,dx$$

where

$f(x)$ is called the integrand,

$a$ = lower limit of integration

$b$ = upper limit of integration

## Simpson's 1/3 Rule

The trapezoidal rule was based on approximating the integrand by a first order polynomial, and then integrating the polynomial over interval of integration. Simpson's 1/3 rule is an extension of Trapezoidal rule where the integrand is approximated by a second order polynomial.

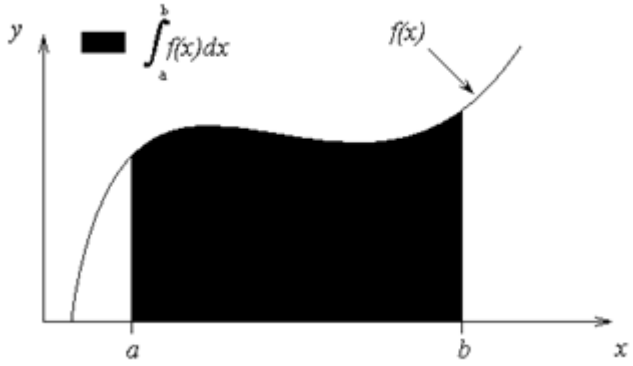



**Figure 1** Integration of a function

Hence

$$I = \int_a^b f(x)\,dx \approx \int_a^b f_2(x)\,dx$$

where $f_2(x)$ is a second order polynomial given by

$$f_2(x) = a_0 + a_1 x + a_2 x^2$$

Choose

$$(a, f(a)),\ \left(\frac{a+b}{2}, f\left(\frac{a+b}{2}\right)\right),\ \text{and}\ (b, f(b))$$

as the three points of the function to evaluate $a_1$ and $a_2$.

$$f(a) = f_2(a) = a_0 + a_1 a + a_2 a^2$$

$$f\left(\frac{a+b}{2}\right) = f_2\left(\frac{a+b}{2}\right) = a_0 + a_1\left(\frac{a+b}{2}\right) + a_2\left(\frac{a+b}{2}\right)^2$$

$$f(b) = f_2(b) = a_0 + a_1 b + a_2 b^2$$

Solving the above three equations for unknowns, $a_0$, $a_1$ and $a_2$ give

$$a_0 = \frac{a^2 f(b) + abf(b) - 4abf\left(\frac{a+b}{2}\right) + abf(a) + b^2 f(a)}{a^2 - 2ab + b^2}$$

$$a_1 = -\frac{af(a) - 4af\left(\frac{a+b}{2}\right) + 3af(b) + 3bf(a) - 4bf\left(\frac{a+b}{2}\right) + bf(b)}{a^2 - 2ab + b^2}$$

$$a_2 = \frac{2\left(f(a) - 2f\left(\frac{a+b}{2}\right) + f(b)\right)}{a^2 - 2ab + b^2}$$

Then

$$\begin{aligned} I &\approx \int_a^b f_2(x)dx \\ &= \int_a^b \left(a_0 + a_1 x + a_2 x^2\right) dx \\ &= \left[a_0 x + a_1 \frac{x^2}{2} + a_2 \frac{x^3}{3}\right]_a^b \\ &= a_0(b-a) + a_1 \frac{b^2 - a^2}{2} + a_2 \frac{b^3 - a^3}{3} \end{aligned}$$

Substituting values of $a_0$, $a_1$ and $a_2$ give

$$\int_a^b f_2(x)dx = \frac{b-a}{6}\left[f(a) + 4f\left(\frac{a+b}{2}\right) + f(b)\right]$$

Since for the single application of Simpson 1/3 rule, the interval $[a, b]$ is broken into 2 segments, the segment width

$$h = \frac{b-a}{2}$$

The Simpson's 1/3 rule can be rewritten as

$$\int_a^b f(x)dx \approx \frac{h}{3}\left[f(a) + 4f\left(\frac{a+b}{2}\right) + f(b)\right]$$

Since the above form has 1/3 in its formula, it is called Simpson's 1/3 rule.

**Example 1**

The distance covered by a rocket in meters from $t = 8s$ to $t = 30s$ is given by

$$x = \int_8^{30} \left(2000 \ln\left[\frac{140000}{140000 - 2100t}\right] - 9.8t\right) dt$$

a) Use Simpson's 1/3 rule to find the approximate value of $x$.

b) Find the true error, $E_t$.

c) Find the absolute relative true error, $|\epsilon_t|$.

**Solution**

a)

$$x \approx \frac{b-a}{6}\left[f(a)+4f\left(\frac{a+b}{2}\right)+f(b)\right]$$

$$a = 8$$

$$b = 30$$

$$\frac{a+b}{2} = 19$$

$$f(t) = 2000\ln\left[\frac{140000}{140000-2100t}\right] - 9.8t$$

$$f(8) = 2000\ln\left[\frac{140000}{140000-2100(8)}\right] - 9.8(8) = 177.27m/s$$

$$f(30) = 2000\ln\left[\frac{140000}{140000-2100(30)}\right] - 9.8(30) = 901.67m/s$$

$$f(19) = 2000\ln\left(\frac{140000}{140000-2100(19)}\right) - 9.8(19) = 484.75m/s$$

$$\begin{aligned} x &\approx \frac{b-a}{6}\left[f(a)+4f\left(\frac{a+b}{2}\right)+f(b)\right] \\ &= \left(\frac{30-8}{6}\right)[f(8)+4f(19)+f(30)] \\ &= \frac{22}{6}[177.27+4\times 484.75+901.67] \\ &= 11065.72m \end{aligned}$$

b) The exact value of the above integral is

$$\begin{aligned} x &= \int_8^{30}\left(2000\ln\left[\frac{140000}{140000-2100t}\right]-9.8t\right)dt \\ &= 11061.34m \end{aligned}$$

So the true error is

$$\begin{aligned} E_t = &\text{ True Value} - \text{Approximate Value} \\ &= 11061.34 - 11065.72 \\ &= -4.38m \end{aligned}$$

c) The absolute relative true error is

$$\begin{aligned}|\epsilon_t| &= \left|\frac{\text{True Error}}{\text{True Value}}\right| \times 100 \\ &= \left|\frac{-4.38}{11061.34}\right| \times 100 \\ &= 0.0396\%\end{aligned}$$

## Appendix: Alternative Methods of Deriving Simpson's 1/3rd rule (Not in syllabus)

Alternative Method 1:

Simpson's 1/3 rule can also be derived by approximating $f(x)$ by a second order polynomial using Newton's divided difference polynomial as

$$f_2(x) = b_0 + b_1(x-a) + b_2(x-a)\left(x - \frac{a+b}{2}\right)$$

where

$$b_0 = f(a)$$

$$b_1 = \frac{f\left(\frac{a+b}{2}\right) - f(a)}{\frac{a+b}{2} - a}$$

$$b_2 = \frac{\frac{f(b) - f\left(\frac{a+b}{2}\right)}{b - \frac{a+b}{2}} - \frac{f\left(\frac{a+b}{2}\right) - f(a)}{\frac{a+b}{2} - a}}{b-a}$$

Integrating Newton's divided difference polynomial gives us

$$\begin{aligned}\int_a^b f(x)dx &\approx \int_a^b f_2(x)dx \\ &= \int_a^b \left[b_0 + b_1(x-a) + b_2(x-a)\left(x - \frac{a+b}{2}\right)\right]dx \\ &= \left[b_0 x + b_1\left(\frac{x^2}{2} - ax\right) + b_2\left(\frac{x^3}{3} - \frac{(3a+b)x^2}{4} + \frac{a(a+b)x}{2}\right)\right]_a^b \\ &= b_0(b-a) + b_1\left(\frac{b^2-a^2}{2} - a(b-a)\right) \\ &\quad + b_2\left(\frac{b^3-a^3}{3} - \frac{(3a+b)\left(b^2-a^2\right)}{4} + \frac{a(a+b)(b-a)}{2}\right)\end{aligned}$$

Substituting values of $b_0$, $b_1$, and $b_2$ into this equation yields the same result as before

$$\begin{aligned}\int_a^b f(x)dx &\approx \frac{b-a}{6}\left[f(a)+4f\left(\frac{a+b}{2}\right)+f(b)\right]\\ &= \frac{h}{3}\left[f(a)+4f\left(\frac{a+b}{2}\right)+f(b)\right]\end{aligned}$$

Alternative Method 2:

One could even use the Lagrange polynomial to derive Simpson's formula. Notice any method of three-point quadratic interpolation can be used to accomplish this task. In this case, the interpolating function becomes

$$\begin{aligned}f_2(x) = &\frac{\left(x-\frac{a+b}{2}\right)(x-b)}{\left(a-\frac{a+b}{2}\right)(a-b)}f(a)+\frac{(x-a)(x-b)}{\left(\frac{a+b}{2}-a\right)\left(\frac{a+b}{2}-b\right)}f\left(\frac{a+b}{2}\right)\\ &+\frac{(x-a)\left(x-\frac{a+b}{2}\right)}{(b-a)\left(b-\frac{a+b}{2}\right)}f(b)\end{aligned}$$

Integrating this function gets

$$\begin{aligned}\int_a^b f_2(x)dx = &\left[\begin{array}{l}\dfrac{\frac{x^3}{3}-\frac{(a+3b)x^2}{4}+\frac{b(a+b)x}{2}}{\left(a-\frac{a+b}{2}\right)(a-b)}f(a)\\ +\dfrac{\frac{x^3}{3}-\frac{(a+b)x^2}{2}+abx}{\left(\frac{a+b}{2}-a\right)\left(\frac{a+b}{2}-b\right)}f\left(\frac{a+b}{2}\right)\\ +\dfrac{\frac{x^3}{3}-\frac{(3a+b)x^2}{4}+\frac{a(a+b)x}{2}}{(b-a)\left(b-\frac{a+b}{2}\right)}f(b)\end{array}\right]_a^b\\ = &\frac{\frac{b^3-a^3}{3}-\frac{(a+3b)\left(b^2-a^2\right)}{4}+\frac{b(a+b)(b-a)}{2}}{\left(a-\frac{a+b}{2}\right)(a-b)}f(a)\\ &+\frac{\frac{b^3-a^3}{3}-\frac{(a+b)\left(b^2-a^2\right)}{2}+ab(b-a)}{\left(\frac{a+b}{2}-a\right)\left(\frac{a+b}{2}-b\right)}f\left(\frac{a+b}{2}\right)\\ &+\frac{\frac{b^3-a^3}{3}-\frac{(3a+b)\left(b^2-a^2\right)}{4}+\frac{a(a+b)(b-a)}{2}}{(b-a)\left(b-\frac{a+b}{2}\right)}f(b)\end{aligned}$$

Believe it or not, simplifying and factoring this large expression yields you the same result as before

$$\begin{aligned}\int_a^b f(x)dx &\approx \frac{b-a}{6}\left[f(a)+4f\left(\frac{a+b}{2}\right)+f(b)\right]\\ &= \frac{h}{3}\left[f(a)+4f\left(\frac{a+b}{2}\right)+f(b)\right]\end{aligned}$$

Alternative Method 3:

Simpson's 1/3 rule can also be derived by the method of coefficients. Assume

$$\int_a^b f(x)dx \approx c_1 f(a) + c_2 f\left(\frac{a+b}{2}\right) + c_3 f(b)$$

Let the right-hand side be an exact expression for the integrals $\int_a^b 1dx$, $\int_a^b xdx$, and $\int_a^b x^2dx$. This implies that the right-hand side will be exact expressions for integrals of any linear combination of the three integrals for a general second order polynomial. Now

$$\int_a^b 1dx = b-a = c_1+c_2+c_3$$

$$\int_a^b xdx = \frac{b^2-a^2}{2} = c_1 a + c_2\frac{a+b}{2} + c_3 b$$

$$\int_a^b x^2dx = \frac{b^3-a^3}{3} = c_1 a^2 + c_2\left(\frac{a+b}{2}\right)^2 + c_3 b^2$$

Solving the above three equations for $c_1$, $c_2$ and $c_3$ give

$$c_1 = \frac{b-a}{6}$$

$$c_2 = \frac{2(b-a)}{3}$$

$$c_3 = \frac{b-a}{6}$$

This gives

$$\begin{aligned}\int_a^b f(x)dx &\approx \frac{b-a}{6}f(a) + \frac{2(b-a)}{3}f\left(\frac{a+b}{2}\right) + \frac{b-a}{6}f(b)\\ &= \frac{b-a}{6}\left[f(a)+4f\left(\frac{a+b}{2}\right)+f(b)\right]\\ &= \frac{h}{3}\left[f(a)+4f\left(\frac{a+b}{2}\right)+f(b)\right]\end{aligned}$$

The integral from the first method

$$\int_a^b f(x)dx \approx \int_a^b (a_0 + a_1 x + a_2 x^2)dx$$

can be viewed as the area under the second order polynomial, while the equation from Method 4

$$\int_a^b f(x)dx \approx \frac{b-a}{6} f(a) + \frac{2(b-a)}{3} f\left(\frac{a+b}{2}\right) + \frac{b-a}{6} f(b)$$

can be viewed as the sum of the areas of three rectangles.

# Chapter 07.03: Simpson's 1/3rd Rule of Integration

Lesson: Composite Simpson's 1/3rd Rule

## Learning Objective

*After successful completion of this lesson, you should be able to:*

1) *derive the composite (also called multiple-segment) Simpson's 1/3rd rule formula,*

2) *use the composite (also called multiple-segment) Simpson's 1/3rd rule of integration to solve problems.*

## Composite Simpson's 1/3 Rule

Just like in composite trapezoidal rule, one can subdivide the interval $[a, b]$ into $n$ segments and apply Simpson's 1/3 rule repeatedly over every two segments. Note that $n$ needs to be even. Divide interval $[a, b]$ into $n$ equal segments, so that the segment width is given by

$$h = \frac{b-a}{n}$$

Now

$$\int_a^b f(x)dx = \int_{x_0}^{x_n} f(x)dx$$

where

$$x_0 = a$$

$$x_n = b$$

$$\int_a^b f(x)dx = \int_{x_0}^{x_2} f(x)dx + \int_{x_2}^{x_4} f(x)dx + \ldots\ldots + \int_{x_{n-4}}^{x_{n-2}} f(x)dx + \int_{x_{n-2}}^{x_n} f(x)dx$$

Apply Simpson's 1/3rd Rule over each interval,

$$\int_a^b f(x)dx \cong (x_2 - x_0)\left[\frac{f(x_0) + 4f(x_1) + f(x_2)}{6}\right] + (x_4 - x_2)\left[\frac{f(x_2) + 4f(x_3) + f(x_4)}{6}\right] + \ldots + (x_{n-2} - x_{n-4})\left[\frac{f(x_{n-4}) + 4f(x_{n-3}) + f(x_{n-2})}{6}\right] + (x_n - x_{n-2})\left[\frac{f(x_{n-2}) + 4f(x_{n-1}) + f(x_n)}{6}\right]$$

Since

$$x_i - x_{i-2} = 2h$$

$$i = 2, 4, \ldots, n$$

then

$$\int_a^b f(x)dx \cong 2h\left[\frac{f(x_0) + 4f(x_1) + f(x_2)}{6}\right] + 2h\left[\frac{f(x_2) + 4f(x_3) + f(x_4)}{6}\right] + \ldots + 2h\left[\frac{f(x_{n-4}) + 4f(x_{n-3}) + f(x_{n-2})}{6}\right] + 2h\left[\frac{f(x_{n-2}) + 4f(x_{n-1}) + f(x_n)}{6}\right]$$

$$= \frac{h}{3}\left[f(x_0) + 4\left\{f(x_1) + f(x_3) + \ldots + f(x_{n-1})\right\} + 2\left\{f(x_2) + f(x_4) + \ldots + f(x_{n-2})\right\} + f(x_n)\right]$$

$$= \frac{h}{3}\left[f(x_0) + 4\sum_{\substack{i=1 \\ i=\text{odd}}}^{n-1} f(x_i) + 2\sum_{\substack{i=2 \\ i=\text{even}}}^{n-2} f(x_i) + f(x_n)\right]$$

$$\int_a^b f(x)dx \cong \frac{b-a}{3n}\left[f(x_0) + 4\sum_{\substack{i=1 \\ \text{i=odd}}}^{n-1} f(x_i) + 2\sum_{\substack{i=2 \\ i=\text{even}}}^{n-2} f(x_i) + f(x_n)\right]$$

**Example 1**

Use composite Simpson's 1/3 rule with 4 segments to approximate the distance covered by a rocket in meters from $t = 8$s to $t = 30$s as given by

$$x = \int_8^{30}\left(2000\ln\left[\frac{140000}{140000 - 2100t}\right] - 9.8t\right)dt$$

a) Use composite Simpson's 1/3rd rule with 4 segments to estimate $x$.

b) Find the true error, $E_t$ for part (a).

c) Find the absolute relative true error, $|\epsilon_t|$ for part (a).

**Solution:**

a) Using composite Simpson's 1/3 rule,

$$x \approx \frac{b-a}{3n}\left[f\left(t_0\right)+4\sum_{\substack{i=1\\ \text{i=odd}}}^{n-1} f\left(t_i\right)+2\sum_{\substack{i=2\\ i=\text{even}}}^{n-2} f\left(t_i\right)+f\left(t_n\right)\right]$$

$$n=4$$

$$a=8$$

$$b=30$$

$$\begin{aligned} h &= \frac{b-a}{n} \\ &= \frac{30-8}{4} \\ &= 5.5 \end{aligned}$$

$$f(t) = 2000\ln\left[\frac{140000}{140000-2100t}\right]-9.8t$$

So

$$f\left(t_0\right) = f(8)$$

$$\begin{aligned} f(8) &= 2000\ln\left[\frac{140000}{140000-2100(8)}\right]-9.8(8) \\ &= 177.27m/s \end{aligned}$$

$$f\left(t_1\right) = f(8+5.5) = f(13.5)$$

$$\begin{aligned} f(13.5) &= 2000\ln\left[\frac{140000}{140000-2100(13.5)}\right]-9.8(13.5) \\ &= 320.25m/s \end{aligned}$$

$$f\left(t_2\right) = f(13.5+5.5) = f(19)$$

$$\begin{aligned} f(19) &= 2000\ln\left(\frac{140000}{140000-2100(19)}\right)-9.8(19) \\ &= 484.75m/s \end{aligned}$$

$$f(t_3) = f(19 + 5.5) = f(24.5)$$

$$\begin{aligned} f(24.5) &= 2000 \ln \left[ \frac{140000}{140000 - 2100(24.5)} \right] - 9.8(24.5) \\ &= 676.05 m/s \end{aligned}$$

$$f(t_4) = f(t_n) = f(30)$$

$$\begin{aligned} f(30) &= 2000 \ln \left[ \frac{140000}{140000 - 2100(30)} \right] - 9.8(30) \\ &= 901.67 m/s \end{aligned}$$

$$\begin{aligned} x &= \frac{b-a}{3n} \left[ f(t_0) + 4 \sum_{\substack{i=1 \\ i=\text{odd}}}^{n-1} f(t_i) + 2 \sum_{\substack{i=2 \\ i=\text{even}}}^{n-2} f(t_i) + f(t_n) \right] \\ &= \frac{30-8}{3(4)} \left[ f(8) + 4 \sum_{\substack{i=1 \\ i=\text{odd}}}^{3} f(t_i) + 2 \sum_{\substack{i=2 \\ i=\text{even}}}^{2} f(t_i) + f(30) \right] \\ &= \frac{22}{12} \left[ f(8) + 4f(t_1) + 4f(t_3) + 2f(t_2) + f(30) \right] \\ &= \frac{11}{6} [f(8) + 4f(13.5) + 4f(24.5) + 2f(19) + f(30)] \\ &= \frac{11}{6} [177.27 + 4(320.25) + 4(676.05) + 2(484.75) + 901.67] \\ &= 11061.64 m \end{aligned}$$

b) The exact value of the above integral is

$$\begin{aligned} x &= \int_8^{30} \left( 2000 \ln \left[ \frac{140000}{140000 - 2100t} \right] - 9.8t \right) dt \\ &= 11061.34 m \end{aligned}$$

So the true error is

$$\begin{aligned} E_t &= \text{ True Value} - \text{Approximate Value} \\ E_t &= 11061.34 - 11061.64 \\ &= -0.30 m \end{aligned}$$

c) The absolute relative true error is

$$
\begin{aligned}
|\epsilon_t| &= \left| \frac{\text{True Error}}{\text{True Value}} \right| \times 100 \\
&= \left| \frac{-0.3}{11061.34} \right| \times 100 \\
&= 0.0027\%
\end{aligned}
$$

**Table 1** Values of composite Simpson's 1/3 rule for Example 1

| n | Approximate Value | $E_t$ | $\lvert\varepsilon_t\rvert$ |
|---|---|---|---|
| 2 | 11065.72 | $-4.38$ | 0.0396% |
| 4 | 11061.64 | $-0.30$ | 0.0027% |
| 6 | 11061.40 | $-0.06$ | 0.0005% |
| 8 | 11061.35 | $-0.02$ | 0.0002% |
| 10 | 11061.34 | $-0.01$ | 0.0001% |

## Error in Composite Simpson's 1/3 Rule

The true error in a single application of Simpson's 1/3rd Rule for $\int_a^b f(x)\text{dx}$ is given by

$$
E_t = -\frac{(b-a)^5}{2880} f^{(4)}(\zeta),\ a < \zeta < b
$$

where $\zeta$ is some point in $[a, b]$ and $f^{(4)}$ stands for the fourth derivative of the function.

What is the error then in the composite Simpson's $1/3^{\text{rd}}$ rule for $\int_a^b f(x)dx$?

$$
E_t = -\frac{(b-a)^5}{180n^4} \overline{f}^{\,(4)}
$$

where

$$
\bar{f}^{(4)} = \frac{\sum_{i=1}^{\frac{n}{2}} f^{(4)}(\zeta_i)}{\frac{n}{2}}
$$

$n$ = number of segments

# Chapter 07.03: Simpson's 1/3rd Rule of Integration

Lesson: Error in Composite Simpson's 1/3rd rule

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *derive the formula for the true error in the composite (also called multiple-segment) Simpson's 1/3 rule of integration.*

## True Error in Composite Simpson's 1/3 rule

The true error in a single application of Simpson's 1/3rd Rule is given by

$$E_t = -\frac{(b-a)^5}{2880} f^{(4)}(\zeta), a < \zeta < b$$

The $f^{(4)}$ in the true error expression stands for the fourth derivative of the function

In composite Simpson's 1/3 rule, the error is the sum of the errors in each application of Simpson's 1/3 rule. The error in the $n$ segments Simpson's 1/3rd Rule is given by

$$
\begin{aligned}
E_1 &= -\frac{(x_2 - x_0)^5}{2880} f^{(4)}(\zeta_1), x_0 < \zeta_1 < x_2 \\
&= -\frac{h^5}{90} f^{(4)}(\zeta_1) \\
E_2 &= -\frac{(x_4 - x_2)^5}{2880} f^{(4)}(\zeta_2), x_2 < \zeta_2 < x_4 \\
&= -\frac{h^5}{90} f^{(4)}\left(\zeta_2\right) \\
&\vdots \\
E_i &= -\frac{\left(x_{2i} - x_{2(i-1)}\right)^5}{2880} f^{(4)}\left(\zeta_i\right), \quad x_{2(i-1)} < \zeta_i < x_{2i} \\
&= -\frac{h^5}{90} f^{(4)}\left(\zeta_i\right) \\
&\vdots \\
E_{\frac{n}{2}-1} &= -\frac{(x_{n-2} - x_{n-4})^5}{2880} f^{(4)}\left(\zeta_{\frac{n}{2}-1}\right), x_{n-4} < \zeta_{\frac{n}{2}-1} < x_{n-2} \\
&= -\frac{h^5}{90} f^{(4)}\left(\zeta_{\frac{n}{2}-1}\right) \\
E_{\frac{n}{2}} &= -\frac{(x_n - x_{n-2})^5}{2880} f^{(4)}\left(\zeta_{\frac{n}{2}}\right), x_{n-2} < \zeta_{\frac{n}{2}} < x_n
\end{aligned}
$$

Hence, the total error in the composite Simpson's 1/3 rule is

$$
\begin{aligned}
&= -\frac{h^5}{90} f^{(4)}\left(\zeta_{\frac{n}{2}}\right) \\
E_t &= \sum_{i=1}^{\frac{n}{2}} E_i \\
&= -\frac{h^5}{90} \sum_{i=1}^{\frac{n}{2}} f^{(4)}\left(\zeta_i\right) \\
&= -\frac{(b-a)^5}{90n^5} \sum_{i=1}^{\frac{n}{2}} f^{(4)}\left(\zeta_i\right) \\
&= -\frac{(b-a)^5}{180n^4} \frac{\sum_{i=1}^{\frac{n}{2}} f^{(4)}(\zeta_i)}{\frac{n}{2}},
\end{aligned}
$$

The term $\dfrac{\sum_{i=1}^{\frac{n}{2}} f^{(4)}(\zeta_i)}{\frac{n}{2}}$ is an approximate average value of $f^{(4)}(x), a < x < b$. Hence

$$
E_t = -\frac{(b-a)^5}{180n^4} \overline{f}^{(4)}
$$

where

$$\bar{f}^{(4)} = \frac{\sum_{i=1}^{\frac{n}{2}} f^{(4)}(\zeta_i)}{\frac{n}{2}}$$

# Multiple Choice Test

## Chapter 07.03 Simpson's 1/3rd Rule of Integration

(1). The highest order of polynomial integrand for which Simpson's 1/3 rule of integration is exact is

(A) first

(B) second

(C) third

(D) fourth

(2). The value of $\int_{0.2}^{2.2} e^x \mathrm{dx}$ by using single-application (2-segments) Simpson's 1/3 rule most nearly is

(A) 7.8036

(B) 7.8423

(C) 8.4433

(D) 10.246

(3). The value of $\int_{0.2}^{2.2} e^x \mathrm{dx}$ by using composite Simpson's 1/3 rule with 4 segments most nearly is

(A) 7.8036

(B) 7.8062

(C) 7.8423

(D) 7.9655

(4). The velocity of a body is given by

$$\begin{aligned} v(t) &= 2t,\ 1 \leq t \leq 5 \\ &= 5t^2 + 3,\ 5 < t \leq 14 \end{aligned}$$

where $t$ is given in seconds, and $v$ is given in $m/s$. Using single-application (two-segment) Simpson's 1/3 rule, the distance in meters covered by the body from $t = 2$ to $t = 9$ seconds most nearly is

(A) 949.33

(B) 1039.7

(C) 1200.5

(D) 1442.0

(5). The value of $\int_3^{19} f(x)\,\mathrm{dx}$ by using single-application (2-segment) Simpson's 1/3 rule is estimated as 702.039. The estimate of the same integral using composite Simpson's 1/3 rule with 4 segments most nearly is

(A) $702.039 + \frac{8}{3}\left[2f(7) - f(11) + 2f(15)\right]$

(B) $\frac{702.039}{2} + \frac{8}{3}\left[2f(7) - f(11) + 2f(15)\right]$

(C) $702.039 + \frac{8}{3}\left[2f(7) + 2f(15)\right]$

(D) $\frac{702.039}{2} + \frac{8}{3}\left[2f(7)\,2f(15)\right]$

(6). The following data of the velocity of a body is given as a function of time.

| $Time\ (s)$ | 4 | 7 | 10 | 15 |
|---|---|---|---|---|
| $Velocity\ (m/s)$ | 22 | 24 | 37 | 46 |

The best estimate of the distance in meters covered by the body from $t = 4$ to $t = 15$ using combined Simpson's 1/3 rule and the trapezoidal rule would be

(A) 354.70

(B) 362.50

(C) 368.00

(D) 378.80

For complete solution, go to

http://nm.mathforcollege.com/mcquizzes/07int/quiz_07int_simpson_solution.pdf

# Problem Set

## Chapter 07.03: Simpson's 1/3rd Rule of Integration

(1). Find the value of $\int_{0.2}^{2.2} xe^x dx$ by using single-application composite Simpson's 1/3 rule.

**Answer:** 12.012

(2). Find the value of $\int_{-3.4}^{2.2} x^2 e^x dx$ by using composite Simpson's 1/3 rule. Use 4 segments.

**Answer:** 24.418

(3). The upward velocity of a rocket is given by

$$v(t) = 200\ln(t+1) - 10t,\ t > 0$$

where $t$ is given in seconds and $v$ is given in m/s.

a) Use single-application Simpson's 1/3 rule to calculate the distance covered by the rocket from $t = 0$ to $t = 5$s.

b) What is the true value of the distance covered by the rocket from $t = 0$ to $t = 5$s?

c) What is the true error in part (a)?

d) What is the relative true error in part (a)?

e) What is the absolute relative true error in percentage for part (a).

f) Use composite Simpson's 1/3rd rule to calculate the distance covered by the rocket from $t = 0$ to $t = 5$s. Use 4 segments.

g) What is the absolute relative approximate error in percentage for part (e), assuming the previous approximation is from part (a)?

h) Based on the answer from part (g), how many significant digits are correct in the answer in part (g).

**Answer:** *a*) 1008.8 m *b*) 1025.1 m *c*) 16.309 m *d*) 0.015910 *e*) 1.5910% *f*) 1022.8 m *g*) 1.3685% *h*) 1

(4). The upward velocity of a body is given by

$$v(t) = \ln\left(\frac{160}{160 - 9t}\right),\ t > 0,$$

where $t$ is given in seconds, and $v$ in m/s. What is the distance covered by the body from $t = 5$ to $t = 9$ seconds? Use composite Simpson's 1/3 rule with 4 segments.

**Answer:** 2.0251 m

(5). The true error for the single-application Simpson's 1/3 rule used to calculate the approximate value of the integral $\int_a^b f(x)dx$ is given by

$$E_t = -\frac{(b-a)^5}{2880} f^{(4)}(\xi), a \leq \xi \leq b$$

For the integral $\int_3^6 4e^{2x}dx$, what is the value of $\xi$?

**Answer:** $\xi = 4.7088$

(6). The true error for the single application Simpson's 1/3 rule used to calculate the approximate value of the integral $\int_a^b f(x)dx$ is given by

$$E_t = -\frac{(b-a)^5}{2880} f^{(4)}(\xi), a \leq \xi \leq b$$

For the integral $\int_3^6 4e^{2x}dx$,

a) find the lower and upper estimate of the true error based on the true error formula.

b) find the exact value of the integral.

c) verify that the estimates of part (a) bracket the true error.

**Answer:** *a)* $E_{\text{upper}} = -878880$, $E_{\text{lower}} = -2178.5$ *b)* Exact Integral = 324700, Approximate Integral = 391140

# Chapter 07.05: Gauss Quadrature Rule of Integration

Lesson: Theory of Gauss Quadrature Rule

## Learning Objective

*After successful completion of this lesson, you should be able to:*

1) *derive the Gauss quadrature method for integration.*

## Introduction

The two-point Gauss quadrature rule is an extension of the trapezoidal rule approximation where the arguments of the function are not predetermined. In single-segment trapezoidal rule for approximating the integral

$$\int_a^b f(x)dx,$$

the end-points $a$ and $b$ are chosen as the quadrature points, but in the two-point Gaussian quadrature rule, two points $x_1$ and $x_2$ somewhere between the end-points $a$ and $b$ are chosen. How do we find these quadrature points is a question, and so is why we wish to find these points? The answer to the "why" question is to gain more accuracy for the same order of computational time over the trapezoidal rule. The derivation and the example following will answer both these questions.

Before we derive the two-point Gaussian quadrature rule, we derive the trapezoidal rule from the method of undetermined coefficients. Why? Because that is the method used to derive the two-point Gaussian quadrature rule as well.

## Trapezoidal Rule Derivation Using Method of Undetermined Coefficients

Choose the integral $\int_a^b f(x)dx$ approximated as follows.

$$\int_a^b f(x)dx \approx c_1 f(a) + c_2 f(b) \qquad (1)$$

The coefficients $c_1$ and $c_2$ are undetermined. We find these coefficients such that right-hand side is exact for integrals of first-order polynomial $a_0 + a_1 x$.

From exact integration

$$\int_a^b (a_0 + a_1 x)\, dx = \left[ a_0 x + a_1 \frac{x^2}{2} \right]_a^b$$
$$= a_0 (b - a) + a_1 \left( \frac{b^2 - a^2}{2} \right) \qquad (2)$$

But we want the right-hand side of Equation (1) to give the same result as Equation (2) for $f(x) = a_0 + a_1 x$ which gives

$$c_1 f(a) + c_2 f(b) = c_1(a_0 + a_1 b) + c_2(a_0 + a_1 b)$$
$$= a_0 (c_1 + c_2) + a_1 (c_1 a + c_2 b) \qquad (3)$$

Hence from Equations (2) and (3),

$$a_0 (b - a) + a_1 \left( \frac{b^2 - a^2}{2} \right) = a_0 (c_1 + c_2) + a_1 (c_1 a + c_2 b) \qquad (4)$$

Since $a_0$ and $a_1$ are arbitrary constants for the chosen general straight line, the coefficients of $a_0$ and $a_1$ need to be equal. That gives

$$c_1 + c_2 = b - a \qquad (5a)$$

$$c_1 a + c_2 b = \frac{b^2 - a^2}{2} \qquad (5b)$$

Multiplying Equation (5a) by $a$ and subtracting from Equation (5b) gives

$$c_2 = \frac{b - a}{2} \qquad (6a)$$

Substituting the value of $c_2$ from Equation (6a) in Equation (5a) gives

$$c_1 = \frac{b - a}{2} \qquad (6b)$$

Therefore

$$\int_a^b f(x) dx \approx c_1 f(a) + c_2 f(b)$$
$$= \frac{b - a}{2} f(a) + \frac{b - a}{2} f(b) \qquad (7)$$

## Two-point Gaussian Quadrature Rule Using Method of Undetermined Coefficients

In the two-point Gauss quadrature rule, an integral instead of Equation (1) is approximated as

$$\int_a^b f(x)dx \approx c_1 f(x_1) + c_2 f(x_2) \qquad (8)$$

The quadrature points are the two additional unknowns. There are four unknowns in Equation (8) – the two quadrature points $x_1$, $x_2$, and the two weights $c_1$ and $c_2$. Hence, these are found by assuming that the formula gives exact results for integrating a general third-order polynomial, $f(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3$.

$$\begin{aligned}\int_a^b f(x)dx &= \int_a^b (a_0 + a_1 x + a_2 x^2 + a_3 x^3)dx \\ &= \left[a_0 x + a_1 \frac{x^2}{2} + a_2 \frac{x^3}{3} + a_3 \frac{x^4}{4}\right]_a^b \\ &= a_0 (b-a) + a_1 \left(\frac{b^2 - a^2}{2}\right) + a_2 \left(\frac{b^3 - a^3}{3}\right) + a_3 \left(\frac{b^4 - a^4}{4}\right) \qquad (9)\end{aligned}$$

But we want the right-hand side of Equation (8) to give the same result as Equation (9) for $f(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3$ which is

$$c_1 f(x_1) + c_2 f(x_2) = c_1 \left(a_0 + a_1 x_1 + a_2 x_1^2 + a_3 x_1^3\right) + c_2 \left(a_0 + a_1 x_2 + a_2 x_2^2 + a_3 x_2^3\right) \qquad (10)$$

Equating Equations (9) and (10) gives

$$\begin{aligned}&a_0(b-a) + a_1(\frac{b^2 - a^2}{2}) + a_2(\frac{b^3 - a^3}{3}) + a_3(\frac{b^4 - a^4}{4}) \\ &= c_1(a_0 + a_1 x_1 + a_2 x_1^2 + a_3 x_1^3) + c_2(a_0 + a_1 x_2 + a_2 x_2^2 + a_3 x_2^3) \\ &= a_0(c_1 + c_2) + a_1(c_1 x_1 + c_2 x_2) + a_2(c_1 x_1^2 + c_2 x_2^2) + a_3(c_1 x_1^3 + c_2 x_2^3) \qquad (11)\end{aligned}$$

Since in Equation (11), the constants $a_0$, $a_1$, $a_2$, and $a_3$ are arbitrary in the $f(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3$, the coefficients of $a_0$, $a_1$, $a_2$, and $a_3$ must be equal. This equating gives us four equations as follows.

$$b - a = c_1 + c_2$$

$$\frac{b^2 - a^2}{2} = c_1 x_1 + c_2 x_2$$

$$\frac{b^3 - a^3}{3} = c_1 {x_1}^2 + c_2 {x_2}^2$$

$$\frac{b^4 - a^4}{4} = c_1 {x_1}^3 + c_2 {x_2}^3 \qquad (12a - d)$$

Without proof, we can find that the above four simultaneous nonlinear equations have only one acceptable solution

$$c_1 = \frac{b-a}{2}$$

$$c_2 = \frac{b-a}{2}$$

$$x_1 = \left(\frac{b-a}{2}\right)\left(-\frac{1}{\sqrt{3}}\right) + \frac{b+a}{2}$$

$$x_2 = \left(\frac{b-a}{2}\right)\left(\frac{1}{\sqrt{3}}\right) + \frac{b+a}{2} \qquad (13)$$

Hence from Equations (1) and (13),

$$\begin{aligned}\int_a^b f(x)dx &\approx c_1 f(x_1) + c_2 f(x_2) \\ &= \frac{b-a}{2} f\left(\frac{b-a}{2}\left(-\frac{1}{\sqrt{3}}\right) + \frac{b+a}{2}\right) \\ &+ \frac{b-a}{2} f\left(\frac{b-a}{2}\left(\frac{1}{\sqrt{3}}\right) + \frac{b+a}{2}\right) \qquad (14)\end{aligned}$$

## One-point Gaussian Quadrature Rule Using Method of Undetermined Coefficients

Since we derived the two-point Gaussian quadrature rule without showing complete proof because of its length, we show the complete proof of the one-point Gauss quadrature rule. The derivation is based on approximating the integral by using the function value at a single point $x_1$ and giving it some weight $c_1$ and is given by

$$\int_a^b f(x)dx \approx c_1 f(x_1) \qquad (15)$$

To derive the values of $c_1$ and $x_1$, we assume the formula in Equation (15) gives exact values for an integral of a linear polynomial of the form $a_0 + a_1 x$.

$$\begin{aligned}\int_a^b (a_0 + a_1 x)dx &= \left[a_0 x + \frac{a_1 x^2}{2}\right]_a^b \\ &= a_0(b - a) + a_1\left(\frac{b^2 - a^2}{2}\right) \qquad (16)\end{aligned}$$

From the assumed formula

$$\begin{aligned} c_1 f(x_1) &= c_1(a_0 + a_1 x_1) \\ &= c_1 a_0 + c_1 a_1 x_1 \end{aligned} \qquad (17)$$

Equating Equations (16) and (17) gives

$$a_0(c_1) + a_1(c_1 x_1) = a_0(b \ - \ a) + a_1 \left( \frac{b^2 - a^2}{2} \right)$$

Since $a_0$ and $a_1$ are arbitrary, we can equate the coefficients of $a_0$ and $a_1$, giving

$$c_1 = b - a \qquad (18)$$

$$c_1 x_1 = \left( \frac{b^2 - a^2}{2} \right) \qquad (19)$$

Using Equation (18) in Equation (19) gives

$$(b - a) x_1 = \frac{b^2 - a^2}{2}$$

$$x_1 = \frac{b + a}{2} \qquad (20)$$

Therefore, one-point Gauss quadrature rule can be expressed as

$$\begin{aligned} \int_a^b f(x)\, dx &\approx c_1 f(x_1) \\ &= (b - a) f(\frac{b + a}{2}) \end{aligned} \qquad (21)$$

# Chapter 07.05: Gauss Quadrature Rule of Integration

Lesson: Applications of Gauss Quadrature Rule

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *use the Gauss quadrature rule to approximate definite integrals.*

## Recap

In the previous lesson, you learned the theory of the Gauss quadrature rule. In this lesson, we apply the formulas of the rule to approximate definite integrals.

### Example 1

The following integral is given

$$\int_{0.1}^{1.3} 5xe^{-2x}dx$$

a) Use the two-point Gauss quadrature rule to estimate the value of the integral.

b) Find the true error for part (a).

c) Find the absolute relative true error for part (a).

**Solution**

a) The two-point Gauss quadrature rule is given by

$$\int_a^b f(x)dx \approx c_1 f(x_1) + c_2 f(x_2)$$

where

$$c_1 = \frac{b-a}{2}$$

$$x_1 = \frac{b-a}{2}\left(-\frac{1}{\sqrt{3}}\right) + \frac{b+a}{2}$$

$$c_2 = \frac{b-a}{2}$$

$$x_2 = \frac{b-a}{2}\left(\frac{1}{\sqrt{3}}\right) + \frac{b+a}{2}$$

For this example

$$f(x) = 5xe^{-2x},\ a = 0.1,\ b = 1.3$$

So

$$c_1 = \frac{1.3-0.1}{2} = 0.6$$

$$x_1 = \frac{1.3-0.1}{2}\left(-\frac{1}{\sqrt{3}}\right) + \frac{1.3+0.1}{2} = 0.35359$$

$$c_2 = \frac{1.3-0.1}{2} = 0.6$$

$$x_2 = \frac{1.3-0.1}{2}\left(\frac{1}{\sqrt{3}}\right) + \frac{1.3+0.1}{2} = 1.0461$$

Then

$$\begin{aligned}\int_{0.1}^{1.3} f(x)dx &\approx c_1 f(x_1) + c_2 f(x_2) \\ &= 0.6f(0.35359) + 0.6f(1.04641)\end{aligned}$$

Finding the values of the function at the two quadrature points

$$\begin{aligned}f(0.35359) &= 5(0.35359)e^{-2(0.35359)} \\ &= 0.87166\end{aligned}$$

$$\begin{aligned}f(1.04641) &= 5(1.04641)e^{-2(1.04641)} \\ &= 0.64531\end{aligned}$$

we get

$$\begin{aligned}\int_{0.1}^{1.3} f(x)dx &\approx 0.6 \times 0.87166 + 0.6 \times 0.64531 \\ &= 0.91018\end{aligned}$$

b) The exact value of the above integral can be found as

$$\int_{0.1}^{1.3} 5xe^{-2x}dx = 0.89386$$

The true error

$$\begin{aligned} E_t &= \text{True Value} - \text{Approximate Value} \\ &= 0.89386 \ - \ 0.91018 \\ &= -0.01632 \end{aligned}$$

c) The absolute relative true error $|\varepsilon_t|$ is

$$\begin{aligned} |\varepsilon_t| &= \left| \frac{\text{True Value} - \text{Approximate Value}}{\text{True Value}} \right| \times 100 \\ &= \left| \frac{0.89386 \ - 0.91018}{0.89386} \right| \times 100 \\ &= 1.8258\% \end{aligned}$$

**Example 2**

The following integral is given

$$\int_{0.1}^{1.3} 5xe^{-2x}dx$$

a) Use the one-point Gaussian quadrature rule to estimate the value of the integral.

b) Find the true error for part (a).

c) Find the absolute relative true error, $|\epsilon_t|$, for part (a).

**Solution**

a)

$$\int_a^b f(x)dx \approx (b-a)f\left(\frac{a+b}{2}\right)$$

where

$$a = 0.1$$

$$b = 1.3$$

$$\begin{aligned}\int_{0.1}^{1.3} f(x)dx &\approx (1.3-0.1)f\left(\frac{1.3+0.1}{2}\right)\\ &= (1.2)\,f\,(0.7)\\ &= (1.2)\left(5\,(0.7)\,e^{-2(0.7)}\right)\\ &= 1.0357\end{aligned}$$

b) The true value is

$$\begin{aligned}E_t &= \text{True value} - \text{Approximate value}\\ &= 0.89386 - 1.0357\\ &= -0.14183\end{aligned}$$

c) The absolute relative true error, $|\epsilon_t|$, would be

$$\begin{aligned}|\epsilon_t| &= \left|\frac{\text{True error}}{\text{True value}}\right| \times 100\\ &= \left|\frac{-0.14183}{0.89386}\right| \times 100\\ &= 15.86\%\end{aligned}$$

**Example 3**

Use two-point Gauss quadrature rule to approximate the distance covered by a rocket from $t = 8$ to $t = 30$ seconds as given by

$$x = \int_8^{30} \left(2000 \ln\left[\frac{140000}{140000 - 2100t}\right] - 9.8t\right) dt$$

Also, find the absolute relative true error.

**Solution**

First, change the limits of integration from $[8, 30]$ to $[-1, 1]$ then

$$\begin{aligned}\int_8^{30} f(t)dt &= \frac{30-8}{2}\int_{-1}^{1} f\left(\frac{30-8}{2}x + \frac{30+8}{2}\right)dx\\ &= 11\int_{-1}^{1} f\,(11x+19)\,dx\end{aligned}$$

The weighting factors and function argument values for the two point rule are,

$$c_1 = 1.000000000$$

$$x_1 = -0.577350269$$

$$c_2 = 1.000000000$$

$$x_2 = 0.577350269$$

Now we can use the Gauss quadrature formula

$$\begin{aligned}
11\int_{-1}^{1} f\left(11x+19\right)\mathrm{dx} &\approx 11\left[c_1 f\left(11x_1+19\right)+c_2 f\left(11x_2+19\right)\right] \\
&= 11\left[f\left(11(-0.5773503)+19\right)+f\left(11(0.5773503)+19\right)\right] \\
&= 11\left[f(12.64915)+f(25.35085)\right] \\
&= 11\left[(296.8317)+(708.4811)\right] \\
&= 11058.44 \text{ m}
\end{aligned}$$

where

$$\begin{aligned}
f(12.64915) &= 2000\ln\left[\frac{140000}{140000-2100(12.64915)}\right]-9.8(12.64915) \\
&= 296.8317
\end{aligned}$$

$$\begin{aligned}
f(25.35085) &= 2000\ln\left[\frac{140000}{140000-2100(25.35085)}\right]-9.8(25.35085) \\
&= 708.4811
\end{aligned}$$

The absolute relative true error, $|\epsilon_t|$, is (True value = 11061.34 m)

$$\begin{aligned}
|\epsilon_t| &= \left|\frac{11061.34-11058.44}{11061.34}\right|\times 100 \\
&= 0.0262\%
\end{aligned}$$

# Chapter 07.05: Gauss Quadrature Rule of Integration

Lesson: Higher-Point Gauss Quadrature Rule

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *apply higher-point Gauss quadrature formulas to estimate integrals*

2) *use a table of abscissas and weights to apply the Gauss quadrature rule.*

## Introduction

We have discussed one-point and two-point Gauss quadrature rules in a previous lesson. So, how do higher-point Gauss quadrature rules work? For example, the three-point Gauss quadrature rule is given by

$$\int_a^b f(x)dx \approx c_1 f(x_1) + c_2 f(x_2) + c_3 f(x_3) \qquad (1)$$

The coefficients (also called weighting factors) $c_1$, $c_2$, and $c_3$, and the function arguments $x_1$, $x_2$ and $x_3$ are calculated by assuming the formula gives exact expressions for integrating a fifth-order polynomial

$$\int_a^b \left(a_0 + a_1 x + a_2 x^2 + a_3 x^3 + a_4 x^4 + a_5 x^5\right) dx$$

.

General $n$ -point rules would approximate an integral as

$$\int_a^b f(x)dx \approx c_1 f(x_1) + c_2 f(x_2) + ..... + c_n f(x_n) \qquad (2)$$

## Arguments and weighing factors for *n*-point Gauss quadrature rules

In handbooks (see Table 1), coefficients and arguments for $n$-point Gauss quadrature rule are not given for integrals of the form $\int_a^b f(x)dx$ but for integrals of the form $\int_{-1}^1 g(x)dx$, that is,

$$\int_{-1}^{1} g(x)dx \approx \sum_{i=1}^{n} c_i g(x_i) \qquad (3)$$

**Table 1** Weighting factors $c$ and function arguments $x$ used in Gauss quadrature formulas

| Points | Weighting Factors | Function Arguments |
|---|---|---|
| 1 | $c_1 = 2.000000000$ | $x_1 = 0.000000000$ |
| 2 | $c_1 = 1.000000000$ | $x_1 = -0.577350269$ |
| | $c_2 = 1.000000000$ | $x_2 = 0.577350269$ |
| 3 | $c_1 = 0.555555556$ | $x_1 = -0.774596669$ |
| | $c_2 = 0.888888889$ | $x_2 = 0.000000000$ |
| | $c_3 = 0.555555556$ | $x_3 = 0.774596669$ |
| 4 | $c_1 = 0.347854845$ | $x_1 = -0.861136312$ |
| | $c_2 = 0.652145155$ | $x_2 = -0.339981044$ |
| | $c_3 = 0.652145155$ | $x_3 = 0.339981044$ |
| | $c_4 = 0.347854845$ | $x_4 = 0.861136312$ |
| 5 | $c_1 = 0.236926885$ | $x_1 = -0.906179846$ |
| | $c_2 = 0.478628670$ | $x_2 = -0.538469310$ |
| | $c_3 = 0.568888889$ | $x_3 = 0.000000000$ |
| | $c_4 = 0.478628670$ | $x_4 = 0.538469310$ |
| | $c_5 = 0.236926885$ | $x_5 = 0.906179846$ |
| 6 | $c_1 = 0.171324492$ | $x_1 = -0.932469514$ |
| | $c_2 = 0.360761573$ | $x_2 = -0.661209386$ |
| | $c_3 = 0.467913935$ | $x_3 = -0.238619186$ |
| | $c_4 = 0.467913935$ | $x_4 = 0.238619186$ |
| | $c_5 = 0.360761573$ | $x_5 = 0.661209386$ |
| | $c_6 = 0.171324492$ | $x_6 = 0.932469514$ |

## So if the table is given for integrals with [-1,1] integration limits, how does one solve for integrals with [a,b] integration limits.

The answer lies in that any integral with limits of $[a, b]$ can be converted into an integral with limits $[-1, 1]$. Let

$$x = mt + c \qquad (4)$$

If $x = a$, then $t = -1$

If $x = b$, then $t = +1$

such that

$$a = m(-1) + c$$
$$b = m(1) + c \qquad (5a, b)$$

Solving the two Equations (5a) and (5b) simultaneously gives

$$m = \frac{b-a}{2}$$
$$c = \frac{b+a}{2} \qquad (6a, b)$$

Hence from Equations (4), (6a), (6b),

$$x = \frac{b-a}{2}t + \frac{b+a}{2}$$
$$dx = \frac{b-a}{2}dt$$

Substituting values of $x$ and $dx$ into the integral gives us

$$\begin{aligned}\int_a^b f(x)dx &= \int_{-1}^{1} f\left(\frac{b-a}{2}t + \frac{b+a}{2}\right)\frac{b-a}{2}dt \\ &= \int_{-1}^{1} f\left(\frac{b-a}{2}x + \frac{b+a}{2}\right)\frac{b-a}{2}dx \qquad (7)\end{aligned}$$

Hence, any integral of the form $\int_a^b f(x)dx$ can be converted to an $\int_{-1}^{1} g(x)dx$, and Table 1 can be used to estimate integrals.

**Example 1**

Use two-point Gauss quadrature rule to approximate the distance in meters covered by a rocket from $t = 8$ s to $t = 30$ s as given by

$$x = \int_8^{30} \left(2000 \ln\left[\frac{140000}{140000 - 2100t}\right] - 9.8t\right) dt$$

Change the limits so that one can use the weights and abscissas given in Table 1. Also, find the absolute relative true error. The true value is given as 11061.34 m

**Solution**

First, change the limits of integration from $[8, 30]$ to $[-1, 1]$ using Equation (7) gives

$$\begin{aligned}\int_8^{30} f(t)dt &= \frac{30-8}{2}\int_{-1}^{1} f\left(\frac{30-8}{2}x + \frac{30+8}{2}\right)dx \\ &= 11\int_{-1}^{1} f\left(11x + 19\right)dx\end{aligned}$$

Next, get weighting factors and function argument values from Table 1 for the two-point rule,

$$
\begin{aligned}
c_1 &= 1.000000000 \\
x_1 &= -0.577350269 \\
c_2 &= 1.000000000 \\
x_2 &= 0.577350269
\end{aligned}
$$

Now we can use the Gauss quadrature formula

$$
\begin{aligned}
11\int_{-1}^{1} f\left(11x+19\right)dx &\approx 11\left[c_1 f\left(11x_1+19\right)+c_2 f\left(11x_2+19\right)\right] \\
&= 11\left[f\left(11(-0.5773503)+19\right)+f\left(11(0.5773503)+19\right)\right] \\
&= 11\left[f(12.64915)+f(25.35085)\right] \\
&= 11\left[(296.8317)+(708.4811)\right] \\
&= 11058.44 \text{ m}
\end{aligned}
$$

since

$$
\begin{aligned}
f(12.64915) &= 2000\ln\left[\frac{140000}{140000-2100(12.64915)}\right]-9.8(12.64915) \\
&= 296.8317 \\
f(25.35085) &= 2000\ln\left[\frac{140000}{140000-2100(25.35085)}\right]-9.8(25.35085) \\
&= 708.4811
\end{aligned}
$$

The absolute relative true error, $|\epsilon_t|$, is (True value = 11061.34 m)

$$
\begin{aligned}
|\epsilon_t| &= \left|\frac{11061.34-11058.44}{11061.34}\right|\times 100 \\
&= 0.0262\%
\end{aligned}
$$

**Example 2**

Use three-point Gauss quadrature rule to approximate the distance in meters covered by a rocket from $t = 8$ to $t = 30$ as given by

$$
x = \int_{8}^{30}\left(2000\ln\left[\frac{140000}{140000-2100t}\right]-9.8t\right)dt
$$

Change the limits so that one can use the weights and abscissas given in Table 1. Also, find the absolute relative true error. The true value is given as 11061.34 m

**Solution**

First, change the limits of integration from $[8, 30]$ to $[-1, 1]$ using Equation (7) gives

$$\begin{aligned}\int_8^{30} f(t)dt &= \frac{30-8}{2}\int_{-1}^{1} f\left(\frac{30-8}{2}x + \frac{30+8}{2}\right)dx \\ &= 11\int_{-1}^{1} f\left(11x+19\right)dx\end{aligned}$$

The weighting factors and function argument values are

$$\begin{aligned}c_1 &= 0.555555556 \\ x_1 &= -0.774596669 \\ c_2 &= 0.888888889 \\ x_2 &= 0.000000000 \\ c_3 &= 0.555555556 \\ x_3 &= 0.774596669\end{aligned}$$

and the formula is

$$\begin{aligned}11\int_{-1}^{1} f\left(11x+19\right)dx &\approx 11\left[c_1 f\left(11x_1+19\right) + c_2 f\left(11x_2+19\right) + c_3 f\left(11x_3+19\right)\right] \\ &= 11\begin{bmatrix}\&0.5555556 f\left(11(-.7745967)+19\right) + 0.8888889 f\left(11(0.0000000)+19\right) \\ +0.5555556 f\left(11(0.7745967)+19\right)\end{bmatrix} \\ &= 11\left[0.55556 f(10.47944) + 0.88889 f(19.00000) + 0.55556 f(27.52056)\right] \\ &= 11\left[0.55556 \times 239.3327 + 0.88889 \times 484.7455 + 0.55556 \times 795.1069\right] \\ &= 11061.31 \text{ m}\end{aligned}$$

since

$$\begin{aligned}f(10.47944) &= 2000\ln\left[\frac{140000}{140000-2100(10.47944)}\right] - 9.8(10.47944) \\ &= 239.3327 \\ f(19.00000) &= 2000\ln\left[\frac{140000}{140000-2100(19.00000)}\right] - 9.8(19.00000) \\ &= 484.7455 \\ f(27.52056) &= 2000\ln\left[\frac{140000}{140000-2100(27.52056)}\right] - 9.8(27.52056) \\ &= 795.1069\end{aligned}$$

The absolute relative true error, $|\epsilon_t|$, is (True value = 11061.34 m)

$$\begin{aligned}|\epsilon_t| &= \left|\frac{11061.34 - 11061.31}{11061.34}\right| \times 100 \\ &= 0.0003\%\end{aligned}$$

## So does Gaussian quadrature require that the integral must be transformed to the integral limit of [-1,1]?

No, the limits do not need to be transformed. Gaussian quadrature rule can be written for any limits of integration.

It is just the weights and abscissas are given for the limits of integration of $[-1, 1]$. So if the $n$-point Gaussian quadrature rule for $[-1, 1]$ limits is given as

$$\int_{-1}^{1} g(x)dx \approx \sum_{i=1}^{n} c_i g(x_i)$$

and we also know that

$$\int_a^b f(x)\,dx = \frac{b-a}{2}\int_{-1}^{1} f\left(\frac{b-a}{2}x + \frac{b+a}{2}\right)dx$$

then the $n$-point Gaussian quadrature rule for $[a, b]$ limits of integration can also be found.

$$\begin{aligned}\int_a^b f(x)\,dx &= \frac{b-a}{2}\int_{-1}^{1} f\left(\frac{b-a}{2}x + \frac{b+a}{2}\right)dx \\ &\approx \frac{b-a}{2}\sum_{i=1}^{n} c_i f\left(\frac{b-a}{2}x_i + \frac{b+a}{2}\right) \\ &= \sum_{i=1}^{n} C_i f(X_i)\end{aligned}$$

where

$$\begin{aligned}C_i &= \frac{b-a}{2}c_i \\ X_i &= \frac{b-a}{2}x_i + \frac{b+a}{2}\end{aligned}$$

## Appendix

### Example A.1

For an integral $\int_{-1}^{1} f(x)dx$, derive the two-point Gauss quadrature rule

$$\int_{-1}^{1} f(x)dx \approx c_1 f(x_1) + c_2 f(x_2)$$

### Solution

Assuming the formula

$$\int_{-1}^{1} f(x)dx = c_1 f(x_1) + c_2 f(x_2) \qquad (A1.1)$$

gives exact values for integrals $\int_{-1}^{1} 1dx$, $\int_{-1}^{1} xdx$, $\int_{-1}^{1} x^2dx$, and $\int_{-1}^{1} x^3dx$ . Then

$$\int_{-1}^{1} 1dx = 2 = c_1 + c_2 \qquad (A1.2)$$

$$\int_{-1}^{1} xdx = 0 = c_1x_1 + c_2x_2 \qquad (A1.3)$$

$$\int_{-1}^{1} x^2dx = \frac{2}{3} = c_1{x_1}^2 + c_2{x_2}^2 \qquad (A1.4)$$

$$\int_{-1}^{1} x^3dx = 0 = c_1{x_1}^3 + c_2{x_2}^3 \qquad (A1.5)$$

Multiplying Equation (E1.3) by ${x_1}^2$ and subtracting from Equation (E1.5) gives

$$c_2x_2\left({x_1}^2 - {x_2}^2\right) = 0 \qquad (A1.6)$$

The solution to the above equation is

$$\begin{aligned} &c_2 = 0, \text{ or/and} \\ &x_2 = 0, \text{ or/and} \\ &x_1 = x_2, \text{ or/and} \\ &x_1 = -x_2. \end{aligned}$$

I. $c_2 = 0$ is not acceptable as Equations (A1.2-A1.5) reduce to $c_1 = 2$, $c_1x_1 = 0$, $c_1x_1^2 = \frac{2}{3}$, and $c_1x_1^3 = 0$. But since $c_1 = 2$, then $x_1 = 0$ from $c_1x_1 = 0$, but $x_1 = 0$ conflicts with $c_1x_1^2 = \frac{2}{3}$.

II. $x_2 = 0$ is not acceptable as Equations (A1.2-A1.5) reduce to $c_1 + c_2 = 2$, $c_1x_1 = 0$, $c_1x_1^2 = \frac{2}{3}$, and $c_1x_1^3 = 0$. Since $c_1x_1 = 0$, then $c_1$ or $x_1$ has to be zero, but this violates $c_1x_1^2 = \frac{2}{3} \neq 0$.

III. $x_1 = x_2$ is not acceptable as Equations (A1.2-A1.5) reduce to $c_1 + c_2 = 2$, $c_1x_1 + c_2x_1 = 0$, $c_1x_1^2 + c_2{x_1}^2 = \frac{2}{3}$, and $c_1x_1^3 + c_2x_1^3 = 0$. If $x_1 \neq 0$, then $c_1x_1 + c_2x_1 = 0$ gives $c_1 + c_2 = 0$, and that violates $c_1 + c_2 = 2$. If $x_1 = 0$, then that violates $c_1x_1^2 + c_2{x_1}^2 = \frac{2}{3} \neq 0$.

That leaves the solution of $x_1 = -x_2$ as the only possible acceptable solution, and in fact, it does not have violations (see it for yourself)

$$x_1 = -x_2 \qquad (A1.7)$$

Substituting (A1.7) in Equation (A1.3) gives

$$c_1 = c_2 \qquad (A1.8)$$

From Equations (A1.2) and (A1.8),

$$c_1 = c_2 = 1 \qquad (A1.9)$$

Equations (A1.4) and (A1.9) gives

$${x_1}^2 + {x_2}^2 = \frac{2}{3} \qquad (A1.10)$$

Since Equation (A1.7) requires that the two results be of opposite sign, we get

$$x_1 = -\frac{1}{\sqrt{3}}$$
$$x_2 = \frac{1}{\sqrt{3}}$$

or

$$x_1 = \frac{1}{\sqrt{3}}$$
$$x_2 = -\frac{1}{\sqrt{3}}$$

Both these solutions though lead to a unique solution as

$$\begin{aligned}\int_{-1}^{1} f(x)dx &= c_1 f(x_1) + c_2 f(x_2) \qquad (A1.11)\\ &= 1f\left(-\frac{1}{\sqrt{3}}\right) + 1f\left(\frac{1}{\sqrt{3}}\right)\\ &= f\left(-\frac{1}{\sqrt{3}}\right) + f\left(\frac{1}{\sqrt{3}}\right)\end{aligned}$$

**Example A.2**

What would be the formula for

$$\int_a^b f(x)dx = c_1 f(a) + c_2 f(b)$$

if you want the above formula to give you exact values of integral $\int_a^b \left(a_0 x + b_0 x^2\right) dx$, that is, a linear combination of $x$ and $x^2$.

**Solution**

If the formula is exact for a linear combination of $x$ and $x^2$, then

$$\int_a^b x dx = \frac{b^2 - a^2}{2} = c_1 a + c_2 b$$
$$\int_a^b x^2 dx = \frac{b^3 - a^3}{3} = c_1 a^2 + c_2 b^2 \qquad (A2.1)$$

Solving the two Equations (A2.1) simultaneously gives

$$\begin{bmatrix} a & b \\ a^2 & b^2 \end{bmatrix} \begin{bmatrix} \mathrm{c}_1 \\ \mathrm{c}_2 \end{bmatrix} = \begin{bmatrix} \frac{b^2-a^2}{2} \\ \frac{b^3-a^3}{3} \end{bmatrix}$$

$$c_1 = -\frac{1}{6}\frac{-ab - b^2 + 2a^2}{a}$$
$$c_2 = -\frac{1}{6}\frac{a^2 + ab - 2b^2}{b} \qquad (A2.2)$$

So

$$\int_a^b f(x)dx = -\frac{1}{6}\frac{-ab - b^2 + 2a^2}{a} f(a) - \frac{1}{6}\frac{a^2 + ab - 2b^2}{b} f(b) \qquad (A2.3)$$

Let us see if the formula works.

Evaluate $\int_2^5 \left(2x^2 - 3x\right) dx$ using Equation (A2.3)

$$\begin{aligned}\int_2^5 \left(2x^2 - 3x\right) dx &\approx c_1 f(a) + c_2 f(b) \\ &= -\frac{1}{6}\frac{-(2)(5) - 5^2 + 2(2)^2}{2}\left[2(2)^2 - 3(2)\right] - \frac{1}{6}\frac{2^2 + 2(5) - 2(5)^2}{5}[2(5)^2 - 3(5)] \\ &= 46.5\end{aligned}$$

The exact value of $\int_2^5 \left(2x^2 - 3x\right) dx$ is given by

$$\begin{aligned}\int_2^5 \left(2x^2 - 3x\right) dx &= \left[\frac{2x^3}{3} - \frac{3x^2}{2}\right]_2^5 \\ &= 46.5\end{aligned}$$

Any surprises?

Now evaluate $\int_2^5 3dx$ using Equation (A2.3)

$$\begin{aligned}\int_2^5 3dx &\approx c_1 f(a) + c_2 f(b) \\ &= -\frac{1}{6}\frac{-2(5) - 5^2 + 2(2)^2}{2}(3) - \frac{1}{6}\frac{2^2 + 2(5) - 2(5)^2}{5}(3) \\ &= 10.35\end{aligned}$$

The exact value of $\int_2^5 3dx$ is given by

$$\begin{aligned}\int_2^5 3dx &= [3x]_2^5 \\ &= 9\end{aligned}$$

Because the formula will only give exact values for linear combinations of $x$ and $x^2$, it does not work exactly even for a simple integral of $\int_2^5 3dx$.

Do you see now why we chose $a_0 + a_1 x$ as the integrand for which the formula

$$\int_a^b f(x)dx \approx c_1 f(a) + c_2 f(b)$$

gives us exact values?

# Multiple Choice Test

## Chapter 07.05 Gauss Quadrature Rule of Integration

(1). $\int_5^{10} f(x)dx$ is exactly

(A) $\int_{-1}^{1} f(2.5x+7.5)\ dx$

(B) $2.5\int_{-1}^{1} f(2.5x+7.5)\ dx$

(C) $5\int_{-1}^{1} f(5x+5)\ dx$

(D) $5\int_{-1}^{1} (2.5x+7.5)f(x)\ dx$

(2). For a definite integral of any third order polynomial, the two-point Gauss quadrature rule will give the same results as the

(A) 1-segment trapezoidal rule

(B) 2-segment trapezoidal rule

(C) 3-segment trapezoidal rule

(D) Simpson's 1/3 rule

(3). The estimated value of $\int_{0.2}^{2.2} xe^x dx$ by using the two-point Gauss quadrature rule is most nearly

(A) 11.672

(B) 11.807

(C) 12.811

(D) 14.633

(4). A scientist uses the one-point Gauss quadrature rule based on getting exact results of integration for functions $f(x) = 1$ and $x$. The one-point Gauss quadrature rule approximation for $\int_a^b f(x)dx$ is

(A) $\dfrac{b-a}{2}\left[f(a)+f(b)\right]$

(B) $(b-a)f\left(\dfrac{a+b}{2}\right)$

(C) $\dfrac{b-a}{2}\left[f\left(\dfrac{b-a}{2}\left\{-\dfrac{1}{\sqrt{3}}\right\}+\dfrac{b+a}{2}\right)+f\left(\dfrac{b-a}{2}\left\{\dfrac{1}{\sqrt{3}}\right\}+\dfrac{b+a}{2}\right)\right]$

(D) $(b-a)f(a)$

(5). A scientist develops an approximate formula for integration as

$$\int_a^b f(x)dx \approx c_1 f(x_1), \text{ where } a \le x_1 \le b$$

The values of $c_1$ and $x_1$ are found by assuming that the formula is exact for functions of the form $a_0x + a_1x^2$. The resulting formula would therefore be exact for integrating

(A) $f(x) = 2$

(B) $f(x) = 2 + 3x + 5x^2$

(C) $f(x) = 5x^2$

(D) $f(x) = 2 + 3x$

(6). You are asked to estimate the water flow rate in a pipe of radius $2m$ at a remote area location with a harsh environment. You already know that velocity varies along the radial location, but you do not know how it varies. The flow rate $Q$ is given by

$$Q = \int_0^2 2\pi r V dr$$

To save money, you are allowed to put only two velocity probes (these probes send the data to the central office in New York, NY via satellite) in the pipe. Radial location, $r$ is measured from the center of the pipe, that is, $r = 0$. Radial location, $r = 2m$ is at the pipe radius. The radial locations you would suggest for the two velocity probes for the most accurate calculation of the flow rate are

(A) 0 m, 2 m

(B) 1 m, 2 m

(C) 0 m, 1 m

(D) 0.42 m, 1.58 m

For the complete solution, go to

http://nm.mathforcollege.com/mcquizzes/07int/quiz_07int_gaussquadrature_solution.pdf

# Problem Set

## Chapter 07.05: Gauss Quadrature Rule of Integration

(1). Find the value of the integral $\int_1^5 7e^{-2t}dt$ using

a) 2-point Gaussian quadrature rule,

b) 3-point Gaussian quadrature rule.

**Answer:** *a*) 0.35285
*b*) 0.45812

(2). The upward velocity of a rocket is given by

$$v(t) = 200\ln(t+1) - 10t,\ t > 0$$

where $t$ is given in seconds and $v$ is given in m/s.

a) Use 2-point Gauss quadrature rule to calculate the distance covered by the rocket from $t = 0$ s to $t = 5$ s.

b) What is the true value of the distance covered by the rocket from $t = 0$ s to $t = 5$ s?

c) What is the true error in part (a)?

d) What is the relative true error in part (a)?

e) What is the absolute relative true error in percentage for part (a).

f) Use 3-point Gauss quadrature rule to calculate the distance covered by the rocket from $t = 0$ s to $t = 5$ s.

**Answer:** *a*) 1034.6 m
*b*) 1025.1 m
*c*) $-9.4458$ m
*d*) $-0.0092144$
*e*) 0.92144%
*f*) 1026.2 m

(3). The velocity of a body is given by

$$v(t) = t + 1,\ 0 < t < 2,$$
$$= 33,\ 2 < t < 5$$

where the velocity, $v(t)$ is given in m/s and $t$ is in seconds.

a) Find the distance covered by the body between $1 < t < 1.9$ s by using the two-point Gauss quadrature rule. Find the absolute relative true error.

b) Find the distance covered by the body between $1 < t < 3.9$ s by using the four-point Gauss quadrature rule. Find the absolute relative true error.

**Answer:** *a*) 2.2049, 0%
*b*) 51.75, 20.62%($exact = 65.2$)

(4). The one-point Gauss quadrature rule is defined as

$$\int_a^b f(x)dx \approx c_1 f(x_1),\ a \le x_1 \le b$$

The values of $c_1$ and $x_1$ are found by assuming that the one-point formula is exact for any first order polynomial. Find $c_1$ and $x_1$ in the above one-point Gauss quadrature rule.

**Answer:** $c_1 = b - a,\ x_1 = \dfrac{b+a}{2}$

(5). A scientist develops an approximate formula for integration as

$$\int_a^b f(x)dx \approx c_1 f(x_1),\ \text{where } a \le x_1 \le b$$

The values of $c_1$ and $x_1$ are found by assuming that the formula is exact for the functions of the form $a_0x + a_1x^2$ polynomial. Find $c_1$ and $x_1$.

**Answer:** $x_1 = \dfrac{2(b^2+a^2+ab)}{3(b+a)}$

$c_1 = \dfrac{3(b-a)(b+a)^2}{4(b^2+a^2+ab)}$

(6). A scientist develops an approximate formula for integration as

$$\int_a^b f(x)dx \approx c_1 f(a) + c_2 f(b).$$

The values of $c_1$ and $c_2$ are found by assuming that the formula is exact for the functions of the form $a_0x + a_1x^2$ polynomial.

a) Find the values of $c_1$ and $c_2$.

b) Verify the formula works exactly for the integral

$$\int_2^7 (3x^2 + 5x)dx$$

c) Verify the formula does not work exactly for the integral of

$$\int_2^7 2dx$$

**Answer:** *a)* $c_1 = -\frac{1}{6}\frac{-ab - b^2 + 2a^2}{a},\ c_2 = -\frac{1}{6}\frac{a^2 + ab - 2b^2}{b}$

b) 447.50 by formula as well as exact method

c) 12.976 by formula; 10 by exact method

# Chapter 07.06: Integrating Functions Given as Discrete Data Points

Lesson: Trapezoidal Rule for Functions Given as Discrete Points

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *derive the formula for the trapezoidal rule with unequal segments*

2) *estimate integrals of functions given at discrete data points.*

## Introduction

Most functions to be integrated may not be given mathematically as continuous functions. They may be only given at discrete data points. Such functions have to be integrated numerically. One technique to do so is an extension of the single-segment trapezoidal rule and is called the "Trapezoidal rule with unequal segments."

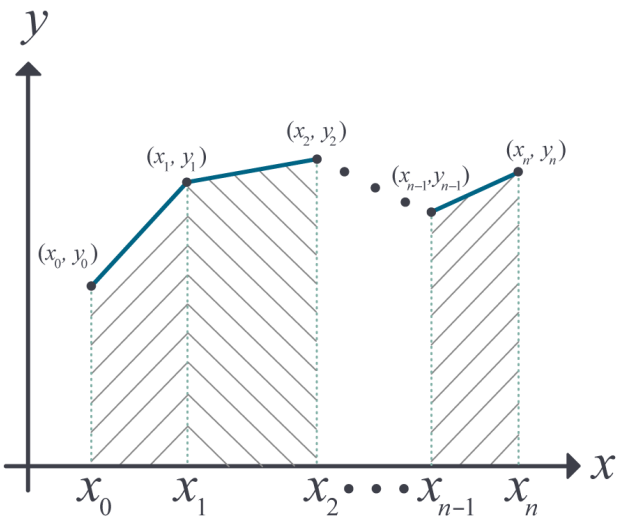


**Figure 1.** Illustration of trapezoidal rule with unequal segments.

For a general case of a function given at $n$ data points $(x_1, f(x_1))$, $(x_2, f(x_2))$, $(x_3, f(x_3)), \ldots, (x_n,\ f(x_n))$, where, $x_1, .x_2, \ldots, x_n$ are in ascending order, the integral $\int_{x_1}^{x_n} f(x)dx$ can be broken into $n-1$ separate integrals

$$\int_{x_1}^{x_n} f(x)dx = \int_{x_1}^{x_2} f(x)dx + \int_{x_2}^{x_3} f(x)dx + ..... + \int_{x_{n-1}}^{x_n} f(x)dx$$

Using a single-segment trapezoidal rule for each integral (Figure 1) gives

$$\begin{aligned}\int_{x_1}^{x_n} f(x)dx \approx (x_2 - x_1)\frac{f(x_1) + f(x_2)}{2} + (x_3 - x_2)\frac{f(x_2) + f(x_3)}{2} + .....\\ ..... + (x_n - x_{n-1})\frac{f(x_{n-1}) + f(x_n)}{2}\\ = \sum_{i=1}^{n-1}(x_{i+1} - x_i)\frac{f(x_i) + f(x_{i+1})}{2} \qquad (1)\end{aligned}$$

Equation (1) estimates an integral of a function given at discrete data points using trapezoidal rule with unequal segments.

**Example 1**

The upward velocity of a rocket is given as a function of time in Table 1.

**Table 1** Velocity as a function of time.

| $t$(s) | $v(t)$(m/s) |
|---|---|
| 0 | 0 |
| 10 | 227.04 |
| 15 | 362.78 |
| 20 | 517.35 |
| 22.5 | 602.97 |
| 30 | 901.67 |

Using the trapezoidal rule with unequal segments, determine the distance, $s$, covered by the rocket from $t = 10$ to $t = 20$ seconds.

**Solution**

To find the distance covered from $t = 10$ to $t = 20$ seconds

$$\int_{10}^{20} v(t) = \int_{10}^{15} v(t)dt + \int_{15}^{20} v(t)dt$$

and applying the trapezoidal rule over each of the above integrals gives

$$\int_{10}^{20} v(t)dt \approx \frac{15-10}{2}[v(10)+v(15)] + \frac{20-15}{2}[v(15)+v(20)]$$
$$= 2.5(227.04+362.78) + 2.5(362.78+517.35)$$
$$= 3674.875 \text{ m}$$

How would you use the trapezoidal rule with unequal segments if the integral to be found has the limits of integration where one or both the limits are not a given data point? An example is given below to solve such a problem.

**Example 2**

The upward velocity of a rocket is given as a function of time in Table 1.

**Table 1** Velocity as a function of time.

| $t$(s) | $v(t)$(m/s) |
|---|---|
| 0 | 0 |
| 10 | 227.04 |
| 15 | 362.78 |
| 20 | 517.35 |
| 22.5 | 602.97 |
| 30 | 901.67 |

Using the trapezoidal rule with unequal segments, determine the distance, $s$, covered by the rocket from $t = 11$ to $t = 16$ seconds.

**Solution** Here, we are asked to estimate the distance covered from $t = 11$ to $t = 16$.

$$\int_{11}^{16} v(t)dt = \int_{11}^{15} v(t)dt + \int_{15}^{16} v(t)dt$$

and applying the trapezoidal rule over each of the above integrals gives

$$\int_{11}^{16} v\,(t)\,dt \approx \frac{15-11}{2}\,[v\,(11)+v\,(15)] + \frac{16-15}{2}[v(15)+v(16)]$$
$$= \frac{15-11}{2}\,(v\,(11)+362.78) + \frac{16-15}{2}\,(362.78+v\,(16))$$

How do we find $v(11)$ and $v(16)$? We use linear interpolation with a simple formula, which is, given $(t_0, v_0)$ and $(t_1, v_1)$, the straight line that passes through the two points is given by

$$v = v_0 + \frac{v_1 - v_0}{t_1 - t_0}(t - t_0),\ t_0 \le t \le t_1$$

To find $v(11)$, we choose the closest two points that bracket $t = 11$, and those two points are $t = 10$ and $t = 15$.

$$\begin{aligned} v(t) &= v(10) + \frac{v(15) - v(10)}{15 - 10}(t - 10),\ 10 \leq\ t \leq 15 \\ &= 227.04 + \frac{362.78 - 227.04}{15 - 10}\left(t - 10\right),\ 10 \leq t \leq 15 \\ &= 227.04 + 27.148\left(t - 10\right),\ 10 \leq t \leq 15 \end{aligned}$$

Then

$$\begin{aligned} v(11) &= 227.04 + 27.148\left(11 - 10\right) \\ &= 254.19 \text{ m/s} \end{aligned}$$

To find $v(16)$, we choose the two closest points that bracket $t = 16$, and those two points are $t = 15$ and $t = 20$.

$$\begin{aligned} v\left(t\right) &= v\left(15\right) + \frac{v\left(20\right) - v\left(15\right)}{20 - 15}\left(t - 15\right), 15 \leq t \leq 20 \\ &= 362.78 + \frac{517.35 - 362.784}{20 - 15}\left(t - 15\right), 15 \leq t \leq 20 \\ &= 362.78 + 30.913\left(t - 15\right), 15 \leq t \leq 20 \end{aligned}$$

Then

$$\begin{aligned} v(16) &= 362.78 + 30.913\left(16 - 15\right) \\ &= 393.69 \text{ m/s} \end{aligned}$$

So

$$\begin{aligned} \int_{11}^{16} v\left(t\right) dt &\approx \frac{15 - 11}{2}\left(v\left(11\right) + 362.78\right) + \frac{16 - 15}{2}(362.78 + v(16)) \\ &= \frac{15 - 11}{2}(254.19 + 362.78) + \frac{16 - 15}{2}(362.78 + 393.69) \\ &= 1612.2 \text{ m} \end{aligned}$$

# Chapter 07.06: Integrating Functions Given as Discrete Data Points

Lesson: Alternative Methods of Integrating Functions Given as Discrete Points

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *integrate functions given at discrete points by several methods.*

## Introduction

In a previous lesson, we used the trapezoidal rule with unequal segments to integrate a function given at discrete data points. In this lesson, we show other methods of integrating a function given at discrete data points. We do this through the same example as the previous lesson and skip the theory as you have already gone through it in previous lessons.

### Example 1

The upward velocity of a rocket is given as a function of time in Table 1.

**Table 1** Velocity as a function of time.

| $t$(s) | $v(t)$(m/s) |
|---|---|
| 0 | 0 |
| 10 | 227.04 |
| 15 | 362.78 |
| 20 | 517.35 |
| 22.5 | 602.97 |
| 30 | 901.67 |

Determine the distance $s$ covered by the rocket from $t = 11$ s to $t = 16$ s using the velocity data provided and use any applicable numerical technique.

**Solution**

The distance covered $s$ is given by

$$\int_{11}^{16} v(t)dt$$

Method: Average Function Method

The velocity of the rocket is not provided at $t = 11$ s and $t = 16$ s, so we will have to use an interval that includes $[11, 16]$ to find the average velocity of the rocket within that domain. In this case, the interval $[10, 20]$ will suffice.

$$\begin{aligned} v(10) &= 227.04 \\ v(15) &= 362.78 \\ v(20) &= 517.35 \end{aligned}$$

The average velocity, $\bar{v}$ is

$$\begin{aligned} \bar{v} &= \frac{v(10) + v(15) + v(20)}{3} \\ &= \frac{227.04 + 362.78 + 517.35}{3} \\ &= 369.06 \text{ m/s} \end{aligned}$$

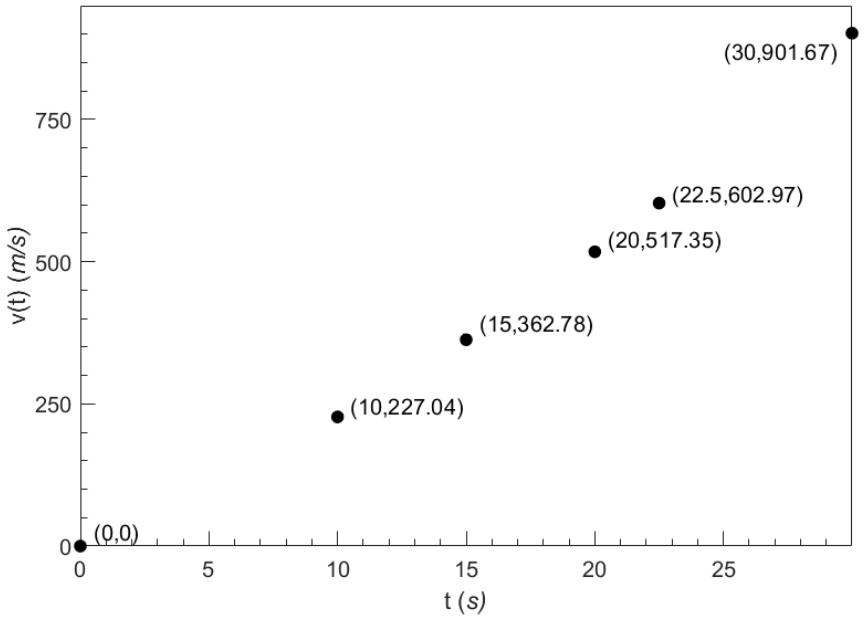



**Figure 1** Velocity vs. time data for the rocket example

Using

$$s = \bar{v}\Delta t,$$

we get

$$\begin{aligned} s &= (369.06)(16 - 11) \\ &= 1845.3m \end{aligned}$$

As you see later, this method should NOT be used. Other methods that are as simple to use and to compute provide more accurate results. The average function method does not account for the differences in the individual segment widths and gives them equal consideration.

Method: Polynomial Interpolation Followed by Integration:

Because we are finding the area under the curve from $t = 11$ to $t = 16$, we choose three points, $t = 10$, $t = 15$, and $t = 20$, to fit a quadratic polynomial through the data. Using polynomial interpolation, our resulting velocity function is (refer to Chapter 05.02 on direct (Vandermonde) method of interpolation)

$$v(t) = 12.05 + 17.733t + 0.3766t^2, \;\; 10 \leq t \leq 20.$$

Now, we simply take the integral of the quadratic within our limits, giving us

$$\begin{aligned} s &\approx \int_{11}^{16} \left(12.05 + 17.733t + 0.3766t^2\right) dt \\ &= \left[12.05t + \frac{17.733t^2}{2} + \frac{0.3766t^3}{3}\right]_{11}^{16} \\ &= 12.05\,(16 - 11) + \frac{17.733}{2}\left(16^2 - 11^2\right) + \frac{0.3766}{3}\left(16^3 - 11^3\right) \\ &= 1604.3 \text{ m} \end{aligned}$$

Method: Spline Interpolation Followed by Integration

Fitting quadratic spline (refer to Chapter 05.05 on quadratic spline interpolation) through the data results in the following set of quadratics

$$\begin{aligned} v(t) &= -0.15667t^2 + 24.271t, \;\; 0 \leq t \leq 10 \\ &= 1.2021t^2 - 2.9053t + 135.88, \;\; 10 \leq t \leq 15 \\ &= -0.44893t^2 + 46.627t - 235.61, \;\; 15 \leq t \leq 20 \\ &= 2.2315t^2 - 60.589t + 836.55, \;\; 20 \leq t \leq 22.5 \\ &= 39.827t - 293.13, \;\; 22.5 \leq t \leq 30 \end{aligned} \qquad (E1..16)$$

The value of the integral would then simply be

$$\begin{aligned} s &= \int_{11}^{16} v(t)\, dt \\ &= \int_{11}^{15} v(t)\, dt + \int_{15}^{16} v(t)\, dt \end{aligned}$$

Substituting proper quadratics based on the limits of integration gives

$$
\begin{aligned}
s &= \int_{11}^{15} 1.2021t^2 - 2.9053t + 135.88 \; dt \\
&\quad + \int_{15}^{16} -0.44893t^2 + 46.627t - 235.61 dt \\
&= \left[ 1.2021\frac{t^3}{3} - 2.9053\frac{t^2}{2} + 135.88t \right]_{11}^{15} \\
&\quad + \left[ -0.44893\frac{t^3}{3} + 46.627\frac{t^2}{2} - 235.61t \right]_{15}^{16} \\
&= 1211.49 + 379.22 \\
&= 1590.7 \text{ m}
\end{aligned}
$$

**Example 2**

What is the absolute relative true error for each of the methods used in Example 1 if the data in Table 1 was obtained from the velocity profile,

$$
v(t) = \left( 2000 \ln \left[ \frac{140000}{140000 - 2100t} \right] - 9.8t \right),
$$

where $v$ is given in m/s and $t$ in s.

**Solution**

The exact distance covered between $t = 11$ s and $t = 16$ s is

$$
\begin{aligned}
s &= \int_{11}^{16} \left( 2000 \ln \left[ \frac{140000}{140000 - 2100t} \right] - 9.8t \right) dt \\
&= 1604.9 \text{ m}
\end{aligned}
$$

Method: Average Function Method

The approximate value obtained using the average function method was 1845.3 m. Hence, the absolute relative true error, $|\epsilon_t|$, is

$$
\begin{aligned}
|\epsilon_t| &= \left| \frac{1604.9 - 1845.3}{1604.9} \right| \times 100\% \\
&= 14.976\%
\end{aligned}
$$

Method: Trapezoidal Rule with Unequal Segments

Although this method was illustrated in a previous lesson, we show it here for completion. The approximate value obtained using the trapezoidal rule with unequal segments was 1612.2 m. Hence, the absolute relative true error, $|\epsilon_t|$, is

$$|\epsilon_t| = \left|\frac{1604.9 - 1612.2}{1604.9}\right| \times 100\%$$
$$= 0.451\%$$

Method: Polynomial Interpolation Followed by Integration

The approximate value obtained using the direct polynomial was 1604.3 m. Hence, the absolute relative true error, $|\epsilon_t|$, is

$$|\epsilon_t| = \left|\frac{1604.9 - 1604.3}{1604.9}\right| \times 100\%$$
$$= 0.037\%$$

Method: Spline Interpolation Followed by Integration

The approximate value obtained using the spline interpolation was 1595.9 m. Hence, the absolute relative true error, $|\epsilon_t|$, is

$$|\epsilon_t| = \left|\frac{1604.9 - 1590.7}{1604.9}\right| \times 100\%$$
$$= 0.885\%$$

**Table 2** Comparison of numerical integration methods for functions given at discrete data points.

| Method | Approximate Value (m) | $\lvert\epsilon_t\rvert$ |
|---|---|---|
| Average Function | 1845.3 | 14.976% |
| Trapezoidal Rule | 1612.2 | 0.451% |
| Polynomial Interpolation | 1604.3 | 0.037% |
| Spline Interpolation | 1595.9 | 0.885% |

The values of the absolute relative true error in Table 2 should be a reason to question the use of the average function method. The trapezoidal rule which is equally simple to apply results in much less true error.

# Multiple Choice Test

Chapter 07.06: Integrating Functions Given as Discrete Data Points

(1). Given below is the discrete data of $y$ vs $x$.

| $x$ | 0 | 10 | 15 | 20 |
|---|---|---|---|---|
| $y$ | 100.00 | 227.04 | 362.78 | 517.35 |

Amongst the following methods, which one would be the best scientific method to use to find the integral $\int_0^{20} y dx$?

(A) Average function method

(B) Trapezoidal rule with unequal segments

(C) $y\,(10) \times (20)$

(D) $y\,(20) \times 20$

(2). Given $(x_1, y_1), (x_2, y_2), (x_3, y_3)$ data pairs where $x_1 < x_2 < x_3$, then $\int_{x_1}^{x_3} y(x)dx$ can be best approximated as

(A) $\dfrac{y_1 + y_2 + y_3}{3}(x_3 - x_1)$

(B) $\dfrac{y_1 + y_2}{2}(x_3 - x_1)$

(C) $(x_2 - x_1)(\dfrac{y_1 + y_2}{2}) + (x_3 - x_2)(\dfrac{y_2 + y_3}{2})$

(D) $y_2(x_3 - x_1)$

(3). The following data of the velocity of a body as a function of time is given.

| Time, $t$ $(s)$ | 0 | 15 | 18 | 22 | 24 |
|---|---|---|---|---|---|
| Velocity, $v$ $(m/s)$ | 22 | 24 | 31.4 | 25 | 123 |

Using the trapezoidal rule with unequal segments, the distance covered in meters by the body from $t$=15 to 22 $s$ is

(A) 171.5

(B) 187.6

(C) 195.9

(D) 204.9

(4). The following data of the velocity of a body as a function of time is given.

| Time, $t$ $(s)$ | 0 | 15 | 18 | 22 | 24 |
|---|---|---|---|---|---|
| Velocity, $v$ $(m/s)$ | 22 | 24 | 37 | 25 | 123 |

Using the trapezoidal rule with unequal segments, the distance covered in meters by the body from $t$=9.3 to 18 $s$ is

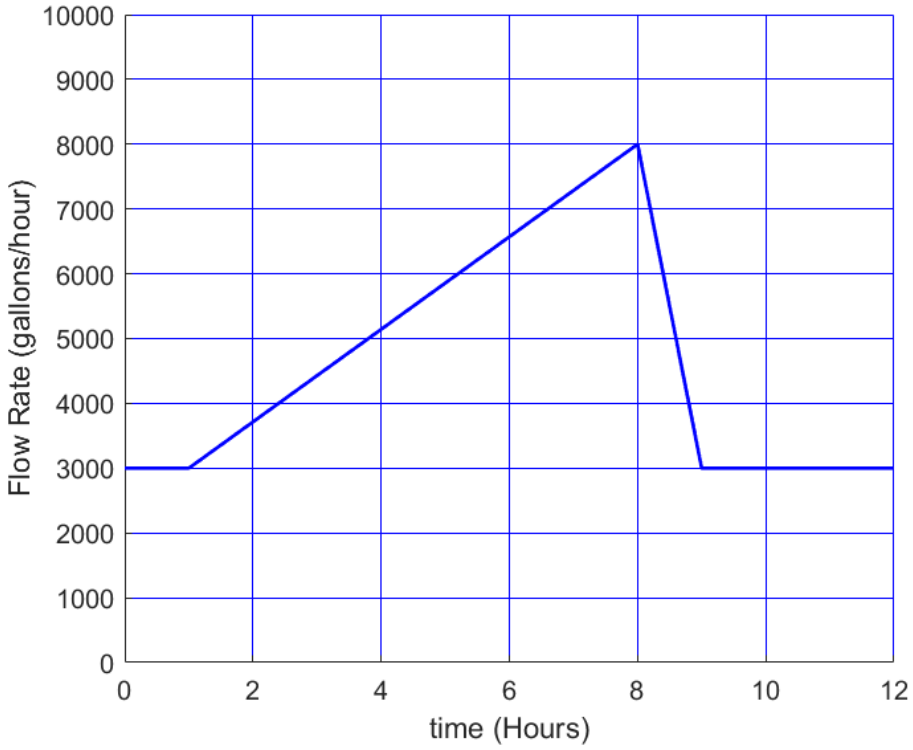
(A) 226.1

(B) 240.7

(C) 262.0

(D) 436.5

(5). The flow rate of oil is given in gallons per hour through a pipeline over a 12-hour period as shown.



The best approximation of the number of gallons of oil that has flowed through the 12-hour period is

(A) 48000

(B) 56000

(C) 66000

(D) 86148

(6). Water is flowing through a circular pipe of 0.5 ft radius, and flow velocity (ft/s) measurements are made from the center to the wall of the pipe as follows.

| Radial Location, $r\ (ft)$ | 0 | 0.17 | 0.33 | 0.50 |
|---|---|---|---|---|
| Velocity, $v\ (ft/s)$ | 10 | 8.8 | 5.6 | 0 |

Note that the volumetric flow rate, $\dot{Q}$ is given by

$$\dot{Q} = \int_0^a 2\pi r v(r) dr$$

where $v(r)$ is the flow velocity of the fluid as a function of the radial location, $r$, and $a$ is the radius of the pipe. The volumetric flow rate in $ft^3/s$ in the pipe by using trapezoidal rule with unequal segments is

(A) 3.226

(B) 3.469

(C) 3.901

(D) 20.27

For complete solution, go to

http://nm.mathforcollege.com/mcquizzes/07int/quiz_07int_discrete_solution.pdf

# Problem Set

Chapter 07.06: Integrating Functions Given as Discrete Data Points

(1). The following data of the velocity of a body as a function of time is given.

| Time $(s)$ | 0 | 15 | 18 | 22 | 24 |
|---|---|---|---|---|---|
| Velocity $(m/s)$ | 22 | 24 | 37 | 25 | 123 |

Use trapezoidal rule with unequal segments OR a better scientific method to find the displacement of the body from $t = 15$ to 22 seconds.

**Answer:** 215.5 $m$

(2). The following data of the velocity of a body as a function of time is given.

| Time $(s)$ | 0 | 15 | 18 | 22 | 24 |
|---|---|---|---|---|---|
| Velocity $(m/s)$ | 22 | 24 | 37 | 25 | 123 |

Use trapezoidal rule with unequal segments OR a better scientific method to find the displacement of the body from $t = 13$ to 18 seconds.

**Answer:** 139.23 $m$ by trapezoidal rule with unequal segments

(3). Water is flowing out at the end of 20 meter long U-shaped channel and falls into a field at the flow rate given below as a function of time.

| Time $(s)$ | 2 | 4 | 7 | 10 |
|---|---|---|---|---|
| Flow rate $(m^3/s)$ | 12 | 15 | 17 | 14 |

Use trapezoidal rule for discrete data to find the amount of water that fell on the field between $t = 3$ and $t = 9$seconds.

**Answer:** 94.25 $m^3$

(4). Water is flowing out at the end of 20 meter long U-shaped channel and falls into a field at the flow rate given below as a function of time.

| Time ($s$) | 2 | 4 | 7 | 10 |
|---|---|---|---|---|
| Flow rate($m^3/s$) | 12 | 15 | 17 | 14 |

Use Simpson's 1/3 rule to find the amount of water that fell on the field between $t = 4$ and $t = 10$ seconds.

**Answer:** 97 $m^3$

(5). Water is flowing through a circular pipe of 1/2 ft radius and flow velocity (ft/s) measurements are made from the center to the wall of the pipe as follows

| Radial Location, $r$ (ft) | 0 | 0.083 | 0.17 | 0.25 | 0.33 | 0.42 | 0.50 |
|---|---|---|---|---|---|---|---|
| Velocity, $v$(ft/$s$) | 10 | 9.72 | 8.88 | 7.5 | 5.6 | 3.1 | 0 |

Estimate the flow rate in the pipe.

**Answer:** $3.8477 ft^3/s$ by trapezoidal rule with unequal segments

(6). Below is the data given for thermal expansion coefficient of steel as a function of temperature.

| Temperature | | | Instantaneous thermal expansion coefficient |
|---|---|---|---|
| $^\circ F$ | $^\circ C$ | $K$ | ($\mu$in/in/$^\circ F$) |
| 80 | 26.67 | 299.67 | 6.47 |
| 60 | 15.56 | 288.56 | 6.36 |
| 40 | 4.44 | 277.44 | 6.24 |
| 20 | −6.67 | 266.33 | 6.12 |
| 0 | −17.78 | 255.22 | 6.00 |
| −20 | −28.89 | 244.11 | 5.86 |
| −40 | −40.00 | 233.00 | 5.72 |
| −60 | −51.11 | 221.89 | 5.58 |
| −80 | −62.22 | 210.78 | 5.43 |
| −100 | −73.33 | 199.67 | 5.28 |
| −120 | −84.44 | 188.56 | 5.09 |
| −140 | −95.56 | 177.44 | 4.91 |
| −160 | −106.67 | 166.33 | 4.72 |
| −180 | −117.78 | 155.22 | 4.52 |
| −200 | −128.89 | 144.11 | 4.30 |
| −220 | −140.00 | 133.00 | 4.08 |
| −240 | −151.11 | 121.89 | 3.83 |
| −260 | −162.22 | 110.78 | 3.58 |
| −280 | −173.33 | 99.67 | 3.33 |
| −300 | −184.44 | 88.56 | 3.07 |
| −320 | −195.56 | 77.44 | 2.76 |
| −340 | −206.67 | 66.33 | 2.45 |

A bascule bridge designer needs to shrink fit a solid circular shaft $A$ of diameter 12.363″ in a hollow cylinder $B$ with inner diameter 12.358″. His plan is to put the solid shaft $A$ in dry-ice/alcohol mixture

to contract its diameter so that it can be slid through the inner diameter of cylinder $B$. He also needs a diametral clearance of at least $0.01''$ so that the shafts do not touch each other before the whole shaft $A$ is slid through. Assume the room temperature is $70°F$ and the dry-ice/alcohol mixture is $-108°F$. Solve the following problems.

a) A consultant on this project assumed the thermal expansion coefficient is constant over the temperature range. He used the value of thermal expansion coefficient at $80°F$ in all his calculations. Find the contraction of diameter in the shaft$A$.

b) In part (a), does he calculate enough contraction in the shaft $A$ diameter?

c) Is his assumption of assuming constant thermal expansion correct? Why or why not? A single sentence answer will do.

d) How would you solve the problem? Solve by at least two different methods you learned in class. Do you calculate enough contraction in shaft $A$ diameter?

e) If you did not get enough contraction in the diameter in part (d), what would you suggest? Give a quantified solution, not just a qualitative answer.

**Answer:** *a*) $-0.14238''$
*b*) No, needed contraction is $0.015''$
*c*) Not a good assumption as the coefficient of thermal expansion is a function of temperature.
*d*) trapezoidal rule with unequal segments$=-0.12872''$; polynomial regression$=-0.012897''$
*e*) cool in liquid nitrogen; you will get an estimated contraction of $-0.023627''$

# Chapter 07.08: Simpson's 3/8 th Rule of Integration

Lesson: Simpson's 3/8 th Rule

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *derive the formula for Simpson's 3/8 rule of integration,*

2) *use Simpson's 3/8 rule it to solve integrals,*

3) *develop the formula for composite (also called multiple-segment) Simpson's 3/8 rule of integration,*

4) *use composite Simpson's 3/8 rule of integration to solve integrals,*

5) *compare true error formulas for composite Simpson's 1/3 rule and composite Simpson's 3/8 rule, and*

6) *use a combination of Simpson's 1/3 rule and Simpson's 3/8 rule to approximate integrals.*

## Introduction

The main objective of this chapter is to develop appropriate formulas for approximating the integral of the form

$$I = \int_a^b f(x)dx \quad (1)$$

Most (if not all) of the developed formulas for integration are based on a simple concept of approximating a given function $f(x)$ by a simpler function (usually a polynomial function) $f_i(x)$, where $i$ represents the order of the polynomial function. In Chapter 07.03, Simpsons 1/3 rule for integration was derived by approximating the integrand $f(x)$ with a $2^{\text{nd}}$ order (quadratic) polynomial function.$f_2(x)$

$$f_2(x) = a_0 + a_1 x + a_2 x^2 \quad (2)$$

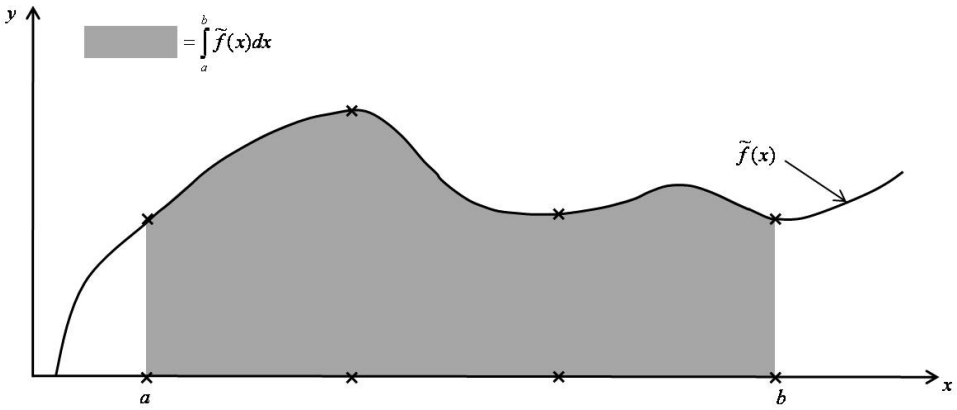



**Figure 1** $\widetilde{f}(x)$ Cubic function.

In a similar fashion, Simpson 3/8 rule for integration can be derived by approximating the given function$f(x)$ with the 3rd order (cubic) polynomial $f_3(x)$

$$\left.\begin{aligned} f_3(x) &= a_0 + a_1 x + a_2 x^2 + a_3 x^3 \\ &= \left\{1, x, x^2, x^3\right\} \times \begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \end{bmatrix} \end{aligned}\right\} \quad (3)$$

which can also be symbolically represented in Figure 1.

Method 1

The unknown coefficients $a_0,\ a_1,\ a_2$ and $a_3$ in Equation (3) can be obtained by substituting 4 known coordinate data points $\{x_0, f(x_0)\}, \{x_1, f(x_1)\}, \{x_2, f(x_2)\}$ and $\{x_3, f(x_3)\}$ into Equation (3) as follows.

$$\left.\begin{aligned} f(x_0) &= a_0 + a_1 x_0 + a_2 x_0^2 + a_3 x_0^2 \\ f(x_1) &= a_0 + a_1 x_1 + a_2 x_1^2 + a_3 x_1^2 \\ f(x_2) &= a_0 + a_1 x_2 + a_2 x_2^2 + a_3 x_2^2 \\ f(x_3) &= a_0 + a_1 x_3 + a_2 x_3^2 + a_3 x_3^2 \end{aligned}\right\} \quad (4)$$

Equation (4) can be expressed in matrix notation as

$$\begin{bmatrix} 1 & x_0 & x_0^2 & x_0^3 \\ 1 & x_1 & x_1^2 & x_1^3 \\ 1 & x_2 & x_2^2 & x_2^3 \\ 1 & x_3 & x_3^2 & x_3^3 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} f(x_0) \\ f(x_1) \\ f(x_2) \\ f(x_3) \end{bmatrix} \quad (5)$$

The above Equation (5) can symbolically be represented as

$$[A]_{4\times 4} \overrightarrow{a}_{4\times 1} = \overrightarrow{f}_{4\times 1} \quad (6)$$

Thus,

$$\overrightarrow{a} = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{bmatrix} = [A]^{-1} \times \overrightarrow{f} \quad (7)$$

Substituting Equation (7) into Equation (3), one gets

$$f_3(x) = \left\{1, x, x^2, x^3\right\} \times [A]^{-1} \times \overrightarrow{f} \quad (8)$$

a) As indicated in Figure 1, one has

b)

$$\begin{array}{ll} x_0 & = a \\ x_1 & = a + h \\ & = a + \frac{b-a}{3} \\ & = \frac{2a+b}{3} \\ x_2 & = a + 2h \\ & = a + \frac{2b-2a}{3} \\ & = \frac{a+2b}{3} \\ x_3 & = a + 3h \\ & = a + \frac{3b-3a}{3} \\ & = b \end{array} \quad (9)$$

With the help from MATLAB *Ref*.2, the unknown vector $\overrightarrow{a}$ (shown in Equation 7) can be solved for symbolically.

Method 2

Using Lagrange interpolation, the cubic polynomial function $f_3(x)$ that passes through 4 data points (see Figure 1) can be explicitly given as

$$\begin{aligned} f_3(x) = & \frac{(x-x_1)(x-x_2)(x-x_3)}{(x_0-x_1)(x_0-x_2)(x_0-x_3)} \times f(x_0) + \frac{(x-x_0)(x-x_2)(x-x_3)}{(x_1-x_0)(x_1-x_2)(x_1-x_3)} \times f(x_1) \\ & + \frac{(x-x_0)(x-x_1)(x-x_3)}{(x_2-x_0)(x_2-x_1)(x_2-x_3)} \times f(x_3) + \frac{(x-x_0)(x-x_1)(x-x_2)}{(x_3-x_0)(x_3-x_1)(x_3-x_2)} \times f(x_3) \end{aligned} \quad (10)$$

## Simpsons 3/8 Rule for Integration

**Substituting the form of $\mathbf{f_3(x)}$ from Method (1) or Method (2),**

$$\begin{aligned} I &= \int_a^b f(x)\,dx \\ &= \int_a^b f_3(x)\,dx \\ &= (b-a) \times \frac{\{f(x_0) + 3f(x_1) + 3f(x_2) + f(x_3)\}}{8} \end{aligned} \quad (11)$$

Since

$$h = \frac{b-a}{3}$$

$$b - a = 3h$$

and Equation (11) becomes

$$I \approx \frac{3h}{8} \times \{f(x_0) + 3f(x_1) + 3f(x_2) + f(x_3)\} \quad (12)$$

Note the 3/8 in the formula, and hence the name of method as the Simpson's 3/8 rule.

The true error in Simpson 3/8 rule can be derived as $Ref.1$

$$E_t = -\frac{(b-a)^5}{6480} \times f''''(\zeta), \text{ where } a \leq \zeta \leq b \quad (13)$$

**Example 1**

The vertical distance in meters covered by a rocket from $t = 8$ to $t = 30$ seconds is given by

$$s = \int_8^{30} \left(2000 \ln \left[\frac{140000}{140000 - 2100t}\right] - 9.8t\right) dt$$

Use Simpson 3/8 rule to find the approximate value of the integral.

**Solution**

$$\begin{aligned} h &= \frac{b-a}{n} \\ &= \frac{b-a}{3} \\ &= \frac{30-8}{3} \\ &= 7.3333 \end{aligned}$$

$$f(t) = 2000 \ln \left[\frac{140000}{140000 - 2100t}\right] - 9.8t$$

$$I \approx \frac{3h}{8} \times \{f(t_0) + 3f(t_1) + 3f(t_2) + f(t_3)\}$$

$$t_0 = 8$$

$$\begin{aligned} f(t_0) &= 2000 \ln \left(\frac{140000}{140000 - 2100 \times 8}\right) - 9.8 \times 8 \\ &= 177.2667 \end{aligned}$$

$$
\begin{aligned}
t_1 &= t_0 + h \\
&= 8 + 7.3333 \\
&= 15.3333
\end{aligned}
$$

$$
\begin{aligned}
f(t_1) &= 2000 \ln\left(\frac{140000}{140000 - 2100 \times 15.3333}\right) - 9.8 \times 15.3333 \\
&= 372.4629
\end{aligned}
$$

$$
\begin{aligned}
t_2 &= t_0 + 2h \\
&= 8 + 2(7.3333) \\
&= 22.6666
\end{aligned}
$$

$$
\begin{aligned}
f(t_2) &= 2000 \ln\left(\frac{140000}{140000 - 2100 \times 22.6666}\right) - 9.8 \times 22.6666 \\
&= 608.8976
\end{aligned}
$$

$$
\begin{aligned}
t_3 &= t_0 + 3h \\
&= 8 + 3(7.3333) \\
&= 30
\end{aligned}
$$

$$
\begin{aligned}
f(t_3) &= 2000 \ln\left(\frac{140000}{140000 - 2100 \times 30}\right) - 9.8 \times 30 \\
&= 901.6740
\end{aligned}
$$

Applying Equation (12), one has

$$
\begin{aligned}
I &= \frac{3}{8} \times 7.3333 \times \{177.2667 + 3 \times 372.4629 + 3 \times 608.8976 + 901.6740\} \\
&= 11063.3104m
\end{aligned}
$$

The exact answer can be computed as

$$
I_{\text{exact}} = 11061.34m
$$

## Composite Simpson 3/8 Rule

Using $n$ = number of equal segments, the width $h$can be defined as

$$
h = \frac{b - a}{n} \quad (14)
$$

The number of segments need to be an integer multiple of 3 as a single application of Simpson 3/8 rule requires 3 segments.

The integral shown in Equation (1) can be expressed as

$$\begin{aligned} I &= \int_a^b f(x)\,dx \\ &\approx \int_a^b f_3(x)\,dx \\ &\approx \int_{x_0=a}^{x_3} f_3(x)\,dx + \int_{x_3}^{x_6} f_3(x)\,dx + ..... + \int_{x_{n-3}}^{x_n=b} f_3(x)\,dx \end{aligned} \quad (15)$$

Using Simpson 3/8 rule (See Equation 12) into Equation (15), one gets

$$I = \frac{3h}{8}\left\{\begin{array}{c} f(x_0) + 3f(x_1) + 3f(x_2) + f(x_3) + f(x_3) + 3f(x_4) + 3f(x_5) + f(x_6) \\ +..... + f(x_{n-3}) + 3f(x_{n-2}) + 3f(x_{n-1}) + f(x_n) \end{array}\right\} \quad (16)$$

$$= \frac{3h}{8}\left\{ f(x_0) + 3\sum_{i=1,4,7,..}^{n-2} f(x_i) + 3\sum_{i=2,5,8,..}^{n-1} f(x_i) + 2\sum_{i=3,6,9,..}^{n-3} f(x_i) + f(x_n) \right\} \quad (17)$$

**Example 2**

The vertical distance in meters covered by a rocket from $t = 8$ to $t = 30$ seconds is given by

$$s = \int_8^{30} \left(2000 \ln\left[\frac{140000}{140000 - 2100t}\right] - 9.8t\right) dt$$

Use composite Simpson 3/8 rule with six segments to estimate the vertical distance.

**Solution**

In this example, one has (see Equation 14):

$$f(t) = 2000 \ln\left[\frac{140000}{140000 - 2100t}\right] - 9.8t$$

$$h = \frac{30 - 8}{6} = 3.6666$$

$$\{t_0, f(t_0)\} = \{8, 177.2667\}$$

$\{t_1, f(t_1)\} = \{11.6666, 270.4104\}$ where $t_1 = t_0 + h = 8 + 3.6666 = 11.6666$

$$\{t_2, f(t_2)\} = \{15.3333, 372.4629\} \text{ where } t_2 = t_0 + 2h = 15.3333$$

$$\{t_3, f(t_3)\} = \{19, 484.7455\} \text{ where } t_3 = t_0 + 3h = 19$$

$$\{t_4, f(t_4)\} = \{22.6666, 608.8976\} \text{ where } t_4 = t_0 + 4h = 22.6666$$

$$\{t_5, f(t_5)\} = \{26.3333, 746.9870\} \text{ where } t_5 = t_0 + 5h = 26.3333$$

$$\{t_6, f(t_6)\} = \{30901.6740\} \text{ where } t_6 = t_0 + 6h = 30$$

Applying Equation (17), one obtains:

$$\begin{aligned} I &= \frac{3}{8}(3.6666)\left\{177.2667 + 3\sum_{i=1,4,..}^{n-2=4} f(t_i) + 3\sum_{i=2,5,..}^{n-1=5} f(t_i) + 2\sum_{i=3,6,..}^{n-3=3} f(t_i) + 901.6740\right\} \\ &= (1.3750)\left\{\begin{array}{c} 177.2667 + 3(270.4104 + 608.8976) \\ +3(372.4629 + 746.9870) + 2(484.7455) + 901.6740 \end{array}\right\} \\ &= 11601.4696m \end{aligned}$$

**Example 3**

Compute

$$I = \int_8^{30} \left\{2000 \ln\left(\frac{140000}{140000 - 2100t}\right) - 9.8t\right\} dt,$$

using Simpson 1/3 rule (with $n_1 = 4$), and Simpson 3/8 rule (with $n_2 = 3$).

**Solution**

The segment width is

$$\begin{aligned} h &= \frac{b-a}{n} \\ &= \frac{b-a}{n_1 + n_2} \\ &= \frac{30-8}{(4+3)} \\ &= 3.1429 \end{aligned}$$

$$f(t) = 2000 \ln\left[\frac{140000}{140000 - 2100t}\right] - 9.8t$$

$$\left.\begin{array}{c} t_0 = a = 8 \\ t_1 = x_0 + 1h = 8 + 3.1429 = 11.1429 \\ t_2 = t_0 + 2h = 8 + 2(3.1429) = 14.2857 \\ t_3 = t_0 + 3h = 8 + 3(3.1429) = 17.4286 \\ t_4 = t_0 + 4h = 8 + 4(3.1429) = 20.5714 \end{array}\right\} \text{Simpson's 1/3rule}$$

$$\begin{aligned} t_5 &= t_0 + 5h = 8 + 5(3.1429) = 23.7143 \\ t_6 &= t_0 + 6h = 8 + 6(3.1429) = 26.8571 \\ t_7 &= t_0 + 7h = 8 + 7(3.1429) = 30 \end{aligned}$$

Now

$$\begin{aligned} f\left(t_0 = 8\right) &= 2000 \ln\left(\frac{140,000}{140,000 - 2100 \times 8}\right) - 9.8 \times 8 \\ &= 177.2667 \end{aligned}$$

Similarly:

$$f\left(t_1\right) = 256.5863$$

$$f\left(t_2\right) = 342.3241$$

$$f\left(t_3\right) = 435.2749$$

$$f\left(t_4\right) = 536.3909$$

$$f\left(t_5\right) = 646.8260$$

$$f\left(t_6\right) = 767.9978$$

$$f\left(t_7\right) = 901.6740$$

For several segments ($n_1 =$ first 4 segments), using composite Simpson 1/3 rule, one obtains (See Equation 19):

$$\begin{aligned} I_1 &= \left(\frac{h}{3}\right)\left\{f\left(t_0\right) + 4\sum_{i=1,3,\ldots}^{n_1-1=3} f\left(t_i\right) + 2\sum_{i=2,\ldots}^{n_1-2=2} f\left(t_i\right) + f\left(t_{n_1}\right)\right\} \\ &= \left(\frac{h}{3}\right)\left\{f\left(t_0\right) + 4\left(f\left(t_1\right) + f\left(t_3\right)\right) + 2f\left(t_2\right) + f\left(t_4\right)\right\} \\ &= \left(\frac{3.1429}{3}\right)\left\{177.2667 + 4\left(256.5863 + 435.2749\right) + 2\left(342.3241\right) + 536.3909\right\} \\ &= 4364.1197 \end{aligned}$$

For several segments ($n_2 =$ last 3 segments), using single application Simpson 3/8 rule, one obtains (See Equation 17):

$$
\begin{aligned}
I_2 &= \left(\frac{3h}{8}\right)\left\{f\left(t_0\right)+3\sum_{i=1,3,\ldots}^{n_2-2=1} f\left(t_i\right)+3\sum_{i=2,\ldots}^{n_2-1=2} f\left(t_i\right)+2\sum_{i=3,6,..}^{n_2-3=0} f\left(t_i\right)+f\left(t_{n_1}\right)\right\} \\
&= \left(\frac{3h}{8}\right)\left\{f\left(t_0\right)+3f\left(t_1\right)+3f\left(t_2\right)+2(\text{no contribution})+f(t_3)\right\} \\
&= \left(\frac{3h}{8}\right)\left\{f\left(t_4\right)+3f\left(t_5\right)+3f\left(t_6\right)+f(t_7)\right\} \\
&= \left(\frac{3}{8}\times 3.1429\right)\left\{536.3909+3\left(646.8260\right)+3\left(767.9978\right)+901.6740\right\} \\
&= 6697.3663
\end{aligned}
$$

The mixed (combined) Simpson 1/3 and 3/8 rules give

$$
\begin{aligned}
I &= I_1 + I_2 \\
&= 4364.1197 + 6697.3663 \\
&= 11061m
\end{aligned}
$$

c) Comparing the truncated error of Simpson 1/3 rule

d)

$$E_t = -\frac{(b-a)^5}{2880}\times f''''\zeta) \quad (18)$$

With Simpson 3/8 rule (See Equation 12), it seems to offer slightly more accurate answer than the former. However, the cost associated with Simpson 3/8 rule (using 3rd order polynomial function) is significantly higher than the one associated with Simpson 1/3 rule (using 2nd order polynomial function).

The number of segments that can be used in the conjunction with Simpson 1/3 rule is 2, 4, 6, 8, ... (any even numbers) for

$$
\begin{aligned}
I &= \int_a^b f(x)dx \\
&\approx \left(\frac{h}{3}\right)\left\{f\left(x_0\right)+4f\left(x_1\right)+f\left(x_2\right)+f\left(x_2\right)+4f\left(x_3\right)+f\left(x_4\right)+...+f\left(x_{n-2}\right)+4f\left(x_{n-1}\right)+f\left(x_n\right)\right\} \\
&= \left(\frac{h}{3}\right)\left\{f\left(x_0\right)+4\sum_{i=1,3,\ldots}^{n-1} f\left(x_i\right)+2\sum_{i=2,4,6\ldots}^{n-2} f\left(x_i\right)+f\left(x_n\right)\right\}
\end{aligned}
$$

However, Simpson 3/8 rule can be used with the number of segments equal to 3,6,9,12,.. (can be certain integers that are multiples of 3).

If the user wishes to use, say 7 segments, then the mixed Simpson 1/3 rule (for the first 4 segments), and Simpson 3/8 rule (for the last 3 segments) would be appropriate.

## Computer Algorithm for Mixed Simpson 1/3 and 3/8 Rule for Integration

Based on the earlier discussion on (single and composite) Simpson 1/3 and 3/8 rules, the following "pseudo" step-by-step mixed Simpson rules for estimating

$$I = \int_a^b f(x)dx$$

can be given as

Step 1

User inputs information, such as

$$f(x) = \text{integrand}$$

$$n_1 = \text{number of segments in conjunction with Simpson 1/3 rule (a multiple of 2 (any even numbers)}$$

$$n_2 = \text{number of segments in conjunction with Simpson 3/8 rule (a multiple of 3)}$$

Step 2

Compute

$$n = n_1 + n_2$$

$$h = \frac{b-a}{n}$$

$$x_0 = a$$

$$x_1 = a + 1h$$

$$x_2 = a + 2h$$

$$\vdots$$

$$x_i = a + ih$$

$$\vdots$$

$$x_n = a + nh = b$$

Step 3

Compute result from composite Simpson 1/3 rule (See Equation 19)

$$I_1 = \left(\frac{h}{3}\right)\left\{f(x_0) + 4\sum_{i=1,3,\ldots}^{n_1-1} f(x_i) + 2\sum_{i=2,4,6\ldots}^{n_1-2} f(x_i) + f(x_{n_1})\right\} \quad (19, repeated)$$

Step 4

Compute result from composite Simpson 3/8 rule (See Equation 17)

$$I_2 = \left(\frac{3h}{8}\right)\left\{f(x_0) + 3\sum_{i=1,4,7\ldots}^{n_2-2} f(x_i) + 3\sum_{i=2,5,8\ldots}^{n_2-1} f(x_i) + 2\sum_{i=3,6,9,\ldots}^{n_2-3} f(x_i) + f(x_{n_2})\right\} \quad (17, repeated)$$

Step 5

$$I \approx I_1 + I_2 \quad (20)$$

and print out the final approximated answer for $I$.

# Multiple Choice Test

## Chapter 07.08 Simpson's Three-Eighth Rule of Integration

(1). Simpson 3/8 rule for integration is mainly based upon the idea of

(A) approximating $f(x)$ in $I = \int_a^b f(x)dx$ by a cubic polynomial

(B) approximating $f(x)$ in $I = \int_a^b f(x)dx$ by a quadratic polynomial

(C) Converting the limit of integral limits $[a, b]$ into $[-1, +1]$

(D) Using similar concepts as Gauss quadrature formula

(2). The exact value of $\int_1^4 (e^{-2x} + 4x^2 - 8)dx$ most nearly is

(A) 6.0067

(B) 5.7606

(C) 60.0675

(D) 67.6075

(3). The approximate value of $\int_1^4 (e^{-2x} + 4x^2 - 8)dx$ by a single application of Simpson's 3/8 rule is

(A) 61.3740

(B) 60.0743

(C) 59.3470

(D) 58.8992

(4). The approximate value of $\int_1^4 (e^{-2x} + 4x^2 - 8)dx$ by a composite Simpson's 3/8 rule with $n$=6 segments is most nearly

(A) 60.8206

(B) 60.6028

(C) 61.0677

(D) 60.0675

(5). The approximate value of $\int_1^4 (e^{-2x} + 4x^2 - 8)dx$ by combination of Simpson's 1/3 rule (n=6 segments) and Simpson's 3/8 rule ($n$=3 segments) most nearly is

(A) 60.0677

(B) 59.0677

(C) 61.0677

(D) 59.7607

(6). Comparing Simpson's 3/8 rule truncated error formula

$$E_t = -\frac{(b-a)^5}{6480} \times f^{(4)}(\zeta), \ a \leq \zeta \leq b,$$

with Simpson's 1/3 rule truncated error formula

$$E_t = -\frac{(b-a)^5}{2880} f^{(4)}(\zeta), \quad a < \zeta < b$$

the following conclusion can be made.

(A) Simpson's 3/8 rule is significantly more accurate than Simpson's 1/3 rule

(B) It is worth it in terms of computational efforts versus accuracy to use Simpson's 3/8 rule instead of Simpson's 1/3 rule.

(C) It is worth it in terms of computational efforts versus accuracy to use Simpson's 3/8 rule instead of Simpson's 1/3 rule.

(D) Simpson's 3/8 rule is less accurate than Simpson's 1/3 rule.

# Problem Set

## Chapter 07.08: Simpson's 3/8th Rule of Integration

(1). Compute the following integral exactly

$$\int_1^4 x^2 e^{(3x)} dx$$

**Answer:** $7.3541 \times 10^5$

/ /

(2). Using composite Simpson 3/8 rule (with $n = 6$ "little" segments), compute the integral

$$\int_1^4 x^2 e^{(3x)} dx$$

**Answer:** $I = 784785.6554$

/ /

(3). Using the MIXED composite Simpson 1/3 rule (with $n_1 = 4$ "little" segments), and Simpson 3/8 rule (with $n_2 = 6$ "little" segments), compute the integral $\int_1^4 x^2 e^{(3x)} dx$

**Answer:** $I = 744173.7172$

/ /

(4). Compute the following integral exactly

$$\int_\pi^{2\pi} x \sin(2x) dx$$

**Answer:** $I = -1.5708$

/ /

(5). Using the single application Simpson 3/8 rule (with $n_2 = 3$ "little" segments), compute the integral $\int_\pi^{2\pi} x \sin(2x) dx$

**Answer:** $I = -1.0684$

/ /

(6). Using the composite Simpson 3/8 rule (with $n_2 = 6$ "little" segments), compute the integral $\int_\pi^{2\pi} x \sin(2x) dx$

**Answer:** $I = -1.6026$

/ /

(7). Using the MIXED, single-application Simpson 1/3 and 3/8 rules (with $n_1 = 2$, and $n_2 = 3$ "little" segments), compute the integral $\int_{\pi}^{2\pi} x \sin(2x) dx$

**Answer:** $I = -1.7357$

# Chapter 08.00: Physical Problem for Ordinary Differential Equations

Lesson: General Engineering

## Summary

To find the temperature of a heated ball as a function of time, a first-order ordinary differential equation must be solved.

## Modeling

You are working for a ball bearing company - Ralph's Bearings. For a long lasting life of some spherical bearings made in this company, after heating them to a high temperature of 1000 K, they need to be quenched for 30 seconds in water that is maintained at a room temperature of 300 K. However, it takes time to take the ball from the furnace to the quenching bath, and its temperature falls. The temperature of the furnace is 1200 K, and it takes 10 seconds to take the ball to the quenching bath. Does the ball need to be in the air for less or more than 10 seconds to get to a temperature of 1000 K?

### Mathematical model

To keep the mathematical simple, one can assume the spherical bearing to be a lumped mass system. What does a lumped system mean? It implies that the internal conduction in the sphere is large enough that the temperature throughout the ball is uniform. This assumption allows us to assume that the temperature is only a function of time and not of the location in the spherical ball. This dependence of temperature only on time for a lumped system means that if a differential equation governs this physical problem, it would be an ordinary differential equation. For a nonlumped system, a partial differential equation would govern. In your heat transfer course, you will learn when a system can be considered lumped or nonlumped. In simplistic terms, this distinction is based on the material, geometry, and heat exchange factors of the ball with its surroundings.

Now assuming a lumped-mass system, let us develop the mathematical model for the above problem. When the ball is taken out of the furnace at the initial temperature of $\theta_0$ and is cooled by radiation to its surroundings at the temperature of $\theta_a$, the rate at which heat is lost to radiation is

$$\text{Rate of heat lost due to radiation} = A\epsilon\sigma\left(\theta^4 - {\theta_a}^4\right) \qquad (1)$$

where

$$A = \text{surface area of the ball, m}^2$$

$$\epsilon = \text{emittance}$$

$$\sigma = \text{Stefan-Boltzmann constant, } 5.67 \times 10^{-8}\ \frac{\text{J}}{\text{s} \cdot \text{m}^2 \cdot \text{K}^4}$$

$$\theta = \text{temperature of the ball at a given time, K}$$

In the above equation for radiation heat loss, emittance $\epsilon$ is defined as the total radiation emitted divided by total radiation that would be emitted by a blackbody at the same temperature. The emittance is always between 0 and 1. A black body is a body that emits and absorbs at any temperature the maximum possible amount of radiation at any given wavelength.

Stefan-Botzmann constant, $\sigma$ was discovered by two Austrian scientists – J. Stefan and L. Boltzmann. Stefan found it experimentally in 1879, and Boltzmann derived it theoretically in 1884.

The energy stored in the mass is given by

$$\text{Energy stored by mass} = mC\theta \qquad (2)$$

where

$$m = \text{mass of the ball, kg}$$

$$C = \text{specific heat of the ball, J}/\left(\text{kg} - \text{K}\right)$$

From an energy balance,

$$\begin{aligned}&\text{Rate at which heat is gained} - \text{Rate at which heat is lost}\\ &= \text{Rate at which heat is stored}\end{aligned}$$

gives

$$-A\epsilon\sigma\left(\theta^4 - \theta_a^4\right) = mC\frac{d\theta}{dt} \qquad (3)$$

Given the

Radius of the ball, $r = 2.0$ cm

Density of the ball, $\rho = 7800\ \text{kg/m}^3$

Specific heat, $C = 420\ \text{J/(kg} \cdot \text{K)}$

Emittance, $\in = 0.85$

Stefan-Boltzmann Constant, $\sigma = 5.67 \times 10^{-8}\ \dfrac{\text{J}}{\text{s} \cdot \text{m}^2 \cdot \text{K}^4}$

Initial temperature of the ball, $\theta\,(0) = 1200$ K,

Ambient temperature, $\theta_a = 300$ K,

we have

Surface area of the ball

$$\begin{aligned} A &= 4\pi r^2 \\ &= 4\pi\,(0.02)^2 \\ &= 5.02654 \times 10^{-3}\text{m}^2 \end{aligned}$$

Mass of the ball

$$\begin{aligned} M &= pV \\ &= \rho\left[\frac{4}{3}\pi r^3\right] \\ &= 7800 \times \left(\frac{4}{3}\right)\pi(0.02)^3 \\ &= 0.261380\ \text{kg} \end{aligned}$$

Hence

$$-A\epsilon\sigma(\theta^4 - {\theta_a}^4) = mC\frac{d\theta}{dt} \qquad (3 - repeated)$$

reduces to

$$-(5.02654 \times 10^{-3})(0.85)(5.67 \times 10^{-8})(\theta^4 - 300^4) = (0.261380)(420)\frac{d\theta}{dt}$$

$$\frac{d\theta}{dt} = -2.20673 \times 10^{-12} \left(\theta^4 - 81 \times 10^8\right) \qquad (4)$$

**An improved mathematical model**

The heat from the ball can also be lost due to convection. The rate of heat lost due to convection is

$$\text{Rate of heat lost due to convection} = hA\left(\theta - \theta_a\right), \qquad (5)$$

where

$$h = \text{the convective cooling coefficient, } \left[\text{W}/\left(\text{m}^2 - \text{K}\right)\right].$$

Hence the heat is lost is due to convection as well as radiation and is given by

$$\text{Rate of heat lost due to convection and radiation} = A \in \sigma\left(\theta^4 - \theta_a^4\right) + hA\left(\theta - \theta_a\right)$$

$$\text{Energy stored by mass} = mC\theta \qquad (6)$$

From an energy balance,

$$\begin{aligned} &\text{Rate at which heat is gained} - \text{Rate at which heat is lost} \\ &= \text{Rate at which heat is stored} \end{aligned}$$

gives

$$-A\epsilon\sigma\left(\theta^4 - \theta_a^4\right) - hA\left(\theta - \theta_a\right) = mC\frac{d\theta}{dt} \qquad (7)$$

Given $h = 350\frac{J}{s-m^2-K}$ and considering both convection and radiation, and substituting the values of the constants given before

$$-(5.02654 \times 10^{-3})(0.85)(5.67 \times 10^{-8})(\theta^4 - 300^4)$$

$$-(350)(5.02654 \times 10^{-3})(\theta - 300) = (0.261380)(420)\frac{d\theta}{dt}$$

$$\frac{d\theta}{dt} = -2.20673 \times 10^{-13}(\theta^4 - 81 \times 10^8) - 1.60256 \times 10^{-2}(\theta - 300) \qquad (8)$$

The solution to the above ordinary differential equation with the initial condition of $\theta\left(0\right) = 1200$ K would give us the temperature of the ball as a function of time. We can then find at what time the ball temperature drops to 1000 K.

## Questions

(1) Note that the ordinary differential equation given by Equation (8) is non-linear. Is there is an exact solution to the problem?

(2) You are asked to solve the inverse problem, that is, when will the dependent variable temperature equal to 1000 K. How would you solve the inverse problem using different numerical methods such as Euler's and Runge-Kutta's methods?

(3) Can you find if convection or radiation can be neglected? How would you quantify the effect of neglecting one or the other?

(4) Find the following at $t = 5$ s

a) Rate of change of temperature,

b) Rate at which heat is lost due to convection,

c) Rate at which heat is lost due to radiation,

d) Rate at which heat is stored in the ball.

# Chapter 08.00: Physical Problem for Ordinary Differential Equations

Lesson: Chemical Engineering

## Summary

Sodium chloride waste control while making soap requires solution of an ordinary differential equation

## Modeling

Soap is prepared through a reaction known as saponification. In saponification, tallow (fats from animals such as cattle) or vegetable fat (e.g. coconut) is reacted with potassium or sodium hydroxide to produce glycerol and fatty acid salt known as "soap". The soap is separated from the glycerol through precipitation by the addition of sodium chloride. Water layer on top of the mixture that contains dissolved sodium chloride is drawn-off the mixture as a waste. This method of soap making is still being practiced in many villages in the developing countries where the price of mass produced soap maybe too expensive for the average villager.

Two chemical engineering students used knowledge of saponification acquired in their organic chemistry class to organize and produce "home made" soap. The local ordinance requires that the minimum concentration level for sodium chloride waste in any liquid that is discharged into the environment must not exceed 11.00 g/L. Sodium chloride laden liquid water is the major waste of the process. The company has only one 15-liter tank for waste storage. On filling the waste tank, the tank contained 15 liters of water and 750 grams of sodium chloride. To continue production and meet local ordinance, it is desired to pump in fresh water into the tank at the rate of 2.0 liters per minute while waste salt water containing 25 grams of salt per liter is added at the rate of 1.5 liters per minute. To keep the solution level at 15 liters, 3.5 liters per minute of the waste is discharged. A sketch Figure 1 of the flows is given below where $A$ represents the waste stream from the process, $B$ is the fresh water stream and $C$ is the discharge stream to the environment. Here, it is assumed that as the two streams, $A$ and $B$ enter into the tank, instantaneously the chloride concentration in the tank changes to the exit concentration,$X_1$.

The material (sodium chloride) balance on the tank system can be written as

$$\text{Accumulation} = \text{input} - \text{output} + \text{removal by reaction} \quad (1)$$

Noting that no chemical reaction occurs in the storage tank (i.e. the third term on the right hand side of (1) is zero), the above equation can be written as

$$\frac{dx_1}{dt} = (25g/L)(1.5L/min) + (0g/L)(2L/min) - (x_1 g/L)(3.5L/min) + 0 \quad (2)$$

Simplifying Equation (2), we obtain

$$\frac{dx_1}{dt} + 3.5x_1 = 37.5 \quad (3)$$

For the initial conditions of the ordinary differential equations in Equation (3), recall that at $t = 0$, the salt concentration in the tank was given as 750 g/15L (50g/L) , that is,

$$t = 0, x_1(0) = \frac{750}{15} = 50g/L \quad (4)$$

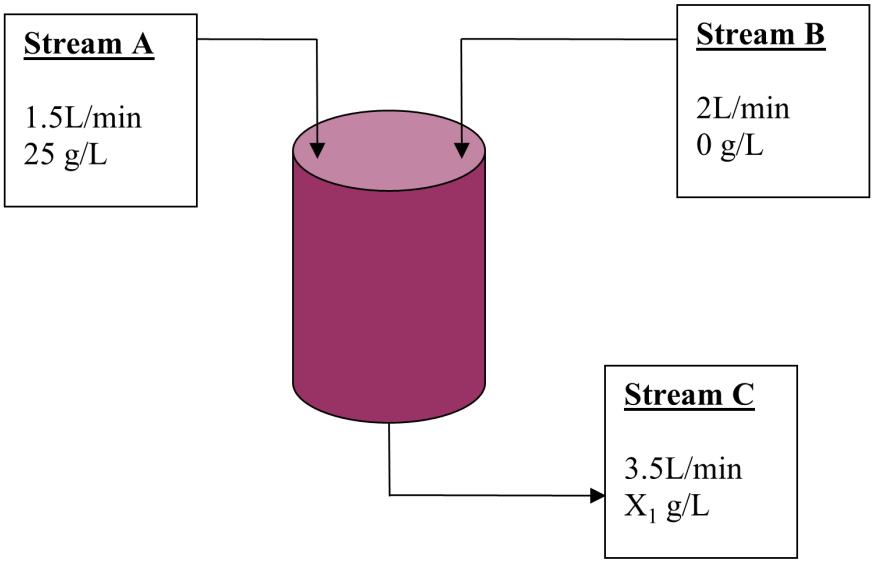



**Figure 1** Sketch of flow for a saponification process.

## Question

(1) Using Equations (3) and (4), numerically determine how the concentration of salt being discharged changes with time.

(2) Plot the solution obtained.

(3) How long did it take to achieve the minimum required local ordinance? At steady state, what is the concentration of salt being discharged from this local soap factory?

# Chapter 08.00: Physical Problem for Ordinary Differential Equations

Lesson: Civil Engineering


## Summary

A physical problem of finding how much time it would take a lake to have safe levels of pollutant. To find the time, the problem is modeled as an ordinary differential equation.


## Modeling

Pollution in lakes can be a serious issue as they are used for recreation use. Pollution resulting from sewage, runoff from suburban yards that is loaded with fertilizers and pesticides to keep the homeowner's associations off your back can be a safety hazard for people. One is generally interested in knowing that if the concentration of a particular pollutant is above acceptable levels, how long will it take for the pollution level to decrease to an acceptable level.

**Figure 1** Pollutant in a lake

$$\text{Mass of pollutant} = \text{Mass of pollutant entering} - \text{Mass of pollutant leaving}$$

which also gives

$$\text{Rate of change of mass of pollutant} = \text{Rate of change of mass of pollutant entering} - \text{Rate of change of mass of pollutant leaving.}$$

If the concentration of pollutant is given by

$$C(t) = \frac{M(t)}{V}$$

where

$$M(t) = \text{mass of pollutant at time } t$$

$$V = \text{Volume of lake.}$$

Rate of change of mass of pollutant entering is $QC_o$, where $Q$ is the flow rate of the water into the lake, and the rate of change of mass of pollutant leaving the lake is $\frac{QM(t)}{V}$.

The above assumes that the flow rate of water going in and out of lake is the same. We also assuming that the pollutant is uniformly distributed in the lake. Also, no reaction is assumed. This gives

$$\frac{dM(t)}{dt} = QC_o - \frac{\dot{Q}M(t)}{V}$$

Now

$$M(t) = VC_o(t)$$

giving

$$V\frac{dC}{dt} = \dot{Q}C_o - \dot{Q}C$$

$$V\frac{dC}{dt} + \dot{Q}C = \dot{Q}C_o$$

Assume a weekly flow rate of fresh water as $1.5 \times 10^6 m^3$. $C_o = 0$ as we are assuming only fresh water coming in. The volume of the lake is $25 \times 10^6 m^3$. If the initial concentration of the pollutant is $10^7 parts/m^3$, and the acceptable level is $5 \times 10^6 parts/m^3$, how much time would it take for the pollutant to reach acceptable levels.

$$25 \times 10^6 \frac{dC}{dt} + 1.5 \times 10^6 C = 0$$

$$\frac{dC}{dt} + 0.06C = 0, C(0) = 10^7$$

# Chapter 08.00: Physical Problem for Ordinary Differential Equations

Lesson: Computer Engineering

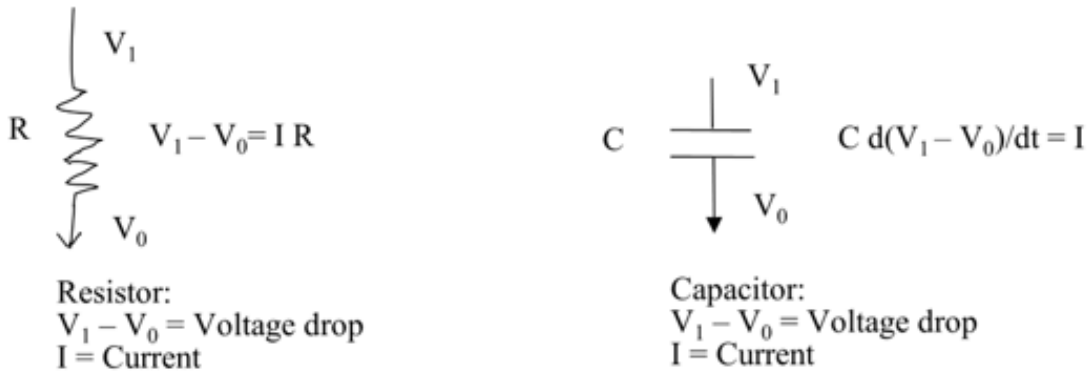
## Summary

Transient analysis of resistor-capacitor requires solution of ordinary differential equations.

## Modeling

Resistors and capacitors are fundamental elements of any circuit. Even the behavior of semiconductor devices in your computer can be modeled employing these basis elements (along with some others). A transistor is comprised of junctions of different kinds of materials, giving rise to interesting electrical properties. The electrical properties at these semiconductor junctions can be characterized using resistors (R) and capacitors (C), giving rise to the name "RC-model". In this module we will consider the electrical behavior of the simplest configuration of the RC elements and see how it can characterized using ordinary differential elements.

Two quantities that are important in electrical circuits are voltage (denoted here by V) and current (denoted here by I). The current through a resistor has a linear relationship with voltage and is defined by $V = IR$, where R, is constant used to quantify the resistor. This is called the Ohms law and you should have seen it one of your physics classes. A capacitor is a bit stranger element for which the rate of change of voltage across it is proportional to the current. Or in other words: $C\frac{dV}{dt} = I$, where C is called the capacitance. Voltage changes slowly across high capacitance. These elements are represented by diagrams as shown below.



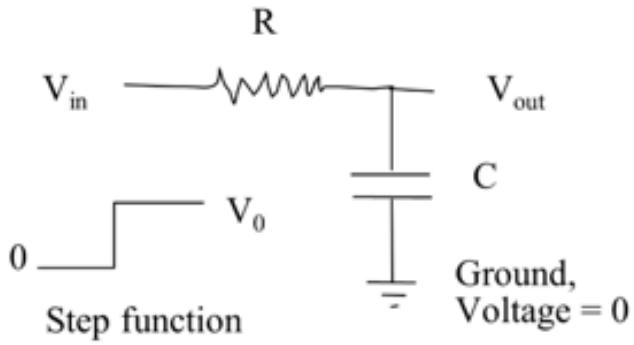



**Figure 1.** A simple RC circuit. As the input voltage is switched on at t=0, what happens to the output voltage.

Consider the simple combination of resistor and capacitor, as shown above. This kind of configuration is not rare and occurs in all power supplies to your computer. There are other components in addition to this, but this is one of the basic arrangements. The voltage at one end of the resistor is denoted by Vin and the voltage at the other end is denoted by Vout. The latter end is also connected to the capacitor, whose other end of which is connected to ground, which can be taken to be at zero voltage. The question we ask is what happens to the output voltage Vout when the input voltage Vin is switched on suddenly?

The voltage and current relationship across the resistor and the capacitor can be expressed, respectively as

$$V_{in} - V_{out} = IR \text{ and}$$

$$C\frac{dV_{out}}{dt} = I$$

where I is the current through the circuit. Eliminating this quantity from the above two equations, we have

$$V_{in} - V_{out} = RC\frac{dV_{out}}{dt}$$

For any input voltage profile we can solve this equation to arrive at the output voltage profile.

**Worked Out Example**

The sudden change in the input voltage can be modeled using the following function

$$V_{in} = \begin{cases} V_0 & \forall t \geq 0 \\ 0 & \forall t < 0 \end{cases}$$

This is also called a "step" function. For this input the differential equation modeling the output voltage for $t > 0$ is given by

$$RC\frac{dV_{out}}{dt} = V_0 - V_{out}$$

The solution of this equation we need some boundary conditions. We know the voltage at $t = 0$ is zero, i.e. $V_{out}(0) = 0$, and the voltage after a long time should be equal to the new input voltage, i.e $V_{out}(\infty) = V_0$. Using boundary conditions, in along with general solution form: $V_{out}(t) = c_o \exp(at) + c_1 \exp(-at) + c_2$, we can arrive at the following solution for the output voltage

$$V_{out}(t) = V_o(1 - \exp(-\frac{t}{RC}))$$

In the Figure 2 we see the plot of the transient output voltage for various values of the product RC. The output starts at zero and gradually converges to V0 = 1. The rate of convergence is dependent on the product RC.

## Questions

(1) If the for $V_{in} = x_1(t)$, the output voltage is given by the function $y_1(t)$, and for $V_{in} = x_2(t)$, the output voltage is given by the function $y_2(t)$, show that the output for the input $V_{in} = x_1(t) + x_2(t)$, is given by $y_1(t) + y_2(t)$.

(2) If the for $V_{in} = x_1(t)$, the output voltage is given by the function $y_1(t)$, show that the output for the input $V_{in} = ax_1(t)$, is given by $ay_1(t)$.

(3) Consider a pulse shaped input that is defined by:

$$V_{in}(t) = \begin{cases} 0 & \forall t < 0 \\ V_0 & 0 \leq t \leq T \\ 0 & t > T \end{cases}$$

What is the form of the output?

Hint: Express this input of in the form $x_1(t) - x_2(t)$ and then use the properties you have derived so far to arrive at the solution.

(4) What can you say about the output as the ratio $\frac{T}{RC}$ is varied?

(5) What can say about the behavior of this circuit when there is a voltage "spike" at the input?

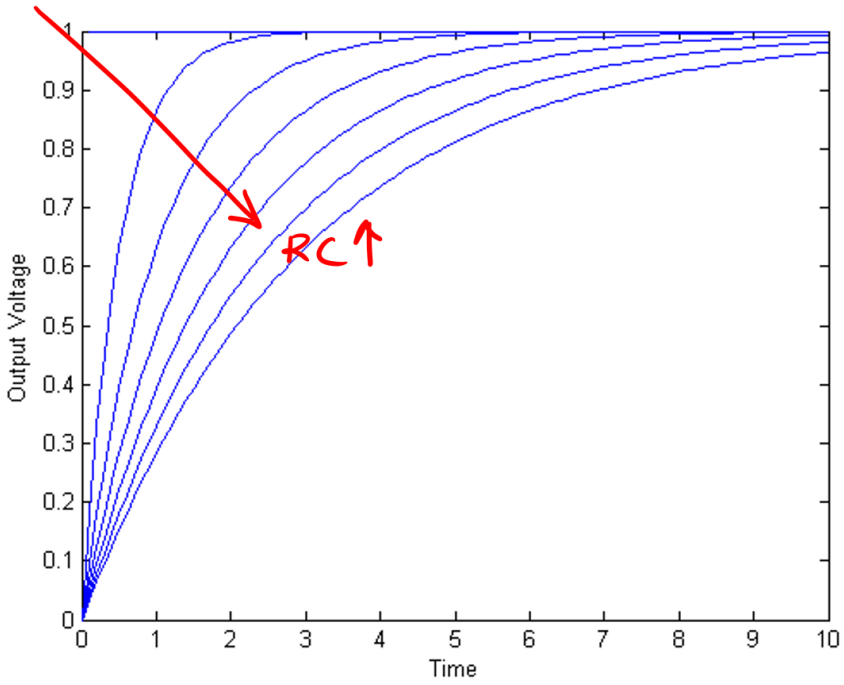



**Figure 2.** Output voltage, Vout(t), for various values of the product RC.

# Chapter 08.00: Physical Problem for Ordinary Differential Equations

Lesson: Electrical Engineering

## Summary

A rectifier-based power supply requires a capacitor to store power temporarily when the rectified waveform from the AC source drops below the target voltage. To size this capacitor properly, a first-order ordinary differential equation must be solved.

## Modeling

Small non-switching power supplies such as AC power bricks are typically built around a small transformer, rectifier, and voltage regulator as shown in Figure 1.

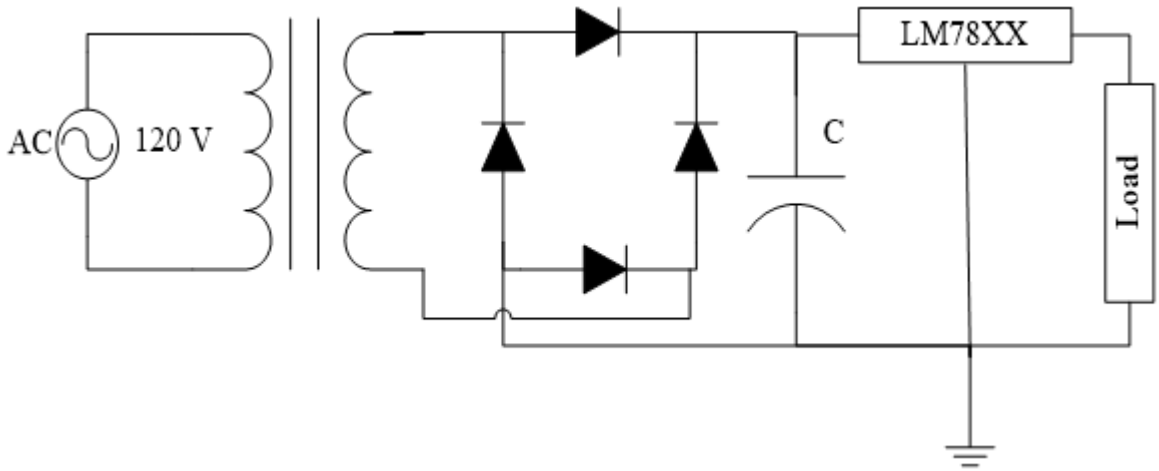



**Figure 1**. Full-Wave Rectified DC Power Supply

The transformer is used to reduce the AC voltage levels to a more reasonable range (e.g. 12 VAC or 18 VAC) and the bridge rectifier will take the negative half cycle of the AC waveform and convert it to a positive half cycle. A typical full-wave rectified waveform is shown in Figure 2. This waveform was generated assuming an ideal diode operating under no-load conditions and can be modeled with the following equation:

$$V_{\text{full}} = \max\left(\left|18 \times \cos\left(t\right) - 1.4\right|, 0\right)$$

For a half-wave rectified system built using only a single diode, the equation would become:

$$V_{\text{half}} = \max\left(18 \times \cos\left(t\right) - 0.7, 0\right)$$

and the waveform would look similar to Figure 2 except every other hump would be replaced with 0.

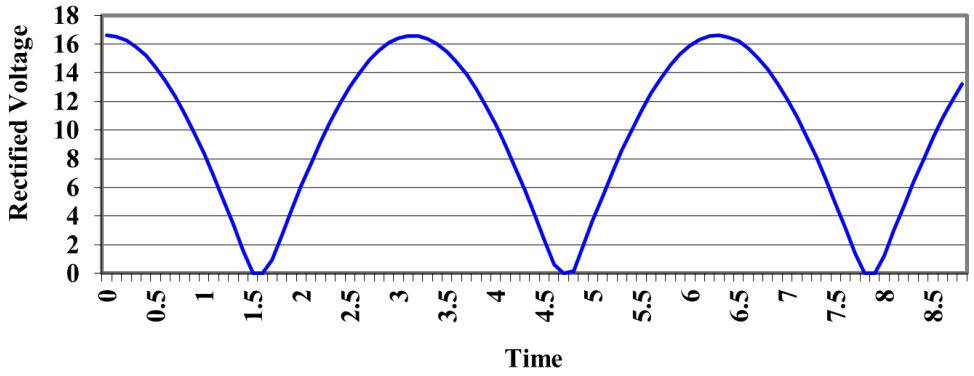



**Figure 2** Full-Wave Rectified AC Signal.

This is clearly not suitable as a DC power source since the load is looking for a constant DC value. This is where the capacitor, C, and the LM78XX voltage regulator of Figure 1 become important. A typical voltage regulator requires that the voltage on the input pin maintain a certain margin above the regulated output voltage. This is typically in the range of 2.5 to 3 V and for the LM7805, which provides 5 VDC, it is 2.5 V [1]. This means that the minimum voltage at the input pin of the LM7805 must not drop below 7.5 V. Clearly, the waveform in Figure 2 does that regularly.

The task of the capacitor, C, in the power supply is to store up charge when the rectifier provides voltage in excess of the required minimum and then support the voltage when the regulated voltage is below the minimum by providing stored charge to the voltage regulator. Thus, the circuit has two distinct phases of operation.

Phase 1

When the rectified voltage is above the voltage across the capacitor $V_c$, then the rectifier provides current to the capacitor and the voltage regulator, and the capacitor charges. The ideal model of a diode will not be adequate here since we need to know how much current the diode can supply. A better model of the diode can be found from its VI characteristic such as the one shown in Figure 3. A model for this will be developed shortly.

Phase 2

When the rectified voltage drops below $V_c$ current ceases to flow between the rectifier diodes and the capacitor. The capacitor then becomes the sole source of current to the voltage regulator by bleeding off its charge. This has the additional effect of continually reducing the voltage across the capacitor. As long as it does not fall below the required threshold, the regulator can successfully do its job.

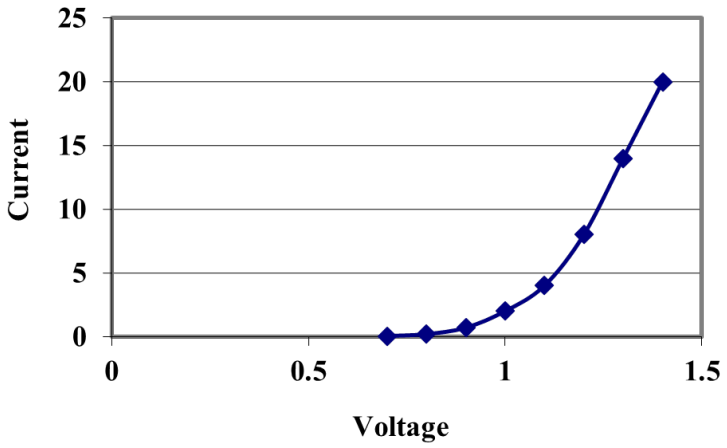



**Figure 3** VI Characteristic of a 1N4001 Diode [2].

Diode Model

The diode of Figure 3 must be modeled to account properly for the current it can supply during the Phase 1 charging phase of the capacitor. It is reasonable to model a diode under these conditions as a voltage drop and a series resistance. This essentially treats the Figure 3 curve as piece-wise linear model and it can only be applied when the diode is on (that is, it does not properly model an under-biased or reverse-biased diode) [3, pp. 67-74]. This model is shown in Figure 4 and for the 1N4001 of Figure 3 suitable values are $V_d = 1.0\ V$ and $R_d = 0.02\ \Omega$. Other diodes will have similar numbers.

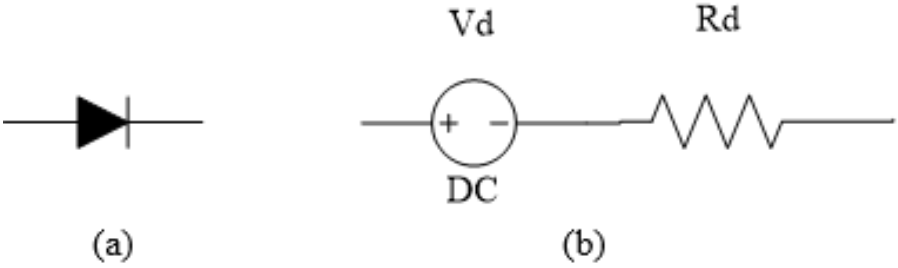



**Figure 4** (a) Ideal Diode and (b) Piece-wise Forward Bias Model.

Load Model

The final piece to the picture is how to model the load represented by the voltage regulator. Rather than get into the details of the internal structure of the voltage regulator it is sufficient to model the worst-case scenario when choosing the appropriate capacitor. This is represented by the peak continuous current that the power-supply is expected to provide. For this problem, it will be assumed to be 100 mA.

Circuit Models

With the Phase 1 charging, and Phase 2 discharging, circuit models of the power supply circuit can now be built. They are shown in Figure 5. In both cases, the circuit must provide 100 mA to the load.

It will be additionally assumed that a transformer will be chosen that provides sufficient AC power at 18 VAC running at 60 Hz.

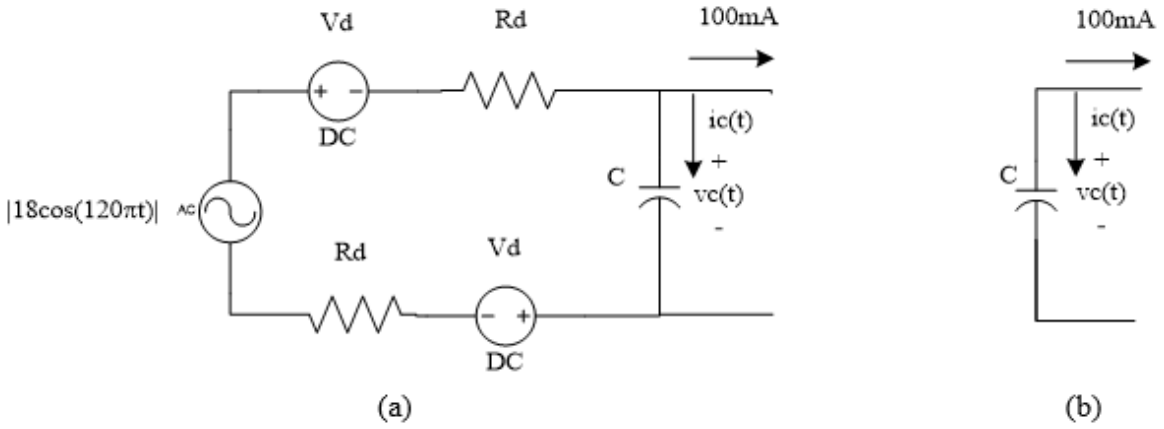



**Figure 5** (a)Phase 1 (Charging) and (b) Phase 2 (Discharging) Circuit Models

Analysis

The analysis of these two circuits is rather straight-forward once the VI characteristic of the capacitor is substituted. In the time-domain, this is represented by the differential equation

$$i_c(t) = C\frac{dv_c(t)}{dt} \text{ subject to initial conditions on } v_c.$$

During the charging of Phase 1, the operative circuit equation is found by writing a Kirchoff's Current Law equation for the node at the top of the capacitor. Doing this yields

$$C\frac{dv_c(t)}{dt} + 100mA = \frac{|18\cos(120\pi t)| - 2V_d - v_c(t)}{2R_d},$$

when

$$v_c(t) \leq |18\cos(120\pi t)| - 2V_d.$$

During the discharging of Phase 2, the circuit is a bit simpler and the equation is

$$C\frac{dv_c(t)}{dt} + 100mA = 0,$$

when

$$v_c(t)\rangle\, |18\cos(120\pi t)| - 2V_d.$$

These are essentially the same differential equation it is just that the forcing function has two cases that must be monitored as the equation is solved. Substituting the previously derived values for $V_d$ and $R_d$ will yield the following equation in standard form.

$$\frac{dv_c\left(t\right)}{\mathrm{dt}} = \frac{1}{C}\left\{-0.1 + \max\left(\frac{\left|18\cos\left(120\pi t\right)\right| - 2 - v_c\left(t\right)}{0.04}, 0\right)\right\}.$$

Assuming that the capacitor is initially discharged (i.e.$v_c\left(0+\right) = 0$) it is simply a matter of trying various standard capacitors (e.g. $100, 150, 220, 330, 470$, and $680\mu F$ or other powers of 10 [4]) into the equation and simulating it for a few cycles of the AC source. Pick the smallest value where $v_c\left(t\right)$ never drops below the required $7.5\ V$.

## Questions

(1) Most capacitors are marked with a tolerance of$\pm 20\%$. How would your analysis handle the case that a $100\ \mu F$capacitor could be as high as $120\ \mu F$ or as low as$80\ \mu F$?

(2) What effect would it have on the differential equation to use a more complex model for the diode?

(3) Is it necessary to have a more detailed model of the voltage regulator? Alternatively, is it suitable to model it simply as a maximum current sink?

(4) Modify the model to account for an AC brownout, that is, what would happen if the AC power source saw a temporary drop in voltage?

## References


[1] LM340/LM78XX Data Sheet, National Semiconductor, http://www.national.com/ds/LM/LM340.pdf, accessed May 11, 2005.

[2] 1N4001 Data Sheet, Fairchild Semiconductor, http://www.fairchildsemi.com/ds/1N/1N4007.pdf, accessed May 12, 2005.

[3] Malvino, A., Electronic Principles, Sixth Edition, Glencoe McGraw-Hill, 1999.

[4] Standard Capacitor Values, RF Café, http://www.rfcafe.com/references/electrical/capacitor_values.htm, accessed May 12, 2005.

# Chapter 08.00: Physical Problem for Ordinary Differential Equations

Lesson: Industrial Engineering

## Summary

Many appliances used in our daily lives as well as numerous applications in industrial automation require controlling speed of a DC motor to function properly. Classical control theory requires solving differential equations to control these types of motors.

## Modeling

In this example we will discuss the closed loop speed control of a DC motor. Figure 1 shows three different DC motors and Figure 2 depicts the inside of a DC motor. Universal Motors which are essentially DC motors are widely used in applications where the speed of a process needs to be controlled. Such applications are encountered frequently in our daily lives such as controlling the speed of fans or controlling the speed of hand-held tools such as drills, and in industrial automation applications such as controlling the speed of conveyor belts.

**Figure 1** A variety of DC motors

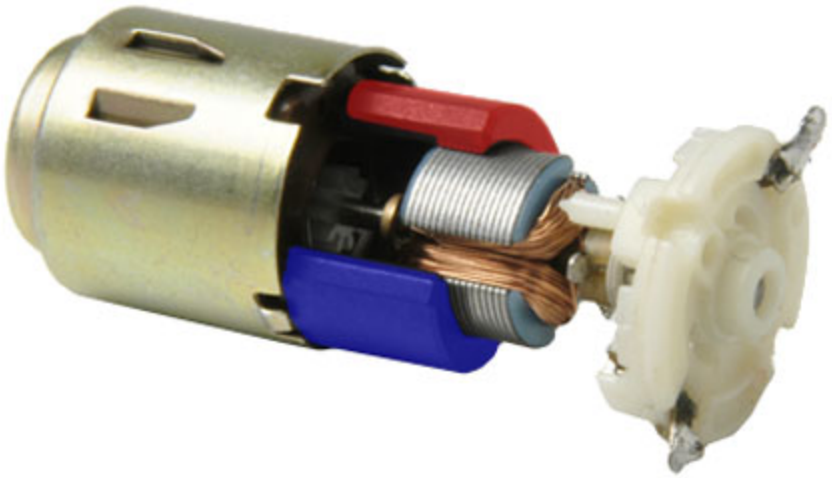


**Figure 2** The inside of a DC motor

Before we can discuss the speed control of a DC motor, it is important to understand the physical time varying relationships that determine the operating characteristics of a DC motor.

Equation (1) below describes the linear relationship between the torque $T$ and the current $i_a$ applied to the motor. The slope of the line is the torque constant $K_t$. Equation (2) describes the linear relationship between the back emf $E_B$ and the armature speed $w$. The slope of the line $K_B$ is the voltage constant. Equation (3) describes the components of the armature voltage $V$ as sum of the back emf and the voltage drop across armature resistance $Ri_a$. Finally, Equation (4) describes the relationship between torque, acceleration and speed of the motor in a no-load system as the sum of the angular acceleration $dw/dt$ multiplied by the inertia of the motor and the load, and the damping of the system $f$ multiplied by the armature speed.

$$T(t) = K_T i_a(t) \quad (1)$$

$$E_B(t) = K_B w(t) \quad (2)$$

$$V(t) = Ri_a(t) + E_B(t) \quad (3)$$

$$T(t) = I\frac{dw}{dt} + fw(t) \quad (4)$$

The equation for the speed of the motor in relation to the voltage input can be derived from these relationships as follows. Substitute the value of back emf from Equation (2) into Equation (3) as:

$$V(t) = Ri_a(t) + K_B w(t) \quad (5)$$

Equating Equations (1) and (4) and solving for ia results in

$$i_a = \frac{I\frac{dw}{dt} + fw(t)}{K_T} \quad (6)$$

Finally substitute the value for ia from Equation (6) into Equation (5) as

$$K_T V(t) = RI\frac{dw}{dt} + (Rf + K_T K_B)w(t) \quad (7)$$

Equation (7) is a first order differential equation which describes the open-loop response of the motor to a voltage input where the output variable system (speed of the motor) is not considered in the control mechanism. In the next section we will consider the closed-loop control of the motor.

**Closed loop control**

Figure 3 illustrates the block diagram of a proportional control system used to control the speed of a DC motor where KP is the proportional gain. The "C" block is the summing point which generates an error term e equal to the difference between the desired speed set by the operator ($w_d$) and the actual speed of the motor. The "DC Motor" block transforms the amplified error input from the "Control Action" block to the output speed of the motor.

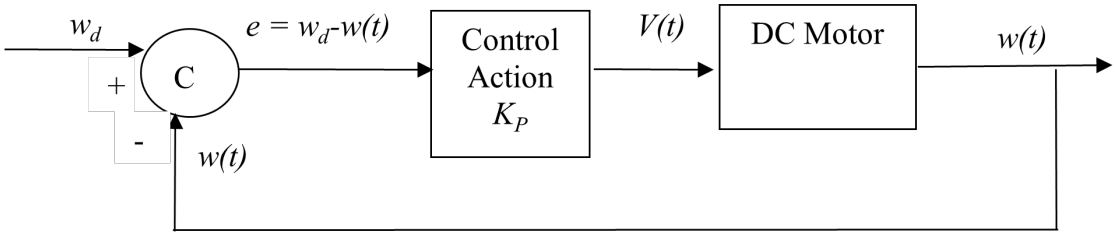



**Figure 3** Block diagram of a DC motor control systems

Based on the control system block diagram

$$V(t) = K_P[w_d - w(t)] \quad (8)$$

Substituting the value of $V(t)$from Equation (8) in to Equation (7) and rearranging terms results in the differential Equation (9) which describes the relationship of the actual speed of the motor to the desired speed.

$$K_T K_P w_d = RI\frac{dw}{dt} + [Rf + K_T(K_B + K_P)]w(t) \quad (9)$$

**Example application**

Consider a machine vision quality inspection system which inspects parts for defects shown in Figure 4. The parts are positioned in fixtures on a conveyor driven by a DC motor. The conveyor is required to ramp up to a certain speed within a specific time (before a part enters the field of view of the camera), maintain a constant speed while the part is under the camera and finally come to a stop once the inspection of the part is completed. Many such applications which utilize DC motors have requirements associated with the steady-state speed and/or a ramp up time for the motor to reach this speed.

Consider a DC motor which has the following specifications:

$$\text{Voltage constant, } K_B = 0.06V.s/rad$$

$$\text{Torque constant, } K_t = 0.06N.m/A$$

$$\text{Armature resistance, } R = 2\Omega$$

$$\text{Moment of inertia, } I = 6 \times 10^{-4} N.m.s^2/rad$$

The open loop response of the motor to a voltage input using Equation (7) and assuming a system without damping (($f$ = 0 N.m.s /rad)) is

$$V(t) = (0.02)\ \frac{dw}{dt} + (0.06)w(t).$$

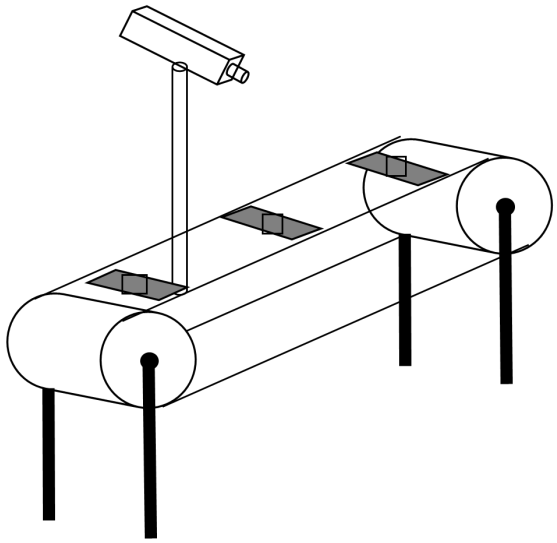


**Figure 4** Machine vision system schematic

The speed of the motor $w\,(t)$ can therefore be controlled by changing the magnitude of this input voltage ($v\,(t)$). Assuming the initial condition $w\,(0) = 0$and a specific input voltage, the above ordinary differential equation can be solved to determine the stead-state speed of the motor as well as the time it takes for the motor to ramp up to this speed. The information obtained from the solution is important

in selection of the appropriate voltage input and/or motor for a particular application to assure that the motor response time and speed is sufficient for the task.

Similarly, the design of a closed loop controller to control the speed of the motor requires the solution to the ordinary differential Equation (9) to determine the user settings to obtain the necessary output from the motor. This application is posed as Exercise Problem 5 below.

## Questions

For the DC motor in the Example Problem, answer the following questions:

(1) Draw the speed response graph $w(t)$ vs $t$ for a step input of 20 Volts without the damping of the system ($f = 0\ N.m.s/rad$)

(2) Draw the speed response graph for a step input of 20 Volts considering the damping of the system where $f = 1.0 \times 10^{-4} N.m.s/rad$.

(3) What is the difference between the steady-state speed of the motor with and without damping?

(4) Consider a closed loop control system with gain $K_P = 10$. What is the closed loop speed response graph of motor to a desired setting of 100 rad/s?

(5) The answer to Question #4 is less than 100 rad/s due to the steady state error present in first order systems. What should be the desired speed setting for the motor speed response to be 100 rad/s?

## References

[1] Boucher, T.O., Computer Automation in Manufacturing: An Introduction, Chapman & Hall, London, 1996.

[2] Bateson, R.N., Introduction to Control System Technology, 6th Edition, Prentice Hall, New Jersey, 1999.

# Chapter 08.00: Physical Problem for Ordinary Differential Equations

Lesson: Mechanical Engineering

## Summary

A physical problem of finding how much time it would take a trunnion to cool down in a refrigerated chamber is presented. To estimate this time, the problem would be modeled as an ordinary differential equation.

## Modeling

To make the fulcrum (Figure 1) of a bascule bridge, a long hollow steel shaft called the trunnion is shrink fit into a steel hub. The resulting steel trunnion-hub assembly is then shrunk-fit into the girder of the bridge.

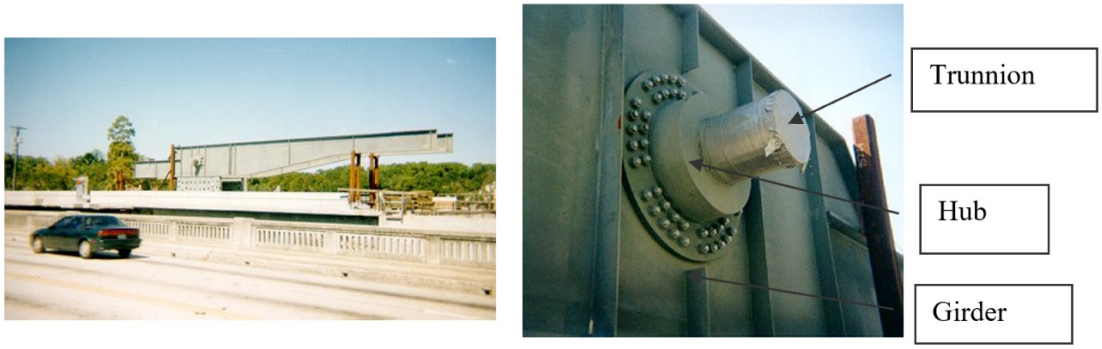



**Figure 1** Trunnion-Hub-Girder (THG) assembly.

This assembly is done by first immersing the trunnion in a cold medium such as liquid nitrogen. After the trunnion reaches the steady-state temperature, that is, the temperature of the cold medium, the trunnion outer diameter contracts. The trunnion is taken out of the medium and slid though the hole of the hub (Figure 1).

When the trunnion heats up, it expands and creates an interference fit with the hub. Because of the large thermal shock that the trunnion undergoes, it is suggested that the trunnion be cooled in stages. First, put the trunnion in a refrigerated chamber, then dip it in a dry-ice/alcohol mixture and then immerse it in a bath of liquid nitrogen. However, this approach will take more time. One is mainly concerned about the cooling time in the refrigerated chamber. Assuming that the room temperature

is 27°C and the refrigerated chamber is set is −33°C, how much time would it take for the trunnion to cool down to −33°C?

Let us do a simplified problem of a solid trunnion and assume the trunnion can be treated as a lumped-mass system. What does a lumped system mean?

Let us develop the mathematical model for the above problem. When the trunnion is placed in the refrigerated chamber, the trunnion loses heat to its surroundings by convection.

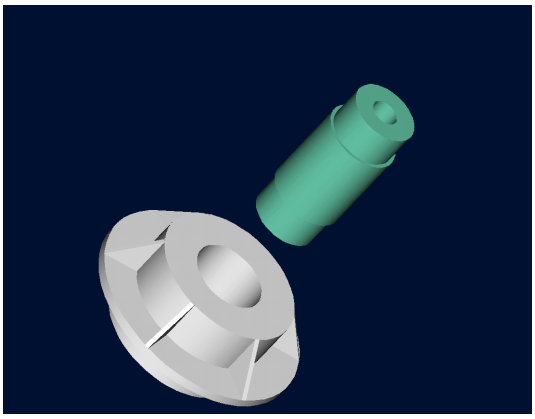


**Figure 2** Trunnion slides through the hub after contracting

$$\text{Rate of heat lost due to convection} = h(\theta) A (\theta - \theta_a). \qquad (1)$$

where

$h(\theta)$ = the convective cooling coefficient, $\text{W/m}^2 \cdot \text{K}$ and is a function of temperature

$A$ = surface area

$\theta_a$ = ambient temperature of the refrigerated chamber

The energy stored in the mass is given by

$$\text{Energy stored by mass} = mC\theta \qquad (2)$$

where

$m$ = mass of the trunnion, kg

$C$ = specific heat of the trunnion, J/(kg-K)

From an energy balance,

Rate at which heat is gained − Rate at which heat is lost
Rate at which heat is stored

gives

$$-h(\theta)\,A\,(\theta-\theta_a) = mC\frac{d\theta}{dt} \qquad (3)$$

Note that the convective cooling coefficient is a function of temperature. Other material parameters such as density (affecting mass), specific heat, and thermal conductivity (affecting whether the system can be considered lumped or not) of the trunnion material are functions of temperature as well, but within our temperature range of 27°C to −33°C these vary by only 10% from the room temperature values. So we assume these parameters to be constant.

Let us now determine the constants needed for the above ordinary differential equation.

Convection coefficient of air

The convection coefficient of air, $h(\theta)$ is not a constant function of temperature and is given by

$$h = \frac{Nu \times k}{D} \qquad (4)$$

where

$Nu$ is the Nusselt number,

$k$ is the thermal conductivity of air,

$D$ is the characteristic diameter and is taken as the outer diameter of the trunnion.

The Nusselt number for a vertical cylinder is given by the empirical formula [1]:

$$Nu = \left(0.825 + \frac{0.387Ra^{\frac{1}{6}}}{\left(1+\left(\frac{0.492}{\mathrm{Pr}}\right)^{\frac{9}{16}}\right)^{\frac{8}{27}}}\right)^2 \qquad (5)$$

where

$Pr$ is the Prandtl number given by

$$\mathrm{Pr} = \frac{\upsilon_k}{\alpha} \qquad (6)$$

$\nu_k$ is the kinematic viscosity of the fluid,

$\alpha$ is the thermal diffusivity,

$R_a$ is the Rayleigh number given by:

$$R_a = G_r \times \mathrm{Pr} \qquad (7)$$

$G_r$ is the Grashoff number given by:

$$Gr = \frac{g\beta(T_{wall} - T_{fluid})D^3}{{\nu_k}^2} \qquad (8)$$

where

$g$ is the gravitational constant,

$\beta$ is the coefficient of volumetric thermal expansion,

$T_{wall}$ is the temperature of the wall,

$T_{fluid}$ is the temperature of the fluid.

To calculate the convection coefficient, we use the following value for kinematics viscosity $\nu$, thermal conductivity $k$, thermal diffusivity $\alpha$, and volumetric coefficient, $\beta$ of air as a function of temperature, $\theta$ [Ref. 1].

**Table 1** Properties of air as a function of temperature.

| $T_{lookup}$ | $\nu$ | $k$ | $\alpha$ | $\beta$ |
|---|---|---|---|---|
| $^\circ C$ | $m^2/s$ | $W/(m-K)$ | $m^2/s$ | $1/K$ |
| $-173.15$ | $2.00 \times 10^{-6}$ | $9.34 \times 10^{-3}$ | $2.54 \times 10^{-6}$ | $1.00 \times 10^{-2}$ |
| $-123.15$ | $4.43 \times 10^{-6}$ | $1.38 \times 10^{-2}$ | $5.84 \times 10^{-6}$ | $6.67 \times 10^{-3}$ |
| $-73.15$ | $7.59 \times 10^{-6}$ | $1.81 \times 10^{-2}$ | $1.03 \times 10^{-5}$ | $5.00 \times 10^{-3}$ |
| $-23.15$ | $1.14 \times 10^{-5}$ | $2.23 \times 10^{-2}$ | $1.59 \times 10^{-5}$ | $4.00 \times 10^{-3}$ |
| 26.85 | $1.59 \times 10^{-5}$ | $2.63 \times 10^{-2}$ | $2.25 \times 10^{-5}$ | $3.33 \times 10^{-3}$ |

Other constants needed are:

$$D = 0.25 \text{ m}$$

$$g = 9.8 \text{ m/s}^2$$

$$T_{fluid} = -33^\circ\text{C}$$

**Table 2** Convection coefficient of air as a function of temperature.

| $T_{wall}$ | $T_{average} = \frac{1}{2}(T\ fluid\ +\ T\ wall)$ | $h$ |
|---|---|---|
| $^\circ C$ | $^\circ C$ | $W/(m^2.K)$ |
| $-33$ | $-16.50$ | $5.846E-02$ |
| $-18$ | $-9.00$ | $4.527E+00$ |
| $-8$ | $-4.00$ | $5.214E+00$ |

| $T_{wall}$ | $T_{average} = \frac{1}{2}(T\ fluid\ +\ T\ wall)$ | $h$ |
|---|---|---|
| 2 | 1.00 | $5.702E+00$ |
| 27 | 13.50 | $6.533E+00$ |

The values in Table 2 are interpolated from Table 1 where temperatures are chosen as the average value for the wall and ambient temperature. The above data is interpolated as:

$$h(\theta) = -3.69 \times 10^{-6}\theta^4 + 2.33 \times 10^{-5}\theta^3 + 1.35 \times 10^{-3}\theta^2 + 5.42 \times 10^{-2}\theta + 5.59 \tag{9}$$

Area, $A$

Outer radius of trunnion, $a = 0.125$ m.

Length of trunnion, $L = 1.36$ m.

gives

$$\begin{aligned} A &= (2\pi a)L + 2\pi a^2 \\ &= 2\pi\,(.125)\,(1.36) + 2\pi(0.125)^2 \\ &= 1.166\ \mathrm{m}^2 \end{aligned}$$

Mass, $m$

Length of the trunnion, $L = 1.36$ m

Density of trunnion material, $\rho = 7800\mathrm{kg/m}^3$

gives

$$\begin{aligned} m &= \rho V \\ &= \rho(\pi r^2 L) \\ &= 7800\pi\,(0.125)^2\,1.36 \\ &= 520.72\ \mathrm{kg} \end{aligned}$$

Other constants

Specific heat, $C = 420$ J/ (kg-K)

Initial temperature of the trunnion, $T(0) = 27^{\circ}\mathrm{C}$,

Ambient temperature, $T_a = -33^{\circ}\mathrm{C}$

The first-order ordinary differential equation:

$$-h\left(\theta\right)A\left(\theta-\theta_a\right)=mC\frac{d\theta}{dt}$$

reduces to

$$-\left(-3.69\times10^{-6}\theta^4+2.33\times10^{-5}\theta^3+1.35\times10^{-3}\theta^2+5.42\times10^{-2}\theta+5.588\right)\times$$
$$(1.166)\times(\theta+33)=(520.72)\times(420)\times\frac{d\theta}{dt}$$

$$\frac{d\theta}{dt}=-5.331\times10^{-6}(-3.69\times10^{-6}\theta^4+2.33\times10^{-5}\theta^3+1.35\times10^{-3}\theta^2+5.42\times10^{-2}\theta+5.588)(\theta+33) \qquad (10)$$

$$\theta\left(0\right)=27^{\circ}C$$

**Is the assumption of the trunnion considered as a lumped system correct?**

To determine whether a system is lumped, we calculate the Biot number $Bi$, which is defined as

$$Bi=\frac{hL}{k_s} \qquad (11)$$

where

$$h=\text{average surface conductance,}$$

$$L=\text{significant length dimension (volume of body/surface area),}$$

$$k_s=\text{thermal conductivity of solid body.}$$

If $B_i<0.1$, the temperature in the body is uniform within 5% error.

In our case:

$$h=4.407\ \frac{\mathrm{W}}{\mathrm{m}^2\cdot\mathrm{K}}$$

$$L=1.36\ \mathrm{m}$$

$$k_s=81\ \frac{\mathrm{W}}{\mathrm{m}\cdot\mathrm{K}} \qquad [Ref.2]$$

$$Bi = \frac{4.407 \times 1.36}{81}$$
$$= 0.074 < 0.1$$

The Biot number is less than 0.1, and one can hence assume the trunnion to be a lumped mass system.

**References**

[1] F. B. Incropera, D. P. Dewitt, Introduction to Heat Transfer, 3rd Ed, John Wiley & Sons, Inc., New York, NY, 2000, Chap. 9.

[2] F. Kreith, M. S. Bohn, Principle of Heat Transfer, 4th Ed, Harper & Row, Publishers, New York, NY, 1993, Appendix 2.

# Chapter 08.01: Prerequisites to Numerical Methods for Solving Ordinary Differential Equations

Lesson: Why Do We Need Ordinary Differential Equations?

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *find why some mathematical models are given by ordinary differential equations*

## Introduction

Differential equations have applications in all areas of science and engineering. A mathematical formulation of most of the physical and engineering problems leads to differential equations. So, it is crucial for engineers and scientists to know how to set up differential equations and solve them.

An equation that consists of derivatives is called a differential equation, and they are of two types

- ordinary differential equations (ODE)

- partial differential equations (PDE)

An ordinary differential equation has only one independent variable. Examples of ordinary differential equations include

$$\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + y = 0, \frac{dy}{dx}(0) = 2,\ y(0) = 4,$$

$$\frac{d^3y}{dx^3} + 3\frac{d^2y}{dx^2} + 5\frac{dy}{dx} + y = \sin x,\ \frac{d^2y}{dx^2}(0) = 12,\ \frac{dy}{dx}(0) = 2,\ y(0) = 4$$

where $x$ is the independent variable, and $y$ is the dependent variable.

Ordinary differential equations are classified in terms of order and degree. The order of an ordinary differential equation is the same as the highest derivative, and the degree of an ordinary differential equation is the power of the highest derivative. Thus the differential equation,

$$x^3\frac{d^3y}{dx^3} + x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} + xy = e^x$$

is of order 3 and degree 1, whereas the differential equation

$$\left(\frac{dy}{dx}+1\right)^2+x^2\frac{dy}{dx}=\sin x$$

is of order 1 and degree 2.

An engineer's approach to differential equations is different from a mathematician. While the latter is interested in the mathematical solution, an engineer should interpret the result physically. So, an engineer's approach can be divided into three phases:

- formulation of a differential equation from a given physical situation,

- solving the differential equation, and

- interpreting the results physically for implementation.

## Formulation of differential equations

So the main question we have in this lesson is why we get differential equations in the first place. As discussed above, the formulation of a differential equation is based on a given physical situation.Let us illustrate it by a spring-mass-damper system.

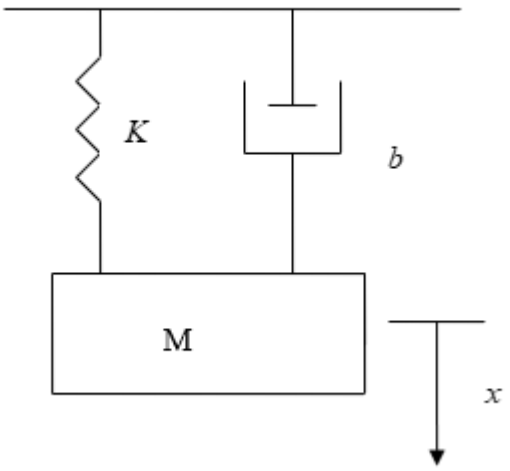



**Figure 1.** Spring-mass damper system

Above is the schematic diagram of a spring-mass-damper system. A block is suspended freely using a spring. As most physical systems involve damping - viscous damping, dry damping, magnetic damping, etc., a damper or dashpot is attached to account for viscous damping.

Let the mass of the block be $M$, the spring constant be $K$, and the damper coefficient be $b$. If we measure displacement from the static equilibrium position, we need not consider gravitational force as it is balanced by tension in the spring at equilibrium.

Below is the free-body diagram of the block at static and dynamic equilibrium. So, the equation of motion is given by

$$Ma = F_S + F_D \qquad (1)$$

where

$F_S$ is the restoring force due to spring.

$F_D$ is the damping force due to the damper.

$a$ is the acceleration.

The restoring force in the spring is given by

$$F_S = -Kx \qquad (2)$$

as the restoring force is proportional to the displacement. The damping force in the damper is given by

$$F_D = -bv \qquad (3)$$

as the damping force is directly proportional to velocity.

Therefore, the equation of motion can be written as

$$Ma = -Kx - bv \qquad (4)$$

At this stage, Equation (4) is a simple equation, where if we know five of the six variables, we can calculate the sixth one. However, such an expectation is not suitable. Assuming that the physical constants, $M$, $K$, and $b$ are fixed and known, we are still left with three variables $a$, $v$, and $x$. Now, again, if we now know two of the three variables, $a$, $v$, and $x$, we can calculate the unknown one. However, such an expectation is still not suitable. But you recognize that

$$v = \frac{dx}{dt}$$

$$a = \frac{d^2x}{dt^2}$$

and hence from Equation (4), we get

$$M\frac{d^2x}{dt^2} = -Kx - b\frac{dx}{dt}$$

$$M\frac{d^2x}{dt^2} + b\frac{dx}{dt} + Kx = 0 \qquad (5)$$

Equation (5) is an ordinary differential equation of second order and degree one. We have reduced Equation (4) from three variables to one at the expense of converting it into an ordinary differential equation. Moreover, two initial conditions, one on $x$ and another on $v = \frac{dx}{dt}$ are expected to be available.

# Chapter 08.01: Prerequisites to Numerical Methods for Solving Ordinary Differential Equations

Lesson: Solving First-Order Ordinary Differential Equations Exactly

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *solve the linear ordinary differential equation of first order with fixed constants by using a classical solution technique.*

## Introduction

In this lesson, we will revisit the classical solution technique used to solve ordinary differential equations. Knowing this technique is a prerequisite to the course.

## Classical Technique

The general form of a linear ordinary differential equation with constant coefficients is given by

$$\frac{d^n y}{dx^n} + k_n \frac{d^{n-1} y}{dx^{n-1}} + ..... + k_3 \frac{d^2 y}{dx^2} + k_2 \frac{dy}{dx} + k_1 y = F(x) \qquad (1)$$

The general solution contains two parts

$$y = y_H + y_P \qquad (2)$$

where

$y_H$ is the homogeneous part of the solution and

$y_P$ is the particular part of the solution.

You can skip to the examples if you are already familiar with the theory.

## Homogenous Part of the Solution

The homogeneous part of the solution $y_H$ is that part of the solution that gives zero when substituted in the left-hand side of the equation. So, $y_H$ is the solution of the equation

$$\frac{d^n y}{dx^n} + k_n \frac{d^{n-1} y}{dx^{n-1}} + ..... + k_3 \frac{d^2 y}{dx^2} + k_2 \frac{dy}{dx} + k_1 y = 0 \qquad (3)$$

Equation (3) can be symbolically written as

$$D^n y + k_n D^{n-1} y + ..... + k_2 D y + k_1 y = 0 \qquad (4)$$

$$(D^n + k_n D^{n-1} + ..... + k_2 D + k_1) y = 0 \qquad (5)$$

where,

$$D^n = \frac{d^n}{dx^n}$$

$$D^{n-1} = \frac{d^{n-1}}{dx^{n-1}}.$$

$$\vdots$$

operating on $y$ is the same as

$$(D - r_1), (D - r_2), \ (D - r_n)$$

operating one after the other in any order, where

$$(D - r_1), (D - r_2), ....., (D - r_n)$$

are factors of

$$D^n + k_n D^{n-1} + ..... + k_2 D + k_1 = 0 \qquad (6)$$

To illustrate

$$(D^2 - 3D + 2) y = 0$$

is same as

$$(D - 2)(D - 1) y = 0$$

$$(D - 1)(D - 2) y = 0$$

Therefore

$$(D^n + k_n D^{n-1} + ..... + k_2 D + k_1)y = 0 \qquad (7)$$

is same as

$$(D - r_n)(D - r_{n-1}).....(D - r_1)y = 0 \qquad (8)$$

operating one after the other in any order.

**Case 1: Roots are real and distinct**

The entire left-hand side becomes zero if $(D - r_1)\, y = 0$. Therefore, the solution to $(D - r_1)\, y = 0$ is a solution to a homogeneous equation. $(D - r_1)\, y = 0$ is called Leibnitz's linear differential equation of first order, and its solution is

$$(D - r_1)\, y = 0 \qquad (9)$$

$$\frac{dy}{dx} = r_1 y \qquad (10)$$

$$\frac{dy}{y} = r_1 dx \qquad (11)$$

Integrating both sides, we get

$$\ln y = r_1 x + c$$

$$y = ce^{r_1 x} \qquad (12)$$

Since any of the $n$ factors can be placed before $y$, there are $n$ different solutions corresponding to $n$ different factors given by

$$C_n e^{r_n x}, C_{n-1} e^{r_{n-1} x}, ....., C_2 e^{r_2 x}, C_1 e^{r_1 x}$$

where

$r_{n,} r_{n-1}, ....., r_2, r_1$ are the roots of Equation (12) and

$C_{n,} C_{n-1}, ....., C_2, C_1$ are constants.

We get the general solution for a homogeneous equation by superimposing the individual Leibnitz's solutions. Therefore

$$y_H = C_1 e^{r_1 x} + C_2 e^{r_2 x} + ..... + C_{n-1} e^{r_{n-1} x} + C_n e^{r_n x} \qquad (13)$$

**Case 2: Roots are real and identical**

If two roots of a homogeneous equation are equal, say $r_1 = r_2$, then

$$(D - r_n)(D - r_{n-1}).....(D - r_1)(D - r_1)y = 0 \qquad (14)$$

Let's work at

$$(D - r_1)(D - r_1)y = 0 \qquad (15)$$

If

$$(D - r_1)y = z \qquad (16)$$

then

$$(D - r_1)z = 0$$

$$z = C_2 e^{r_1 x} \qquad (17)$$

Now substituting the solution from Equation (17) in Equation (16)

$$(D - r_1)y = C_2 e^{r_1 x}$$

$$\frac{dy}{dx} - r_1 y = C_2 e^{r_1 x}$$

$$e^{-r_1 x}\frac{dy}{dx} - r_1 e^{-r_1 x} y = C_2$$

$$\frac{d(e^{-r_1 x} y)}{dx} = C_2$$

$$d(e^{-r_1 x} y) = C_2 dx \qquad (18)$$

Integrating both sides of Equation (18), we get

$$e^{-r_1 x} y = C_2 x + C_1$$

$$y = (C_2 x + C_1)e^{r_1 x} \qquad (19)$$

Therefore, the final homogeneous solution is given by

$$y_H = (C_1 + C_2 x)\, e^{r_1 x} + C_3 e^{r_3 x} + ... + C_n e^{r_n x} \qquad (20)$$

Similarly, if $m$ roots are identical, the solution is given by

$$y_H = \left(C_1 + C_2x + C_3x^2 + ..... + C_mx^{m-1}\right) e^{r_m x} + C_{m+1}e^{r_{m+1}x} + ... + C_n e^{r_n x} \qquad (21)$$

**Case 3: Roots are complex**

If one pair of roots is complex, say $r_1 = \alpha + i\beta$ and $r_2 = \alpha - i\beta$,

where

$$i = \sqrt{-1}$$

then

$$y_H = C_1e^{(\alpha+i\beta)x} + C_2e^{(\alpha-i\beta)x} + C_3e^{r_3x} + ..... + C_ne^{r_nx} \qquad (22)$$

Since

$$e^{i\beta x} = \cos\beta x + i\sin\beta x, \text{ and} \qquad (23a)$$

$$e^{-i\beta x} = \cos\beta x - i\sin\beta x \qquad (23b)$$

then

$$\begin{aligned} y_H &= C_1e^{\alpha x}\left(\cos\beta x + i\sin\beta x\right) + C_2e^{\alpha x}\left(\cos\beta x - i\sin\beta x\right) + C_3e^{r_3x} + .... + C_ne^{r_nx} \\ &= \left(C_1 + C_2\right)e^{\alpha x}\cos\beta x + i\left(C_1 - C_2\right)e^{\alpha x}\sin\beta x + C_3e^{r_3x} + .... + C_ne^{r_nx} \\ &= e^{\alpha x}\left(A\cos\beta x + B\sin\beta x\right) + C_3e^{r_3x} + .... + C_ne^{r_nx} \qquad (24) \end{aligned}$$

where

$$A = C_1 + C_2 \text{ and}$$

$$B = i(C_1 - C_2) \qquad (25)$$

## Particular Part of the Solution

Now, let us look at how the particular part of the solution is found. Consider the general form of the ordinary differential equation

$$\left(D^n + k_nD^{n-1} + k_{n-1}D^{n-2} + ..... + k_1\right)y = F(x) \qquad (26)$$

The particular part of the solution $y_P$ is that part of the solution that gives $F(x)$ when substituted for $y$ in the above equation, that is,

$$\left(D^n + k_n D^{n-1} + k_{n-1} D^{n-2} + ..... + k_1\right) y_P = F(x) \qquad (27)$$

Examples of what the particular part of the solution looks like are given below.

1. When

$$F(x) = e^{ax},$$

the particular part of the solution is of the form

$$Ae^{ax}.$$

We can find $A$ by substituting

$$y_P\,(x) = Ae^{ax} \qquad (28)$$

in the left-hand side of the differential equation and equating coefficient of $e^{ax}$.

2. When

$$F(x) = \sin bx \text{ or } \cos bx.$$

the particular part of the solution is of the form

$$A\sin bx + B\cos bx.$$

We can get $A$and $B$ by substituting

$$y_P\,(x) = A\sin bx + B\cos bx \qquad (29)$$

in the left-hand side of the differential equation and equating the coefficients of $\sin bx$ and/or $\cos bx$.

3. When

$$X = e^{ax}\sin bx \text{ or } e^{ax}\cos bx,$$

the particular part of the solution is of the form

$$e^{ax}\left(A\sin bx + B\cos bx\right).$$

We can get $A$ and $B$ by substituting

$$y_P\,(x) = e^{ax}(A\sin bx + B\cos bx) \qquad (30)$$

in the left-hand side of the differential equation and equating coefficients of $e^{ax}\sin bx$ and $e^{ax}(A\cos bx)$.

The forms of the particular part of the solution for different forcing functions of ordinary differential equations are given below

| $F(x)$ | $y_P(x)$ |
|---|---|
| $a_0 + a_1 x + a_2 x^2$ | $b_0 + b_1 x + b_2 x^2$ |
| $e^{ax}$ | $Ae^{ax}$ |
| $\sin(bx)$ | $A\sin(bx) + B\cos(bx)$ |
| $e^{ax}\sin(bx)$ | $e^{ax}(A\sin(bx) + B\cos(bx))$ |
| $\cos(bx)$ | $A\sin(bx) + B\cos(bx)$ |
| $e^{ax}\cos(bx)$ | $e^{ax}(A\sin(bx) + B\cos(bx))$ |

**Example 1**

Solve

$$3\frac{dy}{dx} + 2y = e^{-x},\ y(0) = 5$$

**Solution**

The homogeneous solution for the above equation is given by

$$(3D + 2)\, y = 0$$

The characteristic equation for the above equation is given by

$$3r + 2 = 0$$

The solution to the equation is

$$r = -0.666667$$

$$y_H = Ce^{-0.666667x}$$

The particular part of the solution is of the form $Ae^{-x}$

$$3\frac{d(Ae^{-x})}{dx} + 2Ae^{-x} = e^{-x}$$

$$-3Ae^{-x} + 2Ae^{-x} = e^{-x}$$

$$-Ae^{-x} = e^{-x}$$

$$A = -1$$

Hence the particular part of the solution is

$$y_P = -e^{-x}$$

The complete solution is given by

$$\begin{aligned} y &= y_H + y_P \\ &= Ce^{-0.666667x} - e^{-x} \end{aligned}$$

The constant $C$ can be obtained by using the initial condition $y(0) = 5$

$$y\left(0\right) = Ce^{-0.666667\times 0} - e^{-0} = 5$$

$$C - 1 = 5$$

$$C = 6$$

The complete solution is

$$y = 6e^{-0.666667x} - e^{-x}$$

**Example 2**

Solve

$$2\frac{dy}{dx} + 3y = e^{-1.5x},\ y(0) = 5$$

**Solution**

The homogeneous solution for the above equation is given by

$$\left(2D + 3\right) y = 0$$

The characteristic equation for the above equation is given by

$$2r + 3 = 0$$

The solution to the equation is

$$r = -1.5$$

$$y_H = Ce^{-1.5x}$$

Based on the forcing function of the ordinary differential equations, the particular part of the solution is of the form $Ae^{-1.5x}$, but since that is part of the form of the homogeneous part of the solution, we need to choose the next independent solution, that is,

$$y_P = Axe^{-1.5x}$$

To find $A$, we substitute this solution in the ordinary differential equation as

$$2\frac{d}{dx}(Axe^{-1.5x}) + 3Axe^{-1.5x} = e^{-1.5x}$$

$$2Ae^{-1.5x} - 3Axe^{-1.5x} + 3Axe^{-1.5x} = e^{-1.5x}$$

$$2Ae^{-1.5x} = e^{-1.5x}$$

$$A = 0.5$$

Hence the particular part of the solution is

$$y_P = 0.5xe^{-1.5x}$$

The complete solution is given by

$$\begin{aligned} y &= y_H + y_P \\ &= Ce^{-1.5x} + 0.5xe^{-1.5x} \end{aligned}$$

The constant $C$ is obtained by using the initial condition $y(0) = 5$.

$$y\,(0) = Ce^{-1.5(0)} + 0.5(0)e^{-1.5(0)} = 5$$

$$C + 0 = 5$$

$$C = 5$$

The complete solution is

$$y = 5e^{-1.5x} + 0.5xe^{-1.5x}$$

# Chapter 08.01: Prerequisites to Numerical Methods for Solving Ordinary Differential Equations

Lesson: Solving Second-Order Differential Equations Exactly

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *solve linear ordinary differential equations of first-order with fixed constants by using classical solution techniques.*

## Applications

In this lesson, we revisit the classical solution technique used to solve ordinary differential equations of the second order. Knowing this technique is a prerequisite to the course.

### Example 1

Solve

$$2\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + 3.125y = \sin x,\ y(0) = 5,\ \frac{dy}{dx}(x = 0) = 3$$

### Solution

The homogeneous equation is given by

$$(2D^2 + 3D + 3.125)y = 0$$

The characteristic equation is

$$2r^2 + 3r + 3.125 = 0$$

The roots of the characteristic equation are

$$
\begin{aligned}
r &= \frac{-3 \pm \sqrt{3^2 - 4 \times 2 \times 3.125}}{2 \times 2} \\
&= \frac{-3 \pm \sqrt{9 - 25}}{4} \\
&= \frac{-3 \pm \sqrt{-16}}{4} \\
&= \frac{-3 \pm 4i}{4} \\
&= -0.75 \pm i
\end{aligned}
$$

Therefore the homogeneous part of the solution is given by

$$y_H = e^{-0.75x}(K_1 \cos x + K_2 \sin x)$$

The particular part of the solution is of the form

$$y_P = A \sin x + B \cos x$$

$$2\frac{d^2}{dx^2}(A \sin x + B \cos x) + 3\frac{d}{dx}(A \sin x + B \cos x) + 3.125(A \sin x + B \cos x) = \sin x$$

$$2\frac{d}{dx}(A \cos x - B \sin x) + 3(A \cos x - B \sin x) + 3.125(A \sin x + B \cos x) = \sin x$$

$$2(-A \sin x - B \cos x) + 3(A \cos x - B \sin x) + 3.125(A \sin x + B \cos x) = \sin x$$

$$(1.125A - 3B) \sin x + (1.125B + 3A) \cos x = \sin x$$

Equating coefficients of $\sin x$ and $\cos x$ on both sides, we get

$$1.125A - 3B = 1$$

$$1.125B + 3A = 0$$

Solving the above two simultaneous linear equations, we get

$$A = 0.109589$$

$$B = -0.292237$$

Hence

$$y_P = 0.109589 \sin x - 0.292237 \cos x$$

The complete solution is given by

$$y = e^{-0.75x}(K_1 \cos x + K_2 \sin x) + (0.109589 \sin x - 0.292237 \cos x)$$

To find $K_1$ and $K_2$ we use the initial conditions

$$y(0) = 5, \ \frac{dy}{dx}(x = 0) = 3$$

From $y(0) = 5$, we get

$$5 = e^{-0.75(0)}(K_1 \cos(0) + K_2 \sin(0)) + (0.109589 \sin(0) - 0.292237 \cos(0))$$

$$5 = K_1 - 0.292237$$

$$K_1 = 5.292237$$

$$\begin{aligned}\frac{dy}{dx} = &-0.75e^{-0.75x}(K_1 \cos x + K_2 \sin x) + e^{-0.75x}(-K_1 \sin x + K_2 \cos x) \\ &+ 0.109589 \cos x + 0.292237 \sin x\end{aligned}$$

From

$$\frac{dy}{dx}(x = 0) = 3,$$

we get

$$\begin{aligned}3 = &-0.75e^{-0.75(0)}(K_1 \cos(0) + K_2 \sin(0)) + \ e^{-0.75(0)}(-K_1 \sin(0) + K_2 \cos(0)) \\ &+ 0.109589 \cos(0) + 0.292237 \sin(0)\end{aligned}$$

$$3 = -0.75K_1 + K_2 + 0.109589$$

$$3 = -0.75(5.292237) + K_2 + 0.109589$$

$$K_2 = 6.859588$$

The complete solution is

$$y = e^{-0.75x}(5.292237 \cos x + 6.859588 \sin x) + 0.109589 \sin x - 0.292237 \cos x$$

**Example 2**

Solve

$$2\frac{d^2y}{dx^2} + 6\frac{dy}{dx} + 3.125y = \cos(x),\ y(0) = 5,\ \frac{dy}{dx}(x = 0) = 3$$

**Solution**

The homogeneous part of the equations is given by

$$(2D^2 + 6D + 3.125)y = 0$$

The characteristic equation is given by

$$2r^2 + 6r + 3.125 = 0$$

$$\begin{aligned} r &= \frac{-6 \pm \sqrt{6^2 - 4(2)(3.125)}}{2(2)} \\ &= \frac{-6 \pm \sqrt{36 - 25}}{4} \\ &= \frac{-6 \pm \sqrt{11}}{4} \\ &= -1.5 \pm 0.829156 \\ &= -0.670844, -2.329156 \end{aligned}$$

Therefore, the homogeneous solution $y_H$ is given by

$$y_H = K_1 e^{-0.670845x} + K_2 e^{-2.329156x}$$

The particular part of the solution is of the form

$$y_P = A\sin x + B\cos x$$

Substituting the particular part of the solution in the differential equation,

$$2\frac{d^2}{dx^2}(A\sin x + B\cos x) + 6\frac{d}{dx}(A\sin x + B\cos x) + 3.125(A\sin x + B\cos x) = \cos x$$

$$2\frac{d}{dx}(A\cos x - B\sin x) + 6(A\cos x - B\sin x) + 3.125(A\sin x + B\cos x) = \cos x$$

$$2(-A\sin x - B\cos x) + 6(A\cos x - B\sin x) + 3.125(A\sin x + B\cos x) = \cos x$$

$$(1.125A - 6B)\sin x + (1.125B + 6A)\cos x = \cos x$$

Equating coefficients of $\cos x$ and $\sin x$ we get

$$1.125B + 6A = 1$$

$$1.125A - 6B = 0$$

The solution to the above two simultaneous linear equations are

$$A = 0.161006$$

$$B = 0.0301887$$

Hence the particular part of the solution is

$$y_P = 0.161006\sin x + 0.0301887\cos x$$

Therefore the complete solution is

$$y = y_H + y_P$$

$$y = (K_1 e^{-0.670845x} + K_2 e^{-2.329156x}) + 0.161006\sin x + 0.0301887\cos x$$

Constants $K_1$ and $K_2$ can be determined using initial conditions. From $y(0) = 5$,

$$y(0) = K_1 + K_2 + 0.0301887 = 5$$

$$K_1 + K_2 = 5 - 0.0301887 = 4.969811$$

Now

$$\begin{aligned}\frac{dy}{dx} = &- 0.670845K_1 e^{-(0.670845)x} - 2.329156K_2 e^{-(2.329156)x} \\ &+ 0.161006\cos x - 0.0301887\sin x\end{aligned}$$

From

$$\frac{dy}{dx}(x = 0) = 3$$

we get

$$-0.670845K_1 - 2.329156K_2 + 0.161006 = 3$$

$$0.670845K_1 + 2.329156K_2 = -3 + 0.161006$$

$$0.670845K_1 + 2.329156K_2 = -2.838994$$

We have two simultaneous linear equations with two unknowns

$$K_1 + K_2 = 4.969811$$

$$0.670845K_1 + 2.329156K_2 = -2.838994$$

Solving the two simultaneous linear equations, we get

$$K_1 = 8.692253$$

$$K_2 = -3.722442$$

The complete solution is

$$\begin{aligned} y =& (8.692253e^{-0.670845x} - 3.722442e^{-2.329156x}) \\ &+ 0.161006 \sin x + 0.0301887 \cos x. \end{aligned}$$

**Example 3**

Solve

$$2\frac{d^2y}{dx^2} + 5\frac{dy}{dx} + 3.125y = e^{-x} \sin x,\ y(0) = 5,\ \frac{dy}{dx}(x = 0) = 3$$

**Solution**

The homogeneous equation is given by

$$(2D^2 + 5D + 3.125)y = 0$$

The characteristic equation is given by

$$2r^2 + 5r + 3.125 = 0$$

$$
\begin{aligned}
r &= \frac{-5 \pm \sqrt{5^2 - 4(2)(3.125)}}{2(2)} \\
&= \frac{-5 \pm \sqrt{25 - 25}}{4} \\
&= \frac{-5 \pm 0}{4} \\
&= -1.25, -1.25
\end{aligned}
$$

Since roots are repeated, the homogeneous solution $y_H$ is given by

$$y_H = (K_1 + K_2 x)e^{(-1.25)x}$$

The particular part of the solution is of the form

$$y_P = e^{-x}(A \sin x + B \cos x)$$

Substituting the particular part of the solution in the ordinary differential equation

$$
\begin{aligned}
2\frac{d^2}{dx^2}e^{-x}(A \sin x + B \cos x) + 5\frac{d}{dx}e^{-x}(A \sin x + B \cos x) \\
+ 3.125e^{-x}(A \sin x + Bcosx) = e^{-x} \sin x
\end{aligned}
$$

$$
\begin{aligned}
&2\frac{d}{dx}\left\{-e^{-x}\left(A \sin x + B \cos x\right) + e^{-x}\left(A \cos x - B \sin x\right)\right\} \\
&+ 5\left\{-e^{-x}\left(A \sin x + B \cos x\right) + e^{-x}\left(A \cos x - B \sin x\right)\right\} \\
&+ 3.125e^{-x}\left(A \sin x + B \cos x\right)
\end{aligned}
$$

$$
\begin{aligned}
= e^{-x} \sin x \; 2\{&e^{-x}(A \sin x + B \cos x) - e^{-x}(A \cos x - B \sin x) - e^{-x}(A \cos x \\
&- B \sin x) - e^{-x}(A \sin x + Bcosx) \\
&+ 5\{-e^{-x}(A \sin x + B \cos x) + e^{-x}(A \cos x - B \sin x)\} + 3.125e^{-x}(A \sin x \\
&+ B \cos x) = e^{-x} \sin x
\end{aligned}
$$

$$-1.875e^{-x}(A \sin x + B \cos x) + e^{-x}(A \cos x - B \sin x) = e^{-x} \sin x$$

$$-1.875(A \sin x + B \cos x) + (A \cos x - B \sin x) = \sin x$$

$$-(1.875A + B) \sin x + (A - 1.875B) \cos x = \sin x$$

Equating coefficients of $\cos x$ and $\sin x$ on both sides, we get

$$A - 1.875B = 0$$

$$1.875A + B = -1$$

Solving the above two simultaneous linear equations, we get

$$A = -0.415224 \text{ and}$$

$$B = -0.221453$$

Hence,

$$y_P = -e^{-x}(0.415224 \sin x + 0.221453 \cos x)$$

Therefore the complete solution is given by

$$y = y_H + y_P$$

$$y = (K_1 + xK_2)e^{-1.25x} - e^{-x}(0.415224 \sin x + 0.221453 \cos x)$$

Constants $K_1$ and $K_2$ can be determined using initial conditions,
From $y(0) = 5$, we get

$$K_1 - 0.221453 = 5$$

$$K_1 = 5.221453$$

Now

$$\begin{aligned}\frac{dy}{dx} = &- 1.25K_1e^{-1.25x} - 1.25K_2xe^{-1.25x} + K_2e^{-1.25x} - \\ &e^{-x}(0.415224 \cos x - 0.221453 \sin x) + e^{-x}(0.415224 \sin x + 0.221453 \cos x)\end{aligned}$$

From $\frac{dy}{dx}(0) = 3$, we get

$$\begin{aligned}&- 1.25K_1e^{-1.25(0)} - 1.25K_2(0)e^{-1.25(0)} + K_2e^{-1.25(0)} \\ &- e^0(0.415224 \cos(0) - 0.221453 \sin(0)) + e^0(0.415224 \sin(0) + 0.221453 \cos(0) = 3\end{aligned}$$

$$-1.25K_1 + K_2 + 0.221453 - 0.415224 = 3$$

$$-1.25K_1 + K_2 = 3.193771$$

$$-1.25(5.221453) + K_2 = 3.193771$$

$$K_2 = 9.720582$$

Substituting

$$K_1 = 5.221453 \text{ and}$$

$$K_2 = 9.720582$$

in the solution, we get

$$y = (5.221453 + 9.720582x)e^{-1.25x} - e^{-x}(0.415224 \sin x + 0.221453 \cos x)$$

# Chapter 08.01: Prerequisites to Numerical Methods for Solving Ordinary Differential Equations

Lesson: Solving an Integral as a First-Order Ordinary Differential Equation

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *recall the two parts of the fundamental theorem of calculus*

2) *verify finding an exact integral using the exact solution of a first-order ordinary differential equation*

## Introduction

In this lesson, we recall the two parts of the fundamental theorem of calculus (sometimes called the first and second fundamental theorems of calculus) to be able to show that the solution to a definite integral

$$\int_a^b f(x)dx$$

can be posed as a solution to an ordinary differential equation. This ability to represent a definite integral as an ordinary differential equation plays a prominent role in showing how we can use numerical methods of ordinary differential equations to conduct numerical integration.

The *first part of the fundamental theorem of calculus* states that if $f(x)$ is a real-valued and continuous function in the open interval $D$, and $a$ is a point in the interval $D$, and if

$$F(x) = \int_a^x f(t)dt \qquad (1)$$

then

$$F'(x) = f(x) \qquad (2)$$

at each point in $D$.

The *second part of the fundamental theorem of calculus* states that if $f(x)$ is a real-valued function in the interval $[a, b]$, and $F(x)$ is the antiderivative of $f(x)$ in $[a, b]$, then

$$\int_a^b f(x)dx = F(b) - F(a) \qquad (3)$$

Hence we can write equation (1) as

$$\frac{dF}{dx} = f(x) \qquad (4)$$

and if we assume $F(a) = 0$, then from Equation (3)

$$\int_a^b f(x)dx = F(b) \qquad (5)$$

So we can pose the value of $\int_a^b f(x)dx$ as solving Equation (4) for $F(b)$ with $F(a) = 0$ as the initial condition. We use the example below to illustrate the equivalency.

**Example 1**

Let us suppose you are asked to find

$$\int_3^9 5x^2 dx \qquad (E1.1)$$

The exact integral using integral calculus is given as

$$\begin{aligned} \int_3^9 5x^2 dx = & \left[\frac{5x^3}{3}\right]_3^9 \\ & = \frac{5(9)^3}{3} - \frac{5(3)^3}{3} \\ & = 1215 - 45 \\ & = 1170 \qquad (E1.2) \end{aligned}$$

Now, if one poses an ordinary differential equation

$$\frac{dy}{dx} = 5x^2 \ , \ y(3) = 0 \qquad (E1.3)$$

then the solution using the classical solution technique is as follows

$$y = y_H + y_P \qquad (E1.4)$$

The homogeneous part of the solution, $y_H$ is found by using the characteristic equation

$$m = 0$$

to give

$$\begin{aligned} y_H &= Ke^{0x} \\ &= K \qquad (E1.5) \end{aligned}$$

The particular part of the solution would be of the form of $Ax^2 + Bx + C$, but since the constant $C$ is part of the homogeneous part of the solution, $y_H = K$, we need to use the next independent solution.

$$\begin{aligned} y_P &= x\left(Ax^2 + Bx + C\right) \\ &= Ax^3 + Bx^2 + Cx \qquad (E1.6) \end{aligned}$$

By substituting $y_P$ in the ordinary differential equation, we get

$$\frac{d}{dx}\left(Ax^3 + Bx^2 + Cx\right) = 5x^2$$

$$3Ax^2 + 2Bx + C = 5x^2 + 0x + 0$$

giving

$$3A = 5$$

$$2B = 0$$

$$C = 0$$

and hence

$$A = \frac{5}{3}$$

$$B = 0$$

$$C = 0$$

So Equation (E1.6) gives

$$y_P = \frac{5}{3}x^3 \qquad (E1.7)$$

The complete solution from Equation (E1.4) is

$$\begin{aligned} y &= y_H + y_P \\ &= K + \frac{5}{3}x^3 \qquad (E1.8) \end{aligned}$$

To find $K$, use the initial condition,

$$y(3) = 0$$

then from Equation (E1.8), we get

$$K + \frac{5}{3}(3)^3 = 0$$

$$K + 45 = 0$$

$$K = -45$$

Hence the solution by rewriting Equation (E1.8) is

$$y = -45 + \frac{5}{3}x^3 \qquad (E1.9)$$

From Equation (E1.9)

$$\begin{aligned} y(9) &= -45 + \frac{5}{3}(9)^3 \\ &= 1170 \qquad (E1.10) \end{aligned}$$

We could have chosen $y\,(3)$ to be any finite number and the value $y\,(9) - y\,(3)$ would still turn out to be 1170.

Why do these values from Equations (E1.2) and (E1.10), one being an integral and another being the solution of an ordinary differential equation, turn out to be the same value 1170. It is not a coincidence. It is because of the second fundamental theorem of calculus, and it states

$$\frac{d}{dx}\int_a^x f\,(t)\,dt = f(x) \qquad (E1.11)$$

Choosing

$$f\,(t) = 5t^2 \text{ and } a = 3$$

$$\frac{d}{dx}\int_3^x 5t^2\ dt = 5x^2 \qquad (E1.12)$$

Calling

$$y\left(x\right)=\int_{3}^{x}5t^{2}\ dt$$

$$\frac{dy}{dx}=5x^{2}$$

By posing the ordinary differential equation as

$$\frac{dy}{dx}=5x^{2},\ y(3)=0 \qquad (E1.13)$$

and finding $y(9)$, we have

$$y\left(3\right)=\int_{3}^{3}5t^{2}\ dt=0$$

$$y\left(9\right)=\int_{3}^{9}5t^{2}\ dt$$

Hence solving

$$\frac{dy}{dx}=5x^{2},\ y(3)=0$$

for $y\left(9\right)$ is the value of the integral

$$\int_{3}^{9}5t^{2}\ dt.$$

# Chapter 08.01: Prerequisites to Numerical Methods for Solving Ordinary Differential Equations

Lesson: Solving Ordinary Differential Equations by Laplace Transforms

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *Find the exact solution of ordinary differential equations by using Laplace transforms*

## Introduction

If $y = f(x)$ is defined at all positive values of $x$, the Laplace transform denoted by $Y(s)$ is given by

$$Y(s) = L\{f(x)\} = \int_0^\infty e^{-sx} f(x) dx \quad (35)$$

where $s$ is a parameter, which can be a real or complex number. We can get back $f(x)$ by taking the inverse Laplace transform of $Y(s)$.

$$L^{-1}\{Y(s)\} = f(x) \quad (36)$$

Laplace transforms are highly useful in solving differential equations. They give the solution directly without the necessity of evaluating arbitrary constants separately.

The following are Laplace transforms of some elementary functions

$$L(1) = \frac{1}{s}$$

$$L(x^n) = \frac{n!}{s^{n+1}}, \text{ where } n = 0, 1, 2, 3...$$

$$L(e^{ax}) = \frac{1}{s-a}$$

$$L(\sin ax) = \frac{a}{s^2 + a^2}$$

$$L(\cos ax) = \frac{s}{s^2 + a^2}$$

$$L(\sinh\ ax) = \frac{a}{s^2 - a^2}$$

$$L(\cosh\ ax) = \frac{s}{s^2 - a^2} \quad (37)$$

The following are the inverse Laplace transforms of some common functions

$$L^{-1}\left(\frac{1}{s}\right) = 1$$

$$L^{-1}\left(\frac{1}{s-a}\right) = e^{ax}$$

$$L^{-1}\left(\frac{1}{s^n}\right) = \frac{x^{n-1}}{(n-1)!}, \text{ where } n = 1, 2, 3.....$$

$$L^{-1}\left(\frac{1}{(s-a)^n}\right) = \frac{e^{ax}x^{n-1}}{(n-1)!}$$

$$L^{-1}\left(\frac{1}{s^2+a^2}\right) = \frac{1}{a}\sin ax$$

$$L^{-1}\left(\frac{s}{s^2+a^2}\right) = \cos ax$$

$$L^{-1}\left(\frac{1}{s^2-a^2}\right) = \frac{1}{a}\sinh\ ax$$

$$L^{-1}\left(\frac{s}{s^2-a^2}\right) = \cosh\ at$$

$$L^{-1}\left(\frac{1}{(s-a)^2+b^2}\right) = \frac{1}{b}e^{ax}\sin bx$$

$$L^{-1}\left(\frac{s-a}{(s-a)^2+b^2}\right) = e^{ax}\cos bx$$

$$L^{-1}\left(\frac{s}{(s^2+a^2)^2}\right) = \frac{1}{2a}x\sin ax \quad (38)$$

## Properties of Laplace transforms

Linear property

If $a,\ b,\ c$ are constants and $f(x),\ g(x)$, and $h(x)$ are functions of $x$ then

$$L[af(x) + bg(x) + ch(x)] = aL(f(x)) + bL(g(x)) + cL(h(x)) \quad (39)$$

Shifting property

If

$$L\{f(x)\} = Y(s) \quad (40)$$

then

$$L\{e^{at}f(x)\} = Y(s-a) \quad (41)$$

Using shifting property we get

$$L\left(e^{ax}x^n\right) = \frac{n!}{(s-a)^{n+1}},\ n \geq 0$$

$$L\left(e^{ax}\sin bx\right) = \frac{b}{(s-a)^2 + b^2}$$

$$L\left(e^{ax}\cos bx\right) = \frac{s-a}{(s-a)^2 + b^2}$$

$$L\left(e^{ax}\sinh bx\right) = \frac{b}{(s-a)^2 - b^2}$$

$$L\left(e^{ax}\cosh bx\right) = \frac{s-a}{(s-a)^2 - b^2} \quad (42)$$

Scaling property

If

$$L\{f(x)\} = Y(s) \quad (43)$$

then

$$L\{f(ax)\} = \frac{1}{a}Y\left(\frac{s}{a}\right) \quad (44)$$

## Laplace transforms of derivatives

If the first $n$ derivatives of $f(x)$ are continuous, then

$$L\{f^n(x)\} = \int_0^\infty e^{-sx}f^n(x)dx \quad (45)$$

Using integration by parts, we get

$$\begin{aligned}\int_0^\infty e^{-sx}f^n(x)dx &= \left[\begin{array}{c} e^{-sx}f^{n-1}(x) - (-s)e^{-sx}f^{n-2}(x) \\ +(-s)^2e^{-sx}f^{n-3}(x) + ..... + (-1)^{n-1}(-s)^{n-1}e^{-sx}f(x)\end{array}\right]_0^\infty \\ &\quad + (-1)^n(-s)^n\int_0^\infty e^{-sx}f(x)dx \\ &= -f^{n-1}(0) - sf^{n-2}(0) - s^2f^{n-3}(0) - ..... - s^{n-1}f(0) + \\ &\quad s^n\int_0^\infty e^{-sx}f(x)dx \\ &= s^nY(s) - f^{n-1}(0) - sf^{n-2}(0) - s^2f^{n-3}(0) - ..... - s^{n-1}f(0) \quad (46)\end{aligned}$$

## Laplace transform technique to solve ordinary differential equations

The following are steps to solve ordinary differential equations using the Laplace transform method

(A) Take the Laplace transform of both sides of ordinary differential equations.

(B) Express $Y(s)$ as a function of $s$.

(C) Take the inverse Laplace transform on both sides to get the solution.

Let us solve Examples 1 through 4 using the Laplace transform method.

### Example 1

Solve

$$3\frac{dy}{dx} + 2y = e^{-x},\ y(0) = 5$$

### Solution

Taking the Laplace transform of both sides, we get

$$L\left(3\frac{dy}{dx} + 2y\right) = L\left(e^{-x}\right)$$
$$3[sY(s) - y(0)] + 2Y(s) = \frac{1}{s+1}$$

Using the initial condition, $y(0) = 5$ we get

$$3[sY(s) - 5] + 2Y(s) = \frac{1}{s+1}$$
$$(3s+2)Y(s) = \frac{1}{s+1} + 15$$
$$(3s+2)Y(s) = \frac{15s+16}{s+1}$$
$$Y(s) = \frac{15s+16}{(s+1)(3s+2)}$$

Writing the expression for $Y(s)$ in terms of partial fractions

$$\frac{15s+16}{(s+1)(3s+2)} = \frac{A}{s+1} + \frac{B}{3s+2}$$
$$\frac{15s+16}{(s+1)(3s+2)} = \frac{3As+2A+Bs+B}{(s+1)(3s+2)}$$
$$15s+16 = 3As+2A+Bs+B$$

Equating coefficients of $s^1$ and $s^0$ gives

$$3A + B = 15$$

$$2A + B = 16$$

The solution to the above two simultaneous linear equations is

$$\begin{aligned} A &= -1 \\ B &= 18 \\ Y(s) &= \frac{-1}{s+1} + \frac{18}{3s+2} \\ &= \frac{-1}{s+1} + \frac{6}{s+0.666667} \end{aligned}$$

Taking the inverse Laplace transform on both sides

$$L^{-1}\{Y(s)\} = L^{-1}\left(\frac{-1}{s+1}\right) + L^{-1}\left(\frac{6}{s+0.666667}\right)$$

Since

$$L^{-1}\left(\frac{1}{s+a}\right) = e^{-at}$$

The solution is given by

$$y(x) = -e^{-x} + 6e^{-0.666667x}$$

**Example 2**

Solve

$$2\frac{dy}{dx} + 3y = e^{-1.5x},\ y(0) = 5$$

**Solution**

Taking the Laplace transform of both sides, we get

$$\begin{aligned} L\left(2\frac{dy}{dx} + 3y\right) &= L\left(e^{-1.5x}\right) \\ 2[sY(s) - y(0)] + 3Y(s) &= \frac{1}{s+1.5} \end{aligned}$$

Using the initial condition $y(0) = 5$, we get

$$\begin{aligned} 2[sY(s) - 5] + 3Y(s) &= \frac{1}{s+1.5} \\ (2s+3)Y(s) &= \frac{1}{s+1.5} + 10 \\ (2s+3)Y(s) &= \frac{10s+16}{s+1.5} \end{aligned}$$

$$
\begin{aligned}
Y(s) &= \frac{10s+16}{(s+1.5)(2s+3)} \\
&= \frac{10s+16}{2(s+1.5)(s+1.5)} \\
&= \frac{10s+16}{2(s+1.5)^2} \\
&= \frac{5s+8}{(s+1.5)^2}
\end{aligned}
$$

Writing the expression for $Y(s)$ in terms of partial fractions

$$
\begin{aligned}
\frac{5s+8}{(s+1.5)^2} &= \frac{A}{s+1.5} + \frac{B}{(s+1.5)^2} \\
\frac{5s+8}{(s+1.5)^2} &= \frac{As+1.5A+B}{(s+1.5)^2} \\
5s+8 &= As+1.5A+B
\end{aligned}
$$

Equating coefficients of $s^1$ and $s^0$ gives

$$
\begin{aligned}
A &= 5 \\
1.5A + B &= 8
\end{aligned}
$$

The solution to the above two simultaneous linear equations is

$$
\begin{aligned}
A &= 5 \\
B &= 0.5 \\
Y(s) &= \frac{5}{s+1.5} + \frac{0.5}{(s+1.5)^2}
\end{aligned}
$$

Taking the inverse Laplace transform on both sides

$$
L^{-1}\{Y(s)\} = L^{-1}\left(\frac{5}{s+1.5}\right) + L^{-1}\left(\frac{0.5}{(s+1.5)^2}\right)
$$

Since

$$
L^{-1}\left(\frac{1}{s+a}\right) = e^{-ax} \text{ and } L^{-1}\left(\frac{1}{(s+a)^2}\right) = xe^{-ax}
$$

The solution is given by

$$
y(x) = 5e^{-1.5x} + 0.5xe^{-1.5x}
$$

**Example 3**

Solve

$$2\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + 3.125y = \sin x,\ y(0) = 5,\ \frac{dy}{dx}(x = 0) = 3$$

**Solution**

Taking the Laplace transform of both sides

$$L\left(2\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + 3.125y\right) = L\left(\sin x\right)$$

and knowing

$$L\left(\frac{d^2y}{dx^2}\right) = s^2Y\left(s\right) - sy\left(0\right) - \frac{dy}{dx}\left(x = 0\right)$$

$$L\left(\frac{dy}{dx}\right) = sY\left(s\right) - y\left(0\right)$$

$$L(\sin x) = \frac{1}{s^2 + 1}$$

we get

$$2\left[s^2Y(s) - sy(0) - \frac{dy}{dx}(x = 0)\right] + 3\left[sY(s) - y(0)\right] + 3.125Y(s) = \frac{1}{s^2 + 1}$$

$$2\left[s^2Y(s) - 5s - 3\right] + 3[sY(s) - 5] + 3.125Y(s) = \frac{1}{s^2 + 1}$$

$$[s(2s + 3) + 3.125]Y(s) - 10s - 21 = \frac{1}{s^2 + 1}$$

$$\left[s\left(2s + 3\right) + 3.125\right]Y(s) = \frac{1}{s^2 + 1} + 10s + 21$$

$$\left[2s^2 + 3s + 3.125\right]Y(s) = \frac{22 + 10s^3 + 10s + 21s^2}{(s^2 + 1)}$$

$$Y(s) = \frac{10s^3 + 21s^2 + 10s + 22}{(s^2 + 1)\left(2s^2 + 3s + 3.125\right)}$$

Writing the expression for $Y(s)$ in terms of partial fractions

$$\frac{As + B}{(2s^2 + 3s + 3.125)} + \frac{Cs + D}{(s^2 + 1)} = \frac{10s^3 + 21s^2 + 10s + 22}{(s^2 + 1)\left(2s^2 + 3s + 3.125\right)}$$

$$\frac{As^3 + As + Bs^2 + B + 2Cs^3 + 3Cs^2 + 3.125Cs + 2Ds^2 + 3Ds + 3.125D}{\left(2s^2 + 3s + 3.125\right)(s^2 + 1)}$$

$$= \frac{10s^3 + 21s^2 + 10s + 22}{(s^2 + 1)\left(2s^2 + 3s + 3.125\right)}$$

$$\frac{(A+2C)\,s^3+(B+3C+2D)\,s^2+(A+3.125C+3D)\,s+(B+3.125D)}{(s^2+1)\,(2s^2+3s+3.125)}$$
$$=\frac{10s^3+21s^2+10s+22}{(s^2+1)\,(2s^2+3s+3.125)}$$

Equating terms of $s^3,\ s^2,\ s^1$ and $s^0$ gives gives four simultaneous linear equations

$$A+2C=10$$
$$B+3C+2D=21$$
$$A+3.125C+3D=10$$
$$B+3.125D=22$$

The solution to the above four simultaneous linear equations is

$$A=10.584474$$
$$B=21.657534$$
$$C=-0.292237$$
$$D=0.109589$$

Hence

$$Y(s)=\frac{10.584474s+21.657534}{2s^2+3s+3.125}+\frac{-0.292237s+0.109589}{s^2+1}$$

$$\left(2s^2+3s+3.125\right)=2\{(s^2+1.5s+0.5625)+1\}=2\{(s+0.75)^2+1\}$$

$$Y(s)=\frac{10.584474(s+0.75)+13.719179}{2\{(s+0.75)^2+1\}}+\frac{-0.292237s+0.109589}{s^2+1}$$
$$=\frac{5.292237(s+0.75)}{\{(s+0.75)^2+1\}}+\frac{6.859589}{\{(s+0.75)^2+1\}}-\frac{0.292237s}{(s^2+1)}+\frac{0.109589}{(s^2+1)}$$
$$L^{-1}Y(s)=5.292237L^{-1}\left(\frac{s+0.75}{(s+0.75)^2+1}\right)+6.859589L^{-1}\left(\frac{1}{(s+0.75)^2+1}\right)$$

Taking the inverse Laplace transform of both sides

$$L^{-1}\left\{Y\left(s\right)\right\}=L^{-1}\left(\frac{5.292237\left(s+0.75\right)}{\{(s+0.75)^2+1\}}\right)+L^{-1}\left(\frac{6.859589}{\{(s+0.75)^2+1}\right)-L^{-1}\left(\frac{0.292237s}{s^2+1}\right)+$$
$$L^{-1}\left(\frac{0.109589}{s^2+1}\right)$$

$$L^{-1}Y(s)=5.292237L^{-1}\left(\frac{s+0.75}{(s+0.75)^2+1}\right)+6.859589L^{-1}\left(\frac{1}{(s+0.75)^2+1}\right)$$
$$-0.292237L^{-1}\left(\frac{s}{s^2+1}\right)+0.109589L^{-1}\left(\frac{1}{s^2+1}\right)$$

Since

$$L^{-1}\left(\frac{s+a}{(s+a)^2+b^2}\right) = e^{-ax}\cos bx$$

$$L^{-1}\left(\frac{b}{(s+a)^2+b^2}\right) = e^{-ax}\sin bx$$

$$L^{-1}\left(\frac{1}{s^2+a^2}\right) = \sin ax$$

$$L^{-1}\left(\frac{s}{s^2+a^2}\right) = \cos ax$$

The complete solution is

$$\begin{aligned} y(x) &= 5.292237e^{-0.75x}cosx + 6.8595859e^{-0.75x}sinx \\ &\quad - 0.292237\cos x + 0.109589\sin x \\ &= e^{-0.75x}\left(5.292237\cos x + 6.859589\sin x\right) - 0.292237\cos x + 0.109589\sin x \end{aligned}$$

**Example 4**

Solve

$$2\frac{d^2y}{dx^2} + 6\frac{dy}{dx} + 3.125y = \cos x,\ y(0) = 5,\ \frac{dy}{dx}(x=0) = 3$$

**Solution**

Taking the Laplace transform of both sides

$$L\left(2\frac{d^2y}{dx^2} + 6\frac{dy}{dx} + 3.125y\right) = L\left(\cos x\right)$$

and knowing

$$L\left(\frac{d^2y}{dx^2}\right) = s^2Y\left(s\right) - sy\left(0\right) - \frac{dy}{dx}\left(x=0\right)$$

$$L\left(\frac{dy}{dx}\right) = sY\left(s\right) - y\left(0\right)$$

$$L(\cos x) = \frac{s}{s^2+1}$$

we get

$$2\left[s^2Y(s) - sy(0) - \frac{dy}{dx}(x=0)\right] + 6\left[sY(s) - y(0)\right] + 3.125Y(s) = \frac{s}{s^2+1}$$

$$2\left[s^2Y(s) - 5s - 3\right] + 6\left[sY(s) - 5\right] + 3.125Y(s) = \frac{s}{s^2+1}$$

$$[s(2s+6)+3.125]\,Y(s) = \frac{s}{s^2+1} + 10s + 36$$

$$\left[2s^2+6s+3.125\right] Y(s) = \frac{36+10s^3+11s+36s^2}{s^2+1}$$

$$Y(s) = \frac{10s^3+36s^2+11s+36}{(s^2+1)\,(2s^2+6s+3.125)}$$

Writing the expression for $Y(s)$ in terms of partial fractions

$$\frac{As+B}{(2s^2+6s+3.125)} + \frac{Cs+D}{(s^2+1)} = \frac{10s^3+36s^2+11s+36}{(s^2+1)\,(2s^2+6s+3.125)}$$

$$\frac{As^3+As+Bs^2+B+2Cs^3+6Cs^2+3.125Cs+2Ds^2+6Ds+3.125D}{(2s^2+6s+3.125)\,(s^2+1)}$$

$$= \frac{10s^3+36s^2+11s+36}{(s^2+1)\,(2s^2+6s+3.125)}$$

$$\frac{(A+2C)\,s^3+(B+6C+2D)\,s^2+(A+3.125C+6D)\,s+(B+3.125D)}{(s^2+1)\,(2s^2+6s+3.125)}$$

$$= \frac{10s^3+36s^2+11s+36}{(s^2+1)\,(2s^2+6s+3.125)}$$

Equating terms of $s^3$, $s^2$, $s^1$ and $s^0$ gives

$$A + 2C = 10$$

$$B + 6C + 2D = 36$$

$$A + 3.125C + 6D = 11$$

$$B + 3.125D = 36$$

The solution to the above four simultaneous linear equations is

$$A = 9.939622$$

$$B = 35.496855$$

$$C = 0.0301886$$

$$D = 0.161006$$

Then

$$Y(s) = \frac{9.939622s+35.496855}{2s^2+6s+3.125} + \frac{0.0301886s+0.161006}{s^2+1}$$

$$\left(2s^2+6s+3.125\right) = 2\{(s^2+3s+2.25)-0.6875\} = 2\{(s+1.5)^2-0.829156^2\}$$

$$
\begin{aligned}
Y(s) &= \frac{9.939622(s+1.5)+20.587422}{2\{(s+1.5)^2-0.829156^2\}} + \frac{0.0301886s+0.161006}{s^2+1} \\
&= \frac{4.969811(s+1.5)}{(s+1.5)^2-0.829156^2} + \frac{10.293711}{(s+1.5)^2-0.829156^2} \\
&\quad + \frac{0.0301886s}{s^2+1} + \frac{0.161006}{s^2+1}
\end{aligned}
$$

Taking the inverse Laplace transform on both sides

$$
\begin{aligned}
L^{-1}Y(s) &= L^{-1}\left(\frac{4.969811(s+1.5)}{(s+1.5)^2-0.829156^2}\right) + L^{-1}\left(\frac{10.293711}{(s+1.5)^2-0.829156^2}\right) \\
&\quad + L^{-1}\left(\frac{0.0301886s}{s^2+1}\right) + L^{-1}\left(\frac{0.161006}{s^2+1}\right) \\
&= 4.969811L^{-1}\left(\frac{(s+1.5)}{(s+1.5)^2-0.829156^2}\right) + 10.293711L^{-1}\left(\frac{1}{(s+1.5)^2-0.829156^2}\right) \\
&\quad + 0.0301886L^{-1}\left(\frac{s}{(s^2+1)}\right) + 0.161006L^{-1}\left(\frac{1}{(s^2+1)}\right)
\end{aligned}
$$

Since

$$
\begin{aligned}
L^{-1}\left(\frac{s+a}{(s+a)^2-b^2}\right) &= e^{-ax}\cosh\, bx \\
L^{-1}\left(\frac{1}{(s+a)^2-b^2}\right) &= \frac{1}{b}e^{-ax}\sinh\, bx \\
L^{-1}\left(\frac{1}{s^2+a^2}\right) &= \frac{1}{a}\sin ax \\
L^{-1}\left(\frac{s}{s^2+a^2}\right) &= \cos ax
\end{aligned}
$$

The complete solution is

$$
\begin{aligned}
y(x) &= 4.969811e^{-1.5x}\cosh(0.829156x) + \frac{10.293711}{0.829156}e^{-1.5x}\sinh(0.829156x) \\
&\quad + 0.0301886\cos x + 0.161006\sin x \\
&= e^{-1.5x}\left(4.969811\left(\frac{e^{0.829156x}+e^{-0.829156x}}{2}\right) + 12.414685\ \left(\frac{e^{0.829156x}-e^{-0.829156x}}{2}\right)\right) \\
&\quad + 0.030188\cos x + 0.161006\sin x \\
&= e^{-1.5x}\left(8.692248e^{0.829156x} - 3.722437e^{-0.829156x}\right) + 0.0301886\cos x \\
&\quad + 0.161006\sin x
\end{aligned}
$$

**Example 5**

Solve

$$2\frac{d^2y}{dx^2} + 5\frac{dy}{dx} + 3.125y = e^{-x}\sin x,\ y(0) = 5, \quad \frac{dy}{dx}(x = 0) = 3$$

**Solution**

Taking the Laplace transform of both sides

$$L\left(2\frac{d^2y}{dx^2} + 5\frac{dy}{dx} + 3.125y\right) = L\left(e^{-x}\sin x\right)$$

knowing

$$L\left(\frac{d^2y}{dx^2}\right) = s^2Y(s) - sy(0) - \frac{dy}{dx}(x = 0)$$

$$L\left(\frac{dy}{dx}\right) = sY(s) - y(0)$$

$$L(e^{-x}\sin x) = \frac{1}{(s+1)^2 + 1}$$

we get

$$2\left[s^2Y(s) - sy(0) - \frac{dy}{dx}(x = 0)\right] + 5[sY(s) - y(0)] + 3.125Y(s) = \frac{1}{(s+1)^2 + 1}$$

$$2\left[s^2Y(s) - 5s - 3\right] + 5[sY(s) - 5] + 3.125Y(s) = \frac{1}{(s+1)^2 + 1}$$

$$\left[s\left(2s + 5\right) + 3.125\right]Y(s) - 10s - 31 = \frac{1}{(s+1)^2 + 1}$$

$$\left[s(2s + 5) + 3.125\right]Y(s) = \frac{1}{(s+1)^2 + 1} + 10s + 31$$

$$\left[2s^2 + 5s + 3.125\right]Y(s) = \frac{63 + 10s^3 + 82s + 51s^2}{s^2 + 2s + 2}$$

$$Y(s) = \frac{10s^3 + 51s^2 + 82s + 63}{\left(s^2 + 2s + 2\right)\left(2s^2 + 5s + 3.125\right)}$$

Writing the expression for $Y(s)$ in terms of partial fractions

$$\frac{As + B}{2s^2 + 5s + 3.125} + \frac{Cs + D}{s^2 + 2s + 2} = \frac{10s^3 + 51s^2 + 82s + 63}{\left(s^2 + 2s + 2\right)\left(2s^2 + 5s + 3.125\right)}$$

$$\frac{2Cs^3 + 5Cs^2 + 3.125Cs + 2Ds^2 + 5Ds + 3.125D + As^3 + 2As^2 + 2As + Bs^2 + 2Bs + 2B}{\left(2s^2 + 5s + 3.125\right)\left(s^2 + 2s + 2\right)}$$

$$= \frac{10s^3 + 51s^2 + 82s + 63}{\left(s^2 + 2s + 2\right)\left(2s^2 + 5s + 3.125\right)}$$

$$\frac{(2C+A)\,s^3+(5C+2D+2A+B)\,s^2+(3.125C+5D+2A+2B)\,s+(3.125D+2B)}{(s^2+2s+2)\,(2s^2+5s+3.125)}$$
$$=\frac{10s^3+51s^2+82s+63}{(s^2+2s+2)\,(2s^2+5s+3.125)}$$

Equating terms of $s^3$, $s^2$, $s^1$ and $s^0$ gives four simultaneous linear equations

$$2C+A=10$$
$$5C+2D+2A+B=51$$
$$3.125C+5D+2A+2B=82$$
$$3.125D+2B=63$$

The solution to the above four simultaneous linear equations is

$$A=10.442906$$
$$B=32.494809$$
$$C=-0.221453$$
$$D=-0.636678$$

Then

$$Y(s)=\frac{10.442906s+32.494809}{2s^2+5s+3.125}+\frac{-0.221453s-0.636678}{s^2+2s+2}$$

$$\left(2s^2+5s+3.125\right)=2\{(s^2+2.5s+1.5625)\}=2(s+1.25)^2$$

$$Y(s)=\frac{10.442906(s+1.25)+19.441176}{2(s+1.25)^2}+\frac{-0.221453(s+1)-0.415225}{(s+1)^2+1}$$
$$=\frac{5.221453(s+1.25)}{(s+1.25)^2}+\frac{9.720588}{(s+1.25)^2}-\frac{0.221453(s+1)}{(s+1)^2+1}-\frac{0.415225}{(s+1)^2+1}$$

Taking the inverse Laplace transform on both sides

$$L^{-1}Y(s)=L^{-1}\left(\frac{5.221453}{(s+1.25)}\right)+L^{-1}\left(\frac{9.720588}{(s+1.25)^2}\right)$$
$$-L^{-1}\left(\frac{0.221453(s+1)}{(s+1)^2+1}\right)-L^{-1}\left(\frac{0.415225}{(s+1)^2+1}\right)$$
$$=5.221453L^{-1}\left(\frac{1}{(s+1.25)}\right)+9.720588L^{-1}\left(\frac{1}{(s+1.25)^2}\right)$$
$$-0.221453L^{-1}\left(\frac{(s+1)}{(s+1)^2+1}\right)-0.415225L^{-1}\left(\frac{1}{(s+1)^2+1}\right)$$

Since

$$L^{-1}\left(\frac{s+a}{(s+a)^2+b^2}\right) = e^{-ax}\cos bx$$

$$L^{-1}\left(\frac{b}{(s+a)^2+b^2}\right) = e^{-ax}\sin bx$$

$$L^{-1}\left(\frac{1}{s+a}\right) = e^{-ax}$$

$$L^{-1}\left(\frac{1}{(s+a)^n}\right) = \frac{e^{-ax}x^{n-1}}{(n-1)!}$$

The complete solution is

$$\begin{aligned} y(x) &= 5.221453e^{-1.25x} + 9.720588e^{-1.25x}x - 0.221453e^{-x}\cos x \\ &\quad - 0.415225e^{-x}\sin x \\ &= e^{-1.25x}\left(5.221453 + 9.720588x\right) + \\ &\quad e^{x}(-0.221453\cos x - 0.415225\sin x) \end{aligned}$$

# Multiple Choice Test

## Chapter 08.01 Prerequisites to Numerical Methods for Solving Ordinary Differential Equations

(1). The ordinary differential equation $2\dfrac{dy}{dx} + x^2 y = 2x + 3,\ y(0) = 5$ is

(A) linear

(B) nonlinear

(C) linear with fixed constants

(D) undeterminable to be linear or nonlinear

(2). A differential equation is considered to be ordinary if it has

(A) one dependent variable

(B) more than one dependent variable

(C) one independent variable

(D) more than one independent variable

(3). Given

$$2\frac{dy}{dx} + 3y = \sin 2x,\ y(0) = 6$$

$y(2)$ most nearly is

(A) 0.17643

(B) 0.29872

(C) 0.32046

(D) 0.58024

(4). The form of the exact solution to

$$2\frac{dy}{dx} + 3y = e^{-x},\ y(0) = 5$$

is

(A) $Ae^{-1.5x} + Be^{-x}$

(B) $Ae^{-1.5x} + Bxe^{-x}$

(C) $Ae^{1.5x} + Be^{-x}$

(D) $Ae^{1.5x} + Bxe^{-x}$

(5). The following nonlinear differential equation can be solved exactly by separation of variables.

$$\frac{d\theta}{dt} = -10^{-6}\left(\theta^2 - 81\right),\ \theta\left(0\right) = 1000$$

The value of $\theta\left(100\right)$ most nearly is

(A) $-99.99$

(B) 909.10

(C) 1000.32

(D) 1111.10

(6). A solid spherical ball taken out of a furnace at 1200 K is allowed to cool in air. Given the following,

radius of the ball = 2 $cm$

density of the ball = $7800\frac{kg}{m^3}$

specific heat of the ball = $420\frac{J}{kg-K}$

emmittance= 0.85

Stefan-Boltzman constant = $5.67 \times 10^{-8}\frac{J}{s-m^2-K^4}$

ambient temperature = 300 $K$

convection coefficient to air = $350\frac{J}{s-m^2-K}$

the differential equation governing the temperature $\theta$ of the ball as a function of time $t$ is given by

(A) $\frac{d\theta}{dt} = -2.2067 \times 10^{-12}\left(\theta^4 - 81 \times 10^8\right), \theta(0) = 1200$

(B) $\frac{d\theta}{dt} = -1.6026 \times 10^{-2}\left(\theta - 300\right), \theta(0) = 1200$

(C) $\frac{d\theta}{dt} = 2.2067 \times 10^{-12}\left(\theta^4 - 81 \times 10^8\right) + 1.6026 \times 10^{-12}\left(\theta - 300\right), \theta(0) = 1200$

(D) $\frac{d\theta}{dt} = -2.2067 \times 10^{-12}\left(\theta^4 - 81 \times 10^8\right) - 1.6026 \times 10^{-2}\left(\theta - 300\right), \theta(0) = 1200$

For complete solution, go to

http://nm.mathforcollege.com/mcquizzes/08ode/quiz_08ode_background_solution.pd

# Problem Set

Chapter 08.01: Prerequisites to Numerical Methods for Solving Ordinary Differential Equations

(1). Solve the ordinary differential equation exactly

$$2\frac{dy}{dx} + 0.4\ y\ =\ 3e^{-x},\ y(0) = 5$$

**Answer:** $y = 6.875e^{-0.2x} - 1.875e^{-x}$

(2). For the ordinary differential equation

$$2\frac{dy}{dx} + 3\ y\ =\ e^{-1.5x},\ y(0) = 5,$$

solve exactly to find

a) $y(0)$ ,

b) $\frac{dy}{dx}$ (0),

c) $y(2.5)$,

d) $\frac{dy}{dx}$ (2.5).

**Answer:** *a*) 5
*b*) $-7$
*c*) 0.14699
*d*) $-0.20872$

(3). Solve the ordinary differential equation exactly

$$2\frac{d^2y}{dx^2} + 10\frac{dy}{dx} + 12y = 6e^x,\ y(0) = 7,\ \frac{dy}{dx}(0) = 11$$

**Answer:** $y = 31e^{-2x} - 24.25e^{-3x} + 0.25e^x$

(4). Solve the ordinary differential equations exactly

$$2\frac{d^2y}{dx^2} + 8\frac{dy}{dx} + 8y = 3\sin(2x),\ y(0) = 7,\ \frac{dy}{dx}(0) = 11$$

**Answer:** $y = e^{-0.75x}(5.292237cosx + 6.859588sinx) + 0.109589sinx - 0.292237cosx$

(5). Solve the ordinary differential equations exactly

$$2\frac{d^2y}{dx^2} + 8\frac{dy}{dx} + 26y = x^2,\ y(0) = 7,\ \frac{dy}{dx}(0) = 11$$

**Answer:** $y = e^{-2x}(6.9986cos3x + 8.3403sin3x) + 0.038462x^2 - 0.023669x + 0.0013655$

(6). A spherical ball is taken out of a furnace at $2500K$. It loses heat due to radiation and convection. Given the following,

Radius of ball, $r = 1.0\ cm$

Density of ball, $\rho = 3000\ kg/m^3$

Specific heat, $C = 1000\ J/(kg \cdot K)$

Emittance, $\in = 0.5$

Stefan-Boltzmann Constant, $\sigma = 5.67 \times 10^{-8} J/(s \cdot m^2 \cdot K^4)$

Initial temperature of the ball, $\theta(0) = 2500\ K$

Ambient temperature, $\theta_a = 300\ K$

Convective cooling coefficient, $h = 500J/(s \cdot m^2 \cdot K)$

the ordinary differential equation that governs the temperature of the ball, $\theta$ as a function of time, $t$ is given by

$$\frac{d\theta}{dt} = -2.8349 \times 10^{-12}(\theta^4 - 81 \times 10^8) - 0.05(\theta - 300).$$

Hint: The above formula was calculated by using the knowledge that

Rate of heat lost due to radiation $= A \in \sigma(\theta^4 - \theta_a^4)$

Rate of heat lost due to convection $= hA(\theta - \theta_a)$

Heat stored $= mc\theta$

a) Note that the above ODE is nonlinear. Is there an exact solution to the problem? If so, find it by hand or by using a mathematical package.

b) The exact solution in part (a) is not explicit. To solve for temperature at a particular time, one needs to solve a nonlinear equation. Find the temperature at $t = 10$ seconds.

c) What are the following at $t = 0s$?

i) temperature,

ii) rate of change of temperature,

iii) rate of heat loss due to radiation,

iv) rate of heat loss due to convection,

v) rate of heat stored.

d) Can either convection or radiation be neglected if we are interested in finding the temperature profile between 0 and 10 s?

**Answer:**

*a*)

$$5.8043 \ln(\theta + 2.696.3) + 7.0370 ln\left(\theta^2 - 2396.4\theta + 6.5478 \times 10^6\right) - 2.104 tan^{-1}\left(\frac{2261.81}{\theta - 1198.2}\right) - 19.878(\theta - 300) = t + 5.1921$$

*b*) 1369.6 $K$

*c*)

*i*) 2500 $K$
*ii*) $-220.72$ $K/s$
*iii*) 1391.3 $W$
*iv*) 1382.3 $W$
*v*) $-2773.6$ $W$

*d*) No, as the rate of heat lost due to convection and radiation is of the same order.

# Chapter 08.02: Euler's Method for Solving Ordinary Differential Equations

Lesson: Derivation of Euler's Method

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *develop Euler's method for solving first-order ordinary differential equations,*

2) *determine how the step size affects the accuracy of a solution, and*

3) *derive Euler's method formula from the Taylor series*

## What is Euler's method?

Euler's method is a numerical technique to solve first-order ordinary differential equations of the form

$$\frac{dy}{dx} = f(x, y), y(x_0) = y_0 \qquad (1)$$

Only first-order ordinary differential equations of the form given by Equation (1) can be solved by using Euler's method. In another lesson, we discuss how Euler's method is used to solve higher-order and coupled (simultaneous) ordinary differential equations.

How does one write a first-order differential equation in the above form?

**Example 1**

Rewrite

$$\frac{dy}{dx} + 2y = 1.3e^{-x}, y\,(0) = 5$$

in

$$\frac{dy}{dx} = f(x, y),\ y(0) = y_0 \text{ form.}$$

**Solution**

$$\frac{dy}{dx} + 2y = 1.3e^{-x}, y(0) = 5$$

$$\frac{dy}{dx} = 1.3e^{-x} - 2y, y(0) = 5$$

In this case

$$f(x, y) = 1.3e^{-x} - 2y$$

**Example 2**

Rewrite

$$e^y \frac{dy}{dx} + x^2 y^2 = 2\sin(3x),\ y(0) = 5$$

in

$$\frac{dy}{dx} = f(x, y),\ y(0) = y_0 \text{ form.}$$

**Solution**

$$e^y \frac{dy}{dx} + x^2 y^2 = 2\sin(3x),\ y(0) = 5$$

$$\frac{dy}{dx} = \frac{2\sin(3x) - x^2 y^2}{e^y},\ y(0) = 5$$

In this case

$$f(x, y) = \frac{2\sin(3x) - x^2 y^2}{e^y}$$

## Derivation of Euler's method

At $x = 0$, we are given the value of $y = y_0$. Let us call $x = 0$ as $x_0$. Now since we know the slope of $y$ with respect to $x$, that is, $f(x, y)$, then at $x = x_0$, the slope is $f(x_0, y_0)$. Both $x_0$ and $y_0$ are known from the initial condition $y(x_0) = y_0$.

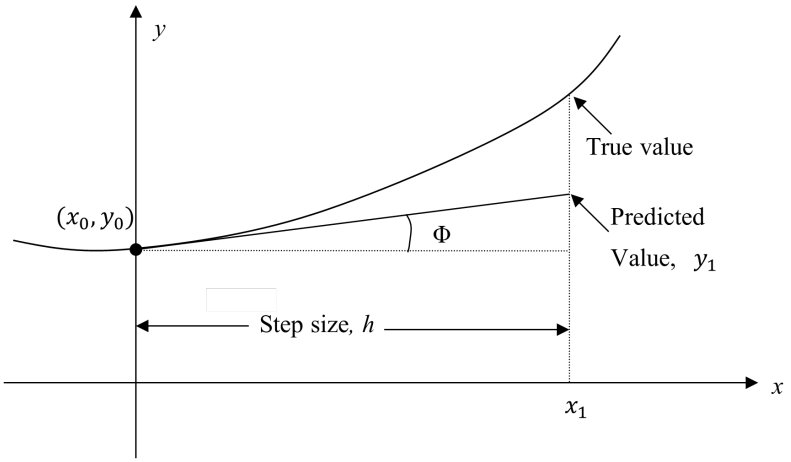



**Figure 1** Graphical interpretation of the first step of Euler's method.

So the slope at $x = x_0$, as shown in Figure 1, is

$$\begin{aligned} \textit{Slope}, \left.\frac{dy}{dx}\right|_{x_0,y_0} &= \frac{\text{Rise}}{\text{Run}} \\ &= \frac{y_1 - y_0}{x_1 - x_0} \\ &= f(x_0, y_0) \end{aligned}$$

So

$$\frac{y_1 - y_0}{x_1 - x_0} = f(x_0, y_0)$$

gives

$$y_1 = y_0 + f(x_0, y_0)(x_1 - x_0)$$

Calling $x_1 - x_0$ the step size $h$, we get

$$y_1 = y_0 + f(x_0, y_0)\, h \qquad (2)$$

One can now use the value of $y_1$ (an approximate value of $y$ at $x = x_1$) to calculate $y_2$, and that would be the predicted value at $x_2$, given by

$$y_2 = y_1 + f(x_1, y_1)\, h$$

$$x_2 = x_1 + h$$

Based on the above equations, if we now know the estimated value of $y$ at $x_i$ as $y_i$, then

$$y_{i+1} = y_i + f(x_i, y_i)\, h \qquad (3)$$

Equation (3) is known as Euler's method and is illustrated graphically in Figure 2. In some textbooks, it is also called the Euler-Cauchy method.

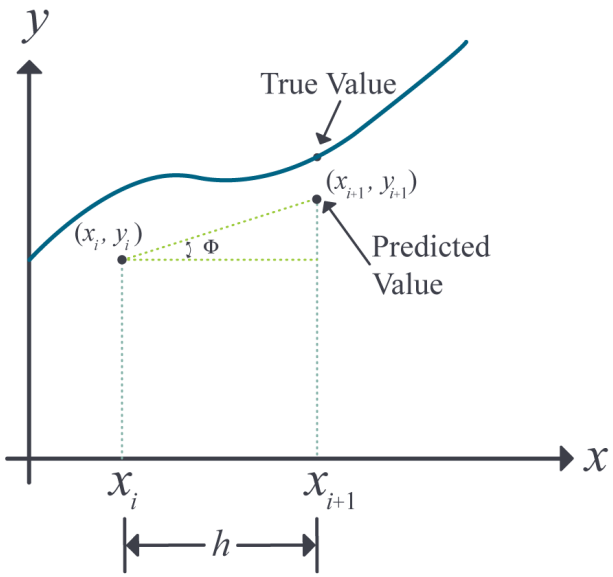



**Figure 2** General graphical interpretation of Euler's method.

## Derivation of Euler's Method from Taylor Series

Euler's method can be derived from the Taylor series as follows.

$$y_{i+1} = y_i + \left.\frac{dy}{dx}\right|_{x_i,y_i}(x_{i+1}-x_i) + \frac{1}{2!}\left.\frac{d^2y}{dx^2}\right|_{x_i,y_i}(x_{i+1}-x_i)^2 + \frac{1}{3!}\left.\frac{d^3y}{dx^3}\right|_{x_i,y_i}(x_{i+1}-x_i)^3 + \ldots \tag{5}$$

Since $\dfrac{dy}{dx} = f(x,y)$

$$y_{i+1} = y_i + f(x_i,y_i)(x_{i+1}-x_i) + \frac{1}{2!}f'(x_i,y_i)(x_{i+1}-x_i)^2 + \frac{1}{3!}f''(x_i,y_i)(x_{i+1}-x_i)^3 + \ldots \tag{6}$$

As you can see, using only the first two terms of the Taylor series in Equation (6)

$$y_{i+1} = y_i + f(x_i,y_i)\,h \qquad (3-repeated)$$

is the Euler's method.

The true error in the Euler's method hence is given by

$$E_t = \frac{f'(x_i, y_i)}{2!}h^2 + \frac{f''(x_i, y_i)}{3!}h^3 + ... \qquad (7)$$

The true error hence is approximately proportional to the square of the step size. That is, as the step size is halved, the true error gets approximately quartered. However, we will observe that as the step size gets halved, the true error instead only gets approximately halved. This reduction in accuracy is because the true error, being proportional to the square of the step size, is the local truncation error, that is, the error from one point to the next. The global truncation error is, however, proportional only to the step size as the error keeps propagating through the $n$ steps taken from the initial value of $x = x_0$ to the final value $x = x_n$.

# Chapter 08.02: Euler's Method for Solving Ordinary Differential Equations

Lesson: Application of Euler's Method

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *Use Euler's Method to solve a first-order ordinary differential equation.*

## Recap of Euler's Method

In the previous lesson, we discussed the theory behind Euler's method of solving an ordinary differential equation of the from $\frac{dy}{dx} = f(x, y)$ where $y(x_0) = y_0$. In this lesson, we illustrate how to apply the algorithm of Euler's method.

### Example 1

A ball at 1200 K is allowed to cool down in air at an ambient temperature of 300 K. Assuming heat is lost only due to radiation, the differential equation for the temperature of the ball is given by

$$\frac{d\theta}{dt} = -2.2067 \times 10^{-12} \left(\theta^4 - 81 \times 10^8\right), \ \theta(0) = 1200 \text{ K} \qquad (E1.1)$$

where $\theta$ is in $K$ and $t$ in seconds. Find the temperature at $t = 480$ seconds using Euler's method. Assume a step size of $h = 240$ seconds. Compare with the exact value.

**Solution**

$$\frac{d\theta}{dt} = -2.2067 \times 10^{-12} \left(\theta^4 - 81 \times 10^8\right)$$

$$f(t, \theta) = -2.2067 \times 10^{-12} \left(\theta^4 - 81 \times 10^8\right)$$

Euler's method formula is

$$\theta_{i+1} = \theta_i + f(t_i, \theta_i) h$$

For $i = 0$, $t_0 = 0$, $\theta_0 = 1200$

$$\begin{aligned}\theta_1 &= \theta_0 + f(t_0, \theta_0)\, h \\ &= 1200 + f(0, 1200) \times 240 \\ &= 1200 + \left(-2.2067 \times 10^{-12} \left(1200^4 - 81 \times 10^8\right)\right) \times 240 \\ &= 1200 + (-4.5579) \times 240 \\ &= 106.09 \ \text{K}\end{aligned}$$

$\theta_1$ is the approximate temperature at

$$t = t_1 = t_0 + h = 0 + 240 = 240 \text{ s}$$

$$\theta_1 = \theta(240) \approx 106.09 \text{ K}$$

For $i = 1$, $t_1 = 240$, $\theta_1 = 106.09$

$$\begin{aligned}\theta_2 &= \theta_1 + f(t_1, \theta_1)\, h \\ &= 106.09 + f(240, 106.09) \times 240 \\ &= 106.09 + \left(-2.2067 \times 10^{-12} \left(106.09^4 - 81 \times 10^8\right)\right) \times 240 \\ &= 106.09 + (0.017595) \times 240 \\ &= 110.32 \ \text{K}\end{aligned}$$

$\theta_2$ is the approximate temperature at

$$t = t_2 = t_1 + h = 240 + 240 = 480 \text{ s}$$

$$\theta_2 = \theta(480) \approx 110.32 \text{ K}$$

The exact solution of the ordinary differential Equation (E1.1) is given without proof as the solution of the nonlinear equation

$$0.92593 \ln \frac{\theta - 300}{\theta + 300} - 1.8519 \tan^{-1}\left(0.333 \times 10^{-2}\theta\right) = -0.22067 \times 10^{-3} t - 2.9282 \qquad (E1.2)$$

The solution to the nonlinear Equation (E1.2) at $t = 480$ s is

$$\theta_{exact} = 647.57 \text{ K}$$

Figure 1 compares the exact solution (647.57 K) with the numerical solution from Euler's method for the step size of $h = 240$.

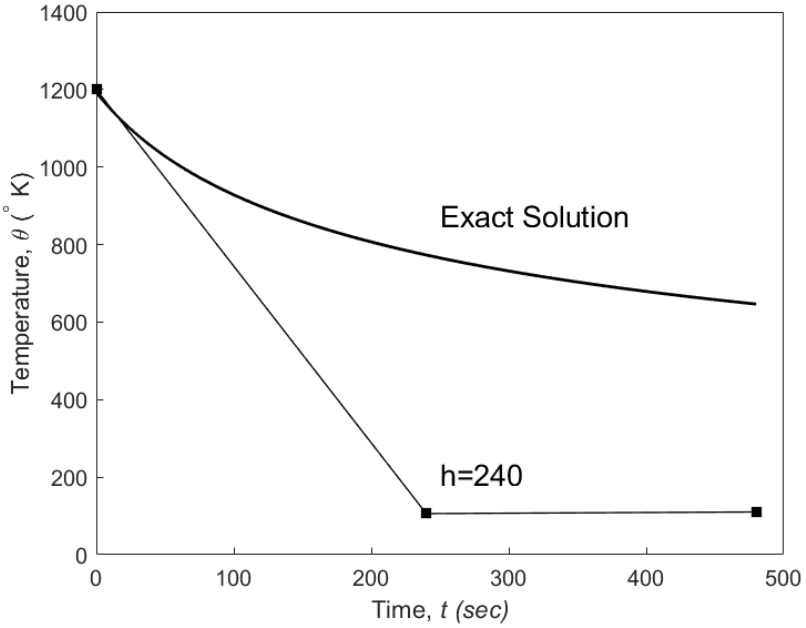



**Figure 1** Comparing the exact and Euler's method solutions.

The problem was solved again using a smaller step size. The results are given below in Table 1.

**Table 1** Temperature at 480 seconds as a function of step size, $h$.

| Step size, $h$ | $\theta\,(480)$ | $E_t$ | $\lvert\epsilon_t\rvert\%$ |
|---|---|---|---|
| 480 | $-987.81$ | 1635.4 | 252.54 |
| 240 | 110.32 | 537.26 | 82.964 |
| 120 | 546.77 | 100.80 | 15.566 |
| 60 | 614.97 | 32.607 | 5.0352 |
| 30 | 632.77 | 14.806 | 2.2864 |

Figure 2 shows how the temperature varies as a function of time for different step sizes.

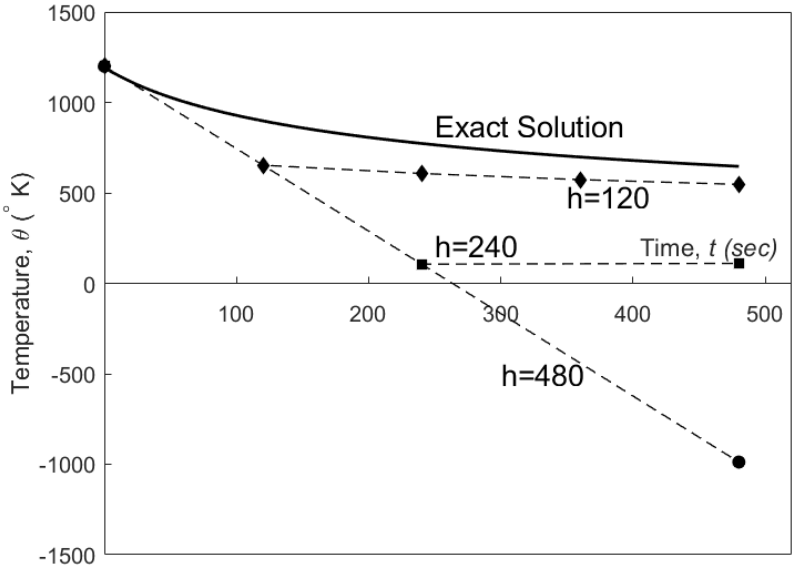



**Figure 2** Comparison of Euler's method with the exact solution for different step sizes.

The values of the calculated temperature at $t = 480$ s as a function of step size are plotted in Figure 3.

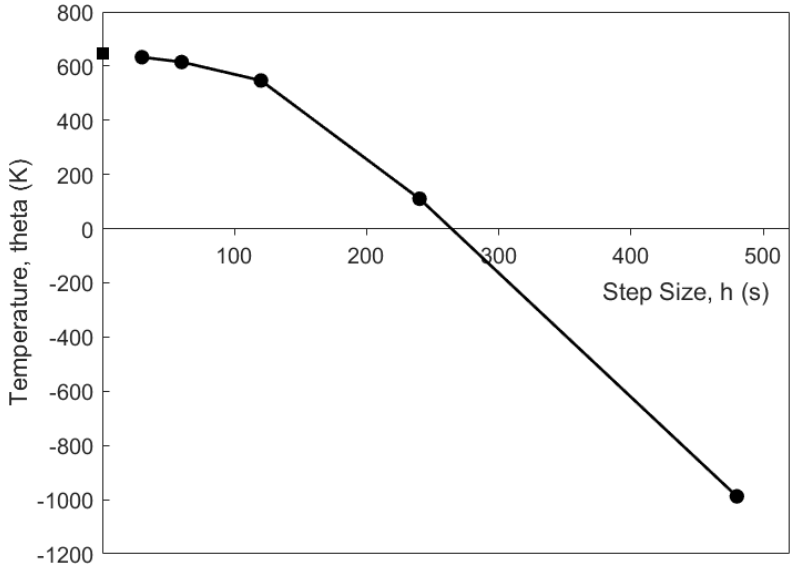



**Figure 3** Effect of step size in Euler's method.

It can be seen that Euler's method results have large errors. In the previous lesson, we found that the true error in the approximation is given by

$$E_t = \frac{f'(x_i, y_i)}{2!}h^2 + \frac{f''(x_i, y_i)}{3!}h^3 + ... \qquad (2)$$

The true error hence is approximately proportional to the square of the step size. That is, as the step size is halved, the true error gets approximately quartered. However, from Table 1, we see that as the step size gets halved, the true error only gets approximately halved (closer to when the step size becomes reasonable). This reduction in accuracy is because the true error, being proportional to the square of the step size, is the local truncation error, that is, the error from one point to the next. The global truncation error is, however, proportional only to the step size as the error keeps propagating through the $n$ steps taken from the initial condition value of $x = x_0$ to $x = x_n$.

# Chapter 08.02: Euler's Method for Solving Ordinary Differential Equations

Lesson: Using Euler's Method to Solve Integrals

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *use Euler's method to find the approximate value of a definite integral.*

## Introduction

In an earlier lesson, when asked to find $\int_a^b f(x)\,dx$, we could rewrite the integral as the solution of an ordinary differential equation (here is where we are using the first part of the fundamental theorem of calculus)

$$\frac{dy}{dx} = f(x),\ y(a) = 0, \qquad (1)$$

where then $y(b)$ (here is where we are using the second part of the fundamental theorem of calculus) gives us the value of the integral $\int_a^b f(x)\,dx$.

### Example 1

Find an approximate value of

$$\int_5^8 6x^3 dx$$

using Euler's method of solving an ordinary differential equation. Use a step size of $h = 1.5$.

**Solution**

Given $\int_5^8 6x^3 dx$, we can rewrite the integral as the solution of an ordinary differential equation

$$\frac{dy}{dx} = 6x^3, y(5) = 0$$

where $y(8)$ gives the value of the integral $\int_5^8 6x^3 dx$.

$$\frac{dy}{dx} = 6x^3 = f(x,y),\ y(5) = 0$$

The Euler's method formula is

$$y_{i+1} = y_i + f(x_i, y_i)\, h$$

Step 1

$$\begin{aligned} i = 0,\ x_0 = 5, y_0 = 0 \\ h = 1.5 \end{aligned}$$

$$\begin{aligned} x_1 &= x_0 + h \\ &= 5 + 1.5 \\ &= 6.5 \end{aligned}$$

$$\begin{aligned} y_1 &= y_0 + f(x_0, y_0)\, h \\ y_1 &= 0 + f(5,0) \times 1.5 \\ &= 0 + f\left(6 \times 5^3\right) \times 1.5 \\ &= 1125 \\ &\approx y(6.5) \end{aligned}$$

Step 2

$$i = 1, x_1 = 6.5, y_1 = 1125$$

$$\begin{aligned} x_2 &= x_1 + h \\ &= 6.5 + 1.5 \\ &= 8 \end{aligned}$$

$$\begin{aligned} y_2 &= y_1 + f(x_1, y_1)\, h \\ &= 1125 + f(6.5, 1125) \times 1.5 \\ &= 1125 + f\left(6 \times 6.5^3\right) \times 1.5 \\ &= 3596.625 \\ &\approx y(8) \end{aligned}$$

Hence

$$\begin{aligned} \int_5^8 6x^3 dx &= y(8) - y(5) \\ &\approx 3596.625 - 0 \\ &= 3596.625 \end{aligned}$$

# Multiple Choice Test

Chapter 08.02 Euler's Method for Solving Ordinary Differential Equations

(1). To solve the ordinary differential equation

$$3\frac{dy}{dx} + 5y^2 = \sin x,\ y(0) = 5$$

by Euler's method, you need to rewrite the equation as

(A) $\frac{dy}{dx} = \sin x - 5y^2,\ y(0) = 5$

(B) $\frac{dy}{dx} = \frac{1}{3}\left(\sin x - 5y^2\right),\ y(0) = 5$

(C) $\frac{dy}{dx} = \frac{1}{3}\left(-\cos x - \frac{5y^3}{3}\right),\ y(0) = 5$

(D) $\frac{dy}{dx} = \frac{1}{3}\sin x,\ y(0) = 5$

(2). Given

$$3\frac{dy}{dx} + 5y^2 = \sin x,\ y(0.3) = 5$$

and using a step size of $h = 0.3$, the value of $y(0.9)$ using Euler's method is most nearly

(A) $-35.318$

(B) $-36.458$

(C) $-658.91$

(D) $-669.05$

(3). Given

$$3\frac{dy}{dx} + \sqrt{y} = e^{0.1x},\ y(0.3) = 5$$

and using a step size of $h = 0.3$, the best estimate of $\frac{dy}{dx}(0.9)$ using Euler's method is most nearly

(A) $-0.37319$

(B) $-0.36288$

(C) $-0.35381$

(D) $-0.34341$

(4). The velocity $(m/s)$ of a body is given as a function of time (seconds) by

$$v(t) = 200 \ln(1+t) - t,\ t \geq 0$$

Using Euler's method with a step size of 5 seconds, the distance traveled in meters by the body from $t = 2$ s to $t = 12$ s is most nearly

(A) 3133.1

(B) 3939.7

(C) 5638.0

(D) 39397

(5). Euler's method can be derived by using the first two terms of the Taylor series of writing the value of $y_{i+1}$, that is, the value of $y$ at $x_{i+1}$, in terms of $y_i$ and all the derivatives of $y$ at $x_i$. If $h = x_{i+1} - x_i$, the explicit expression for $y_{i+1}$ if the first three terms of the Taylor series are chosen to solve the ordinary differential equation

$$2\frac{dy}{dx} + 3y = e^{-5x},\ y(0) = 7$$

would be

(A) $y_{i+1} = y_i + \frac{1}{2}\left(e^{-5x_i} - 3y_i\right)h$

(B) $y_{i+1} = y_i + \frac{1}{2}\left(e^{-5x_i} - 3y_i\right)h - \frac{1}{2}\left(\frac{5}{2}e^{-5x_i}\right)h^2$

(C) $y_{i+1} = y_i + \frac{1}{2}\left(e^{-5x_i} - 3y_i\right)h + \frac{1}{2}\left(-\frac{13}{4}e^{-5x_i} + \frac{9}{4}y_i\right)h^2$

(D) $y_{i+1} = y_i + \frac{1}{2}\left(e^{-5x_i} - 3y_i\right)h - \frac{3}{2}y_i h^2$

(6). A homicide victim is found at 6 : 00 PM in an office building that is maintained at 72°F. When the victim was found, his body temperature was measured to be 85°F. Three hours later, at 9 : 00 PM, his body temperature was recorded at 78°F. Assume the temperature of the body at the time of death is the normal human body temperature of 98.6°F.

The governing equation for the temperature $\theta$ of the body is

$$\frac{d\theta}{dt} = -k(\theta - \theta_a)$$

where

$$\theta = \text{temperature of the body, } ^\circ\text{F}$$

$$\theta_a = \text{ambient temperature, } ^\circ\text{F}$$

$$t = \text{time, hrs}$$

$$k = \text{constant based on thermal properties of the body and air, } 1/\text{s}$$

The estimated time of death most nearly is

(A) 2 : 11 PM

(B) 3 : 13 PM

(C) 4 : 34 PM

(D) 5 : 12 PM

For the complete solution, go to

http://nm.mathforcollege.com/mcquizzes/08ode/quiz_08ode_euler_solution.pdf

# Problem Set

Chapter 08.02: Euler's Method for Solving Ordinary Differential Equations

(1). For the ordinary differential equation

$$2\frac{dy}{dx} + 3xy = e^{-1.5x}, y(0) = 5,$$

use a step size of $h = 2$ and the Euler's method to find $y(6)$.

**Answer:** 329.45

2). For the ordinary differential equation

$$2\frac{dy}{dx} + 3y = e^{-1.5x}, y(0) = 5$$

a) use Euler's method to find $y(2.5)$ using $h = 1.25$,

b) use Euler's method to find $\frac{dy}{dx}(2.5)$ using $h = 1.25$,

c) true value of $y(2.5)$,

d) absolute relative true error for part (a),

e) true value of $\frac{dy}{dx}(2.5)$,

f) absolute relative true error for part (b).

**Answer:** *a*) 3.371
*b*) $-5.0539$
*c*) 0.14699
*d*) 2197.6%
*e*) $-0.20872$
*f*) 2321.4%

(3). Find the approximate value of the integral $\int_3^8 2e^{0.6x}dx$ using Euler's method.

a) Use $h = 2.5$ and compare the value with the exact value.

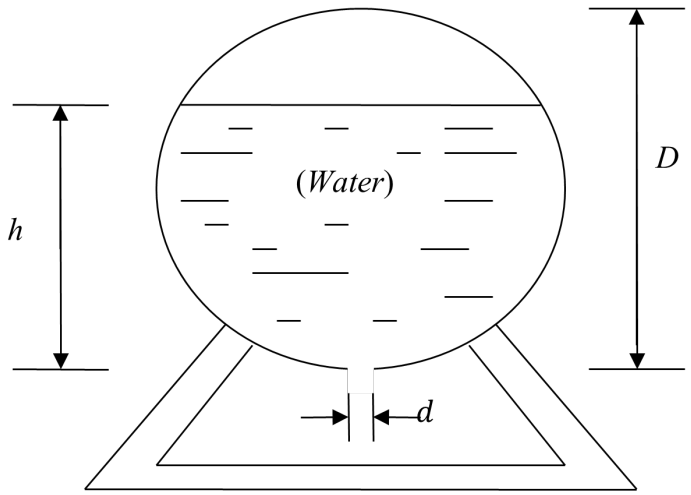
b) Use $h = 1.25$ and compare the value with the exact value.

**Answer:** *a*) 165.81, true value=384.87, true error=219.06
*b*) 258.42, true value=384.87, true error=126.45

(4). From problem (3a), is the value obtained using Euler's method the same as 2-segment LRAM (Left Endpoint Rectangular Approximation), or 2-segment MRAM (Midpoint Rectangular Approximation), or 2-segment RRAM (Right Endpoint Rectangular Approximation) method of integration or the composite trapezoidal rule with 2-segments? Explain.

**Answer:** 165.81, same as LRAM

(5). A water tank with a hole at the top and a circular hole at the bottom is shown in the figure.




Given

$$\text{Diameter of the tank, } D = 6 \text{ m}$$

$$\text{Initial height of the water, } h = 5 \text{ m}$$

$$\text{Diameter of the hole, } d = 4.95 \text{ cm}$$

The flow rate $\dot{Q}$ of water through the bottom hole is given by

$$\dot{Q} = -Av,$$

where

$$A = \text{cross-sectional area of the hole,}$$

$$v = \text{velocity of the water flowing through the hole.}$$

Since

$$v = \sqrt{2gh}$$

where

$$h = \text{height of water from the bottom of the tank,}$$

$$g = \text{acceleration due to gravity.}$$

From this, we get

$$\dot{Q} = -A\sqrt{2gh}$$

Also, since the volume $V$ of the water at height $h$ is given by

$$V = \frac{1}{3}\pi h^2 \left(\frac{3D}{2} - h\right)$$

where

$$\frac{dV}{dt} = \left(\pi hD - \pi h^2\right)\frac{dh}{dt}$$

Since

$$\dot{Q} = \frac{dV}{dt}$$

$$\left(\pi hD - \pi h^2\right)\frac{dh}{dt} = -A\sqrt{2gh}$$

$$\left(\pi hD - \pi h^2\right)\frac{dh}{dt} = -\frac{\pi d^2}{4}\sqrt{2gh}$$

$$\frac{dh}{dt} = -\frac{d^2\sqrt{2gh}}{4\left(hD - h^2\right)}$$

a) Use Euler's method with a step size of 10 minutes to find the height of water at $t = 50$ minutes.

b) Based on the result from part (a), estimate the time at which the height of the water is 4 m.

c) Compare the results of part (a) and part (b) with the exact values.

**Answer:** *a*) 2.7737 m
*b*) 15.965 minutes
*c*) part (a) exact 2.9371 m, $|\epsilon_t| = 5.5623\%$
part (b) exact 19.415 minutes, $|\epsilon_t| = 21.802\%$

(6). A spherical ball is taken out of a furnace at 2500 K. It loses heat due to radiation and convection. Given the following,

Radius of ball, $r = 1.0$ cm

Density of the ball, $\rho = 3000$ kg/m$^3$

Specific heat, $C = 1000$ J/(kg $\cdot$ K)

Emittance, $\in = 0.5$

Stefan-Boltzmann Constant, $\sigma = 5.67 \times 10^{-8}$ J/(s $\cdot$ m$^2$ $\cdot$ K$^4$)

Convective cooling coefficient, $h = 500$ J/(s $\cdot$ m$^2$ $\cdot$ K)

Initial temperature of the ball, $\theta(0) = 2500$ K,

Ambient temperature, $\theta_a = 300$ K,

the ordinary differential equation that governs the temperature of the ball is given by

$$\frac{d\theta}{dt} = -2.8349 \times 10^{-12}(\theta^4 - 81 \times 10^8) - 0.05(\theta - 300).$$

Hint: The above formula was calculated by using the knowledge that

Rate of heat lost due to radiation $= A \in \sigma(\theta^4 - \theta_a^4)$

Rate of heat lost due to convection $= hA(\theta - \theta_a)$

Heat stored $= mc\theta$

where

$A =$ surface area of the ball,

$m =$ mass of the ball.

a) What are following at $t = 0$ s?

i) temperature,

ii) rate of change of temperature,

iii) rate of heat loss due to radiation,

iv) rate of heat loss due to convection,

v) rate of heat stored.

b) Find the following at $t = 10$ s using Euler's method and a step size of $h = 2.5$.

i) temperature,

ii) rate of change of temperature,

iii) rate of heat loss due to radiation,

iv) rate of heat loss due to convection,

v) rate of heat stored.

c) Can either convection or radiation be neglected if we are interested in finding the temperature profile between 0 and 10 s? Base your answer on quantitative reasoning.

d) Find the time when the temperature will be 2000 K using Euler's method with a step size of $h = 2.5$.

**Answer:** *a*) 2500 K, $-220.72$ K/s, 1391.3 W, 1382.3 W, $-2773.6$ W
*b*) 1252.3 K, $-54.568$ K/s, 87.344 W, 598.38 W, $-685.72$ W
*c*) Neither can be ignored, as the rate of heat lost due to convection and radiation are of the same order.
*d*) 2.2654 s using linear interpolation

# Chapter 08.03: Runge-Kutta 2nd-Order Method for Solving Ordinary Differential Equations

Lesson: Theory of Runge-Kutta 2nd-Order Method

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *list the formulas of the Runge-Kutta $2^{nd}$ order method for ordinary differential equations and know how to use them.*

## What is the Runge-Kutta 2nd order method?

The Runge-Kutta 2nd order method is a numerical technique used to solve an ordinary differential equation of the form

$$\frac{dy}{dx} = f(x,y),\ y(x_0) = y_0 \qquad (1)$$

Only first-order ordinary differential equations of the form of Equation (1) can be solved by using the Runge-Kutta 2nd order method. In other sections, we discuss how the Euler and Runge-Kutta methods are used to solve higher-order ordinary or coupled (simultaneous) ordinary differential equations.

How does one write a first-order ordinary differential equation in the above form?

## Runge-Kutta $2^{nd}$ order method

Euler's method is given by

$$y_{i+1} = y_i + f(x_i, y_i)\, h \qquad (2)$$

where

$$y(x_0) = y_0$$

$$h = x_{i+1} - x_i$$

To understand the Runge-Kutta 2nd order method, we need to first derive Euler's method from the Taylor series.

$$y_{i+1} = y_i + \left.\frac{dy}{dx}\right|_{x_i,y_i}(x_{i+1} - x_i) + \frac{1}{2!}\left.\frac{d^2y}{dx^2}\right|_{x_i,y_i}(x_{i+1} - x_i)^2 + \frac{1}{3!}\left.\frac{d^3y}{dx^3}\right|_{x_i,y_i}(x_{i+1} - x_i)^3 + ...$$

Since $\frac{dy}{dx} = f(x, y)$

$$y_{i+1} = y_i + f(x_i, y_i)(x_{i+1} - x_i) + \frac{1}{2!}f'(x_i, y_i)(x_{i+1} - x_i)^2 + \frac{1}{3!}f''(x_i, y_i)(x_{i+1} - x_i)^3 + ... \quad (3)$$

As you can see, the first two terms of the Taylor series in Equation (3)

$$y_{i+1} = y_i + f(x_i, y_i)\, h$$

represent Euler's method and hence can be considered to be a Runge-Kutta 1st order method.

The true error in the approximation is given by

$$E_t = \frac{f'(x_i, y_i)}{2!}h^2 + \frac{f''(x_i, y_i)}{3!}h^3 + ... \quad (4)$$

So what would a 2nd order method formula look like? It would include one more term of the Taylor series as follows.

$$y_{i+1} = y_i + f(x_i, y_i)\, h + \frac{1}{2!}f'(x_i, y_i)\, h^2 \quad (5)$$

Let us take a generic example of a first-order ordinary differential equation

$$\frac{dy}{dx} = e^{-2x} - 3y, y(0) = 5$$

$$f(x, y) = e^{-2x} - 3y$$

Now since $y$ is a function of $x$,

$$\begin{aligned} f'(x, y) &= \frac{\partial f(x, y)}{\partial x} + \frac{\partial f(x, y)}{\partial y}\frac{dy}{dx} \quad (6) \\ &= \frac{\partial}{\partial x}\left(e^{-2x} - 3y\right) + \frac{\partial}{\partial y}\left[\left(e^{-2x} - 3y\right)\right]\left(e^{-2x} - 3y\right) \\ &= -2e^{-2x} + (-3)\left(e^{-2x} - 3y\right) \\ &= -5e^{-2x} + 9y \end{aligned}$$

The 2nd order formula for the above example would be

$$\begin{aligned} y_{i+1} &= y_i + f(x_i, y_i)\, h + \frac{1}{2!} f'(x_i, y_i)\, h^2 \\ &= y_i + \left(e^{-2x_i} - 3y_i\right) h + \frac{1}{2!}\left(-5e^{-2x_i} + 9y_i\right) h^2 \end{aligned}$$

However, we already recognize the pitfall of having to find $f'(x, y)$ in the above method. What Runge and Kutta did was write the 2nd order method as

$$y_{i+1} = y_i + (a_1 k_1 + a_2 k_2)\, h \qquad (7)$$

where

$$k_1 = f(x_i, y_i) \qquad (8a)$$

$$k_2 = f(x_i + p_1 h, y_i + q_{11} k_1 h) \qquad (8b)$$

This form allows one to take advantage of the 2nd order method without having to calculate $f'(x, y)$.

So how do we find the unknowns $a_1$, $a_2$, $p_1$, and $q_{11}$? Without proof, equating Equation (5) and (7), gives three equations.

$$a_1 + a_2 = 1 \qquad (9a)$$

$$a_2 p_1 = \frac{1}{2} \qquad (9b)$$

$$a_2 q_{11} = \frac{1}{2} \qquad (9c)$$

Since we have 3 equations and 4 unknowns, we can assume the value of one of the unknowns. The other three are then determined from the three equations. Generally, the value of $a_2$ is chosen to evaluate the other three constants. The three values used for $a_2$ are $\frac{1}{2}$, 1 and $\frac{2}{3}$, and are known as Heun's Method, the midpoint method, and Ralston's method, respectively.

## Heun's Method

Here $a_2 = \frac{1}{2}$ is chosen, and from Equations (9a)-(9c),

$$a_1 = \frac{1}{2}$$

$$p_1 = 1$$

$$q_{11} = 1$$

resulting in

$$y_{i+1} = y_i + \left(\frac{1}{2}k_1 + \frac{1}{2}k_2\right) h \qquad (10)$$

where

$$k_1 = f\left(x_i, y_i\right) \qquad (11a)$$

$$k_2 = f\left(x_i + h, y_i + k_1 h\right) \qquad (11b)$$

This method is graphically explained in Figure 1.

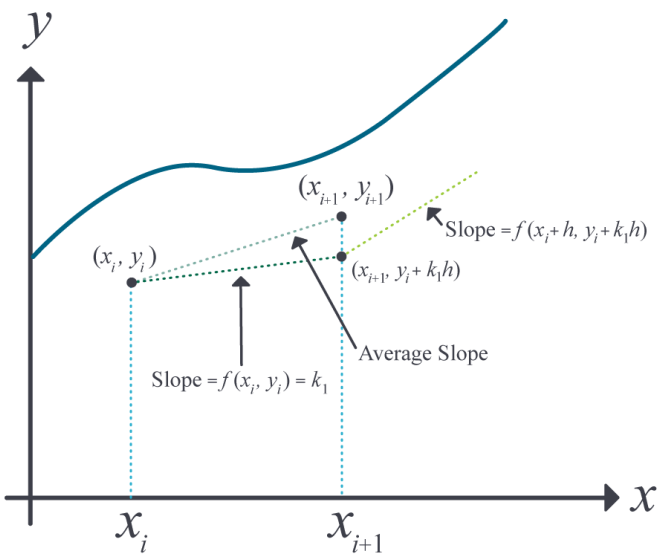



**Figure 1** Runge-Kutta 2nd order method (Heun's method).

## Midpoint Method

Here $a_2 = 1$ is chosen, and from Equations (9a)-(9c),

$$a_1 = 0$$

$$p_1 = \frac{1}{2}$$

$$q_{11} = \frac{1}{2}$$

resulting in

$$y_{i+1} = y_i + k_2 h \qquad (12)$$

where

$$k_1 = f\left(x_i, y_i\right) \qquad (13a)$$

$$k_2 = f\left(x_i + \frac{1}{2}h, y_i + \frac{1}{2}k_1 h\right) \qquad (13b)$$

## Ralston's Method

Here $a_2 = \frac{2}{3}$ is chosen, and from Equations (9a)-(9c),

$$a_1 = \frac{1}{3}$$

$$p_1 = \frac{3}{4}$$

$$q_{11} = \frac{3}{4}$$

resulting in

$$y_{i+1} = y_i + \left(\frac{1}{3}k_1 + \frac{2}{3}k_2\right) h \qquad (14)$$

where

$$k_1 = f\left(x_i, y_i\right) \qquad (15a)$$

$$k_2 = f\left(x_i + \frac{3}{4}h, y_i + \frac{3}{4}k_1 h\right) \qquad (15b)$$

# Chapter 08.03: Runge-Kutta 2nd-Order Method for Solving Ordinary Differential Equations

Lesson: Application of Runge-Kutta 2nd-Order Method

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *Use Runge 2nd order method to solve first-order ordinary differential equations.*

## Recap

In the previous lesson, we discussed the theory behind the Runge-Kutta 2nd order method of solving an ordinary differential equation of the form $dy/dx = f(x, y)$ where $y(x_0) = y_0$. In this lesson, we take an example of how to apply the algorithm of the Runge-Kutta 2nd order method.

### Example 1

A ball at 1200 K is allowed to cool down in air at an ambient temperature of 300 K. Assuming heat is lost only due to radiation, the differential equation for the temperature of the ball is given by

$$\frac{d\theta}{dt} = -2.2067 \times 10^{-12} \left(\theta^4 - 81 \times 10^8\right)$$

where $\theta$ is in K and $t$ in seconds. Find the temperature at $t = 480$ seconds using Runge-Kutta 2nd order method. Assume a step size of $h = 240$ seconds.

**Solution**

$$\frac{d\theta}{dt} = -2.2067 \times 10^{-12} \left(\theta^4 - 81 \times 10^8\right)$$

$$f\left(t, \theta\right) = -2.2067 \times 10^{-12} \left(\theta^4 - 81 \times 10^8\right)$$

As per Heun's method given in the previous lesson for an ordinary differential equation,

$$\frac{d\theta}{dt} = f(t, \theta)$$

Heun's method formula is given by

$$\theta_{i+1} = \theta_i + \left(\frac{1}{2}k_1 + \frac{1}{2}k_2\right) h$$

$$k_1 = f\left(t_i, \theta_i\right)$$

$$k_2 = f\left(t_i + h, \theta_i + k_1 h\right)$$

For Step 1,

$$i = 0,\ t_0 = 0,\ \theta_0 = \theta(0) = 1200 \text{ K}$$

$$\begin{aligned} t_1 &= t_0 + h \\ &= 0 + 240 \\ &= 240 \text{ s} \end{aligned}$$

$$\begin{aligned} k_1 &= f\left(t_0, \theta_o\right) \\ &= f\left(0, 1200\right) \\ &= -2.2067 \times 10^{-12}\left(1200^4 - 81 \times 10^8\right) \\ &= -4.5579 \end{aligned}$$

$$\begin{aligned} k_2 &= f\left(t_0 + h, \theta_0 + k_1 h\right) \\ &= f\left(0 + 240, 1200 + (-4.5579)\, 240\right) \\ &= f\left(240, 106.09\right) \\ &= -2.2067 \times 10^{-12}\left(106.09^4 - 81 \times 10^8\right) \\ &= 0.017595 \end{aligned}$$

$$\begin{aligned} \theta_1 &= \theta_0 + \left(\frac{1}{2}k_1 + \frac{1}{2}k_2\right) h \\ &= 1200 + \left(\frac{1}{2}\left(-4.5579\right) + \frac{1}{2}\left(0.017595\right)\right) 240 \\ &= 1200 + \left(-2.2702\right) 240 \\ &= 655.16 \text{ K} \\ &\approx \theta(240) \end{aligned}$$

For Step 2

$$i = 1, t_1 = 240 \text{ s}, \theta_1 = 655.16 \text{ K}$$

$$
\begin{aligned}
&= t_1 + h \\
&= 240 + 240 \\
&= 480 \text{ s}
\end{aligned}
$$

$$
\begin{aligned}
k_1 &= f(t_1, \theta_1) \\
&= f(240, 655.16) \\
&= -2.2067 \times 10^{-12} \left(655.16^4 - 81 \times 10^8\right) \\
&= -0.38869
\end{aligned}
$$

$$
\begin{aligned}
k_2 &= f(t_1 + h, \theta_1 + k_1 h) \\
&= f(240 + 240, 655.16 + (-0.38869)\, 240) \\
&= f(480, 561.87) \\
&= -2.2067 \times 10^{-12} \left(561.87^4 - 81 \times 10^8\right) \\
&= -0.20206
\end{aligned}
$$

$$
\begin{aligned}
\theta_2 &= \theta_1 + \left(\frac{1}{2}k_1 + \frac{1}{2}k_2\right) h \\
&= 655.16 + \left(\frac{1}{2}(-0.38869) + \frac{1}{2}(-0.20206)\right) 240 \\
&= 655.16 + (-0.29538)\, 240 \\
&= 584.27 \text{ K} \\
&\approx \theta(480)
\end{aligned}
$$

The results from Heun's method are compared with the exact results in Figure 1.

The exact solution of the ordinary differential equation is given by the solution of a nonlinear equation as

$$0.92593 \ln \frac{\theta - 300}{\theta + 300} - 1.8519 \tan^{-1}(0.0033333\theta) = -0.22067 \times 10^{-3} t - 2.9282$$

The solution to this nonlinear equation at $t = 480$ s is

$$\theta(480) = 647.57 \text{ K}$$

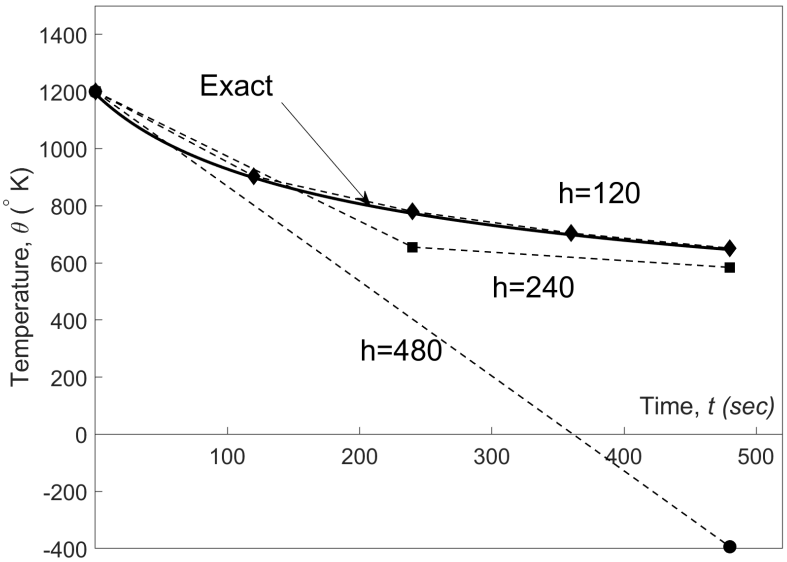



**Figure 1** Heun's method results for different step sizes.

Using a smaller step size would increase the accuracy of the result, as given in Table 1 and Figure 2 below.

**Table 1** Effect of step size for Heun's method

| *Step size*, $h$ | $\theta\,(480)$ | $E_t$ | $\lvert\epsilon_t\rvert\,\%$ |
|---|---|---|---|
| 480 | $-393.87$ | 1041.4 | 160.82 |
| 240 | 584.27 | 63.304 | 9.7756 |
| 120 | 651.35 | $-3.7762$ | 0.58313 |
| 60 | 649.91 | $-2.3406$ | 0.36145 |
| 30 | 648.21 | $-0.63219$ | 0.097625 |

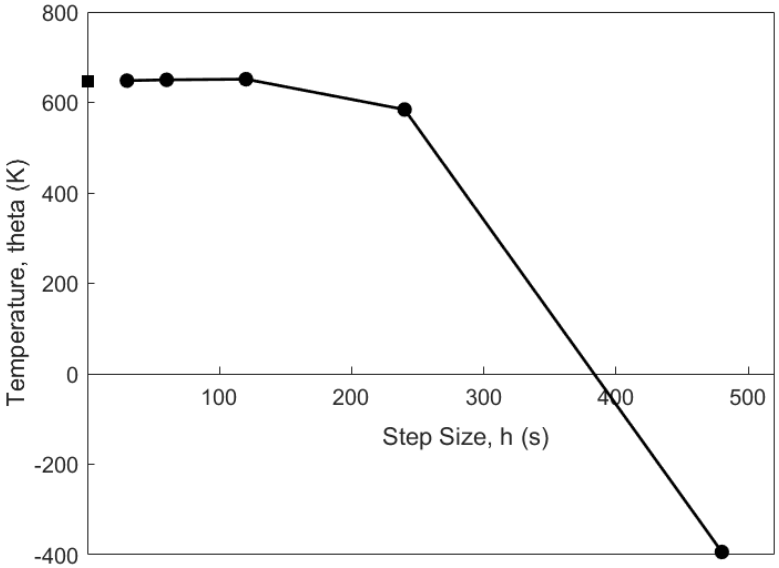



**Figure 2** Effect of step size in Heun's method.

In Table 2, Euler's method and the Runge-Kutta 2nd order method results are shown as a function of step size,

**Table 2** Comparison of Euler and the Runge-Kutta methods

| *Step size*, $h$ | $\theta(480)$ | | | |
|---|---|---|---|---|
| | *Euler* | *Heun* | *Midpoint* | *Ralston* |
| 480 | −987.84 | −393.87 | 1208.4 | 449.78 |
| 240 | 110.32 | 584.27 | 976.87 | 690.01 |
| 120 | 546.77 | 651.35 | 690.20 | 667.71 |
| 60 | 614.97 | 649.91 | 654.85 | 652.25 |
| 30 | 632.77 | 648.21 | 649.02 | 648.61 |

while in Figure 3, the comparison is shown over time.

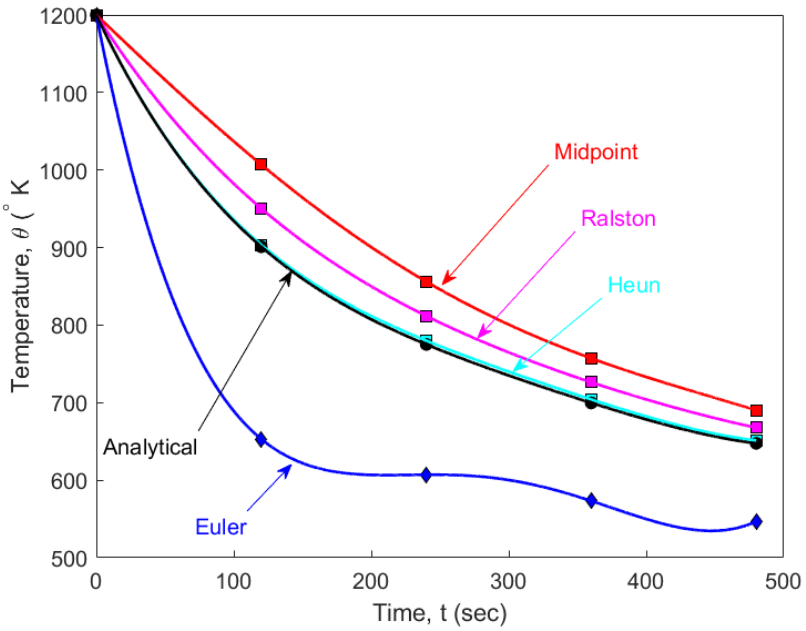



**Figure 3** Comparison of Euler and Runge Kutta methods with exact results over time.

## How do these three methods compare with results obtained if we found $\mathbf{f'(x,y)}$ directly?

Of course, we know that since we are including the first three terms in the series, if the solution is a polynomial of order two or less (that is, quadratic, linear, or constant), any of the three methods are exact. But for any other case, the results are going to be different.

Let us take the example of showing the difference in results.

$$\frac{dy}{dx} = e^{-2x} - 3y, y\left(0\right) = 5.$$

If we directly find $f'\left(x,y\right)$, the first three terms of the Taylor series gives

$$y_{i+1} = y_i + f\left(x_i, y_i\right)h + \frac{1}{2!}f'\left(x_i, y_i\right)h^2$$

where

$$f\left(x,y\right) = e^{-2x} - 3y$$

$$f'\left(x,y\right) = -5e^{-2x} + 9y$$

For a step size of $h = 0.2$, using Heun's method, we find

$$y\left(0.6\right) = 1.0930$$

The exact solution

$$y\left(x\right) = e^{-2x} + 4e^{-3x}$$

gives

$$\begin{aligned} y\left(0.6\right) &= e^{-2(0.6)} + 4e^{-3(0.6)} \\ &= 0.96239 \end{aligned}$$

Then the absolute relative true error is

$$\begin{aligned} |\epsilon_t| &= \left| \frac{0.96239 - 1.0930}{0.96239} \right| \times 100 \\ &= 13.571\% \end{aligned}$$

For the same problem, the results from Euler's method and the three Runge-Kutta methods are given in Table 3.

**Table 3** Comparison of Euler's and Runge-Kutta 2nd order methods

| | *Exact* | *Euler* | *Direct 2nd* | *Heun* | *Midpoint* | *Ralston* |
|---|---|---|---|---|---|---|
| $y(0.6)$ | 0.96239 | 0.4955 | 1.0930 | 1.1012 | 1.0974 | 1.0994 |
| $\|\epsilon_t\|$ % | | 48.514 | 13.571 | 14.423 | 14.029 | 14.236 |

# Chapter 08.03: Runge-Kutta 2nd-Order Method for Solving Ordinary Differential Equations

Lesson: Derivation of Runge-Kutta 2nd-Order Method

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *derive the Runge-Kutta 2nd order method for the first-order ordinary differential equation.*

## How do we get the 2nd order Runge-Kutta method equations?

We wrote the 2nd order Runge-Kutta equations without proof to solve

$$\frac{dy}{dx} = f(x,y),\ y(0) = y_0 \qquad (1)$$

as

$$y_{i+1} = y_i + (a_1 k_1 + a_2 k_2) h \qquad (2)$$

where

$$k_1 = f(x_i, y_i) \qquad (3a)$$

$$k_2 = f(x_i + p_1 h, y_i + q_{11} k_1 h) \qquad (3b)$$

and

$$a_1 + a_2 = 1 \qquad (4a)$$

$$a_2 p_2 = \frac{1}{2} \qquad (4b)$$

$$a_2 q_{11} = \frac{1}{2} \qquad (4c)$$

The advantage of using 2nd order Runge-Kutta method equations is based on not having to find the derivative of $f(x,y)$ symbolically in the ordinary differential equation

So how do we get the above three Equations ((A.4))4a)-(4c)? The question is answered below.

Writing out the first three terms of the Taylor series are

$$y_{i+1} = y_i + \left.\frac{dy}{dx}\right|_{x_i y_i} h + \left.\frac{1}{2!}\frac{d^2y}{dx^2}\right|_{x_i y_i} h^2 + O\left(h^3\right) \tag{5}$$

where

$$h = x_{i+1} - x_i$$

Since

$$\frac{dy}{dx} = f(x,y)$$

we can rewrite the Taylor series as

$$y_{i+1} = y_i + f(x_i, y_i)\, h + \frac{1}{2!} f'(x_i, y_i)\, h^2 + O\left(h^3\right) \tag{6}$$

Now

$$f'(x,y) = \frac{\partial f(x,y)}{\partial x} + \frac{\partial f(x,y)}{\partial y}\frac{dy}{dx}. \tag{7}$$

Hence

$$\begin{aligned} y_{i+1} &= y_i + f(x_i, y_i)\, h + \frac{1}{2!}\left( \left.\frac{\partial f}{\partial x}\right|_{x_i,y_i} + \left.\frac{\partial f}{\partial y}\right|_{x_i,y_i} \times \left.\frac{dy}{dx}\right|_{x_i,y_i} \right) h^2 + O\left(h^3\right) \\ &= y_i + f(x_i, y_i)\, h + \frac{1}{2} \left.\frac{\partial f}{\partial x}\right|_{x_i,y_i} h^2 + \frac{1}{2} \left.\frac{\partial f}{\partial y}\right|_{x_i,y_i} f(x_i, y_i)\, h^2 + O\left(h^3\right) \end{aligned} \tag{8}$$

Now the term used in the Runge-Kutta 2nd order method for $k_2$ can be written as a Taylor series of two variables with the first three terms as

$$\begin{aligned} k_2 &= f(x_i + p_1 h, y_i + q_{11} k_1 h) \\ &= f(x_i, y_i) + p_1 h \left.\frac{\partial f}{\partial x}\right|_{x_i,y_i} + q_{11} k_1 h \left.\frac{\partial f}{\partial y}\right|_{x_i,y_i} + O\left(h^2\right) \end{aligned} \tag{9}$$

Hence

$$
\begin{aligned}
y_{i+1} &= y_i + (a_1 k_1 + a_2 k_2)\, h \\
&= y_i + \left( a_1 f(x_i, y_i) + a_2 \left\{ f(x_i, y_i) + p_1 h \left.\frac{\partial f}{\partial x}\right|_{x_i, y_i} + q_{11} k_1 h \left.\frac{\partial f}{\partial y}\right|_{x_i, y_i} + O\left(h^2\right) \right\} \right) h \\
&= y_i + (a_1 + a_2)\, h f(x_i, y_i) + a_2 p_1 h^2 \left.\frac{\partial f}{\partial x}\right|_{x_i, y_i} + a_2 q_{11} f(x_i, y_i)\, h^2 \left.\frac{\partial f}{\partial y}\right|_{x_i, y_i} + O\left(h^3\right)
\end{aligned} \tag{10}
$$

Equating the terms in Equation (8) and Equation (10), we get

$$a_1 + a_2 = 1$$

$$a_2 p_1 = \frac{1}{2}$$

$$a_2 q_{11} = \frac{1}{2}$$

# Multiple Choice Test

## Chapter 08.03 Runge Kutta Second-Order Method for Solving Ordinary Differential Equations

(1). To solve the ordinary differential equation

$$3\frac{dy}{dx} + xy^2 = \sin x,\ y(0) = 5$$

by the Runge-Kutta 2nd order method, you need to rewrite the equation as

(A) $\frac{dy}{dx} = \sin x - xy^2,\ y(0) = 5$

(B) $\frac{dy}{dx} = \frac{1}{3}\left(\sin x - xy^2\right),\ y(0) = 5$

(C) $\frac{dy}{dx} = \frac{1}{3}\left(-\cos x - \frac{xy^3}{3}\right),\ y(0) = 5$

(D) $\frac{dy}{dx} = \frac{1}{3}\sin x,\ y(0) = 5$

(2). Given

$$3\frac{dy}{dx} + 5y^2 = \sin x,\ y(0.3) = 5$$

and using a step size of $h = 0.3$, the value of $y(0.9)$ using the Runge-Kutta 2nd order Heun's method is most nearly

(A) $-4297.4$

(B) $-4936.7$

(C) $-0.21336 \times 10^{14}$

(D) $-0.24489 \times 10^{14}$

(3). Given

$$3\frac{dy}{dx} + 5\sqrt{y} = e^{0.1x},\ y(0.3) = 5$$

and using a step size of $h = 0.3$, the best estimate of $\frac{dy}{dx}(0.9)$ using the Runge-Kutta 2nd order midpoint method most nearly is

(A) $-2.2473$

(B) $-2.2543$

(C) $-2.6188$

(D) $-3.2045$

(4). The velocity (m/s) of a body is given as a function of time (seconds) by

$$v(t) = 200 \ln(1+t) - t, \; t \geq 0$$

Using the Runge-Kutta 2nd order Ralston method with a step size of 5 seconds, the distance in meters traveled by the body from $t = 2$ to $t = 12$ seconds is estimated most nearly as

(A) 3904.9

(B) 3939.7

(C) 6556.3

(D) 39397

(5). The Runge-Kutta 2nd order method can be derived by using the first three terms of the Taylor series of writing the value of $y_{i+1}$ (that is the value of $y$at $x_{i+1}$) in terms of $y_i$ (that is the value of $y$ at $x_i$) and all the derivatives of $y$at $x_i$. If $h = x_{i+1} - x_i$, the explicit expression for $y_{i+1}$ if the first three terms of the Taylor series are chosen for solving the ordinary differential equation

$$\frac{dy}{dx} + 5y = 3e^{-2x}, y(0) = 7$$

would be

(A) $y_{i+1} = y_i + \left(3e^{-2x_i} - 5y_i\right) h + 5\dfrac{h^2}{2}$

(B) $y_{i+1} = y_i + \left(3e^{-2x_i} - 5y_i\right) h + \left(-21e^{-2x_i} + 25y_i\right) \dfrac{h^2}{2}$

(C) $y_{i+1} = y_i + \left(3e^{-2x_i} - 5y_i\right) h + \left(-6e^{-2x_i}\right) \dfrac{h^2}{2}$

(D) $y_{i+1} = y_i + \left(3e^{-2x_i} - 5y_i\right) h + \left(-6e^{-2x_i} + 5\right) \dfrac{h^2}{2}$

(6). A spherical ball is taken out of a furnace at 1200 K and is allowed to cool in air. You are given the following

$$\text{radius of ball} = 2 \text{ cm}$$

$$\text{specific heat of ball} = 420 \text{ J/kg} \cdot \text{K}$$

$$\text{density of ball} = 7800 \text{ kg/m}^3$$

$$\text{convection coefficient} = 350 \text{ J/s} \cdot \text{m}^2 \cdot \text{K}$$

$$\text{ambient temperature} = 300 \text{ K}$$

The ordinary differential equation that is given for the temperature $\theta$ of the ball is

$$\frac{d\theta}{dt} = -2.20673 \times 10^{-13} \left(\theta^4 - 81 \times 10^8\right)$$

if only radiation is accounted for. If convection is accounted for in addition to radiation, the ordinary differential equation is

(A) $\frac{d\theta}{dt} = -2.20673 \times 10^{-13} \left(\theta^4 - 81 \times 10^8\right) - 1.6026 \times 10^{-2} \left(\theta - 300\right), \theta(0) = 1200 \text{ K}$

(B) $\frac{d\theta}{dt} = -2.20673 \times 10^{-13} \left(\theta^4 - 81 \times 10^8\right) - 4.3982 \times 10^{-2} \left(\theta - 300\right), \theta(0) = 1200 \text{ K}$

(C) $\frac{d\theta}{dt} = -1.6026 \times 10^{-2} \left(\theta - 300\right), \theta(0) = 1200 \text{ K}$

(D) $\frac{d\theta}{dt} = -4.3982 \times 10^{-2} \left(\theta - 300\right), \theta(0) = 1200 \text{ K}$

For complete solution, go to

http://nm.mathforcollege.com/mcquizzes/08ode/quiz_08ode_runge2nd_solution.pdf

# Problem Set

## Chapter 08.03: Runge-Kutta 2nd Order Method for Solving Ordinary Differential Equations

(1). For the ordinary differential equation

$$2\frac{dy}{dx} + 3xy = e^{-1.5x}, y(0) = 5$$

use a step size of $h = 2$ and Runge-Kutta 2nd order Ralston's method to find $y(6)$.

**Answer:** $y(6) \approx -23684$

(2). For the ordinary differential equation

$$2\frac{dy}{dx} + 3y = e^{-1.5x}, y(0) = 5$$

a) use Runge-Kutta 2nd order midpoint method to find $y(2.5)$ using $h = 1.25$,

b) use Runge-Kutta 2nd order midpoint method to find $\frac{dy}{dx}(2.5)$ using $h = 1.25$,

c) true value of $y(2.5)$,

d) absolute relative true error for part (a),

e) true error of $\frac{dy}{dx}(2.5)$,

f) absolute relative true error for part (b).

**Answer:** *a*) $y(2.5) \approx 3.5433$
*b*) $-5.3031$
*c*) 0.14699
*d*) 2310.6%
*e*) $-0.20872$
*f*) 2440.8%

(3). Find the approximate value of the integral $\int_3^8 2e^{0.6x}dx$ using Runge-Kutta 2nd order Heun's method.

a) Use $h = 2.5$ and compare the value with the exact value.

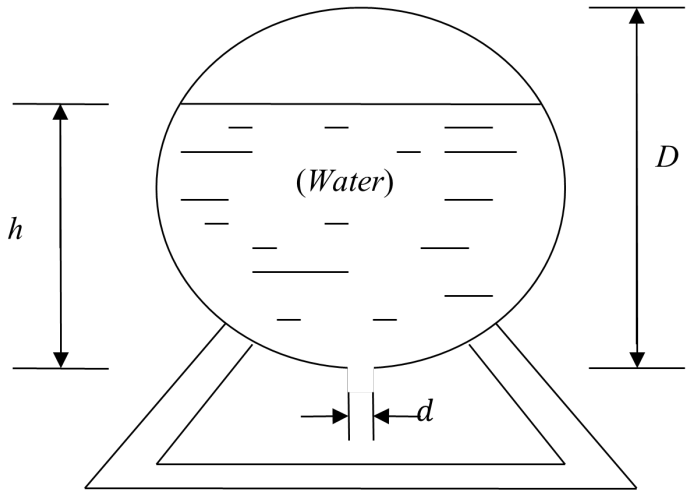
b) Use $h = 1.25$ and compare the value with the exact value.

**Answer:** *a*) 454.46, 18.083%
*b*) 402.74, 4.6441%

(4). From problem (3a), is the value obtained using Heun's Runge-Kutta 2nd order method same as 2-segment LRAM (Left Endpoint Rectangular Approximation), or 2-segment MRAM (Midpoint Rectangular Approximation), or 2-segment RRAM (Right Endpoint Rectangular Approximation) method of integration or composite trapezoidal rule with 2 segments? Explain.

**Answer:** 454.46, MRAM

(5). A water tank with a hole at the top and a circular hole at the bottom is shown in the figure.




Given

$$\text{Diameter of the tank, } D = 6 \text{ m}$$

$$\text{Initial height of the water, } h = 5 \text{ m}$$

$$\text{Diameter of the hole, } d = 4.95 \text{ cm}$$

The flow rate $\dot{Q}$ of water through the bottom hole, is given by

$$\dot{Q} = -Av,$$

where

$$A = \text{cross-sectional area of the hole,}$$

$$v = \text{velocity of the water flowing through the hole.}$$

Since

$$v = \sqrt{2gh}$$

where

$$h = \text{height of the water from the bottom of the tank,}$$

$$g = \text{acceleration due to gravity.}$$

From this, we get

$$\dot{Q} = -A\sqrt{2gh}$$

Also, since the volume $V$ of the water at height $h$ is given by

$$V = \frac{1}{3}\pi h^2 \left(\frac{3D}{2} - h\right)$$

where

$$\frac{dV}{dt} = \left(\pi hD - \pi h^2\right)\frac{dh}{dt}$$

Since

$$\dot{Q} = \frac{dV}{dt}$$

$$\left(\pi hD - \pi h^2\right)\frac{dh}{dt} = -A\sqrt{2gh}$$

$$\left(\pi hD - \pi h^2\right)\frac{dh}{dt} = -\frac{\pi d^2}{4}\sqrt{2gh}$$

$$\frac{dh}{dt} = -\frac{d^2\sqrt{2gh}}{4\left(hD - h^2\right)}$$

a) Use Runge-Kutta 2$^{\text{nd}}$ order Heun's method with a step size of 10 minutes to find the height of water at $t = 50$ minutes.

b) Based on the results from part (a), estimate the time when the height of the water is 4 m.

c) Compare the results of part(a) and part (b) with the exact values.

**Answer:** *a*) 2.9282 m
*b*) 19.248 min

*c*) part (a) $h = 2.9371$ m, 0.303%
part(b) $t = 19.41$ min, 0.835%

(6). For a spherical ball taken out of furnace at 2500 K that loses heat due to radiation and convection, the following is given.

$$\text{Radius of ball, } r = 1.0 \text{ cm},$$

$$\text{Density of the ball, } \rho = 3000 \text{ kg/m}^3,$$

$$\text{Specific heat, } C = 1000 \text{ J/(kg} \cdot \text{K)},$$

$$\text{Emittance, } \in = 0.5,$$

$$\text{Stefan-Boltzmann Constant, } \sigma = 5.67 \times 10^{-8} \text{ J/(s} \cdot \text{m}^2 \cdot \text{K}^4),$$

$$\text{Convective cooling coefficient, } h = 500 \text{ J/(s} \cdot \text{m}^2 \cdot \text{K)},$$

$$\text{Initial temperature of the ball, } \theta(0) = 2500 \text{ K},$$

$$\text{Ambient temperature, } \theta_a = 300 \text{ K},$$

the ordinary differential equation that governs the temperature of the ball is given by

$$\frac{d\theta}{dt} = -2.8349 \times 10^{-12}(\theta^4 - 81 \times 10^8) - 0.05(\theta - 300).$$

Hint: The above formula was calculated by using the knowledge that

$$\text{Rate of heat lost due to radiation } = A \in \sigma(\theta^4 - \theta_a^4)$$

$$\text{Rate of heat lost due to convection } = hA(\theta - \theta_a)$$

$$\text{Heat stored } = mc\theta$$

where

$$A = \text{surface area of the ball},$$

$$m = \text{mass of the ball}.$$

a) What are following at $t = 0$ s?

i) temperature,

ii) rate of change of temperature,

iii) rate of heat loss due to radiation,

iv) rate of heat loss due to convection,

v) rate of heat stored,

b) Find the following at $t = 10$ s using Runge Kutta $2^{\text{nd}}$ order Ralston's method and a step size of $h = 2.5$.

i) temperature,

ii) rate of change of temperature,

iii) rate of heat loss due to radiation,

iv) rate of heat loss due to convection,

v) rate of heat stored,

c) Can either convection or radiation be neglected if we are interested in finding the temperature profile between 0 and 10 s? Base your answer on quantitative reasoning.

d) Find the time when the temperature will be 2000 K using Ralston's method and a step size of $h = 2.5$ s.

**Answer:** *a*) 2500 K, $-220.72$ K/s, 1391.3 W, 1382.3 W, $-2773.6$ W
*b*) 1381.9 K, $-64.41$ K/s, 129.63 W, 679.78 W, $-809.41$ W
*c*) Neither as rate of heat lost due to convection and radiation is of same order.
*d*) 3.1591 s

# Chapter 08.05: On Solving Higher-Order and Coupled Ordinary Differential Equations

Lesson: Solving Higher-Order Ordinary Differential Equations

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *write higher-order ordinary differential equations as simultaneous first-order ordinary differential equations,*

2) *solve higher-order ordinary differential equations numerically*

## Description

In the earlier lessons, we have learned Euler's and Runge-Kutta methods to solve first-order ordinary differential equations of the form

$$\frac{dy}{dx} = f(x,y), y(x_0) = y_0 \qquad (1)$$

What do we do to solve simultaneous (coupled) differential equations or differential equations higher than first order? For example, an $n^{th}$order differential equation of the form

$$a_n \frac{d^n y}{dx^n} + a_{n-1} \frac{d^{n-1} y}{dx^{n-1}} + \ldots + a_1 \frac{dy}{dx} + a_o y = f(x) \qquad (2)$$

with $n$ initial conditions can be solved by assuming

$$y = z_1 \qquad (3.1)$$

$$\frac{dy}{dx} = \frac{dz_1}{dx} = z_2 \qquad (3.2)$$

$$\frac{d^2 y}{dx^2} = \frac{dz_2}{dx} = z_3 \qquad (3.3)$$

$$\vdots$$

$$\frac{d^{n-1}y}{dx^{n-1}} = \frac{dz_{n-1}}{dx} = z_n \qquad (3.n)$$

$$\begin{aligned}\frac{d^n y}{dx^n} &= \frac{dz_n}{dx} \\ &= \frac{1}{a_n}\left(-a_{n-1}\frac{d^{n-1}y}{dx^{n-1}} \ldots - a_1\frac{dy}{dx} - a_0 y + f(x)\right) \\ &= \frac{1}{a_n}\left(-a_{n-1}z_n \ldots - a_1 z_2 - a_0 z_1 + f(x)\right) \quad (3.n+1)\end{aligned}$$

The above Equations from (3.2) to (3.n+1) represent $n$ first-order differential equations as follows

$$\frac{dz_1}{dx} = z_2 = f_1(z_1, z_2, \ldots, x) \qquad (4.1)$$

$$\frac{dz_2}{dx} = z_3 = f_2(z_1, z_2, \ldots, x) \qquad (4.2)$$

$$\vdots$$

$$\frac{dz_n}{dx} = \frac{1}{a_n}\left(-a_{n-1}z_n \ldots - a_1 z_2 - a_0 z_1 + f(x)\right) \qquad (4.n)$$

Each of the $n$ first-order ordinary differential equations should be accompanied by one initial condition. The initial condition should be on the corresponding dependent variable on the left-hand side of the ordinary differential equation. For example, Equation (4.1) would need an initial condition on $z_1$, Equation (4.n) would need an initial condition on $z_n$, etc. These first-order ordinary differential equations (Equations (4.1) thru (4.n)) are simultaneous. Still, they can be solved by the methods used for solving first-order ordinary differential equations that we have already learned in the previous lessons.

**Example 1**

Rewrite the following differential equation as a set of simultaneous first-order differential equations.

$$3\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + 5y = e^{-x}, y(0) = 5,\ y'(0) = 7$$

**Solution**

The ordinary differential equation

$$3\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + 5y = e^{-x}, y(0) = 5,\ y'(0) = 7 \qquad (E1.1)$$

would be rewritten as follows. Assume

$$\frac{dy}{dx} = z, \qquad (E1.2)$$

Then

$$\frac{d^2y}{dx^2} = \frac{dz}{dx} \qquad (E1.3)$$

Substituting Equations (E1.2) and (E1.3) in the given second-order ordinary differential equation gives

$$3\frac{dz}{dx} + 2z + 5y = e^{-x}$$

and rewritten as

$$\frac{dz}{dx} = \frac{1}{3}\left(e^{-x} - 2z - 5y\right) \qquad (E1.4)$$

The set of two simultaneous first-order ordinary differential equations complete with the initial conditions then is

$$\frac{dy}{dx} = z, y\left(0\right) = 5 \qquad (E1.5a)$$

$$\frac{dz}{dx} = \frac{1}{3}\left(e^{-x} - 2z - 5y\right), z\left(0\right) = 7. \qquad (E1.5b)$$

Now one can apply any of the numerical methods used for solving first-order ordinary differential equations.

We write such equations in a subscripted format in a later lesson to make them suitable for matrix and vector representation. In the above example, we would introduce subscripted dependent variables, say, $w_1$ and $w_2$ as

$$w_1 = y \qquad (E1.6)$$

and

$$w_2 = \frac{dy}{dx} \qquad (E1.7)$$

Using Equations (E1.6) and (E1.7), Equations (E1.5a) and (E1.5b) can be rewritten in subscripted form as

$$\frac{dw_1}{dx} = w_2, w_1\left(0\right) = 5 \qquad (E1.8a)$$

$$\frac{dw_2}{dx} = \frac{1}{3}\left(e^{-x} - 2w_2 - 5w_1\right), w_2\left(0\right) = 7. \qquad (E1.8b)$$

Rewriting Equations (E1.8a) and E(1.8b) now as

$$\frac{dw_1}{dx} = 0w_1 + 1w_2, w_1(0) = 5 \qquad (E1.9a)$$

$$\frac{dw_2}{dx} = -\frac{5}{3}w_1 - \frac{2}{3}w_2 + \frac{1}{3}e^{-x}, w_2(0) = 7. \qquad (E1.9b)$$

makes it ready to be written in a matrix form and is called the state-space model.

In the matrix form, the Equations (E1.9a) and (E1.9b) are given as

$$\begin{bmatrix} \frac{dw_1}{dx} \\ \frac{dw_2}{dx} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\frac{5}{3} & -\frac{2}{3} \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{3}e^{-x} \end{bmatrix} \qquad (E1.10)$$

where the initial conditions are given by

$$\begin{bmatrix} w_1(0) \\ w_2(0) \end{bmatrix} = \begin{bmatrix} 5 \\ 7 \end{bmatrix} \qquad (E1.11)$$

The motivation and description of the state-space model are described in detail in the next lesson for higher-order and coupled ordinary differential equations.

**Example 2**

Given

$$\frac{d^2y}{dt^2} + 2\frac{dy}{dt} + y = e^{-t}, y(0) = 1, \frac{dy}{dt}(0) = 2,$$

estimate the following by Euler's method

a) $y(0.75)$

b) the absolute relative true error for part(a), if $y(0.75)|_{exact} = 1.668$

c) $\frac{dy}{dt}(0.75)$

Use a step size of $h = 0.25$.

**Solution**

First, the second-order differential equation is written as two simultaneous first-order differential equations as follows. Assume

$$\frac{dy}{dt} = z$$

then

$$\frac{dz}{dt} + 2z + y = e^{-t}$$

$$\frac{dz}{dt} = e^{-t} - 2z - y$$

So the two simultaneous first-order differential equations are

$$\frac{dy}{dt} = z = f_1(t, y, z), y(0) = 1 \qquad (E2.1)$$

$$\frac{dz}{dt} = e^{-t} - 2z - y = f_2(t, y, z), \ z(0) = 2 \qquad (E2.2)$$

Using Euler's method on Equations (E2.1) and (E2.2), we get

$$y_{i+1} = y_i + f_1(t_i, y_i, z_i) h \qquad (E2.3)$$

$$z_{i+1} = z_i + f_2(t_i, y_i, z_i) h \qquad (E2.4)$$

a) To find the value of $y(0.75)$ and since we are using a step size of 0.25 and starting at $t = 0$, we need to take three steps to find the value of $y(0.75)$.

For $i = 0, t_0 = 0, y_0 = 1, z_0 = 2$,

From Equation (E2.3)

$$\begin{aligned} y_1 &= y_0 + f_1(t_0, y_0, z_0) h \\ &= 1 + f_1(0, 1, 2)(0.25) \\ &= 1 + 2(0.25) \\ &= 1.5 \end{aligned}$$

$y_1$ is the approximate value of $y$ at

$$t = t_1 = t_0 + h = 0 + 0.25 = 0.25$$

$$y_1 = y(0.25) \approx 1.5$$

From Equation (E2.4)

$$\begin{aligned} z_1 &= z_0 + f_2(t_0, y_0, z_0) h \\ &= 2 + f_2(0, 1, 2)(0.25) \\ &= 2 + \left(e^{-0} - 2(2) - 1\right)(0.25) \\ &= 1 \end{aligned}$$

$z_1$ is the approximate value of $z$ (same as $\frac{dy}{dt}$) at $t = 0.25$

$$z_1 = z(0.25) \approx 1$$

For $i = 1, t_1 = 0.25, y_1 = 1.5, z_1 = 1,$

From Equation (E2.3)

$$\begin{aligned} y_2 &= y_1 + f_1(t_1, y_1, z_1)\, h \\ &= 1.5 + f_1(0.25, 1.5, 1)\,(0.25) \\ &= 1.5 + (1)\,(0.25) \\ &= 1.75 \end{aligned}$$

$y_2$ is the approximate value of $y$ at

$$t = t_2 = t_1 + h = 0.25 + 0.25 = 0.50$$

$$y_2 = y(0.5) \approx 1.75$$

From Equation (E2.4)

$$\begin{aligned} z_2 &= z_1 + f_2(t_1, y_1, z_1)\, h \\ &= 1 + f_2(0.25, 1.5, 1)\,(0.25) \\ &= 1 + \left(e^{-0.25} - 2(1) - 1.5\right)(0.25) \\ &= 1 + (-2.7211)\,(0.25) \\ &= 0.31970 \end{aligned}$$

$z_2$ is the approximate value of $z$ at

$$t = t_2 = 0.5$$

$$z_2 = z(0.5) \approx 0.31970$$

For $i = 2, t_2 = 0.5, y_2 = 1.75, z_2 = 0.31970,$

From Equation (E2.3)

$$\begin{aligned} y_3 &= y_2 + f_1(t_2, y_2, z_2)\, h \\ &= 1.75 + f_1(0.50, 1.75, 0.31970)\,(0.25) \\ &= 1.75 + (0.31970)\,(0.25) \\ &= 1.8299 \end{aligned}$$

$y_3$ is the approximate value of $y$ at

$$t = t_3 = t_2 + h = 0.5 + 0.25 = 0.75$$

$y_3 = y(0.75) \approx 1.8299$

From Equation (E2.4)

$$\begin{aligned} z_3 &= z_2 + f_2\left(t_2, y_2, z_2\right) h \\ &= 0.31972 + f_2\left(0.50, 1.75, 0.31970\right)(0.25) \\ &= 0.31972 + \left(e^{-0.50} - 2\left(0.31970\right) - 1.75\right)(0.25) \\ &= 0.31972 + \left(-1.7829\right)(0.25) \\ &= -0.1260 \end{aligned}$$

$z_3$ is the approximate value of $z$ at

$$t = t_3 = 0.75$$

$$z_3 = z\left(0.75\right) \approx -0.12601$$

$$y\left(0.75\right) \approx y_3 = 1.8299$$

b) The exact value of $y\left(0.75\right)$ is

$$y\left(0.75\right)\big|_{exact} = 1.668$$

The absolute relative true error in the result from part (a) is

$$\begin{aligned} \left|\in_t\right| &= \left|\frac{1.668 - 1.8299}{1.668}\right| \times 100 \\ &= 9.7062\% \end{aligned}$$

c)

$$\begin{aligned} \frac{dy}{dx}\left(0.75\right) &= z_3 \\ &\approx -0.12601 \end{aligned}$$

**Example 3**

Given

$$\frac{d^2y}{dt^2} + 2\frac{dy}{dt} + y = e^{-t},\ y(0) = 1,\ \frac{dy}{dt}(0) = 2,$$

estimate the following by Heun's method.

a) $y\left(0.75\right)$

b) $\dfrac{dy}{dx}(0.75)$.

Use a step size of $h = 0.25$.

**Solution**

First, the second-order differential equation is rewritten as two simultaneous first-order differential equations as follows. Assume

$$\frac{dy}{dt} = z$$

then

$$\frac{dz}{dt} + 2z + y = e^{-t}$$

$$\frac{dz}{dt} = e^{-t} - 2z - y$$

So the two simultaneous first-order differential equations with the corresponding initial conditions are

$$\frac{dy}{dt} = z = f_1(t, y, z), y(0) = 1 \qquad (E3.1)$$

$$\frac{dz}{dt} = e^{-t} - 2z - y = f_2(t, y, z), \; z(0) = 2 \qquad (E3.2)$$

Using Heun's method on Equations (E3.1) and (E3.2), we get

$$y_{i+1} = y_i + \frac{1}{2}(k_1^y + k_2^y)h \qquad (E3.3)$$

$$k_1^y = f_1(t_i, \; y_i, \; z_i) \qquad (E3.4a)$$

$$k_2^y = f_1(t_i + h, \; y_i + hk_1^y, \; z_i + hk_1^z) \qquad (E3.4b)$$

$$z_{i+1} = z_i + \frac{1}{2}(k_1^z + k_2^z)h \qquad (E3.5)$$

$$k_1^z = f_2(t_i, \; y_i, \; z_i) \qquad (E3.6a)$$

$$k_2^z = f_2(t_i + h, \; y_i + hk_1^y, \; z_i + hk_1^z) \qquad (E3.6b)$$

For $i = 0, t_o = 0, y_o = 1, z_o = 2$

From Equation (E3.4a)

$$\begin{aligned} k_1^y &= f_1\left(t_0, y_0, z_0\right) \\ &= f_1\left(0, 1, 2\right) \\ &= 2 \end{aligned}$$

From Equation (E3.6a)

$$\begin{aligned} k_1^z &= f_2\left(t_0, y_0, z_0\right) \\ &= f_2\left(0, 1, 2\right) \\ &= e^{-0} - 2\left(2\right) - 1 \\ &= -4 \end{aligned}$$

From Equation (E3.4b)

$$\begin{aligned} k_2^y &= f_1\left(t_0 + h, y_0 + hk_1^y, z_0 + hk_1^z\right) \\ &= f_1\left(0 + 0.25, 1 + \left(0.25\right)\left(2\right), 2 + \left(0.25\right)\left(-4\right)\right) \\ &= f_1\left(0.25, 1.5, 1\right) \\ &= 1 \end{aligned}$$

From Equation (E3.6b)

$$\begin{aligned} k_2^z &= f_2\left(t_0 + h, y_0 + hk_1^y, z_0 + hk_1^z\right) \\ &= f_2\left(0 + 0.25, 1 + \left(0.25\right)\left(2\right), 2 + \left(0.25\right)\left(-4\right)\right) \\ &= f_2\left(0.25, 1.5, 1\right) \\ &= e^{-0.25} - 2\left(1\right) - 1.5 \\ &= -2.7212 \end{aligned}$$

From Equation (E3.3)

$$\begin{aligned} y_1 &= y_0 + \frac{1}{2}\left(k_1^y + k_2^y\right)h \\ &= 1 + \frac{1}{2}\left(2 + 1\right)\left(0.25\right) \\ &= 1.375 \end{aligned}$$

$y_1$ is the approximate value of $y$ at

$$t = t_1 = t_0 + h = 0 + 0.25 = 0.25$$

$$y_1 = y\left(0.25\right) \cong 1.375$$

From Equation (E3.5)

$$
\begin{aligned}
z_1 &= z_0 + \frac{1}{2}\left(k_1^z + k_2^z\right) h \\
&= 2 + \frac{1}{2}(-4 + (-2.7212))(0.25) \\
&= 1.1598
\end{aligned}
$$

$z_1$ is the approximate value of $z$ at

$$
t = t_1 = 0.25
$$

$$
z_1 = z\left(0.25\right) \approx 1.1598
$$

For $i = 1, t_1 = 0.25, y_1 = 1.375, z_1 = 1.1598$

From Equation (E3.4a)

$$
\begin{aligned}
k_1^y &= f_1\left(t_1, y_1, z_1\right) \\
&= f_1\left(0.25, 1.375, 1.1598\right) \\
&= 1.1598
\end{aligned}
$$

From Equation (E3.6a)

$$
\begin{aligned}
k_1^z &= f_2\left(t_1, y_1, z_1\right) \\
&= f_2\left(0.25, 1.375, 1.1598\right) \\
&= e^{-0.25} - 2\left(1.1598\right) - 1.375 \\
&= -2.9158
\end{aligned}
$$

From Equation (E3.4b)

$$
\begin{aligned}
k_2^y &= f_1\left(t_1 + h, y_1 + hk_1^y, z_1 + hk_1^z\right) \\
&= f_1\left(0.25 + 0.25, 1.375 + \left(0.25\right)\left(1.1598\right), 1.1598 + \left(0.25\right)\left(-2.9158\right)\right) \\
&= f_1\left(0.50, 1.6649, 0.43087\right) \\
&= 0.43087
\end{aligned}
$$

From Equation (E3.6b)

$$
\begin{aligned}
k_2^z &= f_2\left(t_1 + h, y_1 + hk_1^y, z_1 + hk_1^z\right) \\
&= f_2\left(0.25 + 0.25, 1.375 + \left(0.25\right)\left(1.1598\right), 1.1598 + \left(0.25\right)\left(-2.9158\right)\right) \\
&= f_2\left(0.50, 1.6649, 0.43087\right) \\
&= e^{-0.50} - 2\left(0.43087\right) - 1.6649 \\
&= -1.9201
\end{aligned}
$$

From Equation (E3.3)

$$\begin{aligned} y_2 &= y_1 + \frac{1}{2}\left(k_1^y + k_2^y\right) h \\ &= 1.375 + \frac{1}{2}\left(1.1598 + 0.43087\right)(0.25) \\ &= 1.5738 \end{aligned}$$

$y_2$ is the approximate value of $y$ at

$$t = t_2 = t_1 + h = 0.25 + 0.25 = 0.50$$

$$y_2 = y\left(0.50\right) \approx 1.5738$$

From Equation (E3.5)

$$\begin{aligned} z_2 &= z_1 + \frac{1}{2}\left(k_1^z + k_2^z\right) h \\ &= 1.1598 + \frac{1}{2}(-2.9158 + (-1.9201))(0.25) \\ &= 0.55533 \end{aligned}$$

$z_2$ is the approximate value of $z$ at

$$t = t_2 = 0.50$$

$$z_2 = z\left(0.50\right) \approx 0.55533$$

For $i = 2, t_2 = 0.50, y_2 = 1.57384, z_2 = 0.55533$

From Equation (E3.4a)

$$\begin{aligned} k_1^y &= f_1\left(t_2, y_2, z_2\right) \\ &= f_1\left(0.50, 1.5738, 0.55533\right) \\ &= 0.55533 \end{aligned}$$

From Equation (E3.6a)

$$\begin{aligned} k_1^z &= f_2\left(t_2, y_2, z_2\right) \\ &= f_2\left(0.50, 1.5738, 0.55533\right) \\ &= e^{-0.50} - 2\left(0.55533\right) - 1.5738 \\ &= -2.0779 \end{aligned}$$

From Equation (E3.4b)

$$
\begin{aligned}
k_2^y &= f_2\left(t_2 + h, y_2 + hk_1^y, z_2 + hk_1^z\right) \\
&= f_1\left(0.50 + 0.25, 1.5738 + (0.25)\,(0.55533), 0.55533 + (0.25)\,(-2.0779)\right) \\
&= f_1\left(0.75, 1.7126, 0.035836\right) \\
&= 0.035836
\end{aligned}
$$

From Equation (E3.6b)

$$
\begin{aligned}
k_2^z &= f_2\left(t_2 + h, y_2 + hk_1^y, z_2 + hk_1^z\right) \\
&= f_2\left(0.50 + 0.25, 1.5738 + (0.25)\,(0.55533), 0.55533 + (0.25)\,(-2.0779)\right) \\
&= f_2\left(0.75, 1.7126, 0.035836\right) \\
&= e^{-0.75} - 2\,(0.035836) - 1.7126 \\
&= -1.3119
\end{aligned}
$$

From Equation (E3.3)

$$
\begin{aligned}
y_3 &= y_2 + \frac{1}{2}\left(k_1^y + k_2^y\right) h \\
&= 1.5738 + \frac{1}{2}\left(0.55533 + 0.035836\right)(0.25) \\
&= 1.6477
\end{aligned}
$$

$y_3$ is the approximate value of $y$ at

$$
t = t_3 = t_2 + h = 0.50 + 0.25 = 0.75
$$

$$
y_3 = y\,(0.75) \approx 1.6477
$$

b) From Equation (E3.5)

$$
\begin{aligned}
z_3 &= z_2 + \frac{1}{2}\left(k_1^z + k_2^z\right) h \\
&= 0.55533 + \frac{1}{2}(-2.0779 + (-1.3119))(0.25) \\
&= 0.13158
\end{aligned}
$$

$z_3$ is the approximate value of $z$ at

$$
t = t_3 = 0.75
$$

$$
z_3 = z\,(0.75) \cong 0.13158
$$

The intermediate and the final results are shown in Table 1.

**Table 1** Intermediate results of Heun's method.

| $i$ | 0 | 1 | 2 |
|---|---|---|---|
| $t_i$ | 0 | 0.25 | 0.50 |
| $y_i$ | 1 | 1.3750 | 1.5738 |
| $z_i$ | 2 | 1.1598 | 0.55533 |
| $k_1^y$ | 2 | 1.1598 | 0.55533 |
| $k_1^z$ | $-4$ | $-2.9158$ | $-2.0779$ |
| $k_2^y$ | 1 | 0.43087 | 0.035836 |
| $k_2^z$ | $-2.7211$ | $-1.9201$ | $-1.3119$ |
| $y_{i+1}$ | 1.3750 | 1.5738 | 1.6477 |
| $z_{i+1}$ | 1.1598 | 0.55533 | 0.13158 |

# Chapter 08.05: On Solving Higher-Order and Coupled Ordinary Differential Equations

Lesson: State-Space Modeling and Higher-Order and Coupled Ordinary Differential Equations

## Learning Objectives

*After successful completion of this lesson, you should be able to:*

1) *set up higher-order and coupled ordinary differential equations as simultaneous first-order ordinary differential equations*

2) *write higher-order and coupled ordinary differential equations in a state-space model form*

3) *solve higher-order and coupled ordinary differential equations numerically using software programs*

## Introduction

In the previous lesson, we showed how to write a higher-order ordinary differential equation as a set of first-order differential equations along with the corresponding initial conditions. In this lesson, we show how we rewrite the set of first-order differential equations in matrix form called as the state-space model. We illustrate this through an example. We show the concept through higher-order and coupled ordinary differential equations.

### Example 1

Write the following ordinary differentiation equation as a state model

$$17\frac{d^3y}{dt^3} + 3\frac{d^2y}{dt^2} + 7\frac{dy}{dt} + 5y = 11e^{-t},$$

$$y(0) = 13,\ \frac{dy}{dt}(0) = 19,\ \frac{d^2y}{dt^2} = 23$$

### Solution

The differential equation is of the third order, and we have three state variables. Let these be named as

$$x_1 = y \qquad (E1.1)$$

$$x_2 = \dot{y} \qquad (E1.2)$$

$$x_3 = \ddot{y} \qquad (E1.3)$$

Note the symbols. The symbol $\dot{y}$ stands for $\frac{dy}{dt}$ and $\ddot{y}$ stands for $\frac{d^2y}{dt^2}$. We are using these symbols as they are the norm in most textbooks.

We also have

$$\dot{x_3} = \dddot{y} \qquad (E1.4)$$

So the given ordinary differential equation

$$17\frac{d^3y}{dt^3} + 3\frac{d^2y}{dt^2} + 7\frac{dy}{dt} + 5y = 11e^{-t},$$

can be written as

$$17\dot{x_3} + 3x_3 + 7x_2 + 5x_1 = 11e^{-t} \qquad (E1.5)$$

Writing equations (E1.2), (E1.3), and (E1.5) with the first derivatives on the left side, we get

$$\dot{x_1} = \dot{y} = x_2$$

$$\dot{x_2} = \ddot{y} = x_3$$

$$\dot{x_3} = \frac{11e^{-t} - 3x_3 - 7x_2 - 5x_1}{17}$$

Rewriting the above three equations with coefficients for all the state variables, we get

$$\dot{x_1} = 0x_1 + 1x_2 + 0x_3 + 0$$

$$\dot{x_2} = 0x_1 + 0x_2 + 1x_3 + 0$$

$$\dot{x_3} = -\frac{5}{17}x_1 - \frac{7}{17}x_2 - \frac{3}{17}x_3 + \frac{11e^{-t}}{17}$$

In the matrix form, the state-space model is given as

$$\begin{bmatrix} \dot{x_1} \\ \dot{x_2} \\ \dot{x_3} \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -\frac{5}{17} & -\frac{7}{17} & -\frac{3}{17} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \frac{11e^{-t}}{17} \end{bmatrix} \qquad (E1.6)$$

where the conditions are given by

$$\begin{bmatrix} x_1(0) \\ x_2(0) \\ x_3(0) \end{bmatrix} = \begin{bmatrix} y(0) \\ \dot{y}(0) \\ \ddot{y}(0) \end{bmatrix} = \begin{bmatrix} y(0) \\ \dfrac{dy}{dt}(0) \\ \dfrac{d^2y}{dt^2}(0) \end{bmatrix} = \begin{bmatrix} 13 \\ 19 \\ 23 \end{bmatrix}$$

and

$$y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

In Equation (E1.6), the $3 \times 3$ matrix below

$$[A] = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -\dfrac{5}{17} & -\dfrac{7}{17} & -\dfrac{3}{17} \end{bmatrix}$$

is the state matrix.

The 3-element column vector in Equation (E1.6)

$$[X] = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

is called the state vector.

The 3-element column vector in Equation (E1.6)

$$[u] = \begin{bmatrix} 0 \\ 0 \\ \dfrac{11e^{-t}}{17} \end{bmatrix}$$

is called the input vector or forcing function vector. The left side of the equation consists of the first derivatives of the state vector.

The general form of the state-space model is

$$[\dot{X}] = [A][X] + [u]$$

**Example 2**

Reduce the following coupled ordinary differential equations to a set of first-order differential equations complete with initial conditions and in the matrix form required to solve them numerically.

$$10\frac{d^2x_1}{dt^2} - 15\left(-2x_1 + x_2\right) = 0$$

$$20\frac{d^2x_2}{dt^2} - 15\left(x_1 - 2x_2\right) = 0$$

$$\frac{dx_1}{dt}(0) = 4,\ \ \frac{dx_2}{dt}(0) = 5,\ \ x_1(0) = 2,\ \ x_2(0) = 3$$

**Solution**

The differential equations given are second-order ordinary differential equations

$$10\frac{d^2x_1}{dt^2} - 15\left(-2x_1 + x_2\right) = 0 \qquad (E2.1)$$

$$20\frac{d^2x_2}{dt^2} - 15\left(x_1 - 2x_2\right) = 0 \qquad (E2.2)$$

The coupled ordinary differential equations (E2.1) and (E2.2) would be rewritten as first-order ordinary differential equations.

Assuming

$$\frac{dx_1}{dt} = x_3, \qquad (E2.3)$$

$$\frac{dx_2}{dt} = x_4 \qquad (E2.4)$$

then

$$\frac{d^2x_1}{dt^2} = \frac{dx_3}{dt}, \qquad (E2.5)$$

$$\frac{d^2x_2}{dt^2} = \frac{dx_4}{dt} \qquad (E2.6)$$

Substituting Equations (E2.3), (E2.4), (E2.5), and (E2.6) with the two new variables $x_3$ and $x_4$ in the given second-order ordinary differential equation (E2.1) gives

$$10\frac{dx_3}{dt} - 15\left(-2x_1 + x_2\right) = 0$$

and then rewriting it gives

$$\frac{dx_3}{dt} = 1.5\left(-2x_1 + x_2\right) \qquad (E2.7)$$

Substituting equations (3), (4), (5), and (6) with the two new variables $x_3$ and $x_4$ in the given second-order ordinary differential equation (2) gives

$$20\frac{dx_4}{dt} - 15\left(x_1 - 2x_2\right) = 0$$

and then rewriting it gives

$$\frac{dx_4}{dt} = 0.75\left(x_1 - 2x_2\right) \qquad (E2.8)$$

Now, writing the equations (E2.3), (E2.4), E(2.7), and E(2.8) along with the initial conditions

$$\frac{dx_1}{dt} = x_3,\ x_1(0) = 4, \qquad (E2.9a)$$

$$\frac{dx_2}{dt} = x_4,\ x_2(0) = 5, \qquad (E2.9b)$$

$$\frac{dx_3}{dt} = 1.5(-2x_1 + x_2),\ x_3(0) = 2, \qquad (E2.9c)$$

$$\frac{dx_4}{dt} = 0.75(x_1 - 2x_2),\ x_4(0) = 3, \qquad (E2.9d)$$

Preparing for the state-space model form, Equations (E2.9a) through (E2.9d) gives

$$\frac{dx_1}{dt} = 0x_1 + 0x_2 + 1x_3 + 0x_4,\ x_1(0) = 4, \qquad (E2.10a)$$

$$\frac{dx_2}{dt} = 0x_1 + 0x_2 + 0x_3 + 1x_4,\ x_2(0) = 5, \qquad (E2.10b)$$

$$\frac{dx_3}{dt} = -3x_1 + 1.5x_2 + 0x_3 + 0x_4,\ x_3(0) = 2, \qquad (E2.10c)$$

$$\frac{dx_4}{dt} = 0.75x_1 - 1.5x_2 + 0x_3 + 0x_4,\ x_4(0) = 3, \qquad (E2.10d)$$

Equations (E2.10a)-(E2.10d) can now be written in state-space model form as

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -3 & 1.5 & 0 & 0 \\ 0.75 & -1.5 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$$

with the initial conditions as

$$\begin{bmatrix} x_1(0) \\ x_2(0) \\ x_3(0) \\ x_4(0) \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ 4 \\ 5 \end{bmatrix}$$

Can you identify the state vector, state matrix, input vector, and the initial conditions on the state vector?

**Example 3**

A suspension system of a bus can be modeled as below.

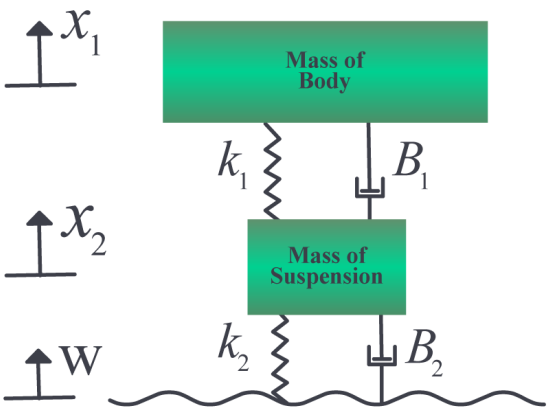



**Figure 1:** a) School bus b) A model of $1/4^{\text{th}}$ of suspension system of a bus (Credit: Model and numbers from <http://www.monografias.com/trabajos-pdf/modeling-bus-suspension-transfer-function/modeling-bus-suspension-transfer-function.pdf>)

Only $1/4^{\text{th}}$ of the bus is modeled. The differential equations that govern the above system can be derived (this is something you will do in your vibrations course) as

$$M_1 \frac{d^2x_1}{dt^2} + B_1 \left( \frac{dx_1}{dt} - \frac{dx_2}{dt} \right) + K_1 (x_1 - x_2) = 0 \qquad (E3.1)$$

$$M_2 \frac{d^2x_2}{dt^2} + B_1 \left( \frac{dx_2}{dt} - \frac{dx_1}{dt} \right) + K_1 (x_2 - x_1) + B_2 \left( \frac{dx_2}{dt} - \frac{dw}{dt} \right) + K_2 (x_2 - w) = 0 \qquad (E3.2)$$

$$x_1(0) = 0, x_3(0) = 0, x_2(0) = 0, x_4(0) = 0$$

where

$$M_1 = \text{body mass}$$

$$M_2 = \text{suspension mass}$$

$$K_1 = \text{spring constant of the suspension system}$$

$$K_2 = \text{spring constant of wheel and tire}$$

$$B_1 = \text{damping constant of the suspension system}$$

$$B_2 = \text{damping constant of wheel and tire}$$

$$x_1 = \text{displacement of the body mass as a function of time}$$

$$x_2 = \text{displacement of the suspension mass as a function of time}$$

$$w = \text{input profile of the road as a function of time}$$

The constants are given as

$$m_1 = 2500 \text{ kg}$$

$$m_2 = 320 \text{ kg}$$

$$k_1 = 80000 \text{ N/m},$$

$$k_2 = 500000 \text{ N/m},$$

$$B_1 = 350 \text{ N-s/m},$$

$$B_2 = 15020 \text{ N-s/m}$$

Reduce the simultaneous differential equations (E3.1) and (E3.2) to simultaneous first-order differential equations and put those in the state variable form complete with corresponding initial conditions.

**Solution**

Substituting the values of the constants in the two differential equations (1) and (2) gives the differential equations (3) and (4), respectively.

$$2500\frac{d^2x_1}{dt^2} + 350\left(\frac{dx_1}{dt} - \frac{dx_2}{dt}\right) + 80000\,(x_1 - x_2) = 0 \qquad (E3.3)$$

$$320\frac{d^2x_2}{dt^2} + 350\left(\frac{dx_2}{dt} - \frac{dx_1}{dt}\right) + 80000\,(x_2 - x_1) + 15020\left(\frac{dx_2}{dt} - \frac{dw}{dt}\right) + 500000(x_2 - w) = 0 \qquad (E3.4)$$

Since $w$ is an input, we take it to the right-hand side to show it as a forcing function and rewrite Equations (E3.4) as

$$320\frac{d^2x_2}{dt^2} + 350\left(\frac{dx_2}{dt} - \frac{dx_1}{dt}\right) + 80000\,(x_2 - x_1) + 15020\left(\frac{dx_2}{dt}\right) + 500000x_2 = 15020\frac{dw}{dt} + 500000w \qquad (E3.5)$$

Now let us start the process of reducing the 2 simultaneous differential equations (Equations (E3.3) and (E3.5)) to 4 simultaneous first-order ordinary differential equations.

Choose

$$\frac{dx_1}{dt} = x_3, \qquad (E3.6)$$

$$\frac{dx_2}{dt} = x_4, \qquad (E3.7)$$

then Equation (E3.3)

$$2500\frac{d^2x_1}{dt^2} + 350\left(\frac{dx_1}{dt} - \frac{dx_2}{dt}\right) + 80000\,(x_1 - x_2) = 0$$

can be written as

$$2500\frac{dx_3}{dt} + 350\,(x_3 - x_4) + 80000\,(x_1 - x_2) = 0$$

$$2500\frac{dx_3}{dt} = -350(x_3 - x_4) - 80000(x_1 - x_2)$$

$$\frac{dx_3}{dt} = -0.14\,(x_3 - x_4) - 32\,(x_1 - x_2) = -32x_1 - 0.14x_3 + 32x_2 + 0.14x_4 \qquad (E3.8)$$

and Equation (E3.5)

$$320\frac{d^2x_2}{dt^2} + 350\left(\frac{dx_2}{dt} - \frac{dx_1}{dt}\right) + 80000\left(x_2 - x_1\right) + 15020\left(\frac{dx_2}{dt}\right) + 500000x_2$$
$$= 15020\frac{dw}{dt} + 500000w$$

can be written as

$$320\frac{dx_4}{dt} + 350\left(x_4 - x_3\right) + 80000\left(x_2 - x_1\right) + 15020x_4 + 500000x_2$$
$$= 15020\frac{dw}{dt} + 500000w$$

$$320\frac{dx_4}{dt} = -350(x_4 - x_3) - 80000(x_2 - x_1) - 15020x_4 -$$
$$500000x_2 + 15020\frac{dw}{dt} + 500000w$$

$$\frac{dx_4}{dt} = -1.09375\left(x_4 - x_3\right) - 250\left(x_2 - x_1\right) - 46.9375x_4 - 1562.5x_2$$
$$+ 46.9375\frac{dw}{dt} + 1562.5w$$
$$= 250x_1 + 1.09375x_3 - 1812.5x_2 - 48.03125x_4$$
$$+ 1562.5w + 46.9375\frac{dw}{dt} \qquad (E3.9)$$

The 4 simultaneous first-order differential equations given by Equations (E3.6) thru (E3.9), complete with the corresponding initial condition, are

$$\frac{dx_1}{dt} = x_3$$
$$= f_1\left(t, x_1, x_2, x_3, x_4\right) \text{ with } x_1\left(0\right) = 0 \qquad (E3.10)$$

$$\frac{dx_2}{dt} = x_4$$
$$= f_2\left(t, x_1, x_2, x_3, x_4\right) \text{with } x_2\left(0\right) = 0 \qquad (E3.11)$$

$$\frac{dx_3}{dt} = -32x_1 + 32x_2 - 0.14x_3 + 0.14x_4$$
$$= f_3\left(t, x_1, x_2, x_3, x_4\right) \text{with } x_3\left(0\right) = 0 \qquad (E3.12)$$

$$\frac{dx_4}{dt} = 250x_1 - 1812.5x_2 + 1.09375x_3 - 48.03125x_4 + 1562.5w + 46.9375\frac{dw}{dt}$$
$$= f_4\left(t, x_1, x_2, x_3, x_4\right) \text{with } x_4\left(0\right) = 0 \qquad (E3.13)$$

The profile of the road is given below.

Assuming that the bus is going at 60 mph, that is, approximately 27 m/s, it takes

$$\frac{6m}{27m/s} = 0.22 \text{ s}$$

to go through one period. So the frequency

$$\begin{aligned} f &= \frac{1}{0.22} \\ &= 4.545 \ \ \text{Hz} \end{aligned}$$

The angular frequency then is

$$\begin{aligned} \omega &= 2 \times \pi \times 4.545 \\ &= 28.6 \text{ rad/s} \end{aligned}$$

giving

$$\begin{aligned} w &= 0.01 \sin(\omega t) \\ &= 0.01 \sin(28.6t) \end{aligned}$$

and

$$\frac{dw}{dt} = 0.286 \cos(28.6t)$$

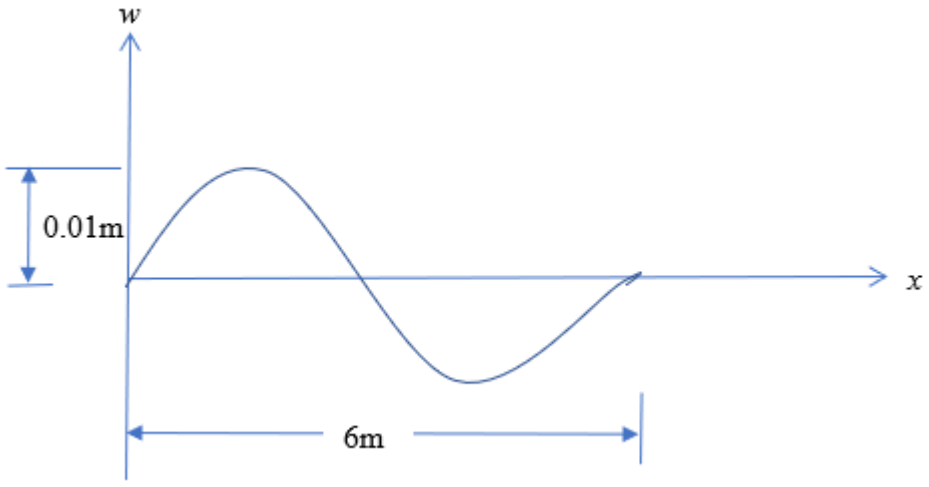



**Figure 2:** Periodic road profile

To put the differential equations given by Equations (E3.10)-(E3.13) in matrix form, we rewrite them as

$$\frac{dx_1}{dt} = x_3 = 0x_1 + 0x_2 + 1x_3 + 0x_4,\ x_1(0) = 0 \qquad (E3.14)$$

$$\frac{dx_2}{dt} = x_4 = 0x_1 + 0x_2 + 0x_3 + 1x_4,\ x_2(0) = 0 \qquad (E3.15)$$

$$\frac{dx_3}{dt} = -32x_1 + 32x_2 - 0.14x_3 + 0.14x_4,\ x_3(0) = 0 \qquad (E3.16)$$

$$\begin{aligned}\frac{dx_4}{dt} =& 250x_1 - 1812.5x_2 + 1.09375x_3 - 48.03125x_4 + \\ & 1562.5w + 46.9375\frac{dw}{dt},\ x_4(0) = 0 \qquad (E3.17)\end{aligned}$$

In state variable matrix form, the differential equations are given by

$$\begin{bmatrix} \frac{dx_1}{dt} \\ \frac{dx_2}{dt} \\ \frac{dx_3}{dt} \\ \frac{dx_4}{dt} \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -32 & 32 & -0.14 & 0.14 \\ 250 & -1812.5 & 1.09375 & -48.03125 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1562.5w + 46.9375\frac{dw}{dt} \end{bmatrix} \qquad (E3.18)$$

where

$$w = 0.01\sin(28.6t) \qquad (E3.19)$$

and the corresponding initial conditions are

$$\begin{bmatrix} x_1(0) \\ x_2(0) \\ x_3(0) \\ x_4(0) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Can you identify the state vector, the state matrix, the input vector, and the initial conditions on the state vector?

State-space models is how numerical solvers expect you to enter the data. For example, the above state-space model would be entered as the following on MATLAB.

First, a function is defined where t is the independent variable, and x is the state-space vector. The names diffx and sysofeqn are the choice of the user. The arguments of the function are the independent variable t and the state vector x. Inside the function, [A] is the state-space matrix from Equation (E3.18), $w$ is the forcing function from Equation (E3.19), [B] is the input vector.

```
function diffx=sysofeqn(t,x)

A=[0  0  1  0; ...

0  0  0  1; ...

-32  32  -0.14  0.14; ...

250   -1812.5  1.09375  -48.03125];

w=0.01*sin(28.6*t);

dw=0.286*cos(28.6*t);

B=[0;  0;  0;   1652.5*w+46.9375*dw];

diffx=A*x+B;

end
```

How is the function used? Assume that you want the output between t=0 and 10. You can use that time span shown as tspan variable. Then the initial conditions of the state vector are entered. The *ode45* routine is a combination of the Runge-Kutta 4th and 5th order methods to solve a system of first-order ordinary differential equations. The inputs of the *ode45* routine are the system of equations, the span in which the output is requested, and the initial conditions.

```
tspan=[0  10];

initial_cond=[0;0;0;0];

[t,x]=ode45('sysofeqn',tspan,initial_cond);

figure (1)

plot(t,x(:,1))
```

A sample of the outputs is shown below with the values of the independent variable t, the state vector x, and the output of x(1) vs t.

t =

| 0 |
|---|
| 0.000003742346606 |
| 0.000007484693212 |
| 0.000011227039818 |
| 0.000014969386423 |
| 0.000033681119453 |
| 0.000052392852482 |

x =

| 0 | 0 | 0 | 0 |
|---|---|---|---|
| 0.000000000000000 | 0.000000000094002 | 0.000000000013164 | 0.000050236522812 |
| 0.000000000000000 | 0.000000000376002 | 0.000000000052670 | 0.000100470633045 |
| 0.000000000000000 | 0.000000000845991 | 0.000000000118540 | 0.000150702329280 |
| 0.000000000000001 | 0.000000001503960 | 0.000000000210794 | 0.000200931610101 |
| 0.000000000000012 | 0.000000007613185 | 0.000000001068580 | 0.000452041733437 |
| 0.000000000000045 | 0.000000018420550 | 0.000000002589166 | 0.000703091259071 |

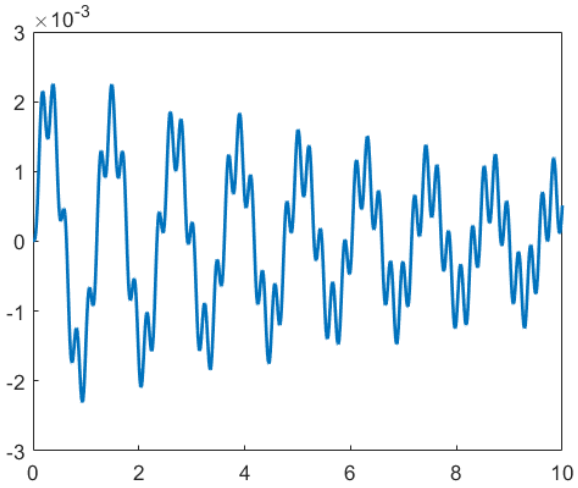



**Figure 3:** Output values and $x_1$ vs. $t$ for the bus-suspension problem.

# Problem Set

Chapter 08.05: Solving Higher-Order and Coupled Ordinary Differential Equations

(1). Reduce the following $2^{\text{nd}}$ order ordinary differential equation to a set of first-order ordinary differential equations complete with initial conditions and in the form required to solve them numerically.

$$5\frac{d^2y}{dt^2} + 3\frac{dy}{dt} + 7y = e^{-t} + t^2,\ y(0) = 6,\ \frac{dy}{dt}(0) = 11.$$

(2). Reduce the following coupled ordinary differential equations to a set of first-order ordinary differential equations complete with initial conditions and in the form required for solving them numerically.

$$10\frac{d^2x_1}{dt^2} - 15(-2x_1 + x_2) = 0,$$

$$20\frac{d^2x_2}{dt^2} - 15(x_1 - 2x_2) = 0,$$

$$x_1(0) = 2,\ x_2(0) = 3,\ \frac{dx_1}{dt}(0) = 4,\ \frac{dx_2}{dt}(0) = 5.$$

(3). The acceleration of a spring-mass system is given by $(t > 0)$

$$\frac{d^2x}{dt^2} + x = e^{-t}$$

where $x$ is the displacement of the mass given in meters and $t$ is the time given in seconds. The initial conditions are given as $x(0) = 3,\ \frac{dx}{dt}(0) = 2$.

a) What is the value of displacement, velocity, and acceleration at $t = 0+$ seconds.

b) Use Euler's method to find the value of displacement, velocity, and acceleration at $t = 0.5$ seconds. Use a step size of 0.25 seconds.

c) What is the exact value of displacement, velocity and acceleration at $t = 0.5$ seconds.

**Answer:** *a*) $3,\ 2,\ -2$
*b*) $3.8750,\ 0.81970,\ -3.2685$
*c*) $3.6958,\ 0.69213,\ -3.0893$

(4). The acceleration of a spring-mass system is given by

$$\frac{d^2x}{dt^2} + x = e^{-t}$$

where $x$ is the displacement of the mass given in meters and $t$ is the time given in seconds. The initial conditions are given as $x(0) = 3$, $\frac{dx}{dt}(0) = 2$. Use Runge Kutta 2$^{\text{nd}}$ order Ralston's method to find the value of displacement, velocity, and acceleration at $t = 0.5$ seconds. Use a step size of 0.25 seconds.

**Answer:** 3.7067 $m$, 0.67811 $m/s$, $-3.1001$ $m/s^2$

(5). Reduce the following ordinary differential equation to the state-space model form. Include the initial conditions in proper form as well.

$$4\frac{d^2y}{dt^2} + 9\frac{dy}{dt} + 12y = 3sin(7x),\ y(0) = 13,\ \frac{dy}{dt}(0) = 19.$$

(6). Reduce the following coupled ordinary differential equations to the state space model form. Include the initial conditions in proper form as well.

$$25\frac{d^2x_1}{dt^2} - 40(-3x_1 + x_2) = 3cos(3t),$$

$$32\frac{d^2x_2}{dt^2} - 12x_1 - 24x_2 = 7e^{-3t},$$

$$x_1(0) = 23,\ x_2(0) = 17,\ \frac{dx_1}{dt}(0) = 37,\ \frac{dx_2}{dt}(0) = 29.$$